

Probability Theory and Random Processes

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McGraw Hill Education (India) Private Limited

Published by McGraw Hill Education (India) Private Limited
P-24, Green Park Extension, New Delhi 110 016

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This edition can be exported from India only by the publishers,
McGraw Hill Education (India) Private Limited.

Print Edition

ISBN (13): 978-93-3290-289-3

ISBN (10): 93-3290-289-5

Ebook Edition

ISBN (13): 978-93-3290-290-9

ISBN (10): 93-3290-290-9

Managing Director: *Kaushik Bellani*

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Typeset at Text-o-Graphics, B-1/56, Aravali Apartment, Sector-34, Noida 201 301, and printed at

Cover Printer:

Dedicated to my Uncle
Late M Krishna Moorthy

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PREFACE

Many signals that we encounter in communication engineering and the data we obtain in computer science engineering are random in nature. Even if the signals are deterministic, they may be corrupted by unwanted signals known as noise. To model and analyze such random signals and their effect on a system's performance, engineers must have knowledge about probability theory and random processes. The book is designed for students of undergraduate level, and aims to introduce the basic concepts of probability theory that are required to understand probability models used in communication and computer science engineering. There are several books on probability and random processes which cover the topics in random processes in depth with less emphasis on problem solving. In this textbook, more emphasis has been given to improve the problem-solving skills of students by providing a large number of solved problems.

Salient Features

- Important topics covered in detail—Bernoulli and Poisson Distributions; Chebyshev and Markov Inequalities; Central Limit Theorem; Linear Regression; Stationary and Ergodic Random Processes; Linear System Theory and Sources of Noise
- Mathematical models explained following step-by-step approach
- Application-based problems discussed aplenty
- Lucid discussions
- Rich pedagogy:
 - ✧ Diagrams: **216**
 - ✧ Solved Examples: **809**
 - ✧ Practice Problems: **247**
 - ✧ Exercise Problems: **255**
 - ✧ Review Questions: **295**
 - ✧ Multiple-Choice Questions: **211**

Chapter Organization

A more detailed description of the chapters is as follows.

Chapter 1 deals with basic concepts in probability, set theory, basic combinatorial analysis, conditional probability, independent events and Bernoulli trials.

Chapter 2 discusses different types of random variables, cumulative distribution functions, probability mass functions of discrete random variables, and probability density functions of continuous random variables. It also introduces different types of discrete and continuous distributions and their applications. These include the Bernoulli distribution, binomial distribution, Poisson distribution, geometric distribution,

hypergeometric distribution, exponential distribution, uniform distribution, normal distribution, Weibull distribution, gamma distribution and beta distributions.

Chapter 3 deals with the concepts of expectation and variance, higher moments, conditional expectation, and the Chebyshev and Markov inequalities. It also gives an insight to topics like characteristic function and moment-generating functions.

Chapter 4 deals with multiple random variables including the joint cumulative distribution function and joint pdf, and conditional distributions. It explains the method of obtaining the pdf of sum of two or more random variables. It also introduces Central Limit Theorem (CLT) which is an important theorem in probability theory.

Chapter 5 deals with operations on multiple random variables. It introduces linear regression and also discusses correlation coefficient, covariance, joint characteristic function and moment-generating functions.

Chapter 6 deals with random processes. It briefly discusses classification of random processes; time-domain characterization of random processes that includes the autocorrelation function, autocovariance function, cross-correlation function and cross-covariance function. It then presents an insight on stationary random processes and ergodic random processes. Random processes like Gaussian processes and Poisson processes are also discussed.

Chapter 7 deals with characterization of random process in the frequency domain. It discusses power spectral density and its relation with autocorrelation function.

Chapter 8 introduces linear system theory and discusses the response of linear systems for random inputs. It also deals with different sources of noise.

Online Learning Centre

The book is accompanied by an online learning center, available at <http://www.mhhe.com/rameshbabu/ptrp> that offers an author video.

Acknowledgements

Firstly, I would like to thank Prof. D Govindarajulu, Principal—Pondicherry Engineering College for his encouragement in writing this book. I am indebted to Dr P Dananjayan, Professor of Electronics and Communication Engineering for his valuable suggestions and guidance. I am grateful to Dr R Anandanatarajan, Professor of Electronics and Instrumentation Engineering, for his constant encouragement and support.

Next, I would like to thank all those reviewers who took out time to review the manuscript and gave important suggestions. Their names are given below.

Babita Tyagi	<i>Galgotias University, Greater Noida, Uttar Pradesh</i>
Deepak Bhatia	<i>University College of Engineering, Rajasthan Technical University (RTU), Kota, Rajasthan</i>
Uma Rathore Bhatt	<i>Institute of Engineering and Technology, Devi Ahilya University, Indore, Madhya Pradesh</i>
Manoranjan Sahoo	<i>Kalinga Institute of Industrial Technology (KIIT) University Bhubaneswar, Odisha</i>

Gajendra Vishwakarma	<i>Indian School of Mines (ISM), Dhanbad, Jharkhand</i>
Nabin Sen	<i>Global Institute of Management and Technology, Kolkata, West Bengal</i>
Ponnusamy Mallappan Sitharselvam	<i>PSNA College of Engineering and Technology, Dindigul, Tamil Nadu</i>
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S Rajender	<i>Methodist College of Engineering and Technology, Osmania University, Hyderabad.</i>

I thank the McGraw Hill Education (India) team, who made this book a reality—Shalini Jha, Koyel Ghosh, Sohini Mukherjee, Satinder Singh, and P L Pandita—for their constant support, and excellent interaction.

Finally, I thank my parents; my wife; children, Surya and Deeksha; for their patience and support during the long effort of writing this book. I have put maximum effort to minimize typographical errors. However, I might have failed to notice some unintentional ones. I request the student and teaching community to come forward and point out the errors in this book and give suggestions for improving the future editions.

P RAMESH BABU

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BASIC CONCEPTS IN PROBABILITY



INTRODUCTION 1.1

The word *probability* literally means chance, a very commonly used word in day-to-day conversation. On observing a cloudy sky we say that there is a chance of rain. At the end of the day, it may rain or may not. In a cricket match between two teams, we often say that the chance of winning for one team is more than that of the other team. The other terms that are very much close to chance are *probably*, *likely*, etc. These terms are used in the context when there is uncertainty in the outcome of an event. Probability is a mathematical measure for uncertainty, likelihood or chance. A mean of evaluating the uncertainty, likelihood and chance of outcome resulting from a statistical experiment is called *theory of probability*.

The origin of theory of probability can be traced back to the seventeenth century. For the first time in 1654, two famous mathematicians, Blair Pascal and Pierre de Format formulated the principle of probability theory. Later, the Dutch scientist Christan Huygens first published book on probability which lead to rapid development of the subject in the 18th century. The scientists Fierre de Laplace, Chebyshev, Markov, Von Mises, and Kolmogov stimulated the development of probability theory by the variety of its applications in different fields like genetics, economics, etc.



PROBABILISTIC MODEL 1.2

To study some observational phenomena, a mathematical model is required. Mathematical models can be divided into two types: (i) deterministic model, and (ii) probabilistic model. Suppose if we want to find current through a simple circuit that consists of a battery in series with a resistor, we use Ohm's law $V = IR$, a mathematical model for a resistor. In this case, the observable flow of current would be $I = V/R$. This model predicts the value of I for the given V and R . If the above experiment were repeated a number of times, we observe the same value of I if V and R are fixed. In nature, many experiments can be modeled with deterministic approach. However, there are also many phenomena which require a different mathematic model known as probabilistic model (or stochastic model). For example, the number of particles emitted by a radioactive material cannot be predicted precisely during a specified time interval even if we know the exact shape, dimensions and chemical composition of the material. That is, there is no mathematical relation that gives a direct relationship between the number of particles emitted and characteristics of radioactive material. Hence, there is no certainty in predicting the emitted number of particles. Hence, we call the event as random. In this case, we must consider a probabilistic model.



RANDOMNESS, RANDOM SIGNALS AND SYSTEMS 1.3

In the above section, we studied that the number of particles emitted by a radioactive material cannot be predicted with certainty. Such events in which the output cannot be predicted with certainty are known as *random events*. Randomness simply means not predictable with certainty. In day-to-day life, we come across many events that are random.

Here are some examples:

- The total number of marks you score in the University examination is a random variable until the results are declared.
- The age of a friend whose birth date you do not know is random,
- The output of a solar detector which depends on weather conditions is also a random variable,

1.3.1 Random Signal

A random signal is a signal characterized by uncertainty before its actual occurrence.

1.3.2 System

A system is one which operates on an input signal and produce another signal as output.



PROBABILITY AND RANDOM PROCESSES 1.4

Many practical problems involve random effects and are too complex to be described accurately using deterministic models. These random processes can be modeled more effectively by using probabilistic models. To obtain probabilistic model of a random process, the basics of probability theory are required. Probability is essential for the study of a random process because the theory of random process is an extension of probability. Thus, probability has to be studied prior to the topics of random process.

The probability and random process theory is applied in applications like (i) communication (ii) signal detection, signal and data processing, estimations and filtering, (iii) quality control, etc.



COMMON TERMS USED IN PROBABILITY THEORY 1.5

Before studying definition of probability, let us know about the terms extensively used in probability theory.

1.5.1 Experiment

An experiment is an act which can be repeated under identical or uniform conditions.

1.5.2 Random Experiment

A random experiment is an experiment that is repeated under same conditions, any number of times, for which the outcome is any one of several possible outcomes that cannot be predicted in advance.

1.5.3 Trial

Any particular performance of a random experiment is called a trial.

1.5.4 Sample Space

The set of all possible outcomes of a random experiment is called sample space which is denoted by S . In tossing a coin experiment, the sample space is

$$S = \{H, T\}$$

The possible outcomes of a sample space are known as sample points. A sample space can be classified as (i) discrete sample space, and (ii) uncountable sample space.



DEFINITION OF PROBABILITY 1.6

Probability can be defined in several ways. In this section, we study three types of definitions.

- Axiomatic definition
- Relative-frequency definition
- Classical definition

1.6.1 Axiomatic Definition

In this approach, the fundamentals of set theory is used to define probability. Consider a random experiment with sample space S . For each event, we assign a non-negative number called probability of the event A denoted as $P(A)$.

The probability $P(A)$ satisfies three axioms:

Axiom 1: $0 \leq P(A) \leq 1$, which means that the probability of A is a non-negative number between 0 and 1 including 0 and 1.

Axiom 2: $P(S) = 1$. The probability of sample space is one.

Axiom 3: For any sequence of mutually exclusive events,

$$P(A_1 \cup A_2 \cup \dots \cup A_n) = P(A_1) + P(A_2) + \dots + P(A_n) \quad (1.1)$$

That is for any sequence of mutually exclusive events, the probability of at least one of these events occurring is the sum of their respective probabilities.

Discrete sample space: A sample space is said to discrete if it contains a countable number of sample points.

Uncountable sample space: A sample space is said to be uncountable if it contains an uncountable number of sample points.

Event In the experiment of a tossing dice, the sample space $S = \{1, 2, 3, 4, 5, 6\}$ and the outcome may be any number from 1 to 6. Thus, there are six possible outcomes for each experiment. The possible outcome of an experiment is called an *event*. The event “the outcome of the toss of a dice is a prime number” is defined by

$$E = \{2, 3, 5\}$$

Note that E is a subset of S . That is the event is always a subset of the sample space S . If the occurrence of an event is inevitable then that event is known as *sure event*. An event that never occurs when an experiment is performed is called an *impossible event*.

Mutually Exclusive Events Two events A and B are said to be mutually exclusive if the occurrence of any one prevents the outcome of another and vice versa. That is, both the events cannot occur simultaneously in a single trial. In this case, $A \cap B = \phi$.

Properties of Probability

- (i) The probability of the complement of A is

$$P(\bar{A}) = 1 - P(A) \quad (1.2)$$

- (ii) The probability of sure event A is almost 1.

$$P(A) \leq 1 \quad (1.3)$$

- (iii) The probability of an impossible event is zero.

$$P(\phi) = 0 \quad (1.4)$$

- (iv) If A is a subset of B then $P(A) \leq P(B)$
 (v) If $A_1, A_2, \dots, A_n$ are mutually exclusive events then

$$P(A_1 \cup A_2 \cup A_3 \cdots \cup A_N) = P(A_1) + P(A_2) + P(A_3) + \cdots + P(A_N) \quad (1.5)$$

- (vi) For any two events A and B ,

$$P(A) = P(A \cap B) + P(A \cap \bar{B}) \quad (1.6)$$

Joint Probability The joint probability of two events A and B is

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) \quad (1.7)$$

If A and B are mutually exclusive events,

$$P(A \cap B) = P(A) + P(B) \quad (1.8)$$

1.6.2 Relative Frequency Definition

Consider a random experiment that is performed n times. If an event occurs $n(A)$ times then the probability of event A ; $P(A)$ is defined as

$$P(A) = \lim_{n \rightarrow \infty} \frac{n(A)}{n} \quad (1.9)$$

The ratio $n(A)/n$ is the relative frequency for this event.

1.6.3 Classical Definition

The probability $P(A)$ of an event A is the ratio of the number of outcomes $n(A)$ of an experiment that are favourable to A to the total number of possible outcomes of the experiment. Note that the total number of possible outcomes is equal to the sample space. Therefore, the probability

$$P(A) = \frac{n(A)}{n(S)} \quad (1.10)$$

Solved Problems



1.1 In each of the following cases, find the simple space:

- (a) A coin tossed three times.
 (b) A coin is tossed. If the result is heads, a dice is thrown. If the dice shows up an even number, the dice is thrown again.

Solution

- (a) A coin is tossed three times. Therefore, the sample space is
 $\{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$
 (b) If the coin shows a tail, the experiment is not conducted. So the sample space is
 $\{T, H1, H21, H22, H23, H24, H25, H26, H3, H41, H42, H43, H45, H46, H5, H61, H62, H63, H64, H65, H66\}$

1.2 From a group of 3 boys and 2 girls, two children are selected at random. Describe the events.

- (a) A: Both selected children are boys.
- (b) B: One boy and one girl.
- (c) C: At least one girl is selected.

Which pairs of events are mutually exclusive?

Solution Let us denote the boys as B_1, B_2 and B_3 and girls as G_1 and G_2 . The sample space S is

$$S = \{B_1 B_2, B_1 G_1, B_1 G_2, B_2 G_1, B_2 G_2, B_3 G_1, B_3 G_2, B_1 B_3, B_2 B_3, G_1 G_2\}$$

- (a) $A = \{B_1 B_2, B_1 B_3, B_2 B_3\}$
- (b) $B = \{B_1 G_1, B_1 G_2, B_2 G_1, B_2 G_2, B_3 G_1, B_3 G_2\}$
- (c) $C = \{G_1 G_2, B_1 G_1, B_1 G_2, B_2 G_1, B_2 G_2, B_3 G_1, B_3 G_2\}$

From the events, we can say that the events A and B , A and C are mutually exclusive.

1.3 A box contains 1 black and 3 red balls. Two balls are drawn at random in succession without replacement. What is the sample space?

Solution The box contain 1 black (B) and 3 red (R) balls.

The sample space S is

$$S = \{BR, RB, RR\}$$



Practice Problems

1.1 A coin is tossed twice. If the second throw is heads, a dice is thrown. Write the sample space.

Ans. $\{TT, HT, TH1, TH2, TH3, TH4, TH5, TH6, HH1, HH2, HH3, HH4, HH5, HH6\}$

1.2 Two dice are thrown. Describe the following events:

- A = The sum is multiple 4.
- B = The sum is greater than 10.
- C = The sum is a prime number.

Ans. $A = \{(3, 1), (2, 2), (6, 2), (1, 3), (5, 3), (4, 4), (3, 5), (2, 6), (6, 6)\}$

$B = \{(6, 5), (5, 6), (6, 6)\}$

$C = \{(1, 1), (2, 1), (4, 1), (6, 1), (1, 2), (3, 2), (5, 2), (2, 3), (4, 3), (1, 4), (3, 4), (2, 5), (6, 5), (1, 6), (5, 6)\}$

Solved Problems



1.4 A card is drawn from a pack of 50 cards numbered 1 to 50. Find the probability of drawing a number which is a square.

Solution Given $n(S) = 50$

Given the event is a number which is a square.

$$A = \{1, 4, 9, 16, 25, 36, 49\}$$

$$n(A) = 7$$

$$\text{Required probability } P(A) = \frac{n(A)}{n(S)} = \frac{7}{50}$$

1.5 A card is drawn from a well-shuffled deck of 52 cards. Find the probability of getting (a) a jack, (b) a red card, (c) a diamond, and (d) a six of hearts.

Solution

- (a) The total number of cards
- $n(S) = 52$

Number of Jacks $n(J) = 4$

$$\text{Required probability} = \frac{n(J)}{n(S)} = \frac{4}{52} = \frac{1}{13}$$

- (b) Number of red cards
- $n(R) = 26$

$$P(R) = \frac{n(R)}{n(S)} = \frac{26}{52} = \frac{1}{2}$$

- (c) Number of diamond cards
- $n(D) = 13$

$$P(D) = \frac{n(D)}{n(S)} = \frac{13}{52} = \frac{1}{4}$$

- (d) Number of cards having six of heart = 1

$$\text{Probability} = \frac{1}{52}$$

1.6 In a single throw of three dice, find the probability of getting a sum of 7.**Solution** In throwing three dice, the total number of possible outcomes is $n(S) = 6^3$ Let A be the event of getting a sum 7. Then

$$A = \{(1, 1, 5), (1, 2, 4), (1, 3, 3), (1, 4, 2), (1, 5, 1), (2, 1, 4), (2, 2, 3), (2, 3, 2), (2, 4, 1), (3, 1, 3), (3, 2, 2), (3, 3, 1), (4, 1, 2), (4, 2, 1), (5, 1, 1)\}$$

$$n(A) = 15$$

$$P(A) = \frac{n(A)}{n(S)} = \frac{15}{6^3}$$

1.7 A coin is tossed four times. Find the chance that heads and tails show alternately**Solution**Sample space $S = \{HHHH, HHHT, HHHT, HHTT, HTHH, HTHT, HTTH, HTTT, THHH, THHT, THTH, THTT, TTHH, TTTH, TTTT\}$

$$n(S) = 16$$

Number of favourable cases = 2

$$\text{Probability} = \frac{2}{16} = \frac{1}{8}$$

**Practice Problem****1.3** Three coins are tossed once. Find the probability of getting

- (a) exactly 2 tails, (b) 3 tails, (c) at least 2 tails, (d) exactly one head, and (e) no heads.

$$(a) \frac{3}{8} \quad (b) \frac{1}{8} \quad (c) \frac{1}{2} \quad (d) \frac{3}{8} \quad (e) \frac{1}{8}$$



BASICS OF SET THEORY 1.7

In order to study the fundamental concepts of probability, we have to study about the basics of set theory.

Definitions

Set A set is a collection of objects. Sets are usually designated by capital letters and specified by the contents of two braces: $\{ \}$. Examples are $A = \{1, 2, 3\}$ and $B = \{2, 3, 5, 6\}$.

The first example describes the set consisting of positive integers 1, 2 and 3, and the second set B consists of positive integers 2, 3, 5 and 6.

Another way of representing a set is by a statement or rule. For example, we might say that A consists of all real numbers between 0 and 1 inclusive. This simply can be written as

$$A = \{x/0 \leq x \leq 1\}$$

Event The individual objects in a set are known as elements denoted by a lowercase letter. When “ a ” is a member of A , we can write $a \in A$ denoting “ a belongs to A ”.

Empty Set A set with no elements is known as an empty set denoted by ϕ .

Universal Set A set with all the elements for the problem under consideration is known as universal set.

Subset If every element of the set A is also an element of the set B then A is said to be a subset of B , denoted as $A \subset B$

Set A is said to be equal to Set B if and only if $A \subset B$ and $B \subset A$

Properties of Empty Sets and Universal Sets

- For every set A , we have $\phi \subset A$
- If U is a universal set then for every set A considered in the context of U , we have $A \subset U$.
- A set is said to be *countable* if its elements can be put in one-to-one correspondence with the natural numbers. If a set is not countable, it is called *uncountable*.
- Two sets are said to be *disjoint* or *mutually exclusive* if they do not have any element in common.
- A set is *finite* if it has finitely many elements, e.g. $A = \{1, 2\}$.
- A set is *infinite* if it has infinitely many elements. A set of all natural numbers is an infinite set.
- Two sets A and B are said to be *equivalent* if there is a one-to-one correspondence between A and B .



BASIC SET OPERATIONS 1.8

1.8.1 Union of Sets

Let A and B be two sets. The union of A and B is the set of all elements that belong to either A or B , or both. It is denoted by $A \cup B$.

$$A \cup B = \{x \mid x \in A \text{ or } x \in B \text{ or both}\}$$

A graphic device known as *Venn diagram* can be used to illustrate the relationships among sets. In this method, the universal set U is denoted by a rectangle. The sets are represented by regions bounded by closed curves. The Venn diagram of $A \cup B$ is shown in Fig. 1.1.

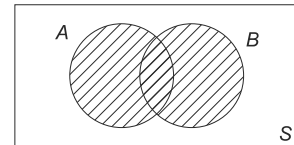


Fig. 1.1 Venn diagram showing $A \cup B$

The operation $\cup$ is commutative and associative.

$$A \cup B = B \cup A \quad (1.11)$$

$$\text{and } A \cup (B \cap C) = (A \cup B) \cap C \quad (1.12)$$

The union of N sets $A_i, i = 1, 2, \dots, N$ is

$$B = A_1 \cup A_2 \dots \cup A_N = \bigcup_{i=1}^N A_i \quad (1.13)$$

1.8.2 Intersection of Sets

The sets of all elements common to A and B is called the intersection of A and B denoted as $A \cap B$.

$$A \cap B = \{x \mid x \in A \text{ and } x \in B\}$$

The Venn diagram of $A \cap B$ is shown in Fig. 1.2.

The operation $\cap$ is commutative and associative.

$$A \cap B = B \cap A \quad (1.14)$$

$$\text{and } A \cap (B \cap C) = (A \cap B) \cap C \quad (1.15)$$

The intersection of N sets $A_i, i = 1, 2, \dots, N$

$$A_1 \cap A_2 \cap \dots \cap A_N = \bigcap_{i=1}^N A_i \quad (1.16)$$

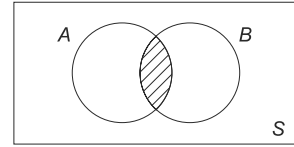


Fig. 1.2 Venn diagram showing $A \cap B$

1.8.3 Difference of Sets

The set of all elements of A that are not in B is called the difference of A and B denoted by $A - B$. It is shown in Fig. 1.3.

1.8.4 Complement of a Set

The set of all elements in the space S but not in A is called the complement of A , denoted as $\bar{A}$. It is equal to $S - A$. The complement of set A is shown in Fig. 1.4.

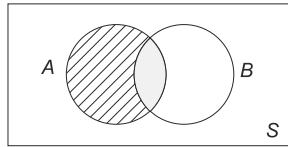


Fig. 1.3 $A - B$ difference of sets

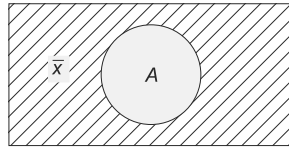


Fig. 1.4 $S - A$ complement of a set

1.8.5 Disjoint Sets (Mutually Exclusive Sets)

Two sets A and B are said to be disjoint or mutually exclusive if they have no elements in common. That is

$$A \cap B = \phi$$

The disjoint sets A and B are shown in Fig. 1.5.

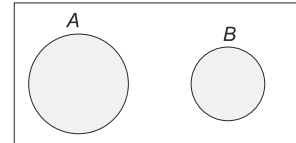


Fig. 1.5 Disjoint sets A and B

1.8.6 Algebra of Sets

There are a number of theorems involving union, intersection and complementation. Among them, the first three important laws are commutative law, distributive law and associative law.

Commutative Law

(a) $A \cup B = B \cup A$ (1.17a)

(b) $A \cap B = B \cap A$ (1.17b)

Distributive Law

(a) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ (1.18a)

(b) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ (1.18b)

Associative Law

(a) $A \cup (B \cup C) = (A \cup B) \cup C$ (1.19a)

(b) $A \cap (B \cap C) = (A \cap B) \cap C$ (1.19b)

Also, we have

(a) $A \cap \phi = \phi$ (1.20a)

(b) $A \cup \phi = A$ (1.20b)

De Morgan's Laws

(a) $\overline{A \cap B} = \bar{A} \cup \bar{B}$ (1.21a)

(b) $\overline{A \cup B} = \bar{A} \cap \bar{B}$ (1.21b)

(c) $\overline{\bar{A}} = A$ (1.21c)

Solved Problems**1.8** State whether the following sets are countable or uncountable, or finite or infinite.

$A = \{2, 3\}, B = \{0 < x \leq 10\} C = \{0 < \text{integers}\}$

$D = \{\text{Number of students in a college}\}$

$E = \{-30 \leq x \leq -1\}$

Solution A and D are countable and finite. C is countable and infinite. B and E are uncountable and infinite.**1.9** Define a set A that contains all integers with magnitude not exceeding 5. Define a second set B having even integers larger than 3 and not larger than 4. Determine if $A \subset B$ and $B \subset A$.**Solution**

$A = \{-5, -4, -3, -2, -1, 0, 1, 2, 3, 4, 5\}$

$B = \{-2, 0, 2, 4\}$

From the above, we know $B \subset A$ **1.10** Two sets are given by

$A = \{-5, -4, -3, -1, 0, 1, 2, 3\}$

and $B = \{-3, -1, 0, 5, 7\}$

Find (a) $A - B$ (b) $B - A$ (c) $A \cup B$ (d) $A \cap B$

Solution $(A - B)$ is the set of all elements of A that are not contained in B .

$$A - B = \{-5, -4, 1, 2, 3\}$$

Similarly,

$$B - A = \{-3, -1, 0\}$$

$$A \cup B = \{-5, -4, -3, -1, 0, 1, 2, 3, 5, 7\}$$

$$A \cap B = \{-3, -1, 0\}$$

1.11 Consider three sets A , B and C

Using Venn diagrams, shade the areas corresponding to the sets

(a) $A \cap B \cap C$, (b) $(A \cup B) - C$, and (c) $\overline{(A \cup B)} \cap C$

Solution

(a)

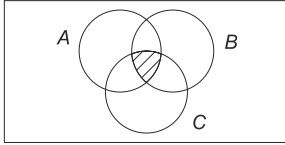


Fig. 1.6

(b)

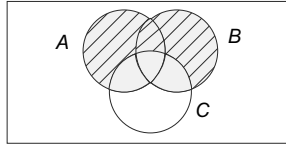


Fig. 1.7

(c)

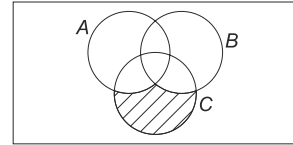


Fig. 1.8

1.12 Use De Morgan's laws to show that

$$(a) \overline{A \cap (B \cup C)} = (\bar{A} \cup \bar{B}) \cap (\bar{A} \cup \bar{C}) \quad (1.22a)$$

$$(b) \overline{A \cap B \cap C} = (\bar{A} \cup \bar{B} \cup \bar{C}) \quad (1.22b)$$

Solution

$$(a) \overline{A \cap (B \cup C)} = \overline{(A \cap B) \cup (A \cap C)}$$

$$= \overline{(A \cap B)} \cap \overline{(A \cap C)}$$

$$= (\bar{A} \cup \bar{B}) \cap (\bar{A} \cup \bar{C})$$

$$(b) \overline{A \cap B \cap C} = \overline{A \cap B} \cup \bar{C}$$

$$= \bar{A} \cup \bar{B} \cup \bar{C}$$

1.13 Prove the following:

$$(a) \overline{A_1 \cap A_2 \cap \dots \cap A_N} = \bar{A}_1 \cup \bar{A}_2 \cup \dots \cup \bar{A}_N \quad (1.23a)$$

$$(b) \overline{A_1 \cup A_2 \cup \dots \cup A_N} = \bar{A}_1 \cap \bar{A}_2 \cap \dots \cap \bar{A}_N \quad (1.23b)$$

Solution

- (a) Let
- $A_1 \cap A_2 = B$

Then

$$\begin{aligned}
 \overline{A_1 \cap A_2 \cap \cdots \cap A_N} &= \overline{B \cap A_3 \cap \cdots \cap A_N} \\
 &= \overline{\bar{B} \cup \overline{A_3 \cap A_4 \cap \cdots \cap A_N}} \\
 &= (\bar{A}_1 \cup \bar{A}_2) \cup \overline{(A_3 \cap \cdots \cap A_N)}
 \end{aligned}$$

Similarly,

- (b) Let
- $A_1 \cup A_2 = B$

$$\overline{A_1 \cap A_2 \cap \cdots \cap A_N} = \bar{A}_1 \cup \bar{A}_2 \cup \bar{A}_3 \cup \bar{A}_4 \cdots \cup \bar{A}_N$$

$$\begin{aligned}
 \overline{A_1 \cup A_2 \cup A_3 \cdots \cup A_N} &= \overline{B \cup A_3 \cdots \cup A_N} \\
 &= \overline{\bar{B} \cap \overline{A_3 \cdots \cup A_N}}
 \end{aligned}$$

$$\overline{A_1 \cup A_2} = \bar{A}_1 \cap \bar{A}_2$$

$$\begin{aligned}
 \Rightarrow \quad \overline{A_1 \cup A_2 \cup A_3 \cdots \cup A_N} &= \bar{A}_1 \cap \bar{A}_2 \cap \overline{(A_3 \cup \cdots \cup A_N)} \\
 &= (\bar{A}_1 \cap \bar{A}_2) \cap \overline{(A_3 \cup A_4 \cup A_5 \cdots \cup A_N)} \\
 &= (\bar{A}_1 \cap \bar{A}_2 \cap \bar{A}_3 \cap \bar{A}_4 \cap \overline{(A_5 \cdots \cup A_N)})
 \end{aligned}$$

If we continue this process

$$\overline{A_1 \cup A_2 \cup A_3 \cdots \cup A_N} = \bar{A}_1 \cap \bar{A}_2 \cap \bar{A}_3 \cdots \cap \bar{A}_N$$

**Practice Problems**

- 1.4**
- Two sets are given by

$$A = \{-3, -2, -1, 0, 1, 2, 3\}$$

$$B = \{-6, -4, -2, 0, 2, 4, 6\}$$

Find (a) $A - B$ (b) $B - A$, (c) $A \cup B$ (iv) $A \cap B$.Ans. (a) $\{-3, -1, 1, 3\}$ (b) $\{-6, -4, 4, 6\}$ (c) $\{-6, -4, -3, -2, -1, 0, 1, 2, 3, 4, 6\}$ (d) $\{-2, 0, 2\}$

- 1.5**
- Using Venn diagrams, prove that

(a) $\overline{A \cup B} = \bar{A} \cap \bar{B}$

(b) $\overline{A \cap B} = \bar{A} \cup \bar{B}$

- 1.6**
- Using a Venn diagram, show that the following identities are true:

(a) $\overline{A \cap B \cap C} = \bar{A} \cup \bar{B} \cup \bar{C}$

(b) $\overline{(A \cup B) \cap C} = C - [(A \cap C) \cup (B \cap C)]$

- 1.7**
- Shade Venn diagrams for the following sets:

(a) $(A \cap \bar{B}) \cup \bar{C}$

(b) $\overline{(A \cap B)} \cup \bar{C}$

(c) $(\bar{A} \cup B) \cup (C \cap \bar{D})$

Solved Problems

1.14 The sample space of an experiment has 10 outcomes given by $S = \{a_1, a_2, \dots, a_N\}$ with equal probability. Three events are defined as

$$A = \{a_1, a_2, a_3, a_5\} \quad B = \{a_2, a_4, a_6\}$$

$$C = \{a_6, a_9\}$$

Find the probability of (a) $A \cup B$ (b) $B \cup \bar{C}$ (c) $A \cap (B \cup C)$

Solution Given the probability of any event = $\frac{1}{10}$

$$(a) \quad P\{A \cup B\} = P\{a_1, a_2, a_3, a_4, a_5, a_6\} = \frac{6}{10} = \frac{3}{5}$$

$$(b) \quad B \cup \bar{C} = \{a_2, a_4, a_6\} \cup \{a_1, a_2, a_3, a_4, a_5, a_7, a_8, a_{10}\} \\ = \{a_1, a_2, a_3, a_4, a_5, a_6, a_7, a_8, a_{10}\}$$

$$P(B \cup \bar{C}) = \frac{9}{10}$$

$$(c) \quad A \cap (B \cup C) = \{a_1, a_2, a_3, a_5\} \cap \{a_2, a_4, a_6, a_9\} \\ = \{a_2\}$$

$$P(A \cap B \cup C) = \frac{1}{10}$$

1.15 Prove that for any two events A and B ,

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) \quad (1.24)$$

Solution

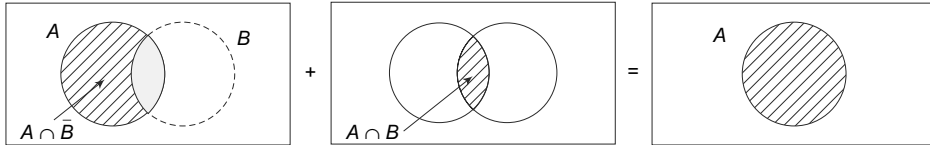


Fig. 1.9

From the Venn diagram shown in Fig. 1.9,

$$A = (A \cap \bar{B}) \cup (A \cap B) \quad (1.25)$$

$$P(A) = P[(A \cap \bar{B}) \cup (A \cap B)] \\ = P[(A \cap \bar{B})] + P(A \cap B) \quad (1.26)$$

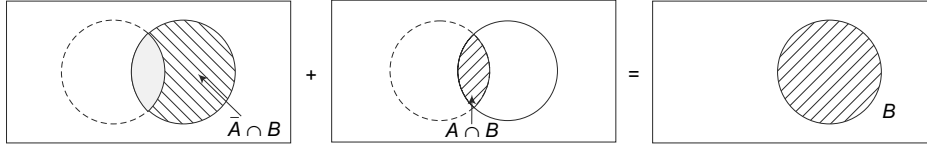


Fig. 1.10

Similarly, from Fig. 1.10,

$$P(B) = P[(\bar{A} \cap B) + P(A \cap B)] \quad (1.27)$$

From the Venn diagram,

$$A \cup B = (A \cap \bar{B}) \cup (\bar{A} \cap B) \cup (A \cap B) \quad (1.28a)$$

$$P(A \cup B) = P(A \cap \bar{B}) + P(\bar{A} \cap B) + P(A \cap B) \quad (1.28b)$$

From Eq. (1.26) and Eq. (1.27), we obtain

$$\begin{aligned} P(A) + P(B) &= P(A \cap \bar{B}) + P(\bar{A} \cap B) + 2P(A \cap B) \\ \Rightarrow P(A \cap \bar{B}) + P(\bar{A} \cap B) &= P(A) + P(B) - 2P(A \cap B) \end{aligned} \quad (1.29)$$

From Eq. (1.28) we can write

$$P(A \cap \bar{B}) + P(\bar{A} \cap B) = P(A \cup B) - P(A \cap B) \quad (1.30)$$

Substituting Eq. (1.30) in Eq. (1.29), we get

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) \quad (1.31)$$

1.16 Show that for any two events A and B ,
 $P(A \cup B) \leq P(A) + P(B)$

Solution We have

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

If A and B are mutually exclusive events, $A \cap B = \phi$

$$P(A \cap B) = 0$$

Therefore, $P(A \cup B) = P(A) + P(B)$

Otherwise, $P(A \cup B) < P(A) + P(B)$

So we can say $P(A \cup B) \leq P(A) + P(B)$

1.17 Prove the following:

If $\bar{A}$ is the complementary event of A then

$$P(A) = 1 - P(\bar{A})$$

Solution We know that the sample space

$$S = A \cup \bar{A}$$

Since $P(S) = 1$, we can write $P(A \cup \bar{A}) = 1$

Also, A and $\bar{A}$ are mutually exclusive events. Hence,

$$P(A \cup \bar{A}) = P(A) + P(\bar{A}) = 1$$

$$\Rightarrow P(A) = 1 - P(\bar{A})$$

1.18 Show that for any two events A and B ,

$$(a) \quad P(\bar{A} \cap B) = P(B) - P(A \cap B) \quad (1.32)$$

$$(b) \quad P(A \cap \bar{B}) = P(A) - P(A \cap B) \quad (1.33)$$

Solution Consider the Venn diagram shown in Fig. 1.11 from which we can find that

$$B = (\bar{A} \cap B) \cup (A \cap B)$$

Since $\bar{A} \cap B$ and $A \cap B$ are mutually exclusive events,

$$\begin{aligned} P(B) &= P[(\bar{A} \cap B) \cup (A \cap B)] \\ &= P(\bar{A} \cap B) + P(A \cap B) \end{aligned}$$

$$\Rightarrow P(\bar{A} \cap B) = P(B) - P(A \cap B)$$

Similarly, we can write

$$A = (A \cap \bar{B}) \cup (A \cap B)$$

$$P(A) = P[(A \cap \bar{B}) \cup (A \cap B)] = P(A \cap \bar{B}) + P(A \cap B)$$

$$P(A \cap \bar{B}) = P(A) - P(A \cap B)$$

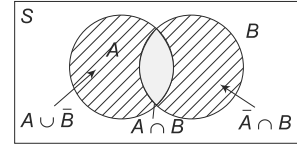


Fig. 1.11

1.19 Prove that if $A \subset B$ then $P(A) \leq P(B)$.

Solution Consider the Venn diagram shown in Fig. 1.12.

From this we can find that A and $\bar{A} \cap B$ are two mutually exclusive events. Also,

$$B = A \cup (\bar{A} \cap B)$$

$$\text{Then } P(B) = P[A \cup (\bar{A} \cap B)]$$

$$= P(A) + P(\bar{A} \cap B)$$

$$P(\bar{A} \cap B) = P(B) - P(A)$$

Since $P(\bar{A} \cap B) \geq 0$; $P(A) \leq P(B)$

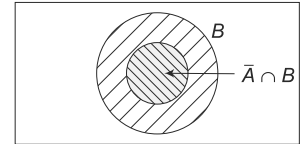


Fig. 1.12



Practice Problem

1.8 Show that

$$(a) P[(A \cap \bar{B}) \cup (B \cap \bar{A})] = P(A)P(B) - 2P(A \cap B)$$

$$(b) P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(A \cap C) + P(A \cap B \cap C)$$

Solved Problem



1.20 Prove that

$$P(\bar{A} \cap \bar{B}) = 1 - P(A) - P(B) + P(A \cap B) \quad (1.34)$$

Solution

$$\begin{aligned} P(\bar{A} \cap \bar{B}) &= P(\overline{A \cup B}) = 1 - P(A \cup B) \\ &= 1 - [P(A) + P(B) - P(A \cap B)] \\ &= 1 - P(A) - P(B) + P(A \cap B) \end{aligned}$$



Practice Problems

1.9 Show that

$$(a) P(\bar{A} \cap \bar{B}) = 1 - P(A \cap B) \quad (b) P(\bar{A} \cup \bar{B}) = 1 - P(A) + P(A \cap B) \quad (c) P(\bar{A} \cap B) = P(B) - P(A \cap B)$$

1.10 Prove that

For any two events A and B

$$P(A - B) = P(A) - P(A \cap B)$$



COMBINATORIAL ANALYSIS 1.9

Many problems in probability theory can be solved by simply counting the number of ways a certain event can occur. The mathematical theory of counting is known as combinatorial analysis.

1.9.1 Basic Principle of Counting

Consider two experiments are to be performed. If one experiment can result in any of m outcomes and if another experiment results in any of n possible outcomes then there are mn possible outcomes of the two experiments. This can be extended to k experiments in which the first one may result in any of n_1 possible outcomes. For each of the possible outcomes, there are n_2 possible outcomes of the second experiment; and for each of the possible outcomes, of the first two experiments, there are n_3 possible outcomes of the third experiment, and so on. Then for k experiments, there are a total of $n_1 n_2 \dots n_k$ outcomes.

Solved Problems



1.21 How many different 6-place license plates are possible if the first three places are English alphabets and the final three are numbers.

Solution From the basic principle of counting the number of license plates, we have $26 \times 26 \times 26 \times 10 \times 10 \times 10 = 17576000$.

1.22 Repeat the above problem under the assumption that no letter or number can be repeated in a single license plate.

Solution In this case, the number of license plates are

$$26 \times 25 \times 24 \times 10 \times 9 \times 8 = 11232000$$

1.23 In a college, a cultural committee is formed with 2 professors, 3 associate professors and 4 assistant professors. A subcommittee of 3, consisting of 1 individual from each group is to be chosen. How many different subcommittees are possible?

Solution Since a single person is to be selected from each group, the number of subcommittees is

$$2 \times 3 \times 4 = 24$$

1.9.2 Permutations

Consider the letters a, b and c . These letters can be arranged in 6 different ways as shown below:

$$abc, acb, bac, bca, cab, cba$$

Each arrangement is known as permutation. The first letter can be chosen from any three letters. Once we choose the first letter, the second letter can be chosen from any two letters, and the third letter can be chosen from the remaining 1.

Therefore, the number of permutations is $3 \times 2 \times 1 = 6$. This can be expanded to n objects. The different permutations of the n objects is

$$n(n-1)(n-2) \dots 3 \cdot 2 \cdot 1 = n! \quad (1.35)$$

For example, in a cricket team with 11 players, the different batting orders are $11! = 39916800$

Note: (i) In n objects of which n_1 are alike, n_2 are alike ... n_r are alike, the number of permutations is

$$\frac{n!}{n_1! n_2! \dots n_r!}$$

(ii) The number of permutations of n objects taking r at a time is $n_{pr} = \frac{n!}{(n-r)!}$ (1.36)

1.9.3 Combinations

For three letters a, b and c , the number of permutations is six. For permutations, the order in which the letters are taken is important. Therefore, abc and acb be counted as two different permutations. However, in combination, the order in which the letters are taken is not considered. Therefore, the number of combinations with letters a, b and c is only one. The number of two-letter combinations that can be formed from a, b and c

are ab, bc, ca . The number of combinations of n things taken r at a time is denoted by n_{cr} or $\binom{n}{r}$

$$\binom{n}{r} = \frac{n!}{(n-r)! r!} \quad (1.37)$$

- If $n_1 + n_2 + \dots + n_r = n$ then the number of possible division of n distinct objects from r distinct groups of respective sizes $n_1, n_2, \dots, n_r$ is

$$\binom{n}{n_1, n_2, \dots, n_r} = \frac{n!}{n_1! n_2! n_3! \dots n_r!} \quad (1.38)$$

- The number of ways of dividing $(p + q)$ items into two groups of p and q items respectively is

$$\frac{(p + q)!}{p! q!} \quad (1.39)$$

- The number of ways of dividing $2p$ items into two equal groups of p each is $\frac{(2p)!}{(p!)^2}$ where the two groups have distinct identity.
- The number of ways of dividing $2p$ items into two equal groups of p each is $\frac{(2p)!}{2!(p!)^2}$ where the groups do not have distinct identity.
- The number of ways in which $(p + q + r)$ things can be divided into three groups containing p , q and r things respectively is

$$\frac{(p + q + r)!}{p! q! r!} \quad (1.40)$$

If $p = q = r$ and three groups are distinct, the number of ways is $\frac{(3p)!}{3!(p!)^3}$

- The number of circular arrangements of n distinct items is $(n - 1)!$ if there is difference between clockwise and anticlockwise arrangements and $(n - 1)!/2$ if there is no difference between clockwise and anticlockwise arrangements.
- For $x_1 + x_2 + \dots + x_n = s$ where $s \geq 0$, the number of positive integer solutions (when $s \geq n$) is $\binom{s - 1}{n - 1}$ and the number of non-negative integral solutions is $\binom{n + s - 1}{n - 1}$.

Solved Problems



1.24 Six boys and six girls sit in a row randomly. Find the probability that

- the six girls sit together
- the boys and girls alternate.

Solution Six boys and six girls are sit randomly in $12!$ ways. If six girls sit together then we take them as a single person and there are $7!$ ways. Among six girls, they can sit in $6!$ ways

The probability that six girls sit together = $\frac{7! 6!}{12!}$

The boys and girls can sit alternately in $2 \times 6! \times 6!$ ways.

Therefore, probability = $\frac{2 \times 6! \times 6!}{12!}$

1.25 If 10 people are to be divided into three committees of sizes 3, 3 and 4, how many divisions are possible?

Hint: The number of possible divisions of n distinct objects into r distinct groups of sizes $n_1, n_2, \dots, n_r$ is

$$\frac{n!}{n_1! n_2! \dots n_r!}$$

Solution Using the above relation,

$$\text{Number of divisions} = \frac{12!}{3!3!4!}$$

1.26 Kumar has 10 friends, of whom he will invite 6 for his birthday party. (a) How many choices has he if 2 of the friends will not attend together? (b) How many choices has he if 2 of his friends will only attend together.

Solution

- (a) Total number of choices = Number of choices when both are not invited
+ Number of choices when one of them is invited

$$\text{Number of choices when both are not invited} = \binom{8}{6}$$

$$\text{Number of choices when one of them is invited} = 2 \binom{8}{5}$$

$$\Rightarrow \text{Total number of choices} = \binom{8}{6} + 2 \binom{8}{5}$$

1.27 In how many ways can Surya who has 10 shirts, 6 pants and 2 shoes be dressed?

Solution Using multiplication rules, the number of ways is equal to $\binom{10}{1} \binom{6}{1} \binom{2}{1} = 120$

1.28 A committee of 8, consisting of 2 professors, 3 associate professors, 3 assistant professors is to be chosen from a group of 5 professors, 5 associate professors and 6 assistant professors. How many committees are possible?

Solution The number of committees is equal to

$$\binom{5}{2} \binom{5}{3} \binom{6}{3} = 2000$$

1.29 In a university examination, a student is to answer 5 out of 8 questions. How many choices does he have? How many does he have if he must answer at least 2 of the first four questions?

Solution (a) From 8 questions, he has to select 5, therefore, he has $\binom{8}{5}$ choices.

- (b) There are three possibilities for a student to satisfy the requirement.
- He can select the first four questions and remaining one question from the rest of four.
 - He can select 3 questions from the first four and 2 questions from the rest of five.
 - He can select 2 questions from the first four and 3 questions from the rest of five.

Therefore, the total number of choices is

$$\binom{4}{4}\binom{4}{1} + \binom{4}{3}\binom{4}{2} + \binom{4}{2}\binom{4}{3} = 4 + 24 + 24 = 52$$

1.30 From 8 consonants and 6 vowels, how many words can be formed consisting of 5 different consonants and 4 different vowels?

Solution From 8 consonants, 5 different consonant can be selected in $\binom{8}{5}$ ways. Similarly, 4 different vowels can be selected from 6 vowels in $\binom{6}{4}$ ways. The resulting 9 different letters can be arranged among themselves in $9!$ ways. Therefore, the number of words

$$= \binom{8}{5}\binom{6}{4}9! = 304819200$$

1.31 How many different letter arrangements can be made from the letters (a) MALAYALAM, and (b) MISSISSIPPI?

Solution

(a) The word MALAYALAM has a total of 9 letters. It consists of 2M, 4A, 2L. Therefore,

$$\text{Number of different letter arrangements} = \frac{9!}{2!4!2!} = 3780$$

(b) The word MISSISSIPPI has a total of 11 letters out of which it has 4S, 4I, and 2P. The number of different letter arrangements is equal to $\frac{11!}{4!4!2!} = 34650$.

1.32 In a university examination, a student has to answer 8 out of 10 questions. How many ways can he answer? If he has to answer at least 3 out of the first five questions then how many choices does he have?

Solution Since the student has to answer 8 out of 10 questions, there are $\binom{10}{8}$ ways.

$$\binom{10}{8} = \frac{10!}{8!2!} = 45$$

For the second case, he has to answer at least three questions out of the first five. There are three possible ways.

(a) He can answer all questions from the first five and select three from the remaining five.

So there are $\binom{5}{5}\binom{5}{3}$ possible ways.

(b) He can answer 4 questions from the first five and select four from the remaining five.

Hence, there are $\binom{5}{4}\binom{5}{4}$ possible ways.

- (c) He can answer 3 questions from the first five and select the remaining five. So there are $\binom{5}{3}\binom{5}{5}$ possible ways.

Hence, the total number of choices is

$$\binom{5}{3}\binom{5}{5} + \binom{5}{4}\binom{5}{4} + \binom{5}{5}\binom{5}{3} = 10 + 25 + 10 = 45$$

1.33 Ramesh has 10 friends, and he decided to invite 6 of them to his birthday party. How many choices does he have if two of the friends will not attend together? How many choices has he if two of his friends will only attend together?

Solution Since two of the friends will not attend together, there are $\binom{8}{6}$ ways of inviting leaving the two and $2\binom{8}{5}$ ways of inviting leaving one of them. Therefore, the total number of choices = $\binom{8}{6} + 2\binom{8}{5} = 140$

In the second case, two of his friends will only attend together. In this case, there are $\binom{8}{6}$ ways of inviting, leaving the two and $\binom{8}{4}$ ways of inviting them. Therefore, the total number of choices = $\binom{8}{6} + \binom{8}{4} = 98$

1.34 For an association, a president, vice president and secretary, all different, are to be chosen from 10 contestants. How many different choices are possible if

- there are no restrictions?
- Ramu and Ravi will not associate together?
- Surya and Harsha will serve together or not at all?
- Guptha will serve only if he is president?

Solution

- Since there are no restrictions the number of choices is $10 \times 9 \times 8 = 720$.
- Without Ramu and Ravi, the number of choices is $8 \times 7 \times 6$ and with one of them in the association the number of choices is $3 \times 2 \times 8 \times 7$. So the total number of choices

$$8 \times 7 \times 6 + (3 \times 2 \times 8 \times 7) = 672$$

- For this case, the association can be formed with both of them or without both. When Harsha and Surya serve together then the number of choices is equal to $6 \times 8 = 48$. When they are not in association, the number of ways to form the association $8 \times 7 \times 6$. Therefore, the total number of ways is

$$8 \times 7 \times 6 + 6 \times 8 = 384$$

- If Guptha is chosen as president then the number of ways to select the remaining posts is $9 \times 8 = 72$. If he is not chosen, the number of ways to select the posts is $9 \times 8 \times 7$. So, therefore, the total number of ways is

$$9 \times 8 + 9 \times 8 \times 7 = 576$$

1.35 How many different 7-place licence plates are possible if the first 2 places are for letters and the other 5 for numbers. How many licence plates would be possible if repetition among letters or numbers were prohibited?

Solution For each letter, we have 26 choices and for each number, we have 10 choices. Therefore, there would be

$$26 \times 26 \times 10 \times 10 \times 10 \times 10 \times 10 = (26)^2(10)^5 = 67600000 \text{ different plates possible.}$$

If repetition is not allowed then for the first-place letter, we have 26 choices and for the second-place letter, we have 25 choices. Similarly, from the first-place number, the number of choices decreases by one for each place. Hence, there would be $26 \times 25 \times 10 \times 9 \times 8 \times 7 \times 6$ different plates possible.

1.36 Show

- In how many ways can 4 boys and 4 girls sit in a row?
- If 4 boys and 4 girls sit in a row in such a way that the boys and girls sit together then how many ways are possible?
- In the above case, if the boys alone must sit together then how many ways are possible?
- In how many ways can they sit if no boys or girls are allowed to sit together?

Solution

- Since there is no restriction, there would be $8!$ ways.
- Since 4 boys and 4 girls each sit together, there are two possible orderings. The boys can sit first and then girls or the girls can sit first and then boys.
Among boys, the number of possible ways is $4!$ and similarly, among girls the number of ways is $4!$. Therefore, the total number of ways is $2(4! \times 4!)$
- There are four possible ways that the boys as a group can occupy the position. Among each group there are $4!$ ways. Therefore, the total number of ways is $4(4! \times 4!)$

B G G G

G B G G

G G B G

G G G B

- The boys and girls can sit like this when the same sexes are not allowed to sit together:

B G B G B G B G

G B G B G B G B

Among boys and girls there are $4!$ ways. Therefore, the total number of ways is $2(4! \times 4!)$

1.37 In how many ways can 3 fiction novels, 2 physics books and 1 mathematics book be arranged?

- The books can be arranged in any order.
- The fiction novels must be together but other books can be arranged in any order.

Solution

- There are $6!$ ways the books can be arranged.
- The fiction novels must be together. So they can be arranged as a group in 4 ways:

F — — —

— F — —

— — F —
 — — — F

Among fiction novels, the number of ways is 6 and among the remaining books, the number of ways is 6. Therefore, total number of ways is $4(6 \times 6) = 144$.

1.38 In how many ways can 10 people be seated in a row if

- (a) there are no restrictions on the seating arrangements?
- (b) persons X and Y must sit next to each other?
- (c) there are 5 men and 5 women and no 2 men or 2 women can sit next to each other?
- (d) there are 6 men and they must sit next to each other?
- (e) there are 5 married couples and each couple must sit together?

Solution

- (a) Since there are no restrictions, the number of ways is $10! = 3628800$
- (b) If X and Y sit in XY order then the number of choices in which they can sit together is $(n - 1)$. Similarly, if they sit in YX order, the number of choices is $(n - 1)$. So the total number of choices they sit together is $2(n - 1)$. For each choice, the number of ways the eight people can sit is $8!$. Hence, the total number of ways is $2(n - 1)8!$

$$2(9)8! = 18 \times 8! = 725760$$

- (c) If men and women sit next to each other, they can sit like this.

$M W M W \dots$
 $W M W M \dots$

Among men, the number of choices is $5!$ and among women the number of choices is $5!$. Therefore, the total number of ways is $2 \times 5! \times 5! = 28,800$

- (d) 6 men occupy the seats in the following ways:

$6M W W W W$
 $W 6M W W W$
 $W W 6M W W$
 $W W W 6M W$
 $W W W W 6M$

Among 6 men, the number of choices $6!$ and among 4 women, the choices is $4!$. Therefore, the total number of ways is $5 \times (6!) \times (4!) = 86,400$

- (e) 5 married couples can be seated as a group in $5!$ ways and among them 2^5 ways. Therefore, the number of ways is $5!(2^5)$

1.39 A single card is drawn from a 52-card deck.

- (a) What is the probability that the card is a king?
- (b) What is the probability that the card will be a 4 or smaller?
- (c) What is the probability that the card is a red five?

Solution

- (a) In a 52-card deck, the number of cards with king is 4. Therefore,

$$P[\text{king}] = \frac{4}{52} = \frac{1}{13}$$

- (b)
- $P(\text{the card is 4 or smaller}) = ?$

The number of cards with 4 or smaller is equal to 4 fours + 4 threes + 4 twos = 12 cards

$$\Rightarrow P(4 \text{ or smaller}) = \frac{12}{52} = \frac{3}{13}$$

- (c)
- $P(\text{red five}) = \frac{2}{52}$

1.40 In a box, there are 250 coloured balls: 50 black, 40 green, 70 red, 60 white and 30 blue. What are the probabilities of selecting a ball of each colour?

Solution Total number of balls = 250

Let B , G , R , W and BL are the events of selecting black, green, red, white and blue balls respectively. Then

$$P(\text{Selecting black ball}) = P(B) = \frac{\text{Number of black balls}}{\text{Total number of balls}} = \frac{50}{250} = \frac{1}{5}$$

Similarly,

$$P(G) = \frac{\text{Number of green balls}}{\text{Total number of balls}} = \frac{40}{250} = \frac{4}{25}$$

$$P(R) = \frac{\text{Number of red balls}}{\text{Total number of balls}} = \frac{70}{250} = \frac{7}{25}$$

$$P(W) = \frac{\text{Number of white balls}}{\text{Total number of balls}} = \frac{60}{250} = \frac{6}{25}$$

$$P(BL) = \frac{\text{Number of blue balls}}{\text{Total number of balls}} = \frac{30}{250} = \frac{3}{25}$$

1.41 An experiment consists of rolling a single dice. Two events are defined as

$A = \{\text{a 6 shows up}\}$ and

$B = \{\text{a 2 or a 5 shows up}\}$

- (a) Find $P(A)$ and $P(B)$. (b) Define a third event C so that, $P(C) = 1 - P(A) - P(B)$.

Solution In rolling a single dice, the sample space

$$S = \{1, 2, 3, 4, 5, 6\}$$

- (a) For the event A , $P(A) = \frac{\text{Occurrence of event } A}{\text{Total sample space}} = \frac{1}{6}$

- (b) For the event B , $P(B) = \frac{\text{Occurrence of 2 or 5}}{\text{Total sample space}} = \frac{2}{6} = \frac{1}{3}$

$$P(C) = 1 - P(A) - P(B) = 1 - \frac{1}{6} - \frac{1}{3} = \frac{1}{2}$$

1.42 A dice is tossed. Find the probabilities of the event $A = \{\text{odd number shows up}\}$, $B = \{\text{number larger than 3 shows up}\}$.

Solution The sample space $S = \{1, 2, 3, 4, 5, 6\}$

Event $A = \{1, 3, 5\}$

$$P(A) = \frac{3}{6} = \frac{1}{2} = \frac{1}{2}$$

Event $B = \{4, 5, 6\}$

$$P(B) = \frac{3}{6} = \frac{1}{2}$$

1.43 When two dice are thrown, determine the probabilities from Axiom 3 for the following events.

(a) $A = \{\text{Sum} = 7\}$, (b) $B = \{8 < \text{sum} \leq 11\}$, (c) $C = \{10 < \text{sum}\}$ and determine (d) $P[B \cap C]$, and (e) $P[B \cup C]$.

Solution When two dice are thrown, the sample space S contains $6^2 = 36$ points as in Solved Problem 1.42. That is, $n(S) = 36$

(a) For Event A , the number of outcomes = 6

Therefore,

$$P(\text{Sum} = 7) = \frac{6}{36} = \frac{1}{6}$$

(b) $B = \{8 < \text{Sum} \leq 11\}$

The possible outcomes for Event B are (3, 6), (4, 5), (5, 4), (6, 3), (4, 6), (5, 5), (4, 6), (6, 5), (5, 6)

That is, $n_B = 9$

$$P(B) = \frac{n_B}{n(s)} = \frac{9}{36} = \frac{1}{4}$$

(c) For Event C , the possible outcomes are (5, 6) (6, 5) (6, 6). That is, $n_c = 3$

$$P(C) = P(10 < \text{sum}) = \frac{3}{36} = \frac{1}{12}$$

(d) For Event $B \cap C$, that $P[8 < \text{sum} \leq 11 \text{ and } 10 < \text{sum}]$, the possible outcomes are (6, 5) and (5, 6).

$$n(B \cap C) = 2$$

$$P(B \cap C) = \frac{2}{36} = \frac{1}{18}$$

(e) For Event $B \cup C$, the number of possible outcomes is 10.

They are (3, 6), (4, 5), (4, 6), (5, 4), (5, 5), (5, 6), (6, 3), (6, 4), (6, 5) and (6, 6)

$$P(B \cup C) = \frac{10}{36} = \frac{5}{18}$$

1.44 If two dies are rolled, what is the probability that the sum of the upturned faces will equal 9?

Solution The sample space S is given by

$$S = \left\{ \begin{array}{cccccc} (1,1) & (1,2) & (1,3) & (1,4) & (1,5) & (1,6) \\ (2,1) & (2,2) & (2,3) & (2,4) & (2,5) & (2,6) \\ (3,1) & (3,2) & (3,3) & (3,4) & (3,5) & (3,6) \\ (4,1) & (4,2) & (4,3) & (4,4) & (4,5) & (4,6) \\ (5,1) & (5,2) & (5,3) & (5,4) & (5,5) & (5,6) \\ (6,1) & (6,2) & (6,3) & (6,4) & (6,5) & (6,6) \end{array} \right\}$$

The number of ways that the sum of the faces is equal to 9 is (3, 6), (4, 5), (5, 4), (6, 3)

$$P(\text{Sum} = 9) = \frac{4}{36} = \frac{1}{9}$$

1.45 Two balls are drawn from an urn containing 5 white and 4 black balls. What is the probability that one of the drawn balls is white and the other black?

Solution The total number of balls is equal to 9. Two balls are drawn, therefore, there would be $\binom{9}{2} = 36$ points in the sample space. Since all outcomes are equally likely, the number of ways a white ball is drawn from 5 white balls is 5. Similarly, the number of ways a black ball is drawn from 4 black balls is 4. Hence, the desired probability is

$$\frac{5 \times 4}{36} = \frac{20}{36} = \frac{5}{9}$$

1.46 A committee of 6 is to be selected from a group of 6 men and 8 women. If the selection is made randomly, find the probability that the committee consists of 3 men and 3 women.

Solution The committee is to be selected from a group of 14 people. A committee of 6 can be selected from 14 persons in $\binom{14}{6}$ possible ways. Since each of the combinations is equally likely, 3 men can be selected from 6 men in $\binom{6}{3}$ combinations, and 3 women can be selected from 8 women in $\binom{8}{3}$ combinations. Now the probability is equal to $\frac{\binom{6}{3}\binom{8}{3}}{\binom{14}{6}} = 0.3729$.

1.47 A bag contains 5 white, 7 red and 3 black balls. If three balls are drawn at random, what is the probability that none of them is red?

Solution 3 balls can be selected from 15 balls in $\binom{15}{3}$ ways. Three balls without red can be drawn like below

(a) All white balls in $\binom{5}{3}$ ways, that is 10 ways.

(b) Two white and one black ball in $\binom{5}{2}\binom{3}{1} = 30$ ways.

(c) One white and two black balls in $\binom{5}{1}\binom{3}{2} = 15$ ways.

(d) All black balls $\binom{3}{3} = 1$ way.

So required probability

$$\frac{10 + 30 + 15 + 1}{\binom{15}{3}} = \frac{56}{\frac{15 \times 14 \times 13}{2!}} = \frac{56}{35 \times 13} = \frac{8}{65}$$

1.48 A box contains 4 point-contact diodes and 6 alloy junction diodes. What is the probability that 3 diodes picked at random contain at least two point contact diodes?

Solution Let p be the event of selecting point contact diode and A be the event of selecting alloy junction diode. The number of ways by which at least two point-contact diodes can be selected from three diodes is

(a) Two point contact diodes and one alloy junction diode.

(b) All three point contact diodes.

$$P(\text{at least two point contact diodes}) = P(p = 2, A = 1) + P(p = 3, A = 0)$$

Three diodes can be selected from 10 diodes in $\binom{10}{3}$ ways.

$$\begin{aligned} P &= \frac{\binom{4}{2}\binom{6}{1}}{\binom{10}{3}} + \frac{\binom{4}{3}\binom{6}{0}}{\binom{10}{3}} \\ &= \frac{36}{120} + \frac{4}{120} = \frac{1}{3} \end{aligned}$$

1.49 Form a group of 3 Indians, 4 Pakistanis and 5 Americans a subcommittee of 4 people is selected by lots. Find the probability that the subcommittee will consist of

(a) 2 Indians and 2 Pakistanis

(b) 1 Indian, 1 Pakistani and 2 Americans

(c) 4 Americans

Solution The group contains 12 people and a subcommittee of 4 people can be selected in $\binom{12}{4}$ ways.

(a) Two Indians and two Pakistanis can be selected in $\binom{3}{2}$ and $\binom{4}{2}$ ways respectively. Therefore, the probability is equal to $\frac{\binom{3}{2}\binom{4}{2}}{\binom{12}{4}} = \frac{18}{495}$

(b) 1 Indian, 1 Pakistani and 2 Americans can be selected in $\binom{3}{1}\binom{4}{1}\binom{5}{2}$ ways. Therefore, the probability = $\frac{\binom{3}{1}\binom{4}{1}\binom{5}{2}}{\binom{12}{4}} = \frac{120}{495}$

(c) Four Americans can be selected from a group of 5 in $\binom{5}{4}$ ways.

Therefore, probability = $\frac{\binom{5}{4}}{\binom{12}{4}} = \frac{5}{495}$

1.50 If each element of a second-order determinant is either zero or one, what is the probability that the value of the determinant is positive?

Solution The number of elements in a 2×2 determinant is 4 and each element can take 2 values. Therefore, the total number of determinants is $2^4 = 16$. Out of 16, the following matrix determinants produce a positive value.

$$\begin{vmatrix} 1 & 0 \\ 1 & 1 \end{vmatrix}, \begin{vmatrix} 1 & 1 \\ 0 & 1 \end{vmatrix}, \begin{vmatrix} 1 & 0 \\ 1 & 1 \end{vmatrix}.$$

Therefore, the probability = $\frac{3}{16}$

1.51 An urn contains 3 red and 6 blue balls. Two balls are drawn at random with replacement. Find the probability of getting (a) 2 red balls, (b) two blue balls, and (c) one red and one blue ball.

Solution

(a) The total number of balls in an urn is 9. The probability of drawing a red ball from 9 balls is $\frac{3}{9} = \frac{1}{3}$.

Since the ball is replaced, the probability of drawing a red ball the second time is also $\frac{1}{3}$.

So the probability of drawing two red balls with replacement is $\left(\frac{1}{3}\right)^2 = \frac{1}{9}$.

- (b) Similarly, the probability of drawing two blue balls with replacement is $\left(\frac{6}{9}\right)^2 = \left(\frac{2}{3}\right)^2 = \frac{4}{9}$
- (c) The probability of drawing one red ball and one blue ball can be done in two ways:
Probability of drawing first red ball and then blue ball + Probability of drawing first blue ball and then red ball

$$= \frac{1}{3} \left(\frac{2}{3}\right) + \frac{2}{3} \left(\frac{1}{3}\right) = \frac{2}{9} + \frac{2}{9} = \frac{4}{9}$$

1.52 A box contains 3 black balls and 6 green balls. One ball at a time is drawn at random, its colour is noted and the ball is replaced in the box for the next draw. Find the probability that the first green ball is drawn on the third draw.

Solution Let G denote drawing a green ball and B denote drawing a black ball. The total number of balls in the sample space is 9. Since the first green ball is drawn in the third draw, the first two draws are black

balls. The probability of drawing a black ball in the first draw is $\frac{\binom{3}{1}}{\binom{9}{1}} = \frac{3}{9} = \frac{1}{3}$.

Since the ball is replaced, the probability of drawing a black ball in the second draw is

$$\frac{\binom{3}{1}}{\binom{9}{1}} = \frac{3}{9} = \frac{1}{3}$$

The probability of drawing a green ball in the third draw is

$$\frac{\binom{6}{1}}{\binom{9}{1}} = \frac{6}{9} = \frac{2}{3}$$

$P(\text{green ball is drawn first time in third draw})$

$$= \frac{1}{3} \cdot \frac{1}{3} \cdot \frac{2}{3} = \frac{2}{27}$$



Practice Problems

1.11 How many different words can be formed by using all the letters of the word “ALLAHABAD”?

(Ans. 7560)

- 1.12** A group consists of 4 girls and 7 boys. In how many ways can a team of 5 members be selected if the team has (i) at least one boy and one girl (ii) at least 3 girls (Ans. 441, 91)
- 1.13** A bag contains 5 black and 6 red balls. Determine the number of ways in which 2 black and 3 red balls can be selected. (Ans. 200)
- 1.14** Among 14 players, 5 are bowlers. In how many ways a team of 11 may be formed with at least 4 bowlers? (Ans. 204)
- 1.15** Five cards are drawn from a well-shuffled pack of 52 cards. Find the probability that all the five cards are hearts. $\left(\text{Ans. } \frac{33}{66640} \right)$
- 1.16** Two dice are thrown together. What is the probability that the sum of two members is divisible by 3 or 4? (Ans. 5/9)
- 1.17** Three letters are dictated to three persons and an envelope is addressed to each of them, the letters are inserted into the envelope at random so tht envelope contains exactly one letter. Find the probability that at least one letter is in its proper envelope. (Ans. 2/3)
- 1.18** A box contains 100 bulbs, 20 of which are defective. 10 bulbs are selected at random, what is the probability that at least one is defective. (Ans. 0.9048)



ADDITION THEOREM 1.10

If A and B are two events associated with a random experiment then

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) \quad (1.41)$$

That is,

$$P(\text{either } A \text{ or } B \text{ or both occur}) = P(A \text{ occurs}) + P(B \text{ occurs}) - P(A \text{ and } B \text{ both occur})$$

since $P(A \text{ and } B \text{ both occur})$ in double counter in

$$P(A \text{ occurs}) + P(B \text{ occurs}).$$

If A and B are mutually exclusive events then

$$P(A \cup B) = P(A) + P(B) \quad (1.42)$$

If A , B and C are three events associated with a random experiment then

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(A \cap C) + P(A \cap B \cap C) \quad (1.43)$$

If A , B and C are mutually exclusive events then

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) \quad (1.44)$$

Consider two events A and B . Then

- Probability of occurrence of A only = $P(A \cap \bar{B})$

We have

$$P(A \cap \bar{B}) = P(A) - P(A \cap B) \quad (1.45)$$

- Probability of occurrence of B only = $P(\bar{A} \cap B)$

$$P(\bar{A} \cap B) = P(B) - P(A \cap B) \quad (1.46)$$

- Probability of occurrence of exactly one of A and B

$$\begin{aligned} &= P(A \cap \bar{B}) + P(\bar{A} \cap B) \\ &= P(A) + P(B) - 2P(A \cap B) \end{aligned} \quad (1.47)$$

Solved Problems

1.53 Two dice are thrown together. Find the probability that the sum of the numbers on the two faces is divisible by 3 or 4.

Solution Let A be the event that the sum be divisible by 3, and B be the event that the sum is divisible by 4.

Two dice can be thrown in $6^2 = 36$ ways. That is, $n(s) = 36$.

Out of 36 possible ways, the number of ways that the sum can be divided by 3 is

$(1, 2), (2, 1), (5, 1), (1, 5), (2, 4), (4, 2), (3, 3), (6, 3), (3, 6), (4, 5), (5, 4), (6, 6) = 12$ ways

The number of ways that the sum is divisible by four $(1, 3), (3, 1), (2, 2), (2, 6), (6, 2), (3, 5), (5, 3), (4, 4), (6, 6) = 9$ ways

The number of ways that the sum is divisible by 3 and 4 is $(6, 6) = 1$ way

We have
$$P(A) = \frac{12}{36} = \frac{1}{3}$$

$$P(B) = \frac{9}{36} = \frac{1}{4}$$

$$P(A \cap B) = \frac{1}{36}$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$= \frac{1}{3} + \frac{1}{4} - \frac{1}{36} = \frac{20}{36} = \frac{5}{9}$$

1.54 Determine the probability of the card being either red or a king when one card is drawn from a regular deck of 52 cards.

Solution Let A be the event of drawing a red card and B be the event of drawing a king. Then $A \cap B$ be the event of drawing a card which is red and a king.

Out of the 52 cards, the number of red cards is 26.

Therefore,
$$P[A] = \frac{26}{52}$$

The number of cards with king is 4. Therefore,

$$P[B] = \frac{4}{52}$$

Also,
$$P[A \cap B] = \frac{2}{52}$$

$$P[A \cup B] = P[A] + P[B] - P[A \cap B]$$

$$= \frac{26}{52} + \frac{4}{52} - \frac{2}{52} = \frac{28}{52}$$



Practice Problems

1.19 One number is chosen from numbers 1 to 200. What is the probability that it is divisible by 4 or 6?

(Ans. 67/200)

1.20 In a class of 60 students 30 opted for computer networks (CN) as an elective, 32 opted for operating systems (OS) and 24 opted for both computer networks and operating systems. If one of these students is selected at random, what is the probability that (i) the student opted for CN or OS (ii) the student opted OS but not CN. (Ans. (a) 19/30, (ii) 2/15)



CONDITIONAL PROBABILITY 1.11

Consider a box containing two white balls and one black ball. Assume that we have already drawn a ball from the box and the ball is not replaced. Now if we draw a ball from the box then the probability of getting a white ball from the box depends on the outcome of the first draw. If we had drawn a white ball on the first draw, the probability of getting a white ball on the second draw is $\frac{1}{2}$ since the box is left with one white ball and one black ball after the first draw. Suppose if we had drawn a black ball on the first draw, the probability of getting a white ball on the second draw is 1. That is, the probability of occurrence of second event depends on the outcome of the first event. If we denote the first event as A and the second event as B then the probability of the event B , given that A is known to have occurred, is denoted by the conditional probability denoted by $P(B|A)$.

Consider another experiment of tossing three coins. The sample space of the experiment is

$$S = \{TTT, TTH, THT, HTT, HHT, HTH, THH, HHH\}$$

The probability of any event is $1/8$. Let A be the event in which at least two tails appear, and B be the event in which the first coin shows a head.

In this case, the sample space of the event A is

$$A = \{TTT, TTH, THT, HTT\}$$

and
$$P(A) = \frac{4}{8} = \frac{1}{2}$$

We can also observe that

$$A \cap B = \{HTT\}$$

and
$$P(A \cap B) = \frac{1}{8}$$

Now if we want to find the probability of Event A with the occurrence of Event B then the sample space reduces from the set S to its subset B . In this sample space, the probability of occurrence of two tails is $\frac{1}{4}$ since there is only one possibility, HTT . That is $P(A|B) = 1/4$.

Note that the elements of B which favour the event A are the common elements of A and B , that is the elements of $A \cap B$. Therefore, conditional probability of A given that B has occurred is defined as

$$\begin{aligned} P(A|B) &= \frac{\text{Number of events favourable to } A \cap B}{\text{Number of events which are favourable to } B} \\ &= \frac{n(A \cap B)}{n(B)} \\ P(A|B) &= \frac{n(A \cap B)/n(s)}{n(B)/n(s)} = \frac{P(A \cap B)}{P(B)} \end{aligned} \quad (1.48)$$

If A and B are mutually exclusive, $A \cap B = \phi$ and $P(A|B) = 0$

The conditional probability satisfies the various postulates of probability:

1. $0 \leq P(A/B) \leq 1$

We know that $A \cap B \subset B$

Therefore, $P(A \cap B) \leq P(B)$

Hence,

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \leq 1, \text{ if } P(B) > 0$$

Also, $P(A \cap B) \geq 0$ and $P(B) > 0$

$$\text{Hence, } \frac{P(A \cap B)}{P(B)} \geq 0$$

Therefore, $0 \leq P(A|B) \leq 1$

2. $P(S|B) = \frac{P(S \cap B)}{P(B)} = \frac{P(B)}{P(B)} = 1$

Let B be an event with sample space S

$$\text{Then } P(S|B) = \frac{P(S \cap B)}{P(B)} = \frac{P(B)}{P(B)} = 1$$

$$\text{Also, } P(B|B) = \frac{P(B \cap B)}{P(B)} = \frac{P(B)}{P(B)} = 1$$

$$P(S|B) = P(B|B) = 1$$

3. $P(A_1 \cup A_2|B) = P(A_1|B) + P(A_2|B)$ if $A_1 \cap A_2 = \phi$ (1.49)

Let A_1 and A_2 be two events associated with a random experiment and S be the sample space.

If B is an event such that $P(B) \neq 0$ then

$$\begin{aligned} P(A_1 \cup A_2 | B) &= \frac{P\{(A_1 \cup A_2) \cap B\}}{P(B)} \\ &= \frac{P\{(A_1 \cap B) \cup (A_2 \cap B)\}}{P(B)} \\ &= \frac{P(A_1 \cap B) + P(A_2 \cap B) - P\{(A_1 \cap B) \cap (A_2 \cap B)\}}{P(B)} \\ &= \frac{P(A_1 \cap B)}{P(B)} + \frac{P(A_2 \cap B)}{P(B)} - \frac{P(A_1 \cap A_2 \cap B)}{P(B)} \\ &= (P(A_1|B) + P(A_2|B) - P(A_1 \cap A_2|B)) \end{aligned} \quad (1.50)$$

If A_1 and A_2 are mutually exclusive

$$P\{(A_1 \cup A_2) | B\} = P(A_1 | B) + P(A_2 | B)$$

Similarly,

$$P(A_1 \cup A_2 \cup \dots | B) = P(A_1|B) + P(A_2|B) + \dots$$

$$\text{if } A_i \cap A_j = \phi \text{ for } i \neq j \quad (1.51)$$

4. For any three events A_1, A_2 and A_3

$$P(A_1 \cap A_2 | A_3) + P(A_1 \cap \bar{A}_2 | A_3) = P(A_1 | A_3) \quad (1.52)$$

$$A_1 \cap A_3 = (A_1 \cap \bar{A}_2 \cap A_3) \cup (A_1 \cap A_2 \cap A_3)$$

$$P(A_1 \cap A_3) = P(A_1 \cap \bar{A}_2 \cap A_3) + P(A_1 \cap A_2 \cap A_3)$$

$$\frac{P(A_1 \cap A_3)}{P(A_3)} = \frac{P(A_1 \cap \bar{A}_2 \cap A_3)}{P(A_3)} + \frac{P(A_1 \cap A_2 \cap A_3)}{P(A_3)}$$

$$P(A_1 | A_3) = P(A_1 \cap \bar{A}_2 | A_3) + P(A_1 \cap A_2 | A_3)$$

The conditional probability $P(A|B)$ can be computed in two ways:

- Considering the probability of A with respect to the reduced sample space B
- Computing $P(A \cap B)$ and $P(B)$ with respect to the original sample space S

Solved Problems



1.55 A dice is rolled. If the outcome is an odd number, what is the probability that it is prime?

Solution Let A be the event of getting a prime number and B be the event of getting an odd number. The universal sample space

$$S = \{1, 2, 3, 4, 5, 6\}$$

$$A = \{2, 3, 5\}$$

$$B = \{1, 3, 5\}$$

$$A \cap B = \{3, 5\}$$

$$P(A) = \frac{3}{6} = \frac{1}{2}; P(B) = \frac{3}{6} = \frac{1}{2}$$

$$P(A \cap B) = \frac{2}{6} = \frac{1}{3}$$

$$P(A | B) = \frac{P(A \cap B)}{P(B)} = \frac{1/3}{1/2} = \frac{2}{3}$$

1.56 A pair of dice is thrown. Find the probability of getting 7 as sum, if it is known that the second dice always exhibits an odd number.

Solution Let A be the event of getting the sum as 7 and B be the event of getting the second dice as an odd number. $n(s) = 36$

$$A = \{(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)\}; n(A) = 6$$

$$P(A) = \frac{n(A)}{n(S)} = \frac{6}{36} = \frac{1}{6}$$

$$B = \{(1, 1), (1, 3), (1, 5), (2, 1), (2, 3), (2, 5), (3, 1), (3, 3), (3, 5), (4, 1), (4, 3), (4, 5), (5, 1), (5, 3), (5, 5), (6, 1), (6, 3), (6, 5)\}$$

$$n(B) = 18$$

$$P(B) = \frac{18}{36} = \frac{1}{2}$$

$$P(A \cap B) = \frac{3}{36} = \frac{1}{12}$$

$$P(A | B) = \frac{P(A \cap B)}{P(B)} = \frac{1/12}{1/2} = \frac{1}{6}$$

1.57 A couple has two children. Find the probability that (a) both the children are boys, if it is known that the older is a boy, and (b) both the children are boys, if it is known that at least one of the children is a boy.

Solution The couple can have two children in four different ways:

$$A = \{B_1B_2, B_1G_2, G_1B_2, G_1G_2\}$$

- (a) Let B be the event of both children being boys

$$B = \{B_1B_2\}$$

Let C be the event of the older being a boy:

$$C = \{B_1B_2, B_1G_1\}$$

$$B \cap C = \{B_1B_2\}$$

$$P(B | C) = \frac{n(B \cap C)}{n(C)} = \frac{1}{2}$$

- (b) Let D be the event that at least one of the children is a boy:

$$D = \{B_1B_2, B_1G_2, G_1B_2\}$$

$$P(B | D) = \frac{P(B \cap D)}{P(D)} = \frac{1/6}{3/6} = \frac{1}{3}$$

1.58 A fair dice is rolled. Consider the events

$$A = \{1, 3, 5\} \quad B = \{1, 2\} \quad \text{and} \quad C = \{1, 3, 4, 5\}$$

Find (a) $P(A | B)$ (b) $P(B | A)$ (c) $P(A | C)$ (d) $P(C | A)$ (e) $P(A \cup B | C)$

Solution The sample space

$$S = \{1, 2, 3, 4, 5, 6\}; \text{ Given } A = \{1, 3, 5\}$$

$$B = \{1, 2\}; C = \{1, 3, 4, 5\}$$

We can find $n(S) = 6, n(A) = 3; n(B) = 2; n(C) = 4$

$$P(A) = \frac{3}{6} = \frac{1}{2}; \quad P(B) = \frac{2}{6} = \frac{1}{3}; \quad P(C) = \frac{4}{6} = \frac{2}{3}$$

$$P(A \cap B) = \frac{1}{6}; \quad P(A \cap C) = \frac{3}{6} = \frac{1}{2}$$

$$P(B \cap C) = \frac{1}{6}; \quad P(A \cup B) = \frac{4}{6} = \frac{2}{3}; \quad P(A \cap B \cap C) = \frac{1}{6}$$

$$(a) \quad P(A | B) = \frac{P(A \cap B)}{P(B)} = \frac{1/6}{1/3} = \frac{1}{2}$$

$$(b) \quad P(B | A) = \frac{P(A \cap B)}{P(A)} = \frac{1/6}{1/2} = \frac{1}{3}$$

$$(c) \quad P(A | C) = \frac{P(A \cap C)}{P(C)} = \frac{1/2}{2/3} = \frac{3}{4}$$

$$(d) \quad P(C | A) = \frac{P(A \cap C)}{P(A)} = \frac{1/2}{1/2} = 1$$

$$\begin{aligned}
 \text{(e)} \quad P(A \cup B | C) &= \frac{P\{(A \cup B) \cap C\}}{P(C)} = \frac{P\{(A \cap C) \cup (B \cap C)\}}{P(C)} \\
 &= \frac{P(A \cap C) + P(B \cap C) - P(A \cap B \cap C)}{P(C)} \\
 &= \frac{\frac{1}{2} + \frac{1}{6} - \frac{1}{6}}{\frac{2}{3}} = \frac{\frac{1}{2}}{\frac{2}{3}} = \frac{3}{4}
 \end{aligned}$$

1.59 A fair dice is rolled. Consider the events

$$A = \{1, 3, 6\} \quad B = \{1, 2, 4\} \quad \text{and} \quad C = \{1, 3, 4, 5\}$$

Find (a) $P(A | B)$ (b) $P(B | A)$ (c) $P(A | C)$ (d) $P(C | A)$ (e) $P(A \cup B | C)$ (f) $P(A \cap B | C)$

Solution

First method The sample space

$$S = \{1, 2, 3, 4, 5, 6\}$$

Given $A = \{1, 3, 6\}; B = \{1, 2, 4\} \quad C = \{1, 3, 4, 5\}$

$$\text{(a)} \quad P(A | B) = \frac{n(A \cap B)}{n(B)} = \frac{1}{3}$$

$$\text{(b)} \quad P(B | A) = \frac{n(A \cap B)}{n(A)} = \frac{1}{3}$$

$$\text{(c)} \quad P(A | C) = \frac{n(A \cap C)}{n(C)} = \frac{2}{4} = \frac{1}{2}$$

$$\text{(d)} \quad P(C | A) = \frac{n(A \cap C)}{n(A)} = \frac{2}{3}$$

$$\text{(e)} \quad P(A \cup B | C) = \frac{n\{A \cup B \cap C\}}{n(C)} = \frac{3}{4}$$

$$\text{(f)} \quad P(A \cap B | C) = \frac{n(A \cap B \cap C)}{n(C)} = \frac{1}{4}$$

$A \cap B = \{1\}$ $A \cap C = \{1, 3\}$ $B \cap C = \{1, 4\}$ $A \cup B = \{1, 2, 3, 4, 6\}$ $A \cup B \cap C = \{1, 3, 4\}$ $A \cap B \cap C = \{1\}$
--

Second method

$$P(A) = \frac{3}{6} = \frac{1}{2}; \quad P(B) = \frac{3}{6} = \frac{1}{2}; \quad P(C) = \frac{4}{6} = \frac{2}{3}$$

$$P(A \cap B) = \frac{1}{6}; \quad P(A \cap C) = \frac{2}{6} = \frac{1}{3}; \quad P(B \cap C) = \frac{2}{6} = \frac{1}{3}$$

$$P(A \cup B) = \frac{5}{6}; \quad P(A \cap B \cap C) = \frac{1}{6}$$

$$\text{(a)} \quad P(A | B) = \frac{P(A \cap B)}{P(B)} = \frac{1/6}{1/2} = \frac{1}{3}$$

$$\text{(b)} \quad P(B | A) = \frac{P(A \cap B)}{P(A)} = \frac{1/6}{1/2} = \frac{1}{3}$$

$$\begin{aligned}
P(A|C) &= \frac{P(A \cap C)}{P(C)} = \frac{1/3}{2/3} = \frac{1}{2} \\
P(C|A) &= \frac{P(A \cap C)}{P(A)} = \frac{1/3}{1/2} = \frac{2}{3} \\
P(A \cup B|C) &= \frac{P\{(A \cup B) \cap C\}}{P(C)} = \frac{P(A \cap C) \cup P(B \cap C)}{P(C)} \\
&= \frac{P(A \cap C) + P(B \cap C) - P(A \cap B \cap C)}{P(C)} \\
&= \frac{\frac{1}{3} + \frac{1}{3} - \frac{1}{6}}{\frac{2}{3}} = \frac{\frac{2}{3}}{\frac{2}{3}} = \frac{3}{4} \\
P(A \cap B|C) &= \frac{P\{(A \cap B) \cap C\}}{P(C)} \\
&= \frac{P(A \cap B \cap C)}{P(C)} = \frac{1/6}{2/3} = \frac{1}{4}
\end{aligned}$$

1.60 One ticket is selected at random from 50 tickets numbered 00, 01, 02, ..., 49. What is the probability that the sum of the digits on the selected ticket is 8 given that the product of these digits is zero?

Solution Let A be the event in which the sum of the digits is 8 and B be the event in which the product of the digits is zero.

The sample space of the event $B = \{00, 01, 02, 03, 04, 05, 06, 07, 08, 09, 10, 20, 30, 40\}$

$$\begin{aligned}
P(B) &= \frac{14}{50}; P(A \cap B) = \frac{1}{50} \\
P(A|B) &= \frac{P(A \cap B)}{P(B)} = \frac{1/50}{14/50} = \frac{1}{14}
\end{aligned}$$

1.61 A mechanical system consists of two subsystems A and B . From the following probabilities, find:

- (a) $P(A \text{ fails} | B \text{ has failed})$
- (b) $P(A \text{ fails alone})$

$$\begin{aligned}
P(A \text{ fails}) &= 0.20, P(B \text{ fails alone}) = 0.15 \\
P(A \text{ and } B \text{ fail}) &= 0.15
\end{aligned}$$

Solution

$$\text{Given, } P(A) = 0.2, P(\bar{A} \cap B) = 0.15; P(A \cap B) = 0.15$$

We have

$$P(\bar{A} \cap B) = P(B) - P(A \cap B)$$

$$\Rightarrow P(B) = P(\bar{A} \cap B) + P(A \cap B) = 0.15 + 0.15 = 0.3$$

$$P(A | B) = \frac{P(A \cap B)}{P(B)} = \frac{0.15}{0.3} = 0.5$$

$$\begin{aligned} P(A \cap \bar{B}) &= P(A) - P(A \cap B) \\ &= 0.20 - 0.15 = 0.05 \end{aligned}$$

1.62 In a school, there are 500 students out of which 230 are girls. It is known that out of 230, 10% of the girls study in Class X. What is the probability that a student chosen at random studies in class X given that the chosen student is a girl.

Solution Let A be the event that a student chosen at random studies in Class X and B be the event of selecting a girl. Now we have to find the probability that the selected student is from Class X given that the chosen student is a girl. That is, $P(A | B)$.

$$\begin{aligned} P(A|B) &= \frac{n(A \cap B)}{n(B)} \\ n(A \cap B) &= 10\% \text{ of } 230 = 23 \\ n(B) &= 230 \\ P(A|B) &= \frac{23}{230} = \frac{1}{10} = 0.1 \end{aligned}$$

1.63 Two cards are drawn from a well-shuffled pack of 52 cards without replacement. What is the probability that one is a red queen and the other is a black king?

Solution Let R_i be the event of drawing red queen in the i^{th} draw and K_i be the event of drawing black king in the i^{th} draw. The king and queen cards can be drawn in two ways. The king can be drawn on the first draw and the queen in the second draw or the queen can be drawn in the first draw and the king in the second draw.

Therefore, the required probability is

$$\begin{aligned} &= P(R_1 \cap K_2) \cup P(K_1 \cap R_2) \\ &= P(R_1 \cap K_2) + P(K_1 \cap R_2) \\ &= P(K_1) P(K_2 | R_1) + P(K_1) P(R_2 | K_1) \end{aligned}$$

Now,

$$P(R_1) = \frac{2}{52}, \quad P(K_2 | R_1) = \frac{2}{51}$$

$$P(K_1) = \frac{2}{52}, \quad P(R_2 | K_1) = \frac{2}{51}$$

$$\text{Probability} = \frac{2}{52} \times \frac{2}{51} + \frac{2}{52} \times \frac{2}{51} = \frac{2}{663}$$

1.64 Cards are numbered 1 to 21. Two cards are drawn one after the other. Find the probability that the first card is a multiple of 6 and the other is a multiple of 10.

Solution Two cards can be drawn in the following mutually exclusive ways.

- (a) First card bears a multiple of 6 and the second bears a multiple of 10
- (b) First card bears a multiple of 10 and the second bears a multiple of 6

Let A_1 be the event of the first card drawn bearing a multiple of 6. A_2 be the event of the second card drawn bearing in multiple of 6. B_1 be the event of the first card drawn bear a multiple of 10. B_2 be the event of the second card drawn bearing multiple of 10.

Therefore, the probability

$$\begin{aligned}
 &= P\{(A_1 \cap B_2) \cup (B_1 \cap A_2)\} \\
 &= P(A_1 \cap B_2) + P(B_1 \cap A_2) \\
 &= P(A_1) P(B_2 | A_1) + P(B_1) P(A_2 | B_1) \\
 P(A_1) &= \frac{3}{21}; P(B_2 | A_1) = \frac{2}{20} \\
 P(B_1) &= \frac{2}{21}; P(A_2 | B_1) = \frac{3}{20} \\
 \Rightarrow \text{Probability} &= \frac{3}{21} \left(\frac{2}{20} \right) + \frac{2}{21} \left(\frac{3}{20} \right) \\
 &= \frac{6}{210} = \frac{2}{70}
 \end{aligned}$$

1.65 From a bag containing 4 white and 6 black balls, two balls are drawn at random. If the balls are drawn one after the other without replacement, find the probability that

- (a) Both balls are white
- (b) Both balls are black
- (c) The first ball is white and the second ball is black
- (d) One ball is white and the other is black

Solution The total number of balls in the bag is equal to ten. The number of white balls is four and the number of black balls is six.

- (a) Let W be the event of drawing the first white ball and W_2 be the event of drawing the second white ball.

$$\begin{aligned}
 P(W_1) &= \frac{\text{Total number of white balls}}{\text{Total number of balls}} = \frac{4}{10} = \frac{2}{5} \\
 P(W_2 | W_1) &= \frac{\binom{3}{1}}{\binom{9}{1}} = \frac{3}{9} = \frac{1}{3} \\
 P(W_1 \cap W_2) &= P(W_1) P(W_2 | W_1) = \frac{4}{10} \left(\frac{3}{9} \right) = \frac{2}{15}
 \end{aligned}$$

- (b) Let B_1 be the event of the drawing first black ball and B_2 be the event of drawing second black ball.

$$P(B_1) = \frac{\binom{6}{1}}{\binom{10}{1}} = \frac{6}{10} = \frac{3}{5}$$

$$P(B_2 | B_1) = \frac{5}{9}$$

$$P(B_1 \cap B_2) = P(B_1) P(B_2 | B_1) = \frac{3}{5} \cdot \frac{5}{9} = \frac{1}{3}$$

- (c) Let W_1 be the event of drawing the first white ball and B_2 be the event of drawing the second ball as black.

$$P(W_1) = \frac{4}{10} = \frac{2}{5}$$

$$P(B_2 | W_1) = \frac{6}{9} = \frac{2}{3}$$

$$P(W_1 \cap B_2) = P(W_1) P(B_2 | W_1) = \frac{2}{5} \left(\frac{2}{3} \right) = \frac{4}{15}$$

- (d) Probability of one ball is white and other is black it can happen in two ways: the first one is white and the second ball is black, or the first one is black and the second one is white.

$$P(W_1 \cap B_2) + P(B_1 \cap W_2)$$

$$P(W_1 \cap B_2) = \frac{4}{15}$$

$$\begin{aligned} P(B_1 \cap W_2) &= P(B_1) P(W_2 | B_1) \\ &= \left(\frac{6}{10} \right) \left(\frac{4}{9} \right) = \frac{4}{15} \end{aligned}$$

$$P(W_1 \cap B_2) + P(B_1 \cap W_2) = \frac{4}{15} + \frac{4}{15} = \frac{8}{15}$$

1.66 A student cannot qualify for the interview if he fails in subjects A and B . The probabilities that he fails in A and B are known to be 0.01 and 0.03 respectively. It is also known that he is more likely to fail in subject B with probability 0.06 if he failed in A .

- (a) What is the probability that he cannot qualify for the interview?
 (b) What is the probability that he fails in A given that he failed in B ?

Solution Given $P(A) = 0.01$ and $P(B) = 0.03$

- (a) $P(\text{he cannot qualify}) = P(\text{fails in } A \cap \text{fails in } B) = P(A \cap B)$

$$\text{Given } P(B | A) = 0.06$$

$$P(B | A) = \frac{P(A \cap B)}{P(A)} \Rightarrow P(A \cap B) = P(B | A) P(A) = 0.06 (0.01) = 0.0006$$

- (b) $P(A | B) = \frac{P(A \cap B)}{P(B)}$

$$\frac{0.0006}{0.03} = 0.02$$

1.67 In a box, there are 100 resistors having resistances and tolerances as shown in Table 1.3. A resistor is chosen with same likelihood of being chosen for the three events: A as “draw a 47 Ω resistor”, B as “draw a 5% tolerance resistor” and C as “draw a 100 Ω resistor”. Determine joint probabilities and conditional probabilities.

Resistance in Ω	Tolerance		
	5%	10%	Total
22	10	14	24
47	28	16	44
100	24	8	32
Total	62	38	100

Solution Given the probabilities are

$$P(A) = P(\text{drawing a } 47 \Omega \text{ resistor}) = \frac{\text{Number of } 47 \Omega \text{ resistors}}{\text{Total number of resistors}} = \frac{44}{100} = 0.44$$

$$P(B) = P(\text{drawing 5\% resistance}) = \frac{62}{100} = 0.62$$

$$P(C) = P(\text{drawing 100 } \Omega \text{ resistance}) = \frac{32}{100} = 0.32$$

The joint probabilities are

$$P(A \cap B) = P(\text{drawing a } 47 \Omega \text{ resistor with 5\% tolerance}) = \frac{28}{100} = 0.28$$

$$P(A \cap C) = P(\text{drawing a } 47 \Omega \text{ resistor with 100 } \Omega \text{ resistor}) = 0$$

$$P(B \cap C) = P(\text{drawing a 100 } \Omega \text{ resistor with 5\% tolerance}) = \frac{24}{100} = 0.24$$

The conditional probabilities are

$$P(A|B) = P(\text{drawing a } 47 \Omega \text{ resistor given that the resistor has 5\% tolerance})$$

$$= \frac{28}{62} = 0.4516$$

Also using the formula,

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{28}{100} \div \frac{62}{100} = \frac{28}{62} = 0.4516$$

Similarly,

$$P(A|C) = P(\text{drawing a } 47 \Omega \text{ resistor from 100 } \Omega \text{ resistors}) = 0$$

$$P(B|C) = P(\text{drawing a 5\% tolerance resistor from a 100 } \Omega \text{ resistors}) = \frac{24}{32} = 0.75$$

Also,
$$P(B|C) = \frac{P(B \cap C)}{P(C)} = \frac{24}{100} \div \frac{32}{100} = 0.75$$

$$P(B|A) = P(\text{drawing a } 5\% \text{ tolerance resistor from } 47 \text{ } \Omega \text{ resistors}) = \frac{28}{44} = 0.636$$

$$P(C|A) = 0$$

$$P(C|B) = (\text{drawing a } 100 \text{ } \Omega \text{ resistors with } 5\% \text{ tolerance resistor}) = \frac{24}{62} = 0.387$$



Practice Problems

1.21 A box contains 5 red and 4 white balls. Two balls are drawn successively from the box without replacement and it is noted that the second one is white. What is the probability that the first is also white. (Ans. 3/8)

1.22 If the probability that a communication system has high selectivity is 0.54 and the probability that it will have high fidelity is 0.81 and the probability that it will have both is 0.18. Find the probability that a system with high fidelity will have high selectivity. (Ans. 0.22)



MULTIPLICATION THEOREM ON PROBABILITY 1.12

If A and B are two events associated with a random experiment then

$$P(A \cap B) = P(A) P(B | A) \text{ if } P(A) \neq 0 \quad (1.53a)$$

or $P(A \cap B) = P(B) P(A | B) \text{ if } P(B) \neq 0 \quad (1.53b)$

Note that $P(B | A) = \frac{P(A \cap B)}{P(A)}$

and $P(A | B) = \frac{P(A \cap B)}{P(B)}$

If A and B are independent events then

$$P(A | B) = P(A) \text{ and } P(B | A) = P(B)$$

$$P(A \cap B) = P(A) P(B)$$

$$P(A \cup B) = 1 - P(\bar{A}) P(\bar{B}) \quad (1.54)$$

If $A_1, A_2, \dots, A_n$ are n events associated with a random experiment then

$$P(A_1 \cap A_2 \cap A_3 \dots \cap A_n) = P(A_1) P(A_2 | A_1) P(A_3 | A_1 \cap A_2) \dots P(A_n | A_1 \cap A_2 \cap A_3 \dots \cap A_{n-1}) \quad (1.55)$$

where $P(A_i | A_1 \cap A_2 \dots \cap A_{i-1})$ represents the conditional probability of the occurrence of the event A_i , given that the events $A_1, A_2, \dots, A_{i-1}$ have already occurred.

Solved Problems



1.68 A bag contains 6 white and 9 black balls. Two balls are drawn in succession without replacement. What is the probability that the first is white and the second is black?

Solution Consider A be the event of getting a white ball in the first draw and B be the event of getting a black ball in the second draw.

The total number of balls is 15 and the probability of getting a white in the first draw is

$$P(A) = \frac{\binom{6}{1}}{\binom{15}{1}} = \frac{6}{15} = \frac{2}{5}$$

Probability of getting black in the second draw when a white ball has already been drawn in the first draw is.

$$P(B | A) = \frac{\binom{9}{1}}{\binom{14}{1}} = \frac{9}{14}$$

The required probability of getting a white ball in the first draw and a black ball in the second draw

$$\begin{aligned} P(A \cap B) &= P(A) P(B | A) \\ &= \frac{2}{5} \cdot \left(\frac{9}{14}\right) = \frac{9}{35} \end{aligned}$$

1.69 Three cards are drawn successively without replacement from a pack of 52 well-shuffled cards. What is the probability that first two cards are kings and the third draw is an ace?

Solution The cards are drawn successively without replacement. Let K_1 and K_2 are the events of drawing a king in the first and second draw respectively and A_3 be the event of drawing an ace in the third draw.

$$P(K_1) = P(\text{first card is king}) = \frac{4}{52}$$

$$P(K_2 | K_1) = P(\text{second card drawn is king given that first card is also king}) = \frac{3}{51}$$

$$P(A_3 | K_1 \cap K_2) = P(\text{third card drawn is ace given that first two cards are kings}) = 4/50$$

$$\begin{aligned} P(K_1 \cap K_2 \cap A_3) &= P(K_1) P(K_2 | K_1) P(A_3 | K_1 \cap K_2) \\ &= \frac{4}{52} \cdot \frac{3}{51} \cdot \frac{4}{50} = \frac{2}{17 \times 13 \times 25} = \frac{2}{5525} \end{aligned}$$

1.70 A jar contains two white and three black balls. A sample of size 4 is made. What is the probability that the sample is in the order {white, black, white, black}?

Solution The jar contains two white and three black balls. We consider the problem without replacement.

Let W be the event of selecting a white ball and B be the event of selecting a black ball. The suffix to the letter W and B is the order in which the ball is drawn. Then

$$P(W_1 \cap B_2 \cap W_3 \cap B_4) = P(W_1) \cdot P(B_2 | W_1) \cdot P(W_3 | B_2 \cap W_1) \cdot P(B_4 | W_1 \cap B_2 \cap W_3)$$

$$P(W_1) = \frac{\text{Number of white balls}}{\text{Total number of balls}} = \frac{2}{5}$$

$$P(B_2 | W_1) = \frac{3}{4}$$

($\because$ since a ball is drawn and not replaced, the total number of balls is 4)

$$P(W_3 | B_2 \cap W_1) = \frac{1}{3}$$

$$P(B_4 | W_1 \cap B_2 \cap W_3) = \frac{2}{2} = 1$$

$$P(W_1 \cap B_2 \cap W_3 \cap B_4) = \frac{2}{3} \left(\frac{3}{4} \right) \left(\frac{1}{3} \right) (1) = 0.1$$

1.71 In a certain group of engineers, 60% have insufficient background of information theory, 50% have inadequate knowledge of probability, and 80% are in either one or both of adequate categories. What is the percentage of people who have adequate knowledge of probability among those who have a sufficient background of information theory?

Solution Let E_1 be the event that the engineers have sufficient knowledge in information theory and E_2 be the event that the engineers have adequate knowledge of probability.

Given:

$$P(\bar{E}_1) = 0.6 \quad \text{and} \quad P(\bar{E}_2) = 0.5$$

$$P(\bar{E}_1 \cup \bar{E}_2) = 0.8$$

Also, we can write $P(E_1) = 0.4$ and $P(E_2) = 0.5$

$$P(E_1 \cap E_2) = 1 - P(\bar{E}_1 \cup \bar{E}_2)$$

$$= 1 - 0.8 = 0.2$$

$$P(E_1 | E_2) = \frac{P(E_1 \cap E_2)}{P(E_2)} = \frac{0.2}{0.5} = 0.4$$

1.72 A bag contains 6 white, 5 red and 7 black balls. If four balls are drawn one by one without replacement, find the probability of getting all white balls.

Solution Let A, B, C and D denote events of getting a white ball in the first, second, third and fourth draws respectively. The required probability is

$$P(A \cap B \cap C \cap D)$$

Using the multiplication theorem, we have

$$P(A \cap B \cap C \cap D) = P(A) P(B | A) P(C | A \cap B) P(D | A \cap B \cap C)$$

$$= \frac{6}{18} = \frac{1}{3}$$

$$P(B | A) = P(\text{drawing a white ball in the second draw given that a white ball is drawn in the first draw})$$

$$= \frac{5}{17}$$

Since the ball is not replaced, the probability of drawing a white ball during the third draw given that two white balls are drawn during the first two draws is

$$P(C | A \cap B) = \frac{4}{16} = \frac{1}{4}$$

Similarly,

$$P(D | A \cap B \cap C) = \frac{3}{15} = \frac{1}{5}$$

$$\begin{aligned} P(A \cap B \cap C \cap D) &= P(A) P(B | A) P(C | A \cap B) P(D | A \cap B \cap C) \\ &= \frac{1}{3} \left(\frac{5}{17} \right) \left(\frac{1}{4} \right) \left(\frac{1}{5} \right) = \frac{1}{204} \end{aligned}$$



Practice Problems

1.23 Four cards are drawn successively, without replacement from a pack of 52 well-shuffled cards. What is the probability that the first two cards are queens, the third card drawn is a king and the fourth one is an ace?

1.24 A box contains eight red balls, seven white balls, and five blue balls. If three balls are drawn successively from the box, find the probability that they are drawn in the order red, white and blue if each ball is replaced after it has been drawn. (Ans. 7/20)

Solved Problems



1.73 A, B and C toss a coin. The one who gets the head first wins. What are their respective chances?

Solution The probability of A winning in the first round is $\frac{1}{2}$. If all of them failed in the first round, the probability of A winning in the second round, is $\left(\frac{1}{2}\right)^3 \frac{1}{2}$. Similarly, A winning in the third round is $\left(\frac{1}{2}\right)^6 \frac{1}{2}$. So the required probability using addition theorem is

$$\frac{1}{2} + \left(\frac{1}{2}\right)^3 \frac{1}{2} + \left(\frac{1}{2}\right)^6 \frac{1}{2} + \dots = \frac{\frac{1}{2}}{1 - \left(\frac{1}{2}\right)^3} = \frac{4}{7}$$

The probability of B winning in the first round is equal to $\left(\frac{1}{2}\right)\left(\frac{1}{2}\right) = \frac{1}{4}$

Similarly, B wins in the second round with probability $\left(\frac{1}{2}\right)^3 \frac{1}{4}$, and so on.

The probability of B winning is

$$\frac{1}{4} + \frac{1}{4} \left(\frac{1}{2}\right)^2 + \frac{1}{4} \left(\frac{1}{2}\right)^5 + \dots = \frac{\frac{1}{4}}{1 - \left(\frac{1}{2}\right)^3} = \frac{2}{7}$$

Similarly, the probability of C winning is

$$\frac{1}{8} + \frac{1}{8} \left(\frac{1}{2}\right)^3 + \frac{1}{8} \left(\frac{1}{2}\right)^6 + \dots = \frac{\frac{1}{8}}{1 - \left(\frac{1}{2}\right)^3} = \frac{1}{7}$$



INDEPENDENT EVENTS 1.13

Two events A and B are independent if and only if

$$P(A \cap B) = P(A) P(B)$$

In case of three events, the condition for independence are

- $P(A_1 \cap A_2) = P(A_1) P(A_2)$
- $P(A_1 \cap A_3) = P(A_1) P(A_3)$
- $P(A_2 \cap A_3) = P(A_2) P(A_3)$
- $P(A_1 \cap A_2 \cap A_3) = P(A_1) P(A_2) P(A_3)$

If all the above conditions are satisfied then the entire set of events are mutually independent.

If A and B are independent events then

$$P(A \cup B) = P(A) + P(B) - P(A) P(B)$$

If A_1, A_2 and A_3 are independent,

$$P[A_1 \cap (A_2 \cap A_3)] = P(A_1) P(A_2 \cap A_3)$$

Solved Problems



1.74 If $P(A) = 0.75$, $P(B) = 0.4$ and $P(A \cap B) = 0.3$, are the events A and B independent?

Solution Two events are independent if

$$P(A \cap B) = P(A) P(B)$$

$$P(A) P(B) = 0.75 (0.4) = 0.3$$

$$P(A \cap B) = 0.3$$

Hence, A and B are independent.

1.75 If A and B are independent events of a random experiment such that $P(A \cap B) = \frac{1}{5}$ and $P(\bar{A} \cap \bar{B}) = \frac{1}{4}$ then find $P(A)$.

Solution Given $P(A \cap B) = \frac{1}{5}$ and $P(\bar{A} \cap \bar{B}) = \frac{1}{4}$

Since A and B are independent,

$$P(A \cap B) = P(A) P(B) = \frac{1}{5}$$

and $P(\bar{A})P(\bar{B}) = \frac{1}{4}$

Let $P(A) = x$ and $P(B) = y$

Then

$$xy = \frac{1}{5}$$

$$P(\bar{A})P(\bar{B}) = [1 - P(A)][1 - P(B)] = (1 - x)(1 - y) = \frac{1}{4}$$

$$1 - x - y + xy = \frac{1}{4}$$

$$1 - x - y = \frac{1}{4} - \frac{1}{5} = \frac{1}{20}$$

$$x + y = \frac{19}{20} = 0.95$$

$$(x - y)^2 = (x + y)^2 - 4xy = \left(\frac{19}{20}\right)^2 - \frac{4}{5}$$

$$x - y = 0.32$$

$$\therefore x = 0.635$$

$$y = 0.315$$

$$\Rightarrow P(A) = 0.635$$

1.76 If A and B are independent events such that $P(B) = \frac{3}{5}$ $P(A \cup \bar{B}) = 0.75$ then find $P(A)$.

Solution Given $P(A \cup \bar{B}) = 0.75$ and $P(B) = 0.6$

$$\Rightarrow P(\bar{B}) = 0.4$$

$$P(A \cup \bar{B}) = P(A) + P(\bar{B}) - P(A \cap \bar{B})$$

$$= P(A) + P(\bar{B}) - P(A)P(\bar{B})$$

$$0.75 = P(A) + 0.4 - P(A)(0.4)$$

$$0.35 = P(A)[1 - 0.4]$$

$$P(A) = 0.583$$

1.77 If A and B are independent events associated with a random experiment then prove that

- $\bar{A}$ and B are independent events
- A and $\bar{B}$ are independent events
- $\bar{A}$ and $\bar{B}$ are independent events.

Solution Since A and B are independent events,

$$P(A \cap B) = P(A)P(B)$$

Consider the Venn diagram shown in Fig. 1.13

From the Venn diagram,

$$B = (\bar{A} \cap B) \cup (A \cap B)$$

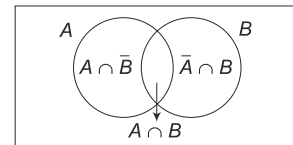


Fig. 1.13

$$\begin{aligned}
P(B) &= P\{(\bar{A} \cap B) \cup (A \cap B)\} \\
&= P(\bar{A} \cap B) + P(A \cap B) \\
&\Rightarrow P(\bar{A} \cap B) = P(B) - P(A \cap B)
\end{aligned}$$

Similarly,

$$\begin{aligned}
P(A \cap \bar{B}) &= P(A) - P(A \cap B) \\
&= P(A) - P(A) P(B) \\
&= P(A) [1 - P(B)] \\
&= P(A) P(\bar{B})
\end{aligned}$$

So, A and $\bar{B}$ are independent events.

We have

$$\begin{aligned}
A &= (A \cap \bar{B}) \cup (A \cap B) \\
P(A) &= P\{(A \cap \bar{B}) \cup (A \cap B)\} \\
&= P(A \cap \bar{B}) + P(A \cap B) \\
P(A \cap \bar{B}) &= P(A) - P(A \cap B) \\
P(A \cap \bar{B}) &= P(A) - P(A) P(B) \\
&= P(A) [1 - P(B)] = P(A) P(\bar{B})
\end{aligned}$$

$$(c) \quad \bar{A} \cap \bar{B} = \overline{A \cup B}$$

$$\begin{aligned}
P(\bar{A} \cap \bar{B}) &= P(\overline{A \cup B}) = 1 - P(A \cup B) \\
&= 1 - \{P(A) + P(B) - P(A \cap B)\} \\
&= 1 - P(A) - P(B) + P(A) P(B) \\
&= [1 - P(B)] - P(A) [1 - P(B)] \\
&= P(\bar{A}) P(\bar{B})
\end{aligned}$$

Therefore, $\bar{A}$ and $\bar{B}$ are also independent.

1.78 A bag contains 4 white, 6 red and 8 black balls. Four balls are drawn one by one with replacement. What is the probability that at least one is white?

Solution Let A_i be the event that the ball drawn in the i^{th} draw is white, $1 \leq i \leq 4$. Since the balls are drawn with replacement, the events A_1, A_2, A_3 and A_4 are independent.

$$P(A_1) = P(A_2) = P(A_3) = P(A_4) = \frac{4}{18} = \frac{2}{9}$$

The probability that at least one is white is

$$P(A_1 \cup A_2 \cup A_3 \cup A_4) = 1 - P(\bar{A}) P(\bar{A}_2) P(\bar{A}_3) P(\bar{A}_4)$$

$$P(\bar{A}) = 1 - P(A_1) = 1 - \frac{2}{9} = \frac{7}{9}$$

Similarly, $P(\bar{A}_2) = P(\bar{A}_3) = P(\bar{A}_4) = \frac{7}{9}$

$$P(A_1 \cup A_2 \cup A_3 \cup A_4) = 1 - \left(\frac{7}{9}\right)^4$$

1.79 What is the probability that the series circuit in Fig. 1.14 with three switches S_1, S_2, S_3 with probabilities $\frac{1}{2}, \frac{1}{4}, \frac{2}{3}$ respectively of functioning will work?

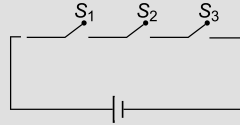


Fig. 1.14

Solution The probability that the switch S_1 works is

$$P(S_1) = \frac{1}{2}$$

Similarly, $P(S_2) = \frac{1}{4}$ and $P(S_3) = \frac{2}{3}$

Current flows through the circuit when all switches work. So the required probability is

$$\begin{aligned} P(S_1 \cap S_2 \cap S_3) &= P(S_1) P(S_2) P(S_3) \\ &= \frac{1}{2} \left(\frac{1}{4}\right) \left(\frac{2}{3}\right) = \frac{1}{12} \end{aligned}$$



Practice Problems

1.25 A dice is thrown thrice. Find the probability of getting an odd number at least once. (Ans. 7/8)

1.26 Given $P(A \cup B) = \frac{5}{6}$, $P(A \cap B) = \frac{1}{3}$ and $P(\bar{B}) = \frac{1}{2}$. Test whether A and B are independent or not (independent).

Solved Problems



1.80 The probabilities that a husband and a wife will be alive 25 years from now are 0.80 and 0.85 respectively. Find the probability that in 25 years (a) both will be alive, (b) neither will be alive, (c) at least one will be alive, and (d) only one will be alive.

Assume that the husband and the wife will be alive independently.

Solution Let A be the event that the husband is alive and B be the event that the wife is alive.

$$P(A) = 0.80, P(B) = 0.85$$

- (a) $P(\text{both alive}) = P(A \cap B) = P(A) P(B) = (0.80)(0.85) = 0.68$
- (b) $P(\text{neither alive}) = P(\bar{A} \cap \bar{B}) = P(\bar{A}) P(\bar{B}) = (1 - 0.80)(1 - 0.85) = 0.03$
- (c) $P(\text{at least one alive}) = P(A \cap \bar{B}) + P(\bar{A} \cap B) + P(A \cap B)$
 $= P(A) P(\bar{B}) + P(\bar{A}) P(B) + P(A) P(B)$
 $= (0.80)(0.15) + (0.20)(0.85) + (0.85)(0.80) = 0.97$
- (d) $P(\text{only one is alive}) = P(A) P(\bar{B}) + P(\bar{A}) P(B)$
 $= 0.80(0.15) + (0.20)0.85$
 $= 0.29$

1.81 A bag contains 6 white, 3 red and 9 black balls. Three balls are drawn one by one with replacement. What is the probability that at least one is white?

Solution Let E_i be the event of drawing a white ball in the i^{th} drawn when $1 \leq i \leq 3$. Since the ball is replaced after every draw, all events E_i , $1 \leq i \leq 3$ are independent. Also, $P(E_i) = \frac{6}{18} = \frac{1}{3}$.

The required probability

$$P(E_1 \cup E_2 \cup E_3) = 1 - P(\bar{E}_1) P(\bar{E}_2) P(\bar{E}_3)$$

$$= 1 - \left(\frac{2}{3}\right)^3$$

1.82 A and B alternately throw a pair of dice. A wins if he throws 6 before B throws 7, and B win if he throws 7 before A throws 6. If A begins, show that his chance of winning is $\frac{30}{61}$.

Solution The number of events in the sample space $= 6^2 = 36$

Let A be the event of throwing the dice whose sum is six. The possible choices are (1, 5), (5, 1), (2, 4), (4, 2) and (3, 3).

$$P(A) = \frac{5}{36}$$

Let B be the event of throwing the dice whose sum is seven.

The possible choices are

(1, 6), (6, 1), (2, 5), (5, 2), (3, 4), (4, 3)

$$P(B) = \frac{6}{36} = \frac{1}{6}$$

If A begins, he can win in any one of first, third, fifth trials. At the same time, B should not win in his trials 2, 4, 6 ...

Probability of A winning the game

$$= P(A) + P(\bar{A} \cap \bar{B} \cap A) + P(\bar{A} \cap \bar{B} \cap \bar{A} \cap \bar{B} \cap A) + \dots$$

$$= P(A) + P(\bar{A}) P(\bar{B}) P(A) + P(\bar{A}) P(\bar{B}) P(\bar{A}) P(\bar{B}) P(A) + \dots$$

$$= \frac{5}{36} + \frac{31}{36} \left(\frac{5}{6}\right) \left(\frac{5}{36}\right) + \frac{31}{36} \left(\frac{5}{6}\right) \left(\frac{31}{36}\right) \left(\frac{5}{6}\right) \left(\frac{5}{36}\right) + \dots$$

$$\begin{aligned}
&= \frac{5}{36} \left[1 + \frac{31 \times 5}{31 \times 6} + \left(\frac{31 \times 5}{36 \times 6} \right)^2 + \dots \right] \\
&= \frac{5}{36} \left[\frac{1}{1 - \frac{31 \times 5}{36 \times 6}} \right] = \frac{5}{36} \left[\frac{36 \times 6}{36 \times 6 - 31 \times 5} \right] = \frac{30}{61}
\end{aligned}$$

1.83 A pair of fair dice is rolled together till a sum of either 4 or 6 is obtained. Find the probability that 6 comes before 4.

Solution Let A denote the event that a sum of 6 occurs, B be the event that the sum of 4 occurs, and C be the event that neither a sum of 4 nor a sum of 6 occurs. We know $n(S) = 36$.

The possible ways of getting 4 are (1, 3), (3, 1) and (2, 2).

Therefore, $P(A) = \frac{3}{36} = \frac{1}{12}$

The possible ways of getting 6 are

(5, 1) (1, 5), (2, 4), (4, 2), (3, 3)

$$P(B) = \frac{5}{36}$$

$$P(C) = 1 - [P(A) + P(B)] = 1 - \left(\frac{1}{12} + \frac{5}{36} \right)$$

$$= \frac{28}{36} = \frac{7}{9}$$

The probability that 6 comes before 4

$$= P(B) + P(C) P(B) + P(C) P(C) P(B) + \dots$$

$$= P(B)[1 + P(C) + P^2(C) + \dots]$$

$$= \frac{P(B)}{1 - P(C)} = \frac{5/36}{1 - \frac{7}{9}} = \frac{5}{8}$$



TOTAL PROBABILITY THEOREM 1.14

The total probability theorem states that if an event A can occur in N mutually exclusive ways, E_1 or E_2 or ... or E_N then the probability of occurrence of A is given by

$$P(A) = P(E_1) P(A|E_1) + P(E_2) P(A|E_2) + \dots + P(E_N) P(A|E_N)$$

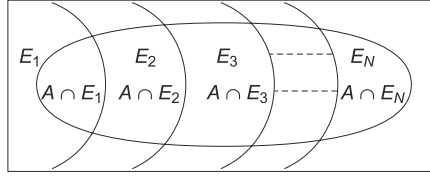
Proof Let S be the sample space $E_1, E_2, \dots, E_n$ are N mutually exclusive and exhaustive events.

These events satisfy

$$E_i \cap E_j = \phi, i \neq j = 1, 2, \dots, N$$

Also, the sample space

$$S = E_1 \cup E_2 \cup E_3 \cup \dots \cup E_N$$

**Fig. 1.15** Total probability theorem

$$\begin{aligned}
 A &= (A \cap E_1) \cup (A \cap E_2) \cup (A \cap E_3) \cup \dots \cup (A \cap E_N) \\
 P(A) &= P(A \cap E_1) + P(A \cap E_2) + P(A \cap E_3) + \dots + P(A \cap E_N) \\
 &= P(E_1) P(A | E_1) + P(E_2) P(A | E_2) + \dots + P(E_N) P(A | E_N) \\
 &= \sum_{i=1}^N P(E_i) P(A | E_i)
 \end{aligned} \tag{1.56}$$

**BAYES' THEOREM 1.15**

Consider an event 'A' of the sample space 'S':

Let $E_1, E_2, \dots, E_N$ be N mutually exclusive and exhaustive events associated with a random experiment. If A is any event which occurs with E_1 or E_2 or ... or E_N then

$$P(E_i | A) = \frac{P(E_i) P(A | E_i)}{\sum_{i=1}^N P(E_i) P(A | E_i)} \quad i = 1, 2, \dots \tag{1.57}$$

Proof From the total probability theorem, we have

$$P(A) = \sum_{i=1}^N P(E_i) P(A | E_i)$$

Using the multiplication theorem of probability, we have

$$\begin{aligned}
 P(A \cap E_i) &= P(A) P(E_i | A) \\
 P(E_i | A) &= \frac{P(A \cap E_i)}{P(A)} = \frac{P(E_i) P(A | E_i)}{P(A)} \\
 &= \frac{P(E_i) P(A | E_i)}{\sum_{i=1}^N P(E_i) P(A | E_i)}
 \end{aligned}$$

The probabilities, $P(E_i)$ and $P(E_i | A)$ are known as *priori* and *posteriori probabilities* respectively.

Solved Problems

1.84 A bag contains 5 red and 4 black balls. A second bag contains 3 red and 5 balls. One bag is selected at random. From the selected bag, one ball is drawn. Find the probability that ball drawn is black.

Solution The black ball can be obtained in two ways:

- (a) Selecting Bag 1 and then drawing a black ball from it
- (b) Selecting Bag 2 and then drawing a black ball from it

Let E_1 and E_2 be the events of selecting Bag 1 and Bag 2 respectively and B be the event of drawing a black ball.

$$P(E_1) = \frac{1}{2} \text{ and } P(E_2) = \frac{1}{2}$$

$$P(B | E_1) = \text{Probability of drawing a black ball when the first bag has been chosen} = \frac{4}{9}$$

Similarly,

$$P(B | E_2) = \frac{5}{8}$$

$$P(\text{black ball}) = P(B) = P(E_1) P(B | E_1) + P(E_2) P(B | E_2)$$

$$\begin{aligned} &= \frac{1}{2} \left(\frac{4}{9} \right) + \frac{1}{2} \left(\frac{5}{8} \right) \\ &= \frac{1}{2} \left[\frac{32 + 45}{72} \right] = \frac{77}{144} \end{aligned}$$

1.85 In a class, 60% of the students are boys and the remaining are girls. It is known that the probability of a boy getting distinction is 0.30 and that of girl getting distinction is 0.35. Find the probability that a student chosen at random will get distinction.

Solution Let E_1 and E_2 represent the events of choosing a boy and girl respectively, and A be the event of getting distinction by a student.

$$P(E_1) = 0.6, P(E_2) = 0.4$$

$$P(A | E_1) = 0.30, P(A | E_2) = 0.35$$

$$\begin{aligned} P(A) &= P(E_1) P(A | E_1) + P(E_2) P(A | E_2) \\ &= (0.6) (0.3) + (0.4) (0.35) = 0.32 \end{aligned}$$

1.86 A letter is known to have come from either TATANAGAR or CALCUTTA. On the envelope, two consecutive letters TA are visible. Find the probability that the letter came from CALCUTTA.

Solution Let E_1 be the event that the letter come from TATANAGAR and E_2 be the event that the letter come from CALCUTTA. Then $P(E_1) = P(E_2) = \frac{1}{2}$.

Let A denote the event that the two consecutive alphabets visible on the envelope are TA.

Now we take TA as a single letter and find the probability of Event A given the Event E_1 .

$$P(A | E_1) = \frac{2}{8}$$

$$\begin{array}{cccccccc} & 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ \textcircled{\text{TA}} & \text{T} & \text{A} & \text{N} & \text{A} & \text{G} & \text{A} & \text{R} & \end{array} \quad (1)$$

$$\begin{array}{cccccccc} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ T & A & \textcircled{TA} & N & A & G & A & R \end{array} \quad (2)$$

Similarly, $P(A | E_2) = \frac{1}{7}$ $\begin{array}{cccccccc} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ C & A & L & C & U & T & \textcircled{TA} \end{array}$ (3)

Probability that the letter has come from Calcutta can be obtained using Bayes' theorem

$$\begin{aligned} P(E_2 | A) &= \frac{P(E_2) P(A | E_2)}{P(E_1) P(A | E_1) + P(E_2) P(A | E_2)} \\ &= \frac{\frac{1}{2} \left(\frac{1}{7} \right)}{\frac{1}{2} \left(\frac{2}{8} \right) + \frac{1}{2} \left(\frac{1}{7} \right)} = \frac{1/14}{\frac{1}{8} + \frac{1}{14}} = \frac{4}{11} \end{aligned}$$



Practice Problem

1.27 A letter is known to have come either from LONDON or CLIFTON. On the postmark, only the two consecutive letters 'ON' are legible. What is the chance that it has come from LONDON? (Ans. 12/17)

Solved Problems



1.87 In a bolt factory, machines A , B and C manufacture 25, 35 and 40% of the total bolts respectively. Of their outputs, 5, 4 and 2 percent respectively are defective bolts. A bolt is drawn at random and found to be defective. What is the probability that the bolt came from machines A , B and C ?

Solution Let E_1 be the bolts manufactured by Machine A , E_2 be the bolts manufactured by Machine B , and E_3 be the bolts manufactured by Machine C .

Then $P(E_1) = \frac{25}{100} = \frac{1}{4}$; $P(E_2) = \frac{35}{100} = \frac{7}{20}$

$$P(E_3) = \frac{40}{100} = \frac{2}{5}$$

Let A be the event of drawing a defective bolt.

Then $P(A | E_1)$ = Probability that the bolt drawn is defective given the condition that it is manufactured by Machine $A = \frac{5}{100} = 0.05$

Similarly,

$$P(A | E_2) = \frac{4}{100} = 0.04$$

and $P(A | E_3) = \frac{2}{100} = 0.02$

A bolt is drawn randomly. The probability that the bolt came from Machine A can be obtained using Bayes' theorem:

$$\begin{aligned}
 P(E_1 | A) &= \frac{P(E_1) P(A | E_1)}{P(E_1) P(A | E_1) + P(E_2) P(A | E_2) + P(E_3) P(A | E_3)} \\
 &= \frac{\frac{1}{4} (0.05)}{\frac{1}{4} (0.05) + \frac{7}{20} (0.04) + (0.04) (0.02)} = \frac{0.0125}{0.0273} = 0.4578
 \end{aligned}$$

Similarly,

$$\begin{aligned}
 P(E_2 | A) &= \frac{P(E_2) P(A | E_2)}{P(E_1) P(A | E_1) + P(E_2) P(A | E_2) + P(E_3) P(A | E_3)} \\
 &= \frac{\frac{7}{20} (0.04)}{\frac{1}{4} (0.05) + \frac{7}{20} (0.04) + (0.04) (0.02)} = \frac{0.014}{0.0273} = 0.5128
 \end{aligned}$$

$$\begin{aligned}
 P(E_3 | A) &= \frac{P(E_3) P(A | E_3)}{P(E_1) P(A | E_1) + P(E_2) P(A | E_2) + P(E_3) P(A | E_3)} \\
 &= \frac{(0.04) (0.02)}{\frac{1}{4} (0.05) + \frac{7}{20} (0.04) + (0.04) (0.02)} = \frac{0.0008}{0.0273} = 0.0293
 \end{aligned}$$

1.88 A binary computer communication channel has the following error probabilities:

$$P(R_1 | S_0) = 0.2, P(R_0 | S_1) = 0.06$$

where

$$S_0 = \{ \text{'0' sent} \}, S_1 = \{ \text{'1' sent} \}$$

$$R_0 = \{ \text{'0' received} \}, R_1 = \{ \text{'1' received} \}$$

Suppose that 0 is sent with a probability of 0.8, find

- The probability that '1' is received
- The probability that '1' was sent given that 1 is received
- The probability that '0' was sent given that '0' is received

Solution

- (a) Given $P(R_1 | S_0) = 0.2$; $P(R_0 | S_1) = 0.06$; $P(S_0) = 0.8$

$$P(R_0 | S_0) = P(\bar{R}_1 | S_0) = 1 - P(R_1 | S_0) = 1 - 0.2 = 0.80$$

$$P(R_1 | S_1) = 1 - P(R_0 | S_1) = 1 - 0.06 = 0.94$$

$$P(S_1) = 1 - P(S_0) = 1 - 0.8 = 0.2$$

$$P(R_1) = P(R_1 | S_0) P(S_0) + P(R_1 | S_1) P(S_1)$$

$$= (0.2) (0.8) + (0.94) (0.2) = 0.348$$

- (b) Using Bayes' rule,

$$P(S_1 | R_1) = \frac{P(R_1 | S_1) P(S_1)}{P(R_1)} = \frac{0.94 (0.2)}{0.348} = 0.54$$

$$\begin{aligned}
 \text{(c)} \quad P(S_0 | R_0) &= \frac{P(R_0 | S_0) P(S_0)}{P(R_0 | S_0) P(S_0) + P(R_0 | S_1) P(S_1)} \\
 &= \frac{(0.80)(0.80)}{(0.80)(0.80) + (0.06)(0.02)} = \frac{0.64}{0.652} = 0.9816
 \end{aligned}$$

1.89 A pack contains 4 white and 2 green pencils, another contains 3 white and 5 green pencils. If one pencil is drawn from each pack, find the probability that

- (a) Both are white, and
 (b) One is white and another is green.

Solution Let the event of selecting the Pack 1 be P_1 and selecting the Pack 2 is P_2 .

Given that Pack 1 contains 4 white (W) and two green (G) pencils and Pack 2 contains 3 white (W) and five green (G) pencils.

- (a) $P(\text{selecting both white}) = P(\text{selecting } W \text{ from Pack 1}) \times P(\text{selecting } W \text{ from Pack 2})$

$$= \frac{4}{6} \times \frac{3}{8} = 0.25$$

- (b) $P(\text{selecting } W \text{ and } G) = P(\text{selecting } W \text{ from Pack 1}) \times P(\text{selecting } G \text{ from Pack 2}) + (\text{selecting } W \text{ from Pack 2}) \times P(\text{selecting } G \text{ from Pack 1})$

$$= \frac{4}{6} \times \frac{5}{8} + \frac{2}{6} \times \frac{3}{8} = 0.542$$

1.90 What is the probability of drawing 3 white and 4 green balls from a bag that contains 5 white and 6 green balls, if 7 balls are drawn simultaneously at random?

Solution Let W be the event of drawing a white ball and G be the event of drawing green ball. The bag contains 5W and 6G balls. The 7 balls are drawn at random.

Out of total of 11 balls, a sample of 7 balls can be chosen in $\binom{11}{7}$ ways. Again, 3 balls can be drawn $\binom{5}{3}$

ways and 4 green balls can be drawn from six green balls $\binom{6}{4}$ ways.

$$\text{The required probability is } \frac{\binom{5}{3} \binom{6}{4}}{\binom{11}{7}} = \frac{10(15)}{330} = \frac{5}{11}$$

1.91 An integer is chosen at random from the first 100 positive integers. What is the probability that the integer chosen is divisible by 6 or 8?

Solution Let A be the event of the number being divisible by 6 and B be the event of a number being divisible by 8.

Between 1 and 100, the total number of integers divisible by 6 is 16. That is $n(A) = 16$

Similarly, $n(B) = 12$

Since 24, 48, 72 and 96 are divisible by both 6 and 8, therefore, $n(A \cap B) = 4$

$$\begin{aligned} n(A \cup B) &= n(A) + n(B) - n(A \cap B) \\ &= 16 + 12 - 4 = 24 \end{aligned}$$

$$P(A \cup B) = \frac{n(A \cup B)}{\text{Total integers}} = \frac{24}{100} = 0.24$$

1.92 From a pack of 52 cards, 1 card is drawn at random. Find the probability of getting a queen.

Solution A pack of 52 cards contains 4 queens.

A queen can be chosen in $\binom{4}{1}$ ways.

The number of ways that a card can be selected from 52 cards is $\binom{52}{1} = 52$

$$P(\text{getting a queen}) = \frac{\binom{4}{1}}{\binom{52}{1}} = \frac{4}{52} = \frac{1}{13}$$

1.93 What is the probability that out of 6 cards, taken from a full pack, 3 will be black and 3 will be red?

Solution A full pack contains 52 cards and six cards can be selected from the pack in $\binom{52}{6}$ ways.

The number of black cards are 26 and three cards can be selected in $\binom{26}{3}$ ways. Similarly, the three red cards can be selected in $\binom{26}{3}$ ways. The total number of ways of choosing 3 black and 3 red cards =

$$\binom{26}{3} \binom{26}{3}.$$

$$\text{The probability} = \frac{\binom{26}{3} \binom{26}{3}}{\binom{52}{6}}$$

1.94 A coin is tossed three times. Find the probability that a head shows up at least once.

Solution Let A be the event that a head shows up at least once.

Since the coin is tossed three items, there are $2^3 = 8$ possible outcomes.

$$P(A) = 1 - P(\text{all tails})$$

$$1 - \frac{1}{8} = \frac{7}{8}$$

1.95 From a pack of 52 cards, Event 'A' is defined as drawing a king card, Event B defined as drawing a jack or queen card, Event C is defined as drawing a heart card. Then find out which of them are statistically independent and dependent events.

Solution Given that A be an event of drawing a king card, B be an event of drawing a jack or queen card, and C be an event of drawing a heart card from a deck of 52 playing cards.

$$P(A) = \frac{4}{52} = \frac{1}{13}; \quad P(B) = \frac{8}{52} = \frac{2}{13}; \quad P(C) = \frac{13}{52}$$

$$(a) \quad \text{Now } P(A)P(B) = \frac{4}{52} \times \frac{8}{52} = \frac{2}{169} = 0.0118$$

But $P(A \cap B) = 0$, since A and B are disjoint

$$\therefore P(A \cap B) \neq P(A)P(B)$$

A and B are not statistically independent.

$$(b) \quad \text{Now } P(B)P(C) = \frac{2}{13} \times \frac{13}{52} = \frac{1}{26} = 0.0384$$

and $P(B \cap C) = \frac{2}{52} = \frac{1}{26}$, since there is one jack of hearts card and one queen of hearts card.

$$\therefore P(B \cap C) = P(B)P(C)$$

So B and C are statistically independent.

$$(c) \quad P(A)P(C) = \frac{1}{13} \times \frac{13}{52} = \frac{1}{52} = 0.01923$$

and $P(A \cap C) = \frac{1}{52}$, since there is one king of hearts card,

$$\therefore P(A \cap C) = P(A)P(C)$$

So A and C are statistically independent.

1.96 Find the probability of 3 coins falling all heads of when tossed simultaneously.

Solution When three coins are tossed simultaneously, the total number of possible outcomes is 8. Out of these eight outcomes, all heads occur only once. Therefore, required probability is

$$= \frac{1}{8} = 0.125$$

1.97 Determine the probability of the card being either a red or a king when one card is drawn from a regular deck of 52 cards.

Solution Let A be the event of drawing a red card and B be the event of drawing a king card.

Since there are 26 red cards,

$$P(A) = \frac{26}{52} = \frac{1}{2} = 0.5$$

Since there are 4 king cards,

$$P(B) = \frac{4}{52} = \frac{1}{13}$$

There are two red king cards (one is king of diamonds and the other is king of hearts)

$$\therefore P(A \cap B) = \frac{2}{52} = \frac{1}{26} = 0.03846$$

$$\begin{aligned} \therefore P(\text{either red or king}) &= P(A \cup B) = P(A) + P(B) - P(A \cap B) \\ &= \frac{1}{2} + \frac{1}{13} - \frac{2}{52} = \frac{28}{52} \end{aligned}$$

1.98 In a box, there are 100 resistors having resistances and tolerances as shown in the table. Let a resistor be selected from the box and assume each resistor has the same likelihood of being chosen. For the three events: A as “draw a 47 Ω resistor”, B as “draw a resistor with 5% tolerance,” and C as “draw a 100 Ω resistor”, calculate the $P(A \cap B)$, $P(A \cap C)$, $P(B \cap C)$.

Number of resistors in a box having resistance and tolerance.

Resistance	Tolerance		
	5%	10%	Total
22	10	14	24
47	28	16	44
100	24	8	32
Total	62	38	100

Solution Total number of resistors = 100

Given A is an event of drawing a 47 Ω resistance. The number of 47 Ω resistors is equal to 44.

$$\text{Therefore, } P(A) = \frac{44}{100} = 0.44$$

$$P(A \cap B) = P(47 \Omega \text{ and } 5\% \text{ tolerance})$$

$$= \frac{28}{100} = 0.28$$

$$P(A \cap C) = P(47 \Omega \text{ and } 100 \Omega) = 0$$

$$P(B \cap C) = P(5\% \cap 100 \Omega) = \frac{24}{100} = 0.24$$

1.99 The coefficients a , b and c of a quadratic equation $ax^2 + bx + c = 0$ are determined by throwing a dice three times. Find the probability that (a) the roots are real, and (b) the roots are complex.

Solution The coefficients can be any value from 1 to 6. For roots to be real, the relation $ac \leq \frac{b^2}{4}$ must be satisfied.

If $b = 1$, no value of a and c satisfies the inequality.

If $b = 2$ then $ac \leq 1$. Then only $a = 1$ and $c = 1$ satisfies the inequality.

If $b = 3$ then $ac \leq \frac{9}{4}$. In this case, (a, c) values can be $(1, 1), (1, 2), (2, 1)$: a total of 3 ways.

If $b = 4$ then $ac \leq 4$. Then (a, c) values can be $(1, 1), (1, 2), (1, 3), (1, 4), (2, 1), (2, 2), (3, 1), (4, 1)$: a total of 8 ways.

If $b = 5$ then $ac \leq \frac{25}{4}$, then (a, c) values can be $(1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (1, 6), (2, 1), (2, 2), (2, 3), (3, 1), (3, 2), (4, 1), (5, 1), (6, 1)$: a total of 14 ways.

If $b = 6$ then $ac \leq 9$. For this case, (a, c) values can be $(1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (1, 6), (2, 1), (2, 2), (2, 3), (2, 4), (3, 1), (3, 2), (3, 3), (4, 1), (4, 2), (5, 1), (6, 1)$: a total of 17 ways.

B	2	3	4	5	6	
Number of ways	1	3	8	14	17	43 total

The total number of ways 43. $n(s) = 6^3 = 216$

$$P(\text{real roots}) = \frac{43}{216}$$

$$P(\text{complex roots}) = \frac{173}{216}$$

1.100 What is the chance that a leap year selected at random will contain 53 Sundays?

Solution A leap year contains 52 weeks and 2 days.

The two days can be any of the following combinations: (a) Sunday and Monday, (b) Monday and Tuesday, (c) Tuesday and Wednesday, (d) Wednesday and Thursday, (e) Thursday and Friday, (f) Friday and Saturday, (g) Saturday and Sunday.

Let A be the event that there are 53 Sundays in a leap year. Out of the possible 7 combinations, the combinations (a) and (g) have Sunday. That is, $n(S) = 7$ and $n(A) = 2$. Therefore, the probability of A is

$$P(A) = \frac{\text{Number of possible outcomes for event } A}{\text{Number of possible outcomes for space } S}$$

$$= \frac{n(A)}{n(S)} = \frac{2}{7}$$

1.101 Four dice are thrown simultaneously. What is the probability that the sum of numbers is exactly 20?

Solution There are six possible outcomes for a single throw of a dice. When four dice are thrown simultaneously, the number of possible outcomes is 64. The following combinations produce the sum as exactly 20.

- (a) Three dice showing 6 and one dice showing 2, (6, 6, 6, 2), (2, 6, 6, 6), (6, 2, 6, 6), (6, 6, 2, 6): a total of 4 outcomes
- (b) Two dice showing 6, one showing 5 and the other 3, that is (6, 6, 5, 3)

The possible outcomes are $\frac{4!}{2!} = 12$

The third possible way is two dice with 6 and two dice with 4. That is, (6, 6, 4, 4). The total number of possible outcomes is $\frac{4!}{2!2!} = 6$

The fourth possible way is two dice with 5, one with six and the other with 4. That is, (5, 5, 6, 4). The total number of possible outcomes is $\frac{4!}{2!} = 12$

The last possible way is with all four dice with 5. That is, (5, 5, 5, 5). There is only one outcome.

$$\frac{4 + 12 + 6 + 12 + 1}{6^4}$$

Hence, the required probability is $\frac{35}{1296}$

1.102 An urn contains 6 white and 9 black balls. If 4 are to be randomly selected without replacement, what is the probability that the first 2 selected are white and the last 3 black?

Solution The total possible way of selecting 4 balls from 15 balls is $\binom{15}{4}$.

The two white balls can be selected in $\binom{6}{2}$ ways and two black balls can be selected in $\binom{9}{2}$ ways.

Therefore, the required probability is

$$\frac{\binom{6}{2} \binom{9}{2}}{\binom{15}{4}} = \frac{15(36)}{15 \times 14 \times 13 \times 3} = \frac{6}{91}$$



Practice Problem

1.28 When two balls are drawn in succession with replacement from a box consisting of 5 white and 7 black balls find the probability that (a) both are white, (b) both are black and (c) the first is white and the second is black.

Solved Problems



1.103 If two dice are thrown, what is the probability of obtaining a sum of at least 10?

Solution The total number of possible outcomes $6^2 = 36$

The number of favorable outcomes for the event “a sum of at least 10” are

$$\{(4, 6), (6, 4), (5, 5), (5, 6), (6, 5), (6, 6)\} = 6$$

$$\therefore \text{the required probability is } = \frac{6}{36} = \frac{1}{6}$$

1.104 When two dice are thrown, find the probability of getting the sums of 10 or 11.

Solution Let A be the event of getting a sum of 10 and B be the event of getting a sum of 11.

The total number of outcomes is $6^2 = 36$

The possible outcomes are $\{(5, 5), (4, 6), (6, 4)\}$

$$\text{Therefore, } P(A) = \frac{3}{36} = 0.0833$$

The possible outcomes for B is $\{(6, 5), (5, 6)\}$

$$P(B) = \frac{2}{36} = 0.0556$$

Since the sets A and B are independent,

$$P(\text{sum of 10 or 11}) = \frac{3}{36} + \frac{2}{36} = \frac{5}{36} = 0.1389$$

1.105 A shipment of components consists of three identical boxes. One box contains 2000 components of which 25% are defective, the second box has 5000 components of which 20% are defective and the third box contains 2000 components of which 600 are defective. A box is selected at random and a component is removed at random from a box. What is the probability that this component is defective? What is the probability that it came from the second box?

Solution Let the events B_1 , B_2 and B_3 are selecting the boxes.

The probability of selecting a box is

$$\therefore P(B_1) = P(B_2) = P(B_3) = \frac{1}{3}$$

Let D be the event of selecting a “defective component”.

Given 25% components are defective in B_1 , therefore,

$$\therefore P(D/B_1) = \frac{25}{100} = 0.25$$

20% components are defective in B_2 . Therefore,

$$P(D/B_2) = \frac{20}{100} = 0.2$$

600 components are defective from 2000 in B_3 . Therefore,

$$P(D/B_3) = \frac{600}{2000} = \frac{3}{10} = 0.3$$

Using the total probability theorem, the probability of getting a defective component is

$$\begin{aligned} P(D) &= P(D/B_1) P(B_1) + P(D/B_2) P(B_2) + P(D/B_3) P(B_3) \\ &= \left(0.25 \cdot \frac{1}{3} + 0.2 \cdot \frac{1}{3} + 0.3 \cdot \frac{1}{3} \right) = 0.25 \end{aligned}$$

$$P(D) = 0.25$$

The probability that the defective component came from the second box is

$$P(B_2/D) = \frac{P(D/B_2)P(B_2)}{P(D)} \text{ (using Bayes' theorem)}$$

$$P(B_2) = \frac{0.2 \times \frac{1}{3}}{0.25} = \frac{20}{75} = 0.267$$

1.106 Three boxes of identical appearance contain two coins each. In one box both are gold, in the second both are silver, and in the third box one is silver and the other is a gold coin. Suppose that a box is selected at random and further that a coin in that box is selected at random. If this coin proves to be gold, what is the probability that the other coin is also gold?

Solution Given 3 boxes, B_1 , B_2 and B_3 are selection randomly.

$$\therefore P(B_1) = P(B_2) = P(B_3) = \frac{1}{3} = 0.33$$

Let the event G is a select a gold coin. Then

$$P(G/B_1) = \frac{2}{2} = 1$$

$$P(G/B_2) = \frac{0}{2} = 0$$

$$P(G/B_3) = \frac{1}{2} = 0.5$$

Probability of selecting a gold coin is

$$P(G) = P(G/B_1) P(B_1) + P(G/B_2) P(B_2) + P(G/B_3) P(B_3)$$

If another coin is also selected, the probability that this coin is also gold is

$$\begin{aligned}
 P(GG) &= P(G/B_1) P(G/GB_1) P(B_1) + P(G/B_2) P(G/GB_2) P(B_2) + P(G/B_3) P(G/GB_3) P(B_3) \\
 &= 1 \times 1 \times \frac{1}{3} + 0 \times 0 \times \frac{1}{3} + \frac{1}{2} \times 0 \times \frac{1}{3} \\
 P(GG) &= \frac{1}{3} + 0 + 0 = \frac{1}{3} = 0.33
 \end{aligned}$$

1.107 In 20 items, 12 are defective and 8 are non-defective. If these items are chosen at random, what is the probability that

- (a) The first two items inspected are defective?
- (b) The first two items inspected are non-defective?
- (c) One is defective and the other is non-defective?

Solution

Number of defective items = 12

Number of non-defective items = 8

Total number of items = 20

Two items can be selected in $\binom{20}{2}$ ways

$$P(\text{both inspected are defective}) = \frac{\binom{12}{2}}{\binom{20}{2}} = \frac{66}{190} = \frac{33}{95}$$

$$P(\text{both inspected are non-defective}) = \frac{\binom{8}{2}}{\binom{20}{2}} = \frac{28}{190} = \frac{14}{95}$$

$$P(\text{one is defective and the other is non-defective}) = \frac{\binom{8}{1} \binom{12}{1}}{\binom{20}{2}} = \frac{96}{190} = \frac{48}{95}$$

1.108 An urn contains four red, three green, five blue and three white balls. What is the probability of selecting a sample size of eight balls containing two red, two green, three blue and one white ball?

Solution

Total number of balls = 4 + 3 + 5 + 3 = 15

The number of combinations for selecting 8 balls from 15 balls is $\binom{15}{8}$.

The number of combinations for selecting two red balls from 4 balls is $\binom{4}{2}$.

The number of combinations for selecting two green balls from three green balls is $\binom{3}{2}$.

The number of combination for selecting three blue balls from five blue balls is $\binom{5}{3}$.

The number of combinations for selecting one white ball from three white balls is $\binom{3}{1}$.

$$\text{The required probability} = \frac{\binom{4}{2} \binom{3}{2} \binom{5}{3} \binom{3}{1}}{\binom{15}{8}} = \frac{6(3)(10)(3)}{6435} = 0.0839$$

1.109 The probability of Ramesh scoring 90% of marks in English, Hindi and Sanskrit are 0.2, 0.3 and 0.5 respectively. If the grades can be regarded as independent events, find the probability that he get 90% in (a) all subjects, (b) in none of the subjects, and (c) exactly in two subjects

Solution Let E_1 , E_2 , and E_3 are the events of Ramesh scoring 90% in English, Hindi and Sanskrit respectively.

$$P(E_1) = 0.2, P(E_2) = 0.3; P(E_3) = 0.5$$

- (a) $P(\text{scoring 90\% in all subjects}) = P(E_1 \cap E_2 \cap E_3)$
 $= P(E_1)P(E_2)P(E_3)$ ($\because$ since E_1, E_2 and E_3 are independent)
 $= (0.2)(0.3)(0.5)$
- (b) $P(\text{scoring 90\% in none of the subjects})$
 $= P(\bar{E}_1 \cap \bar{E}_2 \cap \bar{E}_3)$
 $= P(\bar{E}_1) P(\bar{E}_2) P(\bar{E}_3)$
 $= (0.8)(0.7)(0.5) = 0.28$
- (c) $P(\text{scoring 90\% in exactly two subjects})$
 $= P(E_1 \cap E_2 \cap \bar{E}_3) + P(E_1 \cap \bar{E}_2 \cap E_3) + P(\bar{E}_1 \cap E_2 \cap E_3)$
 $= P(E_1) P(E_2) P(\bar{E}_3) + P(E_1) P(\bar{E}_2) P(E_3) + P(\bar{E}_1) P(E_2) P(E_3)$
 $= (0.2)(0.3)(0.5) + (0.2)(0.7)(0.5) + (0.8)(0.3)(0.2)$
 $= 0.148$

1.110 A , B and C shot to hit a target. If A hits the target 3 times in 5 trials, B hits it 2 times in 3 trials, and C hits 5 times in 8 trials, what is the probability that the target is hit by at least two persons?

Solution Let E_1 , E_2 and E_3 be the events that A , B and C hit the targets respectively.

$$P(E_1) = \frac{3}{5}; P(E_2) = \frac{2}{3}; P(E_3) = \frac{5}{8}$$

The target can be hit at least by two persons in the following ways.

- (a) A and B hit the target and C does not hit
- (b) A and C hit the target and B does not hit
- (c) B and C hit the target and A does not hit
- (d) A , B and C hit the target

The required probability P

$$= P(E_1 \cap E_2 \cap \bar{E}_3) + P(E_1 \cap \bar{E}_2 \cap E_3) + P(\bar{E}_1 \cap E_2 \cap E_3) + P(E_1 \cap E_2 \cap E_3)$$

Since all events are mutually exclusive,

$$\begin{aligned} P &= P(E_1)P(E_2)P(\bar{E}_3) + P(E_1)P(\bar{E}_2)P(E_3) + P(\bar{E}_1)P(E_2)P(E_3) + P(E_1)P(E_2)P(E_3) \\ &= \left(\frac{3}{5}\right)\left(\frac{2}{3}\right)\left(\frac{3}{8}\right) + \frac{3}{5}\left(\frac{1}{3}\right)\left(\frac{5}{8}\right) + \left(\frac{2}{5}\right)\left(\frac{2}{3}\right)\left(\frac{5}{8}\right) + \left(\frac{3}{5}\right)\left(\frac{2}{3}\right)\left(\frac{5}{8}\right) = 0.6916 \end{aligned}$$

1.111 If one card is drawn at random from a pack of cards then show that getting an ace and getting a heart are independent events.

Solution Let A be the event of getting an ace and B be the event of getting a heart.

$$P[A] = P[\text{getting an ace}] = \frac{\binom{4}{1}}{\binom{52}{1}} = \frac{4}{52} = \frac{1}{13}$$

$$P[B] = P[\text{getting a heart}] = \frac{\binom{13}{1}}{\binom{52}{1}} = \frac{13}{52} = \frac{1}{4}$$

$$P[A \cap B] = \frac{\binom{1}{1}}{\binom{52}{1}} = \frac{1}{52} \text{ which is equal to } P(A)P(B).$$

Since $P[A \cap B] = P[A]P[B]$,

The events A and B are independent of each other.

1.112 A technician has two electronic part cabinets with drawn arrangements as below:

3-drawer cabinets	4-drawer cabinets
<i>pnp</i> –transistor	<i>pnp</i> transistor
<i>nnp</i> -transistor	<i>pnp</i> -transistor
<i>nnp</i> -transistor	<i>nnp</i> -transistor
	<i>nnp</i> -transistor

The technician selects one cabinet at random and withdraws a transistor from one of the drawers. Assume that each cabinet and each drawer within the selected cabinet is equally likely to be selected.

- What is the probability that a *pnp*-transistor is chosen?
- Given that an *nnp* transistor is chosen, what is the probability that it comes from the 3-drawer cabinet?

Solution By total probability theorem,

$$P(pnp\text{-transistor}) = P(pnp|3\text{-dr}) P(3\text{-dr}) + P(pnp|4\text{-dr}) P(4\text{-dr})$$

Probability of selecting each draw is $\frac{1}{2}$

$$P(pnp\text{-transistor}) = \frac{1}{3} \cdot \frac{1}{2} + \frac{1}{2} \cdot \frac{1}{2} = \frac{5}{12}$$

- Using Bayes' rule

$$P(3\text{-dr} | npn) = \frac{P(npn|3dr) (P(3-dr))}{P(npn)}$$

$$P(npn) = 1 - P(pnp) = 1 - \frac{5}{12} = \frac{7}{12}$$

$$P(3\text{-dr} | npn) = \frac{\frac{2}{3} \left(\frac{1}{2} \right)}{\frac{7}{12}} = \frac{1}{3} \cdot \frac{12}{7} = \frac{4}{7}$$

1.113 A problem is given to 3 students whose chances of solving it are $\frac{1}{2}$, $\frac{1}{3}$ and $\frac{1}{4}$ respectively. What is the probability that the

- Problem will be solved?
- Exactly two of them will solve the problem.

Solution Let S_1 , S_2 and S_3 be the events that the students A , B and C solve the problem. The $P(S_1) = \frac{1}{2}$, $P(S_2) = \frac{1}{3}$ and $P(S_3) = \frac{1}{4}$

- The problem will be solved if at least one of them solves the problem. That is,

$$P(S_1 \cup S_2 \cup S_3) = 1 - P(\bar{S}_1 \cap \bar{S}_2 \cap \bar{S}_3)$$

$$\begin{aligned}
 &= 1 - P(\bar{S}_1) P(\bar{S}_2) P(\bar{S}_3) \\
 &= 1 - \left[\left(1 - \frac{1}{2}\right) \left(1 - \frac{1}{3}\right) \left(1 - \frac{1}{4}\right) \right] \\
 &= 1 - \left[\left(\frac{1}{2} \cdot \frac{2}{3} \cdot \frac{3}{4}\right) \right] = \frac{3}{4}
 \end{aligned}$$

(b) $P(\text{Exactly two of them solve the problem})$

$$\begin{aligned}
 &= P(S_1 \cap S_2 \cap \bar{S}_3) + P(S_1 \cap \bar{S}_2 \cap S_3) + P(\bar{S}_1 \cap S_2 \cap S_3) \\
 &= P(S_1) P(S_2) P(\bar{S}_3) + P(S_1) P(\bar{S}_2) P(S_3) + P(\bar{S}_1) P(S_2) P(S_3) \\
 &= \left(\frac{1}{2}\right) \left(\frac{1}{3}\right) \left(\frac{3}{4}\right) + \frac{1}{2} \left(\frac{2}{3}\right) \left(\frac{1}{4}\right) + \frac{1}{2} \left(\frac{1}{3}\right) \left(\frac{1}{4}\right) \\
 &= \frac{1}{8} + \frac{1}{12} + \frac{1}{24} = \frac{3+2+1}{24} = \frac{6}{24} = \frac{1}{4}
 \end{aligned}$$

1.114 India plays two matches each with the West Indies and Australia. In any match, the probabilities of India getting 0, 1 and 2 points are 0.45, 0.05 and 0.5 respectively. Assuming that the outcomes are independent, find the probability of India getting at least 7 points.

Solution Let India get two points if it wins, 1 point if the match ends in a tie and '0' points if it loses. India gets at least seven points when it wins all matches or it wins three match with one tie. Therefore,

$$\begin{aligned}
 P(\text{India getting at least seven points}) &= \text{Probability of winning all matches} + \text{Probability of winning three matches and one tie} \\
 &= P(W) P(W) P(W) P(W) + P(W) P(T) P(W) P(W) + P(W) P(W) P(W) P(T) + P(T) P(W) P(W) P(W) + P(W) P(W) P(T) P(W) \\
 &= (0.5)^4 + 4(0.05) \times (0.5)^3 \\
 &= (0.5)^3 [0.5 + 4 \times 0.05] \\
 &= (0.5)^3 (0.7) \\
 &= 0.0875
 \end{aligned}$$



Practice Problem

1.29 An unbiased dice is tossed until a number greater than 4 appears. What is the probability that an even number of tosses is needed for this event? (Ans. 2/5)

Solved Problems



1.115 A boy is throwing stones at a target. The probability of hitting the target at any trial is 1/2. What is the probability of hitting the target 5th time at the 10th throw.

Solution Probability of hitting the target is $\frac{1}{2}$

$$\begin{aligned}
 &P(\text{hitting the target 5}^{\text{th}} \text{ time at the } 10^{\text{th}} \text{ throw}) \\
 &= P(\text{hitting the target 4 times in 9 throws}) \times P(\text{hitting the target in } 10^{\text{th}} \text{ row}) \\
 &= \binom{9}{4} \left(\frac{1}{2}\right)^4 \left(\frac{1}{2}\right)^5 \cdot \frac{1}{2}
 \end{aligned}$$

1.116 When we roll a pair of balanced dice, what are the probability of getting (a) 6, (b) 9, (c) 6, or 9, (d) 4, (e) 2 or 12?

Solution The sample space S is

$$S = \left\{ \begin{array}{l} (1,1) (1,2) (1,3) (1,4) (1,5) (1,6) \\ (2,1) (2,2) (2,3) (2,4) (2,5) (2,6) \\ (3,1) (3,2) (3,3) (3,4) (3,5) (3,6) \\ (4,1) (4,2) (4,3) (4,4) (4,5) (4,6) \\ (5,1) (5,2) (5,3) (5,4) (5,5) (5,6) \\ (6,1) (6,2) (6,3) (6,4) (6,5) (6,6) \end{array} \right\}$$

$$n(S) = 36$$

- (a) Let A be the event when the sum is 6.
Then

$$A = \{(1, 5) (2, 4) (3, 3) (4, 2) (5, 1)\}$$

$$n(A) = 5$$

$$P(A) = \frac{n(A)}{n(S)} = \frac{5}{36}$$

- (b) Let the event B be a sum of 9.
Then

$$B = \{(3, 6) (4, 5) (5, 4) (6, 3)\}$$

$$n(B) = 4$$

$$P(B) = \frac{n(B)}{n(S)} = \frac{4}{36} = \frac{1}{9}$$

- (c) Let C be the event of getting 'sum of 6 or sum of 9'

$$\begin{aligned}
 P(C) &= P(A) + P(B) \\
 &= \frac{5}{36} + \frac{1}{9} = \frac{9}{36} = \frac{1}{4}
 \end{aligned}$$

- (d) Let D be the event of getting a 4,

$$D = \{(1, 3), (2, 2) (3, 1)\}$$

$$n(D) = 3$$

$$P(D) = \frac{3}{36} = \frac{1}{12}$$

(e) Let E be the event of getting 12,

$$E = \{(6, 6)\}$$

$$P(E) = \frac{1}{36}$$

$$P(E) + P(D) = \frac{1}{12} + \frac{1}{36} = \frac{1}{9}$$

1.117 Show that the chances of throwing six with 4, 3, or 2 dice respectively as 1 : 6 : 18.

Solution With four dice, the total number possible combinations is equal to 6^4 .

The number of possible combinations to get six are {1113, 1122, 1311, 3111, 2211, 2112, 1131, 1221, 2121, 1212}

$$\text{So the probability } P_1 = \frac{10}{6^4}$$

With three dice, the total number of possible combinations is equal to 6^3

The number of possible combinations to get six are {114, 141, 411, 123, 132, 312, 213, 321, 231, 222}

$$\text{The probability } P_2 = \frac{10}{6^3}$$

The total number of possible combinations with two dice is 36.

With two dice, the possible combinations to get 6 are {15, 51, 24, 42, 33}

$$\text{The probability } P_3 = \frac{5}{6^2}$$

$$\begin{aligned} P_1 : P_2 : P_3 &= \frac{10}{6^4} : \frac{10}{6^3} : \frac{5}{6^2} \\ &= 10 : 60 : 180 \\ &= 1 : 6 : 18 \end{aligned}$$

1.118 The probability that Ramu passes mathematics is 0.6 and physics is 0.4 and in both is 0.2. What is the probability that he passes in (a) at least in one subject? (b) neither of the subjects? (c) one of the subjects? (d) not in mathematics?

Solution Let A be the event that Ramu passes mathematics, B be the event that Ramu passes physics:

$$P(A) = 0.6; P(B) = 0.4; P(A \cap B) = 0.2$$

(a) $P(\text{at least in one subjects})$

$$= P(A \cup B) = P(A) + P(B) - (P(A \cap B))$$

$$= 0.6 + 0.4 - 0.2 = 0.8$$

(b) $P(\text{neither of the subjects})$

$$= P(\bar{A} \cap \bar{B}) = P(\overline{A \cup B}) = 1 - P(A \cup B) \\ = 1 - 0.8 = 0.2$$

(c) $P(\text{one of the subjects})$

$$= P(\bar{A} \cap B) + P(A \cap \bar{B}) \\ = P(\bar{A})P(B) + P(A)P(\bar{B}) \\ = [1 - P(A)]P(B) + P(A)[1 - P(B)] \\ = (1 - 0.6)(0.4) + (0.6)(1 - 0.4) \\ = 0.16 + 0.36 = 0.52$$

(d) $P(\text{not in mathematics})$

$$P(\bar{A}) = 1 - P(A) \\ = 1 - 0.6 = 0.4$$

1.119 A power plant will shut down if systems S_1 and S_2 or S_1 and S_3 fail simultaneously. The systems S_1 , S_2 and S_3 are independent and their probabilities of failure are 0.02, 0.015, 0.025 respectively.

(a) What is the probability that the plant will shut down?

(b) What is the probability that the plant stays on line given that S_1 failed.

Solution The plant will shut down if S_1 and S_2 or S_1 and S_3 fail simultaneously. The following table shows the cases when the plant will shut down:

F = Fail

W = Working

S_1	S_2	S_3	Plant
F	F	F	Shut down
F	F	W	Shut down
F	W	F	Shut down

$$P(\text{shut down}) = (0.02)(0.015)(0.025) + (0.02)(0.015)(1 - 0.025) + (0.02)(1 - 0.015)(0.025) \\ = 0.0007925$$

$$(b) \quad P(\text{plant is working} \mid S_1 \text{ failed}) = \frac{P(W \cap S_1)}{P(S_1)} = \frac{P(W \mid S_1)P(S_1)}{P(S_1)}$$

If S_1 is failed then the plant works when S_2 and S_3 work

$$P(W \mid S_1) = (1 - 0.015)(1 - 0.025) \\ = 0.985(0.975) \\ = 0.968$$

1.120 Two switches S_1 and S_2 have respectively 95% and 90% chances of working. Find the probability that in the circuit shown in Fig. 1.16 current will flow.

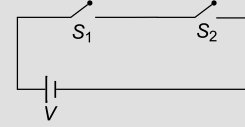


Fig. 1.16

Solution Consider the events S_1 and S_2 correspond to the working of switches S_1 and S_2 respectively. Given $P(S_1) = 0.95$ and $P(S_2) = 0.9$.

The current flows in the circuit if both switches work together. Therefore, the required probability $P(S_1 \cap S_2)$

Since S_1 and S_2 are independent events,

$$P(S_1 \cap S_2) = P(S_1) P(S_2) = (0.95)(0.9) = 0.855$$

1.121 An electric system has four switches arranged as shown in Fig. 1.17. The switches operate independently of one another with a probability p . Find the probability that the current flows through the circuit.

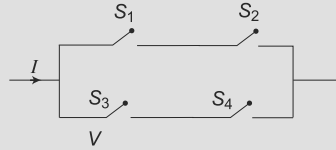


Fig. 1.17

Solution Let $P(S_1)$ be the probability that a switch S_1 will work given $P(S_1) = p$.

The current flows through the circuit when both S_1 and S_2 are closed or when both S_3 and S_4 are closed. So the required probability is

$$P[(S_1 \cap S_2) \cup (S_3 \cap S_4)] = P(S_1 \cap S_2) + P(S_3 \cap S_4) - P[(S_1 \cap S_2) \cap (S_3 \cap S_4)]$$

Since $P(S_i)$ are independent of each other,

$$\begin{aligned} P[S_1 \cap S_2 \cup (S_3 \cap S_4)] &= P(S_1) P(S_2) + P(S_3) P(S_4) - P(S_1) P(S_2) P(S_3) P(S_4) \\ &= P \cdot P + P \cdot P - P \cdot P \cdot P \cdot P \\ &= 2P^2 - P^4 \end{aligned}$$

1.122 Consider the probability that each relay being closed in the circuit shown in Fig. 1.18 is p . If all relays functional independently what is the probability that current flows from A to B?

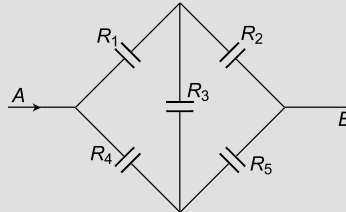


Fig. 1.18

Solution The current flows for the following cases:

- (a) Both R_1 and R_2 function
- (b) R_1, R_3 and R_5 function
- (c) R_4 and R_5 function
- (d) R_4, R_3 and R_2 function

Let E_1 be the event when both R_1 and R_2 function.

E_2 be the event when R_1, R_3 and R_5 function.

E_3 be the event when R_4 and R_5 function.

E_4 be the event when R_4, R_3 and R_2 function.

Now the probability that current flows from A to B is given by

$$\begin{aligned} P(E_1 \cup E_2 \cup E_3 \cup E_4) &= P(E_1) + P(E_2) + P(E_3) + P(E_4) - \{P(E_1 \cap E_2) + P(E_2 \cap E_3) + P(E_3 \cap E_4) + \\ &\quad P(E_1 \cap E_3) + P(E_2 \cap E_4) + P(E_1 \cap E_4)\} + P(E_1 \cap E_2 \cap E_3) + P(E_2 \cap E_3 \cap E_4) \\ &\quad + P(E_1 \cap E_3 \cap E_4) + P(E_1 \cap E_2 \cap E_4) - P(E_1 \cap E_2 \cap E_3 \cap E_4) \end{aligned}$$

Note that $P(E_1) = P(A_1 \cap A_2) = P(A_1) P(A_2) = p^2$

Similarly, $P(E_2) = p^3 = P(E_4)$

$$P(E_3) = p^2$$

$$P(E_1 \cap E_2 \cap E_3) = P(E_2 \cap E_3 \cap E_4) = P(E_4 \cap E_3 \cap E_1) = p^5$$

$$\begin{aligned} P(E_1 \cup E_2 \cup E_3 \cup E_4) &= p^2 + p^3 + p^2 + p^3 - \{p^4 + p^4 + p^4 + p^4 + p^5\} + 4p^5 - p^5 \\ &= 2p^2 + 2p^3 + 5p^4 + 2p^5 \end{aligned}$$



Practice Problems

1.30 In Fig. 1.19 assume that the probability of each relay being closed is p . If all relays function independently, what is the probability that current flows from A to B .

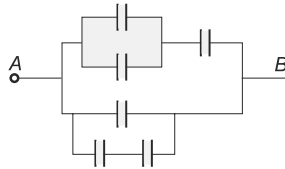


Fig. 1.19

$$(Ans. p + 3p^2 - 4p^3 - p^4 + 3p^5 - p^6)$$

1.31 Each of two persons toss three fair coins. What is the probability they obtain the same number of heads?

$$(Ans. 5/16)$$

1.32 In successive rolls of a pair of fair dice, what is the probability of getting 2 sevens before 6 even numbers.

$$(Ans. 0.555)$$



BERNOULLI TRIALS 1.16

In experiments like tossing a coin, there are only two possible outcomes; head or tail. Similarly, there are events with two possible outcomes like success or failure, hit or miss of the target. If the events are statistically independent that result in a success with probability p and failure with probability $(1 - p)$ then such events are called Bernoulli trials. Now we have to find the probability that an event A occurs exactly k times out of n independent trials.

Let $B_1 = \{A \text{ occurs events } k \text{ times out of } n \text{ trials}\}$

If A occurs ' k ' times out of n trials, then the event A occurs $(n - k)$ times. Since all events are independent,

$$P(B_1) = \underbrace{P(A) P(A) \cdots P(A)}_{k \text{ times}} \underbrace{P(\bar{A}) \cdots P(\bar{A})}_{(n-k) \text{ times}}$$

$$P(A) = p; P(\bar{A}) = 1 - p$$

$$P(B_1) = p^k (1 - p)^{n-k}$$

Since A occurs exactly k times in any order out of n trials,

$$P(B) = \binom{n}{k} p^k (1 - p)^{n-k}$$

1.123 A typist makes an error while typing a letter 0.25% of the time. What is the probability of one error in 10 letters?

Solution Let A be the event of error that occurs exactly once out of 10 letters.

$$\begin{aligned} P(A) &= \binom{10}{1} (0.025)^1 (1 - 0.025)^{10-1} \\ &= 0.199 \end{aligned}$$

1.124 A test consists of 10 multiple-choice questions, with 4 choices. Among the choices, only one is correct and only one can be chosen. A student selects the choices at random. What is the probability that he has 1, 2, 3 correct answers?

Solution Probability of selecting one answer among four choices is equal to $\frac{1}{4}$. That is $p = \frac{1}{4}$

$$\begin{aligned} P(\text{no correct answer}) &= \binom{10}{0} \left(\frac{1}{4}\right)^0 \left(1 - \frac{1}{4}\right)^{10-0} \\ &= 0.0563 \end{aligned}$$

$$\begin{aligned} P(1 \text{ correct answer}) &= \binom{10}{1} \left(\frac{1}{4}\right)^1 \left(1 - \frac{1}{4}\right)^{10-1} \\ &= 0.1877 \end{aligned}$$

$$P(2 \text{ correct answers}) = \binom{10}{2} \left(\frac{1}{4}\right)^2 \left(1 - \frac{1}{4}\right)^{10-2}$$

$$= 0.2816$$

$$P(3 \text{ correct answers}) = \binom{10}{3} \left(\frac{1}{4}\right)^3 \left(1 - \frac{1}{4}\right)^{10-3}$$

$$= 0.25$$

REVIEW QUESTIONS



- Define the following terms:
(i) Sample space (ii) event (iii) trial (iv) sample point
- State the axioms of probability.
- Discuss the following in brief:
(i) Permutation (ii) combination (iii) probability
- Give examples for finite sample space and infinite sample space.
- Define: (a) Continuous sample space (ii) Discrete sample space.
- Define and explain the following with an examples:
(i) Equally likely events (ii) Exhaustive events (iii) Mutually exclusive events
- When are two events said to be mutually exclusive? Explain with an example.
- Distinguish between mutually exclusive events and independent events.
- Give the classical definition of probability.
- Give the axiomatic definition of probability.
- If A and B are independent events, prove that the events $\bar{A}$ and $\bar{B}$, $\bar{A}$ and B , and A and $\bar{B}$ are also independent.
- State and prove the addition theorem of probability.
- Discuss joint and conditional probability.
- Define conditional probability and mention its properties.
- Explain multiplication law of probability.
- Explain about total probability.
State and prove Bayes' theorem of probability.
- Define the terms 'independent events'. State the conditions for independent of (i) any two events, and (ii) any three events A_1 , B and C .
- Define Bernoulli trial.



EXERCISES

Problems

- How many words can be formed using all letters of the word PERMUTATION using each letter exactly once?
(Ans. 39916800)
- There are three copies, each of 4 different books. In how many number of ways can they be arranged on a shelf?
(Ans: $\frac{12!}{(3!)^4}$)

3. From 5 consonant, 4 vowels and 3 capital letter, how many number of words beginning with a capital letter and containing 3 consonants and 2 vowels can formed? (Ans. 816000)
4. How many number of diagonals can be drawn by joining the vertices of an octagon? (Ans. 20)
5. A group consists of 9 married couple. In how many ways can a mixed-doubles game be arranged if no husband and wife play in the same game? (Ans. 1512)
6. Surya has 7 friends. In how many ways can be invite one or more of them to a party. (Ans. 127)
7. In how many ways can a committee of 5 members be selected from 6 men and 5 women consisting of 3 men and 2 women? (Ans. 200)
8. In how many ways can we select a cricket team from 16 players in which 5 players can bowl? Each cricket team must include 2 bowlers. (Ans. 550)
9. How many chords can be drawn through 21 points on a circle? (Ans. 210)
10. What is the probability that a leap year selected at random will contain 53 Saturday? (Ans. 2/7)
11. An integer is chosen from 3 to 16. What is the probability that it is prime? (Ans. 5/14)
12. Two dice are thrown together. Find the probability that the total is 9. (Ans. 1/9)
13. A number is selected at random from 10 to 40. Find the probability that it is divisible by 5.
14. A bag contains 6 red , 5 white, 4 black balls. What is the probability that two balls drawn are red and black? (Ans. 8/35)
15. Prove that $P(\bar{A}) = 1 - P(A)$.
16. Two events A and B are such that $P(A) = \frac{1}{5}$, $P(A/B) = \frac{1}{3}$ and $P(B/A) = \frac{1}{3}$. Find $P(A/\bar{B})$.
17. If A and B are two independent events, show that $P(A/\bar{B}) = 1 - P(\bar{A})P(\bar{B})$.
18. If $P(A \cup B) = \frac{5}{6}$, $P(A \cap B) = \frac{1}{3}$ and $P(\bar{B}) = \frac{1}{2}$, Show that A and B are independent.
19. Given that $P(A) = P(B) = P(C) = \frac{1}{3}$, $P(A \cap B) = P(C \cap B) = 0$ and $P(A \cap C) = \frac{1}{6}$. Evaluate $P(A \cup B \cup C)$:
20. If the probability for A to fail in an examination is 0.2 and that of B is 0.4 then what is the probability that either A or B fails? (Ans. 0.52)
21. From a pack of 52 cards, 1 card is drawn at random. Find the probability of getting a queen. (Ans. 1/13)
22. If $P(A) = 0.4$; $P(B) = 0.7$ and $P(A \cap B) = 0.3$, Find $P(\bar{A} \cap \bar{B})$ and $P(\bar{A} \cup \bar{B})$.
23. A bag contains 12 red balls and 6 white balls. Two balls are drawn one by one without replacement. What is the probability that 60th are white?
24. A box contains 100 tickets numbered 1, 2, ..., 100. Two tickets are chosen at random. It is given that the maximum number on the two chosen tickets is not more than 10. What is the probability that the maximum number on them is 5? (Ans. 1/9)
25. Six boys and six girls sit in a row at random. Find the probability that
(i) six girls sit together (ii) the boys and girls sit alternately. (Ans. (i) 1/132 (ii) 1/142)
26. Ne ticket is selected at random from 100 tickets numbered 00, 01, 02, 98, 99. If X and Y denote sum and product of the digits respectively then find $P(X = \text{even}, Y = \text{odd})$ (Ans. 2/19)
27. There are 10 stations between A and B . A train is to stop at three of these 10 stations. What is the probability that two of these station are consecutive? (Ans. 7/15)
28. Two fair dice are rolled. What is the conditional probability the first are lands on 6 given that the sum of the dice is 11?

29. Five different objects are distributed among 3 persons at random. What is the probability that each person receives at least one objects? (Ans. 50/81)
30. There are 5 letters and 5 addressed envelopes. If the letters are put at random in the envelope, what is the probability that at least one letter may be placed in a wrongly addressed envelope? (Ans. 119/120)
31. In a class, 60% are boys and the rest are girls. 50% of boys and 35% of girls known how to solve a problem. If a student is selected at random and given that the student can solve the problem, What is the probability that the student is a girl? (Ans. 1/4)
32. Raju can hit a target in 2 out of 5 shots and Rithesh can hit the target in 3 out of 4 shots. What is the probability that the target gets hits when both try? (Ans. 17/20)
33. Sita can solve a problem with probability of $3/4$, and Gita can solve the problem with a probability of $5/7$. Find the probability that the problem will be solved by at least one of them. (Ans. 16/21)
34. Find the chances of throwing ten with four dice. (Ans. 5/81)
35. From a bag containing 4 white and 6 black balls, 2 balls are drawn at random. If the balls are drawn one after other without replacement, find the probability that
(i) both balls are white
(ii) both balls are black (Ans. (i) 2/15, (ii) 13)
36. Two students attempt to write a program. Their chances of writing the program successfully are $1/8$ and $1/12$ and the chance of making a common error is $1/1001$. Find the chance that the program is correctly written. (Ans. 13/14)
37. The contents of urns I, II and III are as follows:
(i) 1 white, 2 black and 3 red balls
(ii) 2 white, 1 black and 1 red ball
(iii) 4 white, 5 black and 3 red balls
One urn is chosen at random and two balls are drawn. They happen to be white and red. What is the probability that they came from urns I, II and III? (Ans. 0.2797, 0.466, 0.254)
38. A box contains 5 red and 4 white balls. Two balls are drawn successively from the box without replacement and it is noted that the second one is white. What is the probability that the first is also white? (Ans. 3/8)
39. A urn contains 10 white and 3 black balls. Another urn contains 3 white and 5 black balls. Two balls are drawn at random from the first urn and placed in the second urn and then one ball is taken at random from the latter. What is the probability that it is a white ball? (Ans. 59/130)
40. One integer is chosen at random from the numbers 1, 2, 3, ..., 100. What is the probability that the chosen number is divisible by
(i) 6 or 8 and
(ii) 6 or 8 or both? (Ans. (i) 1/5, (ii) 6/25)
41. Gita goes to office either by car, scooter or bus with probabilities being $\frac{1}{7}, \frac{4}{7}, \frac{2}{7}$, respectively. The probabilities that she reaches office late, if she takes a car, scooter, bus is $\frac{2}{9}, \frac{4}{9}$ and $\frac{1}{3}$ respectively. Given that she reached office in time, what is the probability that she traveled by a car?
42. The probability that a doctor A diagnoses a disease correctly is 0.6. The probability that a patient will die by his treatment after correct diagnosis is 0.4 and the probability of death after wrong diagnosis is 0.7. A patient of Doctor A who had the disease died. What is the probability that his disease was diagnosed correctly? (Ans. 6/13)

43. In a village, a shop distributes only two newspapers A and B . 25% of the village population reads A and 20% reads B , While 8% reads both A and B . It is known that 30% of those who read A and not B look into advertisements and 40% of those who read B but not A look into advertisements while 50% of those who read both A and B look into advertisements. What percentage of the population reads an advertisement? (Ans. 13.9%)
44. Two balls are drawn at random with replacement from a box containing 10 black and 8 red balls. Find the probability that (i) both the balls are red, and (ii) the first balls is black and second is red. (Ans. 16/81, 20/81)
45. The probability that Ravi passes physics is $\frac{2}{3}$ and the probability that he passes both physics and chemistry is $\frac{14}{45}$. The probability that he passes at least one test is $\frac{4}{5}$. What is the probability that he passes chemistry? (Ans. $\frac{4}{9}$)
46. In a single throw of two dice, what is the probability of obtaining a sum of at least 10? (Ans. $\frac{1}{6}$)
47. A couple has two children. Find the probability that both the boys, if it is known that (i) one of the children is a boy, and (ii) the older child is a boy? (Ans. $\frac{1}{3}; \frac{1}{2}$)
48. A fair coin and an unbiased dice are tossed. Let A be the event that a head appears on the coin and B be the event of 3 on the dice. Check whether A and B are independent events or not. (Ans. Independent events)
49. Raju speaks the truth in 60% of cases and Rajesh in 90% of cases. What is the percentage of cases in which they are likely to contradict each other? (Ans. 42%)
50. A man takes a step forward with probability of 0.4 and backwards with probability of 0.6. Find the probability that at the end of eleven steps, he is one step away from the starting point. (Ans. 0.368)

Multiple-Choice Questions

1. $P(A \cap B) + P(A \cap \bar{B}) =$
 (a) $P(A)$ (b) $P(A \cup B)$ (c) $P(B)$ (d) $P(A \cup B)$
2. If A and B are any two events, the probability that exactly one of them occurs is
 (a) $P(A) + P(B) - 2P(A \cap B)$ (b) $P(A) + P(B) - P(A \cup B)$
 (c) $P(A \cap \bar{B}) + P(\bar{A} \cap B)$ (d) $P(\bar{A}) + P(\bar{B})$
3. If two events A and B are such that $P(\bar{A}) = 0.3$, $P(B) = 0.4$ and $P(A \cap B) = 0.5$ then $P(B/A \cap \bar{B}) =$
 (a) $\frac{1}{3}$ (b) $\frac{1}{4}$ (c) $\frac{1}{6}$ (d) $\frac{1}{8}$
5. In two events $P(A \cup B) = P(A) = \frac{6}{5}$; $P(B) = \frac{2}{3}$ then A and B are
 (a) independent (b) mutually exhaustive
 (c) mutually exclusive (d) dependent
5. Let A and B be two events such that $\overline{P(A \cup B)} = \frac{1}{6}$; $P(A \cup B) = \frac{1}{4}$ and $P(\bar{A}) = \frac{1}{4}$ where $\bar{A}$ stands for complement of the event A . The events A and B are
 (a) mutually exclusive and independent (b) independent but not equally likely
 (c) equally likely (d) mutually exclusive

6. If A and B are independent events such that $P(B) = \frac{2}{7}$, $P(A \cup \bar{B}) = 0.8$ then $P(A)$
- (a) 0.1 (b) 0.2 (c) 0.3 (d) 0.4
7. A and B each throw a dice. The probability that A 's throw is not greater than B 's
- (a) $\frac{1}{2}$ (b) $\frac{1}{6}$ (c) $\frac{7}{12}$ (d) $\frac{5}{12}$
8. India and Pakistan play a 5-match one-day series. The probability that India is ahead 2–1 after 3 matches is
- (a) $\frac{3}{5}$ (b) $\left(\frac{1}{2}\right)^2$ (c) $\left(\frac{1}{4}\right)^3$ (d) None of these
9. Let A, B, C be pair-wise independent events with $P(C) > 0$ and $P(A \cap B \cap C) = 0$. Then $P(\bar{A} \cap \bar{B}/C)$ is equal to
- (a) $P(\bar{A}) + P(\bar{B})$ (b) $P(A \cap B)$ (c) $P(A) + P(B)$ (d) $P(\bar{A}) - P(B)$
10. If A and B are mutually exclusive events with $P(B) \neq 1$, then $P(A/\bar{B}) =$
- (a) $\frac{P(A)}{1 - P(B)}$ (b) $\frac{P(A)}{P(\bar{B})}$ (c) $\frac{P(A)}{P(B)}$ (d) $\frac{1 - P(B)}{P(A)}$
11. If A and B are two events such that $P(A) = 0.3$, $P(B) = 0.6$ and $P(B/A) = 0.5$, then $P(A/B) =$
- (a) $\frac{1}{4}$ (b) $\frac{1}{2}$ (c) $\frac{1}{8}$ (d) $\frac{1}{6}$
12. 12 persons are to be seated around a round table. If two particular persons among them are not to be side by side, the total number of arrangements is
- (a) $10!$ (b) $2(10!)$ (c) $8(10!)$ (d) $9(10!)$
13. How many different words can be formed by jumbling the letter in the words MISSISSIPPI in which no two S are adjacent?
- (a) $7 \binom{6}{4} \binom{8}{4}$ (b) $6 \binom{8}{4} \binom{6}{4}$ (c) $8 \binom{8}{4} \binom{6}{4}$ (d) $9 \binom{6}{4} \binom{8}{4}$
14. The total number of four-digit odd numbers that can be formed using 0, 1, 2, 3, 5, 7 are
- (a) 186 (b) 144 (c) 288 (d) 72
15. How many different nine-digit numbers can be from the number 22 23 55 888 by rearranging its digits so that the odd digits occupy even positions?
- (a) 60 (b) 90 (c) 120 (d) 30
16. A fair coin is tossed repeatedly. If the tail appears on the first four tosses then the probability of head appearing on the first toss equals
- (a) $\frac{1}{4}$ (b) $\frac{1}{2}$ (c) $\frac{1}{11}$ (d) $\frac{2}{9}$
17. A box B_1 contains 1 white ball, 3 red balls and 2 black balls. Another box B_2 contains 2 white balls, 3 red balls and 4 black balls. A third box B_3 contains 3 white balls, 4 red balls and 5 black balls. If 1 ball is drawn from each of the three boxes B_1, B_2 and B_3 , the probability that all 3 drawn balls are of the same colour is

- (a) $\frac{110}{648}$ (b) $\frac{82}{648}$ (c) $\frac{164}{648}$ (d) $\frac{180}{648}$

18. A problem in mathematics is given to three students A , B and C and their probabilities of solving the problem are $\frac{1}{2}$, $\frac{1}{3}$, and $\frac{1}{4}$ respectively. The probability that the problem is solved is

(a) $\frac{2}{3}$ (b) $\frac{1}{4}$ (c) $\frac{1}{3}$ (d) $\frac{3}{4}$

19. The number ways in which 5 boys and 5 girls can sit in a row so that all girls sit together is

(a) 43200 (b) 86400 (c) 14400 (d) 15600

20. Three girls are tossed together. The probability that there are exactly two heads is

(a) $\frac{1}{8}$ (b) $\frac{2}{8}$ (c) $\frac{3}{8}$ (d) $\frac{1}{2}$

21. Two events A and B are such that $P(A) = \frac{1}{3}$; $P(B) = \frac{3}{4}$; $P(A \cup B) = \frac{11}{12}$. The value of $P(A/B)$ is

(a) $\frac{1}{9}$ (b) $\frac{1}{7}$ (c) $\frac{1}{8}$ (d) $\frac{1}{6}$

22. Two unbiased dice are thrown and the different between the number of sports turned up is noted. the probability that the difference between numbers is equal to 4 is

(a) $1/9$ (b) $2/9$ (c) $1/3$ (d) 186

23. A man rolls a dice until he gets an even number. The probability that he gets 4 in the last throw is

(a) $\frac{1}{3}$ (b) $\frac{1}{4}$ (c) $\frac{1}{5}$ (d) $\frac{1}{7}$

24. A couple has two children. The probability that both are girls if the eldest is a girl is

(a) $\frac{1}{4}$ (b) $\frac{1}{3}$ (c) $\frac{2}{3}$ (d) $\frac{1}{2}$

25. Two fair dice are rolled. The conditional probability that at least one lands on 6 given that the dice land on different numbers is

(a) $\frac{1}{2}$ (b) $\frac{1}{4}$ (c) $\frac{1}{3}$ (d) $\frac{1}{5}$

26. A dice will be rolled 5 times. The probability that "3" will show up exactly twice is

(a) 0.16 (b) 0.2 (c) 0.32 (d) 0.52

27. A typist makes an error while typing a letter 0.3% of the time. The probability of no error in 10 letters is

(a) 0.82 (b) 0.97 (c) 0.58 (d) 0.65

28. A coin is tossed three times. The probability that head shows up at least once is

(a) $\frac{1}{8}$ (b) $\frac{1}{2}$ (c) $\frac{7}{8}$ (d) $\frac{5}{8}$

29. A dice is rolled 4 times. The probability of 6 coming up at least once is

(a) 0.52 (b) 0.71 (c) 0.42 (d) 0.3

30. A dice is rolled twice and the sum of the number appearing on them is observed to be 7. The conditional probability that the number 2 has appeared at least once is

(a) $\frac{1}{4}$ (b) $\frac{2}{3}$ (c) $\frac{1}{3}$ (d) $\frac{1}{6}$

Answers to Multiple-Choice Questions

1. (a)	2. (a)	3. (b)	4. (c)	5. (b)	6. (c)	7. (c)	8. (d)	9. (d)	10. (a)
11. (a)	12. (d)	13. (a)	14. (b)	15. (a)	16. (b)	17. (b)	18. (d)	19. (b)	20. (c)
21. (b)	22. (a)	23. (a)	24. (d)	25. (c)	26. (a)	27. (b)	28. (c)	29. (a)	30. (c)

2

THE RANDOM VARIABLE



INTRODUCTION 2.1

In the previous chapter, we studied the concept of probability space that completely describes the outcome of a random experiment. We also observed that in many of the examples of random experiments, the result was not a numerical quantity, but in descriptive form. For example, in experiment of tossing a coin, the outcome is head or tail. Similarly, in classifying a manufactured item we used categories like 'defective' and non-defective'. In some experiments, the description of outcomes are sufficient. However in many cases, it is required to record the outcome of the experiment as a number. Even in the experiments where the outcome is descriptive, we assign a number to each outcome of the experiment. For example, in the experiment of tossing a coin, we can assign a number to each outcome. We can assign zero to tail and one to head. The set of all such real numbers associated with an experiment is known as a *random variable*. In this chapter, we will study about the concepts of random variable.



THE RANDOM VARIABLE 2.2

Consider a random experiment E with sample space S . Let s be any outcome of the experiment. Then we can write $s \in S$. We are interested in assigning a real number for every s . Let us consider a function X that assigns real number $X(s)$ to every s . Then $X(s)$ is called a random variable. Thus, the random variable $X(s)$ maps every point in the sample space to a real value.

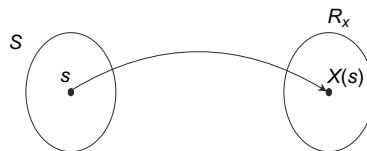


Fig. 2.1 Mapping of a point in the sample space to a real value

The set of all possible values of X is known as range space and denoted by R_X . That is, the random variable X maps the entire sample S to another sample space R_X .

Example 1 In a coin-tossing experiment, the sample space is heads and tails. Now consider a random variable $X(s)$ that maps the sample space to a real value of as follows:

$$X(\text{heads}) = 1; X(\text{tails}) = 0$$

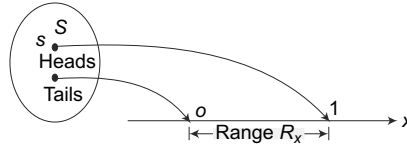


Fig. 2.2 Sample space of coin-tossing experiment mapping to real value

Example 2 Consider another experiment in which we toss two coins.

Then the sample space $S = \{HH, TH, HT, TT\}$

Now define a random variable X to denote the number of heads obtained in the two tosses.

Then $X(HH) = 2$; $X(HT) = X(TH) = 1$ and $X(TT) = 0$. Now the range space $R_X = \{x: x = 0, 1, 2\}$.

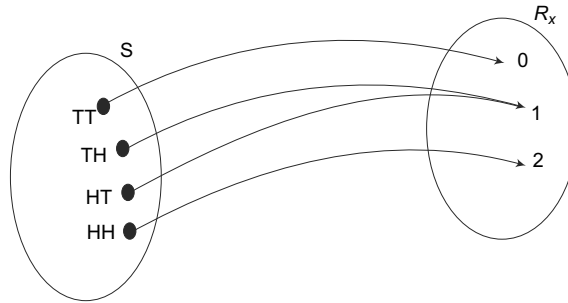


Fig. 2.3 Sample space of two coin experiment mapping to real value

Example 3 Consider an experiment of testing the lifespan of a bulb. Let the lifespan be given in hours denoted by X . Since the manufacturer cannot tell with certainty about the lifespan of a particular bulb, X takes a value which is random. Note that X is a non-negative random variable that takes a value $x \geq 0$.

After considering all the above examples, we can define the random variable as follows.

Definition A random variable is a function that assigns a real number for all the outcomes in the sample space of a random experiment.

The random variable is denoted by uppercase letter X and the measurement value of the random variable is denoted by a lowercase letter x .

REVIEW QUESTIONS



1. Define a random variable and give the concept of random variable.
2. Define and give the concept of random variable.

2.2.1 Assigning Probability to a Random Variable

We know that for each outcome in the sample space S ; a probability may be associated. When we assign a real number for each outcome, the probability associated with the outcome will be assigned to the real number. That is, we assign probability to events associated with R_X in terms of probabilities defined over S .

Consider the experiment discussed in Example 2. The sample space $S = \{TT, HT, TH, HH\}$. Since all events are equally likely, $P(TT) = P(HT) = P(TH) = P(HH) = \frac{1}{4}$. Let X be a random variable that gives the number of heads in each outcome. Then $X(TT) = 0$; $X(HT) = X(TH) = 1$; $X(HH) = 2$. The range $R_X = \{0, 1, 2\}$. Since $P(HT) = P(TH) = \frac{1}{4}$; $P(HT, TH) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$. If we consider the event $(X = 0)$ then we have

$P(X = 0) = \frac{1}{4}$. The event $(X = 1)$ is equivalent to the event (TH, HT) . Therefore, $P(X = 1) = \frac{1}{2}$. Similarly,

$$P(X = 2) = P(HH) = \frac{1}{4}$$

The random variable and the probabilities assigned are shown in Table 2.1.

Table 2.1 Random variable and assigned probabilities

X	0	1	2
$P(X)$	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{4}$

Once the probabilities associated with various outcomes in the range space R_X have been determined, we shall often ignore the original space S .

Solved Problems



2.1 The sample space for an experiment is $S = \{0, 2, 4, 6\}$. List all the possible values of the following random variable

(a) $X = 2s$ (b) $X = 2s^2 - 1$ (c) $X = \frac{1}{2s + 1}$

Solution Given: $S = \{0, 2, 4, 6\}$

(a) $X = 2s = \{0, 4, 8, 12\}$

(b) $X = 2s^2 - 1$; $X = \{-1, 7, 31, 71\}$

(c) $X = \frac{1}{2s + 1}$; $X = \left\{1, \frac{1}{5}, \frac{1}{9}, \frac{1}{13}\right\}$

2.2 In an experiment of rolling a dice and flipping a coin, the random variable X is chosen such that

- (a) A coin heads (H) outcome corresponds to positive values to X that are equal to the numbers that show upon the dice and
- (b) A coin tails (T) outcome corresponds to negative values of X that are equal in magnitude to twice the number shown on the dice. Map the elements of random variable X into points on the real line and explain.

Solution In an experiment of rolling a dice and flipping a coin, the sample space is

$$S = \{(H, 1), (H, 2), (H, 3), (H, 4), (H, 5), (H, 6), (T, 1), (T, 2), (T, 3), (T, 4), (T, 5), (T, 6)\}$$

This sample space is mapped onto the real axis using a random variable X such that when the coin shows heads, the random variable X takes the positive value that shows upon the dice. Therefore,

$$(H, 1) = 1; (H, 2) = 2; (H, 3) = 3; (H, 4) = 4; (H, 5) = 5 \text{ and } (H, 6) = 6$$

Similarly, when the coin shows tails, the random variable takes a value equal to the negative of twice the number the shows upon the dice. Therefore,

$$(T, 1) = -2; (T, 2) = -4; (T, 3) = -6; (T, 4) = -8; (T, 5) = -10 \text{ and } (T, 6) = -12$$

The mapping is shown in Fig. 2.4.

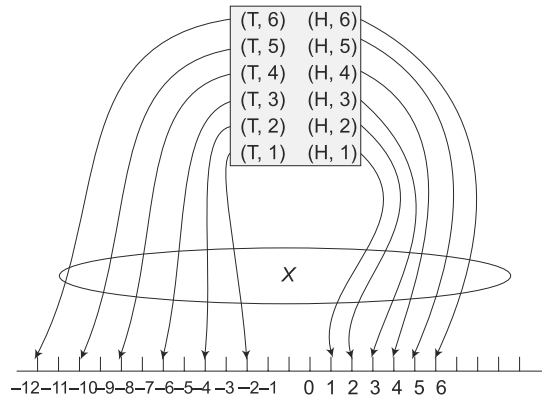


Fig. 2.4 Mapping of a sample space to real numbers

2.3 In an experiment, the pointer on a wheel of chance is spun. The possible outcomes are the numbers from 0 to 12 marked on the wheel. The sample space consists of the numbers in the set $\{0 < s \leq 12\}$ and if the random variable is defined as $X = X(s) = s^2$. Map the elements of random variable on the real line and explain.

Solution In the experiment, the possible outcomes are numbers given by

$$\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$$

The random variable X is given by $X = X(s) = s^2$.

The sample space is $0 < s \leq 12$.

Therefore,

$$X = \{1, 4, 9, 16, 25, 36, 49, 64, 81, 100, 121, 144\}$$

The mapping is shown in Fig. 2.5.

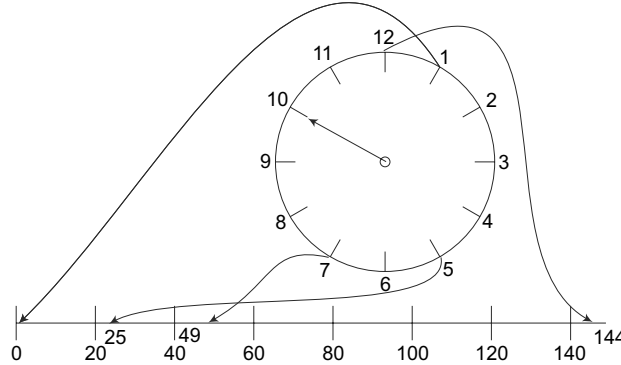


Fig. 2.5 Mapping of a sample space of experiment given in solved problem 2.3

2.2.2 Events Defined by Random Variable

Let X be a random variable and x be a fixed real value. Let us consider an event in which the random variable X takes a value equal to x denoted by $(X = x)$. Then the probability associated with that event is denoted by $P(X = x)$. Similarly, $P(X \leq x)$ is the probability that X takes a value less than or equal to x .

$P(X > x)$ is the probability that X takes a value greater than x . This is equal to $1 - P(X \leq x)$.

$P(x_1 < X < x_2)$ is the probability that X takes a value between x_1 and x_2 .

Example Consider rolling a dice. The sample space $S = \{1, 2, 3, 4, 5, 6\}$. Let Y denotes the random variable that takes even number in the dice thrown. Then the probability that Y takes an even value is

$$P(Y = \text{even}) = \frac{3}{6} = \frac{1}{2}$$

2.2.3 Conditions for a Function to be a Random Variable

The random variable X is a function that maps the sample points in the sample space to real values on the real axis. For a function to be a random variable, it has to satisfy two conditions:

- (i) The set $(X \leq x)$ shall be an event for any real number x . The probability of this event $P(X \leq x)$ is equal to the sum of the probabilities of all the elementary events corresponding to $(X \leq x)$.
- (ii) The probability of the events $(X = \infty)$ and $P(X = -\infty)$ be zero. That is $P(X = -\infty) = 0$; $P(X = \infty) = 0$

REVIEW QUESTION

3. What are the conditions for a function to be a random variable?





Practice Problems

2.1 The sample space for an experiment is $S = (0, 1, 2, 3)$. List all possible values of the following random variables:

(a) $X = 2s$ (b) $X = s^2 - 1$ (c) $X = \frac{1}{1+s}$ (Ans. (a) $X = \{0, 2, 4, 6\}$ (b) $X = \{-1, 0, 3, 8\}$ (c) $X = \{1, 0.5, 0.33, 0.25\}$)

2.2 For the following experiments, determine the possible values of the random variable

- (a) A weighing scale in which the display show only five digits. The random variable is the displayed weight.
- (b) A C program with 1000 lines. The random variable is the number of lines with errors.

2.2.4 Classification of Random Variables

A random variable assigns a real value to each outcome in the sample space of a random experiment. If the range of the random variable takes continuous range of values then the random variable is a continuous random variable. Whenever the random variable takes on only discrete values then the random variable is a discrete random variable. A mixed random variable is one for which some of its values are discrete and some are continuous.

Definition

A continuous random variable is one having a continuous range of values.

A discrete random variable is one that may take on only discrete values.

A mixed random variable is one with both discrete and continuous values.

Solved Problems



2.4 Given that a random variable X has the following values, state if X is discrete, continuous or mixed:

- (a) $\{-5 < x < 10\}$ (b) $\{5, 6, 10 < x < 15, 19, 20\}$ (c) $\{2, 4, 6, 8, 9, 10\}$.

Solution

- (a) $X = \{-5 < x < 10\}$ continuous random variable
- (b) $X = \{5, 6, 10 < x < 15, 19, 20\}$ mixed random variable
- (c) $X = \{2, 4, 6, 8, 9, 10\}$ discrete random variable.



DISTRIBUTION FUNCTION 2.3

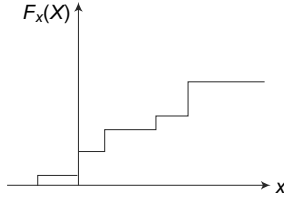
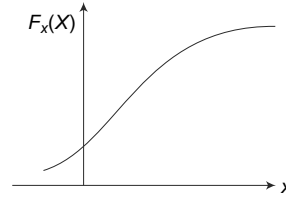
Let X be a random variable and x be a number. Let the event be X taking a value less than or equal to x . Then the Cumulative Distribution Function (CDF) of X is denoted by

$$F_X(x) = P(X \leq x) \quad -\infty < x < \infty \quad (2.1)$$

That is, $F_X(x)$ denotes the probability that the random variable X takes on a value that is less than or equal to x .

It can be shown that the discrete random variable is one having a staircase-type CDF and a continuous random variable is one having an absolutely continuous CDF.

The CDF of discrete random variable and continuous random variable are shown in Fig. 2.6 and Fig. 2.7 respectively.

**Fig. 2.6** CDF of discrete random variable**Fig. 2.7** CDF of continuous random variable

Properties of CDF The properties of CDF are the following

1. $0 \leq F_X(x) \leq 1$ (2.2)

Since PDF is a probability, it must take on values between 0 and 1. That is, $0 \leq F_X(x) \leq 1$.

2. $F_X(-\infty) = 0$ and $F_X(\infty) = 1$ (2.3)

As per distribution of CDF,

(i) $F_X(x) = P(X \leq x)$. Since random vectors are real valued, it can take any value between $-\infty$ and ∞ . $F_X(-\infty) = P(X \leq -\infty)$. Since there are no real numbers less than $-\infty$, $P(X \leq -\infty) = 0$; that is, $F_X(-\infty) = 0$. When $x = \infty$, we include all the values taken by X and their probabilities. Therefore, $F_X(\infty) = 1$.

(ii) $F_X(x)$ is monotonic non-decreasing function.

If we consider two values x_1 and x_2 such that $x_1 < x_2$, then the event $(X \leq x_1)$ is a subset of $(X \leq x_2)$.

Hence, $F_X(x_1) < F_X(x_2)$.

Consider the following probability distribution function of a random variable X shown in table 2.2

Table 2.2

X	1	2	3	4
$P(X)$	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{1}{2}$	$\frac{1}{8}$

From the table, $F_X(2) = \frac{1}{4} + \frac{1}{8} = \frac{3}{8}$ and $F_X(3) = \frac{1}{4} + \frac{1}{8} + \frac{1}{2} = \frac{7}{8}$

$$\therefore F_X(x_1) < F_X(x_2) \quad (2.4)$$

Now consider another probability distribution function shown in table 2.3

Table 2.3

X	1	2	3	4
$P(X)$	$\frac{1}{2}$	$\frac{1}{4}$	0	$\frac{1}{4}$

$$F_X(2) = \frac{1}{2} + \frac{1}{4} = \frac{3}{4} \text{ and } F_X(3) = \frac{1}{2} + \frac{1}{4} + 0 = \frac{3}{4}$$

$$F_X(2) = F_X(3)$$

Therefore, $F_X(x_1) \leq F_X(x_2)$

4. $P(x_1 < X < x_2) = F_X(x_2) - F_X(x_1)$ (2.5)

From the table 2.2

$$P(1 < X \leq 3) = P(X=2) + P(X=3) = \frac{1}{8} + \frac{1}{2} = \frac{5}{8}$$

$$F_X(3) = P(X \leq 3) = P(X=3) + P(X=2) + P(X=1)$$

$$F_X(1) = P(X \leq 1) = P(X=1)$$

$$F_X(3) - F_X(1) = P(X=3) + P(X=2) = \frac{1}{2} + \frac{1}{8} = \frac{5}{8}$$

$$5. \quad P(X > x) = 1 - F_X(x) \quad (2.6)$$

$$6. \quad F_X(x^+) = F_X(x) \quad (2.7)$$

The above property states that $F_X(x)$ is a function continuous from right.

For a discrete random variable X , the *probability mass function (pmf)*, $P_X(x)$ is defined as follows:

$$P_X(x) = P(X=x) \quad (2.8)$$

The CDF of X can be expressed in terms of $P_X(x)$ as follows:

$$F_X(x) = \sum_{i=1}^N P_X[X=x_i] u(x-x_i) \quad (2.9)$$

$$\text{or} \quad F_X(x) = \sum_{K \leq x}^N P_X(k) \quad (2.10)$$

where $u(x-x_i)$ is a step function defined as

$$u(x-x_i) = \begin{cases} 1 & \text{for } x \geq x_i \\ 0 & \text{for } x < x_i \end{cases} \quad (2.11)$$

The CDF of a discrete random variable is a staircase function. The amplitude of a step will be equal to the probability of occurrence of the value of X where the step occurs.

Solved Problems



2.5 The CDF of a random variable X is given by

$$\begin{aligned} F_X(x) &= 0 & x < 0 \\ &= x + \frac{1}{3} & 0 \leq x \leq \frac{2}{3} \\ &= 1 & \text{for } x > \frac{2}{3} \end{aligned}$$

(a) Draw the graph of the CDF.

(b) Complete $P\left(X > \frac{1}{4}\right)$.

Solution

Given: $F_X(x) = 0 \quad x < 0$

$$\begin{aligned}
 &= x + \frac{1}{3} \quad 0 \leq x \leq \frac{2}{3} \\
 &= 1 \quad \text{for } x > \frac{2}{3}
 \end{aligned}$$

The CDF is shown in Fig. 2.8.

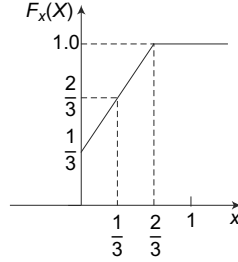


Fig. 2.8 The CDF of random variable X given in Solved Problem 2.5

The probability that $X > \frac{1}{4}$ is given by

$$\begin{aligned}
 P\left[X > \frac{1}{4}\right] &= 1 - P\left[X \leq \frac{1}{4}\right] \\
 &= 1 - \left(\frac{1}{4} + \frac{1}{3}\right) = \frac{5}{12}
 \end{aligned}$$

2.6 Determine whether the following is a valid distribution function

$$F_X(x) = \begin{cases} 1 - e^{-x/2} & x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

Solution

Given
$$F_X(x) = \begin{cases} 1 - e^{-x/2} & x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

A distribution function satisfies the following properties:

(a) $F_X(-\infty) = 0$ (b) $F_X(\infty) = 1$ (c) $0 \leq F_X(x) \leq 1$

The distribution function is zero for $x < 0$. Therefore, it satisfies the condition $F_X(-\infty) = 0$

Also, it satisfies $F_X(\infty) = 1 - e^{-\infty} = 1$ and $0 \leq F_X(x) \leq 1$

Therefore, the given function is a valid distribution function.

2.7 In the experiment of tossing a fair coin three times, the sample space consists of eight equally likely sample points. $S = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$. If X is the random variable giving the number of tails obtained, find (a) $P(X = 2)$ (b) $P(X < 2)$ (c) $P(0 < X < 3)$.

Solution

- (a) The random variable X is associated with the number of tails showing. Therefore, for $X = 2$, the outcomes are $\{HTT, THT, TTH\}$ and the probability $P(X = 2) = \frac{1}{8} + \frac{1}{8} + \frac{1}{8} = \frac{3}{8}$
- (b) Similarly, for $(X < 2)$, the outcomes are $\{HHH, HHT, HTH, THH\}$ and the probability

$$P(X < 2) = \frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8} = \frac{1}{2}$$

- (c) For $0 < X < 3$, the outcomes are $\{HHT, HTH, HTT, THH, THT, TTH\}$ and the probability

$$P(0 < X < 3) = \frac{6}{8} = \frac{3}{4}.$$

2.8 Consider the experiment of throwing two fair dice. Let X be the random variable indicating the sum of numbers that appear. (a) Find the sample space. (b) Find $P(X = 4)$. (c) $P(2 < X \leq 6)$ (d) $P(X \leq 5)$.

Solution

- (a) Since the experiment is of throwing two fair dice, the sample space

$$S = \left\{ \begin{array}{cccccc} (1,1) & (1,2) & (1,3) & (1,4) & (1,5) & (1,6) \\ (2,1) & (2,2) & (2,3) & (2,4) & (2,5) & (2,6) \\ (3,1) & (3,2) & (3,3) & (3,4) & (3,5) & (3,6) \\ (4,1) & (4,2) & (4,3) & (4,4) & (4,5) & (4,6) \\ (5,1) & (5,2) & (5,3) & (5,4) & (5,5) & (5,6) \\ (6,1) & (6,2) & (6,3) & (6,4) & (6,5) & (6,6) \end{array} \right\}$$

and the random variable X can take any value from $\{2, 3, 4, 5, 6, \dots\}$

- (b) The value $X = 4$ can occur in three ways. That is, $(1, 3)$, $(2, 2)$ and $(3, 1)$.

$$\text{Therefore, } P(X = 4) = \frac{3}{36} = \frac{1}{12}$$

- (c) From the sample space, we find that X takes values 3, 4, 5 and 6 in 14 ways.

$$\text{Therefore, } P(2 < X \leq 6) = \frac{14}{36} = \frac{7}{18}$$

- (d) From the sample space, we find that $X \leq 5$ occur in 10 ways. Therefore, $P(X \leq 5) = \frac{10}{36} = \frac{5}{18}$

2.9 Consider the experiment of tossing four fair coins. The random variable X is associated with the number of tails showing. Compute and sketch the CDF of X .

Solution The sample space of the experiment is given by

$$S = \{HHHH, HHHT, HHHT, HHTT, HTHH, HTHT, HTTT, THHH, THHT, THTH, THTT, TTHH, TTHT, TTTH, TTTT\}$$

The random variable X is associated with the number of tails showing.

If the outcome of the experiment is {HHHH} then $X = 0$ and

$$\text{Probability } P(X = 0) = \frac{1}{16}$$

If the outcomes of the experiment are {HHHT, HHTH, HTHH, THHH} then $X = 1$

$$\text{and probability } P(X = 1) = \frac{1}{16} + \frac{1}{16} + \frac{1}{16} + \frac{1}{16} = \frac{4}{16} = \frac{1}{4}$$

If the outcomes of experiments are {HHTT, HTHT, HTTH, THHT, THTH, TTHH}

$$\text{then } X = 2 \text{ and probability } P(X = 2) = \frac{6}{16} = \frac{3}{8}$$

If the outcome of experiments are {HTTT, THTT, TTHT, TTTH}

$$\text{then } X = 3 \text{ and probability } P(X = 3) = \frac{4}{16} + \frac{1}{16} = \frac{5}{16}$$

If the outcome of the experiment is {TTTT} then $X = 4$ and the probability $P(X = 4) = \frac{1}{16}$

The CDF of X is

$$F_X(0) = \frac{1}{16}; F_X(1) = \frac{5}{16}; F_X(2) = \frac{11}{16}; F_X(3) = \frac{15}{16} \text{ and } F_X(4) = 1$$

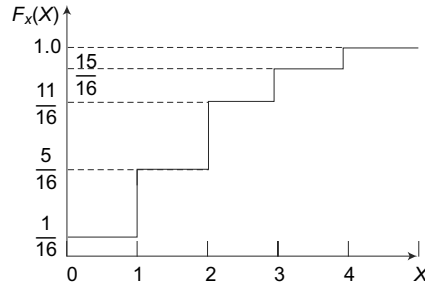


Fig. 2.9

2.10 The random variable X has the discrete variable in the set $\{-1, -0.5, 0.7, 1.5, 3\}$. The corresponding probabilities are assumed to be $\{0.1, 0.2, 0.1, 0.4, 0.2\}$. Plot its distribution function and state whether it is a discrete or continuous distribution function.

Solution The discrete variable X and the corresponding probabilities $P(X)$ are given in the table.

X	-1	-0.5	0.7	1.5	3
$P(X)$	0.1	0.2	0.1	0.4	0.2

The distribution function is given by

$$F_X(x) = P(X \leq x); F_X(-1) = P(X \leq -1) = 0.1$$

$$\begin{aligned} F_X(-0.5) &= P(X \leq -0.5) = P(X = -1) + P(X = -0.5) \\ &= 0.1 + 0.2 = 0.3 \end{aligned}$$

$$\begin{aligned} F_X(0.7) &= P(X \leq 0.7) = P(X = -1) + P(X = -0.5) + P(X = 0.7) \\ &= 0.1 + 0.2 + 0.1 = 0.4 \end{aligned}$$

$$F_X(1.5) = P(X \leq 1.5) = 0.1 + 0.2 + 0.1 + 0.4 = 0.8$$

$$F_X(3) = P(X \leq 3) = 0.1 + 0.2 + 0.1 + 0.4 + 0.2 = 1$$

$$F_X(x) = 1 \text{ for } x \geq 3$$

That is,

$$\begin{aligned} F_X(x) &= 0.1 \text{ for } -1 \leq x < -0.5 \\ &= 0.3 \text{ for } -0.5 \leq x < 0.7 \\ &= 0.4 \text{ for } 0.7 \leq x < 1.5 \\ &= 0.8 \text{ for } 1.5 \leq x < 3 \\ &= 1 \text{ for } x \geq 3 \end{aligned}$$

The plot of $F_X(x)$ is shown in Fig. 2.10.

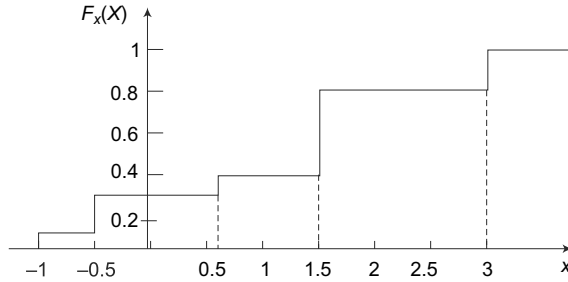


Fig. 2.10

2.11 The probability mass function of X is given in the table.

X	0	1	2	3	4	5	6
$P(X)$	k	$3k$	$5k$	$7k$	$9k$	$11k$	$13k$

Find (a) $P(X < 4)$ (b) $P(X \geq 5)$ and (c) $P(3 < X \leq 6)$ (d) Calculate the minimum value of k such that $P(X \leq 2) > 0.3$.

Solution We know $\sum P(X) = 1$

$$\text{From the table, } \sum P(X) = k + 3k + 5k + 7k + 9k + 11k + 13k = 1$$

$$k = \frac{1}{49}$$

$$(a) \quad P(X < 4) = P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3)$$

$$= k + 3k + 5k + 7k = 16k = \frac{16}{49}$$

$$(b) \quad P(X \geq 5) = P(X = 5) + P(X = 6)$$

$$= 11k + 13k = 24k = \frac{24}{49}$$

$$(c) \quad P(3 < X \leq 6) = P(X = 4) + P(X = 5) + P(X = 6)$$

$$= 9k + 11k + 13k = 33k = \frac{33}{49}$$

(d) Given $P(X \leq 2) > 0.3$

$$P(X \leq 2) = P(X=0) + P(X=1) + P(X=2)$$

$$= k + 3k + 5k = 9k$$

$$9k > 0.3$$

$$\Rightarrow k > \frac{1}{30}$$

2.12 A fair coin is tossed three times and the faces showing up are observed. (a) Write the sample description space (b) If X is the number of heads in each of the outcomes of this experiment, find the probability function. (c) Sketch the CDF and pdf.

Solution The sample space $S = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$

$$\text{The probability } P = \left\{ \frac{1}{8}, \frac{1}{8}, \frac{1}{8}, \frac{1}{8}, \frac{1}{8}, \frac{1}{8}, \frac{1}{8}, \frac{1}{8} \right\}$$

If x_i is the number of heads in each outcome then the probability corresponding to each outcome is shown in the table.

x_i	0	1	2	3
$P(x_i)$	$\frac{1}{8}$	$\frac{1+1+1}{8} = \frac{3}{8}$	$\frac{1+1+1}{8} = \frac{3}{8}$	$\frac{1}{8}$

x_i	0	1	2	3
$P(x_i)$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$

$$F_X(0) = P\{X=0\} = \frac{1}{8}; \quad F_X(1) = P\{X=1\} = \frac{4}{8}$$

$$F_X(2) = P\{X=2\} = \frac{7}{8}; \quad F_X(3) = P\{X=3\} = 1$$

The plot of CDF and pmf are shown in Fig. 2.11

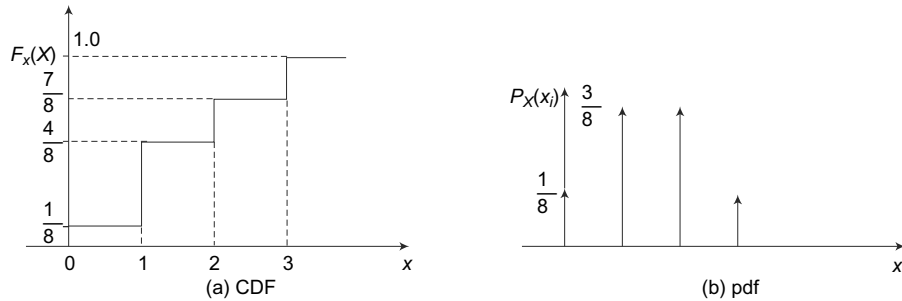


Fig. 2.11

2.13 A random variable X has the pmf given by

$$P_X(x) = \begin{cases} \frac{1}{6} & \text{for } x = 0 \\ \frac{1}{3} & \text{for } x = 1 \\ \frac{3}{10} & \text{for } x = 2 \\ \frac{1}{5} & \text{for } x = 3 \end{cases}$$

Find its CDF.

Solution

We have $F_X(x) = \sum_{k \leq x} P_X(k)$

$$F_X(0) = \sum_{k \leq 0} P_X(k) = P_X(0) = \frac{1}{6}$$

$$F_X(1) = \sum_{k \leq 1} P_X(k) = P_X(0) + P_X(1) = \frac{1}{6} + \frac{1}{3} = \frac{1}{2}$$

$$F_X(2) = \sum_{k \leq 2} P_X(k) = P_X(0) + P_X(1) + P_X(2) = \frac{1}{6} + \frac{1}{3} + \frac{3}{10} = 0.8$$

$$\begin{aligned} F_X(3) &= \sum_{k \leq 3} P_X(k) = P_X(0) + P_X(1) + P_X(2) + P_X(3) \\ &= \frac{1}{6} + \frac{1}{3} + \frac{3}{10} + \frac{1}{5} = 1 \end{aligned}$$

The CDF is shown in Fig. 2.12.

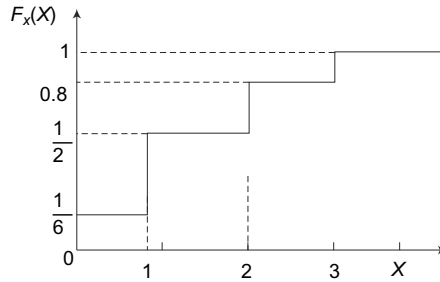


Fig. 2.12

2.14 The pmf of the number of components N of a system that fails is defined by

$$P_N(n) = \begin{cases} \binom{3}{n} (0.3)^n (0.7)^{3-n} & n = 0, 1, 2, 3 \\ 0 & \text{otherwise} \end{cases}$$

- (a) What is the CDF of N ?
 (b) What is the probability that fewer than two components of the system fail?

Solution

$$(a) \quad P_N(0) = \binom{3}{0} (0.3)^0 (0.7)^3 = (0.7)^3 = 0.343$$

$$P_N(1) = \binom{3}{1} (0.3)^1 (0.7)^2 = 3(0.3)(0.7)^2 = 0.441$$

$$P_N(2) = \binom{3}{2} (0.3)^2 (0.7)^1 = 3(0.3)^2(0.7) = 0.189$$

$$P_N(3) = \binom{3}{3} (0.3)^3 (0.7)^0 = (0.3)^3 = 0.027$$

$$F_N(0) = P_N(0) = 0.343$$

$$F_N(1) = P_N(0) + P_N(1) = 0.784$$

$$F_N(2) = P_N(0) + P_N(1) + P_N(2) = 0.973$$

$$F_N(3) = P_N(0) + P_N(1) + P_N(2) + P_N(3) = 1$$

- (b) The probability that fewer than two components of the system fail is

$$\begin{aligned} P_N(k < 2) &= P_N(k = 0) + P_N(k = 1) \\ &= F_N(1) = 0.784 \end{aligned}$$



Practice Problems

2.3 A discrete random variable X has the pmf as given below:

$X = x$	-2	-1	0	1	2	3
$P(X)$	0.1	0.2	0.2	0.3	0.1	0.1

Find $P(X < 2)$; $P(-1 < X \leq 3)$.

(Ans. 0.8, 0.7)

2.4 The CDF of a random variable X is defined by

$$F_X(x) = \begin{cases} 0 & x < 0 \\ 1 - \frac{1}{4}e^{-x}, & x > 0 \end{cases}$$

Determine $P(X = 0)$, and $P(X \geq 0)$.

(Ans. $\frac{3}{4}$; 1)

2.5 Verify that the following is a distribution functions:

$$F_X(x) = \begin{cases} 0 & x < -a \\ \frac{1}{2} \left(\frac{x}{a} + 1 \right) & -a \leq x \leq a \\ 1 & x > a \end{cases} \quad (\text{Ans. Valid distribution function})$$

2.6 $P(X=x) = \begin{cases} \frac{x}{15} & \text{for } x = 1, 2, 3, 4, \text{ and } 5 \\ 0 & \text{otherwise} \end{cases}$

Find (i) $P(X = 1 \text{ or } 2)$ (ii) $P(2 \leq x \leq 4)$.

$$\left(\text{Ans: (i) } \frac{1}{5} \text{ (ii) } \frac{3}{5} \right)$$

2.7 A discrete random variable X has the probability distribution given below:

Values	0	1	2	3	4	5	6	7
$P(x)$	0	a	$2a$	$2a$	$3a$	a^2	$2a^2$	$7a^2 + a$

Find (i) a (ii) $P(X < 6)$.

$$\left(\text{Ans. (i) } \frac{1}{10}, \text{ (ii) } \frac{81}{100} \right)$$

PROBABILITY DENSITY FUNCTION (PDF) 2.4

A probability density function (pdf) provides a simple description of the probabilities associated with a random variable. It is defined as the probability that the random variable X has in an infinitesimal interval about the point $X = x$, normalized by the length of the interval.

Mathematically,

$$f_X(x) = \lim_{dx \rightarrow 0} \frac{P[x \leq X < x + dx]}{dx} \quad (2.12)$$

For continuous random variables

$$P(x \leq X < x + dx) = F_X(x + dx) - F_X(x)$$

Therefore,

$$f_X(x) = \lim_{dx \rightarrow 0} \frac{F_X(x + dx) - F_X(x)}{dx} = \frac{dF_X(x)}{dx} \quad (2.13)$$

That is, the probability density function of a random variable is the derivative of its CDF. Therefore, CDF of a random can be obtained by integrating pdf

$$F_X(x) = \int_{-\infty}^x f_X(u) du \quad (2.14)$$

The properties of pdf are

1. $f_X(x) \geq 0$ (2.15)

From the definition of pdf, it is apparent that the pdf is a non-negative function.

2. $\int_{-\infty}^{\infty} f_X(x) dx = 1$. (2.16)

The area under the density function is unity. This property along with the property 1 can be used to check whether the given function is a valid density function or not.

$$3. \quad F_X(x) = \int_{-\infty}^x f_X(u) du \quad (2.17)$$

The above equation states that the distribution function $F_X(x)$ is equal to the integral of the density function up to the value of x .

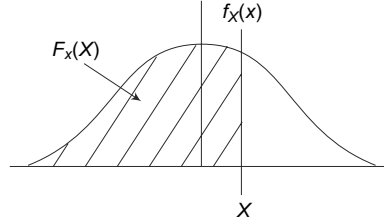


Fig. 2.13 The shaded area is equal to $F_X(x)$

$$4. \quad \int_a^b f_X(x) dx = F_X(b) - F_X(a) = P(a < x \leq b) \quad (2.18)$$

This property states that the probability that the value of x lies between a and b is the area under the density curve from a to b .

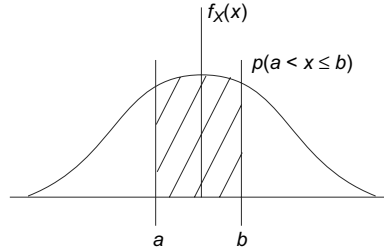


Fig. 2.14 The shaded area is equal to $F_X(b) - F_X(a)$

For a discrete random variable, the density function can be obtained by differentiating Eq. (2.9). The differentiation of unit step is an impulse. Therefore, for a discrete random variable

$$f_X(x) = \sum_{i=1}^N P(x_i) \delta(x - x_i) \quad (2.19)$$

where $\delta(x)$ is known as impulse function defined as

$$\delta(x) = \begin{cases} 1 & \text{for } x = 0 \\ 0 & \text{for } x \neq 0 \end{cases} \quad (2.20)$$

The impulse function shifted by a is defined as

$$\delta(x - a) = \begin{cases} 1 & \text{for } x = a \\ 0 & \text{for } x \neq a \end{cases} \quad (2.21)$$

Thus, the density function for a discrete random variable is the sum of shifted and scaled impulses.

Solved Problems

2.15 Which of the following are valid probability density functions?

$$(a) f_X(x) = \begin{cases} e^{-x} & \text{for } 0 \leq x \leq \infty \\ 0 & \text{otherwise} \end{cases}$$

$$(b) f_X(x) = \begin{cases} xe^{-x^2} & \text{for } 0 \leq x \leq \infty \\ 0 & \text{otherwise} \end{cases}$$

$$(c) f_X(x) = \begin{cases} \frac{1}{4}(x^2 - 1); & |x| < 2 \\ 0 & \text{otherwise} \end{cases}$$

$$(d) f_X(x) = \begin{cases} \frac{1}{5} & 0 \leq x \leq 5 \\ 0 & \text{otherwise} \end{cases}$$

Solution To verify the validity of a pdf, we need to verify whether the function is non-negative and normalized so that the area under the curve is equal to unity.

$$(a) \int_{-\infty}^{\infty} f_X(x) dx = \int_0^{\infty} e^{-x} dx = -e^{-x} \Big|_0^{\infty} = 1$$

Also, the function e^{-x} is non-negative. Hence, it is a valid pdf.

$$(b) f_X(x) = \begin{cases} xe^{-x^2} & \text{for } 0 \leq x \leq \infty \\ 0 & \text{otherwise} \end{cases}$$

$$\begin{aligned} \int_{-\infty}^{\infty} f_X(x) dx &= \int_0^{\infty} xe^{-x^2} dx \\ \Rightarrow \int_0^{\infty} xe^{-x^2} dx &= \frac{1}{2} \int_0^{\infty} e^{-t} dt = \frac{1}{2} e^{-t} \Big|_0^{\infty} = \frac{1}{2} \end{aligned}$$

$\begin{aligned} \text{Let } x^2 &= t \\ 2x dx &= dt \\ x dx &= \frac{dt}{2} \end{aligned}$

Since $\int_{-\infty}^{\infty} f_X(x) dx \neq 1$; it is not a valid pdf

$$(c) f_X(x) = \begin{cases} \frac{1}{4}[x^2 - 1]; & |x| \leq 2 \\ 0 & \text{otherwise} \end{cases}$$

The above pdf takes negative value for $x = 0$. Hence, it is not a valid pdf

$$(d) f_X(x) = \begin{cases} \frac{1}{5} & 0 \leq x \leq 5 \\ 0 & \text{otherwise} \end{cases}$$

$$\int_{-\infty}^{\infty} f_X(x) dx = \int_0^5 \frac{1}{5} dx = \frac{1}{5}(5-0) = 1$$

Also $f_X(x)$ is non-negative. Hence, it is a valid pdf.

2.16 A random variable has a CDF given by

$$F_X(x) = \begin{cases} 1 - e^{-2x} & \text{for } x \leq 0 \\ 0 & \text{otherwise} \end{cases}$$

Then find the pdf.

Solution The pdf is given by

$$\begin{aligned} f_X(x) &= \frac{dF_X(x)}{dx} \\ &= \frac{d}{dx} [1 - e^{-2x}] = -(-2)e^{-2x} = 2e^{-2x} \\ f_X(x) &= 2e^{-2x} \end{aligned}$$

2.17 The probability density function of a random variable is given by

$$f_X(x) = \begin{cases} 1 - |1 - x| & \text{for } 0 < x < 2 \\ 0 & \text{otherwise} \end{cases}$$

Sketch the pdf and verify whether it is valid or not.

Solution Given: $f_X(x) = \begin{cases} 1 - |1 - x| & \text{for } 0 < x < 2 \\ 0 & \text{otherwise} \end{cases}$

The above equation can be written as

$$f_X(x) = \begin{cases} x & \text{for } 0 < x < 1 \\ 2 - x & \text{for } 1 < x < 2 \end{cases}$$

The sketch of the pdf is shown in Fig. 2.15. The $f_X(x)$ is a valid distribution if it satisfies

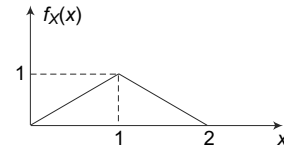


Fig. 2.15

$$\int_{-\infty}^{\infty} f_X(x) dx = 1$$

$$\begin{aligned} \int_{-\infty}^{\infty} f_X(x) dx &= \int_0^1 x dx + \int_1^2 (2-x) dx = \left. \frac{x^2}{2} \right|_0^1 + 2x \Big|_1^2 - \left. \frac{x^2}{2} \right|_1^2 \\ &= \frac{1}{2} + 2 - \left(\frac{4}{2} - \frac{1}{2} \right) = 1 \end{aligned}$$

Hence, $f_X(x)$ is a valid density function.

2.18 In a cost price shop, the amount of rice (in hundreds of kilos) that sells in a day is a random variable with density

$$f_X(x) = \begin{cases} Cx & \text{for } 0 \leq x < 3 \\ C(5-x) & \text{for } 3 \leq x \leq 5 \\ 0 & \text{otherwise} \end{cases}$$

- (a) For what value of C is $f_X(x)$ a valid pdf?
 (b) What is the probability that the number of kilos of rice that will be sold in a day is (i) more than 200 kilos (ii) between 150 and 400 kilos?

Solution

(a) For a valid pdf, $\int_{-\infty}^{\infty} f_X(x) dx = 1$

Given: $f_X(x) = \begin{cases} Cx & \text{for } 0 \leq x < 3 \\ C(5-x) & \text{for } 3 \leq x \leq 5 \\ 0 & \text{otherwise} \end{cases}$

$$\begin{aligned} \int_{-\infty}^{\infty} f_X(x) dx &= \int_0^3 Cx dx + \int_3^5 C(5-x) dx = 1 \\ &= C \left[\frac{x^2}{2} \right]_0^3 + C \left[5x - \frac{x^2}{2} \right]_3^5 = 1 \\ &= C \left[\frac{9}{2} \right] + C[(10-8)] = 1 \\ &= \frac{9C}{2} + 2C = 1 \quad \Rightarrow \quad C = \frac{2}{13} \end{aligned}$$

(b)(i) $P(X > 2) = \int_2^5 f_X(x) dx$

$$\begin{aligned} &= \int_2^3 \frac{2}{13} x dx + \int_3^5 \frac{2}{13} (5-x) dx \\ &= \frac{2}{13} \left[\frac{x^2}{2} \right]_2^3 + \frac{2}{13} \left[5x - \frac{x^2}{2} \right]_3^5 \\ &= \frac{2}{13} \left[\frac{5}{2} \right] + \frac{2}{13} [10-8] = \frac{5}{13} + \frac{4}{13} = \frac{9}{13} \end{aligned}$$

(ii) $P(1.5 < X < 4) = \int_{1.5}^3 \frac{2}{13} x dx + \int_3^4 \frac{2}{13} (5-x) dx$

$$\begin{aligned}
&= \frac{2}{13} \frac{x^2}{2} \Big|_{1.5}^3 + \frac{2}{13} \left[5x - \frac{x^2}{2} \right] \Big|_3^4 \\
&= \frac{2}{13} [4.5 - 1.125] + \frac{2}{13} \left[5 - \frac{7}{2} \right] \\
&= 0.75
\end{aligned}$$

2.19 Consider the probability density function $f_X(x) = ae^{-b|x|}$ where x is a random variable whose allowable values range from $x = -\infty$ to ∞ . Find (a) the CDF of $F_X(x)$, (b) the relationship between a and b , (c) the probability that the outcome x lies between 1 and 2.

Solution

Given: $f_X(x) = a e^{-b|x|}$

That is $f_X(x) = ae^{bx}$ for $x < 0$
 $= ae^{-bx}$ for $x \geq 0$

(a) We know that the CDF is

$$F_X(x) = \int_{-\infty}^x f_X(x) dx$$

For $x < 0$,

$$\begin{aligned}
F_X(x) &= \int_{-\infty}^x a e^{bx} dx \\
&= \frac{a}{b} e^{bx} \Big|_{-\infty}^x = \frac{a}{b} e^{bx}
\end{aligned}$$

For $x \geq 0$,

$$\begin{aligned}
F_X(x) &= \int_{-\infty}^0 a e^{bx} dx + \int_0^x a e^{-bx} dx \\
&= \frac{a}{b} e^{bx} \Big|_{-\infty}^0 + \frac{a}{(-b)} e^{-bx} \Big|_0^x \\
&= \frac{a}{b} - \frac{a}{b} (e^{-bx} - 1) = \frac{a}{b} + \frac{a}{b} - \frac{a}{b} e^{-bx} \\
&= \frac{a}{b} (2 - e^{-bx}) \\
F_X(x) &= \frac{a}{b} e^{bx} \text{ for } x < 0 \\
&= \frac{a}{b} (2 - e^{-bx}) \text{ for } x \geq 0
\end{aligned}$$

(b) We know $\int_{-\infty}^{\infty} f_X(x) dx = 1$

$$\begin{aligned}\int_{-\infty}^{\infty} f_X(x) dx &= \int_{-\infty}^0 a e^{bx} dx + \int_0^{\infty} a e^{-bx} dx \\ &= \frac{a}{b} e^{bx} \Big|_{-\infty}^0 + \left(-\frac{a}{b} \right) e^{-bx} \Big|_0^{\infty} \\ &= \frac{a}{b} - \frac{a}{b}(-1) = \frac{2a}{b}\end{aligned}$$

That is, $\frac{2a}{b} = 1 \Rightarrow \frac{a}{b} = \frac{1}{2}$ or $b = 2a$

The relationship between a and b is given by

$$b = 2a$$

$$\begin{aligned}\text{(c)} \quad P(1 \leq X \leq 2) &= \int_1^2 f_X(x) dx = \int_1^2 a e^{-bx} dx = \frac{a e^{-bx}}{(-b)} \Big|_1^2 \\ &= -\frac{a}{b} [e^{-2b} - e^{-b}] = \frac{a}{b} [e^{-b} - e^{-2b}] \\ &= \frac{1}{2} [e^{-b} - e^{-2b}]\end{aligned}$$

2.20 X is a continuous random variable with the pdf $f_X(x) = \begin{cases} A\{3x - x^2\}, & 0 < x < 2 \\ 0 & \text{otherwise} \end{cases}$

- (a) What is the value of A ?
 (b) Find $P(X > 1)$.

Solution We know $\int_{-\infty}^{\infty} f_X(x) dx = 1$

$$\text{(a) Given: } f_X(x) = \begin{cases} A\{3x - x^2\}, & 0 < x < 2 \\ 0 & \text{otherwise} \end{cases}$$

$$\begin{aligned}\int_{-\infty}^{\infty} f_X(x) dx &= \int_0^2 A(3x - x^2) dx \\ &= A \left\{ \frac{3x^2}{2} \Big|_0^2 - \frac{x^3}{3} \Big|_0^2 \right\} = A \left[6 - \frac{8}{3} \right] = A \left[\frac{10}{3} \right]\end{aligned}$$

That is, $\frac{10A}{3} = 1$ from which $A = \frac{3}{10}$

$$f_X(x) = \begin{cases} \frac{3}{10}\{3x - x^2\}, & 0 < x < 2 \\ 0 & \text{otherwise} \end{cases}$$

$$\begin{aligned} \text{(b)} \quad P(X > 1) &= \int_1^{\infty} f_X(x) dx = \frac{3}{10} \left[\int_1^2 (3x - x^2) dx \right] \\ &= \frac{3}{10} \left[\frac{3x^2}{2} - \frac{x^3}{3} \right]_1^2 = \frac{3}{10} \left[\frac{3}{2}(4 - 1) - \frac{1}{3}(8 - 1) \right] \\ &= \frac{3}{10} \left[\frac{9}{2} - \frac{7}{3} \right] = \frac{13}{20} \end{aligned}$$

2.21 Find the value of the constant k so that

$$f_X(x) = \begin{cases} k x^2(1 - x^3), & 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

Is this a proper density function of a continuous random variable.

Solution For a density function to be legitimate, it has to satisfy the condition

$$\int_{-\infty}^{\infty} f_X(x) dx = 1$$

$$\text{Given:} \quad f_X(x) = \begin{cases} k x^2(1 - x^3), & 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

$$\begin{aligned} \int_{-\infty}^{\infty} f_X(x) dx &= \int_0^1 k x^2(1 - x^3) dx \\ &= k \left[\frac{x^3}{3} \Big|_0^1 - \frac{x^6}{6} \Big|_0^1 \right] \\ &= k \left[\frac{1}{3} - \frac{1}{6} \right] = \frac{k}{6} \end{aligned}$$

$$\text{That is} \quad \frac{k}{6} = 1 \Rightarrow k = 6$$

For $k = 6$, $f_X(x)$ is a proper density function.

2.22 The CDF of a random variable X is defined by

$$F_X(x) = \begin{cases} 0 & x < 1 \\ A(x-1) & 1 \leq x \leq 4 \\ 1 & x \geq 4 \end{cases}$$

(a) What is the value of A ? (b) What is $P[X > 2]$? (c) What is $P[1 \leq X \leq 3]$?

Solution *First method* To find the value of A , we apply the condition $F_X(4) = 1$ from which we have

$$F_X(4) = A(4-1) = 1 \Rightarrow A = \frac{1}{3}$$

$$P(X > 2) = 1 - P(X \leq 2) = 1 - F_X(2)$$

$$= 1 - \frac{1}{3}(2-1) = \frac{2}{3}$$

The probability that X lies between 1 and 3 is

$$\begin{aligned} P(1 \leq X \leq 3) &= F_X(3) - F_X(1) \\ &= \frac{1}{3}(3-1) - \frac{1}{3}(1-1) \\ &= \frac{2}{3} \end{aligned}$$

Second method

We also know

$$f_X(x) = \frac{d}{dx} F_X(x)$$

For the given $F_X(x)$,

$$f_X(x) = \begin{cases} A & \text{for } 1 \leq x \leq 4 \\ 0 & \text{otherwise} \end{cases}$$

$$\text{We know } \int_{-\infty}^{\infty} f_X(x) dx = 1 \Rightarrow \int_1^4 A dx = 1$$

from which we can obtain $3A = 1$

$$\text{That is, } A = \frac{1}{3}$$

$$P(X > 2) = \int_2^4 f_X(x) dx = \frac{1}{3}(4-2) = \frac{2}{3}$$

$$P(1 \leq X \leq 3) = \int_1^3 A dx = \frac{1}{3}(3-1) = \frac{2}{3}$$

2.23 For the following pdfs, find the constant k so that $f_X(x)$ satisfies the conditions of being a pdf of a random variable X . Also calculate $P(-1 < X < 1)$

$$(a) \quad f_X(x) = \begin{cases} k(1-x^2) & 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

$$(b) \quad f_X(x) = \begin{cases} kx^2e^{-2x} & 0 < x < \infty \\ 0 & \text{otherwise} \end{cases}$$

$$(c) \quad f_X(x) = \begin{cases} k(x+1) & -1 < x < 3 \\ 0 & \text{otherwise} \end{cases}$$

Solution

$$(a) \quad \text{Given } f_X(x) = \begin{cases} k(1-x^2) & 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

$$\text{We know } \int_{-\infty}^{\infty} f_X(x) dx = 1 \Rightarrow \int_0^1 k(1-x^2) dx = 1$$

$$k \left(x - \frac{x^3}{3} \right) \Big|_0^1 = k \left[1 - \frac{1}{3} \right] = 1$$

$$\Rightarrow \quad k = \frac{3}{2}$$

$$(b) \quad \int_{-\infty}^{\infty} f_X(x) dx = 1 \Rightarrow \int_0^{\infty} kx^2 e^{-2x} dx = 1$$

$$\int_0^{\infty} kx^2 e^{-2x} dx = \left. \frac{-x^2 e^{-2x}}{2} \right|_0^{\infty} - \left. \frac{xe^{-2x}}{2} \right|_0^{\infty} - \left. \frac{e^{-2x}}{4} \right|_0^{\infty} = \frac{k}{4} = 1$$

$$\Rightarrow \quad k = 4$$

$$P(-1 < X < 1) = \int_{-1}^1 f_X(x) dx = \int_0^1 4x^2 e^{-2x} dx$$

$$= 4 \left[\frac{-x^2 e^{-2x}}{2} - \frac{xe^{-2x}}{2} - \frac{e^{-2x}}{4} \right] \Big|_0^1$$

$$= 4 \left[\left(-\frac{1}{2} e^{-2} - \frac{1}{2} e^{-2} - \frac{e^{-2}}{4} \right) - \left(-\frac{1}{4} \right) \right] = 0.3233$$

$$(c) \quad \int_{-\infty}^{\infty} f_X(x) dx = 1 \Rightarrow \int_{-1}^3 k(x+1) dx = 1$$

$$\lim_{x \rightarrow \infty} x^2 e^{-2x} = \lim_{x \rightarrow \infty} \frac{x^2}{e^{2x}} = 0$$

Use L'Hospital rule,

$$\text{Similarly, } \lim_{x \rightarrow \infty} x e^{-2x} = 0$$

$$k \left(\frac{x^2}{2} + x \right) \Big|_{-1}^3 = k \left[\frac{9-1}{2} + 3 - (-1) \right]$$

$$8k = 1 \Rightarrow k = \frac{1}{8}$$

$$P(-1 < X < 1) = \int_{-1}^1 f_x(x) dx$$

$$= \frac{1}{8} \int_{-1}^1 (x+1) dx = \frac{1}{8} \left(\frac{x^2}{2} + x \right) \Big|_{-1}^1 = \frac{1}{4}$$

2.24 Is the following function a legitimate pdf?

$$f_X(x) = \begin{cases} \frac{3}{64}x^2 & 0 \leq x \leq 4 \\ 0 & \text{otherwise} \end{cases}$$

Solution For $f_X(x)$ to be a legitimate pdf, it has to satisfy the condition

$$\int_{-\infty}^{\infty} f_X(x) dx = 1$$

$$\int_{-\infty}^{\infty} f_X(x) dx = \int_0^4 \frac{3}{64} x^2 dx = \frac{3}{64} \frac{x^3}{3} \Big|_0^4$$

$$= \frac{3}{64} \left[\frac{64}{3} \right] = 1$$

Therefore, $f_X(x)$ is a legitimate pdf.

2.25 The density function $f_X(x)$ is defined as

$$f_X(x) = \begin{cases} \frac{k}{2} & 0 < x \leq 2 \\ \frac{1}{4} & 2 < x \leq 3 \end{cases}$$

Find k and then the distribution function $F_X(x)$.

Solution

Given: $f_X(x) = \begin{cases} \frac{k}{2} & 0 < x \leq 2 \\ \frac{1}{4} & 2 < x \leq 3 \end{cases}$

We know $\int_{-\infty}^{\infty} f_X(x) dx = 1$

$$\int_0^2 \frac{k}{2} dx + \int_2^3 \frac{1}{4} dx = \frac{k}{2}(2-0) + \frac{1}{4}(3-2)$$

$$\Rightarrow k + \frac{1}{4} = 1 \quad \therefore k = \frac{3}{4}$$

$$f_X(x) = \begin{cases} \frac{3}{8}, & 0 < x \leq 2 \\ \frac{1}{4}, & 2 < x \leq 3 \end{cases}$$

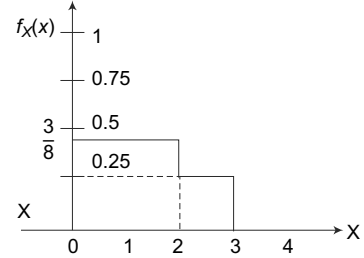


Fig. 2.16

The pdf is shown in Fig. 2.16. From the pdf, we can find that we have to find $F_X(x)$ for four ranges of x given by

- (a) $x < 0$ (b) $0 < x \leq 2$ (c) $2 < x \leq 3$ (d) $x > 3$

(a) Since $f_X(x)$ is zero for $x < 0$, $F_X(x) = 0$

(b) For $0 < x \leq 2$ $F_X(x) = \int_0^x \frac{3}{8} dx = \frac{3}{8}x$

(c) For $2 \leq x \leq 3$ $F_X(x) = \int_0^2 \frac{3}{8} dx + \int_2^x \frac{1}{4} dx = \frac{3}{8}(2) + \frac{1}{4}(x-2) = \frac{1}{4}(x+1)$

(d) For $x > 3$ $F_X(x) = 1$

The sketch of $F_X(x)$ is shown in Fig. 2.17.

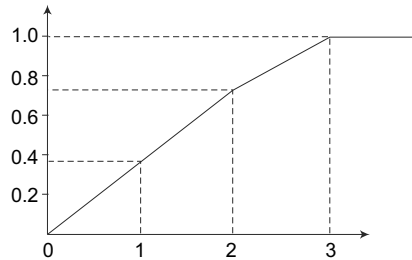


Fig. 2.17

2.26 If the probability density function of a random variable is given by

$$f_X(x) = k(1 - 2x^2); \quad 0 < x < 1$$

find the value k and $F_X(x)$.

Solution Given: $f_X(x) = k(1 - 2x^2)$, $0 < x < 1$

We know that $\int_{-\infty}^{\infty} f_X(x) dx = 1$

$$\Rightarrow \int_0^1 k(1-2x^2)dx = 1$$

$$k \left[x - \frac{2}{3}x^3 \right]_0^1 = 1; \quad k \left(1 - \frac{2}{3} \right) = 1$$

$$k = 3$$

That is $f_X(x) = 3(1-2x^2); 0 < x < 1$

The CDF is given by

$$F_X(x) = \int_{-\infty}^x f_X(x)dx$$

$$= \int_0^x 3(1-2x^2)dx$$

$$= 3 \left(x - \frac{2}{3}x^3 \right) \Big|_0^x$$

$$= 3 \left(x - \frac{2}{3}x^3 \right)$$

Since $f_X(x) = 0$ for $x \leq 0$, $F_X(x) = 0$ for $x < 0$

$$F_X(x) = 0 \quad \text{for} \quad x \leq 0$$

$$= \begin{cases} 3 \left(x - \frac{2}{3}x^3 \right) & \text{for } 0 < x < 1 \\ 0 & \text{for } x > 1 \end{cases}$$



Practice Problems

2.8 A random variable X has the following probability distribution.

x	-2	-1	0	1	2	3
$p(x)$	0.1	k	0.2	$2k$	0.3	$3k$

Find (a) the value of k . (b) Evaluate $P(X < 2)$, $P(-2 < X < 2]$, $P(-2 < X < 3)$ (c) Find the CDF of X . (d) Find the pdf of X

$$\left(\text{Ans. (a) } k = \frac{1}{5} \text{ (b) } \frac{1}{2}, \frac{2}{9}, \frac{7}{10} \right)$$

2.9 A commulative distribution function of a random variable X is $F_X(x) = 1 - (1+x)e^{-x}$, $x > 0$. Find the pdf of x .

$$[\text{Ans. } xe^{-x}, x > 0]$$

2.10 The pdf of a continuous random variable X is $f_X(x) = ke^{-|x|}$. Find k and $F_X(x)$.

$$\left(\text{Ans. } k = \frac{1}{2}, F_X(x) = 0.5e^x \text{ for } x < 0; \frac{1}{2}(2 - e^{-x}) \text{ for } x > 0 \right)$$

2.11 If X has the distribution function

$$F_X(x) = \begin{cases} 0 & \text{for } x < 1 \\ \frac{1}{3} & \text{for } 1 \leq x < 4 \\ \frac{5}{6} & \text{for } 4 \leq x < 10 \\ 1 & \text{for } x \geq 10 \end{cases} \quad \begin{array}{l} \text{Find (a) pdf of } X \\ \text{(b) } P(2 < X < 6) \end{array}$$

$$\left(\text{Ans. (a)} \frac{1}{3} \delta(x-1) + \frac{1}{2} \delta(x-4) + \frac{1}{6} \delta(x-10) \text{ (b)} \frac{1}{6} \right)$$

2.12 Find the cumulative distribution function $F_X(x)$ corresponding to the pdf $f_X(x) = \frac{1}{\pi} \cdot \frac{1}{1+x^2}$ $-\infty < x < \infty$

$$\left(\text{Ans. } F_X(x) = \frac{1}{\pi} \tan^{-1} x + \frac{1}{2} \right)$$

2.13 The pdf of the samples of the amplifier of speech waveform is found to decay exponentially at a rate α , so the following pdf is proposed.

$$f_X(x) = C e^{-\alpha|x|} \quad -\infty < x < \infty$$

Find the constant C , and then find the probability $P(|x| < v)$.

$$\left(\text{Ans. } C = \frac{\alpha}{2}, 1 - e^{-\alpha v} \right)$$



DISCRETE DISTRIBUTIONS 2.5

2.5.1 Binomial Distribution

A trial with only two possible outcomes is called a Bernoulli trial. The outcomes of the trial can be described using terms like *success* or *failure*, *defective* or *non-defective*, 1 or *zero*. One example of Bernoulli trial is the coin-tossing experiment in which the outcome results in heads or tails. In another example, the condition of the parts in a machine may be categorized as defective or non-defective. For the two outcomes, we have to assign a probability whose sum is equal to one. If p is the probability of success then $(1-p)$ is the probability of failure. We can assign a random variable X such that when the outcome of the trial is success, we define $X = 1$ and when the outcome is a failure, we define $X = 0$. The random variable X is called *Bernoulli random variable* and its pmf is given by

$$P_X(x) = \begin{cases} 1-p & x = 0 \\ p & x = 1 \end{cases} \quad (2.22)$$

The pmf is also defined as

$$P_X(x) = p^x (1-p)^{1-x} \quad x = 0, 1 \quad (2.23)$$

Suppose we conduct n independent trials, each of which results in a 'success' with probability p and a failure with probability $(1-p)$. If X represents the number of successes that occur in the n trials, X is said to be binomial random variable with parameters (n, p) . The pmf of a binomial random variable with parameters (n, p) is given by

$$P_X(x) = \binom{n}{x} p^x (1-p)^{n-x} \quad x = 0, 1, \dots, n \quad (2.24)$$

$\binom{n}{x}$ is called a *binomial coefficient* and is defined as $\binom{n}{x} = \frac{n!}{x!(n-x)!}$

The binomial density and distribution functions are also given by

$$f_X(x) = \sum_{k=0}^n \binom{n}{k} p^k (1-p)^{n-k} \delta(x-k) \quad (2.25)$$

and

$$F_X(x) = \sum_{k=0}^n \binom{n}{k} p^k (1-p)^{n-k} u(x-k) \quad (2.26)$$

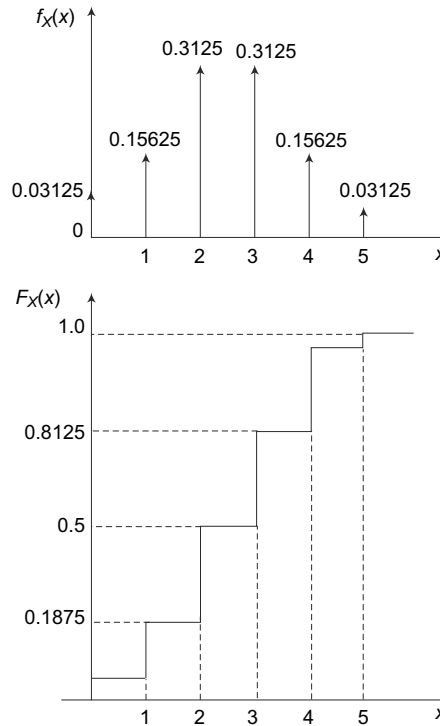


Fig. 2.18 (a) Binomial density function (b) Binomial distribution function for $N = 5$ and $p = 0.5$

The Binomial density and distribution functions are shown in Fig. 2.18(a) and 2.18(b) respectively.

Applications

Binomial distribution can be applied to many problems of repeated independent trials with only two possible outcomes in each trial. The applications include (i) many games of chance, (ii) signal detection in radar and sonar systems, and (iii) inequality control to classify items as defective or non-defective.

REVIEW QUESTIONS



4. Define density function. Find the equation for binomial distribution function.
5. What is binomial density function? Find the equation for binomial distribution function.

Solved Problems

2.27 During October, Chennai has rainfall on an average three days a week. Obtain the probability that (a) rain will fall on at least 2 days of a given week, (b) first three days of a given week will be rainy and the remaining 4 days will be wet.

Solution

(a) Given: $p = \frac{3}{7}$; $n = 7$

Let $P(X = x)$ denote the probability of rain falls

$$\begin{aligned} P(X \geq 2) &= P(X = 2) + P(X = 3) + P(X = 4) + P(X = 5) + P(X = 6) + P(X = 7) \\ &= \binom{7}{2} \left(\frac{3}{7}\right)^2 \left(\frac{4}{7}\right)^5 + \binom{7}{3} \left(\frac{3}{7}\right)^3 \left(\frac{4}{7}\right)^4 + \binom{7}{4} \left(\frac{3}{7}\right)^4 \left(\frac{4}{7}\right)^3 \\ &\quad + \binom{7}{5} \left(\frac{3}{7}\right)^5 \left(\frac{4}{7}\right)^2 + \binom{7}{6} \left(\frac{3}{7}\right)^6 + \left(\frac{4}{7}\right) + \binom{7}{7} \left(\frac{3}{7}\right)^7 \left(\frac{4}{7}\right)^0 \\ &= 0.235 + 0.293 + 0.2203 + 0.099 + 0.024 + 0.00265 = 0.87395 \end{aligned}$$

(b) Probability that the first 3 days will have no rain, and remaining 4 days will be wet is

$$= (1 - p)^3 (p)^4 = \left(\frac{4}{7}\right)^3 \left(\frac{3}{7}\right)^4 = 0.0063$$

2.28 The random variable X has a binomial distribution with $n = 8$ and $p = 0.5$. Determine the following probabilities. (a) $P(X = 4)$ (b) $P(X \leq 2)$ (c) $P(X \geq 7)$ (d) $P(3 \leq X < 5)$.

Solution For a binomial random variable,

$$P(X = x) = \binom{n}{x} p^x q^{n-x} \quad x = 0, \dots, n$$

Given: $n = 8$ and $p = 0.5$, therefore, $q = 0.5$

$$\begin{aligned} \text{(a)} \quad P(X = 4) &= \binom{8}{4} (0.5)^4 (0.5)^4 \\ &= 70(0.5)^8 = 0.273 \end{aligned}$$

$$\begin{aligned} \text{(b)} \quad P(X \leq 2) &= P(X = 1) + P(X = 2) \\ &= \binom{8}{1} (0.5)^1 (0.5)^7 + \binom{8}{2} (0.5)^2 (0.5)^6 \\ &= 0.03125 + 0.1094 = 0.14063 \end{aligned}$$

$$\begin{aligned} \text{(c)} \quad P(X \geq 7) &= P(X = 7) + P(X = 8) \\ &= \binom{8}{7} (0.5)^7 (0.5) + \binom{8}{8} (0.5)^8 (0.5)^0 \end{aligned}$$

$$\begin{aligned}
&= 0.0351 \\
P(3 \leq X < 5) &= P(X=3) + P(X=4) \\
&= \binom{8}{3} (0.5)^3 (0.5)^5 + \binom{8}{4} (0.5)^4 (0.5)^4 \\
&= 0.21875 + 0.273 \\
&= 0.492
\end{aligned}$$

2.29 Determine the CDF of a random variable with $n = 3$ and $P = 1/4$.

Solution

$$P(X=x) = \binom{n}{x} p^x q^{n-x} \quad x = 0, 1, 2, \dots, n$$

Given $n = 3$ and $p = \frac{1}{4}$

$$P(X=0) = \binom{3}{0} (0.25)^0 (0.75)^3 = 0.4219$$

$$P(X=1) = \binom{3}{1} (0.25) (0.75)^2 = 0.4218$$

$$P(X=2) = \binom{3}{2} (0.25)^2 (0.75) = 0.1406$$

$$P(X=3) = \binom{3}{3} (0.25)^3 (0.75)^0 = 0.015625$$

$$F_X(x) = 0 \quad \text{for } x < 0$$

$$F_X(x) = 0.4218 \quad \text{for } 0 \leq x < 1$$

$$F_X(x) = 0.8436 \quad \text{for } 1 \leq x < 2$$

$$F_X(x) = 0.9842 \quad \text{for } 2 \leq x < 3$$

$$F_X(x) = 1 \quad \text{for } x \geq 3$$

2.30 A machine produces items in which 10% are defective. Every hour the mechanic draws a sample of size 20 for inspection. If the sample contains no defective items, the operator does not stop the machine. What is the probability that the machine will not be stopped?

Solution Let X be a random variable that represents the number of defective items in a sample of 20. The probability that an item is defective is $p = 0.1$. X obeys binomial distribution.

$$P(X=x) = \binom{n}{x} p^x (1-p)^{n-x}$$

Given: $n = 20$; $p = 0.1$

$$P(X = x) = \binom{20}{x} (0.1)^x (0.9)^{20-x}$$

The probability that there is no defective item is

$$\begin{aligned} P(X = 0) &= \binom{20}{0} (0.1)^0 (0.9)^{20} \\ &= (0.9)^{20} = 0.1216 \end{aligned}$$

2.31 Find the probability that in tossing a fair coin 5 times, there will appear (a) 3 heads (b) 3 heads and 2 tails, (c) at least 1 heads, (d) not more than 1 tails.

Solution The probability of getting heads/tails in tossing a coin is equal to $1/2$. That is $p = q = 1/2$.

A coin is tossed 5 times.

That is, $n = 5$ and $p = q = \frac{1}{2}$

For a binomial random variable,

$$P(X = x) = \binom{n}{x} p^x q^{n-x}$$

$$(a) \quad P(\text{getting 3 heads}) = \binom{5}{3} \left(\frac{1}{2}\right)^3 \left(\frac{1}{2}\right)^2 = \frac{5}{16}$$

$$(b) \quad P(\text{getting 3 heads and 2 tails}) = \binom{5}{3} \left(\frac{1}{2}\right)^3 \left(\frac{1}{2}\right)^2 = \frac{5}{16}$$

$$\begin{aligned} (c) \quad P(\text{at least 1 heads}) &= 1 - P(\text{no heads}) \\ &= 1 - \binom{5}{0} \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^5 = 1 - \left(\frac{1}{2}\right)^5 \\ &= \frac{31}{32} \end{aligned}$$

$$\begin{aligned} (d) \quad P(\text{not more than 1 tails}) &= P(0 \text{ tails and 5 heads}) + P(1 \text{ tails and 4 heads}) \\ &= \binom{5}{5} \left(\frac{1}{2}\right)^5 \left(\frac{1}{2}\right)^0 + \binom{5}{4} \left(\frac{1}{2}\right)^4 \left(\frac{1}{2}\right)^1 \\ &= \left(\frac{1}{2}\right)^5 + 5 \left(\frac{1}{2}\right)^5 = \frac{6}{32} = \frac{3}{16} \end{aligned}$$

2.32 In 256 sets of 12 tosses of fair coin, how many cases may one expect of 8 heads and 4 tails?

Solution

Given: $n = 12$; $p = q = 1/2$

$P(\text{getting 8 heads and 4 tails})$

$$= \binom{12}{8} \left(\frac{1}{2}\right)^8 \left(\frac{1}{2}\right)^4$$

$$= \binom{12}{8} \left(\frac{1}{2}\right)^{12} = 0.12$$

The number of cases one may expect of 8 heads and 4 tails is

$$(256)(0.12) = 30.72 \approx 31 \text{ times}$$

2.33 A lot contains 2% defective items. What should be the number of items in a random sample so that the probability of finding at least 1 defective item in it is at least 0.9?

Solution Given: $p = 0.02$ and $q = 0.98$

Let X be a random variable which denotes the number of defective items.

Probability of getting at least 1 defective item is given by $P(X \geq 1)$.

$$P(X \geq 1) = 1 - P(X = 0) = 0.9$$

$$P(X = 0) = 0.1$$

$$P(X = 0) = \binom{n}{0} (p)^0 (q)^{n-0}$$

$$\Rightarrow (0.02)^0 (0.98)^n = 0.1$$

$$n \log (0.98) = \log (0.1)$$

$$\Rightarrow n = 113.97$$

Therefore, $n = 114$

2.34 An irregular 6-faced dice is such that the probability that it gives 3 odd numbers in 7 throws is twice the probability that it gives 4 odd numbers in 7 throws. How many sets of exactly 7 trials can be expected to give no odd number out of 5000 sets?

Solution Let X be the event of getting odd numbers. Let p_1 be the probability of getting 3 odd numbers out of 7 throws and p_2 be the probability of getting 4 odd numbers out of 7 throws. Given, $p_1 = 2p_2$.

$$\Rightarrow p(X = 3) = 2p(X = 4)$$

$$p(X = 3) = \binom{7}{3} (p)^3 (q)^{7-3}$$

$$\begin{aligned}
p(X=4) &= \binom{7}{4} p^4 (q)^{7-4} \\
\Rightarrow p^3 q^4 &= 2p^4 q^3 \\
q &= 2p \\
p + q &= 1 \Rightarrow p = \frac{1}{3} \text{ and } q = \frac{2}{3} \\
P(\text{no odd number}) &= p(X=0) = \binom{7}{0} p^0 (q)^{7-0} \\
&= \left(\frac{2}{3}\right)^7
\end{aligned}$$

Total number of sets to give no odd number

$$\begin{aligned}
&= 5000 \left(\frac{2}{3}\right)^7 = 292.64 \\
\Rightarrow &293 \text{ sets}
\end{aligned}$$

2.35 A system is divided into n components, each of which functions independently with a probability p . The system functions effectively if at least one-half of its components operate. For what values of p is a 5-component system more likely to function effectively than a 3-component system?

Solution Let the random variable X denote the number of components that function effectively. That is, we have

$$P(X=x) = \binom{n}{x} p^x q^{n-x} \quad x=0, 1, 2, \dots, 5$$

For a 5-component system, the system functions effectively if 3 or 4 or 5 components function effectively. Hence, we can write

$$\begin{aligned}
P(\text{5-component system can function}) &= P(X=3) + P(X=4) + P(X=5) \\
&= \binom{5}{3} p^3 q^{5-3} + \binom{5}{4} p^4 q^{5-4} + \binom{5}{5} p^5 q^{5-5} \\
&= 10 p^3 q^2 + 5 p^4 q + p^5
\end{aligned}$$

The 3-component system functions effectively, if 2 or 3 components function effectively.

$$\begin{aligned}
P(\text{3-component system can function}) &= P(X=2) + P(X=3) \\
&= \binom{3}{2} p^2 q^{3-2} + \binom{3}{3} p^3 q^{3-3} \\
&= 3 p^2 q + p^3
\end{aligned}$$

$P(5\text{-component system can function}) \geq P(3\text{-component system function})$

$$10p^3 q^2 + 5p^4 q + p^5 \geq 3p^2 q + p^3$$

$$10pq^2 + 5p^2 q + p^3 \geq 3q + p$$

$$\Rightarrow 10p(1-p)^2 + 5p^2(1-p) + p^3 \geq 3(1-p) + p$$

$$10(p)(1+p^2-2p) + 5p^2(1-p) + p^3 \geq 3-2p$$

$$10p + 10p^3 - 20p^2 + 5p^2 - 5p^3 + p^3 \geq 3-2p$$

$$6p^3 - 15p^2 + 10p \geq 3-2p$$

$$6p^3 - 15p^2 + 12p - 3 \geq 0$$

$$2p^3 - 5p^2 + 4p - 1 \geq 0$$

$$\Rightarrow (p-1)^2(2p-1) \geq 0$$

$$\Rightarrow p \geq \frac{1}{2}$$

$$(p-1)^2 \geq 0 \Rightarrow p \geq 1$$

2.36 Data over a noisy communication channel are transmitted in the form of blocks, each block with 16 binary digits. The probability that a received binary digit is in error because of channel noise is 0.05. Find the probability that the number of errors per block is greater than or equal to 4. Assume that the errors occur in various digit positions within a block independently.

Solution Let X be the event that represents the number of errors in a block.

Given: $p = 0.05$; $q = 0.95$

$$n = 16$$

$$P(X \geq 4) = 1 - P(X < 4)$$

$$= 1 - \{P(X=0) + P(X=1) + P(X=2) + P(X=3)\}$$

$$P(X=x) = \binom{n}{x} (p)^x (q)^{n-x}$$

$$P(X=x) = \binom{16}{x} (0.05)^x (0.95)^{n-x}$$

$$P(X \geq 4) = 1 - \left\{ \binom{16}{0} (0.05)^0 (0.95)^{16} + \binom{16}{1} (0.05) (0.95)^{15} \right.$$

$$\left. + \binom{16}{2} (0.05)^2 (0.95)^{14} + \binom{16}{3} (0.05)^3 (0.95)^{13} \right\}$$

$$= 1 - \{0.44 + 0.37 + 0.1463 + 0.036\}$$

$$= 0.00776$$

2.37 In the month of September, on an average, rain falls on 10 days. Find the probability (a) that the rain will fall on just two days of a given week, (b) that in the first three days of a given week, it rains and the remaining four days it won't rain.

Solution

The probability that rain falls on any day is equal to $\frac{10}{30} = \frac{1}{3}$

The probability that rain falls on x days in a week is

$$\binom{7}{x} \left(\frac{1}{3}\right)^x \left(\frac{2}{3}\right)^{7-x}$$

$$P(X=3) = \binom{7}{3} \left(\frac{1}{3}\right)^3 \left(\frac{2}{3}\right)^{7-3} = 0.256$$

$P(\text{first three days rain and next four days no rain})$

$$= \left(\frac{1}{3}\right)^3 \left(\frac{2}{3}\right)^4 = 0.0073$$

2.38 If the probability of success is $\frac{1}{50}$, how many trials are necessary in order that probability of at least one success is greater than $\frac{1}{2}$?

Solution Let the random variable X denote the number of successes

Given: $p = \frac{1}{50}$ and $q = 1 - p = \frac{49}{50}$

$$P(\text{at least one success}) = P(X \geq 1)$$

$$= 1 - P(X = 0)$$

$$P(X = x) = \binom{n}{x} p^x (1 - p)^{n-x}$$

$$P(X = 0) = \binom{n}{0} p^0 (1 - p)^n$$

$$= \left(\frac{49}{50}\right)^n$$

$$1 - P(X = 0) = 1 - \left(\frac{49}{50}\right)^n > \frac{1}{2}$$

$$\left(1 - \frac{1}{50}\right) > (0.98)^n; \quad \frac{1}{2} > (0.98)^n$$

$$\Rightarrow n = 35$$

2.39 A certain airline, having observed that not all persons making reservation show up for the flight sells 125 seats for a flight that holds only 120 seats. The probability that a passenger does not show up is 0.1 and the passengers behave independently.

- (a) What is the probability that every passenger who shows up can take the flight?
 (b) What is the probability that the flight departs with empty seats?

Solution Let X be the number of persons making reservations on a flight and show up for the flight. The random variable, X has binomial distribution with $n = 125$ and $p = 0.9$.

- (a) The probability that every passenger who shows up can take the flight is

$$\begin{aligned} P(X \leq 120) &= 1 - P(X > 120) \\ &= 1 - P(X = 121) - P(X = 122) - P(X = 123) - P(X = 124) - P(X = 125) \\ &= 1 - \left\{ \binom{125}{121} (0.9)^{121} (0.1)^4 + \binom{125}{122} (0.9)^{122} (0.1)^3 + \binom{125}{123} (0.9)^{123} (0.1)^2 \right. \\ &\quad \left. + \binom{125}{124} (0.9)^{124} (0.1) + \binom{125}{125} (0.9)^{125} (0.1)^0 \right\} \\ &= 0.9961 \end{aligned}$$

- (b) The probability that the flight departs with empty seats

$$\begin{aligned} P(X \leq 119) &= P(X \leq 120) - P(X = 120) = 0.9961 - P(X = 120) = 0.9961 - 7.57 \times 10^{-3} \\ &= 0.9885 \end{aligned}$$

2.40 A family has 5 children. Find the probability that there are (a) 3 boys and 2 girls, and (b) fewer girls than boys.

Solution

$$n = 5 \quad p = 1/2$$

$$(a) \quad P(3 \text{ boys}) = \binom{5}{3} \left(\frac{1}{2}\right)^3 \left(\frac{1}{2}\right)^2 = 0.3125$$

There are fewer girls than boys if there are 0, 1 or 2 girls.

Then, $P = P(0 \text{ girls}) + P(1 \text{ girls}) + P(2 \text{ girls})$

$$\begin{aligned} &= \left(\frac{1}{2}\right)^5 + \binom{5}{1} \frac{1}{2} \left(\frac{1}{2}\right)^4 + \binom{5}{2} \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^3 \\ &= \frac{1}{32} [1 + 5 + 10] = 0.5 \end{aligned}$$



Practice Problems

2.14 A random variable X has a binomial distribution with $n = 10$ and $p = 0.1$. Determine the following probabilities.

- (a) $P(X = 4)$ (b) $P(X \leq 2)$ (c) $P(2 \leq X < 5)$ (Ans. (a) 0.0111 (b) 0.9298 (c) 0.2623)

2.15 A multiple-choice test has six questions, each of which has four possible answers. What is the probability that a student will get four or more correct answers by just guessing? (Ans. 0.0376)

2.16 In an electronics laboratory, it is found that 10% of transistors are defective. A random sample of 20 transistors are taken for inspection. (a) What is the probability that all are good? (b) What is the probability that at the most there are 3 defective transistors.

(Ans. (a) 0.1216 (b) 0.8671)

2.17 In a large consignment of electric bulbs 10% are known to be defective. A random sample of 20 is taken for inspection. Find the probability that (i) all are good bulbs (ii) at most 3 are defective bulbs. (Ans. (i) 0.1216 (ii) 0.8666)

2.5.2 Multinomial Distribution

Multinomial distribution may be thought of as a generalization of the binomial distribution. Consider an experiment E , with sample S . Let the sample space S be partitioned into k mutually exclusive events $E_1, E_2, \dots, E_k$. Consider n independent repetitions of E and let $p_i = P(E_i)$ with $\sum_{i=1}^k p_i = 1$. Let X_i be the number of times E_i occurs among n repetitions of E , $i = 1, \dots, k$. Then

$$P(X_1 = n_1, X_2 = n_2, \dots, X_k = n_k) = \frac{n!}{n_1! n_2! \dots n_k!} p_1^{n_1} \dots p_k^{n_k} \quad (2.27)$$

where

$$\sum_{i=1}^k n_i = n$$

Solved Problems



2.41 A lab attendant keeps a large number of resistors in a drawer. About 50 percent of these resistors are 1 kΩ, about 30% are 10 kΩ and the remaining 20% are 1 MΩ. Suppose that 10 resistors are chosen at random. What is the probability that there are exactly five 1 kΩ resistors, four 10 kΩ resistors and one 1 MΩ resistor?

Solution

$$p_1 = P(R = 1 \text{ k}\Omega) = 0.5$$

$$p_2 = P(R = 10 \text{ k}\Omega) = 0.3$$

$$p_3 = P(R = 1 \text{ M}\Omega) = 0.2$$

Also, $n = 10$; $n_1 = 5$; $n_2 = 4$; $n_3 = 1$

Let X_1, X_2 and X_3 be the events of selecting a 1 kΩ resistor, 10 kΩ resistor and 1 MΩ resistor.

$$P(X_1 = 5, X_2 = 4, X_3 = 1) = \frac{10!}{5! 4! 1!} (0.5)^5 (0.3)^4 (0.2)^1 = 0.06378$$

2.5.3 Poisson Distribution

A discrete random variable X taking on one of the values 0, 1, 2 ... is said to be a Poisson random variable with parameter λ , where $\lambda > 0$, if its pmf is of the form

$$p_X(k) = \frac{\lambda^k e^{-\lambda}}{k!} \quad k = 0, 1, 2, \dots \quad (2.28)$$

Equation (2.28) defines the pmf, since

$$\sum_{k=0}^{\infty} p_X(k) = e^{-\lambda} \sum_{k=0}^{\infty} \frac{\lambda^k}{k!} = e^{-\lambda} \left[1 + \frac{\lambda}{1!} + \frac{\lambda^2}{2!} + \dots \right] = e^{-\lambda} e^{\lambda} = 1$$

The CDF of X is given by

$$F_X(k) = P(X \leq k) = \sum_{r=0}^k \frac{\lambda^r e^{-\lambda}}{r!} \quad (2.29)$$

The Poisson random variable density and distribution functions can also be expressed as

$$f_X(k) = e^{-\lambda} \sum_{k=0}^{\infty} \frac{\lambda^k}{k!} \delta(x - k) \quad (2.30)$$

$$\text{and} \quad F_X(k) = e^{-\lambda} \sum_{k=0}^{\infty} \frac{\lambda^k}{k!} u(x - k) \quad (2.31)$$

The Poisson distribution has many applications in diverse areas because it may be used as an approximation for a binomial random variable with parameters (n, p) when n is large and p is small so that their product np is a moderate size. The Poisson distribution can be used in applications such as the number of telephone calls arriving at a switchboard during various intervals of time, number of customers arriving at a bank during various intervals of time, random failure of experiments, number of earthquakes occurring during some fixed time span, and the number of electrons emitted from a heated cathode during a fixed time period are usually modeled by Poisson random variables.

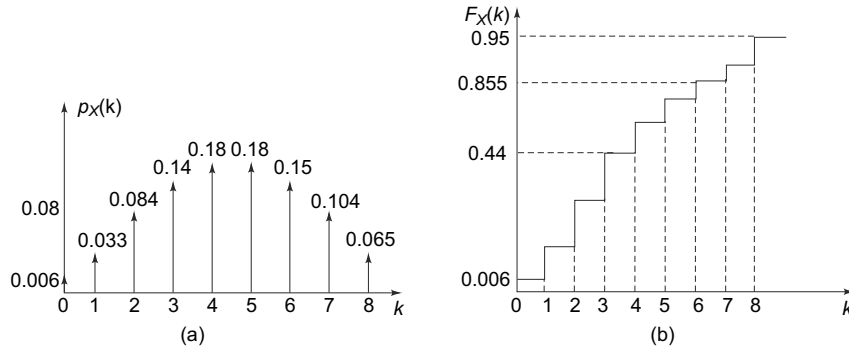


Fig. 2.19 Poisson (a) density, and (b) distribution functions for $\lambda = 5$

The Poisson density and distribution functions are shown in Fig. 2.19.

2.42 A random variable X is known to be Poisson with $\lambda = 4$ (a) Plot the density and distribution functions for this random variable. (b) What is the probability of the event $0 \leq X \leq 5$?

Solution Given: $\lambda = 4$

(a) The density function

$$f_X(x) = e^{-\lambda} \sum_{k=0}^{\infty} \frac{\lambda^k}{k!} \delta(x - k)$$

$$f_X(0) = \frac{e^{-4} (4)^0}{0!} = e^{-4} = 0.0183$$

$$f_X(1) = \frac{e^{-4} (4)^1}{1!} = 0.0732$$

$$f_X(2) = \frac{e^{-4} (4)^2}{2!} = 0.1465$$

$$f_X(3) = \frac{e^{-4} (4)^3}{3!} = 0.1953$$

$$f_X(4) = \frac{e^{-4} (4)^4}{4!} = 0.1953$$

$$f_X(5) = \frac{e^{-4} (4)^5}{5!} = 0.1563$$

$$f_X(6) = \frac{e^{-4} (4)^6}{6!} = 0.1042$$

$$f_X(7) = \frac{e^{-4} (4)^7}{7!} = 0.0595$$

$$f_X(8) = \frac{e^{-4} (4)^8}{8!} = 0.0298$$

$$f_X(9) = \frac{e^{-4} (4)^9}{9!} = 0.01323$$

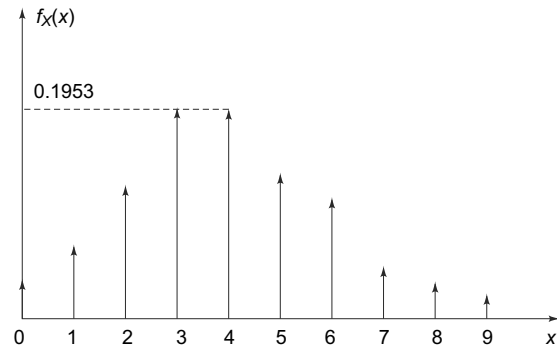


Fig. 2.20 pdf of Poisson random variable

The plot of the density function is shown in Fig. 2.20.

The distribution function is given by

$$F_X(k) = e^{-\lambda} \sum_{k=0}^{\infty} \frac{\lambda^k}{k!} u(x - k)$$

or
$$F_X(k) = P[X \leq k] = \sum_{k=0}^{\infty} \frac{\lambda^k e^{-\lambda}}{k!}$$

$$F_X(0) = \frac{e^{-4}(4)^0}{0!} = 0.0183$$

$$F_X(1) = \frac{e^{-4}(4)^0}{0!} + \frac{e^{-4}(4)^1}{1!} = 0.0183 + 0.0732 = 0.0915$$

$$F_X(2) = \frac{e^{-4}(4)^0}{0!} + \frac{e^{-4}(4)^1}{1!} + \frac{e^{-4}(4)^2}{2!} = 0.0183 + 0.0732 = 0.1465 = 0.238$$

$$F_X(3) = 0.238 + f_X(3) = 0.238 + 0.1953 = 0.4333$$

$$F_X(4) = 0.4333 + f_X(4) = 0.4333 + 0.1953 = 0.6286$$

$$F_X(5) = 0.6286 + f_X(5) = 0.6286 + 0.1563 = 0.7849$$

$$F_X(6) = 0.7849 + f_X(6) = 0.7849 + 0.1042 = 0.8891$$

$$F_X(7) = 0.8891 + f_X(7) = 0.8891 + 0.0595 = 0.9486$$

$$F_X(8) = 0.9486 + f_X(8) = 0.9486 + 0.0298 = 0.9784$$

$$F_X(9) = 0.9784 + f_X(9) = 0.9784 + 0.01323 = 0.9916$$

The plot of the distribution function is shown in Fig. 2.21.

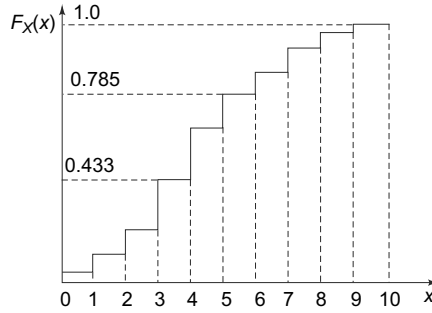


Fig. 2.21 Distribution function of Poisson random variable

(b) $P[0 \leq X \leq 5] = F_X(5) = 0.7849$.

Given a time interval T we define a random variable x which represents the number of outcomes occurring during the time interval T . Clearly, $X = k$ means the number of outcomes in the interval $(0, T)$ equals k . In such a case, the distribution of X is given by

$$P(X = k) = \frac{e^{-\lambda T} (\lambda T)^k}{k!} \quad (2.32)$$

Thus the number of outcomes in an interval of length T is a Poisson distributed random variable with parameter $b = \lambda T$, where λ is the average number of outcomes in the given interval.

2.43 People enter a club at a rate of two for every 4 minutes

- What is the probability that no one enters between 10.00 p.m. and 10.10?
- What is the probability that at least 4 people enter during that time?

Solution Given: $\lambda = 2/4 = \frac{1}{2}$ and $T = 10$ min

$$b = \lambda T$$

$$= \frac{1}{2} (10) = 5$$

$$\begin{aligned}
 \text{(a)} \quad P(X = k) &= \frac{e^{-b}(b)^k}{k!} \\
 P(X = 0) &= \frac{e^{-5}(0.5)^0}{0!} = 0.6065 \\
 \text{(b)} \quad P(X \leq 4) &= 1 - P(X \leq 3) = 1 - (P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3)) \\
 P(X = 1) &= \frac{e^{-5} 5}{1!} = 0.03369 \\
 P(X = 2) &= \frac{e^{-5} (5)^2}{2!} = 0.08422 \\
 P(X = 3) &= \frac{e^{-5} (5)^3}{3!} = 0.1404 \\
 P(X \leq 4) &= 1 - (0.6065 + 0.03369 + 0.08422 + 0.1404) \\
 &= 0.1352
 \end{aligned}$$

REVIEW QUESTIONS



6. What is Poisson random variable? Explain in brief.
7. Prove that for large values of n , binomial distribution can be approximated to Poisson distribution.

Solved Problems



2.44 In a country, the suicide rate is five per million people per month. Find the probability that in a city of a population of 10,00,000, there will be at most five suicides in a month.

Solution Given: $n = 10,00,000$

$$p = \frac{5}{10,00,000} = 5 \times 10^{-6}$$

$$np = 10,00,000 (5 \times 10^{-6}) = 5$$

Since $np < 10$, the Poisson distribution with $\lambda = 5$ can be assumed for this problem.

Using Poisson pdf, we can find

$$\begin{aligned}
 P(X \leq 4) &= \sum_{k=0}^4 e^{-\lambda} \frac{\lambda^k}{k!} \\
 &= \sum_{k=0}^4 \frac{e^{-5} (5)^k}{k!} \\
 &= e^{-5} \left[1 + \frac{5}{1} + \frac{5^2}{2!} + \frac{5^3}{3!} + \frac{5^4}{4!} \right] \\
 &= 0.4404
 \end{aligned}$$

2.45 The accident rate in a year is one per thousand people. Given that an insurance company has insured 5,000 persons from the population, find the probability that at most two persons will incur this accident.

Solution

Given: $p = \frac{1}{1000} = 0.001$
 $n = 5000$
 $np = 5$

Since $np < 10$, we can use Poisson distribution with $\lambda = 5$

$$P[X = k] = \frac{e^{-\lambda} (\lambda)^k}{k!}$$

$$P[X \leq 2] = P[X = 0] + P[X = 1] + P[X = 2]$$

$$= \frac{e^{-5} (5)^0}{0!} + \frac{e^{-5} 5}{1!} + \frac{e^{-5} (5)^2}{2!}$$

$$= \left(1 + 5 + \frac{5^2}{2}\right) e^{-5} = 0.1247$$

2.46 Suppose you buy a lottery ticket in 50 lotteries in each of which your chance of winning a prize is $\frac{1}{100}$. What is the (approximate) probability that you will win a prize (a) at least once? (b) exactly once? (c) at least twice.

Solution

$$\lambda = np = 50 \left(\frac{1}{100} \right) = 0.5$$

$$P(X = k) = \frac{e^{-\lambda} (\lambda)^k}{k!}$$

(a) $P(X \geq 1) = 1 - P(X = 0) = 1 - e^{-0.5} = 0.3935$

(b) $P(X = 1) = \frac{e^{-0.5} (0.5)}{1} = 0.3037$

(c) $P(X \geq 2) = 1 - P(X = 0) - P(X = 1)$
 $= 1 - e^{-0.5} - e^{-0.5} (0.5)$
 $= 0.0902$

Poisson Approximation to the Binomial Distribution

Consider a binomial random variable X with parameters (n, p) and pmf

$$P_X(x) = \binom{n}{x} p^x (1-p)^{n-x}, x = 0, 1, 2, \dots, n \quad (2.33)$$

where $\binom{n}{x}$ is a binomial coefficient given by $\frac{n!}{(n-x)! x!}$. That is, the pmf involves evaluating $n!$ whose value is very high for even moderate values of n . Therefore, it is required to develop an approximate method for large values of n .

Let us define a parameter λ which is equal to the product of n and p . That is, $\lambda = np$ from which we can write $p = \lambda/n$. Substituting this value in the expression for pmf, we obtain

$$\begin{aligned}
 P_X(x) &= \binom{n}{x} \left(\frac{\lambda}{n}\right)^x \left(1 - \frac{\lambda}{n}\right)^{n-x} \\
 &= \frac{n!}{(n-x)!x!} \left(\frac{\lambda}{n}\right)^x \left(1 - \frac{\lambda}{n}\right)^{n-x} \\
 &= \frac{n(n-1)(n-2)\cdots(n-x+1)(n-x)!}{x!(n-x)!n^x} (\lambda)^x \left(1 - \frac{\lambda}{n}\right)^{n-x} \\
 &= \frac{n^x \left(1 - \frac{1}{n}\right)\left(1 - \frac{2}{n}\right)\cdots\left(1 - \frac{x-1}{n}\right) \lambda^x \left(1 - \frac{\lambda}{n}\right)^{n-x}}{x!n^x} \\
 &= \frac{\left(1 - \frac{1}{n}\right)\left(1 - \frac{2}{n}\right)\cdots\left(1 - \frac{x-1}{n}\right) \lambda^x \left(1 - \frac{\lambda}{n}\right)^n \left(1 - \frac{\lambda}{n}\right)^{-x}}{x!}
 \end{aligned} \tag{2.34}$$

For $n \rightarrow \infty$, the above expression reduces to

$$P_X(x) = \frac{\lambda^x}{x!} \lim_{n \rightarrow \infty} \left(1 - \frac{\lambda}{n}\right)^n \left(1 - \frac{\lambda}{n}\right)^{-x}$$

Using the expressions

$$\lim_{n \rightarrow \infty} \left(1 - \frac{\lambda}{n}\right)^n = 1 \quad \text{for } \lambda < n$$

and
$$\lim_{n \rightarrow \infty} \left(1 + \frac{\lambda}{n}\right)^n = e^\lambda$$

we get

$$\lim_{n \rightarrow \infty} P_X(x) = \frac{\lambda^x e^{-\lambda}}{x!} \tag{2.35}$$

which is a Poisson distribution.

2.47 Assume automobile arrives at a gasoline station are Poisson and occur at an average rate of 50/h. If all cars are assumed to require one minute to obtain fuel, what is the probability that a waiting line will occur at the pump?

Solution Average rate at which an automobile arrives at a gasoline station is 50/h.

Therefore,
$$\lambda = \frac{50}{60} \text{ arrivals/minute}$$

$$T = 1 \text{ minute}$$

$$b = \lambda T = \frac{50}{60} (1) = \frac{5}{6}$$

A waiting line occurs if two or more cars arrive in any one-minute interval. Therefore, the probability that a waiting line occurs can be obtained by evaluating $P(X \geq 2)$.

$$\begin{aligned} P(X \geq 2) &= 1 - P(X \leq 1) \\ &= 1 - F_X(1) - F_X(0) \end{aligned}$$

For Poisson distribution,

$$F_X(x) = e^{-b} \sum_{k=0}^{\infty} \frac{b^k}{k!} u(x - k)$$

$$b = \frac{5}{6}$$

$$P(X \geq 2) = 1 - e^{-\frac{5}{6}} \left[1 + \frac{5}{6} \right] = 0.2032$$

2.48 A manufacturer of cotton pins knows that 5% of his products are defective. If he sells pins in boxes of 100 and guarantees that not more than 4 pins will be defective, what is the approximate probability that a box will fail to meet the guaranteed quality?

Solution Given:

$$p = \frac{5}{100} = 0.05 \text{ and } n = 100$$

$$\lambda = np = 100(0.05) = 5$$

We have

$$P(X = x) = \frac{e^{-\lambda} \lambda^x}{x!}$$

The box will fail to meet the guaranteed quality if the box contains more than 4 defective pins.

$$\begin{aligned} P(X > 4) &= 1 - P(X \leq 4) \\ &= 1 - \{P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3) + P(X = 4)\} \\ &= 1 - e^{-5} \left(1 + 5 + \frac{5^2}{2!} + \frac{5^3}{3!} + \frac{5^4}{4!} \right) \\ &= 0.5595 \end{aligned}$$

2.49 Wireless sets are manufactured with 25 soldered joints, each on the average of 1 joint in 500 defective. How many sets can be expected to be free from defective joints in a consignment of 10000 sets?

Solution Given: $n = 25$ and $p = \frac{1}{500}$

Let X be the random variable that represents the number of defective joints in a set.

$$P(X=x) = \frac{e^{-\lambda} \lambda^x}{x!}$$

$$P(X=0) = \frac{e^{-1/20} \left(\frac{1}{20}\right)^0}{0!} = e^{-1/20} = 0.9512$$

The expected number of sets that are free from defects is

$$(0.9512) \times (10,000) = 9512$$

2.50 The number of particles emitted from a radioactive source in a specified interval follows Poisson distribution. If the probability of no emission equals $\frac{1}{3}$, what is the probability that 2 or more emissions occur?

Solution Let X be the random variable that denotes the number of particles emitted. We have

$$P(X=x) = \frac{e^{-\lambda} \lambda^x}{x!}$$

$$P(X=0) = e^{-\lambda} = \frac{1}{3}$$

from which $\lambda = \ln 3$

$$P(2 \text{ or more emissions}) = P(X \geq 2)$$

$$1 - \{P(X=0) + P(X=1)\}$$

$$P(X=1) = \frac{e^{-\lambda} \lambda}{1!} = \frac{1}{3} \ln 3$$

$$\begin{aligned} P(X \geq 2) &= 1 - \left\{ \frac{1}{3} + \frac{1}{3} \ln 3 \right\} = \frac{2}{3} - \frac{1}{3} \ln 3 \\ &= \frac{(2 - \ln 3)}{3} \end{aligned}$$

2.51 A company markets a modem that has a bit error probability of 10^{-4} , and the bit errors are independent. The buyer will test the modem by sending a known message of 10^4 digits and checking the received message. He will reject the modem if the number of errors are more than two. Find the probability that the customer will buy the modem.

Solution Given: $p = 10^{-4}$ and $n = 10^4$

$$\lambda = np = 10^{-4}(10^4) = 1$$

Let X be the random variable for the number of errors.

The customer will buy the modem if $X \leq 2$.

Therefore,

$$P(X \leq 2) = P(X = 0) + P(X = 1) + P(X = 2)$$

$$P(X = x) = \frac{e^{-\lambda} \lambda^x}{x!}$$

$$P(X \leq 2) = e^{-1} \left[1 + 1 + \frac{1^2}{2!} \right] = e^{-1}(2.5) = 0.9197$$

2.52 In a lot of semiconductor diodes, 1 in 400 diodes is defective. If the diodes are packed in boxes of 100, what is the probability that any given box of diodes will contain (a) no defective, (b) 1 or more defective, and (c) less than 2 defectives diodes?.

Solution The probability $p = \frac{1}{400}$

$$n = 100$$

$$\lambda = np = \frac{100}{400} = \frac{1}{4} = 0.25$$

Let X be the random variable that represents the defective diode

$$P(X = x) = \frac{e^{-\lambda} \lambda^x}{x!}$$

$$(a) \quad P(X = 0) = e^{-0.25} = 0.7788$$

$$(b) \quad P(1 \text{ or more defective}) = P(X \geq 1) = 1 - P(X = 0) \\ = 1 - 0.7788 = 0.2212$$

$$(c) \quad P(\text{less than 2 defectives}) = P(X < 2) = P(X = 0) + P(X = 1) \\ = 0.7788 + e^{-0.25} \times 0.25 = 0.9735$$

2.53 The number of page requests that arrive at a Web server is a Poisson random variable with an average of 3000 requests per minute

(a) Find the probability that there are no requests in a 100 ms period.

(b) Find the probability that there are between 2 and 4 requests in a 100 ms period.

Solution Given the number of page requests is 300 per minute, which is equal to 50 requests per second or 5 requests/100 ms.

Therefore, $\lambda = 5$.

$$P[X = x] = \frac{e^{-\lambda} \lambda^x}{x!}$$

$$P(X = 0) = e^{-\lambda} = e^{-5}$$

$$P(2 \leq X \leq 4) = \sum_{x=2}^4 \frac{(5)^x e^{-5}}{x!} \\ = e^{-5} \left[\frac{(5)^2}{2!} + \frac{(5)^3}{3!} + \frac{(5)^4}{4!} \right]$$

$$\begin{aligned}
&= e^{-5} 5^2 \left[\frac{1}{2!} + \frac{5}{3!} + \frac{25}{4!} \right] \\
&= e^{-5} (25) \left[\frac{1}{2} + \frac{5}{6} + \frac{25}{24} \right] \\
&= 0.4
\end{aligned}$$

2.54 Let X be a Poisson random variable with parameter λ . What value of λ maximizes $P(X = k)$?

Solution For a Poisson random variable,

$$P(X = k) = \frac{e^{-\lambda} \lambda^k}{k!}$$

$$\frac{dp}{d\lambda} = \frac{1}{k!} \left[-\lambda^k e^{-\lambda} + k e^{-\lambda} \lambda^{k-1} \right] = 0$$

$$\Rightarrow e^{-\lambda} [\lambda - k] \lambda^{k-1} = 0$$

from which $\lambda = k$

2.55 In an industrial area, the average number of fatal accidents per month is $\frac{1}{3}$. The number of accidents per month is described by a Poisson distribution. What is the probability that 6 months will pass without an accident.

Solution Given: $\lambda = \frac{1}{3}$; $T = 6 \Rightarrow b = \lambda T = \frac{1}{3}(6) = 2$

Let X be the random variable that represents the number of accidents.

$$P(X = x) = \frac{e^{-b} b^x}{x!}$$

$$P(X = 0) = \frac{e^{-2}(2)^0}{0!} = e^{-2} = 0.135$$

2.56 Let X be a Poisson random variable with parameter λ . Show that

$$P(X \text{ is even}) = \frac{1}{2} (1 + e^{-2\lambda})$$

Solution

$$P(X \text{ is even}) = P(X = 0) + P(X = 2) + P(X = 4) + \dots$$

$$P(X = x) = \frac{e^{-\lambda} \lambda^x}{x!}$$

$$P(X \text{ is even}) = e^{-\lambda} + \frac{e^{-\lambda} \lambda^2}{2!} + \frac{e^{-\lambda} \lambda^4}{4!} + \dots$$

$$\begin{aligned}
&= e^{-\lambda} \left(1 + \frac{\lambda^2}{2!} + \frac{\lambda^4}{4!} + \dots \right) \\
&= e^{-\lambda} \frac{(e^{\lambda} e^{-\lambda})}{2} \\
&= \frac{1 + e^{-2\lambda}}{2}
\end{aligned}$$

2.57 The number of errors in a textbook follow a Poisson distribution with a mean of 0.02 error per page. What is the probability that there are two or less errors in 100 pages?

Solution

$$\begin{aligned}
\lambda &= 0.02 \times 100 = 2 \\
P(X \leq 2) &= P(X = 0) + P(X = 1) + P(X = 2) \\
&= e^{-2} + \frac{e^{-2} \cdot 2}{1!} + \frac{e^{-2} (2)^2}{2!} \\
&= e^{-2} (1 + 2 + 2) = 5e^{-2} \\
&= 0.6766
\end{aligned}$$

2.58 A textbook that contains 400 pages has 250 errors. Find the probability that a given page contains (a) 2 errors, and (b) 2 or more errors.

Solution Given: $n = 250$ $p = \frac{1}{400}$

$$\lambda = np = \frac{250}{400} = \frac{5}{8}$$

$$(a) \quad P(2 \text{ errors}) = \frac{\left(\frac{5}{8}\right)^2 e^{-5/8}}{2} = 0.1045$$

$$P(0 \text{ errors}) = \frac{\left(\frac{5}{8}\right)^0 e^{-5/8}}{0!} = 0.535$$

$$(b) \quad P(1 \text{ error}) = \frac{\left(\frac{5}{8}\right)^1 e^{-5/8}}{1!} = 0.3345$$

$$\begin{aligned}
P &= 1 - P(0 \text{ or } 1 \text{ error}) = 1 - P(0 \text{ errors}) - P(1 \text{ error}) \\
&= 1 - (0.535) - 0.3345 \\
&= 0.1305
\end{aligned}$$



Practice Problems

2.18 If customer arrive at a counter in accordance with Poisson with mean of 2 per minute, find the probability that the interval between two consecutive arrivals is more than 1 minute. (Ans. 0.135)

2.19 The number of telephone calls that arrive at a telephone exchange is modelled as a Poisson random variable with 15 average calls per hour

- (a) What is the probability that there are 5 or less calls in one hour?
 (b) What is the probability that there are exactly 20 calls in one hour (Ans. (a) 0.00279; (b) 0.0418)

2.20 A random variable X is known to be Poisson with $\lambda = 5$ (a) plot the density and distribution function for this random variable. (b) What is the probability of the event $(0 \leq X \leq 6)$. (Ans. 0.759)

2.21 In a city the average murders per week is 5 and their occurrences follow a Poisson distribution. What is the probability that there will be 8 or more murders in a given week? (Ans. 0.133)

2.22 In an industry, the average number of fatal accidents per month is 1. The number of accidents per month follows a Poisson distribution. What is the probability that 6 months will pass without a fatal accident. (Ans. 0.00248)

2.23 Find the probability that at most 5 defective fuses will be found in a box of 200 fuses if experience shows that 2% of such fuses are defective. (Ans. 0.781)

2.24 A switchboard can handle only 24 phone calls per minute. If the incoming calls per minute follow a Poisson distribution with parameter 16, find the probability that the switchboard is overloaded in any one minute. (Ans. 0.29)

2.5.4 The Geometric Distribution

Suppose that we perform an experiment A repetitively such that the repetitions are independent and on each repetition the probability of success is p and that of failure is q . Then the pdf of the random variable X for which the number of repetitions required until a first success occurs is given by

$$P(X = n) = p(1 - p)^{n-1}; n = 1, 2, \dots \quad (2.36)$$

In order for X to equal n , the first $(n - 1)$ attempts must result a failure and the n th trial is a success.

Properties of Geometric Distributions

1. The geometric distribution has memoryless property. Suppose that an event A has not occurred in the first n trials. Then the probability it will not occur in the next n trials is same.
2. The sum of geometric random variables is negative binomial.

Solved Problems



2.59 Prove that the geometric distribution possesses memoryless property.

Solution We have $P(X = n) = p(1 - p)^{n-1}; n = 1, 2, \dots$

For memoryless property we have to show that

$$P(X > s + t | X > t) = P(X > s)$$

$$\begin{aligned} P(X > t) &= \sum_{n=t}^{\infty} p(1 - p)^n = \sum_{n=t}^{\infty} pq^n = p(q^t + q^{t+1} + \dots) \\ &= pq^t(1 + q + \dots) = pq^t \frac{1}{1 - q} = \frac{pq^t}{p} = q^t \end{aligned}$$

$$P(X > s + t | X > t) = \frac{P(X > s + t)}{P(X > t)} = \frac{q^{s+t}}{q^t} = q^s$$

We have, $P(X > s) = q^s$

Therefore, $P(X > s + t | X > t) = P(X > s)$

2.60 The probability that rain will occur on any given day during summer (May to June) equals 0.15. Assuming independence from day to day, what is the probability that it rains on May 4?

Solution We have $P(X = n) = pq^{n-1}$. Here, $n = 4$.
Therefore,

$$\begin{aligned} P(X = 4) &= (0.15) (0.85)^3 \\ &= 0.092 \end{aligned}$$

2.61 A dice is thrown repeatedly until 5 appears. What is the probability that it must be thrown more than 4 times?

Solution Given $n = 4$

The probability of getting 5 in throwing a dice is $\frac{1}{6}$

$$p = \frac{1}{6} \text{ and } q = \frac{5}{6}$$

$$P(X = n) = pq^{n-1}$$

$$\begin{aligned} P(X > 4) &= 1 - \{P(X = 1) + P(X = 2) + P(X = 3) + P(X = 4)\} \\ &= 1 - \{p + pq + pq^2 + pq^3\} \\ &= 1 - p(1 + q + q^2 + q^3) \\ &= 1 - p \frac{(1 - q^4)}{1 - q} = q^4 \\ &= \left(\frac{5}{6}\right)^4 \end{aligned}$$

2.62 Suppose that a trainee soldier shoots a target in an independent fashion. The probability that the target is hit on any one shot is 0.7.

- What is the probability that the target would be hit on 10th attempt?
- What is the probability that it takes him less than 4 shots?
- What is the probability that it takes him an even number of shots?

Solution Given: $p = 0.7$ and $q = 0.3$

- $P(\text{target hit on 10th attempt})$

$$= p(X = 10) = pq^9 = (0.7) (0.3)^9 = 0.0000138$$

- $P(\text{target hit in less than 4 shots})$

$$= P(X \leq 4) = P(X = 1) + P(X = 2) + P(X = 3) + P(X = 4)$$

$$= p + pq + pq^2 + pq^3$$

$$= \frac{p(1 - q^3)}{1 - q} = 1 - q^3$$

$$= 1 - (0.3)^3 = 0.973$$

(c) $P(\text{target hit in even number of shots})$

$$= P(X = 2) + P(X = 4) + P(X = 6) + \dots$$

$$= pq + pq^3 + pq^5 + \dots$$

$$= pq(1 + q^2 + q^4 + \dots) = \frac{pq}{1 - q^2}$$

$$= \frac{(0.7)(0.3)}{1 - (0.3)^2} = 0.23077$$

2.63 A student passing his examination on any attempt is 0.75. What is the probability that he will pass the examination (a) in the first attempt, and (b) less than three attempts?

Solution Let X denote the number of attempts.

The probability that the student passes the examination in the third attempt is

$$P(X = 3) = pq^2 = 0.75(0.25)^2 = 0.0469$$

The probability that he will pass in less than three attempts

$$P(X < 3) = P(X = 1) + P(X = 2)$$

$$p + pq = p(1 + q) = 0.75(1 + 0.25)$$

$$= 0.9375$$

2.64 The probability of a successful optical alignment in the assembly of an optical storage product is 0.8. Assume that trials are independent, what is the probability that the first successful alignment requires exactly four trials?

Solution Let X denote the number of trials required to achieve the first successful alignment. Then

$$P(X = 4) = pq^{4-1} = pq^3$$

Given $p = 0.8$ and $q = 0.2$

$$P(X = 4) = 0.8(0.2)^3 = 0.0064$$



Practice Problems

2.25 If the probability that a student passes the examination equals 0.4, what is the probability that he need fewer than 5 attempts before he passes the examination? (Ans. 0.92)

2.26 The probability that a candidate can pass in an examination is 0.6.

(a) What is the probability that he pass in the third trial?

(b) What is the probability that he pass before the third trial?

(Ans. 0.096, 0.84)

Solved Problems

2.65 A coin is tossed until the first head occurs. If the tosses are independent and the probability of a head occurring is p , find the value of p so that the probability that an odd number of tosses is required is equal to 0.75.

Solution Let X be the random variable that represent number of tosses required to get first head.

$P(\text{odd number of tosses required to get first head})$

$$= P(X = 1) + P(X = 3) + P(X = 5) + \dots$$

$$= p + pq^2 + pq^4 + \dots$$

$$= P(1 + q^2 + q^4 + \dots)$$

$$1 + q^2 + q^4 + \dots = \frac{1}{1 - q^2}$$

$$= \frac{p}{1 - q^2} = \frac{P}{(1 - q)(1 + q)} = \frac{p}{p(1 + q)}$$

$$= \frac{1}{1 + q}$$

$$\text{Given } \frac{1}{1 + q} = 0.75 \Rightarrow q = \frac{0.25}{0.75} = \frac{1}{3}$$

2.66 If X is a geometric random variable, show analytically that

$$P(X = n + k | X > n) = P(X = k)$$

Solution

$$P(X = n + k) = pq^{n+k-1}$$

$$P(X > n) = \sum_{k=n+1}^{\infty} pq^{k-1} = pq^n + pq^{n+1} + pq^{n+2} + \dots$$

$$= pq^n[1 + q + q^2 + \dots] = \frac{pq^n}{1 - q} = q^n$$

$$P(X = n + k | X > n) = \frac{P(X = n + k)}{P(X > n)}$$

$$\because P + q = 1$$

$$= \frac{pq^{n+k-1}}{q^n} = pq^{k-1}$$

$$\text{We have } P(X = k) = pq^{k-1}$$

$$\text{Therefore, } P(X = n + k | X > n) = pq^{k-1}$$

2.67 A and B shoot independently until each has his own target. The probability that A and B hit the target is $\frac{2}{3}$ and $\frac{4}{7}$ respectively. Find the probability that B will require more shots than A .

Solution Let A require X trials to first hit the target. Then X follows geometric distribution with pdf

$$P(X = n) = p_1 q_1^{n-1}$$

Given $p_1 = \frac{2}{3}; q_1 = \frac{1}{3}$

$$\Rightarrow P(X = n) = \frac{2}{3} \left(\frac{1}{3} \right)^{n-1} \quad n = 1, 2, \dots$$

Let B require Y trials to first hit the target. Then

$$P(Y = n) = p_2 q_2^{n-1}$$

Given $p_2 = \frac{4}{7} \Rightarrow q_2 = \frac{3}{7}$

$$P(Y = n) = \frac{4}{7} \left(\frac{3}{7} \right)^{n-1}; n = 1, 2, \dots$$

The event that B requires more shots than A occurs when $X = n$ and $Y = n + 1$ or $n + 2 \dots$

Therefore, $P(B \text{ requires more shots than } A)$

$$= \sum_{n=1}^{\infty} P(X = n) \text{ and } P(Y = n + 1, n + 2 \dots)$$

Since A and B shoot independently,

$P(B \text{ requires more shots than } A)$

$$\begin{aligned} &= \sum_{n=1}^{\infty} P(X = n) P(Y = n + 1 \text{ or } Y = n + 2 \text{ or } \dots) \\ &= \sum_{n=1}^{\infty} \frac{2}{3} \left(\frac{1}{3} \right)^{n-1} \left\{ \frac{4}{7} \left(\frac{3}{7} \right)^n + \frac{4}{7} \left(\frac{3}{7} \right)^{n+1} + \dots \right\} \\ &= \sum_{n=1}^{\infty} \frac{2}{3} \left(\frac{1}{3} \right)^{n-1} \cdot \frac{4}{7} \left(\frac{3}{7} \right)^n \left\{ 1 + \frac{3}{7} + \left(\frac{3}{7} \right)^2 + \dots \right\} \\ &= \sum_{n=1}^{\infty} \frac{2}{3} \left(\frac{1}{3} \right)^{n-1} \cdot \frac{4}{7} \left(\frac{3}{7} \right)^n \frac{1}{\left(1 - \frac{3}{7} \right)} \\ &= \sum_{n=1}^{\infty} \frac{2}{3} \left(\frac{1}{3} \right)^{n-1} \left(\frac{3}{7} \right)^n = \frac{2}{7} \sum_{n=1}^{\infty} \left(\frac{1}{7} \right)^{n-1} \\ &= \frac{2}{7} \left(1 + \frac{1}{7} + \dots \right) = \frac{2}{7} \left(\frac{1}{1 - \frac{1}{7}} \right) = \frac{1}{3} \end{aligned}$$

2.68 If the probability that an applicant for a driver's license will pass the road test on any given trial is 0.8, what is the probability that he will finally pass the test

- (a) On the fourth trial, and
- (b) In less than 4 trials?

Solution Let X represent the number of trials required to achieve the first success. Then X is a random variable with geometric distribution

$$P(X = x) = pq^{x-1}, x = 1, 2, 3 \dots \infty$$

Given $p = 0.8 \Rightarrow q = 0.2$

- (a) $P(\text{passing the test in 4}^{\text{th}} \text{ trial}) = P(X = 4)$

$$\begin{aligned} P(X = 4) &= 0.8(0.2)^{4-1} = 0.8(0.2)^3 \\ &= 0.0064 \end{aligned}$$

- (b) $P(\text{passing the test in less than 4-trials})$

$$P(X < 4) = P(X = 1) + P(X = 2) + P(X = 3)$$

$$\begin{aligned} &= \sum_{x=1}^3 pq^{x-1} = \sum_{x=1}^3 (0.8)(0.2)^{x-1} \\ &= 0.8 + (0.8)(0.2) + 0.8(0.2)^2 \\ &= 0.8(1 + 0.2 + 0.2^2) \\ &= 0.992 \end{aligned}$$

2.5.5 Negative Binomial Distribution (Pascal Distribution)

In a geometric distribution, we repeat the trials until the first success occurs. Suppose we repeat the trials until r successes occur and the number of trials taken for r successes is equal to n . Then

$$P(X = n) = \binom{n-1}{r-1} p^r q^{n-r}, n = r, r+1, \dots \quad (2.37)$$

In order to get r th success in n th trial, there must be $(r-1)$ successes in the previous $(n-1)$ trials. The random variable X whose pmf follows Eq. (2.37) is said to be a negative binomial random variable. If $r = 1$, the above equation reduces to geometric distribution.

If X_1 is the number of trials required for the first success, X_2 is the number of additional trials required after the first success for second success and X_3 is the number of additional trials required after the second

success for third success, then $X = \sum_{i=1}^n X_i$ follows negative binomial distribution and X_i follows geometric distribution

In binomial distribution, we fix the number of trials (n) and are concerned with number of successes in n trials. whereas in a negative binomial distribution, we fix the number of successes and then find the number of trials required.

In independent trials, each resulting in a success with probability p , the probability of r successes occurring before m failures is given by

$$\sum_{n=r}^{r+m-1} \binom{n-1}{r-1} p^r (1-p)^{n-r}$$

Solved Problem



2.69 The probability of a hockey player A scoring a goal through penalty corner is 60%. In a match, what is the probability that A scores his third goal in the 5th penalty corner?

Solution

Given: $p = 0.6 \Rightarrow q = 0.4$
 $n = 5, \quad r = 3$

$$\begin{aligned} P(X=5) &= \binom{5-1}{3-1} (0.6)^3 (1-0.6)^{5-3} \\ &= \binom{4}{2} (0.6)^3 (0.4)^2 \\ &= 0.20736 \end{aligned}$$



Practice Problem

2.27 The probability that a cricket player hits a boundary is 0.25. In an over, what is the probability that he hits the 4th boundary in sixth ball?
 (Ans. 0.02197)

2.5.6 Hypergeometric Distribution

Let us consider a box that contains N diodes out of which r diodes are defective and $(N-r)$ are non-defective. If we choose diodes randomly without replacement and the number of defective diodes are represented by X then the probability that we obtain k defective diodes is equal to

$$P(X=k) = \frac{\binom{n}{k} \binom{N-n}{n-k}}{\binom{N}{n}}, \quad k = 0, 1, 2, \dots \quad (2.38)$$

Any random variable X whose pmf is given by Eq. (2.38) is said to be a hypergeometric random variable.

Solved Problems



2.70 A vendor supplies 100 heaters to a shop every month. Before accepting the heaters, the owner selects 10 heaters randomly and accepts if none of these are defective. If one or more defectives are found then all 100 heaters are inspected. If two defectives are found, what is the probability that 100% inspection is required?

Solution Let X be the number of defectives found. Then 100% inspection is required for $X \geq 1$. Therefore,

$$P(X \geq 1) = 1 - P(X = 0)$$

Given: $N = 100, n = 10, k = 0$

From Eq. (2.37)

$$P(X = 0) = \frac{\binom{10}{0} \binom{90}{10}}{\binom{100}{10}} = 0.33$$

$$P(X \geq 1) = 1 - 0.33 = 0.67$$

2.71 In a college, the number of boys and girls are 700 and 300 respectively. Twelve members are drawn randomly to form a committee. What is the probability that committee consists of all girls?

Solution Total number of students = 1000; $n = 300$; $k = 12$

$$P(\text{'0' boys and 12 girls}) = \frac{\binom{300}{12} \binom{700}{0}}{\binom{1000}{12}} = 4.54 \times 10^{-7}$$



Practice Problem

2.28 A purchaser of electronic components buys diodes in lots of 10. He inspects 3 diodes randomly from a lot and accepts the lot only if all 3 are non-defective. If 25% of the lots have 4 defective diodes and 75% have only 1, what proportion of lots does the purchaser reject? (Ans. 43.3 per cent)

2.5.7 Discrete Uniform Distribution

A discrete random variable is said to have a uniform distribution if each of the n values in the range, say $x_1, x_2, \dots, x_n$ has equal probability $1/n$. The pmf of a discrete random variable is given by

$$P_K(k) = \begin{cases} \frac{1}{n} & k = x_1, x_2, \dots, x_n \\ 0 & \text{otherwise} \end{cases} \quad (2.39)$$

The phase of a radio-frequency sinusoid can be modelled as a uniform random variable. Although the transmitter knows the phase of the sinusoid, the receiver may have no information about the phase. In this case, phase at the receiver is modelled as a random variable with uniform distribution over the interval $(0, 2\pi)$. The pmf of a discrete random variable with uniform distribution is shown in Fig. 2.22.

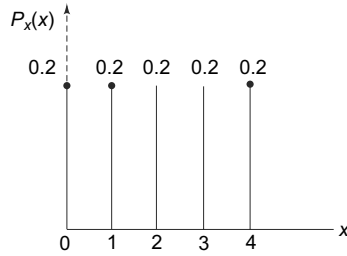


Fig. 2.22 pmf of discrete uniform distribution



CONTINUOUS DISTRIBUTION 2.6

2.6.1 Uniform Distribution

A continuous random variable X is said to be uniformly distributed over the interval $[a, b]$ if the pdf of X is given by

$$f_X(x) = \begin{cases} \frac{1}{b-a} & \text{for } a \leq x \leq b \\ 0 & \text{elsewhere} \end{cases} \quad (2.40)$$

where a and b are real constants with $a < b$.

The CDF of a continuous uniform random variable is

$$\begin{aligned} F_X(x) &= \int_{-\infty}^x f_X(u) du \\ &= \int_a^x \frac{1}{b-a} du = \frac{1}{b-a} (x-a) \\ \Rightarrow F_X(x) &= \frac{x-a}{b-a} \quad \text{for } a \leq x \leq b \end{aligned} \quad (2.41)$$

A plot of the pdf and CDF of a uniform random variable is shown in Fig. 2.23. The uniform random variable with zero mean and unit variance is denoted as $U(0, 1)$.

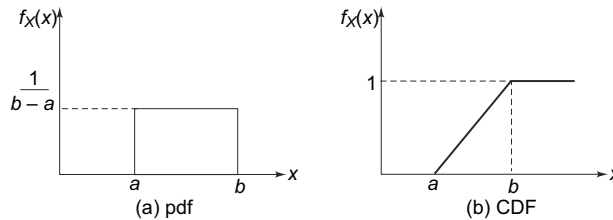


Fig. 2.23 pdf and CDF of a uniform random variable

REVIEW QUESTIONS

8. Mention the applications are uniform, random variable.
9. Draw the pdf and CDF of a uniform random variable.
10. Briefly discuss about uniform distribution.

Solved Problems

2.72 If X is uniformly distributed over $(0, 5)$, calculate the probability that (a) $X < 2$ (b) $X > 4$ (c) $1 < X < 3$.

Solution Given: X is uniformly distributed over $(0, 5)$.

We know for uniform distribution,

$$f_X(x) = \begin{cases} \frac{1}{b-a} & \text{for } a \leq x \leq b \\ 0 & \text{otherwise} \end{cases}$$

Hence, we can find

$$f_X(x) = \begin{cases} \frac{1}{5} & \text{for } 0 \leq x \leq 5 \\ 0 & \text{otherwise} \end{cases}$$

$$P(X < 2) = \int_0^2 \frac{1}{5} dx = \frac{2}{5}$$

$$P(X > 4) = \int_4^{\infty} \frac{1}{5} dx = \int_4^5 \frac{1}{5} dx = \frac{1}{5}$$

$$P(1 < X < 3) = \int_1^3 \frac{1}{5} dx = \frac{1}{5}(2) = \frac{2}{5}$$

2.73 If Y is a random variable uniformly distributed over $(0,5)$, find the probability that the roots of the equation $4x^2 + 4xY + Y + 2 = 0$ are both real.

Solution For uniform distribution

$$f_X(x) = \begin{cases} \frac{1}{b-a} & \text{for } a \leq x \leq b \\ 0 & \text{otherwise} \end{cases}$$

Given: $b = 5$ and $a = 0$. Therefore,

$$f_X(x) = \begin{cases} \frac{1}{5} & \text{for } 0 \leq x \leq 5 \\ 0 & \text{otherwise} \end{cases}$$

The roots of the equation $4x^2 + 4xY + Y + 2 = 0$ are real if it satisfies

$$(4Y)^2 - 4(4)(Y + 2) > 0$$

$$16Y^2 - 16Y - 32 > 0$$

$$Y^2 - Y - 2 > 0$$

$$(Y - 2)(Y + 1) > 0$$

$P[(Y - 2)(Y + 1) > 0]$ is $P[Y > 2 \text{ and } Y > -1]$ or

$P[Y > 2]$ since Y lies between 0 and 5. Hence,

$$P[Y > 2] = \int_2^5 \frac{1}{5} dx = \frac{3}{5}$$

$$4x^2 + 4xY + Y + 2 = 0$$

$$a = 4; b = 4Y$$

$$C = Y + 2$$

$$b^2 - 4ac > 0$$

for roots to be real

2.74 The thickness of a sheet in an automobile component is uniformly distributed between 0.9 and 1.10 millimetres.

- Determine the CDF of sheet thickness.
- Determine the proportion of sheets that exceed 1.0 mm thickness.
- What thickness is exceeded by 20% sheets?

Solution Given: $b = 1.10$ (in mm) and $a = 0.9$ (in mm)

$$f_X(x) = \begin{cases} \frac{1}{b-a} & \text{for } a \leq x \leq b \\ 0 & \text{otherwise} \end{cases}$$

Therefore,

$$f_X(x) = \begin{cases} \frac{1}{1.10 - 0.9} = \frac{1}{0.2} = 5 & \text{for } 0.9 \leq x \leq 1.10 \\ 0 & \text{otherwise} \end{cases}$$

- (a) The CDF is given by

$$F_X(x) = \int_{-\infty}^x f_X(x) dx = \int_0^x 5 dx = 5x$$

- (b) $P[X > 1.0 \text{ mm}] = \int_{1.0}^{1.1} 5 dx = 5[1.1 - 1.0] = 0.5$

- (c) $\int_x^{1.10} 5 dx = 0.2$

$$5[1.1 - x] = 0.2$$

$$1.1 - x = 0.04$$

$$x = 1.1 - 0.04 = 1.06 \text{ mm}$$

2.75 If X has a uniform distribution in $(-a, a)$, $a > 0$, find a such that $P(|X| < 1) = P(|X| > 1)$.

Solution Given X has uniform distribution in the interval $(-a, a)$

$$\Rightarrow f_X(x) = \begin{cases} \frac{1}{2a}, & -a < x < a \\ 0 & \text{otherwise} \end{cases}$$

$$P(|X| < 1) = P(-1 < X < 1)$$

$$\text{and } P(|X| > 1) = 1 - P(-1 < X < 1)$$

$$\text{Given: } P(|X| < 1) = P(|X| > 1)$$

$$\Rightarrow P(-1 < X < 1) = 1 - P(-1 < X < 1)$$

$$2P(-1 < X < 1) = 1 \Rightarrow P(-1 < X < 1) = 0.5$$

$$P(-1 < X < 1) = \int_{-1}^1 f_X(x) dx = \int_{-1}^1 \frac{1}{2a} dx = \frac{1}{2a}(2) = \frac{1}{a}$$

$$\text{That is, } \frac{1}{a} = 0.5 \Rightarrow a = 2$$

2.76 If X is a uniformly distributed random variable with pdf $U(-1, 3)$, find $P(X < 0)$.

Solution For a uniform random variable in the interval $(-1, 3)$, the pdf is given by

$$f_X(x) = \begin{cases} \frac{1}{4} & \text{for } -1 \leq x \leq 3 \\ 0 & \text{otherwise} \end{cases}$$

$$P(X < 0) = 1 - P(X \geq 0)$$

$$P(X \geq 0) = \int_0^3 f_X(x) dx = \int_0^3 \frac{1}{4} dx = \frac{3}{4}$$

$$P(X < 0) = 1 - P(X \geq 0)$$

$$= 1 - \frac{3}{4} = \frac{1}{4}$$

2.77 The number of personal computers sold daily at a CompuWorld is uniformly distributed with a minimum of 2000 PCs and a maximum of 5000 PCs.

- the probability that daily sales will fall between 2,500 and 3,000 PC.
- what is the probability that CompuWorld will exactly sell 2500 PCs?
- what is the probability that CompuWorld will exactly sell 2500 PCs?

Solution Let X be the random variable that represents the number of PCs sold. X follows uniform distribution with $a = 2000$ and $b = 5000$

$$f_X(x) = \begin{cases} \frac{1}{5000 - 2000} = \frac{1}{3000} & \text{for } 2000 \leq x \leq 5000 \\ 0 & \text{otherwise} \end{cases}$$

$$\begin{aligned} \text{(a) } P(2500 \leq x \leq 3000) &= \int_{2500}^{3000} f_X(x) dx = \int_{2500}^{3000} \frac{1}{3000} dx = \frac{1}{3000} (3000 - 2500) \\ &= \frac{1}{3000} (500) = \frac{1}{6} \end{aligned}$$

$$\text{(b) } P(\text{That sales is at least 4000 PCs}) = P(X \geq 4000)$$

$$\begin{aligned} &= \int_{4000}^{5000} f_X(x) dx = \int_{4000}^{5000} \frac{1}{3000} dx \\ &= \frac{1}{3000} (5000 - 4000) = \frac{1000}{3000} = \frac{1}{3} \end{aligned}$$

$$\text{(c) } P(\text{The sales is exactly 2500}) = P(X = 2500) = 0$$

2.78 Starting at 5.00 a.m. every half an hour, there is a flight from San Francisco airport to Los Angeles International airport. Suppose that none of these planes is completely sold out and that they always have room for passengers. A person who wants to fly to Los Angel arrives at the airport at a random time between 8.45 a.m. and 9.45 a.m. Find the probability that she waits

- (a) at most 10 minutes, and
- (b) at least 15 minutes.

Solution There is a flight for every half an hour. Hence, the flight timings are 5.00 a.m., 5.30 a.m. ... 8.30 a.m. 9.00 a.m., 9.30. a.m., 10 a.m. and so on.

A person arrives between 8.45 a.m. and 9.45 a.m. Therefore, her arrival time can be thought of as a random variable X with uniform distribution given by

$$f_X(x) = \begin{cases} \frac{1}{60} & \text{for } 0 < x \leq 60 \\ 0 & \text{otherwise} \end{cases}$$

- (a) The person waits almost 10 minutes if she arrives at the airport between 8.50 a.m. and 9.00 a.m. or between 9.20 a.m. and 9.30 a.m.

Therefore we have to find

$$P(5 < X < 15) + P(35 < X < 45)$$

which is equal to

$$\int_5^{15} \frac{1}{60} dx + \int_{35}^{45} \frac{1}{60} dx = \frac{1}{60} (15 - 5) + \frac{1}{60} (45 - 30) = \frac{1}{3}$$

- (b) In second problem we have to find $P(\text{she waits at least 15 minutes})$

She waits at least 15 minutes if she arrives at the airport between 8.45 a.m. and 9.00 a.m or 9.15 a.m. and 9.30 a.m.

$$P(0 < X < 15) + P(30 < X < 45)$$

$$= \int_0^{15} \frac{1}{60} dx + \int_{30}^{45} \frac{1}{60} dx$$

$$\begin{aligned}
&= \frac{1}{60}(15) + \frac{1}{60}(15) \\
&= \frac{1}{2}
\end{aligned}$$

2.79 You know that your college bus arrives at your boarding point at same time uniformly distributed between 8.00 and 8.20 a.m. (a) What will be the probability that you will have to wait longer than 5 minutes? (b) If at 8.10 a.m. the bus has not yet arrived, what is the probability that you will have to wait at least an additional 5 minutes?

Solution For our convenience, we leave the hours data given in the problem and consider only the time interval in minutes. Since the time interval is 20 minutes and uniformly distributed, we can write

$$f_X(x) = \begin{cases} \frac{1}{20} & \text{for } 0 \leq x \leq 20 \\ 0 & \text{otherwise} \end{cases}$$

(a) The probability that you will have to wait longer than 5 minutes is

$$P[X \geq 5] = \int_5^{20} f_X(x) dx = \int_5^{20} \frac{1}{20} dx = \frac{15}{20} = \frac{3}{4}$$

(b) The probability that you will have to wait at least an additional 5 minutes if the bus has not yet arrived at 8.10 a.m. is

$$P[10 \leq x \leq 15] = \int_{10}^{15} f_X(x) dx = \int_{10}^{15} \frac{1}{20} dx = \frac{1}{20}(15 - 10) = \frac{5}{20} = \frac{1}{4}$$

2.80 A point is chosen at random on the line segment $[0, 5]$. What is the probability that the chosen point lies between 2.5 and 4?

Solution The pdf of the random chosen point has a uniform distribution with pdf

$$f_X(x) = \begin{cases} \frac{1}{5} & \text{for } 0 \leq x \leq 5 \\ 0 & \text{otherwise} \end{cases}$$

The probability that the chosen point lies between 2.5 and 4 is

$$P[2.5 \leq x \leq 4] = \int_{2.5}^4 \frac{1}{5} dx = \frac{1}{5}[1.5] = 0.3$$

2.81 A point is chosen at random on a line segment of length L . Find the probability that the ratio of the shorter to the longer segment is less than 4.

Solution The length of the line segment is L . Since a point can be chosen at random, it can be assumed to have uniform distribution. The density function of the random variable is

$$f_X(x) = \begin{cases} \frac{1}{L} & \text{for } 0 \leq x \leq L \\ 0 & \text{otherwise} \end{cases}$$

A point is selected at random such that the ratio of the shorter to longer segment is less than $1/4$. This is possible when the point lies between 0 and $L/5$ or $4L/5$ and L . Hence, we have to find

$$P\left[0 \leq X \leq \frac{L}{5}\right] + P\left[\frac{4L}{5} \leq X \leq L\right]$$

$$P\left[0 \leq X \leq \frac{L}{5}\right] = \int_0^{\frac{L}{5}} \frac{1}{L} dx = \frac{1}{L} \left[\frac{L}{5}\right] = \frac{1}{5}$$

$$P\left[\frac{4L}{5} \leq X \leq L\right] = \int_{\frac{4L}{5}}^L \frac{1}{L} dx = \frac{1}{L} \left[L - \frac{4L}{5}\right] = \frac{1}{5}$$

$$P\left[0 \leq X \leq \frac{L}{5}\right] + P\left[\frac{4L}{5} \leq X \leq L\right] = \frac{1}{5} + \frac{1}{5} = \frac{2}{5}$$



Practice Problems

2.29 A random variable X has uniform distribution over $(-3, 3)$. Compute (a) $P(X < 2)$ (b) $P(|X| < 2)$ (c) $P(1X - 21 < 2)$ (d) Find k for which $P(X > k) = 1/3$.

$$\left(\text{Ans. (a) } \frac{5}{6} \text{ (b) } \frac{2}{3} \text{ (c) } \frac{1}{2} \text{ (d) } k = 1 \right)$$

2.30 If X is uniformly distributed over $(0, 10)$, calculate the probability that (a) $X > 6$ (b) $3 < X < 8$.

$$\left(\text{Ans. (a) } \frac{4}{10}, \text{ (b) } \frac{1}{2} \right)$$

2.31 Buses arrive at a specified stop at 15-minutes interval starting at 7 a.m. That is, they arrive at 7, 7.15, 7.30, 7.45 and so on. If a passenger arrives at the stop at a random time that is uniformly distributed between 7 and 7.30 a.m. find the probability that he waits (a) less than 5 minutes for a bus, and (b) at least 12 minutes for a bus.

$$(\text{Ans. (a) } 1/3 \text{ (b) } 1/5)$$

2.32 A point is chosen at random on the line segment $[0, 10]$. What is the probability that the chosen point lies between 5 and 7.

$$(\text{Ans. } 0.2)$$

2.33 If the random variable k is uniformly distributed over $(1, 7)$, then what is the probability that the roots of the equation $x^2 + 2kx + (2k + 3) = c$ are (a) real (b) equal.

$$(\text{Ans. } 2/3)$$

2.34 Consider a random variable $X \sim U(a, b)$ where $a < b$. Find α such that

$$P(|X| > \alpha) = P(|X| < \alpha).$$

$$\left(\text{Ans. : } \frac{b-a}{4} \right)$$

2.35 Find the probability that the following equation has real roots,

$$x^2 + 3xY + 1 = 0, \text{ where } Y \sim U(-4, 4).$$

$$\left(\text{Ans. : } \frac{5}{6} \right)$$

2.6.2 Normal Distribution

The normal distribution was first introduced by DeMoivre in 1733, who derived it as a limiting form of the binomial distribution. The result was later extended by Laplace and others. But it has been attributed to Gauss, who first published a reference to it in 1809. Therefore, the normal distribution is also famously known as *Gaussian distribution*.

A continuous random variable X is said to have normal distribution with mean μ_X ($-\infty < \mu_X < \infty$) and standard deviation σ_X , if it has the pdf given by

$$f_X(x) = \frac{1}{\sqrt{2\pi} \sigma_X} e^{-(x-\mu_X)^2/2\sigma_X^2}; -\infty < x < \infty \quad (2.42)$$

The parameter σ_X^2 is referred to as variance. An example of Gaussian pdf is shown in Fig. (2.24). The pdf is a bell-shaped curve that is symmetric about μ_X which is the mean of X and has a width that is proportional to σ_X .

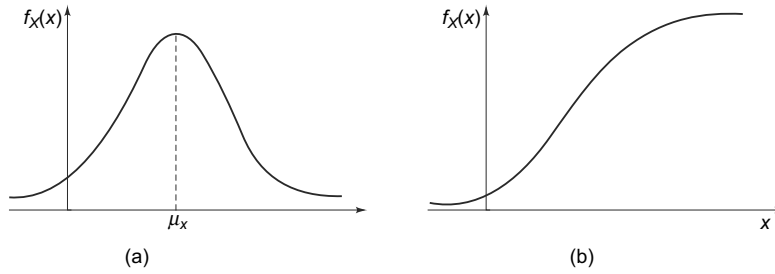


Fig. 2.24 (a) Density and distribution (b) Function of Gaussian random variable

The normal distribution with mean μ_X and variance σ_X^2 is usually designated by the shorthand notation

$$X \sim N(\mu_X, \sigma_X^2). \quad (2.43)$$

This is read as X is distributed normally with mean μ_X and variance σ_X^2 . When $\mu_X = 0$ and $\sigma_X^2 = 1$ then the random variable is called *standard normal random variable* and is designated as $X \sim N(0, 1)$. However, in engineering literature, the term Gaussian is much more common.

Properties of Normal Distributions

The normal distribution has the following properties

- (i) The pdf $f_X(x)$ is a bell-shaped curve that is symmetrical about μ_X , which is the mean of X . It satisfies the relation $f_X(\mu_X + x) = f_X(\mu_X - x)$
- (ii) The maximum value of $f_X(x)$ occurs at $x = \mu_X$.
- (iii) $\lim_{x \rightarrow \infty} f_X(x) = 0$ and $\lim_{x \rightarrow -\infty} f_X(x) = 0$
- (iv) The point of inflection of $f_X(x)$ are at $x = \mu_X \pm \sigma_X$.
- (v) The area under the normal density curve is unity

To show that $f_X(x)$ is a valid pdf, we have to prove

$$\frac{1}{\sqrt{2\pi\sigma_X^2}} \int_{-\infty}^{\infty} e^{-(x-\mu_X)^2/2\sigma_X^2} dx = 1 \quad (2.44)$$

Let us assume $\frac{x-\mu_X}{\sigma_X} = z$, then we obtain $dx = \sigma_X dz$

Substituting the above value in Eq. (2.44) we obtain

$$\frac{1}{\sqrt{2\pi\sigma_X^2}} \int_{-\infty}^{\infty} e^{-(x-\mu_X)^2/2\sigma_X^2} dx = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-z^2/2} dz \quad (2.45)$$

Let us assume

$$I = \int_{-\infty}^{\infty} e^{-z^2/2} dz \quad (2.46)$$

We can also write

$$I = \int_{-\infty}^{\infty} e^{-y^2/2} dy \quad (2.47)$$

Multiply Eq. (2.46) and Eq. (2.47) . Then

$$I^2 = \int_{-\infty}^{\infty} e^{-z^2/2} dz \int_{-\infty}^{\infty} e^{-y^2/2} dy \quad (2.48a)$$

$$= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-(z^2+y^2)/2} dz dy \quad (2.48b)$$

Let $z = r \cos \theta$ and $y = r \sin \theta$, Then

$$dz dy = r d\theta dr$$

$$I^2 = \int_0^{\infty} \int_0^{2\pi} e^{-r^2/2} r d\theta dr$$

$$= 2\pi \int_0^{\infty} r e^{-r^2/2} dr$$

$$= 2\pi$$

$$I = \sqrt{2\pi}$$

Hence, $\frac{1}{\sqrt{2\pi\sigma_X^2}} \int_{-\infty}^{\infty} e^{-(x-\mu_X)^2/2\sigma_X^2} dx = 1$

If X is a random variable with mean μ_X and standard deviation σ_X , then the random variable

$$Z = \frac{X - \mu_X}{\sigma_X} \quad (2.49)$$

is a normal random variable with zero mean and standard deviation 1.

The CDF of X is given by

$$\begin{aligned} F_X(x) &= P[X \leq x] \\ &= \frac{1}{\sqrt{2\pi} \sigma_X^2} \int_{-\infty}^x e^{-(u - \mu_X)^2 / 2\sigma_X^2} du \end{aligned}$$

Let

$$y = \frac{u - \mu_X}{\sigma_X}, \text{ then}$$

$du = \sigma_X dy$ and the CDF of X becomes

$$F_X(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{(x - \mu_X)/\sigma_X} e^{-y^2/2} dy$$

For a standard normal random variable,

$$F_X(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-y^2/2} dy \quad (2.50)$$

which is denoted as $\phi(x)$.

The value of $\phi(x)$ are usually given for non-negative values of x shown in Table (2.4). For negative values of x , $F_X(x)$ can be obtained using the relation

$$\phi(-x) = 1 - \phi(x) \quad -\infty < x < \infty \quad (2.51)$$

This equation states that if Z is a standard normal random variable then

$$P[Z \leq -x] = P[Z > x] \quad -\infty < x < \infty$$

Thus, we can write the CDF of x as

$$\begin{aligned} F_X(x) &= P[X \leq x] = P\left[\frac{X - \mu_X}{\sigma_X} \leq \frac{x - \mu_X}{\sigma_X}\right] \\ &= \phi\left[\frac{X - \mu_X}{\sigma_X}\right] \end{aligned} \quad (2.52)$$

If X is normally distributed with parameters μ_X and σ_X^2 then $Z = \frac{X - \mu_X}{\sigma_X}$ is a standard normal random variable.

Table 2.4 Area $\phi(x)$ under the standard normal curve to the left of x

[illegible]

The Error Function The error function is defined as

$$\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-y^2} dy \quad (2.53)$$

Values of error functions are usually provided in mathematical tables. The complementary error function is defined as

$$\begin{aligned} \operatorname{erfc}(x) &= 1 - \operatorname{erf}(x) \\ &= \frac{2}{\sqrt{\pi}} \int_x^\infty e^{-y^2} dy \end{aligned} \quad (2.54)$$

The relation between error function and $\phi(x)$ is given by

$$\phi(x) = \frac{1}{2} \left[1 + \operatorname{erf} \left(\frac{x}{\sqrt{2}} \right) \right] \quad (2.55)$$

The Q-function The Q -function is defined as

$$Q(x) = \frac{1}{\sqrt{2\pi}} \int_x^\infty e^{-y^2/2} dy \quad (2.56)$$

This is most widely used in electrical engineering

Comparing with Eq. (2.53) we can observe that the Q -function in terms of error function is given by

$$Q(x) = \frac{1}{2} \left[1 - \operatorname{erf} \left(\frac{x}{\sqrt{2}} \right) \right] \quad (2.57)$$

The Q -function satisfies the relation

$$Q(-x) = 1 - Q(x) \quad (2.58)$$

from which we can get

$$F_X(x) = \phi(x) = 1 - Q \left[\frac{x - \mu_X}{\sigma_X} \right] \quad (2.59)$$

Problem-Solving Procedure Consider a random variable $X \sim N(50, 2)$. If we want to find the probability that X is less than or equal to 54, that is $P[X \leq 54] = F[54]$, we first define a random variable

$$Z = \frac{x - \mu_X}{\sigma_X}$$

whose distribution is $N(0, 1)$.

Since we are interested at $x = 54$, we obtain

$$Z = \frac{x - \mu_X}{\sigma_X} = \frac{54 - 50}{2} = 2$$

Now the probability that the Z random variable is less than or equal to 2 is equal to the probability that the original random variable X is less than or equal to 54.

$$F(54) = \phi(2) = 0.9772.$$

REVIEW QUESTIONS

11. Explain in detail about normal distribution
12. Mention the properties of normal distribution
13. Define error function,
14. Define Q -function
15. Draw pdf and CDF of normal distribution.

Solved Problems

2.82 Suppose the current in a diode is assumed to follow a normal distribution with a mean of 10 mA and a variance of 4 mA. What is the probability that a measurement will exceed 12 mA?

Solution Let the diode current in milliamps is represented by a random variable X . The probability that $X > 12$ can be represented by $P(X > 12)$. Let us consider a random variable $Z = \frac{X - \mu_X}{\sigma_X} = \frac{12 - 10}{2} = 1$. We note that $X > 12$ corresponds to $Z > 1$. From the Table, 2.4

$$\begin{aligned} P(X > 12) &= P(Z > 1) = 1 - P[Z \leq 1] \\ &= 1 - 0.8413 \\ &= 0.1587 \end{aligned}$$

2.83 Assume X is normally distributed with mean of 10 and a standard deviation of 2. Determine the following.

(a) $P(X < 13)$ (b) $P(X > 9)$ (c) $P(6 < X < 14)$ (d) $P(2 < X < 4)$

Solution

Let $Z = \frac{X - \mu_X}{\sigma_X} = \frac{X - 10}{2}$

(a) $P(X < 13) = P\left(Z < \frac{13 - 10}{2}\right) = P[Z < 1.5] = 0.9332$

(b) $P(X > 9) = P\left(Z > \frac{9 - 10}{2}\right) = P[Z > -0.5]$

$$\begin{aligned} &= 1 - P(Z \leq -0.5) = 1 - \phi(-0.5) \\ &= 1 - [1 - \phi(0.5)] \\ &= 1 - (1 - 0.6915) = 0.6915 \end{aligned}$$

(c) $P(6 < X < 14)$

$$P\left(\frac{6 - 10}{2} < Z < \frac{14 - 10}{2}\right) = P(-2 < Z < 2)$$

$$\begin{aligned}
&= \phi(2) - \phi(-2) = \phi(2) - [1 - \phi(2)] \\
&= 2\phi(2) - 1 = 2(0.9772) - 1 \\
&= 0.9544
\end{aligned}$$

$$\begin{aligned}
(d) \quad P[2 < X < 4] &= P\left[\frac{2-10}{2} < Z < \frac{4-10}{2}\right] = P(-4 < Z < -3) \\
&= \phi(-3) - \phi(-4) = [1 - \phi(3)] - [1 - \phi(4)] \\
&= (1 - 0.9987) - (1 - 1) \\
&= 0.0013
\end{aligned}$$

2.84 The lifetime of IC chips manufactured by a semiconductor manufacturer is approximately normally distributed with mean $= 5 \times 10^6$ hours and standard deviation of 5×10^6 hours. A mainframe manufacturer requires that at least 95% of a batch should have a lifetime greater than 4×10^6 hours. Will the deal be made?

Solution Given: $\mu_X = 5 \times 10^6$ and $\sigma_X = 5 \times 10^6$

$$\text{Let } Z = \frac{X - \mu_X}{\sigma_X}$$

$$X = 4 \times 10^6$$

$$\begin{aligned}
P(X > 4 \times 10^6) &= P\left(Z > \frac{4 \times 10^6 - 5 \times 10^6}{5 \times 10^6}\right) \\
&= P(Z > -0.2) = 1 - P(Z \leq -0.2) \\
&= 1 - \phi(-0.2) = 1 - (1 - \phi(0.2)) \\
&= 1 - (1 - 0.5793) = 0.5793, \text{ i.e., } 57.93\%
\end{aligned}$$

Hence, the deal cannot be made.

2.85 An analog signal received at the detector (measured in μV) may be modelled as a Gaussian random variable $N(200, 256)$ at a fixed point in time. What is the probability that the signal will exceed $240 \mu\text{V}$? What is the probability that the signal is larger than $240 \mu\text{V}$, given that it is lesser than $210 \mu\text{V}$?

Solution Given: $X \sim N(200, 256)$

$$\text{Let } Z = \frac{X - \mu_X}{\sigma_X} = \frac{X - 200}{16}$$

$$\text{For } X = 240, \quad Z = \frac{240 - 200}{16} = \frac{40}{16} = 2.5$$

$$\begin{aligned}
P(Z > 2.5) &= 1 - P(Z \leq 2.5) \\
&= 1 - \phi(2.5) \quad \text{From Table (2.4)}
\end{aligned}$$

$$= 1 - 0.9938 = 0.0062$$

$$P(X > 240 | X > 210)$$

$$\begin{aligned} &= P\left(Z > \frac{240 - 200}{10} \mid Z > \frac{240 - 200}{16}\right) \\ &= P(Z > 2.5 \mid Z > 0.625) \\ &= \frac{1 - P(Z \leq 2.5)}{1 - P(Z \leq 0.625)} \quad (\text{From Table 2.4}) \\ &= \frac{1 - \phi(2.5)}{1 - \phi(0.625)} \\ &= \frac{0.0062}{1 - 0.73} = \frac{0.0062}{0.27} = 0.023 \end{aligned}$$

2.86 The annual rainfall (in hundreds of mm) of a certain region is normally distributed with $\mu = 5$ and $\sigma = 2$. What is the probability that starting with this year, it will take over 10 years before a year occurs having a rainfall over 700 mm? What assumptions are you making?

Solution Given: $\mu_X = 5$; $\sigma_X = 2$ and $X = 7$

$$Z = \frac{X - \mu_X}{\sigma_X} = \frac{7 - 5}{2} = 1$$

$$P[Z \leq 1] = \phi(1) = 0.8413$$

Probability that starting from this year, it will take over 10 years before a year occurs having a rainfall over 700 mm is given by $(0.8413)^{10}$. The assumption is that the rainfall in each year is independent of other.

2.87 A company conducts tests for job aspirants and passes them if they achieve a score of 500. If the test scores are normally distributed with a mean of 485 and a standard deviation of 30, what percentage of the aspirants pass the test?

Solution Given: $X = 500$ $\mu_X = 485$

$$\sigma_X = 30$$

$$\begin{aligned} P[X \geq 500] &= P\left[Z \geq \frac{500 - 485}{30}\right] \\ &= P\left(Z \geq \frac{1}{2}\right) = 1 - P\left(Z < \frac{1}{2}\right) \\ &= 1 - \phi\left(\frac{1}{2}\right) \quad (\text{from Table 2.4}) \\ &= 1 - 0.6915 \\ &= 0.3085 \end{aligned}$$

That is, 30.85% of aspirants pass the test.

2.88 The output of a light bulb is normally distributed with mean of 2000 end foot candles and a standard deviation of 60 end foot candles. Determine a lower specification limit such that only 4% of the manufactured bulbs will be defective.

Solution Given: $\mu_X = 2000$; $\sigma_X = 60$

$$\text{Let } Z = \frac{X - 2000}{60}$$

$$P\left(Z \leq \frac{x - 2000}{60}\right) = \phi\left(\frac{x - 2000}{60}\right) = 0.04$$

$$\Rightarrow \phi\left(\frac{2000 - x}{60}\right) = 0.96$$

$$\begin{aligned} \frac{2000 - x}{60} &= \phi^{-1}(0.96) \\ &= 1.75 \quad (\text{From Tables 2.4}) \end{aligned}$$

$$x = 2000 - 60 \times 1.75$$

$$= 1895 \text{ fc}$$

$$\boxed{x = 1895 \text{ fc}}$$

2.89 In a test on 2000 electric bulbs it was found that the life of a particular make was normally distributed with an average life of 2040 hours and standard duration of 60 hours. Estimate the number of bulbs likely to burn for (a) more than 2150 hours, (b) less than 1950 hours, and (c) more than 1920 hours but less than 2160 hours.

Solution

$$(a) \quad P(X \geq 2150) = ?$$

$$\text{We have } Z = \frac{X - \mu}{\sigma} = \frac{X - 2040}{60}$$

$$\text{For } X = 2150; Z = \frac{2150 - 2040}{60} = 1.833$$

That is,

$$\begin{aligned} P(X \geq 2150) &= P(Z \geq 1.833) = 1 - P(Z < 1.833) \\ &= 1 - \phi(1.833) = 1 - 0.9664 = 0.0336 \end{aligned}$$

The number of bulbs expected to burn for more than 2150 hours is

$$2000 \times 0.0336 \approx 67$$

$$(b) \quad P(X < 1950) = ?$$

$$\text{We have } Z = \frac{X - \mu}{\sigma} = \frac{1950 - 2040}{60} = -1.5$$

$$\begin{aligned} P(X < 1950) &= P(Z < -1.5) = \phi(-1.5) = 1 - \phi(1.5) \\ &= 1 - 0.9332 = 0.0668 \end{aligned}$$

The number of bulbs expected to burn for less than 1950 hours is equal to $2000 \times 0.0668 \approx 134$

- (c)
- $P(\text{more than 1920 hours but less than 2160 hours})$

$$\begin{aligned}
 P(1920 \leq X \leq 2160) &= P(-2 \leq Z \leq 2) \\
 &= \phi(2) - \phi(-2) \\
 &= \phi(2) - [1 - \phi(2)] \\
 &= 2\phi(2) - 1 \\
 &= 2(0.9712) - 1 \\
 &= 0.9424
 \end{aligned}$$

The number of bulbs with lifetime between 1920 hours and 2160 hours is

$$0.9424 \times 2000 \approx 1885$$

2.90 The average test marks in a class is 80. The standard deviation is 6. If the marks are distributed normally, how many students in a class of 200 receives marks between 70 and 90.

Solution Let X represent the random variable that represents the marks of students

Given: $\mu_X = 80, \sigma_X = 6$

Let $X_1 = 70$ and $X_2 = 90$

$$Z_1 = \frac{X_1 - \mu_X}{\sigma_X} = \frac{70 - 80}{6} = \frac{-10}{6} = -1.67$$

$$Z_2 = \frac{X_2 - \mu_X}{\sigma_X} = \frac{90 - 80}{6} = \frac{10}{6} = 1.67$$

$$\begin{aligned}
 P(Z_1 < Z \leq Z_2) &= P(-1.67 \leq Z < 1.67) \\
 &= \phi(1.67) - \phi(-1.67) \\
 &= 2\phi(1.67) - 1 \\
 &= 2(0.9525) - 1 \\
 &= 0.905
 \end{aligned}$$

The number of students who receive marks between 70 and 90 is $200 \times 0.905 = 181$

2.91 The average life of a certain type electric bulb is 1200 hours. What percentage of this type of bulbs is expected to fail in the first 800 hours of working? What percentage is expected to fail between 800 and 1000 hours? Assume a normal distribution with $\sigma = 200$ hours.

Solution Given: $\mu_X = 1200, \sigma_X = 200$

$$\begin{aligned}
 P(X \leq 800) &= P\left(Z \leq \frac{800 - 1200}{200}\right) = P[Z \leq -2] \\
 &= \phi(-2) = 1 - \phi(2) \quad (\text{from Table 2.4}) \\
 &= 1 - 0.9772 = 0.0228
 \end{aligned}$$

= 2.28% of the bulbs are expected to fail in the first 800 hours.

$$\begin{aligned}
 P[800 \leq X \leq 1000] &= P(-2 < Z \leq -1) = \phi(-1) - \phi(-2) \\
 &= [1 - \phi(1)] - [1 - \phi(2)] \\
 &= \phi(2) - \phi(1) = 0.9772 - 0.8413 \\
 &= 0.1359 \\
 &= 13.59\% \text{ of the bulbs are expected to fail between 800 and 1000 hours.}
 \end{aligned}$$



Practice Problems

2.36 The marks obtained by a number of students in mathematics is normally distributed with mean 65 and SD 5. Find the probability that a student scores above 75. (Ans. 0.0228)

2.37 X is normally distributed and the mean of X is 12 and standard deviation is 4. Find the probability of the following: (a) $X \geq 20$ (b) $0 \leq X \leq 12$ (c) Find x when $P[X > x] = 0.24$. (Ans. (a) 0.0228 (b) 0.49865 (c) 14.84)

2.38 The savings bank account of a customer showed an average balance of ₹ 150 and a standard deviation of ₹ 50. Assuming that the account balance are normally distributed, (2006)

- (a) What percentage of account is over ₹ 200?
- (b) What percentage of account is between ₹ 120 and ₹ 170?
- (c) What percentage of account is less than ₹ 75?

(Ans. (a) 15.87% (b) 38.11% (c) 6.68%)

2.39 A person riding two-wheeler travels on a highway with a mean speed of 60 km/h and a standard deviation of 4 km/h. (a) What is the probability that he travels at a speed between (a) 55 km/h and 65 km/h; (b) more than 65 km/h

(Ans: (a) 0.7888; (b) 0.0054)

2.40 The time required for a professor to evaluate 10 answer scripts follows a normal distribution with average time of one hour and a standard deviation of 5 minutes.

- (a) What is the probability that he will take less than 45 minutes?
- (b) What is the probability that he will take more than 65 minutes?

(Ans.(a) 0.00135 (b) 0.15866)

2.6.3 Normal Approximation to the Binomial Distribution

In Section (2.5.2) we discussed about Poisson approximation to the binomial distribution. In this section, we will discuss about the normal approximation to the binomial distribution. The approximation is given by a well-known theorem known as DeMoivre-Laplace limit theorem, which was originally proven for the special case $p = \frac{1}{2}$ by DeMoivre in 1733 and then extended to general p by Laplace in 1812.

2.6.4 DeMoivre-Laplace Limit Theorem

Let a random variable X have a binomial distribution with parameters n and p . Then for any $a < b$

$$P\left\{a \leq \frac{X - np}{\sqrt{npq}} \leq b\right\} \rightarrow \phi(b) - \phi(a) \quad (2.60)$$

as $n \rightarrow \infty$.

where ϕ is the distribution function of the standard normal distribution. The normal approximation can be used for the values of x satisfying $npq \geq 10$. Figure 2.25 illustrates how the pmf of a binomial random variable becomes more and more normal as n becomes larger and larger.

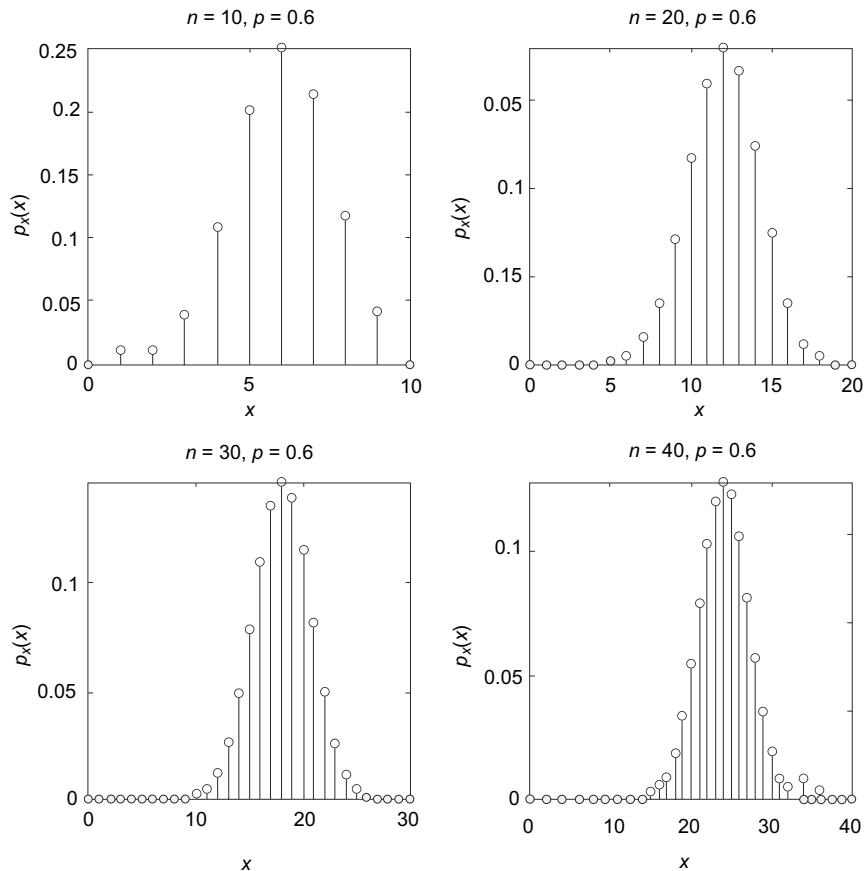


Fig. 2.25 Illustrating how the pmf of binomial distribution approaches to normal distribution as n increases

Solved Problem



2.92 A pair of dice is rolled 900 times. Let X denote the number of times a total of 9 occurs. Find $P(90 \leq X \leq 110)$.

Solution Given: $n = 900$

The probability for 9 to occur when a pair of dice is rolled is given by $p = \frac{4}{36} = \frac{1}{9} \Rightarrow q = \frac{8}{9}$

Using DeMoivre–Laplace limit theorem,

$$np = \frac{1}{9}(900) = 100$$

$$\begin{aligned}
P(90 \leq X \leq 110) &= P\left[\frac{90-100}{\sqrt{900 \times \frac{1}{9} \times \frac{8}{9}}} \leq \frac{X-100}{\sqrt{900 \times \frac{1}{9} \times \frac{8}{9}}} \leq \frac{110-100}{\sqrt{900 \times \frac{1}{9} \times \frac{8}{9}}}\right] \\
&= \phi\left(\frac{10}{9.429}\right) - \phi\left(-\frac{10}{9.428}\right) = \phi(1.06) - \phi(-1.06) \\
&= \phi(1.06) - [1 - \phi(1.06)] = 2\phi(1.06) - 1 \\
&= 2(0.8554) - 1 \\
&= 0.7108
\end{aligned}$$

2.6.5 Exponential Distribution

Exponential distribution is a good model for the time between two consecutive occurrences of independent events such as the time between failures at a constant rate. It has many applications in reliability, computer system and communication systems.

A continuous random variable X is an exponential random variable with parameter $\lambda > 0$ if its density function is given by

$$f_X(x) = \begin{cases} \lambda e^{-\lambda(x-a)} & x \geq a \\ 0 & x < a \end{cases} \quad (2.61)$$

$$\text{For } a = 0, \quad f_X(x) = \begin{cases} \lambda e^{-\lambda x} & x \geq 0 \\ 0 & x < 0 \end{cases} \quad (2.62)$$

The CDF is given by

$$\begin{aligned}
F_X(x) &= P(X \leq x) = \int_{-\infty}^x f_X(u) du \\
&= \int_a^x \lambda e^{-\lambda(u-a)} du
\end{aligned}$$

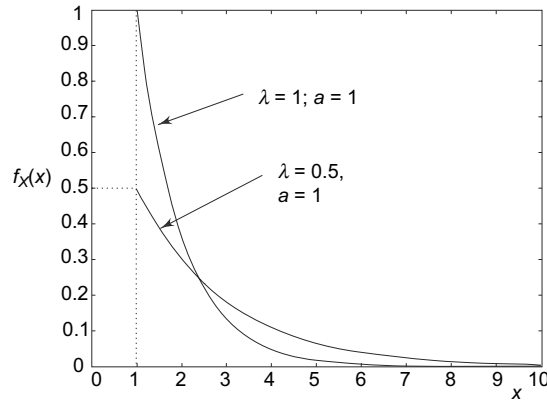


Fig. 2.26 Probability density function of Exponential Distribution

$$\begin{aligned}
&= \lambda e^{\lambda a} \int_a^x e^{-\lambda u} du = -\frac{\lambda e^{\lambda a}}{\lambda} e^{-\lambda u} \Big|_a^x \\
&= e^{-\lambda a} [e^{-\lambda x} - e^{-\lambda a}] \\
&= 1 - e^{-\lambda(x-a)}
\end{aligned} \tag{2.63}$$

For $a = 0$,

$$F_X(x) = P\{X \leq x\} = 1 - e^{-\lambda x} \tag{2.64}$$

Memoryless Property of Exponential Distribution

The exponential distribution with $a = 0$ is a memoryless distribution. Consider the events $[X > t + s]$ and $[X > s]$. Suppose the lifetime of an equipment is a random variable S . Assume that the equipment is already in use for time s . Then the probability that it will work an additional time t given, it has worked for s hours is given by

$$P[X > t + s | X > s] = \frac{P[X > t + s, X > s]}{P[X > s]} \tag{2.65}$$

$$\begin{aligned}
&= \frac{P[X > t + s]}{P[X > s]} = \frac{1 - P[X \leq t + s]}{1 - P[X \leq s]} \\
&= \frac{1 - (1 - e^{-\lambda(t+s)})}{1 - [1 - e^{-\lambda s}]} = \frac{e^{-\lambda(t+s)}}{e^{-\lambda s}} \\
&= e^{-\lambda t}
\end{aligned} \tag{2.66}$$

which is equal to $P[X > t]$.

That is,

$$P[X > (t + s) | X > s] = P[X > t] \tag{2.67}$$

A random variable X that satisfies the above equation is said to have memoryless property.

From the above result, we find that even though the equipment worked for time s , the probability that it will work for an additional time t depends only on t (not on s) and is identical to the probability of working time t for a new piece of equipment. That is, the equipment does not remember that it has been in use for time s .

In other example, if the equipment has not failed by time s , the residual life of the equipment has a pdf that is identical to that of the life of the equipment before s .

Although this cannot be true for any real-world quantity, this distribution is sometimes a good approximation for some practical problems.

Relationship between Exponential and Poisson Distributions The pmf of a Poisson random variable is given by

$$P_X(x) = \frac{e^{-b} b^x}{x!}$$

where $b = \lambda T$. $P_X(x)$ is the distribution of the number of occurrences of an event in the interval $(0, T)$.

Now consider $P(0)$ which is the probability of occurrences in an interval of T seconds. This is given by

$$P(0) = e^{-b} = e^{-\lambda T}.$$

Thus, the probability that there is at least one occurrence in an interval of T seconds is $1 - e^{-\lambda T}$.

For an exponential random variable Y with parameter λ , the probability that an event occurs not later than time T is given by

$$P(Y \leq T) = F_Y(T) = 1 - e^{-\lambda T}$$

Thus, an exponential distributed Y with parameter λ describes the interval between occurrence of events defined by a Poisson random variable with parameter b .

Thus, the relationship between Poisson distribution and exponential distribution can be stated as follows:

If the number of occurrences has a Poisson distribution then the time between successive occurrences has an exponential distribution.

Solved Problems



2.93 The time required to complete a work is an exponential distributed random variable with $\lambda = \frac{1}{2}$. What is the probability that time to complete the work exceeds 2 hours?

Solution Given: $\lambda = \frac{1}{2}$

Let X be a random variable that represents the time to complete a work.

$$\begin{aligned} f_X(x) &= \lambda e^{-\lambda x} \quad \text{for } x \geq 0 \\ &= 0 \quad \text{otherwise} \end{aligned}$$

$$\begin{aligned} \text{Also } F_X(x) &= 1 - e^{-\lambda x} \\ P[X > 2] &= 1 - P[X \leq 2] \\ &= 1 - F_X(2) = 1 - (1 - e^{-2\lambda}) = e^{-2\lambda} \end{aligned}$$

Substituting the value of λ , we get

$$P[X > 2] = e^{-1}$$

2.94 The number of years a washing machine functions is exponentially distributed with $\lambda = \frac{1}{10}$. What is the probability that it will be working after an additional 10 years?

Solution We know that for an exponential distributed random variable,

$$F_X(x) = P[X \leq x] = 1 - e^{-\lambda x}$$

Because of the memoryless property, it is irrelevant how many years the washing machine has been working.

The probability that it works an additional 10 years given that it worked for t years is

$$P[X > (t + 10) | X > t] = P[X > 10]$$

$$= 1 - P[X \leq 10]$$

$$= 1 - F_X(10)$$

$$F_X(x) = 1 - e^{-\lambda x}. \quad \text{Hence}$$

$$1 - F_X(10) = 1 - (1 - e^{-10\lambda}) = e^{-10\lambda}$$

$$\text{For } \lambda = \frac{1}{10},$$

$$P[X > (t + 10) | X > t] = e^{-1}$$

2.95 If the number of kilometres that a car runs before its battery wears out is exponentially distributed with an average value of 10000 km and if the owner desires to take a 5000 km trip, what is the probability that he will be able to complete his trip without having to replace the car battery? Assume that the car has been used for some time.

Solution Let X denote the random variable that represents kilometres that a car runs before its battery wears out. X is exponentially distributed with an average value of 10000 km. That is,

$$\frac{1}{\lambda} = 10000$$

$$f_X(x) = \lambda e^{-\lambda x}, \quad x \geq 0$$

$$\frac{1}{10000} e^{-x/10000}, \quad x \geq 0$$

The owner completes the trip without replacing the car battery if $X > 5000$.

Therefore, we have to find $P(X > 5000)$. That is, $\int_{5000}^{\infty} f_X(x) dx$

$$P(X > 5000) = \int_{5000}^{\infty} \frac{1}{10000} e^{-x/10000} dx$$

$$\text{Let } \frac{x}{10000} = t, \text{ then } dx = 10000 dt$$

$$\text{When } x = 5000, \quad t = 0.5$$

$$\begin{aligned} P(X > 5000) &= \int_{0.5}^{\infty} e^{-t} dt = -e^{-t} \Big|_{0.5}^{\infty} = e^{-0.5} \\ &= 0.6065 \end{aligned}$$

2.96 If X is exponentially distributed with parameter λ , find the value of k such that $\frac{P(X > k)}{P(X \leq k)} = a$.

Solution Since X is exponentially distributed,

$$f_X(x) = \lambda e^{-\lambda x}$$

$$P(X > k) = \int_k^{\infty} \lambda e^{-\lambda x} dx = -\lambda \frac{e^{-\lambda x}}{\lambda} \Big|_k^{\infty} = -\lambda(-e^{-\lambda k})/\lambda = e^{-\lambda k}$$

$$P(X \leq k) = 1 - P(X > k) = 1 - e^{-\lambda k}$$

Given: $\frac{P(X > k)}{P(X \leq k)} = \frac{e^{-\lambda k}}{1 - e^{-\lambda k}} = a$

$$\Rightarrow e^{-\lambda k} + a e^{-\lambda k} = a$$

$$e^{-\lambda k} (1 + a) = a$$

$$e^{-\lambda k} = \frac{a}{1 + a}$$

$$-\lambda k = \ln \left(\frac{a}{1 + a} \right)$$

$$\lambda k = \ln \left(\frac{1 + a}{a} \right)$$

$$k = \frac{1}{\lambda} \ln \left(\frac{1 + a}{a} \right)$$

2.97 The mileage which car owners get with certain kinds of radial tyres is a random variable having an exponential distribution with a mean of 40000 km. Find the probability that one of these tyres will last (a) at least 20,000 km, and (b) at most 30,000 km

Solution Let X be the random variable that represents the mileage obtained with the tyre.

Given: $\lambda = \frac{1}{40000}$

We have $f_X(x) = \lambda e^{-\lambda x} = \frac{1}{40000} e^{-x/40000} \quad x > 0$

$$P(\text{at least } 20000 \text{ km}) = P(X > 20000) = 1 - P(X \leq 20000)$$

$$P(X \leq 20000) = \int_0^{20000} \frac{1}{40000} e^{-x/40000} dx$$

Let $\frac{x}{40000} = t \Rightarrow dx = 40000 dt$

When $x = 20000$; $t = 0.5$

$$P(X \leq 20000) = \int_0^{0.5} e^{-t} dt = -e^{-t} \Big|_0^{0.5} = 0.3935$$

$$P(X > 20000) = 1 - 0.3935 = 0.6065$$

$$P(\text{at most } 30,000 \text{ km}) = P(X < 30,000 \text{ km})$$

$$\begin{aligned}
 P(X < 30000) &= \int_0^{30000} \frac{1}{40000} e^{-x/40000} dx \\
 &= \int_0^{0.75} e^{-t} dt \\
 &= 1 - e^{-0.75} = 0.5270
 \end{aligned}$$

2.98 Suresh has a car whose lifetime mileage is exponentially distributed with $\lambda = \frac{1}{20}$. He has driven the car for 20000 km and sold it to Ramesh. What is the probability that Ramesh would get at least 20000 km out of it? Repeat the above problem if the lifetime mileage is uniformly distributed over (0, 40000). Assume X is a random variable that represents mileage in thousand km.

Solution Since the exponential distribution has memoryless property, we have

$$P[X > (t + s) | X > s] = P[X > t]$$

Given $s = 10$ and $t = 20$

$$P[X > 30 | X > 10] = P[X > 20]$$

$$\begin{aligned}
 P[X > 20] &= 1 - P[X \leq 20] = 1 - (1 - e^{-\lambda(20)}) \\
 &= 1 - \left(1 - e^{-\left(\frac{1}{20}\right) \cdot 20} \right) = e^{-1}
 \end{aligned}$$

If X is a uniformly distributed random variable over (0, 40),

$$f_X(x) = \begin{cases} \frac{1}{40} & \text{for } 0 \leq x \leq 40 \\ 0 & \text{otherwise} \end{cases}$$

$$P[X > 30 | X > 10] = \frac{1 - F_X(30)}{1 - F_X(10)}$$

$$F_X(30) = \int_{-\infty}^{30} f(u) du = \int_0^{30} \frac{1}{40} dx = \frac{3}{4}$$

$$F_X(10) = \int_{-\infty}^{10} f(u) du = \int_0^{10} \frac{1}{40} dx = \frac{1}{4}$$

$$P[X > 30 | X > 10] = \frac{1 - 3/4}{1 - 1/4} = \frac{1/4}{3/4} = \frac{1}{3}$$

REVIEW QUESTIONS



16. Show that the inter-arrival times of a Poisson process with intensity λ obeys an exponential distribution.

17. Explain in detail about the memoryless property of exponential distribution.
18. Describe in detail about exponential distribution.



Practice Problems

2.41 The lifetime of a machine part is exponentially distributed with a mean of 400 hours.

- (a) What is the probability that the machine part fails in less than 100 hours?
- (b) What is the probability that the machine part works for more than 500 hrs before failure?
- (c) If the machine part worked for 400 hours without failure, what is the probability of a failure in the next 100 hours?

(Ans. (a) 0.2212 (b) 0.2865 (c) 0.2212)

2.42 The time between arrivals of suburban trains at Chennai Central railway station is exponentially distributed with a mean of 10 minutes:

- (a) What is the probability that you wait longer than one hour for a train?
- (b) Suppose you have already been waiting for one hour for a train, what is the probability that one arrives within the next 10 minutes.

(Ans. (a) .0025 (b) 0.6321)

2.6.6 Rayleigh Distribution

The Rayleigh density and distribution functions are given by

$$f_X(x) = \begin{cases} \frac{x}{\sigma^2} e^{-x^2/2\sigma^2} & x \geq 0 \\ 0 & x < 0 \end{cases} \quad (2.68)$$

$$F_X(x) = \begin{cases} 1 - e^{-x^2/2\sigma^2} & x \geq 0 \\ 0 & x < 0 \end{cases} \quad (2.69)$$

The Rayleigh pdf is shown in Fig. 2.27.

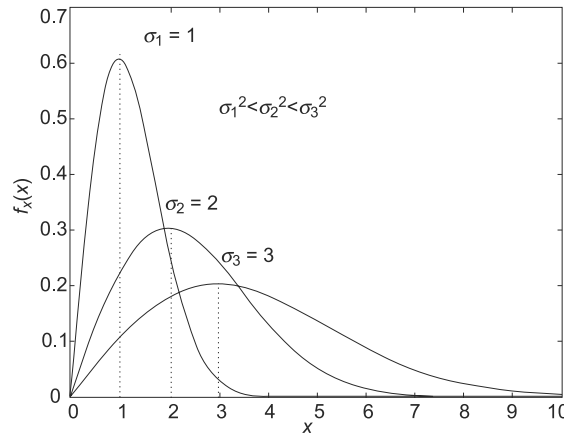


Fig. 2.27 Probability density function of Rayleigh Distribution

Rayleigh random variable is often a good model for measurement error and the amplitude or envelope of a signal with two orthogonal components. It is used to represent the envelope of narrowband noise.

Solved Problem



2.99 The Rayleigh density function is given by

$$f_X(x) = \begin{cases} x e^{-x^2/2} & x \geq 0 \\ 0 & x < 0 \end{cases}$$

- (a) Prove that $f_X(x)$ satisfies the properties of the pdf (i) $f_X(x) \geq 0$ for all x and (ii) $\int_{-\infty}^{\infty} f_X(x) dx = 1$
 (b) Find the distribution function $F_X(x)$.
 (c) Find $P(0.5 < x \leq 2)$.

Solution

- (a) $f_X(x) \geq 0$ for all x , since $f_X(x)$ is defined for only positive values

Consider $\int_0^{\infty} x e^{-x^2/2} dx$

Let $\frac{x^2}{2} = t$
 $x dx = dt$

$$\int_0^{\infty} x e^{-x^2/2} dx = \int_0^{\infty} e^{-t} = e^{-t} \Big|_0^{\infty} = 1$$

(b) $F_X(x) = \int_{-\infty}^x f_u(u) du$

$$= \int_0^x u e^{-u^2/2} du$$

$$= \int_0^{x^2/2} e^{-p} dp = -e^{-p} \Big|_0^{x^2/2} = -[e^{-x^2/2} - 1]$$

$$= 1 - e^{-x^2/2}$$

Let $\frac{u^2}{2} = p$
 $u du = dp$

(c) $P[0.5 < X \leq 2] = F_X(2) - F_X(0.5)$
 $= (1 - e^{-2}) - (1 - e^{-0.125})$
 $= e^{0.125} - e^{-2}$
 $= 0.9978$

REVIEW QUESTIONS



19. Sketch the probability density function and probability distribution function of (a) exponential distribution, (b) uniform distribution, and (c) Gaussian distribution

20. Define and explain the following density functions (a) Binomial (b) Exponential (c) Uniform (d) Rayleigh.
21. Define Rayleigh density and distribution function and explain them with their plots.

2.6.7 Gamma Distribution

The random variable X has a gamma distribution if its pdf is defined by

$$f_X(x) = \begin{cases} \frac{\lambda(\lambda x)^{\alpha-1} e^{-\lambda x}}{\Gamma(\alpha)}, & x > 0 \\ 0 & \text{elsewhere} \end{cases} \quad (2.70)$$

The distribution depends on two parameters, α and λ , both satisfying $\lambda > 0$, $\alpha > 0$. The parameter α is known as *shape parameter* and the parameter λ is known as *scale parameter*. $\Gamma(\alpha)$ is a gamma function given by

$$\begin{aligned} \Gamma(\alpha) &= \int_0^{\infty} x^{\alpha-1} e^{-x} dx \quad \alpha > 0 \\ &= \int_0^{\infty} x^{\alpha-1} e^{-x} dx = -e^{-x} x^{\alpha-1} \Big|_0^{\infty} - \int_0^{\infty} [e^{-x} (\alpha-1) x^{\alpha-2}] dx \\ &= 0 + (\alpha-1) \int_0^{\infty} [e^{-x} x^{\alpha-2}] dx \\ &= (\alpha-1) \Gamma(\alpha-1) \end{aligned} \quad (2.71)$$

Therefore, we can write

$$\begin{aligned} \Gamma(n) &= (n-1) \Gamma(n-1) = (n-1)(n-2) \Gamma(n-2) \\ &= (n-1)(n-2)(n-3) \dots \Gamma(1) \end{aligned}$$

$$\begin{aligned} \text{Since} \quad \Gamma(1) &= \int_0^{\infty} e^{-x} dx = 1 \\ \Gamma(n) &= (n-1)! \end{aligned} \quad (2.72)$$

We also have

$$\Gamma\left(\frac{1}{2}\right) = \int_0^{\infty} x^{-1/2} e^{-x} dx = \sqrt{\pi} \quad (2.73)$$

The amount of time one has to wait until a total of n events occur can be modelled as the gamma distribution. Therefore, it is used to model waiting time of customers in a queuing system and lifetime of devices in reliability studies. If α is an integer then the distribution represents an Erlang distribution.

If $\alpha = 1$, the distribution reduces to the exponential. If $\lambda = \frac{1}{2}$ and $t = \frac{\alpha}{2}$, the distribution is known as Chi-squared distribution with n degrees of freedom.

The gamma distribution is also given by

$$f_X(x) = \frac{1}{\Gamma(\alpha)} \beta^\alpha x^{\alpha-1} e^{-x/\beta}, x > 0 \quad (2.74)$$

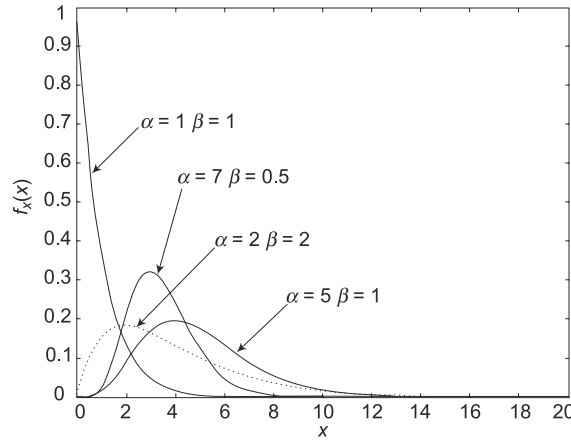


Fig. 2.28 Probability density function of Gamma Distribution

where $\beta = \frac{1}{\lambda}$ is known as *rate parameter*

The gamma pdf for different values of α and β are shown in Fig. 2.28.

Solved Problems



2.100 Show that the pdf of a gamma random variable integrates to one.

Solution For a gamma random variable,

$$f_X(x) = \frac{\lambda(\lambda x)^{\alpha-1} e^{-\lambda x}}{\Gamma(\alpha)}$$

$$\int_0^{\infty} f_X(x) dx = \int_0^{\infty} \frac{\lambda(\lambda x)^{\alpha-1} e^{-\lambda x}}{\Gamma(\alpha)} dx$$

Let $\lambda x = p$

$$\lambda dx = dp \quad \text{or} \quad dx = \frac{dp}{\lambda}$$

$$= \int_0^{\infty} \frac{\lambda (p)^{\alpha-1} e^{-p}}{\Gamma(\alpha)} \frac{dp}{\lambda}$$

$$\begin{aligned}
&= \frac{1}{\Gamma(\alpha)} \int_0^{\infty} p^{\alpha-1} e^{-p} dp \\
&= \frac{1}{\Gamma(\alpha)} \Gamma(\alpha) = 1
\end{aligned}$$

2.101 The daily consumption of water in an apartment in excess of 10,000 litres is approximately distributed as a gamma random variable with parameters $\alpha = 2$ and $\lambda = \frac{1}{5000}$. The apartment has a daily storage of 15,000 litres. What is the probability that the water is insufficient on a particular day?

Solution Let X denote the consumption of water in a day. Then $Y = X - 10000$ denotes the water in excess of 10000 litres which follows a gamma distribution with $\alpha = 2$ and $\lambda = \frac{1}{5000}$.

The pdf of Y is given by

$$f_Y(y) = \left(\frac{1}{5000} \right)^2 \frac{y e^{-y/5000}}{\Gamma(2)} = \frac{1}{(5000)^2} y e^{-y/5000}$$

Since daily storage of water is 15000 litres, the water is insufficient on a particular day if the consumption exceeds 15000 litres. So the required probability is

$$P(X > 15000) = P(Y > 5000)$$

$$= \int_{5000}^{\infty} f_Y(y) dy = \int_{5000}^{\infty} \frac{y e^{-y/5000}}{(5000)^2} dy$$

$$\text{Let } \frac{y}{5000} = t \Rightarrow dy = 5000 dt$$

when $y = 5000$; $t = 1$ and when $y = \infty$; $t = \infty$

$$\begin{aligned}
P(Y > 5000) &= \int_1^{\infty} \frac{(5000 t) e^{-t} 5000 (dt)}{(5000)^2} = \int_1^{\infty} t e^{-t} dt \\
&= -te^{-t} \Big|_1^{\infty} - e^{-t} \Big|_1^{\infty} = e^{-1} + e^{-1} = 2e^{-1}
\end{aligned}$$

2.102 In a post office, the time between constant visit is exponentially distributed with a mean of 5 minutes. What is the probability that the time between the arrival of the third customer and the arrival of the sixth customer is greater than 15 minutes?

Solution We have gamma distribution $f_X(x) = \frac{\lambda(\lambda x)^{\alpha-1} e^{-\lambda x}}{\Gamma(\alpha)}$

The rate at which the customer visits the post office is 5 minutes. That is, $\lambda = \frac{1}{5}$

Relative to the third customer, the sixth customer is the third arrival. Therefore $\alpha = 3$.

We have to find $P(X > 15)$

$$\begin{aligned} P(X > 15) &= \int_{15}^{\infty} \frac{x^2 e^{-x/5}}{5^3 \Gamma(3)} dx \\ &= \frac{1}{5^3 2!} \int_{15}^{\infty} x^2 e^{-x/5} dx \quad \boxed{\Gamma(3) = 2!} \end{aligned}$$

Let $\frac{x}{5} = t \Rightarrow dx = 5 dt$

$$\begin{aligned} P(X > 15) &= \frac{1}{5^3 2!} \int_3^{\infty} (5t)^2 e^{-t} 5 dt = \frac{(5)^3}{250} \int_3^{\infty} t^2 e^{-t} dt \\ &= \frac{1}{2} \left[(-t^2 e^{-t} - 2te^{-t} - 2e^{-t}) \right]_3^{\infty} \\ &= \frac{1}{2} [9 + 6 + 2] e^{-3} = 0.4231 \end{aligned}$$

2.103 The time interval between calls at a certain phone booth is exponentially distributed with a mean of 3 minutes. A arrived at the booth, while B was using the phone. He observed that nobody was present in the queue. Also, B spent 2 minutes on the call before A arrived. By the time B finished his conversation, more than 4 people were waiting behind A . What is the probability that the time between the instant A starts using the phone and the time the third person behind A starts his call is greater than 10 minutes?

Solution Let X be the time interval between calls. Given X is exponentially distributed with $\lambda = \frac{1}{3}$
Also, $\alpha = 3$

$$\begin{aligned} f_X(x) &= \frac{\lambda^3 x^2 e^{-x/3}}{\Gamma(3)} \quad 0 < x < \infty \\ P(X > 10) &= \int_{10}^{\infty} \frac{\left(\frac{1}{3}\right)^3 x^2 e^{-x/3}}{\Gamma(3)} dx \\ &= \frac{1}{(3)^3 2} \int_{10}^{\infty} x^2 e^{-x/3} dx \end{aligned}$$

Let $\frac{x}{3} = t \Rightarrow dx = 3 dt$

$$P(X > 10) = \frac{1}{2} \int_{10/3}^{\infty} t^2 e^{-t} dt = \frac{1}{2} \left[(-t^2 e^{-t} - 2te^{-t} - 2e^{-t}) \right]_{\frac{10}{3}}^{\infty}$$

$$\begin{aligned}
&= \frac{1}{2} \left[\left(\frac{10}{3} \right)^2 + 2 \left(\frac{10}{3} \right) + 2 \right] e^{-10/3} \\
&= 0.353
\end{aligned}$$



Practice Problem

2.43 In a service centre, a TV is repaired by a person at an average rate of one TV for every three days with exponential distribution. What is the probability that four TVs can be repaired within 5 days? (Ans. 0.265)

2.6.8 Weibull Distribution

The pdf of a random variable X that has Weibull distribution is given by

$$f_X(x) = \begin{cases} \alpha \beta x^{\beta-1} e^{-\alpha x^\beta}, & x > 0, \alpha > 0, \beta > 0 \\ 0 & \text{elsewhere} \end{cases} \quad (2.75)$$

The Weibull random variable is a good model for a failure time with time invariance or time-varying failure rate. In a situation where a system consists of many part may fail due to failure in any of the parts. It has been shown that a Weibull distribution provides an excellent model for describing the life time of the system. The distribution function of Weibull random variable X is

$$F_X(x) = \begin{cases} 1 - e^{-\alpha x^\beta}, & x > 0, \alpha > 0, \beta > 0 \\ 0 & x < 0 \end{cases} \quad (2.76)$$

If $\beta = 1$, the Weibull distribution reduces to the exponential distribution.

When $\beta = 2$ and $\alpha = \frac{1}{2\sigma^2}$, the Weibull distribution becomes Rayleigh distribution.

The pdf of Weibull distribution is shown in Fig. 2.29.

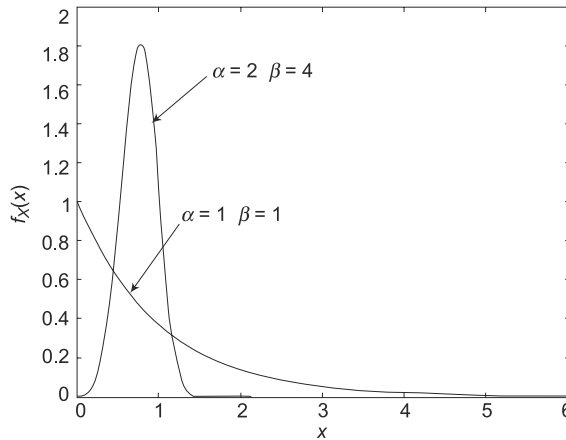


Fig. 2.29 Probability density function of a Weibull Distribution

Solved Problems

2.104 The life time of a semiconductor diode is a random variable having a Weibull distribution with the parameters $\alpha = 0.025$ and $\beta = 0.5$. What is the probability that the diode will still be in operating condition for more than 4000 hours.

Solution Given: X is Weibull random variable

$$f_X(x) = \alpha\beta x^{\beta-1} e^{-\alpha x^\beta}, x > 0, \alpha > 0, \beta > 0$$

$$\alpha = 0.025 \text{ and } \beta = 0.5$$

$$f_X(x) = 0.0125 x^{-1/2} e^{-0.025 x^{1/2}}$$

$$P(X > 4000) = \int_{4000}^{\infty} (0.0125) x^{-1/2} e^{-0.025 x^{1/2}} dx$$

$$\text{Let } 0.025x^{1/2} = p$$

$$\frac{1}{2} (0.025)x^{-1/2} dx = dp; \text{ when } x = 4000, p = 1.58$$

$$\begin{aligned} P(X > 4000) &= \int_{1.58}^{\infty} e^{-p} dp = -e^{-p} \Big|_{1.58}^{\infty} \\ &= e^{-1.58} = 0.2057 \end{aligned}$$

2.105 An electronic circuit that consists of 6 transistors has a life length (in years) can be considered as a random variable that follows Weibull distribution with parameters $\alpha = 20$ and $\beta = 1.5$. If these transistors functions independent of one another what is the probability that no transistor will have to be replaced during the first 3 months of service.

Solution Given: $\alpha = 20, \beta = 1.5$

$$\begin{aligned} f_X(x) &= \alpha\beta x^{\beta-1} e^{-\alpha x^\beta} \quad x > 0 \\ &= 30 x^{1/2} e^{-20 x^{1.5}} \end{aligned}$$

$$P\left(X > \frac{1}{4}\right) = \int_{1/4}^{\infty} 30x^{1/2} e^{-20x^{1.5}} dx$$

$$\text{Let } 20x^{1.5} = p$$

$$\text{when } x = \frac{1}{4}, p = 2.5$$

$$30x^{1/2} dx = dp$$

$$P\left(X > \frac{1}{4}\right) = \int_{2.5}^{\infty} e^{-p} dp = -e^{-p} \Big|_{2.5}^{\infty}$$

$$= e^{-2.5} = 0.0821$$

2.106 Find the probability of failure-free performance of a dc motor over a period of 5000 hours. The life expectancy of the motor is defined by Weibull distribution with parameters $\alpha = 10^{-7}$ and $\beta = 2$.

Solution

The life expectancy of the motor is defined by Weibull distribution with $\alpha = 10^{-7}$ and $\beta = 2$

We have $f_X(x) = \alpha\beta x^{\beta-1} e^{-\alpha x^\beta} = 2 \times 10^{-7} x e^{-10^{-7} x^2}$

$P(\text{failure-free performance of a dc motor})$

$$= 1 - P(X < 5000)$$

$$= 1 - \int_0^{5000} 2 \times 10^{-7} x e^{-\alpha x^2} dx$$

$$= 1 - \int_0^{5000} 2 \times 10^{-7} x e^{-10^{-7} x^2} dx$$

Let $10^{-7} x^2 = t$

$$2 \times 10^{-7} x dx = dt$$

Also at $x = 5000$; $t = 2.5$

$P(\text{failure-free performance of a dc motor})$

$$= 1 - \int_0^{2.5} e^{-t} dt$$

$$= 1 - (-e^{-t}) \Big|_0^{2.5}$$

$$= e^{-2.5} = 0.08208$$

**Practice Problems**

2.44 If the life of a semiconductor is a random variable having a Weibull distribution with parameters $\alpha = 0.01$ and $\beta = 0.5$, what is the probability that the semiconductor will be in operating condition after 5000 hours? (Ans. $e^{-0.707}$)

2.45 The lifetime of a component measured in hours is Weibull distributed with parameters $\alpha = 10^{-3}$ and $\beta = 2$. Find the probability that such a component will last more than 100 hours. (Ans. $e^{-0.1}$)

2.6.9 Beta Distribution

A random variable is said to have a beta distribution if its density function is given by

$$f_X(x) = \frac{1}{B(\alpha, \beta)} x^{\alpha-1} (1-x)^{\beta-1} \quad 0 < x < 1, \alpha > 0, \beta > 0 \quad (2.77)$$

$$= 0 \quad \text{otherwise}$$

where $B(\alpha, \beta) = \int_0^1 x^{\alpha-1} (1-x)^{\beta-1} dx \quad (2.78)$

The relation between beta function and gamma function is given by

$$\beta(\alpha, \beta) = \frac{\Gamma(\alpha) \Gamma(\beta)}{\Gamma(\alpha + \beta)} \quad (2.79)$$

The β density function for different values of α and β are shown in Fig. 2.30.

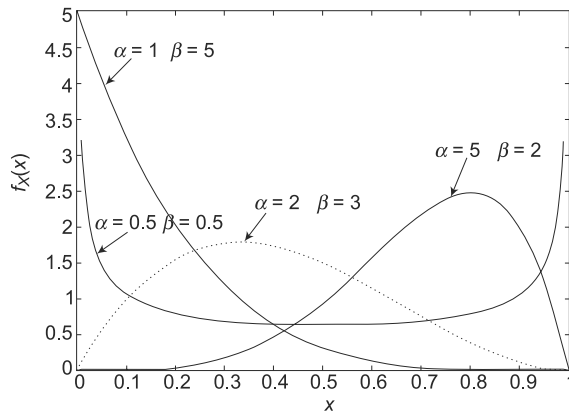


Fig. 2.30 Probability density function of Beta Distribution

2.6.10 Cauchy Distribution

A random variable X is said to have a Cauchy distribution with parameter θ , $-\infty < \theta < \infty$, if its density function is given by

$$f_X(x) = \frac{1}{\pi} \frac{1}{[1 + (x - \theta)^2]} \quad -\infty < x < \infty \quad (2.80)$$

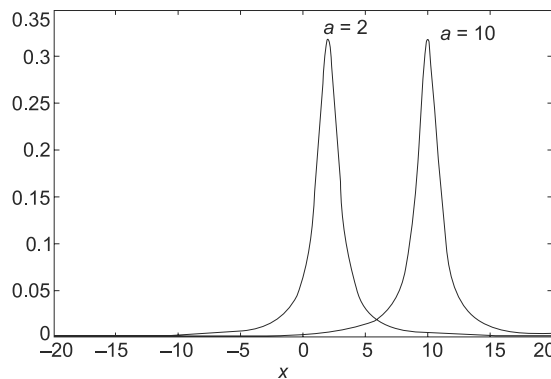


Fig. 2.31 Probability density function of Cauchy Distribution

REVIEW QUESTIONS



22. Write expression for pdf of
 - (a) Gamma random variable
 - (b) β random variable
 - (c) Cauchy random variable
23. Prove that $\Gamma(n) = (n - 1)!$

24. Prove the expression

$$\beta(\alpha_1, \beta) = \frac{\Gamma(\alpha) \pi(\beta)}{\Gamma(\alpha + \beta)}$$

25. Verify that the gamma density function integrates to 1.

26. Compute $\Gamma\left(n + \frac{1}{2}\right)$ for $n = 1, 2$.

27. Verify that the beta density function integrates to one.

2.6.11 Log-normal Distribution

If the logarithm of a random variable is Gaussian, the random variable is said to have the log-normal distribution. The pdf of a log-normal random variable is given by

$$f(x) = \begin{cases} \frac{1}{\sqrt{2\pi}\sigma} x^{-1} e^{-(\ln x - \mu)^2 / (2\sigma^2)} & x > 0 \\ 0 & x \leq 0 \end{cases} \quad (2.81)$$

The log-normal random variable is a good model for problems involving random effects that are Gaussian when measured in a logarithmic scale. The probability distribution function of X is

$$F(x) = \frac{1}{2} [1 + \operatorname{erf}(u)] \quad (2.82)$$

$$\text{where } u = \frac{1}{\sqrt{2}\sigma} \ln\left(\frac{x}{\bar{x}}\right) \quad (2.83)$$

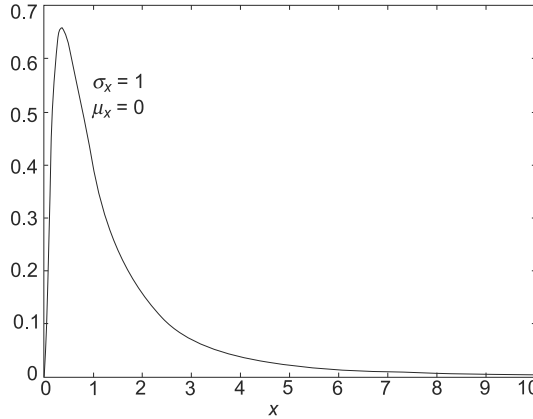


Fig. 2.32 Probability density function of log-normal Distribution

2.6.12 Laplace Distribution

The Laplace density function is defined as

$$f(x) = \frac{1}{\sqrt{2\lambda}} e^{-\frac{|x - \bar{x}|}{\lambda}} \quad -\infty < x < \infty, \quad \lambda > 0 \text{ and } -\infty < \bar{x} < \infty \quad (2.84)$$

If mean $\mu = \bar{x} = 0$ and $\alpha = \frac{1}{\lambda}$ then the density function becomes

$$f(x) = \frac{\alpha}{2} e^{-\alpha|x|} \quad (2.85)$$

The Laplace density function for various values of λ is shown in Fig. 2.33.

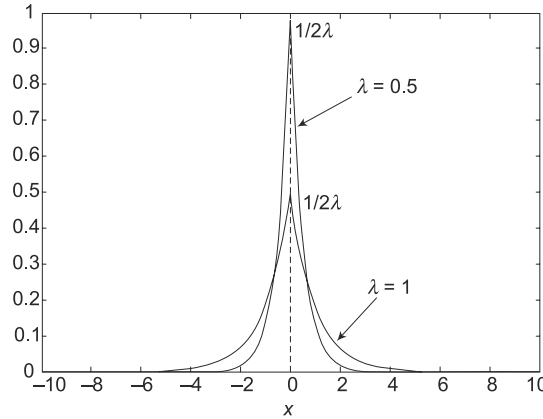


Fig. 2.33 Probability density function of Laplace Distribution

2.6.13 Chi-Square Distribution

If $X_1, X_2 \dots X_n$ are Gaussian random variables with zero mean and variance 1 and if $Y^2 = X_1^2 + X_2^2 + \dots + X_n^2$ then the random variable Y^2 is said to have a Chi-square distribution with n degrees of freedom with pdf.

$$f_Y(Y) = \frac{(y^2)^{\frac{n}{2}-1}}{2^{\frac{n}{2}} \Gamma\left(\frac{n}{2}\right)} e^{-y^2/2} \quad y^2 \geq 0 \quad (2.86)$$

$$= 0 \quad y^2 < 0$$

The gamma function is defined as

$$\Gamma(x) = \int_0^{\infty} u^{x-1} e^{-u} du \quad (2.87)$$

Figure 2.34 shows the pdf of a Chi-square distributed random variable with different degrees of freedom.

Chi-square random variables are used in the Chi-square test which is popular among many statistics tests. This test provides a measure to judge if a random variable contradicts its underlying assumption made regarding its distribution.

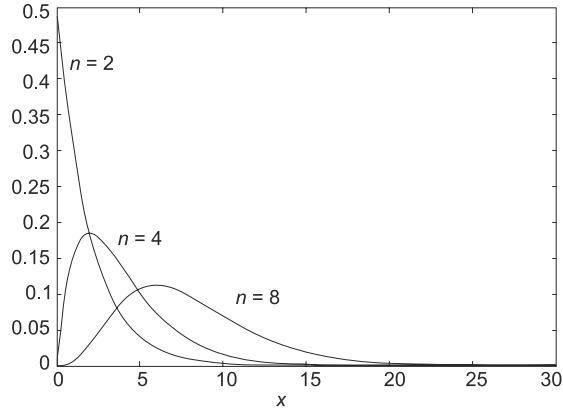


Fig. 2.34 Probability density function of Chi square Distribution

2.6.14 Rice Density Function

The random variable R with density function of the form

$$f(r) = \frac{r}{\sigma^2} e^{-\frac{(r^2 + \lambda)}{2\sigma^2}} I_0 \left(\frac{\sqrt{\lambda}r}{\sigma^2} \right), r \geq 0 \quad (2.88)$$

is known as Rice density function.

The function $I_0(x)$ is the zero-order modified Bessel function given by

$$I_0(x) = \frac{1}{2\pi} \int_0^{2\pi} e^{x \cos \theta} d\theta = \sum_{n=0}^{\infty} \frac{x^{2n}}{2^{2n} (n!)^2} \quad (2.89)$$

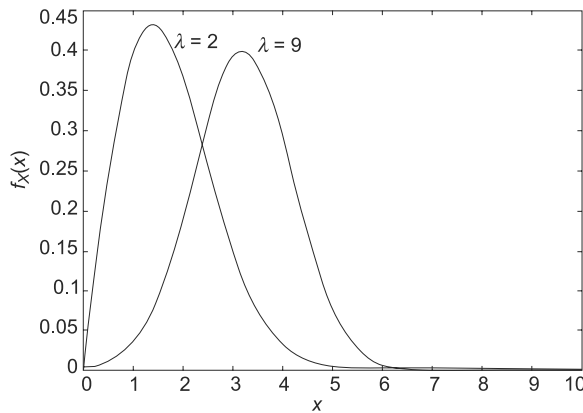


Fig. 2.35 Probability density function of Rice Distribution

2.6.15 Maxwell's Distribution

Let us assume that $X = \sqrt{X_1^2 + X_2^2 + X_3^2}$ where X_1, X_2 and X_3 are Gaussian random variables, each with zero mean and variance σ^2 . Then X has a Chi-square distribution with three degrees of freedom. The distribution of $Y = \sqrt{X_1^2 + X_2^2 + X_3^2}$ is known as the Maxwell's distribution given by

$$f(y) = \begin{cases} \frac{1}{\sigma^3} \sqrt{\frac{2}{\pi}} y^2 e^{-\frac{y^2}{2\sigma^2}} & y \geq 0 \\ 0 & \text{otherwise} \end{cases} \quad (2.90)$$

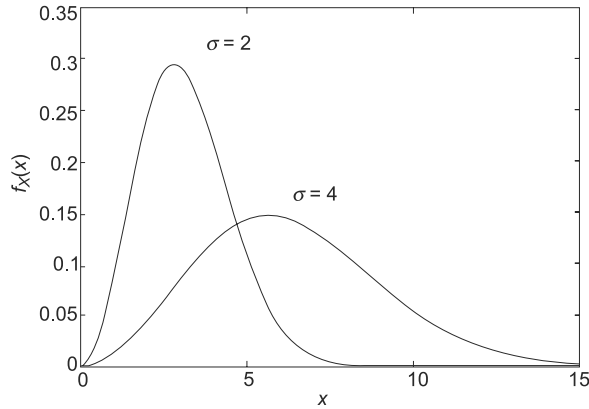


Fig. 2.36 Probability density function of Maxwell's Distribution

REVIEW QUESTIONS



28. Write the pdf of the following distributions:
 - (a) Log-normal
 - (b) Laplace
 - (c) Chi-square
29. For what type of applications are the following distribution are used? (a) Log-normal (b) Chi-square.

CONDITIONAL DISTRIBUTION AND DENSITY FUNCTION

2.7



In Chapter 1, we studied the concept of conditional probability. It is defined as follows. The conditional probability for the event A given the event B with $P(B) \neq 0$ is defined as

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \quad (2.91)$$

We will extend this concept to distribution $F_X(x)$ and densities $f_X(x)$

2.7.1 Conditional Distribution

Let A be an event $[X \leq x]$ for the random variable X . Then the conditional distribution $f_X(x | A)$, given that the event A has occurred is defined as

$$f_X(x | A) = P[X \leq x | A] = \frac{P[X \leq x \cap A]}{P(A)} \quad (2.92)$$

Properties of Conditional Distribution

$$(i) \quad F_X(\infty | A) = 1; \quad F_X(-\infty | A) = 0 \quad (2.93)$$

We know

$$F_X(-\infty | A) = P[X \leq -\infty | A]$$

$$\text{Since} \quad F_X(-\infty) = P[X \leq -\infty] = 0$$

$$F_X(-\infty | A) = 0$$

$$F_X(\infty | A) = 1$$

$$F_X(\infty | A) = P[X \leq \infty | A]$$

$$\text{Since} \quad F_X(\infty) = P[X \leq \infty] = 1$$

$$(i) \quad F_X(\infty | A) = 1 \quad (2.94a)$$

$$(ii) \quad 0 \leq F_X(x | A) \leq 1 \quad (2.94b)$$

Since $0 \leq F_X(x) \leq 1$, $P[X \leq x | A]$ also bounded between the same values.

Hence,

$$0 \leq F_X(x | A) \leq 1$$

$$(iii) \quad F_X(x_1 | A) < F_X(x_2 | A) \text{ if } x_1 < x_2, \quad (2.95)$$

Since $F_X(x_1) < F_X(x_2)$ if $x_1 < x_2$

Therefore

$$F_X(x_1 | A) < F_X(x_2 | A)$$

$$(iv) \quad F_X(x_2 | A) - F_X(x_1 | A) = P[x_1 < X \leq x_2 | A] \quad (2.96)$$

$$(v) \quad F_X(x | A) = \int_{-\infty}^{\infty} f_X(u | A) du \quad (2.97)$$

$$(vi) \quad F_X(x^+ | A) = F_X(x | A) \quad (2.98)$$

Conditional Density Function

The conditional density function is defined as

$$f_X(x | A) = \frac{d}{dx} F_X(x | A) \quad (2.99)$$

$$= \lim_{\Delta x \rightarrow 0} \frac{P[x < X < x + \Delta x | A]}{\Delta x}$$

Properties of Conditional Density Function

The conditional density function satisfies the same properties as the ordinary density function. They are

$$(i) \quad f_X(x | A) \geq 0 \quad (2.100)$$

$$(ii) \quad \int_{-\infty}^{\infty} f_X(x | A) dx = 1 \quad (2.101)$$

$$(iii) \quad F_X(x | A) = \int_{-\infty}^x f_X(s | A) ds \quad (2.102)$$

$$(iv) \quad P[x_1 < X \leq x_2 | A] = \int_{x_1}^{x_2} f_X(x | A) dx \quad (2.103)$$

2.7.2 Methods of Defining Conditioning Event

Conditioning event A depends on X . Define an event A in terms of X . Consider the following cases (i) $A = \{X \leq b_2\}$ (ii) $A = \{b_1 < X \leq b_2\}$

Case (i) $A = \{X \leq b_2\}$

The CDF can be obtained from Eq. (2.91) which is given by

$$F_X(x | A) = \frac{P(X \leq x, X \leq b_2)}{P(X \leq b_2)}$$

where $P[X \leq x, X \leq b_2] = P(X \leq x \cap X \leq b_2)$

Now we consider two cases:

- (a) $x \geq b_2$: since $x \geq b_2$ the joint event $[X \leq x, X \leq b_2] = [X \leq b_2]$.
Therefore,

$$F_X[x | A] = \frac{P(X \leq b_2)}{P(X \leq b_2)} = 1, x \geq b_2$$

- (b) $x < b_2$: Now the joint event

$$[X \leq x, X \leq b_2] = \{X \leq x\}.$$

Hence,

$$F_X(x | A) = \frac{P(X \leq x)}{P(X \leq b_2)} = \frac{F_X(x)}{F_X(b_2)}, x < b_2$$

The corresponding density function is

$$f_X(x | A) = \begin{cases} \frac{f_X(x)}{F_X(b_2)} & x < b_2 \\ 0 & x \geq b_2 \end{cases}$$

Case (ii) $A = [b_1 < x \leq b_2]$ with $b_1 < b_2$

$$F_X(x | A) = \frac{P(X \leq x, b_1 < X \leq b_2)}{P(b_1 < X \leq b_2)}$$

Here, we have three situations:

- (a) $x \geq b_2$

$$F_X(x | A) = \frac{P(b_1 < X \leq b_2)}{P(b_1 < X \leq b_2)} = 1 \quad x \geq b_2$$

(b) $b_1 < x \leq b_2$, then $\{X \leq x, b_1 < x \leq b_2\} = \{b_1 < X \leq x\}$

$$\begin{aligned} F_X(x | A) &= \frac{P(b_1 < X \leq x)}{P(b_1 < X \leq b_2)} \\ &= \frac{F_X(x) - F_X(b_1)}{F_X(b_2) - F_X(b_1)} \quad b_1 < x < b_2 \end{aligned}$$

(c) $x \leq b_1$, then $\{X \leq x, b_1 < X \leq b_2\} = \{\emptyset\}$

$$F_X(x | A) = \frac{P(\emptyset)}{P(b_1 < X \leq b_2)} = 0, \quad x < b_1$$

The corresponding density function is

$$f_X(x | A) = \frac{f_X(x)}{F_X(b_2) - F_X(b_1)} \quad \text{for } b_1 \leq x < b_2$$

REVIEW QUESTIONS



30. Define conditional distribution and density function.
31. Explain the properties of conditional distribution.
32. Explain different methods of defining conditioning event.

Solved Problems



2.107 The random variable X is Poisson with a parameter b . Find the conditional pmf of X given $A = [X \text{ is even}]$.

Solution The pmf of the Poisson random variable is

$$P_X(x) = \frac{b^x e^{-b}}{x!} \quad x = 0, 1, 2, \dots$$

The CDF of X is given by

$$F_X(k) = \sum_{r=0}^k \frac{b^r e^{-b}}{r!}$$

$$A = [X = \text{even}]$$

The probability of A is

$$\begin{aligned} P(A) &= P(X = \text{even}) = \sum_{\substack{r=0 \\ (r=\text{even})}}^{\infty} \frac{b^r e^{-b}}{r!} \\ &= \sum_{r=0}^{\infty} \frac{b^{2r} e^{-b}}{(2r)!} = e^{-b} \sum_{r=0}^{\infty} \frac{b^{2r}}{(2r)!} \end{aligned}$$

$$\begin{aligned} e^x &= 1 + \frac{x}{1!} + \frac{x^2}{2!} + \dots \\ e^{-x} &= 1 - \frac{x}{1!} + \frac{x^2}{2!} - \dots \\ \frac{e^x + e^{-x}}{2} &= 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \dots \end{aligned}$$

$$\begin{aligned}
 P(A) &= e^{-b} \left[1 + \frac{b^2}{2!} + \frac{b^4}{4!} + \dots \right] \\
 &= e^{-b} \left[\frac{e^b + e^{-b}}{2} \right] = \frac{1 + e^{-2b}}{2}
 \end{aligned}$$

If x is even, $[X = x] \subset A$ and $[X = x] \cap A = [X = x]$

If x is odd, $[X = x] \cap A = \phi$. Hence

$$\begin{aligned}
 P_X[x | A] &= \frac{2e^{-b} b^x}{(1 + e^{-2b}) x!} \quad x \text{ even} \\
 &= \frac{P(\phi)}{P(B)} = 0; \quad x \text{ odd}
 \end{aligned}$$

2.108 A discrete random variable X has the probability mass function given below.

x	0	1	2	3	4	5	6
$P_X(x)$	0	a	$2a$	$2a$	a^2	$3a^2$	$2a^2$

Find (a) the value of a , (b) $P[X < 5]$, $P[X \geq 4]$, $P[0 < X < 4]$, (c) the distribution function of X .

Solution

(a) We know $\sum_x P_X(x) = 1$

$$\begin{aligned}
 \Rightarrow \quad & 0 + a + 2a + 2a + a^2 + 3a^2 + 2a^2 = 1 \\
 & 6a^2 + 5a - 1 = 0 \Rightarrow 6a^2 + 6a - a - 1 = 0 \\
 & 6a(a + 1) - 1(a + 1) = 0 \\
 & (6a - 1)(a + 1) = 0
 \end{aligned}$$

$$\Rightarrow \quad a = \frac{1}{6}$$

(b) $P[X < 5] = P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3) + P(X = 4)$

$$= a^2 + 5a = \left(\frac{1}{6}\right)^2 + \frac{5}{6} = \frac{1}{36} + \frac{5}{6} = \frac{1 + 30}{36} = \frac{31}{36}$$

$P[X \geq 4] = P(X = 4) + P(X = 5) + P(X = 6)$

$$a^2 + 3a^2 + 2a^2 = 6a^2$$

$$6a^2 = 6\left(\frac{1}{6}\right)^2 = 6\left(\frac{1}{36}\right) = \frac{1}{6}$$

$P(0 < X < 4) = P(X = 1) + P(X = 2) + P(X = 3)$

$$= a + 2a + 2a = 5a = \frac{5}{6}$$

(c) Distribution of X

X	0	1	2	3	4	5	6
$F_X(x) =$ $P_X(X \leq x)$	0	$a = \frac{1}{6}$	$3a = \frac{1}{2}$	$5a = \frac{5}{6}$	$a^2 + 5a = \frac{31}{36}$	$4a^2 + 5a = \frac{34}{36}$	$= 1$

2.109 The daily consumption of milk in excess of 20,000 gallons is approximately exponentially distributed with $\theta = 3000$. The city has a daily stock of 35,000 gallons. What is the probability that of two days selected at random, the stock is insufficient for both days?

Solution Let X denote the daily consumption of milk and Y denote the excess amount of milk consumed in a day.

Then we can write $Y = X - 20000$

The random variable Y follows exponential distribution with $\theta = 3000$. Therefore,

$$f_Y(y) = \frac{1}{3000} e^{-y/3000}$$

When $X > 35000$ then the stock is insufficient. Therefore, the probability that the stock is insufficient for one day $= P[X > 35000]$

$$P[X > 35000] = P[Y + 20000 > 35000]$$

$$= P[Y > 15000]$$

$$= \int_{15000}^{\infty} \frac{1}{3000} e^{-y/3000} dy$$

$$= -e^{-y/3000} \Big|_{15000}^{\infty} = e^{-5}$$

The probability that of two days selected at random, the stock is insufficient for both days $= e^{-5}(e^{-5}) = e^{-10}$

2.110

The density function of a mixed random variable X is given by

$$f_X(x) = k \{e^x [u(-x-1) - u(-x-4)] + e^{-x} \{u(x-1) - u(x-4)\} + \frac{1}{4} [\delta(x+2) + \delta(x) + \delta(x-2)]\}$$

Find k and $F_X(x)$.

Solution The probability density function $f_X(x)$ is sketched as shown in Fig. 2.37.

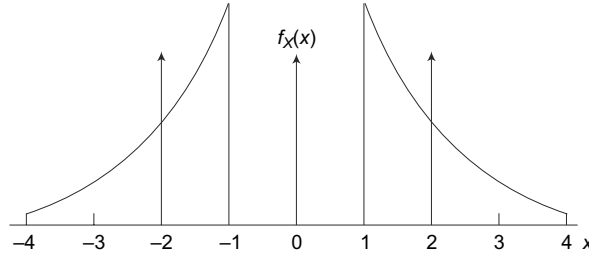


Fig. 2.37

The pdf is defined over the interval $[-4, 4]$. Therefore, to evaluate k , we integrate $f_X(x)$ for the above interval .

$$\begin{aligned}
 \int_{-4}^4 f_X(x) dx &= k \left\{ \int_{-4}^{-1} e^x dx + \int_{-1}^1 e^{-x} dx \right\} + \frac{1}{4} \int \delta(x+2) dx + \frac{1}{4} \int \delta(x) dx + \frac{1}{4} \int \delta(x+2) dx \\
 &= k \{e^{-1} - e^{-4} - e^{-4} + e^{-1}\} + \frac{1}{4} + \frac{1}{4} + \frac{1}{4} \\
 &= 2k [e^{-1} - e^{-4}] + \frac{3}{4}
 \end{aligned}$$

We know $\int_{-\infty}^{\infty} f_X(x) dx = 1$

$$\Rightarrow 2k [e^{-1} - e^{-4}] + \frac{3}{4} = 1$$

from which $k = 0.3576$

The CDF is $F_X(x) = \int_{-\infty}^x f_X(x) dx$

(a) For $x \leq -5$, since $f_X(x) = 0$, $F_X(x) = 0$

(b) For $-4 < x \leq -2$,

$$\begin{aligned}
 F_X(x) &= \int_{-4}^x 0.357 e^x dx = 0.3576 e^x \Big|_{-4}^x \\
 &= 0.3576 (e^x - e^{-4})
 \end{aligned}$$

(c) For $-2 < x \leq -1$

$$f_X(x) = 0.3576 (e^x - e^{-1}) + \frac{1}{4}$$

$$\begin{aligned} F_X(x) &= \int_{-4}^{-1} 0.3576 e^x dx + \frac{1}{4} \\ &= 0.3576 (e^{-1} - e^{-4}) + \frac{1}{4} \\ &= \frac{1}{8} + \frac{1}{4} = \frac{3}{8} \end{aligned}$$
$$F_X(x) = \frac{3}{8} + \frac{1}{4} = \frac{5}{8}$$
$$F_X(x) = \frac{5}{8} + \int_1^x 0.3576 e^{-x} dx$$
$$= \frac{5}{8} + 0.3576 (e^{-1} - e^{-x})$$
$$F_X(x) = \frac{5}{8} + 0.3576(e^{-1} - e^{-x}) + \frac{1}{4}$$
$$F_X(x) = \frac{5}{8} + 0.3576(e^{-1} - e^{-x}) + \frac{1}{4} = 1$$

- Describe the underlying sample space S .
- Find the probability for various values of X .

[illegible]

10	20	20	20	20	20	20	20	20	20	X	110
10	20	20	20	20	20	20	20	20	20	X	110
10	20	20	20	20	20	20	20	20	20	X	110
10	20	20	20	20	20	20	20	20	20	X	110
100	110	110	110	110	110	110	110	110	110	110	X

From the sample space, we can observe

$$P(X = 20) = \frac{9(8)}{10 \times 9} = \frac{72}{90} = \frac{4}{5}$$

$$P(X = 110) = \frac{2 \times 9}{10 \times 9} = \frac{1}{5}$$



Practice Problem

2.46 Repeat Solved Problem (2.111) with replacement

$$\left(\text{Ans. } \frac{81}{100}, \frac{18}{100}, \frac{1}{100} \right)$$

Solved Problems



2.112 Two balls are randomly chosen from an urn containing 10 white, 3 black and 2 green balls. Suppose that we win ₹ 10 for each black ball selected and we lose ₹ 5 for each white ball selected. Let X denote our winning.

Find the probability for different possible values of X . Plot CDF.

Solution

Case 1: $P(\text{Both balls are black})$

When both balls are black, we will win ₹ 20. Therefore,

$$P(X = 20) = \frac{\binom{3}{2}}{\binom{15}{2}} = \frac{3}{105}$$

Case 2: $P(1 \text{ white and } 1 \text{ black})$

$$P(X = 5) = \frac{\binom{10}{1} \binom{3}{1}}{\binom{15}{2}} = \frac{30}{105}$$

Case 3: $P(\text{both balls are given})$

$$P(X=0) = \frac{\binom{2}{2}}{\binom{15}{2}} = \frac{1}{105}$$

Case 4: $P(1 \text{ black and } 1 \text{ green})$

$$P(X=10) = \frac{\binom{3}{1}\binom{2}{1}}{\binom{15}{2}} = \frac{6}{105}$$

Case 5: $P(1 \text{ white and } 1 \text{ green})$

$$P(X=-5) = \frac{\binom{10}{1}\binom{2}{1}}{\binom{15}{2}} = \frac{20}{105}$$

Case 6: $P(\text{Both white})$

$$P(X=-10) = \frac{\binom{10}{2}}{\binom{15}{2}} = \frac{45}{105}$$

X	-10	-5	0	5	10	20
pdf	$\frac{45}{105}$	$\frac{20}{105}$	$\frac{1}{105}$	$\frac{30}{105}$	$\frac{6}{105}$	$\frac{3}{105}$

The pdf and CDF plots are shown in Fig. 2.38

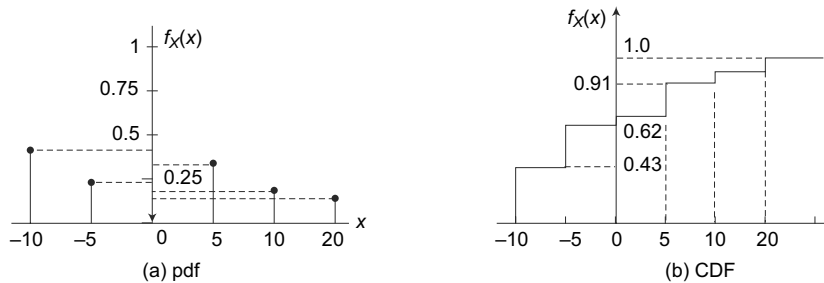


Fig. 2.38

2.113 If the probability density function of a random variable is given by

$$f_X(x) = \begin{cases} C e^{-x/4}, & 0 \leq x < 1 \\ 0 & \text{otherwise} \end{cases}$$

Find the value of and evaluate $F_X(0.5)$

Solution

$$\text{Given} = \begin{cases} f_X(x) = C e^{-x/4} & 0 \leq x < 1 \\ 0 & \text{otherwise} \end{cases}$$

$$\text{For a valid pdf} \quad \int_0^1 C e^{-x/4} dx = \frac{C e^{-x/4}}{\left(-\frac{1}{4}\right)} \Bigg|_0^1 = 1$$

$$\Rightarrow \quad C = \frac{1}{4(1 - e^{-0.25})} = 1.13$$

$$\therefore \quad f_X(x) = \begin{cases} 1.13 e^{-x/4}, & 0 \leq x < 1 \\ 0 & \text{otherwise} \end{cases}$$

$$F_X(0.5) = \int_0^{0.5} 1.13 e^{-x/4} dx = -1.13 (4) e^{-\frac{x}{4}} \Bigg|_0^{0.5} = 0.5311$$

2.114 The CDF of a random variable Y is

$$F_Y(y) = \begin{cases} 1 - e^{(-0.4\sqrt{y})}, & y \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

Evaluate $P(2.5 < Y \leq 6.2)$.

Solution

$$\begin{aligned} P(2.5 < Y \leq 6.2) &= F_Y(6.2) - F_Y(2.5) \\ &= (1 - e^{-0.4\sqrt{6.2}}) - (1 - e^{-0.2\sqrt{2.5}}) \\ &= e^{-0.4\sqrt{2.5}} - e^{-0.4\sqrt{6.2}} \\ &= 0.531 - 0.369 = 0.1616 \end{aligned}$$

2.115 A random variable X has the distribution function

$$F_X(x) = \sum_{n=1}^{12} \frac{n^2}{650} u(x - n)$$

Find the probability: (a) $P(-\infty < x \leq 6.5)$ (b) $P(X > 4)$ (c) $P(6 < X \leq 9)$

Solution The CDF is given by

$$F_X(x) = \sum_{n=1}^{12} \frac{n^2}{650} u(x-n)$$

$$(a) \quad P(-\infty < x \leq 6.5) = F_X(6)$$

$$\begin{aligned} F_X(6) &= \frac{1}{650} \sum_{n=1}^6 n^2 \\ &= \frac{1}{650} \frac{n(n+1)(2n+1)}{6} \Big|_{n=6} = 0.14 \end{aligned}$$

$$(b) \quad P(X > 4) = 1 - P(X \leq 4) = 1 - F_X(4)$$

$$\begin{aligned} &= 1 - \sum_{n=1}^4 \frac{n^2}{650} \\ &= 1 - \frac{1}{650} \frac{n(n+1)(2n+1)}{6} \Big|_{n=4} \\ &= 0.954 \end{aligned}$$

$$\begin{aligned} P(6 < X \leq 9) &= F_X(9) - F_X(6) \\ &= \sum_{n=6}^9 \frac{n^2}{650} = \frac{1}{650} \left[\frac{9 \times 10 \times 19}{6} - \frac{6 \times 7 \times 13}{6} \right] \\ &= \frac{1}{650} [285 - 91] = 0.2985 \end{aligned}$$

2.116 The diameter of an electric cable is assumed to be a continuous random variable X with pdf

$$f_X(x) = 6x(x-1) \quad 0 \leq x \leq 1$$

- (a) Verify that $f_X(x)$ is a valid density function.
 (b) Find $P(0 \leq X \leq 0.5)$.

Solution The pdf of random variable X is given by

$$f_X(x) = 6x(x-1) \quad 0 \leq x \leq 1$$

- (a) For a valid pdf,

$$\int_{-\infty}^{\infty} f_X(x) dx = 1$$

$$\begin{aligned} \text{That is, } \int_0^1 6x(x-1) dx &= 6 \left[\frac{x^3}{3} - \frac{x^2}{2} \right]_0^1 \\ &= 6 \left[\frac{1}{3} - \frac{1}{2} \right] \end{aligned}$$

Therefore, $f_X(x)$ is a valid pdf.

$$\begin{aligned}
 \text{(b) } P(0 \leq x \leq 0.5) &= \int_0^{0.5} 6x(1-x) dx \\
 &= 6 \int_0^{0.5} (x - x^2) dx \\
 &= 6 \left[\frac{x^2}{2} - \frac{x^3}{3} \right]_0^{0.5} \\
 &= \left[\frac{(0.5)^2}{2} - \frac{(0.5)^3}{3} \right] = 0.5 \\
 \therefore P(0 \leq X \leq 0.5) &= 0.5
 \end{aligned}$$

2.117 The continuous random variable X has pdf $f_X(x) = \frac{x}{2}$, $0 \leq x \leq 2$. Two independent determinations of X are made. What is the probability that both of these determined will be greater than one? If three determinations had been made, what is the probability that exactly two of these are larger than one?

Solution Given $f_X(x) = \frac{x}{2}$, $0 \leq x \leq 2$

Let X_1 and X_2 be two independent determinations.

$$P(X_1 > 1) = \int_1^2 \frac{x}{2} dx = \frac{1}{2} \left[\frac{x^2}{2} \right]_1^2 = \frac{3}{4}$$

$$P(X_2 > 1) = \int_1^2 \frac{x}{2} dx = \frac{1}{2} \left[\frac{x^2}{2} \right]_1^2 = \frac{3}{4}$$

$$P(X_1 > 1, X_2 > 1) = \binom{3}{4} \binom{3}{4} = \frac{9}{16}$$

Let X_1, X_2 and X_3 be three independent determinations. We can find that

$$P(X < 1) = \int_0^1 \frac{x}{2} dx = \frac{x^2}{4} \Big|_0^1 = \frac{1}{4}$$

$P(\text{two determinations larger than 1})$

$$\begin{aligned}
 &= P(X_1 > 1, X_2 > 1, X_3 < 1) + P(X_1 > 1, X_2 < 1, X_3 > 1) + P(X_1 < 1, X_2 > 1, X_3 > 1) \\
 &= \left(\frac{3}{4} \right) \left(\frac{3}{4} \right) \left(\frac{1}{4} \right) + \left(\frac{3}{4} \right) \left(\frac{1}{4} \right) \left(\frac{3}{4} \right) + \frac{1}{4} \left(\frac{3}{4} \right) \left(\frac{3}{4} \right)
 \end{aligned}$$

$$= \frac{9}{64} + \frac{9}{64} + \frac{9}{64} = \frac{27}{64}$$

2.118 The continuous random variable X has pdf $f_X(x) = 3x^2$, $-1 \leq x \leq 0$. If b is a number satisfying $-1 < b < 0$, compute $P(X > b | X < b/2)$.

Solution

$$\begin{aligned} F_X(x) &= \int_{-\infty}^x f_X(u) du \\ &= \int_{-1}^x 3u^2 du = u^3 \Big|_{-1}^x = x^3 + 1 \\ F_X(x) &= x^3 + 1 \\ P(X > b | X < b/2) &= \frac{P\left[X > b, X < \frac{b}{2}\right]}{P\left[X < \frac{b}{2}\right]} \\ &= \frac{F_X\left(\frac{b}{2}\right) - F_X(b)}{F_X\left(\frac{b}{2}\right)} = \frac{\left(\frac{b}{2}\right)^3 + 1 - b^3 - 1}{\left(\frac{b}{2}\right)^3 + 1} \\ &= -\frac{7b^3}{b^3 + 8} \end{aligned}$$

2.119 Check whether the following are valid pdfs"

(a) $f_X(x) = \frac{1}{\pi} \frac{1}{1+x^2} \quad -\infty \leq x \leq \infty$

(b) $f_X(x) = \frac{1}{4}(x-1)^3$ if $1 \leq x \leq 3$
 $= 0$ otherwise

Solution

(a) Given: $f_X(x) = \frac{1}{\pi} \frac{1}{1+x^2}; -\infty < x < \infty$

For a valid pdf, $\int_{-\infty}^{\infty} f_X(x) dx = 1$

$$\int_{-\infty}^{\infty} \frac{1}{\pi} \left(\frac{1}{1+x^2} \right) dx = \frac{1}{\pi} \tan^{-1} x \Big|_{-\infty}^{\infty} = \frac{1}{\pi} \left[\frac{\pi}{2} - \left(-\frac{\pi}{2} \right) \right] = 1$$

So, $f_X(x) = \frac{1}{\pi(1+x^2)}$ is a valid pdf

$$(b) \quad f_X(x) = \frac{1}{4}(x-1)^3 \text{ if } 1 \leq x \leq 3$$

$$\begin{aligned} \int_{-\infty}^{\infty} f_X(x) dx &= \frac{1}{4} \int_1^3 (x-1)^3 dx = \frac{1}{4} \left. \frac{(x-1)^4}{4} \right|_1^3 \\ &= \frac{2^4}{(4)^2} = 1 \end{aligned}$$

So, $f_X(x)$ is a valid pdf.

2.120 A random variable X has a pdf $f_X(x) = a^2 x e^{-ax} u(x)$, where a is a constant. Find CDF and evaluate

$$P\left(X \leq \frac{1}{a}\right) \text{ and } P\left[\frac{1}{a} \leq X \leq \frac{2}{a}\right].$$

Solution Given $f_X(x) = a^2 x e^{-ax} u(x)$

$$\begin{aligned} F_X(x) &= \int_{-\infty}^x a^2 x e^{-ax} \\ &= a^2 \int_0^x x e^{-ax} \\ &= a^2 \left[-\frac{x e^{-ax}}{a} \right]_0^x - \frac{e^{-ax}}{a^2} \Big|_0^x \\ &= a^2 \left[-\frac{x e^{-ax}}{a} - \frac{e^{-ax}}{a^2} + \frac{1}{a^2} \right] \\ &= -ax e^{-ax} - e^{-ax} + 1 \\ &= 1 - (ax + 1) e^{-ax} \end{aligned}$$

$$\begin{aligned} P\left(X \leq \frac{1}{a}\right) &= F_X\left(\frac{1}{a}\right) = \left[1 - (ax + 1) e^{-ax}\right]_{x=\frac{1}{a}} \\ &= 1 - (1 + 1) e^{-1} = 1 - 2e^{-1} = 0.264 \end{aligned}$$

$$\begin{aligned} P\left(\frac{1}{a} \leq X \leq \frac{2}{a}\right) &= F\left(\frac{2}{a}\right) - F\left(\frac{1}{a}\right) \\ F\left(\frac{2}{a}\right) &= \left[1 - (ax + 1) e^{-ax}\right]_{x=\frac{2}{a}} = 1 - 3e^{-2} \end{aligned}$$

$$\begin{aligned}\Rightarrow P\left(\frac{1}{a} \leq X \leq \frac{2}{a}\right) &= (1 - 3e^{-2}) - (1 - 2e^{-1}) \\ &= 2e^{-1} - 3e^{-2} = 0.330\end{aligned}$$

2.121 For a random variable with binomial distribution, show that for $0 \leq x < n$,

$$P_X(x+1)/P_X(x) = \frac{(n-x)p}{(x+1)(1-p)}$$

Using the above result, show that

- (a) $P_X(x+1) > P_X(x)$ if $x < np - (1-p)$
- (b) $P_X(x+1) = P_X(x)$ if $x = np - (1-p)$
- (c) $P_X(x+1) < P_X(x)$ if $x > np - (1-p)$

Solution

We know $P_X(x) = \binom{n}{x} p^x (1-p)^{n-x}$

$$P_X(x+1) = \binom{n}{x+1} p^{x+1} (1-p)^{n-x-1}$$

$$\begin{aligned}\frac{P_X(x+1)}{P_X(x)} &= \frac{\binom{n}{x+1} p^{x+1} (1-p)^{n-x-1}}{\binom{n}{x} p^x (1-p)^{n-x}} = \frac{n!}{(x+1)!(n-x-1)!} \cdot \frac{x!(n-x)!}{n!} \cdot \frac{p}{1-p} \\ &= \left(\frac{n-x}{x+1}\right) \left(\frac{p}{1-p}\right)\end{aligned}$$

Given:

$$(a) \quad P_X(x+1) > P_X(x) \Rightarrow \frac{P_X(x+1)}{P_X(x)} > 1$$

$$\text{Hence, } \left(\frac{n-x}{x+1}\right) \left(\frac{p}{1-p}\right) > 1; (n-x)p > (x+1)(1-p)$$

$$np - xp > x + 1 - xp - p$$

$$np > x + 1 - p$$

$$np - 1 + p > x$$

$$\Rightarrow x < np - (1-p)$$

$$(b) \quad \text{Given: } P_X(x+1) = P_X(x) \Rightarrow \frac{P_X(x+1)}{P_X(x)} = 1$$

$$\Rightarrow (n-x)p = (x+1)(1-p)$$

$$\Rightarrow x = np - (1-p)$$

$$\begin{aligned} \text{(c) Given: } P_X(x+1) < P_X(x) &\Rightarrow \frac{P_X(x+1)}{P_X(x)} < 1 \\ &\Rightarrow k > np - (1-p) \end{aligned}$$

2.122 In a noisy channel, the error probability for a digit is 0.05.

- (a) Calculate the probability of more than one error in 20 received digits.
- (b) Repeat part (a) using the Poisson approximation.

Solution Given: $p = 0.05$; $n = 20$

Let X be the random binomial random variable with parameters $(20, 0.05)$.

$$\begin{aligned} \text{(a) } P(X > 1) &= 1 - P(X \leq 1) \\ &= 1 - P(X = 0) - P(X = 1) \\ &= 1 - \binom{20}{0} (0.05)^0 (0.95)^{20} - \binom{20}{1} (0.05)^1 (0.95)^{19} \\ &= 1 - 0.3589 - 0.3773 = 0.263 \\ \text{(b) } \lambda &= np = 20(0.05) = 1 \\ P(X > 1) &= 1 - P(X = 0) - P(X = 1) \\ &= 1 - \frac{e^{-1} (1)^0}{0!} - \frac{e^{-1} (1)}{1!} \\ &= 1 - e^{-1} - e^{-1} = 0.264 \end{aligned}$$

2.123 An urn contains 3 white and 3 black balls. We randomly chose 2 balls. If 1 of them is white and 1 is black, we stop. If not, we replace the balls in the urn and again randomly select 2 balls. This continues until 1 of the 2 chosen is white. What is the probability that we shall make exactly n selections.

Solution Total number of balls in the urn = 6

Two balls can be selected from the urn in $\binom{6}{2}$ ways.

One white ball can be selected from 3 white balls in $\binom{3}{1}$ ways.

Similarly, one black ball is selected from 3 black balls is $\binom{3}{1}$ ways.

$$P(\text{selecting 1 black and 1 white ball}) = \frac{\binom{3}{1} \binom{3}{1}}{\binom{6}{2}} = \frac{3 \times 3}{15} = \frac{3}{5}$$

Probability that n selections are required before first success is

$$\begin{aligned} P(X = n) &= pq^{n-1} \quad (\text{Geometric distribution}) \\ &= \frac{3}{5} \left(\frac{2}{5} \right)^{n-1} \end{aligned}$$

2.124 A salesman is paid ₹ 100 for each sale he makes. The probability that he sells the product in a sale is 0.75.

- What is the probability that he earned his third ₹ 100 on the fifth sale he made?
- If he made five sales per hour, what is the probability that he earned ₹ 900 in two hours?

Solution The probability that the salesman sells the product is 0.75. Let X denote the number of calls up to and including k th success. The pmf of X is given by

$$\begin{aligned} p_X(x) &= \binom{x-1}{k-1} p^k (1-p)^{x-k} \\ &= \binom{x-1}{k-1} (0.75)^k (0.25)^{x-k} : \quad \begin{array}{l} x = k, k+1, \dots \\ k = 1, 2, \dots, x \end{array} \end{aligned}$$

- The probability that the salesman earned his third ₹ 100 on the fifth sale is

$$P_X(x=5) = \binom{5-1}{3-1} (0.75)^3 (0.25)^2 = 0.1582$$

- If he made five sales per hour then the total number of sales in 2 hours is 10. He earns ₹ 900 if he made 9 sales. That is he sells the product 9 times in 10 sales. Therefore, it is a binomial distribution of 9 successes in 10 trials.

That is
$$p = \binom{10}{9} (0.75)^9 (0.25)^1 = 0.1877$$

2.125 A company invited applications to fill three vacancies. Out of 20 applications received, five applicants were found to be qualified for the post. It is observed from past experience that 25% applicants who were offered this kind of post rejected the offer. If the company ranks the selected five applicants according to their performance, what is the probability that fifth ranked applicant will be offered one of the positions?

Solution The probability that an applicant offered a job actually joins is 0.75

Let X be a random variable that denotes the number of candidates offered a job upto and including the k th candidate to accept the job. Then the pmf is given by

$$p_X(x) = \binom{n-1}{k-1} (0.75)^k (0.25)^{n-k} \quad \begin{array}{l} k=1, 2, \dots, n \\ n=k, k=1, \dots \end{array}$$

The probability that the fifth ranked applicant will be offered one of the three positions is the probability that the fifth candidates is either the first, second or third candidate to accept a job. Therefore, the required probability is given by

$$P = (0.75) (0.25)^4 + \binom{4}{1} (0.75)^2 (0.25)^3 + \binom{4}{2} (0.75)^3 (0.25)^2 = 0.1963$$

2.126 A multiple-choice examination has 10 problems, each of which has four possible answers. What is the probability that Hari will get five or more correct answers by just guessing?

Solution Given that each multiple choice questions has four possible answers, therefore, the probability of getting a correct answer by just guessing is $p = \frac{1}{4}$.

Let X be a random variable that denotes the number of questions Hari answers correctly from 10 questions.

The probability that X is greater than or equal to five is

$$\begin{aligned} P(X \geq 5) &= P(X = 5) + P(X = 6) + P(X = 7) + P(X = 8) + P(X = 9) + P(X = 10) \\ &= \binom{10}{5} \left(\frac{1}{4}\right)^5 \left(\frac{3}{4}\right)^5 + \binom{10}{6} \left(\frac{1}{4}\right)^6 \left(\frac{3}{4}\right)^4 + \binom{10}{7} \left(\frac{1}{4}\right)^7 \left(\frac{3}{4}\right)^3 \\ &\quad + \binom{10}{8} \left(\frac{1}{4}\right)^8 \left(\frac{3}{4}\right)^2 + \binom{10}{9} \left(\frac{1}{4}\right)^9 \left(\frac{3}{4}\right) + \binom{10}{10} \left(\frac{1}{4}\right)^{10} \\ &= 0.0584 + 0.0162 + 0.00309 + 3.86 \times 10^{-4} + 2.86 \times 10^{-5} + 9.536 \times 10^{-7} \\ &= 0.0781 \end{aligned}$$

2.127 In a state, studies indicate that 20% of marriages end in divorce. If divorces between couples are independent of each other, find the probability

- that exactly 4 of 12 couples will stay married
- that only Ram and Geetha will stay married

Solution

- The probability that a marriage ends in divorce is 0.2. Therefore, we can say that the probability that a couple stays married is 0.8. Let X denote the number of couples that stay married. Then the probability that exactly 4 out of 12 couples will stay married is given by

$$P(X = 4) = \binom{12}{4} (0.8)^4 (0.2)^8 = 0.00052$$

- The probability that Ram and Geetha only stay married whereas the remaining eleven couples get divorced is given by

$$P = (0.2)^{11} (0.8) = 1.64 \times 10^{-8}$$

2.128 An urn contains B black balls and W white balls. A ball is randomly selected from the urn, its colour is noted and the ball is put back into the urn. The process is repeated until a black ball is selected.

- What is the probability that the experiment stops after n trials?
- What is the probability that the experiment requires at least k trials before it stops?

Solution The total number of balls in an urn = $B + W$

The probability of selecting a black ball is $\frac{B}{B + W}$

Let X be a random variable that denotes the number of trials until a black ball is selected. The pmf of X is given by

$$p_X(x) = pq^{n-1} \quad n = 1, 2, \dots$$

The probability that the experiment stops exactly after n trials is

$$P(x = n) = \frac{B}{B+W} \left(\frac{B}{B+W} \right)^{n-1} \quad n = 1, 2, \dots$$

The probability that the experiment requires at least k trials before it stops is given by

$$\begin{aligned} P(X \geq k) &= 1 - P(X < k) \\ &= 1 - \sum_{n=1}^{k-1} pq^{n-1} \\ &= 1 - \sum_{n=1}^{k-1} p(1-q)^{n-1} \\ &= 1 - p\{1 + (1-p) + \dots + (k-1) \text{ terms}\} \\ &= 1 - p \left\{ \frac{1 - (1-p)^{k-1}}{1 - (1-p)} \right\} \\ &= (1-p)^{k-1} = q^{k-1} = \left(\frac{W}{B+W} \right)^{k-1} \end{aligned}$$

REVIEW QUESTIONS



33. Define random variable and explain the concept of random variable.
34. What are the conditions for a function to be a random variable?
35. Define discrete and continuous random variables with suitable examples.
36. Define distribution function of a random variable.
37. Define probability mass function.
38. Explain the properties of distribution function.
39. What are the important conditions that a pdf should satisfy?
40. Define pdf of continuous random variable X .
41. What are the values of $F_X(-\infty)$ and $F_X(\infty)$?
42. List and explain the properties of discrete probability density function.
43. Explain the relation between CDF and pdf.
44. If X is a continuous random variable prove that $P(X = C) = 0$.
45. What are binomial density and distribution functions?
46. What is Poisson random variable? Explain in brief.
47. When a random variable X is said to have uniform distribution.
48. What is Gaussian random variable? Explain its significance?
49. Draw pdf and CDF Curve of Gaussian random variable.
50. Derive an expression for the error function of the standard normal random variable.

51. Define complementary error function.
52. List the properties of Gaussian curve.
53. Sketch the probability density function of exponential random variable.
54. What are the applications of (a) Poisson random variable (b) exponential random variable.
55. Define and Explain Rayleigh density function.
56. Define Weibull distribution and explain its application.
57. Write equations for pdf of the following distributions (a) Gamma distribution (b) Beta distribution.
58. Define conditional distribution and density functions and explain their properties.



EXERCISES

Problems

1. If X is a discrete random variable having the probability distribution

$X = x$	-2	-1	0	1	2
$P(X = x)$	k	$k/2$	$2k$	k	$k/2$

Find $P(X \leq 1)$

2. If X is a discrete random variable having the pmf

X	1	2	3	4
$p(x)$	k	$2k$	$3k/2$	$k/2$

Find $P(2 \leq X < 3)$.

3. After a coin is tossed two times, if X is the number of heads, find the probability distribution of X .

4. If $P(X = x) = \begin{cases} kx, & x = 1, 2, 3, 4, 5 \\ 0 & \text{otherwise} \end{cases}$

represents pmf of a random variable X , find

(i) k (ii) $P(X \text{ being a prime number})$ (iii) $P(1/2 < X < 5 \mid X > 1)$

$$\left(\text{Ans: (i) } \frac{1}{15} \text{ (ii) } \frac{11}{15} \text{ (iii) } \frac{1}{7} \right)$$

5. If the CDF of a random variable is given by

$$F_X(x) = \begin{cases} 0 & \text{for } x < 0 \\ x^2/16 & \text{for } 0 \leq x \leq 4 \\ 1 & \text{for } 4 < x \end{cases}$$

Find $P(X > 1/X < 3)$

$$\left(\text{Ans: } \frac{8}{9} \right)$$

6. Consider the experiment of tossing four fair coins. The random variable X is associated with the number of tails showing. Compute and sketch CDF of X .
7. The CDF of a random variable X is given by

$$F_X(x) = \begin{cases} 0 & x \leq 0 \\ C(1 - e^{-x}), & x > 0 \end{cases}$$

Find $f_X(x)$.

$$\left(\text{Ans: } f_X(x) = \begin{cases} 0 & x \leq 0 \\ C e^{-x}, & x > 0 \end{cases} \right)$$

8. Determine whether the following is a valid distribution function

$$F_X(x) = \begin{cases} 1 - e^{-x} & \text{for } x \geq 0 \\ 0 & \text{elsewhere} \end{cases}$$

9. Consider the pdf of a random variable X given by

$$f_X(x) = a e^{-bx}, -\infty < x < \infty.$$

Find CDF of X .

$$\left(\text{Ans: } F_X(x) = \begin{cases} \frac{a}{b} e^{bx} & \text{for } x \leq 0 \\ \frac{a}{b} (2 - e^{-bx}) & \text{for } x \geq 0 \end{cases} \right)$$

10. Consider the distribution function for X defined by

$$F_X(x) = \begin{cases} 0, & x < 0 \\ 1 - \frac{1}{4} e^{-x}, & x \geq 0 \end{cases}$$

Find (i) $P(X = 0)$, (ii) $P(X \geq 0)$

$$\left(\text{Ans: (i) } \frac{3}{4}; \text{ (ii) } 1 \right)$$

11. A continuous random variable X has a pdf

$$f_X(x) = \begin{cases} 3x^2 & 0 \leq x \leq 1 \\ 0 & \text{elsewhere} \end{cases}$$

Find a and b such that

- (i) $P(X < a) = (P(X > a))$ and
(ii) $P(X > b) = 0.05$

12. The random variable X has the discrete variable in the set $\{-1, -0.5, 0.7, 1.5, 3\}$, the corresponding probabilities are assumed to be $\{0.1, 0.2, 0.1, 0.4, 0.2\}$. Plot its distribution function.

13. A random variable X is uniformly distributed in the interval $(-2, 2)$

- (a) Find and plot the CDF of X .
(b) Find (i) $P(X > -0.5)$ (ii) $P(-1 \leq x \leq 1)$

14. A random variable X has pdf

$$f_X(x) = \begin{cases} cx(1-x^2) & 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

- (i) Find c (ii) Find $P(0 < X < 0.5)$

$$(\text{Ans: (i) } 4 \text{ (ii) } 0.4375)$$

15. Find the CDF of the Cauchy random variable which has pdf

$$f_X(x) = \frac{1}{\pi(x^2 + 1)} - \infty < x < \infty$$

$$\left(\text{Ans: } \frac{1}{\pi} \left[\tan^{-1}(x) + \frac{\pi}{2} \right] \right)$$

16. The random variable X has a binomial distribution with $n = 5$ and $p = 0.6$.

Find (i) $P(X = 2)$ (ii) $P(X \leq 2)$ (iii) $P(1 \leq X \leq 3)$.

17. In electronic laboratory it is found that 5% of voltmeters are defective. A random sample of 10 voltmeters are taken for inspection. (i) what is the probability that all are good (ii) at most there are two defective voltmeters.
18. In a country the suicide rate is 2 per one lakh people per month. Find the probability that in a city of population 10,00,000 there will be at most two suicides in a month.
19. The number of errors in a textbook follow a Poisson distribution with a mean of 0.05 error per page. What is the probability that there are five or less errors in 50 pages?
20. The number of telephone calls that arrive at a telephone exchange is modelled as a Poisson random variable with 10 average calls per hour.
 - (a) What is the probability that there are three or less calls in one hour?
 - (b) What is the probability that there are exactly 10 calls in one hour?
21. If the probability that a student pass the examination equals 0.6. What is the probability that he needs more than 3 attempts before he pass the examination?
22. A point is chosen at random on the line segment $[0, 8]$. What is the probability that the chosen point lies between 5 and 6?
23. A typist types 3 letters correctly for every 50 letters. What is the probability that the fifth letter typed is the first erroneous letter?
24. In an examination taken by 500 candidates, the mean and standard deviation of marks are found to be 35% and 10% respectively. If the marks saved are normally distributed, find how many will pass if 30% is fixed as minimum?
25. X is normally distributed with mean 10 and standard deviation 5. Find
 - (i) $P(X \geq 15)$ (ii) $P(1 \leq X \leq 12)$.
26. In a cinema hall, the time required for a person to give tickets follows a normal distribution. He gives ten tickets with an average time of 5 minutes and a standard deviation of 1 minute.
 - (i) What is the probability that he will take less than 3 minutes to issue 10 tickets?
 - (ii) What is the probability he will take more than 6 minutes?
27. The number of years a TV function is exponentially distributed with $X = \frac{1}{8}$. What is the probability that it will work after an additional 8 years?

Multiple-Choice Questions

1. Which of the following statement(s) are False?
 - (a) $F_X(x)$ is monotonically increasing on the real line
 - (b) $F_X(x)$ is continuous from the right everywhere
 - (c) $F_X(x) = \int_{-\infty}^{\infty} f_X(x) dx$
 - (d) $P(a \leq X \leq b) = \int_a^b f_X(x) dx$
2. If X is a discrete random variable with geometric distribution
Then $P(X \approx x)$ for $x = 1, 2, \dots, \infty$
 - (a) q^x
 - (b) pq^{x-1}
 - (c) $(1-q)^{x-1}$
 - (d) $\frac{1}{q^x}$

3. A random variable X' has pdf given by

$$f_X(x) = \begin{cases} \frac{x}{a^2} e^{-x^2/2a^2} & x \geq 0 \\ 0 & x < 0 \end{cases}$$

What is the value of $P(X > a)$

- (a) e (b) $e^{-1/2}$ (c) $e^{1/2}$ (d) 2^{-1}
4. $B(a, b) =$
- (a) $\Gamma(a)\Gamma(b)$ (b) $\frac{\Gamma(a)\Gamma(b)}{\Gamma(a) + \Gamma(b)}$ (c) $\frac{\Gamma(a)\Gamma(b)}{\Gamma(a-b)}$ (d) $\frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}$
5. A random variable X has density function given by

$$f_X(x) = \frac{1}{\pi(1+x^2)} \quad -\infty < x < \infty$$

Then the random variable X has the

- (a) Cauchy distribution (b) Laplace distribution
(c) Pascal distribution (d) Weibull distribution
6. If z is a unit normal random variable, then

$$\lim_{x \rightarrow \infty} \frac{P(z \geq x + a/x)}{P(z \geq x)}$$

- (a) a (b) $e^{-a^2/2}$ (c) e^{-a} (d) $e^{-a/2}$
7. If X is an exponential random variable with parameter 1, and $c > 0$, then CX is exponential random variable with parameter
- (a) $\frac{\lambda}{C}$ (b) $C\lambda$ (c) $\lambda + C$ (d) $\lambda - C$

8. If X is a random variable with pdf

$$f_X(x) = \begin{cases} C(1-x^2) & -1 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

The value of C is

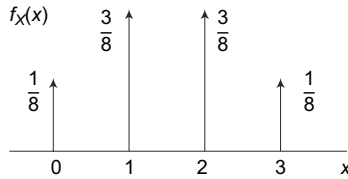
- (a) $\frac{1}{4}$ (b) $\frac{3}{4}$ (c) $\frac{1}{2}$ (d) $\frac{3}{2}$
9. The pdf of a random variable X is given by

$$f_X(x) = \begin{cases} \frac{10}{x^2}, & x > 10 \\ 0 & x \leq 10 \end{cases}$$

What is the value of $P(X > 20)$

- (a) $\frac{1}{4}$ (b) $\frac{1}{2}$ (c) $\frac{1}{8}$ (d) 1

10. The pdf of a discrete random variable is shown in figure.



What is its CDF?

- (a) $\frac{1}{8} \delta(x) + \frac{3}{8} \delta(x-1) + \frac{3}{8} \delta(x-2) + \frac{1}{8} \delta(x-3)$
 (b) $\frac{1}{8} \delta(x) + \frac{3}{8} \delta(x+1) + \frac{3}{8} \delta(x+2) + \frac{1}{8} \delta(x+3)$
 (c) $\frac{1}{8} u(x) + \frac{3}{8} u(x+1) + \frac{3}{8} u(x+2) + \frac{1}{8} u(x+3)$
 (d) $\frac{1}{8} u(x) + \frac{3}{8} u(x-1) + \frac{3}{8} u(x-2) + \frac{1}{8} u(x-3)$
11. A random variable X has a pdf

$$f_X(x) = Ce^{-a|x|}$$

$$P(X < b) =$$

 (a) $1 - e^{-ab}$ (b) $1 - e^{-a}$ (c) $1 - e^{-ab}$ (d) $1 - e^{-b/a}$
12. Which of the random variable satisfies memoryless property?
 (a) Uniform distribution (b) Laplace distribution
 (c) exponential distribution (d) Poisson distribution
13. X is a Poisson random variable with parameter λ . The value of λ maximizes $P(X = k)$ is
 (a) $\frac{k}{2}$ (b) k (c) k^2 (d) $2k$
14. If X is a geometric random variable
 $P(X = n + k | X > n)$ is equal to
 (a) $P(X = n + k)$ (b) $P(X = n)$ (c) $P(X = k)$ (d) $P(X = n - k)$
15. A continuous random variable X has the density function

$$f_X(x) = \frac{C}{1+x^2}, \quad -\infty < x < \infty$$

 The value of C is
 (a) $\frac{1}{\sqrt{\pi}}$ (b) $\frac{1}{\pi}$ (c) $\frac{2}{\pi}$ (d) $\frac{2}{\sqrt{2\pi}}$
16. The pdf of a random variable X is $f_X(x) = a^2 x e^{-ax} u(x)$. the value of $P\left(X < \frac{1}{a}\right)$
 (a) $1 - 2e^{-1}$ (b) $1 - e^{-1}$ (c) $2 - e^{-1}$ (d) $\frac{1}{2} - e^{-1}$

17. The CDF of a random variable X is

$$f_X(x) = 1 - (1+x)e^{-x}, x > 0. \text{ The pdf of } X \text{ is}$$

- (a) $e^{-1}, x > 0$ (b) $\frac{1}{x}e^{-x}, x > 0$
 (c) $x^2e^{-x}, x > 0$ (d) $xe^{-x}, x > 0$
18. $P(x_1 < X \leq x_2)$ is equal to
 (a) $F_X(x_1) - F_X(x_2)$ (b) $F_X(x_2) - F_X(x_1)$ (c) $f_X(x_1) - f_X(x_2)$ (d) $f_X(x_2) - f_X(x_1)$
19. If X follows a Poisson distribution such that $P(X=1) = \frac{1}{4}$ and $P(X=2) = \frac{3}{8}$. Then the value of $P(X=3)$ is
 (a) 1 (b) 0.5 (c) 0.375 (d) 0.625
20. If Y is uniformly distributed over $(-b, b)$, the value of ' b ' that satisfies $P(|Y| < 1)$ is
 (a) 2 (b) 3 (c) 1 (d) 4
21. Consider the function

$$f_X(x) = \begin{cases} 2x & 0 \leq x \leq 6 \\ 0 & \text{otherwise} \end{cases}$$

The value of b for $f_X(x)$ to be a legitimate pdf

- (a) 1 (b) 2 (c) 0.5 (d) 1.5
22. The CDF of the random variable X is defined by

$$F_X(x) = \begin{cases} 0 & x < 2 \\ \frac{1}{4}(x-2) & 2 \leq x \leq 6 \\ 1 & x \geq 6 \end{cases}$$

The value of $P(X > 4)$ is

- (a) 1 (b) $\frac{1}{2}$ (c) $\frac{1}{4}$ (d) $\frac{3}{2}$
23. The CDF of a random variable is given by

$$F_X(x) = \begin{cases} 0 & x < a \\ \frac{x-a}{b-a} & a \leq x < b \\ 1 & x \geq b \end{cases}$$

Then the random variable has

- (a) Laplace distribution (b) Poisson distribution
 (c) Uniform distribution (d) Pascal distribution
24. The CDF of an exponential random variable is

- (a) $\frac{1}{b} \exp\left(-\frac{x}{b}\right)u(x)$ (b) $\left[1 - \exp\left(-\frac{x}{b}\right)\right]u(x)$
 (c) $\frac{1}{b} \exp\left(\frac{x}{b}\right)u(-x)$ (d) $\left[1 - \exp\left(\frac{x}{b}\right)\right]u(x)$

25. The pdf of a Laplace random variable is given by

(a) $\frac{1}{2} \exp\left(\frac{x}{b}\right)$ (b) $1 - \frac{1}{2} \exp\left(-\frac{x}{b}\right)$ (c) $\frac{1}{b} - \exp\left(-\frac{x}{b}\right)$ (d) $\frac{1}{2b} \exp\left(-\frac{|x|}{b}\right)$

26. The probability of a defective diode from a diode manufacturing company is 3%. If a box contains 400 diodes, what is the probability that exactly 3 pens are defective?

(a) 0.000176 (b) 0.00176 (c) 0.0176 (d) 0.176

27. Which of the following distributions has memoryless property?

(a) Exponential distribution (b) Normal distribution
(c) Uniform distribution (d) Pascal distribution

28. The pmf of a random variable is

$$p_K(k) = \begin{cases} \frac{\lambda^k e^{-\lambda}}{k!} & k = 0, 1, 2, \dots \\ 0 & \text{otherwise} \end{cases}$$

The value of $P(k > 1)$ is

(a) $e^{-\lambda}(1 + \lambda)$ (b) $\lambda e^{-\lambda}$ (c) $1 - e^{-\lambda}(1 + \lambda)$ (d) $1 - \lambda e^{-\lambda}$

29. The random variable X has the pmf

$$p_X(x) = \binom{5}{x} (0.1)^x (0.9)^{5-x}, \quad x = 0, 1, \dots, 5$$

The value of $P(X = 1)$

(a) 0.4 (b) 0.33 (c) 0.35 (d) 0.42

30. The pdf of a random variable is

$$f_X(x) = \begin{cases} \frac{3}{4}(1 - x^2) & -1 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

The value of $P(0 < X < 0.5)$ is

(a) 0.344 (b) 0.454 (c) 0.656 (d) 0.546

31. The pdf of a random variable X is given by

$$f_X(x) = \begin{cases} 2e^{-2(x-0.5)} & x \geq 0.5 \\ 0 & x < 0.5 \end{cases}$$

The CDF is

(a) $1 - 2e^{-2x-0.5}, x \geq 0.5$ (b) $1 - e^{-2(x-0.5)}, x \geq 0$
(c) $1 - e^{-(x-0.5)}, x \geq 0$ (d) $1 - e^{-x+0.5}, x \geq 0$

32. If X is a geometric random variable then $P(X = n + k | X > n) =$

(a) $P(X = n)$ (b) $P(X = n + k)$ (c) $P(X = k)$ (d) $P(X = n - k)$

33. The life time of a system in weeks has the pdf

$$f_X(x) = \begin{cases} 0.25 e^{-0.25x} & x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

The probability that the system will not fail within two weeks is

- (a) 0.5025 (b) 0.6065 (c) 0.4035 (d) 0.3935

34. An one ohm resistor with 10% tolerance has pdf

$$f_R(r) = \begin{cases} 5 & 0.9 < r < 1.1 \\ 0 & \text{otherwise} \end{cases}$$

$P(0.95 < R < 1.05) =$

- (a) 0.25 (b) 0.5 (c) 0.75 (d) 0.4

Answers to Multiple-Choice Questions

- | | | | | | | | | | |
|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| 1. (b) | 2. (b) | 3. (b) | 4. (d) | 5. (a) | 6. (c) | 7. (b) | 8. (c) | 9. (a) | 10. (d) |
| 11. (b) | 12. (b) | 13. (b) | 14. (c) | 15. (b) | 16. (c) | 17. (b) | 18. (c) | 19. (a) | 20. (d) |
| 21. (a) | 22. (b) | 23. (c) | 24. (b) | 25. (d) | 26. (b) | 27. (a) | 28. (c) | 29. (b) | 30. (a) |
| 31. (b) | 32. (c) | 33. (b) | 34. (b) | | | | | | |

3

OPERATIONS ON ONE RANDOM VARIABLE



INTRODUCTION 3.1

In Chapter 2, we studied about the concept of random variable and its characterization using distribution and density functions. Also we studied about different types of distributions and discussed the way some of the real, physical-world random phenomenon can be modeled using the above distributions. In this chapter, we will study one of the operations that can be performed on a random variable. This operation known as *expectation* is used to find the characteristics of a random variable.



EXPECTATION 3.2

Consider a variable X that takes values $x_1, x_2, \dots, x_p$ with frequencies $n_1, n_2, \dots, n_p$, then the weighted average of X is given by

$$\bar{X} = (n_1x_1 + n_2x_2 + \dots + n_px_p)/(n_1 + n_2 + \dots + n_p) \quad (3.1)$$

$$\begin{aligned} &= \frac{n_1x_1 + n_2x_2 + \dots + n_px_p}{N} \\ &= \left(\frac{n_1}{N}\right)x_1 + \left(\frac{n_2}{N}\right)x_2 + \dots + \left(\frac{n_p}{N}\right)x_p \\ &= \sum_i \left(\frac{n_i}{N}\right)x_i \end{aligned} \quad (3.2)$$

If X is random then the term $\left(\frac{n_i}{N}\right)$ in the above equation is equal to the probability of X taking the value x_i , which is denoted as $P(X = x_i)$ or $p_X(x_i)$ and $\bar{X}$ is known as the *statistical average* of X . Now we can write

$$\bar{X} = \sum_i p_X(x_i)x_i \quad (3.3)$$

The term $\bar{X}$ is a number used to find the center of the distribution of a random variable. The term *expectation* is used for the process of averaging a random variable.

Therefore, we can write

$$\bar{X} = E[X] = \sum_i p_X(x_i) x_i \quad (3.4)$$

The expected value of X is the weighted average of possible values that X can take where each weight is equal to the probability that X takes that value. The expected value of X is also denoted by μ_X .

Solved Problems



3.1 If X is the outcome when we roll a fair dice, find the expectation of X .

Solution For a fair dice, the sample space is $S = \{1, 2, \dots, 6\}$ each outcome with equal probability $\frac{1}{6}$.

Therefore,

$$\begin{aligned} E[X] &= \mu_X = \sum_i x_i p_X(x_i) \\ &= 1\left(\frac{1}{6}\right) + 2\left(\frac{1}{6}\right) + 3\left(\frac{1}{6}\right) + 4\left(\frac{1}{6}\right) + 5\left(\frac{1}{6}\right) + 6\left(\frac{1}{6}\right) \\ &= \frac{7}{2} \end{aligned}$$

3.2 A random variable X takes two values 0 and 1 with equal probability. Find $E[X]$.

Solution

$$E[X] = \sum_i x_i p_X(x_i)$$

$$\text{Given, } p_X(0) = \frac{1}{2}; p_X(1) = \frac{1}{2}$$

$$\Rightarrow E[X] = 0\left(\frac{1}{2}\right) + 1\left(\frac{1}{2}\right) = \frac{1}{2}$$

3.3 Calculate the expectation of a geometric random variable with PMF $P[X = n] = p(1 - p)^{n-1}$, $n \geq 1$.

Solution

$$\begin{aligned} E[X] &= \sum_n x_n p_X(x_n) \\ &= \sum_n x_n P(X = n) \\ &= \sum_n n p(1 - p)^{n-1} \\ &= p \sum_n n(1 - p)^{n-1} = p \sum_{n=1}^{\infty} n q^{n-1} \text{ where } q = 1 - p \\ &= p \sum_{n=1}^{\infty} \frac{d}{dq} (q^n) \end{aligned}$$

$$\begin{aligned}
&= p \frac{d}{dq} \left[\sum_{n=1}^{\infty} q^n \right] = p \frac{d}{dq} \left[\frac{q}{1-q} \right] \\
&= p \left(\frac{1}{1-q} \right)^2 = \frac{p}{(1-q)^2} = \frac{p}{p^2} = \frac{1}{p} \\
E[X] &= \frac{1}{p}
\end{aligned}$$



EXPECTATION OF A CONTINUOUS RANDOM VARIABLE 3.3

If X is a continuous random variable with probability density function of $f_X(x)$ then the expectation of the random variable is given by

$$E[X] = \mu_X = \int_{-\infty}^{\infty} x f_X(x) dx \quad (3.5)$$

Solved Problem



3.4 If X is a random variable with

$$\begin{aligned}
f_X(x) &= 2x \text{ for } 0 \leq x \leq 1 \\
&= 0 \text{ otherwise}
\end{aligned}$$

find $E[X]$.

Solution

$$E[X] = \int_{-\infty}^{\infty} x f_X(x) dx$$

Given, $f_X(x) = 2x \text{ for } 0 \leq x \leq 1$
 $= 0 \text{ otherwise}$

$$\begin{aligned}
E[X] &= \int_0^1 x(2x) dx \\
&= 2 \int_0^1 x^2 dx = \frac{2}{3} x^3 \Big|_0^1 = \frac{2}{3}
\end{aligned}$$



EXPECTED VALUE OF A FUNCTION OF A RANDOM VARIABLE 3.4

Given a random variable X with probability density function $f_X(x)$ then the expected value of a real function $g(X)$ of that random variable is given by

$$E[g(X)] = \int_{-\infty}^{\infty} g(X) f_X(x) dx \quad (3.6)$$

If X is a discrete random variable then

$$E[g(x)] = \sum_{i=1}^{\infty} g(x_i) p_X(x_i) \quad (3.7)$$

Solved Problem

3.5 In an experiment the input X and output Y are given by the relation

$$Y = g(X) = X^3/2$$

If X is a random variable with probability density function

$$f_X(x) = \begin{cases} \frac{1}{4}; & 2 \leq x \leq 6 \\ 0, & \text{otherwise} \end{cases}$$

then find the expected value of Y .

Solution

$$\begin{aligned} E[Y] &= E[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x) dx \\ &= \int_{-\infty}^{\infty} \frac{x^3}{2} f_X(x) dx = \int_2^6 \frac{x^3}{2} \left(\frac{1}{4} \right) dx \\ &= \frac{1}{8} \left. \frac{x^4}{4} \right|_2^6 = \frac{6^4 - 2^4}{32} = 40 \end{aligned}$$

3.6 In the circuit shown in Fig. 3.1, R_1 is a random variable uniformly distributed on $R_0 - \Delta R$ and $R_0 + \Delta R$.

(a) Find an expression for the power distributed in R_2 for any constant voltage V .

(b) Find the mean value of power when R_1 is random. (c) Find the mean power if $V = 10$ volts, $R_2 = 100 \Omega$, $R_0 = 150 \Omega$ and $\Delta R = 10 \Omega$.

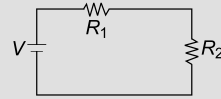


Fig. 3.1 Circuit for Solved Problem 3.6

Solution

$$\text{Power } P_{R_2} = \left(\frac{V}{R_1 + R_2} \right)^2 R_2$$

The pdf of R_1 is

$$f_{R_1}(R_1) = \begin{cases} \frac{1}{2\Delta R} & \text{for } R_0 - \Delta R \leq R_1 \leq R_0 + \Delta R \\ 0 & \text{otherwise} \end{cases}$$

$$\begin{aligned}
\bar{P} &= \int_{-\infty}^{\infty} P_{R_2} f_{R_1}(R_1) dR_1 \\
&= \int_{R_0 - \Delta R}^{R_0 + \Delta R} \frac{1}{2\Delta R} \left(\frac{V}{R_1 + R_2} \right)^2 R_2 dR_1 \\
&= \frac{V^2 R_2}{2\Delta R} \left(\frac{-1}{R_1 + R_2} \right) \Bigg|_{R_0 - \Delta R}^{R_0 + \Delta R} \\
&= -\frac{V^2 R_2}{2\Delta R} \left[\frac{1}{R_0 + \Delta R + R_2} - \frac{1}{R_0 - \Delta R + R_2} \right] \\
&= -\frac{V^2 R_2}{2\Delta R} \left(\frac{-2\Delta R}{(R_0 + R_2)^2 - (\Delta R)^2} \right) \\
&= \frac{V^2 R_2}{[(R_0 + R_2)^2 - (\Delta R)^2]}
\end{aligned}$$

Given: $V = 10$ volts
 $R_0 = 150 \Omega$
 $R_2 = 100 \Omega$
and $\Delta R = 10 \Omega$

$$\bar{P} = \frac{(10)^2(100)}{(250)^2 - 10^2} = 0.16$$



Practice Problems

3.1 Find mean of the discrete random variable X with the following pmf.

$$P_X(x) = \begin{cases} \frac{1}{3}; & x = 3 \\ \frac{2}{3}; & x = 6 \end{cases} \quad (\text{Ans: } 5)$$

3.2 If X is a random variable with the following distribution function

$$F_X(x) = \begin{cases} 0 & x \leq 1 \\ 0.3 & 1 \leq x \leq 2 \\ 0.5 & 2 \leq x \leq 3 \\ 0.8 & 3 \leq x \leq 4 \\ 1 & x \geq 4 \end{cases}$$

- (a) What is the expected value of X ? (Ans: (a) 2.4; (b) $0.3 \delta(x-1) + 0.2 \delta(x-2) + 0.3 \delta(x-3) + 0.2 \delta(x-4)$)
(b) What is the pmf of x ?

3.3 If the random variable X has probability density function

$$f_X(x) = ax^4 \quad 0 \leq x \leq 1$$

(a) What is the value of a ?

(b) What is the expected value of X ?

$$\left(\text{Ans: (a) } 5 \text{ (b) } \frac{5}{6} \right)$$

3.4 A random variable X is given by

X	-2	-1	0	1	2
$p_X(x)$	$\frac{1}{10}$	$\frac{1}{5}$	$\frac{1}{5}$	$\frac{3}{10}$	$\frac{1}{5}$

Find $E[X]$.

$$\left(\text{Ans: } \frac{3}{5} \right)$$

Solved Problems



3.7 Let X be a continuous random variable with

$$\text{pdf } f_X(x) = \frac{8}{x^3}, \quad x > 2. \text{ Find } E[W] \text{ where } W = \frac{X}{3}$$

Solution

$$\text{Given: } W = g(X) = \frac{X}{3}$$

$$f_X(x) = \frac{8}{x^3}; \quad x > 2$$

$$E[W] = E[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x) dx$$

$$= \int_2^{\infty} \frac{x}{3} \left(\frac{8}{x^3} \right) dx = \frac{8}{3} \int_2^{\infty} x^{-2} dx = \frac{-8}{3} \left[\frac{1}{x} \right]_2^{\infty} = \frac{4}{3}$$

$$E[W] = \frac{4}{3}$$

3.8 If X has the probability density function

$$f_X(x) = e^{-x}, \quad x > 0$$

$$= 0 \quad \text{otherwise}$$

find the expected value of $g(X) = e^{3X/4}$

Solution

$$\text{Given } g(x) = e^{3X/4}$$

$$f_X(x) = e^{-x}, \quad x > 0$$

$$= 0; \text{ otherwise}$$

$$\begin{aligned}
 E[g(X)] &= \int_{-\infty}^{\infty} g(x)f_X(x)dx = \int_0^{\infty} e^{-x} e^{3x/4} dx = \int_0^{\infty} e^{-(x/4)} dx \\
 &= \frac{e^{-x/4}}{(-1/4)} \Big|_0^{\infty} = -4(-1) = 4
 \end{aligned}$$

3.9 Find the expected value of the function $g(X) = X^2$, where x is a random variable defined by the density $f_X(x) = ae^{-ax}u(x)$ where 'a' is constant.

Solution

Given: $g(X) = X^2$

and $f_X(x) = ae^{-ax}u(x)$

$$\begin{aligned}
 E[g(X)] &= \int_{-\infty}^{\infty} g(X)f_X(x)dx \\
 &= \int_0^{\infty} x^2 ae^{-ax} dx & \boxed{u(x) = 1 \text{ for } x \geq 0,} \\
 &= a \int_0^{\infty} x^2 e^{-ax} dx & \boxed{= 0 \text{ for } x < 0} \\
 &= a \left[\left(\frac{-1}{a} \right) x^2 e^{-ax} \Big|_0^{\infty} - \left(\frac{2}{a^2} \right) x e^{-ax} \Big|_0^{\infty} - \left(\frac{2}{a^3} \right) e^{-ax} \Big|_0^{\infty} \right] \\
 &= \frac{2}{a^2} e^{-ax} \Big|_0^{\infty} = \frac{2}{a^2} & \boxed{\lim_{x \rightarrow \infty} x^2 e^{-ax} = 0} \\
 & & \boxed{\lim_{x \rightarrow \infty} x e^{-ax} = 0} \\
 E[g(X)] &= \frac{2}{a^2}
 \end{aligned}$$

3.10 The nickel contained in an alloy say X , may be considered as a random variable with the following pdf

$$f_X(x) = \frac{x^2}{100} \quad 0 \leq x \leq 5. \text{ Find } E[X].$$

Solution

$$E[X] = \int_{-\infty}^{\infty} x f_X(x) dx = \int_0^5 x \left(\frac{x^2}{100} \right) dx$$

$$\begin{aligned}
 &= \frac{1}{100} \int_0^5 x^3 dx = \frac{1}{100} \left. \frac{x^4}{4} \right|_0^5 = \frac{1}{100} \left(\frac{5^4}{4} \right) \\
 &= \frac{25}{16}
 \end{aligned}$$

3.11 The pressure P on the surface of an airplane is given by the relationship $P = 0.0025V^2$. The velocity V is a uniformly distributed random variable over the interval $(0, 5)$. Find the expected value of P .

Solution

$$\begin{aligned}
 \text{Given: } P &= 0.0025 V^2 \text{ and } f_V(v) = \frac{1}{5}; & 0 \leq v \leq 5 \\
 &= 0; & \text{otherwise}
 \end{aligned}$$

$$\begin{aligned}
 E[P] &= \int_{-\infty}^{\infty} p f_V(v) dv = \int_0^5 0.0025 V^2 \left(\frac{1}{5} \right) dv \\
 &= \frac{1}{5} 0.0025 \left. \frac{v^3}{3} \right|_0^5 = \frac{0.0025}{5} \left(\frac{125}{3} \right) \\
 &= 0.0208
 \end{aligned}$$

REVIEW QUESTIONS



1. Explain in detail about expectation. How is the expected value calculated for discrete random variable?
2. Give the importance of expected value. How is it calculated for a continuous random variable?



Practice Problems

3.5 The density function of a continuous random variable X is given by

$$\begin{aligned}
 f_X(x) &= \frac{2}{9}x; & 0 \leq x \leq 3 \\
 &= 0; & \text{otherwise}
 \end{aligned}
 \quad (\text{Ans: } 2)$$

Find the expectation of X .

3.6 X is a random variable with density function

$$\begin{aligned}
 f_X(x) &= A e^{-x} & 0 \leq x \leq \infty \\
 &= 0 & \text{otherwise}
 \end{aligned}
 \quad (\text{Ans: (a) } A = 1; \text{ (b) } 1)$$

- (a) Find the value of A , (b) mean of X .



CONDITIONAL EXPECTED VALUE 3.5

The conditional expected value of X , given that an event A has occurred, is given by

$$E[X|A] = \int_{-\infty}^{\infty} x f_X(x|A) dx \quad (3.8)$$

where $f_X(x|A)$ is the conditional density.

If $p_X(A)$ is the probability that the event A occurs then the conditional density is given by

$$f_X(x|A) = \frac{f_X(x)}{p_X(A)} \quad (3.9)$$

Let A be an event that depends on X , defined by $A = \{x \leq a\}$, $-\infty < a < \infty$, then $E[X|A]$ is equal to

$$E[X|x \leq a] = \frac{\int_{-\infty}^a x f_X(x) dx}{F_X(a)} \quad (3.10)$$

where $F_X(a) = \int_{-\infty}^a f_X(x) dx$

and conditional pdf

$$f_X(x|x \leq a) = \begin{cases} \frac{f_X(x)}{p_X(x \leq a)} = \frac{f_X(x)}{F_X(a)}; & \text{for } x \leq a \\ 0 & \text{for } x \geq a \end{cases} \quad (3.11)$$

Solved Problems



3.12 X is a random variable with pdf

$$f_X(x) = \begin{cases} \frac{1}{40} & 10 \leq x \leq 50 \\ 0 & \text{otherwise} \end{cases}$$

Find the conditional probability for the event $A = \{X \leq 40\}$

Solution

Given: $f_X(x) = \begin{cases} \frac{1}{40} & 10 \leq x \leq 50 \\ 0 & \text{otherwise} \end{cases}$

The conditional pdf

$$f_X(x|x \leq 40) = \begin{cases} \frac{f_X(x)}{p_X(X \leq 40)} & x \leq 40 \\ 0 & \text{otherwise} \end{cases}$$

$$p_X[x \leq 40] = F_X(40) = \int_{-\infty}^{40} f_X(x) dx = \int_{10}^{40} \frac{1}{40} dx$$

$$= \frac{1}{40}(30) = \frac{3}{4}$$

$$f_X(x|x \leq 40) = \frac{4}{3} f_X(x) \quad x \leq 40$$

$$= 0 \quad \text{otherwise}$$

$$E[X|x \leq 40] = \frac{\int_{-\infty}^{40} x f_X(x) dx}{F_X(40)} = \frac{4}{3} \int_{10}^{40} \left(\frac{x}{40} \right) dx$$

$$= \frac{1}{30} \left. \frac{x^2}{2} \right|_{10}^{40} = \frac{1500}{30(2)} = \frac{50}{2} = 25$$

3.13 Consider an exponential random variable with

$$f_X(x) = \begin{cases} \frac{1}{b} e^{-x/b} & x \geq 0 \\ 0 & x < 0 \end{cases}$$

Suppose the event is the random variable X taking a value $A = \{X \geq a\}$. Find the conditional expectation.

Solution The conditional pdf is

$$f_X(x|x \geq a) = \begin{cases} \frac{f_X(x)}{P(x \geq a)}; & x \geq a \\ 0; & \text{otherwise} \end{cases}$$

$$P[X \geq a] = \frac{1}{b} \int_a^{\infty} e^{-x/b} dx$$

$$= \frac{1}{b} \left. \frac{e^{-x/b}}{\left(-\frac{1}{b}\right)} \right|_a^{\infty} = e^{-a/b}$$

$$f_X(x|x \geq a) = \frac{\frac{1}{b} e^{-x/b}}{e^{-a/b}} = \frac{e^{a/b}}{b} e^{-x/b} \text{ for } x \geq a$$

$$= 0; \quad \text{otherwise}$$

$$\begin{aligned}
 E[X|x \geq a] &= \int_{-\infty}^{\infty} x f_X(x | x \geq a) dx \\
 &= \frac{1}{b} e^{a/b} \int_a^{\infty} x e^{-x/b} dx \\
 &= \frac{1}{b} e^{a/b} \left[-bx e^{-x/b} \Big|_a^{\infty} - b^2 e^{-x/b} \Big|_a^{\infty} \right] \\
 &= \frac{1}{b} e^{a/b} \left[abe^{-a/b} + b^2 e^{-a/b} \right] \\
 &= a + b
 \end{aligned}$$

or

$$\begin{aligned}
 E[X|x > a] &= \frac{\int_a^{\infty} x f_X(x) dx}{P[x \geq a]} \\
 &= \frac{\int_a^{\infty} x \frac{1}{b} e^{-x/b} dx}{e^{-a/b}} \\
 &= e^{a/b} \frac{1}{b} \int_a^{\infty} x e^{-x/b} dx \\
 &= \frac{1}{b} e^{a/b} \left[-bx e^{-x/b} \Big|_a^{\infty} - b^2 e^{-x/b} \Big|_a^{\infty} \right] \\
 &= \frac{1}{b} e^{a/b} \left[abe^{-a/b} + b^2 e^{-a/b} \right] \\
 &= a + b
 \end{aligned}$$



Practice Problems

3.7 The pdf of a random variable X is given by

$$f_X(x) = \begin{cases} 0.1 & 20 \leq x \leq 30 \\ 0 & \text{otherwise} \end{cases} \quad (\text{Ans: } 22.5)$$

Find the conditional expected value of X , given that $X \leq 25$.

3.8 The pdf of a continuous random variable X is given by

$$f_X(x) = 2e^{-2x}, \quad x \geq 0 \quad (\text{Ans: } 0.4627)$$

Find the conditional expected value of X , given that $X \leq 2$.

3.9 If the pdf of a random variable X is given by $f_X(x) = \frac{2}{9}x$, $0 \leq x \leq 3$, find the conditional expected value of X , given that $X \leq 2$.

$$\left(\text{Ans: } \frac{4}{3} \right)$$

REVIEW QUESTIONS

3. Explain conditional expected value of X , given that an event A has occurred.

**PROPERTIES OF EXPECTATION 3.6****1. Expectation of a constant is equal to the constant itself.**

Proof: Let $X = a$

$$E[X] = E[a] = \int_{-\infty}^{\infty} a f_X(x) dx = a \int_{-\infty}^{\infty} f_X(x) dx \quad \text{Since } \int_{-\infty}^{\infty} f_X(x) dx = 1$$

$$= a$$

2. The expectation of constant times the random variable is equal to the constant times expectation of the random variable.

That is, $E[CX] = CE[X]$

where C is a constant.

Proof:

$$E[CX] = \int_{-\infty}^{\infty} CX f_X(x) dx = C \int_{-\infty}^{\infty} X f_X(x) dx \quad (3.12)$$

$$= CE[X] \quad (3.13)$$

3. If a and b are constants then

$$E[aX + b] = aE[X] + b$$

Proof:

$$\begin{aligned} E[aX + b] &= \int_{-\infty}^{\infty} (ax + b) f_X(x) dx \\ &= \int_{-\infty}^{\infty} ax f_X(x) dx + \int_{-\infty}^{\infty} b f_X(x) dx \\ &= a \int_{-\infty}^{\infty} x f_X(x) dx + b \int_{-\infty}^{\infty} f_X(x) dx \\ &= aE[X] + b \end{aligned}$$

4. If $X \geq 0$ then $E[X] \geq 0$

Proof:

$$E[X] = \int_{-\infty}^{\infty} x f_X(x) dx$$

$$= \int_{-\infty}^0 x f_X(x) dx + \int_0^{\infty} x f_X(x) dx$$

Since $x \geq 0$

$$E[X] = \int_0^{\infty} x f_X(x) dx$$

Since $x \geq 0$ and $f_X(x) \geq 0$, we can write

$$E[X] = \int_0^{\infty} x f_X(x) dx \geq 0$$

$$E[X] \geq 0$$

5. Expectation of the sum of random variables is equal to the sum of expectations.

Proof: If $g_1(x)$ and $g_2(x)$ are two different functions of a random variable X , then

$$E[g_1(x) + g_2(x)] = E[g_1(x)] + E[g_2(x)]$$

$$E[g(x)] = \int_{-\infty}^{\infty} g(x) f_X(x) dx \quad (3.14)$$

$$\begin{aligned} E[g_1(x) + g_2(x)] &= \int_{-\infty}^{\infty} [g_1(x) + g_2(x)] f_X(x) dx \\ &= \int_{-\infty}^{\infty} g_1(x) f_X(x) dx + \int_{-\infty}^{\infty} g_2(x) f_X(x) dx \\ &= E[g_1(x)] + E[g_2(x)] \end{aligned}$$

REVIEW QUESTIONS

4. Explain the properties of expectations
5. Prove that (a) $E[aX + b] = aE[X] + b$
 (b) $E[CX] = CE[X]$
 (c) $E[g_1(x) + g_2(x)] = E[g_1(x)] + E[g_2(x)]$

Solved Problems

3.14 Let X be a random variable defined by the density function

$$f_X(x) = \begin{cases} \left(\frac{\pi}{16}\right) \cos\left(\frac{\pi x}{8}\right) & -4 \leq x \leq 4 \\ 0 & \text{elsewhere} \end{cases}$$

Find $E[3X]$ and $E[X^2]$.

Solution

Given:
$$f_X(x) = \begin{cases} \frac{\pi}{16} \cos \frac{\pi x}{8} & \text{for } -4 \leq x \leq 4 \\ 0 & \text{elsewhere} \end{cases}$$

(a) Using Property 2,

we know that $E[3X] = 3E[X]$

$$\begin{aligned} E[X] &= \int_{-\infty}^{\infty} x f_X(x) dx = \int_{-4}^4 x \frac{\pi}{16} \cos \left(\frac{\pi x}{8} \right) dx \\ &= \frac{\pi}{16} \int_{-4}^4 x \cos \left(\frac{\pi x}{8} \right) dx \\ &= \frac{\pi}{16} \left[\frac{8x}{\pi} \sin \frac{\pi x}{8} \right]_{-4}^4 - \frac{8}{\pi} \int_{-4}^4 \sin \frac{\pi x}{8} dx \\ &= \left[\frac{1}{2} x \sin \frac{\pi x}{8} \right]_{-4}^4 + \frac{4}{\pi} \cos \frac{\pi x}{8} \Big|_{-4}^4 \\ &= 2 \sin \frac{\pi}{2} - (-2) \sin \left(-\frac{\pi}{2} \right) + \frac{4}{\pi} \cos \frac{\pi}{2} - \frac{4}{\pi} \cos \left(-\frac{\pi}{2} \right) \\ &= 0 \quad \left[\because \cos \frac{\pi}{2} = 0 \right] \end{aligned}$$

$$E[3X] = 3E[X] = 0$$

(b)
$$E[X^2] = \int_{-\infty}^{\infty} x^2 f_X(x) dx$$

$$\begin{aligned} &= \int_{-4}^4 x^2 \frac{\pi}{16} \cos \left(\frac{\pi x}{8} \right) dx = \frac{\pi}{16} \int_{-4}^4 x^2 \cos \left(\frac{\pi x}{8} \right) dx \\ &= \frac{\pi}{16} \left[\frac{8}{\pi} x^2 \sin \frac{\pi x}{8} \right]_{-4}^4 + 2x \left(\frac{8}{\pi} \right)^2 \cos \frac{\pi x}{8} \Big|_{-4}^4 - 2 \left(\frac{8}{\pi} \right)^2 \int_{-4}^4 \cos \frac{\pi x}{8} dx \\ &= \frac{1}{2} x^2 \sin \frac{\pi x}{8} \Big|_{-4}^4 + \frac{8x}{\pi} \cos \frac{\pi x}{8} \Big|_{-4}^4 - \frac{64}{\pi^2} \sin \frac{\pi x}{8} \Big|_{-4}^4 \\ &= \frac{1}{2} \left[16 \sin \frac{\pi}{2} - 16 \sin \left(-\frac{\pi}{2} \right) \right] + \frac{8}{\pi} \left[4 \cos \frac{\pi}{2} - 4 \cos \left(-\frac{\pi}{2} \right) \right] - \frac{64}{\pi^2} \left[\sin \frac{\pi}{2} - \sin \left(-\frac{\pi}{2} \right) \right] \end{aligned}$$

$$= \frac{1}{2}(32) - \frac{64}{\pi^2}(2) = 16 - \frac{128}{\pi^2}$$

3.15 The density function of a random variable X is $f_X(x) = \begin{cases} 5e^{-5x} & 0 \leq x \leq \infty \\ 0 & \text{elsewhere} \end{cases}$

Find (a) $E[X]$, (b) $E[(X-1)^2]$, and (c) $E[3X-1]$.

Solution

$$\begin{aligned} \text{(a)} \quad E[X] &= \int_{-\infty}^{\infty} x f_X(x) dx = \int_0^{\infty} x [5e^{-5x}] dx = 5 \int_0^{\infty} x e^{-5x} dx \\ &= 5 \left[-\frac{x}{5} e^{-5x} \Big|_0^{\infty} + \int_0^{\infty} \frac{e^{-5x}}{5} dx \right] = -x e^{-5x} \Big|_0^{\infty} - \frac{1}{5} e^{-5x} \Big|_0^{\infty} \\ &= (0-0) - \frac{1}{5}[0-1] = \frac{1}{5} \end{aligned}$$

$$\lim_{x \rightarrow \infty} x e^{-5x} = 0$$

$$\Rightarrow E[X] = \frac{1}{5}$$

$$\begin{aligned} E[X^2] &= \int_{-\infty}^{\infty} x^2 f_X(x) dx = \int_0^{\infty} x^2 (5e^{-5x}) dx = 5 \int_0^{\infty} x^2 e^{-5x} dx \\ &= 5 \left[\frac{-x^2 e^{-5x}}{5} \Big|_0^{\infty} - \frac{2x}{25} e^{-5x} \Big|_0^{\infty} + \frac{2}{25} \int_0^{\infty} e^{-5x} dx \right] \\ &= 5 \left[(0-0) - (0-0) + \frac{2}{25} \left(-\frac{1}{25} \right) e^{-5x} \Big|_0^{\infty} \right] \\ &= 5 \left[-\frac{2}{125} (0-1) \right] = \frac{2}{25} \end{aligned}$$

$$\lim_{x \rightarrow \infty} x^2 e^{-5x} = 0$$

$$E[X^2] = \frac{2}{25}$$

$$\begin{aligned} \text{(b)} \quad E[(X-1)^2] &= E[X^2 - 2X + 1] \\ &= E[X^2] - E[2X] + E[1] \\ &= E[X^2] - 2E[X] + 1 \\ &= \frac{2}{25} - \frac{2}{5} + 1 = \frac{17}{25} \end{aligned}$$

$$\Rightarrow E[(X-1)^2] = \frac{17}{25}$$

$$\text{(c)} \quad E[3X-1] = 3E[X] - 1$$

$$= 3 \left[\frac{1}{5} \right] - 1 = -\frac{2}{5}$$



Practice Problems

3.10 If X is a random variable with $E[X] = 2$, find

- (a) $E[2X - 4]$
 (b) $E[3X]$

(Ans: (a) 0 (b) 6)

3.11 A random variable X has the following pdf:

$$f_X(x) = \begin{cases} x & 0 \leq x \leq 1 \\ \frac{3}{4} - \frac{x}{4} & 1 \leq x \leq 3 \\ 0 & \text{elsewhere} \end{cases}$$

Find (a) $E[X]$ (b) $E[X^2]$ (c) $E[X - 2]^2$

(Ans: (a) $\frac{7}{6}$ (b) $\frac{7}{4}$ (c) $\frac{13}{12}$)



MOMENTS 3.7

Moments of a random variable X are of two types:

- (i) Moments about origin
 (ii) Central moments

3.7.1 Moments about Origin

Let X is a random variable with pdf $f_X(x)$.

Then the n^{th} order moment about the origin is given by

$$m_n = E[X^n] = \int_{-\infty}^{\infty} x^n f_X(x) dx \quad (3.15)$$

If $n = 0$, we get the area of the function $f_X(x)$ which is equal to 1. While $n = 1$ is equal to $E[X]$. The second moment about origin is known as *mean square* of X (mean square value) given by

$$m_2 = E[X^2] = \int_{-\infty}^{\infty} x^2 f_X(x) dx \quad (3.16)$$

If X is a discrete random variable then n^{th} order moment about origin is given by

$$E[X^n] = \sum_i x_i^n p_X(x_i) \quad (3.17)$$

3.7.2 Central Moments

In central moments, the mean is subtracted from the variable before the moment is taken in order to remove bias in the higher moments due to the mean.

For a random variable X with pdf $f_X(x)$, the n^{th} order central moment is given by

$$\mu_n = E[(X - \mu_X)^n] = \int_{-\infty}^{\infty} (x - \mu_X)^n f_X(x) dx \quad (3.18)$$

For discrete random variable the n^{th} order central moment is given by

$$\mu_n = E[(X - \mu_X)^n] = \sum_i (x_i - \mu_X)^n p_X(x_i) \quad (3.19)$$

For a random variable the first-order central moment is zero. That is

$$\begin{aligned} \mu_1 &= E[(X - \mu_X)] \\ &= E[X] - E[\mu_X] = \mu_X - \mu_X = 0 \end{aligned} \quad (3.20)$$

$$\Rightarrow \boxed{\mu_1 = 0}$$

Therefore, the lowest central moment or any real interest is the second central moment

3.7.3 Variance of a Random Variable

The central moment for the case of $n = 2$ is very important and is known as variance, which is denoted by σ_X^2 . Thus, the variance is given by

$$\sigma_X^2 = E[(X - \mu_X)^2] = \int_{-\infty}^{\infty} (x - \mu_X)^2 f_X(x) dx \quad (3.21)$$

The positive square root σ_X of variance is called *standard deviation*. It is a measure of the spread in a pdf or pmf. If a random variable has a concentrated pdf or pmf, it will have a small variance. If it has a widely spread pdf or pmf, it will have a large variance.

Consider three random variables X_1 , X_2 and X_3 with same mean and different variances. From Fig. 3.2 we find that X_1 has smallest spread about the mean whereas X_3 has largest spread about the mean. In other words, most of the values of X_1 are close to the mean value, whereas the values of X_3 that are close to the mean are very less. In terms of variance, we say that X_1 has smallest variance while X_3 has largest variance.

From Eq. (3.21), we have

$$\begin{aligned} \sigma_X^2 &= E[(X - \mu_X)^2] = E[X^2 - 2X\mu_X + \mu_X^2] \\ &= E[X^2] - 2\mu_X E[X] + E[\mu_X^2] \\ &= E[X^2] - 2\mu_X(\mu_X) + \mu_X^2 \\ &= E[X^2] - \mu_X^2 \\ &= m_2 - m_1^2 \end{aligned} \quad (3.22)$$

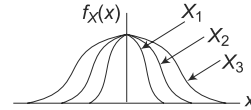


Fig. 3.2 pdf of three random variables

3.7.4 Properties of Variance

1. The variance of a constant is zero.

Proof: Consider a random variable X . The variance of X is defined as

$$\text{Var}(X) = E[(X - \mu_X)^2] \quad (3.23)$$

If X is a constant value ' a ' then the mean is given by $E[X] = \mu_X = a$

$$\text{Var}(a) = E[(X - a)^2] = E[(a - a)^2] = 0 \quad (3.24)$$

2. Variance of a nonrandom constant time random variable plus a constant is equal to the first constant square times the variance of the random variable.

$$\text{Var}(CX) = C^2 \text{Var}(X) \quad (3.25)$$

Proof: Let $Y = CX$.

If μ_X is the mean of random variable X , then the mean of Y is equal to

$$E[Y] = E[CX] = CE[X] = C\mu_X$$

$$\begin{aligned}\text{Var}(Y) &= E[(Y - \mu_Y)^2] \\ &= E[(CX - C\mu_X)^2] \\ &= E[C^2(X - \mu_X)^2] = C^2 E[(X - \mu_X)^2] \\ &= C^2 \text{Var}(X)\end{aligned}$$

$$\Rightarrow \text{Var}(CX) = C^2 \text{Var}(X)$$

3. If X is a random variable and a, b are real constant, then

$$\text{Var}(aX + b) = a^2 \text{Var}(X) \quad (3.26)$$

Proof:

$$\text{Var}(aX + b) = E[\{(aX + b) - \mu_{aX+b}\}^2]$$

$$\mu_{aX+b} = E[aX + b] = a E[X] + b = a\mu_X + b$$

$$\begin{aligned}\Rightarrow \text{Var}(aX + b) &= E[\{aX + b - (a\mu_X + b)\}^2] \\ &= E[a(X - \mu_X)^2] \\ &= a^2 E[(X - \mu_X)^2] = a^2 \text{Var}(X)\end{aligned}$$

REVIEW QUESTIONS



6. If X is random variable, show that $\text{Var}(aX + b) = a^2 \text{Var}(X)$.
7. Define variance and explain its properties.

Solved Problems



3.16 A random variable X denotes the outcome of throwing a fair dice. Find the mean and variance.

Solution When we throw a dice, the sample space is $S = \{1, 2, 3, 4, 5, 6\}$. Let X denotes the outcome. Then the probability of each outcome is $\frac{1}{6}$

$$\begin{aligned}\text{The mean } \mu_X &= E[X] = \sum_i x_i p_X(x_i) \\ &= 1\left(\frac{1}{6}\right) + 2\left(\frac{1}{6}\right) + 3\left(\frac{1}{6}\right) + 4\left(\frac{1}{6}\right) + 5\left(\frac{1}{6}\right) \\ &= \frac{1}{6}[1 + 2 + 3 + 4 + 5 + 6] = \frac{21}{6} = \frac{7}{2} = 3.5 \\ E[X^2] &= \sum_i x_i^2 p_X(x_i)\end{aligned}$$

$$\begin{aligned}
&= 1^2 \left(\frac{1}{6}\right) + 2^2 \left(\frac{1}{6}\right) + 3^2 \left(\frac{1}{6}\right) + 4^2 \left(\frac{1}{6}\right) + 5^2 \left(\frac{1}{6}\right) + 6^2 \left(\frac{1}{6}\right) \\
&= \frac{1}{6} [1^2 + 2^2 + 3^2 + 4^2 + 5^2 + 6^2] \\
&= \frac{91}{6} \\
\sigma_X^2 &= E[X^2] - \{E[X]\}^2 = \frac{91}{6} - \left(\frac{7}{2}\right)^2 = \frac{35}{12}
\end{aligned}$$

3.17 The pmf of a discrete variable is given by

$$p_X(x) = \begin{cases} \frac{1}{4} & x = 0, 2, 4, 6 \\ 0 & \text{otherwise} \end{cases}$$

Find mean and variance.

Solution

$$\begin{aligned}
\mu_X &= E[X] = \sum_i x_i p_X(x_i) \\
&= 0 \left(\frac{1}{4}\right) + 2 \left(\frac{1}{4}\right) + 4 \left(\frac{1}{4}\right) + 6 \left(\frac{1}{4}\right) = 3 \\
E[X^2] &= \sum_i x_i^2 p_X(x_i) = (0^2) \frac{1}{4} + 2^2 \left(\frac{1}{4}\right) + 4^2 \left(\frac{1}{4}\right) + 6^2 \left(\frac{1}{4}\right) = 14 \\
\sigma_X^2 &= E[X^2] - \{E[X]\}^2 \\
&= 14 - (3)^2 = 5
\end{aligned}$$

3.18 When two unbiased dice are thrown simultaneously, find expected value of the sum of numbers shown on the dice.

Solution When two dice are thrown simultaneously, the sample space is

$S = \{(1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (1, 6), (2, 1), (2, 2), (2, 3), (2, 4), (2, 5), (2, 6), (3, 1), (3, 2), (3, 3), (3, 4), (3, 5), (3, 6), (4, 1), (4, 2), (4, 3), (4, 4), (4, 5), (4, 6), (5, 1), (5, 2), (5, 3), (5, 4), (5, 5), (5, 6), (6, 1), (6, 2), (6, 3), (6, 4), (6, 5), (6, 6)\}$

Let X be the random variable which denotes the sum of the numbers shown on the dice. Then we can write

$$\begin{aligned}
P[X=2] &= \frac{1}{36}; \quad P[X=3] = \frac{2}{36}; \quad P[X=4] = \frac{3}{36}; \quad P[X=5] = \frac{4}{36}; \quad P[X=6] = \frac{5}{36}; \\
P[X=7] &= \frac{6}{36}; \quad P[X=8] = \frac{5}{36}; \quad P[X=9] = \frac{4}{36}; \quad P[X=10] = \frac{3}{36}; \\
P[X=11] &= \frac{2}{36}; \quad P[X=12] = \frac{1}{36};
\end{aligned}$$

The above values are tabulated as below.

X	2	3	4	5	6	7	8	9	10	11	12
$p_X(x)$	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$

$$\begin{aligned}
 E[X] &= \sum_{i=2}^{12} x_i p_X(x_i) \\
 &= 2\left(\frac{1}{36}\right) + 3\left(\frac{2}{36}\right) + 4\left(\frac{3}{36}\right) + 5\left(\frac{4}{36}\right) + 6\left(\frac{5}{36}\right) + 7\left(\frac{6}{36}\right) \\
 &\quad + 8\left(\frac{5}{36}\right) + 9\left(\frac{4}{36}\right) + 10\left(\frac{3}{36}\right) + 11\left(\frac{2}{36}\right) + 12\left(\frac{1}{36}\right) \\
 &= \frac{2 + 6 + 12 + 20 + 30 + 42 + 40 + 36 + 30 + 22 + 12}{36} \\
 &= \frac{252}{36} = 7
 \end{aligned}$$

3.19 If X be a discrete random variable with probability mass function

X	-2	-1	0	1	2
$p_X(x)$	$\frac{2}{5}$	$\frac{1}{10}$	$\frac{1}{5}$	$\frac{1}{5}$	$\frac{1}{10}$

Find (i) $E[X]$, (ii) $E[2X + 5]$, (iii) $E[X^2]$, (iv) $E[(X + 2)^2]$.

Solution We know that

$$\begin{aligned}
 \text{(i)} \quad E[X] &= \sum x_i p_X(x_i) \\
 &= -2\left(\frac{2}{5}\right) + (-1)\left(\frac{1}{10}\right) + 0\left(\frac{1}{5}\right) + 1\left(\frac{1}{5}\right) + 2\left(\frac{1}{10}\right) \\
 &= -\frac{4}{5} - \frac{1}{10} + 0 + \frac{1}{5} + \frac{2}{10} = \frac{-8 - 1 + 0 + 2 + 2}{10} \\
 &= -0.5
 \end{aligned}$$

$$\text{(ii)} \quad E[2X + 5] = 2E[X] + 5 = 2[-0.5] + 5 = 4$$

$$\text{(iii)} \quad E[X^2] = \sum x_i^2 p_X(x_i)$$

$$\begin{aligned}
&= (-2)^2 \left(\frac{2}{5}\right) + (-1)^2 \left(\frac{1}{10}\right) + (0)^2 \left(\frac{1}{5}\right) + (1)^2 \left(\frac{1}{5}\right) + (2)^2 \left(\frac{1}{10}\right) \\
&= \frac{8}{5} + \frac{1}{10} + \frac{1}{5} + \frac{4}{10} = \frac{16+1+2+4}{10} = \frac{23}{10} = 2.3
\end{aligned}$$

$$\begin{aligned}
\text{(iv)} \quad E[(X+2)^2] &= E[X^2 + 4X + 4] \\
&= E[X^2] + 4E[X] + E[4] \\
&= 2.3 + 4(-0.5) + 4 \\
&= 4.3
\end{aligned}$$

3.20 The random variable X takes the values $x_n = n$; $n = 1, 2, \dots$ with probability $p_X(x_n) = \left(\frac{1}{2}\right)^n$. Find the mean value of X .

Solution

$$\begin{aligned}
E[X] &= \sum_i x_i p_X(x_i) \\
&= \sum_{n=1}^{\infty} x_n p_X(x_n) = \sum_{n=1}^{\infty} n p_X(x_n)
\end{aligned}$$

$$\text{Given: } p_X(x_n) = \left(\frac{1}{2}\right)^n$$

$$E[X] = \sum_{n=1}^{\infty} n \left(\frac{1}{2}\right)^n$$

$$\text{We know } \sum_{n=1}^{\infty} x^n = \frac{x}{1-x}$$

Differentiating on both sides, we get

$$\sum_{n=1}^{\infty} n x^{n-1} = \frac{1}{(1-x)^2}$$

$$\Rightarrow \sum_{n=1}^{\infty} n x^n = \frac{x}{(1-x)^2}$$

$$\Rightarrow \sum_{n=1}^{\infty} n \left(\frac{1}{2}\right)^n = \frac{\frac{1}{2}}{\left(1 - \frac{1}{2}\right)^2} = 2$$

$$E[X] = 2$$

3.21 Find the mean and variance of the random variable whose pdf is given in Fig. 3.3.

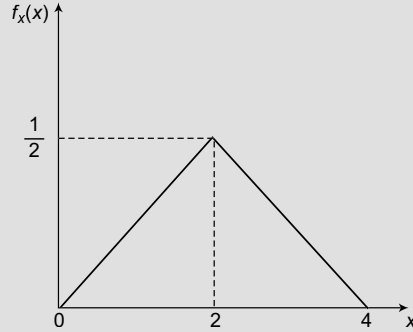


Fig. 3.3 pdf of random variable X for Solved Problem 3.21

Solution

For $0 \leq x \leq 2$

$$f_X(x) = \frac{x}{4}$$

and

For $2 \leq x \leq 4$; $f_X(x) = 1 - \frac{x}{4}$

The mean $\mu_X = E[X] = \int_{-\infty}^{\infty} x f_X(x) dx$

$$= \int_0^2 x \left(\frac{x}{4} \right) dx + \int_2^4 x \left(1 - \frac{x}{4} \right) dx$$

$$= \frac{1}{4} \int_0^2 x^2 dx + \int_2^4 x dx - \frac{1}{4} \int_2^4 x^2 dx$$

$$= \frac{1}{4} \left[\frac{x^3}{3} \right]_0^2 + \frac{1}{2} x^2 \Big|_2^4 - \frac{1}{4} \left[\frac{x^3}{3} \right]_2^4$$

$$= \frac{2}{3} + 6 - \frac{56}{12} = 2$$

The variance $\sigma_X^2 = E[X^2] - \mu_X^2$

$$E[X^2] = \int_{-\infty}^{\infty} x^2 f_X(x) dx$$

For $0 \leq x \leq 2$, the equation for $f_X(x)$ is

$$f_X(x) = \frac{1}{4}x$$

For $2 \leq x \leq 4$, the line passes through the coordinates

$\left(2, \frac{1}{2}\right)$ and $(4, 0)$.

$$f_X(x) - \frac{1}{2} = \frac{-1}{4}(x - 2)$$

$$f_X(x) = 1 - \frac{x}{4}$$

$$\begin{aligned}
 &= \int_0^2 x^2 \left(\frac{x}{4} \right) dx + \int_2^4 x^2 \left(1 - \frac{x}{4} \right) dx \\
 &= -\frac{1}{4} \frac{x^4}{4} \bigg|_0^2 + \frac{x^3}{3} \bigg|_2^4 - \frac{1}{4} \frac{x^4}{4} \bigg|_2^4 \\
 &= 1 + \frac{56}{3} + 15 = \frac{104}{3} \\
 \sigma_X^2 &= E[X^2] - \mu_X^2 \\
 &= \frac{104}{3} - (2)^2 = \frac{92}{3}
 \end{aligned}$$



Practice Problems

- 3.12** Find $E[X^2]$ and variance of discrete random variable of Practice Problem 3.1. (Ans. 27, 2)
- 3.13** Find $E[X^2]$ and variance of discrete random variable of Practice Problem 3.2. (Ans. 7, 1.24)
- 3.14** For the random variable given in Practice Problem 3.3, find $E[X^2]$, $E[X^3]$ and variance. (Ans: $\frac{5}{7}, \frac{5}{8}, 0.0198$)
- 3.15** A random variable has the following pdf:

$$f_X(x) = \begin{cases} x & 0 \leq x \leq 1 \\ \frac{3}{4} - \frac{x}{4} & 1 \leq x \leq 3 \\ 0 & \text{otherwise} \end{cases}$$

Find $E[X]$, $E[X^2]$ and σ_X^2 . (Ans: $\sigma_X^2 = 0.388$)

Solved Problems



3.22 A random variable with Laplace distribution has pdf given by

$$f_X(x) = \frac{\lambda}{2} e^{-\lambda|x|} \quad \lambda > 0 \quad -\infty < x < \infty$$

(i) Find the CDF. (ii) Find the mean and variance.

Solution

Given: $f_X(x) = \frac{\lambda}{2} e^{-\lambda|x|}; \lambda > 0 \quad -\infty < x < \infty$

The CDF $F_X(x) = \int_{-\infty}^x f_X(u) du$

For $x < 0$,

$$\begin{aligned}
 F_X(x) &= \int_{-\infty}^x \frac{\lambda}{2} e^{\lambda u} du \\
 &= \frac{\lambda}{2} \int_{-\infty}^x e^{\lambda u} du = \frac{\lambda}{2} \frac{e^{\lambda u}}{\lambda} \Big|_{-\infty}^x \\
 &= \frac{1}{2} e^{\lambda x}
 \end{aligned}$$

For $x > 0$,

$$\begin{aligned}
 F_X(x) &= \int_{-a}^0 \frac{\lambda}{2} e^{\lambda u} du + \int_0^x \frac{\lambda}{2} e^{-\lambda u} du \\
 &= \frac{\lambda}{2} \left(\frac{1}{\lambda} \right) + \frac{\lambda}{2} \frac{e^{-\lambda u}}{(-\lambda)} \Big|_0^x \\
 &= \frac{1}{2} - \frac{1}{2} (e^{-\lambda x} - 1) = 1 - \frac{1}{2} e^{-\lambda x}
 \end{aligned}$$

$$\begin{aligned}
 \Rightarrow F_X(x) &= \frac{1}{2} e^{\lambda x} \text{ for } x < 0 \\
 &= 1 - \frac{1}{2} e^{-\lambda x} \text{ for } x \geq 0
 \end{aligned}$$

$$\begin{aligned}
 \mu_x = E[X] &= \int_{-\infty}^{\infty} x f_X(x) dx = \int_{-\infty}^0 x \left(\frac{\lambda}{2} e^{\lambda x} \right) dx + \int_0^{\infty} x \left(\frac{\lambda}{2} e^{-\lambda x} \right) dx \\
 &= \frac{\lambda}{2} \int_{-\infty}^0 x e^{\lambda x} dx + \frac{\lambda}{2} \int_0^{\infty} x e^{-\lambda x} dx \\
 &= \frac{\lambda}{2} \left\{ \frac{x}{\lambda} e^{\lambda x} \Big|_{-\infty}^0 - \frac{e^{\lambda x}}{\lambda^2} \Big|_{-\infty}^0 \right\} + \frac{\lambda}{2} \left\{ \frac{-x e^{-\lambda x}}{\lambda} \Big|_0^{\infty} - \frac{e^{-\lambda x}}{\lambda^2} \Big|_0^{\infty} \right\} \\
 &= \frac{\lambda}{2} \left\{ (0 - 0) - \frac{1}{\lambda^2} \right\} + \frac{\lambda}{2} \left\{ (0 - 0) - \left(0 - \frac{1}{\lambda^2} \right) \right\} \\
 &= 0
 \end{aligned}$$

$$E[X^2] = \int_{-\infty}^{\infty} x^2 f_X(x) dx$$

$$\begin{aligned}
&= \int_{-\infty}^0 x^2 \frac{\lambda}{2} e^{\lambda x} dx + \int_0^{\infty} x^2 \frac{\lambda}{2} e^{-\lambda x} dx \\
&= \frac{\lambda}{2} \int_{-\infty}^0 x^2 e^{\lambda x} dx + \frac{\lambda}{2} \int_0^{\infty} x^2 e^{-\lambda x} dx \\
&= \frac{\lambda}{2} \left\{ \frac{x^2 e^{\lambda x}}{\lambda} \Big|_{-\infty}^0 - \frac{2}{\lambda^2} \frac{x e^{\lambda x}}{\lambda^2} \Big|_{-\infty}^0 + \frac{2}{\lambda^3} e^{\lambda x} \Big|_{-\infty}^0 \right\} \\
&\quad + \frac{\lambda}{2} \left\{ \frac{-x^2 e^{-\lambda x}}{\lambda} \Big|_0^{\infty} - \frac{2x e^{-\lambda x}}{\lambda^2} \Big|_0^{\infty} - \frac{2e^{-\lambda x}}{\lambda^3} \Big|_0^{\infty} \right\} \\
&= \frac{\lambda}{2} \left\{ (0-0) - \frac{2}{\lambda^2} (0-0) + \frac{2}{\lambda^3} \right\} + \frac{\lambda}{2} \left\{ (0-0) - \frac{2}{\lambda^2} (0-0) - \frac{2}{\lambda^3} (-1) \right\} \\
&= \frac{2}{\lambda^2}
\end{aligned}$$

3.23 Suppose $f_X(x) = \frac{x}{8}$ for $0 < x < 4$. Determine the mean and variance of X .

Solution

Given: $f_X(x) = \frac{x}{8}$

$$\mu_X = E[X] = \int_{-\infty}^{\infty} x f_X(x) dx = \int_0^4 x \left(\frac{x}{8} \right) dx = \frac{1}{8} \frac{x^3}{3} \Big|_0^4$$

$$= \frac{1}{8} \left[\frac{64-0}{3} \right] = \frac{8}{3}$$

$$E[X^2] = \int_{-\infty}^{\infty} x^2 f_X(x) dx = \int_0^4 x^2 \left(\frac{x}{8} \right) dx = \int_0^4 \frac{x^3}{8} dx = \frac{1}{8} \frac{x^4}{4} \Big|_0^4 = 8$$

$$\sigma_X^2 = \text{Var}(X) = E[X^2] - \{E[X]\}^2$$

$$= 8 - \left(\frac{8}{3} \right)^2 = 0.8889$$

3.24 For the following probability density function

$$\begin{aligned} f_X(x) &= kx, 0 \leq x < 2 \\ &= k(4-x); 2 \leq x \leq 4 \\ &= 0 \quad \text{otherwise} \end{aligned}$$

- (i) Find the value of k for which $f_X(x)$ is a valid pdf.
- (ii) Find the mean and variance of X .
- (iii) Find the CDF.

Solution For a valid pdf, $\int_{-\infty}^{\infty} f_X(x) dx = 1$

$$\int_{-\infty}^{\infty} f_X(x) dx = \int_0^2 kx dx + \int_2^4 k(4-x) dx = 1$$

$$k \frac{x^2}{2} \Big|_0^2 + k 4x \Big|_2^4 - k \frac{x^2}{2} \Big|_2^4 = 1$$

$$2k + 8k - 6k = 1$$

$$k = \frac{1}{4}$$

$$\mu_X = E[X] = \int_{-\infty}^{\infty} x f_X(x) dx$$

$$= \int_0^2 x(kx) dx + \int_2^4 x k(4-x) dx$$

$$= \frac{1}{4} \int_0^2 x^2 dx + \frac{1}{4} \int_2^4 x(4-x) dx$$

$$= \frac{1}{4} \frac{x^3}{3} \Big|_0^2 + \frac{1}{4} \left\{ 4 \frac{x^2}{2} \Big|_2^4 - \frac{x^3}{3} \Big|_2^4 \right\}$$

$$= \frac{2}{3} + \frac{1}{4} \left\{ 24 - \frac{56}{3} \right\} = \frac{2}{3} + \frac{1}{4} \left(\frac{16}{3} \right) = 2$$

$$\mu_X = 2$$

$$E[X^2] = \int_{-\infty}^{\infty} x^2 f_X(x) dx$$

$$= \frac{1}{4} \int_0^2 x^3 dx + \frac{1}{4} \int_2^4 x^2(4-x) dx$$

$$\begin{aligned}
&= \frac{1}{4} \frac{x^4}{4} \Big|_0^2 + \frac{1}{4} \left\{ 4 \frac{x^3}{3} \Big|_2^4 - \frac{x^4}{4} \Big|_2^4 \right\} \\
&= 1 + \frac{1}{4} \left\{ 4 \left(\frac{56}{3} \right) - 60 \right\} \\
&= \frac{14}{3} \\
\sigma_X^2 &= E[X^2] - \{E[X]\}^2 \\
&= \frac{14}{3} - (2)^2 = \frac{2}{3}
\end{aligned}$$

CDF, $F_X(x) = \int_{-\infty}^x f_X(u) du$

$\Rightarrow F_X(x) = 0$ for $x < 0$

For $0 \leq x \leq 2$,

$$F_X(x) = \int_{-\infty}^x \frac{1}{4} u du = \int_0^x \frac{1}{4} u du = \frac{1}{4} \frac{u^2}{2} \Big|_0^x = \frac{x^2}{8}$$

For $2 \leq x \leq 4$

$$\begin{aligned}
F_X(x) &= \int_0^2 \frac{1}{4} x dx + \int_2^x \frac{1}{4} (4-x) dx \\
&= \frac{1}{4} \frac{x^2}{2} \Big|_0^2 + \frac{1}{4} \left\{ 4x \Big|_2^x - \frac{x^2}{2} \Big|_2^x \right\} \\
&= \frac{1}{2} + \frac{1}{4} \left\{ 4(x-2) - \frac{x^2}{2} + 2 \right\} \\
&= \frac{-x^2}{8} + x - 1
\end{aligned}$$

$$F_X(x) = 1 \text{ for } x \geq 4$$

3.25 Find the expected value of the function $g(X) = X^3$ where X is a random variable with pdf $f_X(x) = \frac{1}{2} e^{-x/2} u(x)$.

Solution

$$E[g(x)] = \int_{-\infty}^{\infty} g(x) f_X(x) dx$$

$$\begin{aligned}
&= \int_0^{\infty} \frac{1}{2} e^{-x/2} x^3 dx \\
&= \frac{1}{2} \left\{ -2x^3 e^{-x/2} \Big|_0^{\infty} + 12x^2 e^{-x/2} \Big|_0^{\infty} - 48x e^{-x/2} \Big|_0^{\infty} + 48 \int_0^{\infty} e^{-x/2} dx \right\} \\
&= \frac{1}{2} \left\{ -2(0) + 12(0 - 0) - 48(0 - 0) + 48 \frac{e^{-x/2}}{(-1/2)} \Big|_0^{\infty} \right\} = 48
\end{aligned}$$

3.26 A random variable X represents the value of coins (in rupees) given in change when purchases are made at a particular store. Suppose the probability of ₹ 0.5, ₹ 1, ₹ 2, ₹ 5, ₹ 10, being present in change are $\frac{1}{10}, \frac{1}{5}, \frac{2}{5}, \frac{1}{5}$ and $\frac{1}{10}$ respectively. (i) Write an expression for the pdf of X . (ii) Find the mean of X .

Solution

$$\begin{aligned}
f_X(x) &= 0.1 \delta(x - 0.5) + 0.2 \delta(x - 1) + 0.4 \delta(x - 2) + 0.2 \delta(x - 5) + 0.1 \delta(x - 10) \\
E[X] &= \mu_X = 0.5(0.1) + (0.2)(1) + (0.4)(2) + (0.2)5 + (0.1)10 \\
&= ₹3.05
\end{aligned}$$



Practice Problems

3.16 A random variable X has a pdf

$$f_X(x) = \begin{cases} \frac{1}{2} \cos x & \text{for } -\frac{\pi}{2} < x < \frac{\pi}{2} \\ 0 & \text{otherwise} \end{cases}$$

Find the mean value of the function $g(x) = x^3/2$.

(Ans: zero)

3.17 Repeat the above problem for $g(x) = x^4$.

(Ans: 0.96)

3.18 A random variable X is uniformly distributed on the interval $(-10, 25)$. Another random variable $Y = e^{-X/10}$ is formed. Find $E[Y]$.

(Ans: 0.7532)

3.19 Find the mean and variance for a random variable with the density function

$$f_X(x) = \frac{1}{2b} e^{-|x|/b} \quad (\text{Ans: Zero, } 2b^2)$$

3.20 The pdf of a random variable is given by $f_X(x) = \frac{16/\pi}{(4+x^2)^2} 0 < x < \infty$. Find the mean, second-order moment and variance of X .

(Ans: $\frac{2}{\pi}, 2, 2 - \frac{4}{\pi^2}$)

3.21 A random variable has a probability density

$$f_X(x) = \begin{cases} \frac{4}{3}(1-x^3) & 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases} \quad (\text{Ans: } \frac{2}{5}, \frac{2}{9}, 0.062)$$

Find $E[X]$, $E[X^2]$ and σ_X^2 .

3.22 A random variable has a pdf

$$f_X(x) = \begin{cases} 1 - |x| & -1 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

Find the mean and variance of X .

(Ans. 0, 1/6)

3.23 The distribution function of a random variable X is $F_X(x) = 1 - (1+x)e^{-x}$, $x > 0$. Find the pdf of X . Also find $E[X]$, $E[X^2]$ and σ_X^2 .

(Ans. $f_X(x) = xe^{-x}$, $x > 0$; $E[X] = 2$; $E[X^2] = 6$; $\sigma_X^2 = 2$)

Solved Problems



3.27 The first four moments of a distribution about $x = 4$ are 1, 4, 10 and 45 respectively. Show that the mean is 5, variance is 3, $\mu_3 = 0$ and $\mu_4 = 26$.

Solution

Given: $E[X - 4] = 1$

$$E[(X - 4)^2] = 4$$

$$E[(X - 4)^3] = 10$$

$$E[(X - 4)^4] = 45$$

$$\Rightarrow E[X] = \mu_X = 5 \Rightarrow m_1 = 5$$

$$E[(X - 4)^2] = 4 \Rightarrow E[X^2 - 8X + 16] = 4$$

from which $E[X^2] = 28 \Rightarrow m_2 = 28$

$$E[(X - 4)^3] = 10 \Rightarrow E[X^3 - 12X^2 + 48X - 64] = 10$$

$$\begin{aligned} E[X^3] &= 10 + 12 E[X^2] - 48E[X] + 64 \\ &= 10 + 12(28) - 48(5) + 64 = 170 \end{aligned}$$

$$m_3 = 170$$

$$\Rightarrow E[(X - 4)^4] = 45$$

$$E[X^4] - 16E[X^3] + 96E[X^2] - 256E[X] + 256 = 45$$

$$\begin{aligned} \Rightarrow E[X^4] &= 45 + 16E[X^3] - 96E[X^2] + 256E[X] - 256 \\ &= 45 + 16(170) - 96(28) + 256(5) - 256 \\ &= 1101 \end{aligned}$$

$$\Rightarrow m_4 = 1101$$

$$m_1 = \mu_X = E[X] = 5$$

$$\begin{aligned} \sigma_X^2 &= \text{Var}(X) = E[X^2] - \{E(X)\}^2 \\ &= 28 - (5)^2 = 3 \end{aligned}$$

$$\begin{aligned} \mu_3 &= m_3 - 3\mu_X \sigma_X^2 - m_1^3 = 170 - 3(5)(3) - (5)^3 \\ &= 0 \end{aligned}$$

$$\begin{aligned} \mu_4 &= E[(X - \mu_X)^4] = E[X^4] - 4\mu_X E[X^3] + 6\mu_X^2 E[X^2] - 4\mu_X^3 E[X] + \mu_X^4 \\ &= m_4 - 4m_1 m_3 + 6m_2 m_1^2 - 3m_1^4 \\ &= 1101 - 4(5)(170) + 6(28)(5)^2 - 3(5)^4 = 26 \end{aligned}$$



MEAN AND VARIANCE OF SOME DISTRIBUTION 3.8

3.8.1 Binomial Distribution

The pmf of a binomial random variable with parameters (n, p) is given by

$$p_X(x) = \binom{n}{x} p^x (1-p)^{n-x} \quad x = 0, 1, \dots, n \quad (3.27)$$

Mean The mean or expectation of a discrete random variable is given by

$$E[X] = \sum_{x=0}^n x p_X(x_i) \quad (3.28)$$

For a binomial random variable,

$$\begin{aligned} E[X] &= \sum_{x=0}^n x \binom{n}{x} p^x (1-p)^{n-x} \\ &= \sum_{x=0}^n x \frac{n!}{x!(n-x)!} p^x (1-p)^{n-x} \end{aligned} \quad (3.29)$$

Since the first term in the summation is zero, we can write

$$\begin{aligned} E[X] &= \sum_{x=1}^n \frac{n!}{(x-1)!(n-x)!} p^x (1-p)^{n-x} \\ &= \sum_{x=1}^n \frac{n(n-1)!}{(x-1)!(n-x)!} p \cdot p^{x-1} (1-p)^{n-x} \\ &= np \sum_{x=1}^n \frac{(n-1)!}{(x-1)![(n-1)-(x-1)]!} p^{x-1} (1-p)^{(x-1)-(x-1)} \\ &= np \sum_{x=1}^n \binom{n-1}{x-1} p^{x-1} (1-p)^{(n-1)-(x-1)} \\ &= np \sum_{x=0}^{n-1} \binom{n-1}{x} p^x (1-p)^{n-1-x} \end{aligned} \quad (3.30)$$

We have the binomial expansion

$$(p+q)^n = \sum_{x=0}^n \binom{n}{x} p^x q^{n-x} \quad (3.31)$$

$$\text{Similarly } (p+q)^{n-1} = \sum_{x=0}^{n-1} \binom{n-1}{x} p^x q^{n-1-x} \quad (3.32)$$

Comparing Eq. (3.30) and Eq. (3.32) we get

$$\begin{aligned} E[X] &= np[p + (1-p)]^{n-1} \\ &= np \\ E[X] &= np \end{aligned} \quad (3.33)$$

Variance

$$\sigma_X^2 = \text{Var}(X) = E[X^2] - \{E[X]\}^2$$

$$E[X^2] = \sum_{i=0}^n x_i^2 p[x_i]$$

For the binomial random variable,

$$\begin{aligned} E[X^2] &= \sum_{x=0}^n x^2 \binom{n}{x} p^x (1-p)^{n-x} \\ &= \sum_{x=0}^n [x(x-1) + x] \binom{n}{x} p^x (1-p)^{n-x} \\ &= \sum_{x=0}^n x(x-1) \binom{n}{x} p^x (1-p)^{n-x} + \sum_{x=0}^n x \binom{n}{x} p^x (1-p)^{n-x} \end{aligned}$$

Since the first term in both summations is zero,

$$E[X^2] = \sum_{x=1}^n x(x-1) \binom{n}{x} p^x (1-p)^{n-x} + \sum_{x=1}^n x \binom{n}{x} p^x (1-p)^{n-x}$$

For $x=1$, the first summation is zero. The second summation is equal to $E[X]$.

Therefore,

$$\begin{aligned} E[X^2] &= \sum_{x=2}^n x(x-1) \binom{n}{x} p^x (1-p)^{n-x} + E[X] \\ &= \sum_{x=2}^n x(x-1) \frac{n!}{x!(n-x)!} p^x (1-p)^{n-x} + E[X] \\ &= \sum_{x=2}^n \frac{n!}{(x-2)!(n-x)!} p^x (1-p)^{n-x} + E[X] \\ &= \sum_{x=2}^n \frac{n(n-1)(n-2)!}{(x-2)!(n-x)!} p^x (1-p)^{n-x} + E[X] \\ &= n(n-1) \sum_{x=2}^n \frac{(n-2)!}{(x-2)![(n-2)-(x-2)]!} p^2 p^{x-2} (1-p)^{(n-2)-(x-2)} + E[X] \\ &= n(n-1)p^2 \sum_{x=2}^n \binom{n-2}{x-2} p^{x-2} (1-p)^{(n-2)-(x-2)} + E[X] \\ &= n(n-1)p^2 \sum_{x=0}^{n-2} \binom{n-2}{x} p^x (1-p)^{(n-2)-x} + E[X] \\ &= (n)(n-1)p^2 (p + 1-p)^{n-2} + E[X] \\ E[X] &= np \end{aligned}$$

$$\begin{aligned}
E[X^2] &= n(n-1)p^2 + np = n^2p^2 + np(1-p) \\
\sigma_X^2 &= \text{Var}[X] = E[X^2] - \{E[X]\}^2 \\
&= n^2p^2 + np(1-p) + (np)^2 \\
&= np(1-p) \\
\sigma_X^2 &= np(1-p)
\end{aligned} \tag{3.34}$$

Solved Problems



3.28 If the mean and variance of the binomial distribution are 6 and 1.5 respectively, find $E[X - P[X \geq 3]]$.

Solution Given $E[X] = 6$ and $\sigma_X^2 = 1.5$

$$E[X] = np = 6 \text{ and } \sigma_X^2 = np(1-p) = 1.5$$

$$\frac{np}{np(1-p)} = \frac{6}{1.5} = 4$$

$$\Rightarrow \frac{1}{1-p} = 4 \Rightarrow p = 0.75$$

$$np = 6 \Rightarrow n = \frac{6}{p} = \frac{6}{0.75} = 8$$

$$P[X \geq 3] = 1 - P[X < 3]$$

$$= 1 - \{P[X=0] + P[X=1] + P[X=2]\}$$

$$= 1 - \left\{ \binom{8}{0} p^0 (1-p)^8 + \binom{8}{1} p^1 (1-p)^7 + \binom{8}{2} p^2 (1-p)^6 \right\}$$

$$= 1 - \{(0.75)^0 (0.25)^8 + 8(0.75)(0.25)^7 + 28(0.75)^2 (0.25)^6\}$$

$$= 0.995$$

$$E[X - P[X \geq 3]] = E[X] - E[P(X \geq 3)]$$

$$= 6 - 0.995 = 5.005$$

3.29 A 40 student B. Tech class contains 25 boys and 15 girls. If a teacher ask a question, the probability that boys knowing the answer is $1/5$ and that of girls is $3/5$. Let X denote a random variable that denotes the number of students who know the answer to a question that teacher asks in class. Find μ_X and $\text{Var}(X)$.

Solution The total number of students in the class is 40.

The probability that a randomly selected student is a boy is $P(B) = \frac{25}{40} = \frac{5}{8}$

Similarly the probability that a randomly selected student is a girl is $P(G) = \frac{15}{40} = \frac{3}{8}$

The probability that a randomly selected student knows the answer is

$$P = \frac{1}{5} \left(\frac{5}{8} \right) + \frac{3}{5} \left(\frac{3}{8} \right) \\ = \frac{14}{40} = \frac{7}{20}$$

Let X be a random variable that denotes the number of students who know the answer, then

$$E[X] = np = 40 \left(\frac{7}{20} \right) = 14$$

$$\text{The variance } \sigma_X^2 = npq = 40 \left(\frac{7}{20} \right) \left(\frac{13}{20} \right) \\ = \frac{9}{10} = 9.1$$

3.30 For a binomial distribution mean is 6 and standard deviation is $\sqrt{2}$. Find the first two terms of the distribution.

Solution For binomial distribution mean $= np = 6$

$$\text{Variance} = npq = (\sqrt{2})^2 = 2$$

$$q = \frac{npq}{np} = \frac{2}{6} = \frac{1}{3}$$

$$p = 1 - q = 1 - \frac{1}{3} = \frac{2}{3}$$

$$np = n \left(\frac{2}{3} \right) = 6 \Rightarrow n = 9$$

The first two terms of the distribution is given by

$$P(X = x) = \binom{n}{x} p^x q^{n-x}, x = 0, 1$$

$$P(X = 0) = \binom{9}{0} \left(\frac{2}{3} \right)^0 \left(\frac{1}{3} \right)^9 = \left(\frac{1}{3} \right)^9$$

$$P(X = 1) = \binom{9}{1} \left(\frac{2}{3} \right)^1 \left(\frac{1}{3} \right)^8 = 9.14 \times 10^{-4}$$

3.31 6 dice are thrown 729 times. How many times do you expect atleast three dice to show 5 or 6.

Solution When a dice is thrown, there are 6 possible outcomes. Probability of getting 2 or 3 is given by $\frac{2}{6} = \frac{1}{3}$.

Since $p = \frac{1}{3}$; $q = 1 - p = 1 - \frac{1}{3} = \frac{2}{3}$

Given $n = 6$

$$\begin{aligned} P(\text{getting at least three dice to show 5 or 6}) &= P(X \geq 3) = P(X = 3) + P(X = 4) + P(X = 5) + P(X = 6) \\ &= \binom{6}{3} \left(\frac{1}{3}\right)^3 \left(\frac{2}{3}\right)^3 + \binom{6}{4} \left(\frac{1}{3}\right)^4 \left(\frac{2}{3}\right)^2 + \binom{6}{5} \left(\frac{1}{3}\right)^5 \left(\frac{2}{3}\right) + \binom{6}{6} \left(\frac{1}{3}\right)^6 \left(\frac{2}{3}\right)^0 \\ &= \frac{160 + 60 + 12 + 1}{729} = \frac{233}{729} \end{aligned}$$

The number of times that one can expect three dice to show 5 or 6 is equal to $\frac{233}{729}(729) = 233$ times.

3.32 Four coins were tossed on a table 100 times. The number of heads fallen in each of the 100 times was noted. The results are

x	0	1	2	3	4
f	6	30	32	26	6

Fit a binomial distribution to the data. Compare these frequencies with the expected frequencies.

Solution Let n and p are the parameter of the binomial distribution.

The mean from the data

$$\frac{6(0) + 30(1) + 32(2) + 26(3) + 6(4)}{100} = \frac{196}{100} = 1.96$$

The mean of the binomial distribution is given by np

We have $n = 4$. Therefore $np = 1.96 \Rightarrow p = \frac{1.96}{4} = 0.49$

$$q = 1 - p = 1 - 0.49 = 0.51$$

The expected frequency of heads

$$= 100 \binom{4}{x} (0.49)^x (0.51)^{4-x} \quad x = 0, 1, 2, 3, 4$$

The values can be obtained as follows

$$0.7 \times 100, 0.26 \times 100, 0.37 \times 100, 0.24 \times 100, 0.06 \times 100$$

That is the expected frequencies are

$$7, 26, 37, 24 \text{ and } 6$$



Practice Problem

3.24 50 balls are tossed into 25 boxes. What is the expected number of balls in the fifth box?

(Ans. 2)

REVIEW QUESTION

8. For the binomial density, show that $E[X] = np$ and $\sigma_X^2 = np(1-p)$

3.8.2 Negative Binomial Distribution

The probability mass function of a negative binomial distribution is given by

$$P(X=x) = \binom{x+k-1}{k-1} p^k q^x; x=0, 1, \dots \quad (3.35)$$

The mean value is given by

$$\begin{aligned} \mu_X &= \sum_{x=0}^{\infty} xP(X=x) \\ &= \sum_{x=0}^{\infty} x \binom{x+k-1}{k-1} p^k q^x \\ &= \binom{k}{k-1} p^k q + 2 \binom{k+1}{k-1} p^k q^2 + \dots \\ &= k p^k q + 2 \frac{k(k+1)}{2!} p^k q^2 + \dots \\ &= k p^k q \left[1 + \frac{k+1}{1!} q + \frac{(k+2)(k+1)}{2!} q^2 + \dots \right] \\ &= k p^k q (1-q)^{-(k+1)} \\ &= k p^k q p^{-(k+1)} = \frac{kq}{p} \end{aligned} \quad (3.36)$$

The variance is given by

$$\begin{aligned} \text{Var}(X) &= E[X^2] - \{E[X]\}^2 \\ &= E[X(X-1) + X] - \{E[X]\}^2 \\ &= E[X(X-1)] + E[X] - \{E[X]\}^2 \end{aligned} \quad (3.37)$$

$$\begin{aligned} E[X(X-1)] &= \sum_{x=0}^{\infty} x(x-1)P(X=x) \\ &= \sum_{x=0}^{\infty} x(x-1) \binom{x+k-1}{k-1} p^k q^x \\ &= 2 \binom{k+1}{k-1} p^k q^2 + 3(2) \binom{k+2}{k-1} p^k q^3 + 4(3) \binom{k+3}{k-1} p^k q^4 + \dots \\ &= 2 \frac{k(k+1)}{2!} p^k q^2 + 3(2) \frac{(k+2)(k+1)k}{3!} p^k q^3 + 4(3) \frac{(k+3)(k+2)(k+1)(k)}{4!} p^k q^4 + \dots \\ &= k(k+1) p^k q^2 \left[1 + (k+2)q + \frac{(k+2)(k+3)}{2} q^2 + \dots \right] \\ &= k(k+1) p^k q^2 [(1-q)^{-(k+2)}] \end{aligned} \quad (3.38)$$

$$\begin{aligned}
&= k(k+1) p^k q^2 p^{-(k+2)} \\
&= k(k+1) \left(\frac{q}{p}\right)^2 \\
\text{Var}(X) &= E[X(X-1)] + E[X] - \{E[X]\}^2 \\
&= k(k+1) \left(\frac{q}{p}\right)^2 + \frac{kq}{p} - \frac{k^2 q^2}{p^2} \\
&= \frac{kq}{p} \left[\frac{(k+1)q}{p} + 1 - \frac{kq}{p} \right] \\
&= \frac{kq}{p} \left(\frac{p+q}{p} \right) = \frac{kq}{p^2} \\
&= \frac{k(1-p)}{p^2}
\end{aligned} \tag{3.39}$$

REVIEW QUESTIONS



9. Derive the expressions for mean and Variance of a negative binomial distribution.

3.8.3 Geometric Distribution

The probability mass function of a geometric distribution is given by

$$P(X=x) = q^{x-1} p, x = 1, 2, \dots \tag{3.41}$$

where p is the probability of success and $q = 1 - p$.

The mean value is given by

$$\begin{aligned}
\mu_X &= \sum_i x_i p_X(x_i) \\
&= \sum_{x=1}^{\infty} x p q^{x-1} \\
&= p \sum_{x=1}^{\infty} x q^{x-1} \\
&= p(1-q)^{-1} = \frac{p}{p^2} = \frac{1}{p}
\end{aligned} \tag{3.42}$$

The variance is given by

$$\text{Var}(X) = E(X^2) - \{E[X]\}^2$$

$$\begin{aligned}
E[X(X-1)] &= \sum_{x=1}^{\infty} x(x-1) p q^{x-1} \\
&= \sum_{x=2}^{\infty} x(x-1) p q^{x-1}
\end{aligned}$$

$$\begin{aligned}
 &= 2pq \sum_{x=2}^{\infty} \left[\frac{x(x-1)}{2} q^{x-2} \right] \\
 2pq(1-q)^{-3} &= \frac{2q}{p^2}
 \end{aligned} \tag{3.43}$$

$$\begin{aligned}
 \text{Var}(X) &= E[X^2] - \{E[X]\}^2 = E[X(X-1)] + E[X] - \{E[X]\}^2 \\
 &= \frac{2q}{p^2} + \frac{1}{p} - \frac{1}{p^2} = \frac{1-p}{p^2} = \frac{q}{p^2}
 \end{aligned} \tag{3.44}$$

Solved Problems



3.33 If the probability of success on each trial is 0.2, find the expected number of trials required before first success.

Solution Given: $p = 0.3$ from which we get $q = 1 - p = 1 - 0.2 = 0.8$. Let X represents the number of trials required to get first success. Then X follows geometric distributions. The expected number of trials required before first success is

$$E[X] = \frac{1}{p} = \frac{1}{0.2} = 5$$

3.34 The cost of conducting an experiment is ₹ 500. The experiment is continued until the first successful result is achieved. If the experiment fails, an additional amount of ₹ 150 is required to meet additional cost. If the probability of success is 0.25 and individual trials are independent, what is the expected cost of the entire procedure.

Solution The cost of experiment is ₹ 500.

The additional cost required if the experiment fails is ₹ 150. Let X be the number trials required until the first success occur. Then there will be $(X - 1)$ failures before first success. The total cost

$$\begin{aligned}
 C &= 500(X) + 150(X - 1) \\
 &= 650X - 150
 \end{aligned}$$

$$\begin{aligned}
 \text{Expected cost} &= E(C) = E[650X - 150] \\
 &= 650 E[X] - 150
 \end{aligned}$$

$$\text{The expected number of trials} = \frac{1}{p} = \frac{1}{0.25} = 4$$

$$\begin{aligned}
 \Rightarrow E(C) &= 650(4) - 150 \\
 &= ₹ 2450
 \end{aligned}$$

3.35 Let N be a geometric random variable with $S_N = \{1, 2, \dots\}$. Find (a) $P(N = k | N \leq m)$ (b) probability that N is odd.

Solution For a geometric random variable

$$P(X = x) = pq^{x-1}, x = 1, 2, \dots$$

$$\begin{aligned}
 P(N = k | N \leq m) \\
 &= \frac{P(N = k, N \leq m)}{P(N \leq m)} = \frac{P(N = k)}{P(N \leq m)}
 \end{aligned}$$

$$P(N = k) = p q^{k-1} = p(1-p)^{k-1}$$

$$\begin{aligned}
 P(N \leq m) &= \sum_{k=1}^m p q^{k-1} = \sum_{k=1}^m p(1-p)^{k-1} \\
 &= p + p(1-p) + p(1-p)^2 + \dots m \text{ terms} \\
 &= p[1 + (1-p) + (1-p)^2 + \dots m \text{ terms}] \\
 &= \frac{p[1 - (1-p)^{m+1}]}{1 - (1-p)} = 1 - (1-p)^{m+1}
 \end{aligned}$$

$$\begin{aligned}
 P(N \text{ is odd}) &= P(N = 1) + P(N = 3) + \dots \\
 &= p + p(1-p)^2 + p(1-p)^4 + \dots \\
 &= p[1 + (1-p)^2 + (1-p)^4 + \dots] \\
 &= \frac{p}{1 - (1-p)^2} = \frac{p}{1 - (1 + p^2 - 2p)} = \frac{p}{2p - p^2} = \frac{1}{2 - p} \\
 &= \frac{1}{2 - (1 - q)} = \frac{1}{1 + q} \\
 &= P(N \text{ is odd}) = \frac{1}{1 + q}
 \end{aligned}$$

3.36 In your key bunch there are exactly 5 keys. If you try to open the door using the keys one after the other, what is the expected number of keys you will have to try before the door is opened.

Solution The bunch contains 5 keys. Therefore the probability that a key opens the door on any trial is $1/5$. Let X be a random variable that denotes the number of trials required

$$p_X(x) = A p(1-p)^{x-1} \quad x = 1, 2, \dots, 5$$

The value of A can be obtained by equating

$$\begin{aligned}
 \sum_{x=1}^5 A p(1-p)^{x-1} &= 1 \\
 A p[1 + (1-p) + \dots + (1-p)^4] &= 1 \\
 A p \left\{ \frac{1 - (1-p)^5}{1 - (1-p)} \right\} &= 1
 \end{aligned}$$

$$\begin{aligned}
 \Rightarrow A &= \frac{1}{1 - (1-p)^5} \\
 p_X(x) &= \frac{p(1-p)^{x-1}}{1 - (1-p)^5} \quad x = 1, 2, \dots, 5
 \end{aligned}$$

The expected number of keys is given by

$$\begin{aligned}
 \sum_{x=1}^n x p_X(x) &= \sum_{x=1}^5 \frac{x p(1-p)^{x-1}}{1-(1-p)^5} \\
 &= \frac{p}{1-(1-p)^5} \left\{ \sum_{x=1}^5 x(1-p)^{x-1} \right\} \\
 &= \frac{1/5}{1-\left(\frac{4}{5}\right)^5} \left\{ 1 + 2\left(\frac{4}{5}\right) + 3\left(\frac{4}{5}\right)^2 + 4\left(\frac{4}{5}\right)^3 + 5\left(\frac{4}{5}\right)^4 \right\} = 2.559
 \end{aligned}$$

REVIEW QUESTIONS



10. Derive expressions for the mean and Variance of a geometric distribution.

3.8.4 Poisson Distribution

The pdf of Poisson random variable is given by

$$p_X(x) = \frac{e^{-\lambda} \lambda^x}{x!} \quad (3.45)$$

Mean

$$E[X] = \sum_i x_i p_X[x_i]$$

For a Poisson random variable

$$\begin{aligned}
 E[X] &= \sum_{x=0}^{\infty} x p_X(x) = \sum_{x=0}^{\infty} \frac{x e^{-\lambda} \lambda^x}{x!} = e^{-\lambda} \sum_{x=0}^{\infty} \frac{x \lambda^x}{x!} \\
 &= e^{-\lambda} \sum_{x=1}^{\infty} \frac{\lambda^x}{(x-1)!} = e^{-\lambda} \sum_{x=1}^{\infty} \frac{\lambda(\lambda^{x-1})}{(x-1)!} \\
 &= \lambda e^{-\lambda} \sum_{x=1}^{\infty} \frac{\lambda^{x-1}}{(x-1)!} \\
 &= \lambda e^{-\lambda} \left[1 + \frac{\lambda}{1!} + \frac{\lambda^2}{2!} + \dots \right] = \lambda e^{-\lambda} (e^{\lambda}) \\
 &= \lambda \\
 E[X] &= \lambda \quad (3.46)
 \end{aligned}$$

Variance

$$\begin{aligned}
 \sigma_X^2 &= \text{Var}[X] = E[X^2] - \{E[X]\}^2 \\
 E[X^2] &= \sum_{x=0}^{\infty} x^2 p_X(x) = \sum_{x=0}^{\infty} \frac{x^2 e^{-\lambda} \lambda^x}{x!}
 \end{aligned}$$

$$\begin{aligned}
&= \sum_{x=0}^{\infty} \frac{x(x-1) + x}{x!} e^{-\lambda} \lambda^x \\
&= \sum_{x=0}^{\infty} \frac{x(x-1)}{x!} e^{-\lambda} \lambda^x + \sum_{x=0}^{\infty} \frac{e^{-\lambda} \lambda^x}{x!}
\end{aligned}$$

Since the first two terms in the summation are zero,

$$\begin{aligned}
E[X^2] &= \sum_{x=2}^{\infty} \frac{x(x-1)}{x!} e^{-\lambda} \lambda^x + E[X] \\
E[X^2] &= \sum_{x=2}^{\infty} \frac{x(x-1)}{x(x-1)(x-2)!} e^{-\lambda} \lambda^{x-2} \lambda^2 + E[X] \\
&= \lambda^2 \sum_{x=2}^{\infty} \frac{e^{-\lambda} \lambda^{x-2}}{(x-2)!} + E[X] \\
&= \lambda^2 e^{-\lambda} \left[1 + \frac{\lambda}{1!} + \frac{\lambda^2}{2!} + \dots \right] + E[X] \tag{3.47}
\end{aligned}$$

$$\begin{aligned}
&= \lambda^2 e^{-\lambda} (e^{\lambda}) + \lambda = \lambda^2 + \lambda \\
\sigma_X^2 &= \text{Var}[X] = \lambda^2 + \lambda - \lambda^2 = \lambda \tag{3.48} \\
\sigma_X^2 &= \lambda
\end{aligned}$$

For a Poisson random variable, the mean and variance are same.

REVIEW QUESTIONS



11. Prove that for a Poisson random variable, the mean and variance are same.
12. Derive the expression for mean and variance of a Poisson random variable.

Solved Problems



3.37 The number of customers that enter a bank in an hour is a Poisson random variable and suppose that $P[X = 0] = 0.223$. Determine the mean and variance of X .

Solution For a Poisson random variable,

$$\begin{aligned}
p_X(x) &= \frac{e^{-\lambda} \lambda^x}{x!} \\
P(X=0) &= e^{-\lambda}
\end{aligned}$$

Given: $P[X = 0] = 0.223$

$\Rightarrow e^{-\lambda} = 0.223$. Therefore $\lambda = 1.5$

For a Poisson random variable,

Mean = variance = λ

Hence, mean = variance = 1.5

3.38 If X and Y are independent Poisson random variable such that

$P(X = 1) = P(X = 2)$; $P(Y = 2) = P(Y = 3)$, find the variance of $X - 2Y$

Solution Given: $P(X = 1) = P(X = 2)$

and $P(Y = 2) = P(Y = 3)$

$$P(X = x) = \frac{e^{-\lambda} \lambda^x}{x!}$$

Therefore, $\frac{e^{-\lambda} \lambda^1}{1!} = \frac{e^{-\lambda} \lambda^2}{2!} \Rightarrow \lambda = \frac{\lambda^2}{2}$

$$\lambda^2 - 2\lambda = 0; \lambda(\lambda - 2) = 0$$

$\Rightarrow \lambda = 2$

$$P(Y = y) = \frac{e^{-\lambda} \lambda^y}{y!}$$

$\Rightarrow \frac{e^{-\lambda} \lambda^2}{2!} = \frac{e^{-\lambda} \lambda^3}{3!} \Rightarrow \lambda^2 = \frac{\lambda^3}{3}$

$$\lambda^3 - 3\lambda^2 = 0; \lambda^2(\lambda - 3) = 0$$

$\Rightarrow \lambda = 3$

For a Poisson random variable, variance = λ

Hence $\text{Var}(X) = 2, \text{Var}(Y) = 3$

$$\text{Var}(X - 2Y) = \text{Var}(X) + 4 \text{Var}(Y)$$

$$= 2 + 4(3) = 14$$

3.8.5 Uniform Distribution

The probability density function of uniform random variable X over the interval (a, b) is

$$\begin{aligned} f_X(x) &= \frac{1}{b-a} \text{ for } a \leq x \leq b \\ &= 0 \quad \text{elsewhere} \end{aligned} \quad (3.49)$$

Mean

$$\begin{aligned} \mu_X &= E[X] = \int_{-\infty}^{\infty} x f_X(x) dx \\ &= \int_a^b x \left(\frac{1}{b-a} \right) dx = \frac{1}{b-a} \int_a^b x dx \\ &= \frac{1}{b-a} \left[\frac{x^2}{2} \right]_a^b = \frac{b^2 - a^2}{2(b-a)} = \frac{b+a}{2} \end{aligned}$$

$$E[X] = \frac{b+a}{2} \quad (3.50)$$

Variance

$$\begin{aligned} \sigma_X^2 &= E[X^2] - \{E[X]\}^2 \\ E[X^2] &= \int_{-\infty}^{\infty} x^2 f_X(x) dx \\ &= \int_a^b x^2 \left(\frac{1}{b-a} \right) dx = \frac{1}{b-a} \left. \frac{x^3}{3} \right|_a^b = \frac{b^3 - a^3}{3(b-a)} \\ &= \frac{b^2 + ab + b^2}{3} \end{aligned} \quad (3.51)$$

$$\begin{aligned} \sigma_X^2 &= E[X^2] - \{E[X]\}^2 \\ &= \frac{b^2 + ab + a^2}{3} - \left(\frac{b+a}{2} \right)^2 \\ &= \frac{b^2 + ab + a^2}{3} - \frac{b^2 + 2ab + a^2}{4} \\ &= \frac{b^2 - 2ab + a^2}{12} \\ &= \frac{(b-a)^2}{12} \end{aligned} \quad (3.52)$$

Solved Problems

3.39 Find the n^{th} moment of uniform random variable and hence its mean.

Solution The n^{th} moment of a random variable is given by

$$E[X^n] = \int_{-\infty}^{\infty} x^n f_X(x) dx. \text{ For a uniform random variable,}$$

$$\begin{aligned} f_X(x) &= \frac{1}{b-a} \text{ for } a \leq x \leq b \\ &= 0 \quad \text{otherwise} \end{aligned}$$

Therefore,

$$\begin{aligned} E[X^n] &= \int_a^b x^n \left(\frac{1}{b-a} \right) dx = \frac{1}{b-a} \left[\frac{x^{n+1}}{n+1} \right]_a^b \\ &= \frac{1}{b-a} \left[\frac{(b)^{n+1} - a^{n+1}}{n+1} \right] \end{aligned} \quad (3.53)$$

Mean value can be obtained by substituting $n = 1$ in the above formula.

$$E[X] = \frac{1}{b-a} \left[\frac{b^2 - a^2}{2} \right] = \frac{b+a}{2} \quad (3.54)$$

3.40 Rounding of errors X are uniformly distributed. If the sixth decimal place of a calculator will be rounded find the mean and variance of X . Find the probability that the numerical error is in between 0.000001 and 0.000004.

Solution Since the sixth decimal place of a calculator will be rounded, the X is uniformly distributed over $(-0.000005, 0.000005)$. Then $E[X] = 0$

$$\sigma_X^2 = \frac{(0.000005 + 0.000005)^2}{12} = \frac{(0.00001)^2}{12} = 8.35 \times 10^{-12}$$

$$P[0.000001 \leq X \leq 0.000004] = \frac{0.000004 - 0.000001}{0.00001} = 0.3$$

3.41 A random variable X has a continuous uniform distribution over the interval $(2, 6)$.

- Determine mean, variance and standard deviation of X .
- Find $P[X \leq 2]$.

Solution Given $X \sim U(2, 6)$; $b = 6$ and $a = 2$

$$E[X] = \frac{b+a}{2} = \frac{6+2}{2} = 4$$

$$E[X] = 4$$

$$\sigma_X^2 = \frac{(b-a)^2}{12} = \frac{(6-2)^2}{12} = \frac{16}{12} = 1.33$$

$$\text{Standard deviation} = \sigma_X = \sqrt{1.33}$$

$$P\{X \leq 4\} = \int_2^4 f_X(x) dx = \int_2^4 \frac{1}{6-2} dx = \frac{1}{4}(2) = 0.5$$

3.42 Suppose the time it takes to fill an application form is uniformly between 2 to 3 minutes.

- What is the mean and variance of the time it take to fill the form.
- What is the probability that it will take less than 150 sec to fill the form?
- Determine the cumulative distribution of the time it take to fill the form.

Solution Let X be the random variable that represents the time to fill the form. X is uniformly distributed between 2 to 3 minutes. That is, $b = 3$ min and $a = 2$ min

$$f_X(x) = 1 \text{ for } 2 < x < 3$$

$$= 0 \quad \text{otherwise}$$

$$E[X] = \frac{b+a}{2} = \frac{3+2}{2} = 2.5 \text{ min}$$

$$\sigma_X^2 = \text{Var}[X] = \frac{(b-a)^2}{12} = \frac{1}{12} = 0.0833 \text{ min}$$

$$P[X \leq 2.5] = \int_2^{2.5} (1) dx = 0.5$$

$$F_X(x) = \frac{x}{b-a} - \frac{a}{b-a} = \frac{x-a}{b-a} = \frac{x-2}{3-2} = x-2$$

$$F_X(x) = x-2$$

3.43 Find the mean and variance of the uniform discrete random variable that takes n values in the set $(1, 2, \dots, n)$ with equal probability.

Solution The probability = $\frac{1}{n}$

The mean $\sum_i x_i p_X(x_i) = 1\left(\frac{1}{n}\right) + 2\left(\frac{1}{n}\right) + \dots + n\left(\frac{1}{n}\right)$

$$= \frac{1}{n} [1 + 2 + \dots + n]$$

$$= \frac{1}{n} \frac{n(n+1)}{2} = \frac{n+1}{2}$$

$$E[X] = \frac{n+1}{2}$$

$$E[X^2] = \sum_i x_i^2 p_X(x_i) = 1^2 \left(\frac{1}{n}\right) + 2^2 \left(\frac{1}{n}\right) + 3^2 \left(\frac{1}{n}\right) + \dots + n^2 \left(\frac{1}{n}\right)$$

$$= \frac{1}{n} [1^2 + 2^2 + 3^2 + \dots + n^2]$$

$$= \frac{1}{n} \frac{n(n+1)(2n+1)}{6} = \frac{(n+1)(2n+1)}{6}$$

$$\sigma_X^2 = E[X^2] - \{E[X]^2\}$$

$$= \frac{(n+1)(2n+1)}{6} - \frac{(n+1)^2}{4} = \frac{n+1}{2} \left\{ \frac{2n+1}{3} - \frac{n+1}{2} \right\}$$

$$= \frac{n+1}{2} \left\{ \frac{n-1}{6} \right\} = \frac{n^2-1}{12}$$



Practice Problems

3.25 The pdf of the time it takes to fill a water tank is (a) $f_X(x) = 0.05$; (b) $30 \leq x \leq 50$ minutes. Determine the mean variance of the time to complete the filling of the water tank. (Ans: (a) 40, (b) 33.33)

3.26 Suppose X has a continuous uniform distribution over the interval $[-2, 2]$.

- (a) Determine the mean, variance, and standard deviation of X .
- (b) Determine the value of x such that $P[-x < X < x] = 0.90$. (Ans: (a) Zero (b) 1.33, (c) 1.153, $x = 2.8$)
- (c) Find the mean and variance of the discrete uniform distribution over possible values $x = n, n + 1, \dots, m$.

REVIEW QUESTION



3.13 Derive expressions for mean and variance of a uniform random variable X over the interval (a, b) .

3.8.6 Normal Distribution

The pdf of a random variable is given by

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma_X} e^{-(x-\mu_X)^2/2\sigma_X^2} \quad (3.55)$$

Mean

$$\begin{aligned} \mu_X = E[X] &= \int_{-\infty}^{\infty} x f_X(x) dx \\ &= \int_{-\infty}^{\infty} x \frac{1}{\sqrt{2\pi}\sigma_X} e^{-(x-\mu_X)^2/2\sigma_X^2} dx \\ &= \frac{1}{\sqrt{2\pi}\sigma_X} \int_{-\infty}^{\infty} x e^{-(x-\mu_X)^2/2\sigma_X^2} dx \end{aligned}$$

Let $\frac{x - \mu_X}{\sigma_X} = t \Rightarrow dx = \sigma_X dt$ and $x = \sigma_X t + \mu_X$

$$\begin{aligned} E[X] &= \frac{1}{\sqrt{2\pi}\sigma_X} \int_{-\infty}^{\infty} (\sigma_X t + \mu_X) e^{-t^2/2} (\sigma_X dt) \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \sigma_X t e^{-t^2/2} dt + \frac{1}{\sqrt{2\pi}} \mu_X \int_{-\infty}^{\infty} e^{-t^2/2} dt \\ &= \frac{\sigma_X}{\sqrt{2\pi}} \int_{-\infty}^{\infty} t e^{-t^2/2} dt + \frac{1}{\sqrt{2\pi}} \mu_X \int_{-\infty}^{\infty} e^{-t^2/2} dt \end{aligned}$$

The first integral is zero since $t e^{-t^2/2}$ is an odd function. Hence,

$$E[X] = \frac{1}{\sqrt{2\pi}} \mu_X \int_{-\infty}^{\infty} e^{-t^2/2} dt$$

The integral $\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-t^2/2} dt$ represents the area enclosed by a normal random variable $N(0, 1)$ and is equal to unity.

Therefore,

$$E[X] = \mu_X$$

Variance

$$\sigma_X^2 = E[X^2] - \{E[X]\}^2$$

$$\begin{aligned} E[X^2] &= \int_{-\infty}^{\infty} x^2 f_X(x) dx \\ &= \int_{-\infty}^{\infty} x^2 \frac{1}{\sqrt{2\pi} \sigma_X} e^{-(x-\mu_X)^2/2\sigma_X^2} dx \end{aligned}$$

Let $\frac{x-\mu_X}{\sigma_X} = t$ then $dx = \sigma_X dt$ and $x = \sigma_X t + \mu_X$

$$\begin{aligned} E[X^2] &= \frac{1}{\sqrt{2\pi} \sigma_X} \int_{-\infty}^{\infty} (\sigma_X t + \mu_X)^2 e^{-t^2/2} (\sigma_X dt) \\ &= \frac{\sigma_X^2}{\sqrt{2\pi}} \int_{-\infty}^{\infty} t^2 e^{-t^2/2} dt + \frac{2\sigma_X \mu_X}{\sqrt{2\pi}} \int_{-\infty}^{\infty} t e^{-t^2/2} dt + \frac{\mu_X^2}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-t^2/2} dt \end{aligned}$$

We know $\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-t^2/2} dt = 1$ and $\int_{-\infty}^{\infty} t e^{-t^2/2} dt = 0$

Therefore,

$$E[X^2] = \frac{\sigma_X^2}{\sqrt{2\pi}} \int_{-\infty}^{\infty} t^2 e^{-t^2/2} dt + \mu_X^2$$

Also

$$\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} t^2 e^{-t^2/2} dt = 1$$

$$E[X^2] = \sigma_X^2 + \mu_X^2 \quad (3.56a)$$

The variance

$$\begin{aligned} \text{Var}(X) &= E[X^2] - \{E[X]\}^2 \\ &= \sigma_X^2 + \mu_X^2 - \mu_X^2 = \sigma_X^2 \\ \text{Var}(X) &= \sigma_X^2 \end{aligned} \quad (3.56b)$$

Solved Problem



3.44 In a semester examination, the total score of a student is a Gaussian random variable $X \sim N(\mu_X; \sigma_X^2)$. If the average score is 600 and 15.9% students' scores are about 750, find μ_X , σ_X and $P(750 < X < 900)$.

Solution Given; $\mu_X = 600$ and $P[X > 750] = 0.159$

$$P[X > 750] = P\left[\frac{X - 600}{\sigma_X} > \frac{750 - 600}{\sigma_X}\right] = 0.159$$

$$1 - \phi(x) = 0.159$$

$$\phi(x) = 1 - 0.159 = 0.841$$

From Table, 2.4 we can find $x = 1$

$$\text{This means } \frac{750 - 600}{\sigma_X} = 1 \Rightarrow \sigma = 150$$

$$\begin{aligned} P(750 < X < 900) &= P\left[\frac{750 - 600}{150} < \frac{X - 600}{150} < \frac{900 - 600}{150}\right] = P\left(1 < \frac{X - 600}{150} < 2\right) \\ &= \phi(2) - \phi(1) \\ &= 0.9772 - 0.8413 = 0.136 \end{aligned}$$

3.45 If X is a Gaussian random variable $X \sim N(\mu_X, \sigma_X^2)$, find the mean and variance of $Y = aX + b$ for all $a \neq 0$.

Solution

$$\text{Given: } Y = aX + b$$

$$E[Y] = E[aX + b] = aE[X] + b$$

$$= a\mu_X + b$$

$$\sigma_Y^2 = E[(Y - \mu_Y)^2] = E[aX + b - (a\mu_X + b)^2]$$

$$= E[a^2(X - \mu_X)^2] = a^2 E[(X - \mu_X)^2]$$

$$= a^2 \sigma_X^2 \quad (3.57)$$



Practice Problems

3.27 If X is a normal random variable with $\mu_X = 2$ and $\sigma_X^2 = 16$, find, (i) $P\{2 \leq x \leq 6\}$, (ii) $P[|X - 2| > 6]$.

(Ans: (i) 0.34; (ii) 0.1296)

3.28 The compressive strength of samples of cement can be modelled by a random variable $X \sim N(\mu_X, \sigma_X^2)$ where $\mu_X = 400 \text{ kg/cm}^2$ and $\sigma_X = 80 \text{ kg/cm}^2$.

- What is the probability that a sample's strength is less than 450 kg/cm^2 ?
- What strength is exceeded by 90% of the samples?

REVIEW QUESTION



- Find mean and variance of Gaussian random variable.

3.8.7 Exponential Distribution

The probability density function of an exponential random variable is given by

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x} & \text{for } x \geq 0 \\ 0 & \text{otherwise} \end{cases} \quad (3.58)$$

Mean The mean of a random variable is

$$\begin{aligned} \mu_X = E[X] &= \int_{-\infty}^{\infty} x f_X(x) dx \\ &= \int_0^{\infty} x \lambda e^{-\lambda x} dx = \lambda \int_0^{\infty} x e^{-\lambda x} dx \quad \boxed{\lim_{x \rightarrow \infty} \frac{x}{e^{\lambda x}} = 0} \\ &= \lambda \left\{ \left[\frac{-x e^{-\lambda x}}{\lambda} \right]_0^{\infty} + \int_0^{\infty} \frac{e^{-\lambda x}}{\lambda} dx \right\} \\ &= \lambda \left[(0 - 0) - \frac{1}{\lambda^2} e^{-\lambda x} \Big|_0^{\infty} \right] \\ &= \frac{1}{\lambda} \end{aligned} \quad (3.59)$$

$$\mu_X = E[X] = \frac{1}{\lambda}$$

Variance

$$\begin{aligned} \sigma_X^2 &= \text{Var}[X] = E[X^2] - \{E[X]\}^2 \\ E[X^2] &= \int_{-\infty}^{\infty} x^2 f_X(x) dx \\ &= \int_0^{\infty} x^2 \lambda e^{-\lambda x} dx = \lambda \int_0^{\infty} x^2 e^{-\lambda x} dx \\ &= \lambda \left\{ \left[\frac{-x^2 e^{-\lambda x}}{\lambda} \right]_0^{\infty} - \frac{2x e^{-\lambda x}}{\lambda^2} \Big|_0^{\infty} + 2 \int_0^{\infty} \frac{e^{-\lambda x}}{\lambda^2} dx \right\} \\ &= \lambda \left\{ (0 - 0) - 2(0 - 0) - \frac{2}{\lambda^3} e^{-\lambda x} \Big|_0^{\infty} \right\} \\ &= \lambda \left(\frac{2}{\lambda^3} \right) = \frac{2}{\lambda^2} \end{aligned} \quad (3.60)$$

$$\text{Var}(X) = \sigma_X^2 = E[X^2] - \{E[X]\}^2 = \frac{2}{\lambda^2} - \left(\frac{1}{\lambda} \right)^2 = \frac{1}{\lambda^2} \quad (3.61)$$

Solved Problems

3.46 A random variable X has exponential pdf given by

$$f_X(x) = \begin{cases} 2e^{-2x} & \text{for } x \geq 0 \\ 0 & \text{for } x < 0 \end{cases}$$

Find mean and variance.

Solution

Given: $f_X(x) = \begin{cases} 2e^{-2x} & \text{for } x \geq 0 \\ 0 & \text{for } x < 0 \end{cases}$

Comparing with Eq. (3.58) we get

$$\lambda = 2$$

We know $E[X] = \frac{1}{\lambda} = \frac{1}{2} = 0.5$

$$\sigma_X^2 = \frac{1}{\lambda^2} = \frac{1}{4} = 0.25$$

3.47 The time T in hours between train arrivals at a railway station is a random variable with pdf

$$f_T(t) = 0.5 e^{-0.5t} \quad t \geq 0$$

- (a) What is the expected value of T ?
 (b) What is the variance of T ?

Solution Given: $f_T(t) = 0.5 e^{-0.5t} \quad t \geq 0$

$$E[T] = \frac{1}{0.5} = 2$$

$$\sigma_T^2 = \frac{1}{(0.5)^2} = \frac{1}{0.25} = 4$$

3.48 The pdf of the times T in weeks between employee strikes at a certain company is given by

$$f_T(t) = 0.01 e^{-0.01t} \quad t \geq 0$$

- (a) Find $P(20 < t < 40)$
 (b) What is the expected time between strikes at the company?

Solution Given: $f_T(t) = 0.01 e^{-0.01t} \quad t \geq 0$

$$\begin{aligned} \text{(a)} \quad F_T(t) &= \int_{-\infty}^t f_T(t) dt = \int_0^t 0.01 e^{-0.01t} dt \\ &= 0.01 \frac{e^{-0.01t}}{(-0.01)} \Bigg|_0^t = -(e^{-0.01t} - 1) \\ &= 1 - e^{-0.01t} \end{aligned}$$

3.50

Probability Theory and Random Processes

$$\begin{aligned} p(20 < t < 40) &= F_T(40) - F_T(20) = \{(1 - e^{-4}) - (1 - e^{-2})\} \\ &= e^{-2} - e^{-4} = 0.117 \end{aligned}$$

(b) The expected time between strikes at the company is

$$E[T] = \frac{1}{0.01} = 100$$

3.49 If X follows an exponential distribution with $P(X \leq 2) = P(X > 2)$, find the mean and variance

Solution The pdf of exponential random variable is given by

$$f_X(x) = \lambda e^{-\lambda x}, \quad x > 0$$

Given:

$$P(X \leq 2) = P(X > 2)$$

$$\begin{aligned} P(X \leq 2) &= \int_{-\infty}^2 f_X(x) dx = \int_0^2 \lambda e^{-\lambda x} dx = \lambda \left. \frac{e^{-\lambda x}}{-\lambda} \right|_0^2 \\ &= -(e^{-2\lambda} - 1) = 1 - e^{-2\lambda} \end{aligned}$$

$$P(X > 2) = 1 - P(X \leq 2) = 1 - (1 - e^{-2\lambda}) = e^{-2\lambda}$$

Equating the terms $P(X \leq 2)$ and $P(X > 2)$, we get

$$\begin{aligned} 1 - e^{-2\lambda} &= e^{-2\lambda} \Rightarrow 2e^{-2\lambda} = 1 \text{ and } e^{-2\lambda} = 0.5 \\ -2\lambda &= \ln(0.5); 2\lambda = \ln 2; \lambda = 0.5 \ln 2 \end{aligned}$$

$\Rightarrow$

$$\lambda = 0.3466$$

For exponential random variable

$$\text{Mean} = \frac{1}{\lambda} = \frac{1}{0.3466} = 2.885$$

$$\text{Variance} = \frac{1}{\lambda^2} = \frac{1}{(0.3466)^2} = 8.324$$

3.50 If X has uniform distribution in $(-2, 3)$ and Y has exponential distribution with parameter λ , find λ such that $\text{Var}(X) = \text{Var}(Y)$.

Solution X has uniform distribution in $(-2, 3)$.

$$\text{The pdf of } X \text{ is } f_X(x) = \begin{cases} \frac{1}{5} & \text{for } -2 < x < 3 \\ 0 & \text{otherwise} \end{cases}$$

$$\text{Var}(X) = \frac{(b-a)^2}{12} = \frac{25}{12}$$

Y is exponential random variable with pdf

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

The variance of exponential random variable is $\frac{1}{\lambda^2}$

$$\Rightarrow \frac{1}{\lambda^2} = \frac{25}{12}; \lambda^2 = \frac{12}{25} \Rightarrow \lambda = \frac{2\sqrt{3}}{5}$$

REVIEW QUESTION



15. For a random variable X with exponential distribution, derive expressions for mean and variance.

3.8.8 Rayleigh Distribution

The pdf of a Rayleigh random variable is

$$f_X(x) = \begin{cases} \frac{x}{\sigma^2} e^{-x^2/2\sigma^2} & \text{for } x > 0 \\ 0 & \text{for } x \leq 0 \end{cases} \quad (3.62)$$

Mean

$$\begin{aligned} E[X] &= \mu_X = \int_{-\infty}^{\infty} x f_X(x) dx \\ &= \int_0^{\infty} x \frac{x}{\sigma^2} e^{-x^2/2\sigma^2} dx = \frac{1}{\sigma^2} \int_0^{\infty} x^2 e^{-x^2/2\sigma^2} dx \\ &= \frac{1}{\sigma^2} \int_0^{\infty} x \cdot x e^{-x^2/2\sigma^2} dx = \int_0^{\infty} x \cdot d(-e^{-x^2/2\sigma^2}) \\ &= -x e^{-x^2/2\sigma^2} \Big|_0^{\infty} + \int_0^{\infty} e^{-x^2/2\sigma^2} dx \\ &= \sqrt{2\pi\sigma^2} \int_0^{\infty} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-x^2/2\sigma^2} dx \\ &= \frac{\sqrt{2\pi\sigma^2}}{2} \left[\underbrace{\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-x^2/2\sigma^2} dx}_{=1} \right] \\ &= \sigma \sqrt{\frac{\pi}{2}} \end{aligned}$$

Note. The term in brackets is the area under pdf curve of Gaussian random variable with $N(0, \sigma^2)$

Variance

$$\begin{aligned} \sigma_X^2 &= E[X^2] - \{E[X]\}^2 \\ E[X^2] &= \int_0^{\infty} x^2 \frac{x}{\sigma^2} e^{-x^2/2\sigma^2} dx = \frac{1}{\sigma^2} \int_0^{\infty} x^3 e^{-x^2/2\sigma^2} dx \end{aligned}$$

$$\begin{aligned}
&= \int_0^{\infty} x^2 d(-e^{-x^2/2\sigma^2}) \\
&= -x^2 e^{-x^2/2\sigma^2} \Big|_0^{\infty} + \int_0^{\infty} 2x e^{-x^2/2\sigma^2} dx = 2 \int_0^{\infty} x e^{-x^2/2\sigma^2} dx \\
&= 2\sigma^2 \int_0^{\infty} \frac{x}{\sigma^2} e^{-x^2/2\sigma^2} dx \tag{3.63}
\end{aligned}$$

Let $\frac{x^2}{2\sigma^2} = t \Rightarrow x dx = \sigma^2 dt$

$$\begin{aligned}
E[x^2] &= 2\sigma^2 \int_0^{\infty} e^{-t} dt = 2\sigma^2 \\
\sigma_X^2 &= \text{Var}(X) \\
&= E[X^2] - \{E[X]\}^2 \\
&= 2\sigma^2 - \left(\sigma \sqrt{\frac{\pi}{2}}\right)^2 \\
&= 2\sigma^2 - \left(\frac{\pi}{2}\right)\sigma^2 \\
&= \sigma^2 \left[2 - \frac{\pi}{2}\right] \tag{3.64}
\end{aligned}$$

3.8.9 Gamma Distribution

The probability density function of a random variable X that has a gamma distribution with parameters α and β is given by

$$f_X(x) = \begin{cases} \frac{1}{\beta^\alpha} \frac{x^{\alpha-1}}{\Gamma(\alpha)} e^{-x/\beta}, & x > 0 \\ 0 & \text{elsewhere} \end{cases} \tag{3.65}$$

The mean value is given by

$$\begin{aligned}
E[X] &= \int_{-\infty}^{\infty} x f_X(x) dx \\
&= \int_0^{\infty} x \frac{x^{\alpha-1}}{\beta^\alpha \Gamma(\alpha)} e^{-x/\beta} dx \\
&= \frac{1}{\beta^\alpha \Gamma(\alpha)} \int_0^{\infty} x^{\alpha} e^{-x/\beta} dx \tag{3.66}
\end{aligned}$$

$$\begin{aligned}
\text{Let } \frac{x}{\beta} = t \text{ then } dx &= \beta dt \\
&= \frac{1}{\beta^\alpha \Gamma(\alpha)} \int_0^\infty (\beta t)^\alpha e^{-t} \beta dt \\
&= \frac{\beta}{\Gamma(\alpha)} \int_0^\infty t^\alpha e^{-t} dt
\end{aligned} \tag{3.67}$$

$$\text{We have } \Gamma(\alpha) = \int_0^\infty x^{\alpha-1} e^{-x} dx \tag{3.68}$$

$$\Rightarrow \Gamma(\alpha + 1) = \int_0^\infty x^\alpha e^{-x} dx \tag{3.69}$$

$$E[X] = \frac{\beta}{\Gamma(\alpha)} \Gamma(\alpha + 1) \quad \boxed{\Gamma(n) = (n-1)!}$$

$$\frac{\beta \alpha!}{(\alpha-1)!} = \alpha \beta \tag{3.70}$$

Variance of Gamma Distribution

$$\begin{aligned}
E[X^2] &= \int_{-\infty}^\infty x^2 f_X(x) dx = \int_0^\infty x^2 \frac{1}{\beta^\alpha \Gamma(\alpha)} x^{\alpha-1} e^{-x/\beta} dx \\
&= \int_0^\infty x^{\alpha+1} \frac{1}{\beta^\alpha \Gamma(\alpha)} e^{-x/\beta} dx
\end{aligned}$$

$$\text{Let } \frac{x}{\beta} = t \quad \Rightarrow dx = \beta dt$$

$$\begin{aligned}
\Rightarrow E[X^2] &= \int_0^\infty (\beta t)^{\alpha+1} \frac{1}{\beta^\alpha \Gamma(\alpha)} e^{-t} (\beta dt) \\
&= \int_0^\infty \frac{\beta^2 t^{\alpha+1}}{\Gamma(\alpha)} e^{-t} dt \\
&= \frac{\beta^2}{\Gamma(\alpha)} \int_0^\infty t^{\alpha+1} e^{-t} dt \\
&= \frac{\beta^2 \Gamma(\alpha + 2)}{\Gamma(\alpha)} = \frac{\beta^2 (\alpha + 1)!}{(\alpha - 1)!} \\
&= (\alpha + 1) \alpha \beta^2
\end{aligned} \tag{3.71}$$

$$\begin{aligned}
\text{Var}(X) &= E[X^2] - \{E[X]\}^2 \\
&= \alpha(\alpha + 1) \beta^2 - [\alpha \beta]^2 \\
&= \alpha^2 \beta^2 + \alpha \beta^2 - \alpha^2 \beta^2 \\
&= \alpha \beta^2
\end{aligned} \tag{3.72}$$

3.8.10 Beta Distribution

The pdf of a random variable X with beta distribution is given by

$$f_X(x) = \frac{1}{\beta(\alpha, \beta)} x^{\alpha-1} (1-x)^{\beta-1} \quad \begin{matrix} 0 < x < 1 \\ \alpha > 0, \beta > 0 \end{matrix} \quad (3.73)$$

The mean value is given by

$$\begin{aligned} E[X] &= \int_{-\infty}^{\infty} x f_X(x) dx \\ &= \int_0^1 \frac{x}{\beta(\alpha, \beta)} x^{\alpha-1} (1-x)^{\beta-1} dx \\ &= \frac{1}{\beta(\alpha, \beta)} \int_0^1 x^{\alpha} (1-x)^{\beta-1} dx \end{aligned} \quad \boxed{\beta(\alpha, \beta) = \int_0^1 x^{\alpha-1} (1-x)^{\beta-1} dx}$$

$$\begin{aligned} E[X] &= \frac{1}{\beta(\alpha, \beta)} \beta(\alpha+1, \beta) \\ &= \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \cdot \frac{\Gamma(\alpha+1)\Gamma(\beta)}{\Gamma(\alpha+1+\beta)} \end{aligned} \quad (3.74)$$

We have

$$\Gamma[\alpha] = (\alpha-1)! \quad (3.75)$$

$$\begin{aligned} E[X] &= \frac{(\alpha+\beta-1)!}{(\alpha-1)!(\beta-1)!} \cdot \frac{\alpha!(\beta-1)!}{(\alpha+\beta)!} \\ &= \frac{\alpha}{\alpha+\beta} \end{aligned} \quad (3.76)$$

$$\begin{aligned} E[X^2] &= \int_{-\infty}^{\infty} x^2 f_X(x) dx \\ &= \frac{1}{\beta(\alpha, \beta)} \int_0^1 x^2 x^{\alpha-1} (1-x)^{\beta-1} dx = \frac{1}{\beta(\alpha, \beta)} \int_0^1 x^{\alpha+1} (1-x)^{\beta-1} dx \\ &= \frac{1}{\beta(\alpha, \beta)} \beta(\alpha+2, \beta) \\ &= \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \cdot \frac{\Gamma(\alpha+2)\Gamma(\beta)}{\Gamma(\alpha+\beta+2)} \\ &= \frac{(\alpha+\beta-1)!(\alpha+1)!}{(\alpha-1)!(\alpha+\beta+1)!} = \frac{\alpha(\alpha+1)}{(\alpha+\beta+1)(\alpha+\beta)} \end{aligned} \quad (3.77)$$

$$\begin{aligned}
\text{Var}(X) &= E[X^2] - \{E[X]\}^2 \\
&= \frac{\alpha(\alpha+1)}{(\alpha+\beta+1)(\alpha+\beta)} - \frac{\alpha^2}{(\alpha+\beta)^2} \\
&= \frac{\alpha(\alpha+1)(\alpha+\beta) - \alpha^2(\alpha+\beta+1)}{(\alpha+\beta+1)(\alpha+\beta)^2} \\
&= \frac{\alpha\beta}{(\alpha+\beta+1)(\alpha+\beta)^2} \tag{3.78}
\end{aligned}$$

Means and Variance of Weibull Distribution The pdf of a random variable X with Weibull distribution is given by

$$f_X(x) = \begin{cases} \alpha\beta x^{\beta-1} e^{-\alpha x^\beta}, & x > 0, \alpha > 0 \text{ and } \beta > 0 \\ 0 & \text{otherwise} \end{cases} \tag{3.79}$$

$$\begin{aligned}
E[X] &= \int_{-\infty}^{\infty} x f_X(x) dx \\
&= \int_0^{\infty} \alpha\beta x^\beta e^{-\alpha x^\beta} dx = \alpha\beta \int_0^{\infty} x^\beta e^{-\alpha x^\beta} dx
\end{aligned}$$

$$\text{Let } \alpha x^\beta = t \Rightarrow x = \left(\frac{t}{\alpha}\right)^{1/\beta}$$

$$dx = \frac{1}{\beta} \left(\frac{t}{\alpha}\right)^{\frac{1}{\beta}-1} \left(\frac{1}{\alpha}\right) dt$$

$$E[X] = \alpha\beta \int_0^{\infty} \left(\frac{t}{\alpha}\right) e^{-t} \frac{1}{\alpha\beta} \left(\frac{t}{\alpha}\right)^{\frac{1}{\beta}-1} dt$$

$$= \left(\frac{1}{\alpha}\right)^{\frac{1}{\beta}} \int_0^{\infty} (t)^{\frac{1}{\beta}} e^{-t} dt \quad \boxed{\int_0^{\infty} x^{\alpha-1} e^{-x} dx = \Gamma(\alpha)}$$

$$= \left(\frac{1}{\alpha}\right)^{\frac{1}{\beta}} \int_0^{\infty} (t)^{\frac{1}{\beta}} e^{-t} dt$$

$$= \left(\frac{1}{\alpha}\right)^{\frac{1}{\beta}} \Gamma\left(1 + \frac{1}{\beta}\right) \tag{3.80}$$

$$E[X^2] = \int_{-\infty}^{\infty} x^2 f_X(x) dx$$

$$= \int_0^{\infty} \alpha \beta x^2 x^{\beta-1} e^{-\alpha x^{\beta}} dx = \int_0^{\infty} \alpha \beta x^{\beta+1} e^{-\alpha x^{\beta}} dx$$

Let $\alpha x^{\beta} = t$

Then $x = \left(\frac{t}{\alpha}\right)^{\frac{1}{\beta}}$

$$dx = \frac{1}{\beta} \left(\frac{t}{\alpha}\right)^{\frac{1}{\beta}-1} \left(\frac{1}{\alpha}\right) dt$$

$$E[X^2] = \int_0^{\infty} \alpha \beta \left(\frac{t}{\alpha}\right)^{\frac{\beta+1}{\beta}} e^{-t} \left[\frac{1}{\beta}\right] \left(\frac{t}{\alpha}\right)^{\frac{1}{\beta}-1} \frac{1}{\alpha} dt$$

$$= \int_0^{\infty} \left(\frac{t}{\alpha}\right)^{2/\beta} e^{-t} dt$$

$$= \left(\frac{1}{\alpha}\right)^{\frac{2}{\beta}} \int_0^{\infty} (t)^{2/\beta} e^{-t} dt \quad \boxed{\int_0^{\infty} x^{\alpha-1} e^{-x} dx = \Gamma(\alpha)}$$

$$= \left(\frac{1}{\alpha}\right)^{\frac{2}{\beta}} \Gamma\left(1 + \frac{2}{\beta}\right) \quad (3.81)$$

$$\text{Var}(X) = E[X^2] - \{E[X]\}^2$$

$$= \left(\frac{1}{\alpha}\right)^{\frac{2}{\beta}} \Gamma\left(1 + \frac{2}{\beta}\right) - \left(\frac{1}{\alpha}\right)^{\frac{2}{\beta}} \left[\Gamma\left(1 + \frac{1}{\beta}\right)\right]^2$$

$$= \left(\frac{1}{\alpha}\right)^{\frac{2}{\beta}} \left\{ \Gamma\left(1 + \frac{2}{\beta}\right) - \left[\Gamma\left(1 + \frac{1}{\beta}\right)\right]^2 \right\} \quad (3.82)$$

Mean and Variance of Chi-square Distribution The pdf of X with chi-square distribution is given by

$$f_X(x) = \begin{cases} \frac{x^{\frac{n}{2}-1}}{2^{\frac{n}{2}} \Gamma\left(\frac{n}{2}\right)} e^{-x/2}, & x \geq 0 \\ 0 & \text{otherwise} \end{cases} \quad (3.83)$$

$$E[X] = \int_{-\infty}^{\infty} x f_X(x) dx$$

$$= \int_0^{\infty} x \frac{x^{\frac{n}{2}-1}}{2^{\frac{n}{2}} \Gamma\left(\frac{n}{2}\right)} e^{-x/2} dx = \int_0^{\infty} \frac{x^{\frac{n}{2}}}{2^{\frac{n}{2}} \Gamma\left(\frac{n}{2}\right)} e^{-x/2} dx$$

Let $\frac{x}{2} = t$

Then $E[X] = \int_0^{\infty} \frac{2^{n/2} t^{n/2} e^{-t}}{2^{n/2} \Gamma\left(\frac{n}{2}\right)} (2 dt)$

$$= \frac{2}{\Gamma\left(\frac{n}{2}\right)} \Gamma\left(\frac{n}{2} + 1\right) = \frac{2\left(\frac{n}{2}\right)!}{\left(\frac{n}{2} - 1\right)!} = 2\left(\frac{n}{2}\right) = n$$

$$\boxed{E[X] = n}$$

(3.84)

$$E[X^2] = \int_0^{\infty} x^2 \frac{x^{\frac{n}{2}-1}}{2^{\frac{n}{2}} \Gamma\left(\frac{n}{2}\right)} e^{-\frac{x}{2}} dx = \int_0^{\infty} \frac{x^{\frac{n}{2}+1}}{2^{\frac{n}{2}} \Gamma\left(\frac{n}{2}\right)} e^{-\frac{x}{2}} dx$$

Let $\frac{x}{2} = t$, then

$$E[X^2] = \frac{2}{2^{\frac{n}{2}} \Gamma\left(\frac{n}{2}\right)} \int_0^{\infty} (2)^{\frac{n}{2}+1} (t)^{\frac{n}{2}+1} e^{-t} dt$$

$$= \frac{4 \Gamma\left(\frac{n}{2} + 2\right)}{\Gamma\left(\frac{n}{2}\right)} = 4 \left(\frac{n}{2} + 1\right) \left(\frac{n}{2}\right) = n(n+2)$$

(3.85)

$$\begin{aligned} \text{Var}(X) &= E[X^2] - \{E[X]\}^2 \\ &= n(n+2) - (n)^2 \\ &= 2n \end{aligned}$$

$$\boxed{\text{var}(X) = 2n}$$

(3.86)

REVIEW QUESTION



16. Derive expression for mean and variance of a random variable with Rayleigh distribution.



SKEW AND SKEWNESS COEFFICIENT 3.9

The third central moment μ_3 of a random variable X with pdf $f_X(x)$ is defined as

$$\mu_3 = E[(X - \mu_X)^3] = \int_{-\infty}^{\infty} (x - \mu_X)^3 f_X(x) dx \quad (3.87)$$

For a discrete random variable,

$$\mu_3 = \sum_i (x_i - \mu_X)^3 p_X(x_i) \quad (3.88)$$

A skew is a measure of the asymmetry of the density function of a random variable about its mean. If the pdf is symmetric about $x = \mu_X$ then it has zero skew. In fact, for a random variable that has symmetric pdf about $x = \mu_X$ has $\mu_n = 0$ for all odd values of n .

Skewness Coefficient The normalized third central moment $\frac{\mu_3}{\sigma_X^3}$ is known as the skewness coefficient. It is given by

$$\text{Skewness coefficient} = \frac{\mu_3}{\sigma_X^3} = \frac{E[(X - \mu_X)^3]}{E[(X - \mu_X)^3]} \quad (3.89)$$

The skewness coefficient is a dimensionless quantity. It is positive if the random variable has a pdf skewed to the right as shown in Fig 3.4(a) and negative if skewed to the left as shown in Fig. 3.4(c).

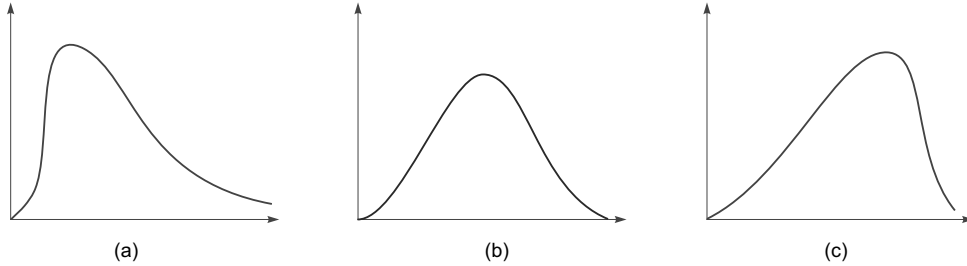


Fig. 3.4 Skewness Coefficient



KURTOSIS 3.10

The fourth central moment is called *kurtosis* and is a measure of the peakedness of a random variable near the mean. The coefficient kurtosis is dimensionless and is given as

$$C_k = \frac{E[(X - \mu_X)^4]}{\sigma_X^4} \quad (3.90)$$

The Kurtosis for a normal distribution is three. The alternate definition for Kurtosis is

$$\frac{E[(X - \mu_X)^4]}{\sigma_X^4} - 3 \quad (3.91)$$

which is also known as *excess Kurtosis*.

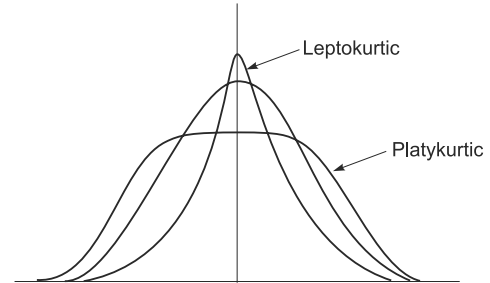


Fig. 3.5 Kurtosis

This definition is used so that the normal distribution has a Kurtosis of zero.

Distributions with negative excess Kurtosis are known as *Platykurtic distribution*. Examples are uniform distribution and Bernoulli distribution. Distributions with positive excess kurtosis are known as *Leptokurtic*. Examples are Laplace distribution, exponential distribution and position distribution.

REVIEW QUESTIONS



17. Define skewness and skewness coefficient.
18. Define Kurtosis.

RELATIONSHIP BETWEEN CENTRAL MOMENTS AND MOMENTS ABOUT ORIGIN

3.11



We know

$$m_n = E[X^n] = \int_{-\infty}^{\infty} x^n f_X(x) dx$$

and
$$\mu_n = E[(X - \mu_X)^n] = \int_{-\infty}^{\infty} (x - \mu_X)^n f_X(x) dx$$

$$\mu_n = E[(X - \mu_X)^n]$$

Using binomial expansion,

$$\begin{aligned} \mu_n &= E \left[\sum_{k=0}^n \binom{n}{k} (X)^k (-\mu_X)^{n-k} \right] \\ &= \sum_{k=0}^n \binom{n}{k} E(X^k) (-\mu_X)^{n-k} \\ &= \sum_{k=0}^n \binom{n}{k} m_k (-\mu_X)^{n-k} \\ m_n &= E[(X - \mu_X + \mu_X)^n] \\ &= E \left[\sum_{k=0}^n (X - \mu_X)^k (\mu_X)^{n-k} \right] \\ &= \sum_{k=0}^n \binom{n}{k} E[(X - \mu_X)^k] \mu_X^{n-k} \\ &= \sum_{k=0}^n \binom{n}{k} \mu_X^k \mu_X^{n-k} \\ \mu_0 &= m_0 \end{aligned} \tag{3.92}$$

$$\begin{aligned}
\mu_1 &= \sum_{k=0}^1 \binom{1}{k} m_k (-\mu_X)^{1-k} \\
&= \binom{1}{0} m_0 (-\mu_X)^{1-0} + \binom{1}{1} m_1 (-\mu_X)^{1-1} \\
&= m_1 - m_0 \mu_X
\end{aligned} \tag{3.93}$$

Mean

$$\begin{aligned}
E[X] &= \mu_1 = m_1 - m_0 m_0 \\
&= m_1(1 - m_0) = 0 \\
\mu_2 &= \sum_{k=0}^1 \binom{2}{k} m_k (-\mu_X)^{2-k} \\
&= \binom{2}{0} m_0 (-\mu_X)^2 + \binom{2}{1} m_1 (-\mu_X) + \binom{2}{2} m_2 (-\mu_X)^0 \\
&= m_0 (-m_1)^2 + 2m_1 (-m_1) + m_2 \\
&= m_1^2 - 2m_1^2 + m_2 \\
&= m_2 - m_1^2 \\
\mu_3 &= \binom{3}{0} m_0 (-\mu_X)^3 + \binom{3}{1} m_1 (-\mu_X)^2 + \binom{3}{2} m_2 (-\mu_X)^1 + \binom{3}{3} m_3 (-\mu_X)^0 \\
&= \binom{3}{0} m_0 (-m_1^3) + \binom{3}{1} m_1 (-m_1)^2 + \binom{3}{2} m_2 (-m_1) + \binom{3}{3} m_3 (-m_1)^0 \\
&= -m_0 m_1^3 + 3m_1^3 - 3m_1 m_2 + m_3 \\
&= m_3 - 3m_1 (m_2 - m_1^2) - m_1^3 \\
\Rightarrow \mu_3 &= m_3 - 3\mu_X \sigma_X^2 - m_1^3
\end{aligned} \tag{3.94}$$

Solved Problems

3.51 Find skew and skewness coefficient for the exponential density function

$$f_X(x) = \begin{cases} \frac{1}{b} e^{-x/b} & \text{for } x \geq 0 \\ 0 & \text{for } x < 0 \end{cases}$$

Solution

Given:
$$f_X(x) = \begin{cases} \frac{1}{b} e^{-x/b} & \text{for } x \geq 0 \\ 0 & \text{for } x < 0 \end{cases}$$

$$\begin{aligned}
E[X] &= \int_{-\infty}^{\infty} x f_X(x) dx = \int_0^{\infty} \frac{x}{b} e^{-x/b} dx \\
&= \frac{1}{b} \left\{ -bx e^{-x/b} \Big|_0^{\infty} + b \int_0^{\infty} e^{-x/b} dx \right\} \\
&= \frac{1}{b} \left\{ b \frac{e^{-x/b}}{(-1/b)} \Big|_0^{\infty} \right\} = \frac{b^2}{b} = b \\
E[X] &= b \tag{3.96}
\end{aligned}$$

$$\begin{aligned}
E[X^2] &= \int_{-\infty}^{\infty} x^2 f_X(x) dx = \frac{1}{b} \int_0^{\infty} x^2 e^{-x/b} dx \\
&= \frac{1}{b} \left\{ -bx^2 e^{-x/b} \Big|_0^{\infty} - 2xb^2 e^{-x/b} \Big|_0^{\infty} + 2b^2 \int_0^{\infty} e^{-x/b} dx \right\} \\
&= \frac{1}{b} \left\{ 2b^2 \frac{e^{-x/b}}{(-1/b)} \Big|_0^{\infty} \right\} = \frac{2b^3}{b} = 2b^2 \tag{3.97}
\end{aligned}$$

$$\begin{aligned}
\sigma_X^2 &= E[X^2] - \{E[X]\}^2 \\
&= 2b^2 - b^2 = b^2
\end{aligned}$$

$$\begin{aligned}
E[X^3] &= \int_{-\infty}^{\infty} x^3 f_X(x) dx = \frac{1}{b} \int_0^{\infty} x^3 e^{-x/b} dx \\
&= \frac{1}{b} \left\{ -bx^3 e^{-x/b} \Big|_0^{\infty} - 3b^2 x^2 e^{-x/b} \Big|_0^{\infty} - 6b^3 x e^{-x/b} \Big|_0^{\infty} + 6b^3 \int_0^{\infty} e^{-x/b} dx \right\} \\
&= 6b^3 \tag{3.98}
\end{aligned}$$

$$\begin{aligned}
\mu_3 &= E[X^3] - 3\mu_X \sigma_X^2 - \mu_X^3 \\
&= 6b^3 - 3(b)(b^2) - b^3 = 2b^3 \tag{3.99}
\end{aligned}$$

$$\text{Skewness coefficient} = \frac{\mu_X}{\sigma_X^3} = \frac{2b^3}{b^3} = 2$$

3.52 Find kurtosis for the exponential density function given in the above problem.

Solution

$$E[X^4] = \int_{-\infty}^{\infty} x^4 f_X(x) dx = \frac{1}{b} \int_0^{\infty} x^4 e^{-x/b} dx$$

$$\begin{aligned}
&= \frac{1}{b} \left\{ -bx^4 e^{-x/b} \Big|_0^\infty - 4x^3 b^2 e^{-x/b} \Big|_0^\infty - 12b^3 x^2 e^{-x/b} \Big|_0^\infty - 24x b^4 e^{-x/b} \Big|_0^\infty + 24b^4 \int_0^\infty e^{-x/b} dx \right\} \\
&= \frac{1}{b} (24b^5) = 24b^4 \\
\text{Kurtosis} &= \frac{E[(X - \mu_X)^4]}{\sigma_X^4} \\
E[(X - \mu_X)^4] &= E[X^4 - 4X^3 \mu_X + 6X^2 \mu_X^2 - 4X \mu_X^3 + \mu_X^4] \\
&= E[X^4] - 4\mu_X E[X^3] + 6\mu_X^2 E[X^2] - 4\mu_X^3 E[X] + \mu_X^4 \\
&= 24b^4 - 4b(6b^3) + 6b^2(2b^2) - 4b^3(b) + b^4 \\
&= 9b^4 \\
\text{Kurtosis} &= \frac{9b^4}{b^4} = 9
\end{aligned}$$

3.53 Consider a random variable with Laplace distribution with a pdf given by

$$f_X(x) = \frac{b}{2} e^{-b|x|}$$

Find the mean, variance, skewness coefficient and kurtosis.

Solution The pdf is symmetrical about $x = 0$. Therefore, the mean is zero and all central moments are equal to moments at origin. Also due to symmetry of pdf, all odd moments are zero. That is $\mu_1 = 0$; $\mu_3 = 0$ and $\mu_5 = 0$.

$$\begin{aligned}
\sigma_X^2 &= E[X^2] - \mu_X^2 = E[X^2] \\
E[X^2] &= \int_{-\infty}^{\infty} x^2 \frac{b}{2} e^{-b|x|} dx = \frac{b}{2} \int_{-\infty}^{\infty} x^2 e^{-b|x|} dx \\
&= 2 \left(\frac{b}{2} \right) \int_0^{\infty} x^2 e^{-bx} dx \\
&= b \left\{ \frac{-x^2 e^{-bx}}{b} \Big|_0^{\infty} - \frac{2x}{b^2} e^{-bx} \Big|_0^{\infty} + \frac{2}{b^2} \int_0^{\infty} e^{-bx} dx \right\} \\
&= b \left\{ \frac{2}{b^2} \left(\frac{e^{-bx}}{-b} \Big|_0^{\infty} \right) \right\} = \frac{2}{b^2} \\
\Rightarrow \quad \sigma_X^2 &= \frac{2}{b^2}
\end{aligned}$$

The skewness coefficient is zero

$$\text{Kurtosis} = \frac{E[(X - \mu_X)^4]}{\sigma_X^4}$$

$$\begin{aligned} E[(X - \mu_X)^4] &= E[X^4] - 4\mu_X E[X^3] + 6\mu_X^2 E[X^2] - 4\mu_X^3 E[X] + \mu_X^4 \\ &= E[X^4] \quad [\because \mu_X = 0] \end{aligned}$$

$$E[X^4] = \int_{-\infty}^{\infty} x^4 f_X(x) dx = \frac{b}{2} \int_{-\infty}^{\infty} x^4 e^{-b|x|} dx$$

$$= b \int_0^{\infty} x^4 e^{-bx} dx$$

$$= b \left\{ \frac{-x^4 e^{-bx}}{b} \Big|_0^{\infty} - \frac{4x^3}{b^2} e^{-bx} \Big|_0^{\infty} - \frac{12x^2}{b^3} e^{-bx} \Big|_0^{\infty} - \frac{24xe^{-bx}}{b^4} \Big|_0^{\infty} + \frac{24}{b^4} \int_0^{\infty} e^{-bx} dx \right\}$$

$$= b \left\{ \frac{24}{b^4} \left(\frac{e^{-bx}}{-b} \right) \Big|_0^{\infty} \right\}$$

$$= \frac{24}{b^4}$$

$$\text{Kurtosis} = \frac{E[(X - \mu_X)^4]}{\sigma_X^4} = \frac{24/b^4}{4/b^4} = 6$$

3.54 The exponential density function is given by

$$f_X(x) = \begin{cases} \frac{1}{b} e^{-(x-a)/b} & x \geq a \\ 0 & x < a \end{cases}$$

Find variance and coefficient of skewness.

Solution

$$\text{Given: } f_X(x) = \begin{cases} \frac{1}{b} e^{-(x-a)/b} & x \geq a \\ 0 & x < a \end{cases}$$

$$\mu_X = E[X] = \int_{-\infty}^{\infty} x f_X(x) dx = \frac{1}{b} \int_a^{\infty} x e^{-(x-a)/b} dx$$

$$= \frac{1}{b} e^{a/b} \int_a^{\infty} x e^{-x/b} dx$$

$$\begin{aligned}
&= \frac{1}{b} e^{a/b} \left\{ -xb e^{-x/b} \Big|_a^\infty + \int_a^\infty b e^{-x/b} dx \right\} \\
&= \frac{1}{b} e^{a/b} \{ ab e^{-a/b} - b^2 (-e^{-a/b}) \} \\
&= \frac{1}{b} e^{a/b} e^{-a/b} (ab + b^2) = a + b \\
E[X] &= a + b \\
E[X^2] &= \int_{-\infty}^\infty x^2 f_X(x) dx = \frac{1}{b} e^{a/b} \int_a^\infty x^2 e^{-x/b} dx \\
&= \frac{1}{b} e^{a/b} \left\{ -bx^2 e^{-x/b} \Big|_a^\infty - 2xb^2 e^{-x/b} \Big|_a^\infty + 2b^2 \int_a^\infty e^{-x/b} dx \right\} \\
&= \frac{1}{b} e^{a/b} \{ e^{-a/b} (a^2 b + 2ab^2 + 2b^3) \} \\
E[X^2] &= a^2 + 2ab + 2b^2 \\
\sigma_X^2 &= E[X^2] - \{E[X]\}^2 = (a^2 + 2ab + 2b^2) - (a^2 + b^2 + 2ab) \\
&= b^2 \\
\boxed{\sigma_X^2} &= b^2 \\
E[X^3] &= \int_{-\infty}^\infty x^3 f_X(x) dx = \frac{1}{b} e^{a/b} \int_a^\infty x^3 e^{-x/b} dx \\
&= \frac{1}{b} e^{a/b} \left\{ -bx^3 e^{-x/b} \Big|_a^\infty - 3b^2 x^2 e^{-x/b} \Big|_a^\infty - 6b^3 x e^{-x/b} \Big|_a^\infty - 6b^4 e^{-x/b} \Big|_a^\infty \right\} \\
&= \frac{1}{b} e^{a/b} \left\{ a^3 b e^{-a/b} + 3b^2 a^2 e^{-a/b} + 6b^3 a e^{-a/b} + 6b^4 e^{-a/b} \right\} \\
&= \frac{1}{b} \{ a^3 b + 3b^2 a^2 + 6b^3 a + 6b^4 \} \\
E[X^3] &= a^3 + 3a^2 b + 6ab^2 + 6b^3 \\
\text{Coefficient of skewness} &= \frac{\mu_3}{\sigma_X^3} \\
\mu_3 &= m_3 - 3\mu_X \sigma_X^2 - m_1^3 \\
&= (a^3 + 3a^2 b + 6ab^2 + 6b^3) - 3(a+b)b^2 - (a+b)^3 \\
&= 2b^3 \\
\text{Coefficient of skewness} &= \frac{\mu_3}{\sigma_X^3} = \frac{2b^3}{b^3} = 2
\end{aligned}$$

3.55 Prove that central moments μ_n are related to moments m_k about the origin by

$$\mu_n = \sum_{k=0}^n \binom{n}{k} (-\mu_X)^{n-k} m_k$$

Solution

$$\begin{aligned} \mu_n &= E[(X - \mu_X)^n] \\ &= E\left[\sum_{k=0}^n \binom{n}{k} X^k (-\mu_X)^{n-k}\right] \\ &= \sum_{k=0}^n \binom{n}{k} E[X^k] (-\mu_X)^{n-k} \\ &= \sum_{k=0}^n \binom{n}{k} (-\mu_X)^{n-k} m_k \end{aligned}$$

3.56 A random variable X has the density function

$$f_X(x) = \begin{cases} k(x^2 - 5x + 8) & 0 \leq x \leq 4 \\ 0 & \text{otherwise} \end{cases}$$

Find the moments m_0, m_1, m_2 and μ_2 .

Solution

We know
$$\int_{-\infty}^{\infty} f_X(x) dx = 1$$

$$\int_0^4 k(x^2 - 5x + 8) dx = 1$$

$$k \left[\frac{x^3}{3} \Big|_0^4 - \frac{5}{2} x^2 \Big|_0^4 + 8x \Big|_0^4 \right] = 1$$

$$\Rightarrow k \left[\frac{64}{3} - 40 + 32 \right] = 1$$

from which, $k = \frac{3}{40}$

So
$$f_X(x) = \begin{cases} \frac{3}{40}(x^2 - 5x + 8) & 0 \leq x \leq 4 \\ 0 & \text{otherwise} \end{cases}$$

$$m_n = \int_{-\infty}^{\infty} x^n f_X(x) dx = \int_0^4 x^n \frac{3}{40} (x^2 - 5x + 8) dx$$

$$\begin{aligned}
&= \frac{3}{40} \left[\int_0^4 (x^{n+2} - 5x^{n+1} + 8x^n) dx \right] \\
&= \frac{3}{40} \left[\frac{4^{n+3}}{n+3} - \frac{5(4^{n+2})}{n+2} + \frac{8(4^{n+1})}{n+1} \right] \\
m_0 &= \frac{3}{40} \left[\frac{4^3}{3} - \frac{5(16)}{2} + 8(4) \right] = \frac{3}{40} \left(\frac{80}{6} \right) = 1 \\
m_1 &= \frac{3}{40} \left[\frac{4^4}{4} - \frac{5(4)^3}{3} + \frac{8(4)^2}{2} \right] = \frac{3}{40} \left(\frac{64}{3} \right) = 1.6 \\
m_2 &= \frac{3}{40} \left[\frac{4^5}{5} - \frac{5(4)^4}{4} + \frac{8(4)^3}{3} \right] = \frac{3}{40} \left(\frac{832}{15} \right) = 4.16 \\
\mu_2 &= m_2 - m_1^2 \\
&= 4.16 - 2.56 = 1.6
\end{aligned}$$

3.57 Find the skew and coefficient of skewness for a Rayleigh random variable

$$\begin{aligned}
f_X(x) &= \frac{2}{b} x e^{-x^2/b} \quad x \geq 0 \\
&= 0 \quad x < 0
\end{aligned}$$

Solution

Given: $f_X(x) = \frac{2}{b} x e^{-x^2/b} \quad x \geq 0$
 $= 0 \quad x < 0$

$$\begin{aligned}
E[X] &= \mu_X = \int_{-\infty}^{\infty} x f_X(x) dx \\
&= \int_0^{\infty} x \left(\frac{2}{b} x e^{-x^2/b} \right) dx = \int_0^{\infty} x d(-e^{-x^2/b}) \\
&= -x e^{-x^2/b} \Big|_0^{\infty} + \int_0^{\infty} e^{-x^2/b} dx \quad \boxed{\lim_{x \rightarrow \infty} x e^{-x^2/b} = 0}
\end{aligned}$$

$$\begin{aligned}
\Rightarrow E[X] &= \int_0^{\infty} e^{-x^2/b} dx = \sqrt{b\pi} \int_0^{\infty} \frac{e^{-x^2/b}}{\sqrt{b\pi}} dx \\
&= \frac{\sqrt{b\pi}}{2} \underbrace{\int_{-\infty}^{\infty} \frac{e^{-x^2/b}}{\sqrt{b\pi}} dx}_{=1}
\end{aligned}$$

$$E[X] = \frac{\sqrt{b\pi}}{2}$$

$$\begin{aligned} E[X^2] &= \int_{-\infty}^{\infty} x^2 f_x(x) dx = \int_0^{\infty} x^2 \left(\frac{2}{b} x e^{-x^2/b} \right) dx \\ &= \int_0^{\infty} x^2 d(-e^{-x^2/b}) \\ &= -x^2 e^{-x^2/b} \Big|_0^{\infty} + \int_0^{\infty} 2x e^{-x^2/b} dx \\ &= \int_0^{\infty} 2x e^{-x^2/b} dx \end{aligned}$$

$$\begin{aligned} E[X^2] &= b \int_0^{\infty} e^{-t} dt \quad \boxed{\text{Let } \frac{x^2}{b} = t \cdot 2x dx = bdt} \\ &= b(-e^{-t}) \Big|_0^{\infty} = b \end{aligned}$$

$$\begin{aligned} \sigma_X^2 &= E[X^2] - \{E[X]\}^2 \\ &= b - \frac{b\pi}{4} = b \left[\frac{4 - \pi}{4} \right] \end{aligned}$$

$$\begin{aligned} E[X^3] &= \int_0^{\infty} x^3 \left(\frac{2}{b} x e^{-x^2/b} \right) dx \\ &= \int_0^{\infty} x^3 d(-e^{-x^2/b}) \\ &= \underbrace{-x^3 e^{-x^2/b}}_{=0} \Big|_0^{\infty} + \int_0^{\infty} 3x^2 e^{-x^2/b} dx \\ &= \int_0^{\infty} 3x^2 e^{-x^2/b} dx = 3 \int_0^{\infty} x (x e^{-x^2/b}) dx \\ &= \frac{3b}{2} \int_0^{\infty} x \left(\frac{2}{b} x e^{-x^2/b} \right) dx = \frac{3b}{2} \int_0^{\infty} x d(-e^{-x^2/b}) \\ &= \frac{3b}{2} \left\{ \underbrace{-x e^{-x^2/b}}_{=0} \Big|_0^{\infty} + \int_0^{\infty} e^{-x^2/b} dx \right\} \\ &= \frac{3b}{2} \sqrt{b\pi} \int_0^{\infty} \frac{e^{-x^2/b}}{\sqrt{b\pi}} dx \end{aligned}$$

$$= \frac{3b}{4} \sqrt{b\pi} \underbrace{\int_{-\infty}^{\infty} \frac{1}{\sqrt{b\pi}} e^{-x^2/b} dx}_{=1}$$

$$E[X^3] = \frac{3b}{4} \sqrt{b\pi}$$

$$\begin{aligned} \mu_3 &= E[(X - \mu_X)^3] \\ &= E[X^3] - 3E[X] E[X^2] + 2\{E[X]\}^3 \\ &= \frac{b^{3/2} 3\sqrt{\pi}}{4} - 3\sqrt{\frac{b\pi}{4}} b + 2 \frac{(b\pi)^{3/2}}{8} \\ &= \frac{\sqrt{\pi}}{4} (\pi - 3) b^{3/2} \end{aligned}$$

$$\begin{aligned} \text{Coefficient of skewness} &= \frac{\mu_3}{\sigma_X^3} = \frac{\sqrt{\frac{\pi}{4}} (\pi - 3) b^{3/2}}{b^{3/2} \left(\frac{4 - \pi}{4}\right)^{3/2}} \\ &= \frac{2\sqrt{\pi} (\pi - 3)}{(4 - \pi)^{3/2}} = 0.6311 \end{aligned}$$



Practice Problem

3.29 Find the skewness and skewness coefficient of random variable X with pdf

$$f_X(x) = \begin{cases} \frac{2}{9}x, & 0 \leq x \leq 3 \\ 0 & \text{otherwise} \end{cases} \quad (\text{Ans: } -0.0707)$$



CHEBYSHEV'S INEQUALITY 3.12

Chebyshev's inequality gives us a way of understanding how the variance measures the probability of deviation about μ_X . It places a bound on the probability that an experimental value of a random variable X will differ from the mean μ_X by more than a fixed number ε . The value $|X - \mu_X|$ is the measure of the amount by which the value of X differs above and below from its mean μ_X . Chebyshev's inequality states that the upper bound for this difference is directly proportional to the variance and inversely proportional to ε^2 . That is, for any $\varepsilon > 0$,

$$P[|X - \mu_X| \geq \varepsilon] \leq \frac{\sigma_X^2}{\varepsilon^2} \quad (3.100)$$

We know

$$\sigma_X^2 = \int_{-\infty}^{\infty} (x - \mu_X)^2 f_X(x) dx$$

$$= \int_{-\infty}^{(\mu_X - \varepsilon)} (x - \mu_X)^2 f_X(x) dx + \int_{\mu_X - \varepsilon}^{\mu_X + \varepsilon} (x - \mu_X)^2 f_X(x) dx + \int_{\mu_X + \varepsilon}^{\infty} (x - \mu_X)^2 f_X(x) dx$$

Since $\int_{\mu_X - \varepsilon}^{\mu_X + \varepsilon} (x - \mu_X)^2 f_X(x) dx > 0$, if we omit the second integral then we can write

$$\sigma_X^2 \geq \int_{-\infty}^{(\mu_X - \varepsilon)} (x - \mu_X)^2 f_X(x) dx + \int_{\mu_X + \varepsilon}^{\infty} (x - \mu_X)^2 f_X(x) dx$$

Now, $(x - \mu_X)^2 \geq \varepsilon^2$ if and only if $|x - \mu_X| \geq \varepsilon$.

Therefore,

$$\begin{aligned} \sigma_X^2 &\geq \int_{-\infty}^{\mu_X - \varepsilon} \varepsilon^2 f_X(x) dx + \int_{\mu_X + \varepsilon}^{\infty} \varepsilon^2 f_X(x) dx \\ &= \varepsilon^2 [P_X(X \leq \mu_X - \varepsilon) + P_X(X \geq \mu_X + \varepsilon)] \end{aligned}$$

$$\Rightarrow P[|X - \mu_X| \geq \varepsilon] \leq \frac{\sigma_X^2}{\varepsilon^2} \quad (3.101)$$

Solved Problems



3.58 A random variable X has a mean of 9 and a variance of 3. Use Chebyshev's inequality to obtain an upper bound for $P[|X - 9| \geq 3]$.

Solution From Chebyshev's inequality

$$P[|X - 9| \geq 3] \leq \frac{\sigma_X^2}{3^2} = \frac{3}{3^2} = \frac{1}{3}$$

3.59 A random variable has an average of 10 and variance 5. Find the probability that X will be between 6 and 14.

Solution By Chebyshev's inequality,

$$\begin{aligned} P[|X - \mu_X| \geq \varepsilon] &\leq \frac{\sigma_X^2}{\varepsilon^2} \\ P[|X - 10| \geq 4] &\leq \frac{\sigma_X^2}{4^2} = \frac{5}{16} \\ P[6 < X < 14] &\geq 1 - \frac{5}{16} = \frac{11}{16} \end{aligned}$$

3.60 Consider the random variable with the distribution given by $P(X = k) = 2^{-k}$, $k = 1, 2, \dots$. Find the upper bound using Chebyshev's inequality and also find the actual probability.

Solution

Given $P[X = k] = \frac{1}{2^k}$

$$\mu_X = E[X] = \sum_i x P[X = k] = \sum_{k=1}^{\infty} k 2^{-k}$$

$$\sum_{k=1}^{\infty} k 2^{-k}$$

$$= \frac{1/2}{\left(1 - \frac{1}{2}\right)^2} = 2$$

$$\Rightarrow \mu_X = 2$$

$$E[X^2] = \sum_i x^2 P[X = k] = \sum_i k^2 2^{-k} = 6$$

$$\text{Var}(X) = E[X^2] - \{E[X]\}^2 = 6 - (2)^2 = 2$$

$$P[|X - 2| > 2] \leq \frac{\text{Var}(X)}{2^2} = \frac{1}{2}$$

Also, $P[|X - 2| > 2] = 1 - P[0 \leq x \leq 4]$

$$= 1 - \left(\frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} + \frac{1}{2^4}\right) = 2^{-4}$$

$$\sum_{k=1}^{\infty} k r^k = \frac{r}{(1-r)^2}$$

$$\sum_{k=1}^{\infty} k^2 r^k = \frac{(1-r)r}{(1-r)^3}$$

3.61 The rainfall in Hyderabad is a normally distributed random variable with 50 cm mean and 9 cm^2 variance. Find a simple upper bound on the probability that rainfall in a particular year will exceed the mean by 5 cm.

Solution

We know $P[|X - \mu| \geq \varepsilon] \leq \frac{\sigma^2}{\varepsilon^2}$

$$P(|X - 50| \geq 5) \leq \frac{9}{25}$$

The actual value of probability

$$\begin{aligned} P(|X - 50| \geq 5) &= 2 P(X < 45) \\ &= 2P\left[\frac{X - \mu_X}{\sigma} < \frac{45 - 50}{3}\right] \\ &= 2P\left[\frac{X - \mu_X}{\sigma} < \left(\frac{-5}{3}\right)\right] \\ &= 2 \phi(-1.667) \\ &= 2[1 - \phi(1.667)] \\ &= 2(1 - 0.95) = 0.1 \end{aligned}$$

3.62 If a dice is thrown 2400 times, show that the probability that the number of sixes lies between 325 and 475 is at least 0.94.

Solution Let X denote the number of sixes.

$$\text{Given: } n = 2400; p = \frac{1}{6}$$

$$E[X] = np = 400$$

$$\text{Var}(X) = npq = 2400 \times \frac{1}{6} \times \frac{5}{6} = \frac{2000}{6} = \frac{1000}{3}$$

Using Chebyshev's inequality,

$$\begin{aligned} P[|X - 400| < 75] &\geq 1 - \frac{\text{Var}(X)}{(75)^2} \\ &\geq 1 - \frac{\left(\frac{1000}{3}\right)}{(75)^2} \\ &\geq 0.94 \end{aligned}$$



MARKOV'S INEQUALITY 3.13

The Markov inequality applies to a random variable that takes only non-negative values. If X is a random variable with non-negative values then for any $a > 0$

$$P[X \geq a] \leq \frac{E[X]}{a} \quad (3.102)$$

We know
$$E[X] = \int_{-\infty}^{\infty} x f_X(x) dx$$

Since X takes only non-negative values,

$$E[X] = \int_0^{\infty} x f_X(x) dx = \int_0^a x f_X(x) dx + \int_a^{\infty} x f_X(x) dx$$

If we omit the first integral, we can write

$$\begin{aligned} E[X] &\geq \int_a^{\infty} x f_X(x) dx \\ &\geq \int_a^{\infty} a f_X(x) dx = a \int_a^{\infty} f_X(x) dx = aP[X \geq a] \end{aligned}$$

$$\Rightarrow P[X \geq a] \leq \frac{E[X]}{a}$$

Solved Problems

3.63 The number of transistors produced in a manufacturing unit during a week is a random variable with mean 100 and variance 25.

- (a) What is the probability that the week's production will exceed 125?
- (b) What is the probability that the production will be between 50 and 150 over one week?
- (c) If the variance of a week's production is equal to 40, then what can be said about the probability that this week's production will be between 80 and 120?

Solution

- (a) Let X be the number of transistors produced in one week.
Using Markov's inequality,

$$\begin{aligned} P(X > 125) &\leq \frac{E[X]}{125} \\ &\leq \frac{100}{125} = \frac{4}{5} = 0.8 \end{aligned}$$

- (b) Using Chebyshev's inequality,

$$\begin{aligned} P[|X - 100| \geq 50] &\leq \frac{\sigma_X^2}{(50)^2} \\ &\leq \frac{25}{(50)^2} = \frac{1}{100} = 0.01 \end{aligned}$$

$$P[|X - 100| < 50] = 1 - 0.01 = 0.99$$

Hence, the probability that production in one week period to be between 50 and 150 is 0.99.

- (c) By Chebyshev's inequality,

$$\begin{aligned} P[|X - 100| \geq 20] &\leq \frac{\sigma_X^2}{20^2} \\ &\leq \frac{40}{(20)^2} = \frac{1}{10} = 0.1 \\ &= 0.9 \end{aligned}$$

So the probability that this week's production will be between 80 and 120 is at least 0.9.

3.64 Two fair dice whose faces are numbered 1 to 6 are thrown. If X is the sum of the numbers shown up, prove that

$$P[|X - 7| \geq 3] \leq \frac{35}{54}$$

Solution From Solved Problem 3.18, we can find that

$$E[X] = 7$$

Also

$$\begin{aligned}
 E[X^2] &= 2^2 \left(\frac{1}{36} \right) + 3^2 \left(\frac{2}{36} \right) + 4^2 \left(\frac{3}{36} \right) + 5^2 \left(\frac{4}{36} \right) + 6^2 \left(\frac{5}{36} \right) \\
 &\quad + (7)^2 \left(\frac{6}{36} \right) + (8)^2 \left(\frac{5}{36} \right) + (9)^2 \left(\frac{4}{36} \right) + (10)^2 \left(\frac{3}{36} \right) + (11)^2 \left(\frac{2}{36} \right) + \frac{(12)^2}{36} \\
 &= \frac{1974}{36} \\
 \sigma_X^2 &= E[X^2] - \{E[X]\}^2 = \frac{1974}{36} - (7)^2 = \frac{210}{36}
 \end{aligned}$$

Using Chebyshev's inequality,

$$\begin{aligned}
 P[|X - 7| \geq 3] &\leq \frac{\sigma_X^2}{(3)^2} \\
 &\leq \frac{210}{(36)(9)} = \frac{35}{54}
 \end{aligned}$$

3.65 A random variable has pdf $f_X(x) = e^{-x}$ for $x \geq 0$. Show that Chebyshev inequality gives

$$P[|X - 1| > 2] < \frac{1}{4}. \text{ Show that actual probability is } e^{-3}.$$

Solution

Given $f_X(x) = e^{-x}$ for $x \geq 0$

$$\begin{aligned}
 E[X] &= \int_0^{\infty} x e^{-x} dx = -x e^{-x} \Big|_0^{\infty} + \int_0^{\infty} e^{-x} dx \\
 &= -e^{-x} \Big|_0^{\infty} = 1
 \end{aligned}$$

$$\begin{aligned}
 E[X^2] &= \int_0^{\infty} x^2 e^{-x} dx \\
 &= -x^2 e^{-x} \Big|_0^{\infty} + 2x e^{-x} \Big|_0^{\infty} + 2 \int_0^{\infty} e^{-x} dx \\
 &= -2x e^{-x} \Big|_0^{\infty} = 2 \\
 \sigma_X^2 &= E[X^2] - \{E[X]\}^2 \\
 &= 2 - 1 = 1
 \end{aligned}$$

$$P[|X - 1| > 2] \leq \frac{\sigma^2}{\varepsilon^2}$$

From the actual probability,

$$\begin{aligned}
 P[|X - 1| > 2] &= \int_2^{\infty} f_X(x) dx \\
 P[|X| > 3] &= P[X < -3 \text{ and } X > 3] \\
 &= \int_{-\infty}^{-3} f_X(x) dx + \int_3^{\infty} f_X(x) dx \\
 &= \int_3^{\infty} f_X(x) dx = \int_3^{\infty} e^{-x} dx = e^{-3}
 \end{aligned}$$



Practice Problems

3.30(a) A random variable has the pdf $f_X(x) = 3e^{-3x}$. Obtain an upper bound for $P(X \geq 1)$ using the Markov inequality.

$$\left(\text{Ans: } \frac{1}{6} \right)$$

3.30(b) A random variable has the pdf $f_X(x) = 3e^{-3x}$, $x \geq 0$. Obtain an upper bound for $P[|X - E[X]| \geq 1]$.

$$\left(\text{Ans: } \frac{1}{9} \right)$$

3.31 A random variable X has a mean of 6 and a variance of 3. Use the Chebyshev inequality to obtain an upper bound $P[|X - 6| \geq 2]$. (Ans: 0.75)

REVIEW QUESTIONS



19. State Chebyshev's inequality. What is its significance?
20. State Markov's inequality.



FUNCTION OF A RANDOM VARIABLE 3.14

In the study of the systems, it is often required to find pdf of the output variable Y if the pdf of input variable X is known. In such cases, we have to know a method to find $f_Y(y)$ given $f_X(x)$. In this section, we will study different methods of obtaining $f_Y(y)$ if $f_X(x)$ is known.

Consider a system shown in Fig. 3.6. The output of the system Y can be expressed as a transformation of X denoted by

$$Y = T[X]$$

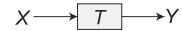


Fig. 3.6 System

That is each value in random variable X maps to another value in random variable Y . Now the problem is to determine either the distribution function $F_Y(y)$ or density function $f_Y(y)$, given the distribution function $F_X(x)$ or density function $f_X(x)$.

To find $f_Y(y)$, we have to find the nature of X and the type of transformation. The input random variable may be a continuous, discrete or mixed random variable and the transformation may be linear, nonlinear, segments, etc. Therefore, we have to consider different cases for study.

Monotonic Transformation A function is said to be monotonic increasing if $T[x_1] < T[x_2]$ for any $x_1 < x_2$. Also let us assume that the function is continuous, one-to-one, and differentiable for all values of x .

A typical function of this form is shown in Fig. 3.7. Under these assumptions, the inverse function $X = T^{-1}[Y]$ exists and is well behaved since there is a one-to-one relation between values of Y , their corresponding values of X . Consider a particular value y_0 corresponding to the particular value x_0 as shown in Fig. 3.7. The values x_0 and y_0 are related by

$$y_0 = T(x_0) \text{ or } x_0 = T^{-1}[x_0] \quad (3.103)$$

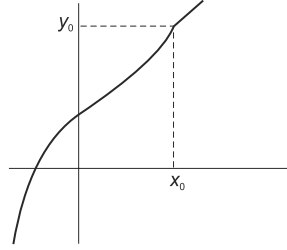


Fig. 3.7 Particular value of x_0

Now the probability that Y takes a value less than or equal to y_0 is given by

$$F_Y(y_0) = P[Y \leq y_0] = P[X \leq T^{-1}(y_0)] = F_X[T^{-1}(y_0)] \quad (3.104)$$

This can also be written as

$$F_X(x_0) = F_Y[T(x_0)] \quad (3.105)$$

Now differentiating Eq. (3.104) with respect to y produces

$$\begin{aligned} \frac{d}{dy} F_Y(y_0) &= \frac{d}{dy} [F_X(T^{-1}(y_0))] \\ \Rightarrow f_Y(y_0) &= f_X[T^{-1}(y_0)] \frac{d}{dy} (T^{-1}y_0) \end{aligned}$$

For any value of X and Y , we can write

$$f_Y(y) = f_X[T^{-1}(y)] \frac{d}{dy} T^{-1}(y) \quad (3.106)$$

Similarly, by differentiating Eq. (3.105), we get

$$\begin{aligned} f_X(x) &= f_Y[T(x)] \frac{dy}{dx} \\ &= f_Y(y) \frac{dy}{dx} \end{aligned} \quad (3.107)$$

$$\text{or } f_Y(y) = \left. \frac{f_X(x)}{dy/dx} \right|_{x=T^{-1}(y)} \quad (3.108)$$

$$\text{or } f_Y(y) = f_X(x) \frac{dx}{dy} \quad (3.109)$$

Now consider a monotonic decreasing function as shown in Fig. 3.8. In this case the event $\{Y \leq y_0\}$ is equivalent to the event $X \geq T^{-1}(y_0)$. Therefore,

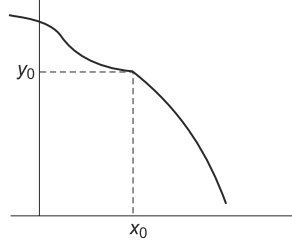


Fig. 3.8 Monotonic decreasing function

$$F_Y(y_0) = P[Y \leq y_0] = P[X \geq T^{-1}(y_0)] = 1 - F_X[T^{-1}(y_0)] = 1 - F_X(x_0)$$

Differentiating with respect to Y gives

$$f_Y(y) = -f_X(x) \left. \frac{dx}{dy} \right|_{x=T^{-1}(y)} \quad (3.110)$$

However the slope of $T^{-1}(y)$ is negative. Therefore, we can write for each type of monotonic transformation

$$\begin{aligned} f_Y(y) &= f_X(x) \left/ \left(\frac{dx}{dy} \right) \right|_{x=T^{-1}(y)} \\ &= \frac{f_X(x)}{\left| \frac{dy}{dx} \right|} \bigg|_{x=T^{-1}(y)} \end{aligned} \quad (3.111)$$

Nonmonotonic Transformation Consider a monotonic function as shown in Fig. 3.9. Let us consider a point $y = y_0$. Corresponding to this, there exists three values of x . That is, there are several values of x that maps to the same point y . In this case, we cannot associate the event $\{Y \leq y_0\}$ with the events of the form $\{X \leq T^{-1}(y_0)\}$ or $\{X \geq T^{-1}(y_0)\}$. To avoid this problem, we calculate the pdf of Y directly rather than first finding the CDF. For this, the function is redrawn as shown in Fig. 3.10. Now consider an event of the form $\{y \leq Y \leq y + dy\}$ for an infinitesimal dy . The probability of this event is $P[y \leq Y \leq y + dy] = f_Y(y)dy$. Corresponding to this event, there exist three events involving the random variable X around x_1 , x_2 and x_3 . The values of x_1 , x_2 , and x_3 can be obtained by solving the equation $y = T(x)$.

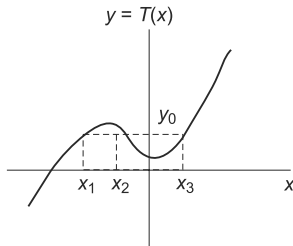


Fig. 3.9 Nonmonotonic transformation

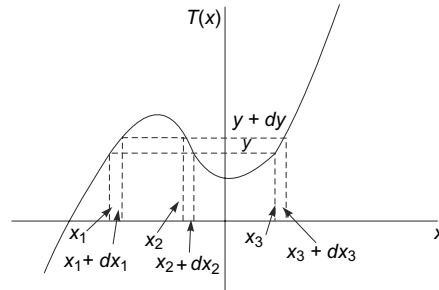


Fig. 3.10 Transformation of Fig. 3.9

The probability of this event $P[y < Y < y + dy]$ is the sum of probability x lying between x_1 and $x_1 + dx_1$, the probability that x lies between $x_2 + dx_2$ and x_2 and the probability that x_3 lies between x_3 and $x_3 + dx_3$. We can write $P[y \leq Y \leq y + dy] = P(x_1 \leq x \leq x_1 + dx_1) + P(x_2 + dx_2 \leq X \leq x_2) + P(x_3 \leq X \leq x_3 + dx_3)$

$$P[x_1 \leq X \leq x_1 + dx_1] = f_X(x_1)dx_1$$

$$P[x_2 + dx_2 \leq X \leq x_2] = f_X(x_2)|dx_2|$$

$$P[x_3 \leq X \leq x_3 + dx_3] = f_X(x_3)dx_3$$

In general,

$$P[y \leq Y \leq y + dy] = f_Y(y)dy = f_X(x_1)|dx_1| + f_X(x_2)|dx_2| + f_X(x_3)|dx_3|$$

The slope of the curve at $x = x_1$ is given by

$$\frac{dy}{dx_1} = T'(x_1) \Rightarrow |dx_1| = \frac{dy}{|T'(x_1)|}$$

Similarly,

$$|dx_2| = \frac{dy}{|T'(x_2)|} \text{ and } |dx_3| = \frac{dy}{|T'(x_3)|}$$

$$\Rightarrow f_Y(y) dy = \frac{f_X(x_1)}{|T'(x_1)|} dy + \frac{f_X(x_2)}{|T'(x_2)|} dy + \frac{f_X(x_3)}{|T'(x_3)|} dy$$

$$f_Y(y) = \frac{f_X(x_1)}{|T'(x_1)|} + \frac{f_X(x_2)}{|T'(x_2)|} + \frac{f_X(x_3)}{|T'(x_3)|}$$

If there are n roots for equation $y = T(x)$ then

$$f_Y(y) = \sum_n \frac{f_X(x_n)}{|T'(x_n)|} \quad (3.112)$$

$$T'(x_n) = \left. \frac{dT(x)}{dx} \right|_{x=x_n} \quad (3.113)$$

Solved Problems



3.66 If X is a random variable with pdf $f_X(x)$ then find pdf of $Y = ax + b$.

Solution Given $Y = aX + b$. The relationship between the values of x and y of random variable X and Y are given by

$$y = T(x) = ax + b$$

from which we can find $T'(x) = a$ and

$$x = \frac{y - b}{a}$$

The pdf of Y is

$$\begin{aligned} f_Y(y) &= \frac{f_X(x_n)}{|T'(x_n)|} = \frac{f_X(x_n)}{|a|} \bigg|_{x=\frac{y-b}{a}} \\ &= \frac{1}{|a|} f_X\left(\frac{y-b}{a}\right) \end{aligned} \quad (3.114)$$

3.67 If X is a uniformly distributed random variable between X_1 and X_2 , find $f_Y(y)$ for the above problem.

Solution

$$f_X(x) = \begin{cases} \frac{1}{X_2 - X_1} & \text{for } X_1 \leq x \leq X_2 \\ 0 & \text{otherwise} \end{cases}$$

$$f_Y(y) = \frac{1}{|a|} f_X\left(\frac{y-b}{a}\right)$$

For $x = X_1$; $y_1 = aX_1 + b$

$x = X_2$; $y_2 = aX_2 + b$

Y is also uniformly distributed in the interval $(aX_1 + b, aX_2 + b)$

3.68 If X is uniformly distributed in $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, find the probability density function of $Y = \tan X$.

Solution Given: $X \sim U\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$. That is, $b = \frac{\pi}{2}$ and $a = -\frac{\pi}{2}$

$$\begin{aligned} f_X(x) &= \frac{1}{(\pi/2) - (-\pi/2)} = \frac{1}{\pi} \quad \text{for } -\frac{\pi}{2} \leq x \leq \frac{\pi}{2} \\ &= 0 \quad \text{otherwise} \end{aligned}$$

Let $y = T(x) = \tan x$

Then $T'(x) = \sec^2 x$

Also, $x = \tan^{-1} y$

When $x = -\frac{\pi}{2}$; $y = \tan x = \tan\left(-\frac{\pi}{2}\right) = -\infty$
When $x = \pi/2$; $y = \tan\left(\frac{\pi}{2}\right) = \infty$

For the range of X , the random variable Y satisfies $-\infty < y < \infty$

The pdf of Y is given by

$$f_Y(y) = \frac{f_X(x_n)}{|T'(x_n)|} = \frac{1/\pi}{\sec^2 x} \bigg|_{x=\tan^{-1} y}$$

$$\begin{aligned}
 &= \frac{1}{\pi(1+y^2)} \\
 \Rightarrow \quad f_Y(y) &= \frac{1}{\pi(1+y^2)}; -\infty < y < \infty
 \end{aligned} \tag{3.115}$$

3.69 If X is a continuous random variable with pdf $f_X(x)$, find distribution function and pdf of random variable $Y = aX^2$ $a > 0$.

Solution If $Y < 0$; $f_Y(y) = 0$

$$\begin{aligned}
 F_Y(y) &= P[Y \leq y] = P[aX^2 \leq y] = P\left[\frac{-\sqrt{y}}{a} \leq X \leq \frac{\sqrt{y}}{a}\right] \\
 &= F_X\left(\sqrt{\frac{y}{a}}\right) - F_X\left(-\sqrt{\frac{y}{a}}\right) \\
 f_Y(y) &= \frac{d}{dy}[F_Y(y)] = \frac{f_X\left(\sqrt{\frac{y}{a}}\right)}{2\sqrt{ay}} + \frac{f_X\left(-\sqrt{\frac{y}{a}}\right)}{2\sqrt{ay}}
 \end{aligned} \tag{3.116}$$

3.70 If the random variable X is uniformly distributed over $(-1, 1)$ find the density function of $Y = \sin\left(\frac{\pi X}{2}\right)$

Solution

Given: $Y = \sin\left(\frac{\pi X}{2}\right)$.

The relationship between the values x and y of random variable X and Y are given by $y = T(x)$, then

$$T(x) = \frac{\pi}{2} \cos\left(\frac{\pi}{2} x\right)$$

Also, $x = \frac{2}{\pi} \sin^{-1}(y)$

Given: $f_X(x) = \frac{1}{2}$ for $-1 \leq x \leq 1$
 0 otherwise

For this range of X , Y also satisfies $-1 \leq y \leq 1$.

The pdf of Y is given by

$$\begin{aligned}
 f_Y(y) &= \frac{f_X(x_n)}{|T'(x_n)|} = \frac{1/2}{\left(\frac{\pi}{2}\right) \cos\left(\frac{\pi}{2}x\right)} \bigg|_{x=\frac{2}{\pi} \sin^{-1}(y)} \\
 &= \frac{1}{\pi \cos(\sin^{-1} y)} \\
 &= \frac{1}{\pi \sqrt{1-y^2}}, -1 \leq y \leq 1
 \end{aligned}
 \tag{3.117}$$

Let $\sin^{-1} y = \theta$
 $y = \sin \theta$
 $\cos(\sin^{-1} y) = \cos \theta$
 $= \sqrt{1-y^2}$

3.71 Let $Y = X^2$. Find the pdf of Y if X is a random variable over $(-1, 3)$.

Solution

$$f_X(x) = \frac{1}{4} \text{ for } -1 \leq x \leq 3$$

$$Y = T(x) = x^2 \Rightarrow T'(x) = 2x$$

Also, $X = \pm \sqrt{Y}$,

For the given range of X , the random variable Y satisfies $1 \leq y \leq 9$

The pdf of Y is given by

$$\begin{aligned}
 f_Y(y) &= \sum_n \frac{f_X(x_n)}{|T'(x_n)|} \\
 &= \frac{1/4}{|2x|} \bigg|_{x=-\sqrt{y}} + \frac{1/4}{|2x|} \bigg|_{x=\sqrt{y}} \\
 &= \frac{1/4}{2\sqrt{y}} + \frac{1/4}{2\sqrt{y}} = \frac{1}{4\sqrt{y}} \\
 \Rightarrow f_Y(y) &= \frac{1}{4\sqrt{y}}; 1 \leq y \leq 9
 \end{aligned}
 \tag{3.118}$$

3.72 If X is a uniformly distributed random variable over $(0, 1)$, find the pdf of the random variable $Y = X^n$ for $0 < y < 1$.

Solution

$$f_X(x) = \begin{cases} 1 & \text{for } 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

Let $y = x^n$ then $x = y^{1/n}$

$$f_Y(y) = \left. \frac{f_X(x)}{T'(x)} \right|_{x=y^{1/n}}$$

$$T'(x) = nx^{n-1}$$

$$f_Y(y) = \left. \frac{1}{x^{n-1}} \right|_{x=y^{1/n}} = \frac{1}{n y^{(n-1)/n}} \quad (3.119)$$



Practice Problems

3.32(a) If X is a random variable with pdf $f_X(x)$ then find pdf of (i) $Y = |X|$ (ii) $Y = e^X$ (iii) $Y = a/X$

$$\left[\text{Ans: (i) } f_X(y) + f_X(-y), \text{ (ii) } \frac{f_X(\ln y)}{y}, \text{ (iii) } \frac{a f_X(a/y)}{y^2} \right]$$

3.32(b) The pdf of a random variable X is given by $f_X(x) = \begin{cases} \frac{x^2}{100} & -5 < x < 5 \\ 0 & \text{otherwise} \end{cases}$
Find pdf of $Y = \frac{1}{2}(12 - X)$

Solved Problems



3.73 Find the pdf of $Y = -\ln(1 - X)$ where X is a uniform random variable over $(0, 1)$.

Solution

Given: $f_X(x) = \begin{cases} 1 & \text{for } 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$

and $Y = -\ln(1 - X)$

The relationship between the values x and y of random variables X and Y are given by

$$Y = T(x) = -\ln(1 - x)$$

$$T'(x) = -\frac{1}{1-x}(-1) = \frac{1}{1-x}$$

and $\ln(1 - x) = -y$

$$1 - x = e^{-y}$$

$$x = 1 - e^{-y}$$

$$f_Y(y) = \left. \frac{f_X(x)}{|T'(x)|} \right|_{x=1-e^{-y}} = \left. \frac{1}{1/(1-x)} \right|_{x=1-e^{-y}} = 1 - (1 - e^{-y}) = e^{-y}$$

$$f_Y(y) = \begin{cases} e^{-y} & 0 \leq y \leq 1 \\ 0 & \text{otherwise} \end{cases}$$



Practice Problems

3.33 If X is a random variable uniformly distributed over the interval $(0, \pi/2)$, find the pdf of (a) $Y = \sin X$ (b) $Z = \cos X$.

$$\left(\text{Ans: } \frac{2}{\pi\sqrt{1-y^2}}, 0 \leq y \leq 1 \right)$$

3.34 If X is a random variable with an exponential pdf of the form $f_X(x) = \frac{1}{3}e^{-x/3}u(x)$. A new random variable is defined using the equation $Y = 2X + 1$. Find $f_Y(y)$.

$$\left(\text{Ans: } \frac{1}{6}e^{-(y-1)/6}; 1 \leq y < \infty \right)$$

3.35 If X is a standard normal random variable, find the pdf of (i) $Y = |X|$ (ii) $Y = X^2$ (iii) $Y = X^3$ (iv) $Y = e^X$.

$$\left(\text{Ans: (ii)} \frac{1}{\sqrt{2\pi y}} e^{-y/2} u(y); \text{(iv)} \frac{1}{\sqrt{2\pi}} e^{-(\ln y)^2/2} \right)$$

3.36 If X is a random variable with pdf given by

$$f_X(x) = \frac{1}{\pi(1+x^2)}$$

find the PDF of $Y = \frac{1}{X}$.

$$\left(\text{Ans: } \frac{1}{\pi(1+y^2)} \right)$$

3.37 If $X \sim U(0, 1)$ find the density of $Y = \ln(1/X)$.

$$(\text{Ans: } e^{-y} u(y))$$

3.38 If X is exponential density function with $\lambda = 1$, show that $Y = \sqrt{X}$ is Rayleigh $(1/\sqrt{2})$.



MOMENT GENERATING FUNCTION 3.15

We know that the random variable X is a function from the sample space to the real numbers. The random variable is characterized by mean, variance and high-order moments which can be evaluated using the probability density function. Now we will study a new function known as *Moment Generating Function* (MGF) which can be used to evaluate the above characteristics of the random variable.

Definition The moment-generating function $M_X(u)$ of the random variable X is defined for all real values of u by

$$M_X(u) = E[e^{uX}]$$

If X is discrete with pmf $p_X(x)$ then

$$M_X(u) = \sum_{i=1}^{\infty} e^{ux_i} p_X(x_i) \quad (3.120)$$

and if X is continuous with pdf $f_X(x)$ then

$$M_X(u) = \int_{-\infty}^{\infty} e^{ux} f_X(x) dx \quad (3.121)$$

In either the discrete or the continuous case, $M_X(u)$ is simply the expected value of e^{uX} .

$M_X(u)$ is called the moment generating function because all the moments of X can be evaluated by successively differentiating $M_X(u)$ and substituting $u = 0$. That is

$$E[X] = M'_X(0) \quad (3.122)$$

$$E[X''] = M''_X(0) \quad (3.123)$$

$$E[X'''] = M'''_X(0) \text{ so on.} \quad (3.124)$$

Consider the equation

$$M_X(u) = \sum_x e^{ux} p_X(x) \quad (3.125)$$

Differentiating with respect to u on both sides, we get

$$\begin{aligned} M'_X(u) &= \sum_x \frac{d}{du} [e^{ux} p_X(x)] \\ &= \sum_x [x e^{ux} p_X(x)] = E[X e^{uX}] \end{aligned} \quad (3.126)$$

$$M'_X(0) = E[X]$$

Similarly,

$$M''_X(u) = \sum_x \frac{d}{dx} [x e^{ux} p_X(x)] \quad (3.127)$$

$$= \sum_x x^2 e^{ux} p_X(x) \quad (3.128)$$

$$= E[X^2 e^{uX}]$$

$$M''_X(0) = E[X^2] \quad (3.129)$$

In general, the n^{th} derivative of $M_X(u)$ is

$$M_X^n(u) = E[X^n e^{uX}] \quad (3.130)$$

The same result can be proved for a continuous random variable.

3.15.1 Properties of MGF

1. The n^{th} derivative of $M_X(u)$ evaluated at $u = 0$ yields $E[X^n]$

Proof: From Eq. (3.130)

$$M_X^n(u) = E[X^n e^{uX}]$$

$$\Rightarrow M_X^n(0) = E[X^n]$$

2. If $M_X(u)$ is the MGF of a random variable X then the MGF of the random variable $Y = kX$ is $M_X(ku)$

Proof:

$$\begin{aligned} M_Y(u) &= E[e^{uY}] \\ &= E[e^{ukX}] = E[e^{X(ku)}] \\ &= M_X(ku) \end{aligned} \quad (3.131)$$

3. If X_1 and X_2 are two independent random variables then

$$M_{(X_1 + X_2)}(u) = M_{X_1}(u) \cdot M_{X_2}(u)$$

Proof:

$$\begin{aligned}
 M_{(X_1 + X_2)}(u) &= E[e^{u(X_1 + X_2)}] \\
 &= E[e^{uX_1} e^{uX_2}] = E[e^{uX_1}] E[e^{uX_2}] \\
 &= M_{X_1}(u) M_{X_2}(u)
 \end{aligned} \tag{3.132}$$

4. If X is a random variable with MGF $M_X(u)$ then MGF of $Y = aX + b$ is $e^{bu} M_X(au)$.

Proof:

$$\begin{aligned}
 Y &= aX + b \\
 M_Y(u) &= E[e^{(aX + b)u}] \\
 &= E[e^{Xau} e^{bu}] = e^{bu} E[e^{Xau}] \\
 &= e^{bu} E[e^{X(au)}] = e^{bu} M_X(au)
 \end{aligned} \tag{3.133}$$

5. If $M_X(u)$ is MGF of the random variable X then MGF of $Y = \frac{X + a}{b}$ is $e^{\frac{a}{b}u} M_X\left(\frac{u}{b}\right)$.

Proof:

$$\begin{aligned}
 M_Y(u) &= E\left[e^{u\left(\frac{X+a}{b}\right)}\right] = E\left[e^{\frac{uX}{b}} e^{ua/b}\right] \\
 &= e^{ua/b} E[e^{uX/b}] \\
 &= e^{\frac{a}{b}u} E[e^{X(u/b)}] \\
 &= e^{\frac{a}{b}u} M_X\left(\frac{u}{b}\right)
 \end{aligned} \tag{3.134}$$

6. If two random variables X and Y have MGFs such that $M_X(u) = M_Y(u)$ then X and Y are said to have identical distributions.

3.15.2 Convergence Property of MGF

The MGFs defined in Eq. (3.120) and in Eq. (3.121) are an infinite series and improper integral respectively. Such a series or integral may not converge to finite value for all values of u . Therefore, MGF is not defined for all values of u . In this section, we find the condition for convergence of MGF. From Eq. (3.121) we have

$$\begin{aligned}
 M_X(u) &= \int_{-\infty}^{\infty} e^{ux} f_X(x) dx \\
 |M_X(u)| &= \left| \int_{-\infty}^{\infty} e^{ux} f_X(x) dx \right|
 \end{aligned}$$

We know that magnitude of an integral is less than or equal to the integral of the magnitude

$$\begin{aligned} \left| \int_{-\infty}^{\infty} e^{ux} f_X(x) dx \right| &\leq \int_{-\infty}^{\infty} |e^{ux} f_X(x)| dx \\ &\leq \int_{-\infty}^{\infty} |e^{ux}| |f_X(x)| dx \end{aligned} \quad (3.135)$$

Since $\int_{-\infty}^{\infty} f_X(x) dx = 1$, $M_X(u)$ is finite if $|e^{ux}|$ is finite.

Solved Problems



3.74 Find the moment-generating function of the random variable having probability density function

$$\begin{aligned} f_X(x) &= x, \quad 0 < x < 1 \\ &= 2-x, \quad 1 < x < 2 \\ &= 0 \quad \text{elsewhere} \end{aligned}$$

Solution

Given:

$$\begin{aligned} f_X(x) &= x \text{ for } 0 \leq x \leq 1 \\ &= 2-x \text{ for } 1 \leq x \leq 2 \\ &= 0 \quad \text{otherwise} \end{aligned}$$

The moment-generating function

$$\begin{aligned} M_X(u) &= E[e^{ux}] = \int_{-\infty}^{\infty} f_X(x) e^{ux} dx \\ &= \int_0^1 x e^{ux} dx + \int_1^2 (2-x) e^{ux} dx \\ &= \left[\frac{x e^{ux}}{u} \right]_0^1 - \left[\frac{e^{ux}}{u^2} \right]_0^1 + \left[\frac{2e^{ux}}{u} \right]_1^2 - \left[\frac{x e^{ux}}{u} \right]_1^2 - \left[\frac{e^{ux}}{u^2} \right]_1^2 \\ &= \frac{e^u}{u} - \frac{(e^u - 1)}{u^2} + \frac{2e^{2u}}{u} - \frac{2e^u}{u} - \left[\frac{2e^{2u} - e^u}{u} - \frac{e^{2u}}{u^2} + \frac{e^u}{u^2} \right] \\ &= \frac{1}{u} [e^u + 2e^{2u} - 2e^u - e^u] + \frac{1}{u^2} [1 - e^u + e^{2u} - e^u] \\ &= \frac{1 - 2e^u + e^{2u}}{u^2} \end{aligned}$$

$$= \left(\frac{1 - e^{2u}}{u} \right)^2.$$

3.75 Prove that the moment-generating function of the random variable X having the pdf

$$f_X(x) = \begin{cases} \frac{1}{3}, & -1 < x < 2 \\ 0 & \text{elsewhere} \end{cases}$$

$$\text{is given by } M_X(u) = \begin{cases} \frac{e^{2u} - e^{-u}}{3u} & u \neq 0 \\ 1 & u = 0 \end{cases}$$

Solution

$$\begin{aligned} M_X(u) &= E[e^{uX}] = \int_{-\infty}^{\infty} e^{ux} f_X(x) dx \\ &= \frac{1}{3} \int_{-1}^2 e^{ux} dx = \frac{1}{3u} e^{ux} \Big|_{-1}^2 = \frac{e^{2u} - e^{-u}}{3u} \end{aligned}$$

when $u = 0$, $M_X(u) = \frac{0}{0}$ form

Applying L Hospital's rule,

$$\left. \frac{2e^{2u} + e^u}{3} \right|_{u=0} = \frac{2+1}{3} = 1 \text{ for } u = 0$$

3.76 The pdf of a random variable is given by

$$\begin{cases} P(1-P)^x & x = 0, 1, 2 \\ 0 & \text{elsewhere} \end{cases}$$

Find the moment-generating function and evaluate mean and variance.

Solution

$$\text{Given: } f_X(x) = \begin{cases} P(1-P)^x & x = 0, 1, 2 \dots \\ 0 & \text{elsewhere} \end{cases}$$

$$\begin{aligned} M_X(u) &= E[e^{uX}] \\ &= \sum_x e^{ux} f_X(x) = \sum_{x=0}^{\infty} e^{ux} P(1-P)^x = P \sum_{x=0}^{\infty} e^{ux} (1-P)^x \\ &= P[1 + (1-P)e^u + (1-P)^2 e^{2u} + (1-P)^3 e^{3u} + \dots] \end{aligned}$$

$$\begin{aligned}
&= P \left[\frac{1}{1 - (1 - P)e^u} \right] = \frac{P}{1 - (1 - P)e^u} \\
E[X] &= M'_X(u) \Big|_{u=0} \\
M'_X(u) &= \frac{P(1 - P)e^u}{\{1 - (1 - P)e^u\}^2} \quad \text{and} \quad E[X] = \frac{P(1 - P)e^u}{\{1 - (1 - P)e^u\}^2} \Big|_{u=0} = \frac{1 - P}{P} \\
E[X] &= \frac{1 - P}{P} \\
E[X^2] &= M''_X(u) \Big|_{u=0} \\
M''_X(u) &= P(1 - P)e^u \frac{1 + (1 - P)e^u}{\{1 - (1 - P)e^u\}^3} \\
E[X^2] &= M''_X(u) \Big|_{u=0} = \frac{P(1 - P)(2 - P)}{P^3} = \frac{(1 - P)(2 - P)}{P^2} \\
\sigma_X^2 &= \text{Var}(X) = E[X^2] - \{E[X]\}^2 \\
&= \frac{(1 - P)(2 - P)}{P^2} - \left(\frac{1 - P}{P} \right)^2 = \frac{1 - P}{P^2} [(2 - P) - (1 - P)] \\
&= \frac{1 - P}{P^2}
\end{aligned}$$

3.77 Find the moment-generating function of the following:

(i) $Y = aX + b$ (ii) $Y = \frac{X + a}{b}$.

Solution

(i) Given: $Y = aX + b$

Let X be a random variable with MGF $M_X(u)$ then MGF of $Y = aX + b$ is

$$\begin{aligned}
M_Y(u) &= E[e^{(aX + b)u}] \\
&= E[e^{Xau} e^{bu}] = e^{bu} E[e^{Xau}] \\
&= e^{bu} E[e^{X(au)}] = e^{bu} M_X(au)
\end{aligned}$$

(ii) Given: $Y = \frac{X + a}{b}$

$$\begin{aligned}
M_Y(u) &= E \left[e^{\left(\frac{X + a}{b} \right) u} \right] = E[e^{Xu/b} e^{au/b}] \\
&= e^{\frac{au}{b}} E[e^{X(u/b)}]
\end{aligned}$$

$$= e^{\frac{au}{b}} M_X\left(\frac{u}{b}\right).$$

3.78 For the random variable X whose density function is $f_X(x) = \begin{cases} \frac{1}{b-a} & a \leq x \leq b \\ 0 & \text{otherwise} \end{cases}$

Determine, (i) Moment generating function, (ii) Mean, (iii) Variance.

Solution

$$\begin{aligned} M_X(u) &= \int_{-\infty}^{\infty} e^{ux} f_X(x) dx = \int_a^b \frac{e^{ux}}{b-a} dx = \frac{1}{b-a} \left. \frac{e^{ux}}{u} \right|_a^b \\ &= \frac{e^{ub} - e^{ua}}{u(b-a)} \end{aligned}$$

The mean is given by

$$\begin{aligned} \mu_X = E[X] &= \int_{-\infty}^{\infty} x f_X(x) dx \\ &= \int_a^b \frac{x}{b-a} dx = \frac{1}{b-a} \left. \frac{x^2}{2} \right|_a^b = \frac{b^2 - a^2}{(b-a)^2} = \frac{b+a}{2} \end{aligned}$$

$$\mu_X = E[X] = \frac{b+a}{2}$$

$$\begin{aligned} E[X^2] &= \int_{-\infty}^{\infty} x^2 f_X(x) dx = \int_a^b \frac{x^2}{b-a} dx = \frac{1}{b-a} \left. \frac{x^3}{3} \right|_a^b \\ &= \frac{b^3 - a^3}{3(b-a)} = \frac{b^2 + ab + a^2}{3} \end{aligned}$$

$$\begin{aligned} \sigma_X^2 &= E[X^2] - \{E[X]\}^2 \\ &= \frac{b^2 + ab + a^2}{3} - \frac{(b+a)^2}{4} = \frac{4b^2 + 4ab + 4a^2 - 3a^2 - 3b^2 - 6ab}{12} \\ &= \frac{b^2 + a^2 - 2ab}{12} = \frac{(b-a)^2}{12} \end{aligned}$$

$$\sigma_X^2 = \frac{(b-a)^2}{12}.$$

3.79 Find the MGF of the two parameter exponential distribution whose density function is given by $f_X(x) = \lambda e^{-\lambda(x-a)}$, $x \geq a$ and hence find mean and variance.

Solution

Given $f_X(x) = \lambda e^{-\lambda(x-a)}, x \geq a$

The MGF $M_X(u) = \int_{-\infty}^{\infty} f_X(x) e^{ux} dx$

$$\begin{aligned}
 &= \int_a^{\infty} \lambda e^{-\lambda(x-a)} e^{ux} dx \\
 &= \lambda e^{a\lambda} \int_a^{\infty} e^{-(\lambda-u)x} dx \\
 &= \lambda e^{a\lambda} \left[\frac{e^{-(\lambda-u)x}}{-(\lambda-u)} \right]_a^{\infty} \\
 &= \frac{-\lambda e^{a\lambda}}{\lambda-u} \{-e^{-(\lambda-u)a}\} \\
 &= \frac{\lambda}{\lambda-u} e^{au}
 \end{aligned}$$

Mean $= E[X] = M'_X(u) \Big|_{u=0}$

$$\begin{aligned}
 &= \frac{d}{du} \left[\frac{\lambda}{\lambda-u} e^{au} \right] \Big|_{u=0} = \lambda \frac{d}{du} \left(\frac{e^{au}}{\lambda-u} \right) \Big|_{u=0} \\
 &= \lambda \left\{ \frac{a(\lambda-u)e^{au} - e^{au}(-1)}{(\lambda-u)^2} \right\} \Big|_{u=0} \\
 &= \lambda \left(\frac{a\lambda+1}{\lambda^2} \right) = \left(\frac{a\lambda+1}{\lambda} \right)
 \end{aligned}$$

$$E[X] = a + \frac{1}{\lambda}$$

$$E[X^2] = M''_X(u) \Big|_{u=0}$$

$$\begin{aligned}
 M'_X(u) &= \frac{\lambda}{(\lambda-u)^2} \{a(\lambda-u)e^{au} + e^{au}\} \\
 E[X^2] &= M''_X(u) \Big|_{u=0} = \lambda \left\{ \frac{d}{du} \left(\frac{a(\lambda-u)e^{au} + e^{au}}{(\lambda-u)^2} \right) \right\} \Big|_{u=0} \\
 &= \frac{\lambda}{(\lambda-u)^4} \{(\lambda-u)^2 [a(-e^{au} + (\lambda-u)ae^{au}) + ae^{au}] \\
 &\quad - 2e^{au}(a\lambda - au + 1)(\lambda-u)(-1)\} \Big|_{u=0}
 \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{\lambda^3} \{ \lambda^2 [a(-1 + a\lambda) + a] - 2(a\lambda + 1)(-\lambda) \} \\
&= \frac{1}{\lambda^3} \{ \lambda^2 (a^2 \lambda) + 2a\lambda^2 + 2\lambda \} = a^2 + \frac{2a}{\lambda} + \frac{2}{\lambda^2} \\
E[X^2] &= a^2 + \frac{2a}{\lambda} + \frac{2}{\lambda^2} \\
\text{Var}[X] &= E[X^2] - \{E[X]\}^2 \\
&= \left(a^2 + \frac{2a}{\lambda} + \frac{2}{\lambda^2} \right) - \left(a + \frac{1}{\lambda} \right)^2 \\
&= a^2 + \frac{2a}{\lambda} + \frac{2}{\lambda^2} - a^2 - \frac{1}{\lambda^2} - \frac{2a}{\lambda} \\
&= \frac{1}{\lambda^2} \\
\text{Var}(X) &= \frac{1}{\lambda^2}.
\end{aligned}$$

3.80 Prove that the moment-generating function of the sum of two independent random variables is the product of their generating function.

Solution Let X_1 and X_2 be two independent random variables and their sum is represented by X . Let $M_{X_1}(u)$ and $M_{X_2}(u)$ are moment generating function of X_1 and X_2 respectively. Then we can write

$$M_{X_1}(u) = E[e^{uX_1}] \text{ and } M_{X_2}(u) = E[e^{uX_2}]$$

The moment-generating function of $X_1 + X_2$ is

$$\begin{aligned}
M_{X_1 + X_2}(u) &= E[e^{u(X_1 + X_2)}] = E[e^{uX_1} e^{uX_2}] \\
&= E[e^{uX_1}] E[e^{uX_2}] \\
&= M_{X_1}(u) M_{X_2}(u)
\end{aligned}$$

3.81 Find the moment-generating function of the random variable whose moments are $m_r = (r + 1)! 2^r$

Solution We know

$$\begin{aligned}
M_X(u) &= E[e^{uX}] = E \left[1 + ux + u^2 \cdot \frac{x^2}{2!} + u^3 \frac{x^3}{3!} + \dots \right] \\
&= 1 + uE[X] + \frac{u^2}{2!} E[X^2] + \dots
\end{aligned}$$

$$= 1 + m_1 u + m_2 \frac{u^2}{2!} + \dots$$

Where $m_1 = E[X]$; $m_2 = E[X^2]$ so on.

Given the moments $m_r = (r+1)! 2^r$

$$\begin{aligned} M_X(u) &= \sum_{r=0}^{\infty} \frac{u^r}{r!} m_r \\ &= \sum_{r=0}^{\infty} \frac{u^r}{r!} (r+1)! 2^r = \sum_{r=0}^{\infty} u^r (r+1) 2^r = \sum_{r=0}^{\infty} (r+1)(2u)^r = \sum_{r=0}^{\infty} r(2u)^r + \sum_{r=0}^{\infty} (2u)^r \end{aligned}$$

We know $\sum_{r=0}^{\infty} (2u)^r = \frac{1}{1-2u}$

$$\sum_{r=0}^{\infty} r(2u)^r = 2u \sum_{r=0}^{\infty} r(2u)^{r-1} = u \sum_{r=0}^{\infty} 2r(2u)^{r-1} = u \sum_{r=0}^{\infty} \frac{d}{du} (2u)^r$$

$$= u \cdot \frac{d}{du} \left[\sum_{r=0}^{\infty} (2u)^r \right] = u \cdot \frac{d}{du} \left[\frac{1}{1-2u} \right] = \frac{2u}{(1-2u)^2}$$

$$M_X(u) = \frac{2u}{(1-2u)^2} + \frac{1}{1-2u} = \frac{1}{(1-2u)^2}$$



CHARACTERISTIC FUNCTION (CF) 3.16

In the previous section, we studied that the MGF can be used to find the moments of a random variable. But the existence of MGF depends on the value of u . An alternative way of finding moments of a random variable is the characteristic function which is defined as

$$\begin{aligned} \phi_X(\omega) &= E[e^{j\omega X}] \\ &= \int_{-\infty}^{\infty} e^{j\omega x} f_X(x) dx \end{aligned} \quad (3.136)$$

From the above definition, we can find that the characteristics function is a Fourier transform of pdf where the sign of ω is positive, instead of negative. The Fourier transform of a signal gives the frequency content of the signal. But from CF, we will not get such information. There is no connection between the frequency variable ' ω ' and physical frequency.

From Eq. (3.136). We can also obtain the pdf using

$$f_X(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \phi_X(\omega) e^{-j\omega x} d\omega$$

If X is a discrete random variable then

$$\phi_X(\omega) = \sum_i e^{j\omega x_i} P_X(x_i) \quad (3.137)$$

3.16.1 Convergence of CF

Consider the equation

$$\phi_X(\omega) = \int_{-\infty}^{\infty} e^{j\omega x} f_X(x) dx$$

The magnitude of $\phi_X(\omega)$ is equal to

$$|\phi_X(\omega)| = \left| \int_{-\infty}^{\infty} e^{j\omega x} f_X(x) dx \right|$$

We know that the value of an integral is less than or equal to the integral of its absolute value.

Applying this condition, we get

$$\begin{aligned} \left| \int_{-\infty}^{\infty} e^{j\omega x} f_X(x) dx \right| &\leq \int_{-\infty}^{\infty} |e^{j\omega x}| |f_X(x)| dx \\ |e^{j\omega x}| &= 1 \\ \Rightarrow \left| \int_{-\infty}^{\infty} e^{j\omega x} f_X(x) dx \right| &\leq \int_{-\infty}^{\infty} |f_X(x)| dx \\ \text{Since } \int_{-\infty}^{\infty} f_X(x) dx &= 1 \\ \left| \int_{-\infty}^{\infty} e^{j\omega x} f_X(x) dx \right| &\leq 1 \end{aligned} \quad (3.138)$$

Thus, for a random variable, CF always exists.

Like MGF, CF also can be used to find the moments of the random variable. Consider the definition of characteristic function

$$\begin{aligned} \phi_X(\omega) &= E[e^{j\omega X}] \\ &= \int_{-\infty}^{\infty} e^{j\omega x} f_X(x) dx \end{aligned}$$

Expanding the term $e^{j\omega x}$, we obtain

$$\begin{aligned} \phi_X(\omega) &= \int_{-\infty}^{\infty} f_X(x) dx \left[1 + \frac{j\omega x}{1!} + \frac{(j\omega x)^2}{2!} + \dots \right] \\ &= \int_{-\infty}^{\infty} f_X(x) dx + j\omega \int_{-\infty}^{\infty} x f_X(x) dx + \frac{(j\omega)^2}{2!} \int_{-\infty}^{\infty} x^2 f_X(x) dx + \dots \end{aligned} \quad (3.139)$$

We know $\int_{-\infty}^{\infty} f_X(x) dx = 1$ and

$$E[X^n] = \int_{-\infty}^{\infty} x^n f_X(x) dx$$

Now $\phi_X(\omega)$ can be written as

$$\phi_X(\omega) = 1 + \frac{j\omega}{1!} E[X] + \frac{(j\omega)^2}{2!} E[X^2] + \frac{(j\omega)^3}{3!} E[X^3] + \dots$$

Differentiating $\phi_X(\omega)$ with respect to ω

$$\frac{d\phi_X(\omega)}{d\omega} = \frac{j}{1!} E[X] + \frac{2j^2\omega}{2!} E[X^2] + \frac{j^3(3\omega^2)}{3!} E[X^3] + \dots + \dots \frac{k(j)^k \omega^{k-1}}{k!} E[X^k] + \dots \quad (3.140)$$

Substitute $\omega = 0$ in the above Eq. (3.140)

$$\begin{aligned} \frac{j}{1!} E[X] &= \left. \frac{d\phi_X(\omega)}{d\omega} \right|_{\omega=0} \\ \Rightarrow E[X] &= \frac{1}{j} \left. \frac{d\phi_X(\omega)}{d\omega} \right|_{\omega=0} \\ &= \frac{1}{j} \phi'_X(0) \end{aligned} \quad (3.141)$$

Similarly, if we differentiate Eq. (3.140), we get

$$\frac{d^2\phi_X(\omega)}{d\omega^2} = \frac{2j^2}{2!} E[X^2] + \frac{(j)^3 6\omega}{3!} E[X^3] + \dots$$

Substituting $\omega = 0$, we get

$$\phi''_X(0) = \frac{2j^2}{2!} E[X^2] \Rightarrow E[X^2] = \left(\frac{1}{j}\right)^2 \phi''_X(0) \quad (3.142)$$

In general,

$$E[X^k] = \left(\frac{1}{j}\right)^k \phi_X^k(0) \quad (3.143)$$

3.16.2 Properties of Characteristic Functions

1. The characteristic function has maximum value 1 at $\omega = 0$

Proof:

$$\begin{aligned} \phi_X(\omega) &= E[e^{j\omega X}] \\ &= \int_{-\infty}^{\infty} e^{j\omega x} f_X(x) dx \end{aligned}$$

$$\phi_X(\omega)\Big|_{\omega=0} = \int_{-\infty}^{\infty} f_X(x) dx = 1$$

2. For real, ω $|\phi_X(\omega)| \leq 1$

Proof:

$$\phi_X(\omega) = \int_{-\infty}^{\infty} e^{j\omega x} f_X(x) dx$$

$$\begin{aligned} |\phi_X(\omega)| &= \left| \int_{-\infty}^{\infty} e^{j\omega x} f_X(x) dx \right| \quad \boxed{|e^{j\omega x}| = 1} \\ &\leq \int_{-\infty}^{\infty} |e^{j\omega x}| |f_X(x)| dx \end{aligned} \quad (3.144)$$

$$\leq \int_{-\infty}^{\infty} |f_X(x)| dx$$

$$\leq 1$$

3. $\phi_X^*(\omega) = \phi_X(-\omega)$

Proof:
$$\phi_X(\omega) = \int_{-\infty}^{\infty} e^{j\omega x} f_X(x) dx$$

$$\begin{aligned} \phi_X^*(\omega) &= \left[\int_{-\infty}^{\infty} e^{j\omega x} f_X(x) dx \right]^* \\ &= \int_{-\infty}^{\infty} e^{-j\omega x} f_X(x) dx = \int_{-\infty}^{\infty} e^{(jx)(-\omega)} f_X(x) dx \end{aligned}$$

$$= \phi_X(-\omega) \quad (3.145)$$

4. If $f_X(-x) = f_X(x)$ then $\phi_X(\omega)$ is real.

Proof:
$$\phi_X(\omega) = \int_{-\infty}^{\infty} f_X(x) e^{j\omega x} dx$$

$$\text{Re} [\phi_X(\omega)] + j \text{Im} [\phi_X(\omega)]$$

$$= \int_{-\infty}^{\infty} f_X(x) \cos \omega x + j \int_{-\infty}^{\infty} f_X(x) \sin \omega x$$

If $f_X(-x) = f_X(x)$, we can write

$$\int_{-\infty}^{\infty} f_X(x) \cos \omega x = 2 \int_0^{\infty} f_X(x) \sin \omega x$$

and

$$\int_{-\infty}^{\infty} f_X(x) \sin \omega x = 0$$

Therefore $\phi_X(\omega)$ is real.

5. $\phi_X(\omega)$ cannot be purely imaginary.

6. If $\phi_X(\omega)$ is a characteristic function of a random variable then $\phi_X(c\omega) = \phi_{cX}(\omega)$.

Proof: We know $\phi_X(\omega) = E[e^{j\omega X}]$ from which $\phi_X(c\omega) = E[e^{j\omega cX}]$

$$= \phi_{cX}(\omega) \quad (3.146)$$

7. If $\phi_X(\omega)$ is characteristic function of a random variables then the characteristic function of $Y = aX + b$ is given by

$$\phi_Y(\omega) = e^{j\omega b} \phi_X(a\omega) \quad (3.147)$$

Proof: $Y = aX + b$, $\phi_Y(\omega) = E[e^{j\omega y}] = E[e^{j\omega(aX+b)}] = e^{j\omega aX} E[e^{j\omega bX}]$ (3.148)

$$= e^{j\omega b} \phi_X(a\omega)$$

8. If X_1 and X_2 are two independent random variables then $\phi_{X_1 + X_2}(\omega) = \phi_{X_1}(\omega) \phi_{X_2}(\omega)$

$$\begin{aligned} \phi_{X_1 + X_2}(\omega) &= E[e^{j\omega(X_1 + X_2)}] = E[e^{j\omega X_1}] E[e^{j\omega X_2}] \\ &= \phi_{X_1}(\omega) \phi_{X_2}(\omega) \end{aligned} \quad (3.149)$$

Solved Problems



3.82 Prove that the characteristic function and probability density function form a Fourier transform pair.

Solution Consider the equation for a Fourier transform pair.

$$F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt \quad (3.150)$$

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{j\omega t} d\omega \quad (3.151)$$

The characteristic function is

$$\phi_X(\omega) = \int_{-\infty}^{\infty} e^{j\omega x} f_X(x) dx \quad (3.152)$$

From Eq. (3.151) and Eq. (3.152), we can observe that both are similar except for the sign exponent.

$$\text{Now} \quad f_X(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \phi_X(\omega) e^{-j\omega x} d\omega \quad (3.153)$$

In this case also the Eq. (3.153) is same as Eq. (3.151) except sign. Therefore we can say that

$$f_X(x) \xrightarrow{FT} \phi_X(\omega)$$

3.83 Find the density function of a random variable X whose characteristic function is

$$\phi_X(\omega) = \frac{1}{2} e^{-|\omega|} \quad -\infty \leq \omega \leq \infty$$

Solution The density function

$$\begin{aligned} f_X(x) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \phi_X(\omega) e^{-j\omega x} d\omega \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{1}{2} e^{-|\omega|} e^{-j\omega x} d\omega \\ &= \frac{1}{2\pi} \left[\int_{-\infty}^0 \frac{1}{2} e^{\omega} e^{-j\omega x} d\omega + \int_0^{\infty} \frac{1}{2} e^{-\omega} e^{-j\omega x} d\omega \right] \\ &= \frac{1}{4\pi} \left[\int_{-\infty}^0 e^{(1-jx)\omega} d\omega + \int_0^{\infty} e^{-(1+jx)\omega} d\omega \right] \\ &= \frac{1}{4\pi} \left[\frac{1}{1-jx} e^{(1-jx)\omega} \Big|_{-\infty}^0 + \frac{1}{-(1+jx)} e^{-(1+jx)\omega} \Big|_0^{\infty} \right] \\ &= \frac{1}{4\pi} \left[\frac{1}{1-jx} + \frac{1}{1+jx} \right] = \frac{2}{4\pi(1+x^2)} = \frac{1}{2\pi(1+x^2)} \end{aligned}$$

3.84 Find the moment generating function and characteristic function of the random variable X which has uniform distribution.

Solution

$$\begin{aligned} M_X(u) &= \int_{-\infty}^{\infty} e^{ux} f_X(x) dx = \frac{1}{b-a} \int_a^b e^{ux} dx = \frac{1}{b-a} \cdot \left(\frac{1}{u} \right) e^{ux} \Big|_a^b \\ &= \frac{1}{b-a} \frac{e^{ub} - e^{-ub}}{u} \\ M_X(u) &= \frac{e^{ub} - e^{-ub}}{u(b-a)} \end{aligned}$$

Characteristics function

$$\phi_X(\omega) = E[e^{j\omega X}] = \int_{-\infty}^{\infty} e^{j\omega x} f_X(x) dx$$

$$\begin{aligned}
 &= \int_a^b \frac{1}{b-a} e^{j\omega x} dx = \frac{1}{(b-a)} \frac{1}{j\omega} e^{j\omega x} \Big|_a^b \\
 &= \frac{e^{j\omega b} - e^{j\omega a}}{j\omega(b-a)}
 \end{aligned}$$

3.85 A random variable X has a characteristic function given by

$$\phi_X(\omega) = \begin{cases} 1 - |\omega| & |\omega| \leq 1 \\ 0 & |\omega| > 1 \end{cases}$$

Solution The density function

$$\begin{aligned}
 f_X(x) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \phi_X(\omega) e^{-j\omega x} d\omega \\
 f_X(x) &= \frac{1}{2\pi} \left[\int_{-1}^0 (1 + \omega) e^{-j\omega x} d\omega + \int_0^1 (1 - \omega) e^{-j\omega x} d\omega \right] \\
 &= \frac{1}{2\pi} \left[\int_{-1}^0 e^{-j\omega x} d\omega + \int_{-1}^0 \omega e^{-j\omega x} d\omega + \int_0^1 e^{-j\omega x} d\omega - \int_0^1 \omega e^{-j\omega x} d\omega \right] \\
 \int_{-1}^0 e^{-j\omega x} d\omega &= \frac{1}{-jx} e^{-j\omega x} \Big|_{-1}^0 = \frac{-1}{jx} [1 - e^{jx}] = \frac{e^{jx} - 1}{jx} \\
 \int_0^1 e^{-j\omega x} d\omega &= \frac{1}{-jx} e^{-j\omega x} \Big|_0^1 = \frac{-1}{jx} [e^{-jx} - 1] = \frac{1 - e^{-jx}}{jx} \\
 \int_{-1}^0 \omega e^{-j\omega x} d\omega &= \frac{-\omega e^{-j\omega x}}{jx} \Big|_{-1}^0 + \frac{1}{x^2} e^{-j\omega x} \Big|_{-1}^0 \\
 &= \frac{-e^{jx}}{jx} + \frac{1}{x^2} [1 - e^{jx}] \\
 \int_0^1 \omega e^{-j\omega x} d\omega &= \frac{-\omega e^{-j\omega x}}{jx} \Big|_0^1 + \frac{1}{x^2} e^{-j\omega x} \Big|_0^1 \\
 &= \frac{-e^{-jx}}{jx} + \frac{e^{-jx} - 1}{x^2}
 \end{aligned}$$

Substituting all integral values, we get

$$f_X(x) = \frac{1}{2\pi} \left[\frac{e^{jx} - 1}{jx} - \frac{e^{jx}}{jx} + \frac{1}{x^2} (1 - e^{jx}) + \frac{(1 - e^{-jx})}{jx} + \frac{e^{-jx}}{jx} - \frac{(e^{-jx} - 1)}{x^2} \right]$$

$$= \frac{1}{2\pi} \left[\frac{1 - e^{jx} - e^{-jx} + 1}{x^2} \right] = \frac{1}{2\pi} \left[\frac{2 - 2\cos x}{x^2} \right] = \frac{1}{\pi x^2} (1 - \cos x)$$

$$\Rightarrow f_X(x) = \frac{1}{\pi x^2} (1 - \cos x)$$

3.86 The characteristic function of a random variable X is given by $\phi_X(\omega) = \frac{1}{(1 - j2\omega)^{N/2}}$. Find the mean and second moments of X .

Solution Mean

$$E[X] = \frac{1}{j} \frac{d}{d\omega} \phi_X(\omega) \Big|_{\omega=0}$$

$$\phi_X(\omega) = \frac{1}{(1 - j2\omega)^{N/2}}$$

$$\frac{d\phi_X(\omega)}{d\omega} = \frac{d}{d\omega} \left[\frac{1}{(1 - j2\omega)^{N/2}} \right]$$

$$= -\frac{N}{2} (1 - j2\omega)^{-\frac{N}{2}-1} (-2j) = jN(1 - j2\omega)^{-\frac{N}{2}-1}$$

$$E[X] = \frac{1}{j} \left[jN(1 - j2\omega)^{-\frac{N}{2}-1} \right] \Big|_{\omega=0} = N$$

Second moment

$$E[X^2] = \left(\frac{1}{j} \right)^2 \frac{d^2}{d\omega^2} \phi_X(\omega) \Big|_{\omega=0}$$

$$= -\frac{d}{d\omega} \left[jN(1 - j2\omega)^{-\frac{N}{2}-1} \right] \Big|_{\omega=0}$$

$$= -jN \left[\left(-\frac{N}{2} - 1 \right) (1 - j2\omega)^{-\frac{N}{2}-2} (-2j) \right] \Big|_{\omega=0}$$

$$= -jN \left[-\frac{N}{2} - 1 \right] [-2j]$$

$$= 2N \left(\frac{N}{2} + 1 \right)$$

3.87 Find the density function of the random variable whose characteristic function is $e^{-\sigma^2 \omega^2 / 2}$.

Solution

$$\begin{aligned}
 f_X(x) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \phi_X(\omega) e^{-j\omega x} d\omega \\
 &= \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-\sigma^2 \omega^2 / 2} e^{-j\omega x} d\omega = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(\sigma^2 \omega^2 + 2j\omega x)} d\omega \\
 &= \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-\frac{1}{2}\left(\sigma^2 \omega^2 + 2j\omega x\right) + \left(\frac{jx}{\sigma}\right)^2 - \left(\frac{jx}{\sigma}\right)^2} d\omega \\
 &= \frac{1}{2\pi} e^{-x^2 / 2\sigma^2} \int_{-\infty}^{\infty} e^{-\frac{1}{2}\left(\sigma\omega + \frac{jx}{\sigma}\right)^2} d\omega
 \end{aligned}$$

$$\text{Let } \sigma\omega + \frac{jx}{\sigma} = t \Rightarrow d\omega = dt/\sigma$$

$$\begin{aligned}
 &= \frac{1}{2\pi\sigma} e^{-x^2 / 2\sigma^2} \int_{-\infty}^{\infty} e^{-t^2 / 2} dt \\
 &= \frac{1}{\sqrt{2\pi}\sigma} e^{-x^2 / 2\sigma^2} \underbrace{\left[\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-t^2 / 2} dt \right]}_{=1}
 \end{aligned}$$

The term $\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-t^2 / 2} dt$ is area enclosed by standard normal pdf. Therefore, it is equal to one.

$$\Rightarrow f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-x^2 / 2\sigma^2}$$



MGF AND CF OF SOME DISTRIBUTIONS 3.17

3.17.1 Binomial Distribution

The pmf of a binomial distribution is given by

$$P_X(x) = \binom{n}{x} p^x (1-p)^{n-x}$$

MGF

$$\begin{aligned}
M_X(u) &= E[e^{uX}] = \sum_{x=0}^n e^{ux} \binom{n}{x} p^x (1-p)^{n-x} \\
&= \sum_{x=0}^n \binom{n}{x} (p \cdot e^u)^x (1-p)^{n-x} \\
&= [p \cdot e^u + (1-p)]^n \\
M_X(u) &= [p \cdot e^u + (1-p)]^n
\end{aligned} \tag{3.154}$$

CF

$$\begin{aligned}
\phi_X(\omega) &= E[e^{j\omega X}] = \sum_{x=0}^n e^{j\omega x} \binom{n}{x} p^x (1-p)^{n-x} \\
&= \sum_{x=0}^n \binom{n}{x} (pe^{j\omega})^x (1-p)^{n-x} \\
&= [p \cdot e^{j\omega} + (1-p)]^n
\end{aligned} \tag{3.155}$$

3.17.2 Poisson Distribution

The pmf of a Poisson random variable is

$$P_X(x) = \frac{e^{-\lambda} \lambda^x}{x!}$$

MGF

$$\begin{aligned}
M_X(u) &= E[e^{uX}] = \sum_{x=0}^{\infty} e^{ux} \frac{e^{-\lambda} \lambda^x}{x!} \\
&= \sum_{x=0}^{\infty} \frac{e^{-\lambda} (\lambda e^u)^x}{x!} \\
&= e^{-\lambda} \sum_{x=0}^{\infty} \frac{(\lambda e^u)^x}{x!} \\
&= e^{-\lambda} \left[1 + \frac{\lambda e^u}{1!} + \frac{(\lambda e^u)^2}{2!} + \dots \right] \\
&= (e^{-\lambda} e^{\lambda e^u}) = e^{\lambda(e^u - 1)}
\end{aligned} \tag{3.156}$$

CF

$$\begin{aligned}
\phi_X(\omega) &= E[e^{j\omega X}] \\
&= \sum_{x=0}^{\infty} e^{j\omega x} \frac{e^{-\lambda} \lambda^x}{x!} = \sum_{x=0}^{\infty} \frac{e^{-\lambda} (\lambda e^{j\omega})^x}{x!}
\end{aligned}$$

$$\begin{aligned}
&= e^{-\lambda} \left[1 + \frac{(\lambda e^{j\omega})}{1!} + \frac{(\lambda e^{j\omega})^2}{2!} + \dots \right] \\
&= e^{-\lambda} e^{\lambda(e^{j\omega})} \\
&= e^{\lambda} e^{(\lambda e^{j\omega} - \lambda)}
\end{aligned} \tag{3.157}$$

3.17.3 Uniform Distribution

The density function of a uniform random variable over the interval (a, b) is given by

$$f_X(x) = \begin{cases} \frac{1}{b-a} & \text{for } a \leq x \leq b \\ 0 & \text{otherwise} \end{cases}$$

MGF

$$\begin{aligned}
M_X(u) &= E[e^{uX}] \\
&= \int_{-\infty}^{\infty} e^{ux} f_X(x) dx = \int_a^b e^{ux} \frac{1}{b-a} dx \\
&= \frac{1}{b-a} \frac{e^{ux}}{u} \Big|_a^b = \frac{e^{ub} - e^{ua}}{(b-a)u}
\end{aligned} \tag{3.158}$$

CF

$$\begin{aligned}
\phi_X(\omega) &= E[e^{j\omega X}] \\
&= \int_{-\infty}^{\infty} e^{j\omega x} f_X(x) dx = \int_a^b e^{j\omega x} \frac{1}{b-a} dx \\
&= \frac{1}{b-a} \frac{e^{j\omega x}}{j\omega} \Big|_a^b = \frac{e^{j\omega b} - e^{j\omega a}}{j\omega(b-a)}
\end{aligned} \tag{3.159}$$

3.17.4 Gaussian/Normal Distribution

The pdf of a Gaussian random variable is given by

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma_X} e^{-(x-\mu_X)^2/2\sigma_X^2} \quad \text{for } -\infty < x < \infty$$

MGF

$$\begin{aligned}
M_X(u) &= E[e^{uX}] \\
&= \int_{-\infty}^{\infty} e^{ux} f_X(x) dx = \int_{-\infty}^{\infty} e^{ux} \frac{1}{\sqrt{2\pi}\sigma_X} e^{-(x-\mu_X)^2/2\sigma_X^2} dx
\end{aligned}$$

Let $\frac{x - \mu_X}{\sigma_X} = t$
Then $x = \sigma_X t + \mu_X$
$dx = \sigma_X dt$

$$\begin{aligned}
&= \frac{1}{\sqrt{2\pi} \sigma_X} \int_{-\infty}^{\infty} e^{u(\sigma_X t + \mu_X)} e^{-t^2/2} \sigma_X dt \\
&= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{\left(\mu_X u + \sigma_X u t - \frac{t^2}{2} + \frac{\sigma_X^2 u^2}{2} - \frac{\sigma_X^2 u^2}{2} \right)} dt \\
&= \frac{1}{\sqrt{2\pi}} e^{\left(\mu_X u + \frac{\sigma_X^2 u^2}{2} \right)} \int_{-\infty}^{\infty} e^{\left(\frac{t^2}{2} + \frac{\sigma_X^2 u^2}{2} - \sigma_X u t \right)} dt \\
&= \frac{1}{\sqrt{2\pi}} e^{\left(\mu_X u + \frac{\sigma_X^2 u^2}{2} \right)} \int_{-\infty}^{\infty} e^{\frac{-(t - \mu_X u)^2}{2}} dt \\
&= \frac{1}{\sqrt{2\pi}} e^{\left(\mu_X u + \frac{\sigma_X^2 u^2}{2} \right)} \int_{-\infty}^{\infty} e^{-\frac{p^2}{2}} dp \\
&= e^{\left(\mu_X u + \frac{\sigma_X^2 u^2}{2} \right)} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-p^2/2} dp
\end{aligned} \tag{3.160}$$

Let $t - \mu_X u = p$
 $dt = dp$

$\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-p^2/2} dp$ represents the area enclosed by a random variable $N(0, 1)$ which is equal to 1. Hence,

$$M_X(u) = e^{\mu_X u + \sigma_X^2 u^2 / 2}$$

CF

$$\begin{aligned}
\phi_X(\omega) &= E[e^{j\omega X}] = \int_{-\infty}^{\infty} e^{j\omega x} f_X(x) dx \\
&= \int_{-\infty}^{\infty} e^{j\omega x} \frac{1}{\sqrt{2\pi} \sigma_X} e^{-(x - \mu_X)^2 / 2\sigma_X^2} dx \\
&= \frac{1}{\sqrt{2\pi} \sigma_X} \int_{-\infty}^{\infty} e^{j\omega(\sigma_X t + \mu_X)} e^{-t^2/2} \sigma_X dt \\
\phi_X(\omega) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{\left(j\omega \mu_X + j\sigma_X \omega t - \frac{t^2}{2} + \frac{\sigma_X^2 (j\omega)^2}{2} - \frac{\sigma_X^2 (j\omega)^2}{2} \right)} dt
\end{aligned}$$

Let $\frac{x - \mu_X}{\sigma_X} = ty$
 $\sigma_X dt$
 $\sigma_X t + \mu_X$

$$\begin{aligned}
&= \frac{1}{\sqrt{2\pi}} e^{(j\omega\mu_X + \frac{\sigma_X^2(j\omega)^2}{2})} \int_{-\infty}^{\infty} e^{-(t - j\omega\sigma_X)^2/2} dt \\
&= \frac{1}{\sqrt{2\pi}} e^{(j\omega\mu_X + \frac{\sigma_X^2(j\omega)^2}{2})} \int_{-\infty}^{\infty} e^{-p^2/2} dp \\
&= e^{(j\omega\mu_X + \frac{\sigma_X^2(j\omega)^2}{2})}
\end{aligned} \tag{3.161}$$

Let $t - j\omega\sigma_X = p$
 $dt = dp$

3.17.5 Exponential Distribution

The density function of an exponential random variable is given by

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x} & \text{for } x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

MGF

$$\begin{aligned}
M_X(u) &= \int_{-\infty}^{\infty} e^{ux} f_X(x) dx = \int_0^{\infty} e^{ux} \lambda e^{-\lambda x} dx \\
&= \lambda \int_0^{\infty} e^{-(\lambda - u)x} dx = \frac{\lambda}{-(\lambda - u)} e^{-(\lambda - u)x} \Big|_0^{\infty} \\
&= \frac{\lambda}{\lambda - u}
\end{aligned} \tag{3.162}$$

$$\Rightarrow M_X(u) = \frac{\lambda}{\lambda - u}$$

CF

$$\begin{aligned}
\phi_X(\omega) &= \int_{-\infty}^{\infty} e^{j\omega x} f_X(x) dx = \int_0^{\infty} e^{j\omega x} \lambda e^{-\lambda x} dx \\
&= \lambda \int_0^{\infty} e^{-(\lambda - j\omega)x} dx = \frac{\lambda}{-(\lambda - j\omega)} e^{-(\lambda - j\omega)x} \Big|_0^{\infty} \\
&= \frac{\lambda}{\lambda - j\omega} \\
\Rightarrow \phi_X(\omega) &= \frac{\lambda}{\lambda - j\omega}
\end{aligned} \tag{3.163}$$

3.17.6 Rayleigh Distribution

The pdf of a Rayleigh random variable is $f_X(x) = \begin{cases} \frac{x}{\sigma^2} e^{-x^2/2\sigma^2} & x \geq 0 \\ 0 & \text{elsewhere} \end{cases}$

MGF

$$\begin{aligned} M_X(u) &= \int_{-\infty}^{\infty} e^{ux} f_X(x) dx = \int_0^{\infty} e^{ux} \frac{x}{\sigma^2} e^{-x^2/2\sigma^2} dx \\ &= \frac{1}{\sigma^2} e^{\sigma^2 u^2/2} \int_0^{\infty} x e^{-\frac{1}{2}\left(\frac{x}{\sigma} - u\sigma\right)^2} dx \end{aligned}$$

Let $\left(\frac{x}{\sigma} - u\sigma\right) = t \Rightarrow x = \sigma(t + u\sigma)$
 $dx = \sigma dt$

$$\begin{aligned} M_X(u) &= \frac{1}{\sigma^2} e^{\sigma^2 u^2/2} \left\{ \int_0^{\infty} \sigma(t + u\sigma) e^{-t^2/2} \sigma dt \right\} \\ &= e^{\sigma^2 u^2/2} \left\{ \int_0^{\infty} t e^{-t^2/2} dt + \int_0^{\infty} u \sigma e^{-t^2/2} dt \right\} \\ &= e^{\sigma^2 u^2/2} \left\{ 1 + u\sigma \sqrt{\frac{\pi}{2}} \right\} \end{aligned} \quad \boxed{\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-t^2/2} dt = 1} \quad (3.164)$$

CF

Characteristic function

$$\begin{aligned} \phi_X(\omega) &= M_X(u) \Big|_{u=j\omega} \\ &= e^{-\sigma^2 \omega^2/2} \left\{ 1 + j\omega \sigma \sqrt{\frac{\pi}{2}} \right\} \end{aligned} \quad (3.165)$$

3.17.7 Gamma Distribution

The pdf of Gama random variable is

$$f_X(x) = \begin{cases} \frac{x^{\alpha-1} e^{-x/\beta}}{\beta^2 \Gamma(\alpha)} & x > 0 \\ 0 & \text{otherwise} \end{cases}$$

MGF

$$M_X(u) = \int_{-\infty}^{\infty} e^{ux} f_X dx \quad (3.166)$$

$$= \int_{-\infty}^{\infty} e^{ux} \frac{x^{\alpha-1} e^{-x/\beta}}{\beta^\alpha \Gamma(\alpha)} dx$$

$$= \frac{1}{\beta^\alpha \Gamma(\alpha)} \int_0^{\infty} x^{\alpha-1} e^{-x/\beta} e^{ux} dx = \frac{1}{\beta^\alpha \Gamma(\alpha)} \int_0^{\infty} x^{\alpha-1} e^{-\left(\frac{1-u\beta}{\beta}\right)x} dx$$

Let $\left(\frac{1-u\beta}{\beta}\right)x = t$, then $dx = \frac{\beta dt}{1-u\beta}$

$$M_X(u) = \frac{1}{\beta^\alpha \Gamma(\alpha)} \int_0^{\infty} \left(\frac{\beta t}{1-u\beta}\right)^{\alpha-1} e^{-t} \left(\frac{\beta dt}{1-u\beta}\right)$$

$$= \frac{1}{\beta^\alpha \Gamma(\alpha)} \frac{\beta^{\alpha-1}}{(1-u\beta)^{\alpha-1}} \left[\int_0^{\infty} t^{\alpha-1} e^{-t} dt \right] \frac{\beta}{1-u\beta}$$

$$= \frac{1}{\Gamma(\alpha)(1-u\beta)^\alpha} \int_0^{\infty} t^{\alpha-1} e^{-t} dt$$

$$= \frac{1}{\Gamma(\alpha)(1-u\beta)^\alpha} \Gamma(\alpha)$$

$$= (1-u\beta)^{-\alpha} \quad (3.167)$$

CF

$$f_X(\omega) = M_X(a) \Big|_{u=j\omega}$$

$$= (1-j\omega\beta)^{-\alpha} \quad (3.168)$$

3.17.8 Chi-Square Distribution

The pdf of a Chi-square random variable is given by

$$f_X(x) = \begin{cases} \frac{x^{\frac{n}{2}-1}}{2^{\frac{n}{2}} \Gamma\left(\frac{n}{2}\right)} e^{-x/2}, & x > 0 \\ 0 & \text{otherwise} \end{cases} \quad (3.169)$$

MGF

$$\begin{aligned}
 M_X(u) &= \int_0^{\infty} e^{ux} \frac{x^{\frac{n}{2}-1}}{2^{\frac{n}{2}} \Gamma\left(\frac{n}{2}\right)} e^{-x/2} dx = \int_0^{\infty} \frac{x^{\frac{n}{2}-1}}{2^{\frac{n}{2}} \Gamma\left(\frac{n}{2}\right)} e^{\left(\frac{1}{2}-u\right)x} dx \\
 &= \frac{1}{2^{\frac{n}{2}} \Gamma\left(\frac{n}{2}\right)} \int_0^{\infty} x^{\frac{n}{2}-1} e^{-\left(\frac{1}{2}-u\right)x} dx
 \end{aligned}$$

$$\text{Let } \left(\frac{1-2u}{2}\right)x = t \Rightarrow dx = \frac{2 dt}{1-2u}$$

$$\begin{aligned}
 M_X(u) &= \frac{1}{2^{\frac{n}{2}} \Gamma\left(\frac{n}{2}\right)} \int_0^{\infty} \frac{2^{\frac{n}{2}-1} t^{\frac{n}{2}-1}}{(1-2u)^{\frac{n}{2}-1}} e^{-t} \frac{2 dt}{1-2u} \\
 &= \frac{1}{\Gamma\left(\frac{n}{2}\right) (1-2u)^{\frac{n}{2}}} \int_0^{\infty} t^{\frac{n}{2}-1} e^{-t} dt \\
 &= \frac{1}{\Gamma\left(\frac{n}{2}\right) (1-2u)^{\frac{n}{2}}} \Gamma\left(\frac{n}{2}\right) \\
 &= \left(\frac{1}{1-2u}\right)^{\frac{n}{2}} \tag{3.170}
 \end{aligned}$$

CF

$$\begin{aligned}
 \phi_X(\omega) &= M_X(u) \Big|_{u=j\omega} \\
 &= \left(\frac{1}{1-j2\omega}\right)^{\frac{n}{2}} \tag{3.171}
 \end{aligned}$$

REVIEW QUESTION

21. Show that the characteristic function of a Poisson random variable is defined by

$$\phi_X(\omega) = \exp(-b(1 - e^{j\omega}))$$

3.17.10 Chernoff's Inequality and Bound

Let X be any random variable with moment-generating function $M_X(u)$. Then

$$P[X \geq a] \leq \min_{u \geq 0} e^{-au} M_X(u) \quad (3.172)$$

Proof:

Consider

$$\begin{aligned} P[X \geq a] &= \int_a^{\infty} f_X(x) dx \\ &= \int_{-\infty}^{\infty} f_X(x) u(x-a) dx \end{aligned}$$

where $u(x-a)$ is a unit step function given by

$$u(x-a) = \begin{cases} 1 & \text{for } x \geq a \\ 0 & \text{for } x < a \end{cases}$$

Figure 3.11 shows the unit step function and an exponential upper bound. From the Fig. 3.11, we can find that

$$\begin{aligned} u(x-a) &\leq e^{u(x-a)} \\ \Rightarrow P[X \geq a] &\leq \int_{-\infty}^{\infty} f_X(x) e^{u(x-a)} dx \\ &\leq e^{-au} \int_{-\infty}^{\infty} f_X(x) e^{ux} dx \\ P[X \geq a] &\leq e^{-au} M_X(u) \end{aligned} \quad (3.173)$$

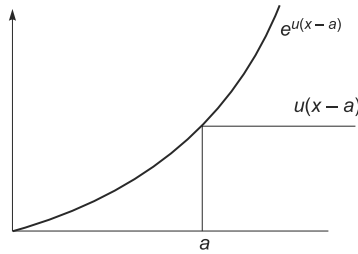


Fig. 3.11 Unit step function and an exponential upper bound

The above equation is known as Chernoff's inequality. The Chernoff's bound can be obtained by minimizing the right-hand side of Eq. (3.173).

Therefore,

$$P[X \geq a] \leq \min_{u \geq 0} e^{-au} M_X(u) \quad (3.174)$$

3.17.11 Cumulants

Consider a random variable X with MGF $M_X(u)$. If we take logarithm of $M_X(u)$, we obtain the generating function of a cumulant $\psi(u)$ given by

$$\begin{aligned} \psi_X(u) &= \log [M_X(u)] \\ &= \log \{E[e^{Xu}]\} = \log \left\{ E\left[1 + Xu + \frac{X^2 u^2}{2!} + \frac{X^3 u^3}{3!} + \dots\right] \right\} \end{aligned} \quad (3.175)$$

$$= \log \left[1 + \frac{u}{1!} E[X] + \frac{u^2}{2!} E[X^2] + \frac{u^3}{3!} E[X^3] + \dots \right] \quad (3.176)$$

We know

$$\begin{aligned} \log(1+x) &= x - \frac{x^2}{2} + \frac{x^3}{3} \dots \\ \Rightarrow \psi_X(u) &= \left\{ \frac{u}{1!} E[X] + \frac{u^2}{2!} E[X^2] + \frac{u^3}{3!} E[X^3] + \dots \right\} \\ &\quad - \frac{1}{2} \left\{ \frac{u}{1!} E[X] + \frac{u^2}{2!} E[X^2] + \frac{u^3}{3!} E[X^3] + \dots \right\}^2 \\ &\quad + \frac{1}{3} \left\{ \frac{u}{1!} E[X] + \frac{u^2}{2!} E[X^2] + \frac{u^3}{3!} E[X^3] + \dots \right\}^3 \\ &= \frac{u}{1!} E[X] + \frac{u^2}{2!} \{E[X^2] - (E[X])^2\} + \frac{u^3}{3!} [E[X^3] - 3E[X]E[X^2]] + \dots \\ &= \psi_1 u + \psi_2 \frac{u^2}{2!} + \psi_3 \frac{u^3}{3!} + \dots \end{aligned} \quad (3.177)$$

where

$$\begin{aligned} \psi_1 &= E[X] = \text{mean} \\ \psi_2 &= E[X^2] - [E[X]]^2 = \text{variance} \\ \psi_3 &= \mu_3; \psi_4 = \mu_4 - 3\psi_2^2 \dots \end{aligned} \quad (3.178)$$

The coefficient of $\frac{u^r}{r!}$ in the above series is known as components denoted by ψ_r . The cumulants are very useful in finding the central moments without finding non-central moments.

If we differentiate $\psi_X(u)$ with respect to u , ' r ' times and then substitute $u = 0$, we get the r^{th} cumulant

That is,

$$\psi_r = \left. \frac{d^r}{du^r} \psi_X(u) \right|_{u=0}$$

If $r = 1$

$$\psi_1 = \left. \frac{d\psi_X(u)}{du} \right|_{u=0}$$

$$\frac{d}{du}(\psi_X(u)) = \frac{d}{du}[\log M_X(u)] = \left. \frac{1}{M_X(u)} M'_X(u) \right|_{u=0}$$

$$M_X(0) = 1; M'_X(0) = m_1$$

$$\text{If } r = 2; \left. \frac{d}{du} \left[\frac{M'_X(u)}{M_X(u)} \right] \right|_{u=0} = \left. \frac{M_X(u)M''_X(u) - \{M'_X(u)\}^2}{[M_X(u)]^2} \right|_{u=0}$$

$$M_X(u)_{u=0} = 1$$

$$\psi_2 = \left. M''_X(u) - \{M'_X(u)\}^2 \right|_{u=0}$$

$$= E[X^2] - \{E[X]\}^2$$

$$= \text{Variance of } X$$

Solved Problems



3.88 A random variable X has probability mass function $p_X(x) = \left(\frac{1}{2}\right)^x$; $x = 1, 2, 3, \dots$

Find the moment generating function, mean and variance.

Solution

$$\text{Given: } p_X(x) = \left(\frac{1}{2}\right)^x; x = 1, 2, 3, \dots$$

$$\begin{aligned} M_X(u) &= \sum_i e^{ux_i} p_X(x_i) \\ &= e^u \left(\frac{1}{2}\right) + e^{2u} \left(\frac{1}{2}\right)^2 + e^{3u} \left(\frac{1}{2}\right)^3 + \dots \\ &= \sum_{n=1}^{\infty} e^{nu} \left(\frac{1}{2}\right)^n = \sum_{n=1}^{\infty} \left(\frac{e^u}{2}\right)^n = \sum_{n=0}^{\infty} \left(\frac{e^u}{2}\right)^n - 1 \\ &= \frac{1}{1 - e^{u/2}} - 1 = \frac{e^{u/2}}{1 - e^{u/2}} \end{aligned} \quad \boxed{\because \sum_{n=0}^{\infty} r^n = \frac{1}{1-r}}$$

$$\begin{aligned}
 E[X] &= \sum_i x_i p_X(x_i) \\
 &= 1\left(\frac{1}{2}\right) + 2\left(\frac{1}{2}\right)^2 + 3\left(\frac{1}{3}\right)^3 + \dots = \sum_{k=1}^{\infty} k \left(\frac{1}{2}\right)^k
 \end{aligned}$$

Consider

$$\begin{aligned}
 \sum_{k=1}^{\infty} k a^k &= a \sum_{k=0}^{\infty} k a^{k-1} = a \sum_{k=0}^{\infty} \frac{d}{da} (a^k) \\
 &= a \frac{d}{da} \left[\sum_{k=0}^{\infty} a^k \right] \quad \boxed{\sum_{k=1}^{\infty} k a^k = \frac{a}{(1-a)^2}} \\
 &= a \frac{d}{da} \left[\frac{1}{1-a} \right] = \frac{a}{(1-a)^2}
 \end{aligned}$$

$$\text{Here, } a = \frac{1}{2}; \quad \sum_{k=1}^{\infty} k \left(\frac{1}{2}\right)^k = \frac{1/2}{\left(1 - \frac{1}{2}\right)} = 2$$

$$\begin{aligned}
 E[X^2] &= \sum_i x_i^2 p_X(x_i) = (1)^2 \frac{1}{2} + (2)^2 \left(\frac{1}{2}\right)^2 + 3^2 \left(\frac{1}{2}\right)^3 + 4^2 \left(\frac{1}{2}\right)^4 + \dots \\
 &= \sum_{k=1}^{\infty} k^2 \left(\frac{1}{2}\right)^k
 \end{aligned}$$

$$\begin{aligned}
 \text{Consider } \sum_{k=1}^{\infty} k^2 a^k &= \sum_{k=0}^{\infty} k^2 a^k = a \sum_{k=0}^{\infty} k \cdot k \cdot a^{k-1} \\
 &= a \sum_{k=0}^{\infty} \frac{d}{da} (k a^k) = a \sum_{k=0}^{\infty} \frac{d}{da} \left[a \frac{d}{da} (a^k) \right] \\
 &= a \frac{d}{da} \left[a \frac{d}{da} \left(\sum_{k=0}^{\infty} a^k \right) \right] \\
 &= a \frac{d}{da} \left[a \frac{1}{(1-a)^2} \right] = a \frac{d}{da} \left[\frac{a}{(1-a)^2} \right] \\
 &= a \left[\frac{(1-a)^2 + 2a(1-a)}{(1-a)^4} \right] = a \left[\frac{1-a+2a}{(1-a)^3} \right] = \frac{a(1+a)}{(1-a)^3} \\
 E[X^2] &= \frac{\frac{1}{2} \left(1 + \frac{1}{2}\right)}{\left(1 - \frac{1}{2}\right)^3} = \frac{\frac{3}{4}}{\frac{1}{8}} = 6
 \end{aligned}$$

$$\text{Var}(X) = E[X^2] - \{E[X]\}^2 = 6 - 2^2 = 2$$

3.89 Find the mean and characteristic function of a binomial distribution.

Solution The binomial distribution is given by

$$\begin{aligned} p_X(x) &= \binom{n}{x} p^x (1-p)^{n-x} \\ E[X] &= \sum_{i=0}^{\infty} x_i p_X(x_i) \\ &= \sum_{x=0}^n x \binom{n}{x} p^x (1-p)^{n-x} = \sum_{x=1}^n x \binom{n}{x} p^x (1-p)^{n-x} \\ &= \sum_{x=1}^n x \frac{n!}{x!(n-x)!} p^x (1-p)^{n-x} \\ &= np \sum_{x=1}^n x \frac{(n-1)!}{x!(n-x)!} p^{x-1} (1-p)^{(n-1)-(x-1)} \\ &= np \sum_{x=1}^n \frac{(n-1)!}{(x-1)!(n-x)!} p^{x-1} (1-p)^{(n-1)-(x-1)} \\ &= np \sum_{x=1}^n \frac{(n-1)!}{(x-1)![(n-1)-(x-1)]!} p^{x-1} (1-p)^{(n-1)-(x-1)} \\ &= np \sum_{x=1}^n \binom{n-1}{x-1} p^{x-1} (1-p)^{(n-1)-(x-1)} \\ &= np(p + (1-p))^{n-1} = np \\ \phi_X(\omega) &= E[e^{j\omega X}] = \sum_{i=1}^n e^{j\omega x_i} p[x_i] = \sum_{x=0}^n e^{j\omega x} \binom{n}{x} p^x (1-p)^{n-x} \\ &= \sum_{x=0}^n \binom{n}{x} (pe^{j\omega})^x (1-p)^{n-x} \\ &= [pe^{j\omega} + (1-p)]^n \end{aligned}$$

3.90 The erlang random variable X has a characteristic function

$$\phi_X(\omega) = \left(\frac{a}{a - j\omega} \right)^N \text{ for } a > 0 \text{ and } N = 1, 2, \dots$$

Show that $\mu_X = \frac{N}{a}$; $E[X^2] = N(N+1)/a^2$ and $\sigma_X^2 = \frac{N}{a^2}$

Solution

$$\begin{aligned}
E[X] &= m_1 = -j \frac{d\phi_X(\omega)}{d\omega} \Big|_{\omega=0} \\
&= -j \left[N \left(\frac{a}{a-j\omega} \right)^{N-1} \frac{-a(-j)}{(a-j\omega)^2} \right] \Big|_{\omega=0} \\
&= -j \left[N \frac{aj}{a^2} \right] = \frac{N}{a} \\
E[X^2] &= m_2 = -\frac{d^2\phi_X(\omega)}{d\omega^2} \Big|_{\omega=0} = -\frac{d}{d\omega} \left[\frac{d\phi_X(\omega)}{d\omega} \right] \Big|_{\omega=0} \\
&= -\frac{d}{d\omega} \left[\frac{jN}{a} \left(\frac{a}{a-j\omega} \right)^{N+1} \right] \Big|_{\omega=0} \\
&= -\frac{jN}{a} \left\{ (N+1) \left(\frac{a}{a-j\omega} \right)^N \frac{-a(-j)}{(a-j\omega)^2} \right\} \Big|_{\omega=0} \\
&= -\frac{jN}{a} \left\{ \frac{(N+1)(aj)}{a^2} \right\} = \frac{N(N+1)}{a^2} \\
\sigma_X^2 &= E[X^2] - \{E[X]\}^2 \\
&= \frac{N(N+1)}{a^2} - \frac{N^2}{a^2} = \frac{N}{a^2} \quad \boxed{\sigma_X^2 = \frac{N}{a^2}}
\end{aligned}$$

3.91 Let the random variable X have the pdf with $f_X(x) = \begin{cases} \frac{1}{2}e^{-x/2}, & x > 0 \\ 0 & \text{otherwise} \end{cases}$

Find the moment-generating function and mean about the variance of X .

Solution The MGF of X is given by

$$\begin{aligned}
M_X(u) &= E[e^{uX}] = \int_{-\infty}^{\infty} e^{ux} f_X(x) dx \\
&= \int_0^{\infty} e^{ux} \left(\frac{1}{2} e^{-x/2} \right) dx = \frac{1}{2} \int_0^{\infty} e^{-\left(\frac{1}{2}-u\right)x} dx \\
&= \frac{1}{2} \left\{ \frac{e^{-\left(\frac{1}{2}-u\right)x}}{-\left(\frac{1}{2}-u\right)} \Big|_0^{\infty} \right\}
\end{aligned}$$

$$\begin{aligned}
 &= \frac{1}{2} \frac{1}{\left(\frac{1}{2} - u\right)} = \frac{1}{1 - 2u} \\
 \text{Mean} = M'_X(0) &= \frac{d}{du} \left[\frac{1}{1 - 2u} \right] \bigg|_{u=0} \\
 &= -\frac{1}{(1 - 2u)^2} (-2) \bigg|_{u=0} \\
 &= 2
 \end{aligned}$$

Variance of X

$$\begin{aligned}
 \sigma_X^2 &= E[X^2] - \{E[X]\}^2 \\
 E[X^2] &= M''_X(0) = \frac{d}{du} M'_X(u) \bigg|_{u=0} \\
 &= \frac{d}{du} \left(\frac{2}{(1 - 2u)^2} \right) \bigg|_{u=0} \\
 &= 2 \left(\frac{-2}{(1 - 2u)^3} (-2) \right) \bigg|_{u=0} \\
 &= 8 \\
 \sigma_X^2 &= E[X^2] - \{E[X]\}^2 \\
 &= 8 - (2)^2 = 4
 \end{aligned}$$

3.92 A discrete random variable has moment generating function

$$M_X(u) = \left(\frac{1}{4} + \frac{3}{4} e^u \right)^5.$$

Find $E(X)$, $\text{Var}(X)$, and $P(X = 2)$.

Solution

$$\begin{aligned}
 \text{Given: } M_X(u) &= \left(\frac{1}{4} + \frac{3}{4} e^u \right)^5 \\
 E[X] &= m'_X(u) \bigg|_{u=0} \\
 &= \left\{ \frac{d}{du} \left(\frac{1}{4} + \frac{3}{4} e^u \right)^5 \right\} \bigg|_{u=0}
 \end{aligned}$$

$$\begin{aligned}
&= 5 \left(\frac{1}{4} + \frac{3}{4} e^u \right)^4 \left(\frac{3}{4} e^u \right) \Big|_{u=0} \\
&= 5 \left(\frac{3}{4} \right) = \frac{15}{4} \\
E[X^2] &= M_X''(u) \Big|_{u=0} \\
&= \frac{d}{du} \left\{ \frac{15}{4} e^u \left(\frac{1}{4} + \frac{3}{4} e^u \right)^4 \right\} \Big|_{u=0} \\
&= \frac{15}{4} \left\{ \left(\frac{1}{4} + \frac{3}{4} e^u \right)^4 e^u + 4 e^u \left(\frac{1}{4} + \frac{3}{4} e^u \right)^3 \left(\frac{3}{4} e^u \right) \right\} \Big|_{u=0} \\
&= \frac{15}{4} \left\{ \left(\frac{1}{4} + \frac{3}{4} \right)^4 + 4 \left(\frac{1}{4} + \frac{3}{4} \right)^3 \frac{3}{4} \right\} \\
&= \frac{15}{4} \{1 + 3\} = 15 \\
\text{Var}(X) &= E[X^2] - \{E[X]\}^2 \\
&= 15 - \left(\frac{15}{4} \right)^2 = \frac{15}{16}
\end{aligned}$$

If the given MGF is of a binomial random variable X ,

$$P[X = x] = \binom{n}{x} p^x q^{n-x}$$

For a binomial random variable,

$$\begin{aligned}
\text{Mean} &= np = \frac{15}{4} \\
n = 5 \quad \Rightarrow \quad p &= \frac{3}{4} \text{ and } q = \frac{1}{4} \\
P[X = 2] &= \binom{5}{2} \left(\frac{3}{4} \right)^2 \left(\frac{1}{4} \right)^3 \\
&= 10 \left(\frac{9}{16} \right) \left(\frac{1}{64} \right) = \frac{415}{512}
\end{aligned}$$

3.93 Show that for the uniform distribution $f_X(x) = \frac{1}{2a} - a < x < a$, the moment-generating function about the origin is $\frac{\sin au}{au}$.

Solution

Given: $f_X(x) = \begin{cases} \frac{1}{2a}; & -a < x < a \\ 0 & \text{otherwise} \end{cases}$

$$\begin{aligned} M_X(u) &= E[e^{uX}] = \int_{-\infty}^{\infty} e^{ux} f_X(x) dx \\ &= \int_{-a}^a e^{ux} \left(\frac{1}{2a} \right) dx = \frac{1}{2a} \int_{-a}^a e^{ux} dx \\ &= \frac{1}{2au} e^{ux} \Big|_{-a}^a = \frac{1}{2au} (e^{au} - e^{-au}) \\ &= \frac{\sinh au}{au} \end{aligned}$$

3.94 The MGF of a uniform distribution for a random variable X is $\frac{1}{u}(e^{5u} - e^{4u})$. Find $E[X]$.

Solution

Given: $M_X(u) = \frac{e^{5u} - e^{4u}}{u}$

For a uniform distribution, the MGF is given by

$$M_X(u) = \frac{1}{b-a} \left[\frac{e^{bu} - e^{au}}{u} \right]$$

Comparing the above equation with the given MGF, we find $b = 5$ and $a = 4$

Therefore, $E[X] = \frac{b+a}{2} = \frac{9}{2}$

3.95 The pdf of a random variable X is given by

$$f_X(x) = \begin{cases} \frac{2^{n+1} x^n e^{-2x}}{n!} & x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

Find the characteristic function, mean and variance.

Solution

Given: $f_X(x) = \begin{cases} \frac{2^{n+1} x^n e^{-2x}}{n!} & x \geq 0 \\ 0 & \text{otherwise} \end{cases}$

The characteristic function

$$\begin{aligned}\phi_X(\omega) &= E[e^{j\omega X}] \\ &= \int_0^{\infty} \frac{e^{j\omega x} 2^{n+1} x^n e^{-2x}}{n!} dx \\ &= \frac{2^{n+1}}{n!} \int_0^{\infty} e^{-(2-j\omega)x} x^n dx\end{aligned}$$

Let $(2-j\omega) = a$,

Then

$$\begin{aligned}\phi_X(\omega) &= \frac{2^{n+1}}{n!} \int_0^{\infty} e^{-ax} x^n dx \\ &= \frac{2^{n+1}}{n!} \frac{n!}{(a)^{n+1}} = \left(\frac{2}{a}\right)^{n+1} \\ &= \left(\frac{2}{2-j\omega}\right)^{n+1} = \left(\frac{2-j\omega}{2}\right)^{-(n+1)} \\ &= \left(1 - \frac{j\omega}{2}\right)^{-(n+1)}\end{aligned}$$

The mean is equal to

$$\begin{aligned}E[X] &= \frac{1}{j} \phi'(\omega) \Big|_{\omega=0} \\ \phi'(\omega) &= \frac{d}{d\omega} \left\{ \left[1 - \frac{j\omega}{2} \right]^{-(n+1)} \right\} \\ &= -(n+1) \left(1 - \frac{j\omega}{2} \right)^{-(n+2)} \left(\frac{-j}{2} \right) \\ \phi'(\omega) \Big|_{\omega=0} &= \frac{j(n+1)}{2} \\ E[X] &= \frac{1}{j} \phi'(\omega) \Big|_{\omega=0} = \frac{1}{j} \frac{j(n+1)}{2} = \frac{n+1}{2} \\ E[X] &= \frac{n+1}{2}\end{aligned}$$

Variance

$$\mu_X = E[X^2] - \{E[X]\}^2$$

$$\begin{aligned}
E[X^2] &= \left(\frac{1}{j}\right)^2 \phi''(\omega) \Big|_{\omega=0} \\
\phi''(\omega) &= -(n+1) [-(n+2)] \left[1 - \frac{j\omega}{2}\right]^{-(n+3)} \left(-\frac{j}{2}\right)^2 \\
\phi''(0) &= \frac{-(n+1)(n+2)}{2^2} \\
E[X^2] &= -\phi''(0) = \frac{(n+1)(n+2)}{2^2} \\
\sigma_X^2 &= E[X^2] - \{E[X]\}^2 \\
&= \frac{(n+1)(n+2)}{2^2} - \frac{(n+1)^2}{2^2} \\
&= \frac{n+1}{4} [(n+2) - (n+1)] = \frac{n+1}{4} \quad \boxed{\sigma_X^2 = \frac{n+1}{4}}
\end{aligned}$$

3.96 The characteristic function of a random variable is given by

$$\phi_X(\omega) = \frac{1}{1 - j\omega a} \quad a > 0$$

Find mean and variance.

Solution

$$\begin{aligned}
\text{Given: } \phi_X(\omega) &= \frac{1}{1 - j\omega a} \\
\text{Mean} &= \frac{1}{j} \phi'_X(\omega) \Big|_{\omega=0} \\
\phi'_X(\omega) &= \frac{-1}{(1 - j\omega a)^2} (-ja) = \frac{ja}{(1 - j\omega a)^2} \\
E[X] &= \frac{1}{j} \frac{ja}{(1 - j\omega a)^2} \Big|_{\omega=0} = a \\
\sigma_X^2 &= \text{Var}(X) = E[X^2] - \{E[X]\}^2 \\
E[X^2] &= \left(\frac{1}{j}\right)^2 \phi''_X(\omega) \Big|_{\omega=0} \\
\phi''_X(\omega) &= \frac{d}{d\omega} \left[\frac{ja}{(1 - j\omega a)^2} \right] = \frac{-2(ja)(-ja)}{(1 - j\omega a)^3} \\
&= \frac{-2a^2}{(1 - j\omega a)^3}
\end{aligned}$$

$$\begin{aligned}
\phi_X''(\omega)|_{\omega=0} &= -2a^2 \\
E[X^2] &= -\phi_X''(\omega)|_{\omega=0} = 2a^2 \\
\sigma_X^2 &= \text{Var}(X) = E[X^2] - \{E[X]\}^2 \\
&= 2a^2 - a^2 = a^2
\end{aligned}$$

3.97 If the random variable X has the moment-generating function $M_X(u) = \frac{2}{2-u}$, determine the variance of X .

Solution

$$\begin{aligned}
\text{Given: } M_X(u) &= \frac{2}{2-u} \\
E[X] &= M_X'(u)|_{u=0} \\
E[X^2] &= M_X''(u)|_{u=0} \\
M_X'(u) &= \frac{d}{du} \left(\frac{2}{2-u} \right) = \frac{2}{(2-u)^2} \\
E[X] &= M_X'(0) = \frac{1}{2} \\
M_X''(u) &= \frac{d}{du} \left[\frac{2}{(2-u)^2} \right] = \frac{4}{(2-u)^3} \\
E[X^2] &= M_X''(0) = \frac{1}{2} \\
\sigma_X^2 &= E[X^2] - \{E[X]\}^2 \\
&= \frac{1}{2} - \left(\frac{1}{2} \right)^2 = \frac{1}{4} \\
\sigma_X^2 &= \frac{1}{4}
\end{aligned}$$

3.98 A random variable X is uniformly distributed on $(0, 6)$. If X is transformed to a new random variable $Y = 2(X-3)^2 - 4$, find $E[Y]$, $\text{Var}[Y]$.

Solution X is a uniformly distributed random variable with $b = 6$ and $a = 0$.

Therefore,

$$f_X(x) = \begin{cases} \frac{1}{6}; & 0 \leq x \leq 6 \\ 0 & \text{otherwise} \end{cases}$$

$$\begin{aligned}
E[X] &= \frac{b+a}{2} = \frac{6}{2} = 3 \\
\sigma_X^2 &= \frac{(b-a)^2}{12} = \frac{(6)^2}{12} = 3 \\
\sigma_X^2 &= E[X^2] - \{E[X]\}^2 \\
\Rightarrow E[X^2] &= \sigma_X^2 + \{E[X]\}^2 \\
&= 3 + (3)^2 = 12 \\
Y &= 2(X-3)^2 - 4 \\
E[Y] &= E[2(X-3)^2 - 4] \\
&= E[2(X^2 - 6X + 9) - 4] \\
&= 2E[X^2] - 12E[X] + 14 \\
&= 2(12) - 12(3) + 14 = 2 \\
E[Y] &= 2 \\
E[Y^2] &= E[\{2(X-3)^2 - 4\}^2] \\
&= E[\{2X^2 - 12X + 14\}^2] \\
&= E[4X^4 + 144X^2 + 196 - 48X^3 - 336X + 56X^2] \\
&= 4E[X^4] - 48E[X^3] + 200E[X^2] - 336E[X] + 196 \\
E[X^4] &= \int_{-\infty}^{\infty} x^4 f_X(x) dx = \frac{1}{6} \int_0^b x^4 dx = \frac{x^5}{5(6)} = \frac{6^4}{5} \\
E[X^3] &= \int_{-\infty}^{\infty} x^3 f_X(x) dx = \frac{1}{6} \int_0^b x^3 dx = \frac{x^4}{4(6)} \Big|_0^6 = \frac{6^3}{4} \\
E[Y^2] &= 4 \left(\frac{6^4}{5} \right) - 48 \left(\frac{6^3}{4} \right) + 200(12) - 336(3) + 196 \\
&= 1036.8 - 2592 + 2400 - 1008 + 196 \\
&= 32.8 \\
\sigma_Y^2 &= E[Y^2] - \{E[Y]\}^2 \\
&= 32.8 - (2)^2 = 28.8
\end{aligned}$$

3.99 A fair coin is tossed until tail appears. Find the mathematical expectation of the number of tosses.

Solution Assume X be the random variable of the event number of tosses. Since a fair coin is tossed, $P(H) = \frac{1}{2}$ and $P(T) = \frac{1}{2}$.

The possible outcomes until a tail appears is $T, HT, HHT, HHHT, HHHHT, HHHHHT$, and so on.

$$\begin{aligned}
P(T) &= \frac{1}{2}; P(HT) = \left(\frac{1}{2}\right)\left(\frac{1}{2}\right) = \frac{1}{2^2}; P(HHT) = \left(\frac{1}{2}\right)^3 \\
P(HHHT) &= \left(\frac{1}{2}\right)^4; P(HHHHT) = \left(\frac{1}{2}\right)^5; P(HHHHHT) = \left(\frac{1}{2}\right)^6 \\
E[X] &= \sum x_i p_X(x_i) \\
&= 1\left(\frac{1}{2}\right) + 2\left(\frac{1}{2}\right)^2 + 3\left(\frac{1}{2}\right)^3 + 4\left(\frac{1}{2}\right)^4 + 5\left(\frac{1}{2}\right)^5 + \dots
\end{aligned}$$

The above expression is of the form

$$\begin{aligned}
&\sum_{k=1}^{\infty} k a^k \text{ where } a = \frac{1}{2} \\
&\sum_{k=1}^{\infty} k a^k = \frac{a}{(1-a)^2} \\
\Rightarrow E[X] &= \frac{1/2}{\left(1 - \frac{1}{2}\right)^2} = 2
\end{aligned}$$

3.100 If the mean and variance of the binomial distribution are 6 and 1.5 respectively, find $E[X - P(X \geq 3)]$.

Solution A random variable 'X' is said to be binomial if its pdf

$$f_X(x) = \binom{n}{x} \begin{cases} p^x q^{n-x} & x = 0, 1, 2, \dots, n \\ 0 & \text{otherwise} \end{cases}$$

The mean and variance of binomial random variable are $E[X] = np$ and $\sigma_X^2 = npq$

Given $np = 6$ and $npq = 1.5$

$$\Rightarrow q = \frac{npq}{np} = \frac{1.5}{6} = 0.25$$

$$\text{Also } p = 1 - q = 1 - 0.25 = 0.75$$

$$np = 6 \Rightarrow n = 6/p = 6/0.75 = 8$$

$$\begin{aligned}
E[X - P(X \geq 3)] &= E[X] - E\{P(X \geq 3)\} \\
&= E[X] - P[X \geq 3]
\end{aligned}$$

$$P(X \geq 3) = 1 - P(X < 3)$$

$$P(X \leq 3) = 1 - P(X = 0) - P(X = 1) - P(X = 2)$$

$$P(X=0) = \binom{8}{0} p^0 q^{8-0} = (0.25)^8 = 15.25 \times 10^{-6}$$

$$P(X=1) = \binom{8}{1} (0.75)^1 (0.25)^7 = 366.2 \times 10^{-6}$$

$$P(X=2) = \binom{8}{2} (0.75)^2 (0.25)^6 = 3.845 \times 10^{-3}$$

$$P(X < 3) = 0.995$$

$$\begin{aligned} E[X - P(X \geq 3)] &= E(X) - P(X \geq 3) \\ &= 6 - 0.995 \\ &= 5.005 \end{aligned}$$

3.101 A Gaussian random variable X with $\mu_X = 0$ and $\sigma_X^2 = 16$ is applied to square law, full-wave diode detector with transfer characteristic $Y = 3X^2$. Find the mean value of the output voltage Y .

Solution

Given: $Y = 3X^2$

$$E[Y] = \int_{-\infty}^{\infty} 3x^2 f_X(x) dx = 3 \int_{-\infty}^{\infty} x^2 \frac{e^{-(x-\mu_X)^2/2\sigma_X^2}}{\sigma_X \sqrt{2\pi}} dx$$

$\mu_X = 0$ and $\sigma_X^2 = 16$

$$E[Y] = 3 \int_{-\infty}^{\infty} x^2 \frac{e^{-x^2/2\sigma_X^2}}{\sigma_X \sqrt{2\pi}} dx \quad \text{Let } \frac{x}{\sigma_X} = \xi$$

$$dx = \sigma_X d\xi$$

$$= 3 \int_{-\infty}^{\infty} \frac{(\sigma_X^2 \xi^2) e^{-\xi^2/2}}{\sigma_X \sqrt{2\pi}} \sigma_X d\xi$$

$$= 3 \sigma_X^2 \int_{-\infty}^{\infty} \underbrace{\frac{\xi^2}{\sqrt{2\pi}}}_{=1} e^{-\xi^2/2} d\xi$$

$$= 3 \sigma_X^2$$

$$= 3(16) = 48$$

3.102 Show that $E[(X-a)^2]$ is minimum when $a = \mu_X$.

Solution Consider

$$Y = E[(X-a)^2] = E[X^2 - 2aX + a^2]$$

$$= E[X^2] - 2aE[X] + a^2$$

$$\frac{dY}{da} = -2aE[X] + 2a = 0$$

$$\Rightarrow E[X] = a$$

3.103 A continuous random variable X has a density function

$$f_X(x) = \begin{cases} \frac{2x}{9} & 0 < x < 3 \\ 0 & \text{otherwise} \end{cases}$$

(a) Find CDF. (b) Find μ_X and σ_X . (c). Find a value of m such that $p_X(X \geq m) = p_X(X \leq m)$.

Solution

$$(a) \quad \text{The CDF } F_X(x) = \int_{-\infty}^x f_X(u) du$$

$$\text{For } x < 0; F_X(x) = 0$$

$$\begin{aligned} \text{For } 0 < x < 3, F_X(x) &= \int_{-\infty}^x \frac{2u}{9} du = \int_0^x \frac{2u}{9} du = \frac{2}{9} \frac{u^2}{2} \Big|_0^x \\ &= \frac{2}{9} \left(\frac{x^2}{2} \right) = \frac{x^2}{9} \end{aligned}$$

$$F_X(x) = 1 \quad \text{for } x > 3.$$

$$\begin{aligned} (b) \quad \mu_X = E[X] &= \int_{-\infty}^{\infty} x f_X(x) dx \\ &= \int_0^3 x \left(\frac{2x}{9} \right) dx = \frac{2}{9} \int_0^3 x^2 dx = \frac{2}{9} \frac{x^3}{3} \Big|_0^3 = 2 \end{aligned}$$

$$\begin{aligned} E[X^2] &= \int_{-\infty}^{\infty} x^2 f_X(x) dx \\ &= \int_{-\infty}^{\infty} x^2 \left(\frac{2x}{9} \right) dx = \frac{2}{9} \int_0^3 x^3 dx \\ &= \frac{2}{9} \frac{x^4}{4} \Big|_0^3 = \frac{9}{2} \end{aligned}$$

$$\sigma_X^2 = E[X^2] - (\mu_X)^2 = \frac{9}{2} - 4 = \frac{1}{2}$$

$$(c) \quad p_X(X \geq m) = \int_m^{\infty} f_X(x) dx = \int_m^3 \frac{2}{9} x dx$$

$$p_X(X \leq m) = \int_0^m f_X(x) dx = \int_0^m \frac{2}{9} x dx$$

$$\text{Given: } p_X(X \geq m) = p_X(X \leq m)$$

$$\Rightarrow \int_m^3 x dx = \int_0^m x dx$$

$$\frac{9}{2} - \frac{m^2}{2} = \frac{m^2}{2} \Rightarrow m^2 = \frac{9}{2}.$$

$$m = \frac{3}{\sqrt{2}}$$

3.104 A random variable X has the density function

$$f_X(x) = \frac{1}{a} e^{-b|x|} \quad -\infty \leq x \leq \infty$$

Find $E[X]$, $E[X^2]$.

Solution

$$\begin{aligned} E[X] &= \int_{-\infty}^{\infty} x f_X(x) dx = \int_{-\infty}^{\infty} x \frac{1}{a} e^{-b|x|} dx \\ &= \int_{-\infty}^0 x \frac{1}{a} e^{bx} dx + \int_0^{\infty} x \frac{1}{a} e^{-bx} dx \\ &= \frac{1}{a} \left[\int_{-\infty}^0 x e^{bx} dx + \int_0^{\infty} x e^{-bx} dx \right] \\ &= \frac{1}{a} \left[\left. \frac{x e^{bx}}{b} \right|_{-\infty}^0 - \int_{-\infty}^0 \frac{e^{bx}}{b} dx + \left. \left(\frac{-x}{b} e^{-bx} \right) \right|_0^{\infty} + \int_0^{\infty} \frac{e^{-bx}}{b} dx \right] \\ &= \frac{1}{a} \left[(0 - 0) - \frac{1}{b^2} + (0 - 0) + \frac{1}{b^2} \right] = 0 \end{aligned}$$

$$\begin{aligned} E[X^2] &= \int_{-\infty}^{\infty} x^2 f_X(x) dx = \int_{-\infty}^{\infty} x^2 \frac{1}{a} e^{-b|x|} dx \\ &= \frac{1}{a} \int_{-\infty}^{\infty} x^2 e^{-b|x|} dx = \frac{1}{a} \left[\int_{-\infty}^0 x^2 e^{bx} dx + \int_0^{\infty} x^2 e^{-bx} dx \right] \\ &= \frac{1}{a} \left[\left. \frac{x^2 e^{bx}}{b} \right|_{-\infty}^0 - \frac{2xe^{bx}}{b^2} \Big|_{-\infty}^0 + \frac{2}{b^3} e^{bx} \Big|_{-\infty}^0 - \left. \frac{x^2 e^{-bx}}{b} \right|_0^{\infty} - \frac{2x}{b^2} e^{-bx} \Big|_0^{\infty} - \frac{2}{b^3} e^{-bx} \Big|_0^{\infty} \right] \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{a} \left[0 - 0 - 2(0 - 0) + \frac{2}{b^3} - (0 - 0) - (0 - 0) - \frac{2}{b^3}(0 - 1) \right] \\
&= \frac{4}{ab^3} \\
E[X^2] &= \frac{4}{ab^3}
\end{aligned}$$

3.105 Let X be a random variable defined by the density function

$$f_X(x) = \begin{cases} \frac{5}{4}(1 - x^4) & 0 < x \leq 1 \\ 0 & \text{elsewhere} \end{cases}$$

Find $E[X]$, $E[X^2]$ and variance.

Solution

Given:
$$f_X(x) = \begin{cases} \frac{5}{4}(1 - x^4) & 0 < x \leq 1 \\ 0 & \text{elsewhere} \end{cases}$$

$$\begin{aligned}
E[X] &= \int_{-\infty}^{\infty} x f_X(x) dx = \int_0^1 \frac{5}{4} x (1 - x^4) dx \\
&= \frac{5}{4} \left[\frac{x^2}{2} - \frac{x^6}{6} \right]_0^1 = \frac{5}{4} \left[\frac{1}{2} - \frac{1}{6} \right] = \frac{5}{4} \left(\frac{3-1}{6} \right) = \frac{5}{12}
\end{aligned}$$

$$\therefore E[X^2] = \frac{5}{12}$$

$$\begin{aligned}
E[X^2] &= \int_{-\infty}^{\infty} x^2 f_X(x) dx = \int_0^1 \frac{5}{4} x^2 (1 - x^4) dx = \frac{5}{4} \left(\frac{x^3}{3} - \frac{x^7}{7} \right) \Big|_0^1 \\
&= \frac{5}{4} \left[\frac{1}{3} - \frac{1}{7} \right] = \frac{5}{4} \left[\frac{4}{21} \right] = \frac{5}{21}
\end{aligned}$$

$$\therefore E[X^2] = \frac{5}{21}$$

$$\begin{aligned}
\text{Variance } \sigma_X^2 &= E[X^2] - \{E[X]\}^2 \\
&= \frac{5}{21} - \left(\frac{5}{12} \right)^2 = \frac{5}{21} - \frac{25}{144} \\
&= 0.6448
\end{aligned}$$

3.106 In a hall, three ACs are operational with probabilities 0.75, 0.85 and 0.95 respectively. The operation of each AC is independent of each other. Let X be the number of ACs that are operational. Find the mean and variance of X .

Solution X is the number of ACs that are in operation. So it takes four values $X = 0, X = 1, X = 2$ and $X = 3$.

$$\text{We know } E[X] = \sum_i x_i p_X(x_i) \text{ and } E[X^2] = \sum_i x_i^2 p_X(x_i)$$

$$P_X(X = 0) = (1 - 0.75)(1 - 0.85)(1 - 0.95) = 0.001875$$

$$P_X(X = 1) = (0.75)(1 - 0.85)(1 - 0.95) + (1 - 0.75)(0.85)(1 - 0.95) + (1 - 0.75)(1 - 0.85)(0.95) \\ = 0.051875$$

$$P_X(X = 2) = (0.75)(0.85)(1 - 0.95) + (0.85)(0.95)(1 - 0.75) + (0.75)(1 - 0.85)(0.95) \\ = 0.340625$$

$$P_X(X = 3) = (0.85)(0.75)(0.95) \\ = 0.605625$$

$$E[X] = (0) P_X(X = 0) + (1) P_X(X = 1) + (2) P_X(X = 2) + (3) P_X(X = 3) \\ = 0(0.001875) + 1(0.051875) + 2(0.340625) + (3)(0.605625) \\ = 2.55$$

$$E[X^2] = (0)^2 P_X(X = 0) + (1)^2 P_X(X = 1) + (2)^2 P_X(X = 2) + (3)^2 P_X(X = 3) \\ = 6.865$$

$$\sigma_x^2 = \text{Var}(X) = E[X^2] - \{E[X]\}^2 \\ = 6.865 - (2.55)^2 = 0.3625$$

3.107 In a village, a sales woman is selling soap packets house to house. The probability that she will sell a soap packet at any house she visit is 0.25.

- If she visited 10 houses, what is the probability that she sold soap packets exactly six of these houses?
- If she visited 10 houses, what is the expected number of packets she sold?
- What is the probability that she sold fourth packet in the eighth house she visited?

Solution Let X denote the number of packets the sales woman sold.

- X is binomial random variable with the probability of success 0.25. The probability that she sell exactly six packets in 10 houses is

$$P(X = 6) = \binom{10}{6} (0.25)^6 (0.75)^4 = 0.0162$$

- The expected number of soap packets that she sold after visiting 10 houses is

$$E[X] = np = 10(0.25) = 2.5$$

- If she sells fourth packet in the eighth house she visit, this means that she sold 3 packets during her visit of seven houses.

Therefore the desired probability is given by

$$\begin{aligned} & \binom{8-1}{4-1} (0.25)^4 (0.75)^4 \\ &= \binom{7}{3} (0.25)^4 (0.75)^4 = 0.043 \end{aligned}$$

3.108 If X and Y are independent Poisson variates, show that conditional distribution of X given $X + Y$ is binomial.

Solution Since X and Y are independent Poisson variates, $X + Y$ is also Poisson random variable.

Let X be distributed with parameter λ_x and Y be distributed with parameter λ_y . Then $X + Y$ is Poisson random variable with parameter $\lambda_x + \lambda_y$.

$$\begin{aligned} P(X = x | X + Y = z) &= \frac{P(X = x \cap Y = z - x)}{P(X + Y = z)} \\ &= \frac{P(X = x) P(Y = z - x)}{P(X + Y = z)} \\ &= \frac{\frac{e^{-\lambda_x} (\lambda_x)^x}{x!} \cdot \frac{e^{-\lambda_y} (\lambda_y)^{z-x}}{(z-x)!}}{\frac{e^{-(\lambda_x + \lambda_y)} (\lambda_x + \lambda_y)^z}{z!}} \\ &= \frac{z!}{x! (z-x)!} \frac{(\lambda_x)^x (\lambda_y)^{z-x}}{(\lambda_x + \lambda_y)^z} = \frac{z!}{x! (z-x)!} \frac{(\lambda_x)^x (\lambda_y)^{z-x}}{(\lambda_x + \lambda_y)^x (\lambda_x + \lambda_y)^{z-x}} \\ &= \binom{z}{x} \left(\frac{\lambda_x}{\lambda_x + \lambda_y} \right)^x \left(\frac{\lambda_y}{\lambda_x + \lambda_y} \right)^{z-x} \\ &= \binom{z}{x} p^x (1-p)^{z-x} \end{aligned}$$

Therefore the conditional distribution is a binomial distribution with $p = \frac{\lambda_x}{\lambda_x + \lambda_y}$

3.109 A random sample of size n is taken from a general Gamma distribution with parameters λ and α . Show that the mean μ_X of the sample also follows a Gamma distribution with parameter λ and $n\alpha$.

Solution The pdf of gamma random variable X is given by

$$f_X(x) = \begin{cases} \frac{\lambda(\lambda x)^{\alpha-1} e^{-\lambda x}}{\Gamma(\alpha)}, & x > 0 \\ 0 & \text{elsewhere} \end{cases}$$

The MGF of X is given by

$$M_X(u) = \left(\frac{\lambda}{\lambda - u} \right)^\alpha$$

The mean of the sample is

$$\mu_X = \frac{X_1 + X_2 + \cdots + X_n}{n}$$

where $X_1, X_2, \dots, X_n$ are the members of the sample.

Since $X_1, X_2, \dots, X_n$ are independent, using reproductive property of Gamma distribution we can write

$$\begin{aligned} M_{\mu_X}(u) &= \frac{M_{X_1 + X_2 + \cdots + X_n}(u)}{n} \\ &= M_{X_1 + X_2 + \cdots + X_n}\left(\frac{u}{n}\right) \\ &= M_{X_1}\left(\frac{u}{n}\right) M_{X_2}\left(\frac{u}{n}\right) \cdots M_{X_n}\left(\frac{u}{n}\right) \\ &= \left(\frac{\lambda}{\lambda - \frac{u}{n}} \right)^\alpha \left(\frac{\lambda}{\lambda - \frac{u}{n}} \right)^\alpha \cdots \left(\frac{\lambda}{\lambda - \frac{u}{n}} \right)^\alpha \\ &= \left(\frac{\lambda}{\lambda - \frac{u}{n}} \right)^{n\alpha} \end{aligned}$$

Therefore μ_X follows Gamma distribution with parameter λ and $n\alpha$.

3.110 Prove additive property of binomial distribution.

Solution If X_1 and X_2 are two independent binomial variables with parameters (n_1, p) and (n_2, p) respectively, then $X_1 + X_2$ is a binomial random variable with parameter $(n_1 + n_2, p)$.

Proof Let $M_{X_1}(u)$ and $M_{X_2}(u)$ are moment generating functions of X_1 and X_2 respectively.

Since X_1 and X_2 are independent

$$M_{X_1 + X_2}(u) = M_{X_1}(u) M_{X_2}(u)$$

we have

$$M_{X_1}(u) = (q + pe^u)^{n_1}$$

and

$$M_{X_2}(u) = (q + pe^u)^{n_2}$$

$$\begin{aligned} M_{X_1 + X_2}(u) &= M_{X_1}(u) M_{X_2}(u) = (q + pe^u)^{n_1} (q + pe^u)^{n_2} \\ &= (q + pe^u)^{n_1 + n_2} \end{aligned}$$

That is $X_1 + X_2$ is also binomial random variable with parameters $(n_1 + n_2, p)$.

3.111 A discrete random variable X has MGF $\left(\frac{1}{3} + \frac{2}{3} e^u\right)^4$. Find $E(X)$, $\text{Var}(X)$ and $P(X=3)$.

Solution Comparing with MGF of binomial random variable, we get $q = \frac{1}{3}$; $p = \frac{2}{3}$ and $n = 4$

$$E[X] = np = 4\left(\frac{2}{3}\right) = \frac{8}{3}$$

$$\text{Var}(X) = npq = 4\left(\frac{1}{3}\right)\left(\frac{2}{3}\right) = \frac{8}{9}$$

$$P(X=x) = \binom{n}{x} p^x q^{n-x}; x=0, 1, 2, \dots, n$$

$$P(X=2) = \binom{4}{2} \left(\frac{2}{3}\right)^2 \left(\frac{1}{3}\right)^2 = 6 \left(\frac{4}{9}\right) \left(\frac{1}{9}\right) = \frac{8}{27}$$

3.112 Prove that for binomial distribution

$$\mu_{r+1} = pq \left(nr \mu_{r-1} + \frac{d}{dp} \mu_r \right)$$

Using the above relation find μ_2 , μ_3 and μ_4

Solution The r^{th} order central moment μ_r is given by

$$\begin{aligned} \mu_r &= E[X - \mu_X]^r = \sum (x - \mu_r)^r p_X(x) \\ &= \sum (x - np)^r \binom{n}{x} p^x (1-p)^{n-x} \end{aligned}$$

Differentiating with respect to p we get,

$$\begin{aligned} \frac{d\mu_r}{dp} &= \frac{d}{dp} \left\{ \sum (x - np)^r \binom{n}{x} p^x (1-p)^{n-x} \right\} \\ &= \sum \binom{n}{x} \{ (x - np)^{r-1} (-rn) p^x (1-p)^{n-x} + (x - np)^r x p^{x-1} (1-p)^{n-x} \\ &\quad + (x - np)^r p^x (n-x)(1-p)^{n-x-1} (-1) \} \\ &= -rn \underbrace{\sum \binom{n}{x} (x - np)^{r-1} p^x (1-p)^{n-x}}_{\mu_{r-1}} + \sum \binom{n}{x} (x - np)^r p^{x-1} (1-p)^{n-x-1} \{ x(1-p) - (p(n-x)) \} \\ &= -rn \mu_{r-1} + \sum \binom{n}{x} (x - np)^r p^{x-1} (1-p)^{n-x-1} (x - np) \end{aligned}$$

$$= -rn \mu_{r-1} + \frac{1}{pq} \underbrace{\sum \binom{n}{x} (x - np)^{r+1} p^x q^{n-x}}_{\mu_{r+1}}$$

$$= -rn \mu_{r-1} + \frac{1}{pq} \mu_{r+1}$$

$$\Rightarrow \mu_{r+1} = pq \left[\frac{d\mu_r}{dp} + nr \mu_{r-1} \right]$$

$$\mu_2 = \mu_{r+1} \Big|_{r=1} = pq \left[\frac{d\mu_1}{dp} + n \mu_0 \right]$$

We know $\mu_1 = 0$ and $\mu_0 = 1$
 $\mu_2 = npq = np(1-p)$

$$\mu_3 = \mu_{r+1} \Big|_{r=2} = pq \left[\frac{d\mu_2}{dp} + 2n \mu_1 \right]$$

$$= pq \left\{ \frac{d}{dp} [np(1-p)] \right\}$$

$$= npq(1-2p)$$

$$\mu_4 = \mu_{r+1} \Big|_{r=3} = npq [1 + 3pq(n-2)]$$

3.113 The MGF of a discrete random variable X taking values $1, 2, \dots, \infty$ is $e^u(5-4e^u)^{-1}$, find the mean and variance of X .

Solution Given:

$$M_X(u) = \frac{e^u}{5-4e^u} = \frac{0.2 e^u}{1-0.8 e^u} \quad (3.179)$$

The MGF of a geometric random variable is given by

$$M_X(u) = \frac{pe^u}{1-qe^u} \quad (3.180)$$

Comparing Eq. (3.179) and Eq. (3.180), we get

$$p = 0.2 \text{ and } q = 0.8$$

$$\text{Mean} = \frac{1}{p} = \frac{1}{0.2} = 5$$

$$\text{Variance} = \frac{q}{p^2} = \frac{0.8}{(0.2)^2} = 20$$

3.114 Find the mean and variance of the distribution given by

$$P(X = r) = \frac{2}{3^r} \quad r = 1, 2, \dots, \infty$$

Solution Given:

$$\begin{aligned} P(X = r) &= 2 \left(\frac{1}{3} \right)^r, \quad r = 1, 2, \dots, \infty \\ &= \frac{2}{3} \left(\frac{1}{3} \right)^{r-1}; \quad r = 1, 2, \dots, \infty \end{aligned}$$

Comparing with geometric distribution

$$P(X = r) = pq^{r-1} \quad r = 1, 2, \dots, \infty$$

We get, $p = \frac{2}{3}$ and $q = \frac{1}{3}$

$$\text{The mean} = \frac{1}{p} = \frac{3}{2} = 1.5$$

$$\text{The variance} = \frac{q}{p^2} = \frac{1/3}{4/9} = \frac{1}{3} \cdot \frac{9}{4} = \frac{3}{4}$$

3.115 If the MGF of a random variable X is of the form $(0.25 e^u + 0.75)^6$, find the MGF of $2X + 1$.

Solution The MGF of binomial random variable X is given by

$$M_X(u) = (q + pe^u)^n$$

Given: $M_X(u) = (0.25e^u + 0.75)^6$

Hence, $p = 0.25$; $q = 0.75$ and $n = 6$

The MGF of $2X + 1$ is given by

$$\begin{aligned} M_{2X+1}(u) &= E[e^{(2X+1)u}] = E[e^{2Xu} e^u] \\ &= e^u E[e^{2Xu}] \\ &= e^u E[e^{X(2u)}] \\ &= e^u (0.2 e^{2u} + 0.75)^6 \end{aligned}$$

3.116 If X follows $B\left(3, \frac{1}{3}\right)$ and Y follows $B\left(5, \frac{1}{3}\right)$. Find $P(X + Y \geq 1)$.

Solution Given:

$$n_1 = 3 \text{ and } p_1 = \frac{1}{3}$$

$$n_2 = 5 \text{ and } p_2 = \frac{1}{3}$$

Since $p_1 = p_2$; $X + Y$ follows binomial distribution with $n = n_1 + n_2 = 8$. Therefore $X + Y$ follows binomial distribution with $n = 8$ and $p = \frac{1}{3}$. We can find that $q = \frac{2}{3}$.

$$\begin{aligned} P(X + Y = x) &= \binom{n}{x} p^x q^{n-x} \\ &= \binom{8}{x} \left(\frac{1}{3}\right)^x \left(\frac{2}{3}\right)^{8-x} \\ P(X + Y \geq 1) &= 1 - P(X + Y = 0) \\ &= 1 - \left\{ \binom{8}{0} \left(\frac{1}{3}\right)^0 \left(\frac{2}{3}\right)^8 \right\} \\ &= 1 - \left(\frac{2}{3}\right)^8 = 0.961 \end{aligned}$$

3.117 If X is a Poisson variable with mean λ , show that $\frac{X - \lambda}{\sqrt{\lambda}}$ is a variable with zero mean and unit variance.

Solution Since X is a Poisson variable, the mean and variance are equal and is given by

$$E[X] = \text{Var}[X] = \lambda.$$

The mean of random variable $\frac{X - \lambda}{\sqrt{\lambda}}$ is given by

$$\begin{aligned} E\left[\frac{X - \lambda}{\sqrt{\lambda}}\right] &= \frac{1}{\sqrt{\lambda}} E[X] - \frac{\lambda}{\sqrt{\lambda}} = \frac{\lambda}{\sqrt{\lambda}} - \frac{\lambda}{\sqrt{\lambda}} = 0 \\ \text{Var}\left[\frac{X - \lambda}{\sqrt{\lambda}}\right] &= \frac{1}{\lambda} \text{Var}(X - \lambda) \\ &= \frac{1}{\lambda} \{\text{Var}(X) - \text{Var}(\lambda)\} = \frac{1}{\lambda} \cdot (\lambda - 0) = 1 \end{aligned}$$

Therefore, $\frac{X - \lambda}{\sqrt{\lambda}}$ is a Poisson random variable with zero mean and unit variance.

3.118 If X is a Poisson variate such that $E[X^2] = 9$, find $E[X]$.

Solution For a Poisson random variable $E[X] = \text{Var}[X] = \lambda$

$$\begin{aligned} \text{Var}(X) &= E[X^2] - \{E[X]\}^2 \\ \lambda &= 9 - \lambda^2 \Rightarrow \lambda^2 + \lambda - 9 = 0 \\ \lambda &= 2.54 \end{aligned}$$

3.119 The MGF of a discrete random variable X is $e^{9(e^u - 1)}$, find $P(X = \lambda + \sigma)$ if λ and σ are the mean and standard deviation of X .

Solution The MGF of a discrete random variable X is

$$M_X(u) = e^{\lambda(e^u - 1)}$$

Comparing with the given $M_X(u) = e^{9(e^u - 1)}$ we obtain $\lambda = 9$

For a Poisson random variable

$$E[X] = \text{Var}(X) = \lambda = 9$$

The standard deviation $\sigma = \sqrt{\text{Var}(X)} = \sqrt{9} = 3$

$$\begin{aligned} P(X = \lambda + \sigma) &= P(X = 12) = \left. \frac{e^{-\lambda} (\lambda)^{12}}{12!} \right|_{\lambda=9} \\ &= \left. \frac{e^{-9} (9)^{12}}{12!} \right| = 0.07276 \end{aligned}$$

3.120 If the random variable X follows an exponential distribution with parameter 2 prove that $Y = X^3$ follows a Weibull distribution with parameters 2 and $1/3$.

Solution Given:

$$\begin{aligned} f_X(x) &= 2e^{-\lambda x}; x \geq 0 \\ &= 0; x < 0 \end{aligned}$$

The new random variable

$$Y = X^3$$

Let $y = x^3$, then $x = y^{1/3}$

$$f_Y(y) = \left. \frac{f_X(x)}{T'(x)} \right|_{x=y^{1/3}}$$

$$T(x) = x^3 \Rightarrow T'(x) = 3x^2$$

$$f_Y(y) = \left. \frac{2e^{-2x}}{3x^2} \right|_{x=y^{1/3}} = \frac{2}{3} y^{-2/3} e^{-2y^{1/3}}$$

$$\Rightarrow \boxed{f_Y(y) = \frac{2}{3} y^{-2/3} e^{-2y^{1/3}}}$$

Which is the Weibull distribution with $\alpha = 2$ and $\beta = \frac{1}{3}$.

3.121 The pmf of a random variable X is given by

$$P(X=r) = \frac{C \lambda^r}{r!}, r = 0, 1, 2, \dots$$

Find (i) $P(X=0)$ (ii) $P(X>2)$

Solution Given:

$$P(X=r) = \frac{C \lambda^r}{r!}, r = 0, 1, 2, \dots$$

We have $\sum_{r=0}^{\infty} P(X=r) = 1$

$\Rightarrow$

$$P(X=0) + P(X=1) + \dots = 1$$

$$C + \frac{C\lambda}{1!} + \frac{C\lambda^2}{2!} + \dots = 1$$

$$C \left(1 + \frac{\lambda}{1!} + \frac{\lambda^2}{2!} + \dots \right) = 1$$

$$C(e^\lambda) = 1 \Rightarrow C = e^{-\lambda}$$

$$P(X=0) = \frac{e^{-\lambda} \lambda^2}{r!} \Big|_{r=0} = e^{-\lambda}$$

$$\begin{aligned} P(X>2) &= 1 - P(X \leq 2) \\ &= 1 - \{P(X=0) + P(X=1) + P(X=2)\} \\ &= 1 - \left\{ e^{-\lambda} + \frac{e^{-\lambda} \lambda}{1!} + \frac{e^{-\lambda} \lambda^2}{2!} \right\} \\ &= 1 - e^{-\lambda} \left(1 + \lambda + \frac{\lambda^2}{2} \right) \end{aligned}$$

3.122 A class has 12 boys and 8 girls. If the teacher randomly selects a group of 12 students to represent the class in a competition. What is the probability that six members of the group are girls? What is the expected number of boys in the group?

Solution The total number of students is 20. A group of 12 students can be selected in $\binom{20}{12}$ ways.

Since six members of the group are girls, remaining six students are boys. 6 girls can be selected in $\binom{8}{6}$ ways and six boys can be selected in $\binom{12}{6}$ ways. Therefore the required probability is given by

$$p = \frac{\binom{12}{6} \binom{8}{6}}{\binom{20}{12}} = 0.2054$$

The probability that a randomly selected student is a boy is $p = \frac{12}{20} = 0.6$.

The expected number of boys in the selected group of 12 students is $12(0.6) = 7.2$

3.123 A purse contains 10 ₹ 2 coins and 2 ₹ 5 coins. Let X be the total amount that results when two coins are drawn from the purse without replacement.

- (a) Plot CDF of random variable X
- (b) Find mean and variance of X .

Solution The purse contains 10 ₹ 2 coins and 2 ₹ 5 coins. When two coins are drawn at random without replacement, the possible outcomes for the total amount are ₹ 4, ₹ 7, and ₹ 10. X represents the total amount for two draws.

Let E_1 be the event of first draw and E_2 be the event of second draw.

The total amount is equal to 4 when we get ₹ 2 coins for the two draws. Therefore,

$$\begin{aligned} P(X=4) &= P(\text{getting ₹ 2 coin in first draw}) \cdot P(\text{getting ₹ 2 coin in second draw}) \\ &= \frac{10}{12} \cdot \left(\frac{9}{11}\right) = \frac{45}{66} \end{aligned}$$

Similarly,

$$\begin{aligned} P(X=7) &= P(E_1=2) \cdot P(E_2=5) + P(E_1=5) P(E_2=2) \\ &= \frac{10}{12} \cdot \frac{2}{12} + \frac{2}{12} \cdot \frac{10}{11} \\ &= \frac{40}{132} \end{aligned}$$

$$\begin{aligned} P(X=10) &= P(E_1=5) P(E_2=5) \\ &= \frac{2}{12} \cdot \frac{1}{11} = \frac{2}{132} \end{aligned}$$

The cdf of random variable X is shown in Fig. 3.12.

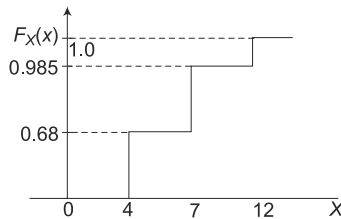


Fig. 3.12

The mean of X is given by

$$\begin{aligned}
 E[X] &= \sum_i x_i p_X(x_i) \\
 &= 4\left(\frac{45}{66}\right) + 7\left(\frac{40}{132}\right) + 10\left(\frac{2}{132}\right) = 5 \\
 E[X^2] &= (4)^2 \frac{45}{66} + (7)^2 \frac{40}{132} + (10)^2 \frac{2}{132} \\
 &= 27.27 \\
 \sigma_X^2 &= E[X^2] - E[X]^2 \\
 &= 27.27 - (5)^2 = 2.2727
 \end{aligned}$$

3.124 The random variable X is uniformly distributed in the interval $(-2, 2)$

(a) Find and plot the CDF of X .

(b) Use the CDF to find $P(X \leq 0)$ and $P\left[\left|X - \frac{1}{2}\right| < 1\right]$

Solution Given: X is uniformly distributed in the interval $(-2, 2)$. The pdf is given by

$$f_X(x) = \frac{1}{2 - (-2)} \quad \text{for } -2 \leq x \leq 2$$

0 otherwise

$$\begin{aligned}
 f_X(x) &= \frac{1}{4} \quad \text{for } -2 \leq x \leq 2 \\
 &= 0 \quad \text{otherwise}
 \end{aligned}$$

The CDF is given by

$$\begin{aligned}
 F_X(x) &= \int_{-2}^x f_X(x) dx = \frac{1}{4} x \Big|_{-2}^x \\
 &= \frac{1}{4} (x + 2); \text{ for } -2 \leq x \leq 2 \\
 F_X(x) &= \frac{1}{4} (x + 2); -2 \leq x \leq 2
 \end{aligned}$$

The plot of CDF is shown in Fig. 3.13.

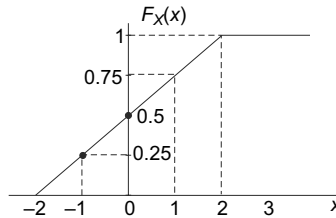


Fig. 3.13

$$\begin{aligned}
P_X(X < 0) &= F_X(0) = \frac{1}{2} \\
&= p \left(\left| X - \frac{1}{2} \right| < 1 \right) \\
&= p \left[-1 < \left(X - \frac{1}{2} \right) < 1 \right] \\
&= p \left(-\frac{1}{2} < X < \frac{3}{2} \right) \\
F_X \left(\frac{3}{2} \right) - F_X \left(-\frac{1}{2} \right) &= \frac{1}{4} \left(\frac{3}{2} + 2 \right) - \frac{1}{4} \left(2 - \frac{1}{2} \right) = \frac{1}{2}
\end{aligned}$$

REVIEW QUESTIONS



22. State and prove properties of variance of a random variable.
23. Explain the following terms; (i) Expectation, (ii) Conditional expected value, (iii) Covariance.
24. Explain the following terms:
(i) Variance (ii) Skew (iii) Kurtosis (iv) Moment
25. Find the n^{th} moment of a uniform random variable and hence its mean.
26. State the Chebyshev's inequality.
27. What is characteristic function?
28. State and prove properties of characteristic functions of a random variable X .
29. Explain about the moment-generating function of a random variable.
30. Explain in detail about transformation.
31. Explain in detail about the nonmonotonic transformation of a continuous random variable.
32. Find the mean and variance of a uniform distribution.
33. Find the mean and variance of a Laplace distribution.
34. Find the mean and variance of a Gaussian distribution.
35. Find the mean and variance of an exponential distribution.
36. Find the mean and variance of a binomial distribution.
37. Find the mean and variance of a geometric distribution.
38. Find MGF and CF of a random variable with following distributions: (i) Uniform (ii) Binomial (iii) Normal (iv) Exponential (v) Gamma (vi) Chi-square
39. If $M_X(u)$ is the MGF of a random variable X about the origin, then show that

$$E[X^r] = \left. \frac{d^r M_X(u)}{du^r} \right|_{u=0}$$

40. Show that $E[(X - a)^2]$ is minimized at $a = E(X)$.
41. If X is an exponential distributions with parameter x_1 , show that $f = x_1/\beta$ has a Weibull distribution.
42. Find the r^{th} moment of a random variable X in terms of its characteristic function.

**EXERCISES****Problems**

1. Given a random variable X and its density function

$$f_X(x) = \begin{cases} 1 & 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

Evaluate $E[X]$.

[Ans. 1/2]

2. If a random variable X is uniformly distributed over $(-a, 3a)$. Find the variance of X . $\left(\text{Ans: } \frac{4}{3} a^2 \right)$
3. For a Poisson random variable X with parameter λ show that $E[X(X-1)] = \lambda^2$ and $E[X(X-1)(X-2)] = \lambda^3$.
4. X is a uniformly distributed random variable over $(-1, 1)$. Find the expected value of

$$Y = \sin\left(\frac{\pi X}{2}\right) \text{ and } Z = |X|.$$

5. The density function of a random variable X is $g_X(x) = \begin{cases} 5e^{-5x} & 0 \leq x < \infty \\ 0 & \text{elsewhere} \end{cases}$

Find, (i) $E[X]$, (ii) $E[(X-1)^2]$, (iii) $E[3X-1]$

$\left(\text{Ans: } \frac{1}{5}, \frac{17}{25}, \frac{2}{5} \right)$

6. The probabilities of students X , Y and Z passing an examination are 0.6, 0.8 and 0.95 respectively. Let N be the number of students passing the examination. Find the mean and variance of N .
7. The temperature of a vessel is measured four times to be $t = 99, 100, 101, 102$ with the probabilities of correct measurement 0.2, 0.3, 0.4 and 0.1 respectively. Find the expected value and variance of the temperature.

8. Find the mean and variance of the random variable X with pdf .

$\left(\text{Ans: } \frac{1}{3}, \frac{2}{9} \right)$

$$f_X(x) = \begin{cases} \frac{1}{2}(x+1) & \text{for } -1 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

9. A random variable X has the density function $f_X(x) = \frac{1}{a} e^{-b|x|} - \infty \leq x \leq \infty$

Find $E[X]$, $E[X^2]$.

[Ans: 0, $4/ab^3$]

10. For the triangular distribution

$$f_X(x) = \begin{cases} x & 0 \leq x \leq 1 \\ 2-x & 1 \leq x \leq 2 \\ 0 & 0 \text{ otherwise} \end{cases}$$

Find the mean, variance and MGF.

$\left(\text{Ans: } 1, \frac{1}{6}, \frac{1}{u^2} (e^u - 1) \right)$

11. The CDF of a random variable X is

$$F_X(x) = 1 - (1+x)e^{-x}. \text{ Find pdf of } X, \text{ mean and variance of } X. \quad [\text{Ans: } x e^{-x}; 2; 2]$$

12. A continuous random variable X has the pdf $f_X(x)$ given by $f_X(x) = C e^{-|x|}$ - $-\infty < x < \infty$. Find the value of C and MGF.

$$\left(\text{Ans: } \frac{1}{2}; \frac{1}{1-u^2} \right)$$

13. Find the MGF of the random variable having the pdf $f_X(x) = \begin{cases} \frac{1}{4} & \text{for } -2 \leq x \leq 2 \\ 0 & \text{otherwise} \end{cases}$.

$$\left(\text{Ans: } \frac{1}{2u} \sin h 2u \right)$$

14. If X is a discrete random variable with probability function $P_X(x) = \frac{1}{k^x}$ $x = 1, 2, 3, \dots$ find
(i) MGF of X , (ii) Mean and variance of X .

$$\left(\text{Ans: } \frac{e^u}{k - e^u}; \frac{k}{(k-1)^2} \right)$$

15. Find the MGF of the random variable X having the pdf $f_X(x) = \begin{cases} \frac{x}{4} e^{-x/2}, & x > 0 \\ 0 & \text{elsewhere} \end{cases}$

$$\text{Also deduce the first four moments about the origin.} \quad \left(\text{Ans: } \frac{1}{(1-2u)^2}, 4, 24, 192, 1920 \right)$$

16. If a random variable has an exponential distribution with parameter 1, find the pdf of $Y = \sqrt{X}$.

$$(\text{Ans: } f_Y(y) = 2y e^{-y^2}; y > 0)$$

17. Given a random variable X with pdf

$$f_X(x) = \begin{cases} 1 - |x| & -1 \leq x \leq 1 \\ 0 & \text{elsewhere} \end{cases}$$

Find the mean and variance of X .

$$(\text{Ans: } 0, 1/6)$$

18. Given an exponential distributed random variable X with parameter $\lambda = 1$. Find the mean of

$$Y = 2X e^{-3X}$$

19. X is a poisson random variable with parameter λ and $P(X = 1) = P(X = 2)$. Find the mean and variance of X and probability $P(X = 3)$.

20. Given a random variable $X \sim U(0, 1)$, determine the constants c and d such that $Y = (X + d)$ is uniformly distributed over (a, b) .

21. The pdf of a random variable X is given by

$$f_X(x) = \frac{1}{3} \text{ for } 2 \leq x \leq 5$$

Find the pdf of $Y = 3X - 5$.

22. The pdf of a random variable X is given as

$$P(X) = \frac{1}{2^x} \quad x = 0, 1, 2, \dots, \infty$$

Find the MGF and $E[X]$.

$$\left(\text{Ans: } \frac{e^{u/2}}{1 - e^{u/2}}, 2, 2 \right)$$

23. Find the characteristic function of a random variable with density function

$$f_X(x) = \begin{cases} \frac{x}{2} & \text{for } 0 \leq x \leq 2 \\ 0 & \text{otherwise} \end{cases} \quad \left(\text{Ans: } \frac{1}{2\omega^2} [e^{j2\omega}(1 - 2j\omega) - 1] \right)$$

24. X is a uniformly distributed random variable over $(-1, 1)$. Let $Y = 4 - X^2$. Find the pdf of Y .

$$\left(\text{Ans: } \frac{1}{2} (4 - y)^{-1/2} \right)$$

25. The current having through a 2Ω resistor is uniformly distributed over the interval $(9, 11)$. Find the pdf of the power P .

$$\left(\text{Ans: } \frac{1}{8} (2p)^{1/2} \right)$$

26. The MGF of a random variable X is $\frac{1}{u} (e^{5u} - e^{4u})$. Find $E[X]$.

27. If X is uniformly distributed in $(-1, 1)$, then find the pdf of $Y = \sin \frac{\pi X}{2}$.

28. X is random variable with $\mu_X = 0$ and variance σ_X^2 . If Y is a random variable defined as $Y = X^3$. Find $f_Y(y)$.

29. X is a random variable with Chi-square distribution with pdf

$$f_X(x) = \frac{\frac{n}{2} x^{\frac{n}{2}-1} e^{-\frac{x}{2}}}{2^{\frac{n}{2}} \Gamma\left(\frac{n}{2}\right)} \quad x > 0$$

$$= 0 \quad \text{elsewhere}$$

Find pdf of $Y = \sqrt{X}$.

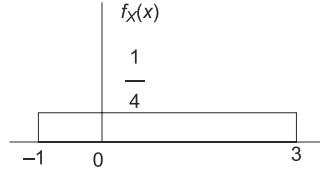
30. X is a Laplace random variable with pdf.

$$f_X(x) = \frac{1}{2a} e^{-|x|/a}. \text{ Find the characteristic function}$$

31. The characteristic function of the Laplace density is given by $\phi_X(\omega) = \frac{e^{-jm\omega}}{(1 + b\omega)^2}$, find mean and variance.

Multiple-Choice Questions

1. For a random variable X , following the pdf $f_X(x)$ shown in Fig. 3.14 the mean and variance are, respectively

**Fig. 3.14**

- (a) $\frac{1}{2}$ and $\frac{2}{3}$ (b) 1 and $\frac{4}{3}$ (c) 1 and $\frac{2}{3}$ (d) 2 and $\frac{4}{3}$
2. X is a uniformly distributed random variable over $(0, 1)$. The distribution of a random variable defined by $Y = X^n$ is
- (a) $Y^{1/n}$ (b) $\frac{1}{n} y^{-(n-1)/n}$ (c) $\frac{1}{y^n} - 1$ (d) $\frac{n-1}{y^n}$
3. If X is uniformly distributed over (a, b) , which of the following represents variance of X distributed over $(0, 1)$.
- (a) $\frac{b+a}{2}$ (b) $\frac{(b-a)^2}{12}$ (c) $\frac{b-a}{2}$ (d) $\frac{(b+a)^2}{12}$
4. If X is a normal random variable with parameters $\mu_X = 3$ $\sigma_X^2 = 9$, $P(X > 5)$ is
- (a) 0.8413 (b) 0.1587 (c) 0.3413 (d) 0.6587
5. If X is uniformly distributed over $(0, 1)$, the density function of $Y = e^X$
- (a) $f_Y(y) = e^{1/y}$ $0 \leq y \leq 1$ (b) $f_Y(y) = \frac{1}{y}$, $1 < y < e$
- (c) $f_Y(y) = y^{1/y}$ $0 \leq y \leq \infty$ (d) $f_Y(y) = e^{-1/y}$ $0 \leq y \leq e$
6. X is uniformly distributed over $(1, 3)$. The pdf of the random variable $Y = 3X + 4$ is
- (a) $f_Y(y) = \frac{1}{5}$, $7 < y < 12$ (b) $f_Y(y) = \frac{1}{6}$, $7 < y < 13$
- (c) $f_Y(y) = \frac{1}{7}$; $7 < y < 14$ (d) $f_Y(y) = \frac{1}{8}$; $7 < y < 15$
7. X is uniformly distributed over $(-1, 1)$. The pdf of $W = |X|$ is
- (a) $f_W(w) = 1$, $0 < w < 1$ (b) $f_W(w) = \frac{1}{2}$ $0 < w < 2$
- (c) $f_W(w) = 2$; $0 < w < 0.5$ (d) $f_W(w) = \frac{1}{3}$ $0 < w < 3$
8. The variance of binomial random variable X is
- (a) np (b) np^2 (c) npq (d) $nq^2 p$

9. The random variable which has equal mean and variance is
 (a) Poisson (b) Exponential (c) Pascal (d) Gamma
10. A random variable X has the density function

$$f_X(x) = \begin{cases} e^{-x} & \text{for } x \geq 0 \\ 0 & \text{for } x < 0 \end{cases}$$

The value of $E[e^{2X/3}]$ is

- (a) 5 (b) 4 (c) 1 (d) 3
11. If a random variable X has the MGF $M_X(u) = \frac{3}{3-u}$, then standard deviation of X is

- (a) $\frac{1}{2}$ (b) $\frac{1}{4}$ (c) $\frac{1}{5}$ (d) $\frac{1}{3}$

12. $\text{Var}(aX + bY) =$
 (a) $a \text{Var}(X) + b \text{Var}(Y) + 2 \text{Cov}(X, Y)$
 (b) $a^2 \text{Var}(X) + b^2 \text{Var}(Y)$
 (c) $a^2 \text{Var}(X) + b^2 \text{Var}(X) + 2ab \text{Cov}(X, Y)$
 (d) $a^2 \text{Var}(X) + b^2 \text{Var}(Y) - 2ab$

13. A continuous random variable X has a pdf

$$f_X(x) = \frac{1}{2} x^2 e^{-x} \text{ for } x > 0$$

The mean and variance are respectively.

- (a) 3 and 2 (b) 3 and 4 (c) 2 and 3 (d) 3 and 3
14. The characteristic function of a Chi-square random variable is given by

- (a) $f_X(\omega) = \frac{1}{1 + 2j\omega}$ (b) $\frac{1}{(1 - 2j\omega)^n}$
 (c) $\frac{1}{(1 - 2j\omega)^{n/2}}$ (d) $\frac{1}{(1 + 2j\omega)^{N/2}}$

15. The MGF of exponential random variable with $\lambda = 1$ is given by

- (a) $\frac{1}{(1-u)^2}$ (b) $\frac{1}{1-u}$ (c) $\frac{1}{1+u}$ (d) $\frac{1}{(1+u)^2}$

16. The PMF of a random variable X is given by

$$p_X(x) = \frac{e^{-\lambda} \lambda^x}{x!}, \lambda = 0, 1, 2, \dots$$

The value of $P(X \geq 2)$ is

- (a) Zero (b) $\lambda^2 + 2\lambda$
 (c) $1 - e^{-\lambda} - \lambda^2 + 2\lambda$ (d) $1 - e^{-\lambda} - \lambda e^{-\lambda} - \frac{\lambda^2 e^{-\lambda}}{2}$

17. The pdf of a random variable X is given by

$$f_X(x) = \begin{cases} \frac{1}{3} e^{-x/3} & x > 0 \\ 0 & \text{otherwise} \end{cases}$$

The value of $P(X > 3)$ is

- (a) $\frac{1}{e^2}$ (b) $\frac{1}{e}$ (c) e^2 (d) $(1 - e^{-1})$
18. If X is uniformly distributed over $(0, 1)$ $E[e^X] =$
- (a) $\frac{1}{1 - e}$ (b) $e - 1$ (c) $\frac{1}{e}$ (d) $\frac{1}{1 - e^2}$
19. The Kurtosis of Gaussian random variable is
- (a) 3 (b) 1 (c) 2 (d) ∞
20. X is a random variable with parameter $\lambda = 1$, then $Y = \sqrt{X}$ has the pdf given by
- (a) Rayleigh $\left(\frac{1}{\sqrt{2}}\right)$ (b) Rayleigh $\left(\frac{1}{e}\right)$
- (c) Uniform over $(0, 1)$ (d) Normal with mean 0 and variance 1
21. The characteristic function of poisson random variable is
- (a) $e^{-\lambda\omega}$ (b) $e^{-\lambda e^{-j\omega}}$ (c) $e^{-\lambda(1 - e^{-j\omega})}$ (d) $e^{-\lambda(e^{j\omega} - 1)}$
22. The pdf of a uniform random variable is given by

$$f_X(x) = \begin{cases} \frac{1}{a} & 0 \leq x \leq a \\ 0 & \text{otherwise} \end{cases}$$

The n^{th} order moment is given by

- (a) $\frac{a^n}{n+1}$ (b) $\frac{a^n}{n}$ (c) $\frac{a^n}{n-1}$ (d) $\frac{a^n}{2}$
23. The pdf of a random variable $X \sim U(-\pi, \pi)$ the pdf of $Y = a \tan X$, $a > 0$, is
- (a) Cauchy distribution (b) Normal distribution
- (c) Laplace distribution (d) Gamma distribution
24. $\text{Var}(aX + b) =$
- (a) $a \text{Var}(X) + b$ (b) $a^2 \text{Var}(X) + b$
- (c) $a^2 \text{Var}(X)$ (d) $a^2 \text{Var}(X) + b^2$
25. The pdf of random variable X is $f_X(x) = \frac{3}{2}(1 - x^2)$, $0 < x < 1$

The variance is given by

- (a) $\frac{9}{120}$ (b) $\frac{19}{320}$ (c) $\frac{1}{110}$ (d) $\frac{18}{160}$

4

MULTIPLE RANDOM VARIABLES



INTRODUCTION 4.1

In the previous chapter we studied the properties of a single random variable defined on a given sample space. But in many random experiments, we may have to deal with two or more random variables defined on the same sample space. For example, in analysing an electric circuit, current (i) and voltage (v) at different points in a circuit may be of our interest and we would consider (i, v) as a single experimental outcome. In another example, the hardness (h) and the tensile (t) strength of a steel piece may be of interest. In this case, the outcome of the experiment can be represented as (h, t) . Let us consider another example in which we collect the details of students. If we concentrate only on the age of students then we deal with a single random variable. On the other hand, if we collect details like age, weight and height then we deal with multiple random variables.

In this chapter, we extend the theory we studied for a single random variable to few or more random variables. We first consider the bivariate random variable and later consider processes with more than two random variables.



VECTOR RANDOM VARIABLES 4.2

Consider two random variables X and Y on a sample space. Let X and Y represent age and height of students respectively. If X and Y represents the age and the height of a specific student then this value (X, Y) can be represented as a random point in the x - y plane. If the number of students is n , there will be n such values $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$ that can be represented in the x - y plane. This order pair of number (x, y) is known as *specific value* of vector random variable where X and Y denote a two-dimensional vector random variable.

Consider another example of finding the number of defects in a semiconductor chip and their location. If n denotes the number of defects and $x_1, x_2, \dots, x_n$ denote the locations then the outcome of the experiment consists of the vector random variable given by $s = \{n, x_1, x_2, \dots, x_n\}$.

In the above two examples, the outcome of an experiment is a vector of random variables. In the first experiment, the outcome is a two-dimensional vector while in the second, it is an n -dimension vector. From these examples, we can define a vector random variable as a function that assigns a vector of real number to each outcome s in S , the sample space of the random experiment.

Figure 4.1 shows the mapping from s -plane to xy -plane. The plane of all points (x, y) in the ranges of X and Y is known as *joint sample space* denoted by S_j .

It is a vector space where the components of any vector are the values of the random variables X and Y . In the above Fig. 4.1, if S represents the sample space of students then $[X(s_1), Y(s_1)]$ represents the age and height of the student s_1 , and $[X(s_2), Y(s_2)]$ represents the age and height of the student s_2 .

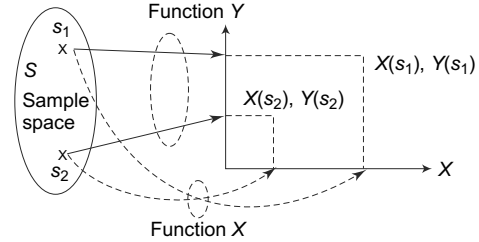


Fig. 4.1 Mapping from the sample space S to the joint sample space S_j .

REVIEW QUESTIONS

1. Define vector random variable.
2. Define two-dimensional random variable.

Solved Problems

4.1 Two events A and B defined on a sample space S are related to a joint sample space through random variables X and Y . The events are defined as $A = \{x_1 \leq X \leq x_2\}$ and $B = \{Y \leq y\}$. Make a sketch of two sample spaces showing areas corresponding to both events and the event $A \cap B = \{x_1 < X \leq x_2, Y \leq y\}$.

Solution The events are defined as $A = \{x_1 \leq X \leq x_2\}$ and $B = \{Y \leq y\}$

The events A and B refer to the sample space s , while events $\{x_1 \leq X \leq x_2\}$ and $\{Y \leq y\}$ refer to the joint sample space S_j . Event A corresponds to all points in S_j for which the X coordinate values lie between x_1 and x_2 , and the event B corresponds to the Y coordinate values in S_j not exceeding y . Our interest is in the event $A \cap B$ in S . This event $A \cap B = \{x_1 \leq X \leq x_2, Y \leq y\}$ defined on S_j is shown cross-hatched in Fig. 4.2.

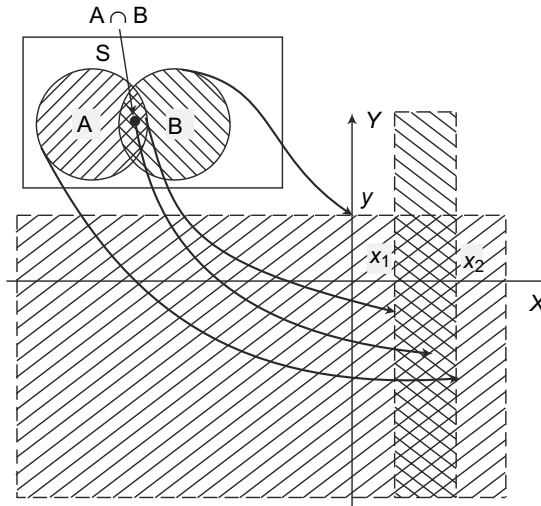


Fig. 4.2 The event $A \cap B$ defined in Solved Problem 4.1

4.2 Repeat the above problem for events

$$A = \{x_1 \leq X \leq x_2\}$$

$$B = \{y_1 \leq Y \leq y_2 \text{ or } y_3 \leq Y \leq y_4\}$$

Solution The events are $A = \{x_1 \leq X \leq x_2\}$, $B = \{y_1 \leq Y \leq y_2 \text{ or } y_3 \leq Y \leq y_4\}$

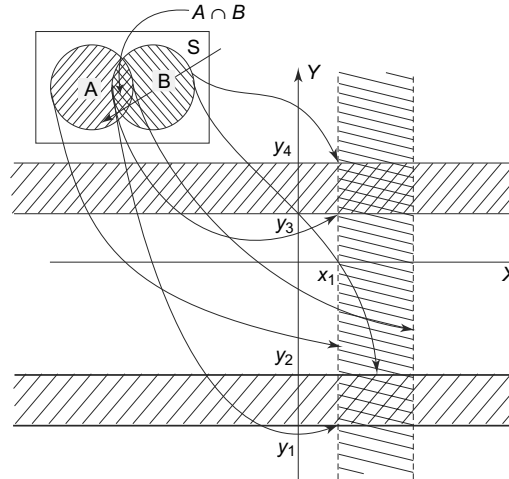


Fig. 4.3 The joint event $A \cap B$ defined on S and corresponding joint event defined on S_j

The joint event $A \cap B$ defined on the S corresponds to the joint event $\{x_1 \leq X \leq x_2 \text{ and } y_1 \leq Y \leq y_2 \text{ or } y_3 \leq Y \leq y_4\}$ defined on S_j . This joint event is shown cross-hatched in Fig. 4.3.

4.3 For the pair of random variables, sketch the region of the plane corresponding to the following events: (a) $\{X + Y > 2\}$ (b) $X/Y < 2$.

Solution

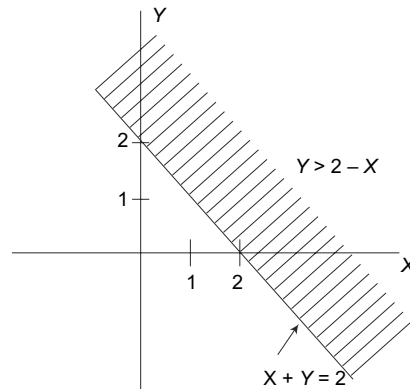


Fig. 4.4 The region of the plane corresponding to the event $X + Y > 2$

- (a) Draw a straight line $X + Y = 2$ on the xy -plane. Then the region of the plane corresponding to the event $X + Y > 2$ lies right to this line as shown in Fig. 4.4.
- (b) Draw a straight line $Y = X/2$ on the xy -plane. Then the region on the plane corresponding to the event $(X/Y) < 2$ lies above this line as shown in Fig. 4.5.

4.4 Sketch the sample space of two random variables X and Y and define the regions that correspond to the events $A = \{Y \leq X\}$, $B = \{X \leq 2\}$, $C = \{Y > -3\}$. Indicate the region defined by $A \cap B \cap C$.

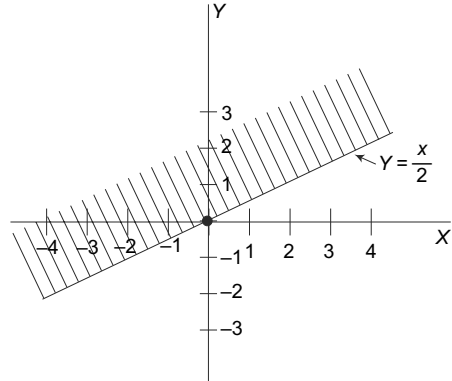


Fig. 4.5 The region of the plane corresponding to the event $(x/y) < 2$

Solution Draw the straight lines $Y = X$, $X = 2$ and $Y = -3$ on the xy -plane. Find the regions corresponding to the events A , B and C . Now the intersection portion of all events is shown in Fig. 4.6.

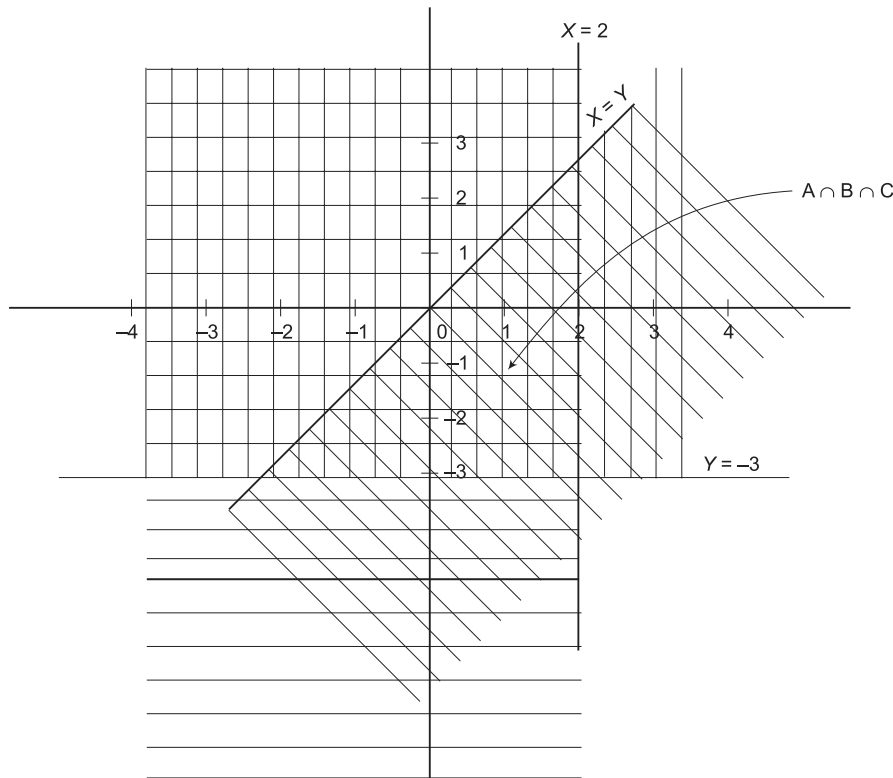


Fig. 4.6 The region of the plane for the event $A \cap B \cap C$, where A , B and C are defined in Solved Problem 4.4



Practice Problem

4.1 Two events A and B defined on the sample spaces S are related to a joint sample space through random variables X and Y and are defined by $A = \{x_1 \leq X \leq x_2\}$ and $B = \{y_1 \leq Y \leq y_2\}$. Draw the sketch of two sample space. Make a sketch of two sample spaces showing areas corresponding to both events and the event $A \cap B$.



JOINT PROBABILITY MASS FUNCTION 4.3

Consider a two-dimensional random variable (X, Y) . Let the possible values of (X, Y) be from a countable set $S = \{x_j, y_k\}$. That is, the possible values of (X, Y) may be represented as (x_j, y_k) $j = 1, 2, \dots, n$, and $k = 1, 2, \dots, m$. For each possible outcome of a two-dimensional random variable, we assume a number $p_{X,Y}(x_j, y_k)$, Probability Mass Function (PMF) given by

$$p_{X,Y}(x_j, y_k) = P(X \leq x_j, Y \leq y_k) \quad (4.1)$$

Thus, the joint pmf gives the possible probability of the occurrence of the pairs (x_j, y_k) . Since the probability of the sample space ' S ' is 1, we can write

$$\sum_{j=1}^n \sum_{k=1}^m p_{X,Y}(x_j, y_k) = 1 \quad (4.2)$$

4.3.1 Marginal pmf

We are also interested in finding the probability of events involving each of the random variables in isolation. The pmf of the random variable X is given by

$$\begin{aligned} p_X(x_j) &= P(X = x_j) \\ &= P(X = x_j, Y = \text{any value}) \\ &= P[(X = x_j \text{ and } Y = y_1) \cup (X = x_j \text{ and } Y = y_2) \dots] \\ &= \sum_{k=1}^{\infty} p_{X,Y}(x_j, y_k) \end{aligned} \quad (4.3)$$

Similarly,

$$p_Y(y_k) = \sum_{j=1}^{\infty} p_{X,Y}(x_j, y_k) \quad (4.4)$$

4.3.2 Properties of Joint pmf

1. The pmf can neither be negative nor exceed unity, that is $0 \leq p_{X,Y}(x, y) \leq 1$

$$\sum_{x \leq a} \sum_{y \leq b} p_{X,Y}(x, y) = 1 \quad (4.5)$$

$$\sum_{x \leq a} \sum_{y \leq b} p_{X,Y}(x, y) = F_{X,Y}(a, b) \quad (4.6)$$

4. If X and Y are independent random variables,

$$p_{X,Y}(x, y) = p_X(x) p_Y(y) \quad (4.7)$$

5. If X has N possible values $x_1, x_2, \dots, x_N$ and Y has M possible values $y_1, y_2, \dots, y_M$, then

$$F_{X,Y} = \sum_{n=1}^N \sum_{m=1}^M P(x_n, y_m) u(x - x_n) u(y - y_m) \quad (4.8)$$

where $u(x)$ is a unit step function defined as

$$u(x) = \begin{cases} 1 & \text{for } x \geq 0 \\ 0 & \text{for } x < 0 \end{cases}$$

The shifted unit step function is defined as

$$u(x - a) = \begin{cases} 1 & \text{for } x \geq a \\ 0 & \text{for } x < a \end{cases}$$



JOINT PROBABILITY MATRIX 4.4

Consider two discrete random variables X and Y with joint probability distribution $P(X, Y)$. Let X take values x_i , $i = 1, 2, \dots, n$ and Y take values y_j , $j = 1, 2, \dots, n$. Then the joint probability distribution can be represented by an $m \times n$ matrix with X representing the rows and Y representing the columns as below.

$$P(X, Y) =$$

$X \backslash Y$	y_1	y_2	---	y_n	$p_X(x_i)$
x_1	$p(x_1, y_1)$	$p(x_1, y_2)$	---	$p(x_1, y_n)$	$p_X(x_1)$
x_2	$p(x_2, y_1)$	$p(x_2, y_2)$	---	$p(x_2, y_n)$	$p_X(x_2)$
	$\vdots$	$\vdots$		$\vdots$	$\vdots$
x_m	$p(x_m, y_1)$	$p(x_m, y_2)$	---	$p(x_m, y_n)$	$p_X(x_m)$
$p_Y(y_j)$	$p_Y(y_1)$	$p_Y(y_2)$	---	$p_Y(y_n)$	1

From the joint distribution matrix, we can find the distribution of individual random variables. The distribution of X can be obtained by summing each row. The distribution of Y can be obtained by summing each column.

From the joint distribution matrix, we can observe.

$$\sum_i \sum_j p(x_i, y_j) = 1$$

$$p_X(x_i) = P(X = x_i) = \sum_j p(x_i, y_j) \quad (4.9)$$

$$p_Y(y_j) = P(Y = y_j) = \sum_i p(x_i, y_j) \quad (4.10)$$

Solved Problems



4.5 The joint probability distribution of two random variables X and Y is represented by a joint probability matrix given by

$$P(X, Y) =$$

$X \backslash Y$	1	2	3	4
1	0.1	0	0.2	0
2	0	0.1	0	0
3	0.2	0	0.3	0
4	0	0	0	0.1

Find the marginal distribution of X and Y . Find $P(X \leq 2, Y \leq 4)$.

Solution Given the joint distribution,

X \ Y	Y				$\sum_j p(x_i, y_j)$
	1	2	3	4	
1	0.1	0	0.2	0	0.3
2	0	0.1	0	0	0.1
3	0.2	0	0.3	0	0.5
4	0	0	0	0.1	0.1
$\sum_i p(x_i, y_j)$	0.3	0.1	0.5	0.1	

$$P(Y = 1) = 0.1 + 0 + 0.2 + 0 = 0.3 \text{ (Sum of first column)}$$

$$P(Y = 2) = 0 + 0.1 + 0 + 0 = 0.1 \text{ (Sum of second column)}$$

$$P(Y = 3) = 0.2 + 0.3 = 0.5 \text{ (Sum of third column)}$$

$$P(Y = 4) = 0 + 0 + 0 + 0.1 = 0.1 \text{ (Sum of fourth column)}$$

$$P(X = 1) = 0.1 + 0 + 0.2 + 0 = 0.3 \text{ (Sum of first row)}$$

$$P(X = 2) = 0 + 0.1 + 0 + 0 = 0.1 \text{ (Sum of second row)}$$

$$P(X = 3) = 0.2 + 0 + 0.3 + 0 = 0.5 \text{ (Sum of third row)}$$

$$P(X = 4) = 0 + 0 + 0 + 0.1 \text{ (Sum of fourth row)}$$

$$P(X \leq 2, Y \leq 4) = 0.1 + 0 + 0 + 0.1 + 0.2 + 0 + 0 + 0 = 0.4 \text{ (Sum of the elements in the rows 1 and 2)}$$

4.6 A fair coin is tossed four times. Let X denote the number of heads obtained in the first two tosses, and let Y denote the number of heads obtained in the last two tosses. Find the joint pmf of X and Y . Show that X and Y are independent random variables.

Solution The sample space and the values of X and Y are shown below.

S	X	Y
TTTT	0	0
TTTH	0	1
TTHT	0	1
TTHH	0	2
THTT	1	0
THTH	1	1
THHT	1	1
THHH	1	2
HTTT	1	0
HTTH	1	1
HTHT	1	1
HTHH	1	2
HHTT	2	0
HHTH	2	1
HHHT	2	1
HHHH	2	2

$$p_{X,Y}(0, 0) = P(X = 0, Y = 0) = P(TTTT) = \frac{1}{16}$$

$$p_{X,Y}(0, 1) = P(X = 0, Y = 1) = P(TTTH) \cup P(TTHT) = \frac{1}{16} + \frac{1}{16} = \frac{1}{8}$$

$$p_{X,Y}(0, 2) = P(X = 0, Y = 2) = P(TTHH) = \frac{1}{16}$$

$$p_{X,Y}(1, 0) = P(X = 1, Y = 0) = P(THTT) + P(HTTT) = \frac{1}{16} + \frac{1}{16} = \frac{1}{8}$$

$$\begin{aligned} p_{X,Y}(1, 1) &= P(X = 1, Y = 1) = P(THTH) + P(THHT) + P(HTTH) + P(HTHT) \\ &= \frac{1}{16} + \frac{1}{16} + \frac{1}{16} + \frac{1}{16} = \frac{1}{4} \end{aligned}$$

$$\begin{aligned} p_{X,Y}(1, 2) &= P(X = 1, Y = 2) \\ &= P(HTHH) + P(THHH) = \frac{1}{16} + \frac{1}{16} = \frac{1}{8} \end{aligned}$$

$$p_{X,Y}(2, 0) = P(X = 2, Y = 0) = P(HHTT) = \frac{1}{16}$$

$$p_{X,Y}(2, 1) = P(X = 2, Y = 1) = P(HHTH) + P(HHHT) = \frac{1}{16} + \frac{1}{16} = \frac{1}{8}$$

$$p_{X,Y}(2, 2) = P(X = 2, Y = 2) = P(HHHH) = \frac{1}{16}$$

The joint pmf of X and Y is tabulated below.

$X \backslash Y$	0	1	2
0	$\frac{1}{16}$	$\frac{1}{8}$	$\frac{1}{16}$
1	$\frac{1}{8}$	$\frac{1}{4}$	$\frac{1}{8}$
2	$\frac{1}{16}$	$\frac{1}{8}$	$\frac{1}{16}$

From the table,

$$p_X(0) = \frac{1}{16} + \frac{1}{8} + \frac{1}{16} = \frac{1}{4}; p_X(1) = \frac{1}{8} + \frac{1}{4} + \frac{1}{8} = \frac{1}{2}$$

$$p_X(2) = \frac{1}{16} + \frac{1}{8} + \frac{1}{16} = \frac{1}{4}$$

Similarly, $p_Y(0) = \frac{1}{4}, p_Y(1) = \frac{1}{2}; p_Y(2) = \frac{1}{4}$

We can also observe

$$p_X(0) p_Y(0) = \left(\frac{1}{4}\right) \left(\frac{1}{4}\right) = \frac{1}{16} = p_{X,Y}(0, 0)$$

$$p_X(0) p_Y(1) = \left(\frac{1}{4}\right)\left(\frac{1}{2}\right) = \frac{1}{8} = p_{X,Y}(0, 1)$$

$$p_X(0) p_Y(2) = \left(\frac{1}{4}\right)\left(\frac{1}{4}\right) = \frac{1}{16} = p_{X,Y}(0, 2)$$

$$p_X(1) p_Y(0) = \left(\frac{1}{2}\right)\left(\frac{1}{4}\right) = \frac{1}{8} = p_{X,Y}(1, 0)$$

$$p_X(1) p_Y(2) = \left(\frac{1}{2}\right)\left(\frac{1}{2}\right) = \frac{1}{4} = p_{X,Y}(1, 1)$$

$$p_X(1) p_Y(2) = \left(\frac{1}{2}\right)\left(\frac{1}{4}\right) = \frac{1}{8} = p_{X,Y}(1, 2)$$

$$p_X(2) p_Y(2) = \left(\frac{1}{4}\right)\left(\frac{1}{4}\right) = \frac{1}{16} = p_{X,Y}(2, 2)$$

For all values of X and Y , the joint pmf satisfies

$$p_{X,Y}(x, y) = p_X(x) p_Y(y)$$

Therefore, X and Y are statistically independent.

4.7 The joint pmf of (X, Y) is given by

$$p_{X,Y}(x_i, y_j) = \begin{cases} k(x_i + y_j) & x_i = 1, 2; y_j = 1, 2, 3 \\ 0 & \text{otherwise} \end{cases}$$

where k is constant.

(a) Find the value of k . (b) Find the marginal pmf of X and Y .

Solution

$$\begin{aligned} \sum_{x_i} \sum_{y_j} p_{X,Y}(x_i, y_j) &= 1 \\ \Rightarrow k \sum_{x_i=1}^2 \sum_{y_j=1}^3 (x_i + y_j) &= k \sum_{x_i=1}^2 (x_i + 1) + (x_i + 2) + (x_i + 3) \\ &= k \sum_{x_i=1}^2 (3x_i + 6) = k \{(3 + 6) + (6 + 6)\} \end{aligned}$$

$$21k = 1 \Rightarrow k = \frac{1}{21}$$

$$\begin{aligned} p_X(x_i) &= \sum_{y_j} p_{X,Y}(x_i, y_j) = \sum_{y_j=1}^3 p_{X,Y}(x_i, y_j) = \sum_{y_j=1}^3 k(x_i + y_j) \\ &= \frac{1}{21} \sum_{y_j=1}^3 (x_i + y_j) = \frac{1}{21} [(x_i + 1) + (x_i + 2) + (x_i + 3)] \end{aligned}$$

$$\begin{aligned}
 &= \frac{3x_i + 6}{21} \quad x_i = 1, 2, 3 \\
 p_Y(y_j) &= \sum_{x_i} p_{X,Y}(x_i, y_j) = \frac{1}{21} \sum_{x_i=1}^2 (x_i + y_j) \\
 &= \frac{(1 + y_j) + (2 + y_j)}{21} = \frac{3 + 2y_j}{21}; y_j = 1, 2
 \end{aligned}$$



Practice Problems

4.2 The joint pmf of two random variables X and Y is given by

$$\begin{aligned}
 p_{X,Y}(x, y) &= k(2x + y); \quad x = 1, 2; y = 1, 2 \\
 &= 0 \quad \text{otherwise}
 \end{aligned}$$

$$\left(\begin{aligned} \text{Ans: } &\frac{1}{18}; p_X(x_i) = \frac{4x_i + 3}{18} \quad x_i = 1, 2 \\ &p_Y(y_i) = \frac{3 + y_i}{9} \quad y_i = 1, 2 \end{aligned} \right)$$

(a) Find the value of k . (b) Find the marginal pmfs of X and Y .

4.3 A fair coin is tossed three times. Let X denote a random variable that takes the value 0 if the first toss is a head and the value 1 if the first toss is a tail. Let Y be a random variable that defines the total number of heads in the three tosses. Find the joint pmf of X and Y . Prove that X and Y are not independent.

$$\left(\begin{aligned} \text{Ans: } &p_{X,Y}(0, 0) = 0; p_{X,Y}(0, 1) = \frac{1}{8}; p_{X,Y}(0, 2) = \frac{1}{4}; p_{X,Y}(0, 3) = \frac{1}{8}; \\ &p_{X,Y}(1, 0) = \frac{1}{8}; p_{X,Y}(1, 1) = \frac{1}{4}; p_{X,Y}(1, 2) = \frac{1}{8}; p_{X,Y}(1, 3) = 0 \end{aligned} \right)$$

Solved Problems



4.8 The joint pmf of a bivariate random variable (X, Y) is given by

$$p_{X,Y}(x_i, y_j) = \begin{cases} k x_i y_j^2 & x_i = 1, 2, 3 \\ 0 & \text{otherwise} \end{cases} \quad y_j = 1, 2, 3$$

where k is a constant.

(a) Find the value of k . (b) Find the marginal pmfs of X and Y . (c) Are X and Y independent?

Solution We know

$$\begin{aligned}
 \sum_{x_i} \sum_{y_j} p_{X,Y}(x_i, y_j) &= 1 \\
 \sum_{x_i=1}^3 \sum_{y_j=1}^3 k x_i y_j^2 &= k \sum_{x_i=1}^3 (x_i + 4x_i + 9x_i) \\
 &= k \sum_{x_i=1}^3 (14x_i) = k[14 + 28 + 42] = 84k \\
 84k &= 1 \Rightarrow k = \frac{1}{84}
 \end{aligned}$$

The marginal pmf of X is

$$\begin{aligned} p_X(x_i) &= \sum_{y_j} p_{X,Y}(x_i, y_j) = \frac{1}{84} \sum_{y_j=1}^3 x_i y_j^2 \\ &= \frac{1}{84} [x_i + 4x_i + 9x_i] = \frac{1}{6} x_i; \quad x_i = 1, 2, 3 \end{aligned}$$

The marginal pmf of Y is

$$\begin{aligned} p_Y(y_j) &= \sum_{x_i} p_{X,Y}(x_i, y_j) \\ &= \frac{1}{84} \sum_{x_i=1}^3 x_i y_j^2 = \frac{y_j^2}{14}; \quad y_j = 1, 2, 3 \end{aligned}$$

4.9 The joint pmf of (X, Y) is given by $F_{X,Y}(x, y) = k(2x + 3y)$; $x = 0, 1, 2$; $y = 1, 2, 3$. Find all marginal probability distributions. Find $(P(X + Y > 3))$.

Solution The joint probability distribution is represented by the probability matrix as shown below:

For $x = 0$; $F_{X,Y}(0, y) = 3yk$ for $y = 1, 2, 3$

$$\Rightarrow F_{X,Y}(0, 1) = 3k; F_{X,Y}(0, 2) = 6k$$

$$F_{X,Y}(0, 3) = 9k.$$

Similarly,

For $x = 1$; $F_{X,Y}(1, y) = k(2 + 3y)$; $y = 1, 2, 3$

$$F_{X,Y}(1, 1) = 5k; F_{X,Y}(1, 2) = 8k; F_{X,Y}(1, 3) = 11k$$

For $x = 2$; $F_{X,Y}(2, y) = k(4 + 3y)$; $y = 1, 2, 3$

$$F_{X,Y}(2, 1) = 7k; F_{X,Y}(2, 2) = 10k$$

$$F_{X,Y}(2, 3) = 13k$$

$X \backslash Y$	1	2	3
0	3k	6k	9k
1	5k	8k	11k
2	7k	10k	13k

We know

$$\sum_i \sum_j p(x_i, y_j) = 1$$

Sum all the elements in the probability matrix,

$$72k = 1$$

$$\Rightarrow k = \frac{1}{72}$$

Substitute $k = \frac{1}{72}$ in the above matrix and tabulate as follows.

X \ Y	Y			
	1	2	3	
0	$\frac{3}{72}$	$\frac{6}{72}$	$\frac{9}{72}$	Sum of row 1 $P(X=0) = \frac{18}{72}$
1	$\frac{5}{72}$	$\frac{8}{72}$	$\frac{11}{72}$	Sum of row 2 $P(X=1) = \frac{24}{72}$
2	$\frac{7}{72}$	$\frac{10}{72}$	$\frac{13}{72}$	Sum of row 3 $P(X=2) = \frac{30}{72}$
	Sum of column 1 $P(Y=1) = \frac{15}{72}$	Sum of column 2 $P(Y=2) = \frac{24}{72}$	Sum of column 3 $P(Y=3) = \frac{33}{72}$	1

$$\begin{aligned}
 P(X+Y > 3) &= P(X+Y=4) + P(X+Y=5). \\
 &= P(X=1, Y=3) + P(X=2, Y=2) + P(X=2, Y=3) \\
 &= \frac{11}{72} + \frac{10}{72} + \frac{13}{72} = \frac{34}{72}
 \end{aligned}$$

4.10 The joint space for two random variables X and Y and corresponding probabilities are shown in the table.

(X, Y)	(1, 1)	(2, 2)	(3, 3)	(4, 4)
p	0.2	0.3	0.35	0.15

Find

- $F_{X,Y}(x, y)$
- Marginal distribution functions of X and Y .
- $P(X \leq 2, Y \leq 2)$
- $P(1 < X \leq 3, Y \geq 3)$

Solution

$$\begin{aligned}
 \text{(a)} \quad F_{X,Y}(x, y) &= P(X \leq x, Y \leq y) = \sum_{x_i \leq x} \sum_{y_j \leq y} P(x = x_i, y = y_j) \\
 F_{X,Y}(4, 4) &= P(X \leq 4, Y \leq 4) \\
 &= P(X=4, Y=4) + P(X=3, Y=3) + P(X=2, Y=2) + P(X=1, Y=1) \\
 &= 0.15 + 0.35 + 0.3 + 0.2 = 1 \\
 F_{X,Y}(3, 3) &= P(X \leq 3, Y \leq 3) \\
 &= \sum_{x_i \leq 3} \sum_{y_j \leq 3} P(x = x_i, y = y_j)
 \end{aligned}$$

$$\begin{aligned}
&= P(X = 1, Y = 1) + P(X = 2, Y = 2) + P(X = 3, Y = 3) \\
&= 0.2 + 0.3 + 0.35 = 0.85
\end{aligned}$$

Similarly,

$$\begin{aligned}
F_{X,Y}(2, 2) &= P(X = 1, Y = 1) + P(X = 2, Y = 2) \\
&= 0.2 + 0.3 = 0.5
\end{aligned}$$

$$F_{X,Y}(1, 1) = P(X = 1, Y = 1) = 0.2$$

(b) Marginal distribution function of X is

$$P(X = x_i) = p_X(x_i) = \sum_{y_j} p_{X,Y}(x_i, y_j)$$

$$\begin{aligned}
P(X = 1) &= p_X(1) = \sum_{y_j=1}^4 p_{X,Y}(1, y_j) \\
&= p_{X,Y}(1, 1) + p_{X,Y}(1, 2) + p_{X,Y}(1, 3) + p_{X,Y}(1, 4) \\
&= 0.2 + 0 + 0 + 0 = 0.2
\end{aligned}$$

$$\begin{aligned}
P(X = 2) &= p_X(2) = \sum_{y_j=1}^4 p_{X,Y}(2, y_j) \\
&= p_{X,Y}(2, 1) + p_{X,Y}(2, 2) + p_{X,Y}(2, 3) + p_{X,Y}(2, 4) \\
&= 0 + 0.3 + 0 + 0 = 0.3
\end{aligned}$$

$$\begin{aligned}
P(X = 3) &= p_X(3) = \sum_{y_j=1}^4 p_{X,Y}(3, y_j) \\
&= p_{X,Y}(3, 1) + p_{X,Y}(3, 2) + p_{X,Y}(3, 3) + p_{X,Y}(3, 4) \\
&= 0 + 0 + 0.35 + 0 = 0.35
\end{aligned}$$

$$\begin{aligned}
P(X = 4) &= p_X(4) = \sum_{y_j=1}^4 p_{X,Y}(4, y_j) \\
&= p_{X,Y}(4, 1) + p_{X,Y}(4, 2) + p_{X,Y}(4, 3) + p_{X,Y}(4, 4) \\
&= 0 + 0 + 0 + 0.15 = 0.15
\end{aligned}$$

Marginal distribution function of Y is

$$P(Y = y_j) = p_Y(y_j) = \sum_{x_i} p_{X,Y}(x_i, y_j)$$

$$\begin{aligned}
P(Y = 1) &= p_Y(1) = \sum_{x_i=1}^4 p_{X,Y}(x_i, 1) \\
&= p_{X,Y}(1, 1) + p_{X,Y}(2, 1) + p_{X,Y}(3, 1) + p_{X,Y}(4, 1) \\
&= 0.2 + 0 + 0 + 0 = 0.2
\end{aligned}$$

$$\begin{aligned}
 P(Y=2) &= p_Y(2) = \sum_{x_i=1}^4 p_{X,Y}(x_i, 2) \\
 &= p_{X,Y}(1, 2) + p_{X,Y}(2, 2) + p_{X,Y}(3, 2) + p_{X,Y}(4, 2) \\
 &= 0 + 0.3 + 0 + 0 = 0.3
 \end{aligned}$$

$$\begin{aligned}
 P(Y=3) &= p_Y(3) = \sum_{x_i=1}^4 p_{X,Y}(x_i, 3) \\
 &= p_{X,Y}(1, 3) + p_{X,Y}(2, 3) + p_{X,Y}(3, 3) + p_{X,Y}(4, 3) \\
 &= 0 + 0 + 0.35 + 0 = 0.35
 \end{aligned}$$

$$\begin{aligned}
 P(Y=4) &= p_Y(4) = \sum_{x_i=1}^4 p_{X,Y}(x_i, 4) \\
 &= p_{X,Y}(1, 4) + p_{X,Y}(2, 4) + p_{X,Y}(3, 4) + p_{X,Y}(4, 4) \\
 &= 0 + 0 + 0 + 0.15 = 0.15
 \end{aligned}$$

$$(c) P(X \leq 2, Y \leq 2)$$

$$\begin{aligned}
 &= P(X=1, Y=1) + P(X=2, Y=2) \\
 &= 0.2 + 0.3 = 0.5
 \end{aligned}$$

$$(d) P(1 < X \leq 3, Y \geq 3)$$

$$\begin{aligned}
 &= P(X=2, Y=3) + P(X=3, Y=3) + P(X=2, Y=4) + P(X=3, Y=4) \\
 &= 0 + 0.35 + 0 + 0 = 0.35
 \end{aligned}$$

4.11 Two discrete random variables X and Y have joint pmf given by the following table:

$Y \backslash X$	1	2	3
1	$\frac{1}{12}$	$\frac{1}{6}$	$\frac{1}{12}$
2	$\frac{1}{6}$	$\frac{1}{4}$	$\frac{1}{12}$
3	$\frac{1}{12}$	$\frac{1}{12}$	0

Compute the probability of each of the following events:

(a) $X \leq 1.5$ (b) XY even (c) Y is even given that X is even.

Solution

$$\begin{aligned}
 (a) \quad P(X \leq 1.5) &= P(X=1) \\
 &= P(X=1, Y=1) + P(X=1, Y=2) + P(X=1, Y=3) \\
 &= \frac{1}{12} + \frac{1}{6} + \frac{1}{12} = \frac{1}{3}
 \end{aligned}$$

$$(b) \quad P(XY = \text{even}) = P(X = 1, Y = 2) + P(X = 2, Y = 1) + P(X = 2, Y = 2) \\ + P(X = 2, Y = 3) + P(X = 3, Y = 2)$$

$$= \frac{1}{6} + \frac{1}{6} + \frac{1}{4} + \frac{1}{12} + \frac{1}{12} = \frac{3}{4}$$

$$(c) \quad P(Y = \text{even} | X = \text{even}) = \frac{P(X = \text{even}, Y = \text{even})}{P(X = \text{even})}$$

$$P(X = \text{even}, Y = \text{even}) = \frac{1}{4}$$

$$P(X = \text{even}) = P(X = 2) = P(X = 2, Y = 1) + P(X = 2, Y = 2) + P(X = 2, Y = 3)$$

$$= \frac{1}{6} + \frac{1}{4} + \frac{1}{12} = \frac{1}{2}$$

$$P(Y = \text{even} | X = \text{even}) = \frac{1/4}{1/2} = \frac{1}{2}$$



Practice Problem

4.4 Find the marginal pmf's for the pair of random variables whose joint pmf is

X \ Y	-1	0	1
-1	$\frac{1}{3}$	$\frac{1}{6}$	0
0	0	$\frac{1}{6}$	$\frac{1}{6}$
1	$\frac{1}{6}$	0	0

Find $P(X = Y)$ and $P(X \geq Y)$.

$$\left(\text{Ans. } \frac{1}{2}, \frac{2}{3} \right)$$

Solved Problems



4.12 In an experiment, 3 balls are randomly selected from an urn containing 3 red, 3 white, and 4 blue balls. If X denotes the number of red balls chosen and Y denotes the number of white balls chosen then find the joint pmf of X and Y . Also find marginal pmf of X and Y .

Solution Let i and j denote the number of red and white balls chosen respectively. Then we can write

$$p(i, j) = P(X = i, Y = j)$$

The total number of balls in the urn is 10. The number of possible ways that 3 balls can be selected from the urn is $\binom{10}{3}$. Now we will find the probability $p(i, j)$ of different events by varying the values of i, j . Let us find $p(0, 0) = P(X = 0, Y = 0)$. In this event, no red and white balls are chosen. Therefore, 3 balls are chosen from 4 blue balls.

Hence,

$$P(0, 0) = P(X = 0, Y = 0) = \frac{\binom{4}{3}}{\binom{10}{3}} = 0.0333$$

Let $i = 0$ and $j = 1$. In this event, only one white ball is selected. Since a red ball is not chosen, the remaining balls must be chosen from blue balls. Hence, we can write

$$P(0, 1) = P(X = 0, Y = 1) = \frac{\binom{3}{1}\binom{4}{2}}{\binom{10}{3}} = 0.15$$

Similarly,

$$p(0, 2) = \frac{\binom{3}{2}\binom{4}{1}}{\binom{10}{3}} = 0.1$$

$$p(0, 3) = \frac{\binom{3}{3}}{\binom{10}{3}} = \frac{1}{120} = 0.00833$$

$$p(1, 0) = \frac{\binom{3}{1}\binom{4}{2}}{\binom{10}{3}} = \frac{3(6)}{120} = 0.15$$

$$p(1, 1) = \frac{\binom{3}{1}\binom{3}{1}\binom{4}{1}}{\binom{10}{3}} = \frac{36}{120} = 0.3$$

$$p(1, 2) = \frac{\binom{3}{1}\binom{3}{2}}{\binom{10}{3}} = \frac{9}{120} = 0.075$$

$$p(2, 0) = \frac{\binom{3}{2}\binom{4}{1}}{\binom{10}{3}} = \frac{3(4)}{120} = 0.1$$

$$p(2, 1) = \frac{\binom{3}{2}\binom{3}{1}}{\binom{10}{3}} = \frac{3(3)}{120} = 0.075$$

$$p(3, 0) = \frac{\binom{3}{3}}{\binom{10}{3}} = 0.00833$$

$\sum_j p(x_i, y_j)$	$X \backslash Y$					$\sum_j p(x_i, y_j)$
		0	1	2	3	
$\sum_j p(x_i, y_j)$	0	0.0333	0.15	0.1	0.00833	0.29163
	1	0.15	0.3	0.075	0	0.525
	2	0.1	0.075	0	0	0.175
	3	0.00833	0	0	0	0.00833
		0.29163	0.525	0.175	0.00833	1
		$P_Y(Y=0)$	$P_Y(Y=1)$	$P_Y(Y=2)$	$P_Y(Y=3)$	

The marginal pmf of X is given by

$$P_X(X=0) = 0.29163; P_X(X=1) = 0.525$$

$$P_X(X=2) = 0.175; P_X(X=3) = 0.00833$$

The marginal pmf of Y is given by

$$P_Y(Y=0) = 0.29163, P_Y(Y=1) = 0.525$$

$$P_Y(Y=2) = 0.175, P_Y(Y=3) = 0.00833$$

4.13 In a colony, 10% of the families have no children, 30% have one child, 40% have two children, and 20% have three children. In a family, B represents the number of boys and G represents the number of girls. If a family is chosen at random from the colony, find the pmf of the family. Assume that in each family each child is equally likely to be a boy or a girl.

Solution Let B = number of boys and G = number of girls.

If $B = 0, G = 0$ then the family has no children.

Therefore, $P(B = 0, G = 0) = \text{Probability (no children)} = 0.1$

If $B = 0$ and $G = 1$ then the family has one girl child from the total of one child.

$$\begin{aligned} P(B = 0, G = 1) &= P(1 \text{ girl and total of 1 child}) \\ &= P(1 \text{ child}) \cdot P(1 \text{ girl}/1 \text{ child}) \\ &= (0.3) (1/2) = 0.15 \end{aligned}$$

$$\begin{aligned} P(B = 0, G = 2) &= P(2 \text{ girls and total of 2 children}) \\ &= P(2 \text{ children}) P(2 \text{ girls}/2 \text{ children}) \\ &= (0.4) \binom{2}{0} \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^0 \\ &= 0.4 \left(\frac{1}{2}\right)^2 = 0.1 \end{aligned}$$

$$\begin{aligned} P(B = 0, G = 3) &= P(3 \text{ girls and total of 3 children}) = P(3 \text{ children}) P(3 \text{ girls}/3 \text{ children}) \\ &= 0.2 \left[\binom{3}{0} \left(\frac{1}{2}\right)^3 \left(\frac{1}{2}\right)^0 \right] \end{aligned}$$

$$= 0.2 \left(\frac{1}{2} \right)^3 = 0.025$$

$$P(B = 1, G = 0) = 0.15, P(B = 2, G = 0) = 0.1 \text{ and } P(B = 3, G = 0) = 0.025$$

$$P(B = 1, G = 1) = P(1 \text{ girl and 1 boy and total of 2 children})$$

$$= P(2 \text{ children}) P(1 \text{ boy and 1 girl})$$

$$= (0.4) \left(\frac{1}{2} \right) = 0.2$$

$$P(B = 1, G = 2) = P(1 \text{ boy and 2 girl and total of three children})$$

$$= P(3 \text{ children}) P(1 \text{ boy and 2 girls})$$

$$= (0.2) \left[\binom{3}{1} \left(\frac{1}{2} \right)^1 \left(\frac{1}{2} \right)^2 \right]$$

$$= 0.2 \left(\frac{3}{8} \right) = 0.075$$

Similarly, $P(B = 2, G = 1) = 0.075$

The joint pmf is shown in the table.

		Girls				
$P(B = i, G = j)$		0	1	2	3	$P[B = i]$
Boys	0	0.1	0.15	0.1	0.025	0.375
	1	0.15	0.2	0.075	0	0.425
	2	0.1	0.075	0	0	0.175
	3	0.025	0	0	0	0.025
	$P[G = j]$	0.375	0.425	0.175	0.025	

4.14 The joint distribution function of two discrete random variables X and Y is given by

$$F_{X,Y}(x, y) = \begin{cases} \frac{1}{4} & x = 1, y = 1 \\ \frac{3}{8} & x = 2, y = 2 \\ \frac{3}{4} & x = 2, y = 1 \\ 1 & x = 2, y = 2 \end{cases}$$

Determine the

- (a) joint pmf of X and Y ; (b) marginal pmf of X (c) marginal pmf of Y

Solution

(a) We know

$$F_{X,Y}(a, b) = \sum_{x \leq a} \sum_{y \leq b} p_{X,Y}(x, y)$$

$$F_{X,Y}(1, 1) = p_{X,Y}(1, 1) = \frac{1}{4}$$

$$F_{X,Y}(1, 2) = p_{X,Y}(1, 1) + p_{X,Y}(1, 2) = \frac{3}{8}$$

$$\Rightarrow p_{X,Y}(1, 2) = \frac{3}{8} - \frac{1}{4} = \frac{1}{8}$$

$$F_{X,Y}(2, 1) = p_{X,Y}(1, 1) + p_{X,Y}(2, 1) = \frac{3}{4}$$

$$\Rightarrow p_{X,Y}(2, 1) = \frac{3}{4} - \frac{1}{4} = \frac{1}{2}$$

$$F_{X,Y}(2, 2) = p_{X,Y}(1, 1) + p_{X,Y}(1, 2) + p_{X,Y}(2, 1) + p_{X,Y}(2, 2) = 1$$

$$p_{X,Y}(2, 2) = 1 - \left(\frac{1}{4} - \frac{1}{8} + \frac{1}{2} \right) = \frac{1}{8}$$

The joint pmf $p_{X,Y}(x, y) = \frac{1}{4}; x = 1, y = 2$

$$= \frac{1}{8}; x = 1, y = 2$$

$$= \frac{1}{2}; x = 2, y = 1$$

$$= \frac{1}{8}; x = 2, y = 2$$

The joint pmf is shown in the table.

$X \backslash Y$	1	2
1	$\frac{1}{4}$	$\frac{1}{8}$
2	$\frac{1}{2}$	$\frac{1}{8}$

From the table,

$$(b) \quad P_X(X = 1) = \frac{1}{4} + \frac{1}{8} = \frac{3}{8}$$

$$P_X(X = 2) = \frac{1}{2} + \frac{1}{8} = \frac{5}{8}$$

$$(c) \quad P_Y(Y = 1) = \frac{1}{4} + \frac{1}{2} = \frac{3}{4}$$

$$P_Y(Y=2) = \frac{1}{8} + \frac{1}{8} = \frac{1}{4}$$



Practice Problem

4.5 The joint pmf of two random variables X and Y is given by

$$p_{X,Y}(x, y) = \begin{cases} 0.20 & x=1, y=1 \\ 0.15 & x=2, y=2 \\ 0.25 & x=3, y=3 \\ 0.40 & x=4, y=4 \\ 0 & \text{otherwise} \end{cases}$$

$$\text{Ans: (a) } F_{X,Y}(1, 1) = 0.2$$

$$F_{X,Y}(2, 2) = 0.35$$

$$F_{X,Y}(3, 3) = 0.6$$

$$F_{X,Y}(4, 4) = 1$$

$$(b) 0.35$$

Find (a) $F_{X,Y}(x, y)$. (b) Find $P(1 \leq X \leq 2, Y \leq 2)$.

Solved Problems



4.15 The joint space for two random variables X and Y and corresponding probabilities are shown in the table. Find and plot (a) $F_{X,Y}(x, y)$, (b) marginal distribution functions of X and Y . (c) Find $P(0.5 < X < 1.5)$ (d) Find $P(X \leq 1, Y \leq 2)$ and (e) Find $P(1 < X \leq 2, Y \leq 3)$.

X, Y	1,1	2,2	3,3	4,4
P	0.05	0.35	0.45	0.15

Solution

(a) The joint distribution function is given by

$$F_{X,Y}(x, y) = P(X \leq x, Y \leq y)$$

$$= \sum_{x_i \leq x} \sum_{y_j \leq y} P(X = x_i, Y = y_j)$$

$$F_{X,Y}(1, 1) = P(X \leq 1, Y \leq 1)$$

$$= P(X = 1, Y = 1) = 0.05$$

$$F_{X,Y}(2, 2) = P(X \leq 2, Y \leq 2)$$

$$= P(X = 1, Y = 1) + P(X = 2, Y = 2)$$

$$= 0.05 + 0.35 = 0.40$$

$$F_{X,Y}(3, 3) = P(X \leq 3, Y \leq 3)$$

$$= P(X = 1, Y = 1) + P(X = 2, Y = 2) + P(X = 3, Y = 3)$$

$$= 0.05 + 0.35 + 0.45 = 0.85$$

$$F_{X,Y}(4, 4) = P(X \leq 4, Y \leq 4)$$

$$= P(X = 1, Y = 1) + P(X = 2, Y = 2) + P(X = 3, Y = 3) + P(X = 4, Y = 4)$$

$$= 0.85 + 0.15 = 1.00$$

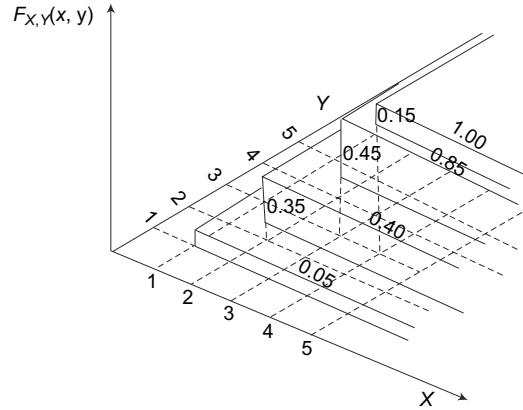


Fig. 4.7 Joint distribution function of (X, Y) given in Solved Problem 4.15

The joint distribution function can be constructed from the above values. First, we can find that for $x < 1$ and/or $y < 1$, $F_{X,Y} = 0$. At the point $(1, 1)$, a step function of amplitude 0.05 exists. This value is maintained for $x \geq 1$ and $y \geq 1$. A step amplitude of 0.35 is added at the point $(2, 2)$ which results $F_{X,Y}(x, y) = 0.40$ for $x \geq 2$ and $y \geq 2$. This value holds until the point $(3, 3)$ where another step signal of amplitude 0.45 is added. From this point, $F_{X,Y}(x, y) = 0.85$. Finally, a fourth stair of amplitude 0.15 is added. The resulting joint distribution function $F_{X,Y}(x, y)$ is shown in Fig. 4.7.

The expression for joint distribution function is

$$F_{X,Y}(x, y) = 0.05 u(x-1) u(y-1) + 0.35 u(x-2) u(y-2) + 0.45 u(x-3) u(y-3) + 0.15 u(x-4) u(y-4)$$

The joint distribution is shown in the table below.

$Y \backslash X$	1	2	3	4	$P_X(i)$
1	0.05	0	0	0	0.05
2	0	0.35	0	0	0.35
3	0	0	0.45		0.45
4	0	0	0	0.15	0.15
$P_Y(j)$	0.05	0.35	0.45	0.15	

- (b) The plot for marginal distribution functions of X and Y are shown in Fig. 4.8.

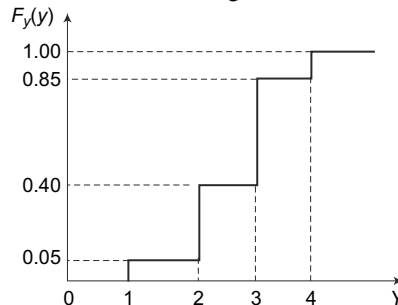
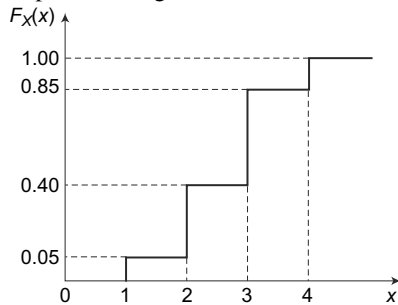


Fig. 4.8 (a) Marginal distribution function of X .

(b) Marginal distribution function of Y .

The marginal distribution function of X is given by

$$\begin{aligned} F_X(x) &= F_{X,Y}(x, \infty) \\ &= 0.05 u(x-1) + 0.35 u(x-2) + 0.45 u(x-3) + 0.15 u(x-4) \end{aligned}$$

The marginal distribution function of Y is given by

$$\begin{aligned} F_Y(y) &= F_{X,Y}(\infty, y) \\ &= 0.05 u(y-1) + 0.35 u(y-2) + 0.45 u(y-3) + 0.15 u(y-4) \end{aligned}$$

- (c) $P(0.5 < x < 1.5) = P(X=1) = p_X(1) = 0.05$
- (d) $P(X \leq 1, Y \leq 2) = P(X=1, Y=1) + P(X=1, Y=2) = 0.05 + 0 = 0.05$
- (e) $P(1 < X \leq 2, Y \leq 3)$
- $$\begin{aligned} &= P(X=2, Y=1) + P(X=2, Y=2) + P(X=2, Y=3) \\ &= 0 + 0.35 + 0 = 0.35 \end{aligned}$$

4.16 Discrete random variables X and Y have a joint distribution function

$$\begin{aligned} F_{X,Y}(x, y) &= 0.1 u(x+4) u(y-1) + 0.15 u(x+3) u(y+5) \\ &\quad + 0.17 u(x+1) u(y-3) + 0.05 u(x) u(y-1) \\ &\quad + 0.18 u(x-2) u(y+2) + 0.23 u(x-3) u(y-4) + 0.12 u(x-4) u(y+3) \end{aligned}$$

(a) Sketch $F_{X,Y}(x, y)$. (b) Find the marginal distribution functions of X and Y . (c) Find $P(-1 \leq X \leq 4, -3 \leq Y \leq 3)$ and $P(X < 1, Y \leq 2)$.

Solution The joint probability matrix is shown in the table below.

$X \backslash Y$	-5	-3	-2	1	3	4
-4				0.1		
-3	0.15					
-1					0.17	
0				0.05		
2			0.18			
3						0.23
4		0.12				

$$F_{X,Y}(-4, -2) = P(X \leq -4, Y \leq -2) = 0$$

$$F_{X,Y}(-4, 1) = P(X \leq -4, Y \leq 1) = 0.1$$

$$F_{X,Y}(-3, -5) = P(X \leq -3, Y \leq -5) = 0.15$$

$$F_{X,Y}(-3, 1) = P(X \leq -3, Y \leq 1) = 0.1 + 0.15 = 0.25$$

$$F_{X,Y}(-1, -3) = 0.15$$

$$F_{X,Y}(-1, 1) = P(X \leq -1, Y \leq 1) = 0.25$$

$$F_{X,Y}(-1, 3) = P(X \leq -1, Y \leq 3) = 0.25 + 0.17 = 0.42$$

$$F_{X,Y}(0, 1) = P(X \leq 0, Y \leq 1) = 0.25 + 0.05 = 0.30$$

$$F_{X,Y}(0, 3) = P(X \leq 0, Y \leq 3) = 0.42 + 0.05 = 0.47$$

$$F_{X,Y}(2, -2) = P(X \leq 2, Y \leq -2) = 0.15 + 0.18 = 0.33$$

$$F_{X,Y}(2, 1) = P(X \leq 2, Y \leq 1) = 0.33 + 0.15 = 0.48$$

$$F_{X,Y}(2, 3) = P(X \leq 2, Y \leq 3) = 0.15 + 0.1 + 0.17 + 0.05 + 0.18 = 0.65$$

$$F_{X,Y}(3, 4) = P(X \leq 3, Y \leq 4) = 0.65 + 0.23 = 0.88$$

$$F_{X,Y}(4, -5) = 0.15$$

$$F_{X,Y}(4, -3) = 0.15 + 0.12 = 0.27$$

$$F_{X,Y}(4, -2) = 0.27 + 0.18 = 0.45$$

$$F_{X,Y}(4, 1) = 0.45 + 0.15 = 0.60$$

$$F_{X,Y}(4, 3) = 0.60 + 0.17 = 0.77$$

$$F_{X,Y}(4, 4) = 0.77 + 0.23 = 1.00$$

Using the above values of $F_{X,Y}(x, y)$, the constructed distribution function is shown in Fig. 4.9.

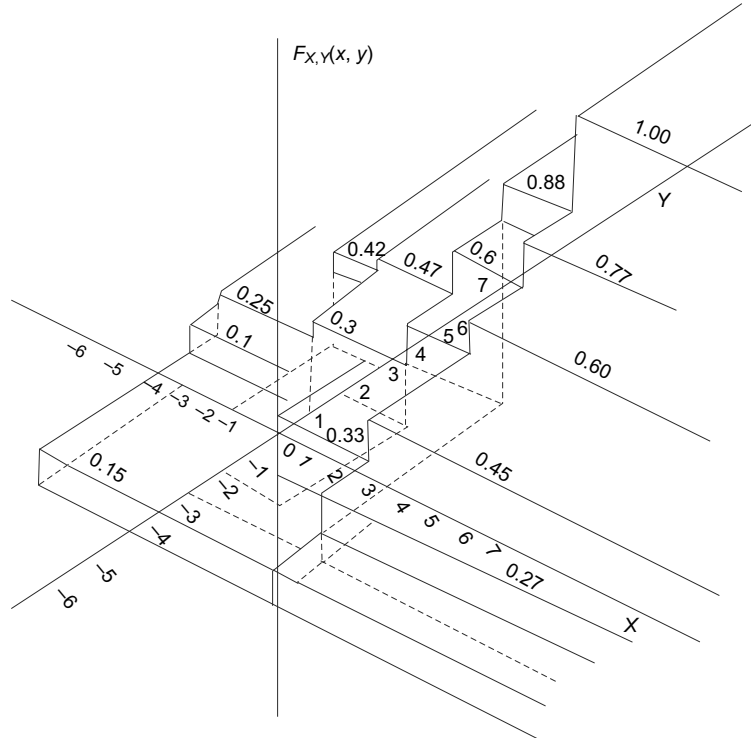


Fig. 4.9 Probability distribution function of (X, Y) .

The marginal distribution function is given by

$$\begin{aligned} F_X(x) &= F_{X,Y}(x, \infty) \\ &= 0.1 u(x+4) + 0.15 u(x+3) + 0.17 u(x+1) + 0.05 u(x) + 0.18 u(x-2) \\ &\quad + 0.23 u(x-3) + 0.12 u(x-4) \end{aligned}$$

The sketch of $F_X(x)$ is shown in Fig. 4.10(a).

The marginal density function of Y is given by

$$\begin{aligned} F_Y(y) &= F_{X,Y}(\infty, y) \\ &= 0.1 u(y-1) + 0.15 u(y+5) + 0.17 u(y-3) + 0.05 u(y-1) \\ &\quad + 0.18 u(y+2) + 0.23 u(y-4) + 0.12 u(y+3) \end{aligned}$$

The sketch of $F_Y(y)$ is shown in Fig. 4.10(b).

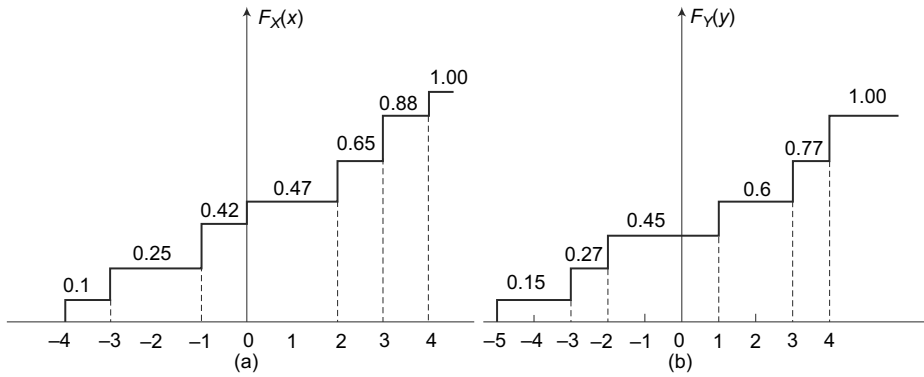


Fig. 4.10 (a) Marginal distribution function of X , (b) Marginal distribution function of Y

(b) $P(-1 < X \leq 4, -3 < Y \leq 3)$

We know

$$P(x_1 < X \leq x_2, y_1 < Y \leq y_2) = F_{X,Y}(x_2, y_2) + F_{X,Y}(x_1, y_1) - F_{X,Y}(x_2, y_1) - F_{X,Y}(x_1, y_2)$$

Therefore,

$$P(-1 < X \leq 4, -3 < Y \leq 3) = F_{X,Y}(4, 3) + F_{X,Y}(-1, -3) - F_{X,Y}(4, -3) - F_{X,Y}(-1, 3)$$

Substituting the value, we get

$$0.77 + 0.15 - 0.27 - 0.42 = 0.23$$

or, the shaded portion in the probability matrix only satisfies the range $-1 < X \leq 4$ and $-3 < Y \leq 3$.

Therefore,

$$P(-1 < X \leq 4, -3 < Y \leq 3) = 0.18 + 0.05 = 0.23$$

$$\begin{aligned} P(X < 1, Y \leq 2) &= F_{X,Y}(-3, -5) + F_{X,Y}(-4, 1) + F_{X,Y}(0, 1) \\ &= 0.15 + 0.1 + 0.05 = 0.3 \end{aligned}$$



Practice Problems

4.6 Discrete random variables X and Y have a joint density function

$$F_{X,Y}(x, y) = 0.1 u(x) u(y-1) + 0.2 u(x-1) u(y) + 0.05 u(x-2) u(y-2) \\ + 0.2 u(x-3) u(y-3) + 0.3 u(x-4) u(y-4) + 0.15 u(x-5) u(y-5)$$

(a) Sketch $F_{X,Y}(x, y)$. (b) Find the marginal distribution functions of X and Y .

$$(Ans: F_X(x) = 0.1 u(x) + 0.2 u(x-1) + 0.05 u(x-2) + 0.2 u(x-3) + 0.3 u(x-4) + 0.15 u(x-5))$$

4.7 The joint probability distribution function of X and Y is given by the following table:

X \ Y	Y		
	1	2	3
2	$\frac{1}{8}$	$\frac{1}{12}$	$\frac{1}{24}$
4	$\frac{1}{4}$	0	$\frac{1}{4}$
6	$\frac{1}{8}$	$\frac{1}{12}$	$\frac{1}{24}$

(a) Find the probability distribution of Y . (b) Are X and Y independent?

$$(Ans: f_Y(y) = 0.5 u(y-1) + \frac{1}{6} u(y-2) + \frac{1}{3} u(y-3))$$



JOINT DISTRIBUTION AND ITS PROPERTIES 4.5

Consider two random variables X and Y . Let us define two events A and B as $A = \{X \leq x\}$ and $B = \{Y \leq y\}$. We already know the probability of these two events can be defined as $F_X(x) = P(X \leq x)$ and $F_Y(y) = P(Y \leq y)$. Now let us define a joint event $\{X \leq x, Y \leq y\}$. The probability of this event is a function of the numbers x and y by a joint probability distribution function $F_{X,Y}(x, y)$ given by $F_{X,Y}(x, y) = P(X \leq x, Y \leq y)$.

1. Since $F_{X,Y}(x, y)$ is a probability, the CDF of two random variables is also bounded between 0 and 1. That is, $0 \leq F_{X,Y}(x, y) \leq 1$ for $-\infty < x < \infty, -\infty < y < \infty$.
2. $F_{X,Y}(x, y)$ is a nondecreasing function of both x and y .
3. $F_{X,Y}(-\infty, -\infty) = 0, F_{X,Y}(-\infty, y) = 0, F_{X,Y}(x, -\infty) = 0, F_{X,Y}(\infty, \infty) = 1$ (4.11)

The joint CDF of two random variables X and Y is

$$F_{X,Y}(x, y) = P(X \leq x, Y \leq y) = \sum_j \sum_k P(x_j, y_k) \quad (4.12)$$

where $P(x_j, y_k)$ is the joint probability

$$F_{X,Y}(-\infty, -\infty) = P(X \leq -\infty, Y \leq -\infty)$$

Since there is no value less than $-\infty$, for both X and Y , the joint probability $P(X \leq -\infty, Y \leq \infty) = 0$

Thus, $F_{X,Y}(-\infty, -\infty) = 0$. Similarly, $F_{X,Y}(x, -\infty) = 0$ and $F_{X,Y}(-\infty, y) = 0$

Also, $F_{X,Y}(\infty, \infty) = P(X \leq \infty, Y \leq \infty)$. Since all values of X and Y are included, $F_{X,Y}(\infty, \infty) = 1$.

4. If $x_1 \leq x_2$ and $y_1 \leq y_2$ then

$$F_{X,Y}(x_1, y_1) \leq F_{X,Y}(x_1, y_2) \leq F_{X,Y}(x_2, y_2) \quad (4.13)$$

Similarly,

$$F_{X,Y}(x_1, y_1) \leq F_{X,Y}(x_2, y_1) \leq F_{X,Y}(x_2, y_2) \quad (4.14)$$

$$5. \quad P(x_1 < X \leq x_2, Y \leq y) = F_{X,Y}(x_2, y) - F_{X,Y}(x_1, y) \quad (4.15)$$

$$6. \quad P(X \leq x, y_1 < Y \leq y_2) = F_{X,Y}(x, y_2) - F_{X,Y}(x, y_1) \quad (4.16)$$

$$7. \quad \text{If } x_1 \leq x_2 \text{ and } y_1 \leq y_2 \\ F_{X,Y}(x_2, y_2) + F_{X,Y}(x_1, y_1) - F_{X,Y}(x_2, y_1) - F_{X,Y}(x_1, y_2) \\ = P(x_1 < X \leq x_2, y_1 < Y \leq y_2) \geq 0 \quad (4.17)$$

$$8. \quad \lim_{x \rightarrow a^+} F_{X,Y}(x, y) = F_{X,Y}(a, y) \quad (4.18)$$

$$9. \quad \lim_{y \rightarrow b^+} F_{X,Y}(x, y) = F_{X,Y}(x, b) \quad (4.19)$$

The marginal CDFs are obtained as

$$F_X(x) = F_{X,Y}(x, \infty) \quad (4.20)$$

$$F_Y(y) = F_{X,Y}(\infty, y) \quad (4.21)$$

REVIEW QUESTIONS



3. Define two-dimensional discrete random variable.
4. Define joint probability mass function.
5. Define marginal pmf of random variables X and Y .
6. Explain in detail about the properties of joint pmf.
7. Define joint distribution of continuous random variables.
8. Explain the properties of joint distribution.
9. Distinguish between joint distribution and marginal distribution.

Solved Problems



4.17 The joint distribution function for two random variables X and Y is

$$F_{X,Y}(x, y) = u(x) u(y) (1 - e^{-ax} - e^{-ay} + e^{-a(x+y)}).$$

Sketch $F_{X,Y}(x, y)$. Assuming $a = 0.6$, find $P(X \leq 1, Y \leq 1)$, $P(1 < X \leq 2)$, $P(-1 < X \leq 2, 1 < Y \leq 2)$.

Solution The joint distribution function is shown in Fig. 4.11 Given $a = 0.6$,

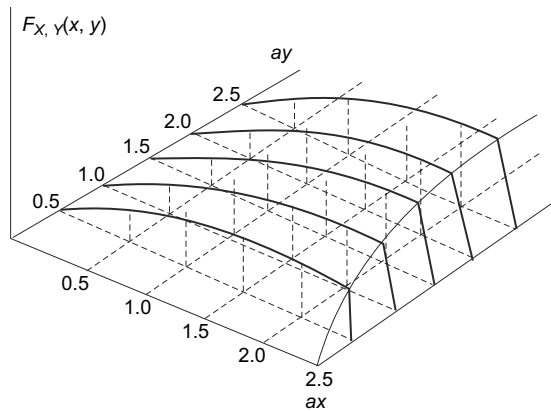


Fig. 4.11 Joint distribution function of (X, Y) .

$$P(X \leq 1, Y \leq 1) = F_{X,Y}(1, 1) \\ = (1 - e^{-a} - e^{-a} + e^{-2a}) = (1 - e^{-0.6} - e^{-0.6} + e^{-1.2}) = 0.2035$$

$$P(1 < X \leq 2) = F_X(2) - F_X(1) \\ = F_{X,Y}(2, \infty) - F_{X,Y}(1, \infty) \\ = (1 - e^{-2(1.6)}) - (1 - e^{-0.6}) \\ = 0.2476$$

$$P(-1 < X \leq 2, 1 < Y \leq 2) = F_{X,Y}(2, 2) + F_{X,Y}(-1, 1) - F_{X,Y}(2, 1) - F_{X,Y}(-1, 2)$$

Since $F_{X,Y}(x, y) = 0$ for $x < 0$ and $y < 0$,

$$F_{X,Y}(-1, 1) = 0 \text{ and } F_{X,Y}(-1, 2) = 0 \\ F_{X,Y}(2, 2) = 1 - e^{-1.2} - e^{-1.2} + e^{-2.4} = 0.4883 \\ F_{X,Y}(2, 1) = 1 - e^{-1.2} - e^{-0.6} + e^{-1.8} = 0.3153$$

$$\Rightarrow P(-1 < X \leq 2, 1 < Y \leq 2) = 0.4883 - 0.3153 = 0.173$$



Practice Problems

4.8 The random variables X and Y have joint CDF given by

$$F_{X,Y}(x, y) = \begin{cases} \left(1 - \frac{1}{x^2}\right)\left(1 - \frac{1}{y^2}\right) & \text{for } x > 1, y > 1 \\ 0 & \text{otherwise} \end{cases}$$

(a) Find the marginal CDF of X and Y ; (b) Find the probability of the events $\{X < 2, Y \leq 4\}$, $\{X > 3, Y > 2\}$.

$$\left(\text{Ans: (a)} F_X(x) = \begin{cases} 1 - \frac{1}{x^2}, & x > 1 \\ 0 & \text{otherwise} \end{cases} ; \text{(b)} \frac{45}{64}, \frac{1}{36} \right)$$

Solved Problems



4.18 The joint CDF of a two-dimensional random variable is $F_{X,Y}(x, y)$. Show that $P(X > a, Y > c) = 1 - F_X(a) - F_Y(c) + F_{X,Y}(a, c)$.

Solution

$$P(X > a, Y > c) = P(a < X < \infty, c < Y < \infty) \\ = P(a < X \leq \infty, Y \leq \infty) - P(a < X \leq \infty, Y \leq c) \\ = F_{X,Y}(\infty, \infty) - F_{X,Y}(a, \infty) - (F_{X,Y}(\infty, c) - F_{X,Y}(a, c)) \\ = F_{X,Y}(\infty, \infty) - F_{X,Y}(a, \infty) - F_{X,Y}(\infty, c) + F_{X,Y}(a, c)$$

We have

$$F_{X,Y}(\infty, \infty) = 1$$

$$F_{X,Y}(a, \infty) = F_X(a)$$

$$F_{X,Y}(\infty, c) = F_Y(c)$$

$$\Rightarrow P(X > a, Y > c) = 1 - F_X(a) - F_Y(c) + F_{X,Y}(a, c)$$

4.19 Let $F_X(x)$ and $F_Y(y)$ be valid one-dimensional CDFs. Show that $F_{X,Y}(x, y) = F_X(x) F_Y(y)$ satisfies the properties of a two-dimensional CDF.

Solution

- (a) $F_{X,Y}$ is a nondecreasing function of both x and y .

If $x_1 \leq x_2$ and $y_1 \leq y_2$ then

$$F_{X,Y}(x_1, y_1) \leq F_{X,Y}(x_2, y_2)$$

$$F_{X,Y}(x_1, y_1) = F_X(x_1) F_Y(y_1) \text{ and } F_{X,Y}(x_2, y_2) = F_X(x_2) F_Y(y_2)$$

Since $F_X(x_1) \leq F_X(x_2)$ and $F_Y(y_1) \leq F_Y(y_2)$, therefore,

$$F_X(x_1) F_Y(y_1) \leq F_X(x_2) F_Y(y_2)$$

- (b) $F_{X,Y}(x_1, -\infty) = 0$

$$F_{X,Y}(x_1, -\infty) = F_X(x_1) F_Y(-\infty) = 0 \text{ since } F_Y(-\infty) = 0$$

$$F_{X,Y}(-\infty, y_1) = 0$$

$$\begin{aligned} F_{X,Y}(-\infty, y_1) &= F_X(-\infty) F_Y(y_1) \\ &= 0 \text{ since } F_X(-\infty) = 0 \end{aligned}$$

$$F_{X,Y}(\infty, \infty) = 1$$

$$F_{X,Y}(\infty) = F_X(\infty) F_Y(\infty) = 1(1) = 1$$

- (c) The marginal CDFs are

$$F_X(x) = F_{X,Y}(x, \infty)$$

$$F_{X,Y}(x, \infty) = F_X(x) F_Y(\infty) = F_X(x)$$

$$F_Y(y) = F_{X,Y}(\infty, y)$$

$$F_{X,Y}(\infty, y) = F_X(\infty) F_Y(y) = F_Y(y)$$

$$\lim_{x \rightarrow a^+} F_{X,Y}(x, y) = \lim_{x \rightarrow a^+} F_X(x) F_Y(y) = F_X(a) F_Y(y) = F_{X,Y}(a, y)$$

$$\lim_{x \rightarrow b^+} F_{X,Y}(x, y) = \lim_{x \rightarrow b^+} F_X(x) F_Y(y) = F_X(b) F_Y(y) = F_{X,Y}(b, y)$$

$$\begin{aligned} P(x_1 < X \leq x_2, y < Y \leq y_2) &= P(x_1 < X \leq x_2) P(y < Y \leq y_2) \\ &= (F_X(x_2) - F_X(x_1)) (F_Y(y_2) - F_Y(y_1)) \end{aligned}$$

$$\begin{aligned}
&= F_X(x_2) F_Y(y_2) - F_X(x_1) F_Y(y_2) - F_X(x_2) F_Y(y_1) + F_X(x_1) F_Y(y_1) \\
&= F_{X,Y}(x_2, y_2) - F_{X,Y}(x_1, y_2) - F_{X,Y}(x_2, y_1) + F_{X,Y}(x_1, y_1)
\end{aligned}$$

4.20 Find whether the function

$G_{X,Y}(x, y) = u(x) u(y) (1 - e^{-(x+y)})$ is a valid distribution function.

Solution The given function

$$G_{X,Y}(x, y) = 0 \text{ for } x < 0, y < 0$$

Therefore, it satisfies the conditions.

$$G_{X,Y}(-\infty, -\infty) = 0; G_{X,Y}(-\infty, y) = 0 \text{ and } G_{X,Y}(x, -\infty) = 0$$

Therefore, Property 1 is satisfied.

$$\text{Also, } G_{X,Y}(\infty, \infty) = 1 - e^{-\infty} = 1$$

Property 2 is satisfied.

$$G_{X,Y}(0, 0) = 0 \text{ and } G_{X,Y}(\infty, \infty) = 1$$

So it satisfies the condition

$$0 < G_{X,Y}(x, y) \leq 1$$

The third property is satisfied.

$$\frac{d}{dx} [G_{X,Y}(x, y)] = \frac{d}{dx} [1 - e^{-(x+y)}] = e^{-x} e^{-y} \quad x > 0, y > 0$$

Similarly,

$$\frac{d}{dy} [G_{X,Y}(x, y)] = \frac{d}{dy} [1 - e^{-(x+y)}] = e^{-x} e^{-y} \quad x > 0, y > 0$$

Note that $e^{-(x+y)}$ is non-negative for all x and y

The fourth property is satisfied.

Now assume the values of X and Y such that

$$0 < x_1 < x_2 \text{ and } 0 < y_1 < y_2$$

$$P(x_1 < x \leq x_2, y_1 < y \leq y_2)$$

$$= G_{X,Y}(x_2, y_2) + G_{X,Y}(x_1, y_1) - G_{X,Y}(x_1, y_2) - G_{X,Y}(x_2, y_1)$$

$$G_{X,Y}(x_2, y_2) = (1 - e^{-x_2 - y_2})$$

$$G_{X,Y}(x_1, y_1) = 1 - e^{-x_1 - y_1}$$

$$G_{X,Y}(x_1, y_2) = 1 - e^{-x_1 - y_2}$$

$$G_{X,Y}(x_2, y_1) = 1 - e^{-x_2 - y_1}$$

$$\Rightarrow P(x_1 < X \leq x_2, y_1 < Y \leq y_2)$$

$$= (1 - e^{-x_2 - y_2}) + (1 - e^{-x_1 - y_1}) - (1 - e^{-x_1 - y_2}) - (1 - e^{-x_2 - y_1})$$

$$\begin{aligned}
&= e^{-x_1 - y_2} + e^{-x_2 - y_1} - e^{-x_2 - y_2} - e^{-x_1 - y_1} \\
&= e^{-x_1} e^{-y_2} + e^{-x_2} e^{-y_1} - e^{-x_2} e^{-y_2} - e^{-x_1} e^{-y_1} \\
&= e^{-x_1} (e^{-y_2} - e^{-y_1}) - e^{-x_2} (e^{-y_2} - e^{-y_1}) \\
&= (e^{-x_1} - e^{-x_2}) (e^{-y_2} - e^{-y_1})
\end{aligned}$$

Note that $(e^{-x_1} - e^{-x_2}) > 0$ since $x_1 < x_2$

But $(e^{-y_2} - e^{-y_1}) < 0$ since $y_1 < y_2$

Therefore,

$$P(x_1 < X \leq x_2, y_1 < Y \leq y_2) < 0$$

which does not satisfy the condition 5 of a valid distribution function.

Therefore, the given function is not a valid distribution function.

4.21 Check whether the function

$$\begin{aligned}
G_{X,Y}(x, y) &= 0 \quad x < y \\
&= 1 \quad x \geq y
\end{aligned}$$

is a valid joint distribution function.

Solution Given:

$$\begin{aligned}
G_{X,Y}(x, y) &= 0 \quad \text{for } x < y \\
&= 1 \quad \text{for } x \geq y
\end{aligned}$$

Since the function is zero for $x < y$, we can write

$$G_{X,Y}(-\infty, -\infty) = 0; G_{X,Y}(-\infty, y) = 0$$

But $G_{X,Y}(x, -\infty) = 1$ for all values of x except $x = -\infty$.

Therefore, $G_{X,Y}(x, y)$ is not a valid distribution function. Also, consider a point such that

$$x_1 < y_1 \leq x_2 < y_2, \text{ then}$$

$$\begin{aligned}
P(x_1 < X \leq x_2, y_1 < Y \leq y_2) \\
&= G_{X,Y}(x_1, y_1) + G_{X,Y}(x_2, y_2) - G_{X,Y}(x_2, y_1) - G_{X,Y}(x_1, y_2)
\end{aligned}$$

Now,

$$G_{X,Y}(x_1, y_1) = 0 \quad \text{since } x_1 < y_1$$

$$G_{X,Y}(x_2, y_2) = 0 \quad \text{since } x_2 < y_2$$

$$G_{X,Y}(x_1, y_2) = 0 \quad \text{since } x_1 < y_2$$

$$G_{X,Y}(x_2, y_1) = 1 \quad \text{since } x_2 > y_1$$

Therefore,

$$P(x_1 < X \leq x_1, y_1 < Y \leq y_2) = 0 + 0 - 1 - 0 = -1$$

which is less than zero and violates the property of distribution function. Hence, the given $G_{X,Y}(x, y)$ is not a valid joint pdf.



Practice Problems

4.9 Find whether the function

$$G_{X,Y}(x, y) = \begin{cases} 0 & \text{for } x + y < 1 \\ 1 & \text{for } x + y \geq 1 \end{cases}$$

is a valid distribution function.

(Ans. Not a valid distribution function)

4.10 Check whether the function

$$G_{X,Y}(x, y) = \begin{cases} 1 - \frac{1}{x^2 y^2} & x > 1, y \geq 1 \\ 0 & \text{elsewhere} \end{cases}$$

is a valid distribution function.

(Ans. Not a valid distribution function)



JOINT DENSITY FUNCTION AND ITS PROPERTIES 4.6

The joint probability density function of a pair of random variables (X, Y) evaluated at the point (x, y) is

$$f_{X,Y}(x, y) = \lim_{\substack{\Delta x \rightarrow 0 \\ \Delta y \rightarrow 0}} \frac{P(x \leq X < x + \Delta x, y \leq Y < y + \Delta y)}{\Delta x \Delta y} \quad (4.22)$$

We know

$$\begin{aligned} P(x \leq X < x + \Delta x, y \leq Y < y + \Delta y) \\ = F_{X,Y}(x + \Delta x, y + \Delta y) - F_{X,Y}(x, y + \Delta y) - F_{X,Y}(x + \Delta x, y) + F_{X,Y}(x, y) \end{aligned} \quad (4.23)$$

Dividing the above Eq. (4.23) by Δx and taking the limit $\Delta x \rightarrow 0$, we get

$$\begin{aligned} \lim_{\Delta x \rightarrow 0} \left\{ \frac{P(x \leq X < x + \Delta x, y \leq Y < y + \Delta y)}{\Delta x} \right\} \\ = \lim_{\Delta x \rightarrow 0} \left\{ \frac{F_{X,Y}(x + \Delta x, y + \Delta y) - F_{X,Y}(x, y + \Delta y)}{\Delta x} \right\} \\ - \lim_{\Delta x \rightarrow 0} \left\{ \frac{F_{X,Y}(x + \Delta x, y) - F_{X,Y}(x, y)}{\Delta x} \right\} \\ = \frac{\partial}{\partial x} F_{X,Y}(x, y + \Delta y) - \frac{\partial}{\partial x} F_{X,Y}(x, y) \end{aligned} \quad (4.24)$$

Now dividing the above Eq. (4.24) with Δy and taking $\Delta y \rightarrow 0$,

$$\begin{aligned} \lim_{\substack{\Delta x \rightarrow 0, \\ \Delta y \rightarrow 0}} \frac{P(x \leq X < x + \Delta x, y \leq Y < y + \Delta y)}{\Delta x \Delta y} \\ = \lim_{\Delta y \rightarrow 0} \frac{\frac{\partial}{\partial x} F_{X,Y}(x, y + \Delta y)}{\Delta y} - \lim_{\Delta y \rightarrow 0} \frac{\frac{\partial}{\partial x} F_{X,Y}(x, y)}{\Delta y} \\ = \frac{\partial^2}{\partial x \partial y} F_{X,Y}(x, y) \end{aligned} \quad (4.25)$$

That is, the joint pdf of $f_{X,Y}(x, y)$ can be obtained from the joint CDF $F_{X,Y}(x, y)$ by taking a partial derivative with respect to each variable.

The joint pdf of two random variables X and Y is defined as the second derivative of the joint distribution function whenever it exists.

$$f_{X,Y}(x, y) = \frac{\partial^2 F_{X,Y}(x, y)}{\partial x \partial y} \quad (4.26)$$

The joint CDF can be obtained in terms of joint pdf using the equation

$$F_{X,Y}(x, y) = \int_{-\infty}^y \int_{-\infty}^x f_{X,Y}(u, v) du dv \quad (4.27)$$

The joint pdf has the following properties:

$$1. \text{ For all } x \text{ and } y, f_{X,Y}(x, y) \geq 0 \quad (4.28)$$

$$2. \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx dy = 1 \quad (4.29)$$

Using Eq. (4.27) and together with the fact $F_{X,Y}(\infty, \infty) = 1$, we can prove the above equation.

$$3. F_{X,Y}(x, y) = \int_{-\infty}^y \int_{-\infty}^x f_{X,Y}(u, v) du dv \quad (4.30)$$

$$4. F_X(x) = \int_{-\infty}^x \int_{-\infty}^{\infty} f_{X,Y}(u, v) du dv \quad (4.31)$$

$$5. F_Y(y) = \int_{-\infty}^y \int_{-\infty}^{\infty} f_{X,Y}(u, v) du dv \quad (4.32)$$

$$6. f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dy \quad (4.33)$$

Using Eq. (4.20),

$$F_X(x) = F_{X,Y}(x, \infty) = \int_{-\infty}^x \int_{-\infty}^{\infty} f_{X,Y}(u, v) dv du$$

$$\frac{d}{dx} F_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, v) dv$$

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, v) dv = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dy \quad (4.34)$$

$$7. f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx \quad (4.35)$$

$$8. f_{X,Y}(x, y) \text{ is continuous for all except possibly finite values of } x \text{ and } y.$$

9. $P(x_1 < X \leq x_2, y_1 < Y \leq y_2)$

$$= \int_{y_1}^{y_2} \int_{x_1}^{x_2} f_{X,Y}(x, y) dx dy \quad (4.36)$$

10. If X and Y are statistically independent random variables.

$$f_{X,Y}(x, y) = f_X(x) f_Y(y) \quad (4.37)$$

The joint density function for any two discrete random variables can be obtained by finding the derivative of Eq. (4.8) which is given by

$$f_{X,Y}(x, y) = \sum_{n=1}^N \sum_{m=1}^M P(x_n, y_m) \delta(x - x_n) \delta(y - y_m) \quad (4.38)$$

Note that $\delta(x)$ is known as unit sample sequence defined by

$$\delta(x) = \begin{cases} 1 & \text{for } x = 0 \\ 0 & \text{for } x \neq 0 \end{cases}$$

It is also known as unit impulse sequence

The shifted unit impulse sequence is defined as

$$\delta(x - a) = \begin{cases} 1 & \text{for } x = a \\ 0 & \text{for } x \neq a \end{cases}$$

If we differentiate unit step function we get an impulse.



PMF, PDF FOR N RANDOM VARIABLES 4.7

In this section, we extend the concepts of pmf, CDF and pdf for N random variables

For a set of N discrete random variables $X_1, X_2, \dots, X_N$, the joint pmf is given by

$$P_X(x_1, x_2, \dots, x_N) = P(X_1 = x_1, X_2 = x_2, \dots, X_N = x_N) \quad (4.39)$$

For a set of N continuous random variables $X_1, X_2, \dots, X_N$,

$$F_X(x_1, x_2, x_3, \dots, x_N) = P(X_1 \leq x_1, X_2 \leq x_2, \dots, X_N \leq x_N) \quad (4.40)$$

and the pdf

$$f_X(x_1, x_2, \dots, x_N) = \frac{\partial^N}{\partial x_1 \partial x_2 \dots \partial x_N} F_X(x_1, x_2, \dots, x_N) \quad (4.41)$$

REVIEW QUESTIONS



10. Define joint pdf of two random variables X and Y .
11. Explain the properties of joint pdf.
12. Define pmf for N random variables.

Solved Problems

4.22 If the probability density function is given by

$$f_{X,Y}(x, y) = \begin{cases} \frac{6}{5}(x + y^2) & 0 \leq x \leq 1, 0 \leq y \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

obtain the marginal pdf of X and that of Y .

Hence, find $P\left(\frac{1}{4} \leq y \leq \frac{3}{4}\right)$.

Solution Given:

$$f_{X,Y}(x, y) = \begin{cases} \frac{6}{5}(x + y^2) & \text{for } 0 \leq x \leq 1, 0 \leq y \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

The marginal pdf of X is

$$\begin{aligned} f_X(x) &= \int_{-\infty}^{\infty} f_{X,Y}(x, y) dy \\ &= \frac{6}{5} \int_0^1 (x + y^2) dy = \frac{6}{5} \left[x + \frac{y^3}{3} \right]_0^1 \\ &= \frac{6}{5} \left(x + \frac{1}{3} \right) \quad 0 \leq x \leq 1 \end{aligned}$$

The marginal pdf of Y is

$$\begin{aligned} f_Y(y) &= \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx \\ &= \frac{6}{5} \int_0^1 (x + y^2) dx = \frac{6}{5} \left[\frac{x^2}{2} + y^2 x \right]_0^1 \\ f_Y(y) &= \frac{6}{5} \left(y^2 + \frac{1}{2} \right); 0 \leq y \leq 1 \end{aligned}$$

$$\begin{aligned} P\left(\frac{1}{4} \leq y \leq \frac{3}{4}\right) &= \int_{1/4}^{3/4} f_Y(y) dy = \int_{1/4}^{3/4} \frac{6}{5} \left(y^2 + \frac{1}{2} \right) dy \\ &= \frac{6}{5} \left[\left(\frac{y^3}{3} + \frac{y}{2} \right) \right]_{1/4}^{3/4} \\ &= \frac{6}{5} \left\{ \frac{1}{3} \left(\frac{27}{64} - \frac{1}{64} \right) + \frac{1}{2} \left(\frac{3}{4} - \frac{1}{4} \right) \right\} \end{aligned}$$

$$\begin{aligned}
 &= \frac{6}{5} \left\{ \frac{1}{3} \left(\frac{26}{64} \right) + \frac{1}{2} \left(\frac{1}{2} \right) \right\} \\
 &= 0.4625
 \end{aligned}$$

4.23 Find K if the joint probability density function of the bivariate random variable (X, Y) is given by

$$f_{X,Y}(x, y) = \begin{cases} K(1-x)(1-y), & \text{if } 0 < x < 1, 0 < y < 1 \\ 0 & \text{otherwise} \end{cases}$$

Solution If $f_{X,Y}(x, y)$ is a valid pdf then

$$\begin{aligned}
 \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx dy &= 1 \\
 \int_0^1 \int_0^1 k(1-x)(1-y) dx dy &= 1 \\
 k \int_0^1 (1-x) dx \int_0^1 (1-y) dy &= 1 \\
 k \left[\left(x - \frac{x^2}{2} \right) \Big|_0^1 \right] \left[\left(y - \frac{y^2}{2} \right) \Big|_0^1 \right] &= 1 \\
 k \left(\frac{1}{2} \right) \left(\frac{1}{2} \right) &= 1 \\
 k &= 4
 \end{aligned}$$

4.24 Joint probabilities of two random variables X and Y are given in the table:

$X \backslash Y$	1	2
1	0.2	0.15
2	0.1	0.2
3	0.2	0.15

- Find out joint and marginal distribution functions.
- Plot joint and marginal density functions.

Solution

$$\begin{aligned}
 F_{X,Y}(1, 1) &= P(X \leq 1, Y \leq 1) = P(X = 1, Y = 1) = 0.2 \\
 F_{X,Y}(2, 1) &= P(X \leq 2, Y \leq 1) = P(X = 1, Y = 1) + P(X = 2, Y = 1) = 0.2 + 0.1 = 0.3 \\
 F_{X,Y}(3, 1) &= P(X \leq 3, Y \leq 1) = P(X = 1, Y = 1) + P(X = 2, Y = 1) + P(X = 3, Y = 1) \\
 &= 0.2 + 0.1 + 0.2 = 0.5 \\
 F_{X,Y}(1, 2) &= P(X \leq 1, Y \leq 2) = P(X = 1, Y = 1) + P(X = 1, Y = 2) = 0.2 + 0.15 = 0.35 \\
 F_{X,Y}(2, 2) &= P(X \leq 2, Y \leq 2) \\
 &= P(X = 1, Y = 1) + P(X = 1, Y = 2) + P(X = 2, Y = 1) + P(X = 2, Y = 2)
 \end{aligned}$$

$$\begin{aligned}
&= 0.2 + 0.15 + 0.1 + 0.2 = 0.65 \\
F_{X,Y}(3, 2) &= P(X \leq 3, Y \leq 2) \\
&= P(X = 1, Y = 1) + P(X = 1, Y = 2) + P(X = 2, Y = 1) + P(X = 2, Y = 2) \\
&\quad + P(X = 3, Y = 1) + P(X = 3, Y = 2) \\
&= 0.2 + 0.15 + 0.1 + 0.2 + 0.2 + 0.15 = 1.00
\end{aligned}$$

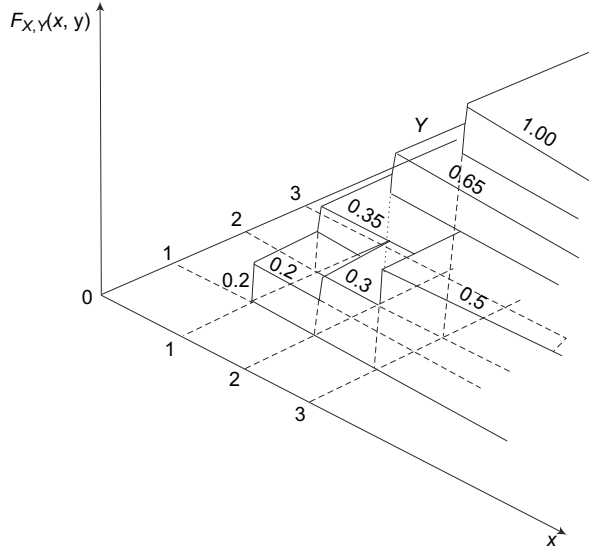
The joint distribution function can be constructed as follows. From the above calculated values of $F_{X,Y}(x, y)$, we can observe that for any $x < 1$ and/or $y < 1$, $F_{X,Y}(x, y)$ has no elements. Only at the point (1, 1) does a function assumes a step value of amplitude 0.2. This value is maintained for all $x \geq 1$ and $y \geq 1$. For large X and Y , the point (2, 1) produces a step amplitude of 0.1 which adds to the first, resulting in a step amplitude of 0.3. Similarly, at the point (3, 1), a step of amplitude 0.2 is added. So we have a step amplitude of 0.5 from the point (3, 1). At the point (1, 2) a step amplitude of 0.15 is added which results in a step amplitude of 0.35 from this point. If we repeat this process of adding step amplitudes at the given points shown in the table, we get $F_{X,Y}(x, y)$ as shown in Fig. 4.12(a).

The expression for joint distribution function is given by

$$\begin{aligned}
F_{X,Y}(x, y) &= 0.2 u(x-1) u(y-1) + 0.1 u(x-2) u(y-1) + 0.2 u(x-3) u(y-1) \\
&\quad + 0.15 u(x-1) u(y-2) + 0.2 u(x-2) u(y-2) + 0.15 u(x-3) u(y-2) \\
F_X(x) &= F_{X,Y}(x, \infty) \\
&= 0.2 u(x-1) + 0.1 u(x-2) + 0.2 u(x-3) + 0.15 u(x-1) + 0.2 u(x-2) + 0.15 u(x-3) \\
&= 0.35 u(x-1) + 0.3 u(x-2) + 0.35 u(x-3) \tag{4.42}
\end{aligned}$$

$$\begin{aligned}
F_Y(y) &= F_{X,Y}(\infty, y) \\
&= 0.2 u(y-1) + 0.1 u(y-1) + 0.2 u(y-1) + 0.15 u(y-2) + 0.2 u(y-2) + 0.15 u(y-2) \\
&= 0.5 u(y-1) + 0.5 u(y-2) \tag{4.43}
\end{aligned}$$

The sketch of marginal distribution functions are shown in Fig. 4.12(b) and 4.12(c).



(a)

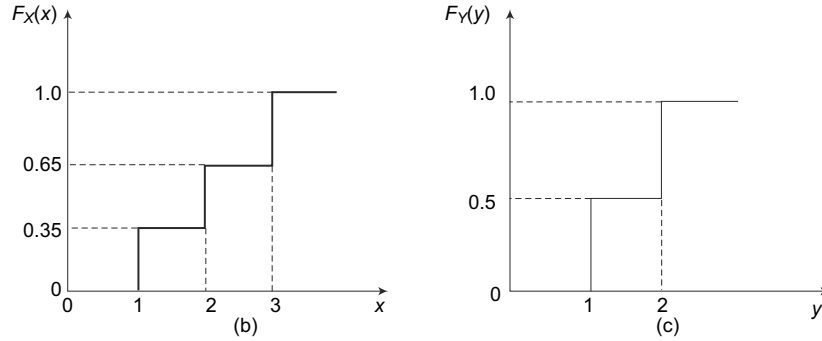


Fig. 4.12 (a) Joint distribution of (X, Y) (b) Marginal distribution function of X (c) Marginal distribution function of Y

From Eq. (4.42), we can observe that for $x < 1$, $F_X(x) = 0$. Only at the point $x = 1$, a step amplitude of 0.35 is added. This value is maintained for all $x \geq 1$ and at $x = 2$, a step amplitude of 0.2 adds to the function which results in a step amplitude of 0.65. At $x = 3$, a step function of 0.35 is added. The resultant $F_X(x)$ is shown in Fig. 4.12(b)

From Eq. (4.43), we find that $F_Y(y)$ has two step functions of amplitude 0.5 at $y = 1$ and $y = 2$. This results in $F_Y(y)$ shown in Fig. 4.12(c).

The joint pdf of the random variables X and Y can be obtained by using

$$f_{X,Y}(x, y) = \frac{\partial^2 F_{X,Y}(x, y)}{\partial x \partial y}$$

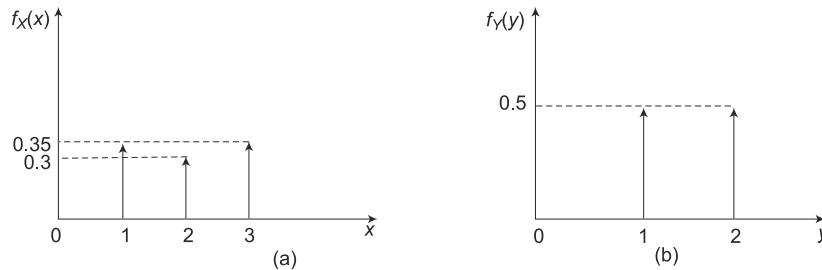
We know if we differentiate a unit step function, we get an impulse. Therefore, $f_{X,Y}(x, y)$ can be written as $f_{X,Y}(x, y) = 0.2 \delta(x-1) \delta(y-1) + 0.1 \delta(x-2) \delta(y-1) + 0.2 \delta(x-3) \delta(y-1) + 0.15 \delta(x-1) \delta(y-2) + 0.2 \delta(x-2) \delta(y-2) + 0.15 \delta(x-3) \delta(y-2)$

Similarly, by differentiating $F_X(x)$ and $F_Y(y)$, we get,

$$f_X(x) = 0.35 \delta(x-1) + 0.3 \delta(x-2) + 0.35 \delta(x-3) \text{ and}$$

$$f_Y(y) = 0.5 \delta(y-1) + 0.5 \delta(y-2).$$

The plots of marginal density functions and joint density functions are shown in Fig. 4.13.



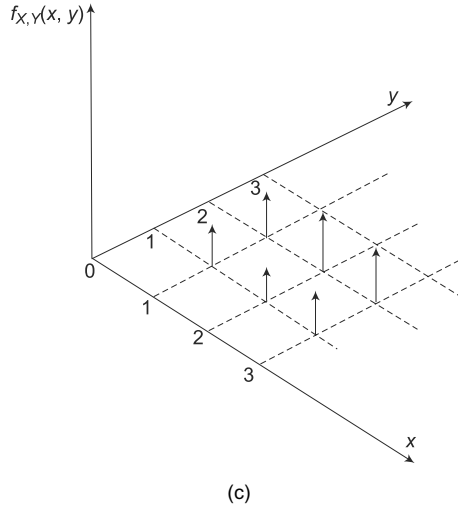


Fig. 4.13 (a) Marginal density function of X , (b) Marginal density function of Y , (c) Joint density function of (X, Y)



Practice Problems

4.11 The random variables X and Y have joint pdf

$$f_{X,Y}(x, y) = k(x + y) \text{ for } 0 \leq x \leq 1, 0 \leq y \leq 1$$

(a) Find k (b) Find the marginal pdfs of X and Y . (c) Find the joint pdf of (X, Y) .

(Ans. (a) $k = 1$, (b) $x + (1/2)$, $y + (1/2)$, (c) $F_{X,Y}(x, y) = xy(x + y)/2$ for $0 \leq x \leq 1, 0 \leq y \leq 1$)

4.12 Two random variables X and Y have the joint pdf given by

$$f_{X,Y}(x, y) = \begin{cases} \frac{1}{2} e^{-2y} & 0 \leq x \leq 4, y \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

Find $f_Y(y)$.

(Ans: $2e^{-2y}$, $y \geq 0$)

Solved Problems



4.25 A random vector (X, Y) is uniformly distributed ($f_{X,Y}(x, y) = k$) in the region shown in Fig. 4.14 and zero elsewhere. (a) Find the value of k . (b) Find the marginal pdfs of X and Y .

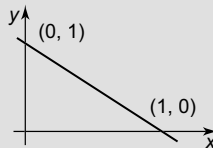
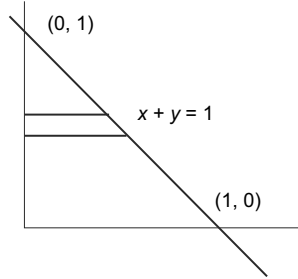


Fig. 4.14

Solution**Fig. 4.15**

We know that

$$\begin{aligned} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx dy &= 1 \\ \Rightarrow \int_0^1 \int_0^{1-y} k dx dy &= 1 \\ &= k \int_0^1 (1-y) dy = k \left(y - \frac{y^2}{2} \right) \bigg|_0^1 = k \left(\frac{1}{2} \right) = 1 \\ \Rightarrow k &= 2 \end{aligned}$$

$$\begin{aligned} f_X(x) &= \int_{-\infty}^{\infty} f_{X,Y}(x, y) dy \\ \Rightarrow \int_0^{1-x} 2 dy &= 2(1-x); 0 < x < 1 \end{aligned}$$

and

$$f_Y(y) = \int_0^{1-y} 2 dx = 2(1-y); 0 < y < 1$$

4.26 The joint pmf of two random variables X and Y is given by

$$p_{X,Y}(x, y) = \begin{cases} k(x + 2y); & x = 1, 2; y = 1, 2 \\ 0 & \text{otherwise} \end{cases}$$

where k is a constant.

(a) What is the value of k ? (b) Find the marginal pmfs of X and Y . (c) Are X and Y independent?

Solution

(a) We know $\sum_x \sum_y p_{X,Y}(x, y) = 1$

$$\begin{aligned}
\Rightarrow \quad & \sum_{x=1}^2 \sum_{y=1}^2 k(x+2y) = 1 \\
& = k \sum_{x=1}^2 (x+2) + (x+4) = 1 \\
\Rightarrow \quad & k \sum_{x=1}^2 (2x+6) = 1 \\
& k(8+10) = 1 \Rightarrow k = \frac{1}{18}
\end{aligned}$$

(b) The marginal pmf of X is given by

$$\begin{aligned}
p_X(x) &= \sum_y p_{X,Y}(x, y) \\
&= \frac{1}{18} \sum_{y=1}^2 (x+2y) = \frac{1}{18} [x+2+x+4] = \frac{1}{18} (2x+6) = \frac{x+3}{9}
\end{aligned}$$

$$p_X(x) = \frac{1}{9} (x+3); x=1, 2$$

$$\begin{aligned}
p_Y(y) &= \sum_x p_{X,Y}(x, y) = \frac{1}{18} \sum_{x=1}^2 (x+2y) \\
&= \frac{(2y+1) + (2+2y)}{18} = \frac{4y+3}{18}
\end{aligned}$$

$$P_y(y) = \frac{1}{18} [4y+3] \quad y=1, 2$$

Since $p_{XY}(x, y) \neq p_X(x) p_Y(y)$, X and Y are not independent.

4.27 The joint density function of two random variables X and Y is

$$\begin{aligned}
f_{X,Y}(x, y) &= 0.1 \delta(x) \delta(y) + 0.12 \delta(x-4) \delta(y) + 0.05 \delta(x) \delta(y-1) \\
&\quad + 0.25 \delta(x-2) \delta(y-1) + 0.3 \delta(x-2) \delta(y-3) + 0.18 \delta(x-4) \delta(y-3)
\end{aligned}$$

Find and plot the joint density and distribution functions of X and Y , and the marginal function distribution of X and Y .

Solution The distribution function can be obtained by integrating density function. The integral of an impulse function is equal to the step function. Therefore, we can write

$$\begin{aligned}
F_{X,Y}(x, y) &= 0.1 u(x) u(y) + 0.12 u(x-4) u(y) + 0.05 u(x) u(y-1) \\
&\quad + 0.25 u(x-2) u(y-1) + 0.3 u(x-2) u(y-3) + 0.18 u(x-4) u(y-3)
\end{aligned}$$

The marginal distribution function of X is

$$\begin{aligned} F_X(x) = F_{X,Y}(x, \infty) &= 0.1 u(x) + 0.12 u(x-4) + 0.05 u(x) + 0.25 u(x-2) + 0.3 u(x-2) + 0.18 u(x-4) \\ &= 0.15 u(x) + 0.55 u(x-2) + 0.3 u(x-4) \end{aligned}$$

The marginal density function of Y is

$$\begin{aligned} F_Y(y) = f_{X,Y}(\infty, y) &= 0.1 u(y) + 0.12 u(y) + 0.05 u(y-1) + 0.25 u(y-1) \\ &\quad + 0.3 u(y-3) + 0.18 u(y-3) \\ &= 0.22 u(y) + 0.3 u(y-1) + 0.48 u(y-3) \end{aligned}$$

We write the joint density function as a table:

X \ Y	Y			
	0	1	2	3
0	0.1	0.05		
1				
2		0.25		0.3
3				
4	0.12			0.18

From the table, we can find

$$\begin{aligned} F_{X,Y}(0, 0) &= P(X \leq 0, Y \leq 0) = P(X = 0, Y = 0) = 0.1 \\ F_{X,Y}(4, 0) &= P(X \leq 4, Y \leq 0) = P(X = 0, Y = 0) + P(X = 1, Y = 0) \\ &\quad + P(X = 2, Y = 0) + P(X = 3, Y = 0) + P(X = 4, Y = 0) \\ &= 0.1 + 0 + 0 + 0 + 0.12 = 0.22 \end{aligned}$$

Similarly,

$$\begin{aligned} F_{X,Y}(0, 1) &= P(X \leq 0, Y \leq 1) = P(X = 0, Y = 0) + P(X = 0, Y = 1) \\ &= 0.1 + 0.05 = 0.15 \end{aligned}$$

$$F_{X,Y}(2, 1) = 0.1 + 0.05 + 0.25 = 0.4$$

$$F_{X,Y}(2, 3) = 0.1 + 0.05 + 0.25 + 0.3 = 0.70$$

$$F_{X,Y}(4, 3) = 0.1 + 0.05 + 0.25 + 0.3 + 0.12 + 0.18 = 1.00$$

$$F_{X,Y}(4, 1) = P(X \leq 4, Y \leq 1) = 0.1 + 0.05 + 0.25 + 0.12 = 0.52$$

$$F_{X,Y}(4, 2) = 0.1 + 0.05 + 0.25 + 0.12 = 0.52$$

$$F_{X,Y}(3, 3) = 0.1 + 0.05 + 0.25 + 0.3 = 0.7$$

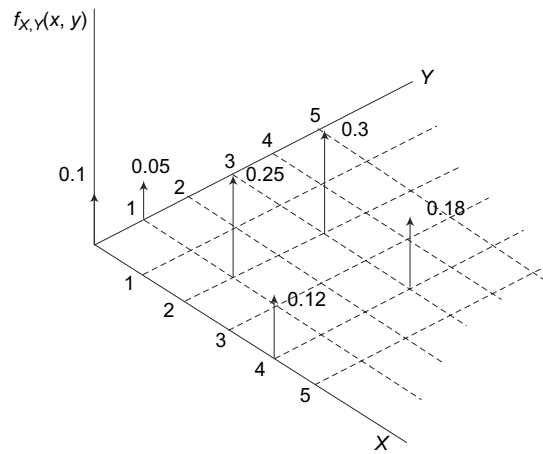


Fig. 4.16 Joint pdf of (X, Y)

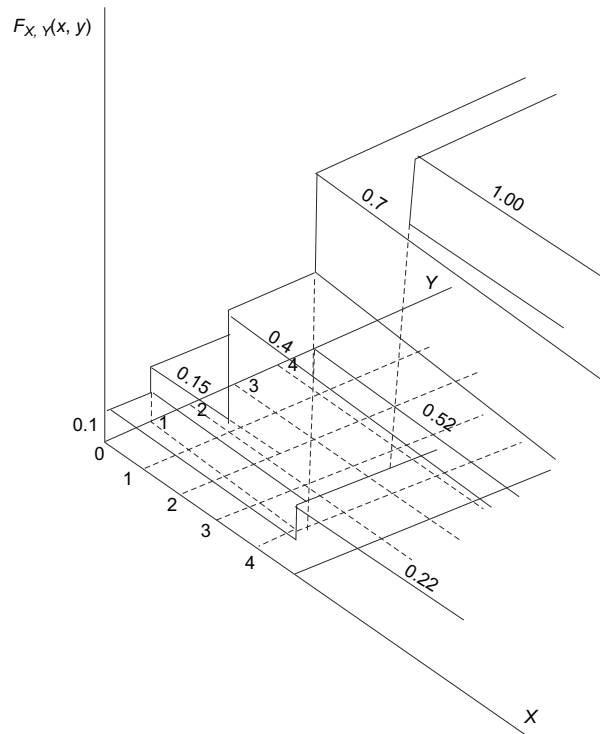


Fig. 4.17 Joint distribution function of (X, Y)

The marginal distribution functions of X and Y are shown in Fig. 4.18.

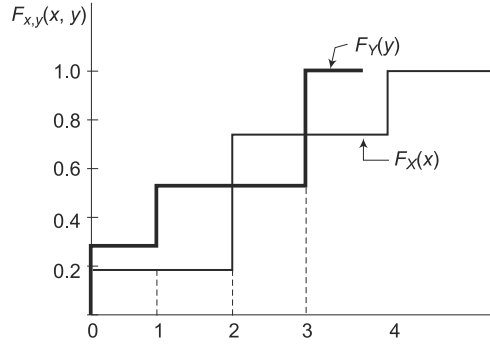


Fig. 4.18

4.28 The joint CDF of a bivariate random variable is given by

$$F_{XY}(x, y) = \begin{cases} (1 - e^{-ax})(1 - e^{-by}) & x \geq 0, y \geq 0; a, b > 0 \\ 0 & \text{otherwise} \end{cases}$$

- Find marginal CDFs of X and Y .
- Find whether X and Y are independent.
- Find $P(X \leq 1, Y \leq 1)$, $P(X \leq 1)$, $P(Y > 1)$, $P(X > x, Y > y)$.

Solution

- Given: $F_{XY}(x, y) = (1 - e^{-ax})(1 - e^{-by})$

The marginal CDF of $F_X(x) = F_{X,Y}(x, \infty)$

$$F_X(x) = (1 - e^{-ax})(1 - e^{-b(\infty)}) = 1 - e^{-ax}$$

Similarly,

$$\begin{aligned} F_Y(y) &= F_{XY}(\infty, y) \\ &= (1 - e^{-a(\infty)})(1 - e^{-by}) \\ &= 1 - e^{-by} \end{aligned}$$

$$\begin{aligned} F_X(x) &= (1 - e^{-ax}) \quad \text{for } x \geq 0 \\ &= 0 \quad \text{for } x < 0 \end{aligned}$$

$$\begin{aligned} F_Y(y) &= (1 - e^{-by}) \quad \text{for } y \geq 0 \\ &= 0 \quad \text{for } y < 0 \end{aligned}$$

- It can be easily observed that

$$F_{X,Y}(x, y) = F_X(x) F_Y(y)$$

Therefore, X and Y are independent.

$$P(X \leq 1, Y \leq 1) = F_{X,Y}(1, 1) = (1 - e^{-a})(1 - e^{-b})$$

$$P(X \leq 1) = F_X(1) = (1 - e^{-a})$$

$$\begin{aligned} P(Y > 1) &= 1 - P(Y \leq 1) = 1 - F_Y(1) \\ &= 1 - (1 - e^{-b}) = e^{-b} \end{aligned}$$

$$P(X > x, Y > y) = P(x < X \leq \infty, y < Y \leq \infty)$$

Using Eq. (4.17), we can write

$$\begin{aligned} P(x < X \leq \infty, y < Y \leq \infty) &= F_{X,Y}(\infty, \infty) - F_{X,Y}(x, \infty) - F_{X,Y}(\infty, y) + F_{X,Y}(x, y) \\ &= 1 - F_X(x) - F_Y(y) + F_{X,Y}(x, y) \\ &= 1 - (1 - e^{-ax}) - (1 - e^{-by}) + (1 - e^{-by})(1 - e^{-ax}) \\ &= e^{-ax} - 1 + e^{-by} + 1 - e^{-by} + e^{-ax} e^{-by} - e^{-ax} \\ &= e^{-ax} e^{-by} \end{aligned}$$



Practice Problems

4.13 The joint density function of random variables X, Y, Z is

$$F_{X,Y,Z}(x, y, z) = 8xyz \quad 0 < x, y, z < 1$$

Find $P(X < Y < Z)$.

(Ans. 1/6)

4.14 The joint pdf of two random variables X and Y is given by

$$f_{X,Y}(x, y) = xe^{-y(1+x)}, \quad x > 0, y > 0$$

Find the marginal pdfs of X and Y .

$$\left(\text{Ans: } f_X(x) = e^{-x}, f_Y(y) = \frac{1}{(1+y)^2} \right)$$

Solved Problems



4.29 The joint pdf of random variables (X, Y) is

$$\begin{aligned} f_{X,Y}(x, y) &= abe^{-(ax+by)} \quad x > 0, y > 0 \\ &= 0 \quad \text{elsewhere} \end{aligned}$$

Find $P(X > Y)$.

Solution

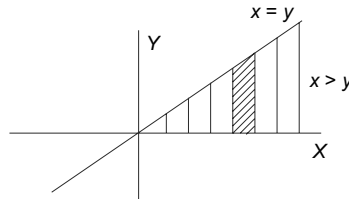


Fig. 4.19

$$\begin{aligned}
P(X > Y) &= \int_0^{\infty} \int_0^x abe^{-(ax+by)} dy dx \\
&= \int_0^{\infty} ab e^{-ax} \left(\int_0^x e^{-by} dy \right) dx \\
&= \int_0^{\infty} \frac{ab}{(-b)} e^{-ax} \left(e^{-by} \Big|_0^x \right) dx \\
&= a \int_0^{\infty} e^{-ax} (1 - e^{-bx}) dx \\
&= a \left[\int_0^{\infty} e^{-ax} dx - \int_0^{\infty} e^{-(a+b)x} dx \right] \\
&= a \left[\frac{e^{-ax}}{-a} \Big|_0^{\infty} - \frac{e^{-(a+b)x}}{-(a+b)} \Big|_0^{\infty} \right] \\
&= a \left[-\frac{1}{a}(-1) + \frac{1}{a+b}(-1) \right] \\
&= a \left[\frac{1}{a} - \frac{1}{a+b} \right] = \frac{ab}{a(a+b)} \\
&= \frac{b}{a+b}
\end{aligned}$$

4.30 The joint pdf of (X, Y) is given by

$$f_{X,Y}(x, y) = \begin{cases} K e^{-(x+2y)} & x > 0, y > 0 \\ 0 & \text{elsewhere} \end{cases}$$

where k is a constant

- (a) Find the value of k ,
 (b) Find $P(X > 1, Y < 1)$, $P(X < Y)$ and $P(X \leq 2)$.

Solution

$$f_{X,Y}(x, y) = \begin{cases} k e^{-(x+2y)} & x > 0, y > 0 \\ 0 & \text{elsewhere} \end{cases}$$

For a valid pdf,

$$\begin{aligned}
\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx dy &= 1 \\
\Rightarrow \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} k e^{-(x+2y)} dx dy &= k \int_0^{\infty} \int_0^{\infty} e^{-x} e^{-2y} dx dy
\end{aligned}$$

$$\begin{aligned}
 &= -k \int_0^{\infty} e^{-2y} (e^{-x}) \Big|_0^{\infty} dy = k \int_0^{\infty} e^{-2y} dy \\
 &= \frac{-k}{2} e^{-2y} \Big|_0^{\infty} = \frac{k}{2}
 \end{aligned}$$

$$k/2 = 1 \Rightarrow k = 2$$

$$\begin{aligned}
 P(X > 1, Y < 1) &= \int_1^{\infty} \int_0^1 2 e^{-(x+2y)} dy dx \\
 &= 2 \int_1^{\infty} e^{-x} \left(\int_0^1 e^{-2y} dy \right) dx \\
 &= - \int_1^{\infty} e^{-x} \left(e^{-2y} \Big|_0^1 \right) dx \\
 &= - \int_1^{\infty} e^{-x} (e^{-2} - 1) dx \\
 &= 0.846 \int_1^{\infty} e^{-x} dx \\
 &= 0.846 e^{-1} = 0.3178
 \end{aligned}$$

$$\begin{aligned}
 P(X < Y) &= \int_0^{\infty} \int_x^{\infty} 2 e^{-(x+2y)} dy dx \\
 &= 2 \int_0^{\infty} e^{-x} \int_x^{\infty} e^{-2y} dy dx \\
 &= - \int_0^{\infty} e^{-x} \left(e^{-2y} \Big|_x^{\infty} \right) dx = \int_x^{\infty} e^{-3x} dx = \frac{-1}{3} e^{-3x} \Big|_0^{\infty} \\
 &= \frac{1}{3}
 \end{aligned}$$

$$P(X \leq 2) = ?$$

$$\begin{aligned}
 f_X(x) &= \int_0^{\infty} 2 e^{-(x+2y)} dy = 2e^{-x} \frac{e^{-2y}}{-2} \Big|_0^{\infty} \\
 &= -e^{-x}(-1) = e^{-x}
 \end{aligned}$$

$$\begin{aligned}
 P(X \leq 2) &= \int_0^2 e^{-x} dx = \frac{e^{-x}}{-1} \Big|_0^2 \\
 &= -(e^{-2} - 1) = 1 - e^{-2} = 0.864
 \end{aligned}$$

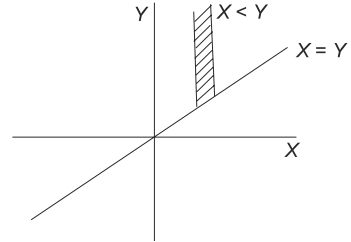


Fig. 4.20

4.31 Find the joint density and both marginal density functions of the distribution function of Solved Problem (4.17).

Solution

$$F_{X,Y}(x, y) = u(x) u(y) (1 - e^{-ax} - e^{-ay} + e^{-a(x+y)})$$

We know

$$\begin{aligned} f_{X,Y}(x, y) &= \frac{\partial^2 F_{X,Y}(x, y)}{\partial x \partial y} \\ &= \frac{\partial^2}{\partial x \partial y} \{u(x) u(y) (1 - e^{-ax} - e^{-ay} + e^{-a(x+y)})\} \\ &= \frac{\partial^2}{\partial x \partial y} \{(1 - e^{-ax}) u(x) (1 - e^{-ay}) u(y)\} \\ &= \frac{\partial}{\partial x} \{(1 - e^{-ax}) u(x) [(1 - e^{-ay}) \delta(y) + u(y) a e^{-ay}]\} \end{aligned}$$

at $y = 0$; $1 - e^{-ay} = 0$, therefore

$$f_{X,Y}(x, y) = \frac{\partial}{\partial x} \{(1 - e^{-ax}) u(x)\} u(y) a e^{-ay}$$

$$= u(y) a e^{-ay} \{(1 - e^{-ax}) \delta(x) + u(x) a e^{-ax}\}$$

At $x = 0$, $(1 - e^{-ax}) \delta(x) = 0$

Therefore, $f_{X,Y}(x, y) = a^2 e^{-a(x+y)} u(x) u(y)$

The marginal density function of X is given by

$$\begin{aligned} f_X(x) &= \int_0^\infty a^2 e^{-a(x+y)} dy \\ &= a^2 e^{-ax} \left. \frac{e^{-ay}}{-a} \right|_0^\infty = a e^{-ax}; \quad x > 0 \end{aligned}$$

$$\begin{aligned} f_Y(y) &= \int_0^\infty a^2 e^{-a(x+y)} dx \\ &= a^2 e^{-ay} \left. \frac{e^{-ax}}{-a} \right|_0^\infty = a e^{-ay} \quad y > 0 \end{aligned}$$

$$f_X(x) = a e^{-ax} \quad x > 0$$

$$f_Y(y) = a e^{-ay} \quad y > 0$$

$\delta(y)$	$= 1$ at $y = 0$ $= 0$ at $y \neq 0$
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4.32 If the function

$$f_{X,Y}(x, y) = \begin{cases} b e^{-2x} \cos\left(\frac{y}{2}\right), & 0 \leq x \leq 1, 0 \leq y \leq \pi \\ 0 & \text{elsewhere} \end{cases}$$

where b is a positive constant, is a valid joint density function, find b .

Solution If $f_{XY}(x, y)$ is a valid density function then

$$\begin{aligned} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx dy &= 1 \\ \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx dy &= \int_0^1 \int_0^{\pi} b e^{-2x} \cos\left(\frac{y}{2}\right) dx dy = 1 \\ &= b \int_0^{\pi} \cos\left(\frac{y}{2}\right) \left(\frac{e^{-2x}}{(-2)} \right) \bigg|_0^1 dy \\ &= \frac{b}{2} (1 - e^{-2}) \int_0^{\pi} \cos\left(\frac{y}{2}\right) dy = \frac{b(1 - e^{-2})}{2} \frac{\sin\left(\frac{y}{2}\right)}{\left(\frac{1}{2}\right)} \bigg|_0^{\pi} \\ &= b(1 - e^{-2})(1) = 1 \\ \Rightarrow \quad b &= \frac{1}{1 - e^{-2}} \end{aligned}$$



Practice Problems

4.15 The joint pdf of (X, Y) is given by

$$f_{X,Y}(x, y) = \begin{cases} k x^2 (4 - y) & x < y < 2x, 0 < x < 2 \\ 0 & \text{elsewhere} \end{cases} \quad \left(\text{Ans: } \frac{5}{32} \right)$$

where k is a constant.

(a) Find the value of k . (b) Find the marginal pdfs of X and Y .

4.16 Two random variables X, Y have joint density function

$$f_{X,Y}(x, y) = c e^{-4x^2 - 6xy - 9y^2}$$

Find (a) c ; (b) $f_X(x)$; (c) $f_Y(y)$.

$$\left(\text{Ans: (a) } c = \frac{3\sqrt{3}}{\pi}; \text{ (b) } \sqrt{\frac{3}{\pi}} e^{-2x^2}; \text{ (c) } \frac{3}{2} \sqrt{\frac{3}{\pi}} e^{-27y^2/4} \right)$$

4.17 Find a value of the constant b so that the function

$$f_{X,Y}(x, y) = b x^2 y e^{-2xy} u(x-1) u(y-2) \quad \left(\text{Ans: } \frac{8e^4}{3} \right)$$

is a valid joint probability density.

Solved Problems

4.33 Find the value of b for the function given below to be a valid pdf.

$$f_{X,Y}(x, y) = \begin{cases} b(x^2 + 4y^2) & 0 \leq |x| < 1 \text{ and } 0 \leq y < 2 \\ 0 & \text{elsewhere} \end{cases}$$

Solution For a valid pdf,

$$\begin{aligned} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx dy &= 1 \\ \Rightarrow \int_0^2 \int_{-1}^1 b(x^2 + 4y^2) dx dy &= b \int_0^2 \left(\frac{x^3}{3} \Big|_{-1}^1 + 4y^2 x \Big|_{-1}^1 \right) dy \\ &= b \int_0^2 \left(\frac{2}{3} + 8y^2 \right) dy = b \left[\frac{2}{3} y \Big|_0^2 + \frac{8}{3} y^3 \Big|_0^2 \right] \\ &= b \left[\frac{4}{3} + \frac{64}{3} \right] = \frac{68}{3} b \end{aligned}$$

For a valid pdf, $\frac{68}{3} b = 1 \Rightarrow b = \frac{3}{68}$

CONDITIONAL DISTRIBUTION AND DENSITY FUNCTIONS

4.8



The conditional distribution function of a random variable X , given some event A , is defined as

$$\begin{aligned} F_X(x | A) &= P(X \leq x | A) \\ &= \frac{P(X \leq x \cap A)}{P(A)} \end{aligned} \quad (4.44)$$

with $P(A) \neq 0$.

The conditional density function is given by

$$f_X(x | A) = \frac{d}{dx} F_X(x | A) \quad (4.45)$$

4.8.1 Conditional Distribution and Density-Point Conditioning

Consider two random variables X and Y . The distribution function of a random variable X at a given condition that the second variable Y has some specific value y is known as point conditioning.

Consider an event A defined as

$$A = \{y - \Delta y < Y \leq y + \Delta y\} \quad (4.46)$$

where Δy is a small quantity that tends to zero.

Then using Eq. (4.44) we can write

$$\begin{aligned}
 F_X(x | y - \Delta y < Y \leq y + \Delta y) &= P(X \leq x | y - \Delta y < Y \leq y + \Delta y) \\
 &= \frac{P(X \leq x \cap (y - \Delta y < Y \leq y + \Delta y))}{P(y - \Delta y < Y \leq y + \Delta y)} \\
 &= \frac{F_{X,Y}(x, (y - \Delta y < y \leq y + \Delta y))}{F_X(y - \Delta y < Y \leq y + \Delta y)}
 \end{aligned} \tag{4.47}$$

Using Eq. (4.36), we can write

$$F_X(x | y - \Delta y < Y \leq y + \Delta y) = \frac{\int_{y - \Delta y}^{y + \Delta y} \int_{-\infty}^x f_{X,Y}(u, v) du dv}{\int_{y - \Delta y}^{y + \Delta y} f_X(v) dv} \tag{4.48}$$

If X and Y are continuous random variables for some value of Δy , Eq. (4.48) can be written as

$$\begin{aligned}
 F_X(x | y - \Delta y < Y \leq y + \Delta y) &= \frac{\int_{-\infty}^x f_{X,Y}(u, y) du \int_{-\Delta y}^{\Delta y} dv}{f_Y(y) 2 \Delta y} \\
 &= \frac{\int_{-\infty}^x f_{X,Y}(u, y) du}{f_Y(y)} \left(\frac{2 \Delta y}{2 \Delta y} \right)
 \end{aligned} \tag{4.49}$$

In the limit $\Delta y \rightarrow 0$,

$$F_X(x | Y = y) = \frac{\int_{-\infty}^x f_{X,Y}(u, y) du}{f_Y(y)} \tag{4.50}$$

Taking differentiation on both sides

$$f_X(x | Y = y) = \frac{f_{X,Y}(x, y)}{f_Y(y)} \tag{4.51}$$

for every y such that $f_Y(y) \neq 0$.

Similarly, we can also show that

$$f_Y(y | x) = \frac{f_{X,Y}(x, y)}{f_X(x)} \tag{4.52}$$

4.8.2 Discrete Random Variables

Consider both X and Y discrete random variables with values $x_i, i = 1, 2, \dots, N$ and $y_j, j = 1, 2, \dots, M$ respectively. Let the probability of these values be denoted by $p(x_i)$ and $p(y_j)$.

We know that the density functions can be expressed as

$$f_X(x) = \sum_{i=1}^N p(x_i) \delta(x - x_i) \quad (4.53)$$

$$f_Y(y) = \sum_{j=1}^M p(y_j) \delta(y - y_j) \quad (4.54)$$

If the probability of joint occurrence of x_i and y_j is denoted by $p(x_i, y_j)$ then

$$f_{X,Y}(x, y) = \sum_{i=1}^N \sum_{j=1}^M p(x_i, y_j) \delta(x - x_i) \delta(y - y_j) \quad (4.55)$$

The conditional distribution function is given by

$$\begin{aligned} F_X(x \mid y - \Delta y < Y \leq y + \Delta y) \\ = \frac{\sum_{i=1}^N \sum_{j=1}^M p(x_i, y_j) u(x - x_i) u(y - y_j)}{\sum_{j=1}^M p(y_j) u(y - y_j)} \end{aligned} \quad (4.56)$$

For a specific value of $y_j = y_k$ and $\Delta y \rightarrow 0$,

$$F_X(x \mid y_k) = \frac{\sum_{i=1}^N p(x_i, y_k) u(x - x_i)}{p(y_k)} \quad (4.57)$$

After differentiation,

$$f_X(x \mid y_k) = \sum_{i=1}^N \frac{p(x_i, y_k)}{p(y_k)} \delta(x - x_i) \quad (4.58)$$

$$\text{Similarly, } F_Y(y \mid x_k) = \sum_{j=1}^M \frac{p(x_k, y_j)}{p(x_k)} u(y - y_j) \quad (4.59)$$

$$\text{and } f_Y(y \mid x_k) = \sum_{j=1}^M \frac{p(x_k, y_j)}{p(x_k)} \delta(y - y_j) \quad (4.60)$$

Interval Conditioning Consider an event A defined in the interval $A = \{y_1 < Y \leq y_2\}$ where y_1 and y_2 are real numbers. We assume that $p(A) = P(y_1 < Y \leq y_2) \neq 0$. Then using Eq. (4.44),

$$\begin{aligned} F_X(x \mid (y_1 < Y \leq y_2)) &= P(X \leq x \mid (y_1 < Y \leq y_2)) \\ &= \frac{P(X \leq x \cap (y_1 < Y \leq y_2))}{P(y_1 < Y \leq y_2)} \end{aligned} \quad (4.61)$$

$$= \frac{F_{X,Y}(x, y_1 < Y \leq y_2)}{F_Y(y_1 < Y \leq y_2)} \quad (4.62)$$

$$= \frac{F_{X,Y}(x, y_2) - F_{X,Y}(x, y_1)}{F_Y(y_2) - F_Y(y_1)} \quad (4.63)$$

$$\begin{aligned}
 & \int_{y_1}^{y_2} \int_{-\infty}^x f_{X,Y}(u, y) du dy \\
 &= \frac{\int_{y_1}^{y_2} \int_{-\infty}^{\infty} f_{X,Y}(u, y) du dy}{\int_{y_1}^{y_2} \int_{-\infty}^{\infty} f_{X,Y}(u, y) du dy}
 \end{aligned} \tag{6.64}$$

After differentiation, we can get the conditional density function as

$$f_X(x | y_1 < Y \leq y_2) = \frac{\int_{y_1}^{y_2} f_{X,Y}(x, y) dy}{\int_{y_1}^{y_2} \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx dy} \tag{4.65}$$

Properties of Conditional Density Functions

$$1. \quad f_X(x | A) \geq 0 \tag{4.66}$$

$$2. \quad \int_{-\infty}^{\infty} f_X(x | A) dx = 1 \tag{4.67}$$

$$3. \quad F_X(x_1 | A) = \int_{-\infty}^{x_1} f_X(x | A) dx \tag{4.68}$$

$$4. \quad P(x_1 < A < x_2 | A) = \int_{x_1}^{x_2} f_X(x | A) dx \tag{4.69}$$

REVIEW QUESTIONS



13. Define conditional distribution and density functions of two random variables X and Y .
14. Define and explain conditional probability density function. Give its properties.

Solved Problems



4.34 A product is classified according to the number of defects it contains (X_1) and the factors that produce it (X_2). The joint probability distribution is

$X_2 \backslash X_1$	1	2
0	1/8	1/16
1	1/16	1/16
2	3/16	1/8
	1/8	1/4

- (a) Find the marginal distribution function of X_1 .
- (b) Find the conditional distribution function of X_1 when X_2 is equal to 1.

Solution

- (a) The marginal distribution function
- X_1
- can be obtained by finding the sum of each row.

$$P(X_1 = 0) = \frac{1}{8} + \frac{1}{16} = \frac{3}{16}; P(X_1 = 1) = \frac{1}{16} + \frac{1}{16} = \frac{1}{8}$$

$$P(X_1 = 2) = \frac{3}{16} + \frac{1}{8} = \frac{5}{16}; P(X_1 = 3) = \frac{1}{8} + \frac{1}{4} = \frac{3}{8}$$

$$(b) P(X_1 = 0 | X_2 = 1) = \frac{P(X_1 = 0, X_2 = 1)}{P(X_2 = 1)}$$

$$P(X_2 = 1) = \text{Sum of first column}$$

$$= \frac{1}{8} + \frac{1}{16} + \frac{3}{16} + \frac{1}{8} = \frac{1}{2}$$

$$P(X_1 = 0 | X_2 = 1) = \frac{1/8}{1/2} = \frac{1}{4}$$

$$P(X_1 = 1 | X_2 = 1) = \frac{1/16}{1/2} = \frac{1}{8}$$

$$P(X_1 = 2 | X_2 = 1) = \frac{P(X_1 = 2, X_2 = 1)}{P(X_2 = 1)} = \frac{3/16}{1/2} = \frac{3}{8}$$

$$P(X_1 = 3 | X_2 = 2) = \frac{P(X_1 = 3, X_2 = 2)}{P(X_2 = 2)} = \frac{1/4}{1/2} = \frac{1}{2}$$

4.35 The input and output of a communication system are two random variables X and Y respectively. The joint probability mass function of X and Y is given by a matrix below.

$$P(X, Y) = \begin{matrix} & \begin{matrix} Y \\ X \end{matrix} & \begin{matrix} -1 & 0 & 1 \end{matrix} \\ \begin{matrix} -1 \\ 0 \\ 1 \end{matrix} & \begin{bmatrix} \frac{1}{8} & 0 & \frac{1}{4} \\ 0 & \frac{1}{16} & \frac{7}{16} \\ \frac{1}{16} & 0 & \frac{1}{16} \end{bmatrix} \end{matrix}$$

Find (a) $P(Y = 1 | X = 1)$ (b) $P(X = 1 | Y = 1)$.

Solution

$$(a) P(Y = 1 | X = 1) = \frac{P(X = 1, Y = 1)}{P(X = 1)}$$

$$P(X = 1, Y = 1) = \frac{1}{16}$$

$$P(X = 1) = \text{Sum of third row of the matrix}$$

$$= \frac{1}{16} + 0 + \frac{1}{16} = \frac{1}{8}$$

$$P(Y = 1 | X = 1) = \frac{1/16}{1/8} = \frac{1}{2}$$

$$(b) \quad P(X = 1 | Y = 1) = \frac{P(X = 1, Y = 1)}{P(Y = 1)}$$

$P(Y = 1)$ = Sum of third column of the matrix

$$= \frac{1}{4} + \frac{7}{16} + \frac{1}{16} = \frac{3}{4}$$

$$P(X = 1 | Y = 1) = \frac{1/16}{3/4} = \frac{1}{12}$$

4.36 The joint pmf of random variables (X, Y) is shown in the table. Find (a) $P(X = 1)$ (b) $P(Y = 1)$ (c) $P(X = 1 | Y = 1)$

$X \backslash Y$	-1	0	1
-1	$\frac{1}{4}$	0	0
0	$\frac{1}{8}$	$\frac{1}{4}$	$\frac{1}{8}$
1	0	0	$\frac{1}{4}$

Solution The pmf of X can be obtained by computing the row sums, whereas the pmf of Y can be obtained by computing the column sums

$$P(X = 1) = 0 + 0 + \frac{1}{4} = \frac{1}{4} \quad (\text{Sum of third row})$$

$$P(Y = 1) = 0 + \frac{1}{8} + \frac{1}{4} = \frac{3}{8} \quad (\text{Sum of third column})$$

$$P(X = 1 | Y = 1) = \frac{P(X = 1, Y = 1)}{P(Y = 1)} = \frac{1/4}{3/8} = \frac{2}{3}$$

4.37 Find the marginal pmfs for the pair of random variables with the joint pmf shown below.

$X \backslash Y$	-1	0	1
-1	$\frac{1}{6}$	0	$\frac{1}{6}$
0	0	$\frac{1}{3}$	0
1	$\frac{1}{6}$	0	$\frac{1}{6}$

Find the probability of the events

$$A = \{X \leq 0\}, B = \{X = -Y\}, F_X(x), F_Y(y)$$

Solution The pmf of X can be obtained by computing the row sum, and pmf of Y can be obtained by computing the column sum.

$$P(X = -1) = \frac{1}{6} + 0 + \frac{1}{6} = \frac{1}{3}$$

$$P(X = 0) = 0 + \frac{1}{3} + 0 = \frac{1}{3}$$

$$P(X = 1) = \frac{1}{6} + 0 + \frac{1}{6} = \frac{1}{3}$$

Similarly, $P(Y = -1) = \frac{1}{6} + 0 + \frac{1}{6} = \frac{1}{3}$

$$P(Y = 0) = 0 + \frac{1}{3} + 0 = \frac{1}{3}$$

$$P(Y = 1) = \frac{1}{6} + 0 + \frac{1}{6} = \frac{1}{3}$$

$$P(X \leq 0) = \text{Sum of first two rows of the table}$$

$$= 0 + \frac{1}{3} + 0 + \frac{1}{6} + 0 + \frac{1}{6} = \frac{2}{3}$$

$$P(X = -Y) = P(X = 0, Y = 0) + P(X = -1, Y = 1) + P(X = 1, Y = -1)$$

$$= \frac{1}{3} + \frac{1}{6} + \frac{1}{6} = \frac{2}{3}$$

The joint distribution function can be written as

$$F_{X,Y}(x, y) = \frac{1}{6} u(x+1) u(y+1) + \frac{1}{6} u(x+1) u(y-1) \\ + \frac{1}{3} u(x) u(y) + \frac{1}{6} u(x-1) u(y+1) + \frac{1}{6} u(x-1) u(y-1)$$

$$F_X(x) = F_{X,Y}(x, \infty)$$

$$= \frac{1}{6} u(x+1) + \frac{1}{6} u(x+1) + \frac{1}{3} u(x) + \frac{1}{6} u(x-1) + \frac{1}{6} u(x-1)$$

$$= \frac{1}{3} u(x+1) + \frac{1}{3} u(x) + \frac{1}{3} u(x-1)$$

$$F_Y(y) = F_{X,Y}(\infty, y)$$

$$= \frac{1}{6} u(y+1) + \frac{1}{6} u(y-1) + \frac{1}{3} u(y) + \frac{1}{6} u(y+1) + \frac{1}{6} u(y-1)$$

$$= \frac{1}{3} u(y+1) + \frac{1}{3} u(y) + \frac{1}{3} u(y-1)$$

4.38 Two discrete random variables X and Y have the joint probability density function

$$P(X, Y) = \frac{\lambda^x e^{-\lambda} p^y (1-p)^{x-y}}{y! (x-y)!}, \quad y = 0, 1, 2, \dots, x = 0, 1, 2, \dots$$

where λ, p are constants with $\lambda > 0$ and $0 < p < 1$. Find (a) marginal probability density functions of X and Y . (b) The conditional distribution function of Y for a given X and of X for a given Y .

Solution

(a) The marginal distribution function of X is

$$\begin{aligned} p_X(x) &= \sum_y P(X, Y) \\ &= \sum_{y=0}^x \frac{\lambda^x e^{-\lambda} p^y (1-p)^{x-y}}{y! (x-y)!} \\ &= \frac{\lambda^x e^{-\lambda}}{x!} \sum_{y=0}^x \frac{x! p^y (1-p)^{x-y}}{y! (x-y)!} = \frac{\lambda^x e^{-\lambda}}{x!} \sum_{y=0}^x \binom{x}{y} p^y (1-p)^{x-y} \\ &= \frac{\lambda^x e^{-\lambda}}{x!} (1-p+p)^x \\ &= \frac{\lambda^x e^{-\lambda}}{x!} \quad x = 0, 1, 2, \dots \end{aligned}$$

Note that X is a random variable with Poisson distribution.

The marginal distribution function of Y is

$$\begin{aligned} p_Y(y) &= \sum_x P(X, Y) \\ &= \sum_{x=y}^{\infty} \frac{\lambda^x e^{-\lambda} p^y (1-p)^{x-y}}{y! (x-y)!} \\ &= \frac{e^{-\lambda} p^y \lambda^y}{y!} \sum_{x=y}^{\infty} \frac{\lambda^{x-y} (1-p)^{x-y}}{y! (x-y)!} \\ &= \frac{(\lambda p)^y e^{-\lambda}}{y!} \sum_{x=y}^{\infty} \frac{[\lambda(1-p)]^{x-y}}{(x-y)!} \\ &= \frac{(\lambda p)^y e^{-\lambda}}{y!} e^{\lambda(1-p)} = \frac{e^{-p\lambda} (\lambda p)^y}{y!}; \quad y = 0, 1, 2 \end{aligned}$$

Note that Y is a random variable with Poisson distribution with parameter λp .

(b) The conditional probability distribution of Y given X is

$$\begin{aligned} P(Y = y | X = x) &= \frac{p_{X,Y}(x, y)}{p_X(x)} = \frac{\lambda^x e^{-\lambda} p^y (1-p)^{x-y}}{y! (x-y)!} \frac{x!}{\lambda^x e^{-\lambda}} \\ &= \binom{x}{y} p^y (1-p)^{x-y}, \quad x > y \end{aligned}$$

The conditional probability distribution of X given Y is

$$\begin{aligned}
 P(X=x | Y=y) &= \frac{p_{X,Y}(x, y)}{p_Y(y)} \\
 &= \frac{\lambda^x e^{-\lambda} p^y (1-p)^{x-y}}{y! (x-y)!} \cdot \frac{y!}{e^{-p\lambda} (\lambda p)^y} \\
 &= \frac{e^{-(1-p)\lambda} \lambda^{x-y} (1-p)^{x-y}}{(x-y)!} \\
 &= \frac{e^{-(1-p)\lambda} [\lambda(1-p)]^{x-y}}{(x-y)!}
 \end{aligned}$$

4.39 A binary communication channel carries data as one of the two types of signals denoted by 0 and 1. Due to noise, a transmitted '0' is sometimes received as 1 and a transmitted 1 is sometimes received as '0'. For a given channel, assume a probability of 0.94 that a transmitted '0' is correctly received as a '0' and a probability of 0.91 that a transmitted 1 is received as 1. Further, assume a probability of 0.45 of transmitting a '0'. If a signal is sent, determine (a) probability that a 1 is received, (b) probability that a '0' is received, (c) probability that a 1 was transmitted, given that a 1 was received, (d) probability that a '0' was transmitted, given that a '0' was received, and (e) probability of error.

Solution Let us denote the transmitted message as X and received message as Y .

Probability that a transmitted '0' is correctly received as '0' is $P(Y=0 | X=0) = 0.94$

Similarly, $P(Y=1 | X=0) = 1 - 0.94 = 0.06$

$$P(Y=1 | X=1) = 0.91$$

$$P(Y=0 | X=1) = 0.91 = 0.09$$

$$P(X=0) = 0.45; P(X=1) = 1 - 0.45 = 0.55$$

$$\begin{aligned}
 P(X=0, Y=0) &= P(Y=0 | X=0) P(X=0) \\
 &= (0.94) (0.45) = 0.423
 \end{aligned}$$

$$\begin{aligned}
 P(X=0, Y=1) &= P(Y=1 | X=0) P(X=0) \\
 &= (0.06) (0.45) = 0.027
 \end{aligned}$$

$$\begin{aligned}
 P(X=1, Y=0) &= P(Y=0 | X=1) P(X=1) \\
 &= (0.09) (0.55) = 0.0495
 \end{aligned}$$

$$\begin{aligned}
 P(X=1, Y=1) &= P(Y=1 | X=1) P(X=1) \\
 &= (0.91) (0.55) = 0.5005
 \end{aligned}$$

The joint probability matrix is

		Y	
		0	1
X	0	0.423	0.027
	1	0.0495	0.5005

- (a) $P(Y = 0) = 0.423 + 0.0495 = 0.4725$
 (b) $P(Y = 1) = 0.027 + 0.5005 = 0.5275$
 (c) Probability that a 1 is transmitted and 1 is received

$$P(X = 1 | Y = 1) = \frac{P(X = 1, Y = 1)}{P(Y = 1)} = \frac{0.5005}{0.5275} = 0.9488$$

$$(d) \quad P(X = 0 | Y = 0) = \frac{P(X = 0, Y = 0)}{P(Y = 0)} = \frac{0.423}{0.4725} = 0.89523$$

- (e) Probability of error =

$$\begin{aligned} P(Y = 1 | X = 0) P(X = 0) + P(Y = 0 | X = 1) P(X = 1) \\ = (0.06)(0.45) + (0.09)(0.55) = 0.0765 \end{aligned}$$

4.40 The joint pdf of three random variables X , Y and Z is

$$f_{X,Y,Z}(x, y, z) = K(x + y + z) \quad \text{for} \quad \begin{aligned} 0 \leq x \leq 1 \\ 0 \leq y \leq 1 \\ 0 \leq z \leq 1 \end{aligned}$$

- (a) Find k .
 (b) Find $f_X(x | y, z)$ and $f_Z(z | x, y)$.
 (c) Find $f_X(x)$, $f_Y(y)$, $f_Z(z)$.

Solution

- (a) For a valid pdf,

$$\begin{aligned} \int_0^1 \int_0^1 \int_0^1 k(x + y + z) dx dy dz &= 1 \\ &= k \int_0^1 \int_0^1 \left(\frac{1}{2} + y + z \right) dy dx = k \int_0^1 \left(\frac{1}{2} + \frac{1}{2} + z \right) dz \\ &= k \left(1 + \frac{z^2}{2} \Big|_0^1 \right) = 1 \quad \Rightarrow \quad k = \frac{2}{3} \end{aligned}$$

$$\Rightarrow f_{X,Y,Z}(x, y, z) = \frac{2}{3}(x + y + z)$$

$$(b) \quad f_X(x | y, z) = \frac{f_{X,Y,Z}(x, y, z)}{f_{Y,Z}(y, z)}$$

$$f_{Y,Z}(y, z) = \frac{2}{3} \int_0^1 (x + y + z) dx = \frac{2}{3} \left(y + z + \frac{1}{2} \right)$$

$$f_X(x | y, z) = \frac{\frac{2}{3}(x + y + z)}{\frac{2}{3} \left(y + z + \frac{1}{2} \right)} = \frac{x + y + z}{y + z + \frac{1}{2}}$$

$$f_{X,Y}(x, y) = \frac{2}{3} \int_0^1 (x + y + z) dz = \frac{2}{3} \left(x + y + \frac{1}{2} \right)$$

$$f_Z(z | x, y) = \frac{f_{X,Y,Z}(x, y, z)}{f_{X,Y}(x, y)} = \frac{x + y + z}{x + y + \frac{1}{2}}$$

$$\begin{aligned} \text{(c)} \quad f_X(x) &= \int_{-\infty}^{\infty} f_{X,Y}(x, y) dy \\ &= \frac{2}{3} \int_0^1 \left(x + y + \frac{1}{2} \right) dy = \frac{2}{3} \left(x + \frac{1}{2} + \frac{1}{2} \right) = \frac{2}{3} (x + 1) \\ f_Y(y) &= \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx = \frac{2}{3} \int_0^1 \left(x + y + \frac{1}{2} \right) dx = \frac{2}{3} (y + 1) \\ f_Z(z) &= \int_{-\infty}^{\infty} f_{Y,Z}(y, z) dy \\ &= \frac{2}{3} \int_0^1 \left(y + z + \frac{1}{2} \right) dy = \frac{2}{3} (z + 1) \end{aligned}$$

4.41 Show that

$$f_{X,Y,Z}(x, y, z) = f_Z(z | x, y) f_Y(y | x) f_X(x).$$

Solution We know

$$f_Z(z | x, y) = \frac{f_{X,Y,Z}(x, y, z)}{f_{X,Y}(x, y)}$$

$$\Rightarrow f_{X,Y,Z}(x, y, z) = f_Z(z | x, y) f_{X,Y}(x, y)$$

$$\text{Also, } f_Y(y | x) = \frac{f_{X,Y}(x, y)}{f_X(x)}$$

$$\Rightarrow f_{X,Y}(x, y) = f_Y(y | x) f_X(x)$$

$$\Rightarrow f_{X,Y,Z}(x, y, z) = f_Z(z | x, y) f_Y(y | x) f_X(x)$$

4.42 The joint density function of the random variables X and Y is given by

$$f_{XY}(x, y) = 8xy \quad 0 < x < 1; 0 < y < x$$

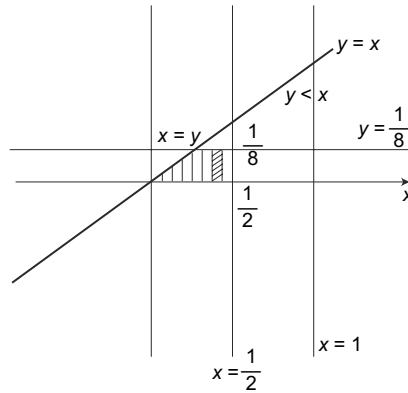
Find $P\left(Y < \frac{1}{8} \mid X < \frac{1}{2}\right)$. Also find the conditional density function $f_Y(Y/X)$.

Solution

Given: $f_{X,Y}(x, y) = 8xy$ $0 < x < 1$; $0 < y < x$
 $= 0$ elsewhere

$$P\left(Y < \frac{1}{8} \mid X < \frac{1}{2}\right) = \frac{P\left(X < \frac{1}{2}, Y < \frac{1}{8}\right)}{P\left(X < \frac{1}{2}\right)}$$

$$P\left(X < \frac{1}{2}, Y < \frac{1}{8}\right) = ?$$

**Fig. 4.21**

From the figure we can find the interval of integration for x and y are $\left(y, \frac{1}{2}\right)$ and $\left(0, \frac{1}{8}\right)$ respectively.

$$\begin{aligned} P\left(X < \frac{1}{2}, Y < \frac{1}{8}\right) &= \int_0^{1/8} \int_y^{1/2} 8xy \, dx \, dy \\ &= 8 \int_0^{1/8} \left(\frac{x^2}{2} \Big|_y^{1/2} \right) y \, dy \\ &= 8 \int_0^{1/8} \frac{1}{2} \left(\frac{1}{4} - y^2 \right) y \, dy = 4 \int_0^{1/8} \left(\frac{y}{4} - y^3 \right) dy \\ &= 4 \left\{ \frac{1}{4} \frac{y^2}{2} \Big|_0^{1/8} - \frac{y^4}{4} \Big|_0^{1/8} \right\} = 4 \left(\frac{1}{5/2} - \frac{1}{4(4096)} \right) \\ &= \frac{1}{128} - \frac{1}{4096} = \frac{31}{4096} \end{aligned}$$

$$= \frac{31}{4096}$$

The marginal density function of X is given by

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dy$$

$$= \int_0^x 8xy dy = 8x \left. \frac{y^2}{2} \right|_0^x$$

$$= 4x^3, 0 < x < 1$$

$$P\left(X < \frac{1}{2}\right) = \int_0^{1/2} 4x^3 dx = 4 \left. \frac{x^4}{4} \right|_0^{1/2} = \frac{1}{16}$$

$$P\left(Y < \frac{1}{8} \mid X < \frac{1}{2}\right) = \frac{P\left(X < \frac{1}{2}, Y < \frac{1}{8}\right)}{P\left(X < \frac{1}{2}\right)}$$

$$= \left(\frac{31}{4096}\right) \left/\left(\frac{1}{16}\right)\right.$$

$$= \frac{31}{256}$$

$$F(Y/X) = \frac{F_{X,Y}(x, y)}{f_X(x)}$$

$$= \frac{8xy}{4x^3}$$

$$= 2y/x^2$$



Practice Problems

4.18 The joint density function of random variables X and Y is

$$f_{X,Y}(x, y) = \frac{2}{(1+x+y)^3}, x > 0, y > 0$$

Find (a) $F_{X,Y}(x, y)$, (b) $f_X(x)$ and (c) $f_Y(y|X=x)$.

$$\left(\text{Ans: (a) } 1 - \frac{1}{1+x} + \frac{1}{1+x+y} - \frac{1}{1+y}, \text{ (b) } \frac{1}{(1+x)^2}, \text{ (c) } \frac{2(1+x)^2}{(1+x+y)^3} \right)$$

Solved Problem

4.43 The joint pdf of a two-dimensional random variable (X, Y) is given by

$$f_{X,Y}(x, y) = \begin{cases} x^2 + \frac{xy}{3}, & 0 < x < 1, \quad 0 < y < 2 \\ 0 & \text{elsewhere} \end{cases}$$

Find (a) $P\left(X > \frac{1}{2}\right)$ (b) $P(Y < X)$ (c) $P\left(Y < \frac{1}{2} \mid X < \frac{1}{2}\right)$

Check whether conditional density functions are valid.

Solution

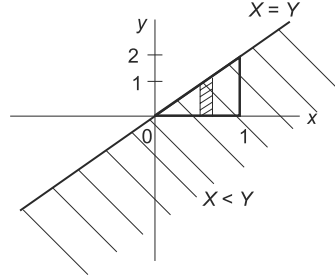
(a) $f_{X,Y}(x, y) = x^2 + \frac{xy}{3}$

$$\begin{aligned} f_X(x) &= \int_{-\infty}^{\infty} f_{X,Y}(x, y) dy = \int_0^2 \left(x^2 + \frac{xy}{3}\right) dy \\ &= x^2(y) \Big|_0^2 + \frac{x}{3} \frac{y^2}{2} \Big|_0^2 \\ &= 2x^2 + \frac{2x}{3} \quad 0 < x < 1 \end{aligned}$$

The marginal density function of Y is

$$\begin{aligned} f_Y(y) &= \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx \\ &= \int_0^1 \left(x^2 + \frac{xy}{3}\right) dx \\ &= \frac{1}{3} + \frac{y}{3} \left(\frac{1}{2}\right) = \frac{1}{3} + \frac{y}{6}; \quad 0 < y < 1 \end{aligned}$$

$$\begin{aligned} P\left(X > \frac{1}{2}\right) &= \int_{1/2}^{\infty} f_X(x) dx = \int_{1/2}^1 \left(2x^2 + \frac{2x}{3}\right) dx \\ &= \frac{2}{3} x^3 \Big|_{1/2}^1 + \frac{1}{3} x^2 \Big|_{1/2}^1 \\ &= \frac{2}{3} \left(\frac{7}{8}\right) + \frac{1}{3} \left(\frac{3}{4}\right) = \frac{14+6}{24} = \frac{5}{6} \end{aligned}$$

(b) $P(Y < X) = ?$ **Fig. 4.22**

From the figure, we can find that the interval of integration for y is from 0 to x ; whereas for x , it is from 0 to 1.

$$\begin{aligned}
 P(Y < X) &= \int_0^1 \int_0^x \left(x^2 + \frac{xy}{3} \right) dy dx \\
 &= \int_0^1 \left(x^3 + \frac{x^3}{6} \right) dx = \frac{7}{6} \frac{x^4}{4} \Big|_0^1 = \frac{7}{24}
 \end{aligned}$$

(c) $P\left[Y < \frac{1}{2} \mid X < \frac{1}{2}\right]$

$$= \frac{P\left(X < \frac{1}{2}, Y < \frac{1}{2}\right)}{P\left(X < \frac{1}{2}\right)}$$

$$\begin{aligned}
 P\left(X < \frac{1}{2}, Y < \frac{1}{2}\right) &= \int_0^{1/2} \int_0^{1/2} \left(x^2 + \frac{xy}{3} \right) dx dy \\
 &= \int_0^{1/2} \left(\frac{1}{24} + \frac{y}{24} \right) dy \\
 &= \frac{1}{48} + \frac{1}{24} \frac{y^2}{2} \Big|_0^{1/2} = \frac{4+1}{192} = \frac{5}{192}
 \end{aligned}$$

$$P\left[X < \frac{1}{2}\right] = \int_0^{1/2} \left(2x^2 + \frac{2x}{3} \right) dx = \frac{2}{3} \left(\frac{1}{8} \right) + \frac{2}{3} \left(\frac{1}{6} \right) = \frac{1}{6}$$

$$\begin{aligned}
 p\left[Y < \frac{1}{2} \mid X < \frac{1}{2}\right] &= \frac{P\left(X < \frac{1}{2}, Y < \frac{1}{2}\right)}{P\left(X < \frac{1}{2}\right)} \\
 &= \frac{5/192}{1/6} = \frac{5}{32}
 \end{aligned}$$

The conditional density function

$$\begin{aligned}
 f_Y(y/x) &= \frac{f_{X,Y}(x, y)}{f_X(x)} \\
 &= \frac{x^2 + \frac{xy}{3}}{2x^3 + \frac{2x}{3}} = \frac{3x + y}{6x + 2}
 \end{aligned}$$

The conditional density function must satisfy

$$\begin{aligned}
 \int_{-\infty}^{\infty} f_Y(y/x) dy &= 1 \\
 \Rightarrow \int_0^2 f_Y(y/x) dy &= \int_0^2 \left(\frac{3x + y}{6x + 2} \right) dy = \frac{1}{6x + 2} \int_0^2 (3x + y) dy \\
 &= \frac{1}{6x + 2} \left\{ 3x y \Big|_0^2 + \frac{y^2}{2} \Big|_0^2 \right\} = \frac{6}{6x + 2} \left[6x + \frac{2^2}{2} \right] = 1 \\
 f_X(x/y) &= \frac{f_{X,Y}(x, y)}{f_Y(y)} \\
 &= \frac{x^2 + \frac{xy}{3}}{\frac{1}{3} + \frac{y}{6}} = \frac{3x^2 + xy}{3} \left(\frac{6}{6 + 3y} \right) = \frac{2(3x^2 + xy)}{y + 2} \\
 &= \frac{6x^2 + 2xy}{y + 2} \\
 \int_{-\infty}^{\infty} f_X(x/y) dx &= \int_0^1 \left(\frac{6x^2 + 2xy}{y + 2} \right) dx \\
 &= \frac{1}{y + 2} \left[6 \frac{x^3}{3} \Big|_0^1 + 2y \frac{x^2}{2} \Big|_0^1 \right] \\
 &= \frac{1}{y + 2} [2 + y] = 1
 \end{aligned}$$



Practice Problems

4.19 The joint density function of random variables X and Y is

$$f_{X,Y}(x, y) = 4xy; x > 0, y < 1$$

Find (a) $f_X(x)$ (b) $f_Y(y)$ (c) $f_X(x|Y=y)$ (d) $f_Y(y|X=x)$.

(Ans. (a) $2x$ (b) $2y$ (c) $2x$ (d) $2y$)

4.20 The joint density function of X and Y is given by

$$f_{X,Y}(x, y) = y e^{-y(x+1)}; x > 0, y > 0$$

Find $f_X(x)$, $f_Y(y)$, $f_X(x|Y=y)$

(Ans. $\frac{1}{(1+x)^2}; e^{-y}; 1 - e^{-xy}$)

4.21 The joint density function of X and Y is

$$f_{X,Y}(x, y) = C(x^2 - y^2)e^{-x}; 0 \leq x < \infty, -x \leq y \leq x$$

Find $f_Y(y | X=x)$.

(Ans. $0.5 + 0.75 \frac{y}{x} - 0.25 \frac{y^3}{x^3}$)



STATISTICAL INDEPENDENCE 4.9

In Section (1.13) we studied that two events A and B are statistically independent if and only if

$$P(A \cap B) = P(A) P(B) \quad (4.70)$$

Let us define two events $A = \{X \leq x\}$ and $B = \{Y \leq y\}$ where X and Y are two random variables. The two random variables X and Y are statistically independent if $P(X \leq x, Y \leq y) = P(X \leq x) P(Y \leq y)$

In terms of distribution function, we can express Eq. (4.70) as

$$F_{X,Y}(x, y) = F_X(x) F_Y(y) \quad (4.71)$$

By differentiation, if X and Y are independent, we can write

$$f_{X,Y}(x, y) = f_X(x) f_Y(y) \quad (4.72)$$

This is the condition for two random variables X and Y to be statistically independent.

The conditional distribution function for an independent event is

$$\begin{aligned} F_X(x | Y \leq y) &= \frac{P(X \leq x, Y \leq y)}{P(Y \leq y)} \\ &= \frac{F_{X,Y}(x, y)}{F_Y(y)} \end{aligned} \quad (4.73)$$

By using the condition for independence, we get

$$F_X(x | Y \leq y) = \frac{F_X(x) F_Y(y)}{F_Y(y)} = F_X(x) \quad (4.74)$$

Similarly, we can also show that

$$F_Y(y | X \leq x) = F_Y(y) \quad (4.75)$$

Similarly, for independent random variables,

$$f_X(x | Y \leq y) = f_X(x) \quad (4.76)$$

$$f_Y(y | X \leq x) = f_Y(y) \quad (4.77)$$

Solved Problems



4.44 The joint density function of two random variables X and Y is given by

$$F_{X,Y}(x, y) = a^2 e^{-a(x+y)} u(x) u(y).$$

(a) Find the conditional density functions $f_X(x | Y = y)$ and $f_Y(y | X = x)$. (b) Are the random variables X and Y statistically independent.

Solution

(a) Given $f_{X,Y}(x, y) = a^2 e^{-a(x+y)} u(x) u(y)$

The conditional density function

$$f_X(x | Y = y) = \frac{f_{X,Y}(x, y)}{f_Y(y)}$$

From Solved Problem (4.31), we have

$$f_X(x) = a e^{-ax} u(x) \text{ and}$$

$$f_Y(y) = a e^{-ay} u(y)$$

$$f_X(x | Y = y) = \frac{a^2 e^{-a(x+y)} u(x) u(y)}{a e^{-ay} u(y)} = a e^{-ax} u(x)$$

$$f_Y(y | X = x) = \frac{f_{X,Y}(x, y)}{f_X(x)} = \frac{a^2 e^{-a(x+y)} u(x) u(y)}{a e^{-ax} u(x)} = a e^{-ay} u(y)$$

(b) Since $f_X(x | Y = y) = f_X(x)$

and

$$f_Y(y | X = x) = a e^{-ay} u(y)$$

X and Y are statistically independent

4.45 The joint pdf of the random variable (X, Y) is given by $f_{X,Y}(x, y) = K xy e^{-(x^2+y^2)}$, $x > 0, y > 0$. Find the value of K and prove also that X and Y are independent.

Solution We know for a valid pdf.

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx dy = 1$$

Since $x > 0$ and $y > 0$, we can write

$$\int_0^{\infty} \int_0^{\infty} K xy e^{-(x^2+y^2)} dx dy = 1$$

$$K \left[\int_0^{\infty} x e^{-x^2} dx \int_0^{\infty} y e^{-y^2} dy \right] = 1$$

Consider $\int_0^{\infty} x e^{-x^2} dx$

Let $x^2 = t \Rightarrow 2x dx = dt$

$$\Rightarrow \int_0^{\infty} x e^{-x^2} dx = \frac{1}{2} \int_0^{\infty} e^{-t} dt = \frac{1}{2} (-e^{-t}) \Big|_0^{\infty} = \frac{1}{2}$$

Similarly, $\int_0^{\infty} y e^{-y^2} dy = \frac{1}{2}$

$$\Rightarrow K \left(\frac{1}{2} \right) \left(\frac{1}{2} \right) = 1$$

Therefore, $K = 4$

The marginal pdf of X is given by

$$\begin{aligned} f_X(x) &= \int_0^{\infty} f_{X,Y}(x, y) dy \\ &= \int_0^{\infty} 4xy e^{-(x^2+y^2)} dy \\ &= 4x e^{-x^2} \int_0^{\infty} y e^{-y^2} dy \\ &= 4x e^{-x^2} \left(\frac{1}{2} \right) = 2x e^{-x^2} \quad x > 0 \end{aligned}$$

The marginal density function of Y is given by

$$\begin{aligned} f_Y(y) &= \int_0^{\infty} f_{X,Y}(x, y) dx \\ &= 4y e^{-y^2} \int_0^{\infty} x e^{-x^2} dx \\ &= 4y e^{-y^2} \left(\frac{1}{2} \right) = 2y e^{-y^2} \quad y > 0 \end{aligned}$$

The joint pdf satisfies

$$f_{X,Y}(x, y) = f_X(x) f_Y(y)$$

Therefore, X and Y are independent.

4.46 The joint pdf of the random variables X and Y is

$$f_{X,Y}(x, y) = \begin{cases} K e^{-(x+y)} & \text{for } 0 \leq x \leq \infty, 0 \leq y \leq \infty \\ 0 & \text{otherwise} \end{cases}$$

(a) Find K . (b) Find marginal densities of X and Y . (c) Find $P(0 \leq x \leq 2, 2 \leq y \leq 3)$. (d) Check X and Y for independence.

Solution

(a) For joint pdf, we know

$$\begin{aligned} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx dy &= 1 \\ \Rightarrow \int_0^{\infty} \int_0^{\infty} K e^{-(x+y)} dx dy &= K \int_0^{\infty} e^{-y} \left(\frac{e^{-x}}{(-1)} \Big|_0^{\infty} \right) dy = K \int_0^{\infty} e^{-y} dy = -K e^{-y} \Big|_0^{\infty} K \\ \Rightarrow \boxed{k=1} \end{aligned}$$

(b) The marginal density function $f_X(x)$ is given by

$$\begin{aligned} f_X(x) &= \int_{-\infty}^{\infty} f_{X,Y}(x, y) dy \\ &= \int_0^{\infty} e^{-(x+y)} dy = e^{-x} \frac{e^{-y}}{-1} \Big|_0^{\infty} = e^{-x} \\ f_Y(y) &= \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx \\ &= \int_0^{\infty} e^{-(x+y)} dx = e^{-y} \frac{e^{-x}}{(-1)} \Big|_0^{\infty} = e^{-y} \\ \Rightarrow f_X(x) &= \begin{cases} e^{-x} & \text{for } x \geq 0 \\ 0 & \text{elsewhere} \end{cases} \\ f_Y(y) &= \begin{cases} e^{-y} & \text{for } y \geq 0 \\ 0 & \text{elsewhere} \end{cases} \\ P(0 \leq x \leq 2, 2 \leq y \leq 3) &= \int_2^3 \int_0^2 e^{-(x+y)} dx dy = (1 - e^{-2}) \int_2^3 e^{-y} dy = (1 - e^{-2}) \frac{e^{-y}}{-1} \Big|_2^3 = (1 - e^{-2}) (e^{-2} - e^{-3}) \end{aligned}$$

(d) From the given $f_{X,Y}(x, y)$ and marginal pdfs, we can easily observe that

$$f_{X,Y}(x, y) = f_X(x) f_Y(y)$$

Hence, X and Y are independent.



Practice Problem

4.22 Find a constant b so that the function

$$f_{X,Y}(x,y) = \begin{cases} b e^{-(x+y)} & 0 < x < a \text{ and } 0 < y < \infty \\ 0 & \text{elsewhere} \end{cases}$$

is a valid joint density function. Check whether X and Y are independent.

(Ans. $b = (1 - e^{-a})^{-1}$)

Solved Problems



4.47 Radha and Mohan decide to meet at a park between 5.00 p.m. and 6 p.m. They arrive independently and their arrival time is uniformly distributed. Find the probability that the first to arrive has to wait longer than 15 minutes.

Solution Let R and M denote the time past 5 that Radha and Mohan arrive at the park. Since Radha and Mohan meet at a park between 5.00 p.m. and 6.00 p.m. and arrival time is uniformly distributed, we can write the pdf of arrival times of both Radha and Mohan as

$$\begin{aligned} f_R(r) &= \frac{1}{60} \text{ for } 0 \leq r \leq 60 \\ &= 0 \text{ otherwise} \\ f_M(m) &= \frac{1}{60} \text{ for } 0 \leq m \leq 60 \\ &= 0 \text{ otherwise} \end{aligned}$$

Here, we consider two cases (a) Mohan arrives at the park before Radha (b) Radha arrives at the park before Mohan.

In the first case, the probability can be expressed as $P(M + 15 < R)$, and for second case, the probability can be expressed as $P(R + 15 < M)$.

Therefore, the desired probability is $P(M + 15 < R) + P(R + 15 < M)$.

$$\begin{aligned} P(M + 15 < R) &= \iint_{m+15 < r} f_{M,R}(m,r) dr dm \\ &= \int_{15}^{60} \int_0^{r-15} f_M(m) f_R(r) dm dr \\ &= \left(\frac{1}{60}\right)^2 \int_{15}^{60} \int_0^{r-15} dr dm = \left(\frac{1}{60}\right)^2 \int_{15}^{60} (r-15) dr \\ &= \left(\frac{1}{60}\right)^2 \left[\frac{r^2}{2} \Big|_{15}^{60} - 15r \Big|_{15}^{60} \right] \\ &= \left(\frac{1}{60}\right)^2 \left[\frac{(60)^2 - (15)^2}{2} - 15(60 - 15) \right] = 0.28125 \end{aligned}$$

Similarly,

$$\begin{aligned}
 P(R + 15 < M) &= \iint_{r+15 < m} f_{M,R}(m, r) dr dm \\
 &= \int_{15}^{60} \int_0^{m-15} f_M(m) f_R(r) dr dm \\
 &= \left(\frac{1}{60}\right)^2 \int_{15}^{60} (m-15) dm \\
 &= \left(\frac{1}{60}\right)^2 \left[\frac{m^2}{2} \Big|_{15}^{60} - 15m \Big|_{15}^{60} \right] = 0.28125
 \end{aligned}$$

$\Rightarrow$ The probability

$$\begin{aligned}
 &= 0.28125 + 0.28125 \\
 &= 0.5625
 \end{aligned}$$

4.48 The joint pdf of random variables (X, Y) is

$$f_{X,Y}(x, y) = \frac{1}{4} e^{-|x| - |y|} \quad \begin{matrix} -\infty < x \leq \infty \\ -\infty < y \leq \infty \end{matrix}$$

- Are X and Y statistically independent random variables?
- Calculate the probability for $X \leq 1$ and $Y \leq 0$.

Solution

Given: $f_{X,Y}(x, y) = \frac{1}{4} e^{-|x| - |y|} \quad \begin{matrix} -\infty < x \leq \infty \\ -\infty < y \leq \infty \end{matrix}$

- The marginal density function of X is

$$\begin{aligned}
 f_X(x) &= \int_{-\infty}^{\infty} f_{X,Y}(x, y) dy \\
 &= \int_{-\infty}^{\infty} \frac{1}{4} e^{-|x| - |y|} dy \\
 &= \frac{1}{4} e^{-|x|} \left[\int_{-\infty}^{\infty} e^{-|y|} dy \right] = \frac{1}{4} e^{-|x|} \left[\int_{-\infty}^0 e^y dy + \int_0^{\infty} e^{-y} dy \right] \\
 &= \frac{1}{4} e^{-|x|} \left[e^y \Big|_{-\infty}^0 - e^{-y} \Big|_0^{\infty} \right] \\
 &= \frac{1}{2} e^{-|x|}
 \end{aligned}$$

$$\begin{aligned}
 f_Y(y) &= \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx \\
 &= \int_{-\infty}^{\infty} \frac{1}{4} e^{-|x| - |y|} dx = \frac{1}{4} e^{-|y|} \left\{ \int_{-\infty}^{\infty} e^x dx + \int_0^{\infty} e^{-x} dx \right\} \\
 &= \frac{1}{2} e^{-|y|}
 \end{aligned}$$

From the marginal and joint density functions, we can observe

$$f_{X,Y}(x, y) = f_X(x) f_Y(y)$$

Hence, X and Y are independent.

$$\begin{aligned}
 \text{(b) } P(X \leq 1, Y \leq 0) &= \int_{-\infty}^0 \int_{-\infty}^1 f_{X,Y}(x, y) dx dy \\
 &= \int_{-\infty}^0 \int_{-\infty}^1 \frac{1}{4} e^{-|x| - |y|} dx dy \\
 &= \frac{1}{4} \int_{-\infty}^0 e^{-|y|} \left[\int_{-\infty}^0 e^x dx + \int_0^1 e^{-x} dx \right] dy \\
 &= \frac{1}{4} \int_{-\infty}^0 e^{-|y|} [2 - e^{-1}] dy \\
 &= \frac{1}{4} (2 - e^{-1}) \left[\int_{-\infty}^0 e^y dy \right] = \frac{1}{4} (2 - e^{-1}) \\
 P(X \leq 1, Y \leq 0) &= \frac{1}{4} (2 - e^{-1})
 \end{aligned}$$

4.49 The joint pdf of a bivariate random variable (X, Y) is given by

$$f_{X,Y}(x, y) = \begin{cases} k e^{-(ax+by)} & x > 0, y > 0 \\ 0 & \text{otherwise} \end{cases}$$

where a, b and k are constants.

(a) Determine the value of k . (b) Are X and Y independent? (c) Find out $P(X > Y)$.

Solution

$$\text{Given: } f_{X,Y}(x, y) = \begin{cases} k e^{-(ax+by)} & x > 0, y > 0 \\ 0 & \text{otherwise} \end{cases}$$

We know,

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx dy = 1$$

$$\begin{aligned}
\int_0^{\infty} \int_0^{\infty} k e^{-(ax+by)} dx dy &= k \int_0^{\infty} \int_0^{\infty} e^{-ax} e^{-by} dx dy \\
&= k \int_0^{\infty} e^{-ax} \left(\frac{e^{-by}}{-b} \Big|_0^{\infty} \right) dx \\
&= \frac{k}{b} \int_0^{\infty} e^{-ax} dx = \frac{k}{b} \frac{e^{-ax}}{(-a)} \Big|_0^{\infty} = \frac{k}{ab}
\end{aligned}$$

We know $\frac{k}{ab} = 1 \Rightarrow k = ab$

The marginal density function of X is given by

$$\begin{aligned}
f_X(x) &= \int_{-\infty}^{\infty} f_{X,Y}(x, y) dy \\
&= \int_0^{\infty} ab e^{-(ax+by)} dy \\
&= ab e^{-ax} \int_0^{\infty} e^{-by} dy = ab e^{-ax} \frac{e^{-by}}{-b} \Big|_0^{\infty} \\
&= ab e^{-ax} \left(\frac{1}{b} \right) = a e^{-ax}
\end{aligned}$$

$$\begin{aligned}
\Rightarrow f_X(x) &= a e^{-ax} \quad x > 0 \\
&= 0 \text{ otherwise}
\end{aligned}$$

$$\begin{aligned}
f_Y(y) &= \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx \\
&= \int_{-\infty}^{\infty} ab e^{-ax} e^{-by} dx = ab e^{-by} \frac{e^{-ax}}{-a} \Big|_0^{\infty} \\
&= b e^{-by}
\end{aligned}$$

$$\begin{aligned}
\Rightarrow f_Y(y) &= b e^{-by} \text{ for } y > 0 \\
&= 0 \text{ otherwise}
\end{aligned}$$

Given: $f_{X,Y}(x, y) = ab e^{-ax} e^{-by}$

Since $f_{X,Y}(x, y) = f_X(x) f_Y(y)$,

X and Y are independent.

$$\begin{aligned}
 (c) \quad P(X > Y) &= \int_0^{\infty} \int_0^x ab e^{-(ax+by)} dy dx \\
 &= ab \int_0^{\infty} e^{-ax} \left(\frac{e^{-by}}{-b} \bigg|_0^x \right) dx \\
 &= \frac{ab}{b} \int_0^{\infty} e^{-ax} (1 - e^{-bx}) dx \\
 &= a \left[\frac{e^{-ax}}{-a} \bigg|_0^{\infty} - \frac{e^{-(a+b)x}}{-(a+b)} \bigg|_0^{\infty} \right] \\
 &= 1 - \frac{1}{a+b} = \frac{b}{a+b} \\
 P(X > Y) &= \frac{b}{a+b}
 \end{aligned}$$

4.50 (X, Y) is a bivariate random variable in which X and Y are independent. If X is a uniform random variable over $(0, 0.5)$ and Y is an exponential random variable with parameter $\lambda = 4$, find the joint pdf of (X, Y) .

Solution X is a random variable with uniform distribution with $b = 0.5$ and $a = 0$. Hence,

$$f_X(x) = \frac{1}{0.5 - 0} = 2 \quad \text{for } 0 \leq x \leq 0.5$$

Y is an exponential random variable with $\lambda = 4$

$$\begin{aligned}
 f_Y(y) &= (4)^2 e^{-4y} \quad y > 0 \\
 &= 16 e^{-4y} \quad y > 0
 \end{aligned}$$

Since X and Y are independent,

$$\begin{aligned}
 f_{X,Y}(x, y) &= f_X(x) f_Y(y) \\
 &= 2(16 e^{-4y})
 \end{aligned}$$

$$\Rightarrow f_{X,Y}(x, y) \begin{cases} = 32 e^{-4y} & 0 < x < 0.5, y > 0 \\ = 0 & \text{otherwise} \end{cases}$$

DISTRIBUTION AND DENSITY OF SUM OF RANDOM VARIABLES 4.10



Consider two continuous random variables X and Y with pdfs $f_X(x)$ and $f_Y(y)$ respectively. Let Z be a random variable defined as $Z = X + Y$. Now we are interested in finding the pdf and CDF of Z .

The CDF of Z is defined as

$$\begin{aligned} F_Z(z) &= P(Z \leq z) \\ &= P(X + Y \leq z) = \iint_S f_{X,Y}(x, y) dx dy \end{aligned} \quad (4.78)$$

Since X and Y are independent,

$$F_Z(z) = \iint_S f_X(x) f_Y(y) dx dy \quad (4.79)$$

where S is the area to the left of the line $X + Y = Z$

$$\begin{aligned} F_Z(z) &= \iint_{X+Y \leq Z} f_X(x) f_Y(y) dx dy \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{z-y} f_X(x) f_Y(y) dx dy \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{z-y} f_X(x) dx f_Y(y) dy \\ &= \int_{-\infty}^{\infty} F_X(z-y) f_Y(y) dy \end{aligned} \quad (4.80)$$

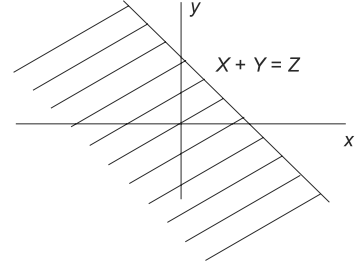


Fig. 4.23

The pdf of z is obtained by differentiating the CDF. The cumulative distribution is called the convolution of the distributions $F_X(x)$ and $F_Y(y)$. Differentiating Eq. (4.80), we get

$$\begin{aligned} f_Z(z) &= f_{X+Y}(z) \\ &= \frac{d}{dz} \left[\int_{-\infty}^{\infty} F_X(z-y) f_Y(y) dy \right] \\ &= \int_{-\infty}^{\infty} \frac{d}{dz} F_X(z-y) f_Y(y) dy \\ &= \int_{-\infty}^{\infty} f_X(z-y) f_Y(y) dy \end{aligned} \quad (4.81)$$

$$= \int_{-\infty}^{\infty} f_X(z-y) f_Y(y) dy \quad (4.82)$$

The above expression is known as *convolution integral*. That is, the density function of the sum of two statistically independent random variables is the convolution of their individual density functions.

4.10.1 Sum of Several Random Variables

Consider the sum of independent random variables $X_1, X_2, X_3, \dots, X_N$ which is expressed as

$$Y = X_1 + X_2 + \dots + X_N$$

Let $Y_1 = X_1 + X_2$

Then the CDF of Y_1 is the convolution of the CDFs of X_1 and X_2 . That is,

$$f_{Y_1}(y_1) = f_{X_1}(x_1) * f_{X_2}(x_2)$$

Next, consider the random variable

$$Y_3 = X_3 + X_2 + X_1 = X_3 + Y_2$$

Since X_3 is independent of X_2 and X_1 , we can find

$$\begin{aligned} f_{Y_3}(y_3) &= f_{X_3}(x_3) * f_{Y_2}(y_2) \\ &= f_{X_3}(x_3) * f_{X_2}(x_2) * f_{X_1}(x_1) \end{aligned}$$

For the sum of N random variables,

$$f_Y(y) = f_{X_1}(x_1) * f_{X_2}(x_2) \dots * f_{X_N}(x_N) \quad (4.83)$$

4.10.2 pmf of Sum of Two Discrete Random Variables

Consider two independent discrete random variables X and Y each of which may assume non-negative integer values. Let $p_X(x)$ and $p_Y(y)$ are pmf of X and Y respectively. Then the pmf of $Z = X + Y$ is given by

$$p_Z(z) = \sum_{k=0}^z p_X(k) p_Y(z-k)$$

Proof: We have

$$p_X(x) = P(X=x), x = 0, 1, 2 \dots$$

$$\text{and } p_Y(y) = P(Y=y), y = 0, 1, 2 \dots$$

$$\text{Given: } Z = X + Y$$

$$\begin{aligned} \text{Let } p_Z(z) &= P(Z=z) \\ &= p_Z(X=0, Y=z \text{ or } X=1, Y=z-1, \dots X=z, Y=0) \\ &= \sum_{k=0}^z p_Z(X=k, Y=z-k) \\ &= \sum_{k=0}^z p_X(k) p_Y(z-k) \end{aligned}$$

$$\text{or } p_Z(z) = \sum_{k=0}^z p_Y(k) p_X(z-k) \quad (4.84)$$

REVIEW QUESTIONS



15. What is the probability density function of the sum of two random variables?
16. Explain the method of finding the distribution and density functions for a sum of statistically independent random variables.

Solved Problems



4.51 A continuous random signal X is transmitted over a channel. At the receiver, the received signal Y consists of additive noise N . That is, $Y = X + N$. Find the distribution and density of Y if X and N are jointly continuous random variables.

Solution The distribution of the received signal Y is given by

$$\begin{aligned}
 f_Y(y) &= P(Y \leq y) \\
 &= P(X + N \leq y) \\
 F_Y(y) &= \int_{-\infty}^{\infty} \int_{-\infty}^{y-x} f_{X,N}(x, n) dn dx \\
 f_Y(y) &= \frac{\partial}{\partial y} \int_{-\infty}^{\infty} \int_{-\infty}^{y-x} f_{X,N}(x, n) dn dx \\
 &= \int_{-\infty}^{\infty} \frac{\partial}{\partial y} \int_{-\infty}^{\infty} f_{X,N}(x, n) dn dx \\
 &= \int_{-\infty}^{\infty} f_{X,N}(x, y-x) dx
 \end{aligned}$$

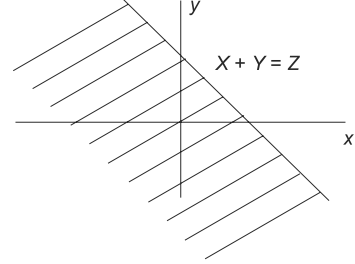


Fig. 4.24

4.52 Random variables X and Y have respective density functions

$$f_X(x) = 0.1 \delta(x-1) + 0.2 \delta(x-2) + 0.4 \delta(x-3) + 0.3 \delta(x-4)$$

$$f_Y(y) = 0.4 \delta(y-5) + 0.5 \delta(y-6) + 0.1 \delta(y-7)$$

Find the density function of $W = X + Y$ if X and Y are independent.

Solution

Given: $f_X(x) = 0.1 \delta(x-1) + 0.2 \delta(x-2) + 0.4 \delta(x-3) + 0.3 \delta(x-4)$

$$f_Y(y) = 0.4 \delta(y-5) + 0.5 \delta(y-6) + 0.1 \delta(y-7)$$

$$\begin{aligned}
 f_Z(z) &= \int_{-\infty}^{\infty} f_X(x) f_Y(z-x) dx \\
 &= \int_{-\infty}^{\infty} \{0.1 \delta(x-1) + 0.2 \delta(x-2) + 0.4 \delta(x-3) + 0.3 \delta(x-4)\} \\
 &\quad \{0.4 \delta(z-x-5) + 0.5 \delta(z-x-6) + 0.1 \delta(z-x-7)\} dx
 \end{aligned}$$

Consider the integral of type

$$\int_{-\infty}^{\infty} k_1 \delta(x-x_m) k_2 \delta(z-x-x_k) dx = k \int_{-\infty}^{\infty} \delta(x-x_m) \delta(z-x-x_k) dx$$

where x_m and x_k are constants. In the above integral, the impulses coincide when $z = x_m + x_k$. Therefore, the integral equals $k \delta(z - x_m - x_k)$ where $k = k_1 k_2$

$$\begin{aligned}
 f_Z(z) &= 0.04 \delta(z-6) + 0.08 \delta(z-7) + 0.16 \delta(z-8) + 0.12 \delta(z-9) \\
 &\quad + 0.05 \delta(z-7) + 0.1 \delta(z-8) + 0.2 \delta(z-9) + 0.15 \delta(z-10) \\
 &\quad + 0.01 \delta(z-8) + 0.02 \delta(z-9) + 0.04 \delta(z-10) + 0.03 \delta(z-11)
 \end{aligned}$$

$$= 0.04 \delta(z - 6) + 0.13 \delta(z - 7) + 0.27 \delta(z - 8) + 0.34 \delta(z - 9) \\ + 0.19 \delta(z - 10) + 0.03 \delta(z - 11)$$

4.53 Find the density of $W = X + Y$ where the densities of X and Y are assumed to be

$$f_X(x) = u(x) - u(x - 1); f_Y(y) = u(y) - u(y - 1)$$

Solution The pdfs of random variables X and Y are shown in Fig. 4.25(a)

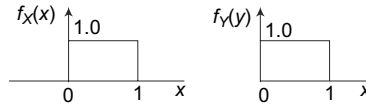


Fig. 4.25(a)

Given: $W = X + Y$

The pdf of the sum of two random variables is the convolution of their individual density functions. That is,

$$f_W(w) = \int_{-\infty}^{\infty} f_Y(y) f_X(w - y) dy$$

The sketch of $f_X(-y)$ and $f_X(w - y)$ are shown in Fig. 4.25(b).

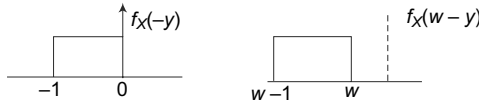


Fig. 4.25(b)

Sketch $f_Y(y)$ and $f_X(w - y)$ on the same axis as below.

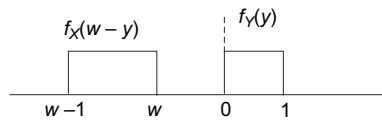


Fig. 4.25(c)

From Fig. 4.25(c) observe we that $f_W(w) = 0$ for $w < 0$, since the functions $f_X(w - y)$ and $f_Y(y)$ do not overlap.

For $0 < w < 1$, the function $f_X(w - y)$ and $f_Y(y)$ are drawn in Fig. 4.25(d).

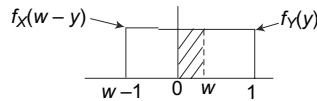


Fig. 4.25(d)

The overlapping interval is from 0 to w . Therefore,

$$f_W(w) = \int_0^w dw = w$$

For $1 < w < 2$, the functions $f_Y(y)$ and $f_X(w-y)$ are sketched as in Fig. 4.25(e)

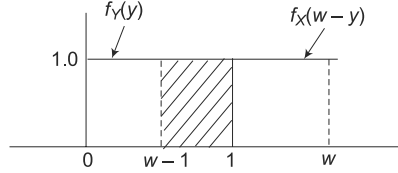


Fig. 4.25 (e)

The overlapping integral is from $w-1$ to 1.

$$f_W(w) = \int_{w-1}^1 dw = 1 - (w-1) = 2 - w$$

For $w > 2$, the functions $f_X(w-y)$ and $f_Y(y)$ do not overlap. Therefore,

$$f_W(w) = 0 \text{ for } w > 2$$

$$\Rightarrow f_W(w) = 0 \text{ for } w < 0$$

$$f_W(w) = w \text{ for } 0 < w < 1$$

$$f_W(w) = 2 - w \text{ for } 1 < w < 2$$

$$= 0 \text{ for } w > 2$$

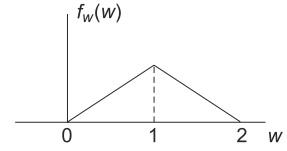


Fig. 4.25(f)

The $f_W(w)$ is shown in Fig. 4.25(f).

4.54 If X and Y are two random variables which are Gaussian, if a random variable Z is defined as $Z = X + Y$, find $f_Z(z)$.

Solution Given: $X \sim N(0, 1)$ and $Y \sim N(0, 1)$

That is, $\mu_X = 0$, $\sigma_X^2 = 1$ and $\mu_Y = 0$, $\sigma_Y^2 = 1$

$$f_X(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}; \quad f_Y(y) = \frac{1}{\sqrt{2\pi}} e^{-y^2/2}$$

$$f_Z(z) = f_X(x) * f_Y(y) = \int_{-\infty}^{\infty} f_Y(y) f_X(z-y) dy$$

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-(z-y)^2/2} e^{-y^2/2} dy$$

$$\begin{aligned}
&= \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-(z^2 - 2yz + y^2 + y^2)/2} dy = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-(z^2 - 2yz + 2y^2)/2} dy \\
&= \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-z^2/4} e^{-(\sqrt{2}y - z/\sqrt{2})^2/(\sqrt{2})^2} e^{z^2/4} dy \\
&= \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-z^2/4} e^{-(\sqrt{2}y - z/\sqrt{2})^2/2} dy \\
&= \frac{1}{\sqrt{2\pi}} \left(\frac{1}{\sqrt{2}} \right) \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-z^2/4} e^{-p^2/2} dp \\
&= \frac{e^{-z^2/4}}{\sqrt{2\pi}} \frac{1}{\sqrt{2\pi} \sqrt{2}} \int_{-\infty}^{\infty} e^{-p^2/2} dp \\
&= \frac{1}{\sqrt{2\pi}} \frac{1}{\sqrt{2}} e^{-z^2/4} \underbrace{\left\{ \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-p^2/2} dp \right\}}_{=1} \\
&= \frac{1}{\sqrt{2\pi}} \frac{1}{\sqrt{2}} e^{-z^2/4} = \frac{1}{\sqrt{2\pi} \sqrt{2}} e^{-z^2/2(\sqrt{2})^2}
\end{aligned}$$

Let

$$\sqrt{2} y - \frac{z}{\sqrt{2}} = p$$

$$\sqrt{2} dy = dp$$

$$dy = \frac{dp}{\sqrt{2}}$$

That is, $\sigma = \sqrt{2}$. So Z is also a Gaussian random variable with zero mean and variance 2.

4.55 If X and Y are independent random variables with density functions $f_X(x) = e^{-x} u(x)$ and $f_Y(y) = e^{-2y} u(y)$, find the density function of $Z = X + Y$.

Solution Given:

$$f_X(x) = e^{-x} u(x); \quad f_Y(y) = e^{-2y} u(y)$$

$$\begin{aligned}
f_Z(z) &= \int_{-\infty}^{\infty} f_X(x) f_Y(z-x) dx \\
&= \int_0^z e^{-x} e^{-2(z-x)} dx \\
&= e^{-2z} \int_0^z e^x dx = e^{-2z} e^x \Big|_0^z = e^{-2z} (e^z - 1) \\
&= e^{-z} - e^{-2z} \\
f_Z(z) &= e^{-z} - e^{-2z} \quad \text{for } z \geq 0 \\
&= 0 \text{ elsewhere}
\end{aligned}$$

4.56 Two independent random variables X and Y are having their densities as $f_X(x) = e^{-x} u(x)$ and $f_Y(y) = e^{-y} u(y)$. Find $P(X + Y \leq 1)$.

Solution Given: $f_X(x) = e^{-x} u(x)$; $f_Y(y) = e^{-y} u(y)$

Let $X + Y = z$.

Then

$$P(X + Y \leq 1) = P(z \leq 1) = \int_0^1 f_Z(z) dz$$

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z-x) dx$$

Since $f_X(x) = 0$ for $x < 0$ and

$$f_Y(y) = 0 \text{ for } y < 0$$

$$f_Z(z) = \int_0^z f_X(x) f_Y(z-x) dx$$

$$f_Z(z) = \int_0^z e^{-x} e^{-(z-x)} dx = e^{-z} \int_0^z dz = ze^{-z}$$

$$\begin{aligned} P(z \leq 1) &= \int_0^1 f_Z(z) dz = \int_0^1 ze^{-z} dz \\ &= -ze^{-z} \Big|_0^1 - e^{-z} \Big|_0^1 \\ &= 1 - 2e^{-1} \end{aligned}$$

4.57 Two independent random variables X and Y have the probability density functions respectively as

$$f_X(x) = xe^{-x}, x > 0 \quad f_Y(y) = 1 \quad 0 \leq y \leq 1$$

Calculate the probability distribution and density functions of the random variable $Z = X + Y$.

Solution

$$f_X(x) = xe^{-x}, x > 0$$

$$f_Y(y) = 1, 0 \leq y \leq 1$$

$$= 0 \text{ otherwise}$$

Given: $Z = X + Y$

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z-x) dx$$

For $z < 0$, $f_Z(z) = 0$

For $0 < z < 1$

$$\begin{aligned} f_Z(z) &= \int_0^z x e^{-x} dx \\ &= -x e^{-x} \Big|_0^z - e^{-x} \Big|_0^z \\ &= 1 - e^{-z}(z + 1) \end{aligned}$$

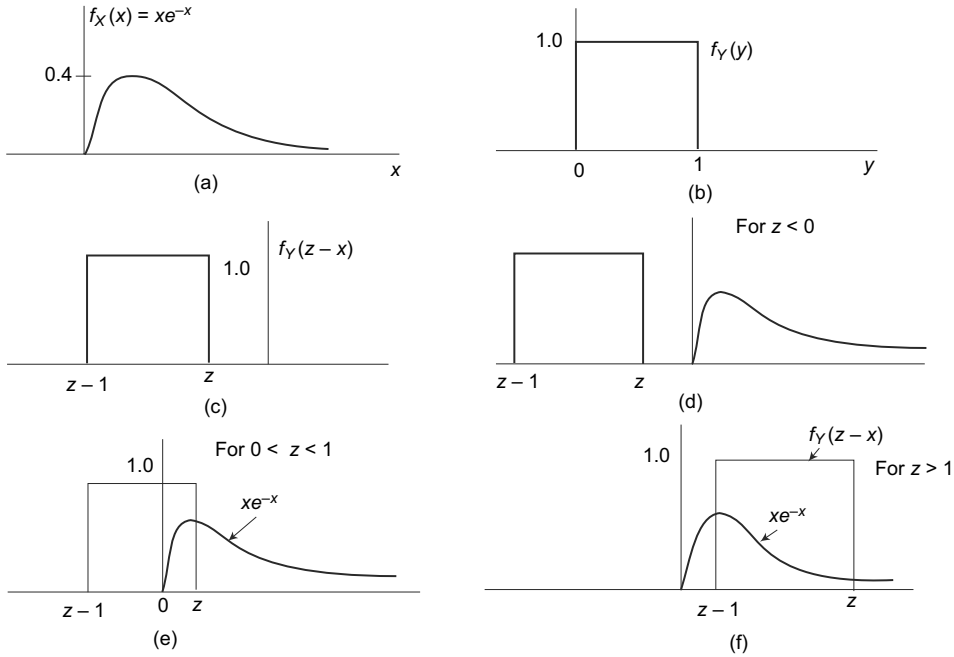


Fig. 4.26

For $z > 1$,

$$\begin{aligned} f_Z(z) &= \int_{z-1}^z x e^{-x} dx = -x e^{-x} \Big|_{z-1}^z - e^{-x} \Big|_{z-1}^z \\ &= -z e^{-z} + (z-1) e^{-(z-1)} - e^{-z} + e^{-(z-1)} \\ &= -z e^{-z} + z e^{-(z-1)} - e^{-(z-1)} - e^{-z} + e^{-(z-1)} \\ &= z e^{-(z-1)} - e^{-z}(z+1) \end{aligned}$$

4.58 X and Y are two independent random variables with uniform density over $(-2, 2)$ and $(-1, 1)$ respectively.

Find $P(Z < -1)$ where $Z = X + Y$.

Solution The pdfs of X and Y random variables are shown in Fig. 4.27(a) and (b).

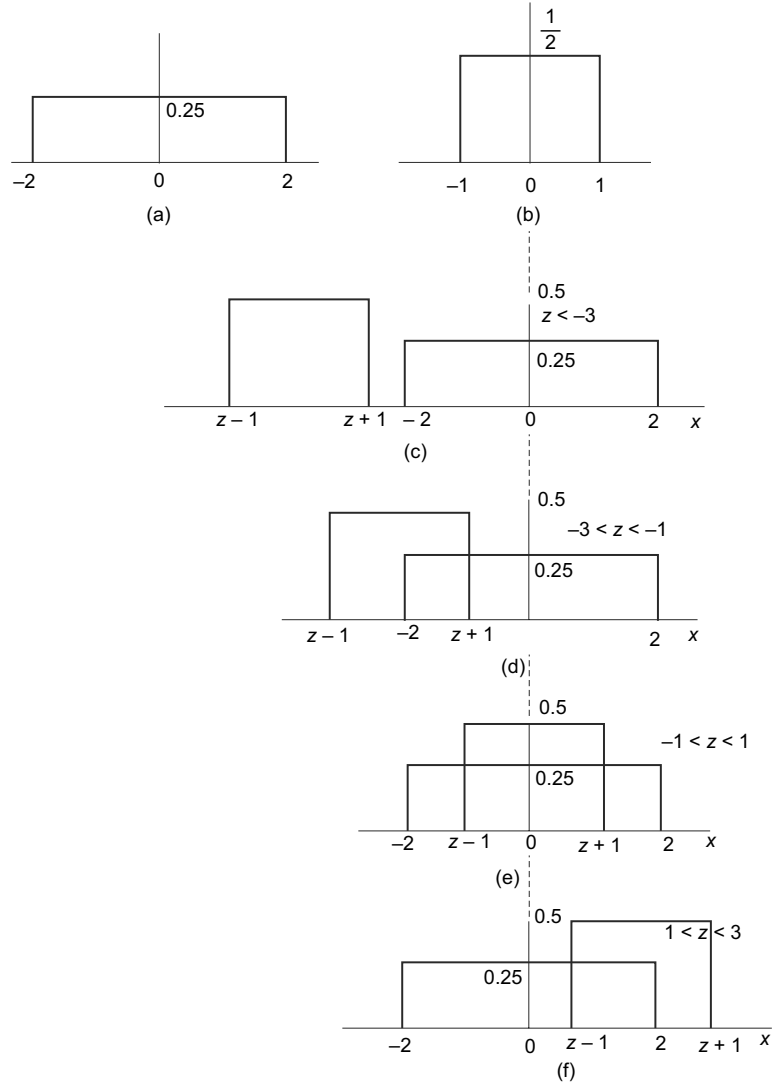


Fig. 4.27

From Fig. 4.27(c) for $z < -3$,

$$f_Z(z) = 0$$

For $-3 < z < -1$,

$$f_Z(z) = \int_{-2}^{z+1} \frac{1}{2} \left(\frac{1}{4} \right) dx = \frac{1}{8} [(z+1) - (-2)] = \frac{z+3}{8}$$

For $-1 < z < 1$

$$f_Z(z) = \int_{z-1}^{z+1} \frac{1}{2} \left(\frac{1}{4} \right) dx = \frac{1}{8} [(z+1) - (z-1)] = \frac{1}{4}$$

For $1 < z < 3$

$$f_Z(z) = \int_{z-1}^2 \frac{1}{2} \left(\frac{1}{4} \right) dx = \frac{1}{8} [2 - (z-1)] = \frac{3-z}{8}$$

$$\begin{aligned} P(Z < -1) &= \int_{-3}^{-1} \frac{z+3}{8} dx \\ &= \frac{1}{8} \left[\frac{z^2}{2} \Big|_{-3}^{-1} + 3z \Big|_{-3}^{-1} \right] \\ &= \frac{1}{8} \left[\frac{(-1)^2 - (-3)^2}{2} + 3(-1) - 3(-3) \right] \\ &= \frac{1}{4} \end{aligned}$$

4.59 The random variables X and Y have density functions $f_X(x) = \frac{1}{a} (u(x) - u(x-a))$ and $f_Y(y) = be^{-by} u(y)$ where $a > 0$. Find the density function of $W = X + Y$ if X and Y are statistically independent.

Solution Given: $f_X(x) = \frac{1}{a} (u(x) - u(x-a))$

$$f_Y(y) = be^{-by} u(y)$$

The pdfs are shown in Fig. 4.29(a) and 4.29 (b).

$$f_Z(z) = \int_{-\infty}^{\infty} f_Y(y) f_X(z-y) dy$$

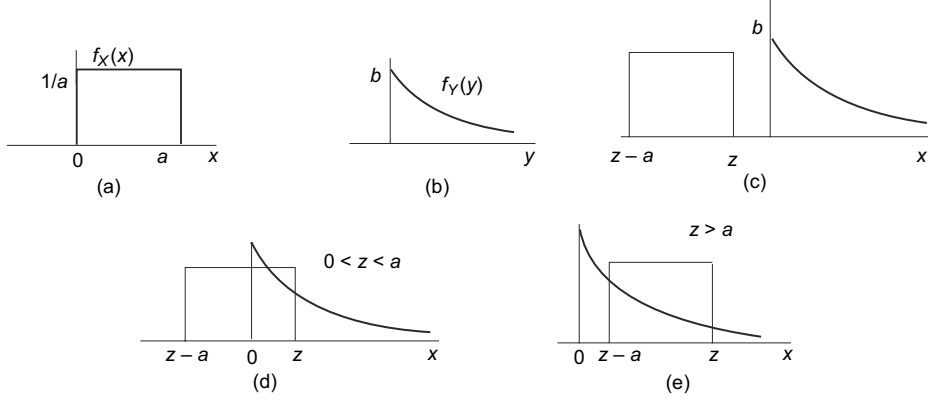


Fig. 4.28

For $z < 0$,

$$f_Z(z) = 0$$

For $0 < z < a$,

$$f_Z(z) = \int_0^z \frac{1}{a} b e^{-by} dy = \frac{-1}{a} (e^{-bz} - 1) = \frac{1 - e^{-bz}}{a}$$

For $z > a$,

$$\begin{aligned} f_Z(z) &= \frac{b}{a} \int_{z-a}^z e^{-by} dy = -\frac{1}{a} e^{-by} \Big|_{z-a}^z = \frac{-1}{a} [e^{-bz} - e^{-b(z-a)}] \\ &= \frac{e^{-bz}}{a} (e^{ba} - 1) \end{aligned}$$

4.60 Two independent random variables X and Y have densities $f_X(x) = 5e^{-5x} u(x)$ and $f_Y(y) = 2e^{-2y} u(y)$. Find the density of the sum $Z = X + Y$.

Solution Given: $f_X(x) = 5e^{-5x} u(x)$; $f_Y(y) = 2e^{-2y} u(y)$

The density function of $Z = X + Y$ is a convolution of the density functions $f_X(x)$ and $f_Y(y)$. Therefore,

$$\begin{aligned} f_Z(z) &= f_X(x) * f_Y(y) \\ &= \int_{-\infty}^{\infty} f_X(x) f_Y(z-x) dx \end{aligned}$$

Since $f_X(x)$ and $f_Y(y)$ are zero for $x < 0$ and $y < 0$ respectively, we can write

$$\begin{aligned} f_Z(z) &= \int_0^z f_X(x) f_Y(z-x) dx \\ &= \int_0^z 5e^{-5x} 2e^{-2(z-x)} dx \end{aligned}$$

$$\begin{aligned}
 &= 10 e^{-2z} \int_0^z e^{-3x} dx = 10 e^{-2z} \left. \frac{e^{-3x}}{-3} \right|_0^z \\
 &= \frac{-10}{3} e^{-2z} [e^{-3z} - 1] = \frac{-10}{3} (e^{-5z} - e^{-2z}) \\
 f_Z(z) &= \frac{10}{3} (e^{-2z} - e^{-5z}) u(z)
 \end{aligned}$$

4.61 The probability density functions of statistically independent random variables X and Y are

$$f_X(x) = \frac{1}{2} e^{-(x-1)/2} u(x-1); f_Y(y) = \frac{1}{4} e^{-(y-3)/4} u(y-3).$$

Find the pdf of the $z = X + Y$.

Solution Given: $f_X(x) = \frac{1}{2} e^{-(x-1)/2} u(x-1); f_Y(y) = \frac{1}{4} e^{-(y-3)/4} u(y-3)$

From the given pdfs, we can find $f_X(x) = 0$ for $x < 1$ and $f_Y(y) = 0$ for $y < 3$. Since $f_X(x)$ and $f_Y(y)$ are pdfs of independent random variables X and Y we can write the pdf of $X + Y$ as a convolution of $f_X(x)$ and $f_Y(y)$.

$$\begin{aligned}
 f_Z(z) &= f_X(x) * f_Y(y) \\
 &= \int_{-\infty}^{\infty} f_X(x) f_Y(z-x) dx
 \end{aligned}$$

The rough sketches of $f_X(x)$ and $f_Y(y)$ are shown in Fig. 4.29(a) and (b).

The sketches of $f_X(x)$ and $f_Y(z-x)$ are shown on the same axis in Fig. 4.29(c).

From Fig. 4.29(d), we can observe that for $z-3 < 1$ or $z < 4$, there is no overlap of $f_X(x)$ and $f_Y(z-x)$ and hence, the convolution of $f_X(x)$ and $f_Y(z-x)$ is zero. That is,

$$f_Z(z) = 0 \text{ for } z < 4$$

For $z > 4$, both functions $f_X(x)$ and $f_Y(z-x)$ are shown in Fig. 4.29(e).

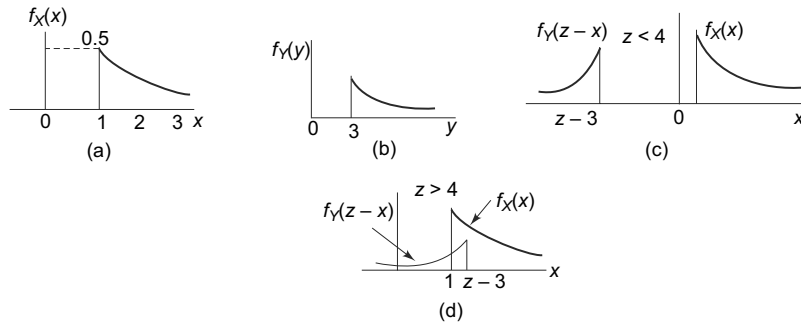


Fig. 4.29

From the figure we observe that the curves overlap in the interval between 1 and $z - 3$. Therefore,

$$\begin{aligned}
 f_Z(z) &= \int_1^{z-3} f_X(x) f_Y(z-x) dx \\
 &= \int_1^{z-3} \frac{1}{2} e^{-(x-1)/2} \frac{1}{4} e^{-(z-x-3)/4} dx \\
 &= \frac{1}{8} \int_1^{z-3} e^{-x/2} e^{1/2} e^{-z/4} e^{x/4} e^{3/4} dx \\
 &= \frac{1}{8} e^{(5-z)/4} \int_1^{z-3} e^{-x/4} dx \\
 &= \frac{1}{8} e^{(5-z)/4} \left. \frac{e^{-x/4}}{(-1/4)} \right|_1^{z-3} \\
 &= \frac{-1}{2} e^{(5-z)/4} [e^{-(z-3)/4} - e^{-1/4}] \\
 &= \frac{1}{2} [e^{-(z-4)/4} - e^{-(z-4)/2}]
 \end{aligned}$$

Since $f_Z(z) = 0$ for $z < 4$, we can write

$$f_Z(z) = \frac{1}{2} [e^{-(z-4)/4} - e^{-(z-4)/2}] u(z-4)$$

4.62 Three statistically independent random variables X_1, X_2 and X_3 all have the same density function

$$f_{X_i}(x_i) = \frac{1}{a} [u(x_i) - u(x_i - a)]; i = 1, 2, 3$$

Find and sketch the density function of $Y = X_1 + X_2 + X_3$ if $a > 0$ is constant.

Solution Given: $Y = X_1 + X_2 + X_3$

Let $X_1 + X_2 = z$,

Then $f_Z(z) = f_{X_1}(x_1) * f_{X_2}(x_2)$

$$f_Z(z) = \int_{-\infty}^{\infty} f_{X_1}(x_1) f_{X_2}(z-x_1) dx_1$$

The pdf of x_i is shown in Fig. 4.30(a).

The functions $f_{X_1}(x_1)$ and $f_{X_2}(z-x)$ are shown in Fig. 4.30(b).

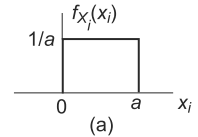


Fig. 4.30 (a)

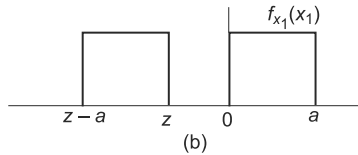


Fig. 4.30 (b)

For $z < 0$, the convolution of $f_{X_1}(x_1)$ and $f_{X_2}(z - x)$ is zero.

For $0 < z < a$,

$$\begin{aligned} f_Z(z) &= \int_0^z f_{X_1}(x_1) f_{X_2}(z - x) dx_1 \\ &= \int_0^z \frac{1}{a^2} dx_1 = \frac{z}{a^2} \end{aligned}$$

For $a < z < 2a$,

$$\begin{aligned} f_Z(z) &= \int_{z-a}^a \frac{1}{a^2} dx_1 \\ &= \frac{1}{a^2} [a - (z - a)] \\ &= \frac{2a - z}{a^2} \end{aligned}$$

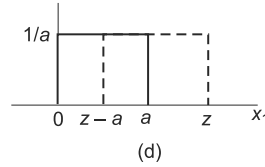
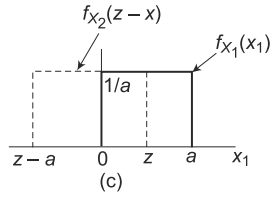


Fig. 4.30 (c), (d)

For $z > 2a$,

the functions $f_{X_1}(x_1)$ and $f_{X_2}(z - x)$ do not overlap hence, $f_Z(z) = 0$. The sketch of $f_Z(z)$ is shown in Fig. 4.30(e).

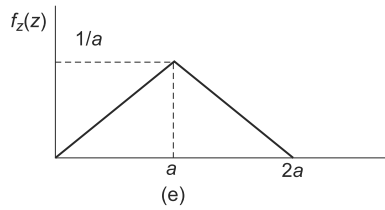


Fig. 4.30 (e)

Now $Y = Z + X_3$

So $f_Y(y) = f_Z(z) * f_{X_3}(x_3)$

The sketch for $f_{X_3}(x_3)$ is shown in Fig. 4.30(f).

$$f_Y(y) = \int_{-\infty}^{\infty} f_Z(z) f_{X_3}(y-z) dz$$

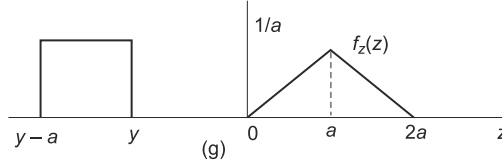


Fig. 4.30(g)

For $y < 0$, the functions $f_Z(z)$ and $f_{X_3}(y-z)$ do not overlap. Therefore, $f_Y(y) = 0$

For $0 < y < a$,

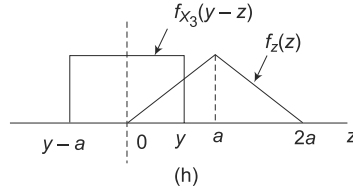


Fig. 4.30(h)

$$f_Y(y) = \int_0^y f_Z(z) f_{X_3}(y-z) dz = \int_0^y \frac{z}{a^2} \left(\frac{1}{a} \right) dz = \frac{1}{a^3} \frac{z^2}{2} \Big|_0^y = \frac{y^2}{2a^3}$$

For $a < y < 2a$,

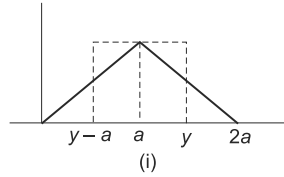


Fig. 4.30(i)

$$\begin{aligned} f_Y(y) &= \int_{y-a}^y f_Z(z) f_{X_3}(y-z) dz \\ &= \int_{y-a}^a f_Z(z) f_{X_3}(y-z) dz + \int_a^y f_Z(z) f_{X_3}(y-z) dz \end{aligned}$$

$$\begin{aligned}
&= \int_{y-a}^a \frac{z}{a^2} \left(\frac{1}{a} \right) dz + \int_a^y \left(\frac{2a-z}{a^2} \right) \frac{1}{a} dz \\
&= \frac{1}{a^3} \left[\frac{z^2}{2} \right]_{y-a}^a + \frac{1}{a^3} \left[2az \Big|_a^y - \frac{z^2}{2} \Big|_a^y \right] \\
&= \frac{1}{2a^3} [a^2 - (y-a)^2] + \frac{1}{a^3} \left[2a(y-a) - \frac{(y^2 - a^2)}{2} \right] \\
&= \frac{1}{2a^3} [a^2 - (y^2 + a^2 - 2ay)] + \frac{1}{a^3} \left[2ay - 2a^2 - \frac{y^2}{2} + \frac{a^2}{2} \right] \\
&= \frac{1}{2a^3} [2ay - y^2] + \frac{1}{a^3} \left[2ay - \frac{3}{2}a^2 - \frac{y^2}{2} \right] \\
&= \frac{(2ay - y^2) + (4ay - 3a^2 - y^2)}{2a^3} \\
&= \frac{6ay - 2y^2 - 3a^2}{2a^3}
\end{aligned}$$

For $2a < y < 3a$,

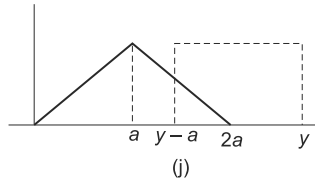


Fig. 4.30(j)

$$\begin{aligned}
f_Y(y) &= \int_{y-a}^{2a} f_Z(z) f_{X_3}(y-z) dz \\
&= \int_{y-a}^{2a} \left(\frac{2a-z}{a^2} \right) \left(\frac{1}{a} \right) dz \\
&= \frac{1}{a^3} \int_{y-a}^{2a} (2a-z) dz \\
&= \frac{1}{a^3} \left[2a(2a-y+a) - \frac{z^2}{2} \Big|_{y-a}^{2a} \right]
\end{aligned}$$

$$\begin{aligned}
&= \frac{1}{a^3} \left\{ 2a(3a - y) - \left[\frac{(2a)^2 - (y - a)^2}{2} \right] \right\} \\
&= \frac{1}{a^3} \left\{ 6a^2 - 2ay - \frac{(4a^2 - y^2 - a^2 + 2ay)}{2} \right\} \\
&= \frac{1}{2a^3} [9a^2 - 6ay + y^2] = \frac{1}{2a^3} (3a - y)^2
\end{aligned}$$

For $y > 3a$,

$$f_Y(y) = 0$$

4.63 If X and Y are random variables with respective parameters (a_1, b) and (a_2, b) then prove that $X + Y$ is gamma random variable with $(a_1 + a_2, b)$.

Solution

Given: $f_X(x) = \frac{b^{a_1} x^{a_1-1} e^{-bx}}{\Gamma(a_1)} u(x)$ and $f_Y(y) = \frac{b^{a_2} y^{a_2-1} e^{-by}}{\Gamma(a_2)} u(y)$

Let $Z = X + Y$

Then

$$\begin{aligned}
f_Z(z) &= \int_{-\infty}^{\infty} f_X(z-y) f_Y(y) dy \\
&= \int_0^z \frac{b^{a_1} (z-y)^{a_1-1} e^{-b(z-y)} b^{a_2} y^{a_2-1} e^{-by}}{\Gamma(a_1) \Gamma(a_2)} dy
\end{aligned}$$

$\because f_X(x) = 0 \text{ for } x < 0$ $f_Y(y) = 0 \text{ for } y < 0$

Let $x = \frac{y}{z} \Rightarrow dy = x dz$

$$\begin{aligned}
f_Z(z) &= \int_0^1 \frac{b^{a_1+a_2} e^{-bz} z^{a_1-1} (1-x)^{a_1-1} z^{a_2-1} x^{a_2-1}}{\Gamma(a_1) \Gamma(a_2)} z dx \\
&= \frac{b^{a_1+a_2} e^{-bz}}{\Gamma(a_1) \Gamma(a_2)} \int_0^1 z^{a_1-1} (1-x)^{a_1-1} z^{a_2-1} x^{a_2-1} z dx \\
&= \frac{b^{a_1+a_2} z^{a_1+a_2-1} e^{-bz}}{\Gamma(a_1) \Gamma(a_2)} \int_0^1 (1-x)^{a_1-1} x^{a_2-1} dx \\
&= \frac{b^{a_1+a_2} z^{a_1+a_2-1} e^{-bz}}{\Gamma(a_1) \Gamma(a_2)} B(a_2, a_1) \\
&= \frac{b^{a_1+a_2} z^{a_1+a_2-1} e^{-bz}}{\Gamma(a_1) \Gamma(a_2)} \frac{\Gamma(a_1) \Gamma(a_2)}{\Gamma(a_1 + a_2)}
\end{aligned}$$

$$= \frac{b^{a_1+a_2} e^{-bz} z^{a_1+a_2-1}}{\Gamma(a_1+a_2)}$$

which is gamma distribution with parameters $(a_1 + a_2, b)$.

4.64 If $X_1, X_2, \dots, X_n$ are n independent exponential random variables each having parameter b then prove that $X_1 + X_2 + \dots + X_n$ is gamma random variable with parameter (n, b) .

Solution Let $Y = X_1 + X_2 + \dots + X_n$ and

Let $Z = X_1 + X_2$;

Given: $f_{X_1}(x_1) = be^{-bx_1} u(x_1)$ and

$$f_{X_2}(x_2) = be^{-bx_2} u(x_2)$$

$$\begin{aligned} f_Z(z) &= \int_{-\infty}^{\infty} f_{X_1}(z-x_2) f_{X_2}(x_2) dx_2 \\ &= b^2 \int_0^z e^{-b(z-x_2)} e^{-bx_2} dx_2 = b^2 e^{-bz} \int_0^z dx_2 = b^2 z e^{-bz} \text{ for } z \geq 0 \end{aligned}$$

Let $t = X_1 + X_2 + X_3 = Z + X_3$; we know $f_{X_3}(x_3) = be^{-bx_3} u(x_3)$

$$\begin{aligned} f_T(t) &= f_Z(z) * f_{X_3}(x_3) \\ &= \int_{-\infty}^{\infty} f_{X_3}(t-z) f_Z(z) dz \\ &= b^3 \int_0^t e^{-b(t-z)} z e^{-bz} dz = b^3 \int_0^t z e^{-bt} dz = \frac{b^3 e^{-bt} z^2}{2} \end{aligned}$$

If we repeat the above process, we get

$$f_Y(y) = \frac{b^n e^{-by} y^{n-1}}{(n-1)!} = \frac{b^n e^{-by} y^{n-1}}{\Gamma(n)}$$

which is a gamma random variable with parameter (n, b) .

4.65 The joint density function of two random variables X and Y is

$$f_{X,Y}(x, y) = c x^{n_1-1} (y-x)^{n_2-1} e^{-y} \text{ for } 0 < x < y < \infty$$

Find (a) the constant c , and (b) the marginal distribution function X and Y .

Solution We know,

$$f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx$$

$$\begin{aligned}
&= \int_0^y c x^{n_1-1} (y-x)^{n_2-1} e^{-y} dx \\
&= f_Y(Y) = \int_0^1 c (yz)^{n_1-1} (1-z)^{n_2-1} y^{n_2-1} e^{-y} y dz \quad \boxed{\text{Let } x = yz} \\
&\quad \boxed{dx = y dz} \\
&= c y e^{-y} \int_0^1 (y)^{n_1+n_2-2} (z)^{n_1-1} (1-z)^{n_2-1} dz \\
&= c y^{n_1+n_2-1} e^{-y} B(n_1, n_2) \\
&= c y^{n_1+n_2-1} e^{-y} \frac{\Gamma(n_1) \Gamma(n_2)}{\Gamma(n_1+n_2)}
\end{aligned}$$

We know,

$$\begin{aligned}
&\int_0^\infty f_Y(y) dy = 1 \\
\Rightarrow &c \int_0^\infty y^{n_1+n_2-1} e^{-y} \frac{\Gamma(n_1) \Gamma(n_2)}{\Gamma(n_1+n_2)} dy = 1 \\
\Rightarrow &c \Gamma(n_1) \Gamma(n_2) \underbrace{\int_0^\infty \frac{y^{n_1+n_2-1} e^{-y}}{\Gamma(n_1+n_2)} dy}_{=1} = 1 \Rightarrow c = \frac{1}{\Gamma(n_1) \Gamma(n_2)} \\
f_X(x) &= \int_x^\infty c x^{n_1-1} (y-x)^{n_2-1} e^{-y} dy = c x^{n_1-1} \int_x^\infty (y-x)^{n_2-1} e^{-y} dy \\
\text{Let } y-x &= p \Rightarrow dy = dp \\
f_X(x) &= c x^{n_1-1} \int_0^\infty p^{n_2-1} e^{-(p+x)} dp = c x^{n_1-1} e^{-x} \int_0^\infty p^{n_2-1} e^{-p} dp \\
&= c \Gamma(n_2) x^{n_1-1} e^{-x} \underbrace{\int_0^\infty \frac{p^{n_2-1} e^{-p}}{\Gamma(n_2)} dp}_{=1} \\
&= \frac{1}{\Gamma(n_1) \Gamma(n_2)} \Gamma(n_2) x^{n_1-1} e^{-x} \\
&= \frac{x^{n_1-1} e^{-x}}{\Gamma(n_1)}
\end{aligned}$$

4.66 If X and Y are independent Poisson random variables with parameters λ_1 and λ_2 , compute the distribution of $X + Y$.

Solution

Given: $p_X(x) = \frac{e^{-\lambda_1} \lambda_1^x}{x!}$ and $p_Y(y) = \frac{e^{-\lambda_2} \lambda_2^y}{y!}$

$$\begin{aligned}
 p_{X+Y}(z) &= P(X+Y=z) = \sum_{k=0}^z P(X=k, Y=z-k) \\
 &= \sum_{k=0}^z P(X=k) P(Y=z-k) \quad \text{Since } X \text{ and } Y \text{ are independent} \\
 &= \sum_{k=0}^z \frac{e^{-\lambda_1} (\lambda_1)^k}{k!} \frac{e^{-\lambda_2} (\lambda_2)^{z-k}}{(z-k)!} \\
 &= e^{-(\lambda_1+\lambda_2)} \sum_{k=0}^z \frac{(\lambda_1)^k (\lambda_2)^{z-k}}{k! (z-k)!} \\
 &= \frac{e^{-(\lambda_1+\lambda_2)}}{z!} \sum_{k=0}^z \frac{z!}{k! (z-k)!} (\lambda_1)^k (\lambda_2)^{z-k} \\
 &= \frac{e^{-(\lambda_1+\lambda_2)}}{z!} \sum_{k=0}^z \binom{z}{k} (\lambda_1)^k (\lambda_2)^{z-k} \\
 &= \frac{e^{-(\lambda_1+\lambda_2)}}{z!} (\lambda_1 + \lambda_2)^z
 \end{aligned}$$

Therefore, z is a Poisson random variable with parameter $(\lambda_1 + \lambda_2)$.

**Practice Problems**

4.23 If X and Y are independent Rayleigh random variables with common parameter σ^2 , determine the density of

$Z = X/Y$.

4.24 The random variables X and Y are independent with experimental densities

$$f_X(x) = ae^{-ax}u(x) \text{ and } f_Y(y) = be^{-by}u(y)$$

Find the density (a) $\max(X, Y)$ (b) $2X + Y$.

$$\left(\text{Ans: } \frac{2z}{(1+z^2)^2}, 0 \leq z \leq \infty \right)$$

$$\left(\text{Ans: (a) } \{ae^{-az}(1-e^{-bz}) + be^{-bz}(1-e^{-az})\} u(z) \right. \\ \left. \text{(b) } \frac{ab}{a-2b} (e^{-bz} - e^{-az/2}) u(z) \right)$$

**CENTRAL LIMIT THEOREM 4.11**

According to the Central Limit Theorem (CLT), the distribution of a random process which is a cumulative effect of a large number of independent noise sources can be assumed to be Gaussian. For example, in communication systems, the noise is always modelled as a random variable with Gaussian distribution. This is a valid assumption since the noise in a communication system is the cumulative effect of many random noise sources.

4.11.1 Unequal Distribution

The central limit theorem states that the probability distribution function of the sum of a large number variables of random with arbitrary distribution approaches a Gaussian distribution. Let $X_1, X_2, \dots, X_n$ be independent random variables with mean $\bar{X}_i$ and variance $\sigma_{X_i}^2$ respectively. Let $X_i, i = 1, 2, \dots, N$ have arbitrary probability densities. The central limit theorem states that the sum $Y_N = X_1 + X_2 + \dots + X_N$ which has mean $\bar{Y}_N = \bar{X}_1 + \bar{X}_2 + \dots + \bar{X}_N$ and variance $\sigma_{Y_N}^2 = \sigma_{X_1}^2 + \sigma_{X_2}^2 + \dots + \sigma_{X_N}^2$ has probability distribution that asymptotically approaches Gaussian as $N \rightarrow \infty$. The sufficient condition for the theorem's validity.

$$\begin{aligned} \sigma_{X_i}^2 &> B_1 > 0, \quad i = 1, 2, \dots, N \\ E(|X_i - \bar{X}_i|^3) &< B_2, \quad i = 1, 2, \dots, N \end{aligned} \quad (4.85)$$

where B_1 and B_2 are positive constants.

4.11.2 Equal Distribution

Let $X_1, X_2, \dots, X_N$ be a sequence of independent and identically distributed (*i. i. d.*) random variables with $E(X_i) = \mu_X$ and variance σ_X^2 and same probability density function $f_{X_k}(x), k = 1, 2, \dots, N$. Let $Y_N = X_1 + X_2 + \dots + X_N$. Since the sum has an infinite variance as $N \rightarrow \infty$, we normalize the random variable. The new random variable W_N has zero mean and unit variance. That is,

$$W_N = \frac{Y_N - \bar{Y}_N}{\sigma_{Y_N}} \quad (4.86)$$

We know $Y_N = \sum_{i=1}^N X_i; \bar{Y}_N = \sum_{n=1}^N \bar{X}_n$ and $\sigma_{Y_N}^2 = \sum_{n=1}^N \sigma_{X_n}^2$

$$W_N = \frac{\sum_{n=1}^N X_n - \sum_{n=1}^N \bar{X}_n}{\left\{ \sum_{n=1}^N \sigma_{X_n}^2 \right\}^{1/2}} \quad (4.87)$$

Since all random variables have the same distribution, we can write $\sigma_{X_n}^2 = \sigma_X^2$ and $\bar{X}_n = \mu_X$ from which

$$W_N = \frac{\sum_{n=1}^N (X_n - \mu_X)}{\left\{ \sum_{n=1}^N \sigma_X^2 \right\}^{1/2}} = \frac{\sum_{n=1}^N (X_n - \mu_X)}{(N \sigma_X^2)^{1/2}} \quad (4.88)$$

$$= \frac{1}{\sigma_X \sqrt{N}} \sum_{n=1}^N (X_n - \mu_X) = \frac{Y_N - N \mu_X}{\sigma_X \sqrt{N}} \quad (4.89)$$

According to the central limit theorem, the variate W_N has a distribution that approaches the Gaussian distribution with mean $N \mu_X$ and variance $N \sigma_X^2$ as $N \rightarrow \infty$ provided the moment generation function exists.

Proof: To prove that the random variable has Gaussian density function, we should prove that the MGF of W is $e^{u^2/2}$.

The moment-generating function of W is

$$\begin{aligned} M_W(u) &= E(e^{uW}) \\ &= E\left[e^{u \sum_{n=1}^N (X_n - \bar{X})/\sigma_X \sqrt{N}}\right] \end{aligned} \quad (4.90)$$

Since $X_n, n = 1, 2, \dots, N$ are independent with identical distribution functions, we can write

$$M_W(u) = E\left[e^{u(X_n - \bar{X})/\sigma_X \sqrt{N}}\right]^N \quad (4.91)$$

Using Taylor's series expansion, we can write

$$e^x = 1 + x + \frac{x^2}{2!} + \dots \quad (4.92)$$

$$\text{Therefore, } M_W(u) = E\left[1 + \frac{u}{\sigma_X \sqrt{N}} (X_n - \bar{X}) + \left(\frac{u}{\sigma_X \sqrt{N}}\right)^2 \frac{(X_n - \bar{X})^2}{N} + \frac{R}{N}\right]^N \quad (4.93)$$

where $\frac{R}{N}$ is the remainder term.

$$\begin{aligned} M_W(u) &= \left\{1 + \frac{u}{\sigma_X \sqrt{N}} E[X_n - \bar{X}] + \frac{u^2}{N\sigma_X^2} \frac{E[X_n - \bar{X}]^2}{2} + E\left[\frac{R}{N}\right]\right\}^N \\ &= \left\{1 + \frac{u}{\sigma_X \sqrt{N}} (E[X_n] - E[\bar{X}]) + \frac{u^2}{N\sigma_X^2} \frac{\sigma_X^2}{2} + E\left[\frac{R}{N}\right]\right\}^N \\ &= \left\{1 + \frac{u}{\sigma_X \sqrt{N}} (\bar{X} - \bar{X}) + \frac{u^2}{2N} + E\left[\frac{R}{N}\right]\right\}^N \\ \log(M_W(u)) &= N \log\left\{1 + \frac{u^2}{2N} + E\left[\frac{R}{N}\right]\right\} \\ \log(1+x) &= x + \frac{x^2}{2} + \frac{x^3}{3} + \dots \\ \log(M_W(u)) &= N \left\{ \frac{u^2}{2N} + E\left[\frac{R}{N}\right] + \frac{1}{2} \left(\frac{u^2}{2N} + E\left[\frac{R}{N}\right] \right)^2 + \dots \right\} \\ &= \frac{u^2}{2} + \frac{E[R]}{N} + \frac{N}{2} \left(\frac{u^2}{2N} + \frac{E[R]}{N} \right)^2 + \dots \end{aligned} \quad (4.94)$$

$E(R)$ approaches zero as $N \rightarrow \infty$. Therefore, we have

$$\begin{aligned} \lim_{N \rightarrow \infty} \log[M_W(u)] &= \frac{u^2}{2} \\ \lim_{N \rightarrow \infty} [M_W(u)] &= e^{u^2/2} \end{aligned} \quad (4.95)$$

Therefore, as $N \rightarrow \infty$, the distribution of W tends to standard normal distribution.

Let $X_1, X_2, \dots, X_n$ be a random sample from a random variable X . Then we write the sample mean $\bar{X}$ as

$$\bar{X} = \frac{X_1 + X_2 + \dots + X_N}{N} = \frac{Y_N}{N} \quad (4.96)$$

Since the sample mean is also a random variable, it has a mean value which is given by

$$E[\bar{X}] = E\left[\frac{1}{N} \sum_{i=1}^N X_i\right] = \frac{1}{N} \sum_{i=1}^N E[X_i] = \frac{1}{N} \sum_{i=1}^N \mu_X = \mu_X \quad (4.97)$$

That is, the mean value of the sample mean is equal to the true value. The variance of the sample mean is equal to

$$\begin{aligned} \text{Var}[\bar{X}] &= \text{Var}\left[\frac{1}{N} \sum_{i=1}^N X_i\right] = \frac{1}{N^2} \text{Var}\left[\sum_{i=1}^N X_i\right] = \frac{1}{N} \sum_{i=1}^N \text{Var}(X_i) \\ &= \frac{1}{N^2} (N \sigma_X^2) = \frac{\sigma_X^2}{N} \end{aligned} \quad (4.98)$$

Since the sample mean is the sum of random variables, the CLT says that it tends to be asymptotically normal regardless of the distribution of the random variable X_i , $i = 1, 2, \dots, N$. If we define the standard normal score of the sample mean

$$Z = \frac{\bar{X} - \mu_X}{\sigma_X / \sqrt{N}}. \quad \text{Then } F_{\bar{X}}(x) = P(\bar{X} < x) = \phi\left(\frac{x - \mu_X}{\sigma_X / \sqrt{N}}\right) \quad ((4.99))$$

Let us consider $X_1, X_2, \dots, X_k$ be a sequence of independent and identically distributed uniform random variables. Consider the sum $Z = X_1 + X_2$. From Solved Problem (4.63), we can find that the pdf of Z has a triangular pdf as shown in Fig. (4.30). In the same problem, we observed that the pdf of $Y = X_1 + X_2 + X_3$ has a pdf with parabolic shape which is quite close to Gaussian pdf. Figure (4.31) shows how fast the sum of n independent random variables tends to be a Gaussian random variables.

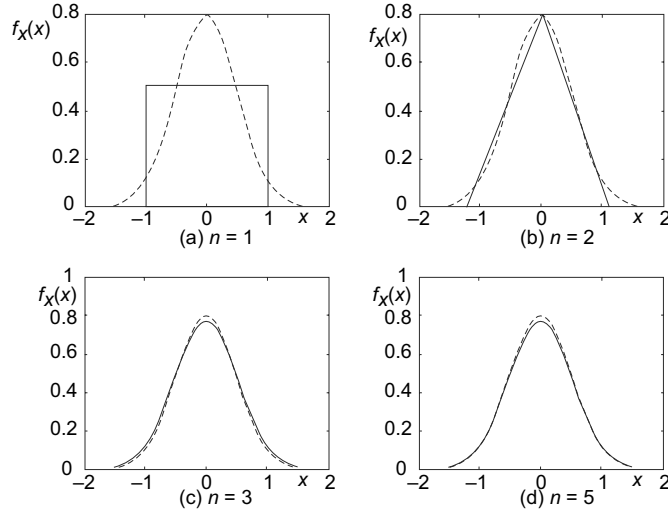


Fig. 4.31

REVIEW QUESTIONS

17. State and prove the central limit theorem.
18. State the central limit theorem for equal and unequal distributions.

Solved Problems

4.67 Ten dice are thrown. Find the approximate probability that the sum obtained is between 40 and 50.

Solution Let X_i be the event of throwing a dice. Then we can define the sum as a random variable Y_N

$= \sum_{i=1}^{10} X_i$. We know that when we throw a dice, the possible outcomes are 1, 2, 3, 4, 5 and 6, each with

probability $\frac{1}{6}$. Therefore, we can find

$$E(X_i) = \sum_i x_i p_{X_i}(x_i) = \frac{1}{6}(1 + 2 + 3 + 4 + 5 + 6) = \frac{7}{2}$$

$$\text{and } E(X_i^2) = \sum_i x_i^2 p_{X_i}(x_i) = \frac{1}{6}(1^2 + 2^2 + 3^2 + 4^2 + 5^2 + 6^2) = \frac{91}{6}$$

$$\sigma_{X_i}^2 = E[X_i^2] - \{E[X_i]\}^2 = \frac{91}{6} - \left(\frac{7}{2}\right)^2 = \frac{35}{12}$$

$$\text{Also } E(Y_N) = 10 E(X_i) = 35$$

$$\sigma_{Y_N}^2 = 10[\sigma_{X_i}^2] = \frac{175}{6}$$

If the sum approximates the normal random variable then the CDF of the normal random variable Y_N is

$$P(Y_N \leq y) = F_{Y_N}(y) = \phi\left[\frac{y - E[Y_N]}{\sigma_{Y_N}}\right] = \phi\left(\frac{y - 35}{\sqrt{175/6}}\right)$$

$$P(50 \leq Y_N \leq 40) = F_{Y_N}(50) - F_{Y_N}(40)$$

$$= \phi\left(\frac{50 - 35}{\sqrt{175/6}}\right) - \phi\left(\frac{40 - 35}{\sqrt{175/6}}\right)$$

$$= \phi(2.78) - \phi(0.93)$$

$$= 0.9973 - 0.8238 = 0.1735$$

4.68 A distribution with unknown mean μ has variance 1.5. Using CLT, find how large a sample should be taken from the distribution in order that the probability will be at least 0.95 and the probability sample mean will be within 0.5 of the population mean.

Solution Let $\bar{X}$ be the sample mean of a sample size of n . Given, $E(X_i) = \mu$ and $\sigma^2 = 1.5$

The variance of the sample mean is σ^2/N and the mean of $\bar{X} = \mu$. That is, $\bar{X} \sim N(\mu, \sigma^2/N)$

$$\text{Let } Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{N}}$$

$$P\left(\frac{|\bar{X} - \mu|}{\sigma/\sqrt{N}} \leq \frac{0.5}{\sigma/\sqrt{N}}\right) = P\left(\frac{-0.5}{\sigma/\sqrt{N}} \leq Z \leq \frac{0.5}{\sigma/\sqrt{N}}\right) \geq 0.95$$

$$\Rightarrow 2\phi\left(\frac{0.5\sqrt{N}}{\sqrt{1.5}}\right) - 1 = 0.95$$

$$\text{from which } \phi\left(\frac{0.5\sqrt{N}}{\sqrt{1.5}}\right) = 0.975$$

From table (2.4) we can find

$$\frac{0.5\sqrt{N}}{\sqrt{1.5}} = 1.96 \Rightarrow N = \frac{(1.96)^2 (1.5)}{0.25} = 4.44$$

$$\Rightarrow N \approx 5$$

4.69 The lifetime of a certain brand of an electric bulb may be considered a random variable with mean 1200 and SD 250. Find the probability using CLT that the average lifetime of 60 bulbs exceeds 1250 hours.

Solution Given sample size = 60; $\mu = 1200$; $\sigma = 250$. Let $\bar{X}$ denote the mean lifetime of 60 bulbs. We know $E[\bar{X}] = \mu = 1200$

$$\sigma_{\bar{X}}^2 = \frac{\sigma^2}{N} = \frac{(250)^2}{60} = 1041.67$$

$$\sigma_{\bar{X}} = 32.27$$

We have to find $P(\bar{X} \geq 1250)$

$$\text{Let } Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{N}}$$

$$\text{For } \bar{X} = 1250; Z = \frac{1250 - 1200}{32.27} = 1.55$$

$$\begin{aligned} P(\bar{X} \geq 1250) &= P(Z \geq 1.55) = 1 - F(Z \leq 1.55) \\ &= 1 - \phi(1.55) = 1 - 0.9394 = 0.0606 \end{aligned}$$

4.70 Suppose that orders at a restaurant are identically independent random variables with mean $\mu = ₹ 8$ and standard deviation ₹ 2. Estimate, (a) The probability that the first 100 customers spend a total of more than ₹ 840, i.e. $P(X_1 + X_2 + X_3 + \dots + X_{100} > 840)$ (b) $P(780 < X_1 + X_2 + X_3 + \dots + X_{100} < 820)$.

Solution Let $X_i, i = 1, 2, \dots, 100$ are *i. i. d.* random variables. Given, $E(X_i) = 8$ and $\sigma_{X_i}^2 = 4$

Let us define

$$Y_N = X_1 + X_2 + X_3 + \dots + X_{100}$$

$$E(Y_N) = N E(X_i) = 100 \times 8 = 800$$

$$\sigma_{Y_N}^2 = N \sigma_{X_i}^2 = 100 \times 4 = 400 \Rightarrow \sigma_{Y_N} = 20$$

Let
$$W_N = \frac{Y_N - E[Y_N]}{\sigma_{Y_N}} = \frac{Y_N - 800}{20}$$

$$\begin{aligned} P(Y_N \geq 840) &= P\left(\frac{Y_N - E[Y_N]}{\sigma_{Y_N}} \geq \frac{840 - 800}{20}\right) \\ &= P(W_N \geq 2) = 1 - \phi(2) \\ &= 1 - 0.9772 = 0.0228 \end{aligned}$$

$$\begin{aligned} P(780 < Y_N < 820) &= P\left(\frac{780 - 800}{20} \leq W_N \leq \frac{820 - 800}{20}\right) \\ &= P(-1 \leq W_N \leq 1) = 2\phi(1) - 1 \\ &= 2(0.8413) - 1 \\ &= 0.6826 \end{aligned}$$

4.71 The round-off error to the second decimal place has the uniform distribution on the interval $[-0.05, 0.05]$. What is the probability that the absolute error in the sum of 1000 numbers is less than 1?

Solution Since error is uniformly distributed on $(-0.05, 0.05)$, the mean value is zero and variance $\frac{[0.05 - (-0.05)]^2}{12} = 8.33 \times 10^{-4}$. Let Y_N be a random variable which is the sum of 1000 numbers. Let the standard normal variable

$$Z = \frac{Y_N - E[Y_N]}{\sigma_{Y_N}} = \frac{Y_N}{\sigma_{Y_N}}$$

$$E(Y_N) = 0 \text{ and } \sigma_{Y_N}^2 = 1000 \times 8.33 \times 10^{-4} = 0.833 \Rightarrow \sigma_{Y_N} = 0.913$$

$$\begin{aligned} P(|Y_N - E(Y_N)| < 1) &= P(|Y_N| < 1) \\ &= P\left(|Z| < \frac{1}{0.91}\right) = P(|Z| < 1.099) \\ &= 2\phi(1.099) - 1 \\ &= 2(0.8643) - 1 = 0.7286 \end{aligned}$$

4.72 A random sample of size 100 is taken from a population whose mean is 60 and variance is 400. Using CLT, find what probability can we assert that the mean of the sample will not differ from $\mu = 60$ by more than 4.

Solution Given sample size $N = 100$

Mean $\mu = 60$

Variance $\sigma^2 = 400 \Rightarrow \sigma = 20$

Given that the sample mean $\bar{X}$ will not differ from μ by more than 4, that is $|\bar{X} - \mu| \leq 4$ from which we can write $|\bar{X} - 60| \leq 4$. Let $Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{N}} = \frac{\bar{X} - 60}{20/\sqrt{100}} = \frac{\bar{X} - 60}{2}$

$$\begin{aligned} P(|\bar{X} - 60| \leq 4) &= P\left(|Z| \leq \frac{\bar{X} - \mu}{\sigma/\sqrt{N}}\right) \\ &= P\left(|Z| \leq \frac{4}{2}\right) = P(|Z| < 2) \\ &= P(-2 \leq Z \leq 2) = 2\phi(2) - 1 \\ &= 2(0.9772) - 1 = 0.9544 \end{aligned}$$

4.73 Consider a random variable Y_N which is the sum of 36 independent experimental values of the random variable X whose pdf is given by

$$f_X(x) = \begin{cases} \frac{1}{5} & 1 \leq x \leq 6 \\ 0 & \text{otherwise} \end{cases}$$

Find the probability that Y_N lies in the range $110 \leq Y_N \leq 130$.

Solution From the given pdf, we can find that

$$E(X) = \frac{6+1}{2} = 3.5$$

$$\text{and } \sigma_X^2 = \frac{(6-1)^2}{12} = \frac{25}{12}$$

The mean and variance of Y_N are given by

$$E(Y_N) = N E(X) = 36(3.5) = 126$$

$$\sigma_{Y_N}^2 = N \sigma_X^2 = 36\left(\frac{25}{12}\right) = 75$$

If the sum approximates the normal random variable, the CDF of the normalized random variable of Y_N becomes

$$\begin{aligned} P(Y_N \leq y) &= F_{Y_N}(y) = \phi\left(\frac{y - E[Y_N]}{\sigma_{Y_N}}\right) \\ &= \phi\left(\frac{y - 126}{\sqrt{75}}\right) \end{aligned}$$

The probability that Y_N lies in the range $110 \leq Y_N \leq 130$ is given by

$$P(110 \leq Y_N \leq 130) = F_{Y_N}(130) - F_{Y_N}(110)$$

$$\begin{aligned}
&= \phi\left(\frac{130-126}{\sqrt{75}}\right) - \phi\left(\frac{110-126}{\sqrt{75}}\right) \\
&= \phi(0.46) - \phi(-1.85) \\
&= \phi(0.46) - (1 - \phi(1.85)) \\
&= 0.6722 - 0.0322 \\
&= 0.64
\end{aligned}$$

4.74 A random sample of size 25 is taken from a normal population with mean 49 and variance 1.44. Using CLT, find the probability that the mean of this sample falls between 48.52 and 49.6.

Solution Given the mean $\mu = 49$ and $\sigma^2 = 1.44 \Rightarrow \sigma = 1.2$

Let $\bar{X}$ be sample mean. Then $E[\bar{X}] = \mu = 49$

$$\text{Let } Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{N}}$$

$$\begin{aligned}
P(48.52 \leq \bar{X} \leq 49.6) &= P\left(\frac{48.52 - 49}{1.2/\sqrt{25}} \leq Z \leq \frac{49.6 - 49}{1.2/\sqrt{25}}\right) \\
&= P(-2 \leq Z \leq 2.5) = \phi(2.5) - \phi(-2) \\
&= \phi(2.5) - (1 - \phi(2)) \\
&= 0.9938 - (1 - 0.9772) \\
&= 0.971
\end{aligned}$$

4.75 In an elevator, a caution reacts ‘capacity 600 kg or 10 persons’. Assume a standard deviation of 15 kg for the weight of a man drawn at random from all the people who might ride the elevator and calculate approximately his expected weight. It is given that the probability that a full load of 10 persons will be more than 600 kg is 0.25.

Solution Given: $N = 10$

Let X_i ($i = 1, 2, \dots, 10$) denote the individual weight. The total weight is $Y_N = \sum_{i=1}^{10} X_i$. Let $E(X_i) = \mu$ and given $\sigma_{X_i} = 15$ kg.

$$E(Y_N) = 10\mu \text{ and } \sigma_{Y_N} = \sigma_{X_i} \sqrt{N} = 15\sqrt{10}$$

$$\text{Let } W_N = \frac{Y_N - E[Y_N]}{\sigma_{Y_N}} = \frac{Y_N - 10\mu}{\sigma_{X_i} \sqrt{N}}$$

$$P(Y_N > 600) = P\left(W_N > \frac{600 - 10\mu}{15\sqrt{10}}\right) = 0.25$$

$$\Rightarrow P\left(W_N \leq \frac{600 - 10\mu}{15\sqrt{10}}\right) = 0.75$$

From the table (2.4), we can find

$$\frac{600 - 10\mu}{15\sqrt{10}} = 0.67$$

$$\mu = 56.8$$

So the expected weight is 56.8 kg.



Practice Problems

4.25 A transistor has a lifetime T which is exponentially distributed with $\alpha = 0.5$. If 100 transistors are tested, what is the probability that $175 < \bar{T} < 225$? (Ans: 0.7888)

4.26 The lifetime of a bulb is an exponential random variable with a mean of 50 hours. If 20 lightbulbs are tested and their lifetimes are measured, estimate the probability that the sum of the lifetimes is less than 1050 hours. (Ans: 0.5871)

Solved Problems



4.76 Two random variables X and Y have the following joint probability density function

$$f_{XY}(x, y) = \begin{cases} 2 - x - y & \text{for } 0 \leq x \leq 1 \text{ and } 0 \leq y \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

Find the marginal probability density functions of X and Y .

Solution The marginal density function of X is given by

$$\begin{aligned} f_X(x) &= \int_{-\infty}^{\infty} f_{X,Y}(x, y) dy \\ &= \int_0^1 (2 - x - y) dy = \left(2y - xy - \frac{y^2}{2} \right) \Big|_0^1 \\ &= \frac{3}{2} - x \end{aligned}$$

The marginal density function of Y is given by

$$\begin{aligned} f_Y(y) &= \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx \\ &= \int_0^1 (2 - x - y) dx = \left(2x - \frac{x^2}{2} \right) \Big|_0^1 - yx \Big|_0^1 = \frac{3}{2} - y \end{aligned}$$

4.77 Two independent random variables X and Y have their densities as

$$f_X(x) = e^{-x} u(x) \text{ and } f_Y(y) = e^{-y} u(y)$$

Find $P(X + Y \leq 1)$.

Solution Given: $f_X(x) = e^{-x} u(x)$ and $f_Y(y) = e^{-y} u(y)$

$$\begin{aligned}
 P(X + Y \leq 1) &= \int_0^1 \int_{y=0}^{1-x} f_{X,Y}(x, y) dy dx && \begin{cases} X + Y = 1 \\ Y = 1 - x \end{cases} \\
 &= \int_0^1 \int_0^{1-x} e^{-x} e^{-y} dy dx \\
 &= \int_0^1 e^{-x} \left[(-e^{-y}) \Big|_0^{1-x} \right] dx = \int_0^1 e^{-x} [1 - e^{-(1-x)}] dx \\
 &= \int_0^1 (e^{-x} - e^{-1}) dx = -e^{-x} \Big|_0^1 - e^{-1} x \Big|_0^1 \\
 &= \boxed{1 - 2e^{-1}}
 \end{aligned}$$

4.78 The joint density function of two random variables (X, Y) is

$$f_{X,Y}(x, y) = \begin{cases} 1/ab & 0 < x < a \quad \text{and} \quad 0 < y < b \\ 0 & \text{elsewhere} \end{cases}$$

- (a) Find and sketch $F_{X,Y}(x, y)$
 (b) If $a < b$, find $P\left(X + Y \leq \frac{3a}{4}\right)$

Solution

$$\begin{aligned}
 \text{(a)} \quad F_{X,Y}(x, y) &= \int_0^x \int_0^y \frac{1}{ab} dx dy \\
 &= \frac{xy}{ab} \quad 0 \leq x \leq a, \quad 0 \leq y \leq b
 \end{aligned}$$

$$F_{X,Y}(0, 0) = 0$$

If $x \geq a$ and $0 \leq y \leq b$,

$$F_{X,Y}(x, y) = \frac{a}{a} \left(\frac{y}{b} \right) = \frac{y}{b}$$

If $0 \leq x \leq a$ and $y \geq b$ then, $F_{X,Y}(x, y) = \frac{x}{a} \left(\frac{b}{b} \right) = \frac{x}{a}$

If $x \geq a$ and $y \geq b$, then

$$F_{X,Y}(x, y) = \frac{ab}{ab} = 1$$

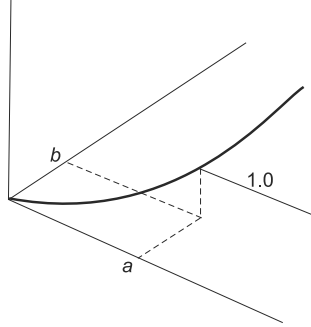


Fig. 4.32

The sketch of $F_{X,Y}(x, y)$ is shown in Fig. 4.32 and the distribution function is

$$F_{X,Y}(x, y) = \begin{cases} 0 & \text{for } x < 0, \quad y < 0 \\ \frac{xy}{ab} & 0 < x < a \quad \text{and} \quad 0 < y < b \\ \frac{x}{a} & 0 < x < a \quad \text{and} \quad y \geq b \\ \frac{y}{b} & 0 < y < b \quad \text{and} \quad x \geq a \\ 1 & x \geq a \quad \text{and} \quad y \geq b \end{cases}$$

- (b) If $a < b$ and $X + Y \leq \frac{3a}{4}$

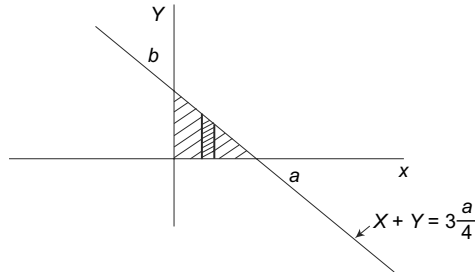


Fig. 4.33

The intervals of integration for x and y are $\left(0, \frac{3a}{4}\right)$ and $\left(0, \frac{3a}{4} - x\right)$ respectively.

$$P\left[X + Y \leq \frac{3a}{4}\right] = \int_0^{3a/4} \int_0^{\frac{3a}{4} - x} \frac{1}{ab} dy dx$$

$$\begin{aligned}
&= \int_0^{3a/4} \int_0^{\frac{3a}{4}-x} \frac{1}{ab} dy dx \\
&= \frac{1}{ab} \int_0^{3a/4} \left(\frac{3a}{4} - x \right) dx = \frac{1}{ab} \left[\frac{3a}{4} x \Big|_0^{3a/4} - \frac{x^2}{2} \Big|_0^{3a/4} \right] \\
&= \frac{1}{ab} \left[\frac{9a^2}{16} - \frac{9a^2}{32} \right] = \frac{18a^2 - 9a^2}{32ab} \\
&= \frac{9a^2}{32ab} = \frac{9a}{32b}
\end{aligned}$$

4.79 The joint pdf $f_{X,Y}(x, y)$ is given by $f_{X,Y}(x, y) = \begin{cases} k(6 - x - y) & 0 \leq x \leq 2, \quad 0 \leq y \leq 2 \\ 0 & \text{otherwise} \end{cases}$

Find $F_{X,Y}(x, y)$ and $f_{X,Y}(x, y)$ for different values of X and Y .
Also find $P(0.5 \leq X \leq 1.5, 1 \leq Y \leq 1.5)$.

Solution

$$\text{Given: } f_{X,Y}(x, y) = \begin{cases} k(6 - x - y) & 0 \leq x \leq 2, \quad 0 \leq y \leq 2 \\ 0 & \text{otherwise} \end{cases}$$

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx dy = 1$$

$$\Rightarrow \int_0^2 \int_0^2 k(6 - x - y) dx dy = 1$$

$$k \int_0^2 \left[6x \Big|_0^2 - \frac{x^2}{2} \Big|_0^2 - yx \Big|_0^2 \right] dy = 1$$

$$k \int_0^2 [10 - 2y] dy = 1 \Rightarrow k(20 - 4) = 1$$

$$\text{from which we can obtain } k = \frac{1}{16}$$

In the region – $V, x < 0, y < 0$

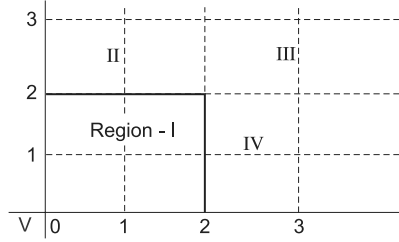


Fig. 4.34

Therefore, $F_{X,Y}(x, y) = 0$

In the region III, $x > 2, y > 2$

Therefore, $F_{X,Y}(x, y) = 1$

In the region I, $0 < x < 2$ and $0 < y < 2$

$$F_{X,Y}(x, y) = \frac{1}{16} \int_0^x \int_0^y (6 - u - v) dv du$$

(u and v are dummy variables)

$$\begin{aligned} F_{X,Y}(x, y) &= \frac{1}{16} \int_0^x \left[6v \Big|_0^y - uv \Big|_0^y - \frac{v^2}{2} \Big|_0^y \right] du \\ &= \frac{1}{16} \int_0^x \left(6y - uy - \frac{y^2}{2} \right) du \\ &= \frac{1}{16} \left[6yu \Big|_0^x - y \frac{u^2}{2} \Big|_0^x - \frac{y^2}{2} u \Big|_0^x \right] \\ &= \frac{1}{16} \left[6xy - \frac{x^2 y}{2} - \frac{xy^2}{2} \right] \end{aligned}$$

Region II, $0 \leq x \leq 2; y > 2$

We get marginal distribution function of x .

$$\begin{aligned} F_X(x) = F_{X,Y}(x, 2) &= \frac{1}{16} \left[6x(2) - \frac{x^2}{2}(2) - \frac{x}{2}(2)^2 \right] \\ &= \frac{1}{16} (10x - x^2) \quad 0 \leq x \leq 2 \end{aligned}$$

Region IV $0 \leq y \leq 2; x > 2$

$$F_Y(y) = F_{X,Y}(2, y)$$

$$\begin{aligned}
&= \frac{1}{16} \left[6y(2) - \frac{y^2}{2}(2) - \frac{y}{2}(2)^2 \right] \\
&= \frac{1}{16} (10y - y^2) \quad 0 \leq y \leq 2
\end{aligned}$$

Region III $x > 2, y > 2$

$$F_{X,Y}(x, y) = 1$$

The joint density functions can be obtained by differentiating the distribution function. That is,

$$f_{X,Y}(x, y) = 0 \quad x \leq 1 \text{ or } y \leq 1$$

$$f_{X,Y}(x, y) = \frac{1}{16} (6 - x - y), \quad 0 \leq x \leq 2, 0 \leq y \leq 2$$

$$f_{X,Y}(x, y) = \frac{d}{dx} \left[\frac{1}{16} (10x - x^2) \right] = \frac{1}{16} (10 - 2x); \quad 0 \leq x \leq 2, y > 2$$

$$f_{X,Y}(x, y) = \frac{d}{dy} \left[\frac{1}{16} (10y - y^2) \right] = \frac{1}{16} (10 - 2y); \quad x > 2; 0 \leq y \leq 2$$

$$f_{X,Y}(x, y) = 0 \quad \text{for } x > 2, y > 2$$

$$P(0.5 \leq X \leq 1.5, 1 \leq Y \leq 1.5)$$

$$\begin{aligned}
&= \int_1^{1.5} \int_{0.5}^{1.5} \frac{1}{16} (6 - x - y) dx dy = \frac{1}{16} \int_1^{1.5} \left(6x \Big|_{0.5}^{1.5} - xy \Big|_{0.5}^{1.5} - \frac{y^2}{2} \Big|_{0.5}^{1.5} \right) dy \\
&= \frac{1}{16} \int_1^{1.5} (5 - y) dy = \frac{1}{16} \left[5y \Big|_1^{1.5} - \frac{y^2}{2} \Big|_1^{1.5} \right] \\
&= \frac{1}{16} \left[5(0.5) - \frac{1}{2} ((1.5)^2 - 1) \right] \\
&= \frac{1}{16} [2.5 - 0.625] = 0.1172
\end{aligned}$$

or

$$P(0.5 \leq X \leq 1.5, 1 \leq Y \leq 1.5)$$

$$= F_{X,Y}(1.5, 1.5) + F_{X,Y}(0.5, 1) - F_{X,Y}(1.5, 1) - F_{X,Y}(0.5, 1.5)$$

$$F_{X,Y}(1.5, 1.5) = \frac{1}{16} \left[6(1.5)(1.5) - \frac{(1.5)^2(1.5)}{2} - \frac{1.5(1.5)^2}{2} \right] = 10.125$$

$$F_{X,Y}(0.5, 1) = \frac{1}{16} \left[6(0.5)(1) - \frac{(0.5)^2(1)}{2} - \frac{(0.5)(1)^2}{2} \right] = 2.625$$

$$F_{X,Y}(1.5, 1) = \frac{1}{16} \left[6(1.5)(1) - \frac{(1.5)^2(1)}{2} - \frac{1.5(1)^2}{2} \right] = 7.125$$

$$F_{X,Y}(0.5, 1.5) = \frac{1}{16} \left[6(0.5)(1.5) - \frac{(0.5)^2(1.5)}{2} - \frac{0.5(1.5)^2}{2} \right] = 3.75$$

$$P(0.5 \leq X \leq 1.5, 1 \leq Y \leq 1.5))$$

$$= \frac{1}{16} (10.125 + 2.625 - 7.125 - 3.75) = 0.1172$$

4.80 Find the joint distribution function if $f_{X,Y}(x, y) = \begin{cases} \frac{1}{1 - e^{-a}} e^{-(x+y)}, & 0 < x < a, 0 < y < \infty \\ 0 & \text{elsewhere} \end{cases}$

Solution

Given: $f_{X,Y}(x, y) = \frac{1}{1 - e^{-a}} e^{-(x+y)} \quad 0 < x < a, 0 < y < \infty$

$$F_{X,Y}(x, y) = 0 \text{ for } x < 0, y < 0$$

$$F_{X,Y}(x, y) = \int_0^x \int_0^y \frac{1}{1 - e^{-a}} e^{-(u+v)} dv du$$

$$= \frac{1}{1 - e^{-a}} \int_0^x \int_0^y e^{-u} e^{-v} dv du$$

$$= \frac{1}{1 - e^{-a}} \left[-e^{-u} \Big|_0^x \right] \left[-e^{-v} \Big|_0^y \right]$$

$$= \frac{(1 - e^{-x})(1 - e^{-y})}{1 - e^{-a}}; 0 < x < a, 0 < y < \infty$$

$$F_{XY}(a, y) = \frac{(1 - e^{-a})(1 - e^{-y})}{1 - e^{-a}} = (1 - e^{-y})$$

$$\Rightarrow F_{X,Y}(x, y) = 1 - e^{-y} \text{ for } x \geq a \text{ and } 0 \leq y \leq \infty$$

4.81 Let X and Y be two random variables with joint pdf as

$$f_{X,Y}(x, y) = \frac{1}{2\pi\sigma^2} e^{-(x^2 + y^2)/2\sigma^2} \text{ for } -\infty < x < \infty \quad -\infty < y < \infty$$

Find $P(x^2 + y^2 \leq a^2)$.

Solution

Given: $f_{X,Y}(x,y) = \frac{1}{2\pi\sigma^2} e^{-(x^2+y^2)/2\sigma^2}$

$$P(X^2 + Y^2 \leq a^2) = ?$$

The region under consideration is a circle of radius a as shown in Fig. 4.35. Therefore, we use polar coordinates to solve the above problem.

Let $x = r \cos \theta$; $y = r \sin \theta$.

Then

$$dx dy = r dr d\theta$$

Also, θ varies from 0 to 2π , and r varies from 0 to a

$$\begin{aligned} P(X^2 + Y^2 \leq a^2) &= \int_{r=0}^a \int_{\theta=0}^{2\pi} \frac{1}{2\pi\sigma^2} e^{-r^2/2\sigma^2} r dr d\theta \\ &= \frac{1}{2\pi\sigma^2} \int_0^{a^2/2\sigma^2} \int_0^{2\pi} e^{-p} \sigma^2 dp d\theta \\ &= \frac{1}{2\pi\sigma^2} \int_0^{2\pi} \left(\int_0^{a^2/2\sigma^2} e^{-p} \sigma^2 dp \right) d\theta \\ &= \frac{1}{2\pi} \int_0^{2\pi} \left(\int_0^{a^2/2\sigma^2} e^{-p} dp \right) d\theta \\ &= \frac{1}{2\pi} \int_0^{2\pi} \left\{ (-e^{-p}) \Big|_0^{a^2/2\sigma^2} \right\} d\theta \\ &= \frac{1}{2\pi} \int_0^{2\pi} (1 - e^{-a^2/2\sigma^2}) d\theta \\ &= 1 - e^{-a^2/2\sigma^2} \end{aligned}$$

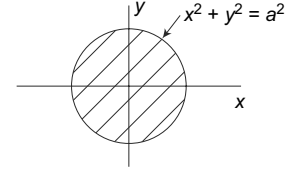


Fig. 4.35

Let $r^2/2\sigma^2 = p \Rightarrow r dr = \sigma^2 dp$
 When $r = 0$; $p = 0$; $r = a$; $p = a^2/2\sigma^2$



Practice Problem

4.27 Find whether the random variables X and Y given in Solved Problem (4.80) are independent or not.

(Ans: Independent)

Solved Problems



4.82 In tossing-a-die experiment, the random variable X denotes the number of full pairs and the random variable Y represents the remaining dots. Find the joint probability distribution and pmf. Sketch the distribution function.

Solution The outcomes of the event and their probabilities are shown below

S	1	2	3	4	5	6
X	0	1	1	2	2	3
Y	1	0	1	0	1	0
$P(X, Y)$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$

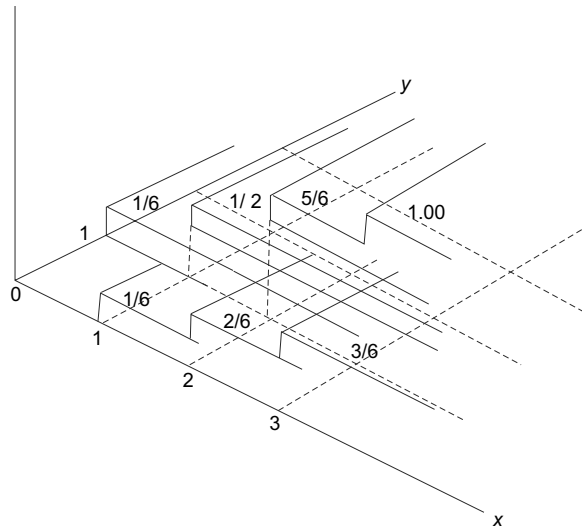


Fig. 4.36

The joint distribution function is sketched as shown in Fig. 4.36.

		Y	
		0	1
X	0	0	$\frac{1}{6}$
	1	$\frac{1}{6}$	$\frac{1}{6}$
	2	$\frac{1}{6}$	$\frac{1}{6}$
	3	$\frac{1}{6}$	0

$$P(X=0) = \frac{1}{6}$$

$$P(X=1) = \frac{1}{6} + \frac{1}{6} = \frac{1}{3}$$

$$P(X=2) = \frac{1}{6} + \frac{1}{6} = \frac{1}{3}$$

$$P(X=3) = \frac{1}{6}$$

$$P(Y=0) = \frac{1}{6} + \frac{1}{6} = \frac{1}{6} = \frac{1}{2}$$

$$P(Y=1) = \frac{1}{6} + \frac{1}{6} = \frac{1}{6} = \frac{1}{2}$$

4.83 Find the marginal pmf for the pairs of random variables with the indicated joint pmf

$X \backslash Y$	-1	0	1
-1	$\frac{1}{6}$	$\frac{1}{6}$	0
0	0	0	$\frac{1}{3}$
1	$\frac{1}{6}$	$\frac{1}{6}$	0

Find (i) $P(X \geq Y)$ (ii) $P(X = -Y)$.

Solution The expression for the joint distribution function is

$$F_{X,Y}(x,y) = \frac{1}{6} u(x+1) u(y+1) + \frac{1}{6} u(x+1) u(y) + \frac{1}{3} u(x) u(y-1) \\ + \frac{1}{6} u(x-1) u(y+1) + \frac{1}{6} u(x-1) u(y)$$

The marginal pmf $F_X(x)$ is

$$F_X(x) = F_{X,Y}(x, \infty) \\ = \frac{1}{6} u(x+1) + \frac{1}{6} u(x+1) + \frac{1}{3} u(x) + \frac{1}{6} u(x-1) + \frac{1}{6} u(x-1) \\ = \frac{1}{3} u(x+1) + \frac{1}{3} u(x) + \frac{1}{3} u(x-1)$$

$$F_Y(y) = F_{X,Y}(\infty, y) \\ = \frac{1}{6} u(y+1) + \frac{1}{6} u(y) + \frac{1}{3} u(y-1) + \frac{1}{6} u(y+1) + \frac{1}{6} u(y) \\ = \frac{1}{3} u(y-1) + \frac{1}{3} u(y) + \frac{1}{3} u(y+1)$$

$$P(X \geq Y) = P(X=-1, Y=-1) + P(X=1, Y=-1) + P(X=1, Y=1) \\ = \frac{1}{6} + \frac{1}{6} + \frac{1}{6} = \frac{1}{2}$$

$$P(X = -Y) = P(X = 0, Y = 0) + P(X = -1, Y = 1) + P(X = 1, Y = -1) = 0 + 0 + \frac{1}{6} = \frac{1}{6}.$$

4.84 If the joint pdf of a two-dimensional random variable (X, Y) is given by

$$f_{X,Y}(x, y) = \begin{cases} k(6 - x - y), & 0 < x < 2, 2 < y < 4 \\ 0 & \text{otherwise} \end{cases}$$

Find (a) the value of k , (b) $P(X < 1, Y < 3)$, (c) $P(X + Y < 3)$, and (d) $P(X < 1/Y < 3)$.

Solution

We know
$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx dy = 1$$

$$\Rightarrow \int_2^4 \int_0^2 k(6 - x - y) dx dy = 1$$

$$k \left[\int_2^4 \left(6x - \frac{x^2}{2} - yx \right) \Big|_0^2 dy \right] = 1$$

$$k \left[\int_2^4 (12 - 2 - 2y) dy \right] = 1$$

$$k \int_2^4 (10 - 2y) dy = 1$$

$$k [10(4 - 2) - y^2 \Big|_2^4] = 1$$

$$k(20 - 12) = 1$$

$$k = \frac{1}{8}$$

$$\begin{aligned} P(X < 1, Y < 3) &= \int_{-\infty}^1 \int_{-\infty}^3 f_{X,Y}(x, y) dy dx \\ &= \int_0^1 \int_2^3 \frac{1}{8} (6 - x - y) dy dx \\ &= \frac{1}{8} \int_0^1 \left\{ 6y \Big|_2^3 - xy \Big|_2^3 - \frac{y^2}{2} \Big|_2^3 \right\} dx \\ &= \frac{1}{8} \int_0^1 \left[6 - x - \frac{5}{2} \right] dx \end{aligned}$$

$$\begin{aligned}
 &= \frac{1}{8} \int_0^1 \left(\frac{7}{2} - x \right) dx = \frac{1}{8} \left(\frac{7}{2} x \Big|_0^1 - \frac{x^2}{2} \Big|_0^1 \right) \\
 &= \frac{1}{8} \left[\frac{7}{2} - \frac{1}{2} \right] = \frac{3}{8}
 \end{aligned}$$

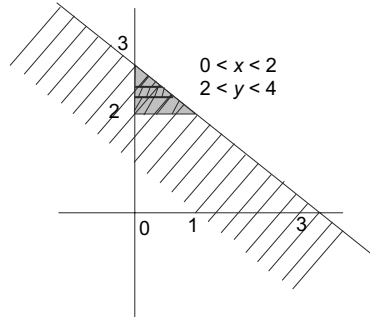


Fig. 4.37

$$P(X + Y < 3) = \int_2^3 \int_0^{3-y} f_{X,Y}(x, y) dx dy \quad (\text{See Fig. 4.37 to find the interval of integration})$$

$$\begin{aligned}
 &= \int_2^3 \int_0^{3-y} \frac{1}{8} (6 - x - y) dx dy \\
 &= \frac{1}{8} \int_2^3 \left\{ 6x \Big|_0^{3-y} - \frac{x^2}{2} \Big|_0^{3-y} - yx \Big|_0^{3-y} \right\} dy \\
 &= \frac{1}{8} \int_2^3 \left\{ 6(3-y) - \frac{1}{2}(3-y)^2 - y(3-y) \right\} dy \\
 &= \frac{1}{8} \int_2^3 \left\{ 18 - 6y - \frac{1}{2}(9 + y^2 - 6y) - 3y + y^2 \right\} dy \\
 &= \frac{1}{8} \int_2^3 \left\{ 18 - 6y - \frac{9}{2} - \frac{y^2}{2} + 3y - 3y + y^2 \right\} dy \\
 &= \frac{1}{8} \int_2^3 \left(\frac{y^2}{2} - 6y + \frac{27}{2} \right) dy = \frac{1}{8} \left(\frac{y^3}{6} - 3y^2 \Big|_2^3 + \frac{27}{2} y \Big|_2^3 \right) \\
 &= \frac{1}{8} \left(\frac{19}{6} - 15 + \frac{27}{2} \right) = \frac{1}{8} \left(\frac{19 - 90 + 81}{6} \right) = \frac{5}{24}
 \end{aligned}$$

$$(iv) P(X < 1 | Y < 3) = \frac{P(X < 1, Y < 3)}{P(Y < 3)}$$

$$P(Y < 3) = \int_2^3 f_Y(y) dy$$

$$f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx$$

$$\begin{aligned} &= \int_0^2 \frac{1}{8} (6 - x - y) dx = \frac{1}{8} \left[6x \Big|_0^2 - \frac{x^2}{2} \Big|_0^2 - yx \Big|_0^2 \right] \\ &= \frac{1}{8} (12 - 2 - 2y) = \frac{(5 - y)}{4} \end{aligned}$$

$$P(Y < 3) = \int_2^3 f_Y(y) dy = \int_2^3 \frac{5 - y}{4} dy = \frac{1}{4} \int_2^3 (5 - y) dy$$

$$= \frac{1}{4} \left[5y \Big|_2^3 - \frac{y^2}{2} \Big|_2^3 \right]$$

$$= \frac{1}{4} \left[5 - \frac{5}{2} \right] = \frac{5}{8}$$

$$P(X < 1 | Y < 3) = \frac{P(X < 1, Y < 3)}{P(Y < 3)}$$

$$= \frac{3/8}{5/8} = \frac{3}{5}$$

4.85 If the joint density function of two random variables X and Y are

$$f_{X,Y}(x, y) = e^{-(x+y)}, x \geq 0, y \geq 0$$

= 0 otherwise

Find (a) $P(X < 1)$ (b) $P(X + Y < 1)$.

Solution The marginal density function of X is given by

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dy$$

$$f_X(x) = \int_0^{\infty} e^{-(x+y)} dy = e^{-x} \int_0^{\infty} e^{-y} dy$$

$$= e^{-x} (e^{-y}) \Big|_0^{\infty} = e^{-x} \quad x \geq 0$$

$$f_X(x) = e^{-x}, x \geq 0$$

$$\begin{aligned} P(X < 1) &= \int_0^1 f_X(x) dx = \int_0^1 e^{-x} dx = -e^{-x} \Big|_0^1 \\ &= 1 - e^{-1} = 0.632 \end{aligned}$$

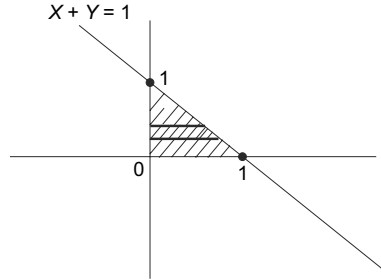


Fig. 4.38

Using Fig. 4.38, we can find

$$\begin{aligned} P(X + Y < 1) &= \int_0^1 \int_0^{1-y} e^{-(x+y)} dx dy \\ &= \int_0^1 e^{-y} \left[\int_0^{1-y} e^{-x} dx \right] dy \\ &= \int_0^1 e^{-y} (1 - e^{-(1-y)}) dy \\ &= \int_0^1 (e^{-y} - e^{-1}) dy \\ &= -e^{-y} \Big|_0^1 - e^{-1} y \Big|_0^1 \\ &= (1 - e^{-1}) - e^{-1} = 0.264 \end{aligned}$$

4.86 The joint probability density function of two random variables X and Y is given by

$$f_{X,Y}(x, y) = \begin{cases} a(2x + y^2), & 0 \leq x \leq 2, \quad 2 \leq y \leq 4 \\ 0 & \text{elsewhere} \end{cases}$$

Find (a) the value of a , and (b) $P(x < 1, y > 3)$.

Solution

$$\text{Given: } f_{X,Y}(x, y) = \begin{cases} a(2x + y^2), & 0 \leq x \leq 2, \quad 2 \leq y \leq 4 \\ 0 & \text{otherwise} \end{cases}$$

We have

$$\begin{aligned} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx dy &= 1 \\ \Rightarrow \int_2^4 \int_0^2 a(2x + y^2) dx dy &= 1 \\ a \int_2^4 \left[x^2 \Big|_0^2 + y^2 x \Big|_0^2 \right] dy &= 1 \\ a \int_2^4 (4 + 2y^2) dy &= 1 \\ a \left\{ 4y \Big|_2^4 + 2 \frac{y^3}{3} \Big|_2^4 \right\} &= 1 \\ a \left\{ 4(4 - 2) + \frac{2}{3}(4^3 - 2^3) \right\} &= 1 \\ a \left\{ 8 + \frac{2}{3}(56) \right\} &= 1 \end{aligned}$$

$$\begin{aligned} \Rightarrow a &= \frac{3}{136} \\ P(X \leq 1, Y > 3) &= \int_0^1 \int_3^4 \frac{3}{136} (2x + y^2) dy dx \\ &= \frac{3}{136} \left\{ \int_0^1 \left[2xy \Big|_3^4 + \frac{y^3}{3} \Big|_3^4 \right] dx \right\} \\ &= \frac{3}{136} \left[\int_0^1 \left(2x + \frac{37}{3} \right) dx \right] \\ &= \frac{3}{136} \left[x^2 \Big|_0^1 + \frac{37}{3} x \Big|_0^1 \right] \\ &= \frac{3}{136} \left(\frac{40}{3} \right) = \frac{5}{17} \end{aligned}$$

4.87 Let X and Y be jointly continuous random variables with joint density function

$$f_{X,Y}(x,y) = \begin{cases} xy e^{-(x^2+y^2)/2} & x > 0, y > 0 \\ 0 & \text{otherwise} \end{cases}$$

Check whether X and Y are independent.

Find (a) $P(X \leq 1, Y \leq 1)$, and (b) $P(X + Y \leq 1)$.

Solution

Given $f_{X,Y}(x,y) = \begin{cases} xy e^{-(x^2+y^2)/2} & x > 0, y > 0 \\ 0 & \text{otherwise} \end{cases}$

The marginal pdf of X is

$$\begin{aligned} f_X(x) &= \int_{-\infty}^{\infty} f_{X,Y}(x,y) dy \\ &= \int_0^{\infty} xy e^{-(x^2+y^2)/2} dy \\ &= x e^{-x^2/2} \int_0^{\infty} y e^{-y^2/2} dy \\ \int_0^{\infty} y e^{-y^2/2} dy &= ? \end{aligned}$$

Let $\frac{y^2}{2} = t \quad \Rightarrow y dy = dt$

$$\Rightarrow \int_0^{\infty} y e^{-y^2/2} dy = \int_0^{\infty} e^{-t} dt = -e^{-t} \Big|_0^{\infty} = 1$$

$$\Rightarrow \boxed{\int_0^{\infty} y e^{-y^2/2} dy = 1}$$

$$\Rightarrow f_X(x) = x e^{-x^2/2}$$

Similarly, the marginal pdf of Y is

$$\begin{aligned} f_Y(y) &= \int_{-\infty}^{\infty} f_{X,Y}(x,y) dx \\ &= \int_0^{\infty} y e^{-y^2/2} dx x e^{-x^2/2} dx \end{aligned}$$

$$= y e^{-y^2/2} \int_0^{\infty} x e^{-x^2/2} dx = y e^{-y^2/2}$$

Since $f_{X,Y}(x, y) = f_X(x) f_Y(y)$, X and Y are independent,

$$\begin{aligned} P(X \leq 1, Y \leq 1) &= \int_0^1 \int_0^1 x y e^{-(x^2+y^2)/2} dx dy \\ &= \int_0^1 x e^{-x^2/2} dx \int_0^1 y e^{-y^2/2} dy \end{aligned}$$

$$\int_0^1 x e^{-x^2/2} dx = ?$$

Let $x^2/2 = t \Rightarrow x dx = dt$

$$\Rightarrow \int_0^1 x e^{-x^2/2} dx = \int_0^{1/2} e^{-t} dt = -e^{-t} \Big|_0^{1/2} = (1 - e^{-1/2})$$

$$\text{Similarly, } \int_0^1 y e^{-y^2/2} dy = 1 - e^{-1/2}$$

$$P(X \leq 1, Y \leq 1) = (1 - e^{-1/2})(1 - e^{-1/2})$$

when	$x = 0;$	$t = 0$
when	$x = 1$	$t = \frac{1}{2}$

4.88 Two 10-ohm statistically independent and uniformly distributed resistors with 10% tolerance are connected in series in a circuit. Find the probability that the net resistance is less than or equal to 21 ohms.

Solution Let the two resistors be R_1 and R_2 . The net resistance is equal to $R = R_1 + R_2$.

R_1 and R_2 are uniformly distributed with 10% tolerance. That is,

$$f_{R_1}(r_1) = 0.5 \quad \text{for } 9 < r_1 < 11$$

$$f_{R_2}(r_2) = 0.5 \quad \text{for } 9 < r_2 < 11$$

$$P(R_1 + R_2 \leq 21) = 1 - P(R_1 + R_2 > 21)$$

$$= 1 - \iint_{R_1 + R_2 > 21} f_{R_1}(r_1) f_{R_2}(r_2) dr_1 dr_2$$

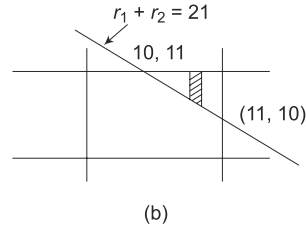
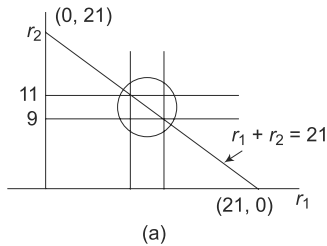


Fig. 4.39

The interval of integration can be obtained from Fig. 4.39(b)

$$\begin{aligned}
 P(R_1 + R_2 > 21) &= \int_{10}^{11} \int_{21-r_1}^{11} (0.5)(0.5) dr_2 dr_1 \\
 &= 0.25 \int_{10}^{11} (11 - 21 + r_1) dr_1 \\
 &= 0.25 \int_{10}^{11} (r_1 - 10) dr_1 = 0.25 \left[\frac{r_1^2}{2} - 10r_1 \right]_{10}^{11} \\
 &= 0.25 \left[\frac{21}{2} - 10 \right] = 0.125
 \end{aligned}$$

$$P(R_1 + R_2 \leq 21) = 1 - 0.125 = 0.875$$

4.89 In an experiment of tossing a fair coin three times, (X, Y) is a random variable, where X denotes number of heads on the first two tosses and Y denotes the number of heads on the third toss.

- Find the range of X .
- Find the range of Y .
- Find the range of (X, Y) .
- Find (i) $P(X \leq 2, Y \leq 1)$ (ii) $P(X \leq 1, Y \leq 1)$ and (iii) $P(X \leq 0, Y \leq 0)$.

Solution The sample space of the experiment is $S = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$

- Since X denotes the number of heads in the first two tosses, the range of X is

$$R_X = \{0, 1, 2\}$$

- Since Y denotes number of heads on the third toss,

$$R_Y = \{0, 1\}$$

- The range of (X, Y) is $\{(0, 0), (0, 1), (1, 0), (1, 1), (2, 0), (2, 1)\}$

- (i) $P(X \leq 2, Y \leq 1) = ?$

From the sample space, we can find each and every event satisfying the joint event $(X \leq 2, Y \leq 1)$.

Therefore, $P(X \leq 2, Y \leq 1) = 1$.

- (ii) $P(X \leq 1, Y \leq 1)$

The events that satisfy the above joint event are

$(HTH, HTT, THH, THT, TTH, TTT)$

$$\text{Therefore, } P(X \leq 1, Y \leq 1) = \frac{6}{8} = \frac{3}{4}$$

- (iii) $P(X \leq 0, Y \leq 0)$

The event that satisfies the above joint event is (TTT)

$$\text{Therefore, } P(X \leq 0, Y \leq 0) = \frac{1}{8}$$

4.90 Two random variables X and Y have the joint pdf

$$f_{X,Y}(x,y) = xye^{-y^2/4} \quad \begin{array}{l} 0 \leq x \leq 1 \\ y \geq 0 \end{array}$$

Find the marginal pdfs of X and Y . Determine if X and Y are statistically independent.

Solution Given:

$$f_{X,Y}(x,y) = xye^{-y^2/4} \quad \begin{array}{l} 0 \leq x \leq 1 \\ y \geq 0 \end{array}$$

The marginal pdf is given by

$$f_X(x) = \int_0^{\infty} xye^{-y^2/4} dy$$

$$\begin{array}{l} \text{let } \frac{y^2}{4} = t \\ ydy = 2dt \end{array}$$

$$f_X(x) = x \int_0^{\infty} ye^{-y^2/4} dy = x \int_0^{\infty} 2e^{-t} dt = -2x e^{-t} \Big|_0^{\infty} = 2x$$

$$f_Y(y) = \int_0^1 xye^{-y^2/4} dx$$

$$= ye^{-y^2/4} \int_0^1 x dx = ye^{-y^2/4} \frac{x^2}{2} \Big|_0^1 = \frac{y}{2} e^{-y^2/4}$$

$$= f_{X,Y}(x,y) = xye^{-y^2/4}$$

Since, $f_{X,Y}(x,y) = f_X(x)f_Y(y)$,

The random variables X and Y are statistically independent.

4.91 In a communication system, a two-dimensional signal (X, Y) is given by

$$X = r \cos(2\pi\theta/4); Y = r \sin(2\pi\theta/4)$$

where θ is a discrete uniform random variable in the set. $\{0, 1, 2, 3\}$

- Find (a) joint pmf of X and Y .
- marginal pmf of X and Y .
- $P(X=0)$, $P(X < r)$.

Solution Substitute $\theta = 0, 1, 2, 3$, and tabulate the values of (X, Y)

θ	0	1	2	3
(X, Y)	$(r, 0)$	$(0, r)$	$(-r, 0)$	$(0, -r)$

The joint pmf matrix is shown below.

X \ Y	Y		
	-r	0	r
-r	0	$\frac{1}{4}$	0
0	$\frac{1}{4}$	0	$\frac{1}{4}$
r	0	$\frac{1}{4}$	0

$$P_X(-r) = \frac{1}{4}; P_X(0) = \frac{1}{2}; P_X(r) = \frac{1}{4}$$

$$P_Y(-r) = \frac{1}{4}; P_Y(0) = \frac{1}{2}; P_Y(r) = \frac{1}{4}$$

$$P(X=0) = \frac{1}{2}; P(X < r) = \frac{3}{4}$$

REVIEW QUESTIONS



19. Describe vector random variables.
20. Define and explain joint distribution function and joint density function of two random variables X and Y .
21. Distinguish between joint distribution and marginal distribution functions.
22. Explain the properties of joint distribution function.
23. Define joint probability mass function.
24. Define marginal pmf.
25. Write a note on joint probability matrix.
26. Define conditional distribution and density function of two random variables X and Y .
27. Explain the properties of conditional density function.
28. Define pmf of N random variables.
29. What is the condition for statistical independence of random variables X and Y ?
30. What is the probability distribution function of sum of two random variables?
31. Explain the method of finding the distribution and density functions for a sum of statistically independent random variables.
32. Find the pdf of sum of several random variables.
33. State and prove the central limit theorem.



EXERCISES

Problems

1. Two jointly exponential random variables X and Y have the joint distribution function

$$F_{X,Y}(x, y) = \begin{cases} 1 - e^{-4x} - e^{-3y} + e^{-(4x+3y)} & x > 0, y \geq 0 \\ \text{otherwise} & \end{cases}$$

- (a) Find $f_X(x)$ and $f_Y(y)$
- (b) Find $P(X \leq 0.5, Y \leq 1)$.

$$(Ans. (a) f_X(x) = \begin{cases} 4e^{-4x} & x \geq 0 \\ 0 & x < 0 \end{cases}, f_Y(y) = \begin{cases} 3e^{-3y} & y \geq 0 \\ 0 & y < 0 \end{cases}; (b) 0.8216)$$

2. The joint pmf of random variables X and Y is given by the matrix

$X \backslash Y$	1	2	3
1	$\frac{1}{4}$	$\frac{1}{8}$	0
2	$\frac{1}{8}$	0	$\frac{1}{8}$
3	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$

$$\left(\begin{array}{l} Ans: P_X(x=1) = \frac{5}{12}; P_X(x=2) = \frac{1}{4} P_X(x=3) = \frac{3}{8} \\ P_Y(y=1) = \frac{1}{2}; P_Y(y=2) = \frac{1}{4}; P_Y(y=3) = \frac{1}{4} \end{array} \right)$$

Find marginal pmfs of X and Y .

3. Two random variables X and Y have the joint pdf

$$f_{X,Y}(x, y) = \begin{cases} 2xy & 0 \leq x \leq 1, 0 \leq y \leq \sqrt{2} \\ 0 & \text{elsewhere} \end{cases}$$

Find (a) $P(X \leq Y)$; (b) $P(X \geq Y)$

4. Find $f_X(x)$ and $f_Y(y)$ if

$$f_{X,Y}(x, y) = \begin{cases} \frac{1}{2} e^{-2y} & 0 \leq x \leq 4, y > 0 \\ 0 & \text{elsewhere} \end{cases}$$

$$\left(\begin{array}{l} Ans: f_X(x) = \frac{1}{4} 0 \leq x \leq 4 \\ f_Y(y) = 2e^{-2y}, y \geq 0 \end{array} \right)$$

5. If the joint pdf of X and Y is

$$f_{X,Y}(x, y) = \begin{cases} e^{-(x+y)} & x > 0, y > 0 \\ 0 & \text{otherwise} \end{cases}$$

Check whether X and Y are independent.

(Ans: Independent)

6. The joint pdf of random variables X and Y is given by

$$f_{X,Y}(x, y) = \begin{cases} 8xy/9, & 1 \leq x \leq y \leq 2 \\ 0 & \text{otherwise} \end{cases}$$

$$\left(Ans: \frac{2y}{4-x^2}, x \leq y \leq 2 \right)$$

Find the conditional density functions of Y given $X = x$.

7. If $X_1, X_2, X_3, \dots, X_n$ are uniform random variables with means = 2.5 and variance = 3/4, use CLT to estimate $P(108 \leq S_n \leq 126)$, where $S_n = X_1 + X_2 + X_3 + \dots + X_n, n = 48$. (Ans: 0.9275)

8. X and Y are independent with a common pdf

$$f_X(x) = \begin{cases} e^{-x} & x \geq 0 \\ 0 & x < 0 \end{cases} \quad \text{and} \quad f_Y(y) = \begin{cases} e^{-y} & y \geq 0 \\ 0 & y < 0 \end{cases} \quad \left(Ans: f_Z(z) = \begin{cases} ze^{-z} & \text{for } z \geq 0 \\ 0 & \text{elsewhere} \end{cases} \right)$$

Find the pdf for $Z = X + Y$.

9. The joint pdf of a bivariate random variable (X, Y) is given as

$$f_{X,Y}(x, y) = \begin{cases} \frac{kx}{y} & 1 < x < 2, 1 < y < 2 \\ 0 & \text{otherwise} \end{cases}$$

$$\left(Ans: \frac{2}{3 \ln 2}, \text{Independent} \right)$$

where k is a constant.

- (a) Determine k .
 (b) Are X and Y independent?
10. Two fair dice are rolled. Find the joint pmf of X and Y when X is the smallest and Y is the largest value obtained on the dice.

11. The joint density function of X and Y is

$$f_{X,Y}(x, y) = C(x^2 - y^2)e^{-x}; 0 \leq x < \infty, -x \leq y \leq x$$

Find the conditional distribution function of Y given $X = x$.

$$\left(\text{Ans. } \frac{1}{2} + \frac{3}{4} \frac{y}{x} - \frac{1}{4} \frac{y^3}{x^3} \right)$$

12. The joint density function of (X, Y) is

$$f_{X,Y}(x, y) = \begin{cases} b(x+y)^2 & \text{for } -2 < x < 2 \text{ and } -3 < y < 3 \\ 0 & \text{elsewhere} \end{cases}$$

Find $b, f_X(x)$ and $f_Y(y)$

$$\left(\text{Ans. } b = \frac{1}{104} \right)$$

13. If the joint density function of (X, Y) is

$$f_{X,Y} = \begin{cases} \frac{5}{6} x^2 y & \text{for } 0 < y < x < 2 \\ 0 & \text{elsewhere} \end{cases}$$

$$\text{Find } f_X(x), f_Y(y). \quad \left(\text{Ans. } f_X(x) = \begin{cases} \frac{5}{32} x^4, & 0 < x < 2 \\ 0 & \text{elsewhere} \end{cases} \quad f_Y(y) = \begin{cases} \frac{5}{16} y \left(\frac{8-y^3}{3} \right), & 0 < y < x \\ 0 & \text{elsewhere} \end{cases} \right)$$

14. Determine the constant b such that the function

$$f(x, y) = 3xy \text{ for } 0 < x < 1, 0 < y < b \text{ else } 0, \text{ is a valid joint density function.} \quad \left(\text{Ans. } \frac{2}{\sqrt{3}} \right)$$

15. Find the conditional density functions for the joint distribution

$$f_{X,Y}(x, y) = 4xy e^{-(x^2+y^2)} u(x) u(y)$$

16. Find the marginal densities of X and Y using the joint density

$$f_{X,Y}(x, y) = 2u(x) u(y) \exp \left[-\left(4y + \frac{x}{2} \right) \right]. \quad \left(\text{Ans. } f_X(x) = \frac{1}{2} e^{-x/2} u(x); f_Y(y) = 4e^{-4y} u(y) \right)$$

17. The joint pdf is $f_{X,Y}(x, y) = \frac{1}{18} e^{-(x/6+y/3)}$ for $x \geq 0, y \geq 0$.

Show that X and Y are statistically independent random variables

19. Let X and Y be the two random variables with joint pdf given by

$$f_{X,Y}(x, y) = \begin{cases} e^{-(x+y)} & ; x, y > 0 \\ 0 & \text{otherwise} \end{cases}$$

(a) Find the marginal densities of X and Y .

(b) Are they independent?

(c) Find $P(X \leq 1, Y \leq 1)$ and $P(X + Y \leq 1)$

$$(\text{Ans. (a) } f_X(x) = e^{-x}, x > 0; f_Y(y) = e^{-y}, y > 0; \text{(b) Independent; (c) } (1 - e^{-1})^2, (1 - 2e^{-1}))$$

20. The joint pdf of (X, Y) is

$$f_{X,Y}(x, y) = ye^{-x/2} e^{-y^2}; x > 0; y > 0$$

Find $f_X(x), f_Y(y)$. (Ans. $\frac{1}{2} e^{-x/2}; 2y e^{-y^2}$)

21. Let X and Y be independent random variables that are confirmly distributed in $(0, 1)$. Find

(i) $P\left(XY < \frac{1}{2}\right)$ (ii) $P\left(\max(X, Y) < \frac{1}{3}\right)$

22. A coin is tossed ten times. What is the probability of getting 3 or 4 or 5 heads? Use central limit theorem. (Ans. 0.5684)

23. In a circuit, three independent resistors are committed in series. The resistors have uniform distribution function in the range 5.95 to 6.05 ohms. Find the resistance of the circuit using normal approximation.

24. Two independent random variables X and Y have pdfs.

$$f_X(x) = \begin{cases} \frac{3}{32} (4 - x^2) & -2 \leq x \leq 2 \\ 0 & \text{elsewhere} \end{cases}$$

$$f_Y(y) = \frac{1}{2} [u(y+1) - u(y-1)]$$

Find pdf of $Z = X + Y$.

25. Two independent random variables X and Y have respective densities

$$f_X(x) = 5e^{-5x} u(x)$$

$$f_Y(y) = 3e^{-3y} u(y)$$

$$\left(\text{Ans: } f_Z(z) = \begin{cases} \frac{15}{2} (e^{-3z} - e^{-5z}), & z \geq 0 \\ 0 & \text{elsewhere} \end{cases} \right)$$

Find the density of $Z = X + Y$.

26. Two discrete random variables X and Y have joint pmf given by the following table

$X \downarrow$	1	2	3	$Y \leftarrow$
1	1/12	1/6	1/12	
2	1/6	1/4	1/12	
3	1/12	1/12	0	

Compute the probability of each of the following events: (a) $X \leq \frac{3}{2}$, (b) X, Y is even. (c) Y is even given that X is even.

27. Let X and Y be jointly continuous random variables with joint density function

$$f(x, y) = \begin{cases} xy \exp(-1/2(x^2 + y^2)), & x > 0, y > 0 \\ 0 & \text{otherwise} \end{cases}$$

Check whether X and Y are independent. Find (a) $P(X \leq 1, Y \leq 1)$ and $(X + Y \leq 1)$

28. If $f_{X,Y}(X, Y) = 0.5 \exp(-|X| - |Y|)$, where X and y are two random variables, if $Z = X + Y$, find $f_Z(Z)$.

29. If two random variables have the joint probability density

$$f(x_1, x_2) = \begin{cases} \frac{2}{3}(x_1 + 2x_2) & \text{for } 0 < x_1 < 1, 0 < x_2 < 1 \\ 0 & \text{elsewhere} \end{cases}$$

Find

(a) the marginal density of x_2

(b) conditional density of the first given that the second takes on the value x_2 .

30. Let X and Y be jointly continuous random variables with joint density function

$$f(x, y) = \begin{cases} xy \exp[-(x^2 + y^2)] & ; \quad x > 0, y > 0 \\ 0 & \text{otherwise} \end{cases}$$

Check whether X and Y are independent. Find

(a) $P(X \leq 1, Y < 1)$ and

(b) $P(X + Y < 1)$

31. If the function $f(x, y) = \begin{cases} be^{-2x} \cos(y/2) & 0 \leq x \leq 1; \quad 0 \leq y \leq \pi \\ 0 & \text{elsewhere} \end{cases}$, $\left(\text{Ans: } b = \frac{1}{1 - e^{-2}} \right)$

where ' b ' is a positive constant, is a valid joint probability density function, find b .

32. The joint probability density function of two random variables X and Y is given by $f(x, y) = \begin{cases} C(2x + y) & 0 \leq x \leq 1, 0 \leq y \leq 2 \\ 0 & \text{elsewhere} \end{cases}$.

Find (a) the value of C

$$\left(\text{Ans: } C = \frac{1}{4}; f_X(x) = \begin{cases} x + \frac{1}{2} & \text{for } 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases} \quad f_Y(y) = \begin{cases} \frac{1}{4}(1 + y) & \text{for } 0 \leq y \leq 2 \\ 0 & \text{elsewhere} \end{cases} \right)$$

(b) marginal distribution functions of X and Y .

33. The joint probability density function of two random variables X and Y is given by

$$f(x, y) = \begin{cases} a(2x + y^2) & 0 \leq x \leq 2, \quad 2 \leq y \leq 4 \\ 0 & \text{elsewhere} \end{cases}$$

Find (a) value of a

(b) $P(X \leq 1, Y > 3)$.

34. Discrete random variables X and Y have a joint distribution function

$$F_{X,Y}(x, y) = 0.1u(x + 4)u(y - 1) + 0.15u(x + 3)u(y + 5) + 0.17u(x + 1)u(y - 3) \\ + 0.05u(x)u(y - 1) + 0.18u(x - 2)u(y + 2) + 0.23u(x - 3)u(y - 4) \\ + 0.12u(x - 4)u(y + 3).$$

Find

(a) sketch $F_{X,Y}(x, y)$

(b) marginal distribution functions of X and Y

(c) $P(-1 < X \leq 4, -3 < Y \leq 3)$

(d) $P(X < 1, Y \leq 2)$.

35. Given the function $f(x, y) = \begin{cases} (x^2 + y^2) / 8\pi & x^2 + y^2 < b \\ 0 & \text{elsewhere} \end{cases}$

- (a) Find the constant b so that this is a valid joint density function.
 (b) Find $P(0.5b < X^2 + Y^2 < 0.8b)$. (Ans: $b = 4, 0.39$)
36. A random vector (X, Y) has pdf given by

$$f_{X,Y}(x, y) = k$$
 in the region given in Fig. 4.40.

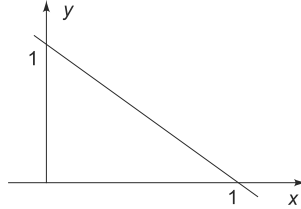


Fig. 4.40

- Find $k, f_X(x), f_Y(y)$ [Ans: $2; 2(1-x); 0 < x < 1, 2(1-y); 0 < y < 1$]
37. In a communication system, two signals (X, Y) are transmitted using the relation

$$X = r \cos(2\pi\theta/6); Y = Y \sin(2\pi\theta/6)$$
 where $\theta = \{0, 1, 2, \dots, 5\}$
 (a) Find the joint pmf of X and Y
 (b) Find the marginal pmf of X and of Y .

Multiple-Choice Questions

1. If $f_{X,Y}(x, y) = \begin{cases} \frac{1}{8}(6 - x - y) & ; \quad 0 < x < 2, 2 < y < 4 \\ 0 & \text{elsewhere} \end{cases}$
 What is the value of $P(X + Y < 3)$?
 (a) $\frac{5}{12}$ (b) $\frac{5}{24}$ (c) $\frac{5}{6}$ (d) $\frac{5}{36}$
2. The joint pdf of (X, Y) is $f_{X,Y}(x, y) = \begin{cases} 1 - (x + y), & 0 < x < 2, 0 < y < 2 \\ 0 & \text{elsewhere} \end{cases}$
 (a) $\frac{1}{4}$ (b) $\frac{1}{2}$ (c) $\frac{1}{8}$ (d) $\frac{1}{16}$
3. The joint pdf of (X, Y) is

$$f_{X,Y}(x, y) = bxy \begin{cases} 0 \leq x \leq 0.5 & \text{and } 0 \leq y \leq 2 \\ 0 & \text{elsewhere} \end{cases}$$
 The value of b for valid pdf is
 (a) 12 (b) 1 (c) 9 (d) 4
4. $P\{x_1 < X \leq x_2, y_1 < Y \leq y_2\} =$
 (a) $F(x_2, y_2) + F(x_1, y_1) - F(x_1, y_2) - (F(x_2, y_1))$
 (b) $F(x_2, y_2) - F(x_1, y_1) + F(x_1, y_2) - (F(x_2, y_1))$

- (c) $F(x_2, y_2) + F(x_1, y_1) + F(x_1, y_2) + (F(x_2, y_1))$
 (d) $F(x_2, y_2) - F(x_1, y_1) - F(x_1, y_2) + (F(x_2, y_1))$
5. The marginal CDF of X is given by
 (a) $F_X(x) = F_{X,Y}(\infty, y)$
 (b) $F_X(x) = F_{X,Y}(x, \infty)$
 (c) $F_X(x) = F_{X,Y}(\infty, Y) + F_{X,Y}(x, \infty)$
 (d) None of the above
6. The marginal pdf of $f_X(x)$ given $f_{X,Y}(x, y) = \begin{cases} 2xy, & 0 < x \leq 1, & 0 \leq y \leq \sqrt{2} \\ 0 & \text{elsewhere} \end{cases}$ is
 (a) $f_X(x) = \begin{cases} 2x & 0 \leq x \leq 2 \\ 0 & \text{elsewhere} \end{cases}$ (b) $f_X(x) = \begin{cases} 2x & 0 < x \leq 1 \\ 0 & \text{elsewhere} \end{cases}$
 (c) $f_X(x) = \begin{cases} x & 0 < x \leq \sqrt{2} \\ 0 & \text{elsewhere} \end{cases}$ (d) $f_X(x) = \begin{cases} x/2 & 0 \leq x \leq 4 \\ 0 & \text{elsewhere} \end{cases}$
7. The joint density function of X and Y is given by

$$f_{X,Y}(x, y) = \begin{cases} 2e^{-x}e^{-2y} & 0 < x < \infty, 0 < y < \infty \\ 0 & \text{otherwise} \end{cases}$$

 The value of $P(X < Y)$ is
 (a) $\frac{2}{3}$ (b) $\frac{1}{9}$ (c) $\frac{1}{6}$ (d) $\frac{1}{3}$
8. The random variables X, Y and Z are uniformly distributed over $(0, 1)$, the value of $P(X \geq YZ)$
 (a) $\frac{1}{4}$ (b) $\frac{3}{4}$ (c) $\frac{34}{4}$ (d) $\frac{5}{4}$
9. X and Y have the joint pdf given by
- | | | | |
|---|-----|-----|-----|
| $\begin{matrix} & Y \\ X \backslash & 0 & 1 & 2 \end{matrix}$ | 0 | 1 | 2 |
| 0 | 0.1 | 0.4 | 0.1 |
| 1 | 0.2 | 0.2 | 0 |
- The value of $P(Y = 1/X = 1)$
 (a) 0.3 (b) $\frac{1}{3}$ (c) $\frac{1}{2}$ (d) $\frac{1}{4}$
10. If $f_{X,Y}(x, y) = \begin{cases} \frac{1}{\pi} & x^2 + y^2 \leq 1 \\ 0 & \text{otherwise} \end{cases}$
 $f_X(x) = ?$
 (a) $\frac{2}{\pi} \sqrt{1-x^2}; 0 \leq x \leq 1$ (b) $\frac{2}{\pi^2} \sqrt{1-x^2}; 0 \leq x \leq \frac{1}{\sqrt{2}}$

(c) $\frac{2}{\pi} \sqrt{x^2 - 1}; 0 \leq x \leq 1$

(d) $\frac{2}{\pi} \sqrt{1 + x^2} \quad 0 \leq x \leq 1$

11. If $f_{X,Y}(x, y) = \begin{cases} 2e^{-(x+y)}, & x \geq 0, y \geq x \\ 0 & \text{elsewhere} \end{cases}$

The value of $P(Y < 2X)$ is

(a) $\frac{1}{4}$

(b) $\frac{1}{2}$

(c) $\frac{1}{3}$

(d) $\frac{1}{5}$

12. If the joint pdf of (X, Y) is

$$f_{X,Y}(x, y) = \begin{cases} \frac{1}{24}(x+y), & 0 < x < 3, \quad x < y < x+2 \\ 0 & \text{elsewhere} \end{cases}$$

The value of $P(Y > 2)$ is

(a) $\frac{5}{6}$

(b) $\frac{1}{9}$

(c) $\frac{7}{18}$

(d) $\frac{5}{9}$

13. If $f_X(x)$ and $f_Y(y)$ are pdfs of two random variables, X and Y then the pdf of $Z = X + Y$ is

(a) $f_Z(z) = \int_{-\infty}^{\infty} f_Y(y) f_X(z-y) dy$

(b) $f_Z(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z-x) dx$

(c) $f_Z(z) = \int_{-\infty}^{\infty} f_Y(y) f_X(z-y) dy$

(d) $f_Z(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(x-y) dx$

14. The pdfs of X and Y are shown in Fig. 4.41.

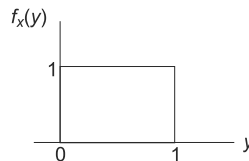
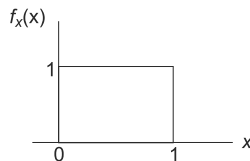
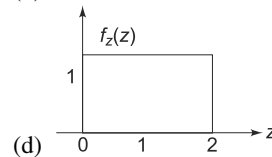
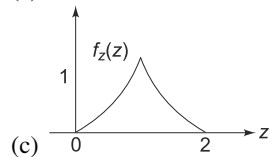
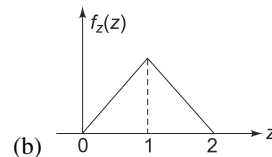
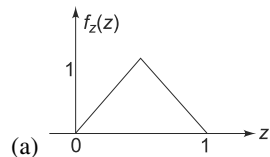


Fig. 4.41

The pdfs of $z = X + Y$ is



15. The joint pdf of random variables (X, Y) is

$$f_{X,Y}(x, y) = xy^2 + \frac{x^2}{8}; 0 \leq x \leq 2, 0 \leq y < 1$$

The value of $P(X + Y \leq 1)$ is

- (a) $\frac{13}{480}$ (b) $\frac{15}{240}$ (c) $\frac{12}{289}$ (d) $\frac{16}{480}$

16. The value of b for $f_{X,Y}(x, y)$ to be a valid pdf

$$f_{X,Y}(x, y) = 3xy \quad 0 < x < 1 \quad \text{and} \quad 0 < y < b$$

elsewhere

- (a) $\frac{1}{\sqrt{3}}$ (b) $\frac{2}{\sqrt{3}}$ (c) $\frac{\sqrt{3}}{2}$ (d) $\frac{4}{\sqrt{3}}$

17. For two independent random variables X and Y

$$P(Y \leq X) =$$

- (a) $\int_{-\infty}^{\infty} f_X(x) f_Y(y) dx$ (b) $\int_{-\infty}^{\infty} f_X(x) f_Y(x - y) dx$
 (c) $\int_{-\infty}^{\infty} f_Y(y) f_X(x) dx$ (d) $\int_{-\infty}^{\infty} f_Y(y) f_Y(y) dy$

18. The joint distribution function of (X, Y) is

$$f_{X,Y}(x, y) = [1 - e^{-ax} - e^{-ay} + e^{-a(x+y)}] u(x) u(y)$$

The value of $f_X(x|Y=y)$

- (a) $e^{-ax} u(x)$ (b) $e^{-ay} u(y)$ (c) $ae^{-ax} u(x)$ (d) $ae^{-ay} u(y)$

19. The joint pdf of random variable $s(X, Y)$ is given by

$$f_{X,Y}(x, y) = \begin{cases} 2e^{-x}e^{-y} & 0 \leq y \leq x < \infty \\ 0 & \text{elsewhere} \end{cases}$$

The value of $P(X + Y \leq 1)$

- (a) $\frac{2}{e}$ (b) $1 - 2e^{-1}$ (c) $1 - 2e$ (d) $1 + 2e^{-1}$

20. X and Y are two independent random variables uniformly distributed in $[-1, 1]$. The value of

$$P\left(XY < \frac{1}{2}\right)$$

- (a) 0.85 (b) 0.92 (c) 0.78 (d) 0.62

21. $P(x_1 < X \leq x_2, Y \leq y) =$

- (a) $F_{X,Y}(x_1, y_1) - F_{X,Y}(x_2, \infty)$ (b) $F_{X,Y}(x_1, x_2) - F_{X,Y}(x_1, y)$
 (c) $F_{X,Y}(x_1, y) - F_{X,Y}(x_2, y)$ (d) $F_{X,Y}(x_2, y) - F_{X,Y}(x_1, y)$

$$f_{X,Y}(x,y) = \begin{cases} (x^2 + y^2) & |8\pi \quad x^2 + y^2 < b \\ 0 & \text{elsewhere} \end{cases}$$

(a) 4
(b) 16
(c) 2
(d) 9

1. (b) 2. (a) 3. (d) 4. (a) 5. (b) 6. (b) 7. (d) 8. (b) 9. (c) 10. (a)
11. (c) 12. (a) 13. (a) 14. (b) 15. (a) 16. (b) 17. (c) 18. (c) 19. (b) 20. (a)
21. (d) 22. (a)

5

OPERATIONS ON MULTIPLE RANDOM VARIABLES



INTRODUCTION 5.1

In Chapter 3, we studied the computation of expectation and variance for a single random variable. In Chapter 4, we studied the joint occurrence of two random variables and their distribution and density functions. Now we extend the concept of expectation and variance to two random variables. We will also study concepts like covariance, correlation, correlation coefficient and their significance.



EXPECTATION OF A FUNCTION OF RANDOM VARIABLES 5.2

Let us consider two random variables X and Y with joint density function $f_{X,Y}(x, y)$. Consider $g(X, Y)$ which is some function of random variables X and Y . Then the expected value of $g(X, Y)$ is given by

$$E[g(X, Y)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x, y) f_{X,Y}(x, y) dx dy \quad (5.1)$$

For discrete random variables, the equivalent expression in terms of the joint pmf is

$$E[g(X, Y)] = \sum_m \sum_n g(x_m, y_n) p_{X,Y}(x_m, y_n) \quad (5.2)$$

If $g(X_1, X_2, \dots, X_n)$ is some function of N random variables $X_1, X_2, \dots, X_N$ then the expected value of the function is equal to

$$E[g(X_1, X_2, \dots, X_n)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(X_1, X_2, \dots, X_n) f_{X_1, X_2, \dots, X_n}(x_1, x_2, \dots, x_n) dx_1 dx_2 \dots dx_n \quad (5.3)$$

Consider a linear function of two random variables X and Y given by $g(X, Y) = aX + bY$, where a and b are constants. The expectation of $g(X, Y)$ is

$$\begin{aligned} E[g(X, Y)] &= E[aX + bY] \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (aX + bY) f_{X,Y}(x, y) dx dy \end{aligned}$$

$$\begin{aligned}
&= a \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x f_{X,Y}(x, y) dx dy + b \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} y f_{X,Y}(x, y) dx dy \\
&= a \int_{-\infty}^{\infty} x \left[\int_{-\infty}^{\infty} f_{X,Y}(x, y) dy \right] dx + b \int_{-\infty}^{\infty} y \left[\int_{-\infty}^{\infty} f_{X,Y}(x, y) dx \right] dy \\
&= a \int_{-\infty}^{\infty} x f_X(x) dx + b \int_{-\infty}^{\infty} y f_Y(y) dy \\
&= aE[X] + bE[Y]
\end{aligned} \tag{5.4}$$

Solved Problems



5.1 If X and Y are independent variables such that

$$X = 1 \text{ with a probability } \frac{1}{2}$$

$$= 0 \text{ with a probability } \frac{1}{3}$$

$$= -1 \text{ with a probability } \frac{1}{6}$$

and $Y = 1 \text{ with a probability } \frac{1}{4}$

$$= 0 \text{ with a probability } \frac{1}{2}$$

$$= -1 \text{ with a probability } \frac{1}{4}$$

Find (a) $E[2X + 3Y]$ (b) $E[X^2 - Y^2]$ (c) $E[XY]$

Solution Since X and Y are independent random variables,

$$P[X = x_i, Y = y_j] = P[X = x_i] P[Y = y_j]$$

Using the above relation, we can find the joint probability mass function as below:

$$P(X = -1, Y = -1) = P(X = -1) P(Y = -1) = \frac{1}{6} \left(\frac{1}{4} \right) = \frac{1}{24}$$

$$P(X = -1, Y = 0) = P(X = -1) P(Y = 0) = \frac{1}{6} \left(\frac{1}{2} \right) = \frac{1}{12}$$

$$P(X = -1, Y = 1) = P(X = -1) P(Y = 1) = \frac{1}{6} \left(\frac{1}{4} \right) = \frac{1}{24}$$

$$P(X = 0, Y = -1) = P(X = 0) P(Y = -1) = \frac{1}{3} \left(\frac{1}{4} \right) = \frac{1}{12}$$

$$P(X=0, Y=0) = P(X=0) P(Y=0) = \frac{1}{3} \left(\frac{1}{2} \right) = \frac{1}{6}$$

$$P(X=0, Y=1) = P(X=0) P(Y=1) = \frac{1}{3} \left(\frac{1}{4} \right) = \frac{1}{12}$$

$$P(X=1, Y=-1) = P(X=1) P(Y=-1) = \frac{1}{2} \left(\frac{1}{4} \right) = \frac{1}{8}$$

$$P(X=1, Y=0) = P(X=1) P(Y=0) = \frac{1}{2} \left(\frac{1}{2} \right) = \frac{1}{4}$$

$$P(X=1, Y=1) = P(X=1) P(Y=1) = \frac{1}{2} \left(\frac{1}{4} \right) = \frac{1}{8}$$

The values are tabulated as shown below.

X \ Y	-1	0	1
-1	$\frac{1}{24}$	$\frac{1}{12}$	$\frac{1}{24}$
0	$\frac{1}{12}$	$\frac{1}{6}$	$\frac{1}{12}$
1	$\frac{1}{8}$	$\frac{1}{4}$	$\frac{1}{8}$

$$E[X] = \sum_i x_i p_X(x_i) = (-1) \left(\frac{1}{6} \right) + 0 \left(\frac{1}{3} \right) + 1 \left(\frac{1}{2} \right) = \frac{1}{2} - \frac{1}{6} = \frac{1}{3}$$

$$E[Y] = \sum_j y_j p_Y(y_j) = (-1) \left(\frac{1}{4} \right) + 0 \left(\frac{1}{2} \right) + 1 \left(\frac{1}{4} \right) = 0$$

$$E[X^2] = \sum_i x_i^2 p_X(x_i) = (-1)^2 \frac{1}{6} + (0)^2 \frac{1}{3} + (1)^2 \frac{1}{2} = \frac{2}{3}$$

$$E[Y^2] = \sum_j y_j^2 p_Y(y_j) = (-1)^2 \left(\frac{1}{4} \right) + (0)^2 \frac{1}{2} + (1)^2 \left(\frac{1}{4} \right) = \frac{1}{2}$$

$$E[XY] = \sum_j \sum_i x_i y_j p_{XY}(x_i, y_j)$$

$$= (-1)(-1) \left(\frac{1}{24} \right) + (-1)(0) \left(\frac{1}{12} \right) + (-1)(1) \left(\frac{1}{24} \right) + 0 + 0 + 0$$

$$+ 1(-1) \left(\frac{1}{8} \right) + 1(0) \left(\frac{1}{4} \right) + 1(1) \left(\frac{1}{8} \right)$$

$$= \frac{1}{24} - \frac{1}{24} - \frac{1}{8} + \frac{1}{8} = 0$$

$$E[2X + 3Y] = 2E[X] + 3E[Y]$$

$$= 2 \left(\frac{1}{3} \right) + 3(0) = \frac{2}{3}$$

$$E[X^2 - Y^2] = E[X^2] - E[Y^2] = \frac{2}{3} - \frac{1}{2} = \frac{4-3}{6} = \frac{1}{6}$$

5.2 The joint density function of random variables X_1, X_2, X_3 and X_4 is given by

$$f_{X_1, X_2, X_3, X_4}(x_1, x_2, x_3, x_4) = \begin{cases} \frac{1}{abcd} & 0 < x_1 < a, 0 < x_2 < b, 0 < x_3 < c, 0 < x_4 < d \\ 0 & \text{elsewhere} \end{cases}$$

$$\text{Find } E[X_1^n X_2^n X_3^n X_4^n]$$

Solution

$$\text{Let } g(X_1, X_2, X_3, X_4) = X_1^n X_2^n X_3^n X_4^n$$

$$\begin{aligned} E[X_1^n X_2^n X_3^n X_4^n] &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x_1^n x_2^n x_3^n x_4^n f_{X_1, X_2, X_3, X_4}(x_1, x_2, x_3, x_4) dx_1 dx_2 dx_3 dx_4 \\ &= \int_0^d \int_0^c \int_0^b \int_0^a \frac{x_1^{n_1} x_2^{n_2} x_3^{n_3} x_4^{n_4}}{abcd} dx_1 dx_2 dx_3 dx_4 \\ &= \frac{1}{abcd} \left[\int_0^d \int_0^c \int_0^b \left(\frac{x_1^{n_1+1}}{n_1+1} \right) \Big|_0^a dx_2 dx_3 dx_4 \right] \\ &= \frac{a^{n_1+1}}{(n_1+1)abcd} \int_0^d \int_0^c \frac{x_2^{n_2+1}}{n_2+1} \Big|_0^b dx_3 dx_4 \\ &= \frac{a^{n_1+1} b^{n_2+1}}{abcd(n_1+1)(n_2+1)} \int_0^d \left(\frac{x_3^{n_3+1}}{n_3+1} \right) \Big|_0^c dx_4 = \frac{a^{n_1+1} b^{n_2+1} c^{n_3+1}}{abcd(n_1+1)(n_2+1)(n_3+1)} \int_0^d x_4^{n_4} dx_4 \\ &= \frac{a^{n_1+1} b^{n_2+1} c^{n_3+1} d^{n_4+1}}{abcd(n_1+1)(n_2+1)(n_3+1)(n_4+1)} \\ &= \frac{a^{n_1} b^{n_2} c^{n_3} d^{n_4}}{(n_1+1)(n_2+1)(n_3+1)(n_4+1)} \end{aligned}$$

5.3 Three statistically independent random variables X_1, X_2 and X_3 have mean value $\mu_{X_1} = 3, \mu_{X_2} = 6$ and $\mu_{X_3} = -2$. Find the mean values of the following functions:

- $g(X_1, X_2, X_3) = X_1 + 3X_2 + 4X_3$
- $g(X_1, X_2, X_3) = X_1 X_2 X_3$
- $g(X_1, X_2, X_3) = -2X_1 X_2 - 3X_1 X_3 + 4X_2 X_3$
- $g(X_1, X_2, X_3) = X_1 + X_2 + X_3$

Solution Given: $\mu_{X_1} = 2$; $\mu_{X_2} = 6$ and $\mu_{X_3} = -2$

(a) $g(X_1, X_2, X_3) = X_1 + 3X_2 + 4X_3$

$$\begin{aligned} E[g(X_1, X_2, X_3)] &= E[X_1 + 3X_2 + 4X_3] = E[X_1] + 3E[X_2] + 4E[X_3] \\ &= 3 + 3(6) + 4(-2) = 13 \end{aligned}$$

(b) $g(X_1, X_2, X_3) = X_1 X_2 X_3$

$$\begin{aligned} E[g(X_1, X_2, X_3)] &= E[X_1 X_2 X_3] = E[X_1] E[X_2] E[X_3] \quad \text{Since } X_1, X_2 \text{ and } X_3 \text{ are independent} \\ &= (3)(6)(-2) \\ &= -36 \end{aligned}$$

(c) $g(X_1, X_2, X_3) = -2X_1 X_2 - 3X_1 X_3 + 4X_2 X_3$

$$\begin{aligned} E[g(X_1, X_2, X_3)] &= -2E[X_1] E[X_2] - 3E[X_1] E[X_3] + 4E[X_2] E[X_3] \\ &= -2(3)(6) - 3(3)(-2) + 4(6)(-2) \\ &= -36 + 18 - 48 = -66 \end{aligned}$$

(d) $g(X_1, X_2, X_3) = X_1 + X_2 + X_3$

$$\begin{aligned} E[g(X_1, X_2, X_3)] &= E[X_1 + X_2 + X_3] \\ &= E[X_1] + E[X_2] + E[X_3] \\ &= 3 + 6 - 2 = 7 \end{aligned}$$



Practice Problems

5.1 If X and Y are independent random variables with pdfs $f_X(x) = \frac{8}{x^3}, x > 2$ and $f_Y(y) = 2y; 0 < y < 1$, find $E[XY]$.
(Ans. 8/3)

5.2 Two statistically independent random variables X_1 and X_2 have mean value $\mu_{X_1} = 5$ and $\mu_{X_2} = 10$. Find the mean value of the following functions:

(a) $g(X_1, X_2) = X_1 + 3X_2$ (b) $g(X_1, X_2) = -X_1 X_2 + 3X_1 + X_2$ (Ans. (a) 35 (b) -25)

Solved Problems



5.4 Prove that the mean value of a weighted sum of random variables equal to the weighted sum of the mean values.

Solution Consider N random variables $X_1, X_2, \dots, X_N$.

Let $g(X_1, X_2, \dots, X_N) = a_1 X_1 + a_2 X_2 + \dots + a_N X_N$

$E[g(X_1, X_2, \dots, X_N)] = E[a_1 X_1 + a_2 X_2 + \dots + a_N X_N]$

$$= E \left[\sum_{i=1}^N a_i X_i \right]$$

$$\begin{aligned}
&= \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} \sum_{i=1}^N a_i x_i f_{X_1, X_2, \dots, X_N}(x_1, x_2, \dots, x_N) dx_1 dx_2 \cdots dx_N \\
&= \sum_{i=1}^N \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} a_i x_i f_{X_1, X_2, \dots, X_N}(x_1, x_2, \dots, x_N) dx_1 dx_2 \cdots dx_N \\
&= a_1 \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} x_1 f_{X_1, \dots, X_N}(x_1, x_2, \dots) dx_1 dx_2 \cdots dx_N \\
&\quad + a_2 \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} x_2 f_{X_1, \dots, X_N}(x_1, x_2, \dots) dx_1 dx_2 \cdots dx_N + \cdots \\
&= a_1 \int_{-\infty}^{\infty} x_1 f_{X_1}(x_1) dx_1 + a_2 \int_{-\infty}^{\infty} x_2 f_{X_2}(x_2) dx_2 + \cdots \\
&= a_1 E[X_1] + a_2 E[X_2] + \cdots + a_N E[X_N]
\end{aligned}$$

REVIEW QUESTIONS



1. Define the expectation of function of a random variable
2. Prove that the mean value of a weighted sum of random variables is equal to the weighted sum of mean values.



JOINT MOMENTS ABOUT THE ORIGIN 5.3

The $(n + k)^{\text{th}}$ order joint moment of random variables X and Y about the origin is given by

$$m_{nk} = E[X^n Y^k] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x^n y^k f_{X,Y}(x, y) dx dy \quad (5.5)$$

If $n = 1$ and $k = 0$, m_{nk} is equal to the mean value of the random variable X and if $n = 0$ and $k = 1$, m_{nk} is equal to the mean value of the random variable Y . In both cases, $n + k = 1$, thus there are two first order moments. The total number of second order moments are three. They are

- (i) $m_{20} = E[X^2]$ the mean square value of X
- (ii) $m_{02} = E[Y^2]$ the mean square value of Y
- (iii) $m_{11} = E[XY]$ the correlation between X and Y .

Also, $m_{n0} = E[X^n]$, the n^{th} order moment of X , and $m_{0k} = E[Y^k]$, the k^{th} order moment of Y .

5.3.1 Correlation

The correlation between two random variables X and Y is defined as

$$\rho_{XY} = E[XY] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xy f_{X,Y}(x, y) dx dy \quad (5.6)$$

If the correlation between two random variables X and Y is zero then they are said to be orthogonal. If $E[XY] = E[X] E[Y]$ then the random variables are uncorrelated. The random variables that are statistically

independent are uncorrelated. But all the uncorrelated random variables are not statistically independent. However, uncorrelated Gaussian random variables are also independent. If $\rho_{XY} = 0$ for two random variables X and Y , they are called *orthogonal*.

For N random variables $X_1, X_2, \dots, X_N$, the $(n_1 + n_2 + \dots + n_N)$ order joint moments are defined by

$$\begin{aligned} m_{n_1 n_2 \dots n_N} &= E \left[X_1^{n_1} X_2^{n_2} \dots X_N^{n_N} \right] \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} X_1^{n_1} X_2^{n_2} \dots X_N^{n_N} f_{X_1 X_2 \dots X_N}(x_1, \dots, x_N) dx_1 \dots dx_N \end{aligned} \quad (5.7)$$

where $n_1, n_2, \dots, n_N$ are all integers.

5.3.2 Joint Central Moments

The $(n + k)^{\text{th}}$ order joint central moment of two random variables X and Y is given by

$$\begin{aligned} \mu_{nk} &= E[(X - \mu_X)^n (Y - \mu_Y)^k] \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (x - \mu_X)^n (y - \mu_Y)^k f_{X,Y}(x, y) dx dy \end{aligned} \quad (5.8)$$

If $n = 2$ and $k = 0$, then

$$\mu_{20} = E[(X - \mu_X)^2] = \sigma_X^2 \quad (5.9)$$

is variance of X .

Similarly, if $n = 0$ and $k = 2$ then

$$\mu_{02} = E[(Y - \mu_Y)^2] = \sigma_Y^2 \quad (5.10)$$

is variance of Y .

Since $n + k = 2$, for the above cases, μ_{20} and μ_{02} are known as *second order moments*. The other second-order joint moment μ_{11} is given by

$$\mu_{11} = E[(X - \mu_X)(Y - \mu_Y)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (x - \mu_X)(y - \mu_Y) f_{X,Y} dx dy \quad (5.11)$$

5.3.3 Covariance

Consider two random variables X and Y with mean values μ_X and μ_Y respectively. Then the second-order moment defined in Eq. (5.11), which is also known as covariance, is given by

$$\text{Cov}(X, Y) = \sigma_{XY} = E[(X - \mu_X)(Y - \mu_Y)] \quad (5.12)$$

Expanding the right side of the equation yields

$$\begin{aligned} \text{Cov}(X, Y) &= E[XY - \mu_X Y - X \mu_Y + \mu_X \mu_Y] \\ &= E[XY] - \mu_X E[Y] - \mu_Y E[X] + \mu_X \mu_Y \\ &= E[XY] - \mu_X \mu_Y - \mu_X \mu_Y + \mu_X \mu_Y \\ &= E[XY] - \mu_X \mu_Y \end{aligned} \quad (5.13)$$

If X and Y are independent random variables then

$$\text{Cov}(X, Y) = E[X] E[Y] - \mu_X \mu_Y = 0 \quad (5.14)$$

That is, for independent random variables, $\text{Cov}(X, Y) = 0$.

However, the converse is not true. That is, if the covariance of X and Y is zero, it does not mean that X and Y are independent random variables. If the covariance of two random variables is zero, we define the two random variables to be uncorrelated.

Properties of Covariance

1. The variance of sum of two random variables X and Y is given by

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2 \text{Cov}(X, Y) \quad (5.15)$$

Proof:

$$\begin{aligned} \text{Var}(X + Y) &= E[(X + Y)^2] - \{E[X + Y]\}^2 \\ &= E[X^2 + Y^2 + 2XY] - \{E[X] + E[Y]\}^2 \\ &= E[X^2 + Y^2 + 2XY] - \mu_X^2 - \mu_Y^2 - 2E[X]E[Y] \\ &= E[X^2] - \mu_X^2 + E[Y^2] - \mu_Y^2 + 2\{E[XY] - E[X]E[Y]\} \\ &= \text{Var}(X) + \text{Var}(Y) + 2 \text{Cov}(X, Y) \end{aligned}$$

2. If X and Y are independent random variables then $\text{Cov}(X, Y) = 0$

Proof: $\text{Cov}(X, Y) = E[XY] - \mu_X \mu_Y$

For independent random variables,

$$\begin{aligned} E[XY] &= E[X]E[Y] \\ &= \mu_X \mu_Y \end{aligned} \quad (5.16)$$

$$\Rightarrow \text{Cov}(X, Y) = 0$$

3. If X and Y are independent random variables then

$$\text{Var}(X + Y) = \text{Var}(X - Y) = \text{Var}(X) + \text{Var}(Y) \quad (5.17)$$

Proof: From Eq. (5.15) we know

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2 \text{Cov}(X, Y)$$

Similarly,

$$\text{Var}(X - Y) = \text{Var}(X) + \text{Var}(Y) - 2 \text{Cov}(X, Y)$$

For independent random variables, $\text{Cov}(X, Y) = 0$. Hence

$$\text{Var}(X + Y) = \text{Var}(X - Y) = \text{Var}(X) + \text{Var}(Y) \quad (5.18)$$

4. $\text{Cov}(X, Y) \leq \sigma_X \sigma_Y$

$$(5.19)$$

Proof: For any real constant a , $E[\{(Y - \mu_Y) - a(X - \mu_X)\}^2] \geq 0$

$$\Rightarrow E[(Y - \mu_Y)^2 + a^2(X - \mu_X)^2 - 2a(Y - \mu_Y)(X - \mu_X)] \geq 0$$

$$E[(Y - \mu_Y)^2] + a^2 E[(X - \mu_X)^2] - 2a E[(Y - \mu_Y)(X - \mu_X)] \geq 0$$

$$\sigma_Y^2 + a^2 \sigma_X^2 - 2a \text{Cov}(X, Y) \geq 0$$

$$\sigma_Y^2 + \sigma_X^2 \left(a^2 - \frac{2a \text{Cov}(X, Y)}{\sigma_X^2} \right) \geq 0$$

$$\sigma_Y^2 + \sigma_X^2 \left[\left(a - \frac{\text{Cov}(X, Y)}{\sigma_X^2} \right)^2 - \left\{ \frac{\text{Cov}(X, Y)}{\sigma_X^2} \right\}^2 \right] \geq 0$$

$$\sigma_Y^2 - \frac{\{\text{Cov}(X, Y)\}^2}{\sigma_X^2} + \sigma_X^2 \left(a - \frac{\text{Cov}(X, Y)}{\sigma_X^2} \right)^2 \geq 0$$

$$\frac{\sigma_X^2 \sigma_Y^2 - \{\text{Cov}(X, Y)\}^2}{\sigma_X^2} + \sigma_X^2 \left(a - \frac{\text{Cov}(X, Y)}{\sigma_X^2} \right)^2 \geq 0$$

In the above sum, the second term is always positive. Therefore, to satisfy the above inequality,

$$\sigma_X^2 \sigma_Y^2 - \{\text{Cov}(X, Y)\}^2 \geq 0$$

$$\sigma_X^2 \sigma_Y^2 \geq \{\text{Cov}(X, Y)\}^2$$

or $\text{Cov}(X, Y) \leq \sigma_X \sigma_Y$

Solved Problems



5.5 Prove the following:

(a) $\text{Cov}(aX, bY) = ab \text{Cov}(X, Y)$ (5.20)

(b) $\text{Var}(aX + bY) = a^2 \text{Var}(X) + b^2 \text{Var}(Y) + 2ab \text{Cov}(X, Y)$ (5.21)

Solution

(a) $\text{Cov}(aX, bY) = E\{[aX - E(aX)][bY - E(bY)]\}$
 $= E\{[aX - aE(X)][bY - bE(Y)]\}$
 $= ab E\{[X - E(X)][Y - E(Y)]\}$
 $= ab \text{Cov}(X, Y)$

(b) $\text{Var}(aX + bY) = E[(aX + bY)^2] - \{E[aX + bY]\}^2$
 $= E[a^2X^2 + b^2Y^2 + 2abXY] - \{aE[X] + bE[Y]\}^2$
 $= a^2 E[X^2] + b^2 E[Y^2] + 2ab E[XY] - \{a^2 E^2[X] + b^2 E^2[Y] + 2ab E[X]E[Y]\}$
 $= a^2 \{E[X^2] - (E[X])^2\} + b^2 \{E[Y^2] - (E[Y])^2\} + 2ab \{E[XY] - E[X]E[Y]\}$
 $= a^2 \text{Var}[X] + b^2 \text{Var}[Y] + 2ab \text{Cov}(X, Y)$

5.6 If $X_1, X_2, \dots, X_n$ are random variables then prove that

$$\text{Var} \left[\sum_{i=1}^n X_i \right] = \sum_{k=1}^n \text{Var}[X_k] + 2 \sum_{j \neq k} \text{Cov}(X_j, X_k) \quad (5.22)$$

Solution

$$\text{Var}[X] = E[X^2] - \{E[X]\}^2$$

So $\text{Var} \left[\sum_{k=1}^n X_k \right] = E \left[\left(\sum_{k=1}^n X_k \right)^2 \right] - \left\{ E \left[\sum_{k=1}^n X_k \right] \right\}^2$

$$\begin{aligned}
&= E[(X_1 + X_2 + X_3 + \dots + X_n)^2] - \{E[X_1] + E[X_2] + \dots + E[X_n]\}^2 \\
&= E[X_1^2 + X_2^2 + \dots + X_n^2 + 2X_1 X_2 + 2X_2 X_3 \dots + 2X_n X_1] \\
&\quad - \{E^2[X_1] + E^2[X_2] + \dots + E^2[X_n] + 2E[X_1] E[X_2] + \dots + 2E[X_n] E[X_1]\} \\
&= E[X_1^2] - \{E[X_1]\}^2 + E[X_2^2] - \{E[X_2]\}^2 + \dots + E[X_n^2] - \{E[X_n]\}^2 \\
&\quad + 2\{E[X_1 X_2] - E[X_1] E[X_2]\} + 2\{E[X_2 X_3] - E[X_2] E[X_3]\} \dots \\
&= \text{Var}[X_1] + \text{Var}[X_2] + \dots + \text{Var}[X_n] + 2 \text{Cov}(X_1, X_2) + 2 \text{Cov}(X_2, X_3) + \dots \\
&= \sum_{k=1}^n \text{Var}[X_k] + 2 \sum_{j \neq k} \text{Cov}(X_j, X_k)
\end{aligned}$$



Practice Problem

5.3 If X and Y are two random variables, find (i) $E[(X + Y)^2]$. (ii) Find $\text{Var}(X + Y)$. (iii) Under what condition is the variance of the sum equal to the sum of the individual variance? ($\text{Cov}(X, Y) = 0$)

REVIEW QUESTIONS



3. Define joint moments about the origin for two random variables.
4. Define two joint central moments for two-dimensional random variables X and Y .
5. Explain in detail about the properties of covariance.

Solved Problems



5.7 Two random variables X and Y have the joint density function

$$f_{X,Y}(x, y) = \frac{1}{24}, 0 < x < 6; 0 < y < 4, \text{ what is the expected value of the function } g(X, Y) = (XY)^2$$

Solution

$$\text{Given: } f_{X,Y}(x, y) = \begin{cases} \frac{1}{24} & 0 < x < 6; \\ & 0 < y < 4 \end{cases}$$

$$\begin{aligned}
E[XY]^2 &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (XY)^2 f_{XY}(x, y) dx dy \\
&= \int_0^4 \int_0^6 \frac{x^2 y^2}{24} dx dy = \frac{1}{24} \int_0^4 y^2 dy \cdot \frac{x^3}{3} \bigg|_0^6 \\
&= 3 \int_0^4 y^2 dy = 3 \frac{y^3}{3} \bigg|_0^4 = 64
\end{aligned}$$

5.8 Find the covariance of the two random variables whose pdf is given by

$$\begin{aligned} f_{X,Y}(x, y) &= 2 \text{ for } x > 0, y > 0, x + y < 1 \\ &= 0 \text{ otherwise} \end{aligned}$$

Solution The covariance is given by

$$\text{Cov}(X, Y) = E[XY] - E[X] E[Y]$$

$$\begin{aligned} E[XY] &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xy f_{X,Y}(x, y) dx dy \\ &= \int_0^1 \int_0^{1-y} 2xy dx dy \\ &= \int_0^1 y \left(x^2 \Big|_0^{1-y} \right) dy = \int_0^1 y(1-y)^2 dy \\ &= \int_0^1 (y + y^3 - 2y^2) dy = \left. \frac{y^2}{2} + \frac{y^4}{4} - \frac{2}{3} y^3 \right|_0^1 \\ &= \frac{1}{2} + \frac{1}{4} - \frac{1}{3} = \frac{1}{12} \end{aligned}$$

$$\begin{aligned} E[X] &= m_{10} = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x f_{X,Y}(x, y) dx dy \\ &= \int_0^1 \int_0^{1-x} 2x dy dx = 2 \int_0^1 x \left(\int_0^{1-x} dy \right) dx \\ &= 2 \int_0^1 x(1-x) dx = 2 \int_0^1 (x - x^2) dx \\ &= 2 \left(\left. \frac{x^2}{2} \right|_0^1 - \left. \frac{x^3}{3} \right|_0^1 \right) = \frac{1}{3} \end{aligned}$$

$$\begin{aligned} E[Y] &= m_{01} = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} y f_{X,Y}(x, y) dx dy \\ &= \int_0^1 \int_0^{1-y} 2y dx dy = \int_0^1 2y \int_0^{1-y} dx dy \\ &= \int_0^1 2y(1-y) dy = \frac{1}{3} \end{aligned}$$

$$\text{Cov}(X, Y) = E[XY] - E[X] E[Y]$$

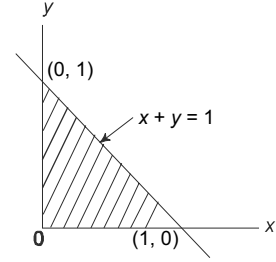


Fig. 5.1

$$= \frac{1}{12} - \frac{1}{9} = -\frac{1}{36}$$

5.9 Random variables X and Y have the joint density function

$$f_{X,Y}(x, y) = \frac{(x+y)^2}{40} \quad -1 < x < 1, -3 < y < 3$$

Find all the third-order moments for X and Y .

Solution The joint moment of X and Y about the origin is defined as

$$m_{nk} = E[X^n Y^k] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x^n y^k f_{X,Y}(x, y) dx dy$$

The third-order moments of X and Y are m_{30} , m_{03} , m_{21} and m_{12}

$$\begin{aligned} m_{30} &= E[X^3 Y^0] = E[X^3] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x^3 f_{X,Y}(x, y) dx dy \\ &= \int_{-3}^3 \int_{-1}^1 \frac{x^3(x+y)^2}{40} dx dy = \frac{1}{40} \int_{-3}^3 \int_{-1}^1 x^3(x^2 + y^2 + 2xy) dx dy \\ &= \frac{1}{40} \left[\int_{-3}^3 \int_{-1}^1 (x^5 + x^3 y^2 + 2x^4 y) dx dy = \frac{1}{40} \int_{-3}^3 \left[\frac{x^6}{6} \Big|_{-1}^1 + y^2 \frac{x^4}{4} \Big|_{-1}^1 + 2y \frac{x^5}{5} \Big|_{-1}^1 \right] dy \right] \\ &= \frac{1}{40} \int_{-3}^3 \left[2y \left(\frac{2}{5} \right) dy = \frac{1}{50} \frac{y^2}{2} \Big|_{-3}^3 \right] dy = 0 \\ m_{03} &= E[X^0 Y^3] = E[Y^3] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} y^3 f_{X,Y}(x, y) dx dy \\ &= \int_{-3}^3 \int_{-1}^1 \frac{y^3(x+y)^2}{40} dx dy = \frac{1}{40} \int_{-3}^3 \int_{-1}^1 y^3(x^2 + y^2 + 2xy) dx dy \\ &= \frac{1}{40} \int_{-3}^3 \int_{-1}^1 (y^3 x^2 + y^5 + 2xy^4) dx dy \\ &= \frac{1}{40} \int_{-3}^3 \left[y^3 \frac{x^3}{3} \Big|_{-1}^1 + y^5 x \Big|_{-1}^1 + y^4 x^2 \Big|_{-1}^1 \right] dy \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{40} \left[\frac{2}{3} \int_{-3}^3 y^3 dy + 2 \int_{-3}^3 y^5 dy + 2y^4(0) \right] \\
&= \frac{1}{40} \left[\frac{2}{3} \frac{y^4}{4} \Big|_{-3}^3 + \frac{1}{3} y^6 \Big|_{-3}^3 \right] = 0 \\
m_{21} = E[X^2 Y] &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x^2 y f_{X,Y}(x, y) dx dy = \int_{-3}^3 \int_{-1}^1 \frac{x^2 y (x+y)^2}{40} dx dy \\
&= \int_{-3}^3 \int_{-1}^1 x^2 y \left(\frac{x^2 + y^2 + 2xy}{40} \right) dx dy = \frac{1}{40} \int_{-3}^3 \int_{-1}^1 (x^4 y + x^2 y^3 + 2x^3 y^2) dx dy \\
&= \frac{1}{40} \left[\int_{-3}^3 \frac{2}{5} \cdot y dy + \int_{-3}^3 \frac{2}{3} y^3 dy + \int_{-3}^3 \frac{2}{4} x^4 \Big|_{-1}^1 dy \right] \\
&= \frac{1}{40} \left[\frac{2}{5} \frac{y^2}{2} \Big|_{-3}^3 + \frac{2}{3} \cdot \frac{y^4}{4} \Big|_{-3}^3 \right] = 0 \\
m_{12} = E[XY^2] &= \int_{-3}^3 \int_{-1}^1 xy^2 \frac{(x+y)^2}{40} dx dy = \frac{1}{40} \int_{-3}^3 \int_{-1}^1 xy^2 (x^2 + y^2 + 2xy) dx dy \\
&= \frac{1}{40} \int_{-3}^3 \int_{-1}^1 (x^3 y^3 + xy^4 + 2x^2 y^3) dx dy = \frac{1}{40} \int_{-3}^3 \frac{2}{3} y^3 x^3 \Big|_{-1}^1 dy \\
&= \frac{1}{30} \frac{y^4}{4} \Big|_{-3}^3 = 0
\end{aligned}$$

All the third-order moments are zero.

5.10 Statistically independent random variables X and Y have moments $m_{10} = 2$, $m_{20} = 14$, $m_{02} = 12$ and $m_{11} = -6$. Find the moment m_{22} .

Solution We know $m_{nk} = E[X^n Y^k]$. Therefore,

$$m_{20} = E[X^2 Y^0] = E[X^2] = 14$$

$$m_{10} = E[X^1 Y^0] = E[X] = 2$$

$$m_{02} = E[X^0 Y^2] = E[Y^2] = 12$$

$$m_{11} = E[X^1 Y^1] = E[XY] = -6$$

Since X and Y are statistically independent,

$$E[XY] = E[X] E[Y] = -6$$

$$E[X] = 2 \Rightarrow E[Y] = -6/2 = -3$$

$$m_{22} = E[(X - \mu_X)^2 (Y - \mu_Y)^2]$$

$$\begin{aligned}
&= \{E[X^2] - \mu_X^2\} \{E[Y^2] - \mu_Y^2\} \\
&\mu_X = E[X] = 2; \mu_Y = E[Y] = -3 \\
\Rightarrow \quad m_{22} &= [14 - (2)^2] [12 - (-3)^2] \\
&= (10) (3) = 30 \\
\Rightarrow \quad &\boxed{m_{22} = 30}
\end{aligned}$$

5.11 A joint density function is given by

$$f_{X,Y}(x, y) = \begin{cases} x(y+1.5) & 0 < x < 1 \text{ and } 0 < y < 1 \\ 0 & \text{elsewhere} \end{cases}$$

Solution The joint moments m_{nk} are given by

$$\begin{aligned}
E[X^n Y^k] &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x^n y^k f_{X,Y}(x, y) dx dy \\
&= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x^n y^k x(y+1.5) dx dy \\
&= \int_0^1 \int_0^1 x^{n+1} y^k (y+1.5) dx dy \\
&= \int_0^1 \int_0^1 (x^{n+1} y^{k+1} + 1.5 x^{n+1} y^k) dx dy \\
&= \int_0^1 \left[\frac{x^{n+2}}{n+2} \Big|_0^1 y^{k+1} + 1.5 \frac{x^{n+2}}{n+2} \Big|_0^1 y^k \right] dy \\
&= \frac{1}{n+2} \left[\int_0^1 y^{k+1} dy + 1.5 \int_0^1 y^k dy \right] \\
&= \frac{1}{n+2} \left[\frac{y^{k+2}}{k+2} \Big|_0^1 + 1.5 \frac{y^{k+1}}{k+1} \Big|_0^1 \right] \\
&= \frac{1}{n+2} \left[\frac{1}{k+2} + \frac{3}{2} \frac{1}{k+1} \right] \\
m_{00} = E[X^0 Y^0] &= \frac{1}{2} \left[\frac{1}{2} + \frac{3}{2} \right] = 1 \\
m_{10} = E[XY^0] &= E[X] = \frac{1}{3} \left[\frac{1}{2} + \frac{3}{2} \right] = \frac{2}{3}
\end{aligned}$$

$$m_{01} = E[X^0 Y^1] = E[Y] = \frac{1}{2} \left[\frac{1}{3} + \frac{3}{2} \cdot \frac{1}{2} \right] = \frac{13}{24}$$

$$m_{11} = E[XY] = \frac{1}{3} \left[\frac{1}{3} + \frac{3}{2} \left(\frac{1}{2} \right) \right] = \frac{13}{36}$$

5.12 X and Y are two independent random variables such that $E[X] = \lambda_1$, variance of $X = \sigma_1^2$, $E[Y] = \lambda_2$, Variance of $Y = \sigma_2^2$. Prove that $\text{Var}[XY] = \sigma_1^2 \sigma_2^2 + \lambda_1^2 \sigma_2^2 + \lambda_2^2 \sigma_1^2$.

Solution Given: $E[X] = \lambda_1$, $\text{Var}[X] = \sigma_1^2$

$$E[Y] = \lambda_2, \text{Var}[Y] = \sigma_2^2$$

$$\text{Var}[XY] = E[(XY)^2] - \{E[XY]\}^2$$

Since X and Y are independent,

$$E[XY] = E[X] E[Y]$$

$$\text{Var}[XY] = E[X^2 Y^2] - \{E[X] E[Y]\}^2$$

$$= E[X^2] E[Y^2] - \{E[X] E[Y]\}^2$$

$$E[X^2] = \text{Var}[X] + \{E[X]\}^2 = \sigma_1^2 + \lambda_1^2$$

$$E[Y^2] = \text{Var}[Y] + \{E[Y]\}^2 = \sigma_2^2 + \lambda_2^2$$

$$\begin{aligned} \Rightarrow \text{Var}[XY] &= (\sigma_1^2 + \lambda_1^2)(\sigma_2^2 + \lambda_2^2) - (\lambda_1 \lambda_2)^2 \\ &= \sigma_1^2 \sigma_2^2 + \lambda_1^2 \sigma_2^2 + \sigma_1^2 \lambda_2^2 + \lambda_1^2 \lambda_2^2 - \lambda_1^2 \lambda_2^2 \\ &= \sigma_1^2 \sigma_2^2 + \lambda_1^2 \sigma_2^2 + \lambda_2^2 \sigma_1^2 \end{aligned}$$

$$\boxed{\text{Var}[XY] = \sigma_1^2 \sigma_2^2 + \lambda_1^2 \sigma_2^2 + \lambda_2^2 \sigma_1^2}$$

5.13 Statistically independent random variables X and Y have moments $m_{10} = 2$, $m_{20} = 16$; $m_{02} = 30$; $m_{11} = -10$. Find the moment μ_{22} .

Solution

$$m_{10} = E[X^1 Y^0] = E[X] = 2; m_{11} = E[X^1 Y^1] = -10$$

$$m_{11} = E[XY] = E[X] E[Y] = 2E[Y] = -10$$

$$\Rightarrow E[Y] = m_{01} = -5$$

$$\mu_{20} = \sigma_X^2 = E[X^2] - \{E[X]\}^2$$

$$= m_{20} - (m_{10})^2$$

$$= 16 - (2)^2 = 12$$

$$\mu_{02} = \sigma_Y^2 = m_{02} - (m_{01})^2$$

$$= 30 - (-5)^2 = 30 - 25 = 5$$

$$\mu_{22} = E[X^2 Y^2] = E[X^2] E[Y^2]$$

$$= \mu_{20} \mu_{02} = 12(5) = 60$$

5.14 Show that two random variables X_1 and X_2 with joint pdf

$$f_{X_1, X_2}(x_1, x_2) = \begin{cases} \frac{1}{16} & |x_1| < 4, 2 < x_2 < 4 \\ 0 & \text{elsewhere} \end{cases}$$

are independent and orthogonal.

Solution The joint pdf is given by

$$f_{X_1, X_2}(x_1, x_2) = \begin{cases} \frac{1}{16} & |x_1| < 4, 2 < x_2 < 4 \\ 0 & \text{elsewhere} \end{cases}$$

The marginal density function

$$\begin{aligned} f_{X_1}(x_1) &= \int_{-\infty}^{\infty} f_{X_1, X_2}(x_1, x_2) dx_2 \\ &= \int_2^4 \frac{1}{16} dx_2 = \frac{1}{16}(4-2) = \frac{1}{8} \end{aligned}$$

$$\Rightarrow f_{X_1}(x_1) = \frac{1}{8} \quad |x_1| < 4$$

The marginal density function

$$\begin{aligned} f_{X_2}(x_2) &= \int_{-\infty}^{\infty} f_{X_1, X_2}(x_1, x_2) dx_1 \\ &= \int_{-4}^4 \frac{1}{16} dx_1 = \frac{1}{16}(8) = \frac{1}{2} \\ f_{X_2}(x_2) &= \frac{1}{2} \quad 2 < x_2 < 4 \end{aligned}$$

To check independence, find the product of $f_{X_1}(x_1)$ and $f_{X_2}(x_2)$ and compare it with $f_{X_1, X_2}(x_1, x_2)$

Since $f_{X_1, X_2}(x_1, x_2) = f_{X_1}(x_1) f_{X_2}(x_2)$ the two random variables are statistically independent.

For orthogonality,

$$\begin{aligned} E[X_1 X_2] &= 0 \\ E[X_1 X_2] &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x_1 x_2 f_{X_1, X_2}(x_1, x_2) dx_1 dx_2 \\ &= \int_{-4}^4 \int_2^4 x_1 x_2 \frac{1}{16} dx_1 dx_2 \end{aligned}$$

$$= \frac{1}{16} \int_{-4}^4 x_2 \, dx_2 \left(\frac{x_1^2}{2} \Big|_{-4}^4 \right) = 0$$

$\therefore X_1$ and X_2 are orthogonal.

5.15 The joint pdf of the random variables X and Y is defined as

$$f_{X,Y}(x, y) = 8e^{-4y} \quad 0 < x < 0.5, y > 0$$

$$= 0 \quad \text{otherwise}$$

- (a) Find the marginal pdfs of X and Y
 (b) What is the covariance of X and Y ?

Solution

Given: $f_{X,Y}(x, y) = 8e^{-4y} \quad 0 < x < 0.5, y > 0$
 $= 0 \quad \text{otherwise}$

- (a) The marginal pdf of X is

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) \, dy = \int_0^{\infty} 8e^{-4y} \, dy$$

$$= 8 \frac{e^{-4y}}{(-4)} \Big|_0^{\infty} = -2[-1] = 2$$

$$f_X(x) = \begin{cases} 2 & \text{for } 0 < x < 0.5 \\ 0 & \text{otherwise} \end{cases}$$

$$f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) \, dx = \int_0^{0.5} 8e^{-4y} \, dx$$

$$= 8e^{-4y}(0.5) = 4e^{-4y}$$

- (b) $\text{Cov}[XY] = E[XY] - E[X]E[Y]$

$$E[XY] = \int_0^{\infty} \int_0^{0.5} xy \, f_{X,Y}(x, y) \, dx \, dy = \int_0^{\infty} \int_0^{0.5} 8e^{-4y} \, dx \, dy$$

$$= \int_0^{\infty} 4e^{-4y} \, dy = 4 \frac{e^{-4y}}{-4} \Big|_0^{\infty} = 1$$

$$E[X] = \int_0^{0.5} xf_X(x) \, dx = \int_0^{0.5} x(2) \, dx = x^2 \Big|_0^{0.5} = 0.25$$

$$E[Y] = \int_0^{\infty} yf_Y(y) \, dy = \int_0^{\infty} y(4e^{-4y}) \, dy = 4 \left\{ -ye^{-4y} \Big|_0^{\infty} + \int_0^{\infty} e^{-4y} \, dy \right\} = 4$$

$$\begin{aligned}\text{Cov}[X, Y] &= E[XY] - E[X] E[Y] \\ &= 1 - 0.25(4) = 0\end{aligned}$$

5.16 The joint pdf of two continuous random variables X and Y is given by

$$f_{X,Y}(x, y) = \begin{cases} e^{-(x+y)} & 0 \leq x \leq \infty, 0 \leq y \leq \infty \\ 0 & \text{otherwise} \end{cases}$$

Are X and Y are independent?

Solution

$$\text{Given: } f_{X,Y}(x, y) = \begin{cases} e^{-(x+y)} & 0 \leq x \leq \infty, 0 \leq y \leq \infty \\ 0 & \text{otherwise} \end{cases}$$

The marginal pdf of X is given by

$$\begin{aligned}f_X(x) &= \int_{-\infty}^{\infty} f_{X,Y}(x, y) dy \\ &= \int_{-\infty}^{\infty} e^{-(x+y)} dy = e^{-x} \int_0^{\infty} e^{-y} dy \\ &= e^{-x} \left(-e^{-y} \Big|_0^{\infty} \right) = e^{-x}, x \geq 0\end{aligned}$$

The marginal pdf of Y is given by

$$\begin{aligned}f_Y(y) &= \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx \\ &= \int_0^{\infty} e^{-(x+y)} dx = e^{-y} \int_0^{\infty} e^{-x} dx \\ &= e^{-y} \left(-e^{-x} \Big|_0^{\infty} \right) = e^{-y}, y \geq 0\end{aligned}$$

Now we can write

$$f_{X,Y}(x, y) = f_X(x) f_Y(y)$$

Therefore, X and Y are independent.

5.17 Let X and Y be random variables having joint density function

$$f_{X,Y}(x, y) = \begin{cases} x + y & \text{for } 0 \leq x \leq 1, 0 \leq y \leq 1 \\ 0 & \text{elsewhere} \end{cases}$$

Find covariance of X and Y .

Solution The covariance of X and Y is given by

$$C_{XY} = E[XY] - E[X] E[Y]$$

$$\begin{aligned} E[XY] &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xy f_{X,Y}(x, y) dx dy \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xy (x + y) dx dy = \int_0^1 \int_0^1 (x^2 y + xy^2) dx dy \\ &= \int_0^1 \left(\frac{y}{3} + \frac{y^2}{2} \right) dy = \frac{1}{3} \left(\frac{1}{2} \right) + \frac{1}{2} \left(\frac{1}{3} \right) = \frac{1}{3} \end{aligned}$$

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dy = \int_0^1 (x + y) dy = x + \frac{1}{2}$$

$$E[X] = \int_{-\infty}^{\infty} x f_X(x) dx = \int_0^1 x \left(x + \frac{1}{2} \right) dx = \frac{1}{3} + \frac{1}{4} = \frac{7}{12}$$

$$f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx = \int_0^1 (x + y) dx = y + \frac{1}{2}$$

$$E[Y] = \int_0^1 y f_Y(y) dy = \int_0^1 y \left(y + \frac{1}{2} \right) dy = \frac{1}{3} + \frac{1}{4} = \frac{7}{12}$$

$$\begin{aligned} C_{XY} &= E[XY] - E[X] E[Y] \\ &= \frac{1}{3} - \frac{7}{12} \left(\frac{7}{12} \right) = -\frac{1}{144} \end{aligned}$$

5.18 X and Y are independent random variables having density function

$$\begin{aligned} f_X(x) &= 2e^{-2x} \text{ for } x \geq 0 \\ &= 0 \text{ otherwise} \end{aligned}$$

And
$$\begin{aligned} f_Y(y) &= 2e^{-2y} \text{ for } y \geq 0 \\ &= 0 \text{ otherwise} \end{aligned}$$

Find (a) $E[X + Y]$ (b) $E[X^2 + Y^2]$ (c) $E[XY]$

Solution

$$(a) \quad E[X + Y] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (X + Y) f_{X,Y}(x, y) dx dy$$

$$\text{We have } f_{X,Y}(x, y) = f_X(x) f_Y(y) = 2e^{-2x} \cdot 2e^{-2y}$$

$$E[X + Y] = 4 \int_0^{\infty} \int_0^{\infty} (x + y) e^{-2x} e^{-2y} dx dy$$

$$\begin{aligned}
&= 4 \int_0^\infty \left[\int_0^\infty x e^{-2x} dx + \int_0^\infty y e^{-2x} dx \right] e^{-2y} dy \\
&= 4 \int_0^\infty \left[\frac{-x}{2} e^{-2x} \Big|_0^\infty - \frac{1}{2} \int_0^\infty e^{-2x} dx + y \frac{e^{-2x}}{-2} \Big|_0^\infty \right] e^{-2y} dy \\
&= 4 \int_0^\infty \left(\frac{1}{4} + \frac{y}{2} \right) e^{-2y} dy = 4 \left[\frac{1}{4} \int_0^\infty e^{-2y} dy + \frac{1}{2} \int_0^\infty y e^{-2y} dy \right] \\
&= 4 \left[\frac{1}{8} + \frac{1}{2} \left(\frac{1}{4} \right) \right] = 1
\end{aligned}$$

$$(b) \quad E[X^2 + Y^2] = E[X^2] + E[Y^2]$$

$$\begin{aligned}
E[X^2] &= \int_{-\infty}^\infty x^2 f_X(x) dx = 2 \int_0^\infty x^2 e^{-2x} dx \\
&= 2 \left[-\frac{x^2}{2} e^{-2x} \Big|_0^\infty - \frac{x e^{-2x}}{2} \Big|_0^\infty - \frac{1}{4} e^{-2x} \Big|_0^\infty \right] = 2 \left(\frac{1}{4} \right) = \frac{1}{2}
\end{aligned}$$

Similarly,

$$E[Y^2] = \frac{1}{2}$$

$$E[X^2 + Y^2] = \frac{1}{2} + \frac{1}{2} = 1$$

$$(c) \quad E[XY] = E[X] E[Y]$$

$$\begin{aligned}
E[X] &= \int_{-\infty}^\infty x f_X(x) dx \\
&= 2 \int_0^\infty x e^{-2x} dx \\
&= 2 \left[\frac{x}{2} e^{-2x} \Big|_0^\infty - \frac{1}{2} \int_0^\infty e^{-2x} dx \right] \\
&= 2 \left[\frac{1}{4} \right] = \frac{1}{2}
\end{aligned}$$

Similarly,

$$E[Y] = \frac{1}{2}$$

$$E[XY] = E[X] E[Y] = \frac{1}{2} \left(\frac{1}{2} \right) = \frac{1}{4}$$

5.19 Two random variables X and Y have the following joint probability density function

$$\begin{aligned} f_{X,Y}(x, y) &= 2 - x - y && \text{for } 0 \leq x \leq 1 \text{ and } 0 \leq y \leq 1 \\ &= 0 && \text{elsewhere} \end{aligned}$$

Find $\text{var}(X)$ and $\text{var}(Y)$.

Solution

$$\begin{aligned} \text{Given: } f_{X,Y}(x, y) &= 2 - x - y && \text{for } 0 \leq x \leq 1 \text{ and } 0 \leq y \leq 1 \\ &= 0 && \text{elsewhere} \end{aligned}$$

$$\begin{aligned} f_X(x) &= \int_{-\infty}^{\infty} f_{X,Y}(x, y) dy = \int_0^1 (2 - x - y) dy \\ &= (2 - x) y \Big|_0^1 - \frac{y^2}{2} \Big|_0^1 = 2 - x - \frac{1}{2} \end{aligned}$$

$$\begin{aligned} f_Y(y) &= \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx \\ &= \int_0^1 (2 - x - y) dx = \frac{3}{2} - y \end{aligned}$$

$$\begin{aligned} E[X^2] &= \int_0^1 x^2 f_X(x) dx = \int_0^1 x^2 \left(\frac{3}{2} - x \right) dx \\ &= \frac{3}{2} \frac{x^3}{3} \Big|_0^1 - \frac{x^4}{4} \Big|_0^1 = \frac{1}{2} - \frac{1}{4} = \frac{1}{4} \end{aligned}$$

$$\begin{aligned} E[X] &= \int_0^1 x f_X(x) dx = \int_0^1 x \left(\frac{3}{2} - x \right) dx \\ &= \frac{3}{2} \frac{x^2}{2} \Big|_0^1 - \frac{x^3}{3} \Big|_0^1 = \frac{3}{4} - \frac{1}{3} = \frac{5}{12} \end{aligned}$$

$$\begin{aligned} \sigma_X^2 &= E[X^2] - \{E[X]\}^2 \\ &= \frac{1}{4} - \left(\frac{5}{12} \right)^2 = \frac{11}{144} \end{aligned}$$

Similarly $\sigma_Y^2 = \frac{11}{144}$

TRANSFORMATIONS OF MULTIPLE RANDOM VARIABLES 5.4



Consider a random variable $Z = g(X, Y)$ which is a function of two random variables X and Y . The expectation of $g(X, Y)$ can be obtained using Eq. (5.1) which is given by

$$Z = E[g(X, Y)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(X, Y) f_{X,Y}(x, y) dx dy \quad (5.23)$$

where $f_{X,Y}(x, y)$ is the joint pdf of X and Y . In the above equation, to find expectation of $g(X, Y)$, we used the joint pdf of X and Y . That is, to find expectation of $g(X, Y)$, it is not necessary to find the density function of the new random variable Z . However, in some practical problems, it may be required to determine the density function of the transformed variable Z , given the joint pdf of X and Y . In this section, we will study the method of finding the density function for a single functional transformation of more than one random variable. Consequently, we also find the pdf of two functions of two random variables.

5.4.1 Single Function of Several Random Variables

Consider a random variable $Y = g(X_1, X_2, \dots, X_N)$. The probability distribution function of Y is given by

$$\begin{aligned} F_Y(y) &= P(Y \leq y) = P[g(X_1, X_2, \dots, X_N) \leq y] \\ &= \int_{\{g(x_1, x_2, \dots, x_N) \leq y\}} \dots \int f_{X_1, X_2, \dots, X_N}(x_1, x_2, \dots, x_N) dx_1 dx_2 \dots dx_N \end{aligned} \quad (5.24)$$

The pdf is $f_Y(y) = \frac{dF_Y(y)}{dy}$

Let us consider one function of two random variables given by $Z = g(X, Y)$. Then

$$\begin{aligned} F_Z(z) &= P(Z \leq z) = P(g(X, Y) \leq z) = P\{(X, Y) \in R_z\} \\ &= \iint_{x, y \in R_z} f_{X,Y}(x, y) dx dy \end{aligned} \quad (5.25)$$

where R_z in the xy plane represents the region where the inequality $g(X, Y) \leq z$ is satisfied

$$f_Z(z) = \frac{d}{dz} F_Z(z)$$

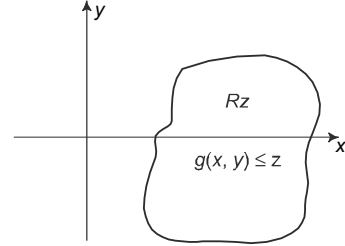


Fig. 5.2 R_z in the xy plane

Solved Problems



5.20 Consider the random variables X and Y with pdfs $f_X(x)$ and $f_Y(y)$. Determine the pdf of $f_Z(z)$ if (a) $Z = X + Y$, (b) $Z = X - Y$, (c) $Z = \sqrt{X^2 + Y^2}$, (d) $Z = \max(X, Y)$, (e) $Z = \min(X, Y)$, and (6) $Z = X/Y$.

Solution

- (a) Given $Z = X + Y$. We know $F_Z(z) = P(Z \leq z) = P(X + Y \leq z)$
we know

$$F_Z(z) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx dy$$

To find the region R_z that corresponds to $x + y \leq z$, draw a straight line $x + y = z$ in the xy -plane as shown in Fig. 5.3. Then the shaded area to the left of the line $x + y = z$ corresponds to the region $x + y \leq z$. The area of the entire shaded region can be obtained by integrating over the horizontal strip along the x -axis first, followed by sliding that strip along the y -axis from $-\infty$ to ∞ .

Therefore,

$$F_Z(z) = \int_{-\infty}^{\infty} \int_{-\infty}^{z-y} f_{X,Y}(x, y) dx dy$$

$$f_z(z) = \frac{d}{dz} F_Z(z) = \int_{-\infty}^{\infty} f_{X,Y}(z-y, y) dy \quad (5.26)$$

If X and Y are independent

$$f_{X,Y}(x, y) = f_X(x) f_Y(y)$$

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(z-y) f_Y(y) dy \quad (5.27)$$

Alternatively,

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z-x) dx \quad (5.28)$$

(b)

$$Z = X - Y$$

$$F_Z(z) = P(Z \leq z) = P(X - Y \leq z)$$

$$= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx dy$$

The region R_Z in the xy -plane where $x - y \leq z$ is shown in Fig. 5.4 (a). The area of the shaded region can be obtained by integrating over the horizontal strip along the x -axis first followed by sliding that strip along the y -axis from $-\infty$ to ∞ .

$$F_Z(z) = \int_{-\infty}^{\infty} \int_{-\infty}^{y+z} f_{X,Y}(x, y) dy$$

Differentiating

$$f_Z(z) = \frac{d}{dz} F_Z(z) = \int_{-\infty}^{\infty} f_{X,Y}(y+z, y) dy \quad (5.29)$$

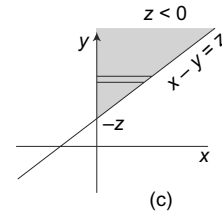
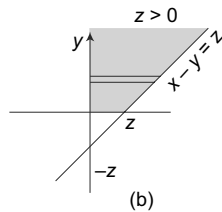
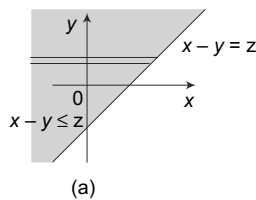


Fig. 5.4

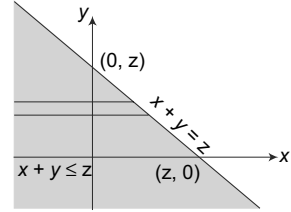


Fig. 5.3

If X and Y are independent,

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(z+y) f_Y(y) dy \quad (5.30)$$

$$= f_X(-z) * f_Y(y) \quad (5.31)$$

which is the convolution of $f_X(-z)$ with $f_Y(z)$.

If $f_X(x) = 0$ for $x < 0$ and $f_Y(y) = 0$ for $y < 0$

then from Fig. 5.4 (b), for $z \geq 0$

$$F_Z(z) = \int_0^{\infty} \int_0^{z+y} f_{X,Y}(x, y) dx dy \text{ for } z \geq 0.$$

For $z < 0$, from Fig. 5.4(c)

$$F_Z(z) = \int_{-z}^{\infty} \int_0^{z+y} f_{X,Y}(x, y) dx dy$$

$$\Rightarrow f_Z(z) = \int_0^{\infty} f_{X,Y}(z+y, y) dy, z \geq 0 \quad (5.32)$$

$$= \int_{-z}^{\infty} f_{X,Y}(z+y, y) dy, z < 0 \quad (5.33)$$

(c) $Z = X^2 + Y^2$

The above equation represents the area of a circle with radius $\sqrt{z}$

$$F_Z(z) = P(x^2 + y^2 \leq z) = \iint_{x^2 + y^2 \leq z} f_{X,Y}(x, y) dx dy$$

The area of the shaded region can be obtained by integrating over the horizontal strip along the x -axis from $-\sqrt{z-y^2}$ to $\sqrt{z-y^2}$ followed by sliding the strip along y -axis from $-\sqrt{z}$ to $\sqrt{z}$.

$$F_Z(z) = \int_{-\sqrt{z}}^{\sqrt{z}} \int_{-\sqrt{z-y^2}}^{\sqrt{z-y^2}} f_{X,Y}(x, y) dx dy$$

$$\Rightarrow f_Z(z) = \int_{-\sqrt{z}}^{\sqrt{z}} \frac{f_{X,Y}(\sqrt{z-y^2}, y) + f_{X,Y}(-\sqrt{z-y^2}, y)}{2\sqrt{z-y^2}} dy \quad (5.34)$$

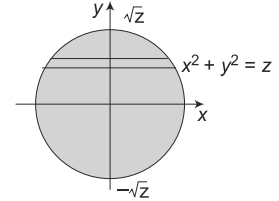


Fig. 5.5

(d) $Z = \max(X, Y)$

If $X > Y$, $\max(X, Y)$ is equal to X and if $X \leq Y$, then $\max(X, Y)$ is equal to Y . Therefore, we can write $Z = \max(X, Y) = X$ for $X > Y$

$$= Y \text{ for } X \leq Y$$

The distribution function of Z is given by

$$F_Z(z) = P(\max(X, Y) \leq z)$$

$$= P(X \leq z, X > Y) + P(Y \leq z, X \leq Y)$$

The region satisfying the equality $P(X \leq z, X > Y)$ is shown in Fig. 5.6. To obtain this region draw two straight lines $x = z$ and $x = y$. The intersection of these two lines is at the point (z, z) .

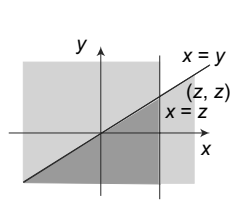


Fig. 5.6

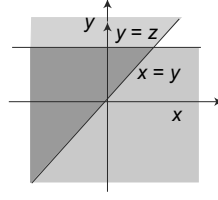


Fig. 5.7

Figures 5.6 and 5.7 show the region satisfying the inequalities

$$P(X \leq z, X > Y) \text{ and } P(Y \leq z, X \leq Y)$$

This region can be obtained by drawing two straight lines $x = y$ and $y = z$. The intersection of these two lines is at point (z, z) . The total region is shown in Fig. 5.8 from which we get

$$F_Z(z) = P(X \leq z, Y \leq z) = F_{X,Y}(z, z)$$

If X and Y are independent,

$$F_Z(z) = F_X(z) F_Y(z)$$

$$f_Z(z) = \frac{d}{dz} F_Z(z)$$

$$= F_X(z) f_Y(z) + F_Y(z) f_X(z)$$

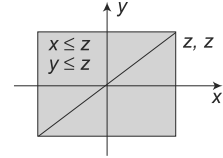


Fig. 5.8

(5.35)

(e) $Z = \min(X, Y)$

If $x \leq y$ then $\min(x, y) = x$ and if $x > y$, $\min(x, y) = y$.

That is,

$$Z = Y \text{ for } X > Y$$

$$= X \text{ for } X \leq Y$$

$$F_Z(z) = P(\min(X, Y) \leq z)$$

$$= P(Y \leq z, X > Y) + P(X \leq z, X \leq Y)$$

(5.36)

The regions satisfying the inequality

$P(Y \leq z, X > Y)$ and $P(X \leq z, X \leq Y)$ are shown in Figs 5.9 and 5.10.

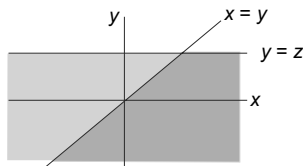


Fig. 5.9

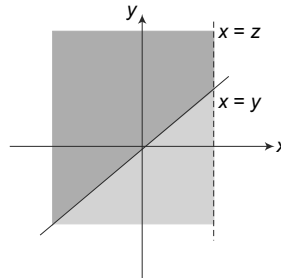


Fig. 5.10

The total region is shown in Fig. 5.11. From Eq. (5.36),

$$\begin{aligned} F_Z(z) &= P(\min(X, Y) \leq z) \\ &= P(Z \leq z) \\ &= 1 - P(Z > z) \\ &= 1 - P(X > z, Y > z) \end{aligned}$$

We know

$$P(x_1 < x \leq x_2, y_1 < y \leq y_2) \quad (5.37)$$

$$= F_{X,Y}(x_2, y_2) - F_{X,Y}(x_1, y_2) - F_{X,Y}(x_2, y_1) + F_{X,Y}(x_1, y_1) \quad (5.38)$$

$$P(X > z, Y > z) = P(z < X < \infty, z < Y < \infty) \quad (5.39)$$

Comparing Eq. (5.39) and Eq. (5.40), we obtain

$$x_1 = y_1 = z; \quad x_2 = y_2 = \infty$$

Therefore,

$$P(X > z, Y > z) = F_{X,Y}(\infty, \infty) - F_{X,Y}(z, \infty) - F_{X,Y}(\infty, z) + F_{X,Y}(z, z)$$

We know

$$F_{X,Y}(\infty, \infty) = 1$$

$$F_{X,Y}(z, \infty) = F_X(z)$$

$$F_{X,Y}(\infty, z) = F_Y(z)$$

$$F_Z(z) = 1 - \{1 - F_X(z) - F_Y(z) + F_{X,Y}(z, z)\} = F_X(z) + F_Y(z) - F_{X,Y}(z, z)$$

If X and Y are independent random variables,

$$F_Z(z) = F_X(z) + F_Y(z) - F_X(z) F_Y(z) \quad (5.40)$$

$$f_Z(z) = \frac{d}{dz} F_Z(z)$$

$$= f_X(z) + f_Y(z) - f_X(z) F_Y(z) - F_X(z) f_Y(z) \quad (5.41)$$

(f) $Z = X/Y$

$$f_Z(z) = P(Z \leq z) = P(X/Y \leq z)$$

The inequality $\frac{x}{y} \leq z$ can be rewritten as

$X \leq Yz$ if $Y > 0$ and $X \geq Yz$ if $Y < 0$. Therefore,

$$\begin{aligned} P(X/Y \leq z) &= P\left(\frac{X}{Y} \leq z, y > 0\right) + P(X/Y \leq z, y < 0) \\ &= P(X \leq Yz, y > 0) + P(X \geq Yz, y < 0) \end{aligned}$$

Consider the first term

$$P(X \leq Yz, Y > 0)$$

The area corresponding to the first term is shown in Fig. 5.12 from which we can find that the range of interval for Y is from 0 to ∞ and for x ; $-\infty$ to yz .

Similarly, for the second term x varies from yz to ∞ and y varies from $-\infty$ to 0 (see Fig. 5.13).

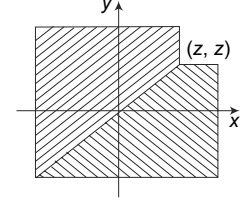


Fig. 5.11

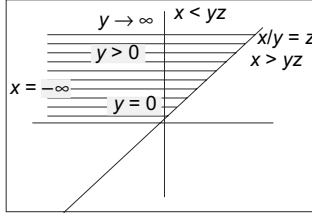


Fig. 5.12

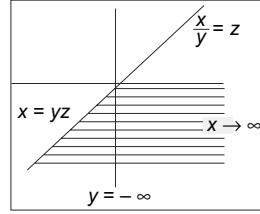


Fig. 5.13

$$F_Z(z) = \int_0^{yz} \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx dy + \int_{-\infty}^0 \int_{yz}^{\infty} f_{X,Y}(x, y) dx dy$$

Differentiating the above equation, we get

$$\begin{aligned} f_Z(z) &= \int_0^{\infty} y f_{X,Y}(yz, y) dy + \int_{-\infty}^0 (-y) f_{X,Y}(yz, y) dy \\ &= \int_{-\infty}^{\infty} |y| f_{X,Y}(yz, y) dy \end{aligned} \quad (5.42)$$

5.21 Find $F_{W,Z}(w, z)$ in terms of $F_{X,Y}(X, Y)$. $W = \max(X, Y)$, $Z = \min(X, Y)$.

Solution We know

$$F_{W,Z}(w, z) = P(W \leq w, Z \leq z) = P\{\max(X, Y) \leq w, \min(X, Y) \leq z\}$$

From Fig. (5.14), we can find that $W \leq w$ if and only if (X, Y) lies in the south-west region. Similarly $Z \leq z$ if and only if (X, Y) lies in the region shown in Fig. 5.15. Therefore, $W \leq w$ and $Z \leq z$ if and only if (X, Y) lies in the intersection of the two regions if $w < z$, the intersection, is the smallest subsection.

$$\begin{aligned} P(W \leq w, Z \leq z) &= P(W \leq w) \\ &= F_{X,Y}(w, w) \quad \text{for } w < z \end{aligned}$$

If $w \geq z$

$$\begin{aligned} P(w \leq w, z \leq z) &= F_{X,Y}(w, w) - P(z \leq X \leq w, z \leq Y \leq w) \\ &= F_{X,Y}(w, w) - [F_{X,Y}(w, w) - F_{X,Y}(w, z) - F_{X,Y}(z, w) + F_{X,Y}(z, z)] \\ &= F_{X,Y}(z, w) + F_{X,Y}(w, z) - F_{X,Y}(z, z) \end{aligned} \quad (5.43)$$

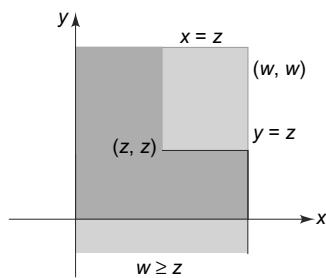


Fig. 5.14

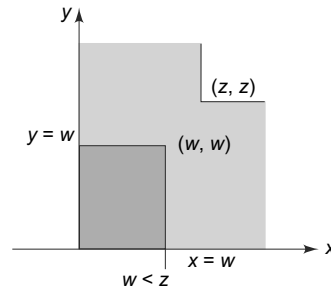


Fig. 5.15

5.22 Consider two random variables X and Y with pdfs $f_X(x) = e^{-x} u(x)$ and $f_Y(y) = e^{-y} u(y)$. If X and Y are independent and identically distributed, find the pdf of (a) $Z = X + Y$, (b) $Z = X - Y$, (c) $Z = \max(X, Y)$, and (d) $Z = \min(X, Y)$.

Solution Given: $f_X(x) = e^{-x} u(x)$; $f_Y(y) = e^{-y} u(y)$. Also, $f_{X,Y}(x, y) = e^{-(x+y)} u(x) u(y)$

(a) $Z = X + Y$

From Eq. (5.26), we have

$$f_Z(z) = \int_{-\infty}^{\infty} f_{X,Y}(z-y, y) dy$$

Since $f_X(x) = 0$ for $x < 0$ and $f_Y(y) = 0$ for $y < 0$,

$$\begin{aligned} f_Z(z) &= \int_0^z f_{X,Y}(z-y, y) dy \\ &= \int_0^z e^{-(z-y+y)} dy = \int_0^z e^{-z} dy = e^{-z} z \Big|_0^z \\ &= ze^{-z}; z > 0 \end{aligned}$$

(b) $Z = X - Y$

From Eqs 5.32 and 5.33

$$\begin{aligned} f_Z(z) &= \int_0^{\infty} f_{X,Y}(z+y, y) dy, z \geq 0 \\ &= \int_{-z}^{\infty} f_{X,Y}(z+y, y) dy, z < 0 \\ f_Z(z) &= \int_0^{\infty} e^{-(z+y+y)} dy = e^{-z} \int_0^{\infty} e^{-2y} dy \\ &= e^{-z} \frac{e^{-2y}}{-2} \Big|_0^{\infty} = -\frac{1}{2} e^{-z} (-1) = \frac{1}{2} e^{-z}; z \geq 0 \\ f_Z(z) &= \int_{-z}^{\infty} f_{X,Y}(z+y, y) dy, z < 0 \\ &= \int_{-z}^{\infty} e^{-(z+y+y)} dy = e^{-z} \frac{e^{-2y}}{-2} \Big|_{-z}^{\infty} = \frac{-e^{-z}}{2} (e^{2z}) = \frac{1}{2} e^{2z-z} \\ &= \frac{1}{2} e^z; z < 0 \end{aligned}$$

(c) $Z = \max(X, Y); F_Z(z) = P\{\max(X, Y) \leq z\} = P\{X \leq z, Y \leq z\}$

From Eq. (5.35)

$$f_Z(z) = F_X(z) f_Y(z) + F_Y(z) f_X(z)$$

Given $f_X(x) = e^{-x} u(x); f_Y(y) = e^{-y} u(y)$

$$F_X(x) = \int_0^x f_X(x) dx = \int_0^x e^{-x} dx = -(e^{-x} - 1) = 1 - e^{-x}$$

Similarly,

$$f_Y(y) = 1 - e^{-y}$$

$$\begin{aligned} f_Z(z) &= F_X(z) f_Y(z) + f_X(z) F_Y(z) \\ &= (1 - e^{-z}) e^{-z} + (1 - e^{-z}) e^{-z} \\ &= 2(1 - e^{-z}) e^{-z} \end{aligned}$$

(d) $Z = \min(X, Y)$

$$\begin{aligned} F_Z(z) &= P\{\min(X, Y) \leq z\} \\ &= 1 - P\{X > z; Y > z\} \\ &= 1 - [1 - F_X(z)] [1 - F_Y(z)] \\ &= F_X(z) + F_Y(z) - F_X(z) F_Y(z) \\ &= 1 - e^{-z} + 1 - e^{-z} - (1 - e^{-z})(1 - e^{-z}) \\ &= 2(1 - e^{-z}) - (1 - 2e^{-z} + e^{-2z}) \\ &= 2 - 2e^{-z} - 1 + 2e^{-z} - e^{-2z} \\ &= 1 - e^{-2z} \end{aligned}$$

$$\begin{aligned} f_Z(z) &= \frac{d}{dz} (F_Z(z)) \\ &= \frac{d}{dz} [1 - e^{-2z}] = 2e^{-z} u(z) \end{aligned}$$

5.23 If X and Y are independent identically distributed random variables with $X \sim N(0, \sigma^2)$ and $Y \sim N(0, \sigma^2)$, find the pdf of (a) $Z = \sqrt{X^2 + Y^2}$, (b) $Z = X^2 + Y^2$.

Solution Given: $X \sim N(0, \sigma^2)$ and $Y \sim N(0, \sigma^2)$

$$f_X(x) = \frac{1}{\sqrt{2\pi} \sigma} e^{-x^2/2\sigma^2}; f_Y(y) = \frac{1}{\sqrt{2\pi} \sigma} e^{-y^2/2\sigma^2}$$

$$F_{X,Y}(x, y) = f_X(x) f_Y(y) = \frac{1}{2\pi \sigma^2} e^{-(x^2 + y^2)/2\sigma^2}$$

(a) $Z = \sqrt{X^2 + Y^2}$
 $X^2 + Y^2 = Z^2$

The above equation represents the equation of a circle with radius Z as shown in Fig. 5.16.

$$F_Z(z) = \int_{-z}^z \int_{-\sqrt{z^2-y^2}}^{\sqrt{z^2-y^2}} f_{X,Y}(x, y) dx dy \quad (5.44)$$

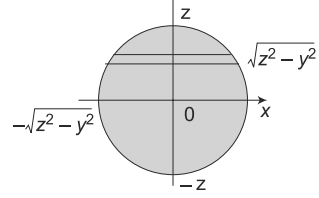


Fig. 5.16

Differentiating, the above equation, we get

$$\begin{aligned} f_Z(z) &= \int_{-z}^z \frac{z}{\sqrt{z^2-y^2}} \{f_{X,Y}(\sqrt{z^2-y^2}, y) + f_{X,Y}(-\sqrt{z^2-y^2}, y)\} dy \\ &= \frac{1}{2\pi\sigma^2} \int_{-z}^z \frac{2z}{\sqrt{z^2-y^2}} e^{-z^2/2\sigma^2} dy \\ &= \frac{1}{2\pi\sigma^2} 2ze^{-z^2/2\sigma^2} \int_{-z}^z \frac{1}{\sqrt{z^2-y^2}} dy \quad \text{Let } y = z \sin \theta, dy = dz \cos \theta \\ &= \frac{2ze^{-z^2/2\sigma^2}}{2\pi\sigma^2} \int_{-\pi/2}^{\pi/2} d\theta \\ &= \frac{z}{\sigma^2} e^{-z^2/2\sigma^2} \text{ for } z \geq 0 \end{aligned} \quad (5.45)$$

$$\Rightarrow F_Z(z) = \frac{z}{\sigma^2} e^{-z^2/2\sigma^2} u(z)$$

(b) Given $z = x^2 + y^2$

The above equation represents the equation of a circle with radius $\sqrt{z}$ as shown in Fig. 5.17.

$$F_Z(z) = \int_{-\sqrt{z}}^{\sqrt{z}} \int_{-\sqrt{z-y^2}}^{\sqrt{z-y^2}} f_{X,Y}(x, y) dx dy \quad (5.46)$$

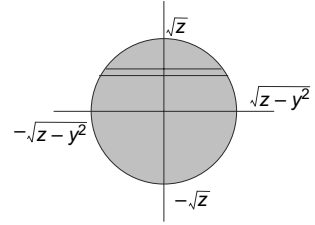


Fig. 5.17

Differentiating the above equation,

$$\begin{aligned} f_Z(z) &= \int_{-\sqrt{z}}^{\sqrt{z}} \frac{1}{2\sqrt{z-y^2}} \{f_{X,Y}(\sqrt{z-y^2}, y) + f_{X,Y}(-\sqrt{z-y^2}, y)\} dy \\ f_{X,Y}(x, y) &= \frac{1}{2\pi\sigma^2} e^{-(x^2+y^2)/2\sigma^2} \\ \Rightarrow f_Z(z) &= \int_{-\sqrt{z}}^{\sqrt{z}} \frac{1}{2\sqrt{z-y^2}} \left\{ \frac{e^{-z/2\sigma^2}}{2\pi\sigma^2} + \frac{e^{-z/2\sigma^2}}{2\pi\sigma^2} \right\} dy \quad \text{Let } y = \sqrt{z} \sin \theta; dy = \sqrt{z} \cos \theta d\theta \end{aligned} \quad (5.47)$$

$$\begin{aligned}
&= \frac{e^{-z/2\sigma^2}}{(2\pi\sigma^2)} \int_{-\sqrt{z}}^{\sqrt{z}} \frac{1}{\sqrt{z-y^2}} dy = \frac{e^{-z/2\sigma^2}}{2\pi\sigma^2} \int_{-\pi/2}^{\pi/2} d\theta \\
&= \frac{e^{-z/2\sigma^2}}{2\sigma^2} \text{ for } z \geq 0 \\
f_Z(z) &= \frac{e^{-z/2\sigma^2}}{2\sigma^2} u(z)
\end{aligned}$$

5.24 Given

$$f_{X,Y}(x, y) = \begin{cases} x + y & 0 \leq x \leq 1, 0 \leq y \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

Find the density of $Z = X + Y$.

Solution Given: $Z = X + Y$

$$\text{and } f_{X,Y}(x, y) = \begin{cases} x + y & 0 \leq x \leq 1, 0 \leq y \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

From Fig. 5.18, the area of the region-I can be obtained by sliding the horizontal strip along the y -axis from 0 to z and along x -axis from 0 to $z - y$. Therefore,

$$\begin{aligned}
F_Z(z) &= \int_0^z \int_0^{z-y} (x + y) dx dy \\
&= \int_0^z \left(\frac{x^2}{2} + yx \right) \Big|_0^{z-y} dy \\
&= \int_0^z \left\{ \frac{(z-y)^2}{2} + y(z-y) \right\} dy \\
&= \int_0^z \left\{ \frac{z^2 + y^2 - 2yz + 2yz - 2y^2}{2} \right\} dy \\
&= \int_0^z \left(\frac{z^2 - y^2}{2} \right) dy = \frac{z^2}{2} y \Big|_0^z - \frac{1}{6} y^3 \Big|_0^z \\
&= \frac{z^3}{2} - \frac{z^3}{6} = \frac{2z^3}{6} = \frac{z^3}{3} \\
f_Z(z) &= \frac{d}{dz} F_Z(z) = \frac{d}{dz} \left(\frac{z^3}{3} \right) = z^2
\end{aligned}$$

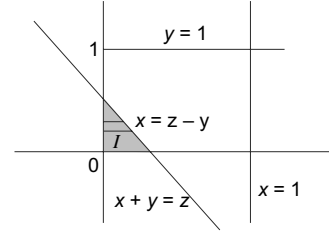


Fig. 5.18

For $1 < z < 2$

$$F_Z(z) = 1 - P(Z > z)$$

From Fig. 5.19

$$\begin{aligned} F_Z(z) &= 1 - \int_{z-1}^1 \int_{z-y}^1 (x+y) dx dy \\ &= 1 - \int_{z-1}^1 \left(\frac{x^2}{2} \Big|_{z-y}^1 + yx \Big|_{z-y}^1 \right) dy \\ &= 1 - \int_{z-1}^1 \left(\frac{1 - (z-y)^2}{2} + y(1-z+y) \right) dy \\ &= 1 - \frac{1}{2} \int_{z-1}^1 (1 - z^2 + y^2 + 2y) dy \\ &= 1 - \frac{1}{2} \left[y \Big|_{z-1}^1 - z^2 y \Big|_{z-1}^1 + \frac{1}{3} y^3 \Big|_{z-1}^1 + y^2 \Big|_{z-1}^1 \right] \\ &= 1 - \left(\frac{4}{3} - z^2 + \frac{z^3}{3} \right) \\ &= -\frac{1}{3} + z^2 - \frac{z^3}{3} \end{aligned}$$

$$f_Z(z) = \frac{d}{dz} F_Z(z) = 2z - z^2 = z(2-z); 1 < z < 2$$

$$f_Z(z) = \begin{cases} z^2 & 0 \leq z \leq 1 \\ z(2-z) & 1 < z < 2 \end{cases}$$

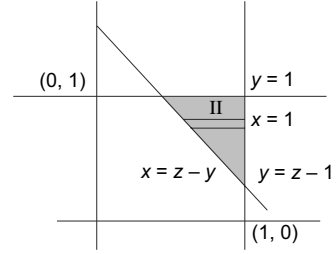


Fig. 5.19



Practice Problems

5.4 Consider two random variables X and Y with joint pdf

$$f_{X,Y}(x,y) = \begin{cases} 1 & 0 \leq x \leq 2, 0 \leq y < 1, 2y \leq x \\ 0 & \text{otherwise} \end{cases}$$

$$\left(\begin{array}{l} \text{Ans. } f_{X,Y}(x,y) = \begin{cases} \left(\frac{1}{3}\right)z & 0 < z < 2 \\ 2 - \left(\frac{2}{3}\right)z & 2 < z < 3 \\ 0 & \text{otherwise} \end{cases} \end{array} \right)$$

Find the pdf of $Z = X + Y$.

5.5 The joint pdf of X and Y is $f_{X,Y}(x, y) = e^{-(x+y)}$. Find the pdf of $U = X - Y$ (Ans. $f_U(u) = \frac{1}{2} e^{-|u|}$)

5.6 If X and Y are independent random variables with joint pdf $f_{X,Y}(x, y)$, find the density function $Z = X/Y$.

$$(Ans. \int_{-\infty}^{\infty} |y| f_X(yz) f_Y(y) dy)$$

5.4.2 Two Functions of Two Random Variables

Consider two random variables X and Y with joint pdf $f_{X,Y}(x, y)$. Let us define two random variables Z and W which arise as functions of X and Y . That is, $Z = g(X, Y)$ and $W = h(X, Y)$. Let us assume that the functions Z and W satisfy the following conditions.

1. The equations $z = g(x, y)$ and $w = h(x, y)$ can be uniquely solved for x and y in terms of z and w . The solutions are given by $X = g(z, w)$ and $y = h(z, w)$.
2. The functions g and h have continuous partial derivatives at all points (x, y) .
3. The determinant

$$J(x, y) = \begin{vmatrix} \frac{\partial g}{\partial x} & \frac{\partial g}{\partial y} \\ \frac{\partial h}{\partial x} & \frac{\partial h}{\partial y} \end{vmatrix} \quad (5.48)$$

$$= \frac{\partial g}{\partial x} \frac{\partial h}{\partial y} - \frac{\partial g}{\partial y} \frac{\partial h}{\partial x} \neq 0 \text{ at all points } (x, y)$$

If the above conditions are satisfied then the random variables Z and W are jointly continuous with density function

$$f_{Z,W}(z, w) = f_{X,Y}(x, y) |J(x, y)|^{-1} \quad (5.49)$$

where $x = g(z, w)$ and $y = h(z, w)$

In case of multiple solutions like $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$

$$f_{Z,W}(z, w) = \frac{f_{X,Y}(x_1, y_1)}{|J(x_1, y_1)|} + \frac{f_{X,Y}(x_2, y_2)}{|J(x_2, y_2)|} + \dots + \frac{f_{X,Y}(x_n, y_n)}{|J(x_n, y_n)|} \quad (5.50)$$

REVIEW QUESTIONS

6. Explain how you obtain the pdf of a single function of several random variables.
7. Explain how you obtain the pdf of two functions of two random variables.

Solved Problems

5.25 Let X and Y be jointly continuous random variables with pdf $f_{X,Y}(x, y)$. Let $Z = X + 2Y$ and $W = X - 2Y$. Find the joint pdf of Z and W in terms of $f_{X,Y}$.

Solution Let $g(X, Y) = X + 2Y$

and $h(X, Y) = X - 2Y$

$$\text{Then } J(X, Y) = \begin{vmatrix} \frac{\partial g}{\partial x} & \frac{\partial g}{\partial y} \\ \frac{\partial h}{\partial x} & \frac{\partial h}{\partial y} \end{vmatrix} = \begin{vmatrix} 1 & 2 \\ 1 & -2 \end{vmatrix} = -2 - 2 = -4$$

The solution of equations

$$z = x + 2y \text{ and } w = x - 2y$$

$$x = \frac{z+w}{2} \text{ and } y = \frac{z-w}{4}$$

The density function is

$$f_{Z,W}(z, w) = \frac{1}{4} f_{X,Y}\left(\frac{z+w}{2}, \frac{z-w}{4}\right)$$

5.26 Let X_1 and X_2 be jointly continuous random variables with pdf $f_{X_1, X_2}(x_1, x_2)$. Let $Y_1 = aX_1 + bX_2$ and $Y_2 = cX_1 + dX_2$ are new random variables, where a, b, c and d are real constants.

Find the joint pdf of X_1 and X_2 in terms of f_{X_1, X_2} .

Solution

Let $Y_1 = g(X_1, X_2) = aX_1 + bX_2$

$$Y_2 = h(X_1, X_2) = cX_1 + dX_2$$

$$J(X_1, X_2) = \begin{vmatrix} \frac{\partial g}{\partial X_1} & \frac{\partial g}{\partial X_2} \\ \frac{\partial h}{\partial X_1} & \frac{\partial h}{\partial X_2} \end{vmatrix} = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

The solution of equations

$$y_1 = ax_1 + bx_2$$

$$y_2 = cx_1 + dx_2$$

is given by

$$x_1 = \frac{dy_1 - by_2}{ad - bc} \text{ and } x_2 = \frac{ay_2 - cy_1}{ad - bc}$$

The joint pdf is $\frac{f_{X_1, X_2}\left(\frac{dy_1 - by_2}{ad - bc}, \frac{ay_2 - cy_1}{ad - bc}\right)}{|ad - bc|}$

Auxiliary Variables Consider the transformation $z = g(X, Y)$ where X and Y are two random variables. To determine the pdf of Z , first we define an auxiliary variable $W = X$ or $W = Y$ and then obtain the joint density of Z and W using Eq. (5.49). Next, we can obtain $f_Z(z)$ from $f_{Z,W}(z, w)$ by proper integration.

5.27 If X and Y are independent random variables each following $N(0, 2)$, find the pdf of $Z = 2X + 3Y$.

Solution Given $Z = 2X + 3Y$

We define the auxiliary random variable $W = Y$

Then the solution of the equations

$$z = 2x + 3y$$

$$w = y$$

$$\text{are } x = \frac{z - 2w}{2} \text{ and } y = w$$

The Jacobian of the transformation is

$$J(x,y) = \begin{vmatrix} \frac{\partial z}{\partial x} & \frac{\partial z}{\partial y} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} \end{vmatrix} = \begin{vmatrix} 2 & 3 \\ 0 & 1 \end{vmatrix} = 2$$

The joint density function of Z and W is

$$f_{Z,W}(z, w) = \frac{1}{2} f_{X,Y} \left(\frac{z - 3w}{2}, w \right)$$

Given $X \sim N(0, 2)$ and $Y \sim N(0, 2)$

That is, $\mu_X = \mu_Y = 0$; $\sigma_X^2 = \sigma_Y^2 = 2$

The pdf of X is $f_X(x) = \frac{1}{\sqrt{2\pi}(\sqrt{2})} e^{-x^2/4}$

and the pdf of Y is $f_Y(y) = \frac{1}{\sqrt{2\pi} \sqrt{2}} e^{-y^2/4}$

Since X and Y are independent,

$$\begin{aligned} f_{X,Y}(x, y) &= f_X(x) f_Y(y) \\ &= \left[\frac{1}{\sqrt{2\pi} \sqrt{2}} e^{-x^2/4} \right] \left[\frac{1}{\sqrt{2\pi} \sqrt{2}} e^{-y^2/4} \right] \\ &= \frac{1}{4\pi} e^{-(x^2 + y^2)/4} \end{aligned}$$

$$\begin{aligned} f_{Z,W}(z, w) &= \frac{1}{2} \left[\frac{1}{4\pi} e^{-\left[\frac{(z-3w)^2}{4} + w^2 \right] / 4} \right] \\ &= \frac{1}{8\pi} e^{-\left[\frac{1}{4} (z-3w)^2 + w^2 \right] / 4} \end{aligned}$$

$$\begin{aligned} f_Z(z) &= \int_{-\infty}^{\infty} f_{Z,W}(z, w) dw \\ &= \frac{1}{8\pi} \int_{-\infty}^{\infty} e^{-\frac{1}{16} [(z-3w)^2 + 4w^2]} dw \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{8\pi} \int_{-\infty}^{\infty} e^{-\frac{1}{16}(z^2 + 9\omega^2 - 6\omega z + 4\omega^2)} d\omega \\
&= \frac{1}{8\pi} e^{-z^2/16} \int_{-\infty}^{\infty} e^{-(13w^2 - 6wz)/16} dw \\
&= \frac{1}{8\pi} e^{-z^2/16} \int_{-\infty}^{\infty} e^{-\frac{13}{16}(w^2 - \frac{6wz}{13})} dw \\
&= \frac{1}{8\pi} e^{-z^2/16} e^{9z^2/(13 \times 16)} \int_{-\infty}^{\infty} e^{-\frac{13}{16}(w - \frac{3z}{13})^2} dw \\
&= \frac{1}{8\pi} e^{-z^2/52} \int_{-\infty}^{\infty} e^{-\frac{13}{16}(w - \frac{3z}{13})^2} dw \\
&= \frac{1}{8\pi} \sqrt{\frac{16}{13}} \left(\frac{1}{2}\right) e^{-z^2/52} \int_{-\infty}^{\infty} e^{-p} p^{-\frac{1}{2}} dp \\
&= \sqrt{\frac{16}{52}} \left(\frac{1}{8\pi}\right) \sqrt{\pi} e^{-z^2/52} = \frac{1}{8} \frac{1}{\sqrt{\pi(2\sqrt{13})}} e^{-z^2/52}
\end{aligned}$$

$$\begin{aligned}
\text{Let } \frac{13}{16} \left(w - \frac{3z}{13}\right)^2 &= p \\
2 \left(w - \frac{3z}{13}\right) dw &= \frac{16}{13} dp \\
2 \sqrt{\frac{16p}{13}} dw &= \frac{16}{13} dp \\
d\omega &= \frac{1}{2} \sqrt{\frac{16}{13}} p^{-1/2}
\end{aligned}$$

$$\int_{-\infty}^{\infty} e^{-p} p^{-1/2} dp = \sqrt{\pi}$$

5.28 The random variables X and Y have joint density function

$$f_{X,Y}(x, y) = \begin{cases} \frac{xy}{18} & \text{for } 0 < x < 3, 1 < y < 3 \\ 0 & \text{otherwise} \end{cases}$$

Find the density function of $Z = X + 2Y$.

Solution Given: $Z = X + 2Y$

Consider an auxiliary random variables $W = Y$

Now we solve the equations $z = x + 2y$

$$W = y$$

To get the solutions $y = w$

$$x = z - 2w$$

The Jacobian of the transformation is

$$J(x, y) = \begin{vmatrix} \frac{\partial z}{\partial x} & \frac{\partial z}{\partial y} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} \end{vmatrix} = \begin{vmatrix} 1 & 2 \\ 0 & 1 \end{vmatrix} = 1 \quad \Rightarrow |J(x, y)| = 1$$

The joint density function of Z and W is

$$\begin{aligned}
f_{Z,W}(z, w) &= f_{X,Y}(x, y) \Big|_{\substack{x=z-2w \\ y=w}} \\
&= f_{X,Y}(z - 2w, w)
\end{aligned}$$

$$= \begin{cases} \frac{(z-2w)w}{18}, & 0 < (z-2w) < 3 \\ 0 & \text{elsewhere} \end{cases}$$

The density function of z is

$$f_Z(z) = \int_{-\infty}^{\infty} f_{Z,W}(z, w) dw$$

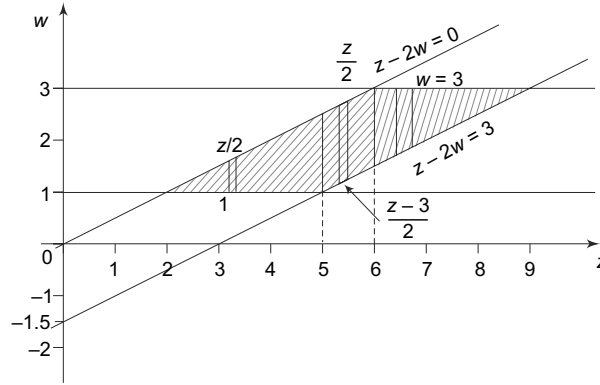


Fig. 5.20

From Fig. 5.20, we can observe that

$$\begin{aligned} f_Z(z) &= \int_1^{z/2} f_{Z,W}(z, w) dw \quad \text{for } 2 \leq z \leq 5 \\ &= \int_1^{z/2} \left(\frac{z-2w}{18} \right) w dw = \frac{1}{18} \int_1^{z/2} (z-2w) w dw \\ &= \frac{1}{18} \left\{ z \frac{w^2}{2} \Big|_1^{z/2} - 2 \frac{w^3}{3} \Big|_1^{z/2} \right\} \\ &= \frac{1}{18} \left\{ \frac{z}{2} \left(\frac{z^2}{4} - 1 \right) - \frac{2}{3} \left(\frac{z^3}{8} - 1 \right) \right\} \\ &= \frac{1}{18} \left\{ \frac{z^3}{8} - \frac{z}{2} - \frac{z^3}{12} + \frac{2}{3} \right\} \\ &= \frac{1}{18} \left\{ \frac{z^3}{24} - \frac{z}{2} + \frac{2}{3} \right\} \end{aligned}$$

$$\begin{aligned}
f_Z(z) &= \int_{\frac{z-3}{2}}^{z/2} f_{Z,W}(z, w) dw \quad \text{for } 5 \leq z \leq 6 \\
&= \frac{1}{18} \int_{\frac{z-3}{2}}^{z/2} (z-2w) w dw \\
&= \frac{1}{18} \left\{ \frac{z}{2} \left[\frac{z^2}{4} - \frac{(z-3)^2}{4} \right] - \frac{2}{3} \left[\frac{z^3}{8} - \frac{(z-3)^3}{8} \right] \right\} \\
&= \frac{1}{18} \left\{ \frac{z}{2} \left(\frac{z^2}{4} - \frac{z^2}{4} + \frac{3}{2}z - \frac{9}{4} \right) - \frac{2}{3} \left(\frac{z^3}{8} - \frac{z^3}{8} + \frac{27}{8} + \frac{9z^2}{8} - \frac{27z}{8} \right) \right\} \\
&= \frac{1}{18} \left\{ \frac{z}{2} \left(\frac{3}{2}z - \frac{9}{4} \right) - \frac{2}{3} \left(\frac{9z^2}{8} - \frac{27z}{8} + \frac{27}{8} \right) \right\} \\
&= \frac{1}{18} \left(\frac{9}{18}z - \frac{9}{4} \right) = \frac{z-2}{36} \\
f_Z(z) &= \int_{\frac{z-3}{2}}^3 f_{Z,W}(z, w) dw \quad \text{for } 6 \leq z \leq 9 \\
&= \frac{1}{18} \int_{\frac{z-3}{2}}^3 (z-2w) w dw \\
&= \frac{1}{18} \left\{ \frac{z}{2} w^2 \Big|_{\frac{z-3}{2}}^3 - \frac{2}{3} w^3 \Big|_{\frac{z-3}{2}}^3 \right\} \\
&= \frac{1}{18} \left\{ \frac{z}{2} \left(9 - \frac{(z-3)^2}{4} \right) - \frac{2}{3} \left(27 - \frac{(z-3)^3}{8} \right) \right\} \\
&= \frac{1}{18} \left\{ \frac{z}{2} \left(\frac{27 - z^2 + 6z}{4} \right) - \frac{2}{3} \left(\frac{243 - z^3 + 9z^2 - 27z}{8} \right) \right\} \\
&= \frac{1}{18} \left\{ \frac{z}{8} (27 - z^2 + 6z) - \frac{1}{12} (243 - z^3 + 9z^2 - 27z) \right\} \\
&= \frac{1}{18} \left\{ \frac{27z - z^3 + 6z^2}{8} - \frac{243 - z^3 + 9z^2 - 27z}{12} \right\}
\end{aligned}$$

$$\begin{aligned}
&= \frac{1}{18} \left\{ \frac{3(27z - z^3 + 6z^2) - 2(243 - z^3 + 9z^2 - 27z)}{24} \right\} \\
&= \frac{1}{18} \left\{ \frac{(81z - 3z^3 + 18z^2) - (486 - 2z^3 + 18z^2 - 54z)}{24} \right\} \\
&= \frac{1}{18} \left\{ \frac{-z^3 + 135z - 486}{24} \right\}
\end{aligned}$$

5.29 If X and Y are Independent random variables with density functions $f_X(x) = e^{-x}u(x)$, $f_Y(y) = 2e^{-y}u(y)$; find the density function of $Z = X + Y$.

Solution Given $f_X(x) = e^{-x}u(x)$; $f_Y(y) = 2e^{-y}u(y)$

Since X and Y are Independent

$$f_{X,Y}(x, y) = f_X(x)f_Y(y) = 2e^{-(x+y)}u(x)u(y)$$

Also, $Z = X + Y$

Consider an auxiliary variable

$$W = Y$$

The solution of the equations

$$z = x + y$$

$$w = y$$

is $y = w$ and $x = z - w$

The Jacobian of the transformation is

$$J(x, y) = \begin{vmatrix} \frac{\partial z}{\partial x} & \frac{\partial z}{\partial y} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} \end{vmatrix} = \begin{vmatrix} 1 & 1 \\ 0 & 1 \end{vmatrix} = 1; |J(x, y)| = 1$$

The joint density function of Z and W is

$$\begin{aligned}
f_{Z,W}(z, w) &= \frac{f_{X,Y}(x, y)}{J(X, Y)} = f_{X,Y}(z - w, w) \\
&= 2e^{-(z-w+2w)} \\
&= 2e^{-(z+w)}
\end{aligned}$$

The range space of (Z, W) can be obtained as follows:

Since $x \geq 0$ and $y \geq 0$, we get $w \geq 0$, $z - w \geq 0$

$$\Rightarrow z \geq w$$

$$f_Z(z) = \int_{-\infty}^{\infty} f_{Z,W}(z, w) dw$$

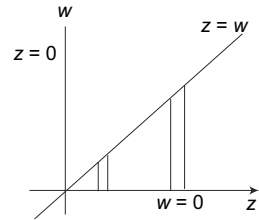


Fig. 5.21

From Fig. 5.21, we can obtain

$$\begin{aligned}
 f_Z(z) &= \int_0^z f_{Z,W}(z, w) dw \\
 &= \int_0^z 2e^{-(z+w)} dw \\
 &= 2e^{-z} \left(\frac{e^{-w}}{-1} \right) \bigg|_0^z = -2e^{-z}(e^{-z} - 1) \\
 &= 2(e^{-z} - e^{-2z}) \\
 \Rightarrow f_Z(z) &= 2(e^{-z} - e^{-2z}); z \geq 0
 \end{aligned}$$



Practice Problems

5.7 X and Y are two independent random variables with density $f_X(x) = 2e^{-2x}$ for $x > 0$ and $f_Y(y) = 2e^{-2y}$ for $y > 0$. Find the density of the random variable $Z = X - Y$.
 (Ans. $f_Z(z) = e^{2z}$ for $z < 0$
 $= e^{-2z}$ for $z > 0$)

Solved Problems



5.30 Let (X, Y) be a two-dimensional non-negative continuous random variable having the density

$$f_{X,Y}(x, y) = \begin{cases} 4xy e^{-(x^2 + y^2)}, & x \geq 0, y \geq 0 \\ 0 & \text{elsewhere} \end{cases}$$

Find the density function $Z = \sqrt{X^2 + Y^2}$

Solution

Given: $Z = \sqrt{X^2 + Y^2}$

Consider an auxiliary random variable $W = X$

Consider the equations $z = \sqrt{x^2 + y^2}$ and $w = x$

The solution is $x = w$ and $y = \sqrt{z^2 - w^2}$

The Jacobian of the transformation is

$$J(x, y) = \begin{vmatrix} \frac{\partial z}{\partial x} & \frac{\partial z}{\partial y} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} \end{vmatrix} = \begin{vmatrix} \frac{x}{\sqrt{x^2 + y^2}} & \frac{y}{\sqrt{x^2 + y^2}} \\ 1 & 0 \end{vmatrix}$$

$$|J(x, y)| = \left| -\frac{y}{\sqrt{x^2 + y^2}} \right| = \frac{y}{\sqrt{x^2 + y^2}}$$

$$f_{Z,W}(z, w) = \frac{f_{X,Y}(w, \sqrt{z^2 - w^2})}{\frac{y}{\sqrt{x^2 + y^2}}} = \frac{\sqrt{x^2 + y^2}}{y} 4w \sqrt{z^2 - w^2} e^{-z^2} = \frac{z}{y} 4wy e^{-z^2} = 4wz e^{-z^2}$$

The range of z and w can be obtained as below. Given $x \geq 0$ and $y \geq 0$. $z = \sqrt{x^2 + y^2}$ and $w = x$
 $x \geq 0 \Rightarrow w \geq 0$
and $y \geq 0 \Rightarrow z^2 - w^2 \geq 0$ or $z > w$

$$\begin{aligned} f_Z(z) &= \int_{-\infty}^{\infty} f_{Z,W}(z, w) dw \\ &= \int_0^z 4wz e^{-z^2} dw \\ &= 4z e^{-z^2} \frac{w^2}{2} \Big|_0^z = 2z^3 e^{-z^2} \quad z \geq 0 \end{aligned}$$

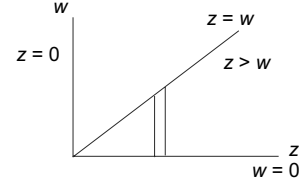


Fig. 5.22

5.31 Consider two independent random variables X and Y with identical uniform distribution in $(0, 1)$. (i) Find the joint density function of $Z = X + Y$ and $W = X - Y$ (ii) Find the density function of Z and W .

Solution Given $X \sim U(0, 1)$ and $Y \sim U(0, 1)$

$$f_X(x) = \begin{cases} 1 & \text{for } 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

$$f_Y(y) = \begin{cases} 1 & \text{for } 0 \leq y \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

Since X and Y are independent

$$f_{X,Y}(x, y) = f_X(x)f_Y(y) = 1 \text{ for } 0 \leq x \leq 1, 0 \leq y \leq 1$$

The transformation is given by $Z = X + Y$ and $W = X - Y$, from which we can find

$$x = \frac{z + w}{2}$$

$$y = \frac{z - w}{2}$$

The Jacobian of the transformation is

$$J(x, y) = \begin{vmatrix} \frac{\partial z}{\partial x} & \frac{\partial z}{\partial y} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} \end{vmatrix} = \begin{vmatrix} 1 & 1 \\ 1 & -1 \end{vmatrix} = -2$$

$$|J(x, y)| = 2$$

The joint density function of Z and W is

$$f_{Z,W}(z, w) = \frac{1}{2} \quad \text{for} \quad 0 \leq \frac{z+w}{2} \leq 1$$

$$0 \leq \frac{z-w}{2} \leq 1$$

The range of Z and W is $0 \leq z + w \leq 2$ and $0 \leq z - w \leq 2$

From Fig. 5.23, we find that in Region I,

$$f_Z(z) = \int_{-z}^z f_{Z,W}(z, w) dw \quad \text{for } 0 \leq z \leq 1$$

$$f_Z(z) = \int_{z-2}^{2-z} f_{Z,W}(z, w) dw \quad \text{for } 1 \leq z \leq 2$$

$$f_Z(z) = \int_{-z}^z \frac{1}{2} dw = \frac{1}{2} w \Big|_{-z}^z = z$$

For $1 \leq z \leq 2$

$$f_Z(z) = \int_{z-2}^{2-z} \frac{1}{2} dw = \frac{1}{2} w \Big|_{z-2}^{2-z} = \frac{1}{2} [2 - z - (z - 2)] = 2 - z$$

$$f_Z(z) = z; 0 \leq z \leq 1$$

$$= 2 - z; 1 \leq z \leq 2$$

From Fig. 5.24 we find that, in Region I,

$$f_W(w) = \int_w^{2-w} f_{Z,W}(z, w) dz \quad \text{for } 0 \leq w \leq 1$$

$$= \frac{1}{2} z \Big|_w^{2-w} = 1 - w$$

$$f_W(w) = \int_{-w}^{w+2} f_{Z,W}(z, w) dz \quad \text{for } -1 \leq w \leq 0$$

$$= \frac{1}{2} z \Big|_{-w}^{w+2} = w + 1$$

$$\Rightarrow f_W(w) = 1 - w \quad \text{for } 0 \leq w \leq 1$$

$$= w + 1 \quad \text{for } -1 \leq w \leq 0$$

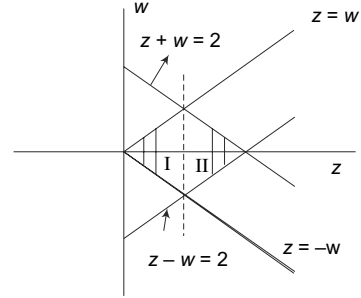


Fig. 5.23

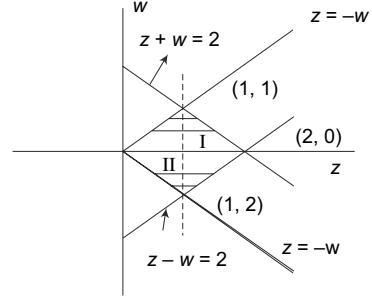


Fig. 5.24

5.32 If X and Y are independent uniform random variables on $(0, 1)$, find the distribution of XY and X/Y .

Solution Given $X \sim U(0, 1)$ and $Y \sim U(0, 1)$

Since X and Y are independent,

$$f_{X,Y}(x, y) = 1 \text{ for } 0 \leq x \leq 1, 0 \leq y \leq 1$$

Let $Z = XY$ and auxiliary random variable $W = X$

The transformation is $z = xy$

$$w = x \text{ from which we get}$$

$$x = w \text{ and } y = z/w$$

The Jacobian of the transformation is

$$J(x, y) = \begin{vmatrix} \frac{\partial z}{\partial x} & \frac{\partial z}{\partial y} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} \end{vmatrix} = \begin{vmatrix} y & x \\ 1 & 0 \end{vmatrix} = -x \Rightarrow |J(x, y)| = x = w$$

$$f_{Z,W}(z, w) = \frac{1}{w} f_{X,Y}(w, z/w) = \frac{1}{w}$$

The range of z and w is

$$0 \leq x \leq 1 \Rightarrow 0 \leq w \leq 1$$

$$0 \leq y \leq 1 \Rightarrow 0 \leq z/w \leq 1$$

or $0 \leq z \leq w$

$$f_Z(z) = \int_{-\infty}^{\infty} f_{Z,W}(z, w) dw = \int_z^1 \frac{1}{w} dw = \log w \Big|_z^1 = -\log z$$

Let $Z = X/Y$ and auxiliary random variable $w = y$. From the transformation $z = x/y$ and $w = y$

The solution for x and y is $y = w$ and $x = zw$

The Jacobian of the transformation is given by

$$J(x, y) = \begin{vmatrix} \frac{\partial z}{\partial x} & \frac{\partial z}{\partial y} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} \end{vmatrix} = \begin{vmatrix} \frac{1}{y} & -\frac{x}{y^2} \\ 0 & 1 \end{vmatrix} = \frac{1}{y}$$

The joint pdf of z and w is

$$f_{Z,W}(z, w) = \frac{f_{X,Y}(x, y)}{|J(x, y)|} \Big|_{\substack{x=zw \\ y=w}} = \frac{f_{X,Y}(zw, w)}{1/y} = w$$

The range of z and w can be obtained as follows

$$\text{For } 0 \leq x \leq 1 \quad 0 \leq zw \leq 1$$

$$\text{For } 0 \leq y \leq 1 \quad 0 \leq w \leq 1$$

In Region R_1 of Fig. 5.26, $0 \leq z \leq 1$

$$f_Z(z) = \int_0^1 f_{Z,W}(z, w) dw = \int_0^1 w dw = \frac{w^2}{2} \Big|_0^1 = \frac{1}{2}$$

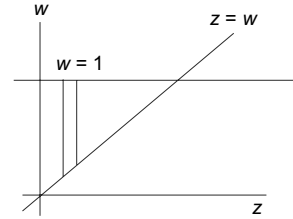


Fig. 5.25

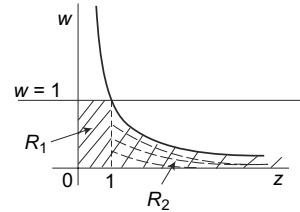


Fig. 5.26

In Region R_2 , $1 \leq z \leq \infty$

$$\begin{aligned}
 f_Z(z) &= \int_0^{1/z} f_{Z,W}(z, w) dw \\
 &= \int_0^{1/z} w dw = \frac{w^2}{2} \Big|_0^{1/z} = \frac{1}{2z^2} \\
 \Rightarrow f_Z(z) &= \begin{cases} \frac{1}{2} & \text{for } 0 \leq z \leq 1 \\ \frac{1}{2z^2}, & \text{for } 1 \leq z \leq \infty \end{cases}
 \end{aligned}$$

5.33 If the joint probability density of X and Y is given by

$$f_{XY}(x, y) = \begin{cases} e^{-(x+y)}; & \text{for } x \geq 0, y \geq 0 \\ 0 & \text{elsewhere} \end{cases}$$

Find the pdf of $Z = X/(X + Y)$.

Solution Given: $Z = X/(X + Y)$

Define the auxiliary random variable $W = X + Y$

For the transformation

$$z = x/(x + y)$$

$$w = x + y$$

The solution is

$$x = wz$$

and

$$z = w/(w + y) \Rightarrow zw + zy = w$$

$$y = \frac{w(1 - z)}{z}$$

$\Rightarrow$

$$x = wz$$

$$y = w(1 - z)/z$$

The Jacobian of the transformation is

$$J(x, y) = \begin{vmatrix} \frac{\partial z}{\partial x} & \frac{\partial z}{\partial y} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} \end{vmatrix} = \begin{vmatrix} \frac{(x + y) - x}{(x + y)^2} & \frac{-x}{(x + y)^2} \\ 1 & 0 \end{vmatrix} = \frac{x}{(x + y)^2} = \frac{z^2}{w}$$

$$f_{Z,W}(z, w) = \frac{f_{XY}(x, y)}{|J(x, y)|} = \frac{w}{z^2} e^{-\left[w + \frac{w(1 - z)}{z}\right]} = \frac{w}{z^2} e^{-w/z}$$

The range space of Z and W is given by

$$x \geq 0, \Rightarrow w \geq 0$$

$$y \geq 0 \Rightarrow \frac{w(1-z)}{z} \geq 0 \Rightarrow 1 \geq z$$

from which $z \leq 1$.

This is $w \geq 0$ and $z \leq 1$

$$\begin{aligned} f_Z(z) &= \int_0^{\infty} f_{Z,W}(z, w) dw \\ &= \int_0^{\infty} \frac{w}{z^2} e^{-w/z} dw = \frac{1}{z^2} \int_0^{\infty} w e^{-w/z} dw \\ &= \frac{1}{z^2} \left[-wz e^{-w/z} \Big|_0^{\infty} + z \int_0^{\infty} e^{-w/z} dw \right] \\ &= \frac{1}{z^2} \left[0 + z \frac{e^{-w/z}}{(-1/z)} \Big|_0^{\infty} \right] \\ &= \frac{1}{z^2} (z^2) = 1 \\ f_Z(z) &= 1 \text{ for } 0 \leq z \leq 1 \end{aligned}$$



Practice Problems

5.8 If X and Y are independent exponential random variables with common parameter λ , show that $X/(X + Y)$ is a uniformly distributed random variable in $(0, 1)$.

5.9 If X and Y are independent exponential random variables each having parameter λ , find the joint density function of

$$Z = X + Y \text{ and } W = e^X. \quad \left(\text{Ans. } \frac{\lambda^2 e^{-\lambda z}}{w}, w \geq 1, z \geq \log w \right)$$

5.10 If X and Y be two independent normal variables with mean 0 and variance σ^2 , find the joint density function of V and W , where $V = X \cos \theta + Y \sin \theta$, $W = X \sin \theta - Y \cos \theta$, where θ is a constant angle.

$$\left(\text{Ans. } f_{V,W}(v, w) = \frac{1}{2\pi\sigma^2} e^{-(x^2 + y^2)/2\sigma^2} \right)$$

5.11 If X and Y are two independent and identically distributed random variables, having the density function $f_X(x) = xe^{-x}$, $x \geq 0$ then prove that U and V are independent given $U = X/Y$ and $V = X + Y$.

5.12 Given $U = X/Y$ and $V = Y$. Then give a relation between pdf of (X, Y) and (U, V)

$$\text{Ans. } f_{U,V}(u, v) = |v| f_{X,Y}(x, y)$$

Solved Problem

5.34 x and y are independent and identically distributed normal random variables with zero mean and common variance σ^2 . Find the pdf of $z = \sqrt{x^2 + y^2}$

Solution

$$z = \sqrt{x^2 + y^2}; x \sim N(0, \sigma^2) \text{ and } Y \sim N(0, \sigma^2)$$

$$\Rightarrow f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-x^2/2\sigma^2}; f_Y(y) = \frac{1}{\sqrt{2\pi}\sigma} e^{-y^2/2\sigma^2}$$

Since x and y are independent,

$$\begin{aligned} f_{X,Y}(x, y) &= f_X(x) f_Y(y) \\ &= \frac{1}{2\pi\sigma^2} e^{-(x^2+y^2)/2\sigma^2} \end{aligned}$$

The transformation is given by

$$z = \sqrt{x^2 + y^2} \Rightarrow z^2 = x^2 + y^2$$

Define an auxiliary variable $w = y$

$$J(x, y) = \begin{vmatrix} \frac{x}{z} & \frac{y}{z} \\ 0 & 1 \end{vmatrix} = \frac{x}{z} = \frac{\sqrt{z^2 - w^2}}{z}$$

The solutions are given by

$$y = w$$

$$x = \pm \sqrt{z^2 - y^2} = \pm \sqrt{z^2 - w^2}$$

$$\begin{aligned} f_{Z,W}(z, w) &= \frac{f_{X,Y}(x_1, y_1)}{|J(x_1, y_1)|} + \frac{f_{X,Y}(x_2, y_2)}{|J(x_2, y_2)|} \\ &= \frac{z}{\sqrt{z^2 - w^2}} \left[\frac{2e^{-z^2}}{2\pi\sigma^2} \right] = \frac{ze^{-z^2}}{\pi\sigma^2\sqrt{z^2 - w^2}} \end{aligned}$$

5.35 The joint pdf of X and Y is given by $f_{X,Y}(x, y) = e^{-(x+y)}$, $x > 0$, $y > 0$. Find the pdf of $Z = (X + Y)/2$.

Solution Given: $f_{X,Y}(x, y) = e^{-(x+y)}$, $x > 0$, $y > 0$

Let $Z = (X + Y)/2$ and $W = Y$

The transformation is $z = (x + y)/2$ and $w = y$ from which we get $x = 2z - w$ and $y = w$

The Jacobian of the transformation is given by

$$J(x, y) = \begin{vmatrix} \frac{\partial z}{\partial x} & \frac{\partial z}{\partial y} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} \end{vmatrix} = \begin{vmatrix} \frac{1}{2} & \frac{1}{2} \\ 0 & 1 \end{vmatrix} = \frac{1}{2}$$

The joint pdf of the random variable Z and W is

$$f_{Z,W}(z, w) = \frac{f_{X,Y}(x, y)}{|J(x, y)|} = 2 e^{-(2z - w + w)} = 2 e^{-2z}$$

The range of z and w is obtained as follows:

Since $y > 0$; $w > 0$

$y = 0$ maps to $w = 0$

$x = 0$ maps to $w = 2z$ or $z = w/2$

For $x > 0$; $2z - w > 0$ or $z > w/2$

Therefore, the range of w is $0 < w < 2z$

$$f_Z(z) = \int_{-\infty}^{\infty} f_{Z,W}(z, w) dw = \int_0^{2z} 2 e^{-2z} dw = 2 e^{-2z} (2z) = 4z e^{-2z}$$

5.36 If X and Y are independent random variables each following $N(0, \sigma^2)$, find the density function of $R = \sqrt{X^2 + Y^2}$ and $\theta = \tan^{-1} (Y/X)$.

Solution

Given: $f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-x^2/2\sigma^2}$

$$f_Y(y) = \frac{1}{\sigma\sqrt{2\pi}} e^{-y^2/2\sigma^2}$$

Since X and Y are independent,

$$\begin{aligned} f_{X,Y}(x, y) &= f_X(x) f_Y(y) \\ &= \frac{1}{2\pi\sigma^2} e^{-(x^2 + y^2)/2\sigma^2} \end{aligned}$$

Let us consider $x = r \cos \theta$ and $y = r \sin \theta$

Then $\sqrt{x^2 + y^2} = r$ and $\theta = \tan^{-1} y/x$

The Jacobian transformation is

$$\begin{aligned} J &= \begin{vmatrix} \frac{\partial r}{\partial x} & \frac{\partial r}{\partial y} \\ \frac{\partial \theta}{\partial x} & \frac{\partial \theta}{\partial y} \end{vmatrix} = \begin{vmatrix} \frac{x}{\sqrt{x^2 + y^2}} & \frac{y}{\sqrt{x^2 + y^2}} \\ \frac{-y}{x^2 + y^2} & \frac{x}{x^2 + y^2} \end{vmatrix} \\ &= \frac{1}{\sqrt{x^2 + y^2} (x^2 + y^2)} = \frac{1}{r^3} \end{aligned}$$

$$f_{R,\theta}(r, \theta) = \frac{f_{X,Y}(r, \theta)}{|J|} = \frac{|r|}{2\pi r^2} e^{-r^2/2\sigma^2} \quad \begin{matrix} r \geq 0 \\ 0 \leq \theta \leq 2\pi \end{matrix}$$

5.37 Let X and Y be independent Gaussian random variables that have zero mean and unit variance. Show that $Z = X/Y$ is a Cauchy random variable.

Solution Let $Z = X/Y$ and auxiliary random variable $W = Y$. From the transform $z = x/y$ and $w = y$, the solution for x and y is $y = w$ and $x = wz$.

The Jacobian for the transformation is given by

$$J(x, y) = \begin{vmatrix} \frac{\partial z}{\partial x} & \frac{\partial z}{\partial y} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} \end{vmatrix} = \begin{vmatrix} \frac{1}{y} & \frac{x}{y^2} \\ 0 & 1 \end{vmatrix} = \frac{1}{y}$$

$$\begin{aligned} f_{Z,W}(z, w) &= \frac{f_{X,Y}(x, y)}{|J(x, y)|} = |y| f_{XY}(zw, w) \\ &= |w| f_{X,Y}(zw, w) \end{aligned}$$

$$f_Z(z) = \int_{-\infty}^{\infty} |w| f_{X,Y}(zw, w) dw$$

Given $f_X(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}; f_Y(y) = \frac{1}{\sqrt{2\pi}} e^{-y^2/2}$

$$f_{X,Y}(x, y) = \frac{1}{2\pi} e^{-(x^2+y^2)/2}$$

$$f_{X,Y}(zw, w) = \frac{1}{2\pi} e^{-(z^2w^2 + w^2)/2}$$

$$f_Z(z) = \int_0^{\infty} |w| \frac{1}{2\pi} e^{-(z^2w^2 + w^2)/2} dw$$

$$= \frac{2}{2\pi} \int_0^{\infty} w e^{-(z^2w^2 + w^2)/2} dw$$

$$= \frac{1}{\pi} \int_0^{\infty} w e^{-(z^2w^2 + w^2)/2} dw$$

$$f_Z(z) = \frac{1}{\pi} \int_0^{\infty} w e^{-w^2(1+z^2)/2} dw$$

Let $(1 + z^2)/2 = a$

Then $f_Z(z) = \frac{1}{\pi} \int_0^{\infty} w e^{-aw^2} dw$

Let $aw^2 = t$ $2aw dw = dt$ $w dw = \frac{dt}{2a}$

$$\begin{aligned}
 \Rightarrow f_Z(z) &= \frac{1}{\pi} \int_0^{\infty} e^{-t} \frac{dt}{2a} \\
 &= \frac{-1}{2a\pi} e^{-t} \Big|_0^{\infty} \\
 &= \frac{1}{2a\pi} = \frac{1}{2 \left(\frac{1+z^2}{2} \right) \pi} = \frac{1}{\pi(1+z^2)}
 \end{aligned}$$

which is a Cauchy distribution.

5.38 If X and Y are independent random variables with pdf e^{-x} , $x \geq 0$ and e^{-y} , $y \geq 0$, find the density functions of $Z = X/(X + Y)$ and $W = X + Y$. Are Z and W independent?

Solution

Given: $Z = X/(X + Y)$

and $W = X + Y$

$$f_X(x) = e^{-x}, x \geq 0 \text{ and } f_Y(y) = e^{-y}, y \geq 0$$

Since X and Y are independent,

$$f_{X,Y}(x, y) = f_X(x) f_Y(y) = e^{-(x+y)} \quad x > 0, y > 0$$

The transformation is

$$z = x/(x + y); w = x + y$$

from which $x = wz; y = w(1 - z)$

Since $x \geq 0$ and $y \geq 0$,

$$w = x + y \geq 0$$

$$y \geq 0 \Rightarrow w(1 - z) = 0 \Rightarrow w \geq 0 \text{ and } z \leq 1$$

$$x = wz \geq 0 \Rightarrow z \geq 0. \text{ Hence, } 0 \leq z \leq 1$$

$$f_{Z,W}(z, w) = \frac{f_{X,Y}(z, w)}{|J(x, y)|}$$

$$J(x, y) = \begin{vmatrix} \frac{\partial z}{\partial x} & \frac{\partial z}{\partial y} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} \end{vmatrix} = \begin{vmatrix} \frac{y}{(x+y)^2} & \frac{-x}{(x+y)^2} \\ 1 & 1 \end{vmatrix} = \frac{x+y}{(x+y)^2} = \frac{1}{x+y} = \frac{1}{w}$$

$$f_{Z,W}(z, w) = we^{-(wz + w - wz)} = we^{-w}$$

$$f_W(w) = \int_0^1 we^{-w} dz = we^{-w}$$

$$f_Z(z) = \int_0^{\infty} w e^{-w} dw = -w e^{-w} \Big|_0^{\infty} e^{-w} \Big|_0^{\infty} = 1$$

$$= f_{Z,W}(z, w) = w e^{-w}$$

$$f_Z(z) = 1 \text{ and } f_W(w) = w e^{-w}$$

Since $f_{ZW}(z, w) = f_Z(z) f_W(w)$

Z and W are independent



Practice Problems

5.13 X and Y are independent random variables with $f_X(x) = e^{-x} u(x)$ and $f_Y(y) = 3e^{-3y} u(y)$. Find the density function of $Z = X/Y$. (Ans. $3/(z+3)^2$)

5.14 If $X \sim U(0, 1)$ and $Y \sim U(0, 1)$, find the joint pdf of $X + Y$ and $X - Y$. Assume X and Y are independent.

(Ans. $f_{Z,W}(z, w) = 1/2, 0 < |w| < z < 2$)

5.15 Show that the convolution of two Cauchy densities is a Cauchy density.

Solved Problems



5.39 If X and Y are independent $\gamma(t_1, \lambda)$ and $\gamma(t_2, \lambda)$ variates respectively, find the distribution of $X + Y$ and $X/(X + Y)$.

Solution Given

$$f_X(x) = \frac{\lambda e^{-\lambda x} (\lambda x)^{t_1-1}}{\Gamma(t_1)}, x \geq 0$$

$$= 0 \quad x < 0$$

$$f_Y(y) = \frac{\lambda e^{-\lambda y} (\lambda y)^{t_2-1}}{\Gamma(t_2)}, y \geq 0$$

$$= 0, y < 0$$

Since X and Y are independent,

$$f_{X,Y}(x, y) = \frac{(\lambda)^{t_1+t_2} x^{t_1-1} e^{-\lambda x} e^{-\lambda y} y^{t_2-1}}{\Gamma(t_1) \Gamma(t_2)}; x > 0, y > 0$$

Let $z = x + y$ and $w = \frac{x}{x+y}$

Solving for x and y, we get $x = wz$ and $y = z(1-w)$

$$J(x, y) = \begin{vmatrix} \frac{\partial z}{\partial x} & \frac{\partial z}{\partial y} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} \end{vmatrix} = \begin{vmatrix} 1 & 1 \\ y & -x \end{vmatrix} = \frac{-1}{x+y} = \frac{-1}{z}$$

$$|J(x, y)| = \frac{1}{z}$$

$$\begin{aligned} f_{Z,W}(z, w) &= \frac{f_{X,Y}(x, y)}{|J(x, y)|} = z f_{X,Y}(x, y) \\ &= z f_X(x) f_Y(y) \\ &= z f_X(wz) f_Y[z(1-w)] \\ &= \frac{z \lambda e^{-\lambda wz} (\lambda wz)^{t_1-1}}{\Gamma(t_1)} \frac{\lambda e^{-\lambda z(1-w)} [\lambda z(1-w)]^{t_2-1}}{\Gamma(t_2)} \\ &= \frac{(\lambda)^{t_1+t_2} z^{t_1+t_2-1} e^{-\lambda z} (w)^{t_1-1} (1-w)^{t_2-1}}{\Gamma(t_1) \Gamma(t_2)} \\ &= \frac{(\lambda)^{t_1+t_2} z^{t_1+t_2-1} e^{-\lambda z} (w)^{t_1-1} (1-w)^{t_2-1}}{\Gamma(t_1+t_2) B(t_1, t_2)} \\ &= \left(\frac{(\lambda)^{t_1+t_2}}{\Gamma(t_1+t_2)} (z)^{t_1+t_2-1} e^{-\lambda z} \right) \left(\frac{1}{B(t_1, t_2)} (w)^{t_1-1} (1-w)^{t_2-1} \right) \end{aligned}$$

$$\frac{\Gamma(t_1) \Gamma(t_2)}{\Gamma(t_1+t_2)} = B(t_1, t_2)$$

That is, Z and W are independent where Z is a $\chi(t_1+t_2, \lambda)$ and W is a $B_1(t_1, t_2)$ variate.

5.40 If X and Y are two independent chi-square random variables with n_1 and n_2 degrees of freedom then prove that X/Y is a $\beta_2\left(\frac{n_1}{2}, \frac{n_2}{2}\right)$ random variable.

Solution

$$f_X(x) = \frac{1}{\Gamma(n_1/2)} \left(\frac{1}{2}\right)^{\frac{n_1}{2}} x^{\frac{n_1}{2}-1} e^{-x/2}; x > 0$$

$$f_Y(y) = \frac{1}{\Gamma\left(\frac{n_2}{2}\right)} \left(\frac{1}{2}\right)^{\frac{n_2}{2}} y^{\frac{n_2}{2}-1} e^{-y/2}, y > 0$$

We know $f_{X,Y}(x, y) = f_X(x) f_Y(y)$

$$f_{X,Y}(x, y) = \left[\frac{1}{\Gamma(n_1/2)} \left(\frac{1}{2}\right)^{\frac{n_1}{2}} x^{\frac{n_1}{2}-1} e^{-\frac{x}{2}} \right] \left[\frac{1}{\Gamma(n_2/2)} \left(\frac{1}{2}\right)^{\frac{n_2}{2}} (y)^{\frac{n_2}{2}-1} e^{-\frac{y}{2}} \right]$$

Let $z = x/y$ and $w = y$

Then $x = zw$ and $y = w$

$$J(x, y) = \begin{vmatrix} \frac{\partial z}{\partial x} & \frac{\partial z}{\partial y} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} \end{vmatrix} = \begin{vmatrix} \frac{1}{y} & -\frac{x}{y^2} \\ 0 & 1 \end{vmatrix} = \frac{1}{y} = \frac{1}{w}$$

$$f_{Z,W}(z, w) = w f_{X,Y}(x, y)$$

$$\begin{aligned} f_{Z,W}(z, w) &= \frac{w}{2^{(n_1+n_2)/2} \Gamma\left(\frac{n_1}{2}\right) \Gamma\left(\frac{n_2}{2}\right)} (zw)^{\frac{n_1}{2}-1} e^{-\frac{wz}{2}} (w)^{\frac{n_2}{2}-1} e^{-w/2} \\ &= \frac{w}{2^{(n_1+n_2)/2} \Gamma\left(\frac{n_1}{2}\right) \Gamma\left(\frac{n_2}{2}\right)} e^{-\frac{w}{2}(1+z)} (wz)^{\frac{n_1}{2}-1} w^{\frac{n_2}{2}-1} \end{aligned}$$

$$\begin{aligned} f_Z(z) &= \int_{-\infty}^{\infty} f_{Z,W}(z, w) dw \\ &= \frac{(z)^{\frac{n_1}{2}-1}}{2^{(n_1+n_2)/2} \Gamma\left(\frac{n_1}{2}\right) \Gamma\left(\frac{n_2}{2}\right)} \int_0^{\infty} w^{\left(\frac{n_1+n_2}{2}-1\right)} e^{-\frac{w}{2}(1+z)} dw \end{aligned}$$

$$\text{Let } \frac{w}{2}(1+z) = t$$

$$\Rightarrow dw = \frac{2dt}{1+z}$$

$$\begin{aligned} f_Z(z) &= \frac{(z)^{\frac{n_1}{2}-1}}{2^{(n_1+n_2)/2} \Gamma\left(\frac{n_1}{2}\right) \Gamma\left(\frac{n_2}{2}\right)} \int_0^{\infty} \left(\frac{2t}{1+z}\right)^{\left(\frac{n_1+n_2}{2}-1\right)} e^{-t} \frac{2dt}{1+z} \\ &= \frac{(z)^{\frac{n_1}{2}-1}}{\Gamma\left(\frac{n_1}{2}\right) \Gamma\left(\frac{n_2}{2}\right) (1+z)^{(n_1+n_2)/2}} \int_0^{\infty} t^{\frac{n_1+n_2}{2}-1} e^{-t} dt \\ &= \frac{(z)^{\frac{n_1}{2}-1}}{\Gamma\left(\frac{n_1}{2}\right) \Gamma\left(\frac{n_2}{2}\right) (1+z)^{(n_1+n_2)/2}} \Gamma\left(\frac{n_1+n_2}{2}\right) \end{aligned}$$

$$= \frac{(z)^{\frac{n_1}{2}-1}}{B\left(\frac{n_1}{2}, \frac{n_2}{2}\right)(1+z)^{(n_1+n_2)/2}}$$

$$\Rightarrow Z \text{ is } \beta_2\left(\frac{n_1}{2}, \frac{n_2}{2}\right) \text{ random variable}$$



Practice Problem

5.16 If X and Y are two independent chi-square random variables with degrees of freedom n_1 and n_2 , show that $X/(X+Y)$ is a $\beta_1\left(\frac{n_1}{2}, \frac{n_2}{2}\right)$ variate.



CORRELATION COEFFICIENT 5.5

The correlation coefficient is a measure of the degree of linearity (similarity) between two random variables X and Y .

The correlation coefficient of two random variables denoted by ρ_{XY} is defined as

$$\rho_{XY} = E\left[\left(\frac{X - \mu_X}{\sigma_X}\right)\left(\frac{Y - \mu_Y}{\sigma_Y}\right)\right] \quad (5.51)$$

where $\frac{X - \mu_X}{\sigma_X}$ and $\frac{Y - \mu_Y}{\sigma_Y}$ are normalized random variables.

From Eq. (5.51)

$$\rho_{XY} = E\left[\frac{(X - \mu_X)(Y - \mu_Y)}{\sigma_X \sigma_Y}\right] = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} = \frac{\sigma_{XY}}{\sigma_X \sigma_Y}$$

or

$$\rho_{XY} = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)} \sqrt{\text{Var}(Y)}} \quad (5.52)$$

The correlation coefficient has the property that

$$-1 \leq \rho_{XY} \leq 1$$

We can prove the above equation as follows.

Suppose that X and Y have variances given by σ_X^2 and σ_Y^2 , respectively.

Then

$$0 \leq \text{Var}\left(\frac{X}{\sigma_X} + \frac{Y}{\sigma_Y}\right)$$

$$\leq \frac{\text{Var}(X)}{\sigma_X^2} + \frac{\text{Var}(Y)}{\sigma_Y^2} + \frac{2 \text{Cov}(X, Y)}{\sigma_X \sigma_Y} \quad (5.53)$$

$$\Rightarrow -1 \leq \rho_{XY} \quad (5.54)$$

On the other hand,

$$\begin{aligned}
 0 &\leq \text{Var} \left(\frac{X}{\sigma_X} - \frac{Y}{\sigma_Y} \right) \\
 0 &\leq \frac{\text{Var}(X)}{\sigma_X^2} + \frac{\text{Var}(Y)}{\sigma_Y^2} - \frac{2 \text{Cov}(X, Y)}{\sigma_X \sigma_Y} \\
 0 &\leq 2(1 - \rho_{XY})
 \end{aligned} \tag{5.55}$$

$$\Rightarrow \rho_{XY} \leq 1$$

Therefore $-1 \leq \rho_{XY} \leq 1$

or

From Eq. (5.55) we have

$$\begin{aligned}
 \sigma_X^2 \sigma_Y^2 &\geq \{\text{Cov}(X, Y)\}^2 \\
 1 &\geq \rho_{XY}^2 \\
 \rho_{XY}^2 \leq 1 &\Rightarrow -1 \leq \rho_{XY} \leq 1
 \end{aligned} \tag{5.56}$$

A value of ρ_{XY} close to +1 or -1 indicates high degree of linearity (similarity) between X and Y , whereas a value near zero indicates lack of such linearity. A positive value of ρ_{XY} indicates that Y tends to increase with increasing X , whereas a negative value indicates that Y tends to decrease with increasing X . If $\rho_{XY} = 0$ then X and Y are said to be uncorrelated.

The scatter diagram can be used to study the independence of random variables X and Y . In the scatter diagram shown in Fig. 5.27(a), the points are aligned over a straight line. So X and Y are correlated and $\rho_{XY} = 1$. In Fig. 5.27(b) the points are aligned on a straight line with negative slope. Therefore, $\rho_{XY} = -1$. In Fig. 5.27(c) the points are scattered closely around straight line, therefore ρ_{XY} is large where in Fig. 5.27(d) the points are scattered loosely around a straight line and hence, ρ_{XY} is small. In Fig. 5.27 the points are scattered throughout the plane without a pattern, therefore, X and Y are independent.

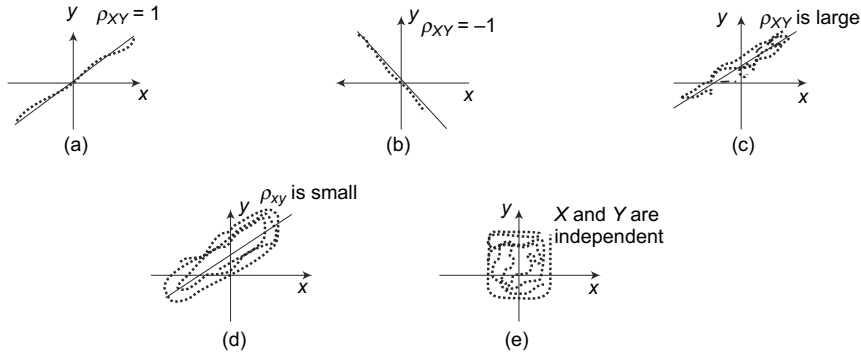


Fig. 5.27

Summary

- (i) Two random variables are said to be uncorrelated if $E[XY] = E[X] E[Y]$. Uncorrelated means not linearly related. In this case, $\text{Cov}(X, Y) = 0$ or $\rho_{XY} = 0$.
- (ii) Two random variables are said to be *orthogonal* if $E[XY] = 0$

- (iii) Two random variables X and Y are said to be independent if for every pair (x, y)
 $f_{X,Y}(x, y) = f_X(x) f_Y(y)$
 or $F_{X,Y}(x, y) = F_X(x) F_Y(y)$
- (iv) If either X or Y has zero mean then X and Y are uncorrelated $\Rightarrow X$ and Y orthogonal
- (v) X and Y are independent $\Rightarrow X$ and Y are uncorrelated: X and Y are uncorrelated $\nRightarrow X$ and Y are independent

REVIEW QUESTIONS



8. Define correlation coefficient.
9. Prove that correlation coefficient lies between -1 and 1 .
10. What is the condition for orthogonality of random variables?

Solved Problems



5.41 Calculate the correlation coefficient for the following heights (in inches) of fathers X and their sons Y .

X	65	66	67	67	68	69	70	72
Y	67	68	65	68	72	72	69	71

Solution The correlation coefficient $\rho_{XY} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$
 where $\text{Cov}(X, Y) = E[XY] - E[X] E[Y]$

Now we find the values in the above equation using the following table:

X	Y	XY	X^2	Y^2
65	67	4355	4225	4489
66	68	4488	4356	4624
67	65	4355	4489	4225
67	68	4556	4489	4624
68	72	4896	4624	5184
69	72	4968	4761	5184
70	69	4830	4900	4761
72	71	5112	5184	5041
544	552	37560	37028	38132

$$\mu_X = E[X] = \frac{\sum X}{n} = \frac{544}{8} = 68$$

$$\mu_Y = E[Y] = \frac{\sum Y}{n} = \frac{552}{8} = 69$$

$$\sigma_X^2 = E[X^2] - \{E[X]\}^2 = \frac{\sum X^2}{n} - \mu_X^2$$

$$\begin{aligned}
&= \frac{37028}{8} - (68)^2 = 4.5 \Rightarrow \sigma_X = 2.121 \\
\sigma_Y^2 &= E[Y^2] - \{E[Y]\}^2 = \frac{\sum Y^2}{n} - \mu_Y^2 \\
&= \frac{38132}{8} - (69)^2 = 5.5 \Rightarrow \sigma_Y = 2.345 \\
\text{Cov}(X, Y) &= E[XY] - E[X] E[Y] \\
E[XY] &= \frac{\sum XY}{n} = \frac{37560}{8} = 4695 \\
\text{Cov}(X, Y) &= 4695 - 68(69) = 3 \\
\rho_{XY} &= \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} = \frac{3}{(2.121)(2.345)} = 0.603
\end{aligned}$$

5.42 Two random variables X and Y are defined as $Y = 4X + 9$. Find the correlation coefficient between X and Y .

Solution The correlation coefficient

$$\begin{aligned}
\rho_{XY} &= \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} \\
\text{Cov}(X, Y) &= E[XY] - E[X] E[Y] \\
&= E[X(4X + 9)] - E[X] E[4X + 9] \\
&= E[4X^2 + 9X] - E[X] [4E[X] + 9] \\
&= 4E[X^2] + 9E[X] - 4\{E[X]\}^2 - 9E[X] \\
&= 4\{E[X^2] - \{E[X]\}^2\} = 4\sigma_X^2 \\
\sigma_Y^2 &= E[Y^2] - \{E[Y]\}^2 = E\{4X + 9\}^2 - \{E[4X + 9]\}^2 \\
&= E[16X^2 + 72X + 81] - \{4E[X] + 9\}^2 \\
&= 16E[X^2] + 72E[X] + 81 - 16\{E[X]\}^2 - 81 - 72E[X] \\
&= 16\{E[X^2] - \{E[X]\}^2\} = 16\sigma_X^2 \\
\Rightarrow \sigma_Y &= 4\sigma_X \\
\rho_{XY} &= \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} = \frac{4\sigma_X^2}{\sigma_X (4\sigma_X)} = 1
\end{aligned}$$

$$\boxed{\rho_{XY} = 1}$$

5.43 Consider two random variables X and Y are independent, zero mean with variance 49 and 25 respectively. Find the correlation coefficient between $(X + Y)$ and $(X - Y)$.

Solution Given $\mu_X = 0$ and $\mu_Y = 0$

Also $\sigma_X^2 = 49$ and $\sigma_Y^2 = 25$

Let $U = X + Y$ and $V = X - Y$

The correlation coefficient between U and V is given by

$$\rho_{UV} = \frac{C_{UV}}{\sigma_U \sigma_V} = \frac{\text{Cov}[U, V]}{\sigma_U \sigma_V}$$

Since $\mu_X = \mu_Y = 0$ we have $E[U] = E[X] + E[Y] = 0$ and $E[V] = E[X] - E[Y] = 0$

and $\text{Cov}(U, V) = E[UV] - E[U] E[V]$

$$= E[UV] = E[(X + Y)(X - Y)]$$

$$= E[X^2 - Y^2] = E[X^2] - E[Y^2]$$

$$E[X^2] = \sigma_X^2 = 49 \text{ and } E[Y^2] = \sigma_Y^2 = 25$$

$$C_{UV} = 49 - 25 = 24$$

$$\sigma_U^2 = E[(X + Y)^2] = E[X^2] + E[Y^2] + 2E[X] E[Y] = E[X^2] + E[Y^2]$$

$$= 49 + 25 = 74$$

$$\sigma_V^2 = E[(X - Y)^2] = E[X^2] + E[Y^2] - 2E[X] E[Y] = E[X^2] + E[Y^2]$$

$$= 49 + 25 = 74$$

$$\begin{array}{l} \because E[X] = 0 \\ E[Y] = 0 \end{array}$$

$$\rho_{UV} = \frac{24}{\sqrt{74} \sqrt{74}} = \frac{24}{74} = 0.324$$



Practice Problem

5.17 If the independent random variables X and Y have the variance 36 and 16 respectively, find the correlation coefficient between $X + Y$ and $X - Y$ assuming X and Y are zero mean. (Ans. 5/13)

Solved Problem



5.44 For the following bivariate distribution, determine the correlation coefficient.

X	0	1	2	3
Y				
1	$\frac{5}{48}$	$\frac{7}{48}$	—	—
2	$\frac{9}{48}$	$\frac{5}{48}$	$\frac{5}{24}$	—
3	—	$\frac{1}{12}$	$\frac{1}{16}$	$\frac{5}{48}$

Solution From the table, $P_X(X=0) = \frac{14}{48}$, $p_X(X=1) = \frac{16}{48}$; $p_X(X=2) = \frac{13}{48}$; $p_X(X=3) = \frac{5}{48}$

$$P_Y(Y=1) = \frac{12}{48}, P_Y(Y=2) = \frac{24}{48}; P_X(Y=3) = \frac{12}{48}$$

$$\begin{aligned} E[X] &= \sum_i x_i P_X(x_i) \\ &= 0\left(\frac{14}{48}\right) + 1\left(\frac{16}{48}\right) + 2\left(\frac{13}{48}\right) + 3\left(\frac{5}{48}\right) = \frac{19}{16} \end{aligned}$$

$$E[Y] = \sum_j y_j P_Y(y_j) = 1\left(\frac{12}{48}\right) + 2\left(\frac{24}{48}\right) + 3\left(\frac{12}{48}\right) = 2$$

$$E[X^2] = (0)^2 \left(\frac{14}{48}\right) + (1)^2 \frac{16}{48} + (2)^2 \frac{13}{48} + (3)^2 \frac{5}{48} = \frac{113}{48}$$

$$E[Y^2] = (1)^2 \frac{12}{48} + (2)^2 \frac{24}{48} + (3)^2 \frac{12}{48} = \frac{216}{48}$$

$$\begin{aligned} E[XY] &= \sum_j \sum_i x_i y_j P_{X,Y}(x_i, y_j) \\ &= \frac{33}{12} \end{aligned}$$

$$\sigma_X^2 = \frac{113}{48} - \left(\frac{14}{16}\right)^2 = 0.944$$

$$\sigma_Y^2 = \frac{216}{48} - (2)^2 = 0.5$$

$$\begin{aligned} \rho_{X,Y} &= \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} \\ &= \frac{E[XY] - E[X]E[Y]}{\sigma_X \sigma_Y} \\ &= \frac{3/8}{(0.97)(0.707)} = 0.545 \end{aligned}$$



Practice Problem

5.18 Find the correlation coefficient $\rho_{X,Y}$ for the bivariate random variable (X, Y) having the joint pdf

$$f_{X,Y}(x, y) = \begin{cases} 2xy & 0 < x < 1, 0 < y < 1 \\ 0 & \text{otherwise} \end{cases} \quad (\text{Ans. } 0.8)$$

Solved Problems



5.45 If X , Y and Z are uncorrelated random variables with the same variance, find the correlation coefficient between $(X + Y)$ and $(Y + Z)$.

Solution Let $U = X + Y$

and $V = Y + Z$

The correlation coefficient

$$\rho_{UV} = \frac{\text{Cov}[U, V]}{\sigma_U \sigma_V}$$

$$\begin{aligned}\text{Cov}[U, V] &= E[(X + Y)(Y + Z)] - E[X + Y]E[Y + Z] \\ &= E[XY + Y^2 + XZ + YZ] - \{E[X]E[Y] + \{E[Y]\}^2 + E[X]E[Z] + E[Y]E[Z]\}\end{aligned}$$

Since X, Y and Z are uncorrelated we have

$$E[XY] = E[X]E[Y]$$

$$E[YZ] = E[Y]E[Z]$$

$$E[XZ] = E[X]E[Z]$$

$$\Rightarrow \text{Cov}[U, V] = E[Y^2] - \{E[Y]\}^2 = \text{Var}[Y]$$

$$\text{Also } \text{Var}[U] = \text{Var}[X] + \text{Var}[Y]$$

$$\text{Var}[V] = \text{Var}[Y] + \text{Var}[Z]$$

Since all random variables have same variance,

$$\text{Var}[U] = 2 \text{Var}[Y]$$

$$\text{Var}[V] = 2 \text{Var}[Y]$$

$$\sigma_U = \sqrt{2 \text{Var}[Y]} \text{ and } \sigma_V = \sqrt{2 \text{Var}[Y]}$$

$$\begin{aligned}\rho_{UV} &= \frac{\text{Cov}[U, V]}{\sigma_U \sigma_V} = \frac{\text{Var}[Y]}{\sqrt{2 \text{Var}[Y]} \sqrt{2 \text{Var}[Y]}} \\ &= \frac{1}{2}\end{aligned}$$

5.46 The joint probability mass function of X and Y is given below

X \ Y	0	1
-1	$\frac{1}{8}$	$\frac{2}{8}$
1	$\frac{3}{8}$	$\frac{2}{8}$

Find the correlation coefficient.

Solution We redraw the table and find marginal functions of X and Y .

X \ Y	0	1	$p_X(x)$
-1	$\frac{1}{8}$	$\frac{2}{8}$	$\frac{3}{8}$
1	$\frac{3}{8}$	$\frac{2}{8}$	$\frac{5}{8}$
$p_Y(y)$	$\frac{1}{2}$	$\frac{1}{2}$	

From the table,

$$P_X(X = -1) = \frac{3}{8}; P_X(X = 1) = \frac{5}{8}$$

$$P_X(Y=0) = \frac{1}{2}; P_Y(Y=1) = \frac{1}{2}$$

$$E[X] = \sum_i x_i p_X(x_i) = (-1) \frac{3}{8} + 1 \left(\frac{5}{8} \right) = \frac{1}{4}; E[Y] = \sum_j y_j p_Y(y_j) = 0 \left(\frac{1}{2} \right) + 1 \left(\frac{1}{2} \right) = \frac{1}{2}$$

$$E[X^2] = \sum_i x_i^2 p_X(x_i) = (-1)^2 \frac{3}{8} + (1)^2 \frac{5}{8} = 1; E[Y^2] = \sum_j y_j^2 p_Y(y_j) = (0)^2 \left(\frac{1}{2} \right) + (1)^2 \frac{1}{2} = \frac{1}{2}$$

$$E[XY] = \sum_j \sum_i x_i y_j p_{XY}(x_i, y_j) \\ = (-1)(0) \left(\frac{1}{8} \right) + (-1)(1) \frac{2}{8} + (1)(0) \frac{3}{8} + 1(1) \frac{2}{8} = 0$$

$$\sigma_X^2 = E[X^2] - \{E[X]\}^2 = 1 - \left(\frac{1}{4} \right)^2 = \frac{15}{16} \Rightarrow \sigma_X = \frac{\sqrt{15}}{4}$$

$$\sigma_Y^2 = E[Y^2] - \{E[Y]\}^2 = \frac{1}{2} - \left(\frac{1}{2} \right)^2 = \frac{1}{4} \Rightarrow \sigma_Y = \frac{1}{2}$$

$$\text{Cov}(X, Y) = E[XY] - E[X] E[Y] = 0 - \frac{1}{4} \left(\frac{1}{2} \right) = -\frac{1}{8}$$

$$\rho = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} = \frac{-1/8}{\left(\frac{\sqrt{15}}{4} \right) \left(\frac{1}{2} \right)} = -0.258$$

5.47 Let the joint distribution of X and Y be given by

$X \backslash Y$	-1	0	1
-1	$\frac{1}{6}$	0	$\frac{1}{6}$
0	$\frac{1}{3}$	0	0
1	$\frac{1}{6}$	0	$\frac{1}{6}$

Show that the covariance is zero even though the two random variables are not independent. (AU 2008)

Solution Let us redraw the table and find marginal functions of X and Y

$X \backslash Y$	-1	0	1	$p_X(x)$
-1	$\frac{1}{6}$	0	$\frac{1}{6}$	$\frac{1}{3}$
0	$\frac{1}{3}$	0	0	$\frac{1}{3}$
1	$\frac{1}{6}$	0	$\frac{1}{6}$	$\frac{1}{3}$
$p_Y(y)$	$\frac{2}{3}$	0	$\frac{1}{3}$	

$$\begin{aligned}
\mu_X = E[X] &= \sum x_i p(x_i) \\
&= (-1) \left(\frac{1}{3} \right) + 0 \left(\frac{1}{3} \right) + 1 \left(\frac{1}{3} \right) = 0 \\
\mu_Y = E[Y] &= \sum y_i p(y_i) \\
&= (-1) \left(\frac{2}{3} \right) + (0)(0) + 1 \left(\frac{1}{3} \right) = -\frac{1}{3} \\
E[XY] &= \sum_j \sum_i x_i y_j p(x_i, y_j) \\
&= (-1)(-1) \left(\frac{1}{6} \right) + (-1)(0)(0) + (-1)(1) \frac{1}{6} + 0(-1) \left(\frac{1}{3} \right) \\
&\quad + 0(0)(0) + (0)(1)(0) + 1(-1) \frac{1}{6} + 1(0)(0) + 1(1) \left(\frac{1}{6} \right) \\
&= \frac{1}{6} - \frac{1}{6} - \frac{1}{6} + \frac{1}{6} = 0 \\
\text{Cov}(X, Y) &= E[XY] - E[X] E[Y] = 0 - (0) \left(-\frac{1}{3} \right) = 0
\end{aligned}$$

For independence $p_{X,Y}(x,y) = p_X(x) p_Y(y)$

$$P[X = -1, Y = -1] = \frac{1}{6}; P_X(X = -1) = \frac{1}{3}, P_Y(Y = -1) = \frac{2}{3}$$

Since

$$P(X = -1, Y = -1) \neq P_X(X = -1) (P_Y(Y = -1))$$

So they are not independent.

5.48 For two random variables X and Y , $E[X] = 5$

$$E[Y] = 10; E[XY] = 75, E[X^2] = 41; E[Y^2] = 149$$

Find covariance and correlation coefficient.

Solution Given: $E[X] = 5; E[Y] = 10; E[XY] = 75; E[X^2] = 41$ and $E[Y^2] = 149$

$$\begin{aligned}
\text{Cov}(X, Y) &= E[XY] - E[X] E[Y] \\
&= 75 - 5(10) = 25
\end{aligned}$$

The correlation coefficient

$$\begin{aligned}
\rho_{XY} &= \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} \\
\sigma_X^2 &= E[X^2] - \{E[X]\}^2 \\
&= 41 - (5)^2 = 16 \Rightarrow \sigma_X = 4 \\
\sigma_Y^2 &= E[Y^2] - \{E[Y]\}^2 = 149 - (10)^2 = 49 \Rightarrow \sigma_Y = 7 \\
\rho &= \frac{25}{4(7)} = \frac{25}{28} = 0.893
\end{aligned}$$

5.49 For random variables X and Y having $\mu_X = 2$; $\mu_Y = 3$; $\sigma_X^2 = 9$; $\sigma_Y^2 = 16$ and $\rho_{XY} = \frac{2}{3}$

Find (i) the covariance of X and Y , (ii) the correlation of X and Y , and (iii) $E[X^2]$ and $E[Y^2]$.

Solution Given: $\mu_X = 2$; $\mu_Y = 3$; $\sigma_X^2 = 9$; $\sigma_Y^2 = 16$ and $\rho_{XY} = \frac{2}{3}$

$$\rho_{XY} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

$$\begin{aligned} \Rightarrow \text{Cov}(X, Y) &= \rho_{XY} \sigma_X \sigma_Y \\ &= \frac{2}{3} (3) (4) = 8 \end{aligned}$$

$$\text{Cov}(X, Y) = E[XY] - E[X] E[Y]$$

$$E[XY] = 8 + 2(3) = 16$$

$$\begin{aligned} E[X^2] &= \sigma_X^2 + \mu_X^2 \\ &= 9 + (2)^2 = 13 \end{aligned}$$

$$\begin{aligned} E[Y^2] &= \sigma_Y^2 + \mu_Y^2 \\ &= 16 + (3)^2 = 25 \end{aligned}$$

5.50 X and Y are zero-mean random variables with $\sigma_X^2 = 9$ and $\sigma_Y^2 = 25$. Their correlation coefficient is $\rho_{XY} = -0.6$. If $W = (aX + 3Y)^2$ (a) Find a value for the parameter a that minimizes the mean value of W . (b) Find the minimum value.

Solution Given: $W = (aX + 3Y)^2$; $\mu_X = 0$, $\mu_Y = 0$, $\sigma_X^2 = E[X^2] = 9$ and $\sigma_Y^2 = E[Y^2] = 25$

$$\rho_{XY} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} = \frac{\text{Cov}(X, Y)}{3(5)}$$

$$\text{Cov}(X, Y) = -0.6 (15) = -9$$

$$\begin{aligned} \text{Cov}(X, Y) &= E[XY] - \mu_X \mu_Y \\ &= E[XY] = -9 \end{aligned}$$

$$\begin{aligned} E[W] &= E\{(aX + 3Y)^2\} \\ &= E[a^2 X^2 + 9Y^2 + 6aXY] \\ &= a^2 E[X^2] + 9E[Y^2] + 6aE[XY] \\ &= 9a^2 - 54a + 225 = 0 \end{aligned}$$

$$\frac{d}{dW} E[W] = 0 \Rightarrow 18a - 54 = 0, a = 3$$

$$E[W]_{\min} = 9(3)^2 - 54(3) + 225 = 144$$

5.51 Two random variable X and Y have means 1 and 2 respectively and variances 4 and 1 respectively. Their correlation coefficient is 0.4. New random variables W and V are defined as

$$V = -X + 2Y; W = X + 3Y$$

Find (a) means, (b) variances, (c) correlations, (d) correlation coefficient of V and W .

Solution Given: $E[X] = 1$ and $E[Y] = 2$

$$\sigma_X^2 = 4 \text{ and } \sigma_Y^2 = 1$$

$$\rho_{XY} = 0.4$$

Given: $V = -X + 2Y$; $W = X + 3Y$

$$E[V] = -E[X] + 2E[Y] = -1 + 2(2) = 3$$

$$E[W] = E[X] + 3E[Y] = 1 + 3(2) = 7$$

The correlation coefficient

$$\rho_{XY} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} = 0.4$$

$$\Rightarrow \text{Cov}(X, Y) = 0.4 \sigma_X \sigma_Y \\ = 0.4 (2)(1) = 0.8$$

$$\text{Cov}(X, Y) = 0.8$$

$$\text{Var}[V] = \text{Var}[-X + 2Y] = \text{Var}[X] + 2^2 \text{Var}[Y] + 2(-1) 2 \text{Cov}(X, Y) \\ = 4 + 4 - 4(0.8) = 4.8$$

$$\text{Var}[W] = \text{Var}[X + 3Y] = \text{Var}[X] + 3^2 \text{Var}[Y] + 2(1)(3) \text{Cov}(X, Y) \\ = 4 + 9(1) + 6(0.8) \\ = 13 + 4.8 = 17.8$$

$$\text{Var}[V] = 4.8$$

$$\text{Var}[W] = 17.8$$

Correlation between V and W is

$$E[VW] = E[(-X + 2Y)(X + 3Y)] \\ = -E[X^2] - E[XY] + 6E[Y^2] \\ E[X^2] = \sigma_X^2 + \{E[X]\}^2 = 4 + 1 = 5 \\ E[Y^2] = \sigma_Y^2 + \{E[Y]\}^2 = 1 + 2^2 = 5$$

$$\text{Cov}[X, Y] = \rho_{XY}(\sigma_X \sigma_Y) \\ = E[XY] - E[X] E[Y] = 0.8$$

$$\Rightarrow E[XY] = 0.8 + (1)(2) = 2.8$$

$$E[VW] = -5 - 2.8 + 6 \times 5 = 22.2$$

$$\rho_{VW} = \frac{E[VW] - E[V]E[W]}{\sigma_V \sigma_W} = \frac{22.2 - 3(7)}{\sqrt{4.8} \sqrt{17.8}} = 0.13$$



Practice Problems

5.19 X , Y and Z are uncorrelated random variables with zero mean and standard deviation 5, 12 and 9 respectively. If $U = X + Y$ and $V = Y + Z$, find the correlation coefficient between U and V . (Ans. 48/65)

5.20 Two random variables X and Y are related as $Y = 3X + 5$. Find the correlation coefficient between X and Y . (Ans. 1)

Solved Problems

5.52 The joint density function of X and Y is given by

$$f_{X,Y}(x, y) = \frac{1}{y} e^{-(y+x/y)}, \quad x > 0, y > 0$$

Find $E[X]$, $E[Y]$, and show that $\text{Cov}(X, Y) = 1$.

Solution

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$$

$$\begin{aligned} E[X] &= \int_0^\infty \int_0^\infty x f_{X,Y}(x, y) dx dy \\ &= \int_0^\infty \int_0^\infty \frac{x}{y} e^{-(y+x/y)} dx dy = \int_0^\infty \int_0^\infty \frac{x}{y} e^{-y} e^{-x/y} dx dy \\ &= \int_0^\infty \frac{1}{y} e^{-y} \left[\int_0^\infty x e^{-x/y} dx \right] dy \\ &= \int_0^\infty \frac{1}{y} e^{-y} \left[-xy e^{-x/y} \Big|_0^\infty - y^2 e^{-x/y} \Big|_0^\infty \right] dy = \int_0^\infty \frac{1}{y} (y^2) e^{-y} dy \\ &= \int_0^\infty y e^{-y} dy = -y e^{-y} \Big|_0^\infty - e^{-y} \Big|_0^\infty = 1 \end{aligned}$$

$$E[X] = 1$$

$$\begin{aligned} E[Y] &= \int_0^\infty \int_0^\infty y f_{X,Y}(x, y) dx dy = \int_0^\infty \int_0^\infty e^{-(y+x/y)} dx dy \\ &= \int_0^\infty e^{-y} \int_0^\infty e^{-x/y} dx dy = \int_0^\infty e^{-y} \left[-y e^{-x/y} \Big|_0^\infty \right] dy \\ &= \int_0^\infty y e^{-y} dy = -y e^{-y} \Big|_0^\infty + \int_0^\infty e^{-y} dy = -e^{-y} \Big|_0^\infty = 1 \end{aligned}$$

$$\begin{aligned} E[XY] &= \int_0^\infty \int_0^\infty xy f_{X,Y}(x, y) dx dy \\ &= \int_0^\infty \int_0^\infty x e^{-(y+x/y)} dx dy = \int_0^\infty e^{-y} \left[\int_0^\infty x e^{-x/y} dx \right] dy \\ &= \int_0^\infty e^{-y} \left[-xy e^{-x/y} \Big|_0^\infty + \int_0^\infty y e^{-x/y} dx \right] dy \end{aligned}$$

$$\begin{aligned}
&= \int_0^{\infty} e^{-y} \left[-xy e^{-x/y} \Big|_0^{\infty} - y^2 e^{-x/y} \Big|_0^{\infty} \right] dy \\
&= \int_0^{\infty} y^2 e^{-y} dy = -y^2 e^{-y} \Big|_0^{\infty} - 2ye^{-y} \Big|_0^{\infty} + \int_0^{\infty} 2e^{-y} dy \\
&= -2e^{-y} \Big|_0^{\infty} = -2(-1) = 2
\end{aligned}$$

$$\begin{aligned}
\text{Cov}[X, Y] &= E[XY] - E[X] E[Y] \\
&= 2 - (1)(1) = 1
\end{aligned}$$

$$\boxed{\text{Cov}(X, Y) = 1}$$

5.53 The random variables X and Y have a joint density function given by

$$f_{X,Y}(x, y) = \begin{cases} \frac{2}{x} e^{-2x} & 0 \leq x < \infty, 0 \leq y \leq x \\ 0 & \text{otherwise} \end{cases}$$

Compute $\text{Cov}(X, Y)$.

Solution

$$\begin{aligned}
E[XY] &= \int_0^{\infty} \int_0^x xy f_{X,Y}(x, y) dy dx \\
&= \int_0^{\infty} \int_0^x xy \frac{2}{x} e^{-2x} dy dx \\
&= 2 \int_0^{\infty} e^{-2x} \left[\int_0^x y dy \right] dx = 2 \int_0^{\infty} e^{-2x} \left(\frac{x^2}{2} \right) dx \\
&= \int_0^{\infty} x^2 e^{-2x} dx = \frac{-x^2 e^{-2x}}{2} \Big|_0^{\infty} - \frac{-2x e^{-2x}}{4} \Big|_0^{\infty} - \frac{1}{2} \int_0^{\infty} e^{-2x} dx \\
&= \frac{1}{2} \frac{e^{-2x}}{-2} \Big|_0^{\infty} = \frac{-1}{4} [-1] = \frac{1}{4} \\
E[X] &= \int_0^{\infty} \int_0^x x f_{X,Y}(x, y) dy dx = \int_0^{\infty} 2e^{-2x} \int_0^x dy dx \\
&= \int_0^{\infty} 2e^{-2x} x dx = 2 \left[\frac{-x}{2} e^{-2x} \Big|_0^{\infty} - \frac{e^{-2x}}{4} \Big|_0^{\infty} \right] \\
&= 2 \left(\frac{1}{4} \right) = \frac{1}{2}
\end{aligned}$$

$$\begin{aligned}
 E[Y] &= \int_0^\infty \int_0^x y \frac{2}{x} e^{-2x} dy dx = \int_0^\infty \frac{2}{x} e^{-2x} \left[\int_0^x y dy \right] dx \\
 &= \int_0^\infty \frac{2}{x} e^{-2x} \left(\frac{x^2}{2} \right) dx = \int_0^\infty x e^{-2x} dx = \frac{-x}{2} e^{-2x} \Big|_0^\infty - \frac{e^{-2x}}{4} \Big|_0^\infty \\
 &= -\frac{1}{4}(-1) = \frac{1}{4}
 \end{aligned}$$

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$$

$$= \frac{1}{4} - \left(\frac{1}{2} \right) \left(\frac{1}{4} \right) = \frac{1}{4} - \frac{1}{8} = \frac{1}{8}$$

$$\boxed{\text{Cov}(X, Y) = \frac{1}{8}}$$

5.54 Discrete random variables X and Y have the joint density

$$f_{X,Y}(x, y) = 0.3 \delta(x + \alpha) \delta(y - 1) + 0.15 \delta(x - 2) \delta(y - 2) + 0.15 \delta(x - \alpha) \delta(y - \alpha) + 0.4 \delta(x - 1) \delta(y - 1)$$

Determine the value of a that minimizes the correlation between X and Y and find minimum correlation. Are X and Y orthogonal.

Solution

$$\text{Correlation} = E[XY] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xy f_{X,Y}(x, y) dx dy$$

$$\begin{aligned}
 &\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xy \delta(x - a) \delta(y - b) dx dy \\
 &= ab
 \end{aligned}$$

$$\begin{aligned}
 &= 0.3(-\alpha)(1) + 0.15(2)(2) + 0.15(\alpha)(\alpha) + 0.4(1)(1) \\
 &= -0.3\alpha + 0.6 + 0.15\alpha^2 + 0.4 \\
 &= 0.15\alpha^2 - 0.3\alpha + 1
 \end{aligned}$$

This is $R_{XY} = 0.15\alpha^2 - 0.3\alpha + 1$

$$\begin{aligned}
 \frac{d R_{XY}}{d\alpha} &= 0.3\alpha - 0.3 = 0 \\
 \alpha &= 1
 \end{aligned}$$

At $\alpha = 1$ $R_{XY} = 0.15(1)^2 - 0.3(1) + 1 = 0.85$

X and Y are not orthogonal since $R_{XY} \neq 0$.



Practice Problem

5.21 For two discrete random variables X and Y ,

$$f_{X,Y}(x, y) = 0.2 \delta(x - \alpha) \delta(y - \alpha) + 0.4 \delta(x + \alpha) \delta(y - 1) + 0.18 \delta(x + 1) \delta(y + 2) + 0.22 \delta(x - 2) \delta(y - 1)$$

Determine the value of α , if any, that minimizes the correlation of X and Y . Find minimum correlation. Are X and Y uncorrelated?

(Ans: $\alpha = 1$, ; $R_{XY} = 0.6$, NO

Solved Problems

5.55 Two random variables V and W are defined as

$$V = X + aY$$

$$W = X - aY$$

where a is a real number and X and Y are random variables. (a) Determine a in terms of moments of X and Y such that V and W are orthogonal (b) If X and Y are Gaussian, show that V and W are statistically independent if $a^2 = \sigma_X^2 / \sigma_Y^2$, where σ_X^2 and σ_Y^2 are the variances of X and Y respectively.

Solution

(a) Given: $V = X + aY$

$$W = X - aY$$

For orthogonality, $E[VW] = 0$

$$\begin{aligned} E[VW] &= E[(X + aY)(X - aY)] \\ &= E[X^2 - a^2 Y^2] = E[X^2] - a^2 E[Y^2] \end{aligned}$$

$$\Rightarrow E[X^2] - a^2 E[Y^2] = 0$$

$$|a| = \sqrt{\frac{E[X^2]}{E[Y^2]}}$$

$$\text{In terms of moments, } |a| = \sqrt{\frac{m_{20}}{m_{02}}}$$

(b) If X and Y are Gaussian, V and W are also Gaussian since V and W are linear transformations of X and Y .

$$\mu_V = E[V] = E[X] + aE[Y] = \mu_X + a\mu_Y$$

$$\mu_W = E[W] = E[X] - aE[Y] = \mu_X - a\mu_Y$$

$$\begin{aligned} \text{Cov}(V, W) &= E[\{V - \mu_V\}\{W - \mu_W\}] \\ &= E[\{X + aY - \mu_X - a\mu_Y\}\{X - aY - \mu_X + a\mu_Y\}] \\ &= E[\{(X - \mu_X) + a(Y - \mu_Y)\}\{(X - \mu_X) - a(Y - \mu_Y)\}] \\ &= E[(X - \mu_X)^2 - a^2(Y - \mu_Y)^2] \\ &= E[(X - \mu_X)^2] - a^2 E[(Y - \mu_Y)^2] \\ &= \sigma_X^2 - a^2 \sigma_Y^2 \end{aligned}$$

$$\text{Since } a^2 = \sigma_X^2 / \sigma_Y^2$$

$$\text{Cov}(V, W) = \sigma_X^2 - \frac{\sigma_X^2}{\sigma_Y^2} (\sigma_Y^2) = 0$$

Therefore, V and W are uncorrelated.

Uncorrelated Gaussian random variables are statistically independent.

5.56 Let $X_1, X_2, \dots, X_n$ be independent and identically distributed random variables having mean μ_X and variance σ_X^2 , and let $\bar{X} = \sum_{i=1}^n X_i / n$. Show that (i) $E[\bar{X}] = \mu_X$, (ii) $\text{Var}(\bar{X}) = \frac{\sigma_X^2}{n}$, and (c) $E\left[\sum_{i=1}^n (X_i - \bar{X})^2\right] = (n-1)\sigma_X^2$.

Solution Given: $\bar{X} = \sum_{i=1}^n X_i / n$

$$\begin{aligned} E[\bar{X}] &= E\left[\sum_{i=1}^n (X_i / n)\right] \\ &= \sum_{i=1}^n E[X_i] / n = \frac{1}{n} \sum_{i=1}^n \mu_X \\ &= \frac{1}{n} (n \mu_X) = \mu_X \end{aligned}$$

$$E[\bar{X}] = \mu_X$$

$$\begin{aligned} \text{Var}(\bar{X}) &= \text{Var}\left[\sum_{i=1}^n X_i / n\right] \\ &= \frac{1}{n^2} \left[\sum_{i=1}^n \text{Var}(X_i)\right] = \frac{1}{n^2} \sum_{i=1}^n \sigma_X^2 = \frac{n\sigma_X^2}{n^2} = \frac{\sigma_X^2}{n} \end{aligned}$$

$$\begin{aligned} E\left[\sum_{i=1}^n (X_i - \bar{X})^2\right] &= E\left[\sum_{i=1}^n (X_i^2 + \bar{X}^2 - 2X_i \bar{X})\right] \\ &= E\left[\sum_{i=1}^n X_i^2 + \sum_{i=1}^n \bar{X}^2 - 2\left(\sum_{i=1}^n X_i\right) \bar{X}\right] \\ &= E\left[\sum_{i=1}^n X_i^2 + \sum_{i=1}^n \bar{X}^2 - 2(n\bar{X}) \bar{X}\right] \quad \boxed{\because \bar{X} = \sum_{i=1}^n X_i / n} \\ &= \sum_{i=1}^n E[X_i^2] + E[n\bar{X}^2] - 2E[n\bar{X}^2] \\ &= \sum_{i=1}^n (\sigma_X^2 + \mu_X^2) - nE[\bar{X}^2] \\ &= n(\sigma_X^2 + \mu_X^2) - n\left(\frac{\sigma_X^2}{n} + \mu_X^2\right) \end{aligned}$$

$$= n\sigma_X^2 - \sigma_X^2 = \sigma_X^2(n-1)$$

5.57 Prove

- (a) $\text{Cov}(a + bX, c + dY) = bd \text{Cov}(X, Y)$
 (b) $\text{Cov}(X + Y, Z) = \text{Cov}(X, Z) + \text{Cov}(Y, Z)$

Solution

- (a) $\text{Cov}(a + bX, c + dY) = E[\{(a + bX) - (a + b\mu_X)\}\{(c + dY) - (c + d\mu_Y)\}]$
 $= E[(a + bX)(c + dY) - (a + b\mu_X)(c + dY) - (c + d\mu_Y)(a + bX) + (a + b\mu_X)(c + d\mu_Y)]$
 $= E[bdXY - bd\mu_XY - bdX\mu_Y + bd\mu_X\mu_Y]$
 $= bd E[(X - \mu_X)(Y - \mu_Y)]$
 $= bd \text{Cov}(X, Y)$
- (b) $\text{Cov}(X + Y, Z) = E[(X + Y - \mu_X - \mu_Y)(Z - \mu_Z)]$
 $= E[XZ + YZ - \mu_XZ - \mu_YZ - X\mu_Z - Y\mu_Z + \mu_X\mu_Z + \mu_Y\mu_Z]$
 $= E[Z(X - \mu_X) - \mu_X(X - \mu_X) + Z(Y - \mu_Y) - \mu_Z(Y - \mu_Y)]$
 $= E[(Z - \mu_Z)(X - \mu_X) + (Y - \mu_Y)(Z - \mu_Z)]$
 $= E[(X - \mu_X)(Z - \mu_Z)] + E[(Y - \mu_Y)(Z - \mu_Z)]$
 $= \text{Cov}(X, Z) + \text{Cov}(Y, Z)$

5.58 If X and Y are identically distributed, not necessarily independent, show $\text{Cov}(X + Y, X - Y) = 0$.

Solution

$$\begin{aligned} \text{Cov}(X + Y, X - Y) &= E[\{(X + Y) - (\mu_X + \mu_Y)\}\{(X - Y) - (\mu_X - \mu_Y)\}] \\ &= E[\{(X - \mu_X) + (Y - \mu_Y)\}\{(X - \mu_X) - (Y - \mu_Y)\}] \\ &= E[(X - \mu_X)^2 - (Y - \mu_Y)^2] \\ &= E\{(X - \mu_X)^2\} - E\{(Y - \mu_Y)^2\} \\ &= \sigma_X^2 - \sigma_Y^2 = 0 \end{aligned}$$

since X and Y are identically distributed.

**Practice Problems**

5.22 If $Y = a + bX$, show that

$$\begin{aligned} \rho_{XY} &= 1 \text{ for } b > 0 \\ &= -1 \text{ for } b < 0 \end{aligned}$$

5.23 Show that $\text{Cov}(X, E[Y|X]) = \text{Cov}(X, Y)$

5.24 Let X and Y be dependent random variables with zero mean and unit variance and correlation coefficient ρ . Show that $E[\max(X^2, Y^2)] \leq 1 + \sqrt{1 - \rho^2}$.

Solved Problem

5.59 A random voltage $V \sim U(0, 6 \text{ V})$ is applied to a resistor whose resistance R is a binary random variable on a value of either 5Ω or 10Ω with equal probability and is independent of V .

- (a) Find the expected power dissipated by R
 (b) R_{vp} , R_{rp} and σ_{vp}^2

Solution Given: $V \sim U(0, 6 \text{ V})$ implies

$$f_V(v) = \frac{1}{6} \quad \text{for } 0 \leq v \leq 6$$

$$= 0 \quad \text{elsewhere}$$

Since R is a binary random variable with value of 5Ω or 10Ω with equal probability, we can write

$$f_R(r) = \frac{1}{2} \delta(r - 5) + \frac{1}{2} \delta(r - 10)$$

$$E[V^2] = \int_{-\infty}^{\infty} v^2 f_V(v) dv = \int_0^6 v^2 \left(\frac{1}{6}\right) dv = \frac{1}{6} \left[\frac{v^3}{3}\right]_0^6 = \frac{(6)^3}{6 \times 3} = 12$$

$$E\left[\frac{1}{R}\right] = \sum_{i=1}^2 \frac{1}{r_i} P\{R = r_i\} = \frac{1}{5} \left(\frac{1}{2}\right) + \left(\frac{1}{10}\right) \left(\frac{1}{2}\right) = \frac{2+1}{20} = \frac{3}{20}$$

Since V and R are independent V^2 and $\frac{1}{R}$ are independent and uncorrelated,

$$E[P] = E[V^2/R] = E[V^2] E\left[\frac{1}{R}\right] = 12 \left(\frac{3}{20}\right) = 1.8 \text{ W}$$

$$R_{vp} = E[VP] = E[V \cdot V^2/R] = E\left[\frac{1}{R}\right] E[V^3]$$

$$= \frac{3}{20} E[V^3] = \frac{3}{20} \int_0^6 \left(\frac{1}{6}\right) v^3 dv = \frac{1}{40} \left[\frac{v^4}{4}\right]_0^6 = 8.1$$

$$R_{rp} = E[R V^2/R] = E[V^2] = 12$$

$$E[V] = \int_0^6 v f_V(v) dv = \frac{1}{6} \left[\frac{v^2}{2}\right]_0^6 = 3$$

$$\sigma_{vp}^2 = R_{vp} - E[V] E[P]$$

$$= 8.1 - (3)(1.8) = 2.7$$



Practice Problems

5.25 Given the joint pdf of X and Y ,

$$f_{X,Y}(x, y) = \begin{cases} A \sin(x + y); & 0 \leq x \leq \frac{\pi}{2}, 0 \leq y \leq \frac{\pi}{2} \\ 0 & \text{elsewhere} \end{cases}$$

Find (a) A , (b) $E[X]$ and $E[Y]$, (c) σ_X^2 and σ_Y^2 and (d) ρ_{XY} .

$$\left(\text{Ans. (a) } \frac{1}{2} \text{ (b) } \frac{\pi}{4}, \frac{\pi}{4} \text{ (c) } \frac{\pi^2}{16} + \frac{\pi}{2}, \text{ (d) } \frac{\pi^2}{16} + \frac{\pi}{2} \right)$$

5.26 Two independent random variables X and Y have pdfs given by

$$f_X(x) = \begin{cases} 4ax; & 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

$$f_Y(y) = \begin{cases} 4by; & 0 \leq y \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

Find the correlation coefficient between $X + Y$ and $X - Y$.

(Ans. zero)

Solved Problem



5.60 Let X and Y be the random variables having joint pdf

$$f_{X,Y}(x, y) = \begin{cases} \frac{3}{2}(x^2 + y^2) & 0 \leq x \leq 1, 0 \leq y \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

Find the correlation coefficient.

Solution The correlation coefficient

$$\rho_{XY} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$$

$$\begin{aligned} E[XY] &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xy f_{X,Y}(x, y) dx dy \\ &= \int_0^1 \int_0^1 xy \frac{3}{2}(x^2 + y^2) dx dy = \frac{3}{2} \int_0^1 \int_0^1 xy(x^2 + y^2) dx dy \\ &= \frac{3}{2} \left[\int_0^1 \left[y \left(\frac{x^4}{4} \right) \Big|_0^1 + y^3 \left(\frac{x^2}{2} \right) \Big|_0^1 \right] dy = \frac{3}{2} \left\{ \int_0^1 \left(\frac{y}{4} + \frac{y^3}{2} \right) dy \right\} \right. \\ &= \frac{3}{2} \left[\frac{y^2}{8} \Big|_0^1 + \frac{y^4}{8} \Big|_0^1 \right] = \frac{3}{2} \left[\frac{1}{8} + \frac{1}{8} \right] = \frac{3}{8} \end{aligned}$$

The marginal pdf of X is given by

$$\begin{aligned} f_X(x) &= \int_{-\infty}^{\infty} f_{X,Y}(x, y) dy = \int_0^1 \frac{3}{2}(x^2 + y^2) dy \\ &= \frac{3}{2} \left(x^2 + \frac{1}{3} \right) = \frac{(3x^2 + 1)}{2} \end{aligned}$$

The marginal pdf of Y is given by

$$\begin{aligned} f_Y(y) &= \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx = \int_0^1 \frac{3}{2}(x^2 + y^2) dx \\ &= \frac{3}{2} \left(\frac{1}{3} + y^2 \right) = \frac{(3y^2 + 1)}{2} \end{aligned}$$

$$\begin{aligned} E[X] &= \int_{-\infty}^{\infty} x f_X(x) dx = \int_0^1 x \frac{(3x^2 + 1)}{2} dx \\ &= \frac{1}{2} \left[3 \frac{x^4}{4} + \frac{x^2}{2} \right]_0^1 = \frac{1}{2} \left[\frac{3}{4} + \frac{1}{2} \right] = \frac{5}{8} \end{aligned}$$

$$\begin{aligned} E[Y] &= \int_{-\infty}^{\infty} y f_Y(y) dy = \int_0^1 y \frac{(3y^2 + 1)}{2} dy \\ &= \frac{1}{2} \left[3 \frac{y^4}{4} + \frac{y^2}{2} \right]_0^1 = \frac{1}{2} \left[\frac{3}{4} + \frac{1}{2} \right] = \frac{5}{8} \end{aligned}$$

$$\text{Cov}(X, Y) = E[XY] - E[X] E[Y]$$

$$= \frac{3}{8} - \left(\frac{5}{8} \right) \left(\frac{5}{8} \right) = \frac{24 - 25}{64} = \frac{-1}{64}$$

$$\begin{aligned} E[X] &= \int_{-\infty}^{\infty} x^2 f_X(x) dx \\ &= \int_0^1 x^2 \left(\frac{3x^2 + 1}{2} \right) dx = \frac{1}{2} \left[3 \frac{x^5}{5} + \frac{x^3}{3} \right]_0^1 \\ &= \frac{1}{2} \left[\frac{3}{5} + \frac{1}{3} \right] = \frac{7}{15} \end{aligned}$$

Similarly, $E[Y^2] = \int_0^1 y^2 \left(\frac{3y^2 + 1}{2} \right) dy = \frac{7}{15}$

$$\text{Var}(X) = E[X^2] - \{E[X]\}^2 = \frac{7}{15} - \left(\frac{5}{8}\right)^2 = \frac{7}{15} - \frac{25}{64} = \frac{73}{960}$$

$$\text{Var}(Y) = E[Y^2] - \{E[Y]\}^2 = \frac{7}{15} - \left(\frac{5}{8}\right)^2 = \frac{73}{960}$$

$$\sigma_X = \sigma_Y = \sqrt{\frac{73}{960}} = 0.2757$$

$$\rho_{XY} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} = \frac{(-1/64)}{\sqrt{\frac{73}{960}} \sqrt{\frac{73}{960}}} = \frac{(-1/64)}{(73/960)} = -0.20547$$



Practice Problems

5.27 If X and Y are independent random variables and U and V are defined by $U = X \cos \theta + Y \sin \theta$, $V = Y \cos \theta - X \sin \theta$, show that the coefficient of correlation between X and Y is given by

$$\frac{(\sigma_Y^2 - \sigma_X^2) \sin 2\theta}{\sqrt{(\sigma_Y^2 - \sigma_X^2) \sin^2 2\theta + 4\sigma_Y^2 \sigma_X^2}}$$

5.28 The continuous random variables X and Y have the pdf $f_{X,Y}(x, y) = e^{-(x+y)}$, $0 < x < \infty$, $0 < y < \infty$. Find the ρ_{XY} .

(Ans. zero)



JOINTLY GAUSSIAN RANDOM VARIABLES 5.6

Two random variables X and Y are said to be jointly Gaussian if their joint pdf is given by

$$f_{X,Y}(x, y) = \frac{1}{2\pi \sigma_X \sigma_Y \sqrt{1 - \rho_{XY}^2}} \exp \left[-\frac{1}{2(1 - \rho_{XY}^2)} \left\{ \frac{(x - \mu_X)^2}{\sigma_X^2} + \frac{(y - \mu_Y)^2}{\sigma_Y^2} + \frac{2\rho(x - \mu_X)(y - \mu_Y)}{\sigma_X \sigma_Y} \right\} \right] \quad (5.57)$$

where μ_X , μ_Y , σ_X^2 , σ_Y^2 and ρ_{XY} are the mean, variance and Correlation coefficient of X and Y . The Equation (5.57) also known as bivariate Gaussian density which is denoted by the shorthand

$$(X, Y) \sim N(\mu_X, \mu_Y, \sigma_X^2, \sigma_Y^2, \rho_{XY}) \quad (5.58)$$

The pdf of jointly Gaussian random variables for different correlation coefficients are shown in Fig. 5.28. Note that if $\rho_{XY} = 0$, then Eq. (5.57) can be written as

$$\begin{aligned} f_{X,Y}(x, y) &= \frac{1}{2\pi \sigma_X \sigma_Y} \exp \left\{ -\frac{1}{2} \left[\frac{(x - \mu_X)^2}{\sigma_X^2} + \frac{(y - \mu_Y)^2}{\sigma_Y^2} \right] \right\} \\ &= \frac{1}{2\pi \sigma_X \sigma_Y} \exp \left\{ -\left(\frac{x - \mu_X}{\sqrt{2} \sigma_X} \right)^2 \right\} \exp \left\{ -\left(\frac{y - \mu_Y}{\sqrt{2} \sigma_Y} \right)^2 \right\} \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{\sqrt{2\pi\sigma_X^2}} \exp\left\{-\frac{(x-\mu_X)^2}{2\sigma_X^2}\right\} \frac{1}{\sqrt{2\pi\sigma_Y^2}} \exp\left\{-\frac{(y-\mu_Y)^2}{2\sigma_Y^2}\right\} \\
&= f_X(x) f_Y(y)
\end{aligned}$$

where
$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma_X^2}} e^{-(x-\mu_X)^2/2\sigma_X^2}$$

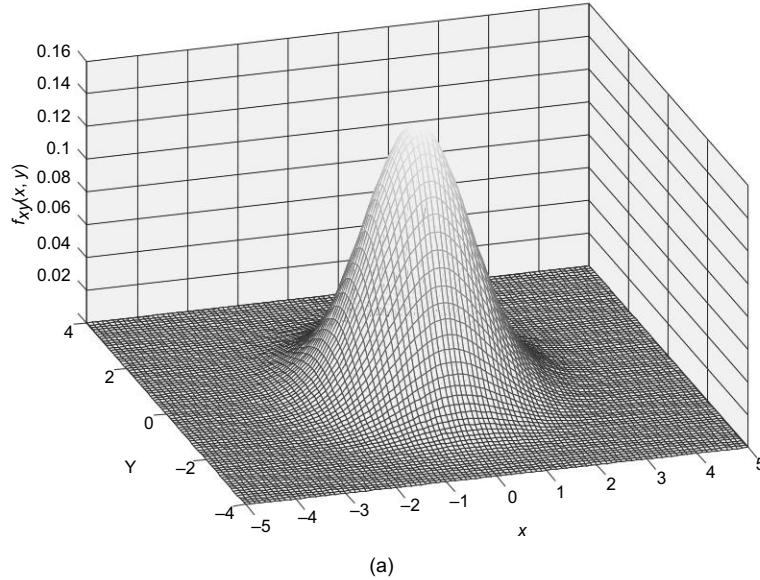
and
$$f_Y(y) = \frac{1}{\sqrt{2\pi\sigma_Y^2}} e^{-(y-\mu_Y)^2/2\sigma_Y^2}$$

From the above discussion, we can find that any uncorrelated Gaussian random variables are also statistically independent.

The loci of constant values of the Gaussian pdf are ellipse; since the following equation describes an ellipse

$$\frac{(x-\mu_X)^2}{\sigma_X^2} + \frac{(y-\mu_Y)^2}{\sigma_Y^2} + \frac{2\rho_{XY}(x-\mu_X)(y-\mu_Y)}{\sigma_X\sigma_Y} = r^2 \quad (5.59)$$

If $\rho_{XY}=0$. The loci is a circle of radius r .



(Continued)

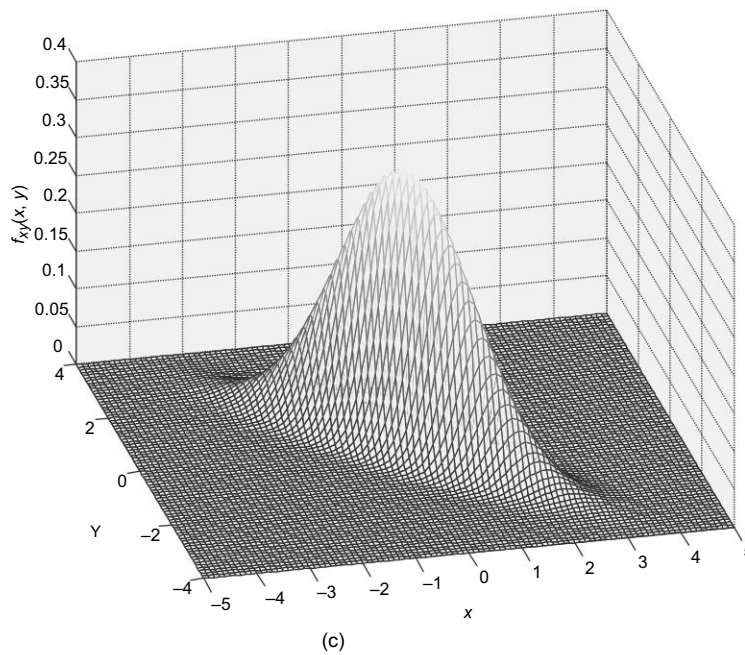
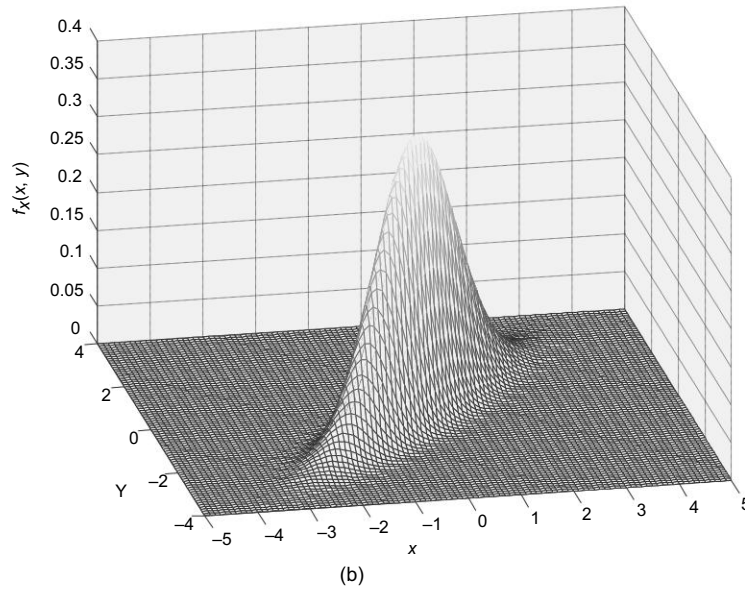


Fig.5.28 The joint density function of normal random variable with (a) $\rho = 0$ (b) $\rho = 1$ (c) $\rho = -1$

5.6.1 Random Variables

The random variables $X_1, X_2, \dots, X_N$ are called jointly Gaussian if their joint pdf can be written as

$$f_{X_1, X_2, \dots, X_N}(x_1, x_2, \dots, x_N) = \frac{1}{(2\pi)^{N/2} |C_X|^{1/2}} \exp \left[\frac{-1}{2} (x - \mu_X)^T C_X^{-1} (x - \mu_X) \right] \quad (5.60)$$

where the matrices

$$[x - \mu_X] = \begin{bmatrix} x_1 - \mu_{X_1} \\ x_2 - \mu_{X_2} \\ \vdots \\ x_N - \mu_{X_N} \end{bmatrix} \text{ and } [C_X] = \begin{bmatrix} C_{11} & C_{12} & \cdots & C_{1N} \\ C_{21} & C_{22} & \cdots & C_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ C_{N1} & C_{N2} & \cdots & C_{NN} \end{bmatrix}$$

The matrix $[C_X]$ is called *covariance matrix* of N random variables given by

$$\begin{aligned} C_{ij} &= E[(X_i - \mu_{X_i})(X_j - \mu_{X_j})] \\ &= \sigma_{X_i}^2 \quad i = j \\ &= C_{X_i X_j} \quad i \neq j \end{aligned}$$

Let us consider two Gaussian random variables X_1 and X_2 , Then the matrix

$$[X - \mu] = \begin{bmatrix} X_1 - \mu_{X_1} \\ X_2 - \mu_{X_2} \end{bmatrix}$$

and the covariance matrix

$$\begin{aligned} C &= \begin{bmatrix} \sigma_{X_1}^2 & \text{Cov}(X_1, X_2) \\ \text{Cov}(X_1, X_2) & \sigma_{X_2}^2 \end{bmatrix} \\ &= \begin{bmatrix} \sigma_{X_1}^2 & \rho_{X_1 X_2} \sigma_{X_1} \sigma_{X_2} \\ \rho_{X_1 X_2} \sigma_{X_1} \sigma_{X_2} & \sigma_{X_2}^2 \end{bmatrix} \end{aligned}$$

where $\rho_{X_1 X_2} = \frac{\text{Cov}(X_1, X_2)}{\sigma_{X_1} \sigma_{X_2}}$

The determinant of C is

$$\begin{aligned} |C| &= \sigma_{X_1}^2 \sigma_{X_2}^2 - \rho_{X_1 X_2}^2 \sigma_{X_1}^2 \sigma_{X_2}^2 \\ &= (1 - \rho_{X_1 X_2}^2) \sigma_{X_1}^2 \sigma_{X_2}^2 \end{aligned}$$

The inverse of the matrix

$$\begin{aligned} C^{-1} &= \frac{\text{Adj } C}{|C|} \\ \text{Adj } C &= \begin{bmatrix} \sigma_{X_2}^2 & -\rho_{X_1, X_2} \sigma_{X_1} \sigma_{X_2} \\ -\rho_{X_1, X_2} \sigma_{X_1} \sigma_{X_2} & \sigma_{X_1}^2 \end{bmatrix} \end{aligned}$$

$$C^{-1} = \frac{Adj}{|C|} = \frac{1}{1 - \rho_{X_1 X_2}^2} \begin{bmatrix} \frac{1}{\sigma_{X_1}^2} & \frac{-\rho}{\sigma_{X_1} \sigma_{X_2}} \\ \frac{-\rho}{\sigma_{X_1} \sigma_{X_2}} & \frac{1}{\sigma_{X_2}^2} \end{bmatrix}$$

Consider the term

$$\begin{aligned} (X - \mu_X)^T C^{-1} (X - \mu_X) &= \frac{1}{(1 - \rho_{X_1 X_2}^2)} \begin{bmatrix} x_1 - \mu_{X_1} & x_2 - \mu_{X_2} \end{bmatrix} \begin{bmatrix} \frac{1}{\sigma_{X_1}^2} & \frac{-\rho}{\sigma_{X_1} \sigma_{X_2}} \\ \frac{-\rho}{\sigma_{X_1} \sigma_{X_2}} & \frac{1}{\sigma_{X_2}^2} \end{bmatrix} \begin{bmatrix} x_1 - \mu_{X_1} \\ x_2 - \mu_{X_2} \end{bmatrix} \\ &= \frac{1}{1 - \rho_{X_1 X_2}^2} \left[\frac{(x_1 - \mu_{X_1})^2}{\sigma_{X_1}^2} - \frac{2\rho(x_1 - \mu_{X_1})(x_2 - \mu_{X_2})}{\sigma_{X_1} \sigma_{X_2}} + \frac{(x_2 - \mu_{X_2})^2}{\sigma_{X_2}^2} \right] \\ f_{X_1, X_2}(x_1, x_2) &= \frac{1}{2\pi |C|^{1/2}} \exp \left[\frac{(x_1 - \mu_{X_1})^2}{\sigma_{X_1}^2} - \frac{2\rho(x_1 - \mu_{X_1})(x_2 - \mu_{X_2})}{\sigma_{X_1} \sigma_{X_2}} + \frac{(x_2 - \mu_{X_2})^2}{\sigma_{X_2}^2} \right] \\ &= \frac{1}{2\pi \sqrt{1 - \rho_{X_1 X_2}^2} \sigma_{X_1} \sigma_{X_2}} \exp \left[\frac{(x_1 - \mu_{X_1})^2}{\sigma_{X_1}^2} - \frac{2\rho(x_1 - \mu_{X_1})(x_2 - \mu_{X_2})}{\sigma_{X_1} \sigma_{X_2}} + \frac{(x_2 - \mu_{X_2})^2}{\sigma_{X_2}^2} \right] \end{aligned} \quad (5.61)$$

5.6.2 Properties of Gaussian Random Variables

- (i) Gaussian random variables are completely characterized by their mean and covariance matrix. That is first and second-order moments are enough to completely define Gaussian random variables.
- (ii) The sum of many uniformly small and negligible random effects tends to be Gaussian.
- (iii) If the Gaussian random variables are uncorrelated, they are also statistically independent.
- (iv) The weighted sum of jointly Gaussian random variables are jointly Gaussian.
- (v) Joint conditional density of jointly Gaussian random variables given the values of some other components is also Gaussian. That is, the conditional density

$$f_{X_1, X_2, \dots, X_k}(x_1, x_2, \dots, x_k | X_{k+1} = x_{k+1}, \dots, X_N = x_N)$$

is Gaussian for every $k < N$.

REVIEW QUESTIONS



11. What are the properties of Gaussian random variables?
12. Give expression for joint pdf of two Gaussian random variables

Solved Problem

5.61 The joint density function two Gaussian random variables X and Y is

$$f_{X,Y}(x, y) = \frac{1}{2\pi \sigma_X \sigma_Y \sqrt{1 - \rho^2}} \exp \left\{ -\frac{1}{2\pi(1 - \rho^2)} \left[\frac{(x - \mu_X)^2}{\sigma_X^2} + \frac{2\rho(x - \mu_X)(y - \mu_Y)}{\sigma_X \sigma_Y} + \frac{(y - \mu_Y)^2}{\sigma_Y^2} \right] \right\}$$

Find the conditional density function $f_X(x|Y=y)$ and $f_Y(y|X=x)$ and show that they are also Gaussian.

Solution

$$\begin{aligned} f_X(x|Y=y) &= \frac{f_{X,Y}(x, y)}{f_Y(y)} \\ &= \frac{\frac{1}{2\pi \sigma_X \sigma_Y \sqrt{1 - \rho^2}} \exp \left\{ -\frac{1}{2(1 - \rho^2)} \left[\frac{(x - \mu_X)^2}{\sigma_X^2} + \frac{2\rho(x - \mu_X)(y - \mu_Y)}{\sigma_X \sigma_Y} + \frac{(y - \mu_Y)^2}{\sigma_Y^2} \right] \right\}}{\frac{1}{\sqrt{2\pi} \sigma_Y} \exp \left\{ -\frac{(y - \mu_Y)^2}{2 \sigma_Y^2} \right\}} \\ &= \frac{1}{\sigma_X \sqrt{2\pi(1 - \rho^2)}} \exp \left\{ -\left[\frac{(x - \mu_X)^2}{2\sigma_X^2 (1 - \rho^2)} + \frac{2\rho(x - \mu_X)(y - \mu_Y)}{2\sigma_X \sigma_Y (1 - \rho^2)} + \frac{(y - \mu_Y)^2}{2\sigma_Y^2 (1 - \rho^2)} - \frac{(y - \mu_Y)^2}{2\sigma_Y^2} \right] \right\} \\ &= \frac{1}{\sigma_X \sqrt{2\pi(1 - \rho^2)}} \exp \left\{ -\left[\frac{(x - \mu_X)^2 \sigma_Y^2 + 2\rho \sigma_X \sigma_Y (x - \mu_X)(y - \mu_Y) + (y - \mu_Y)^2 \sigma_X^2 \rho^2}{2\sigma_X^2 \sigma_Y^2 (1 - \rho^2)} \right] \right\} \\ &= \frac{1}{\sigma_X \sqrt{2\pi(1 - \rho^2)}} \exp \left\{ -\frac{(x - \mu_X)^2 + 2\rho \frac{\sigma_X}{\sigma_Y} (x - \mu_X)(y - \mu_Y) + (y - \mu_Y)^2 \frac{\sigma_X^2}{\sigma_Y^2} \rho^2}{2\sigma_X^2 (1 - \rho^2)} \right\} \\ &= \frac{1}{\sigma_X \sqrt{2\pi(1 - \rho^2)}} \exp \left\{ -\frac{\left(x - \mu_X - \frac{(y - \mu_Y)}{\sigma_Y} \sigma_X \rho \right)^2}{2\sigma_X^2 (1 - \rho^2)} \right\} \\ &= \frac{1}{\sigma_X \sqrt{2\pi(1 - \rho^2)}} \exp \left\{ -\frac{\left[x - \left(\mu_X + \rho \frac{\sigma_X}{\sigma_Y} (y - \mu_Y) \right) \right]^2}{2\sigma_X^2 (1 - \rho^2)} \right\} \end{aligned}$$

The above equation represents Gaussian pdf with mean $\mu_X + \rho \frac{\sigma_X}{\sigma_Y} (y - \mu_Y)$ and variance $\sigma_X^2 (1 - \rho^2)$. Similarly,

$$\begin{aligned}
 f_Y(y|X=x) &= \frac{f_{X,Y}(x,y)}{f_X(x)} \\
 &= \frac{1}{2\pi \sigma_X \sigma_Y (1 - \rho^2)} \exp \left\{ \frac{-1}{2((1 - \rho^2))} \left[\frac{(x - \mu_X)^2}{\sigma_X^2} - \frac{2\rho(x - \mu_X)(y - \mu_Y)}{\sigma_X \sigma_Y} + \frac{(y - \mu_Y)^2}{\sigma_Y^2} \right] \right\} \\
 &= \frac{1}{\sqrt{2\pi} \sigma_X} \exp \left\{ -\frac{(x - \mu_X)^2}{2\sigma_X^2} \right\} \\
 &= \frac{1}{\sigma_Y \sqrt{2\pi(1 - \rho^2)}} \exp \left\{ -\frac{\left[y - (\mu_Y + \rho \frac{\sigma_Y}{\sigma_X} (x - \mu_X)) \right]^2}{2\sigma_Y^2 (1 - \rho^2)} \right\}
 \end{aligned}$$

The above equation represents a Gaussian PDF with mean $\mu_Y + \rho \frac{\sigma_Y}{\sigma_X} (x - \mu_X)$ and variance $\sigma_Y^2 (1 - \rho^2)$.



Practice Problem

5.29 If X and Y are jointly Gaussian distributed random variables with PDF

$$f_{X,Y}(x,y) = \frac{1}{2\pi \sigma_X \sigma_Y (1 - \rho^2)} \exp \left\{ -\frac{1}{2(1 - \rho^2)} + \frac{(y - \mu_Y)^2}{\sigma_Y^2} - \frac{2\rho(x - \mu_X)(y - \mu_Y)}{\sigma_X \sigma_Y} \right\}$$

- Show that the conditional distribution of Y given $X = x$, is Gaussian with mean $\mu_Y + \rho \frac{\sigma_Y}{\sigma_X} (x - \mu_X)$ and variance $\sigma_Y^2 (1 - \rho^2)$.
- Show that X and Y are independent if and only if $\rho = 0$.

Solved Problems



5.62 If z is a standard Gaussian random variable and Y is defined by $Y = a + bz + cz^2$. Show that

$$\rho_{YZ} = \frac{b}{\sqrt{b^2 + 2c^2}}$$

Solution Given: $Z \sim N(0, 1)$

and $Y = a + bz + cz^2$

That is $\mu_z = 0$ and $\sigma_z^2 = 1$

Also $E[z^2] = 1$

$$E[z^3] = 0; E[z^4] = 3$$

$$\begin{aligned} E[Y] &= E[a + bz + cz^2] \\ &= a + bE[z] + cE[z^2] \end{aligned}$$

$$= a + c$$

$$\sigma_Y^2 = E[Y^2] - \{E[Y]\}^2$$

$$\begin{aligned} E[Y^2] &= E[(a + bz + cz^2)^2] \\ &= E[a^2 + b^2 z^2 + c^2 z^4 + 2abz + 2bcz^3 + 2acz^2] \\ &= a^2 + b^2 E[z^2] + cE[z^4] + 2ab E[z] + 2bc E[z^3] + 2ac E[z^2] \\ &= a^2 + b^2 + 3c^2 + 2ac \\ \sigma_Y^2 &= (a^2 + b^2 + 3c^2 + 2ac) - (a + c)^2 \\ &= b^2 + 2c^2 \end{aligned}$$

$$\begin{aligned} \text{Cov}(Y, Z) &= E[(Y - \mu_Y)(z - \mu_Z)] \\ &= E[Y - a - c](z) \\ &= E[Yz] - aE[z] - cE[z] \\ &= E[Yz] \\ &= E[(a + bz + cz^2)z] \\ &= a E(z) + b E(z^2) + E(z^3) \\ &= b \end{aligned}$$

$$\rho_{YZ} = \frac{\text{Cov}(Y, Z)}{\sigma_Y \sigma_Z} = \frac{b}{\sqrt{b^2 + 2c^2}}$$

5.63 Given $Z = (3X + bY)^2$ where X and Y are zero mean random variables with variance $\sigma_X^2 = 25$ and $\sigma_Y^2 = 9$. Their correlation coefficient is $\rho = -0.6$.

- Find a value for the parameter a that minimizes the mean value of Z .
- Find the minimum mean value.

Solution Given $Z = (3X + bY)^2$; $\mu_X = \mu_Y = 0$

$$\sigma_X^2 = 25, \sigma_Y^2 = 9 \text{ and } \rho = -0.6$$

$$\begin{aligned} E[Z] &= \mu_Z = E[(3X + bY)^2] \\ &= E[9X^2 + b^2 Y^2 + 6bXY] \end{aligned}$$

$$= 9 E[X^2] + b^2 E[Y^2] + 6b E[XY]$$

Since μ_X and μ_Y are zero

$$E[X^2] = \sigma_X^2 = 25 \text{ and } E[Y^2] = \sigma_Y^2 = 9$$

Also $\rho = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} = -0.6$

$$\Rightarrow R_{XY} = -0.6(5)(3) = -9$$

$$\begin{aligned} \mu_Z = E[Z] &= 9(25) + b^2(9) + 6b(-9) \\ &= 9b^2 - 54b + 144 \end{aligned}$$

$$\frac{d\mu_z}{db} = 18b - 54 = 0 \Rightarrow b = 3$$

$$\{E[Z]\}_{\min} = 9(3)^2 - 54(3) + 144 = 63$$

5.64 For the two random variables X and Y

$$\begin{aligned} f_{X,Y}(x, y) &= 0.15 \delta(x+1) \delta(y) + 0.1 \delta(x) \delta(y) + 0.1 \delta(x) \delta(y-2) + 0.4 \delta(x-1) \delta(y+2) \\ &\quad + 0.2 \delta(x-1) \delta(y-1) + 0.05 \delta(x-1) \delta(y-3) \end{aligned}$$

Find (a) the correlation (b) the covariance and (c) the correlation coefficient of X and Y (d) Are X and Y either uncorrelated or orthogonal.

Solution The correlation is given by

$$\begin{aligned} R_{XY} &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xy f_{X,Y}(x, y) dx dy \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xy \left\{ 0.15 \delta(x+1) \delta(y) + 0.1 \delta(x) \delta(y) + 0.1 \delta(x) \delta(y-2) \right. \\ &\quad \left. + 0.4 \delta(x-1) \delta(y+2) + 0.2 \delta(x-1) \delta(y-1) + 0.05 \delta(x-1) \delta(y-3) \right\} dx dy \end{aligned}$$

Using $\int_{-\infty}^{\infty} xy \delta(x-k_1) \delta(x-k_2) dx dy = k_1 k_2$

we can simplify R_{XY} as

$$\begin{aligned} R_{XY} &= 0.15(-1)(0) + 0.1(0)(0) + 0.1(0)(2) + 0.4(1)(-2) + 0.2(1)(1) + 0.05(1)(3) \\ &= -0.8 + 0.2 + 0.15 = -0.45 \end{aligned}$$

$$\boxed{R_{XY} = 0.9}$$

$$E[X] = m_{10} = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x f_{X,Y}(x, y) dx dy$$

$$\boxed{\int_{-\infty}^{\infty} \delta(y-k) dy = 1}$$

$$\begin{aligned}
&= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x \{0.15 \delta(x+1) \delta(y) + 0.1 \delta(x) \delta(y) + 0.1 \delta(x) \delta(y-2) \\
&\quad + 0.4 \delta(x-1) \delta(y+2) + 0.2 \delta(x-1) \delta(y-1) + 0.05 \delta(x-1) \delta(y-3)\} dx dy \\
&= (-1)(0.15) + 0.1(0) + 0.1(0) + 0.4(1) + 0.2(1) + (0.05)(1) \\
&= -0.15 + 0.4 + 0.2 + 0.05 = 0.5
\end{aligned}$$

$$\begin{aligned}
E[Y] &= m_{01} = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} y f_{X,Y}(x, y) dx dy \\
&= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \{y 0.15 \delta(x+1) \delta(y) + 0.1 \delta(x) \delta(y) + 0.1 \delta(x) \delta(y-2) + 0.4 \delta(x-1) \delta(y+2) \\
&\quad + 0.2 \delta(x-1) \delta(y-1) + 0.05 \delta(x-1) \delta(y-3)\} dx dy \\
&= 0.15(0) + 0.1(0) + 0.1(2) + 0.4(-2) + 0.2(1) + 0.05(3) \\
&= -0.25
\end{aligned}$$

$$\int_{-\infty}^{\infty} \delta(x-k) dx = 1$$

$$C_{XY} = R_{XY} - E[X] E[Y] = -0.45 - (0.5)(-0.28) = -0.325$$

The correlation coefficient $\rho_{XY} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$

$$\begin{aligned}
E[X^2] &= m_{20} = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x^2 f_{X,Y}(x, y) dx dy \\
&= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x^2 \{0.15 \delta(x+1) \delta(y) + 0.1 \delta(x) \delta(y) + 0.1 \delta(x) \delta(y-2) + 0.4 \delta(x-1) \delta(y+2) \\
&\quad + 0.2 \delta(x-1) \delta(y-1) + 0.05 \delta(x-1) \delta(y-3)\} dx dy \\
&= 0.15(-1)^2 + 0.1(0)^2 + 0.1(0)^2 + 0.4(1)^2 + 0.2(1)^2 + 0.05(1)^2 = 0.8
\end{aligned}$$

$$E[Y^2] = m_{02} = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} y^2 f_{X,Y}(x, y) dx dy$$

$$= 0.15(0)^2 + 0.1(0)^2 + 0.1(2)^2 + 0.4(-2)^2 + 0.2(1)^2 + 0.05(3)^2 = 2.65$$

$$\sigma_X^2 = E[X^2] - \mu_X^2 = 0.8 - (0.5)^2 = 0.55$$

$$\sigma_Y^2 = E[Y^2] - \mu_Y^2 = 2.65 - (-0.25)^2 = 2.5875$$

$$\rho_{XY} = \frac{-0.325}{\sqrt{0.55} \sqrt{2.5875}} = -0.272$$

Since $C_{XY} \neq 0$, the random variable X and Y are correlated.

Also $R_{XY} \neq 0$. Therefore X and Y are not orthogonal.

5.65 X is a random variable with mean $\mu_X = 3$, $\sigma_X^2 = 2$.

- (a) Determine the second-order moment of X about the origin.
- (b) Determine the mean of random variable Y where $Y = -6X + 22$.

Solution Given $\mu_X = 3$; $\sigma_X^2 = 2$

- (a) Second-order moment of X about the origin

$$\begin{aligned} E[X^2] &= \sigma_X^2 + \{E[X]\}^2 \\ &= 2 + (3)^2 = 11 \\ E[X^2] &= 11 \end{aligned}$$

- (b) Given: $Y = -6X + 22$

$$\begin{aligned} E[Y] &= -6 E[X] + 22 \\ &= -6[3] + 22 = 4 \end{aligned}$$

$$\boxed{E[Y] = 4}$$

5.66 Consider two random variables X and Y such that $Y = -3X + 15$, the mean value and the variance of X are 5 and 3 respectively. Find the covariance.

Solution Given: $\mu_X = 5$; $\sigma_X^2 = 3$ and $Y = -3X + 15$

$$\begin{aligned} E[Y] &= E[-3X + 15] = -3E[X] + 15 \\ &= -3(5) + 15 = 0 \end{aligned}$$

$$\begin{aligned} E[X^2] &= \sigma_X^2 + \{E[X]\}^2 \\ &= 3 + (5)^2 = 28 \end{aligned}$$

$$E[Y^2] = E[(-3X + 15)^2] = E[9X^2] + E[-90X] + 225 = 9(28) - 90(5) + 225 = 27$$

$$\text{Cov}[X, Y] = E[XY] - E[X]E[Y]$$

$$\begin{aligned} E[XY] &= E[X(-3X + 15)] = E[-3X^2 + 15X] \\ &= -3E[X^2] + 15E[X] \\ &= -3(28) + 15(5) = -84 + 75 = -9 \end{aligned}$$

$$\text{Cov}[XY] = -9$$

- 5.67** (a) For the random variable X and Y with joint density function given in the solved problem find the second-order moments of X and Y .
 (b) Find the variables of X and Y .
 (c) Find the correlation coefficient.

Solution Given:

$$f_{X,Y}(x, y) = \begin{cases} (x + y)^2 / 40 & -1 \leq x \leq 1, -3 \leq y \leq 3 \\ 0 & \text{elsewhere} \end{cases}$$

The second-order moments are m_{20} , m_{11} , and m_{02}

$$m_{nk} = E[X^n Y^k] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x^n y^k f_{X,Y}(x, y) dx dy$$

$$\begin{aligned} m_{20} &= E[X^2] = \int_{-3}^3 \int_{-1}^1 x^2 (x + y)^2 / 40 dx dy \\ &= \frac{1}{40} \int_{-3}^3 \int_{-1}^1 x^2 (x^2 + 2xy + y^2) dx dy \\ &= \frac{1}{40} \int_{-3}^3 \int_{-1}^1 (x^4 + 2x^3 y + x^2 y^2) dx dy \\ &= \frac{1}{40} \int_{-3}^3 \left\{ \frac{x^5}{5} \Big|_{-1}^1 + y \frac{x^4}{2} \Big|_{-1}^1 + y^2 \frac{x^3}{3} \Big|_{-1}^1 \right\} dy \\ &= \frac{1}{40} \int_{-3}^3 \left(\frac{2}{5} + y(0) + \frac{2}{3} y^2 \right) dy \\ &= \frac{1}{40} \left\{ \frac{2}{5} y \Big|_{-3}^3 + \frac{2}{3} \frac{y^3}{3} \Big|_{-3}^3 \right\} \\ &= \frac{1}{40} \left\{ \frac{12}{5} + 12 \right\} = \frac{1}{40} \cdot \frac{72}{5} = \frac{9}{25} \end{aligned}$$

$$\begin{aligned} m_{11} &= E[XY] = \int_{-3}^3 \int_{-1}^1 xy \frac{(x + y)^2}{40} dx dy \\ &= \int_{-3}^3 \int_{-1}^1 \frac{xy}{40} (x^2 + y^2 + 2xy) dx dy \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{40} \int_{-3}^3 \int_{-1}^1 (x^3 y + xy^3 + 2x^2 y^2) dx dy \\
&= \frac{1}{40} \int_{-3}^3 \left\{ y \frac{x^4}{4} \Big|_{-1}^1 + y^3 \frac{x^2}{2} \Big|_{-1}^1 + \frac{2}{3} y^2 x^3 \Big|_{-1}^1 \right\} dy \\
&= \frac{1}{40} \int_{-3}^3 \frac{4}{3} y^2 dy = \frac{4}{40(9)} y^3 \Big|_{-3}^3 = 0.6 \\
m_{02} = E[Y^2] &= \int_{-3}^3 \int_{-1}^1 y^2 \frac{(x+y)^2}{40} dx dy \\
&= \frac{1}{40} \int_{-3}^3 \int_{-1}^1 y^2 (x^2 + y^2 + 2xy) dx dy = \frac{1}{40} \int_{-3}^3 \int_{-1}^1 (y^2 x^2 + y^4 + 2xy^3) dx dy \\
&= \frac{1}{40} \int_{-3}^3 \left\{ y^2 \frac{x^3}{3} \Big|_{-1}^1 + y^4 x \Big|_{-1}^1 + 2y^3 \frac{x^2}{2} \Big|_{-1}^1 \right\} dy \\
&= \frac{1}{40} \int_{-3}^3 \left(\frac{2}{3} y^2 + 2y^4 \right) dy = \frac{1}{40} \left\{ \frac{2}{3} \frac{y^3}{3} \Big|_{-3}^3 + \frac{2}{5} y^5 \Big|_{-3}^3 \right\} \\
&= \frac{1}{40} (12 + 194.4) = 5.16 \\
\sigma_X^2 &= E[X^2] - \{E[X]\}^2 \\
E[X] = m_{10} &= \int_{-3}^3 \int_{-1}^1 x \frac{(x+y)^2}{40} dx dy = \frac{1}{40} \int_{-3}^3 \int_{-1}^1 (x^3 + xy^2 + 2x^2 y) dx dy \\
&= \frac{1}{40} \int_{-3}^3 \frac{4}{3} y dy = \frac{1}{30} \frac{y^2}{2} \Big|_{-3}^3 = 0
\end{aligned}$$

Similarly,

$$\begin{aligned}
\sigma_Y^2 &= E[Y^2] - \{E[Y]\}^2 \\
E[Y] = m_{01} &= \int_{-3}^3 \int_{-1}^1 y \frac{(x+y)^2}{40} dx dy \\
&= \frac{1}{40} \int_{-3}^3 \int_{-1}^1 (x^2 y + y^3 + xy^2) dx dy \\
&= \frac{1}{40} \int_{-3}^3 \left(\frac{2}{3} y + 2y^3 \right) dy = 0
\end{aligned}$$

$$\Rightarrow \sigma_X^2 = E[X^2] = m_{20} = 0.36$$

$$\sigma_Y^2 = E[Y^2] = m_{02} = 5.16$$

The correlation coefficient

$$\rho_{XY} = \frac{E[(X - \mu_X)(Y - \mu_Y)]}{\sqrt{E[X^2] E[Y^2]}} \quad (\mu_X = \mu_Y = 0)$$

$$= \frac{E[XY]}{\sqrt{E[X^2] E[Y^2]}} = \frac{m_{11}}{\sqrt{m_{20} m_{02}}}$$

$$= \frac{0.6}{\sqrt{0.36 (5.16)}} = 0.44$$

5.68 Find the correlation coefficient for the following data:

X	10	14	18	22	26	30
Y	18	22	24	6	30	36

Solution The correlation coefficient $\rho_{xy} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$

$$\text{Cov}(X, Y) = E[XY] - E[X] E[Y]$$

X	Y	XY	X ²	Y ²
10	18	180	100	324
14	12	168	196	144
18	24	432	324	576
22	6	132	484	36
26	30	780	676	900
30	36	1080	900	1296
120	126	2772	2680	3276

$$\mu_X = E[X] = \frac{\sum X}{n} = \frac{120}{6} = 20$$

$$\mu_Y = E[Y] = \frac{\sum Y}{n} = \frac{126}{6} = 21$$

$$E[XY] = \frac{\sum XY}{n} = \frac{2772}{6} = 462$$

$$\sigma_X^2 = E[X^2] - \mu_X^2; \sigma_Y^2 = E[Y^2] - \mu_Y^2$$

$$E[X^2] = \frac{\sum X^2}{n} = \frac{2680}{6} = 446.667$$

$$E[Y^2] = \frac{\Sigma Y^2}{n} = \frac{3276}{6} = 546$$

$$\sigma_X^2 = (446.667) - (20)^2 = 46.667 \Rightarrow \sigma_X = 6.8313$$

$$\sigma_Y^2 = (546) - (21)^2 = 105 \Rightarrow \sigma_Y = 10.247$$

$$\begin{aligned} \rho &= \frac{E[XY] - \mu_X \mu_Y}{\sigma_X \sigma_Y} \\ &= \frac{462 - 20(21)}{(6.8313)(10.247)} \\ &= 0.6 \end{aligned}$$

5.69 Two random variables X and Y have the joint probability density function given by

$$f_{X,Y}(x, y) = \begin{cases} k(1 - x^2 y) & 0 \leq x \leq 1, 0 \leq y \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Find the value of k .
- (b) Obtain the marginal probability density function of X and Y .
- (c) Find the correlation coefficient between X and Y .

Solution

(a) We know $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx dy = 1$

$$\Rightarrow \int_0^1 \int_0^1 k(1 - x^2 y) dx dy = 1$$

$$\begin{aligned} k \int_0^1 \left[x \Big|_0^1 - y \frac{x^3}{3} \Big|_0^1 \right] dy &= k \int_0^1 \left(1 - \frac{y}{3} \right) dy \\ &= k \left[y \Big|_0^1 - \frac{1}{3} \frac{y^2}{2} \Big|_0^1 \right] = k \left[1 - \frac{1}{6} \right] = k \left(\frac{5}{6} \right) \\ &= k \left(\frac{5}{6} \right) = 1 \Rightarrow k = \frac{6}{5} \end{aligned}$$

- (b) The marginal density function of X is

$$\begin{aligned} f_X(x) &= \int f_{X,Y}(x, y) dy = \int_0^1 \frac{6}{5} (1 - x^2 y) dy \\ &= \frac{6}{5} \left[\left(y - x^2 \frac{y^2}{2} \right) \Big|_0^1 \right] = \frac{6}{5} \left(1 - \frac{x^2}{2} \right); 0 \leq x \leq 1 \end{aligned}$$

The marginal density function of Y is

$$\begin{aligned} f_Y(y) &= \int_0^1 f_{X,Y}(x, y) dx = \int_0^1 \frac{6}{5} (1 - x^2 y) dx \\ &= \frac{6}{5} \left(x - y \frac{x^3}{3} \right) \Big|_0^1 = \frac{6}{5} \left(1 - \frac{y}{3} \right); 0 \leq y \leq 1 \end{aligned}$$

(c) Correlation coefficient

$$\rho_{XY} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

$$\text{Cov}[X, Y] = E[XY] - E[X] E[Y]$$

$$\begin{aligned} E[X] &= \int_{-\infty}^{\infty} x f_X(x) dx = \int_0^1 x \frac{6}{5} \left(1 - \frac{x^2}{2} \right) dx \\ &= \frac{6}{5} \int_0^1 \left(x - \frac{x^3}{2} \right) dx = \frac{6}{5} \left[\frac{x^2}{2} \Big|_0^1 - \frac{x^4}{8} \Big|_0^1 \right] \\ &= \frac{6}{5} \left[\frac{1}{2} - \frac{1}{8} \right] = \frac{9}{20} \end{aligned}$$

$$\begin{aligned} E[Y] &= \int_{-\infty}^{\infty} y f_Y(y) dy = \int_0^1 y \frac{6}{5} \left(1 - \frac{y}{3} \right) dy \\ &= \frac{6}{5} \int_0^1 \left(y - \frac{y^2}{3} \right) dy = \frac{6}{5} \left[\frac{y^2}{2} \Big|_0^1 - \frac{y^3}{9} \Big|_0^1 \right] \\ &= \frac{6}{5} \left[\frac{1}{2} - \frac{1}{9} \right] = \frac{7}{15} \end{aligned}$$

$$\begin{aligned} E[XY] &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xy f_{X,Y}(x, y) dx dy \\ &= \frac{6}{5} \int_0^1 \int_0^1 xy (1 - x^2 y) dx dy = \frac{6}{5} \int_0^1 \int_0^1 (xy - x^3 y^2) dx dy \\ &= \frac{6}{5} \int_0^1 \left(\frac{y}{2} - \frac{y^2}{4} \right) dy = \frac{6}{5} \left[\frac{1}{4} - \frac{1}{12} \right] = \frac{1}{5} \end{aligned}$$

$$E[X^2] = \int_0^1 x^2 \left(1 - \frac{x^2}{2} \right) dx = \frac{6}{5} \left[\left(\frac{x^3}{3} - \frac{x^5}{10} \right) \Big|_0^1 \right] = \frac{6}{5} \left[\frac{1}{3} - \frac{1}{10} \right]$$

$$= \frac{6}{5} \left[\frac{7}{30} \right] = \frac{7}{25}$$

$$E[Y^2] = \frac{6}{5} \int y^2 \left(1 - \frac{y}{3} \right) dy = \frac{6}{5} \left[\frac{y^3}{3} \Big|_0^1 - \frac{y^4}{12} \right] = \frac{6}{5} \left[\frac{1}{3} - \frac{1}{12} \right] = \frac{3}{10}$$

$$\sigma_X^2 = E[X^2] - \{E[X]\}^2 = \frac{7}{25} - \left(\frac{9}{20} \right)^2 = \frac{31}{400} \Rightarrow \sigma_X = \frac{\sqrt{31}}{20}$$

$$\sigma_Y^2 = E[Y^2] - \{E[Y]\}^2 = \frac{3}{10} - \left(\frac{7}{15} \right)^2 = \frac{37}{450} \Rightarrow \sigma_Y = \sqrt{\frac{37}{450}}$$

$$\text{Cov}(X, Y) = E[XY] - E[X] E[Y]$$

$$= \frac{1}{5} - \frac{9}{20} \left(\frac{7}{15} \right) = \frac{-3}{300} = -0.01$$

$$\rho_{XY} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} = -\frac{0.01}{\left(\frac{\sqrt{31}}{20} \right) \left(\sqrt{\frac{37}{450}} \right)} = -0.13$$

5.70 Two random variables X and Y are related by the expression $Y = aX + b$, where a and b are constants.

(a) Show that the correlation coefficient is

$$\rho = 1 \text{ for } a > 0, \text{ any } b$$

$$= -1 \text{ for } a < 0, \text{ any } b$$

(b) Show that the covariance is $C_{XY} = a \sigma_X^2$, where σ_X^2 is the variance of X .

Solution The correlation coefficient

$$\rho_{XY} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

$$C_{XY} = \text{Cov}(X, Y) = E[XY] - E[X] E[Y]$$

Given: $Y = aX + b$

$$E[XY] = E[X(aX + b)]$$

$$= aE[X^2] + bE[X]$$

$$E[Y] = aE[X] + b$$

$$E[X] E[Y] = E[X] \{aE[X] + b\}$$

$$= a\{E[X]\}^2 + bE[X]$$

$$\text{Cov}(X, Y) = aE[X^2] + bE[X] - a\{E[X]\}^2 - bE[X]$$

$$\begin{aligned}
&= a\sigma_X^2 \\
\sigma_Y^2 &= E[(aX + b)^2] - \{E[aX + b]\}^2 \\
&= E[a^2X^2 + 2abX + b^2] - \{aE[X] + b\}^2 \\
&= a^2E[X^2] + 2abE[X] + b^2 - a^2\{E[X]\}^2 - b^2 - 2abE[X] = a^2\{E[X^2] - (E[X])^2\} \\
&= a^2\sigma_X^2 \\
\Rightarrow \quad \sigma_Y &= |a| \sigma_X \\
\rho &= \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} = \frac{a\sigma_X^2}{|a| \sigma_X^2} = \frac{a}{|a|}
\end{aligned}$$

If a is positive, $\rho = 1$

and if a is negative, $\rho = -1$

and $\boxed{C_{XY} = a\sigma_X^2}$

5.71 X and Y are the random variables with variance σ_X and σ_Y respectively. Find the value of k if $U = X + KY$ and $V = X + \frac{\sigma_X}{\sigma_Y} Y$ are uncorrelated.

Solution Two random variables U and V are uncorrelated if

$$\text{Cov}(U, V) = 0$$

We know

$$\text{Cov}(U, V) = E[UV] - E[U]E[V]$$

$$\begin{aligned}
E[UV] &= E[(X + KY)(X + \frac{\sigma_X}{\sigma_Y} Y)] \\
&= E\left[X^2 + KYX + X \frac{\sigma_X}{\sigma_Y} Y + KY^2 \frac{\sigma_X}{\sigma_Y}\right] \\
&= E[X^2] + KE[XY] + \frac{\sigma_X}{\sigma_Y} E[XY] + K \frac{\sigma_X}{\sigma_Y} E[Y^2]
\end{aligned}$$

$$E[U] = E[X + KY] = E[X] + KE[Y]$$

$$E[V] = E[X + \frac{\sigma_X}{\sigma_Y} Y] = E[X] + \frac{\sigma_X}{\sigma_Y} E[Y]$$

$$\begin{aligned}
E[U]E[V] &= \{E[X] + KE[Y]\} \{E[X] + \frac{\sigma_X}{\sigma_Y} E[Y]\} \\
&= \{E[X]\}^2 + KE[X]E[Y] + \frac{\sigma_X}{\sigma_Y} E[X]E[Y] + K \frac{\sigma_X}{\sigma_Y} \{E[Y]\}^2
\end{aligned}$$

$$E[UV] - E[U] E[V]$$

$$= E[X^2] + KE[XY] + \frac{\sigma_X}{\sigma_Y} E[XY] + K \frac{\sigma_X}{\sigma_Y} E[Y^2]$$

$$- \{E[X]\}^2 - KE[X] E[Y] - \frac{\sigma_X}{\sigma_Y} E[X] E[Y] - K \frac{\sigma_X}{\sigma_Y} \{E[Y]\}^2 = 0$$

$$\Rightarrow E[X^2] - \{E[X]\}^2 + K \frac{\sigma_X}{\sigma_Y} [E[Y^2] - \{E[Y]\}^2] + \left(K + \frac{\sigma_X}{\sigma_Y}\right) \{E[XY] - E[X] E[Y]\} = 0$$

$$\Rightarrow \sigma_X^2 + K \frac{\sigma_X}{\sigma_Y} \sigma_Y^2 + \left(K + \frac{\sigma_X}{\sigma_Y}\right) \text{Cov}(X, Y) = 0$$

$$\sigma_X^2 + K \sigma_X \sigma_Y + \left(K + \frac{\sigma_X}{\sigma_Y}\right) \text{Cov}(X, Y) = 0$$

Divide throughout with σ_X , we get

$$(\sigma_X + K \sigma_Y) + \frac{(\sigma_X + K \sigma_Y)}{\sigma_X \sigma_Y} \text{Cov}(X, Y) = 0$$

$$(\sigma_X + K \sigma_Y) \left[1 + \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}\right] = 0$$

From which $K = -\frac{\sigma_X}{\sigma_Y}$

5.72 Find the mean value of the function

$$G(X, Y) = X^2 + Y^2$$

where X and Y are random variables defined by the density function

$$f_{X,Y}(x, y) = \frac{e^{-(x^2 + y^2)/2\sigma^2}}{2\pi \sigma^2}$$

with σ^2 as constant.

Solution Given:

$$f_{X,Y}(x, y) = \frac{e^{-(x^2 + y^2)/2\sigma^2}}{2\pi \sigma^2}$$

and $g(X, Y) = X^2 + Y^2$

$$E[g(X, Y)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(X, Y) f_{X,Y}(x, y) dx dy$$

$$= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (x^2 + y^2) \frac{e^{-(x^2 + y^2)/2\sigma^2}}{2\pi \sigma^2} dx dy$$

$$\begin{aligned}
&= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \left\{ \frac{x^2 \cdot e^{-(x^2+y^2)/2\sigma^2}}{2\pi\sigma^2} + \frac{y^2 \cdot e^{-(x^2+y^2)/2\sigma^2}}{2\pi\sigma^2} \right\} dx dy \\
&= \int_{-\infty}^{\infty} \frac{x^2 \cdot e^{-x^2/2\sigma^2}}{\sqrt{2\pi}\sigma^2} dx \int_{-\infty}^{\infty} \frac{e^{-y^2/2\sigma^2}}{\sqrt{2\pi}\sigma^2} dy + \int_{-\infty}^{\infty} \frac{e^{-x^2/2\sigma^2}}{\sqrt{2\pi}\sigma^2} dx \int_{-\infty}^{\infty} \frac{y^2 \cdot e^{-y^2/2\sigma^2}}{\sqrt{2\pi}\sigma^2} dy
\end{aligned}$$

The pdf of a Gaussian random variable with zero mean and σ^2 variance is $\frac{e^{-x^2/2\sigma^2}}{\sqrt{2\pi}\sigma^2}$. Therefore,

$$\int_{-\infty}^{\infty} \frac{e^{-x^2/2\sigma^2}}{\sqrt{2\pi}\sigma^2} dx = 1 \quad . \text{ Similarly } \int_{-\infty}^{\infty} \frac{e^{-y^2/2\sigma^2}}{\sqrt{2\pi}\sigma^2} dy = 1$$

$$\text{Also, } E[X^2] = \int_{-\infty}^{\infty} x^2 \frac{e^{-x^2/2\sigma^2}}{\sqrt{2\pi}\sigma^2} dx$$

$$\text{and } E[Y^2] = \int_{-\infty}^{\infty} y^2 \frac{e^{-y^2/2\sigma^2}}{\sqrt{2\pi}\sigma^2} dy$$

Since mean is zero $E[X^2] = \sigma_X^2$ and $E[Y^2] = \sigma_Y^2$ Substituting all values in the equation, we get

$$E[g(X,Y)] = \sigma_X^2 + \sigma_Y^2$$

$$\text{Since } \sigma_X^2 = \sigma_Y^2 = \sigma^2$$

$$E[g(X,Y)] = 2\sigma^2$$

5.73 Two random variables X and Y have joint pdf

$$f_{X,Y}(x,y) = \begin{cases} \frac{xy}{96} & 0 < x < y, 1 < y < 5 \\ 0 & \text{otherwise} \end{cases}$$

Find $E[X]$, $E[Y]$, $E[XY]$, $E[2X + 3Y]$, $\text{Var}(X)$, $\text{Var}(Y)$, $\text{Cov}(X,Y)$. What can you infer from $\text{Cov}(X, Y)$?

Solution Given:

$$f_{X,Y}(x,y) = \begin{cases} \frac{xy}{96} & 0 < x < 4, 1 < y < 5 \\ 0 & \text{otherwise} \end{cases}$$

The marginal distribution function of X is given by

$$\begin{aligned}
f_X(x) &= \int_{-\infty}^{\infty} f_{X,Y}(x,y) dy = \int_1^5 \frac{xy}{96} dy = \frac{x}{96} \left. \frac{y^2}{2} \right|_1^5 \\
&= \frac{x}{96} (12) = \frac{x}{8}, 0 < x < 4
\end{aligned}$$

$$f_X(x) = \frac{x}{8}; 0 < x < 4$$

$$f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx = \int_0^4 \frac{xy}{96} dx = \frac{y}{96} \frac{x^2}{2} \bigg|_0^4 = \frac{y}{12}$$

$$f_Y(y) = \frac{y}{12}; 1 < y < 5$$

$$E[X] = \int_{-\infty}^{\infty} x f_X(x) dx = \int_0^4 x \left(\frac{x}{8} \right) dx = \frac{1}{8} \frac{x^3}{3} \bigg|_0^4 = \frac{8}{3}$$

$$E[Y] = \int_{-\infty}^{\infty} y f_Y(y) dy = \int_1^5 y \left(\frac{y}{12} \right) dy = \frac{1}{12} \frac{y^3}{3} \bigg|_1^5 = \frac{31}{9}$$

$$\begin{aligned} E[XY] &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xy f_{X,Y}(x, y) dx dy \\ &= \int_1^5 \int_0^4 xy \left(\frac{xy}{96} \right) dx dy = \frac{1}{96} \int_1^5 \int_0^4 x^2 y^2 dx dy \\ &= \frac{1}{96} \int_1^5 y^2 \left\{ \int_0^4 x^2 dx \right\} dy = \frac{1}{96} \int_1^5 y^2 \left(\frac{64}{3} \right) dy = \frac{2}{9} \int_1^5 y^2 dy \\ &= \frac{2}{9} \frac{y^3}{3} \bigg|_1^5 = \frac{2}{9} \left(\frac{124}{3} \right) = \frac{248}{27} \end{aligned}$$

$$E[2X + 3Y] = 2E[X] + 3E[Y] = 2 \left(\frac{8}{3} \right) + 3 \left(\frac{31}{9} \right) = \frac{16}{3} + \frac{31}{3} = \frac{47}{3}$$

$$\text{Var}(X) = E[X^2] - \{E[X]\}^2$$

$$E[X^2] = \int_{-\infty}^{\infty} x^2 f_X(x) dx = \int_0^4 x^2 \left(\frac{x}{8} \right) dx = \frac{1}{8} \frac{x^4}{4} \bigg|_0^4 = \frac{1}{8} \left[\frac{256}{4} \right] = 8$$

$$\text{Var}(X) = 8 - \left(\frac{8}{3} \right)^2 = 8 - \frac{64}{9} = \frac{8}{9}$$

$$E[Y^2] = E[Y^2] - \{E[Y]\}^2$$

$$E[Y^2] = \int_{-\infty}^{\infty} y^2 f_Y(y) dy = \int_1^5 y^2 \left(\frac{y}{12} \right) dy = \frac{1}{12} \frac{y^4}{4} \bigg|_1^5$$

$$= \frac{1}{12} \left(\frac{624}{4} \right) = 13$$

$$\text{Var}(Y) = 13 - \left(\frac{31}{9} \right)^2 = 13 - \frac{961}{81} = \frac{92}{81}$$

$$\text{Cov}(X, Y) = E[XY] - E[X] E[Y]$$

$$= \frac{248}{27} - \frac{8}{3} \left(\frac{31}{9} \right) = 0$$

Since $\text{Cov}(X, Y) = 0$, X and Y are uncorrelated.

5.74 X and Y are two independent random variables. X is a uniform random variable in the interval $(0, 5)$ and Y is a zero-mean unit variance Gaussian random variable. Find $E[e^X Y^2]$

Solution Given: $X \sim U(0, 5)$ and $Y \sim N(0, 1)$

Since X and Y are independent, we can write

$$E[e^X Y^2] = E[e^X] E[Y^2]$$

Y is a zero mean unit variance random variable.

Therefore, $E[Y^2] = 1$

$$E[e^X] = \int_{-\infty}^{\infty} e^x f_X(x) dx = \int_0^5 \frac{e^x}{5} dx = \frac{1}{5} e^x \Big|_0^5 = \frac{e^5 - 1}{5}$$

5.75 For the pmf of random variables X and Y given in below Table, find the correlation and covariance, and indicate whether the random variables are independent, orthogonal or uncorrelated.

X \ Y	Y		
	-1	0	1
-1	$\frac{1}{6}$	0	$\frac{1}{6}$
0	0	$\frac{1}{3}$	0
1	$\frac{1}{6}$	0	$\frac{1}{6}$

Solution Given the joint pmf

From the table, we can find that

$$P(X = -1) = \frac{1}{3}; P(X = 0) = \frac{1}{3}, P(X = 1) = \frac{1}{3}, P(Y = -1) = \frac{1}{3}; P(Y = 0) = \frac{1}{3}; P(Y = 1) = \frac{1}{3}$$

$$\begin{aligned} E[X] &= \sum_i x_i p_X(x_i) \\ &= (-1) \frac{1}{3} + 0 \left(\frac{1}{3} \right) + 1 \left(\frac{1}{3} \right) = 0 \end{aligned}$$

$$E[Y] = \sum_j y_j p_Y(y_j) = (-1) \frac{1}{3} + (0) \frac{1}{3} + 1 \left(\frac{1}{3} \right) = 0$$

$$\begin{aligned} E[XY] &= \sum_j \sum_i x_i y_j p_{XY}(x_i, y_j) \\ &= (-1)(-1) \left(\frac{1}{6} \right) + (-1)(0)(0) + (-1)(1) \left(\frac{1}{6} \right) + (0)(-1)(0) \\ &\quad + (0)(0) \frac{1}{3} + 0(1)(0) + 1(-1) \left(\frac{1}{6} \right) + 1(0)(0) + 1(1) \frac{1}{6} \\ &= \frac{1}{6} - \frac{1}{6} - \frac{1}{6} + \frac{1}{6} = 0 \end{aligned}$$

$E[XY] = 0$. So X and Y are orthogonal

$\text{Cov}(X, Y) = E[XY] - E[X] E[Y] = 0$. Therefore uncorrelated

$$P[X = -1, Y = -1] = \frac{1}{6}$$

$$P[X = -1] = \frac{1}{3}; P[Y = -1] = \frac{1}{3}$$

$$P[X = -1] P[Y = -1] = \frac{1}{9}$$

$$P[X = -1, Y = -1] \neq P[X = -1] P[Y = -1]$$

Therefore, not independent.



Practice Problem

5.30 For the joint pmf of random variable (X, Y) given in solved problem 4.11, find the correlation, covariance, and indicate whether the random variables are independent, orthogonal and uncorrelated.

(Ans. Correlated, not independent, not orthogonal)

Solved Problems



5.76 Discrete random variables X and Y have the joint density

$$f_{X,Y}(x, y) = 0.1 \delta(x+2) \delta(y-\alpha) + 0.2 \delta(x-1) \delta(x-2) + 0.3 \delta(x+\alpha) \delta(y-2) + 0.4 \delta(x-\alpha) \delta(y-\alpha)$$

Determine the value of α , if any, that minimizes the correlation between X and Y and find the minimum correlation. Are X and Y orthogonal?

Solution Given:

$$f_{X,Y}(x, y) = 0.1 \delta(x+2) \delta(y-\alpha) + 0.2 \delta(x-1) \delta(x-2) + 0.3 \delta(x+\alpha) \delta(y-2) + 0.4 \delta(x-\alpha) \delta(y-\alpha)$$

We have

$$\int_{-\infty}^{\infty} x \delta(x - k_1) dx = k_1$$

$$\begin{aligned}
\int_{-\infty}^{\infty} y \delta(y - k_2) dy &= k_2 \\
\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xy \delta(x - k_1) \delta(y - k_2) dx dy &= k_1 k_2 \\
R_{XY} = E[XY] &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xy f_{X,Y}(x, y) dx dy \\
&= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xy \{0.1 \delta(x+2) \delta(y-\alpha) + 0.2 \delta(x-1) \delta(x-2) + 0.3 \delta(x+\alpha) \delta(y-2) + 0.4 \delta(x-\alpha) \delta(y-\alpha)\} \\
&= 0.1 \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xy \delta(x+2) \delta(y-\alpha) dx dy + 0.2 \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xy \delta(x-1) \delta(x-2) dx dy \\
&\quad + 0.3 \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xy \delta(x+\alpha) \delta(y-2) dx dy + 0.4 \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xy \delta(x-\alpha) \delta(y-\alpha) dx dy \\
&= 0.1 (-2) (\alpha) + 0.2 (1)(2) + 0.3 (-\alpha) (2) + 0.4 (\alpha) (\alpha) = -0.2\alpha + 0.4 - 0.6\alpha + 0.4\alpha^2 \\
&= 0.4\alpha^2 - 0.8\alpha + 0.4 \\
R_{XY} &= 0.4\alpha^2 - 0.8\alpha + 0.4 \\
\frac{dR_{XY}}{d\alpha} &= 0.8\alpha - 0.8 = 0
\end{aligned}$$

For $\alpha = 1$, the correlation is minimum.

The minimum correlation

$$R_{XY}|_{\min} = 0.4(1)^2 - 0.8(1) + 0.4 = 0$$

Since $R_{XY} = 0$; X and Y are orthogonal.

5.77 For two random variables X and Y ,

$$\begin{aligned}
f_{X,Y}(x, y) &= 0.15 \delta(x-1) \delta(y) + 0.2 \delta(x) \delta(y) + 0.1 \delta(x-1) \delta(y-2) + 0.3 \delta(x+1) \delta(y+2) \\
&\quad + 0.2 \delta(x-1) \delta(y-1) + 0.05 \delta(x+1) \delta(y-3)
\end{aligned}$$

Find (i) correlation (ii) covariance, and (iii) correlation coefficient of X and Y . (iv) Are X and Y either uncorrelated or orthogonal?

Solution The distribution function is represented by the probability matrix as given below:

X \ Y	Y					
	-2	-1	0	1	2	3
-1	0.3					0.05
0			0.2			
1			0.15	0.2	0.1	

Correlation is

$$\begin{aligned} R_{XY} = E[XY] &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xy f_{X,Y}(x, y) dx dy = (0.3)(-1)(-2) + 0.05(-1)(3) + 0.2(0)(0) \\ &\quad + 0.15(1)(0) + 0.2(1)(1) + 0.1(1)(2) \\ &= 0.6 - 0.15 + 0.2 + 0.2 = 0.85 \end{aligned}$$

$$\text{Cov}(X, Y) = E[XY] - E[X] E[Y]$$

We have

$$\begin{aligned} E[X] = m_{10} = E[X] &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x f_{X,Y}(x, y) dx dy \quad \boxed{\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x \delta(x - k_1) \delta(x - k_2) dx dy = k_1} \\ &= 0.3(-1) + 0.05(-1) + 0.2(0) + 0.15(1) + (0.2)(1) + (0.1)(1) = 0.1 \end{aligned}$$

$$\begin{aligned} m_{01} = E[Y] &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} y f_{X,Y}(x, y) dx dy \\ &= 0.3(-2) + 0.2(0) + 0.15(0) + 0.2(1) + (0.1)2 + (0.05)3 = -0.05 \end{aligned}$$

$$\text{Cov}(X, Y) = E[XY] - E[X] E[Y]$$

$$= 0.85 - (0.1)(-0.05)$$

$$= 0.855$$

$$\boxed{\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} y \delta(x - k_1) \delta(x - k_2) dx dy = k_2}$$

Correlation coefficient

$$\rho_{XY} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

$$\text{Var}(X) = E[X^2] - \{E[X]\}^2$$

$$\boxed{\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x^2 \delta(x - K_1) \delta(x - K_2) dx dy = K_1^2}$$

$$\begin{aligned} E[X^2] = m_{20} &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x^2 f_{X,Y}(x, y) dx dy = 0.3(-1)^2 + 0.05(-1)^2 + (0.15)(1)^2 + (0.2)(1)^2 + (0.1)(1)^2 \\ &= 0.80 \end{aligned}$$

$$\begin{aligned} E[Y^2] = m_{02} &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} y^2 f_{X,Y}(x, y) dx dy \\ &= 0.3(-2)^2 + 0.2(1)^2 + 0.1(2)^2 + 0.05(3)^2 \\ &= 2.25 \end{aligned}$$

$$\boxed{\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} y^2 \delta(x - K_1) \delta(x - K_2) dx dy = K_2^2}$$

$$\sigma_Y^2 = \text{Var}(Y) = 2.25 - (-0.05)^2 = 2.2475$$

$$\sigma_X^2 = \text{Var}(X) = 0.80 - (0.1)^2 = 0.79$$

$$\rho_{XY} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} = \frac{0.855}{\sqrt{2.2475} \sqrt{0.79}} = 0.6416$$

Since $\rho_{XY} \neq 0$, X and Y are correlated
and $\rho_{XY} \neq 0$, X and Y are not orthogonal

5.78 Let X and Y be two random variables each having three values $-1, 0, 1$ and having the following joint probability distribution

$X \backslash Y$	-1	0	1
-1	0	0.2	0
0	0.1	0.2	0.1
1	0.1	0.2	0.1

Prove that X and Y have different expectations. Also prove that X and Y are uncorrelated and find $\text{Var}(X)$ and $\text{Var}(Y)$.

Solution Redraw the table and sum the rows and columns to find marginal distribution of X and Y .

$X \backslash Y$	-1	0	1	Total	
-1	0	0.2	0	0.2	$P(X = -1)$
0	0.1	0.2	0.1	0.4	$P(X = 0)$
1	0.1	0.2	0.1	0.4	$P(X = 1)$
	0.2	0.6	0.2	1	
	$P(Y = -1)$	$P(Y = 0)$	$P(Y = 1)$		

From the table, we find

$$P(X = -1) = 0.2; P(X = 0) = 0.4; P(X = 1) = 0.4$$

$$P(Y = -1) = 0.2; P(Y = 0) = 0.6; P(Y = 1) = 0.2$$

$$E[X] = \sum_i x_i p_X(x_i) = (-1)(0.2) + 0(0.4) + 1(0.4) = 0.2$$

$$E[Y] = \sum_j y_j p_Y(y_j) = (-1)(0.2) + 0(0.6) + 1(0.2) = 0$$

X and Y have different expectations.

$$\begin{aligned} E[XY] &= \sum_i \sum_j p_{XY}(X = x_i, Y = y_j) \\ &= (-1)(-1)(0) + 0(-1)(0.1) + 1(-1)(0.1) + (-1)(0)(0.2) \\ &\quad + 0(0)(0.2) + 1(0)(0.2) + (-1)(1)(0) + 0(1)(0.1) + 1(1)(0.1) \\ &= -0.1 + 0.1 = 0 \end{aligned}$$

$$\begin{aligned} \text{Cov}(X, Y) &= E[XY] - E[X]E[Y] \\ &= 0 - (0.2)(0) = 0 \end{aligned}$$

Therefore, X and Y are uncorrelated.

$$\text{Var}(X) = E[X^2] - \{E[X]\}^2$$

$$\begin{aligned}
 E[X^2] &= \sum_i x_i^2 p_x(x_i) \\
 &= (-1)^2 (0.2) + (0)^2 (0.4) + (1)^2 (0.4) \\
 &= 0.2 + 0.4 = 0.6
 \end{aligned}$$

$$\text{Var}(X) = 0.6 - (0.2)^2 = 0.56$$

$$\text{Var}(Y) = E[Y^2] - \{E[Y]\}^2$$

$$\begin{aligned}
 E[Y^2] &= \sum_j y_j^2 p_y(y_j) \\
 &= (-1)^2 (0.2) + (0)^2 (0.6) + (1)^2 (0.2) \\
 &= 0.2 + 0.2 = 0.4
 \end{aligned}$$

$$\text{Var}(Y) = 0.4 - (0)^2 = 0.4$$

5.79 The density function of two random variables X and Y is

$$\begin{aligned}
 f_{X,Y}(x, y) &= \frac{xy}{9} & 0 < x < 2; 0 < y < 3 \\
 &= 0 & \text{elsewhere}
 \end{aligned}$$

- (a) Show that X and Y are statistically independent.
- (b) Show that X and Y are uncorrelated.

Solution

(a) Given

$$\begin{aligned}
 f_{X,Y}(x, y) &= \frac{xy}{9} & 0 < x < 2; 0 < y < 3 \\
 &= 0 & \text{elsewhere}
 \end{aligned}$$

The marginal density function of X is

$$\begin{aligned}
 f_X(x) &= \int_{-\infty}^{\infty} f_{X,Y}(x, y) dy \\
 &= \int_0^3 \frac{xy}{9} dy = \frac{x}{9} \frac{y^2}{2} \Big|_0^3 = \frac{x}{2} \text{ for } 0 < x < 2
 \end{aligned}$$

The marginal density function of Y is

$$\begin{aligned}
 f_Y(y) &= \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx \\
 &= \int_0^2 \frac{xy}{9} dx = \frac{y}{9} \frac{x^2}{2} \Big|_0^2 = \frac{2y}{9} \text{ for } 0 < y < 3
 \end{aligned}$$

$$f_{X,Y}(x, y) = \frac{xy}{9}$$

$$f_X(x)f_Y(y) = \frac{x}{2} \left(\frac{2y}{9} \right) = \frac{xy}{9}$$

$$f_{XY}(x, y) = f_X(x)f_Y(y)$$

Therefore, X and Y are statistically independent.

$$\begin{aligned} \text{(b)} \quad E[XY] &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xy f_{X,Y}(x, y) dx dy = \int_0^2 \int_0^3 \frac{(xy)^2}{9} dy dx \\ &= \frac{1}{9} \int_0^2 x^2 \left(\int_0^3 y^2 dy \right) dx = \frac{1}{9} \int_0^2 x^2 \left(\int_0^3 y^2 dy \right) dx \\ &= \frac{9}{9} \int_0^2 x^2 dx = \frac{1}{3} x^3 \Big|_0^2 = \frac{8}{3} \end{aligned}$$

$$\begin{aligned} E[X] &= \int_{-\infty}^{\infty} x f_X(x) dx \\ &= \int_0^2 x (x/2) dx = \frac{1}{2} \frac{x^3}{3} \Big|_0^2 = \frac{8}{6} \end{aligned}$$

$$\begin{aligned} E[Y] &= \int_{-\infty}^{\infty} y f_Y(y) dy \\ &= \int_0^3 y^2 \left(\frac{2y}{9} \right) dy = \frac{2}{9} \frac{y^3}{3} \Big|_0^3 = 2 \end{aligned}$$

$$E[X] E[Y] = \frac{8}{6} (2) = \frac{8}{3}$$

Since $E[XY] = E[X] E[Y]$, X and Y are uncorrelated.

5.80 Consider two statistically independent random variables X and Y with $\mu_X = 3/4$; $E[X^2] = 4$; $\mu_Y = 1$ and $E[Y^2] = 5$. For a random variable $W = X - 2Y + 1$, find (a) R_{XY} , (b) R_{XW} , (c) R_{YW} , and (d) $\text{Cov}(X, Y)$.

Solution Given: $\mu_X = \frac{3}{4}$; $\mu_Y = 1$

$$\text{(a)} \quad R_{XY} = E[XY]$$

Since X and Y are statistically independent,

$$R_{XY} = E[XY] = E[X] E[Y] = \mu_X \mu_Y = \left(\frac{3}{4} \right) (1) = \frac{3}{4}$$

$$\begin{aligned} \text{(b)} \quad R_{XW} &= E[X(X - 2Y + 1)] \\ &= E[X^2] - 2E[XY] + E[X] \end{aligned}$$

$$= 4 - 2 \left(\frac{3}{4} \right) + \frac{3}{4} = 3.25$$

$$\begin{aligned} \text{(c)} \quad R_{YW} &= E[Y(X - 2Y + 1)] \\ &= E[XY] - 2E[Y^2] + E[Y] \\ &= \frac{3}{4} - 2(5) + 1 = -8.25 \end{aligned}$$

$$\begin{aligned} \text{(d)} \quad \text{Cov}(X, Y) &= E[XY] - \mu_X \mu_Y \\ \text{Since } X \text{ and } Y \text{ are statistically independent,} \\ E[XY] &= E[X] E[Y] \\ \Rightarrow \text{Cov}(X, Y) &= 0 \end{aligned}$$

5.81 Two random variables have a uniform density on a circular region defined by

$$f_{X,Y}(x, y) = \begin{cases} 1/\pi r^2 & x^2 + y^2 \leq r^2 \\ 0 & \text{elsewhere} \end{cases}$$

Find the mean value of the function

$$g(X, Y) = X^2 + Y^2$$

Solution

$$\text{Given} \quad g(X, Y) = X^2 + Y^2$$

$$\text{and} \quad f_{X,Y}(x, y) = \begin{cases} 1/\pi r^2 & x^2 + y^2 \leq r^2 \\ 0 & \text{elsewhere} \end{cases}$$

$$\begin{aligned} E[g(X, Y)] &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(X, Y) f_{X,Y}(x, y) dx dy \\ &= \int_{-r}^r \int_{-\sqrt{r^2-x^2}}^{\sqrt{r^2-x^2}} \frac{x^2 + y^2}{\pi r^2} dy dx \\ &= \frac{1}{\pi r^2} \int_{-r}^r \left[2x^2 \sqrt{r^2 - x^2} + \frac{2}{3} (r^2 - x^2)^{3/2} \right] dx \\ &= \frac{2}{\pi r^2} \left[\int_{-r}^r x^2 \sqrt{r^2 - x^2} dx + \frac{1}{3} \int_{-r}^r (r^2 - x^2)^{3/2} dx \right] \end{aligned}$$

$$\text{Let } x = r \sin \theta \Rightarrow dx = r \cos \theta d\theta$$

$$\text{when } x = -r; \theta = -\frac{\pi}{2}$$

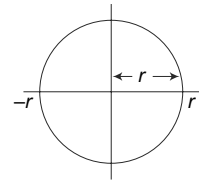


Fig. 5.29

$$X = r, \theta = \frac{\pi}{2}$$

$$\begin{aligned}
 E[g(X,Y)] &= \frac{2}{\pi r^2} \left\{ \int_{-\pi/2}^{\pi/2} r^2 \sin^2 \theta r^2 \cos^2 \theta d\theta + \frac{1}{3} \int_{-\pi/2}^{\pi/2} r^3 \cos^3 \theta (r \cos \theta d\theta) \right\} \\
 &= \frac{2(r^4)}{\pi r^2} \left\{ \int_{-\pi/2}^{\pi/2} \sin^2 \theta \cos^2 \theta d\theta + \frac{1}{3} \int_{-\pi/2}^{\pi/2} \cos^4 \theta d\theta \right\} \\
 &= \frac{2r^2}{\pi} \left\{ \int_{-\pi/2}^{\pi/2} \frac{(1 - \cos 2\theta)}{2} \frac{(1 + \cos 2\theta)}{2} d\theta + \frac{1}{3} \int_{-\pi/2}^{\pi/2} \frac{1}{8} (3 + 4 \cos 2\theta + \cos 4\theta) d\theta \right\} \\
 &= \frac{2r^2}{\pi} \left\{ \int_{-\pi/2}^{\pi/2} \left(\frac{1 - \cos^2 2\theta}{4} \right) d\theta + \frac{1}{24} \int_{-\pi/2}^{\pi/2} (3 + 4 \cos 2\theta + \cos 4\theta) d\theta \right\} \\
 &= \frac{2r^2}{\pi} \left\{ \int_{-\pi/2}^{\pi/2} \left(\frac{1 - \cos 4\theta}{8} \right) d\theta + \frac{1}{24} \int_{-\pi/2}^{\pi/2} (3 + 4 \cos 2\theta + \cos 4\theta) d\theta \right\} \\
 &= \frac{2r^2}{\pi} \left[\frac{1}{8} \theta \Big|_{-\pi/2}^{\pi/2} + \frac{1}{8} \theta \Big|_{-\pi/2}^{\pi/2} \right] \\
 &= \frac{2r^2}{\pi} \left[\frac{\pi}{4} \right] \\
 &= \frac{r^2}{2}
 \end{aligned}$$



JOINT CHARACTERISTIC FUNCTION 5.7

The joint characteristic function is similar to two-dimensional Fourier transform of the joint pdf. It can be used to compute joint moments of a random variable.

Consider two random variables X and Y with a joint pdf $f_{X,Y}(x, y)$. The joint characteristic function of X and Y is given by

$$\phi_{X,Y}(\omega_1, \omega_2) = E \left[e^{j(\omega_1 X + \omega_2 Y)} \right] \quad (5.62)$$

$$= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X,Y}(x, y) e^{j(\omega_1 x + \omega_2 y)} dx dy \quad (5.63)$$

The above expression is equal to two-dimensional Fourier transform of the joint density function with signs of ω_1 and ω_2 reversed.

Using the inverse Fourier transform, we can write

$$f_{X,Y}(x, y) = \frac{1}{(2\pi)^2} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \phi_{X,Y}(\omega_1, \omega_2) e^{-j\omega_1 x} e^{-j\omega_2 y} d\omega_1 d\omega_2 \quad (5.64)$$

If we set $\omega_1 = 0$ or $\omega_2 = 0$ in Eq. (5.62), the result is a marginal characteristic function. That is

$$\phi_X(\omega_1) = \phi_{X,Y}(\omega_1, 0) \quad (5.65)$$

$$\phi_Y(\omega_2) = \phi_{X,Y}(0, \omega_2) \quad (5.66)$$

The various joint moments can be calculated from the joint characteristic function using the following relationship;

$$m_{nk} = (-j)^{n+k} \left. \frac{\partial^{n+k} \phi_{X,Y}(\omega_1, \omega_2)}{\partial \omega_1^n \partial \omega_2^k} \right|_{\omega_1=0, \omega_2=0} \quad (5.67)$$

5.7.1 Properties of Joint Characteristic Functions

(i) Given

$$\phi_{X,Y}(\omega_1, \omega_2) = E[e^{j(\omega_1 X + \omega_2 Y)}]$$

The marginal characteristic functions are given by

$$\phi_X(\omega_1) = \phi_{X,Y}(\omega_1, 0) = E[e^{j\omega_1 X}]$$

$$\phi_Y(\omega_2) = \phi_{X,Y}(0, \omega_2) = E[e^{j\omega_2 Y}]$$

(ii) For independent random variables X and Y , the joint characteristic function is equal to the product of their individual characteristic function.

$$\phi_{X,Y}(\omega_1, \omega_2) = \phi_X(\omega_1) \phi_Y(\omega_2) \quad (5.68)$$

The joint characteristic function is given by

$$\phi_{X,Y}(\omega_1, \omega_2) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{j(\omega_1 x + \omega_2 y)} f_{X,Y}(x, y) dx dy$$

If X and Y are statistically independent,

$$\phi_{X,Y}(x, y) = \phi_X(x) \phi_Y(y)$$

$$\begin{aligned} \phi_{X,Y}(\omega_1, \omega_2) &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{j\omega_1 x} e^{j\omega_2 y} f_X(x) f_Y(y) dx dy \\ &= \int_{-\infty}^{\infty} e^{j\omega_1 x} f_X(x) dx \int_{-\infty}^{\infty} e^{j\omega_2 y} f_Y(y) dy \\ &= \phi_X(\omega_1) \phi_Y(\omega_2) \end{aligned}$$

(iii) If X and Y are independent random variables then

$$\phi_{X+Y}(\omega) = \phi_X(\omega) \phi_Y(\omega)$$

$$\phi_{X+Y}(\omega) = E[e^{j(X+Y)\omega}] = E[e^{j\omega X} e^{j\omega Y}] \quad (5.69)$$

$$= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{j\omega x} e^{j\omega y} f_X(x) f_Y(y) dx dy$$

$$\begin{aligned}
&= \int_{-\infty}^{\infty} e^{j\omega x} f_X(x) dx \int_{-\infty}^{\infty} e^{j\omega y} f_Y(y) dy \\
&= \phi_X(\omega) \phi_Y(\omega)
\end{aligned}$$

- (iv) If X and Y are two random variables, the joint moments can be derived from the joint characteristic function as

$$m_{nk} = (-j)^{n+k} \left. \frac{\partial^{n+k} \phi_{XY}(\omega_1, \omega_2)}{\partial \omega_1^n \partial \omega_2^k} \right|_{\substack{\omega_1=0 \\ \omega_2=0}}$$

REVIEW QUESTIONS



13. Define joint characteristic function.
14. What are the properties of a characteristic function?



JOINT MOMENT GENERATING FUNCTION 5.8

The joint moment-generating function of two random variables X and Y is defined as

$$\begin{aligned}
M_{X,Y}(u_1, u_2) &= E[e^{u_1 X + u_2 Y}] \\
&= E[e^{u_1 X} e^{u_2 Y}] \\
&= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{u_1 x} e^{u_2 y} f_{X,Y}(x, y) dx dy
\end{aligned} \tag{5.70}$$

For discrete random variables,

$$M_{X,Y}(u_1, u_2) = \sum_m \sum_n e^{u_1 x_m + u_2 y_n} p_{XY}(x_m, y_n) \tag{5.71}$$

The marginal moment-generating functions are given by

$$M_X(u_1) = M_{X,Y}(u_1, 0) = E[e^{u_1 X}] \tag{5.72}$$

$$M_Y(u_2) = M_{X,Y}(0, u_2) = E[e^{u_2 Y}] \tag{5.73}$$

If X and Y are statistically independent random variables then

$$M_{X,Y}(u_1, u_2) = M_X(u_1) M_Y(u_2) \tag{5.74}$$

If X and Y are two random variables, then the joint moments can be derived from the joint moment-generating function as

$$m_{nk} = \left. \frac{\partial^{n+k} M_{X,Y}(u_1, u_2)}{\partial u_1^n \partial u_2^k} \right|_{\substack{u_1=0 \\ u_2=0}} \tag{5.75}$$

Solved Problems

5.82 Two random variables X and Y have the joint characteristic function

$$\phi_{X_1, X_2}(\omega_1, \omega_2) = [(1 - j2\omega_1)(1 - j2\omega_2)]^{-N/2}, \quad N > 0$$

and integer.

- (a) Find the correlation and moments m_{20} and m_{02} .
- (b) Determine the means of X_1 and X_2 .
- (c) What is the correlation coefficient?

Solution Given $\phi_{X_1, X_2}(\omega_1, \omega_2) = [(1 - j2\omega_1)(1 - j2\omega_2)]^{-N/2}$

From Eq. (5.67), we have

$$\begin{aligned} m_{nk} &= (-j)^{n+k} \frac{\partial^{n+k} \phi_{X_1, X_2}(\omega_1, \omega_2)}{\partial \omega_1^n \partial \omega_2^k} \\ m_{20} &= (-j)^2 \frac{\partial^2 \phi_{X_1, X_2}(\omega_1, \omega_2)}{\partial \omega_1^2} \bigg|_{\omega_1=0; \omega_2=0} \\ &= (-j)^2 \frac{\partial}{\partial \omega_1} \left[-\frac{N}{2} (1 - j2\omega_2)^{-N/2} (1 - j2\omega_1)^{-\frac{N}{2}-1} (-2j) \right] \bigg|_{\omega_1=0; \omega_2=0} \\ &= - \left[-\frac{N}{2} (1 - j2\omega_2)^{-N/2} \left(-\frac{N}{2} - 1 \right) (1 - j2\omega_1)^{-\frac{N}{2}-2} (-2j)^2 \right] \bigg|_{\omega_1=0; \omega_2=0} \\ &= N(N+2) \\ m_{02} &= (-j)^2 \frac{\partial^2 \phi_{X_1, X_2}(\omega_1, \omega_2)}{\partial \omega_2^2} \bigg|_{\omega_1=0; \omega_2=0} \\ &= \left[-\frac{N}{2} (1 - j2\omega_1)^{-N/2} \left(-\frac{N}{2} - 1 \right) (1 - j2\omega_2)^{-\frac{N}{2}-2} (-2j)^2 \right] \bigg|_{\omega_1=0; \omega_2=0} \\ &= N(N+2) \end{aligned}$$

Correlation

$$\begin{aligned} E[X_1 X_2] &= m_{11} \\ &= (-j)^2 \frac{\partial^2 \phi_{X_1, X_2}(\omega_1, \omega_2)}{\partial \omega_1 \partial \omega_2} \bigg|_{\omega_1=0; \omega_2=0} \end{aligned}$$

$$\begin{aligned}
&= -\frac{\partial}{\partial \omega_2} \left[-\frac{N}{2} (1 - j2\omega_1)^{-\frac{N}{2}-1} (-j2) (1 - j2\omega_2)^{-N/2} \right]_{\substack{\omega_1=0 \\ \omega_2=0}} \\
&= -\left[\left(-\frac{N}{2} \right)^2 (-j2)^2 (1 - j2\omega_1)^{-\frac{N}{2}-1} (1 - j2\omega_2)^{-\frac{N}{2}-1} \right]_{\substack{\omega_1=0 \\ \omega_2=0}} \\
&= N^2 \\
E[X_1] = m_{10} &= -j \frac{\partial}{\partial \omega_1} \phi_{X_1 X_2}(\omega_1, \omega_2) \Big|_{\substack{\omega_1=0 \\ \omega_2=0}} \\
&= -j \left[-\frac{N}{2} (1 - j2\omega_1)^{-\frac{N}{2}-1} (-j2) (1 - j2\omega_2)^{-N/2} \right]_{\substack{\omega_1=0 \\ \omega_2=0}} = N
\end{aligned}$$

Similarly, $m_{01} = N$

$$\begin{aligned}
\sigma_X^2 &= E[X^2] - \{E[X]\}^2 \\
&= m_{20} - (m_{10})^2 \\
&= N(N+2) - (N)^2 \\
&= 2N
\end{aligned}$$

Similarly $\sigma_Y^2 = 2N$

Correlation coefficient

$$\begin{aligned}
\rho_{X_1 X_2} &= \frac{E[X_1 X_2] - E[X_1] E[X_2]}{\sigma_X \sigma_Y} \\
&= \frac{N^2 - N(N)}{\sqrt{4N^2}} = 0
\end{aligned}$$

5.83 If X and Y are independent Poisson random variables with parameters λ_1 and λ_2 , show that the conditional density function of X given $(X + Y)$ is binomial.

Solution Given X and Y are independent Poisson random variables. The MGF of a Poisson random variable is $e^{\lambda(e^u - 1)}$. That is

$$M_X(u) = e^{\lambda_1(e^u - 1)} \text{ and } M_Y(u) = e^{\lambda_2(e^u - 1)}$$

Let $X + Y = Z$ then

$$\begin{aligned}
M_Z(u) &= M_{X+Y}(u) = M_X(u) M_Y(u) \\
&= e^{(\lambda_1 + \lambda_2)(e^u - 1)}
\end{aligned}$$

That is, z is also a Poisson random variable with parameter $(\lambda_1 + \lambda_2)$. That is, $P(X = x) = \frac{e^{-\lambda_1} (\lambda_1)^x}{x!}$;
 $P(Y = y) = \frac{e^{-\lambda_2} (\lambda_2)^y}{y!}$ and $P(X + Y = z) = \frac{e^{-(\lambda_1 + \lambda_2)} (\lambda_1 + \lambda_2)^z}{z!}$

The conditional density function

$$\begin{aligned}
 P(X|Z) &= \frac{P(X = x, Z = z)}{P(Z = z)} \\
 &= \frac{P(X = x, X + Y = z)}{P(X + Y = z)} \\
 &= \frac{P(X = x) P(Y = z - x)}{P(X + Y = z)} \\
 &= \frac{e^{-\lambda_1} (\lambda_1)^x}{x!} \frac{e^{-\lambda_2} (\lambda_2)^{z-x}}{(z-x)!} \\
 &= \frac{e^{-(\lambda_1 + \lambda_2)} (\lambda_1 + \lambda_2)^z}{z!} \\
 &= \binom{z}{x} \frac{(\lambda_1)^x (\lambda_2)^{z-x}}{(\lambda_1 + \lambda_2)^z} = \binom{z}{x} \left(\frac{\lambda_1}{\lambda_1 + \lambda_2} \right)^x \left(\frac{\lambda_2}{\lambda_1 + \lambda_2} \right)^{z-x} \\
 \Rightarrow Z \text{ is a binomial with parameters } z, \frac{\lambda_1}{\lambda_1 + \lambda_2}
 \end{aligned}$$

5.84 If X and Y are zero mean independent normal random variables with variances σ_1^2 and σ_2^2 then prove that the random variable $z = ax + by + c$; $c \neq 0$ is also a normal random variable.

Solution Given: $X \sim N(0, \sigma_1^2)$ and

$$Y \sim N(0, \sigma_2^2)$$

$$Z = ax + by + c$$

$$\begin{aligned}
 \phi_z(\omega) &= E[e^{j\omega z}] = E[e^{j\omega(ax + by + c)}] \\
 &= E[e^{j\omega a x}] E[e^{j\omega b y}] E[e^{j\omega c}] \\
 &= e^{j\omega c} \phi_X(a\omega) \phi_Y(b\omega)
 \end{aligned}$$

For a normal random variable,

$$\phi_X(\omega) = e^{j\omega\mu_X + \sigma_X^2(j\omega)^2/2}$$

$$\mu_X = 0 \text{ and } \sigma_X^2 = \sigma_1^2$$

$$\phi_X(\omega) = e^{-\sigma_1^2 \omega^2 / 2}$$

$$\phi_Y(\omega) = e^{-\sigma_2^2 \omega^2 / 2}$$

$$\begin{aligned}
 \Rightarrow \phi_Z(\omega) &= e^{-a^2 \sigma_1^2 \omega^2 / 2} e^{-b^2 \sigma_2^2 \omega^2 / 2} e^{j\omega c} \\
 &= e^{j\omega c - (a^2 \sigma_1^2 + b^2 \sigma_2^2) \omega^2 / 2}
 \end{aligned}$$

On comparing, we can find

$$Z \sim N(c, a^2 \sigma_1^2 + b^2 \sigma_2^2)$$

5.85 Show that the joint characteristic function of N independent random variables X_i , having characteristic function $\phi_{X_i}(\omega_i)$ is

$$\phi_{X_1, X_2, \dots, X_N}(\omega_1, \omega_2, \dots, \omega_N) = \prod_{i=1}^N \phi_{X_i}(\omega_i)$$

Solution

$$\begin{aligned} \phi_{X_1, \dots, X_N}(\omega_1, \dots, \omega_N) &= E \left[e^{j(\omega_1 X_1 + \dots + \omega_N X_N)} \right] \\ &= \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} f_{X_1, \dots, X_N}(x_1, \dots, x_N) e^{j(\omega_1 x_1 + \dots + \omega_N x_N)} dx_1 \dots dx_N \end{aligned}$$

Since $X_1, X_2, \dots, X_N$ are independent,

$$f_{X_1, \dots, X_N}(x_1, \dots, x_N) = f_{X_1}(x_1) \dots f_{X_N}(x_N)$$

$$\begin{aligned} \phi_{X_1, \dots, X_N}(\omega_1, \dots, \omega_N) &= \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} f_{X_1}(x_1) \dots f_{X_N}(x_N) e^{j\omega_1 x_1} \dots e^{j\omega_N x_N} dx_1, \dots dx_N \\ &= \int_{-\infty}^{\infty} f_{X_1}(x_1) e^{j\omega_1 x_1} dx_1 \dots \int_{-\infty}^{\infty} f_{X_N}(x_N) e^{j\omega_N x_N} dx_N \\ &= \phi_{X_1}(\omega_1) \dots \phi_{X_N}(\omega_N) \\ &= \prod_{i=1}^N \phi_{X_i}(\omega_i) \end{aligned}$$

5.86 For two zero-mean Gaussian random variables X and Y , show that their joint characteristic function is $\phi_{X,Y}(\omega_1, \omega_2) = \exp \left\{ -\frac{1}{2} (\sigma_X^2 \omega_1^2 + 2\rho\sigma_X\sigma_Y \omega_1\omega_2 + \sigma_Y^2 \omega_2^2) \right\}$.

Solution

$$\begin{aligned} \phi_{X,Y}(\omega_1, \omega_2) &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X,Y}(x, y) e^{j\omega_1 x + j\omega_2 y} dx dy \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \frac{\exp \left[\frac{-1}{2(1-\rho^2)} \left(\frac{x^2}{\sigma_X^2} - \frac{2\rho xy}{\sigma_X \sigma_Y} + \frac{y^2}{\sigma_Y^2} \right) \right]}{2\pi \sigma_X \sigma_Y (1-\rho^2)} \exp(j\omega_1 x + j\omega_2 y) dx dy \\ &= \int_{-\infty}^{\infty} \frac{\exp \left[\frac{-1}{2(1-\rho^2)} \left(\frac{x^2}{\sigma_X^2} + j\omega_1 x \right) \right]}{2\pi \sigma_X \sqrt{1-\rho^2}} \underbrace{\int_{-\infty}^{\infty} \frac{1}{\sigma_Y} \exp \left[\frac{-1}{2(1-\rho^2)} \left(\frac{y^2}{\sigma_Y^2} - \frac{2\rho xy}{\sigma_X \sigma_Y} \right) + j\omega_2 y \right] dy}_{I_1} dx \end{aligned}$$

Let
$$I_1 = \int_{-\infty}^{\infty} \frac{1}{\sigma_Y} \exp \left[\frac{-1}{2(1-\rho^2)} \left(\frac{y^2}{\sigma_Y^2} - \frac{2\rho xy}{\sigma_X \sigma_Y} - j2(1-\rho^2) \omega_2 y \right) \right] dy$$

Let
$$\frac{1}{2(1-\rho^2)} = a$$

Let $\frac{y}{\sigma_Y} = t$
 $dy = \sigma_Y dt$

$$\begin{aligned} I_1 &= \int_{-\infty}^{\infty} \exp \left[-a \left(t^2 - \frac{2\rho x}{\sigma_X} t - j2\omega_2(1-\rho^2) \sigma_Y t \right) \right] dt \\ &= \int_{-\infty}^{\infty} e^{-at^2} e^{\left[\frac{2\rho x}{\sigma_X} + j2\omega_2(1-\rho^2) \sigma_Y \right] at} dt \end{aligned}$$

Let
$$b = \left[\frac{2\rho x}{\sigma_X} + j2\omega_2(1-\rho^2) \sigma_Y \right] a$$

Then

$$I_1 = \int_{-\infty}^{\infty} e^{-at^2 + bt} dt = e^{b^2/4a} \int_{-\infty}^{\infty} e^{-a \left(t - \frac{b}{2a} \right)^2} dt$$

Let $\sqrt{a} \left(t - \frac{b}{2a} \right) = z / \sqrt{2}$ then $dt = \frac{dz}{\sqrt{2a}}$

$$I_1 = \frac{1}{\sqrt{2a}} e^{b^2/4a} \int_{-\infty}^{\infty} e^{-z^2/2} dz$$

$$\int_{-\infty}^{\infty} e^{-z^2/2} dz = \sqrt{2\pi}$$

$$I_1 = \sqrt{\frac{\pi}{a}} e^{b^2/4a} = \sqrt{2\pi(1-\rho^2)} \exp \left\{ \left[\frac{\rho x}{\sigma_X} + j\omega_2(1-\rho^2) \sigma_Y \right]^2 / 2(1-\rho^2) \right\}$$

$$\begin{aligned} \phi_{X,Y}(\omega_1, \omega_2) &= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi} \sigma_X} \exp \left[\frac{-1}{2(1-\rho^2)} \left(\frac{x^2}{\sigma_X^2} \right) + j\omega_1 x \right] \exp \left\{ \left[\frac{\rho x}{\sigma_X} + j\omega_2(1-\rho^2) \sigma_Y \right]^2 / 2(1-\rho^2) \right\} dx \\ &= \frac{\exp \left[-\frac{1}{2} (1-\rho^2) \omega_2^2 \sigma_Y^2 \right]}{\sqrt{2\pi}} \underbrace{\int_{-\infty}^{\infty} \frac{1}{\sigma_X} \exp \left[\frac{-x^2}{2\sigma_X^2} + \frac{j\omega_2 \sigma_Y \rho x}{\sigma_X} + j\omega_1 x \right] dx}_{I_2} \end{aligned}$$

Let
$$I_2 = \int_{-\infty}^{\infty} \frac{1}{\sigma_X} \exp \left[\frac{-x^2}{2\sigma_X^2} + \frac{j\omega_2 \sigma_Y \rho x}{\sigma_X} + j\omega_1 x \right] dx$$

Let $\frac{x}{\sigma_X} = m$; then $dx = \sigma_X dm$

$$I_2 = \int_{-\infty}^{\infty} \exp \left[\frac{-m^2}{2} + j\omega_2 \sigma_Y m\rho + jm\sigma_X \omega_1 \right] dm = \int_{-\infty}^{\infty} \exp \left[-\frac{m^2}{2} + j(\rho\omega_2 \sigma_Y + \omega_1 \sigma_X)m \right] dm$$

$$\text{Let } j(\rho\omega_2 \sigma_Y + \omega_1 \sigma_X) = n$$

$$I_2 = \int_{-\infty}^{\infty} \exp \left\{ -\frac{1}{2}(m^2 - 2mn + n^2) \right\} e^{n^2/2} dm$$

$$= e^{n^2/2} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(m-n)^2} dm$$

$$= e^{n^2/2} (\sqrt{2\pi})$$

$$\phi_{X,Y}(\omega_1, \omega_2) = \frac{\exp \left[-\frac{1}{2}(1-\rho^2) \omega_2^2 \sigma_Y^2 \right]}{\sqrt{2\pi}} \exp \left[\frac{\{j(\rho\omega_2 \sigma_Y + \omega_1 \sigma_X)\}^2}{2} \right] (\sqrt{2\pi})$$

$$= \exp \left[-\frac{1}{2}(1-\rho^2) \omega_2^2 \sigma_Y^2 \right] \exp \left[-\frac{1}{2}(\rho^2 \omega_2^2 \sigma_Y^2 + \omega_1^2 \sigma_X^2 + 2\rho\omega_1 \omega_2 \sigma_X \sigma_Y) \right]$$

$$= \exp \left[-\frac{1}{2}(\omega_2^2 \sigma_Y^2 + \omega_1^2 \sigma_X^2 + 2\rho\omega_1 \omega_2 \sigma_X \sigma_Y) \right]$$

5.87 If X and Y are independent gamma random variables with common parameters a and b , find the pdf of (a) $X + Y$, and (b) X/Y .

Solution

(a) Given: $Z = X + Y$

The characteristic function of Gamma random variable is given by

$$\phi_X(\omega) = \frac{1}{(1 - j\omega b)^a}$$

Similarly,

$$\phi_Y(\omega) = \frac{1}{(1 - j\omega b)^a}$$

$$\phi_Z(\omega) = \phi_{X+Y}(\omega) = \phi_X(\omega) \phi_Y(\omega)$$

$$= \frac{1}{(1 - j\omega b)^a} \frac{1}{(1 - j\omega b)^a} = \frac{1}{(1 - j\omega b)^{2a}}$$

Therefore, $X + Y$ is a gamma random variable with parameters $2a$ and b .

$$(b) \text{ Given: } Z = X/Y \quad f_X(x) = \frac{x^{a-1} e^{-x/b}}{\Gamma(a) b^a} \text{ and } f_Y(y) = \frac{y^{a-1} e^{-y/b}}{\Gamma(a) b^a}$$

From Eq. (5.49)

$$f_Z(z) = \int_{-\infty}^{\infty} |y| f_{X,Y}(yz, y) dy$$

For Gamma random variable

$$\begin{aligned} f_Z(z) &= \int_0^{\infty} y f_{X,Y}(yz, y) dy \\ &= \int_0^{\infty} \frac{y(yz)^{a-1} e^{-yz/b}}{\Gamma(a)b^a} \frac{(y)^{a-1} e^{-y/b}}{\Gamma(a)b^a} dy \\ &= \frac{(z)^{a-1}}{[\Gamma(a)b^a]^2} \int_0^{\infty} y^{2(a-1)} e^{-y(1+z)/b} dy \\ &= \frac{(z)^{a-1}}{[\Gamma(a)b^a]^2} = \int_0^{\infty} y^{2a-1} e^{-y(1+z)/b} dy \end{aligned}$$

Let $(1+z)y/b = u$

$$\Rightarrow dy = \frac{b du}{1+z}$$

$$\begin{aligned} \Rightarrow f_Z(z) &= \frac{(z)^{a-1}}{[\Gamma(a)b^a]^2} \int_0^{\infty} \left(\frac{bu}{1+z} \right)^{2a-1} e^{-u} \left(\frac{b}{1+z} \right) du \\ &= \frac{(z)^{a-1}}{[\Gamma(a)b^a]^2} \left(\frac{b}{1+z} \right)^{2a} \Gamma(2a) \\ &= \frac{(z)^{a-1}}{[\Gamma(a)]^2} \frac{\Gamma(2a)}{(1+z)^{2a}} \end{aligned}$$



Practice Problem

5.31 If X and Y are independent gamma random variables with parameters (α_1, β) and (α_2, β) then find the pdfs of the random variables.

(a) $Z = X + Y$ (b) $Z = X/(X + Y)$ (c) $Z = X/Y$.

(Ans. (a) Gamma distribution (b) beta distribution with parameter $(\alpha_1 + \alpha_2, \beta)$)

Solved Problems



5.88 X and Y are independent, identically distributed binomial random variables with parameters n and p . Show that $Z = X + Y$ is also a binomial random variable.

Solution The moment-generating function of a binomial random variable is

$$M_X(u) = [pe^u + (1-p)]^n$$

Similarly,

$$M_Y(u) = [pe^u + (1-p)]^n$$

$$M_{X+Y}(u) = M_X(u) M_Y(u) = [pe^u + (1-p)]^{2n}$$

Therefore, $(X + Y)$ is a binomial random variable with parameters $2n$ and P

5.89 If X and Y are independent random variables with geometric pmf.

$$P(X = k) = pq^k \quad k = 0, 1, 2, \dots$$

$$P(Y = m) = pq^m \quad m = 0, 1, 2, \dots$$

Find pmf of $X + Y$.

Solution Given $P(X = k) = pq^k \quad k = 0, 1, 2, \dots$

$$P(Y = m) = pq^m \quad m = 0, 1, 2, \dots$$

$$P(Z = z) = \sum_{k=0}^z pq^{z-k} pq^k \quad \boxed{\sum_{k=0}^n 1 = n+1}$$

$$= p^2 \sum_{k=0}^z q^z = p^2 q^z (z+1), \quad z = 0, 1, 2, \dots$$

5.90 If X and Y are independent exponential random variables with common parameter λ , show that $X/(X + Y)$ is a $U(0, 1)$.

Solution We know $f_X(x) = \lambda e^{-\lambda x}$ and $f_Y(y) = \lambda e^{-\lambda y}$

$$\text{Let } z = x/(x + y)$$

Let us define an auxiliary variable $w = y$

Solving for x and y , we get

$$x = (wz)/(1 - z)$$

$$\text{and } y = w$$

$$J(x, y) = \begin{vmatrix} \partial z / \partial x & \partial z / \partial y \\ \partial w / \partial x & \partial w / \partial y \end{vmatrix} = \begin{vmatrix} \frac{y}{(x+y)^2} & \frac{-x}{(x+y)^2} \\ 0 & 1 \end{vmatrix}$$

$$\Rightarrow J(x, y) = \frac{y}{(x+y)^2} = \frac{-w}{\left(\frac{w}{1-z}\right)^2} = \frac{(1-z)^2}{w}$$

$$f_{Z,W}(z, w) = \frac{f_{X,Y}(x, y)}{|J(x, y)|}$$

$$\begin{aligned}
&= \frac{f_X(x) f_Y(y)}{|J(x, y)|} = \frac{\lambda^2 e^{-\lambda(x+y)}}{|J(x, y)|} \\
&= \frac{w \lambda^2}{(1-z)^2} e^{-\lambda w(1-z)} \\
f_Z(z) &= \int_0^\infty \frac{w \lambda^2}{(1-z)^2} e^{-\lambda w(1-z)} dw \\
&= \int_0^\infty \frac{\lambda^2}{(1-z)^2} w e^{-\lambda w/(1-z)} dw & \text{Let } \frac{\lambda w}{1-z} = t \\
& & \frac{\lambda}{(1-z)} dw = dt \\
&= \int_0^\infty t e^{-t} dt \\
&= 1 \\
\Rightarrow f_Z(z) &= 1 \text{ for } 0 \leq z \leq 1
\end{aligned}$$

5.91 Consider random variables Y_1 and Y_2 related to arbitrary random variables X and Y by the coordinate rotation.

$$Y_1 = X \cos \theta + Y \sin \theta, Y_2 = -X \sin \theta + Y \cos \theta$$

Find the covariance of Y_1 and Y_2 . For what value of θ are the random variables Y_1 , and Y_2 uncorrelated?

Solution Given: $Y_1 = X \cos \theta + Y \sin \theta$. If μ_X and μ_Y are mean of X and Y respectively then $\mu_{Y1} = \mu_X \cos \theta + \mu_Y \sin \theta$.

Similarly,

$$\mu_{Y2} = -\mu_X \sin \theta + \mu_Y \cos \theta$$

$$\text{Cov}(Y_1, Y_2) = E[(Y_1 - \mu_{Y1})(Y_2 - \mu_{Y2})]$$

$$= E[\{X \cos \theta + Y \sin \theta - \mu_X \cos \theta - \mu_Y \sin \theta\} \{-X \sin \theta + Y \cos \theta + \mu_X \sin \theta - \mu_Y \cos \theta\}]$$

$$= E\{[(X - \mu_X) \cos \theta + (Y - \mu_Y) \sin \theta] [-(X - \mu_X) \sin \theta + (Y - \mu_Y) \cos \theta]\}$$

$$\begin{aligned}
&= E[-(X - \mu_X)^2 \cos \theta \sin \theta + (Y - \mu_Y)^2 \sin \theta \cos \theta - (X - \mu_X)(Y - \mu_Y) \sin^2 \theta \\
&\quad + (X - \mu_X)(Y - \mu_Y) \cos^2 \theta]
\end{aligned}$$

$$= -\sigma_X^2 \cos \theta \sin \theta + \sigma_Y^2 \sin \theta \cos \theta - \text{cov}(X, Y) \sin^2 \theta + \text{cov}(X, Y) \cos^2 \theta$$

$$= \frac{1}{2}(\sigma_Y^2 - \sigma_X^2) \sin 2\theta + \text{cov}(X, Y) (\cos^2 \theta - \sin^2 \theta) \quad (5.76)$$

$$= \frac{1}{2}(\sigma_Y^2 - \sigma_X^2) \sin 2\theta + \text{cov}(X, Y) \cos 2\theta$$

For Y_1 and Y_2 to be uncorrelated, $\text{Cov}(Y_1, Y_2) = 0$

$$\Rightarrow \text{Cov}(X, Y) \cos 2\theta + \frac{1}{2}(\sigma_Y^2 - \sigma_X^2) \sin 2\theta = 0$$

$$\tan 2\theta = \frac{\text{Cov}(X, Y)}{\frac{1}{2}(\sigma_X^2 - \sigma_Y^2)} = \frac{2\rho\sigma_X\sigma_Y}{\sigma_X^2 - \sigma_Y^2} \quad (5.77)$$

$$\Rightarrow \theta = \frac{1}{2} \tan^{-1} \left(\frac{2\rho\sigma_X\sigma_Y}{\sigma_X^2 - \sigma_Y^2} \right) \quad (5.78)$$

5.92 Two Gaussian random variables X and Y have variances $\sigma_X^2 = 9$ and $\sigma_Y^2 = 4$ respectively and correlation coefficient ρ . It is known that a coordinate rotation by an angle $\frac{\pi}{8}$ results in new random variables Y_1 and Y_2 that are uncorrelated. What is ρ ?

Solution Since the rotation results in new random variables Y_1 and Y_2 , the relation between covariance of X , Y and Y_1 , Y_2 is

$$\text{Cov}(Y_1, Y_2) = \frac{1}{2}(\sigma_Y^2 - \sigma_X^2) \sin 2\theta + \text{Cov}(X, Y) \cos 2\theta$$

$$\text{Given: } \sigma_X^2 = 9; \sigma_Y^2 = 4 \Rightarrow \sigma_X = 3 \text{ and } \sigma_Y = 2$$

$$\text{Also } \theta = \frac{\pi}{8}$$

$$\begin{aligned} \text{Cov}(Y_1, Y_2) &= \frac{1}{2}(4 - 9) \sin 2\left(\frac{\pi}{8}\right) + \rho\sigma_X\sigma_Y \cos 2\left(\frac{\pi}{8}\right) \\ &= -\frac{5}{2} \sin\left(\frac{\pi}{4}\right) + \rho(6) \cos \frac{\pi}{4} \end{aligned}$$

$$\therefore \text{ for uncorrelated random variables, } \text{Cov}(Y_1, Y_2) = 0$$

$$\begin{aligned} \Rightarrow \rho &= \frac{\frac{5}{2} \sin\left(\frac{\pi}{4}\right)}{6 \cos \frac{\pi}{4}} = \frac{5}{12} \tan\left(\frac{\pi}{4}\right) \\ &= \frac{5}{12} \end{aligned}$$

5.93 Two Gaussian random variables X and Y have variances $\sigma_X^2 = 16$ and $\sigma_Y^2 = 9$ respectively with correlation coefficient ρ . It is known that a coordinate rotation by an angle $\pi/6$ results in new random variables Y_1 and Y_2 that are uncorrelated. What is ρ ?

Solution Using Eq. (5.78), can get

$$\rho = \frac{\sigma_X^2 - \sigma_Y^2}{2\sigma_X\sigma_Y} \tan 2\theta \quad (5.79)$$

$$\begin{aligned}
&= \frac{16-9}{2(4)(3)} \tan 2\left(\frac{\pi}{6}\right) \\
&= \frac{7}{24} \tan\left(\frac{\pi}{3}\right) \\
&= 0.505
\end{aligned}$$

5.94 X and Y are random variables with joint Gaussian pdf with $\sigma_X^2 = \sigma_Y^2$ and $\rho = 1$. Find a transformation matrix such that new random variables Y_1 and Y_2 are statistically independent.

Solution The angle of rotation

$$\theta = \frac{1}{2} \tan^{-1} \left(\frac{2\rho \sigma_X \sigma_Y}{\sigma_X^2 - \sigma_Y^2} \right) = \frac{1}{2} \tan^{-1} \frac{2\sigma_X \sigma_Y}{\sigma_X^2 - \sigma_Y^2} = \frac{1}{2} \tan^{-1}(\infty) = \frac{1}{2} \left(\frac{\pi}{2} \right)$$

$$\Rightarrow \theta = \frac{\pi}{4}$$

The transformation matrix

$$[T] = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

5.95 The two Gaussian random variables X and Y have first and second-order moments $\mu_X = 1$; $E[X^2] = 2.5$, $\mu_Y = 1.8$, $E[Y^2] = 3.6$ and $R_{XY} = 2.2$. Find (a) $\text{Cov}(X, Y)$, and (b) ρ . Also, find the angle θ of a coordinate rotation that will generate new random variables that are statistically independent.

Solution

$$\begin{aligned}
\text{(a)} \quad \text{Cov}(X, Y) &= E[(X - \mu_X)(Y - \mu_Y)] \\
&= E[XY] - \mu_X \mu_Y \\
&= 2.2 - 1(1.8) = 0.4
\end{aligned}$$

$$\begin{aligned}
\text{(b)} \quad \sigma_X^2 &= E[X^2] - (\mu_X)^2 = 2.5 - (1)^2 = 1.5 \\
\sigma_Y^2 &= E[Y^2] - (\mu_Y)^2 = 3.6 - (1.8)^2 = 0.36
\end{aligned}$$

$$\rho = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} = \frac{0.4}{\sqrt{1.5} \sqrt{0.36}} = 0.544$$

$$\theta = \frac{1}{2} \tan^{-1} \left(\frac{2\rho \sigma_X \sigma_Y}{\sigma_X^2 - \sigma_Y^2} \right) = \frac{1}{2} \tan^{-1} \left(\frac{2(0.544)\sqrt{1.5} \sqrt{0.36}}{1.5 - 0.36} \right) = 0.304 \text{ rad}$$

5.96 The random variable X_1 and X_2 are statistically independent with $U(0, 1)$. Prove that the random variables Y_1 and Y_2 generated by the following transformation are $N(0, 1)$

$$\begin{aligned}
Y_1 &= \sqrt{-2 \ln(X_1)} \cos(2\pi X_2) \\
Y_2 &= \sqrt{-2 \ln(X_1)} \sin(2\pi X_2)
\end{aligned}$$

Solution Given:

$$f_{X_1, X_2}(x_1, x_2) = 1 \text{ for } 0 < x_1 < 1, 0 < x_2 < 1 \\ = 0 \text{ otherwise}$$

$$Y_1 = \sqrt{-2 \ln(X_1)} \cos(2\pi X_2)$$

$$Y_2 = \sqrt{-2 \ln(X_1)} \sin(2\pi X_2)$$

$$Y_1^2 + Y_2^2 = -2 \ln X_1 \Rightarrow X_1 = e^{-(Y_1^2 + Y_2^2)/2}$$

$$f_{Y_1, Y_2}(y_1, y_2) = \frac{1}{|J|} f_{X_1, X_2}(x_1, x_2)$$

$$J = \begin{vmatrix} \frac{\partial y_1}{\partial x_1} & \frac{\partial y_1}{\partial x_2} \\ \frac{\partial y_2}{\partial x_1} & \frac{\partial y_2}{\partial x_2} \end{vmatrix} = \begin{vmatrix} -\frac{\cos(2\pi x_2) (-2 \ln x_1)^{-1/2}}{x_1} & 2\pi \sqrt{-2 \ln x_1} \sin 2\pi x_2 \\ -\frac{\sin(2\pi x_2) (-2 \ln x_1)^{-1/2}}{x_1} & 2\pi \sqrt{-2 \ln x_1} \cos 2\pi x_2 \end{vmatrix}$$

$$= -\frac{2\pi}{x_1} = \frac{-2\pi}{e^{-(y_1^2 + y_2^2)/2}}$$

$$f_{Y_1, Y_2}(y_1, y_2) = \frac{1}{2\pi} e^{-(y_1^2 + y_2^2)/2} f_{X_1, X_2}(x_1, x_2)$$

$$\Rightarrow f_{Y_1, Y_2}(y_1, y_2) = \frac{1}{2\pi} e^{-(y_1^2 + y_2^2)/2}$$

From the above equation, we can find that the joint distribution of Y_1 and Y_2 is normal.



CONDITIONAL MEAN AND VARIANCES 5.9

5.9.1 Discrete Random Variables

Consider two discrete random variables X and Y with pmf $p_{X,Y}(x, y)$. The conditional expected value of Y given that $X = x$, is defined by

$$E[Y/X = x] = \sum_y yp(y/x) \quad (5.80)$$

The conditional variance Y , given $X = x$ is given by

$$\sigma_{Y/X}^2 = E[Y^2/X = x] - \{E[Y/X = x]\}^2 \quad (5.81)$$

Similarly, the conditional expected value of X given that $Y = y$ is given by

$$E[X/Y = y] = \sum_x xp(y/x)$$

and the conditional variance of X given $Y = y$ is given by

$$\sigma_{X/Y}^2 = E[X^2/Y = y] - \{E[X/Y = y]\}^2 \quad (5.82)$$

5.9.2 Continuous Random Variables

Consider two continuous random variables X and Y with the joint pdf $f_{X,Y}(x, y)$. The conditional expected value of Y given that $X = x$ is given by

$$E[Y|X = x] = \int_{-\infty}^{\infty} y f(y/x) dy \quad (5.83)$$

and the conditional variance of Y given $X = x$ is

$$\sigma_{Y|X}^2 = E[Y^2|X = x] - \{E[Y|X = x]\}^2 \quad (5.84)$$

Solved Problems



5.97 The joint density function of X and Y is given by

$$f_{X,Y}(x, y) = \frac{1}{y} e^{-x/y} e^{-y}; 0 < x < \infty; 0 < y < \infty$$

Find $E[X^2/Y = y]$.

Solution We know

$$f(x/y) = \frac{f_{X,Y}(x, y)}{f_Y(y)}$$

$$f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx = \int_0^{\infty} \frac{1}{y} e^{-x/y} e^{-y} dx$$

$$= \frac{1}{y} e^{-y} \left. \frac{e^{-x/y}}{-\frac{1}{y}} \right|_0^{\infty} = e^{-y}$$

$$f(x/y) = \frac{\frac{1}{y} e^{-x/y} e^{-y}}{e^{-y}} = \frac{1}{y} e^{-x/y}$$

$$E[X^2/Y = y] = \int_0^{\infty} x^2 \left(\frac{1}{y} e^{-x/y} \right) dx$$

$$= \frac{1}{y} \int_0^{\infty} x^2 e^{-x/y} dx$$

$$= -x^2 y e^{-x/y} \Big|_0^{\infty} - 2xy^2 e^{-x/y} \Big|_0^{\infty} + \int_0^{\infty} 2y^2 e^{-x/y} dx$$

$$= -2y^3 e^{-x/y} \Big|_0^{\infty} = 2y^3$$

5.98 The joint density function of X and Y is given by

$$f_{X,Y}(x, y) = \frac{1}{y} e^{-y} \quad 0 < x < y, 0 < y < \infty$$

Compute $E[X^3|Y=y]$.

Solution

$$f(x|y) = \frac{f_{X,Y}(x, y)}{f_Y(y)}$$

$$f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx = \frac{1}{y} e^{-y} \int_0^y dx = e^{-y}$$

$$f(x|y) = \frac{\frac{1}{y} e^{-y}}{e^{-y}} = \frac{1}{y}$$

$$\begin{aligned} E[X^2|Y=y] &= \int_{-\infty}^{\infty} x^3 f(x|y) dx \\ &= \int_0^y \frac{x^3}{y} dx = \frac{1}{y} \frac{x^4}{4} \Big|_0^y = \frac{y^3}{4} \end{aligned}$$



LINEAR TRANSFORMATION OF GAUSSIAN RANDOM VARIABLES 5.10

Consider a set of N Gaussian random variables $X_1, X_2, \dots, X_N$ with joint density function

$$f_{X_1, X_2, \dots, X_N}(x_1, x_2, \dots, x_N) = \frac{1}{(2\pi)^{N/2} |C_X|^{1/2}} \exp \left\{ -\frac{(x - \bar{X})^T C_X^{-1} (x - \bar{X})}{2} \right\} \quad (5.85)$$

Let these random variables be linearly transformed to a new set of random variables $Y_1, Y_2, \dots, Y_N$ using the following relation

$$\begin{aligned} Y_1 &= a_{11} X_1 + a_{12} X_2 + \dots + a_{1N} X_N \\ &\vdots \quad \quad \quad \vdots \quad \quad \quad \vdots \quad \quad \quad \vdots \quad \quad \quad \vdots \\ Y_N &= a_{N1} X_1 + a_{N2} X_2 + \dots + a_{NN} X_N \end{aligned} \quad (5.86)$$

where $a_{ij}, i = 1, \dots, N, j = 1, N$ are real numbers.

Equation (5.86) can be represented in matrix form as

$$\begin{bmatrix} Y_1 \\ \vdots \\ Y_N \end{bmatrix} = [T] \begin{bmatrix} X_1 \\ \vdots \\ X_N \end{bmatrix}$$

or $[Y] = [T] [X]$ (5.87)

or $[Y] = \begin{bmatrix} a_{11} & \cdots & a_{1N} \\ a_{21} & \cdots & a_{2N} \\ a_{N1} & \cdots & a_{NN} \end{bmatrix}$ is nonsingular.

and whose inverse is given by

$$[T]^{-1} = \begin{bmatrix} a^{11} & \cdots & a^{1N} \\ a^{21} & \cdots & a^{2N} \\ a^{N1} & \cdots & a^{NN} \end{bmatrix} \quad (5.88)$$

Now, we can express $[X]$ as

$$[X] = [T]^{-1} [Y] \quad (5.89)$$

$$\begin{bmatrix} X_1 \\ \vdots \\ X_N \end{bmatrix} = \begin{bmatrix} a^{11} & \cdots & a^{1N} \\ a^{21} & \cdots & a^{2N} \\ a^{N1} & \cdots & a^{NN} \end{bmatrix} \begin{bmatrix} Y_1 \\ \vdots \\ Y_N \end{bmatrix} \quad (5.90)$$

$$\begin{aligned} X_1 &= a^{11} Y_1 + \cdots + a^{1N} Y_N \\ \vdots & \quad \quad \quad \vdots \\ X_N &= a^{N1} Y_1 + \cdots + a^{NN} Y_N \end{aligned} \quad (5.91)$$

From Eq. (5.91), we can find

$$\frac{\partial X_1}{\partial Y_1} = a^{11}; \frac{\partial X_1}{\partial Y_2} = a^{12} \dots \frac{\partial X_N}{\partial Y_1} = a^{N1}$$

$$\frac{\partial X_N}{\partial Y_N} = a^{NN}$$

In general, we can write

$$\frac{\partial X_i}{\partial Y_i} = a^{ij} \quad (5.92)$$

where a^{ij} represents ij^{th} element of $[T]^{-1}$

We can write

$$X_i - \bar{X}_i = a^{11} (Y_1 - \bar{Y}_1) + \cdots + a^{iN} (Y_N - \bar{Y}_N)$$

The Jacobian of the inverse transformation is

$$J = \begin{vmatrix} \frac{\partial x_1}{\partial y_1} & \frac{\partial x_1}{\partial y_2} & \dots & \frac{\partial x_1}{\partial y_N} \\ \frac{\partial x_2}{\partial y_1} & \frac{\partial x_2}{\partial y_2} & \dots & \frac{\partial x_2}{\partial y_N} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial x_N}{\partial y_1} & \frac{\partial x_N}{\partial y_2} & \dots & \frac{\partial x_N}{\partial y_N} \end{vmatrix} = \begin{vmatrix} a^{11} & a^{12} & \dots & -a^{iN} \\ a^{21} & a^{22} & \dots & a^{2N} \\ \vdots & \vdots & \ddots & \vdots \\ a^{N1} & a^{N2} & \dots & a^{NN} \end{vmatrix} = |[T]^{-1}| \quad (5.93)$$

$$\Rightarrow |J| = |[T]^{-1}| = \frac{1}{|[T]|} \quad (5.94)$$

If $[C_X]$ represents the covariance matrix of random variables $X_1, X_2, \dots, X_N$ then ij^{th} element of the covariance matrix $C_{X_i X_j}$ can be obtained as

$$\begin{aligned} C_{X_i X_j} &= E[(X_i - \bar{X}_i)(X_j - \bar{X}_j)] \\ &= E\left[\{a^{i1}(Y_1 - \bar{Y}) + \dots + a^{iN}(Y_N - \bar{Y}_N)\} \{a^{j1}(Y_1 - \bar{Y}_1) + \dots + a^{jN}(Y_N - \bar{Y}_N)\}\right] \\ &= a^{i1} a^{j1} E[(Y_1 - \bar{Y}_1)(Y_1 - \bar{Y}_1)] + \dots + a^{iN} a^{j1} E[(Y_N - \bar{Y}_N)(Y_1 - \bar{Y}_1)] \\ &\quad + \dots + a^{i1} a^{jN} E[(Y_1 - \bar{Y}_1)(Y_N - \bar{Y}_N)] + \dots + a^{iN} a^{jN} E[(Y_N - \bar{Y}_N)(Y_N - \bar{Y}_N)] \\ &= \sum_{k=1}^N a^{ik} \sum_{m=1}^N a^{jm} E[(Y_k - \bar{Y}_k)(Y_m - \bar{Y}_m)] \\ &= \sum_{k=1}^N a^{ik} \sum_{m=1}^N a^{jm} C_{Y_k Y_m} \end{aligned}$$

$$\Rightarrow [C_X] = [T]^{-1} [C_Y] ([T]^t)^{-1} \quad (5.95)$$

$$\text{from which } [C_X]^{-1} = [T]^t [C_Y]^{-1} [T] \quad (5.96)$$

The determinant of $[C_X]^{-1}$ is

$$|[C_X]^{-1}| = |[C_Y]^{-1}| |[T]|^2$$

The joint density function

$$\begin{aligned} f_{X_1, X_2, \dots, X_N}(x_1, \dots, x_N) \\ = \frac{|[T]| |[C_Y]^{-1}|^{1/2}}{(2\pi)^{N/2}} \exp \left[-\frac{[x - \bar{X}]^t [T]^t [C_Y]^{-1} [T] [x - \bar{X}]}{2} \right] \end{aligned} \quad (5.97)$$

$$f_{Y_1 \dots Y_N}(y_1, \dots, y_N) = \frac{|[C_Y]^{-1}|^{1/2}}{(2\pi)^{N/2}} \exp \left\{ -\frac{(y - \bar{Y})^T \cdot [C_Y]^{-1} [y - \bar{Y}]}{2} \right\} \quad (5.98)$$

Note that

$$[C_Y] = [T][C_X][T]^t$$

Solved Problems



5.99 Zero-mean Gaussian random variables X_1 and X_2 having a covariance matrix

$$[C_X] = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$$

are transformed to new variables

$$Y_1 = 5X_1 + 2X_2$$

$$Y_2 = -X_1 + 3X_2$$

(a) Find covariance matrix of Y_1 and Y_2 .

(b) Find an expression for $f_{Y_1, Y_2}(y_1, y_2)$

Solution

$$[T] = \begin{bmatrix} 5 & 2 \\ -1 & 3 \end{bmatrix}; \quad C_X = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$$

$$C_Y = \begin{bmatrix} 5 & 2 \\ -1 & 3 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 5 & -1 \\ 2 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} 8 & -1 \\ -5 & 7 \end{bmatrix} \begin{bmatrix} 5 & -1 \\ 2 & 3 \end{bmatrix} = \begin{bmatrix} 38 & -11 \\ -11 & 26 \end{bmatrix}$$

$$|[C_Y]| = 38(26) - (-11)^2 = 867$$

$$|[C_Y]|^{-1} = 1.1534 \times 10^{-3}$$

$$[C_Y]^{-1} = \frac{1}{867} \begin{bmatrix} 26 & 11 \\ 11 & 38 \end{bmatrix}$$

$$f_{Y_1, Y_2}(y_1, y_2) = \frac{1.154 \times 10^{-3}}{2\pi} \exp \left\{ -\frac{1}{2} [y_1 \ y_2] [C_Y^{-1}] \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} \right\}$$

5.100 Gaussian random variables X_1 and X_2 for which $\bar{X}_1 = 1$; $\sigma_{X_1}^2 = 4$; $\bar{X}_2 = 2$; $\sigma_{X_2}^2 = 9$ and $C_{X_1 X_2} = -2$ are transformed to new random variables Y_1 and Y_2 according to

$$Y_1 = 2X_1 + X_2$$

$$Y_2 = -X_1 + X_2$$

Find (a) $E[X_1^2]$, (b) $E[X_2^2]$, (c) $\rho_{X_1 X_2}$, (d) $\sigma_{Y_1}^2$ (e), and $\sigma_{Y_2}^2$ (f) $C_{Y_1 Y_2}$.

Solution

$$(a) \quad E[X_1^2] = \sigma_{X_1}^2 + (\bar{X}_1)^2 = 4 + (1)^2 = 5$$

$$(b) \quad E(X_2^2) = \sigma_{X_2}^2 + (\bar{X}_2)^2 = 9 + (2)^2 = 13$$

$$(c) \quad \rho_{X_1 X_2} = \frac{C_{X_1 X_2}}{\sigma_{X_1} \sigma_{X_2}} = \frac{-2}{2(3)} = \frac{-1}{3}$$

$$(d) \quad [C_Y] = [T] [C_X] [T]^t$$

$$T = \begin{bmatrix} 2 & 1 \\ -1 & 1 \end{bmatrix}; \quad T^t = \begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix}$$

$$(e) \quad [C_X] = \begin{bmatrix} \sigma_{X_1}^2 & C_{X_1 X_2} \\ C_{X_1 X_2} & \sigma_{X_2}^2 \end{bmatrix} = \begin{bmatrix} 4 & -2 \\ -2 & 9 \end{bmatrix}$$

$$(f) \quad C_Y = \begin{bmatrix} 2 & 1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 4 & -2 \\ -2 & 9 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 6 & 5 \\ -6 & 11 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 17 & -1 \\ -1 & 17 \end{bmatrix}$$

$$\Rightarrow \quad \sigma_{Y_1}^2 = 17; \sigma_{Y_2}^2 = 17; C_{Y_1 Y_2} = -1$$

**INTRODUCTION TO STATISTICS 5.11**

A population is any entire collection of people, manufacturing items or objects from which we want to make some inference. Since the population is too large to study, to obtain the hidden information about a population, a random sample, which is a subset of the population is generally selected. The sample thus selected should truly represent the population. That is, any inference made from the sample mean and sample variance of a set of data would give information about the average population mean and variance.

For example, to study infant health in a state in 2010, if we consider all children born in that year then we are studying about the entire population. On the other hand, if we consider only the babies born in a particular month then we are considering a sample from the entire population. In this case, we assume that the sample is a representation of the general population. Similarly, the rainfall in a city obtained for all possible years is a population. If we consider the rainfall for five years then it is a sample.

Consider a random variable X with population size N . In the population, let us select a sample, a set of random variables $\{X_1, X_2, X_3, \dots, X_n\}$ with sample size n . The sample that we select truly represent the population if it satisfies three basic requirements: (i) The sample is random. That is, each element of the sample has an equal probability as that of the population. (ii) Each element in the sample has the same distribution as that of the population. (iii) All elements are independent of each other. In other words, the random variables $X_1, X_2, X_3, \dots, X_n$ are independent and identically distributed (i.i.d.)

Let the random variables X under consideration be the number of phone calls arriving at a telephone exchange on Monday between 10 a.m. and 11 a.m. In order to attain a random sample, we choose Mondays at random and record the variables $X_1, X_2, X_3, \dots, X_n$. At the same time, we should not select a Monday on which it appears to be Gandhi Jayanthi.

The capital letters are used to represent random variables in a sample, whereas a particular value taken by each of the random variable is denoted by $x_1, x_2, x_3, \dots, x_n$.

The random samples from a population can be obtained in two ways. (i) with replacement, and (ii) without replacement. In the sampling with replacement, the items that are drawn are put back into the population. Therefore, the random variables $X_1, X_2, X_3, \dots, X_n$ have the same distribution function. If the successive drawings are independent then the random variables $X_1, X_2, X_3, \dots, X_n$ are i.i.d.

In case of sampling without replacement, the probability distribution of random variables $X_1, X_2, \dots, X_n$ are no longer i.i.d. For example, if there is one defective resistor among N resistors then the probability of getting a defective resistor during the first draw is $1/N$. Since the resistor is replaced after every draw, the probability remains same for subsequent draws and hence, the probability distribution of random variables $X_1, X_2, \dots, X_n$ is i.i.d. On the other hand, in sampling without replacement, the probability of drawing a defective resistor during the first draw is $1/N$. If the resistor is not replaced, after the first draw and one that is drawn in the first draw is not defective then the probability of drawing a defective resistor during the second draw is $\frac{1}{N-1}$.

For small values of N , the difference would be significant, hence the distribution of random variables $X_1, \dots, X_n$ is not i.i.d. However, if the population size is large, we can approximate the distribution to be i.i.d.

5.11.1 Sample Mean and Sample Variance

Sample Mean Consider a random variable X that represents the population, with mean μ and variance σ^2 . Let $(X_1, X_2, \dots, X_n)$ be the elements of a random sample X . Then, the sample mean is defined as the average value of the sample.

Mathematically

$$\bar{X} = \frac{1}{N} \sum_{i=1}^n X_i \quad (5.99)$$

Since each X_i in the sample is a random variable, the sample mean is also a random variable. Therefore, it is also characterized by mean and variance.

The expected value of the sample mean is given by

$$E[\bar{X}] = E\left[\frac{1}{n} \sum_{i=1}^n X_i\right] \quad (5.100)$$

$$= \frac{1}{n} \sum_{i=1}^n E[X_i] = \frac{1}{n} [n \mu] = \mu \quad (5.101)$$

That is, the sample mean equals the true mean on the average.

$$\text{Var}(\bar{X}) = \text{Var}\left[\frac{1}{n} \sum_{i=1}^n X_i\right] \quad (5.102)$$

$$= \frac{1}{n^2} \sum_{i=1}^n \text{Var}(X_i) = \frac{1}{n^2} (n \sigma^2) = \frac{\sigma^2}{n} \quad (5.103)$$

That is, the variance of the sample mean is reduced by increasing the sample size n and variance of sample mean goes to zero as $n \rightarrow \infty$.

Applying Chebyshev inequality to the random variable $\bar{X}$, we get

$$P[|\bar{X} - \mu| < \varepsilon] = 1 - \frac{\sigma^2}{\varepsilon^2 n} \quad (5.104)$$

which tends to 1 as $n \rightarrow \infty$ for any finite $\varepsilon > 0$ and finite σ^2 . That is, as $n \rightarrow \infty$, the sample mean $\bar{X}$ converges to μ with probability 1.

Since the sample mean is the sum of independent random variables, for large values of n , the sample mean is approximately Gaussian distributed.

Sample Variance The sample variance of a random sample $(X_1, X_2, \dots, X_n)$ is defined as

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2 \quad (5.105)$$

Sample variance is also a random variable. The expectation of a sample variance is given by

$$\begin{aligned} E[S^2] &= \frac{1}{n-1} E \left[\sum_{i=1}^n (X_i - \bar{X})^2 \right] \\ &= \frac{1}{n-1} \sum_{i=1}^n E[X_i - \bar{X}]^2 \\ &= \frac{1}{n-1} \sum_{i=1}^n E[(X_i - \mu) - (\bar{X} - \mu)]^2 \\ &= \frac{1}{n-1} \left[\sum_{i=1}^n E[(X_i - \mu)^2 + E[(\bar{X} - \mu)^2]] - 2 E[(X_i - \mu)(\bar{X} - \mu)] \right] \\ &= \frac{1}{n-1} \left[\sum_{i=1}^n \left(\sigma^2 + \frac{\sigma^2}{n} \right) - 2 E[(\bar{X} - \mu) \left(\sum_{i=1}^n (X_i - \mu) \right)] \right] \\ &= \frac{1}{n-1} \left[\sum_{i=1}^n \left(\sigma^2 + \frac{\sigma^2}{n} \right) - 2 E[(\bar{X} - \mu) [(X_1 - \mu) + \dots + (X_n - \mu)]] \right] \\ &= \frac{1}{n-1} \left[\sum_{i=1}^n \left(\sigma^2 + \frac{\sigma^2}{n} \right) - 2 E[(\bar{X} - \mu) (n\bar{X} - n\mu)] \right] = \frac{1}{n-1} \left[\sum_{i=1}^n \left(\sigma^2 + \frac{\sigma^2}{n} \right) - 2n E[(\bar{X} - \mu)^2] \right] \\ &= \frac{1}{n-1} \left[\sum_{i=1}^n \left\{ \sigma^2 + \frac{\sigma^2}{n} \right\} - 2\sigma^2 \right] \quad \because E[(\bar{X} - \mu)^2] = \frac{\sigma^2}{n} \\ &= \frac{1}{n-1} [n\sigma^2 - \sigma^2] = \frac{1}{n-1} (n-1)\sigma^2 \\ &= \sigma^2 \end{aligned}$$

That is, the sample variance is equal to the true variance of the population on the average.

5.11.2 Weak Law of Large Numbers

If the random variable X_i are independent and identically distributed, each having finite mean $E[X] = \mu$, then

$$\lim_{n \rightarrow \infty} P[|\bar{X} - \mu| < \varepsilon] = 1 \quad \text{for all } \varepsilon > 0 \quad (5.107)$$

The above expression is known as weak law of large numbers.

5.11.3 Strong Law of Large Numbers

If the random variable X_i , are identically distributed, it states that as $n \rightarrow \infty$

$$P\left\{\lim_{n \rightarrow \infty} \bar{X} = \mu\right\} = 1 \quad (5.108)$$

Solved Problems



5.101 Let X be a random variable with pdf $f_X(x)$ and CDF $F_X(x)$. Let $(X_1, X_2, \dots, X_n)$ be a random sample X . Let us define

$$W = \max(X_1, X_2, \dots, X_n)$$

$$\text{and} \quad Z = \min(X_1, X_2, \dots, X_n).$$

Then find the pdf of W and Z .

Solution

Given $W = \max(X_1, X_2, \dots, X_n); F_X(x) = P[X \leq x]$

The CDF of W is

$$F_W(w) = P[W \leq w]$$

The event $(W \leq w)$ is equivalent to the event $(X_i \leq w \text{ for all } i)$

Since X_i are independent,

$$\begin{aligned} F_W(w) &= P(W \leq w) \\ &= P[X_1 \leq w, X_2 \leq w, \dots, X_n \leq w] \\ &= P[X_1 \leq w] P[X_2 \leq w] \dots P[X_n \leq w] \\ &= F_X(x) F_X(x) \dots F_X(x) \\ &= [F_X(x)]^n \end{aligned} \quad (5.109)$$

$$\begin{aligned} f_W(w) &= n[F_X(x)]^{n-1} \frac{d}{dx} F_X(x) \\ &= n[F_X(x)]^{n-1} f_X(x) \end{aligned}$$

$$Z = \min[X_1, \dots, X_n]$$

The CDF

$$\begin{aligned} F_Z(z) &= P(Z \leq z) = 1 - P(Z > z) \\ &= 1 - \{P(X_1 > z, X_2 > z, \dots, X_n > z)\} \end{aligned}$$

$$\begin{aligned}
&= 1 - P(X_1 > z) P(X_2 > z) \dots P(X_n > z) \\
&\quad (\text{since } X_1, X_2, \dots, X_n \text{ are i.i.d}) \\
&= 1 - \{(1 - F_X(x)) (1 - F_X(x)) \dots (1 - F_X(x))\} \\
&= 1 - \{1 - F_X(x)\}^n \\
f_Z(z) &= \frac{-d}{dZ} [(1 - F_X(x))^n] \\
f_Z(z) &= n[1 - F_X(x)]^{n-1} f_X(x)
\end{aligned} \tag{5.110}$$

5.102 If $X_1, \dots, X_n$ are independent random variable, and if X has exponential distribution with parameter α then prove that $Z = \min [X_1, \dots, X_n]$ has exponential distribution with parameter $n\alpha$.

Solution Given: $Z = \min[X_1, \dots, X_n]$

Let $F_Z(z)$ is CDF of Z . Then

$$F_Z(z) = P(Z \leq z) = 1 - P[Z > z]$$

Using Eq. (5.110) we can write

$$F_Z(z) = 1 - [1 - F_X(x)]^n$$

We know for exponential distribution

$$F_X(x) = 1 - e^{-\alpha x}$$

$$\begin{aligned}
\text{Hence, } F_Z(z) &= 1 - [1 - (1 - e^{-\alpha x})]^n \\
&= 1 - e^{-n\alpha x}
\end{aligned}$$

$$\Rightarrow f_Z(z) = n\alpha e^{-n\alpha x}$$



Practice Problem

5.32 If X_1, X_2, X_3, X_4 are i.i.d exponential random variables with the parameters 1, compute $P\{\min(X_1, \dots, X_4) < a\}$ and $P\{\max(X_1, \dots, X_4) \leq a\}$.

[Ans: $4e^{-4x}$

and $4[1 - e^{-x}]^3 e^{-x}$]



COMPLEX RANDOM VARIABLE 5.12

A complex random variable Z is defined as

$$Z = X + j Y \tag{5.111}$$

where X and Y are real random variables and $j = \sqrt{-1}$. If the joint density function of random variables X and Y is $f_{X,Y}(x, y)$ then the expected value of some function of Z is given by

$$E[g(z)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(z) f_{X,Y}(x, y) dx dy \tag{5.112}$$

The mean value of Z is

$$\begin{aligned} E[Z] &= E[X + j Y] = E[X] + j E[Y] \\ &= \mu_X + j \mu_Y \end{aligned} \quad (5.113)$$

The variance of Z is

$$\text{Var}(Z) = E[|Z - E[Z]|^2] \quad (5.114)$$

Consider two complex random variables

$$Z_1 = X_1 + j Y_1$$

and

$$Z_2 = X_2 + j Y_2$$

Let the joint density function be $f_{X_1, Y_1, X_2, Y_2}(x_1, y_1, x_2, y_2)$

If Z_1 and Z_2 are independent random variables,

$$f_{X_1, Y_1, X_2, Y_2}(x_1, y_1, x_2, y_2) = f_{X_1, Y_1}(x_1, y_1) f_{X_2, Y_2}(x_2, y_2)$$

The correlation between Z_1 and Z_2 is given by

$$R_{Z_1 Z_2} = E[Z_1^* Z_2] \quad (5.115)$$

and covariance is

$$C_{Z_1 Z_2} = E[\{Z_1 - E[Z_1]\}^* \{Z_2 - E[Z_2]\}] \quad (5.116)$$

If Z_1 and Z_2 are uncorrected random variables then

$$R_{Z_1 Z_2} = E[Z_1^*] E[Z_2] \quad (5.117)$$

Solved Problems



5.103 A complex random variable Z is defined by $Z = \cos X + j \sin Y$ where X and Y are independent real random variables uniformly distributed from $-\pi$ to π .

- Find the mean value of Z .
- Find the variance of Z .

Solution

- We have $Z = \cos X + j \sin Y$

The mean value of Z is

$$E[Z] = E[\cos X + j \sin Y] = E[\cos X] + j E[\sin Y]$$

$$E[\cos X] = \int_{-\infty}^{\infty} \cos x f_X(x) dx$$

Given $f_X(x)$ is uniformly distributed from $-\pi$ to π .

Therefore,

$$\begin{aligned} f_X(x) &= \frac{1}{2\pi} && \text{for } -\pi \text{ to } \pi \\ &= 0 && \text{otherwise} \end{aligned}$$

$$E[\cos X] = \int_{-\pi}^{\pi} \frac{1}{2\pi} \cos x dx = \frac{1}{2\pi} \sin x \Big|_{-\pi}^{\pi} = 0$$

Similarly,

$$\begin{aligned}
 E[\sin Y] &= \int_{-\pi}^{\pi} \frac{1}{2\pi} \sin y \, dy = \frac{-1}{2\pi} \cos y \Big|_{-\pi}^{\pi} = 0 \\
 \Rightarrow E[Z] &= 0 \\
 \text{(b) } \text{Var}(Z) &= E[Z^2] - \{E[Z]\}^2 \\
 &= E[Z^2] = E[\cos^2 X + \sin^2 Y] \\
 &= E[\cos^2 X] + E[\sin^2 Y] \\
 E[\cos^2 X] &= E\left[\frac{1 + \cos 2X}{2}\right] = \frac{1}{2} + E\left[\frac{\cos 2X}{2}\right] \\
 E[\cos 2X] &= \frac{1}{2\pi} \int_{-\pi}^{\pi} \cos 2x \, dx = 0 \\
 \Rightarrow E[\cos^2 X] &= \frac{1}{2} \\
 E[\sin^2 Y] &= \frac{1}{2} E[1 - \cos 2Y] \\
 &= \frac{1}{2} - \frac{1}{2} E[\cos 2Y] \\
 &= \frac{1}{2} \\
 \text{Var}[Z] &= \frac{1}{2} + \frac{1}{2} = 1
 \end{aligned}$$



REGRESSION 5.13

The term *regression* is used to analyze the statistical relation between two or more variables. Figure 5.30 shows the relation between two random variables X and Y .

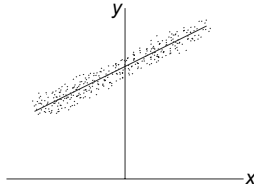


Fig. 5.30

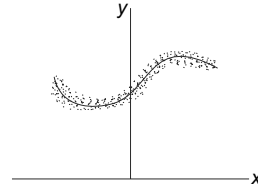


Fig. 5.31

In Fig. 5.30 the sample points are clustered around a straight line. So a straight line gives the best estimate to the value of one variable for any specific value of the other variable. When we fit the data with a straight line then it is known as *line of regression*. In Fig. 5.31, the sample points are clustered around a curve. Therefore, a curve gives the best estimate which is called *curve of regression*.

5.13.1 Line of Regression

In a line of regression, the relation between two variables fitted with a linear function given by

$$Y = a + bX \quad (5.118)$$

Equation (5.118) is known as regression line. Here, Y is a dependent variable and X is an independent random variable. The parameter a is the intercept of the true regression line and b is the slope of the true regression line. The values of a and b are unknown and they must be obtained from the data. Since the data points are scattered around the straight line, Eq. (5.118) will not satisfy all data points. Therefore, we assume a linear regression model with error quantity given by

$$Y = a + bX + \varepsilon \quad (5.119)$$

For a single observation, Eq. (5.119) can be written as

$$y_i = a + bx_i + \varepsilon_i \quad (5.120)$$

where ε_i is the difference between the observed value of Y and the value on the true regression line. The ε_i are independent of each other and follow normal distribution with zero mean and variance σ^2 . Now the goal is to determine the regression model that minimizes the sum of the square error. This line is called the least squares line given by

$$\hat{y}_i = \hat{a} + \hat{b}x_i \quad (5.121)$$

The line of regression can be obtained in two ways.

In the first case, the sum of square of error between the observed $\hat{y}$ and the value on the true-regression line is minimized. That is, we minimize sum of square of errors parallel to y-axis which is known as regression equation of Y and X . Similarly, if we minimize the sum of squares of errors parallel to X -axis then we obtain regression equation of X on Y . For this case, the line equation is given by

$$\hat{x}_i = \hat{a}_i + \hat{b}_1 y_i \quad (5.122)$$

5.13.2 Regression Line of Y on X

For all sample points, Eq. (5.121) can be written as

$$\hat{Y} = \hat{a} + \hat{b}X \quad (5.123)$$

The parameters $\hat{a}$ and $\hat{b}$ are estimated using the method of least square, where the sum of squared error $\sum_{i=1}^n \varepsilon_i^2$ is minimized.

The squared sum of the error is given by

$$\varepsilon = \sum_{i=1}^N (y_i - \hat{y}_i)^2 = \sum_{i=1}^N (y_i - \hat{a} - \hat{b}x_i)^2 \quad (5.124)$$

To find the value of $\hat{a}$ and $\hat{b}$, differentiate the above equation with respect to $\hat{a}$ and $\hat{b}$.

$$\begin{aligned} \frac{\partial E}{\partial \hat{a}} &= 2 \sum_{i=1}^N (y_i - \hat{a} - \hat{b}x_i) (-1) = 0 \\ \Rightarrow \sum_{i=1}^N y_i &= \hat{a}N + \hat{b} \sum_{i=1}^N x_i \end{aligned} \quad (5.125)$$

$$\frac{\partial E}{\partial \hat{b}} = 2 \sum_{i=1}^N (y_i - \hat{a} - \hat{b}x_i)(-x_i) = 0$$

$$\sum_{i=1}^N x_i y_i - \sum_{i=1}^N \hat{a} x_i - \sum_{i=1}^N \hat{b} x_i^2 = 0 \quad (5.126)$$

$$\begin{aligned} \Rightarrow \sum_{i=1}^N x_i y_i &= \sum_{i=1}^N \hat{a} x_i + \sum_{i=1}^N \hat{b} x_i^2 \\ &= \hat{a} \sum_{i=1}^N x_i + \hat{b} \sum_{i=1}^N x_i^2 \end{aligned} \quad (5.127)$$

$$\begin{aligned} \text{From Eq. (5.125), } \hat{a} &= \frac{1}{N} \sum_{i=1}^N y_i - \hat{b} \frac{1}{N} \sum_{i=1}^N x_i \\ &= \mu_Y - \hat{b} \mu_X \end{aligned} \quad (5.128)$$

$$\begin{aligned} \sum_{i=1}^N x_i y_i &= (\mu_Y - \hat{b} \mu_X) \sum_{i=1}^N x_i + \hat{b} \sum_{i=1}^N x_i^2 \\ &= \mu_Y \sum_{i=1}^N x_i - \hat{b} \mu_X \sum_{i=1}^N x_i + \hat{b} \sum_{i=1}^N x_i^2 \end{aligned}$$

$$\begin{aligned} \sum_{i=1}^N x_i y_i - N \mu_X \mu_Y &= \hat{b} \left[\sum_{i=1}^N x_i^2 - N \mu_X^2 \right] \\ \hat{b} &= \frac{\sum_{i=1}^N x_i y_i - N \mu_X \mu_Y}{\sum_{i=1}^N x_i^2 - N \mu_X^2} \end{aligned} \quad (5.129)$$

$$\begin{aligned} &= \frac{\sum_{i=1}^N (x_i - \mu_X)(y_i - \mu_Y)}{\sum_{i=1}^N (x_i - \mu_X)^2} = \frac{\text{Cov}(X, Y)}{\sigma_X^2} \end{aligned} \quad (5.130)$$

$$\hat{b} = \frac{\sum XY - N \frac{\sum X \sum Y}{N^2}}{\sum X^2 - N \frac{1}{N^2} (\sum X)^2} = \frac{N \sum XY - \sum X \sum Y}{N \sum X^2 - (\sum X)^2} \quad (5.131)$$

$$\text{We have } \rho = \frac{\text{Cov}(x, y)}{\sigma_X \sigma_Y}$$

$$\begin{aligned} \Rightarrow \hat{b} &= \rho \frac{\sigma_Y}{\sigma_X} \\ \hat{y}_i &= (\bar{y} - \hat{b} \bar{x}) + \hat{b} x_i \end{aligned}$$

$$\begin{aligned}
 (\hat{y}_i - \bar{y}) &= \hat{b}(x_i - \bar{x}) \\
 &= \rho \frac{\sigma_Y}{\sigma_X} (x_i - \bar{x})
 \end{aligned} \tag{5.132}$$

Since we are predicting the value of y for any given x . Equation (5.132) is known as *regression line of y on x* .

5.13.3 Regression Line of X on Y

In the above section, we discussed the prediction of Y given a value of X . Some times it may be required to predict the value of X given Y . In order to do so, we define a new regression line equation given by

$$\hat{X} = \hat{a}_1 + \hat{b}_1 Y \tag{5.133}$$

For a sample point, the equation can be written as

$$\hat{x}_i = \hat{a}_1 + \hat{b}_1 y_i \tag{5.134}$$

where $\hat{x}_i$ is predicted value of x_i . The parameters $\hat{a}_1$ and $\hat{b}_1$ can be estimated by minimizing the sum of squared errors.

The error is given by $\varepsilon_i = x_i - \hat{x}_i$

The sum of square errors

$$\varepsilon = \sum_{i=1}^N (x_i - \hat{x}_i)^2 \tag{5.135}$$

$$\sum_{i=1}^N (x_i - \hat{a}_1 - \hat{b}_1 y_i)^2$$

$$\frac{\partial \varepsilon}{\partial \hat{a}_1} = 2 \sum_{i=1}^N (x_i - \hat{a}_1 - \hat{b}_1 y_i) (-1) = 0$$

$$\Rightarrow \sum_{i=1}^N x_i = \hat{a}_1 (N) + \hat{b}_1 \sum_{i=1}^N y_i \tag{5.136}$$

$$\Rightarrow \hat{a}_1 = \bar{x} - \hat{b}_1 \bar{y} \tag{5.137}$$

$$\frac{\partial \varepsilon}{\partial \hat{b}_1} = 2 \sum_{i=1}^N (x_i - \hat{a}_1 - \hat{b}_1 y_i) (-y_i) = 0$$

$$= \sum_{i=1}^N x_i y_i - \sum_{i=1}^N \hat{a}_1 y_i - \sum_{i=1}^N \hat{b}_1 y_i^2 = 0$$

$$= \sum_{i=1}^N x_i y_i - \hat{a}_1 N \bar{y} - \hat{b}_1 \sum_{i=1}^N y_i^2 = 0 \tag{5.138}$$

$$\begin{aligned}
&= \sum_{i=1}^N x_i y_i - N(\bar{x} - \hat{b}_1 \bar{y}) \bar{y} - \hat{b}_1 \sum_{i=1}^N y_i^2 = 0 \\
\hat{b}_1 &= \frac{\sum_{i=1}^N x_i y_i - N \bar{x} \bar{y}}{\sum_{i=1}^N y_i^2 - N \bar{y}^2} = \sum_{i=1}^N \frac{(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^N (y_i - \bar{y})^2}
\end{aligned} \tag{5.139}$$

$$= \frac{\text{Cov}(X, Y)}{\sigma_Y^2} \tag{5.140}$$

$$= \frac{\rho \sigma_X}{\sigma_Y}$$

$$(\hat{x}_i - \bar{x}) = \frac{\rho \sigma_X}{\sigma_Y} (y_i - \bar{y}) \tag{5.141}$$

$$\hat{b}_1 = \frac{N \Sigma XY - \Sigma X \Sigma Y}{N \Sigma Y^2 - (\Sigma Y)^2} \tag{5.142}$$

5.13.4 General Form of Regression (Equations)

In general, we can write the regression equation as below.

The regression line of y on x is

$$(y - \bar{y}) = \frac{\rho \sigma_Y}{\sigma_X} (x - \bar{x}) \tag{5.143}$$

$$= b_{YX}(x - \bar{x})$$

$$\Rightarrow b_{yx} = \rho \frac{\sigma_Y}{\sigma_X} \tag{5.144}$$

The regression line of X on Y is

$$(x - \bar{x}) = \frac{\rho \sigma_X}{\sigma_Y} (y - \bar{y}) \tag{5.145}$$

$$= b_{xy} (y - \bar{y}) \tag{5.146}$$

$$b_{xy} = \frac{\rho \sigma_X}{\sigma_Y}$$

where b_{yx} is the coefficient of x where b_{xy} is the coefficient of y

$$\begin{aligned}
b_{yx} b_{xy} &= \rho^2 \\
\rho &= \pm \sqrt{b_{yx} b_{xy}}
\end{aligned} \tag{5.147}$$

Solved Problems

5.104 The following distribution gives the likely price of a commodity in two cities x and y .

	x	y
Mean	65	67
SD	2.5	3.5

Coefficient of correlation between x and y is 0.8. Find

- Regression line of y on x
- The likely price of y when $x = 70$.

Solution

Given: $\bar{x} = 65$; $\bar{y} = 67$; $\sigma_x = 2.5$ and $\sigma_y = 3.5$

Also, $\rho = 0.8$

Regression line of y on x is given by

$$(y - \bar{y}) = \rho \frac{\sigma_y}{\sigma_x} (x - \bar{x})$$

$$(y - \bar{y}) = 0.8 \left(\frac{3.5}{2.5} \right) (x - \bar{x})$$

$$(y - 67) = 1.12 (x - 65)$$

$$y = 1.12x - 5.8$$

$$\text{When } x = 70 \quad y = 1.12(70) - 5.8 = 72.6$$

5.105 Calculate the correlation coefficient and the lines of regression from the following data.

x	62	64	65	69	70	71	72	74
y	126	125	139	145	165	152	180	208

Solution

x	y	xy	x^2	y^2
62	126	7812	3844	15876
64	125	8000	4096	15625
65	139	9035	4225	19321
69	145	10005	4761	21025
70	165	11550	4900	27225
71	152	10792	5041	23104
72	180	12960	5184	32400
74	208	15392	5476	43264
$\Sigma x = 547$	$\Sigma y = 1240$	$\Sigma xy = 85546$	$\Sigma x^2 = 37527$	$\Sigma y^2 = 197840$

$$\bar{x} = 68.375; \bar{y} = 155; \frac{1}{N} \Sigma xy = 10893.25$$

$$\sigma_x^2 = \frac{1}{N} \Sigma x^2 - \bar{x}^2 = \frac{1}{8} (37527) - (68.375)^2 = 15.73$$

$$\sigma_y^2 = \frac{1}{N} \Sigma y - \bar{y}^2 = \frac{1}{8} (197840) - (155)^2 = 705$$

$$\rho_{xy} = \frac{\frac{1}{N} \Sigma xy - \bar{x} \bar{y}}{\sigma_x \sigma_y} = \frac{\frac{85546}{8} - (68.375)(155)}{(3.96)(26.55)} = 0.905$$

The regression line of x on y is

$$x - \bar{x} = \rho \frac{\sigma_x}{\sigma_y} (y - \bar{y})$$

$$(x - 68.375) = 0.905 \left(\frac{3.96}{26.55} \right) (y - 155) = 0.135 (y - 155)$$

$$x = 0.135 y + 47.45$$

The regression line of y on x is

$$(y - \bar{y}) = \rho \frac{\sigma_y}{\sigma_x} (x - \bar{x})$$

$$(y - 155) = 0.905 \left(\frac{26.55}{3.96} \right) (x - 68.375) = 6.076 (x - 68.375)$$

$$y = 6.076x - 260.44$$

5.106 The following table gives the data on rainfall and discharge in a certain river. Obtain the line of regression of y and x .

Rainfall (inches) X	1.53	1.78	2.60	2.95	3.42
Discharge 100 c.c. Y	33.5	36.3	40.0	45.8	53.5

Solution

Rainfall X (inches)	Discharge Y (100 cc)	xy	x^2	y^2
1.53	33.5	51.255	2.34	1122.25
1.78	36.3	64.14	3.17	1317.69
2.60	40.0	104	6.76	1600
2.95	45.8	135.11	8.7025	2097.64
3.42	53.5	182.97	11.696	2862.25
$\Sigma x = 12.28$	$\Sigma y = 209.1$	$\Sigma xy = 537.949$	$\Sigma x^2 = 32.668$	$\Sigma y^2 = 8999.83$

$$\bar{x} = \frac{1}{N} \sum x = 2.456; \quad \bar{y} = \frac{1}{N} \sum y = \frac{209}{5} = 41.8$$

$$\sigma_x^2 = \frac{1}{N} \sum x^2 - \bar{x}^2 = \frac{32.668}{5} - (2.456)^2 = 0.502$$

$$\sigma_y^2 = \frac{1}{N} \sum y^2 - \bar{y}^2 = \frac{8999.83}{5} - (41.8)^2 = 52.726$$

$$\sigma_x = 0.7085; \quad \sigma_y = 7.261$$

The regression line of y on x is

$$y - \bar{y} = \rho \frac{\sigma_y}{\sigma_x} (x - \bar{x})$$

$$\rho = \frac{\frac{1}{N} \sum xy - \bar{x} \bar{y}}{\sigma_x \sigma_y} = \frac{\frac{1}{5} (537.949) - (2.456)(41.8)}{(0.7085)(7.261)} = \frac{4.929}{5.144} = 0.958$$

$$(y - 41.8) = \frac{0.958(7.261)}{0.7085} (x - 2.456)$$

$$y - 41.8 = 9.81(x - 2.456)$$

$$y = 9.81x + 17.706$$

5.107 Can $Y = 5 + 3.8X$ and $X = 3 - 0.5Y$ be the estimated regression equations of Y on X and X on Y respectively? Explain your answer.

Solution Given: $X = 3 - 0.5Y$ and $Y = 5 + 2.8X$ from which we can get $b_{xy} = -9.5$ and $b_{yx} = 2.8$
We know

$$\rho^2 = b_{xy} b_{yx} = (-0.5)(2.8) = -1.4$$

$$\rho = \sqrt{-1.4} \text{ which is imaginary}$$

Therefore, the lines cannot be the estimated regression equations.

5.108 Heights of fathers and son are given in centimeters

Heights of fathers (x)	150	152	155	157	160	161	164	166
Heights of sons (y)	154	156	158	159	160	162	161	164

Find the two lines of regression and calculate the expected average height of the son when the height of the father is 154 cm.

Solution Let us assume 160 is mean of both x and y .

x	y	$X = x - 160$	$Y = y - 160$	X^2	Y^2	XY
150	154	-10	-6	100	36	60
152	156	-8	-4	64	16	32
155	158	-5	-2	25	4	10

(Contd.)

(Contd.)

157	159	-3	-1	19	1	3
160	160	0	0	0	0	0
161	162	1	2	1	4	2
164	161	4	1	16	1	4
166	164	6	4	36	16	24
$\Sigma x = 1265$	$\Sigma y = 1274$	$\Sigma x = -15$	$\Sigma Y = -6$	$\Sigma X^2 = 251$	$\Sigma Y^2 = 78$	$\Sigma XY = 135$

$$\bar{X} = -\frac{15}{8} = -1.875$$

$$\bar{Y} = -\frac{6}{8} = -0.75$$

$$\bar{x} = \bar{X} + 160 = 158.125$$

$$\bar{y} = \bar{Y} + 160 = 160 - 0.75 = 159.25$$

$$b_{yx} = \frac{N \Sigma XY - \Sigma X \Sigma Y}{N \Sigma X^2 - (\Sigma X)^2}$$

$$= \frac{8(135) - (-15)(-6)}{8(251) - (-15)^2} = \frac{990}{1783} = 0.555$$

$$b_{xy} = \frac{N \Sigma XY - \Sigma X \Sigma Y}{N \Sigma Y^2 - (\Sigma Y)^2}$$

$$= \frac{8(135) - (-15)(-6)}{8(78) - (-6)^2} = \frac{990}{588} = 1.683$$

The regression line of y on x is

$$(y - \bar{y}) = b_{yx}(x - \bar{x})$$

$$y - 159.25 = 0.555(x - 158.125)$$

$$y = 0.555x + 71.49$$

when

$$x = 154$$

$$y = 156.96$$

The regression line of x on y is

$$(x - \bar{x}) = b_{xy}(y - \bar{y})$$

$$(x - 158.125) = 1.683(y - 159.25)$$

$$x = 1.683y - 109.76$$

5.109 Show that the angle between the lines of regression lines is $\tan^{-1} \left(\frac{1 - \rho^2}{\rho} \frac{\sigma_X \sigma_Y}{\sigma_X^2 + \sigma_Y^2} \right)$

Solution The regression line of y on x is

$$y - \bar{y} = \rho \frac{\sigma_y}{\sigma_x} (x - \bar{x}) \quad (5.148)$$

and the regression line of x on y is

$$x - \bar{x} = \rho \frac{\sigma_x}{\sigma_y} (y - \bar{y}) \quad (5.149)$$

The slope of two lines are $\rho \frac{\sigma_y}{\sigma_x}$ and $\frac{\sigma_y}{\rho \sigma_x}$

Let θ_1 is angle made by regression line shown in Eq. (5.148)

$$\text{Then } \tan \theta_1 = \frac{\rho \sigma_y}{\sigma_x}$$

Let θ_2 is angle made by regression line shown in Eq. (5.149)

$$\text{Then } \tan \theta_2 = \frac{\sigma_y}{\rho \sigma_x}$$

The angle between two regression lines is $\theta_1 \sim \theta_2$

We can write $\theta = \theta_1 \sim \theta_2$

$$\begin{aligned} \tan \theta &= \tan (\theta_1 \sim \theta_2) = \frac{\tan \theta_1 \sim \tan \theta_2}{1 + \tan \theta_1 \tan \theta_2} \\ &= \frac{\frac{\rho \sigma_y}{\sigma_x} \sim \frac{\sigma_y}{\rho \sigma_x}}{1 + \left(\frac{\rho \sigma_y}{\sigma_x} \right) \left(\frac{\sigma_y}{\rho \sigma_x} \right)} \\ &= \frac{\rho^2 \sigma_y \sim \sigma_y / \rho \sigma_x}{\frac{(\sigma_x^2 + \sigma_y^2)}{\sigma_x^2}} = \frac{\sigma_y (1 - \rho^2) \sigma_x}{\rho (\sigma_x^2 + \sigma_y^2)} \quad (\text{Since } \rho^2 \leq 1) \\ &= \frac{(1 - \rho^2) (\sigma_x \sigma_y)}{\rho (\sigma_x^2 + \sigma_y^2)} \end{aligned} \quad (5.150)$$

5.110 Let x_1, x_2 and x_3 be uncorrelated random variables each having the same standard deviation. Find the correlation coefficient between $(x_1 + x_2)$ and $(x_2 + x_3)$.

Solution Since x_1, x_2 and x_3 are uncorrelated,

$$E[x_1 x_2] = E[x_1] E[x_2]$$

$$E[x_1 x_3] = E[x_1] E[x_3]$$

$$E[x_2 x_3] = E[x_2] E[x_3]$$

Also, $\sigma_{x_1}^2 = \sigma_{x_2}^2 = \sigma_{x_3}^2 = \sigma_x^2$

Let $u = x_1 + x_2$ and $v = x_2 + x_3$

$$\rho_{uv} = \frac{\text{Cov}(u, v)}{\sigma_u \sigma_v}$$

$$\begin{aligned} \text{Cov}(u, v) &= E[uv] - E[u] E[v] \\ &= E[(x_1 + x_2)(x_2 + x_3)] - E[x_1 + x_2] E[x_2 + x_3] \\ &= E[x_1 x_2 + x_2^2 + x_1 x_3 + x_2 x_3] - \{E[x_1] + E[x_2]\} \{E[x_2] + E[x_3]\} \\ &= E[x_1] E[x_2] + E[x_2^2] + E[x_1] E[x_3] + E[x_2] E[x_3] \\ &\quad - E[x_1] E[x_2] - \{E[x_2]\}^2 - E[x_1] E[x_3] - E[x_2] E[x_3] \\ &= E[x_2^2] - \{E[x_2]\}^2 \\ &= \sigma_{x_2}^2 = \sigma_x^2 \\ \sigma_u^2 &= E[(x_1 + x_2)^2] - \{E[x_1 + x_2]\}^2 \\ &= E[x_1^2] + E[x_2^2] + 2E[x_1] E[x_2] - \{E[x_1]\}^2 - \{E[x_2]\}^2 \\ &\quad - 2E[x_1] E[x_2] \\ &= \sigma_{x_1}^2 + \sigma_{x_2}^2 = 2\sigma_x^2 \end{aligned}$$

Similarly, $\sigma_v^2 = 2\sigma_x^2$

$$\rho_{uv} = \frac{\sigma_x^2}{\sqrt{2\sigma_x^2} \sqrt{2\sigma_x^2}} = \frac{1}{2}$$

5.111 For ten observations in price X and supply Y , the following data were obtained.

$\Sigma x = 130$; $\Sigma y = 220$; $\Sigma x^2 = 2288$; $\Sigma y^2 = 5536$ and $\Sigma xy = 3467$. Obtain the line of regression of Y on X and estimate the supply when the price is 16 units.

Solution

$$\bar{x} = \frac{\Sigma x}{n} = \frac{130}{10} = 13; \quad \bar{y} = \frac{\Sigma y}{n} = \frac{220}{10} = 22$$

$$\sigma_x^2 = \frac{1}{N} \Sigma x^2 - (\bar{x})^2 = \frac{2288}{10} - (13)^2 = 59.8$$

$$\sigma_x = 7.73$$

$$\sigma_y^2 = \frac{1}{N} \Sigma y^2 - (\bar{y})^2 = \frac{5536}{10} - (22)^2 = 66.6 \Rightarrow \sigma_y = 8.16$$

$$\rho = \frac{\frac{1}{N} \Sigma xy - \bar{x} \bar{y}}{\sigma_x \sigma_y} = \frac{\frac{3467}{10} - (13)(22)}{(7.73)(8.16)} = 0.964$$

Regression line of y on x -axis

$$\begin{aligned}(y - \bar{y}) &= \rho \frac{\sigma_y}{\sigma_x} (x - \bar{x}) \\(y - 22) &= 0.964 \left(\frac{8.16}{7.73} \right) (x - 13) \\y - 22 &= 1.018 (x - 13) \\y &= 1.018x + 8.766\end{aligned}$$

For $x = 16$; $y = 25.054$

5.112 If the lines of regression of x on y and y on x are $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ respectively, prove that $a_2b_1 < a_1b_2$.

Solution The regression of x on y is

$$\begin{aligned}a_1x + b_1y + c_1 &= 0 \\ \Rightarrow a_1x &= -b_1y - c_1 \\ x &= -\frac{b_1}{a_1}y - \frac{c_1}{a_1}\end{aligned}\tag{5.151}$$

The regression of y on x is

$$\begin{aligned}a_2x + b_2y + c_2 &= 0 \\ b_2y &= -a_2x - c_2 \\ y &= -\frac{a_2}{b_2}x - \frac{c_2}{b_2}\end{aligned}\tag{5.152}$$

From Eq. (5.151), $b_{xy} = -\frac{b_1}{a_1}$

And from Eq. (5.152), $b_{yx} = -\frac{a_2}{b_1}$

$$\begin{aligned}\text{We have } \rho^2 &= b_{xy} b_{yx} = \left(-\frac{b_1}{a_1} \right) \left(-\frac{a_2}{b_2} \right) \\ &= \frac{a_2 b_1}{a_1 b_2}\end{aligned}$$

We have $\rho^2 \leq 1$

$$\Rightarrow \frac{a_2 b_1}{a_1 b_2} \leq 1 \Rightarrow a_2 b_1 \leq a_1 b_2$$

5.113 If $y = 2x - 3$ and $y = 5x + 7$ are the two regression lines, find the mean values of x and y . Find the correlation coefficient between x and y . Find an estimate of x when $y = 1$.

Solution

Let $\bar{x}$ and $\bar{y}$ are the mean values of x and y respectively. Since the mean values satisfy the regression lines, we can write

$$\bar{y} = 2\bar{x} - 3 \Rightarrow 2\bar{x} - \bar{y} = 3$$

and $\bar{y} = 5\bar{x} + 7 \Rightarrow 5\bar{x} - \bar{y} = -7$

Solving for $\bar{x}$ and $\bar{y}$, we get

$$\bar{x} = -\frac{10}{3}$$

$$5\left(-\frac{10}{3}\right) - \bar{y} = -7$$

$$\bar{y} = \frac{-50}{3} + 7 = -\frac{29}{3}$$

Let the equation of line of regression of x and y is

$$y = 2x - 3$$

$$\Rightarrow x = \frac{1}{2}y + 1.5$$

From which $b_{xy} = \frac{1}{2}$

Similarly the equation of line of regression of y on x is

$$y = 5x + 7$$

$$\Rightarrow b_{yx} = 5$$

We have $\rho^2 = b_{xy} b_{yx} = \frac{5}{2}$ from which $\rho = \sqrt{\frac{5}{2}}$ which is greater than one. Therefore, we have to change the assumption of equations.

Now let the equation of line of regression of y on x is

$$y = 2x - 3$$

$$\Rightarrow b_{yx} = 2$$

Let the equation of line of regression of x on y is

$$y = 5x + 7$$

$$x = \frac{y}{5} - \frac{7}{5}$$

$$b_{xy} = \frac{1}{5}$$

$$\rho^2 = b_{xy} b_{yx} = \frac{2}{5}$$

$$\rho = \sqrt{\frac{2}{5}} = 0.6324$$

The estimate of x when $y = 1$ is

$$x = \frac{1}{5} - \frac{7}{5} = -\frac{6}{5}$$

5.114 The joint pdf of two random variables X and Y is given by $f_{X,Y}(x, y) = 8xy$ $0 \leq x \leq y \leq 1$ and zero otherwise obtain the regression curve of X on Y .

Solution

Given: $f_{X,Y}(x, y) = 8xy$ $0 \leq x \leq y \leq 1$

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dy$$

and
$$f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx$$

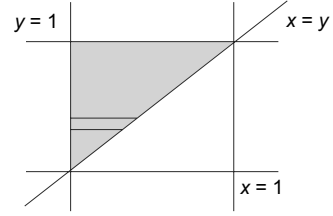


Fig. 5.32

From Fig. 5.32, we can find that in the shaded region x varies from 0 to y and y varies from x to 1.

$$\Rightarrow f_X(x) = \int_x^1 8xy dy = 4x(1 - x^2) \quad 0 \leq x \leq 1$$

$$f_Y(y) = \int_0^y 8xy dx = 8y \frac{x^2}{2} \Big|_0^y = 4y^3 \quad 0 \leq y \leq 1$$

$$E[X] = \int_{-\infty}^{\infty} x f_X(x) dx = \int_0^1 x \cdot 4x(1 - x^2) dx = 4 \frac{x^3}{3} \Big|_0^1 - \frac{4}{5} x^5 \Big|_0^1 = \frac{8}{15}$$

$$E[Y] = \int_{-\infty}^{\infty} y f_Y(y) dy = \int_0^1 4y^4 dy = \frac{4}{5} y^5 \Big|_0^1 = \frac{4}{5}$$

$$E[XY] = \int_0^1 \int_0^y 8x^2 y^2 dx dy = \frac{8}{3} \int_0^1 y^5 dy = \frac{4}{9}$$

$$E[Y^2] = \int_0^1 y^2 \cdot 4y^3 dy = \frac{2}{3}$$

$$\sigma_Y^2 = E[Y^2] - \{E[Y]\}^2 = \frac{2}{3} - \frac{16}{25} = \frac{2}{75}$$

$$\text{Cov}(X, Y) = E[XY] - E[X] E[Y]$$

$$= \frac{4}{9} - \frac{8}{15} \left(\frac{4}{5} \right) = \frac{4}{225}$$

$$b_{xy} = \frac{\text{Cov}(X, Y)}{\sigma_Y^2} = \frac{4/225}{2/75} = \frac{2}{3}$$

The regression line of X on Y is

$$\begin{aligned} \{x - E[X]\} &= b_{xy} \{y - E[Y]\} \\ \left(x - \frac{8}{15}\right) &= \frac{2}{3} \left(y - \frac{4}{5}\right) \\ \Rightarrow 3x - 2y &= 0 \end{aligned}$$



Practice Problem

5.33 The joint pdf of two random variables X and Y is given by $f_{X,Y}(x, y) = 2$ for $0 < x < y < 1$ and zero otherwise. Find the regression curve of X on Y .
(Ans. $2x - y = 0$)

Solved Problems



5.115 The regression lines Y on X and of X on Y are respectively $x + 3y = 0$ and $3x + 2y = 0$. Find the regression line of V on U where $U = X + Y$ and $V = X - Y$.

Solution Regression line Y on X is

$$\begin{aligned} x + 3y &= 0 \\ \Rightarrow y &= -\frac{x}{3} \end{aligned} \quad (5.153)$$

The regression line of Y on X is

$$(y - \bar{y}) = b_{yx} (x - \bar{x}) \quad (5.154)$$

Comparing Eq. (5.153) and Eq. (5.154), we get

$$\bar{y} = E[y] = 0 \text{ and } \bar{x} = E[X] = 0$$

$$\text{Also, } b_{yx} = \frac{\text{Cov}(X, Y)}{\sigma_X^2} = -\frac{1}{3}$$

$$\Rightarrow \text{Cov}(X, Y) = -\frac{\sigma_X^2}{3} \quad (5.155)$$

The regression line of X on Y is

$$\begin{aligned} 3x + 2y &= 0 \\ \Rightarrow x &= -\frac{2}{3}y \end{aligned}$$

The regression line of X on Y is

$$(x - \bar{x}) = b_{xy} (y - \bar{y})$$

$$\text{We have } b_{xy} = -\frac{2}{3}$$

$$\frac{\text{Cov}(X, Y)}{\sigma_y^2} = \frac{-2}{3}$$

$$\text{Cov}(X, Y) = -\frac{2\sigma_Y^2}{3} \quad (5.156)$$

From Eq. (5.155) and Eq. (5.156),

$$\sigma_X^2 = 2\sigma_Y^2$$

Also, $E[XY] = \text{Cov}(X, Y) = -\frac{2\sigma_Y^2}{3} = -\frac{\sigma_X^2}{3}$

Given

$$U = X + Y$$

and

$$V = X - Y$$

$$E[U] = E[X] + E[Y] = 0$$

$$E[V] = E[X] - E[Y] = 0$$

$$\text{Cov}(U, V) = E[UV] - E[U]E[V] = E[UV]$$

$$= E[(X + Y)(X - Y)] = E[X^2] - E[Y^2]$$

$$= \sigma_X^2 - \sigma_Y^2$$

(Since $E[X] = E[Y] = 0$)

$$\sigma_U^2 = E[U^2] - \{E[U]\}^2$$

$$= E[(X + Y)^2] = E[X^2] + E[Y^2] + 2E[XY]$$

$$= \sigma_X^2 + \sigma_Y^2 + 2\left(-\frac{2\sigma_Y^2}{3}\right)$$

$$= 3\sigma_Y^2 - \frac{4\sigma_Y^2}{3} = \frac{5\sigma_Y^2}{3}$$

The regression line of V on U is

$$(v - \bar{v}) = \frac{\text{Cov}(U, V)}{\sigma_U^2} (u - \bar{u})$$

$$v = \frac{\sigma_X^2 - \sigma_Y^2}{\frac{5}{3}\sigma_Y^2} (u)$$

$$= \frac{2\sigma_Y^2 - \sigma_Y^2}{\frac{5}{3}\sigma_Y^2} u$$

$$\Rightarrow \frac{5}{3}v = u$$

$$= \boxed{3u - 5v = 0}$$

5.116 The two regression lines are $4x - 5y + 33 = 0$ and $20x - 9y = 107$ and variance of $x = 25$. Find the mean of x and y . Also, find the value of x .

Solution Let the mean values of x and y be $\bar{x}$ and $\bar{y}$ respectively. The mean values $\bar{x}$ and $\bar{y}$ satisfy the regression line equations.

Therefore we can write

$$4\bar{x} - 5\bar{y} = -33$$

and $20\bar{x} - 9\bar{y} = 107$

Solving for $\bar{x}$ and $\bar{y}$, we get

$$\bar{x} = 13 \text{ and } \bar{y} = 17$$

Let us assume that the equation of line of regression of x and y is

$$4x = 5y - 33$$

$$\Rightarrow x = \frac{5}{4}y - \frac{33}{4} \quad (5.157)$$

and the equation of line of regression of y on x

$$9y = 20x - 107$$

from which

$$y = \frac{20}{9}x - \frac{107}{9} \quad (5.158)$$

From Eq. (5.157) and Eq. (5.158),

$$b_{xy} = \frac{5}{4} \text{ and } b_{yx} = \frac{20}{9}$$

$$\rho^2 = b_{xy} b_{yx} = \frac{100}{4 \times 9}$$

$$\rho = \frac{10}{6} = 1.66 > 1$$

Therefore, our assumption is wrong.

Now we assume that the equation of line of regression of y on x .

$$y = \frac{4}{5}x + \frac{33}{5}$$

$$\Rightarrow b_{yx} = \frac{4}{5}$$

The equation of line of regression of x on y .

$$x = \frac{9}{20}y + \frac{107}{20}$$

$$\Rightarrow b_{xy} = \frac{9}{20}$$

$$\rho^2 = b_{yx} b_{xy} = \frac{4}{5} \left(\frac{9}{20} \right) = \frac{36}{100}$$

$$\rho = \frac{6}{10} = 0.6$$



Practice Problems

5.34 The two regression lines (i) y on x is $7x - 16y + 9 = 0$ and (ii) x on y is $5y - 4x - 3 = 0$. Calculate $\bar{x}$, $\bar{y}$ and correlation coefficient.

$$\left(\text{Ans. } \bar{x} = \frac{-3}{29}; \bar{y} = \frac{15}{29}; \rho = 0.734 \right)$$

5.35 The two lines of regression are

$$8x - 10y + 66 = 0$$

$$40x - 18y - 214 = 0$$

The variance of x is 9. Find $\bar{x}$, $\bar{y}$ and ρ .

(Ans. 13, 17, 0.6)

Solved Problems



5.117 Two random variables have the joint pdf.

$$f_{X,Y}(x, y) = \frac{1}{3}(x + y), 0 \leq x \leq 1, 0 \leq y \leq 2$$

Find

- (a) Correlation coefficient.
- (b) The two regression lines.

Solution Given:

$$f_{X,Y}(x, y) = \frac{1}{3}(x + y), 0 \leq x \leq 1, 0 \leq y \leq 2$$

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dy = \frac{1}{3} \int_0^2 (x + y) dy$$

$$= \frac{1}{3} \left\{ xy \Big|_0^2 + \frac{y^2}{2} \Big|_0^2 \right\} = \frac{1}{3} [2x + 2] = \frac{2}{3} (1 + x); 0 \leq x \leq 1$$

$$f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx$$

$$= \frac{1}{3} \int_0^1 (x + y) dx = \frac{1}{3} \left[\frac{x^2}{2} \Big|_0^1 + yx \Big|_0^1 \right] = \frac{1}{3} \left[\frac{1}{2} + y \right] = \frac{1+2y}{6} \quad 0 < y < 2$$

$$\bar{x} = E[X] = \int_{-\infty}^{\infty} x f_X(x) dx = \int_0^1 \frac{2}{3} (1+x)x dx = \frac{2}{3} \left[\frac{x^2}{2} \Big|_0^1 + \frac{x^3}{3} \Big|_0^1 \right] = \frac{5}{9}$$

$$\begin{aligned} E[X^2] &= \int_{-\infty}^{\infty} x^2 f_X(x) dx = \int_0^1 \frac{2}{3} (1+x)x^2 dx = \frac{2}{3} \left[\left(\frac{x^3}{3} \right)_0^1 + \frac{x^4}{4} \Big|_0^1 \right] \\ &= \frac{2}{3} \left[\frac{1}{3} + \frac{1}{4} \right] = \frac{7}{18} \end{aligned}$$

$$\sigma_X^2 = E[X^2] - \{E[X]\}^2 = \frac{7}{18} - \left(\frac{5}{9} \right)^2 = \frac{13}{162}$$

$$\begin{aligned} \bar{y} = E[Y] &= \int_{-\infty}^{\infty} y f_Y(y) dy = \int_0^2 \frac{y(1+2y)}{6} dy = \frac{1}{6} \left[\frac{y^2}{2} \Big|_0^2 + \frac{2}{3} y^3 \Big|_0^2 \right] \\ &= \frac{1}{6} \left[2 + \frac{16}{3} \right] = \frac{11}{9} \end{aligned}$$

$$\begin{aligned} E[Y^2] &= \int_{-\infty}^{\infty} y^2 f_Y(y) dy = \int_0^2 \frac{y^2(1+2y)}{6} dy = \frac{1}{6} \left[\frac{y^3}{3} \Big|_0^2 - \frac{1}{2} y^4 \Big|_0^2 \right] \\ &= \frac{1}{6} \left[\frac{8}{3} + 8 \right] = \frac{16}{9} \end{aligned}$$

$$\sigma_Y^2 = E[Y^2] - \{E[Y]\}^2 = \frac{16}{9} - \left(\frac{11}{9} \right)^2 = \frac{23}{81}$$

$$\begin{aligned} E[XY] &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xy f_{X,Y}(x, y) dx dy \\ &= \frac{1}{3} \int_0^2 \int_0^1 xy(x+y) dx dy \\ &= \frac{1}{3} \int_0^2 \left(\frac{y}{3} + \frac{y^2}{2} \right) dy = \frac{1}{3} \left[\frac{1}{3} \frac{y^2}{2} \Big|_0^2 + \frac{1}{2} \frac{y^3}{3} \Big|_0^2 \right] \\ &= \frac{1}{3} \left[\frac{2}{3} + \frac{4}{3} \right] = \frac{2}{3} \end{aligned}$$

We know

$$\rho = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$$

$$= \frac{2}{3} - \frac{5}{9} \left(\frac{11}{9} \right) = -\frac{1}{81}$$

$$\rho = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} = \frac{\left(-\frac{1}{81} \right)}{\sqrt{\frac{13}{162} \times \frac{23}{81}}} = -0.082$$

Regression line of x and y

$$(x - \bar{x}) = b_{xy} (y - \bar{y})$$

$$\begin{aligned} \Rightarrow \left(x - \frac{5}{9} \right) &= \frac{\text{Cov}(X, Y)}{\sigma_Y^2} (y - \bar{y}) \\ &= \frac{-\frac{1}{81}}{\frac{23}{81}} \left(y - \frac{11}{9} \right) \\ &= -\frac{1}{23} \left(y - \frac{11}{9} \right) \end{aligned}$$

Regression line of y on x

$$y - \bar{y} = b_{yx} (x - \bar{x})$$

$$\begin{aligned} y - \frac{11}{9} &= \frac{\text{Cov}(X, Y)}{\sigma_X^2} \left(x - \frac{5}{9} \right) \\ &= \frac{-1/81}{13/162} \left(x - \frac{5}{9} \right) \\ y - \frac{11}{9} &= \frac{-2}{13} \left(x - \frac{5}{9} \right) \end{aligned}$$

REVIEW QUESTIONS



15. Define the expectation of a function of random variables.
16. Prove that the mean value of a weighted sum of random variables is equal to the weighted sum of mean values.
17. Define joint moments about the origin.
18. Define correlation.
19. Explain joint central moment in detail.
20. Define covariance and explain the properties of covariance.
21. Explain how you get the pdf of a single function of several random variables.
22. Explain how you obtain the pdf of several functions of several random variables.
23. Define correlation coefficient.
24. Give expression for pdf of two jointly Gaussian random variables.
25. What are the properties of Gaussian random variables?

26. Define joint characteristic function.
27. Define joint moment-generating function.
28. What are the properties of characteristic function?
29. Define conditional mean and variance.
30. Define regression.
31. Define sample mean and sample variance.
32. Write equation of regression line of y on x .
33. Write equation of regression line of x on y .
34. What is the angle between the regression lines?



EXERCISES

Problems

1. If X and Y are independent variables such that

$$X = -1 \text{ with a probability } \frac{1}{4}$$

$$= 0 \text{ with a probability } \frac{1}{4}$$

$$= 1 \text{ with a probability } \frac{1}{2}$$

and $Y = -1 \text{ with a probability } \frac{1}{3}$

$$= 0 \text{ with a probability } \frac{1}{4}$$

$$= 1 \text{ with a probability } \frac{5}{12}$$

$$\left(\text{Ans: (a) } \frac{37}{24}; \text{ (b) } 0 \right)$$

Find (a) $E[X^2 + 2XY + Y^2]$ (b) $E[X^2 - Y^2]$

2. Two statistically independent random variables X_1 and X_2 have mean value $\mu_{X_1} = 2$, $\mu_{X_2} = 5$. Find the mean values of

$$(a) g(X_1, X_2) = X_1^2 + X_1 X_2$$

$$(b) g(X_1, X_2) = X_1 + X_2$$

$$(\text{Ans: (a) } 14; \text{ (b) } 7)$$

3. Two random variables X and Y have joint pdf

$$f_{X,Y}(x, y) = \frac{1}{24} \quad 0 < x < 6; \quad 0 < y < 4$$

Find all third-order moments of X and Y .

$$(\text{Ans: } 54, 16, 16, 24)$$

4. Statistically independent random variables X and Y have moments $m_{10} = 5$; $m_{20} = 25$; $m_{02} = 6$; $m_{11} = 10$. Find the moment m_{22} .

Ans: Zero

5. The joint pdf of the random variables X and Y is defined as

$$f_{X,Y}(x, y) = \begin{cases} 8e^{-4y} & 0 < x < 0.5 \quad y > 0 \\ 0 & \text{otherwise} \end{cases}$$

Find m_{22} .

$$\left(\text{Ans: } \frac{1}{96} \right)$$

6. Two random variables X and Y have means 2 and 0 respectively and variances 4 and 9 respectively. Their correlation coefficient is 0.6. New random variables W and V are defined as

$$W = X - 2Y \text{ and } V = X + Y.$$

Find (a) mean, (b) variance and (c) correlation coefficient of V and W .

7. If the joint pdf of a random variables X and Y is given by $f_{X,Y}(x, y) = 2 - x - y$, in $0 \leq x \leq y < 1$, find $E(X)$ and $E(Y)$.

$$\left(\text{Ans: } \frac{5}{12}, \frac{5}{12} \right)$$

8. Show that $\text{Cov}(aX, bY) = ab \text{Cov}(X, Y)$.

9. X and Y are independent random variables with variance 2 and 3. Find the variance of $3X + 4Y$.

(Ans. 66)

10. The random variables X and Y have joint pdf given by

$$f_{X,Y}(x, y) = \begin{cases} 4xy, & 0 < x < 1, \quad 0 < y < 1 \\ 0 & \text{otherwise.} \end{cases}$$

Find the joint pdf of $V = X^2$ and $W = XY$.

$$\left(\text{Ans: } f_{V,W}(v, w) = \frac{2W}{V}; w^2 < v < 1; 0 < w < 1 \right)$$

11. The two random variables X and Y jointly distributed with density function $f_{X,Y}(x, y) = x + y$, $0 \leq x < 1$, $0 \leq y \leq 1$. Find $\text{Cov}(X, Y)$ and correlation coefficient.

$$\left(\text{Ans: } -\frac{1}{144}, -\frac{1}{11} \right)$$

12. If X is a random variable with $E[X] = 1$ and $E[X(X - 1)] = 4$

Find $\text{Var}(X)$ and $\text{Var}(2 - 3X)$.

(Ans. 4, 36)

13. If X and Y are random variables with $\sigma_X^2 = 2$, $\sigma_Y^2 = 4$ and covariance $\text{Cov}(X, Y) = -2$, find the variance of the random variables $Z = 3X - 4Y + 8$.

(Ans. 130)

14. The joint pdf of two random variables X and Y

$$f_{X,Y}(x, y) = \begin{cases} xy / 96 & 0 \leq x \leq 4, 1 \leq y \leq 5 \\ 0 & \text{elsewhere} \end{cases}$$

Find $\text{Var}(2X + 3Y)$.

15. Given joint pdf of X and Y

$$f_{X,Y}(x, y) = \begin{cases} 4.8y(2 - x) & 0 < x \leq 1, \quad 0 < y < x \\ 0 & \text{elsewhere} \end{cases}$$

Find the correlation, and correlation coefficient. Are X and Y are independent?

16. Show that X and Y are independent random variables if and only if $\rho = 0$.

17. Let X and Y be independent Cauchy random variables with parameters 4 and 9 respectively. Let $Z = X + Y$.

(a) Find the characteristic function of Z .

(b) Find the pdf of Z .

18. X and Y are two independent random variables with parameters 1 and 4 respectively. Let $Z = X + Y$.

(a) Find the characteristic function of Z

(b) Find pdf of Z from the characteristic function found in part (a).

19. Given $Z = X - 2Y$, where X and Y are two independent random variables.
 (a) Find the characteristic function of Z .
 (b) Find the mean and variance of Z from the characteristic function.
20. The joint pdf of two random variables X and Y is given by

$$f_{X,Y}(x, y) = \frac{1}{x^2 y^2}; x \geq 1, y \geq 1$$

Find the joint pdf of $U = XY$, $V = X/Y$.

21. Prove that $\text{Cov}(X + Y, Z) = \text{Cov}(X, Z) + \text{Cov}(Y, Z)$.
 22. If X and Y are identically distributed, not necessarily independent, show $\text{Cov}(X + Y, X - Y) = 0$.
 23. Prove that
 $E[(X - Y)^2] = E[X^2] - E[Y^2]$
 if $Y = E[X/Z]$
 24. Discrete random variables X and Y have the joint density

$$\begin{aligned} f_{X,Y}(x, y) = & 0.2(\delta(x + 2) \delta(y - 1) + 0.25 \delta(x + 1) \delta(y) \\ & + 0.15 \delta(x) \delta(y + 1) + 0.3 \delta(x - 2) \delta(y - 2). \\ & + 0.1 \delta(x - 3) \delta(y)) \end{aligned}$$

Find $m_{01}, m_{10}, m_{11}, m_{20}, m_{02}$, correlation coefficient.

25. Two Gaussian random variables X_1 and X_2 are defined by the mean and covariance matrices

$$[\bar{X}] = \begin{bmatrix} 1 \\ 0 \end{bmatrix} : [C_X] = \begin{bmatrix} 1 & -3 \\ -3 & 2 \end{bmatrix}$$

Two new random variables Y_1 and Y_2 are formed using the transformation

$$[T] = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

Find the matrices (a) $[\bar{Y}]$, and (b) $[C_Y]$. (c) Also find $\rho_{Y_1 Y_2}$.

26. Zero-mean Gaussian random variables X_1, X_2 and X_3 having a covariance matrix

$$[C_X] = \begin{bmatrix} 1 & -1 & 2 \\ -1 & 1 & -1 \\ 2 & -1 & 1 \end{bmatrix}$$

are transformed to new variables

$$\begin{aligned} Y_1 &= X_1 + X_2 - X_3 \\ Y_2 &= X_1 + 2X_2 + 2X_3 \\ Y_3 &= 2X_1 - X_2 + 2X_3 \end{aligned}$$

Find the covariance matrix of Y_1, Y_2 and Y_3

27. The joint pdf of random variable X and Y is given by

$$f_{X,Y}(x, y) = \frac{1}{2\pi C} e^{-(2x^2 + y^2/2)} \text{ for all } x, y$$

Find $\text{Var}(X)$, $\text{Var}(Y)$ and $\text{Cov}(X, Y)$.

$$\left(\text{Ans. } \frac{1}{4}, 1, 0 \right)$$

28. X and Y are independent random variables with $f_X(x) = e^{-x}u(x)$ and $f_Y(y) = 3e^{-3y}u(y)$. Find the density of $Z = \frac{X}{Y}$. (Ans. $f_Z(z) = \frac{3}{(z+3)^2}$)

29. If X and Y are independent random variables with density function $f_X(x) = 1$ for $1 \leq x \leq 2$ and $f_Y(y) = \frac{y}{6}$, $2 \leq y \leq 4$, find the density of $Z = XY$.

$$\left(\text{Ans: } f_Z(z) = \begin{cases} \frac{1}{6}(z-2) & \text{for } 2 \leq z \leq 4 \\ \frac{1}{6}\left(4 - \frac{z}{2}\right) & \text{for } 4 \leq z \leq 8 \end{cases} \right)$$

30. The pdf of random variables X and Y are $f_X(x) = 2e^{-2x}$, $x > 0$ and $f_Y(y) = 2e^{-2y}$ for $y > 0$. Find the pdf of $Z = X + Y$.

$$\left(\text{Ans: } f_Z(z) = \begin{cases} e^{2z} & \text{for } z < 0 \\ e^{-2z} & \text{for } z > 0 \end{cases} \right)$$

31. Given the joint pdf of two random variables X and Y

$$f_{X,Y}(x, y) = \begin{cases} \frac{1}{24}(1 + xy) & 0 \leq x \leq 2, \quad 0 \leq y \leq 4 \\ \text{elsewhere} \end{cases}$$

Find $E[2X + 5Y]$ and ρ .

32. Two random variables X and Y have the joint pdf

$$f_{X,Y}(x, y) = \frac{6}{7}x \quad 1 \leq x + y \leq 2, \quad x \geq 0, y \geq 0$$

Determine if X and Y are independent.

33. Let X_1 , X_2 and X_3 be independent and uniformly distributed in $[-1, 1]$. Find the pdf of $Y = X_1 + X_2$.
34. The joint pdf of X and Y is given by $f_{X,Y}(x, y) = e^{-(x+y)}$, $x > 0$, $y > 0$. Find the probability density

function of $U = \frac{X+Y}{2}$. (Ans. $f_U(u) = \begin{cases} \frac{u}{2}e^{-2u}, & u \geq 0 \\ 0 & \text{otherwise} \end{cases}$)

35. If X and Y are independent exponential distributions with parameter 1, find the pdf of $U = X - Y$.

$$(\text{Ans. } f_U(u) = \begin{cases} \frac{1}{2}e^u & \text{for } u < 0 \\ \frac{1}{2}e^{-u} & \text{for } u > 0 \end{cases})$$

36. If X and Y are independent random variables, each following $N(0, 2)$, find the pdf of $Z = 2X + 3Y$

$$f_Z(z) = \frac{1}{4\sqrt{13}\pi} e^{-z^2/104}$$

37. The joint pdf of two random variable X and Y is given by

$$f_{X,Y}(x, y) = \begin{cases} 2 & 0 < x < y < 1 \\ 0 & \text{otherwise} \end{cases}$$

Find the density function of the random variable $U = \frac{X}{Y}$.

(Ans: $f_{U,V}(u, v) = 2v, 0 < u < 1, 0 < v < 1$)

38. The joint density function of X and Y as

$$f_{X,Y}(x, y) = \begin{cases} \frac{1}{2} x e^{-y}, & 0 < x < 1, y > 0 \\ 0 & \text{elsewhere} \end{cases}$$

Find the distribution of $X + Y$.

$$\text{Ans. } f_U(u) = \begin{cases} \frac{1}{2}(e^u + u - 1) & 0 < u < 2 \\ \frac{1}{2}(e^{-u} + e^{2-u}) & 2 < u < \infty \\ 0 & \text{elsewhere} \end{cases}$$

39. If X and Y are independent random variables with pdf $e^{-x}, x \geq 0$ and $e^{-y}, y \geq 0$ respectively, find the density function of $U = \frac{X}{X+Y}$ and $V = X + Y$.

(Ans: $f_U(u) = 1$ and $f_V(v) = ve^{-v}$)

40. The joint pdf of X and Y is given by

$$f_{X,Y}(x, y) = e^{-(x+y)}, x > 0, y > 0$$

Find the pdf of $U = \frac{X+Y}{2}$.

(Ans: $f_U(u) = 4ue^{-2u}, u > 0$)

41. The joint pdf of two random variables X and Y is given by

$$f_{X,Y}(x, y) = \begin{cases} cx^2(1-y) & 0 \leq x, y \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

Find the covariance.

(Ans: 0)

42. Show that two random variables X_1 and X_2 with joint pdf

$$f_{X_1, X_2}(x_1, x_2) = \frac{1}{16} |x_1| < 4, 2 < x_2 < 4$$

are independent and orthogonal

43. Random variables Z and W are defined by $Z = X + aY$, $W = X - aY$ where a is a real number. Determine a such that Z and W are orthogonal.

44. Find the correlation coefficient for the following data.

X	10	14	18	22	26	30
Y	18	22	24	26	30	36

(Ans: 0.6)

45. Find the correlation coefficient between industrial production and export using the following data.

Production	55	56	58	59	60	60	62
Export	35	38	37	39	44	43	44

(Ans: 0.9)

46. Calculate the correlation coefficient and the lines of regression from the following data.

x	62	64	65	69	70	71	72	74
y	126	125	139	145	165	152	180	208

(Ans: $\rho = 0.9$)

47. If $\bar{x} = 970$, $\bar{y} = 18$, $\sigma_x = 38$, $\sigma_y = 2$ and $\rho = 0.6$, find the line of regression of x and y .

(Ans: $11.4y + 764.8 = x$)

48. The regression equations are $3x + 2y = 6$ and $6x + y = 31$. Find the correlation coefficient between X and Y .

49. Prove that

$$\rho(a_1X + b, a_2Y + b_2) = \frac{a_1a_2}{|a_1||a_2|} \rho(X, Y)$$

where $a_1 \neq 0$, b_1 and b_2 are constants.

Multiple-Choice Questions

- Two random variables X and Y have mean and variance as 2, 4 and 0, 9 respectively. Then the mean and variance of $Z = X + 3Y$ are
 (a) 2, 31 (b) 2, 85 (c) 2, 247 (d) 4, 85
- Two random variables X and Y are related as $Y = 4X + 9$. The correlation coefficient between X and Y is
 (a) 1 (b) 0.8 (c) 0.6 (d) 0.5
- $\text{Cov}(aX, bY) =$
 (a) $a^2b^2 \text{Cov}(X, Y)$ (b) $ab \text{Cov}(X, Y)$ (c) $\frac{a}{b} \text{Cov}(X, Y)$ (d) $(a + b) \text{Cov}(X, Y)$
- If $\text{Var}(X_1) = 5$, $\text{Var}(X_2) = 6$ and $E(X_1) = 0$; $E(X_2) = 0$. $\text{Cov}(X_1, Y_2) = 4$. Then $\text{Var}(Y)$ given $Y = 2X_1 - 3X_2$ is
 (a) 20 (b) 26 (c) 25 (d) 24
- If $f_{X,Y}(x, y) = 2x$, $0 \leq x \leq y \leq 1$, then $E[XY] =$
 (a) $\frac{1}{15}$ (b) $\frac{2}{15}$ (c) $\frac{1}{5}$ (d) $\frac{4}{15}$
- $\text{Cov}(X + Y, X - Y) =$
 (a) $\text{Var}(X) + \text{Var}(Y)$ (b) $\text{Var}(X) - \text{Var}(Y)$ (c) $\text{Var}(X^2 - Y^2)$ (d) $\text{Var}(X^2 + Y^2)$
- Two random variables X and Y have joint pdf

$$f_{X,Y}(x, y) = \begin{cases} \frac{xy}{96} & 0 < x < 4, \quad 1 < y < 5 \\ 0 & \text{otherwise} \end{cases}$$

The value of $E[XY]$ is

- $\frac{248}{27}$ (b) 8 (c) $\frac{8}{9}$ (d) $\frac{31}{9}$
- The joint pdf of two random variables X and Y is
 $f_{X,Y}(x, y) = 0.2 \delta(x - \alpha) \delta(y - \alpha) + 0.4 \delta(x + \alpha) \delta(y - 3) + 0.35 \delta(x - 1) \delta(y - 1)$
 $+ 0.05 \delta(x - 2) \delta(y - 2)$

The value of α for minimum $\text{Cov}(X, Y)$ is

- (a) 3 (b) 0.6 (c) 0.534 (d) 0.75

9. Two random variables X and Y are related by the equation

$$Y = 3 - 3X$$

The value of ρ_{XY} is

- (a) 1 (b) -1 (c) 0.6 (d) -0.6

10. X and Y are two independent random variables with same standard deviation 1. The correlation between X and $X + Y$ is

- (a) 1 (b) $\frac{1}{\sqrt{2}}$ (c) -1 (d) $-\frac{1}{\sqrt{2}}$

11. Angle between two regression lines is

- (a) $\tan^{-1} \left(\frac{1 - \rho^2}{\rho} \frac{\sigma_X \sigma_Y}{\sigma_X^2 + \sigma_Y^2} \right)$ (b) $\tan^{-1} \left(\frac{1 - \rho^2}{\rho} \frac{\sigma_X^2 + \sigma_Y^2}{\sigma_X + \sigma_Y} \right)$
 (c) $\tan^{-1} \left(\frac{1 - \rho^2}{\rho} \frac{\sigma_X + \sigma_Y}{\sigma_X \sigma_Y} \right)$ (d) $\tan^{-1} \left(\frac{1 - \rho^2}{\rho} \frac{\sigma_X \sigma_Y}{\sigma_X + \sigma_Y} \right)$

12. Given $U = X + KY$ and $V = X + \frac{\sigma_X}{\sigma_Y} Y$. If $\rho_{UV} = 0$, the value of K is

- (a) $-\frac{\sigma_X}{\sigma_Y}$ (b) $\frac{\sigma_X}{\sigma_Y}$ (c) $\frac{\sigma_Y}{\sigma_X}$ (d) $-\frac{\sigma_Y}{\sigma_X}$

13. The random variables X , Y and Z are uncorrelated and have same variance. The correlation between $(X + Y)$ and $(Y + Z)$ is

- (a) $\frac{1}{2}$ (b) $\frac{1}{8}$ (c) -1 (d) 1

14. The line of regression of x on y is

- (a) $x - \bar{x} = \rho \frac{\sigma_X}{\sigma_Y} (y - \bar{y})$ (b) $y - \bar{y} = \rho \frac{\sigma_Y}{\sigma_X} (x - \bar{x})$
 (c) $x - \bar{x} = \rho \frac{\sigma_Y}{\sigma_X} (y - \bar{y})$ (d) $y - \bar{y} = \rho \frac{\sigma_X}{\sigma_Y} (x - \bar{x})$

15. Two random variables X and Y have the joint pdf given by

$$f_{X,Y}(x, y) = x + y, 0 \leq x \leq 1, 0 \leq y \leq 1$$

The correlation coefficient is

- (a) $-\frac{1}{11}$ (b) $\frac{1}{144}$ (c) $\frac{1}{11}$ (d) $-\frac{1}{144}$

16. Two lines of regression coincide if and only if

- (a) $\rho_{xy} = 0$ (b) $\rho_{xy} = \pm \frac{1}{\sqrt{2}}$ (c) $\rho_{xy} = \pm 1$ (d) $\rho_{xy} = \pm \frac{1}{2}$

17. The two regressions for random variables X and Y are $Y = 2X$ and $X = \frac{Y}{8}$. The correlation coefficient is

(a) $\frac{1}{2}$ (b) $\frac{3}{4}$ (c) 1 (d) 0

18. The joint characteristic function of two random variables X and Y is given by

$$\phi_{X,Y}(\omega_1, \omega_2) = e^{-2\omega_1^2 - 8\omega_2^2}$$

The mean value of X is given by

(a) 1 (b) 0 (c) -1 (d) 2

19. The joint pdf of two random variables X and Y is

$$f_{X,Y}(x, y) = kxy$$

The value of $E[X^2 + Y^2]$ is

(a) 0 (b) 1 (c) 2 (d) 3

20. The pdf of a random variable X is $f_X(x) = \frac{1}{2}e^{-|x|}$, $-\infty < x < \infty$

The variance of X is given by

(a) $\frac{1}{4}$ (b) $\frac{1}{2}$ (c) -1 (d) 1

21. The random variable X has the pdf

$$f_X(x) = \frac{1}{\sqrt{\pi}} e^{-x^2 + x - \frac{1}{4}}$$

The mean and variance are given by

(a) $\frac{1}{2}$ (b) $1, \frac{1}{2}$ (c) $\frac{1}{2}, \frac{1}{2}$ (d) $0, \frac{1}{2}$

22. Two random variables are related by

$$Y = aX + b$$

The covariance is given by

(a) $a\sigma_X^2$ (b) $a^2\sigma_X^2$ (c) $2a\sigma_X^2$ (d) $0.5a\sigma_X^2$

23. Two random variable X and Y are independent with moments $m_{10} = 2$, $m_{20} = 14$ and $m_{11} = -6$. The value of μ_{22} is

(a) 10 (b) 20 (c) 30 (d) 40

24. If X and Y are independent random variables with $U(0, 1)$ the distribution of $U = XY$ is

(a) $-\log u$, $0 \leq u \leq 1$ (b) $\log u$; $0 \leq u \leq 1$ (c) e^{-u} , $0 \leq u \leq 1$ (d) e^u ; $u < 0$

25. The joint pdf of two random variables X and Y is

$$f_{X,Y}(x, y) = \begin{cases} x + y & 0 \leq x \leq 1, \quad 0 \leq y \leq 1 \\ 0 & \text{elsewhere} \end{cases}$$

The density of random variable $Z = XY$ in $0 \leq z < 1$ is

- (a) $\frac{1-z}{z}$ (b) $\frac{z-1}{z+1}$ (c) $2(1-z)$ (d) $\frac{1}{1-z}$

26. Uncorrelated Gaussian random variables are

- (a) independent (b) not independent (c) orthogonal (d) nonlinear

27. The random variables X and Y have joint pdf

$$f_{X,Y}(x, y) = \begin{cases} 2e^{-2x}/x & 0 \leq x < \infty, \quad 0 \leq y \leq x \\ 0 & \text{otherwise} \end{cases}$$

The value of $\text{Cov}(X, Y)$ is

- (a) 1 (b) $\frac{1}{8}$ (c) $\frac{1}{10}$ (d) -1

Answers to Multiple-Choice Questions

- | | | | | | | | | | |
|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| 1. (b) | 2. (a) | 3. (b) | 4. (b) | 5. (b) | 6. (b) | 7. (a) | 8. (a) | 9. (b) | 10. (b) |
| 11. (a) | 12. (a) | 13. (a) | 14. (a) | 15. (a) | 16. (c) | 17. (a) | 18. (b) | 19. (b) | 20. (b) |
| 21. (c) | 22. (a) | 23. (c) | 24. (a) | 25. (c) | 26. (a) | 27. (b) | | | |

6

RANDOM PROCESSES—TEMPORAL CHARACTERISTICS



INTRODUCTION 6.1

In chapters 1 and 5, we studied in detail about probability theory. In those chapters, we were concerned with outcomes of random experiments and the random variable used to represent them. We know that a random variable is a mapping of an event $s \in S$ where S is the sample space to some real number $X(s)$. The random-variable approach is applied to random problems that are not functions of time. However, in certain random experiments, the outcome may be a function of time. Especially in engineering, many random problems are time dependent. For example, speech signal which is a variation in voltage due to speech utterance is a function of time. Similarly, in a communication system, a set of messages that are to be transmitted over a channel is also a function of time. Such time functions are called *random processes*. In communication systems, a desired deterministic signal is often accompanied by undesired random waveform known as noise which limits the performance of the system. Since the noise is a function of time and cannot be represented by a mathematical equation, it can be treated as a random process. In this chapter, we will study such random processes which may be viewed as a collection of random variables with t as a parameter. That is, instead of a single number $X(s)$, we deal with $X(s, t)$ where $t \in T$ and T is called the parameter set of the process. For a random process $X(s, t)$, the sample space is a collection of time function. Figure 6.1 shows a few members of collection. A realization of $X(s, t)$ is a time function, also called a sample function and member function.

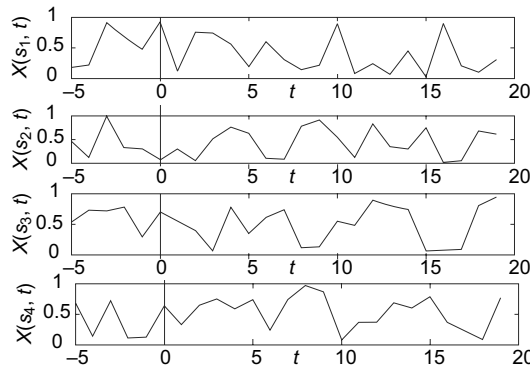


Fig. 6.1 Few members of collection of a random process

A random process becomes a random variable when time is fixed at some particular value. For example, if t is fixed to a value t_1 then the value of $X(s, t)$ at t_1 is a random variable $X(s, t_1)$. On the other hand, if we fix the sample points, $X(t)$ is some real function of time. Thus, for a fixed S , $X(s, t)$ can be viewed as a collection of time functions.

1. $X(s, t_1)$ is a random variable for a fixed time.
2. $X(s_1, t)$ is a sample realization for any points s , in the sample space S .
3. $X(s_1, t_1)$ is a number.
4. $X(s, t)$ is a collection of realization and is called a random process.

Example 1 Consider a random experiment in which we toss a dice and observe the dots on the top face. We know that the possible outcomes of this random experiment are as shown in Fig. 6.2(a) and these outcomes are known as sample space. For these outcomes, let us assign the following time functions shown in Fig. 6.2(b).

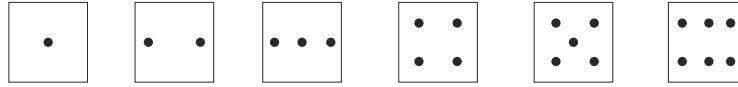


Fig. 6.2 (a) Outcomes of a dice toss

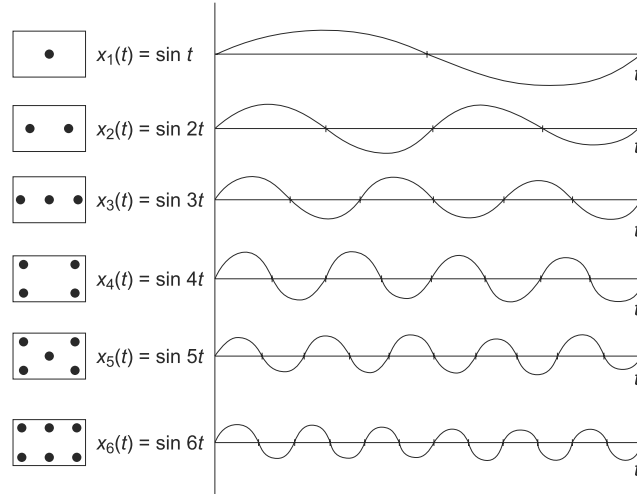


Fig. 6.2 (b) Time functions for dice toss

The set of the waveforms $\{x_1(t), x_2(t), \dots, x_6(t)\}$ represents the random process.

Example 2 In communication systems, the carrier signal is often modelled as a sinusoid with random phase which is given by

$$x(t) = A \cos(2\pi f t + \theta)$$

The phase θ is a random variable with uniform distribution between $-\pi$ and π . The reason for using random phase is that the receiver does not know the time when the transmitter was turned ON or the distance from the transmitter to the receiver. Figure 6.3 shows the realization of the process.

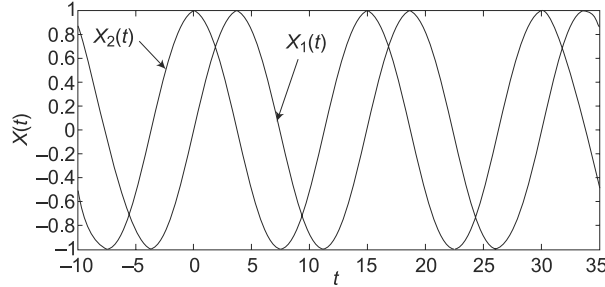


Fig. 6.3 Realisations of sinusoid process with random phase

Example 3 A random telegraph signal $X(t)$ takes the possible states either 0 or 1. The time interval in which the signal remains in one of the possible states is an experimental random variable. Figure 6.4 shows one possible realization for the telegraph signal. Initially, at time $t = 0$, the signal is in the zero state for a time interval of T_1 and switches to the state 1. The signal remains in that state for a time interval T_2 and then switches to zero state again. The process of switching between two states continues depending on the time interval specified by the sequence of exponential random variables.

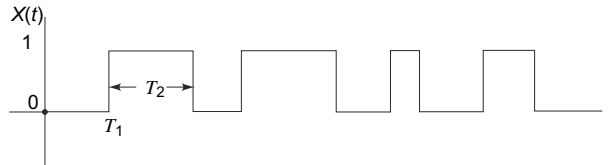


Fig. 6.4 Realization of telegraph signal



CLASSIFICATION OF RANDOM PROCESSES 6.2

In general, a random process or stochastic process $X(s, t)$ can be defined as a random time function assigned to each outcome. Depending on the way the values t and s are defined, the random process can be classified into

- (i) Continuous random process
- (ii) Discrete random process
- (iii) Continuous random sequence
- (iv) Discrete random sequence

6.2.1 Continuous Random Process

Definition A continuous random process is one in which the random variable X is continuous and time t can have any of a continuum of values. The examples of continuous random process are thermal noise in conductors, shot noise in transistors and wind velocity.

Figure 6.5 shows a typical sample function of a continuous random process.

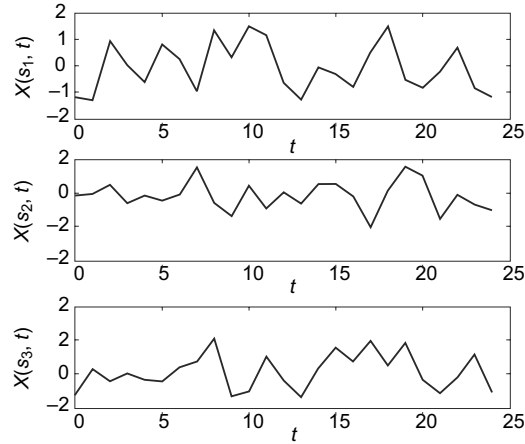


Fig. 6.5 A continuous random process

Let us consider the amplitude of A of the sine wave $X(t) = A \sin(\omega t + \phi)$. If A takes any value in the interval $[-2, 2]$ then the resulting process is a continuous random process $X(s, t)$ given by

$$X(s, t) = s \cos(2\pi t)$$

The realization of this process is shown in Fig. 6.6.

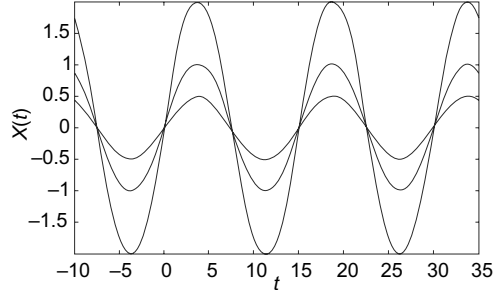


Fig. 6.6 Sine wave random process with A as random variable

6.2.2 Discrete Random Process

A discrete random process is one in which the random variable X assumes discrete values while t is continuous. The random opening and closing of a switch that connects either 0 or 50 volts is a sample function of a discrete random process.

A random telegraph process is an example for discrete random process. At any time instant, the random telegraph signal takes one of two possible states, either 0 or 1. Figure 6.7 shows one possible realization of random telegraph signal in which the signal takes the value 1 for the time interval T_1 and 0 for the time interval T_2 .

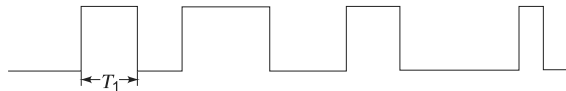


Fig. 6.7 A discrete random process

6.2.3 Continuous Random Sequence

In a continuous random sequence, the random variable X is continuous but the time has only discrete values. A continuous random sequence can be obtained by sampling the continuous random process. Since the continuous random sequence is defined at discrete instants of time, it is also known as a discrete-time random process. It can be represented as a set of random variables $\{X(t_x)\}$ for sample times $t_x = kT_s$, $k = 0, \pm 1, \pm 2, \dots$. The value T_s is known as *sampling time*.

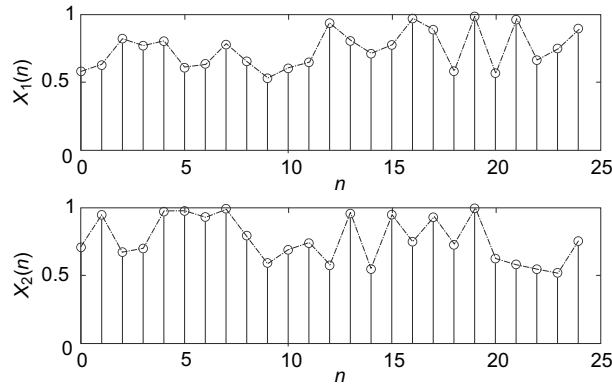


Fig. 6.8 A continuous random sequence

6.2.4 Discrete Random Sequence

In a discrete random sequence, both random variables X and time t are discrete. The discrete random sequence can be obtained by sampling the discrete random process or by quantizing the continuous random sequence. This process is also known as *digital process* and is mostly used in digital-signal processing applications. An example of discrete-time random sequence is shown in Fig. 6.9.

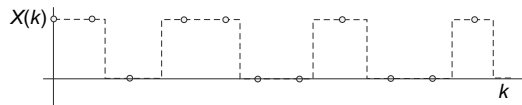


Fig. 6.9 A discrete random sequence



DETERMINISTIC AND NON-DETERMINISTIC PROCESS 6.3

A random process is said to be a non-deterministic random process if future values of any sample function cannot be exactly predicted from the observed past values. Almost all natural random processes are non-deterministic.

A random process is said to be *deterministic* if future values of any sample function can be predicted from past values.

A deterministic random process is

$$X(t) = A \cos(\omega t + \phi)$$

where A and ω are constants and ϕ is a random variable with interval $(-\pi, \pi)$.

A second example for deterministic process is with a sample function

$$X(t) = Ae^{-\beta t} \text{ in which } A \text{ and } \beta \text{ are random units.}$$

REVIEW QUESTIONS



1. Define random process with suitable examples.
2. What are the classification of random process?
3. Explain deterministic and non-deterministic processes.
4. Give some examples for the following random processes.
 - (a) Continuous random process
 - (b) Discrete random process
 - (c) Continuous random sequence
 - (d) Discrete random sequence
5. What is the difference between random sequence and random process?

Solved Problems



6.1 A two-level semirandom binary process is defined as $X(t) = 1$ or 0 for $(n-1)T < t < nT$ where the levels 1 and 0 occur with equal probability, T is a positive constant and $n = 0, \pm 1, \pm 2, \dots$

- (a) Sketch a typical sample function.
- (b) Classify the process.
- (c) Is the process deterministic?

Solution

- (a) Given $X(t) = 0$ for $(n-1)T < t < nT$. A sample function is shown in Fig. 6.10.
- (b) This is a discrete random process.
- (c) The process is not deterministic.

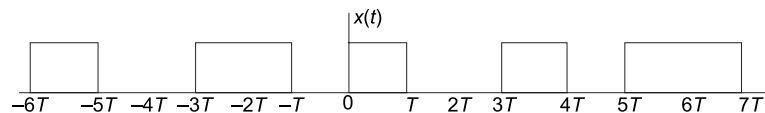


Fig. 6.10 A sample function of two-level semi-random binary process.

6.2 A random process $X(t)$ is defined as $X(t) = A$ where A is a continuous random variable uniformly distributed on $(0,1)$.

- (a) Sketch any two sample functions.
- (b) Classify the process.
- (c) Is it deterministic?

Solution

- (a) A is a uniform random variable on $(0,1)$. Two sample functions for different values of A are shown in Fig. 6.11.

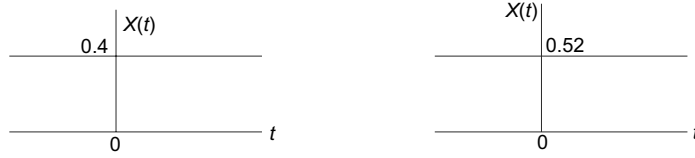


Fig. 6.11

- (b) The process is a continuous random process.
- (c) Since each sample function is a constant with time, the random process is deterministic.

6.3 Repeat solved problem (6.2) for $X(t) = At$ where $A \sim U(0, 1)$.

Solution

- (a) Here, A represents the slope of a straight line passing through the origin. Two different sample functions are shown in Fig. 6.12.

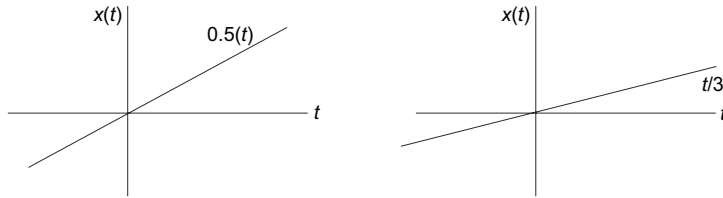


Fig. 6.12

- (b) The process is a continuous random process.
- (c) It is deterministic.



Practice Problem

6.1 A discrete-time random process is defined as follows. A coin is tossed. If the outcome is heads then $X_n = (-1)^n$ and if the outcome is tails $X_n = (-1)^{n+1}$ for all n . Sketch some sample functions of the process.



DISTRIBUTION AND DENSITY FUNCTIONS 6.4

A random process is completely characterized by the joint pdf. The value of a random process $X(t)$ at a particular time t_1 is a random variable $X(t_1)$. Let us define the distribution function associated with this random variable as

$$F_X(x_1; t_1) = P[X(t_1) \leq x_1] \quad (6.1)$$

for any real number x_1 . Similarly, we can write the distribution function for other random variables $x_1, x_2, \dots, x_n$ as

$$F_X(x_2; t_2) = P[X(t_2) \leq x_2]$$

$$\begin{aligned}
F_X(x_3; t_2) &= P[X(t_3) \leq x_3] \\
&\vdots \\
F_X(x_n; t_n) &= P[X(t_n) \leq x_n]
\end{aligned} \tag{6.2}$$

where $0 < t_1 < t_2 \dots < t_n$

The function $F_X(x_1, t_1)$ is known as *first order distribution function* $X(t)$. For two random variables, $X(t_1) = X_1$ and $X(t_2) = X_2$, the joint distribution function known as *second-order joint distribution function* of the random process $X(t)$ is given by

$$F_X(x_1, x_2; t_1, t_2) = P[X(t_1) \leq x_1, X(t_2) \leq x_2] \tag{6.3}$$

Similarly, for N random variables, $X_i = X(t_i)$, the N^{th} order joint distribution function

$F_X(x_1, \dots, x_N, t_1 \dots t_N) = P[X(t_1) \leq x_1, \dots, X(t_N) \leq x_N]$ for all N completely describes the random process.



JOINT DENSITY FUNCTION OF A RANDOM PROCESS 6.5

The joint density function of a random process can be obtained by differentiating the corresponding distribution functions:

(i) The first order density function is

$$f_X(x_1; t_1) = \frac{d}{dx_1} F_X(x_1; t_1) \tag{6.4}$$

(ii) The second order density function is

$$f_X(x_1, x_2; t_1, t_2) = \frac{\partial^2}{\partial x_1 \partial x_2} F_X(x_1, x_2; t_1, t_2) \tag{6.5}$$

(iii) The N^{th} order density function is

$$f_X(x_1, x_2, \dots, x_N; t_1, t_2, \dots, t_N) = \frac{\partial^n}{\partial x_1 \partial x_2 \dots \partial x_N} F_X(x_1, x_2, \dots, x_N; t_1, t_2, \dots, t_N) \tag{6.6}$$

If $X(t)$ is a discrete random process then it is specified by a collection of pmfs

$$p_X(x_1, x_2, \dots, x_n; t_1, t_2, \dots, t_n) = P[X_1 = x_1, X_2 = x_2, \dots, X_n = x_n] \tag{6.7}$$



MEAN AND AUTOCORRELATION FUNCTION OF A RANDOM PROCESS 6.6

The mean value of a random process $X(t)$ is an ensemble average of the random process $X(t)$. It is a function of time and is denoted by

$$\mu_X(t) = E[X(t)] \tag{6.8}$$

where $E[X(t)]$ is the expected value of $X(t)$.

If the probability density function of a random process is $f_X(x, t)$ then the mean of $X(t)$ is

$$\mu_X(t) = E[X(t)] = \int x f_X(x, t) dx \tag{6.9}$$



AUTOCORRELATION FUNCTION 6.7

Autocorrelation provides the similarity between two observations of a random process. Consider a random process $X(t)$ for different times t_1 and t_2 . Let the joint density function of these two random variables be $f_X(x_1, x_2, t_1, t_2)$. The autocorrelation functions of X_1 and X_2 are defined as

$$R_{XX}(t_1, t_2) = E[X(t_1)X(t_2)] \quad (6.10)$$

$$\text{or} \quad R_{XX}(t_1, t_2) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x_1 x_2 f_X(x_1, x_2; t_1, t_2) dx_1 dx_2 \quad (6.11)$$

If we assume the time $t_1 = t$ and $t_2 = t + \tau$

$$\text{then} \quad R_{XX}(t, t + \tau) = E[X(t_1)X(t_2)] \quad (6.12)$$

The parameter τ is called the *delay time* or *lag time*.



STATIONARY AND STATISTICAL INDEPENDENCE 6.8

A random process is said to be stationary if all its statistical properties such as mean value, variance, etc., do not change with time. For such processes, the marginal and joint density function are invariant for any time shift τ .

Consider a random process $X(t)$ with random variable $X(t_1) = x_i$, $i = 1, \dots, N$ with pdf $f_X(x_1, x_2, \dots, x_N; t_1, t_2, \dots, t_N)$. The above process is said to be stationary if it satisfies the relation

$$f_X(x_1, x_2, \dots, x_N; t_1, t_2, \dots, t_N) = f_X(x_1, x_2, \dots, x_N; t_1 + \tau, t_2 + \tau, \dots, t_N + \tau) \quad (6.13)$$

If any of the probability density functions do change with the choice of time origin, then the process is non-stationary. For such processes, the mean value and variance also depend on time.



STATISTICAL INDEPENDENCE 6.9

Consider the random processes $X(t)$ and $Y(t)$ with random variables $X(t_1), X(t_2), \dots, X(t_N)$ and $Y(t'_1), Y(t'_2), \dots, Y(t'_N)$. Let the joint density function of the random variable $X(t)$ be $f_X(x_1, x_2, \dots, x_N; t_1, t_2, \dots, t_N)$ and the joint density function of the random variable $Y(t)$ be $f_Y(y_1, y_2, \dots, y_N; t'_1, t'_2, \dots, t'_N)$. The two random processes $X(t)$ and $Y(t)$ are said to be strictly independent if the joint density function of random process $X(t)$ and $Y(t)$ is equal to product of the individual joint density function of $X(t)$ and $Y(t)$.

That is,

$$f_{X,Y}(x_1, \dots, x_N, y_1, \dots, y_N; t_1, \dots, t_N; t'_1, \dots, t'_N) = f_X(x_1, \dots, x_N; t_1, \dots, t_N) f_Y(y_1, \dots, y_N; t'_1, \dots, t'_N) \quad (6.14)$$



FIRST-ORDER STATIONARY 6.10

A random process is first-order stationary if its first-order distribution and density functions do not change with shift in time.

That is, $F_X(x; t) = F_X(x; t + \tau)$

$$\text{and} \quad f_X(x, t) = f_X(x, t + \tau) \quad (6.15)$$

When a random process satisfies the above condition, the mean value of the process does not change with time shift. The mean value of the random process $X(t)$ is

$$E[X(t_1)] = \int_{-\infty}^{\infty} x_1 f_X(x_1; t_1) dx_1 \quad (6.16)$$

Similarly, the mean value of the random variable $X(t_2)$ is

$$E[X(t_2)] = \int_{-\infty}^{\infty} x_1 f_X(x_1; t_2) dx_1 \quad (6.17)$$

Since $t_2 = t_1 + \tau$, we have

$$E[X(t_2)] = \int_{-\infty}^{\infty} x_1 f_X(x_1; t_1 + \tau) dx_1$$

Using Eq. (6.15), we can write

$$\begin{aligned} E[X(t_2)] &= \int_{-\infty}^{\infty} x_1 f_X(x_1; t_1) dx_1 \\ &= E[X(t_1)] \end{aligned} \quad (6.18)$$

That is, the mean value of the random process is constant and does not change with a shift in time origin.



SECOND-ORDER AND WIDE-SENSE STATIONARY 6.11

A random process is second-order stationary if its second-order distribution and density functions do not change with shift in time. That is,

$$F_X(x_1, x_2; t_1, t_2) = F_X[x_1, x_2; t_1 + \tau, t_2 + \tau] \quad (6.19)$$

and

$$f_X(x_1, x_2; t_1, t_2) = f_X[x_1, x_2; t_1 + \tau, t_2 + \tau] \quad (6.20)$$

For a second-order stationary process, the mean value is constant and does not change with change in time shift. The autocorrelation function of a second-order stationary process is a function of time difference and not absolute time.

Consider a random process $X(t)$ with second-order density function $f_X(x_1, x_2; t_1, t_2)$.

The autocorrelation function of the random process is given by

$$E[X(t_1) X(t_2)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x_1 x_2 f_X(x_1, x_2; t_1, t_2) dx_1 dx_2 \quad (6.21)$$

For a stationary process, the above expression is equal to

$$E[X(t_1) X(t_2)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x_1 x_2 f_X(x_1, x_2; t_1 + \Delta, t_2 + \Delta) dx_1 dx_2 \quad (6.22)$$

For an arbitrary $\Delta = -t_1$,

$$E[X(t_1) X(t_2)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x_1 x_2 f_X(x_1, x_2; 0, t_2 - t_1) dx_1 dx_2$$

Now let $t_2 - t_1 = \tau$ and $t_1 = t$

$$\begin{aligned} E[X(t) X(t + \tau)] &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x_1 x_2 f_X(x_1, x_2; \tau) dx_1 dx_2 \\ &= R_{XX}(\tau) \end{aligned} \quad (6.23)$$

The second-order stationary process is also known as Wide Sense Stationary (WSS), process.

Definition A random process is Wide-Sense Stationary if the mean function and autocorrelation function are independent to time shift.

That is,

$$\mu_X(t) = \mu_X = \text{Constant} \quad (6.24)$$

$$R_{YY}(t, t + \tau) = R_{XX}(\tau) \quad (6.25)$$

A strict-sense stationary process is also WSS provided that mean and autocorrelation functions exist. Thus, the WSS process does not necessarily be stationary in the strict-sense.



JOINTLY WIDE-SENSE STATIONARY 6.12

Two random processes $X(t)$ and $Y(t)$ are said to be jointly wide-sense stationary if their cross correlation function $R_{XY}(t, t + \tau)$ is function of the time difference τ . That is,

$$R_{XY}(t, t + \tau) = E[X(t) X(t + \tau)] = R_{XY}(\tau) \quad (6.26)$$

REVIEW QUESTIONS



6. Define distribution and density functions.
7. Define the mean and auto correlation function of a random process.
8. What is meant by first-order and second-order stationary?
9. When are the random processes are said to be statistical independence?
10. Define second order and a wide-sense stationary process. When the random process is said to be jointly wide-sense stationary. Give an example for WSS process?
11. What is the difference between a SSS and WSS process?
12. When are the processes $X(t)$ and $Y(t)$ said to be jointly stationary in the wide sense?

Solved Problems



6.4 In the fair-coin experiment, a random process $X(t)$ is defined as follows: $X(t) = \cos \pi t$ if heads occur, $X(t) = t$ if tails occur. (a) Find $E[X(t)]$. (b) Find $F_X(x, t)$ for $t = 0.25, 0.5, 1$.

Solution The random process $X(t)$ takes the values $\cos \pi t$ and t with same probability $1/2$. Therefore,

$$\begin{aligned} E[X(t)] &= \sum_i p_X(x) X_i(t) = \frac{1}{2} \cos \pi t + \frac{1}{2} (t) \\ &= 0.5 \cos \pi t + 0.5t \end{aligned}$$

$$\begin{aligned}
X(t, \text{heads}) &= \cos \pi t = \frac{1}{\sqrt{2}} && \text{for } t = 0.25 \\
&= 0 && \text{for } t = 0.5 \\
&= -1 && \text{for } t = 1 \\
X(t, \text{tails}) &= t = 0.25 && \text{for } t = 0.25 \\
&= 0.5 && \text{for } t = 0.5 \\
&= 1 && \text{for } t = 1
\end{aligned}$$

The distribution functions $F_X(x, t)$ are shown in Fig. 6.13.

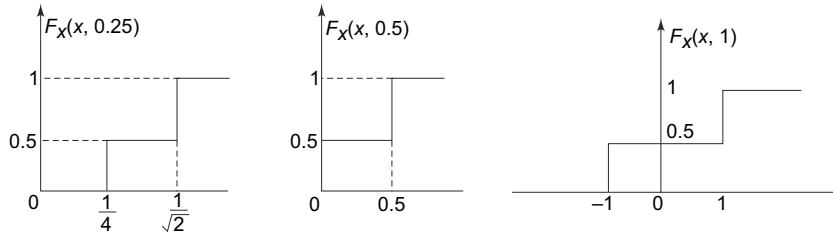


Fig. 6.13

6.5 A random process $X(t)$ is determined by tossing two dice to determine which of the following sinusoids to pick.

$$X(t) = \begin{cases} x_1(t) = \sin t & \text{if the sum is 3 or 7} \\ x_2(t) = \cos 2t & \text{if the sum is 2 or 12} \\ x_3(t) = \sin 2t & \text{otherwise} \end{cases}$$

Find $\mu_X\left(\frac{\pi}{4}\right), \mu_X\left(\frac{\pi}{2}\right), \sigma_X^2\left(\frac{\pi}{4}\right), \sigma_X^2\left(\frac{\pi}{2}\right)$,

Solution In throwing two dice experiments, the number of possible outcomes is 36.

The sum of two dice equal to three can be obtained in two ways. That is, (1, 2) and (2, 1).

The sum of two dice equal to seven can be obtained in six ways. They are (1,6), (2,5), (3,4), (4,3), (5,2), (6,1). Therefore, the probability

$$P[X(t) = x_1(t)] = \frac{2}{36} + \frac{6}{36} = \frac{2}{9}$$

The only way that the sum of two dice is equal to 2 is (1, 1).

The only way that the sum of two dice is equal to 12 is (6, 6).

Therefore, the probability $P[X(t) = x_2(t)] = \frac{1}{36} + \frac{1}{36} = \frac{1}{18}$

$$P[X(t) = x_3(t)] = 1 - \frac{2}{9} - \frac{1}{18} = \frac{13}{18}$$

$$\begin{aligned}
\mu_X(t) &= E[X(t)] = \sum_{i=1}^n x_i(t) P[X(t) = x_i(t)] \\
&= \sum_{i=1}^3 x_i(t) P[X(t) = x_i(t)] \\
&= x_1(t) P[X(t) = x_1(t)] + x_2(t) P[X(t) = x_2(t)] + x_3(t) P[X(t) = x_3(t)] \\
&= \frac{2}{9} \sin t + \frac{1}{18} \cos 2t + \frac{13}{18} \sin 2t \\
\mu_X\left(\frac{\pi}{4}\right) &= \frac{2}{9} \sin \frac{\pi}{4} + \frac{1}{18} \cos \frac{\pi}{2} + \frac{13}{18} \sin \frac{\pi}{2} \\
&= \frac{2}{9} \left(\frac{1}{\sqrt{2}}\right) + \frac{1}{18} (0) + \frac{13}{18} (1) \\
&= 0.8793 \\
\mu_X\left(\frac{\pi}{2}\right) &= \frac{2}{9} \sin\left(\frac{\pi}{2}\right) + \frac{1}{18} \cos 2\left(\frac{\pi}{2}\right) + \frac{13}{18} \sin 2\left(\frac{\pi}{2}\right) \\
&= \frac{2}{9} (1) + \frac{1}{18} (-1) + \frac{13}{18} (0) = \frac{1}{6} \\
E[X^2(t)] &= \sum_{i=1}^3 x_i^2(t) P[X(t) = x_i(t)] \\
&= \frac{2}{9} \sin^2(t) + \frac{1}{18} \cos^2 2(t) + \frac{13}{18} \sin^2 2(t) \\
E\left[X^2\left(\frac{\pi}{4}\right)\right] &= \frac{2}{9} \sin^2\left(\frac{\pi}{4}\right) + \frac{1}{18} \cos^2 2\left(\frac{\pi}{4}\right) + \frac{13}{18} \sin^2 2\left(\frac{\pi}{4}\right) \\
&= \frac{2}{9} \left(\frac{1}{2}\right) + \frac{1}{18} (0) + \frac{13}{18} (1) \\
&= \frac{5}{6} \\
E\left[X^2\left(\frac{\pi}{2}\right)\right] &= \frac{2}{9} \sin^2\left(\frac{\pi}{2}\right) + \frac{1}{18} \cos^2 2\left(\frac{\pi}{2}\right) + \frac{13}{18} \sin^2 2\left(\frac{\pi}{2}\right) \\
&= \frac{2}{9} + \frac{1}{18} (-1)^2 + \frac{13}{18} (0) \\
&= \frac{5}{18} \\
\sigma_X^2\left(\frac{\pi}{4}\right) &= E\left[X^2\left(\frac{\pi}{4}\right)\right] - \left[\mu_X\left(\frac{\pi}{4}\right)\right]^2
\end{aligned}$$

$$\begin{aligned}
&= \frac{5}{6} - (0.8793)^2 \\
&= 0.0602 \\
\sigma_x^2\left(\frac{\pi}{2}\right) &= E\left[X^2\left(\frac{\pi}{2}\right)\right] - \left[\mu_x\left(\frac{\pi}{2}\right)\right]^2 \\
&= \frac{5}{18} - \frac{1}{36} \\
&= \frac{1}{4}
\end{aligned}$$

6.6 If $X(t) = Y \cos t + Z \sin t$ for all t where Y and Z are independent binary random variables, each of which assumes the values -1 and 2 with probabilities $\frac{2}{3}$ and $\frac{1}{3}$ respectively. Prove that $\{X(t)\}$ is wide-sense stationary.

Solution Given Y and Z are independent binary random variables which assumes the values -1 and 2 with probability $\frac{2}{3}$ and $\frac{1}{3}$ respectively. We can write

y	-1	2
$p_Y(y)$	$\frac{2}{3}$	$\frac{1}{3}$

z	-1	2
$p_Z(z)$	$\frac{2}{3}$	$\frac{1}{3}$

From which,

$$E[Y] = \sum_i y_i p_Y(y_i) = (-1)\left(\frac{2}{3}\right) + 2\left(\frac{1}{3}\right) = 0$$

$$E[Z] = \sum_i z_i p_Z(z_i) = (-1)\left(\frac{2}{3}\right) + 2\left(\frac{1}{3}\right) = 0$$

Similarly,

$$\begin{aligned}
E[Y^2] &= \sum_i y_i^2 p_Y(y_i) = (-1)^2\left(\frac{2}{3}\right) + (2)^2\left(\frac{1}{3}\right) \\
&= \left(\frac{2}{3}\right) + \left(\frac{4}{3}\right) = 2
\end{aligned}$$

$$\begin{aligned}
 E[Z^2] &= \sum_i z_i^2 p_Z(z_i) = (-1)^2 \left(\frac{2}{3}\right) + (2)^2 \left(\frac{1}{3}\right) \\
 &= \left(\frac{2}{3}\right) + \left(\frac{4}{3}\right) = 2
 \end{aligned}$$

Since Y and Z are independent, $E[YZ] = E[Y] E[Z] = 0$

$$\begin{aligned}
 E[X(t)] &= E[Y \cos t + Z \sin t] \\
 &= E[Y] \cos t + E[Z] \sin t \\
 &= 0
 \end{aligned}$$

$$\begin{aligned}
 R_{XX}(t, t + \tau) &= E[X(t) X(t + \tau)] \\
 &= E[(Y \cos t + Z \sin t)(Y \cos(t + \tau) + Z \sin(t + \tau))] \\
 &= E[Y^2 \cos t \cos(t + \tau) + YZ \sin t \cos(t + \tau) + YZ \cos t \sin(t + \tau) + Z^2 \sin t \sin(t + \tau)] \\
 &= E[Y^2] \cos t \cos(t + \tau) + E[YZ] \sin t \cos(t + \tau) \\
 &\quad + E[YZ] \cos t \sin(t + \tau) + E[Z^2] \sin t \sin(t + \tau)
 \end{aligned}$$

We know $E[YZ] = 0$ and $E[Y^2] = E[Z^2] = 2$

$$\begin{aligned}
 R_{XX}(t, t + \tau) &= 2 \cos t \cos(t + \tau) + 2 \sin t \sin(t + \tau) \\
 &= 2[\cos t \cos(t + \tau) + \sin t \sin(t + \tau)] \\
 &= 2 \cos(t + \tau - t) \\
 &= 2 \cos \tau
 \end{aligned}$$

Hence, $X(t)$ is a WSS process.



Practice Problems

6.2 If $X(t) = \cos(\omega t + \phi)$ where ϕ is uniformly distributed in $(-\pi, \pi)$, show that $X(t)$ is stationary in wide-sense.
 (Ans: $R_{XX}(t, t + \tau) = \frac{\cos \omega \tau}{2}$)

6.3 If $X(t) = A \cos \omega t + B \sin \omega t$, where A and B are independent random variables and ω is a constant, then prove that $X(t)$ is a WSS.



CORRELATION FUNCTIONS 6.13

Correlation functions are used to measure the similarity between two waveforms. Consider the waveforms shown in Fig. 6.14.

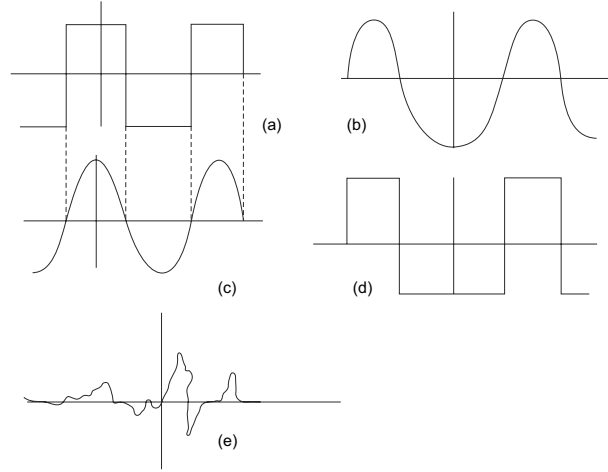


Fig. 6.14 Correlation of waveform

We can easily observe from the figure that the correlation between the waveforms (b) and (c) are strong but negative. The correlation between (a) and (d) is also strong but negative. The correlation between the waveform shown in Fig. 6.14(e) with other waveforms is very weak. The correlation between the waveform (a) and (c), (b) and (d) are medium and positive.

Correlation of Deterministic Waveforms The autocorrelation function of a deterministic power signal is defined as

$$\gamma_{xx}(\tau) = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T x(t)x(t+\tau)dt \quad (6.27)$$

Here, $\gamma_{xx}(\tau)$ measures the similarity between a signal and its time-shifted version.

The cross-correlation function of two deterministic power waveforms $x(t)$ and $y(t)$ is defined as

$$\gamma_{xy}(\tau) = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T x(t)y(t+\tau)dt \quad (6.28)$$

Here γ_{xy} measures the similarity between $x(t)$ and shifted $y(t)$.

6.13.1 Correlation of Random Process

The auto-correlation function between two random processes is given by $R_{XX}(t, t + \tau) = E[X(t) X(t + \tau)]$ (6.29)

The cross-correlation function between two random process is given by

$$R_{X,Y}(t, t + \tau) = E[X(t) Y(t + \tau)]$$

6.13.2 Properties of Autocorrelation Function

In this section, we will study important properties of autocorrelation function $R_{XX}(\tau)$ of a real-valued wide-sense stationary random process $X(t)$.

1. Autocorrelation function is bounded by its value at the origin. That is, $|R_{XX}(\tau)| \leq R_{XX}(0)$. In other words, autocorrelation function $R_{XX}(\tau)$ is maximum at $\tau = 0$. Consider the inequality

$$\begin{aligned}
 E\{[X(t+\tau) \pm X(t)]^2\} &\geq 0 \\
 E[X^2(t+\tau) \pm 2X(t+\tau)X(t) + X^2(t)] &\geq 0 \\
 \Rightarrow R_{XX}(0) \pm 2R_{XX}(\tau) + R_{XX}(0) &\geq 0 \\
 2R_{XX}(0) \pm 2R_{XX}(\tau) &\geq 0 \\
 \Rightarrow 2R_{XX}(0) &\geq \pm 2R_{XX}(\tau) \\
 R_{XX}(0) &\geq |R_{XX}(\tau)| \\
 \text{or } |R_{XX}(\tau)| &\leq R_{XX}(0)
 \end{aligned} \tag{6.30}$$

2. Autocorrelation function $R_{XX}(\tau)$ is an even function, that is

$$\begin{aligned}
 R_{XX}(\tau) &= R_{XX}(-\tau) \\
 R_{XX}(\tau) &= E[X(t)X(t+\tau)] \\
 R_{XX}(-\tau) &= E[X(t)X(t-\tau)] \\
 &= E[X(t+\tau)X(t)] \\
 &= E[X(t)X(t+\tau)] = R_{XX}(\tau)
 \end{aligned} \tag{6.31}$$

3. The autocorrelation function $R_{XX}(\tau)$, at $\tau = 0$ is equal to mean square value. That is,

$$\begin{aligned}
 R_{XX}(0) &= E[X^2(t)] \\
 R_{XX}(\tau) &= E[X(t)X(t+\tau)] \\
 R_{XX}(0) &= E[X(t)X(t)] = E[X^2(t)]
 \end{aligned} \tag{6.32}$$

The mean square value is also known as *average power* of the process.

4. If $X(t)$ has no periodic components and is ergodic, and $\mu_X \neq 0$, then

$$\begin{aligned}
 \lim_{|\tau| \rightarrow \infty} R_{XX}(\tau) &= (\mu_X)^2 \\
 R_{XX}(\tau) &= E[X(t)X(t+\tau)]
 \end{aligned} \tag{6.33}$$

If $X(t)$ has no periodic component then $X(t)$ and $X(t+\tau)$ are uncorrelated as $\tau \rightarrow \infty$. Therefore, we can write

$$\begin{aligned}
 \lim_{|\tau| \rightarrow \infty} E[X(t)X(t+\tau)] &= E[X(t)]E[X(t+\tau)] \\
 E[X(t)] &= E[X(t+\tau)] = \mu_X
 \end{aligned}$$

Therefore,

$$\lim_{|\tau| \rightarrow \infty} R_{XX}(\tau) = \mu_X^2$$

5. If $X(t)$ is ergodic, zero mean and has no periodic component then

$$\lim_{|\tau| \rightarrow \infty} R_{XX}(\tau) = 0$$

6. If $X(t)$ has a periodic component then $R_{XX}(\tau)$ will have a periodic component with the same period.

$$R_{XX}(\tau) = E[X(t)X(t+\tau)]$$

Consider

$$R_{XX}(\tau \pm T) = E[X(t) X(t + \tau \pm T)]$$

Since $X(t)$ is periodic with T , we can write

$$X(t + \tau \pm T) = X(t + \tau)$$

$$\Rightarrow R_{XX}(\tau \pm T) = E[X(t) X(t + \tau)] = R_{XX}(\tau)$$

Since $R_{XX}(\tau \pm T) = R_{XX}(\tau)$

$R_{XX}(\tau)$ is periodic.

7. If $X(t)$ has a dc component then $R_{XX}(\tau)$ will have a dc component.

Thus, if $X(t) = k + N(t)$ where k is a constant then $R_{XX}(\tau) = k^2 + R_{NN}(\tau)$

$$X(t) = k + Nt$$

$$R_{XX}(\tau) = E[X(t) X(t + \tau)]$$

$$= E[\{k + N(t)\} \{k + N(t + \tau)\}]$$

$$= E[k^2 + kN(t) + kN(t + \tau) + N(t)N(t + \tau)]$$

$$= k^2 + kE[N(t)] + kE[N(t + \tau)] + E[N(t)N(t + \tau)]$$

If $E[N(t)] = 0$, then $R_{XX}(\tau) = k^2 + R_{NN}(\tau)$

8. The autocorrelation function $R_{XX}(\tau)$ cannot have an arbitrary shape.
 9. For a random process, its autocorrelation function (if exists) is uniquely determined by the joint probability density function. But the pdf cannot, in general, be determined uniquely by the ACF.
 10. If $X(t)$ is a differentiable random process then the autocorrelation function of the derivative of the random process $\dot{X}(t)$ is the negative second-order derivative of that of $X(t)$

$$R_{\dot{X}\dot{X}}(\tau) = E[\dot{X}(t)\dot{X}(t + \tau)] = \frac{d}{d\tau} R_{XX}(\tau) \quad (6.34)$$

$$R_{\dot{X}\dot{X}}(\tau) = E[\dot{X}(t)\dot{X}(t + \tau)] = -\frac{d^2}{d\tau^2} R_{XX}(\tau) \quad (6.35)$$

(for proof, refer Solved Problem (6.60).)

11. An additive deterministic term of a random process has no effect on its autocovariance.
 Let $X(t)$ be a WSS process and $x(t)$ be a deterministic function. Then

$$Y(t) = X(t) + x(t)$$

$$\mu_Y = E[Y(t)] = x(t) + E[X(t)] = x(t) + \mu_X \quad (6.36)$$

The autocovariance function of $X(t)$ is $C_{XX}(\tau)$. The autocovariance function of $Y(t)$ is

$$C_{YY}(\tau) = R_{YY}(\tau) - E[Y(t)]E[Y(t + \tau)]$$

$$R_{YY}(\tau) = E[\{X(t) + x(t)\} \{X(t + \tau) + x(t + \tau)\}]$$

$$= R_{XX}(\tau) + x(t)\mu_X + x(t + \tau)\mu_X + x(t)x(t + \tau) \quad (6.37)$$

$$E[Y(t)] E[Y(t + \tau)]$$

$$= E[X(t) + x(t)]E[X(t + \tau) + x(t + \tau)]$$

$$= [\mu_X + x(t)][\mu_X + x(t + \tau)]$$

$$= \mu_X^2 + x(t)\mu_X + x(t + \tau)\mu_X + x(t)x(t + \tau)$$

$$\begin{aligned}
\Rightarrow C_{YY}(\tau) &= R_{XX}(\tau) + \mu_X x(t) + \mu_X x(t+\tau) + x(t)x(t+\tau) \\
&\quad - \mu_X^2 - \mu_X x(t) - \mu_X x(t+\tau) - x(t)x(t+\tau) \\
&= R_{XX}(\tau) - \mu_X^2 \\
C_{YY}(\tau) &= C_{XX}(\tau)
\end{aligned} \tag{6.38}$$

Therefore, the deterministic $x(t)$ has no effect on the covariance of $Y(t)$.

12. A multiplicative deterministic term of a random process acts as a scaling factor. Let $x(t)$ be a deterministic signal. Define a random process $Y(t) = x(t)X(t)$ where $X(t)$ is a WSS process.

$$\begin{aligned}
E[Y(t)] &= E[x(t)X(t)] = \mu_X x(t) \\
R_{YY}(t, t+\tau) &= E[Y(t)Y(t+\tau)] = E[x(t)X(t)x(t+\tau)X(t+\tau)] \\
&= x(t)x(t+\tau)R_{XX}(\tau) \\
E[Y(t)]E[Y(t+\tau)] &= E[x(t)X(t)]E[x(t+\tau)X(t+\tau)] \\
&= x(t)\mu_X x(t+\tau)\mu_X \\
&= x(t)x(t+\tau)\mu_X^2 \\
C_{YY}(t, t+\tau) &= R_{YY}(t, t+\tau) - E[Y(t)]E[Y(t+\tau)] \\
&= x(t)x(t+\tau)R_{XX}(\tau) - x(t)x(t+\tau)\mu_X^2 \\
&= x(t)x(t+\tau)[R_{XX}(\tau) - \mu_X^2] \\
&= x(t)x(t+\tau)C_{XX}(\tau)
\end{aligned} \tag{6.39}$$

Therefore, a multiplicative deterministic term of a random process acts as a scaling factor for its covariance.

6.13.3 Properties of Cross Correlation Functions

In this section we will study the properties of cross correlation function of two jointly wide-sense stationary random processes $X(t)$ and $Y(t)$.

1. The cross correlation function $R_{XY}(\tau)$ is not an even function of τ . But it satisfies the condition

$$\begin{aligned}
R_{XY}(\tau) &= R_{YX}(-\tau) \\
R_{XY}(\tau) &= E[X(t)Y(t+\tau)] = E[X(P-\tau)Y(P)] \\
&= R_{YX}(-\tau)
\end{aligned}$$

$$\Rightarrow R_{XY}(\tau) = R_{YX}(-\tau)$$

The above equation represents that $R_{XY}(\tau)$ and $R_{YX}(\tau)$ are mirror images of each other as depicted in Figure (6.15).

2. Let $X(t)$ and $Y(t)$ are two random processes with respective autocorrelation functions $R_{XX}(\tau)$ and $R_{YY}(\tau)$. Then $|R_{XY}(\tau)| \leq \sqrt{R_{XX}(0)R_{YY}(0)}$.

That is cross correlation is bounded by geometric mean of $R_{XX}(0)$ and $R_{YY}(0)$.

3. The cross-correlation is bounded by arithmetic average of $R_{XX}(0)$ and $R_{YY}(0)$

$$|R_{XY}(\tau)| \leq \frac{R_{XX}(0) + R_{YY}(0)}{2}$$

4. If $X(t)$ and $Y(t)$ are differentiable random processes then

$$R_{XY}(\tau) = E[X(t)\dot{Y}(t+\tau)] = \frac{d}{d\tau} R_{XY}(\tau) \text{ and}$$

$$R_{\dot{X}\dot{Y}}(\tau) = E[\dot{X}(t)\dot{Y}(t+\tau)] = -\frac{d^2}{d\tau^2} R_{XY}(\tau)$$

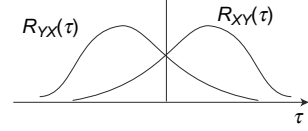


Fig. 6.15 Cross correlation function

5. If $X(t)$ and $Y(t)$ are uncorrelated random processes, then

$$\begin{aligned} R_{XY}(\tau) &= E[X(t)]E[Y(t+\tau)] \\ &= E[X(t)]E[Y(t)] \end{aligned}$$

6. If $X(t)$ and $Y(t)$ are orthogonal random processes, then $R_{XY}(t, t+\tau) = 0$.

REVIEW QUESTIONS



13. State and prove the properties of autocorrelation functions.
14. Prove that the autocorrelation function is an even function.
15. Prove that autocorrelation function value at $\tau = 0$ is equal to mean square value.
16. Prove that $|R_{XX}(\tau)| \leq R_{XX}(0)$.
17. If $X(t)$ has no periodic components and is ergodic then prove that $\lim_{|\tau| \rightarrow \infty} R_{XX}(\tau) = (\mu_X)^2$.
18. Prove that if $X(t)$ has a periodic component then $R_{XX}(\tau)$ will have a periodic component with the same period.
19. Prove that if $X(t)$ has a dc component then $R_{XX}(\tau)$ will have a dc component.
20. If $X(t)$ is differentiable then prove that

$$R_{X\dot{X}}(\tau) = \frac{d}{d\tau} R_{XX}(\tau)$$

$$R_{\dot{X}\dot{X}}(\tau) = -\frac{d^2}{d\tau^2} R_{XX}(\tau)$$

21. Prove that an additive deterministic term of a random process has no effect on its autocovariance.
22. Prove that a multiplicative deterministic term of a random process acts as a scaling factor for its covariance.
23. Consider a random process $Y(t) = x(t) + X(t)$ where $x(t)$ is a deterministic signal and $X(t)$ is WSS process. Prove that $Y(t)$ is not stationary if $x(t)$ is time-varying (refer solved problem 6.54).
24. Explain the properties of cross-correlation function.

Solved Problems



6.7 Prove that $|R_{XY}(\tau)| \leq \sqrt{R_{XX}(0)R_{YY}(0)}$.

Solution Consider the inequality

$$E[\{Y(t+\tau) + \alpha X(t)\}^2] \geq 0 \quad (6.41)$$

Expanding the above equation, we obtain

$$\begin{aligned}
 E[\{Y^2(t+\tau) + \alpha^2 X^2(t) + 2\alpha X(t)Y(t+\tau)\}] &\geq 0 \\
 E[Y^2(t+\tau)] + \alpha^2 E[X^2(t)] + 2\alpha E[X(t)Y(t+\tau)] &\geq 0 \\
 E[Y^2(t+\tau)] &= R_{YY}(0) \\
 E[X^2(t)] &= R_{XX}(0) \\
 E[X(t)Y(t+\tau)] &= R_{XY}(\tau)
 \end{aligned} \tag{6.42}$$

Substituting the above values in Eq. (6.42), we get

$$R_{YY}(0) + \alpha^2 R_{XX}(0) + 2\alpha R_{XY}(\tau) \geq 0 \tag{6.43}$$

The above equation is non-negative with α if the roots are not real. The roots can be imaginary or double roots. The roots are imaginary or double roots if

$$\begin{aligned}
 (2R_{XY}(\tau))^2 - 4R_{XX}(0)R_{YY}(0) &\leq 0 \\
 \Rightarrow 4R_{XY}^2(\tau) &\leq 4R_{XX}(0)R_{YY}(0) \\
 \Rightarrow R_{XY}(\tau) &\leq 2\sqrt{R_{XX}(0)R_{YY}(0)}
 \end{aligned} \tag{6.44}$$

6.8 Prove that $|R_{XY}(\tau)| \leq \frac{R_{XX}(0) + R_{YY}(0)}{2}$.

Solution From the solved problem 6.7, we found that $|R_{XY}(\tau)| \leq \sqrt{R_{XX}(0)R_{YY}(0)}$

Since geometric mean of two positive numbers cannot exceed their arithmetic mean, we can write

$$\sqrt{R_{XX}(0)R_{YY}(0)} \leq \frac{R_{XX}(0) + R_{YY}(0)}{2}$$

$$\text{Therefore, we can write } |R_{XY}(\tau)| \leq \frac{R_{XX}(0) + R_{YY}(0)}{2} \tag{6.45}$$

6.9 Prove that

$$R_{X\dot{Y}}(\tau) = \frac{d}{d\tau} R_{XY}(\tau) \tag{6.46}$$

$$R_{\dot{X}Y}(\tau) = -\frac{d^2}{d\tau^2} R_{XY}(\tau) \tag{6.47}$$

Solution We know $\dot{X}(t) = \lim_{\varepsilon \rightarrow 0} \frac{X(t+\varepsilon) - X(t)}{\varepsilon}$ and $\dot{Y}(t) = \lim_{\varepsilon \rightarrow 0} \frac{Y(t+\varepsilon) - Y(t)}{\varepsilon}$

$$\begin{aligned}
 R_{X\dot{Y}}(t) &= E[X(t)\dot{Y}(t+\tau)] \\
 &= E\left[X(t) \lim_{\varepsilon \rightarrow 0} \left\{ \frac{Y(t+\tau+\varepsilon) - Y(t+\tau)}{\varepsilon} \right\}\right]
 \end{aligned}$$

$$\begin{aligned}
&= \lim_{\varepsilon \rightarrow 0} \left\{ \frac{E[X(t)Y(t+\tau+\varepsilon)] - E[X(t)Y(t+\tau)]}{\varepsilon} \right\} \\
&= \lim_{\varepsilon \rightarrow 0} \left\{ \frac{R_{XY}(\tau+\varepsilon) - R_{XY}(\tau)}{\varepsilon} \right\} = \frac{d}{d\tau} R_{XY} \\
R_{\dot{X}\dot{Y}}(\tau) &= E[\dot{X}(t)\dot{Y}(t+\tau)] \\
&= E\left[\lim_{\varepsilon \rightarrow 0} \left(\frac{X(t+\varepsilon) - X(t)}{\varepsilon} \right) \dot{Y}(t+\tau)\right] \\
&= E\left[\lim_{\varepsilon \rightarrow 0} \left(\frac{X(t+\varepsilon)\dot{Y}(t+\tau) - X(t)\dot{Y}(t+\tau)}{\varepsilon} \right)\right] \\
&= \lim_{\varepsilon \rightarrow 0} \left[\frac{R_{XY}(\tau+\varepsilon) - R_{XY}(\tau)}{\varepsilon} \right] = -\frac{d}{d\tau} R_{XY}(\tau) \\
&= -\frac{d}{d\tau} \left[\frac{d}{d\tau} R_{XY}(\tau) \right] = -\frac{d^2}{d\tau^2} R_{XY}(\tau)
\end{aligned}$$



Practice Problem

6.4 If $X^{(n)}(t) = \frac{d^n X(t)}{dt^n}$ then prove that $R_{X^{(n)}Y^{(m)}}(\tau) = (-1)^m \frac{d^{n+m} R_{XY}(\tau)}{d\tau^{n+m}}$.



RELATIONSHIP BETWEEN TWO RANDOM PROCESSES 6.14

Consider the random processes $X(t)$ and $Y(t)$

- I. They are jointly wide-sense stationary if $X(t)$ and $Y(t)$ are both WSS and their cross-correlation $R_{XY}(t, t+\tau)$ and $R_{YX}(t, t+\tau)$ depend only on the time difference τ .

$$R_{XY}(t, t+\tau) = R_{XY}(\tau) \quad R_{YX}(t, t+\tau) = R_{YX}(\tau)$$

- II. The two random processes are uncorrelated if their cross-correlation is equal to the product of their mean functions

$$\begin{aligned}
R_{XY}(t, t+\tau) &= R_{XY}(\tau) = E[X(t)]E[Y(t+\tau)] \\
&= E[X(t)]E[Y(t)]
\end{aligned} \tag{6.48}$$

or Two random processes are uncorrelated if their covariance $C_{XY}(t, t+\tau) = 0$ (6.49)

- III. Two random processes are orthogonal if $R_{XY}(t, t+\tau) = 0$



DISCRETE-TIME RANDOM PROCESSES 6.15

So far we are dealing with continuous-time random processes. In this section, we are going to apply the concepts we have studied so far for discrete-time random processes. Note that the discrete-time random processes can be obtained by sampling a continuous-time random process.

That is, if the sampling time is equal to T_s then the discrete-time random process is given by

$$X(n) = X(nT_s)$$

Mean of the Random Processes

The mean of discrete-time random processes $X(n)$ is given by

$$\mu_X(n) = E[X(n)]$$

Autocorrelation of DT Random Processes

The autocorrelation function of DT random process is given by

$$R_{XX}(n, n+k) = E[X(n)X(n+k)] \quad (6.50)$$

For a WSS process $\mu_X(n) = \mu_X$ a constant and $R_{XX}(n, n+k) = R_{XX}(k)$

Auto Covariance of DT Random Processes

The autocovariance function of $X(n)$ is given by

$$\begin{aligned} C_{XX}(n, n+k) &= E[\{X(n) - E[X(n)]\} \{X(n+k) - E[X(n+k)]\}] \\ &= R_{XX}(n, n+k) - E[X(n)]E[X(n+k)] \end{aligned} \quad (6.51)$$

If $X(n)$ and $Y(n)$ are two random processes then

$$\begin{aligned} C_{XY}(n, n+k) &= E[\{X(n) - E[X(n)]\} \{Y(n+k) - E[Y(n+k)]\}] \\ &= R_{XY}(n, n+k) - E[X(n)]E[Y(n+k)] \end{aligned}$$

For WSS processes,

$$R_{XX}(n, n+k) = R_{XX}(k) \quad (6.52)$$

$$R_{YY}(n, n+k) = R_{YY}(k) \quad (6.53)$$

$$C_{XX}(n, n+k) = R_{XX}(k) - (\mu_X)^2 \quad (6.54)$$

$$C_{YY}(n, n+k) = R_{YY}(k) - (\mu_Y)^2 \quad (6.55)$$

$$R_{XY}(n, n+k) = R_{XY}(k) \quad (6.56)$$

$$R_{YX}(n, n+k) = R_{YX}(k) \quad (6.57)$$

$$C_{XY}(n, n+k) = R_{XY}(k) - \mu_X \mu_Y \quad (6.58)$$

$$C_{YX}(n, n+k) = R_{YX}(k) - \mu_Y \mu_X \quad (6.59)$$

Solved Problems



6.10 Given the random process $X(t) = A \sin(\omega t + \theta)$ where A and ω are constants and θ is uniformly distributed random variable between $-\pi$ and π . Define a new random process $Y(t) = X^2(t)$. (a) Are $X(t)$ and $Y(t)$ WSS? (b) Are $X(t)$ and $Y(t)$ jointly WSS.

Solution Given: $X(t) = A \sin(\omega t + \theta)$

The new random process $Y(t) = X^2(t) = A^2 \sin^2(\omega t + \theta)$

$$E[X(t)] = E[A \sin(\omega t + \theta)]$$

Since A is a constant, we can write

$$E[X(t)] = AE[\sin(\omega t + \theta)]$$

θ is a uniformly distributed random variable between $-\pi$ and π . Therefore,

$$f_{\Theta}(\theta) = \begin{cases} \frac{1}{2\pi} & \text{for } -\pi \leq \theta \leq \pi \\ 0 & \text{otherwise} \end{cases}$$

$$\begin{aligned} E[\sin(\omega t + \theta)] &= \int_{-\infty}^{\infty} \sin(\omega t + \theta) f_{\Theta}(\theta) d\theta \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} \sin(\omega t + \theta) d\theta = \frac{-1}{2\pi} \cos(\omega t + \theta) \Big|_{-\pi}^{\pi} \\ &= -\frac{1}{2\pi} [\cos(\omega t + \pi) - \cos(\omega t - \pi)] = 0 \end{aligned}$$

$$\begin{aligned} R_{XX}(t, t + \tau) &= E[X(t) X(t + \tau)] \\ &= E[A \sin(\omega t + \theta) A \sin(\omega(t + \tau) + \theta)] \\ &= \frac{A^2}{2} E[\cos \omega \tau - \cos(2\omega t + \omega \tau + 2\theta)] \\ &= \frac{A^2}{2} E[\cos \omega \tau] - \frac{A^2}{2} E[\cos(2\omega t + \omega \tau + 2\theta)] \end{aligned}$$

Since θ is a uniform distributed random variable,

$$E[\cos(2\omega t + \omega \tau + 2\theta)] = 0$$

Also, ω is a constant. Therefore,

$$R_{XX}(t, t + \tau) = \frac{A^2}{2} \cos \omega \tau$$

$\therefore X(t)$ is WSS

$$E[Y(t)] = E[A^2 \sin^2(\omega t + \theta)] = \frac{A^2}{2} E[1 - \cos(2\omega t + 2\theta)]$$

$$= \frac{A^2}{2} - \frac{A^2}{2} E[\cos(2\omega t + 2\theta)]$$

$$\begin{aligned} E[\cos(2\omega t + 2\theta)] &= \int_{-\infty}^{\infty} \cos(2\omega t + 2\theta) f_{\Theta}(\theta) d\theta \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} \cos(2\omega t + 2\theta) d\theta = -\frac{1}{4\pi} \sin(2\omega t + 2\theta) \Big|_{-\pi}^{\pi} \\ &= -\frac{1}{4\pi} [\sin(2\omega t + 2\pi) - \sin(2\omega t - 2\pi)] \\ &= 0 \end{aligned}$$

$$\begin{aligned}
\Rightarrow E[Y(t)] &= \frac{A^2}{2}, \text{ a constant} \\
R_{YY}(t, t + \tau) &= E[Y(t)Y(t + \tau)] = E[A^2 \sin^2(\omega t + \theta) A^2 \sin^2(\omega(t + \tau) + \theta)] \\
&= A^4 E[\{\sin(\omega t + \theta) \sin(\omega(t + \tau) + \theta)\}^2] \\
&= \frac{A^4}{4} E[(\cos(\omega\tau) - \cos(2\omega t + \omega\tau + 2\theta))^2] \\
&= \frac{A^4}{4} E[\cos^2 \omega\tau + \cos^2(2\omega t + \omega\tau + 2\theta) - 2\cos(\omega\tau)\cos(2\omega t + \omega\tau + 2\theta)] \\
&= \frac{A^4}{4} \cos^2 \omega\tau + \frac{A^4}{4} E[\cos^2(2\omega t + \omega\tau + 2\theta)] - 2 \frac{A^4}{4} E[\cos \omega\tau \cos(2\omega t + \omega\tau + 2\theta)] \\
&= \frac{A^4}{4} \cos^2 \omega\tau + \frac{A^4}{4} E\left[\frac{1 - \cos(4\omega t + 2\omega\tau + 4\theta)}{2}\right] - \frac{A^4}{2} \cos(\omega\tau) E[\cos(2\omega t + \omega\tau + 2\theta)]
\end{aligned}$$

Since $\theta \sim U(-\pi, \pi)$

$$E[\cos(2\omega t + \omega\tau + 2\theta)] = 0$$

and

$$E[\cos(4\omega t + 2\omega\tau + 4\theta)] = 0$$

$$\begin{aligned}
\Rightarrow R_{YY}(t, t + \tau) &= \frac{A^4}{4} \cos^2 \omega\tau + \frac{A^4}{4} \cdot \frac{1}{2} \\
&= \frac{A^4}{4} \left[\cos^2 \omega\tau + \frac{1}{2} \right]
\end{aligned}$$

$\Rightarrow Y(t)$ is also WSS.

6.11 A random process $Y(t) = X(t) - X(t + \tau)$ is defined in terms of a process $X(t)$ that is at least wide-sense stationary.

- Show that mean value of $Y(t)$ is zero even if $X(t)$ has a non-zero mean value.
- Show that $\sigma_Y^2 = 2[R_{XX}(0) - R_{XX}(\tau)]$.
- If $Y(t) = X(t) + X(t + \tau)$, find $E[Y(t)]$ and σ_Y^2 .

Solution The given random process is $Y(t) = X(t) - X(t + \tau)$ where $X(t)$ is a wide-sense stationary random process. The mean value of the random process $Y(t)$ is

$$\begin{aligned}
\mu_Y(t) &= E[Y(t)] = E[X(t) - X(t + \tau)] \\
&= E[X(t)] - E[X(t + \tau)] \\
E[X(t)] &= \int_{-\infty}^{\infty} x f_X(x, t) dx \\
E[X(t + \tau)] &= \int_{-\infty}^{\infty} x f_X(x, t + \tau) dx
\end{aligned}$$

For a stationary process, $f_X(t) = f_X(t + \tau)$.

Therefore, $E[X(t)] = E[X(t + \tau)]$ and hence $E[Y(t)] = 0$. Thus, the mean value of the random process $Y(t)$ is zero, even if $X(t)$ has a non-zero mean value.

The variance of the random process

$$\begin{aligned}\sigma_Y^2 &= E[Y^2(t)] - \{E[Y(t)]\}^2 \\ &= E[Y^2(t)] \quad \therefore E[Y(t)] = 0\end{aligned}$$

$$\begin{aligned}E[Y^2(t)] &= E[(X(t) - X(t + \tau))^2] \\ &= E[X^2(t) + X^2(t + \tau) - 2X(t)X(t + \tau)] \\ &= E[X^2(t)] + E[X^2(t + \tau)] - 2E[X(t)X(t + \tau)]\end{aligned}$$

$$E[X^2(t)] = \int_{-\infty}^{\infty} x^2 f_X(x, t) dx$$

$$\begin{aligned}E[X^2(t + \tau)] &= \int_{-\infty}^{\infty} x^2 f_X(x, t + \tau) dx \\ &= \int_{-\infty}^{\infty} x^2 f_X(x, t) dx = E[X^2(t)]\end{aligned}$$

$$\Rightarrow \sigma_Y^2 = 2E[X^2(t)] - 2R_{XX}(t, t + \tau)$$

Since $X(t)$ is a wide-sense stationary process,

$$R_{XX}(t, t + \tau) = R_{XX}(\tau)$$

$$\text{Also, } E[X^2(t)] = R_{XX}(0)$$

$$\begin{aligned}\text{Hence, } \sigma_Y^2 &= 2R_{XX}(0) - 2R_{XX}(\tau) \\ &= 2[R_{XX}(0) - R_{XX}(\tau)]\end{aligned}$$

$$\begin{aligned}\text{(c) If } Y(t) &= X(t) + X(t + \tau) \\ E[Y(t)] &= E[X(t)] + E[X(t + \tau)] = 2E[X(t)]\end{aligned}$$

$$\text{If } E[X(t)] = 0 \quad E[Y(t)] = 0$$

If $E[X(t)]$ is non-zero with a constant C then

$$E[Y(t)] = 2C$$

$$\begin{aligned}\sigma_Y^2 &= E[Y^2(t)] - \{E[Y(t)]\}^2 \\ &= E[(X(t) + X(t + \tau))^2] - (2C)^2 \\ &= E[X^2(t)] + E[X^2(t + \tau)] + 2E[X(t)X(t + \tau)] - 4C^2\end{aligned}$$

We know

$$E[X^2(t)] = E[X^2(t + \tau)] = R_{XX}(0)$$

and $E[X(t) X(t + \tau)] = R_{XX}(\tau)$

Therefore,

$$\sigma_Y^2 = 2R_{XX}(0) + 2R_{XX}(\tau) - 4C^2$$

For zero mean

$$\sigma_Y^2 = 2R_{XX}(0) + 2R_{XX}(\tau)$$

6.12 A random variable process $X(t) = \sin(\omega t + \phi)$ where ϕ is a random variable uniformly distributed in the interval $(0, 2\pi)$.

Prove that $\text{cov}[t, t + \tau] = R_{XX}(t, t + \tau) = \frac{\cos \omega \tau}{2}$.

Solution

$$\text{Given: } f_{\Phi}(\phi) = \begin{cases} \frac{1}{2\pi} & 0 \leq \phi \leq 2\pi \\ 0 & \text{otherwise} \end{cases}$$

$$\begin{aligned} R_{XX}(t, t + \tau) &= E[X(t) X(t + \tau)] \\ &= E[\sin(\omega t + \phi) \sin(\omega(t + \tau) + \phi)] \\ &= \frac{1}{2} E[\cos \omega \tau - \cos(2\omega t + \omega \tau + 2\phi)] \\ &= \frac{1}{2} \cos \omega \tau - \frac{1}{2} E[\cos(2\omega t + \omega \tau + 2\phi)] \\ E[\cos(2\omega t + \omega \tau + 2\phi)] &= \frac{1}{2\pi} \int_0^{2\pi} \cos(2\omega t + \omega \tau + 2\phi) d\phi = 0 \end{aligned}$$

$$R_{XX}(t, t + \tau) = \frac{1}{2} \cos \omega \tau = R_{XX}(\tau)$$

$$E[X(t)] = E[\sin(\omega t + \phi)] = \frac{1}{2\pi} \int_0^{2\pi} \sin(\omega t + \phi) d\phi = 0$$

$$\text{Similarly, } E[X(t + \tau)] = \frac{1}{2\pi} \int_0^{2\pi} \sin(\omega t + \omega \tau + \phi) d\phi = 0$$

$$\begin{aligned} \text{cov}(t, t + \tau) &= R_{XX}(t, t + \tau) - E[X(t)] E[X(t + \tau)] \\ &= R_{XX}(t, t + \tau) = \frac{1}{2} \cos \omega \tau \end{aligned}$$

6.13 A dice is tossed and corresponding to the dots $S = \{1, 2, 3, 4, 5, 6\}$, a random process $X(t)$ is formed with the following time functions

$$X(1, t) = -3; X(2, t) = 1; X(3, t) = 1 - t; X(4, t) = 1 + t; X(5, t) = 2 - t; X(6, t) = t - 2$$

Find $\mu_X(t)$, $E[X^2]$, $\sigma_X^2(t)$, $R_X(t_1, t_2)$, $C_X(t_1, t_2)$.

Solution The probability for the events is same and equal to $\frac{1}{6}$.

$$\begin{aligned}\mu_X(t) &= E[X(t)] = \sum_i p_i X_i(t) = \frac{1}{6} \sum_{i=1}^6 X_i(t) \\ &= \frac{1}{6} [-3 + 1 + 1 - t + 1 + t + 2 - t + t - 2] \\ &= 0\end{aligned}$$

Since mean is zero,

$$\begin{aligned}\sigma_X^2(t) &= E[X^2(t)] = \frac{1}{6} \sum_{i=1}^6 X_i^2(t) \\ &= \frac{1}{6} [(-3)^2 + 1 + (1-t)^2 + (1+t)^2 + (2-t)^2 + (t-2)^2] \\ &= \frac{1}{6} [9 + 1 + 1 + t^2 - 2t + 1 + t^2 + 2t + 4 + t^2 - 4t + t^2 - 4t + 4] \\ &= \frac{1}{6} [20 + 4t^2 - 8t]\end{aligned}$$

$$\begin{aligned}R_{XX}(t_1, t_2) &= E[X(t_1) X(t_2)] \\ &= \frac{1}{6} [(-3)(-3) + (1)(1) + (1-t_1)(1-t_2) + (1+t_1)(1+t_2) + (2-t_1)(2-t_2) + (t_1-2)(t_2-2)] \\ &= \frac{1}{6} [20 - 4t_1 - 4t_2 + 4t_1 t_2] \\ C_{XX}(t_1, t_2) &= R_{XX}(t_1, t_2)\end{aligned}$$

6.14 A random process consists of three sample functions $X(s_1, t_1) = 1$; $X(s_2, t_1) = 2 \sin t$ and $X(t_3, t) = 3 \cos t$, each occurring with equal probability. Is the process stationary in any sense?

Solution Let $x_1 = 1$; $x_2 = 2 \sin t$ and $x_3 = 3 \cos t$

We know

$$f_X(x) = \sum_{i=1}^N p_X(x_i) \delta(x - x_i)$$

Since each sample function occurs with equal probability, $p_X[x_i] = \frac{1}{3}$

$$\begin{aligned}f_X(x) &= \sum_{i=1}^3 p_X(x_i) \delta(x - x_i) \\ &= \frac{1}{3} \delta(x - x_1) + \frac{1}{3} \delta(x - x_2) + \frac{1}{3} \delta(x - x_3)\end{aligned}$$

$$\begin{aligned}
E[X(t)] &= \int_{-\infty}^{\infty} x f_X(x) dx \\
&= \int_{-\infty}^{\infty} x \left[\frac{1}{3} \delta(x-x_1) + \frac{1}{3} \delta(x-x_2) + \frac{1}{3} \delta(x-x_3) \right] dx \\
&= \frac{1}{3} x_1 + \frac{1}{3} x_2 + \frac{1}{3} x_3 \\
&= \frac{1}{3} + \frac{1}{3} 2 \sin t + \frac{1}{3} (3 \cos t) \\
&= \frac{1}{3} + \frac{2}{3} \sin t + \cos t
\end{aligned}$$

$$\int_{-\infty}^{\infty} x \delta(x-x_1) dx = x_1$$

Since the mean value is time dependent, $X(t)$ is not stationary in any sense.

6.15 A random process is defined by $X(t) = A$, where A is a continuous random variable uniformly distributed on $(0, 1)$. Find the mean and autocorrelation of the process.

Solution Given: $X(t) = A$; $A \sim U(0, 1)$

Therefore,

$$\begin{aligned}
E[X(t)] &= E[A] = \int_{-\infty}^{\infty} a f_A(a) da \\
&= \int_0^1 a da = \frac{a^2}{2} \Big|_0^1 = \frac{1}{2} = \frac{1}{2} \\
\mu_X &= \frac{1}{2} \\
R_{XX}(t, t + \tau) &= E[X(t)X(t + \tau)] \\
&= E[A.A] = E[A^2] \\
E[A^2] &= \int_{-\infty}^{\infty} a^2 f_A(a) da = \int_0^1 a^2 da = \frac{a^3}{3} \Big|_0^1 = \frac{1}{3}
\end{aligned}$$

6.16 For the random process described in the above problem, find first-order and second-order density functions.

Solution Given: $X(t) = A$

Let $f_A(a)$ be the density function of A . Since A is constant, the sample function does not change with time and

$$f_X(x_1; t_1) = f_A(x_1)$$

Also, $X(t_2) = X(t_1)$ then

$$f_X(x_2; t_2 | x_1; t_1) = \delta(x_2 - x_1)$$

$$\begin{aligned}
 f_X(x_1, x_2; t_1, t_2) &= \frac{f_X(x_2; t_2 | x_1; t_1)}{f_X(x_1; t_1)} \\
 &= f_A(x_1) \delta(x_2 - x_1)
 \end{aligned}$$

6.17 Given a random process $X(t) = Ae^{-at}$, where a is constant and $A \sim U(0, 1)$, find the mean and autocorrelation of $X(t)$.

Solution

- (a) Given: $X(t) = Ae^{-at}$ where a is constant. Also, $E[A] = 0$ and $\sigma_A^2 = 1$. Hence, $E[A^2] = 1$

$$\mu_X(t) = E[X(t)] = E[Ae^{-at}]$$

$$E[A] = \frac{b+a}{2} = \frac{1+0}{2} = 0.5$$

$$\text{Therefore, } \mu_X(t) = 0.5 e^{-at}$$

- (b) Autocorrelation

$$\begin{aligned}
 R_{XX}(t, t + \tau) &= E[X(t)X(t + \tau)] \\
 &= E[Ae^{-at}Ae^{-a(t+\tau)}] \\
 &= E[A^2 e^{-2at-a\tau}] = E[A^2] e^{-2at-a\tau} = 1 \cdot e^{-2at-a\tau} \\
 &= e^{-(2at+a\tau)}
 \end{aligned}$$

6.18 Find the mean, autocorrelation, variance and autocovariance of the random process $X(t) = tU$ where $U \sim U(0, 1)$.

Solution Given $X(t) = tU$. Also, $E[U] = 0$ and $\sigma_U^2 = 1$. Hence, $E[U^2] = 1$

$$\text{The mean } \mu_X(t) = \int_{-\infty}^{\infty} tU f(u) du$$

$$\begin{aligned}
 f(u) &= 1 \quad \text{for } 0 \leq u \leq 1 \\
 &= 0 \quad \text{otherwise}
 \end{aligned}$$

$$\mu_X(t) = \int_0^1 tU du = t \frac{u^2}{2} \Big|_0^1 = \frac{1}{2}t$$

$$E[X^2(t)] = \int_{-\infty}^{\infty} t^2 U^2 f(u) du = \int_0^1 t^2 U^2 du = t^2 \frac{u^3}{3} \Big|_0^1 = \frac{t^2}{3}$$

$$\begin{aligned}
 \sigma_X^2 &= E[X^2(t)] - [\mu_X(t)]^2 \\
 &= \frac{t^2}{3} - \left(\frac{t}{2}\right)^2 = \frac{t^2}{3} - \frac{t^2}{4} = \frac{t^2}{12}
 \end{aligned}$$

Autocorrelation

$$R_{XX}(t, t + \tau) = E[X(t)X(t + \tau)]$$

$$\begin{aligned}
&= E[t U(t + \tau) U] \\
&= E[U^2] E[t(t + \tau)] \\
&= t(t + \tau) \\
C_{XX}(t, t + \tau) &= R_{XX}(t, t + \tau) - \mu_X(t) \mu_X(t + \tau) \\
&= t(t + \tau) - \frac{1}{2} t \frac{(t + \tau)}{2} \\
&= \frac{3}{4} t(t + \tau)
\end{aligned}$$

6.19 A stochastic process is described by $X(t) = A \sin t + B \cos t$ where A and B are independent random variables with zero mean and equal standard deviation. Show that the process is stationary of the second order.

Solution Given: $X(t) = A \sin t + B \cos t$

And also

$$E[A] = E[B] = 0 \quad \sigma_A^2 = \sigma_B^2 \Rightarrow E[A^2] = E[B^2]$$

$$\begin{aligned}
E[X(t)] &= E[A \sin t + B \cos t] \\
&= E[A] \sin t + E[B] \cos t \\
&= 0
\end{aligned}$$

$$\begin{aligned}
R_{XX}(t, t + \tau) &= E[X(t) X(t + \tau)] = E[(A \sin t + B \cos t)(A \sin(t + \tau) + B \cos(t + \tau))] \\
&= E[A^2] \sin t \sin(t + \tau) + E[AB] \cos t \sin(t + \tau) + E[AB] \sin t \cos(t + \tau) + E[B^2] \cos t \cos(t + \tau)
\end{aligned}$$

We know $E[AB] = 0$

$$\begin{aligned}
E[A^2] &= E[B^2] = \sigma_A^2 = \sigma_B^2 = \sigma^2 \\
\Rightarrow R_{XX}(t, t + \tau) &= \sigma^2 \cos t \cos(t + \tau) + \sigma^2 \sin t \sin(t + \tau) \\
&= \sigma^2 \cos(t + \tau - t) \\
&= \sigma^2 \cos \tau
\end{aligned}$$

6.20 Consider a random process $X(t)$ defined by $X(t) = U \cos t + (V + 1) \sin t$, where U and V are independent random variables for which $E[U] = E[V] = 0$, $E[U^2] = E[V^2] = 1$

- Find the autocovariance function of $X(t)$.
- Is $X(t)$ wide sense stationary? Explain your answer.

Solution Given: $X(t) = U \cos t + (V + 1) \sin t$

Also, $E[U] = E[V] = 0$ and $E[U^2] = E[V^2] = 1$

Since U and V are independent,

$$\begin{aligned}
E[UV] &= E[U] E[V] = 0 \\
E[X(t)] &= E[U \cos t + (V + 1) \sin t]
\end{aligned}$$

$$\begin{aligned}
&= E[U] \cos t + E[V] \sin t + E[\sin t] \\
&= \sin t
\end{aligned}$$

That is, $E[X(t)]$ is a function of time t . Hence, $X(t)$ is not first-order stationary.

The autocovariance function

$$\begin{aligned}
C_{XX}(t, t + \tau) &= R_{XX}(t, t + \tau) = E[X(t) X(t + \tau)] \\
&= E\{[U \cos t + (V + 1) \sin t][U \cos(t + \tau) + (V + 1) \sin(t + \tau)]\} \\
&= E[U^2] \cos t \cos(t + \tau) + E[UV] \sin t \cos(t + \tau) + E[U] \sin t \cos(t + \tau) \\
&\quad + E[UV] \cos t \sin(t + \tau) + E[U] \cos t \sin(t + \tau) + E[(V + 1)^2] \sin t \sin(t + \tau) \\
&= \cos t \cos(t + \tau) + \sin t \sin(t + \tau) + \sin t \sin(t + \tau) \\
&= \cos \tau + \sin t \sin(t + \tau)
\end{aligned}$$

Since autocorrelation function depends on t , $X(t)$ is not WSS.



Practice Problems

6.5 Consider a random process $X(t) = B \cos(50t + \phi)$ where B and ϕ are independent random variables. B is a random variable with zero mean and unit variance. ϕ is uniformly distributed in the interval $(-\pi, \pi)$. Find the mean and autocorrelation of the process.

6.6 Given a stationary random process $X(t) = 10 \cos(100t + \theta)$ where $\theta \in (-\pi, \pi)$ follows uniform distribution. Find the mean and autocorrelation functions of the process.

Solved Problems



6.21 Consider two random processes

$$Y_1(t) = X(t) \cos(\omega_0 t)$$

$$Y_2(t) = Y(t) \cos(\omega_0 t + \theta)$$

where $X(t)$ and $Y(t)$ are jointly wide-sense stationary processes.

- If θ is non-random, is there any value of θ that will make $Y_1(t)$ and $Y_2(t)$ orthogonal?
- If θ is a uniform random variable, statistically independent of $X(t)$ and $Y(t)$, are there any conditions on θ that will make $Y_1(t)$ and $Y_2(t)$ orthogonal?

Solution

- (a) Given: $Y_1(t) = X(t) \cos(\omega_0 t)$

$$Y_2(t) = Y(t) \cos(\omega_0 t + \theta)$$

Since $X(t)$ and $Y(t)$ are jointly wide-sense stationary,

$$\begin{aligned}
R_{XY}(t, t + \tau) &= R_{XY}(\tau) \\
&= R_{Y_1 Y_2}(t, t + \tau) = E[Y_1(t) Y_2(t + \tau)] \\
&= E[X(t) \cos \omega_0 t Y(t + \tau) \cos(\omega_0 t + \omega_0 \tau + \theta)]
\end{aligned}$$

$$\begin{aligned}
&= E[X(t) Y(t + \tau) \cos \omega_0(t) \cos(\omega_0 t + \omega_0 \tau + \theta)] \\
&= \frac{R_{XY}(\tau)}{2} \{E[\cos(\omega_0 \tau + \theta)] + E[\cos(2\omega_0 t + \omega_0 \tau + \theta)]\}
\end{aligned} \tag{6.60}$$

Since θ is non-random, $E[\cos(\omega_0 \tau + \theta)] = \cos(\omega_0 \tau + \theta)$

and

$$E[\cos(2\omega_0 t + \omega_0 \tau + \theta)] = \cos(2\omega_0 t + \omega_0 \tau + \theta)$$

$$\Rightarrow R_{Y_1 Y_2}(t, t + \tau) = \frac{R_{XY}(\tau)}{2} [\cos(\omega_0 \tau + \theta) + \cos(2\omega_0 t + \omega_0 \tau + \theta)]$$

Since θ is a constant, there is no value of θ that makes

$$\cos(\omega_0 \tau + \theta) = -\cos(2\omega_0 t + \omega_0 \tau + \theta)$$

Hence, $Y_1(t)$ and $Y_2(t)$ are not orthogonal for any value of θ .

(b) Let θ is a uniform random variable on (θ_1, θ_2) . Then

$$\begin{aligned}
f_{\Theta}(\theta) &= \frac{1}{\theta_2 - \theta_1} & \theta_1 \leq \theta \leq \theta_2 \\
&= 0 & \text{otherwise}
\end{aligned}$$

From Eq. (6.60),

$$\begin{aligned}
R_{Y_1 Y_2}(t, t + \tau) &= \frac{R_{XY}(\tau)}{2} \{E[\cos(\omega_0 \tau + \theta)] + E[\cos(2\omega_0 t + \omega_0 \tau + \theta)]\} \\
&= \frac{R_{XY}(\tau)}{2} \left\{ \int_{-\infty}^{\infty} \cos(\omega_0 \tau + \theta) f_{\Theta}(\theta) d\theta + \int_{-\infty}^{\infty} \cos(2\omega_0 t + \omega_0 \tau + \theta) f_{\Theta}(\theta) d\theta \right\} \\
&= \frac{R_{XY}(\tau)}{2} \left\{ \frac{1}{\theta_2 - \theta_1} \int_{\theta_1}^{\theta_2} \cos(\omega_0 \tau + \theta) d\theta + \frac{1}{\theta_2 - \theta_1} \int_{\theta_1}^{\theta_2} \cos(2\omega_0 t + \omega_0 \tau + \theta) d\theta \right\} \\
&= \frac{R_{XY}(\tau)}{2(\theta_2 - \theta_1)} \{ \sin(\omega_0 \tau + \theta) \Big|_{\theta_1}^{\theta_2} + \sin(2\omega_0 t + \omega_0 \tau + \theta) \Big|_{\theta_1}^{\theta_2} \} \\
&= \frac{R_{XY}(\tau)}{2(\theta_2 - \theta_1)} \{ \sin(\omega_0 \tau + \theta_2) - \sin(\omega_0 \tau + \theta_1) + \sin((2\omega_0 t + \omega_0 \tau + \theta_2)) - \sin((2\omega_0 t + \omega_0 \tau + \theta_1)) \} \\
&= \frac{R_{XY}(\tau)}{2(\theta_2 - \theta_1)} \left\{ 2 \cos \left[\omega_0 \tau + \frac{(\theta_1 + \theta_2)}{2} \right] \sin \left[\frac{(\theta_2 - \theta_1)}{2} \right] + 2 \cos \left[2\omega_0 t + \omega_0 \tau + \frac{(\theta_1 + \theta_2)}{2} \right] \sin \left(\frac{(\theta_2 - \theta_1)}{2} \right) \right\}
\end{aligned}$$

The above equation is zero,

$$\text{when } \sin \left(\frac{(\theta_2 - \theta_1)}{2} \right) = 0$$

$$\Rightarrow \left(\frac{(\theta_2 - \theta_1)}{2} \right) = \pi \quad \Rightarrow \theta_2 - \theta_1 = 2\pi$$

That is, $\theta \sim U(0, 2\pi)$ then $Y_1(t)$ and $Y_2(t)$ are orthogonal.

6.22 Consider a random process $Y(t) = X(t) \cos(\omega_0 t + \theta)$ where $X(t)$ is second-order stationary, and θ is a random variable independent of $X(t)$ and uniform on $(0, 2\pi)$. The above process is applied to a square-law device and produces an output $W(t)$.

Find (a) $E[W(t)]$, (b) $R_{WW}(t, t + \tau)$, and (c) whether or not $W(t)$ is wide-sense stationary.

Solution Given: $W(t) = Y^2(t) = X^2(t) \cos^2(\omega_0 t + \theta)$

$$E[W(t)] = E[X^2(t) \cos^2(\omega_0 t + \theta)]$$

$$X(t) \text{ is uniform on } (0, 2\pi) \Rightarrow f_{\Theta}(\theta) = \begin{cases} \frac{1}{2\pi} & 0 \leq \theta \leq 2\pi \\ 0 & \text{otherwise} \end{cases}$$

Also, $X(t)$ is independent of θ .

$$E[W(t)] = E[X^2(t)] E[\cos^2(\omega_0 t + \theta)]$$

$$E[X(t)] = \frac{2\pi}{2} = \pi; \quad \sigma_X^2 = \frac{(2\pi - 0)^2}{12} = \frac{\pi^2}{3}$$

$$E[X^2(t)] = \sigma_X^2 + \{E[X]\}^2 = \frac{\pi^2}{3} + \pi^2 = \frac{4\pi^2}{3}$$

$$\begin{aligned} E[W(t)] &= \frac{4\pi^2}{3} E[\cos^2(\omega_0 t + \theta)] \\ &= \frac{4\pi^2}{3} \int_{-\infty}^{\infty} \cos^2(\omega_0 t + \theta) f_{\Theta}(\theta) d\theta \\ &= \frac{4\pi^2}{3} \left(\frac{1}{2\pi} \right) \int_0^{2\pi} \cos^2(\omega_0 t + \theta) d\theta \\ &= \frac{2\pi}{3} \int_0^{2\pi} \left(\frac{1 + \cos 2(\omega_0 t + \theta)}{2} \right) d\theta \\ &= \frac{\pi}{3} \left\{ [2\pi] + \underbrace{\int_0^{2\pi} \cos 2(\omega_0 t + \theta) d\theta}_0 \right\} \\ &= \frac{2\pi^2}{3} \end{aligned}$$

$$R_{WW}(t, t + \tau) = E[W(t) W(t + \tau)]$$

$$= E[X^2(t) \cos^2(\omega_0 t + \theta) X^2(t + \tau) \cos^2(\omega_0 t + \omega_0 \tau + \theta)]$$

$$= E[X^2(t) X^2(t + \tau)] E[\cos^2(\omega_0 t + \theta) \cos^2(\omega_0 t + \omega_0 \tau + \theta)]$$

Consider

$$\begin{aligned}
 & E[\cos^2(\omega_0 t + \theta) \cos^2(\omega_0 t + \omega_0 \tau + \theta)] \\
 &= E\left\{\left[\frac{1 + \cos 2(\omega_0 t + \theta)}{2}\right] \left[\frac{1 + \cos 2(\omega_0 t + \omega_0 \tau + \theta)}{2}\right]\right\} \\
 &= \frac{1}{4} \left\{ \begin{aligned} & 1 + E[\cos 2(\omega_0 t + \theta)] + E[\cos 2(\omega_0 t + \omega_0 \tau + \theta)] \\ & + E[\cos 2(\omega_0 t + \theta) \cos 2(\omega_0 t + \omega_0 \tau + \theta)] \end{aligned} \right\} \\
 &= \frac{1}{4} \{1 + E[\cos 2(\omega_0 t + \theta) \cos 2(\omega_0 t + \omega_0 \tau + \theta)]\} \quad \boxed{E[\cos 2(\omega_0 t + \theta)] = E[\cos 2(\omega_0 t + \omega_0 \tau + \theta)] = 0} \\
 &= \frac{1}{4} \left[1 + \frac{1}{2} E[\cos 2\omega_0 \tau + (\cos 4\omega_0 t + 2\omega_0 \tau + 2\theta)] \right] \\
 &= \frac{1}{4} \left[1 + \frac{1}{2} \cos 2\omega_0 \tau \right] \quad \text{Since } E[\cos 4\omega_0 t + 2\omega_0 \tau + 2\theta] = 0 \\
 &= \frac{1}{4} \left[1 + \frac{1}{2} (2 \cos^2 \omega_0 \tau - 1) \right] \\
 &= \frac{1}{4} \left[\frac{1}{2} + \cos^2 \omega_0 \tau \right] \\
 \Rightarrow \quad R_{WW}(t, t + \tau) &= E[X^2(t) X^2(t + \tau)] \left\{ \frac{1}{4} \left(\frac{1}{2} + \cos^2 \omega_0 \tau \right) \right\} \\
 &= \frac{1}{4} E[X^2(t) X^2(t + \tau)] \left\{ \frac{1}{2} + \cos^2 \omega_0 \tau \right\}
 \end{aligned}$$

Since $R_{WW}(t, t + \tau)$ depends on τ , the process $W(t)$ is WSS.

Solved Problem



6.23 If $X(t)$ is a stationary random process having a mean value $E[X(t)] = 3$ and autocorrelation function $R_{XX}(\tau) = 9 + e^{-2(\tau)}$, find the mean value and the variance of the random variable $Y = \int_0^2 X(t) dt$.

Solution Given: $Y = \int_0^2 X(t) dt$

$$E[Y] = E\left[\int_0^2 X(t) dt\right] = \int_0^2 E[X(t)] dt$$

$$\begin{aligned}
&= \int_0^2 3 dt = 3t \Big|_0^2 = 6 \\
E[Y^2] &= E \left[\int_0^2 X(t) dt \int_0^2 X(u) du \right] \\
&= \int_0^2 \int_0^2 E[X(t) X(u)] du dt = \int_0^2 \int_0^2 R_{XX}[t-u] dt du \\
&= \int_0^2 \int_0^2 9 + 2e^{-|t-u|} dt du = 36 + 2 \int_0^2 \int_0^2 e^{-|t-u|} dt du \\
&= 36 + 2 \left\{ \int_0^2 \int_0^t e^{-(t-u)} du dt + \int_0^2 \int_t^2 e^{t-u} dt du \right\} \\
&= 36 + 2(1 + e^{-2} + e^{-2}) \\
&= 40 + 4e^{-2} \\
\sigma_Y^2 &= E[Y^2] - \{E[Y]\}^2 \\
&= 40 + 4e^{-2} - (6)^2 \\
&= 4[1 + e^{-2}]
\end{aligned}$$



Practice Problems

- 6.7** A random process is given by $Z(t) = At^3 + B$ where A and B are independent random variables. Find $\mu_z(t)$.
- 6.8** Let $X(t) = A \cos \omega t + B \sin \omega t$ where A and B are Gaussian random variables with zero mean and variance σ^2 . Find the mean and autocorrelation of $X(t)$.
- 6.9** If $X(t) = A \sin(\omega t + \theta)$ where A and ω are constants and θ is a random variable uniformly distributed over $(-\pi, \pi)$, find the autocorrelation function of $X^2(t)$.

$$\left(\text{Ans: } R_{XX}(t_1, t_2) = \frac{A^2}{8} [2 + \cos 2\omega(t_1 - t_2)] \right)$$



TIME AVERAGES AND ERGODICITY 6.16

For a random process $X(t)$, the mean and autocorrelation functions are calculated using the ensemble average. To obtain ensemble average, we have to consider all possible realizations. But in most of the processes we may not be able to observe all realizations and in fact, we may have to be satisfied with a Single realization. This will happen in the experiments where it may not be possible to take replicate measurements. In these cases, the mean and autocorrelation function are obtained using a single realization of $X(t)$, and these values are assigned to statistical average of the ensemble.

In some other random processes, the mean and autocorrelation functions of every sample function of the ensemble is same as the ensemble. This implies that it is possible to determine the statistical behaviour of the ensemble from a sample function. In both cases, statistical behaviour of the ensemble can be obtained from only one sample function. These processes where the time average and statistical average are equal are known as *ergodic processes*. Thus, for an ergodic process, time average and statistical average are equal. That is, $\mu_x = \mu_X$ and $R_{xx}(\tau) = R_{XX}(\tau)$.

6.16.1 Time Average Function

The time average of a quantity is defined as

$$A[.] = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T [.] dt \quad (6.61)$$

where the operator A denote times average in a manner analogous to E for the statistical average.

The time average function of the sample function $x(t)$ is defined as

$$\mu_x = \bar{x} = A[x(t)] = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T x(t) dt \quad (6.62)$$

The time correlation function of the sample function $x(t)$ is defined as

$$\begin{aligned} R_{xx}(\tau) &= A[x(t)x(t+\tau)] \\ &= \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T x(t)x(t+\tau) dt \end{aligned} \quad (6.63)$$

For any random process $X(t)$, the values $\bar{x}(t)$ and $R_{xx}(\tau)$ are random variables. If we consider all sample functions then the ensemble average μ_X can be obtained by finding expectation of $\bar{x}$.

$$\begin{aligned} \mu_X = \bar{X} &= E[\bar{x}] = E \left[\lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T x(t) dt \right] \\ &= \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T E[x(t)] dt = \bar{x} \end{aligned} \quad (6.64)$$

Similarly, the ensemble autocorrelation function is obtained from the expectation of $R_{xx}(\tau)$. That is,

$$R_{XX}(\tau) = E[R_{xx}(\tau)] \quad (6.65)$$

$$= E \left[\lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T x(t)x(t+\tau) dt \right] = R_{xx}(\tau) \quad (6.66)$$

If the variances of $\bar{x}$ and $R_{XX}(\tau)$ are made to zero then the ensemble averages are equal to time average. Then we could write

$$\mu_x = \mu_X \quad (6.67)$$

$$R_{xx}(t) = R_{XX}(\tau) \quad (6.68)$$

The ergodic theorem allows the validity of the above two equations which states that “all time averages are equal to the corresponding statistical average”. The random processes that satisfy the ergodic theorem are called *ergodic processes*.

Two random processes are called *jointly ergodic* if they are individually ergodic and their time cross-correlation function is equal to the statistical correlation function

$$\begin{aligned} R_{xy}(\tau) &= \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T x(t) y(t + \tau) dt \\ &= R_{xy}(\tau) \end{aligned} \quad (6.69)$$

Definitions

Ergodic Process A random process is said to be ergodic if all time averages are equal to the corresponding statistical averages.

Jointly Ergodic Processes Two random processes are jointly ergodic if they are individually ergodic and also have a time correlation function that equals the statistical cross-correlation function.

$$R_{xy}(\tau) = \frac{1}{2T} \int_{-T}^T x(t) y(t + \tau) dt = R_{xy}(\tau)$$

6.16.2 Mean Ergodic Processes

A random process $X(t)$ is said to be mean ergodic or ergodic in mean if time average of a sample function μ_x is equal to the statistical average μ_X with probability 1 for all $x(t)$. That is, $\mu_x = \mu_X$

To prove the above statement, we assume

- (a) $X(t)$ has a finite constant mean $\bar{X}$ for all t .
- (b) $X(t)$ is bounded, that is $|X(t)| < \infty$.
- (c) $\lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T E[|X(t)|] dt < \infty$.
- (d) $X(t)$ is a regular process, that is $E[|X(t)|^2] = R_{XX}(t, t) < \infty$.

Consider the time average of the random process given by

$$\bar{A}_X = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T X(t) dt \quad (6.70)$$

Since A_X is a random variable, we can find the mean value as

$$\begin{aligned} \bar{A}_X &= E[A_X] \\ &= E \left[\lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T X(t) dt \right] \\ &= \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T \bar{X} dt = \bar{X} \end{aligned} \quad (6.71)$$

Consider the Chebyshev's inequality which is stated as

$$P[|A_X - E[A_X]| < \varepsilon] \geq 1 - \frac{\text{Var}(A_X)}{\varepsilon^2} \quad \text{any } \varepsilon > 0 \quad (6.72)$$

where $\text{Var}[A_X]$ is variance of A_X .

$$\begin{aligned} \lim_{T \rightarrow \infty} P[|A_X - \bar{A}_X| < \varepsilon] &\geq \lim_{T \rightarrow \infty} 1 - \frac{\text{Var}(A_X)}{\varepsilon^2} \\ &= 1 - \frac{\lim_{T \rightarrow \infty} \text{Var}(A_X)}{\varepsilon^2} \end{aligned} \quad (6.73)$$

When $\lim_{T \rightarrow \infty} \text{Var}(A_X) \rightarrow 0$, we get

$$P\left[\lim_{T \rightarrow \infty} |A_X - \bar{A}_X| < \varepsilon\right] \geq 1 \quad (6.74)$$

That is, any $\varepsilon > 0$, if $\text{Var}(A_X) = 0$ then $A_X = \bar{A}_X$ with probability 1.

So to prove that $X(t)$ is mean ergodic, we have to prove that

$$\lim_{T \rightarrow \infty} \text{Var}[A_X(t)] = 0 \quad (6.75)$$

$$\text{Var}(A_X) = E[(A_X - \bar{A}_X)^2] \quad (6.76)$$

$$\begin{aligned} &= E\left\{\left[\lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T [X(t) - \bar{X}] dt\right]^2\right\} \\ &= E\left[\lim_{T \rightarrow \infty} \left(\frac{1}{2T}\right)^2 \int_{-T}^T \int_{-T}^T [X(t) - \bar{X}][X(t_1) - \bar{X}] dt dt_1\right] \\ &= \lim_{T \rightarrow \infty} \left(\frac{1}{2T}\right)^2 \int_{-T}^T \int_{-T}^T C_{XX}(t, t_1) dt dt_1 \end{aligned} \quad (6.77)$$

Thus, the condition $\lim_{T \rightarrow \infty} \text{Var}[A_X(t)] = 0$ implies

$$\lim_{T \rightarrow \infty} \frac{1}{4T^2} \int_{-T}^T \int_{-T}^T C_{XX}(t, t_1) dt dt_1 = 0 \quad (6.78)$$

If $X(t)$ is a WSS process then

$$C_{XX}(t, t_1) = C_{XX}(\tau)$$

where $\tau = t_1 - t$

We have $d\tau = dt_1$

and when $t_1 = -T$, $\tau = -T - t$; $t_1 = T$, $\tau = T - t$

Therefore, Eq. (6.78) can be written as

$$\text{Var}(A_X) = \lim_{T \rightarrow \infty} \frac{1}{4T^2} \int_{t=-T}^T \int_{\tau=-T-t}^{T-t} C_{XX}(\tau) d\tau dt \quad (6.79)$$

The above double integral is evaluated over a region shown in Fig. 6.16. It consists of two vertical lines at $t = T$ and $t = -T$. Similarly, $\tau = -T - t$ and $\tau = T - t$ are two straight lines with slope equal to -1 .

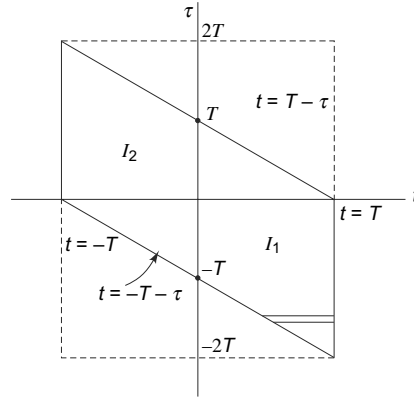


Fig. 6.16 Evaluating the double integral

The total area can be divided into two regions I_1 and I_2 . The area I_1 can be obtained by moving the horizontal strip from bottom to x -axis where $\tau_{\min} = -2T$ and $\tau_{\max} = 0$. At the same time, t varies from $(-T - \tau)$ to T . Similarly, for the area I_2 , τ varies from 0 to $2T$ and t varies from $-T$ to $(T - \tau)$. Hence we can write

$$\text{Var}(A_X) = \lim_{T \rightarrow \infty} \frac{1}{4T^2} \left\{ \int_{\tau=-2T}^0 \int_{t=-T-\tau}^T C_{XX}(\tau) dt d\tau + \int_{\tau=0}^{2T} \int_{t=-T}^{T-\tau} C_{XX}(\tau) dt d\tau \right\} \quad (6.80)$$

Consider I_1 ,

$$\begin{aligned} I_1 &= \int_{\tau=-2T}^0 \int_{t=-T-\tau}^T C_{XX}(\tau) dt d\tau \\ &= \int_{\tau=-2T}^0 [T - (-T - \tau)] C_{XX}(\tau) d\tau \\ &= \int_{-2T}^0 (2T + \tau) C_{XX}(\tau) d\tau \end{aligned}$$

Consider I_2 ,

$$I_2 = \int_0^{2T} \int_{-T}^{T-\tau} C_{XX}(\tau) dt d\tau$$

$$\begin{aligned}
&= \int_0^{2T} C_{XX}(\tau)[T - \tau + T]d\tau \\
&= \int_0^{2T} C_{XX}(\tau)[2T - \tau]d\tau \\
\text{Var}(A_X) &= \frac{1}{4T^2} \left[\int_{-2T}^0 (2T + \tau) C_{XX}(\tau) d\tau + \int_0^{2T} (2T - \tau) C_{XX}(\tau) d\tau \right] \\
&= \frac{2T}{4T^2} \left[\int_{-2T}^0 \left(1 + \frac{\tau}{2T}\right) C_{XX}(\tau) d\tau + \int_0^{2T} \left(1 - \frac{\tau}{2T}\right) C_{XX}(\tau) d\tau \right] \quad (6.81)
\end{aligned}$$

Since $C_{XX}(\tau) = C_{XX}(-\tau)$, we can write

$$\text{Var}(A_T) = \frac{2}{2T} \left[\int_0^{2T} \left(1 - \frac{|\tau|}{2T}\right) C_{XX}(\tau) d\tau \right]$$

Thus, the sufficient condition for mean ergodicity of a process $X(t)$ is

$$\sigma_{A_X}^2 = \text{Var}(A_X) = \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^{2T} \left[1 - \frac{|\tau|}{2T}\right] C_{XX}(\tau) d\tau = 0 \quad (6.82)$$

which also can be stated as

$$\lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-2T}^{2T} \left[1 - \frac{|\tau|}{2T}\right] C_{XX}(\tau) d\tau = 0 \quad (6.83)$$

A discrete random process is said to be mean ergodic if the statistical average of the sequence and time average of the samples are equal with probability 1. The condition for mean ergodic is

$$\lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-2N}^{2N} \left[1 - \frac{|n|}{2N+1}\right] C_{XX}(\tau) = 0 \quad (6.84)$$

6.16.3 Correlation Ergodic Processes

A stationary continuous random process $X(t)$ with autocorrelation function $R_{XX}(\tau)$ is said to be correlation ergodic or ergodic in the autocorrelation if and only if

$$\lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T X(t)X(t+\tau) = R_{XX}(\tau) \quad \text{for all } \tau \quad (6.85)$$

Let us define a process $Y(t) = X(t)X(t+\lambda)$, where λ is a time offset

$$E[Y(t)] = E[X(t)X(t+\lambda)] = R_{XX}(\lambda) \quad (6.86)$$

and

$$\begin{aligned}
R_{YY}(\tau) &= E[Y(t)Y(t+\tau)] \\
&= E[X(t)X(t+\lambda)X(t+\tau)X(t+\tau+\lambda)] \quad (6.87)
\end{aligned}$$

$$\begin{aligned}
C_{YY}(\tau) &= E[Y(t)Y(t+\tau)] - E[Y(t)]E[Y(t+\tau)] \\
&= R_{YY}(\tau) - \{E[Y(t)]\}^2 \\
&= R_{YY}(\tau) - R_{XX}^2(\lambda)
\end{aligned} \tag{6.88}$$

Thus, $X(t)$ is autocorrelation ergodic if the integral

$$\lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-2T}^{2T} \left[1 - \frac{|\tau|}{2T}\right] C_{YY}(\tau) d\tau = 0 \tag{6.89}$$

From which we can write

$$\lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-2T}^{2T} \left[1 - \frac{|\tau|}{2T}\right] [R_{YY}(\tau) - R_{XX}^2(\lambda)] d\tau = 0 \tag{6.90}$$

REVIEW QUESTIONS



25. Define ensemble average and time average of a random process.
26. State mean ergodic theorem.
27. When is the process said to be ergodic in mean?
28. When is a random process said to be correlation ergodic?

Solved Problems



6.24 Is the process $X(t)$ with

$$R_{XX}(t) = \begin{cases} 0, & |\tau| > 1 \\ 4(1-|\tau|), & |\tau| < 1 \end{cases}$$

mean ergodic?

Solution

$$\begin{aligned}
\text{Var}(A_X) &= \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-2T}^{2T} \left(1 - \frac{|\tau|}{2T}\right) R_{XX}(\tau) d\tau \\
&= \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-2T}^{2T} \left(1 - \frac{|\tau|}{2T}\right) 4(1-|\tau|) d\tau \\
&= \lim_{T \rightarrow \infty} \frac{1}{2T} \left\{ \int_{-2T}^0 4\left(1 + \frac{\tau}{2T}\right)(1+\tau) d\tau + \int_0^{2T} 4\left(1 - \frac{\tau}{2T}\right)(1-\tau) d\tau \right\} \\
&= \lim_{T \rightarrow \infty} \frac{1}{2T} \left\{ 4 \int_{-1}^0 (1+\tau)^2 d\tau + 4 \int_0^1 (1-\tau)^2 d\tau \right\}
\end{aligned}$$

$$\begin{aligned}
&= \lim_{T \rightarrow \infty} \frac{1}{2T} \left\{ 4 \int_{-1}^0 (1 + \tau^2 + 2\tau) d\tau + 4 \int_0^1 (1 + \tau^2 - 2\tau) d\tau \right\} \\
&= \lim_{T \rightarrow \infty} \frac{1}{2T} \left\{ 4 \left(\tau + \frac{\tau^3}{3} + \tau^2 \right) \Big|_{-1}^0 + 4 \left(\tau + \frac{\tau^3}{3} - \tau^2 \right) \Big|_0^1 \right\} \\
&= \lim_{T \rightarrow \infty} \frac{1}{2T} \left\{ 4 \left(1 + \frac{1}{3} - 1 \right) + 4 \left[1 + \frac{1}{3} - 1 \right] \right\} \\
&= \lim_{T \rightarrow \infty} \frac{1}{2T} \left(\frac{8}{3} \right) = 0
\end{aligned}$$

Therefore $X(t)$ is mean/ergodic.

6.25 $X(t) = A + B \sin(\omega_0 t + \phi)$ where B , A and ϕ are independent random variables. $\phi \sim U(0, 2\pi)$, A and B are uniformly distributed over $(0, 1)$ and $(0, 2)$ respectively. Check whether $X(t)$ is mean ergodic.

Solution Given: $X(t) = A + B \sin(\omega t + \phi)$

$$\begin{aligned}
\mu_X &= E[X(t)] = E[A] + E[B \sin(\omega t + \phi)] \\
&= E[A] + E[B] E[\sin(\omega t + \phi)]
\end{aligned}$$

Since $\phi \sim U(0, 2\pi)$

$$E[\sin(\omega t + \phi)] = 0$$

$$\Rightarrow \mu_X = E[A] = 0$$

$$\begin{aligned}
R_{XX}(t, t + \tau) &= E[X(t) X(t + \tau)] \\
&= E[\{A + B \sin \omega t + \phi\} \{A + B \sin(\omega t + \omega \tau + \phi)\}] \\
&= E[A^2] + E[\underbrace{AB \sin(\omega t + \phi)}_{=0}] + E[\underbrace{AB \sin(\omega t + \omega \tau + \phi)}_{=0}] \\
&\quad + E[B^2 \sin(\omega t + \phi) \sin(\omega t + \omega \tau + \phi)]
\end{aligned}$$

$$= E[A^2] + \frac{1}{2} E[B^2] E[\cos \omega \tau - \cos(\omega t + \omega \tau + 2\phi)]$$

$$= E[A^2] + \frac{1}{2} E[B^2] \cos \omega \tau$$

Hence, the process $X(t)$ is stationary

$$\begin{aligned}
C_{XX}(\tau) &= R_{XX}(\tau) - E[X(t)] E[X(t + \tau)] \\
&= R_{XX}(\tau) - \{E[X(t)]\}^2 \\
&= E[A^2] + \frac{1}{2} E[B^2] \cos \omega \tau
\end{aligned}$$

$$\begin{aligned}
&= \sigma_A^2 + \frac{1}{2} E[B^2] \cos \omega \tau \\
\text{Var}(A_x) &= \lim_{T \rightarrow \infty} \frac{1}{2T} \left\{ \int_{-2T}^{2T} C_{XX}(\tau) \left[1 - \frac{|\tau|}{2T} \right] d\tau \right\} \\
&= \lim_{T \rightarrow \infty} \frac{1}{2T} \left\{ \int_{-2T}^{2T} \left[\sigma_A^2 + \frac{1}{2} E[B^2] \cos \omega \tau \right] \left[1 - \frac{|\tau|}{2T} \right] d\tau \right\} \\
&= \lim_{T \rightarrow \infty} \frac{1}{2T} \left\{ \int_{-2T}^0 \left[\sigma_A^2 + \frac{1}{2} E[B^2] \cos \omega \tau \right] \left[1 + \frac{\tau}{2T} \right] d\tau \right. \\
&\quad \left. + \int_0^{2T} \left[\sigma_A^2 + \frac{1}{2} E[B^2] \cos \omega \tau \right] \left[1 - \frac{\tau}{2T} \right] d\tau \right\} \\
&= \lim_{T \rightarrow \infty} \frac{1}{2T} \left\{ \sigma_A^2 4T + \frac{1}{2} E[B^2] \frac{\sin 2T\omega}{\omega} - \sigma_A^2 T + \frac{1}{4T} E[B^2] \frac{\sin^2 \omega T}{\omega^2} \right. \\
&\quad \left. - \sigma_A^2 T + \frac{1}{4T} E[B^2] \frac{\sin^2 \omega T}{\omega^2} \right\} \\
&= \lim_{T \rightarrow \infty} \frac{1}{2T} \left[2T \sigma_A^2 + \frac{1}{2T} E[B^2] \frac{\sin^2 \omega T}{\omega^2} \right] \\
&= \lim_{T \rightarrow \infty} \left[\sigma_A^2 + \frac{1}{4T^2} E[B^2] \frac{\sin^2 \omega T}{\omega^2} \right] \\
&= \lim_{T \rightarrow \infty} \left[\sigma_A^2 + \frac{1}{4} E[B^2] \frac{\sin^2 \omega T}{\omega^2 T^2} \right] \\
&= \sigma_A^2
\end{aligned}$$

Therefore, $X(t)$ is not mean-ergodic

6.26 Let $N(t)$ be a zero mean wide-sense stationary noise process for which $R_{NN}(\tau) = \left(\frac{N_0}{2} \right) \delta(\tau)$ where $N_0 > 0$ is a finite constant. Determine if $N(t)$ is mean ergodic.

Solution Given: $\mu_N = 0$. Therefore, $C_{NN}(\tau) = R_{NN}(\tau)$.

The process $N(t)$ is mean ergodic if the variance $\sigma_{A_n}^2 = 0$; where A_n is time average of the process $W(t)$.

The variance $\sigma_{A_n}^2$ is given by

$$\begin{aligned}\sigma_{A_n}^2 &= \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-2T}^{2T} \left(1 - \frac{|\tau|}{2T}\right) C_{NN}(\tau) d\tau \\ &= \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-2T}^{2T} \left(1 - \frac{|\tau|}{2T}\right) \frac{N_0}{2} \delta(\tau) d\tau \\ &= \lim_{T \rightarrow \infty} \frac{1}{2T} \left[\int_{-2T}^{2T} \frac{N_0}{2} \delta(\tau) d\tau - \int_{-2T}^{2T} \frac{|\tau| N_0}{2} \delta(\tau) d\tau \right]\end{aligned}$$

The second term in the above equation is zero at $\tau = 0$.

Hence, we can write

$$\sigma_{A_n}^2 = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-2T}^{2T} \frac{N_0}{2} \delta(\tau) d\tau = 0$$

Therefore, $N(t)$ is mean ergodic.



Practice Problems

6.10 Is the process with autocorrelation function

$$R_{XX}(\tau) = \begin{cases} 0 & |\tau| > 1 \\ 5(1 - |\tau|) & |\tau| \leq 1 \end{cases}$$

mean ergodic?

6.11 Consider a random process $X(t)$ with two sample functions $X(s_1, t) = k$, $X(s_2, t) = -k$ where k is a constant. Show the process is strict sense stationary but not ergodic.

Solved Problems



6.27 A random process is defined by $X(t) = X_0 + vt$ where X_0 and v are statistically independent random variables, uniformly distributed on intervals $[X_{01}, X_{02}]$ and $[v_1, v_2]$ respectively. Find (a) Mean (b) Autocorrelation (c) Autocovariance function of $X(t)$ (d) Is $X(t)$ stationary in any sense?

Solution Given: $X(t) = X_0 + vt$

The mean $\mu_X(t) = E[X(t)] = E[X_0 + vt]$

Since X_0 and v are statically independent, $\mu_X(t) = E[X_0] + E[v]t$

Both X_0 and v are uniformly distributed random variables. From the given data, we find

$$f_{X_0}(x_0) = \begin{cases} \frac{1}{X_{02} - X_{01}} & \text{for } X_{01} \leq X_0 \leq X_{02} \\ 0 & \text{otherwise} \end{cases}$$

$$f_v(v) = \frac{1}{v_2 - v_1} \text{ for } v_1 \leq v \leq v_2$$

$$\begin{aligned} E[X_0] &= \int_{-\infty}^{\infty} x_0 f_{x_0}(x_0) dx_0 = \frac{1}{X_{02} - X_{01}} \int_{X_{01}}^{X_{02}} x_0 dx_0 \\ &= \frac{1}{X_{02} - X_{01}} \left. \frac{x_0^2}{2} \right|_{X_{01}}^{X_{02}} = \frac{X_{02} + X_{01}}{2} \end{aligned}$$

Similarly, $E[v] = \frac{v_1 + v_2}{2}$

$$\mu_X(t) = \frac{X_{02} + X_{01}}{2} + \left(\frac{v_1 + v_2}{2} \right) t$$

The autocorrelation function of $X(t)$ is

$$\begin{aligned} R_{XX}(t, t + \tau) &= E[X(t)X(t + \tau)] \\ &= E[(X_0 + vt)(X_0 + v(t + \tau))] = E[X_0^2 + X_0vt + X_0v\tau + v^2t^2 + x_0v\tau + v^2t\tau] \\ &= E[X_0^2 + X_0v(2t + \tau) + v^2t^2 + v^2t\tau] \end{aligned}$$

$$\begin{aligned} E[X_0^2] &= \int_{-\infty}^{\infty} x_0^2 f_{x_0}(x_0) dx_0 \\ &= \frac{1}{X_{02} - X_{01}} \int_{X_{01}}^{X_{02}} x_0^2 dx_0 = \frac{1}{3(X_{02} - X_{01})} [X_{02}^3 - X_{01}^3] \\ &= \frac{X_{02}^2 + X_{01}^2 + X_{01}X_{02}}{3} \end{aligned}$$

$$E[V^2] = \int_{-\infty}^{\infty} v^2 f_v(v) dv = \int_{v_1}^{v_2} v^2 dv = \frac{1}{3} \left[\frac{v_2^3 - v_1^3}{v_2 - v_1} \right]$$

$$E[V^2] = \left[\frac{v_2^2 + v_1^2 + v_1v_2}{3} \right]$$

$$E[X_0V] = E[X_0]E[V] = \left(\frac{X_{02} + X_{01}}{2} \right) \left(\frac{v_2 + v_1}{2} \right)$$

$$R_{XX}(\tau) = E[X_0^2] + (2t + \tau)E[X_0v] + t\tau E[v^2] + t^2 E[v^2]$$

$$= \frac{X_{0_2}^2 + X_{0_1}^2 + X_{0_1}X_{0_2}}{3} + (2t + \tau) \left(\frac{X_{0_2} + X_{0_1}}{2} \right) \left(\frac{v_2 + v_1}{2} \right) \\ + (t^2 + t\tau) \left(\frac{v_2^2 + v_1^2 + v_2v_1}{3} \right)$$

$$C_{XX}(\tau) = R_{XX}(\tau) - \mu_X(t)\mu_X(t + \tau)$$

$$= \frac{X_{0_2}^2 + X_{0_1}^2 + X_{0_1}X_{0_2}}{3} + (2t + \tau) \left(\frac{X_{0_2} + X_{0_1}}{2} \right) \left(\frac{v_2 + v_1}{2} \right) + (t^2 + t\tau) \left(\frac{v_2^2 + v_1^2 + v_2v_1}{3} \right) \\ - \frac{1}{4} [X_{0_2} + X_{0_1} + (v_2 + v_1)t] [X_{0_2} + X_{0_1} + (v_2 + v_1)(t + \tau)]$$

6.28 Statistically independent zero mean random process $X(t)$ and $Y(t)$ have auto correlation functions $R_{XX}(\tau) = e^{-|\tau|}$ and $R_{YY}(\tau) = \cos(2\pi\tau)$ respectively.

- Find the autocorrelation of the sum $W_1(t) = X(t) + Y(t)$
- Find the auto correlation function of the difference $W_2(t) = X(t) - Y(t)$
- Find the cross-correlation function of $W_1(t)$ and $W_2(t)$

Solution Given: $R_{XX}(\tau) = e^{-|\tau|}$ and $R_{YY}(\tau) = \cos(2\pi\tau)$

Also, $X(t)$ and $Y(t)$ are independent.

- The autocorrelation function of $W_1(t)$ is

$$R_{W_1W_1}(t, t + \tau) = E[W_1(t)W_1(t + \tau)] \\ = E[\{X(t) + Y(t)\}\{X(t + \tau) + Y(t + \tau)\}] \\ = E[\{X(t)X(t + \tau)\} + E[\{Y(t)X(t + \tau)\}] + E[\{X(t)Y(t + \tau)\}] + E[\{Y(t)Y(t + \tau)\}]] \\ = R_{XX}(\tau) + R_{XY}(\tau) + R_{YX}(\tau) + R_{YY}(\tau)$$

Since $X(t)$ and $Y(t)$ are statistically independent,

$$R_{XY}(\tau) = R_{YX}(\tau) = 0$$

$$R_{W_1W_1}(t, t + \tau) = R_{XX}(\tau) + R_{YY}(\tau)$$

- Given $W_2(t) = X(t) - Y(t)$

$$R_{W_2W_2}(t, t + \tau) = E[W_2(t)W_2(t + \tau)] \\ = E[\{X(t) - Y(t)\}\{X(t + \tau) - Y(t + \tau)\}] \\ = R_{XX}(\tau) - R_{XY}(\tau) - R_{YX}(\tau) + R_{YY}(\tau)$$

Since $R_{XY}(\tau) = R_{YX}(\tau)$

$$R_{W_2W_2}(t, t + \tau) = 0$$

$$R_{W_2W_2}(t, t + \tau) = R_{XX}(\tau) + R_{YY}(\tau)$$

(c) The cross-correlation function of

$$\begin{aligned}
 R_{w_1 w_2}(t) &= E[W_1(t)W_2(t+\tau)] \\
 &= E[\{X(t)+Y(t)\}\{X(t+\tau)-Y(t+\tau)\}] \\
 &= E[\{X(t)X(t+\tau)\}+E[Y(t)X(t+\tau)]-E[\{X(t)Y(t+\tau)\}-E[Y(t)Y(t+\tau)]] \\
 &= R_{XX}(\tau)-R_{YY}(\tau) \\
 &= e^{-|\tau|}-\cos(2\pi\tau)
 \end{aligned}$$

6.29 Assume a random process $X(t)$ with four sample functions

$$\begin{aligned}
 X(s_1, t) &= \cos t; & X(s_2, t) &= -\cos t \\
 X(s_3, t) &= \sin t & X(s_4, t) &= -\sin t
 \end{aligned}$$

which are equally likely. Show that it is a wide-sense stationary process.

Solution Since all sample functions are equally likely, their probability $p_X(x) = \frac{1}{4}$

Therefore,

$$\begin{aligned}
 E[X] &= \sum_{i=1}^4 X(s_i, t)p_X(x) = \frac{1}{4} \sum_{i=1}^4 X(s_i, t) \\
 &= \frac{1}{4}(\cos t - \cos t + \sin t - \sin t) \\
 &= 0
 \end{aligned}$$

$$\begin{aligned}
 R_{XX}(t, t+\tau) &= E[X(t)X(t+\tau)] \\
 &= \frac{1}{4} \sum_{i=1}^4 X(s_i, t)X(s_i, t+\tau) \\
 &= \frac{1}{4}[\cos t \cos(t+\tau) + \cos t \cos(t+\tau) + \sin t \sin(t+\tau) + \sin t \sin(t+\tau)] \\
 &= \frac{1}{2}[\cos t \cos(t+\tau) + \sin t \sin(t+\tau)] \\
 &= \frac{1}{2} \cos \tau
 \end{aligned}$$

$$\because [\cos t \cos(t+\tau) + \sin t \sin(t+\tau)] = \cos(t+\tau-t)$$

which is a function of time difference. The process $X(t)$ is a WSS process.



Practice Problem

6.12 A random process $X(t)$ is characterized by four sample functions

$$\begin{aligned}
 X(s_1, t) &= -1; & X(s_2, t) &= -2; \\
 X(s_3, t) &= 3 & X(s_4, t) &= t
 \end{aligned}$$

which are equally likely. Check whether the process is a WSS process.

(Ans. Not a WSS process)

Solved Problems

6.30 Consider a WSS random process $X(t)$ with zero mean and autocorrelation function $R_{XX}(t) = e^{-2|t|}$. This random process is modulating the carrier wave $\cos(\omega t + \phi)$ where ϕ is uniformly distributed in the interval $(0, 2\pi)$. The resulting process is given by $Y(t) = X(t) \cos(\omega t + \phi)$. The carrier wave is independent of $X(t)$. Find the mean, variance and autocorrelation of $Y(t)$. Find whether the process is WSS.

Solution Given: $Y(t) = X(t) \cos(\omega t + \phi)$

$$\mu_Y(t) = E[Y(t)] = E[X(t) \cos(\omega t + \phi)]$$

Since $X(t)$ and $\cos(\omega t + \phi)$ are independent, we can write

$$\mu_Y(t) = E[X(t)] E[\cos(\omega t + \phi)]$$

Given that $E[X(t)] = 0$

Also, we can find $E[\cos(\omega t + \phi)] = \int_{-\infty}^{\infty} \cos(\omega t + \phi) f(\phi) d\phi$

$$\begin{aligned} f(\phi) &= \frac{1}{2\pi} \text{ for } 0 \leq \phi \leq 2\pi \\ &= 0 \quad \text{otherwise} \end{aligned}$$

$$E[\cos(\omega t + \phi)] = \frac{1}{2\pi} \int_0^{2\pi} \cos(\omega t + \phi) f(\phi) d\phi = 0$$

$$\mu_Y(t) = 0$$

The variance

$$\begin{aligned} \sigma_Y^2 &= E[Y^2(t)] - \{E[Y(t)]\}^2 \\ &= E[Y^2(t)] \end{aligned}$$

$$\begin{aligned} E[Y^2(t)] &= E[X^2(t) \cos^2(\omega t + \phi)] \\ &= E[X^2(t)] E[\cos^2(\omega t + \phi)] \end{aligned}$$

$$E[X^2(t)] = R_{XX}(0) = 1$$

$$\begin{aligned} E[\cos^2(\omega t + \phi)] &= E\left[\frac{1 + \cos(2\omega t + \phi)}{2}\right] \\ &= \frac{1}{2} + \frac{1}{2} E[\cos(2\omega t + \phi)] = \frac{1}{2} \end{aligned}$$

$$\sigma_X^2 = \frac{1}{2}$$

$$\begin{aligned} R_{YY}(t) &= E[\{Y(t)Y(t+\tau)\}] \\ &= E[\{X(t)X(t+\tau)\cos(\omega t + \phi)\cos(\omega t + \omega\tau + \phi)\}] \\ &= E[X(t)X(t+\tau)] \frac{1}{2} E[\cos \omega t + \cos(2\omega t + \omega\tau + 2\phi)] \end{aligned}$$

$$\begin{aligned}
&= \frac{R_{XX}(\tau)}{2} \left\{ E(\cos \omega t) + \underbrace{E[\cos(2\omega t + \omega\tau + 2\phi)]}_0 \right\} \\
R_{YY}(\tau) &= \frac{1}{2} R_{XX}(\tau) \cos \omega\tau \\
&= \frac{1}{2} e^{-2|\tau|} \cos \omega\tau
\end{aligned}$$

6.31 The process $X(t)$ is WSS and normal with $E[X(t)] = 0$ and $R_{XX}(t) = 4e^{-2|t|}$. Find $E\{[X(t+1) - X(t-1)]^2\}$.

Solution The random variable is a normal random variable with zero mean. The autocorrelation function is $R_{XX}(t) = 4e^{-2|t|}$

We have $R_{XX}(0) = E[X^2(t)] = 4$

Therefore, the variance $\sigma_X^2 = E[X^2(t)] = 4$

$$\begin{aligned}
&E[X(t+1)^2 - X(t-1)^2] \\
&= E[X^2(t+1)] + E[X^2(t-1)] \\
&\quad - 2E[X(t+1)X(t-1)] \\
&= 2E[X^2(t)] - 2R_{XX}(\tau) \\
&= 2[R_{XX}(0) - R_{XX}(2)] \\
&= 2[4 - 4e^{-4}] \\
&= 8[1 - e^{-4}]
\end{aligned}$$

6.32 Consider random process $X(t) = A \cos(\omega_0 t + \theta)$, $Y(t) = B \cos(\omega_1 t + \phi)$ where A , B , ω_0 and ω_1 are constants, while θ and ϕ are statistically independent random variables, each uniform on $(0, 2\pi)$.

(a) Show that $X(t)$ and $Y(t)$ are jointly wide-sense stationary. (b) If $\theta = \phi$ show that $X(t)$ and $Y(t)$ are not jointly wide-sense stationary unless $\omega_1 = \omega_0$.

Solution Given: $X(t) = A \cos(\omega_0 t + \theta)$

$$Y(t) = B \cos(\omega_1 t + \phi)$$

where A , B , ω_0 and ω_1 are constants, while θ and ϕ are independent random variables with $U(0, 2\pi)$. Therefore,

$$\begin{aligned}
f(\theta) &= \frac{1}{2\pi} && \text{for } 0 \leq \theta \leq 2\pi \\
&= 0 && \text{otherwise}
\end{aligned}$$

and

$$\begin{aligned}
f(\phi) &= \frac{1}{2\pi} && \text{for } 0 \leq \phi \leq 2\pi \\
&= 0 && \text{otherwise}
\end{aligned}$$

$$\begin{aligned}
E[X(t)] &= E[A \cos(\omega_0 t + \theta)] = AE[\cos(\omega_0 t + \theta)] \\
&= A \int_0^{2\pi} \cos(\omega_0 t + \theta) f(\theta) d\theta = \frac{A}{2\pi} \int_0^{2\pi} \cos(\omega_0 t + \theta) d\theta \\
&= \frac{A}{2\pi} \sin(\omega_0 t + \theta) \Big|_0^{2\pi} = 0
\end{aligned}$$

$$E[X(t)] = 0$$

Similarly, $E[Y(t)] = 0$

The autocorrelation function of the random process $X(t)$ is

$$\begin{aligned}
R_{XX}(t, t + \tau) &= E[X(t)X(t + \tau)] \\
&= E[A \cos(\omega_0 t + \theta) A \cos(\omega_0(t + \tau) + \theta)] \\
&= E \left[\frac{A^2}{2} (\cos(2\omega_0 t + \omega_0 \tau + \theta) + \cos \omega_0 \tau) \right] \\
E[\cos(2\omega_0 t + \omega_0 \tau + \theta)] &= \frac{1}{2\pi} \int_0^{2\pi} \cos(2\omega_0 t + \omega_0 \tau + \theta) d\theta \\
&= \frac{1}{2\pi} [\sin(2\omega_0 t + \omega_0 \tau + \theta)]_0^{2\pi} \\
&= \frac{1}{2\pi} [\sin(2\omega_0 t + \omega_0 \tau + 2\pi) - \sin(2\omega_0 t + \omega_0 \tau)] \\
&= 0 \\
E \left[\frac{A^2}{2} \cos \omega_0 \tau \right] &= \frac{A^2}{2} E[\cos \omega_0 \tau] = \frac{A^2}{2} \cos \omega_0 \tau
\end{aligned}$$

$$\Rightarrow R_{XX}(t, t + \tau) = \frac{A^2}{2} \cos \omega_0 \tau$$

Similarly,

$$\begin{aligned}
R_{XX}(t, t + \tau) &= E[Y(t)Y(t + \tau)] \\
&= E[B \cos(\omega_1 t + \phi) B \cos(\omega_1(t + \tau) + \phi)] \\
&= E \left[\frac{B^2}{2} (\cos(2\omega_1 t + \omega_1 \tau + \phi) + \cos(\omega_1 \tau)) \right]
\end{aligned}$$

$$E[\cos(2\omega_1 t + \omega_1 \tau + \phi)] = 0$$

$$\Rightarrow R_{YY}(t, t + \tau) = \frac{B^2}{2} \cos \omega_1 \tau$$

The cross-correlation function between $X(t)$ and $Y(t)$ is given by

$$\begin{aligned} R_{XY}(t, t + \tau) &= E[X(t)Y(t + \tau)] \\ &= E[A \cos(\omega_0 t + \theta) B \cos(\omega_1(t + \tau) + \phi)] \\ &= \frac{AB}{2} E[\cos((\omega_0 + \omega_1)t + \omega_1 \tau + \theta + \phi) + \cos((\omega_0 - \omega_1)t - \omega_1 \tau + \theta - \phi)] \end{aligned}$$

If $\theta = \phi$ then

$$\begin{aligned} R_{XY}(t, t + \tau) &= \frac{AB}{2} E[\cos((\omega_0 + \omega_1)t + \omega_1 \tau + 2\theta) + \cos((\omega_0 - \omega_1)t - \omega_1 \tau)] \\ E[\cos((\omega_0 + \omega_1)t + \omega_1 \tau + 2\theta)] &= 0 \\ R_{XY}(t, t + \tau) &= \frac{AB}{2} E[\cos((\omega_0 - \omega_1)t - \omega_1 \tau)] \end{aligned}$$

If $\omega_0 \neq \omega_1$, the autocorrelation function is a function of t and τ . Therefore, $X(t)$ and $Y(t)$ are not jointly wide-sense stationary.

If $\omega_0 = \omega_1$, $R_{XY} = \frac{AB}{2} E[\cos \omega_1 \tau]$ which is only a function of τ . Thus, the given $X(t)$ and $Y(t)$ are jointly wide-sense stationary.

6.33 A random process $X(t)$ is described by $X(t) = A$ where $X(t)$ is a continuous random variable uniformly distributed over $(0, 1)$. Classify the process.

Solution Given $X(t) = A$ is a random variable uniformly distributed over $(0, 1)$. Therefore,

$$\begin{aligned} f(A) &= 1 \quad \text{for } 0 \leq A \leq 1 \\ &= 0 \quad \text{otherwise} \end{aligned}$$

For uniform distribution,

$$\begin{aligned} f_x(x) &= \frac{1}{b-a}; a \leq x \leq b \\ &= 0 \quad \text{otherwise} \\ \mu_A &= \frac{b+a}{2} = \frac{1+0}{2} = 0.5 \end{aligned}$$

$$\text{and} \quad \sigma_A^2 = \frac{(b-a)^2}{12} = \frac{(1-0)^2}{12} = \frac{1}{12}$$

The autocorrelation function

$$\begin{aligned} R_{XX}(t, t + \tau) &= E[X(t)X(t + \tau)] \\ &= E[A.A] = E[A^2] \end{aligned}$$

$$E[A^2] = \int_{-\infty}^{\infty} A^2 f(A) dA = \int_0^1 A^2 dA = \left[\frac{A^3}{3} \right]_0^1 = \frac{1}{3}$$

Since the autocorrelation function is not a function of t , $X(t)$ is a stationary random process.

6.34 A random process $X(t) = At$, where A is a continuous random variable uniformly distributed in $(0,1)$. Find (a) $E[X(t)]$, (b) $R_{XX}(t, t + \tau)$ (c) Is the process stationary in any sense?

Solution Given: $X(t) = At$ and $A \sim U(0, 1)$

$$f(A) = 1 \quad \text{for } 0 \leq A \leq 1$$

$$= 0 \quad \text{otherwise}$$

$$E[X(t)] = E[At] = \int_{-\infty}^{\infty} At f(A) dA = \int_0^1 At dA = \left. \frac{tA^2}{2} \right|_0^1 = \frac{t}{2}$$

$$E[X(t)] = \frac{t}{2}$$

The autocorrelation function

$$R_{XX}(t, t + \tau) = E[X(t)X(t + \tau)]$$

$$= E[At A(t + \tau)]$$

$$= t(t + \tau)E[A^2]$$

$$E[A^2] = \int_{-\infty}^{\infty} A^2 f(A) dA = \int_0^1 A^2 dA = \left. \frac{A^3}{3} \right|_0^1 = \frac{1}{3}$$

$$\Rightarrow R_{XX}(t, t + \tau) = \frac{1}{3}t(t + \tau)$$

Since mean and autocorrelation functions are function of t , the random process $X(t)$ is not a stationary random process.

6.35 Compute the variance of the random process $X(t)$ whose autocorrelation function is given by

$$R_{XX}(\tau) = 16 + \frac{2}{1 + 2\tau^2}.$$

Solution Using Eq. (6.33) we have

$$\mu_X^2 = \lim_{|\tau| \rightarrow \infty} R_{XX}(\tau)$$

$$= \lim_{|\tau| \rightarrow \infty} 16 + \frac{2}{1 + 2\tau^2} = 16$$

$$\Rightarrow \mu_X = 4$$

Also, we know

$$E[X^2(t)] = R_{XX}(0)$$

$$= 16 + 2 = 18$$

Thus, the variance is given by

$$\begin{aligned}\sigma_X^2 &= E[X^2(t)] - \mu_X^2 \\ &= 18 - (4)^2 = 2\end{aligned}$$

6.36 For a random process $X(t)$, the mean value $\mu_X(t) = 6$ and $R_{XX}(t, t + \tau) = 36 + 25e^{-|\tau|}$. Indicate which of the following statements are true based on what is known with certainty $X(t)$, (a) is first-order stationary (b) has total average power of 61W, (c) is ergodic, (d) is wide-sense stationary, (e) has a periodic component, (f) has an ac power of 36W.

Solution Given: $\mu_X(t) = 6$ and $R_{XX}(t, t + \tau) = 36 + 25e^{-|\tau|}$

- (a) For a first-order stationary process, the mean value is constant. For the given process $X(t)$, the mean value is 6. Therefore, $X(t)$ is first order stationary. Hence, it is true.
 (b) The average power of a random process is

$$P = R_{XX}(0)$$

Given $R_{XX}(t, t + \tau) = R_{XX}(\tau) = 36 + 25e^{-|\tau|}$

$$P = [36 + 25e^{-|\tau|}]_{\tau=0} = 61\text{W}$$

The average power of $X(t)$ is 61W. Hence, it is true.

- (c) From the properties of autocorrelation, we know if $\mu_X(t) \neq 0$ and $X(t)$ is ergodic with no periodic components. Then

$$\lim_{|\tau| \rightarrow \infty} R_{XX}(\tau) = (\mu_X)^2$$

$$R_{XX}(t, t + \tau) = 36 + 25e^{-|\tau|}$$

$$R_{XX}(\tau) = \lim_{|\tau| \rightarrow \infty} [36 + 25e^{-|\tau|}] = 36$$

which is equal to μ_X^2 .

Since the condition is satisfied that $X(t)$ is ergodic.

- (d) A random process is wide-sense stationary if its mean is constant and $R_{XX}(t, t + \tau) = R_{XX}(\tau)$

Since $\mu_X(t) = 6$ and $R_{XX}(t, t + \tau) = 36 + 25e^{-|\tau|}$, a function of τ , the random process $X(t)$ is wide-sense stationary. Hence, it is true.

- (e) It is not having any periodic component. Hence, it is false.
 (f) The ac power of a random process is its variance

$$\sigma_X^2 = E[X^2(t)] - \{E[X(t)]\}^2$$

We know $R_{XX}(\tau) = E[X(t)X(t + \tau)]$. If $\tau = 0$, then

$$R_{XX}(0) = E[X^2(t)]$$

$$\sigma_X^2 = R_{XX}(0) - (\mu_X)^2$$

$$R_{XX}(\tau) = 36 + 25e^{-|\tau|}$$

$$R_{XX}(0) = 61$$

$$\sigma_X^2 = 61 - (6)^2 = 25$$

The last statement that ac power is 36 W is false.

6.37 Given $X(t) = A \cos \omega_0 t + B \sin \omega_0 t$, where A and B are random variables and ω_0 is a constant. Show that $X(t)$ is WSS if A and B are uncorrelated zero-mean random variables having different density functions but the same variance σ^2 .

Solution Given: $X(t) = A \cos \omega_0 t + B \sin \omega_0 t$

$$\begin{aligned} E[X(t)X(t+\tau)] &= E[(A \cos \omega_0 t + B \sin \omega_0 t)(A \cos \omega_0(t+\tau) + B \sin \omega_0(t+\tau))] \\ &= E \left[\begin{aligned} &A^2 \cos \omega_0 t \cos \omega_0(t+\tau) + AB \sin \omega_0 t \cos \omega_0(t+\tau) \\ &+ AB \cos \omega_0 t \sin \omega_0(t+\tau) + B^2 \sin \omega_0 t \sin \omega_0(t+\tau) \end{aligned} \right] \end{aligned}$$

If A and B are uncorrelated with zero mean and variance σ^2 , $E[A] = E[B] = 0$ and $E[A^2] = E[B^2] = \sigma^2$. Also $E[AB] = 0$.

$$\begin{aligned} E[X(t)X(t+\tau)] &= E[A^2 \cos \omega_0 t \cos \omega_0(t+\tau) + B^2 \sin \omega_0 t \sin \omega_0(t+\tau)] \\ &\quad + E[AB][\cos \omega_0 t \sin \omega_0(t+\tau) + \sin \omega_0 t \cos \omega_0(t+\tau)] \\ &= \frac{\sigma^2}{2} [\cos(2\omega_0 t + \tau) + \cos \omega_0 \tau + \cos \omega_0 \tau - \cos(2\omega_0 t + \tau)] \\ &= \sigma^2 \cos \omega_0 \tau \end{aligned}$$

Since the autocorrelation function depends only on τ , the process $x(t)$ is wide-sense stationary.

6.38 A random process $X(t)$ is defined by $X(t) = A \cos t + B \sin t - \infty < t < \infty$ where A and B are independent random variables each of which has a value -1 with a probability $2/3$ and a value 2 with probability $1/3$. Show that $X(t)$ is a wide-sense stationary process.

Solution From the given data the probability mass function of random variable A and B are shown in Fig. 6.17.

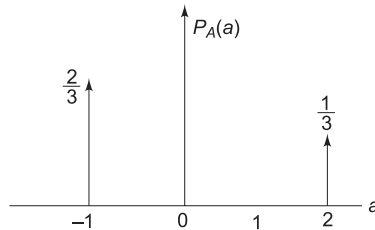


Fig. 6.17

The mean value of A is $E[A] = E[B] = (2/3)(-1) + (1/3)(2) = 0$

$$E[A^2] = E[B^2] = (2/3)(-1)^2 + (1/3)(2)^2$$

$$= (2/3) + (4/3) = 2$$

Since A and B are independent, $E[AB] = E[A] E[B] = 0$

$$\begin{aligned} R_{XX}(t, t + \tau) &= E[X(t)X(t + \tau)] \\ &= E[(A \cos t + B \sin t)(A \cos(t + \tau) + B \sin(t + \tau))] \\ &= E[A^2 \cos t \cos(t + \tau) + AB \sin t \cos(t + \tau) + AB \cos t \sin(t + \tau) + B^2 \sin t \sin(t + \tau)] \\ &= E[A^2] \cos t \cos(t + \tau) + E[AB][\cos(t + \tau) \sin t + \cos t \sin(t + \tau)] + E[B^2] \sin t \sin(t + \tau) \\ &= 2[\cos t \cos(t + \tau) + \sin t \sin(t + \tau)] \\ &= 2 \cos(t - (t + \tau)) = 2 \cos \tau \end{aligned}$$

Hence the process is wide-sense stationary.



Practice Problem

6.13 Repeat the Solved Problem 6.25 with $A \sim U(1, 2)$, $B \sim U(0, 1)$ and $\theta \sim U\left(0, \frac{\pi}{2}\right)$.

Solved Problem



6.39 Consider a random process $X(t) = A \sin \omega t + B \cos \omega t$ $-\infty < t < \infty$, where ω is constant and A and B are random variables.

Find the condition for $X(t)$ to be stationary.

Show that $X(t)$ is WSS if and only if A and B are uncorrelated with equal variance.

Solution Given: $X(t) = A \sin \omega t + B \cos \omega t$

For $X(t)$ to be stationary, the mean must be independent of t . The mean value of $X(t)$ is

$$\begin{aligned} \mu_X(t) &= E[X(t)] = E[A \sin \omega t + B \cos \omega t] \\ &= E[A] \sin \omega t + E[B] \cos \omega t \end{aligned}$$

The mean is independent of t when $E[A] = E[B] = 0$

The auto-correlation function

$$\begin{aligned} R_{XX}(t, t + \tau) &= E[X(t)X(t + \tau)] \\ &= E\{(A \sin \omega t + B \cos \omega t)(A \sin(\omega(t + \tau)) + B \cos(\omega(t + \tau)))\} \\ &= E[A^2] \sin \omega t \sin \omega(t + \tau) + E[AB] \cos \omega t \sin \omega(t + \tau) \\ &\quad + E[AB] \sin \omega t \cos \omega(t + \tau) + E[B^2] \cos \omega t \cos \omega(t + \tau) \\ &= E[A^2] \sin \omega t \sin \omega(t + \tau) + E[AB] \sin(2\omega t + \omega \tau) + E[B^2] \cos \omega t \cos \omega(t + \tau) \end{aligned}$$

The above function depends on τ alone when $E[A^2] = E[B^2] = \sigma^2$ and $E[AB] = 0$

That is, $R_{XX}(t, t + \tau) = \sigma^2 \cos \omega \tau$

Hence, $X(t)$ is WSS if $E[A^2] = E[B^2] = \sigma^2$ and $E[AB] = 0$

6.40 A stationary random process $X(t)$ has autocorrelation $R_{XX}(\tau) = 16 + 4 \cos(2\tau) + 5e^{-4|\tau|}$. Find (a) the average power of its dc component, ac components, non-dc components and non-periodic components, (b) average power of $X(t)$ and (c) the variance of $X(t)$.

Solution Given: $R_{XX}(\tau) = 16 + 4 \cos(2\tau) + 5e^{-4|\tau|}$

The autocorrelation is composed of periodic and non-periodic components.

The periodic component is $4 \cos(2\tau)$ and the non-periodic component is $16 + 5e^{-4|\tau|}$

Let $R_{XX}(\tau) = R_{pc}(\tau) + R_{nc}(\tau)$. Then $R_{pc}(\tau) = 4 \cos 2\tau$ and $R_{nc}(\tau) = 16 + 5e^{-4|\tau|}$

The periodic component $R_{pc}(\tau) = 4 \cos 2\tau$. The average power is $R_{pc}(0) = 4$

The average power of the non-periodic component is $R_{nc}(0) = 16 + 5 = 21$

For the non-periodic component,

$$\lim_{\tau \rightarrow \infty} R_{nc}(\tau) = \mu_X^2$$

$$\mu_X = \pm \sqrt{\lim_{\tau \rightarrow \infty} R_{nc}(\tau)} = \pm \sqrt{\lim_{\tau \rightarrow \infty} (16 + 5e^{-4|\tau|})} = \pm 4$$

The power of non-periodic component

$$\lim_{\tau \rightarrow \infty} R_{nc}(\tau) = \mu_X^2 = 16$$

The average power

$$E[X^2] = R_{XX}(0) = 16 + 4 + 5 = 25$$

The variance

$$\sigma_X^2 = E[X^2] - (\mu_X)^2 = 25 - (4)^2 = 9$$

which is the average power of the non-dc components.

6.41 For the following random processes $X(t) = A_1 \cos(\omega_1 t + \theta)$ and $Y(t) = A_2 \cos(\omega_2 t + \phi)$, find the cross-correlation function if (a) $\theta \neq \phi$, and (b) $\theta = \phi$. The values of A_1 , A_2 , ω_1 and ω_2 are constants. The random variables θ and ϕ are independent and uniformly distributed over $(-\pi, \pi)$.

Solution Given: $X(t) = A_1 \cos(\omega_1 t + \theta)$ and

$$Y(t) = A_2 \cos(\omega_2 t + \phi)$$

The cross-correlation function $R_{XY} = E[X(t)Y(t + \tau)]$

$$R_{XY} = E[A_1 \cos(\omega_1 t + \theta) A_2 \cos(\omega_2 (t + \tau) + \phi)]$$

Since A_1 and A_2 are constants,

$$R_{XY} = A_1 A_2 E[\cos(\omega_1 t + \theta) \cos(\omega_2 (t + \tau) + \phi)]$$

Also ϕ and θ are independent, hence

$$R_{XY} = A_1 A_2 E[\cos(\omega_1 t + \theta)] E[\cos(\omega_2 (t + \tau) + \phi)]$$

$$E[\cos(\omega_1 t + \theta)] = \int_{-\infty}^{\infty} \cos(\omega_1 t + \theta) f(\theta) d\theta$$

θ is uniformly distributed over $(-\pi, \pi)$. Therefore, $f(\theta) = \frac{1}{2\pi}$ for $-\pi \leq \theta \leq \pi$

$$\begin{aligned} \text{Now } E[\cos(\omega_1 t + \theta)] &= \frac{1}{2\pi} \int_{-\pi}^{\pi} \cos(\omega_1 t + \theta) d\theta \\ &= \frac{1}{2\pi} [\sin(\omega_1 t + \theta)]_{-\pi}^{\pi} = 0 \end{aligned}$$

Since $E[\cos(\omega_1 t + \theta)] = 0$, $R_{XY} = 0$

If $\theta = \phi$

$$\text{then } R_{XY} = E[A_1 \cos(\omega_1 t + \theta) A_2 \cos(\omega_2 (t + \tau) + \theta)]$$

$$R_{XY} = \frac{A_1 A_2}{2} E\{\cos[(\omega_1 + \omega_2)t + \omega_2 \tau + 2\theta] + \cos[(\omega_2 - \omega_1)t + \omega_2 \tau]\}$$

Since ω_1, ω_2 are constants,

$$\begin{aligned} R_{XY} &= \frac{A_1 A_2}{2} \cos[(\omega_2 - \omega_1)t + \omega_2 \tau] + \underbrace{E\{\cos[(\omega_1 + \omega_2)t + \omega_2 \tau + 2\theta]\}}_0 \\ R_{XY} &= \frac{A_1 A_2}{2} \cos[(\omega_2 - \omega_1)t + \omega_2 \tau] \end{aligned}$$

6.42 Show that if a random process $X(t)$ is strict-sense stationary then it is also wide-sense stationary.

Solution Given: $X(t)$ is strict-sense stationary. It satisfies the condition that the first-order and second-order density functions are invariant for a time shift. That is,

$$f_X(x, t) = f_X(x, t + \tau)$$

Then we have the mean value

$$\mu_X(t) = \mu_X(t + \tau)$$

And the auto-correlation function

$$R_{XX}(t_1, t_2) = R_{XX}(t_1 + \tau, t_2 + \tau)$$

That is, the auto-correlation function depends on time points t_1 and t_2 only through the difference $|t_1 - t_2|$. Thus, $X(t)$ is WSS.

6.43 Consider two random processes $X(t) = A \cos \omega t + B \sin \omega t$ and $Y(t) = B \cos \omega t - A \sin \omega t$ where A and B are uncorrelated, zero mean random variables with same variance and ω is a constant. Show that $X(t)$ and $Y(t)$ are joint stationary.

Solution Given: $X(t) = A \cos \omega t + B \sin \omega t$

$$Y(t) = B \cos \omega t - A \sin \omega t$$

Since A and B are of zero mean,

$$\begin{aligned} E[X(t)] &= \mu_X = E[A \cos \omega t + B \sin \omega t] \\ &= \cos \omega t E[A] + \sin \omega t E[B] \\ &= 0 \end{aligned}$$

Similarly,

$$\begin{aligned} E[Y(t)] &= \mu_Y = E[B \cos \omega t - A \sin \omega t] \\ &= E[B] \cos \omega t - E[A] \sin \omega t \\ &= 0 \end{aligned}$$

Also,

$$\begin{aligned} E[A^2] &= \sigma_A^2 + \{E[A]\}^2 = \sigma_A^2 \\ E[B^2] &= \sigma_B^2 + \{E[B]\}^2 = \sigma_B^2 \end{aligned}$$

Since $\sigma_A^2 = \sigma_B^2$, we have $E[A^2] = E[B^2]$

The cross-correlation function

$$\begin{aligned} R_{XY}(t, t + \tau) &= E[X(t)Y(t + \tau)] \\ &= E[(A \cos \omega t + B \sin \omega t)(B \cos(\omega t + \omega \tau) - A \sin(\omega t + \omega \tau))] \\ &= E[AB] \cos \omega t \cos(\omega t + \omega \tau) + E[B^2] \sin \omega t \cos(\omega t + \omega \tau) \\ &\quad - E[A^2] \cos \omega t \sin(\omega t + \omega \tau) - E[AB] \sin \omega t \sin(\omega t + \omega \tau) \end{aligned}$$

Since A and B are uncorrelated,

$$E[AB] = E[A]E[B] = 0$$

Also, $E[A^2] = E[B^2]$

$$\begin{aligned} \Rightarrow R_{XY}(t, t + \tau) &= E[A^2] \{ \cos(\omega t + \omega \tau) \sin \omega t - \cos \omega t \sin(\omega t + \omega \tau) \} \\ &= -E[A^2] \sin \omega \tau \end{aligned}$$

Since $R_{XY}(t, t + \tau)$ is independent of time t , $X(t)$ and $Y(t)$ are a jointly stationary process.

6.44 Let $X(t)$ be random process with mean 5 and autocorrelation $R_{XX}(t_1, t_2) = 25 + 4e^{-0.1|t_1 - t_2|}$. Determine the mean, variance and covariance of the random variables $Z = X(5)$ and $W = X(9)$.

Solution Given: $E[X(t)] = \mu_X = 5$ and $R_{XX}(t_1, t_2) = 25 + 4e^{-0.1|t_1 - t_2|}$

From the mean and autocorrelation, we can find that $X(t)$ is a WSS process. Therefore,

$$E[Z] = E[X(5)] = 5 \quad \text{and} \quad E[W] = E[X(9)] = 5$$

$$E[Z^2] = E[W^2] = E[X^2] = R_{XX}(0) = 29$$

$$\text{Var}(Z) = \text{Var}(W) = E(X^2) - \{E[X]\}^2 = 29 - 25 = 4$$

$$\begin{aligned}
\text{Cov}(Z, W) &= R(Z, W) - E(Z)E(W) \\
&= R(9, 5) - E[X(5)]E[X(9)] \\
&= 25 + 4e^{-0.1(9-5)} - (5)(5) \\
&= 4e^{-0.4}
\end{aligned}$$

6.45 If a random process $X(t) = \sin(\omega t + y)$ where y is a uniformly distributed random variable in the interval $(0, 2\pi)$. Prove that $\text{Cov}(t_1, t_2) = \frac{1}{2} \cos \omega(t_1 - t_2)$.

Solution Given: $X(t) = \sin(\omega t + y)$ and y is uniformly distributed in the interval $(0, 2\pi)$

Therefore,

$$f_Y(y) = \begin{cases} \frac{1}{2\pi} & \text{for } 0 \leq y \leq 2\pi \\ 0 & \text{otherwise} \end{cases}$$

$$E[X(t)] = E[\sin(\omega t + y)] = \int_{-\infty}^{\infty} \sin(\omega t + y) f_Y(y) dy$$

$$= \frac{1}{2\pi} \int_0^{2\pi} \sin(\omega t + y) dy = 0$$

$$\text{Cov}(t_1, t_2) = R_{XX}(t_1, t_2) - E[X(t_1)]E[X(t_2)]$$

$$\text{Since } E[X(t)] = 0; E[X(t_1)] = E[X(t_2)] = 0$$

$$\Rightarrow \text{Cov}(t_1, t_2) = R_{XX}(t_1, t_2)$$

$$R_{XX}(t_1, t_2) = E[X(t_1)X(t_2)] = E[\sin(\omega t_1 + y)\sin(\omega t_2 + y)]$$

$$= \frac{1}{2} E[\cos \omega(t_1 - t_2) - \cos(\omega t_1 + \omega t_2 + 2y)]$$

$$= \frac{1}{2} E[\cos \omega(t_1 - t_2)] - \frac{1}{2} E[\cos(\omega t_1 + \omega t_2 + 2y)]$$

$$E[\cos(\omega t_1 + \omega t_2 + 2y)] = \int_{-\infty}^{\infty} \cos(\omega t_1 + \omega t_2 + 2y) f_Y(y) dy$$

$$= \frac{1}{2\pi} \int_0^{2\pi} \cos(\omega t_1 + \omega t_2 + 2y) dy = 0$$

$$\text{Also, } E[\cos(\omega t_1 - \omega t_2)] = \cos \omega(t_1 - t_2)$$

$$\Rightarrow R_{XX}(t_1, t_2) = \frac{1}{2} \cos \omega(t_1 - t_2)$$

Hence, $\text{Cov}(t_1, t_2) = \frac{1}{2} \cos \omega(t_1 - t_2)$

Hence proved.

6.46 A complex random process $Z(t) = X(t) + jY(t)$ is defined by jointly stationary real processes $X(t)$ and $Y(t)$. Show that

$$E\{|Z(t)|^2\} = R_{XX}(0) + R_{YY}(0).$$

Solution

$$\begin{aligned} E\{|Z^*(t)|^2\} &= E[Z^*(t)Z(t)] \\ &= E\{[X(t) - jY(t)][X(t) + jY(t)]\} \\ &= E[X^2(t)] - jE[X(t)Y(t)] + jE[X(t)Y(t)] + E[Y^2(t)] \\ &= E[X^2(t)] + E[Y^2(t)] \\ &= R_{XX}(0) + R_{YY}(0) \end{aligned}$$

$$\begin{aligned} R_{XX}(\tau) &= E[X(t)X(t+\tau)] \\ R_{XX}(0) &= E[X^2(t)] \end{aligned}$$

6.47 Consider the complex random processes $Z_1(t) = X_1(t) + jY_1(t)$; $Z_2(t) = X_2(t) - jY_2(t)$ formed by the real processes $X_1(t)$, $X_2(t)$, $Y_1(t)$ and $Y_2(t)$. Find the expression for the crosscorrelation function of $Z_1(t)$ and $Z_2(t)$ if

- (a) All the real processes are correlated.
- (b) They are uncorrelated.

Solution Given: $Z_1(t) = X_1(t) + jY_1(t)$; $Z_2(t) = X_2(t) - jY_2(t)$

- (a) The cross-correlation

$$\begin{aligned} R_{Z_1 Z_2}(t, t+\tau) &= E[Z_1^*(t)Z_2(t+\tau)] \\ &= E\{[X_1(t) - jY_1(t)][X_2(t+\tau) - jY_2(t+\tau)]\} \\ &= E[X_1(t)X_2(t+\tau)] - jE[X_2(t+\tau)Y_1(t)] - jE[X_1(t)Y_2(t+\tau)] - E[Y_1(t)Y_2(t+\tau)] \\ &= R_{X_1 X_2}(t, t+\tau) - R_{Y_1 Y_2}(t, t+\tau) - jR_{Y_1 X_2}(t, t+\tau) - jR_{X_1 Y_2}(t, t+\tau) \end{aligned}$$

- (b) If the processes are uncorrelated,

$$E[X_1(t)X_2(t+\tau)] = E[X_1(t)]E[X_2(t+\tau)]; E[Y_1(t)Y_2(t+\tau)] = E[Y_1(t)]E[Y_2(t+\tau)]$$

Now

$$\begin{aligned} R_{Z_1 Z_2}(t, t+\tau) &= E[X_1(t)]E[X_2(t+\tau)] - jE[X_2(t+\tau)]E[Y_1(t)] - jE[X_1(t)]E[Y_2(t+\tau)] \\ &\quad - E[Y_1(t)]E[Y_2(t+\tau)] \end{aligned}$$

$$\begin{aligned}
&= E\{[X_1(t) - jY_1(t)]\}E\{X_2(t+\tau) - jY_2(t+\tau)\} \\
&= E[Z_1^*(t)]E[Z_2(t+\tau)]
\end{aligned}$$

6.48 Consider two zero-mean jointly wide-sense stationary random processes $X(t)$ and $Y(t)$ with $\sigma_X^2 = 4$ and $\sigma_Y^2 = 12$. Explain why each of the following functions cannot apply to the processes if they have no periodic components.

- (a) $R_{XX}(\tau) = e^{-3\tau}u(\tau)$ (b) $R_{XX}(\tau) = -\sin(3\tau)e^{-|\tau|}$
(c) $R_{XY}(\tau) = 9(2 + 2\tau^2)^{-1}$ (d) $R_{YY}(\tau) = 3\left[\frac{\sin(2\tau)}{2\tau}\right]^2$
(e) $R_{YY}(\tau) = 5 + 3\left[\frac{\sin(5\tau)}{5\tau}\right]$

Solution Given: $\mu_X = 0, \mu_Y = 0, \sigma_X^2 = 4$ and $\sigma_Y^2 = 12$

(a) $R_{XX}(\tau) = e^{-3\tau}u(\tau)$

The above function does not have even symmetry.

(b) $R_{XX}(\tau) = -\sin(3\tau)e^{-|\tau|}$

The function takes negative value.

(c) $R_{XY}(\tau) = 9(2 + 2\tau^2)^{-1}$

$$\begin{aligned}
|R_{XY}(\tau)| &\leq \sqrt{R_{XX}(0)R_{YY}(0)} \\
&\leq \sqrt{(4)(12)} \\
&\leq \sqrt{48}
\end{aligned}$$

The function does not satisfy the above condition.

(d) $R_{YY}(\tau) = 3\left[\frac{\sin(2\tau)}{2\tau}\right]^2$

From the above equation, $R_{YY}(0) = 3$

But $R_{YY}(0) = \sigma_Y^2 = 12$

(e) $R_{YY}(\tau) = 5 + 3\left[\frac{\sin(5\tau)}{5\tau}\right]$

The function has a constant term which does not satisfy $Y(t)$ and $Y(t)$ has zero mean.

6.49 Consider a random process $Y(t) = X(t)\cos(\omega_0 t + \theta)$ where $X(t)$ is a wide-sense stationary random process, θ is a random variable independent of $X(t)$ and is distributed uniformly in $(-\pi, \pi)$ and ω_0 is a constant. Prove that $Y(t)$ is wide-sense stationary.

Solution Given: $Y(t) = X(t)\cos(\omega_0 t + \theta)$

Since $X(t)$ is a wide-sense stationary process, its mean is constant and the autocorrelation function is independent of time t . That is, $R_{XX}(t, t + \tau) = R_{XX}(\tau)$.

θ is a random variable with uniformly distributed in $(-\pi, \pi)$. Hence, we can write

$$f_{\theta}(\theta) = \frac{1}{2\pi} \quad \text{for } -\pi \leq \theta \leq \pi$$

$$= 0 \quad \text{otherwise}$$

Also, θ is independent of $X(t)$.

$$\begin{aligned} E[Y(t)] &= E[X(t)\cos(\omega_0 t + \theta)] \\ &= E[X(t)]E[\cos(\omega_0 t + \theta)] \\ &= E[X(t)] \int_{-\infty}^{\infty} \cos(\omega_0 t + \theta) f_{\theta}(\theta) d\theta \\ &= \frac{\mu_X}{2\pi} \int_{-\pi}^{\pi} \cos(\omega_0 t + \theta) d\theta \\ &= \frac{\mu_X}{2\pi} \left[\sin(\omega_0 t + \theta) \right]_{-\pi}^{\pi} \\ &= \frac{\mu_X}{2\pi} [\sin(\pi + \omega_0 t) + \sin(-\pi + \omega_0 t)] \\ &= \frac{\mu_X}{2\pi} [-\sin \omega_0 t + \sin \omega_0 t] = 0 \end{aligned}$$

$$\begin{aligned} R_{YY}(t, t + \tau) &= E[Y(t)Y(t + \tau)] \\ &= E[X(t)\cos(\omega_0 t + \theta)X(t + \tau)\cos(\omega_0 t + \omega_0 \tau + \theta)] \\ &= E[X(t)X(t + \tau)]E[\cos(\omega_0 t + \theta)\cos(\omega_0 t + \omega_0 \tau + \theta)] \\ &= \frac{R_{XX}(\tau)}{2} E[\cos(\omega_0 \tau) + \cos(2\omega_0 t + \omega_0 \tau + 2\theta)] \\ &= \frac{R_{XX}(\tau)}{2} \left[\frac{1}{2\pi} \int_{-\pi}^{\pi} \cos(\omega_0 \tau) d\theta + \frac{1}{2\pi} \int_{-\pi}^{\pi} \cos(2\omega_0 t + \omega_0 \tau + 2\theta) d\theta \right] \\ &= \frac{R_{XX}(\tau)}{2} \left[\cos \omega_0 \tau + \frac{1}{2\pi} \underbrace{\left[\frac{\sin(2\omega_0 t + \omega_0 \tau + 2\theta)}{2} \right]_{-\pi}^{\pi}}_0 \right] \\ &= \frac{1}{2} R_{XX}(\tau) \cos \omega_0 \tau \end{aligned}$$

Since $E[Y(t)]$ is a constant and $R_{YY}(t, t + \tau)$ depends on τ , $Y(t)$ is a WSS process.

6.50 A wide-sense stationary random process $Y(t)$ has a power of $E[Y^2(t)] = 4$. Give at least one reason why each of the following expressions cannot be its autocorrelation function.

- (a) $R_{YY}(t, t + \tau) = -4e^{-|\tau|}$
- (b) $R_{YY}(t, t + \tau) = \frac{4\tau}{1 + 2\tau^2 + 2\tau^4}$
- (c) $R_{YY}(t, t + \tau) = 6e^{-\tau^2 - |\tau|}$
- (d) $R_{YY}(t, t + \tau) = 4 \frac{\sin[4(\tau - 2)]}{4(\tau - 2)}$
- (e) $R_{YY}(t, t + \tau) = \frac{\cos^2(5\tau)}{2 + \cos 4t}$

Solution (a) Given: $R_{YY}(t, t + \tau) = -4e^{-|\tau|}$
from which we have $R_{YY}(t, t) = -4$.

Since $R_{YY}(t, t)$ cannot be negative, the above expression cannot be an autocorrelation function.

(b) Given: $R_{YY}(t, t + \tau) = \frac{4\tau}{1 + 2\tau^2 + 2\tau^4}$

The above function is not an even function of θ . Therefore, the above expression cannot be ACF.

(c) Given: $R_{YY}(t, t + \tau) = 6e^{-\tau^2 - |\tau|}$

$R_{YY}(t, t) = 6$ which is not equal to 4 as given in the problem. Therefore, the above expression is not a valid ACF.

(d) Given $R_{YY}(t, t + \tau) = 4 \frac{\sin[4(\tau - 2)]}{4(\tau - 2)}$

The above function attains maximum value at $\tau = 2$ and not at $\tau = 0$. Hence, it is not a valid ACF.

(e) Given $R_{YY}(t, t + \tau) = \frac{\cos^2(5\tau)}{2 + \cos 4t}$

ACF cannot be function of t for a wide-sense stationary random process. Hence, the above expression is not a valid ACF.

6.51 Consider a random process $X(t) = A \sin \omega t + B \cos \omega t$; $-\infty < t < \infty$; where ω is a constant and A and B are random variables

(a) Find the condition for $X(t)$ to be stationary.

(b) Show that $X(t)$ is WSS if and only if A and B are correlated with equal variance.

Solution Given: $X(t) = A \sin \omega t + B \cos \omega t$

(a) For $X(t)$ to be stationary, the mean must be independent of t .

The mean value of $X(t)$ is

$$\mu_X(t) = E[X(t)] = E[A \sin \omega t + B \cos \omega t]$$

Since ω is constant,

$$\mu_X(t) = E[A] \sin \omega t + E[B] \cos \omega t$$

If $E[A] = E[B] = \text{constant}$ then $\mu_X(t)$ depends on time t . Therefore, for $\mu_X(t)$ to be independent of t , $E[A] = E[B] = 0$.

(b) The autocorrelation function

$$\begin{aligned} R_{XX}(t, t + \tau) &= E[X(t)X(t + \tau)] \\ &= E[(A \sin \omega t + B \cos \omega t)(A \sin \omega(t + \tau) + B \cos \omega(t + \tau))] \\ &= E[A^2] \sin \omega t \sin \omega(t + \tau) + E[AB] \cos \omega t \sin \omega(t + \tau) + E[AB] \sin \omega t \cos \omega(t + \tau) \\ &\quad + E[B^2] \cos \omega t \cos \omega(t + \tau) \end{aligned}$$

If $E[A^2] = E[B^2] = \sigma^2$, then

$$\begin{aligned} R_{XX}(t, t + \tau) &= \sigma^2 [\sin \omega t \sin \omega(t + \tau) + \cos \omega t \cos \omega(t + \tau)] \\ &\quad + E[AB] \{\sin \omega(t + \tau) \cos \omega t + \sin \omega t \cos \omega(t + \tau)\} \\ &= \sigma^2 \cos \omega \tau + E[AB] \{\sin(2\omega t + \omega \tau)\} \end{aligned}$$

The above autocorrelation function will be a function of τ only if $E[AB] = 0$. That is, A and B are uncorrelated. Therefore, $X(t)$ is WSS if A and B are uncorrelated with equal variance.

6.52 Consider a random process $X(t) = A \cos \pi t$ where A is a Gaussian random variable with zero mean and variance σ_A^2 . Is $X(t)$ stationary in any sense?

Solution Given: $X(t) = A \cos \pi t$ $A \sim U(0, \sigma_A^2)$

$$\begin{aligned} E[X(t)] &= E[A \cos \pi t] \\ &= E[A] \cos \pi t = 0 \end{aligned}$$

Since $E[A] = 0$, $E[X(t)] = 0$

$$\begin{aligned} E[X^2(t)] &= E[A^2 \cos^2 \pi t] = E[A^2] E\left[\frac{1 + \cos 2\pi t}{2}\right] \\ &= E[A^2] \left[\frac{1 + \cos 2\pi t}{2}\right] = \sigma_A^2 \left[\frac{1 + \cos 2\pi t}{2}\right] \end{aligned}$$

Since $E[X^2(t)]$ depends on time, $X(t)$ is not wide-sense stationary.

6.53 Consider two random processes $X_1(t) = p_1(t + \varepsilon)$ and $X_2(t) = p_2(t + \varepsilon)$ where $p_1(t)$ and $p_2(t)$ are both periodic waveforms with period T . ε is a uniform random variable on the interval $(0, T)$. Find $R_{X_1 X_2}(t, t + \tau)$.

Solution Given: $X_1(t) = p_1(t + \varepsilon)$

$X_2(t) = p_2(t + \varepsilon)$ and $\varepsilon \sim U(0, T)$

$$\begin{aligned} R_{X_1 X_2}(t, t + \tau) &= E[X_1(t)X_2(t + \tau)] \\ &= E[p_1(t + \varepsilon)p_2(t + \varepsilon + \tau)] \end{aligned}$$

$$\begin{aligned} f_\varepsilon(\varepsilon) &= \frac{1}{T} \text{ for } 0 \leq \varepsilon \leq T \\ &= 0 \text{ otherwise} \end{aligned}$$

$$\begin{aligned} R_{X_1 X_2}(t, t + \tau) &= \int_{-\infty}^{\infty} p_1(t + \varepsilon)p_2(t + \varepsilon + \tau)f_\varepsilon(\varepsilon)d\varepsilon \\ &= \frac{1}{T} \int_0^T p_1(t + \varepsilon)p_2(t + \varepsilon + \tau) d\varepsilon \end{aligned}$$

Let $t + \varepsilon = v \Rightarrow dv = d\varepsilon$

Then at $\varepsilon = 0$; $v = t$

At $\varepsilon = T$; $v = t + T$

$$R_{X_1 X_2}(t, t + \tau) = \frac{1}{T} \int_t^{t+T} p_1(v)p_2(v + \tau)dv$$

Since both $p_1(t)$ and $p_2(t)$ are periodic the above expression can be written as

$$R_{X_1 X_2}(t, t + \tau) = \frac{1}{T} \int_0^T p_1(v)p_2(v + \tau) dv = R_{X_1 X_2}(\tau)$$

6.54 Let $X(t)$ be the sum of a deterministic signal $x(t)$ and a wide-sense stationary noise process $N(t)$. Find the mean value, autocorrelation and autocovariance of $X(t)$. Discuss the stationarity of $X(t)$.

Solution Given: $X(t) = x(t) + N(t)$

The mean value $\mu_X(t) = E[X(t)] = E[x(t) + N(t)]$

$$\begin{aligned} &= x(t) + E[N(t)] \\ &= x(t) + \mu_N(t) = x(t) + \mu_N \end{aligned}$$

where μ_N is the mean value of the noise process.

$$\begin{aligned} R_{XX}(t, t + \tau) &= E[X(t)X(t + \tau)] \\ &= E[\{x(t) + N(t)\}\{x(t + \tau) + N(t + \tau)\}] \\ &= E[x(t)x(t + \tau) + N(t)x(t + \tau) + x(t)N(t + \tau) + N(t)N(t + \tau)] \\ &= x(t)x(t + \tau) + x(t + \tau)E[N(t)] + x(t)E[N(t + \tau)] + E[N(t)N(t + \tau)] \\ &= x(t)x(t + \tau) + \mu_N x(t + \tau) + \mu_N x(t) + R_{NN}(\tau) \end{aligned}$$

$$\begin{aligned}
C_{XX}(t, t + \tau) &= R_{XX}(t, t + \tau) - E[x(t) + N(t)]E[x(t + \tau) + N(t + \tau)] \\
&= R_{XX}(t, t + \tau) - \{x(t)x(t + \tau) + x(t)\mu_N + x(t + \tau)\mu_N + \mu_N^2\} \\
&= R_{NN}(\tau) - (\mu_N)^2
\end{aligned}$$

$X(t)$ is not stationary in any sense because the mean is time dependent.

6.55 Find the autocorrelation function of a random process with periodic sample function $p(t) = A \sin^2\left(\frac{2\pi t}{T}\right)$ where A and $T > 0$ are constants.

Solution For a random process with periodic sample function $p(t)$,

$$\begin{aligned}
R_{XX}(t, t + \tau) &= \frac{1}{T} \int_0^T p(v)p(v + \tau) dv \\
&= \frac{1}{T} \int_0^T A \sin^2\left(\frac{2\pi v}{T}\right) A \sin^2\left(\frac{2\pi(v + \tau)}{T}\right) dv \\
&= \frac{A^2}{T} \int_0^T \left[\frac{1 - \cos\left(\frac{4\pi v}{T}\right)}{2} \right] \left[\frac{1 - \cos\left(\frac{4\pi(v + \tau)}{T}\right)}{2} \right] dv \\
&= \frac{A^2}{4T} \int_0^T \left[1 - \cos\frac{4\pi v}{T} - \cos\frac{4\pi(v + \tau)}{T} + \cos\frac{4\pi v}{T} \cos\frac{4\pi(v + \tau)}{T} \right] dv \\
&= \frac{A^2}{4T} \int_0^T \left[1 - \cos\frac{4\pi v}{T} - \cos\frac{4\pi(v + \tau)}{T} + \frac{1}{2} \cos\left(\frac{4\pi\tau}{T}\right) + \frac{1}{2} \cos\left(\frac{8\pi v + 4\pi\tau}{T}\right) \right] dv \\
&= \frac{A^2}{4T} \left[T + \frac{T}{2} \cos\left(\frac{4\pi\tau}{T}\right) \right] = \frac{A^2}{4} \left[1 + \frac{1}{2} \cos\left(\frac{4\pi\tau}{T}\right) \right] \\
&= \frac{A^2}{8} \left[2 + \cos\left(\frac{4\pi\tau}{T}\right) \right] \quad \boxed{\because \int_0^T \cos\frac{4\pi v}{T} dv = 0}
\end{aligned}$$

6.56 A discrete-time random process is defined by $X_n = s^n$ for $n \geq 0$, where s is $U(0, 2)$

- Sketch a sample function of the process.
- Find the CDF of X_n
- Find the joint CDF of X_n and X_{n+1}

Solution Given: $X_n = s^n$ and $s \sim U(0, 1)$

A sample function of the process is shown in Fig. 6.18.

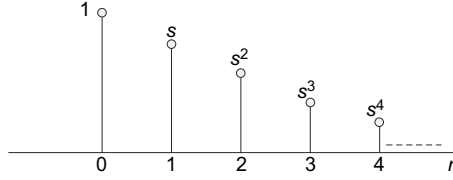


Fig. 6.18

The CDF of X_n is

$$F_{X_n}(x) = P(X_n \leq x) = P(s^n \leq x) = P(s \leq x^{1/n})$$

Since s is a uniform random variable between 0 and 1, we get

$$P(s \leq x^{1/n}) = x^{1/n}$$

If $x \sim U(0, 1)$

$$F_x(x) = \int_{-\infty}^x f_x(x) dx = \int_0^1 f_x(x) dx = \int_0^1 dx = x$$

That is, for a uniform random variable

$$F_x(x) = x$$

$$P(s < x^{1/n}) = x^{1/n}$$

$$\Rightarrow F_{X_n}(x) = x^{1/n} \quad 0 \leq x \leq 1$$

Consider the joint CDF of X_n and X_{n+1}

Let $X = X_n$ and $Y = X_{n+1}$

$$\begin{aligned} F_{X,Y}(x, y) &= P(X \leq x, Y \leq y) = P(X_n \leq x, X_{n+1} \leq y) \\ &= P(s^n \leq x, s^{n+1} \leq y) \\ &= P(s \leq x^{1/n}, s \leq y^{1/(n+1)}) \\ &= P[s \leq \min(x^{1/n}, y^{1/(n+1)})] \\ &= \min(x^{1/n}, y^{1/(n+1)}) \end{aligned}$$

6.57 For the Solved Problem 6.56, find mean and autocovariance of X_n . Check whether the process is mean ergodic.

Solution From the Solved Problem 6.56, we find

$$F_{X_n}(x) = x^{1/n} \quad 0 \leq x \leq 1$$

$$\text{and} \quad f_{X_n}(x) = \begin{cases} 1 & \text{for } 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

$$\begin{aligned}
 E[X_n] &= \int_{-\infty}^{\infty} s^n f_{X_n}(s) ds = \int_0^1 s^n ds = \left. \frac{s^{n+1}}{n+1} \right|_0^1 = \frac{1}{n+1} \\
 E[X_n X_{n+k}] &= E[s^n s^{n+k}] = E[s^{2n+k}] = \frac{1}{2n+k+1} \\
 C_X(n, n+k) &= E[X_n X_{n+k}] - E[X_n] E[X_{n+k}] \\
 &= \frac{1}{2n+k+1} - \frac{1}{n+1} \cdot \frac{1}{n+k+1}
 \end{aligned}$$

Since $C_X(n, n+k)$ is a function of n , X_n is not WSS.

Therefore, X_n is not mean ergodic.

6.58 Explain: (a) Are orthogonal random processes uncorrelated?
(b) Are uncorrelated processes orthogonal?

Solution

(a) Consider two random processes $X(t)$ and $Y(t)$.

If $X(t)$ and $Y(t)$ are orthogonal random processes, then

$$E[X(t_1), Y(t_2)] = 0$$

If $X(t)$ and $Y(t)$ are uncorrelated then

$$\text{Cov}[X(t_1), Y(t_2)] = 0$$

We have

$$\text{Cov}[X(t_1), Y(t_2)] = E[\{X(t_1) - E[X(t_1)]\} \{Y(t_2) - E[Y(t_2)]\}]$$

If either $E[X(t_1)]$ or/and $E[Y(t_2)]$ is zero then

$$\text{Cov}[X(t_1), Y(t_2)] = E[X(t_1) Y(t_2)] = 0$$

and hence $X(t)$ and $Y(t)$ are uncorrelated.

Therefore, two orthogonal random processes are uncorrelated when one of the random processes has zero mean.

(b) For uncorrelated processes,

$$E[X(t_1) Y(t_2)] = E[X(t_1)] E[Y(t_2)]$$

For orthogonal processes $E[X(t_1) Y(t_2)] = 0$

That is, $X(t)$ and $Y(t)$ will be orthogonal if $X(t)$ and/or $Y(t)$ are zero mean.

6.59 Consider a random process $Y_n = X_n + C(n)$ where X_n is a zero-mean unit variance random process and $C(n)$ is deterministic function. (a) Find the mean and variance of Y_n . (b) Find the joint CDF of Y_n and Y_{n+1} . (c) Find the mean and autocovariance function.

Solution

$$Y_n = X_n + C(n)$$

$$E[Y_n] = E[X_n + C(n)] = E[X_n] + C(n) = C(n)$$

$$\text{Var}(X_n) = \text{Var}[X_n + C(n)] = C^2(n) + \text{Var}(X_n)$$

$$= C^2(n) + 1$$

The CDF of Y_n is

$$\begin{aligned} F_{Y_n}(x) &= P(Y_n \leq x) = P[X_n + C(n) \leq x] \\ &= P(X_n \leq x - C(n)) = F_{X_n}[x - C(n)] \end{aligned}$$

The joint CDF of Y_n and Y_{n+1}

$$\begin{aligned} F_{Y_n, Y_{n+1}}[x_1, x_2] &= P(Y_n \leq x_1, Y_{n+1} \leq x_2) \\ &= P(X_n \leq x_1 - C(n), X_{n+1} \leq x_2 - C(n+1)) \\ &= F_{X_n, X_{n+1}}[x_1 - C(n), x_2 - C(n+1)] \end{aligned}$$



Practice Problems

6.14 Consider the random process $X(t) = \cos(t + \phi)$, where ϕ is a random variable with density function $f_\phi(\phi) = \frac{1}{\pi}$, $-\frac{\pi}{2} < \phi < \frac{\pi}{2}$... check whether the process is wide-sense stationary.

6.15 Examine whether the random process $X(t) = A \cos(\omega t + \theta)$ is wide-sense stationary if A and ω are constants and $\theta \sim u(0, 2\pi)$.

6.16 If $X(t) = \sin(\omega t + Y)$ where Y is uniformly distributed in $(0, 2\pi)$, show that $X(t)$ is a wide-sense stationary WSS process.

6.17 Show that the random process $X(t) = \cos(t + \phi)$ where $\phi \sim U(0, 2\pi)$ is

- (a) first-order stationary
- (b) Stationary in wide-sense
- (c) Ergodic in mean.

6.18 Find the mean, mean-square value and variance of the process $X(t)$ for which

$$R_{XX}(\tau) = 50 e^{-10|\tau|} + 25 \cos(5\tau) + 10 \quad (\text{Ans. } \pm 3.16, 85, 75)$$

6.19 The random process $X(t)$ is stationary with $E[X(t)] = 1$ and $R_{XX}(t) = 1 + e^{-2\tau}$, Find the mean and variance of

$$S = \int_0^1 X(t) dt \quad \left[\text{Ans. } 1, \frac{1}{2}(1 + e^{-2}) \right]$$

Solved Problem



6.60 If $\dot{X}(t)$ is time-derivative of a stationary continuous random process $X(t)$ then

- (a) Show that $E[\dot{X}(t)] = 0$
- (b) Find $R_{\dot{X}\dot{X}}(\tau)$ in terms of $R_{XX}(\tau)$
- (c) Find $R_{\dot{X}X}(\tau)$ in terms of $R_{XX}(\tau)$

Solution Let us define $\dot{X}(t) = \lim_{\varepsilon \rightarrow 0} \frac{X(t + \varepsilon) - X(t)}{\varepsilon}$

$$E[\dot{X}(t)] = E\left[\lim_{\varepsilon \rightarrow 0} \frac{X(t+\varepsilon) - X(t)}{\varepsilon}\right]$$

Assuming the order of limit and expectation operations can be interchanged, we get

$$(a) \quad E[\dot{X}(t)] = \lim_{\varepsilon \rightarrow 0} \left\{ \frac{E[X(t+\varepsilon)] - E[X(t)]}{\varepsilon} \right\}$$

$$E[X(t+\varepsilon)] = E[X(t)] = \mu_X$$

$$\text{Hence, } E[\dot{X}(t)] = \lim_{\varepsilon \rightarrow 0} \frac{\mu_X - \mu_X}{\varepsilon} = 0$$

$$(b) \quad R_{\dot{X}\dot{X}}(\tau) = E[X(t)\dot{X}(t+\tau)]$$

$$\begin{aligned} &= E\left[X(t) \left\{ \lim_{\varepsilon \rightarrow 0} \frac{X(t+\tau+\varepsilon) - X(t+\tau)}{\varepsilon} \right\}\right] \\ &= E\left[\lim_{\varepsilon \rightarrow 0} \left\{ \frac{X(t)X(t+\tau+\varepsilon) - X(t)X(t+\tau)}{\varepsilon} \right\}\right] \\ &= \left\{ \lim_{\varepsilon \rightarrow 0} \frac{E[X(t)X(t+\tau+\varepsilon)] - E[X(t)X(t+\tau)]}{\varepsilon} \right\} \\ &= \left\{ \lim_{\varepsilon \rightarrow 0} \frac{R_{XX}(\tau+\varepsilon) - R_{XX}(\tau)}{\varepsilon} \right\} \\ &= \frac{dR_{XX}(\tau)}{d\tau} \end{aligned}$$

$$(c) \quad R_{\dot{X}\dot{X}}(t+\tau) = E[\dot{X}(t)\dot{X}(t+\tau)]$$

$$\begin{aligned} &= E\left\{ \lim_{\varepsilon \rightarrow 0} \left(\frac{X(t+\varepsilon) - X(t)}{\varepsilon} \right) \dot{X}(t+\tau) \right\} \\ &= E\left\{ \lim_{\varepsilon \rightarrow 0} \left[\frac{X(t+\varepsilon)\dot{X}(t+\tau) - X(t)\dot{X}(t+\tau)}{\varepsilon} \right] \right\} \\ &= \lim_{\varepsilon \rightarrow 0} \left\{ \frac{E[X(t+\varepsilon)\dot{X}(t+\tau)] - E[X(t)\dot{X}(t+\tau)]}{\varepsilon} \right\} \\ &= \lim_{\varepsilon \rightarrow 0} \left\{ \frac{R_{X\dot{X}}(\tau+\varepsilon) - R_{X\dot{X}}(\tau)}{\varepsilon} \right\} \\ &= -\frac{dR_{X\dot{X}}(\tau)}{d\tau} = -\frac{d}{d\tau} \left[\frac{dR_{XX}(\tau)}{d\tau} \right] = -\frac{d^2 R_{XX}(\tau)}{d\tau^2} \end{aligned}$$



Poisson random process is a discrete random process. It is widely used to model the processes that count the number of occurrences of an event in the interval $(0, t)$. The event might be the number of phone calls arriving at a switchboard, customers entering at a bank or supermarket; the random failures of equipment in a system or the emission of electrons from the surface of a photodetector. To study about the Poisson process in detail, we first consider some basic definitions.

6.17.1 Counting Process

In all the above examples, the random process $X(t)$ represents the total number of events that occurred in the time interval $(0, t)$. Such types of processes are known as *counting processes*. So we can define the counting process as a random process $X(t)$ that represents the total number of events that have occurred in the interval $(0, t)$.

A counting process satisfies the following conditions.

- (a) The counting of events begins at time $t = 0$; that is, $X(0) = 0$.
- (b) $X(t)$ is an integer value and also $X(t) \geq 0$ which means that it has non-negative values.
- (c) $X(t)$ is a nondecreasing function of time. That is, if $t_1 < t_2$ then $X(t_1) \leq X(t_2)$.
- (d) $X(t_2) - X(t_1)$ represents the number of events occurred in the interval (t_1, t_2) .

The sample function of a counting process is shown in Fig. 6.19.

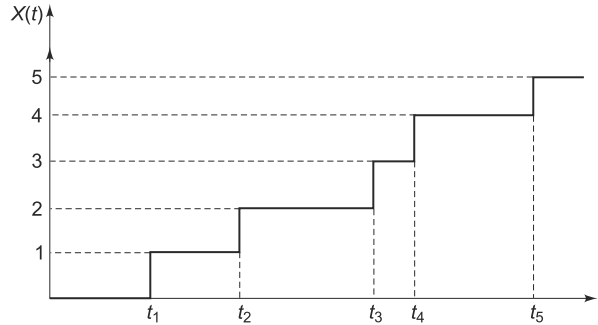


Fig. 6.19 Sample function of a counting process

Independent Increment Processes

The sample function of a counting process shown in Fig. 6.19. Let us consider two disjoint time intervals $(0, t_1)$ and (t_2, t_3) . Let the number of events occurring in the time interval $(0, t_1)$ be E_1 and the number of events occurring in the time interval (t_2, t_3) is E_2 . If E_1 is independent of E_2 then the process is an independent increment process.

Definition A counting process is said to be independent increment process if the number of events occurring in two disjoint time intervals is an independent random vector.

Stationary Increments Consider the sampling function of a counting process shown in Fig. 6.19. For every set of time instants $t_0 = 0 < t_1 < t_2, \dots, t_n$ if the increments $X(t_1) - X(t_0), X(t_2) - X(t_1), \dots, X(t_n) - X(t_{n-1})$ are identically distributed then the process is said to possess stationary increments.

Definition of a Poisson Process

Let $X_i(t)$ be defined as the number of events that occur during a specified time interval $(0, t)$. The random variable $X_i(t)$ defined above may assume the values 0, 1, 2,

The following five assumptions will now be made:

- (a) The number of events occurring during non-overlapping time intervals are independent random variables.
- (b) If $X_i(t)$ is the number of events occurred during $(0, t_1)$ and $Y_i(t)$ is the number of events occurred during $(t_1, t_1 + t)$ then for any $t_1 > 0$, the random variables X_i and Y_i have the same probability distribution.
- (c) If the interval is infinitely small, say Δt then the probability of one event during the interval is directly proportional to the length of that interval. That is,

$$p_1(\Delta t) = P[X_i(t + \Delta t) - X_i(t) = 1] = \lambda \Delta t + O(\Delta t) \quad (6.91)$$

where $O(\Delta t)$ is a function of Δt that goes to zero faster than Δt does. That is,

$$\lim_{\Delta t \rightarrow 0} \frac{O(\Delta t)}{\Delta t} = 0$$

We can write

$$p_1(\Delta t) = P[X_i(t + \Delta t) - X_i(t)] \sim \lambda \Delta t \quad (6.92)$$

- (d) The probability of obtaining two or more events during a sufficiently small interval is negligible.

$$\sum_{k=2}^{\infty} p_k(\Delta t) = O(\Delta t) \quad (6.93)$$

In other words

$$P[X(t + \Delta t) - X(t) \geq 2] = O(\Delta t) \quad (6.94)$$

- (e) Probability that no event in the interval Δt is

$$p_0(\Delta t) = P[X(t + \Delta t) - X(t) = 0] = 1 - \lambda \Delta t + O(\Delta t) \quad (6.95)$$

Consider the expression

$$\sum_{k=0}^{\infty} p_k(\Delta t) = p_0(\Delta t) + p_1(\Delta t) + p_2(\Delta t) + \dots = 1 \quad (6.96)$$

$$\Rightarrow p_0(\Delta t) = 1 - p_1(\Delta t) - p_2(\Delta t) - \dots$$

Using the assumption (c) and (d)

$$p_0(\Delta t) = 1 - [\lambda \Delta t + O(\Delta t)] \sim (1 - \lambda \Delta t) \quad (6.97)$$

We can also write

$$\begin{aligned} p_0(t + \Delta t) &= P[X_i(t + \Delta t) = 0] \\ &= P[X(t) = 0 \text{ and } (X(t + \Delta t) - X(t) = 0)] \end{aligned}$$

$$\begin{aligned} p_0(t)p_0(\Delta t) \\ = p_0(t)[1 - \lambda\Delta t] \end{aligned} \quad (6.98)$$

$$p_0(t + \Delta t) - p_0(t) = -\lambda p_0(t)\Delta t \quad (6.99)$$

From which we can write

$$\frac{p_0(t + \Delta t) - p_0(t)}{\Delta t} = -\lambda p_0(t) \quad (6.100)$$

$$\frac{p_0'(t)}{p_0(t)} = -\lambda$$

Integrating on both sides,

$$\ln p_0(t) = -\lambda t + c$$

If $t = 0$ $p_0(0) = 1 \Rightarrow c = 0$

$$p_0(t) = e^{-\lambda t} \quad (6.101)$$

$p_k(t)$ for $k \geq 1$ can be obtained by considering

$$p_k(t + \Delta t) = P[X(t + \Delta t) = k] \quad (6.102)$$

$$= P[k \text{ occurrences in the interval } (0, t) \text{ and zero occurrences in the interval } (t, t + \Delta t)]$$

$$+ P[k - 1 \text{ occurrences in the interval } (0, t) \text{ and one occurrence in the interval } (t, t + \Delta t)]$$

$$= P[X_i(t) = k; X_i(\Delta t) = 0] + P[X_i(t) = k - 1; X_i(\Delta t) = 1] \quad (6.103)$$

Using the assumption 1, we can write

$$\begin{aligned} p_k(t + \Delta t) &= P[X_i(t) = k]P[X_i(\Delta t) = 0] + P[X_i(t) = k - 1]P[X_i(\Delta t) = 1] \\ &= P[X_i(t) = k]p_0(\Delta t) + P[X_i(t) = k - 1]p_1(\Delta t) \end{aligned} \quad (6.104)$$

Using the relation

$$p_0(\Delta t) \sim (1 - \lambda\Delta t)$$

and

$$p_1(\Delta t) = \lambda\Delta t$$

We can write

$$\begin{aligned} p_k(t + \Delta t) &= P[X_i(t) = k](1 - \lambda\Delta t) + P[X_i(t) = k - 1]\lambda\Delta t \\ &= p_k(t)[1 - \lambda\Delta t] + p_{k-1}(t)\lambda\Delta t \end{aligned} \quad (6.105)$$

$$\begin{aligned} \frac{p_k(t + \Delta t) - p_k(t)}{\Delta t} &= -\lambda p_k(t) + \lambda p_{k-1}(t) \\ \lim_{\Delta t \rightarrow 0} \frac{p_k(t + \Delta t) - p_k(t)}{\Delta t} &= -\lambda p_k(t) + \lambda p_{k-1}(t) \end{aligned} \quad (6.106)$$

$$p_k'(t) + \lambda p_k(t) = \lambda p_{k-1}(t) \quad (6.107)$$

For $k = 1$,

$$p_1'(t) + \lambda p_1(t) = \lambda p_0(t) = \lambda e^{-\lambda t}$$

$$e^{\lambda t} p_1'(t) + \lambda e^{\lambda t} p_1(t) = \lambda$$

$$\frac{d}{dt} [e^{\lambda t} p_1(t)] = \lambda$$

Integrating on both sides, we get

$$e^{\lambda t} p_1(t) = \lambda t + c_1$$

$$\text{At } t = 0; p_1(0) = 0 \Rightarrow c_1 = 0$$

Therefore,

$$p_1(t) = \lambda t e^{-\lambda t} \quad (6.108)$$

For $k = 2$

$$p_2'(t) + \lambda p_2(t) = \lambda p_1(t) = \lambda^2 t e^{-\lambda t}$$

$$e^{\lambda t} p_2'(t) + \lambda e^{\lambda t} p_2(t) = \lambda^2 t$$

$$\frac{d}{dt} [e^{\lambda t} p_2(t)] = \lambda^2 t$$

Integrating on both sides, we get

$$e^{\lambda t} p_2(t) = \frac{\lambda^2 t^2}{2} + c_2$$

$$\text{For } t = 0; p_2(t) = 0 \Rightarrow c_2 = 0$$

$$\Rightarrow p_2(t) = \frac{\lambda^2 t^2}{2} e^{-\lambda t} = \frac{\lambda^2 t^2}{2!} e^{-\lambda t} \quad (6.109)$$

Similarly, we can prove that

$$p_k(t) = \frac{(\lambda t)^k}{k!} e^{-\lambda t}; \quad k = 0, 1, \dots, 2 \quad (6.110)$$

The probability density of the number of occurrences

$$f_X(x) = \sum_{k=0}^{\infty} \frac{(\lambda t)^k}{k!} e^{-\lambda t} \delta(x - k) \quad (6.111)$$

6.17.2 Joint Probability Density Function

To find the pdf of the Poisson process in the interval $0 < t_1 < t_2$, first we consider the probability of k_1 event occurrences over $(0, t_1)$ which is

$$P[X(t) = k_1] = \frac{(\lambda t_1)^{k_1}}{k_1!} e^{-\lambda t_1}, \quad k_1 = 0, 1, 2, \dots \quad (6.112)$$

If k_2 is the number of occurrences over $(0, t_2)$ then the number of occurrences over (t_1, t_2) is $k_2 - k_1$.

The probability that $(k_2 - k_1)$ events occurred over (t_1, t_2) is

$$P[X(t_2) = k_2 | X(t_1) = k_1] = \frac{[\lambda(t_2 - t_1)]^{k_2 - k_1} e^{-\lambda(t_2 - t_1)}}{(k_2 - k_1)!} \quad (6.113)$$

The joint probability of k_2 occurrences at time t_2 and k_1 occurrences at time t_1 is

$$\begin{aligned} P(k_1, k_2) &= P[X(t_2) = k_2 | X(t_1) = k_1] \cdot P[X(t_1) = k_1] \\ &= \frac{(\lambda t_1)^{k_1} [\lambda(t_2 - t_1)]^{k_2 - k_1} e^{-\lambda t_2}}{k_1! (k_2 - k_1)!} \end{aligned} \quad (6.114)$$

The joint density function is

$$f_Y(x_1, x_2) = \sum_{k_1=0}^{\infty} \sum_{k_2=0}^{\infty} P(k_1, k_2) \delta(x_1 - k_1) \delta(x_2 - k_2) \quad (6.115)$$

Mean of Poisson Process

Let $X(t)$ be a Poisson process.

$$\begin{aligned} \text{Then } \mu_X(t) &= E[X(t)] = \sum_{k=0}^{\infty} k p_k(t) = \sum_{k=0}^{\infty} k \frac{(\lambda t)^k}{k!} e^{-\lambda t} \quad k = 0, 1, 2, \dots \\ &= \sum_{k=0}^{\infty} \frac{(\lambda t)^k e^{-\lambda t}}{(k-1)!}; \quad k = 0, 1, 2, \dots \\ &= (\lambda t) e^{-\lambda t} \sum_{k=1}^{\infty} \frac{(\lambda t)^{k-1}}{(k-1)!} = (\lambda t) e^{-\lambda t} \left[1 + \frac{\lambda t}{1!} + \frac{(\lambda t)^2}{2!} + \dots \right] \\ &= \lambda t e^{-\lambda t} [e^{\lambda t}] = \lambda t \end{aligned}$$

$$\Rightarrow E[X(t)] = \lambda t \quad (6.116)$$

Variance of Poisson Process

$$\begin{aligned} E[X^2(t)] &= \sum_{k=0}^{\infty} k^2 p_k(t) = \sum_{k=0}^{\infty} \frac{k^2 e^{-\lambda t} (\lambda t)^k}{k!} \\ &= \sum_{k=1}^{\infty} \left[k(k-1) + k \right] \frac{e^{-\lambda t} (\lambda t)^k}{k!} \\ &= \sum_{k=1}^{\infty} \frac{k(k-1) e^{-\lambda t} (\lambda t)^k}{k!} + \sum_{k=1}^{\infty} \frac{k e^{-\lambda t} (\lambda t)^k}{k!} \\ &= e^{-\lambda t} \sum_{k=2}^{\infty} \frac{(\lambda t)^k}{(k-2)!} + \lambda t \\ &= e^{-\lambda t} \sum_{k=2}^{\infty} (\lambda t)^2 \frac{(\lambda t)^{k-2}}{(k-2)!} + \lambda t \end{aligned}$$

$$\begin{aligned}
&= e^{-\lambda t} (\lambda t)^2 \sum_{k=2}^{\infty} \frac{(\lambda t)^{k-2}}{(k-2)!} + \lambda t \\
&= e^{-\lambda t} (\lambda t)^2 e^{\lambda t} + \lambda t = (\lambda t)[\lambda t + 1] \\
\text{Var}[X(t)] &= E[X^2(t)] - \{E[X(t)]\}^2 \\
&= \lambda t [\lambda t + 1] - (\lambda t)^2 = \lambda t
\end{aligned} \tag{6.117}$$

Autocorrelation Function of Poisson Process

The autocorrelation function is given by

$$\begin{aligned}
R_{XX}(t_1, t_2) &= E[X(t_1)X(t_2)] \\
&= E[X(t_1)\{X(t_1) + (X(t_2) - X(t_1))\}] \\
&= E[X^2(t_1)] + E[X(t_1)\{X(t_2) - X(t_1)\}]
\end{aligned}$$

Using assumption 1 and assuming that $t_1 < t_2$ we get

$$\begin{aligned}
R_{XX}(t_1, t_2) &= E[X^2(t_1)] + E[X(t_1)]E[X(t_2) - X(t_1)] \\
&= E[X^2(t_1)] + E[X(t_1)]\{E[X(t_2)] - E[X(t_1)]\} \\
&= (\lambda t_1)^2 + \lambda t_1 + \lambda t_1[\lambda t_2 - \lambda t_1] \\
&= \lambda^2 t_1 t_2 + \lambda t_1
\end{aligned} \tag{6.118}$$

The autocovariance function is given by

$$\begin{aligned}
C_{XX}(t_1, t_2) &= R_{XX}(t_1, t_2) - E[X(t_1)]E[X(t_2)] \\
&= \lambda^2 t_1 t_2 + \lambda t_1 - (\lambda t_1)(\lambda t_2) = \lambda t_1
\end{aligned} \tag{6.119}$$

If $t_2 < t_1$ then $X(t_2)$ represents the number of events in the interval $(0, t_2)$ and $X(t_1)$ represents the number of events in $(0, t_1)$.

$X(t_1) - X(t_2)$ is the number of events in the interval (t_2, t_1) .

The autocorrelation function is

$$\begin{aligned}
R_{XX}(t_2, t_1) &= E[X(t_2)X(t_1)] \\
&= E[X(t_2)\{X(t_2) + (X(t_1) - X(t_2))\}] \\
&= E[X^2(t_2)] + E[X(t_2)\{X(t_1) - X(t_2)\}]
\end{aligned}$$

$$\begin{aligned}
&= E[X^2(t_2)] + E[X(t_2)]E[\{X(t_1) - X(t_2)\}] \\
&= (\lambda t_2)^2 + \lambda t_2 + \lambda t_2[\lambda t_1 - \lambda t_2] \\
&= \lambda^2 t_2 t_1 + \lambda t_2
\end{aligned} \tag{6.120}$$

$$\begin{aligned}
C_{XX}(t_2, t_1) &= R_{XX}(t_2, t_1) - E[X_1(t)]E[X_2(t)] \\
&= \lambda t_2 + \lambda^2 t_1 t_2 - (\lambda t_1)(\lambda t_2) = \lambda t_2
\end{aligned} \tag{6.121}$$

In general, we can write

$$R_{XX}(t_1, t_2) = \lambda^2 t_1 t_2 + \lambda [\min(t_1, t_2)] \tag{6.122}$$

$$\text{and } C_{XX}(t_1, t_2) = \lambda \min(t_1, t_2) \tag{6.123}$$

Correlation coefficient of Poisson Process

$$\rho = \frac{C_{XX}(t_1, t_2)}{\sigma_{X(t_1)} \sigma_{X(t_2)}} = \frac{\lambda \min(t_1, t_2)}{\sqrt{\lambda t_1} \sqrt{\lambda t_2}}$$

For $t_2 > t_1$,

$$\rho = \frac{\lambda t_1}{\sqrt{\lambda^2 t_1 t_2}} = \sqrt{\frac{t_1}{t_2}} \tag{6.124}$$

and for $t_1 > t_2$,

$$\rho = \frac{\lambda t_2}{\sqrt{\lambda^2 t_2 t_1}} = \sqrt{\frac{t_2}{t_1}} \tag{6.125}$$



Practice Problems

6.20 Show that the autocorrelation function of the Poisson process is

$$\begin{aligned}
R_{XX}(t_1, t_2) &= \lambda t_1 (1 + \lambda t_1); t_1 < t_2 \\
&= \lambda t_2 (1 + \lambda t_1); t_1 > t_2
\end{aligned}$$

6.21 Determine the autocovariance of the Poisson process.

6.22 Prove that Poisson process is not a stationary process.

Properties of the Poisson Process

1. Sum of two independent Poisson processes is also a Poisson process

Let $X_1(t)$ and $X_2(t)$ be two independent Poisson random processes with parameters λ_1 and λ_2 respectively, and then we can write $E[X_1(t)] = \lambda_1 t$ and $E[X_2(t)] = \lambda_2 t$. Let us define a process $X(t)$ which is sum of $X_1(t)$ and $X_2(t)$. That is

$$\begin{aligned}
X(t) &= X_1(t) + X_2(t) \\
\mu_X(t) &= E[X(t)] = E[X_1(t) + X_2(t)] = E[X_1(t)] + E[X_2(t)]
\end{aligned}$$

$$= \lambda_1 t + \lambda_2 t = (\lambda_1 + \lambda_2) t \quad (6.126)$$

The mean of the process $X(t)$ is $(\lambda_1 + \lambda_2)$.

$$\begin{aligned} E[X^2(t)] &= E\left[\{X_1(t) + X_2(t)\}^2\right] \\ &= E[X_1^2(t) + X_2^2(t) + 2X_1(t)X_2(t)] \\ &= E[X_1^2(t)] + E[X_2^2(t)] + 2E[X_1(t)X_2(t)] \\ &= \lambda_1 t(1 + \lambda_1 t) + \lambda_2 t(1 + \lambda_2 t) + 2(\lambda_1 t)(\lambda_2 t) \\ &= \lambda_1^2 t^2 + \lambda_1 t + \lambda_2 t + \lambda_2^2 t^2 + 2\lambda_1 \lambda_2 t^2 \\ &= (\lambda_1 t + \lambda_2 t)^2 + (\lambda_1 + \lambda_2)t \\ &= (\lambda_1 + \lambda_2)^2 t^2 + (\lambda_1 + \lambda_2)t \\ \text{Var}[X(t)] &= E[X^2(t)] - \{E[X(t)]\}^2 \\ &= (\lambda_1 + \lambda_2)^2 t^2 + (\lambda_1 + \lambda_2)t - (\lambda_1 + \lambda_2)^2 t^2 \\ &= (\lambda_1 + \lambda_2)t \end{aligned} \quad (6.127)$$

From the expressions for $\mu_X(t)$ and $\text{Var}[X(t)]$ we can find $X(t)$ is a Poisson process.

2. Difference of two independent Poisson processes is not a Poisson process

Let $X(t) = X_1(t) - X_2(t)$

We know $E[X_1(t)] = \lambda_1 t$ and $E[X_2(t)] = \lambda_2 t$

$$\begin{aligned} X(t) &= X_1(t) - X_2(t) \\ E[X(t)] &= E[X_1(t) - X_2(t)] = E[X_1(t)] - E[X_2(t)] \\ &= (\lambda_1 - \lambda_2)t \\ E[X^2(t)] &= E\left[\{X_1(t) - X_2(t)\}^2\right] \\ &= E[X_1^2(t) + X_2^2(t) - 2X_1(t)X_2(t)] \\ &= E[X_1^2(t)] + E[X_2^2(t)] - 2E[X_1(t)]E[X_2(t)] \\ &= \lambda_1 t(1 + \lambda_1 t) + \lambda_2 t(1 + \lambda_2 t) - 2(\lambda_1 t)(\lambda_2 t) \\ &= \lambda_1^2 t^2 + \lambda_2^2 t^2 - 2\lambda_1 \lambda_2 t^2 + (\lambda_1 + \lambda_2)t \end{aligned}$$

$$= (\lambda_1 - \lambda_2)^2 t^2 + \underbrace{(\lambda_1 + \lambda_2)t}_{\text{not } (\lambda_1 - \lambda_2)} \quad (6.128)$$

Since the underlined parameter is not $(\lambda_1 - \lambda_2)$, the process $X(t)$ is not a Poisson process.

3. The Poisson process is not a stationary process

The mean of a Poisson process $X(t)$ with parameter λ is given by

$$E[X(t)] = \lambda t \quad (6.129)$$

Since the mean is a function of time, the Poisson process is not a stationary process.

4. The interval between two successive occurrences of Poisson process with parameter λ follows an exponential distribution with mean $1/\lambda$.

Let T be the time at which the first event occurs. Since T is continuous random variable the CDF of T is

$$\begin{aligned} F_T(t) &= P(T \leq t) \\ &= 1 - P(T > t) \\ &= 1 - P[\text{no event occurs in } (0, t)] \\ &= 1 - P[X(t) = 0] \end{aligned}$$

$$\begin{aligned} P[X(t) = 0] &= p_0(t) = \frac{e^{-\lambda t} (\lambda t)^0}{0!} = e^{-\lambda t} \\ \Rightarrow F_T(t) &= 1 - e^{-\lambda t} \end{aligned}$$

Probability Density Function

$$f_T(t) = \frac{d}{dt} F_T(t) = \frac{d}{dt} (1 - e^{-\lambda t}) = \lambda e^{-\lambda t} \quad (6.130)$$

Thus, T is an exponential random variable.

REVIEW QUESTIONS



29. If the process $\{X(t); t > 0\}$ is a Poisson process with parameter λ , obtain $P[X(t) = n]$. Is the process first order stationary?
30. State the postulates of a Poisson process and derive the probability distribution. Also prove that the sum of two independent Poisson processes is a Poisson process.
31. Derive Poisson process with rate λ and hence mean. Is Poisson process stationary? Explain.
32. Derive expression for autocorrelation of a Poisson process.
33. Prove that the interval between two successive occurrences of Poisson process with parameter λ follows an exponential distribution with mean $\frac{1}{\lambda}$.

Solved Problems



6.61 The customers arrive at a bank according to Poisson process with mean rate 5 per minute. Find the probability that during a 1-minute interval, no customer arrives.

Solution Given: $t = 1$ and $\lambda = 5 \Rightarrow \lambda t = 5$

We know

$$p_k(t) = P[X(t) = k] = \frac{e^{-\lambda t} (\lambda t)^k}{k!} \quad k = 0, 1, 2, \dots$$

$$p_0(t) = P[X(t) = 0] = \frac{e^{-5} (5)^0}{0!} = e^{-5}$$

6.62 The particles are emitted from a radioactive source at the rate of 50 per hour. Find the probability that exactly 10 particles are emitted during a 10-minute period.

Solution Given: $\lambda = 50$ per hour

$$t = 10 \text{ minute} = 1/6 \text{ h}$$

$$\lambda t = \frac{50}{6} = \frac{25}{3}$$

Given $k = 10$

$$p_k(t) = P[X(t) = k] = \frac{(\lambda t)^k e^{-\lambda t}}{k!}$$

$$p\left[X\left(\frac{1}{6}\right) = 10\right] = \frac{\left(\frac{25}{3}\right)^{10} e^{-25/3}}{10!} = 0.1064$$

6.63 Telephone calls are initiated through an exchange at the average rate of 75 per minute and are described by a Poisson process. Find the probability that more than 3 calls are initiated in any 5-second period.

Solution Given: $\lambda = 75$ per minute

$$= \frac{75}{60} \text{ per second}$$

$$t = 5 \text{ seconds}$$

$$\lambda t = \frac{75}{60} (5) = \frac{75}{12}$$

$$P[X(t) > 3] = 1 - P[X(t) \leq 3]$$

$$= 1 - \{P[X(t) = 3] + P[X(t) = 2] + P[X(t) = 1] + P[X(t) = 0]\}$$

$$P[X(t) = k] = \frac{e^{-\lambda t} (\lambda t)^k}{k!}$$

$$= 1 - \left\{ \frac{\left(\frac{75}{12}\right)^3}{3!} + \frac{\left(\frac{75}{12}\right)^2}{2!} + \frac{\left(\frac{75}{12}\right)}{1!} + \frac{\left(\frac{75}{12}\right)^0}{0!} \right\} e^{-75/12} = 1 - 0.1302 = 0.8698$$

6.64 A machine goes out of order, whenever a component fails. The failure of this part follows a Poisson process with a mean rate of 1 per week. Find the probability that 2 weeks have elapsed since the last failure. If there are 5 spare parts of this component in an inventory and that the next supply is not due in 10 weeks, find the probability that the machine will not be out of order in the next 10 weeks.

Solution Given: $\lambda = 1$; $t = 2$

$$\Rightarrow \lambda t = 2$$

$P[2 \text{ weeks have elapsed since last failure}]$

$$= P[X(2) = 0] = \frac{e^{-2}(2)^0}{0!} = e^{-2} = 0.135$$

Since there are only 5 spare parts, with these parts, the machine can fail 5 numbers of times. Hence, we can write

$$\begin{aligned} P[X(10) \leq 5] &= \sum_{k=0}^5 \frac{e^{-10}(10)^k}{k!} \\ &= e^{-10} \left[1 + \frac{10}{1!} + \frac{10^2}{2!} + \frac{10^3}{3!} + \frac{10^4}{4!} + \frac{10^5}{5!} \right] = 0.0671 \end{aligned}$$

6.65 Aircraft arrive at an airport according to a Poisson process at a rate of 12 per hour. All aircraft are handled by one air-traffic controller. If the controller takes a 2-minute break, what is the probability that he/she will miss one or more arriving aircraft?

Solution Given: $\lambda = 12$ per hour

$$= \frac{12}{60} \text{ per minute} = \frac{1}{5} \text{ per minute}$$

$$t = 2 \text{ minutes}$$

$$\lambda t = \frac{2}{5} = 0.4$$

The probability that he will miss one or more arriving crafts.

$$= 1 - p \text{ (he miss zero aircraft)}$$

$$= 1 - \frac{(\lambda t)^0 e^{-\lambda t}}{0!}$$

$$= 1 - e^{-0.4} = 0.3297$$



Practice Problems

- 6.23** In a village road, buses cross a particular place at a Poisson rate 4 per hour. If a boy starts counting at 9.00 a.m.,
- What is the probability that his count is 1 by 9.30 a.m.
 - What is the probability that his count is 3 by 11.00 a.m.
- (Ans. (i) 0.3088, (ii) 0.522)
- 6.24** Let $X(t)$ a Poisson process with arrival rate λ . Find $E\{[X(t) - X(s)]^2\}$ for $t > s$.
 (Ans. $(t - s)^2 \lambda^2 + (t - s)$)
- 6.25** Find the first-order characteristic function of a Poisson process
 (Ans: $e^{-\lambda}(1 - e^{-j\omega})$)
- 6.26** A radioactive source emits particles at the rate of 10 per minute in accordance with Poisson process. Each particle emitted has a probability of $1/3$ of being rewarded. Find the probability that at least 6 particles are recorded in a 6-minute period.



SINE WAVE RANDOM PROCESS 6.18

A sine wave random process is represented by $X(t) = A \sin(\omega t + \phi)$ where A is the amplitudes, ω is frequency and ϕ is the phase. Figure 6.20(a) shows different sample realizations of sine wave random process if A is a random variable between 0 and 1. Similarly, if ω is a random variable, the different sample realizations are shown in Fig. 6.20(b). If ϕ is a random variable between 0 and π , the sample realizations are shown in Figure 6.20(c).

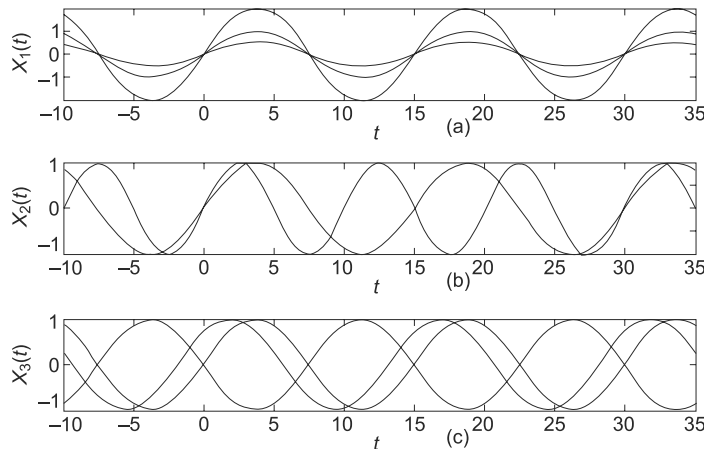


Fig. 6.20 Sine wave random process

Solved Problems



- 6.66** Find the average and time autocorrelation function of a random process $X(t) = A \cos(\omega_0 t + \theta)$ where A and ω_0 are constants and $\theta \sim U(0, 2\pi)$.

Solution Consider a sample function $X(t) = A \cos(\omega_0 t + \theta)$

The time average is given by

$$\begin{aligned}\bar{X} &= \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T X(t) dt \\ &= \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T A \cos(\omega_0 t + \theta) dt = 0\end{aligned}$$

The time autocorrelation function is

$$\begin{aligned}R_{XX}(t, t+T) &= \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T A \cos(\omega_0 t + \theta) A \cos(\omega_0 t + \omega_0 \tau + \theta) dt \\ &= \frac{A^2}{2} \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T}^T \left\{ \frac{\cos(\omega_0 \tau) + \cos(2\omega_0 t + \omega_0 \tau + 2\theta)}{2} \right\} dt \\ &= \frac{A^2}{2} \left\{ \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T (\cos \omega_0 \tau) dt + \underbrace{\lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T \cos(2\omega_0 t + \omega_0 \tau + 2\theta) d\theta}_0 \right\} \\ &= \frac{A^2}{2} \cos(\omega_0 \tau) \left\{ \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T dt \right\} \\ &= \frac{A^2}{2} \cos(\omega_0 \tau) \\ R_{XX}(\tau) &= \frac{A^2}{2} \cos(\omega_0 \tau)\end{aligned}$$

6.67 Determine whether the random process $X(t) = A \cos(\omega_0 t + \theta)$ is wide-sense stationary or not, where A , ω_0 are constants and θ is a uniformly distributed random variable on the interval $(0, 2\pi)$.

Solution For a uniformly distributed random variable on the interval (a, b) , the mean is $\frac{b+a}{2}$ and the variance is $\frac{(b-a)^2}{12}$.

Since θ is a uniformly distributed random variable on the interval

$$\begin{aligned}f(\theta) &= \frac{1}{2\pi} \quad \text{for} \quad 0 \leq \theta \leq 2\pi \\ &= 0 \quad \text{otherwise}\end{aligned}$$

$$E[\theta] = \pi \text{ and } \sigma_\theta^2 = \frac{(2\pi - 0)^2}{12} = \frac{\pi^2}{3}$$

A process $X(t)$ is said to be wide-sense stationary if $E[X(t)] = \text{constant}$ and $E[X(t)X(t+\tau)] = R_{XX}(\tau)$

$$\begin{aligned}
E[X(t)] &= E[A \cos(\omega_0 t + \theta)] = \int_{-\infty}^{\infty} A \cos(\omega_0 t + \theta) f(\theta) d\theta \\
&= \frac{1}{2\pi} \int_0^{2\pi} A \cos(\omega_0 t + \theta) d\theta = \frac{A}{2\pi} \sin(\omega_0 t + \theta) \Big|_0^{2\pi} \\
&= \frac{A}{2\pi} [\sin(\omega_0 t + 2\pi) - \sin \omega_0 t] = 0
\end{aligned}$$

$$E[A \cos(\omega_0 t + \theta)] = 0$$

The auto-correlation function of a random process is given by

$$\begin{aligned}
R_{XX}(t, t + \tau) &= E[A \cos(\omega_0 t + \theta) A \cos(\omega_0 t + \omega_0 \tau + \theta)] \\
&= \frac{1}{2} E[A^2 \cos(2(\omega_0 t + \theta) + \omega_0 \tau) + A^2 \cos(\omega_0 \tau)] \\
&= \frac{A^2}{2} \{E[\cos(2\omega_0 t + \theta) + \omega_0 \tau] + E[\cos \omega_0 \tau]\} \\
E[\cos(2\omega_0 t + \theta + \omega_0 \tau)] &= \int_0^{2\pi} \cos(2\omega_0 t + \theta + \omega_0 \tau) \frac{1}{2\pi} d\theta \\
&= \frac{1}{2\pi} [\sin(2\omega_0 t + \theta + \omega_0 \tau)]_0^{2\pi} \\
&= \frac{1}{2\pi} [\sin(2\omega_0 t + 2\pi + \omega_0 \tau) - \sin(2\omega_0 t + \omega_0 \tau)] = 0 \\
R_{XX}(t, t + \tau) &= \frac{A^2}{2} E[\cos \omega_0 \tau]
\end{aligned}$$

$$\Rightarrow E[\cos \omega_0 \tau] = \cos \omega_0 \tau$$

$$R_{XX}(t, t + \tau) = \frac{A^2}{2} \cos(\omega_0 \tau)$$

Since autocorrelation function depends on the shift τ , the random process $X(t)$ is wide-sense stationary.

6.68 A random process is defined by $X(t) = A \sin \omega t$, $t \geq 0$, where ω is a constant. The amplitude A is uniformly distributed between 0 and 1. Determine the following: (a) $E[X(t)]$ (b) The autocorrelation of $X(t)$ (c) The autocovariance function of $X(t)$.

Solution A is uniformly distributed between 0 and 1.

$$\text{Therefore } E[A] = \frac{0+1}{2} = 0.5$$

(a) The mean of $X(t)$ is given by

$$E[X(t)] = E[A \sin \omega t] = E[A] \sin \omega t = 0.5 \sin \omega t \quad [\because \omega \text{ is a constant}]$$

$$E[X(t)] = 0.5 \sin \omega t$$

The variance of A is given by

$$\sigma_A^2 = \frac{(1-0)^2}{2} = \frac{1}{12}$$

$$E[A^2] = \sigma_A^2 + \{E[A]\}^2 = \frac{1}{12} + \left(\frac{1}{2}\right)^2 = \frac{1}{3}$$

(b) The autocorrelation function of $X(t)$ is given by

$$\begin{aligned} R_{XX}(\tau) &= E[X(t)X(t+\tau)] \\ &= E[A \sin \omega t \ A \sin \omega(t+\tau)] \\ &= E[A^2] \sin \omega t \sin \omega(t+\tau) \\ &= \frac{1}{3} \sin \omega t \sin \omega(t+\tau) \end{aligned}$$

(c) The autocovariance function of $X(t)$ is given by

$$\begin{aligned} C_{XX}(\tau) &= R_{XX}(\tau) - E[X(t)] E[X(t+\tau)] \\ &= \frac{1}{3} \sin \omega t \sin \omega(t+\tau) - 0.5 \sin \omega t (0.5 \sin \omega(t+\tau)) \\ &= \frac{1}{12} \sin \omega t \sin \omega(t+\tau) \end{aligned}$$

6.69 Consider a random process $X(t) = A \cos(\omega_0 t + \theta)$, where ω_0 is a known constant, A is a random variable uniformly distributed over $(0,1)$ and θ is a random variable distributed over $(0, \pi)$. Assume that A and θ are independent. Find the autocorrelation function.

Solution Given: $X(t) = A \cos(\omega_0 t + \theta)$

A is a uniformly distributed random variable over $(0,1)$.

Therefore, $E[A] = 0.5$ and $\sigma_A^2 = \frac{1}{12}$

$$E[A^2] = \sigma_A^2 + [E(A)]^2 = \frac{1}{12} + \frac{1}{4} = \frac{1}{3}$$

The autocorrelation function

$$\begin{aligned} R_{XX}(t, t+\tau) &= E[X(t)X(t+\tau)] \\ &= E[A \cos(\omega_0 t + \theta) A \cos(\omega_0(t+\tau) + \theta)] \\ &= E[A^2 \cos(\omega_0 t + \theta) \cos(\omega_0(t+\tau) + \theta)] \\ &= E[A^2] E[\cos(\omega_0 t + \theta) \cos(\omega_0(t+\tau) + \theta)] \\ &= \frac{1}{2} \left(\frac{1}{3}\right) \{E[\cos \omega_0 \tau] + E[\cos(2\omega_0 t + \omega_0 \tau + 2\theta)]\} \end{aligned}$$

Since θ is a random variable uniformly distributed over $(0, 2\pi)$,

$$E[\cos(2\omega_0 t + \omega_0 \tau + 2\theta)] = \int_{-\infty}^{\infty} \cos(2\omega_0 t + \omega_0 \tau + 2\theta) f(\theta) d\theta$$

$$f(\theta) = \begin{cases} \frac{1}{2\pi} & \text{for } 0 \leq \theta \leq 2\pi \\ 0 & \text{otherwise} \end{cases}$$

$$E[\cos(2\omega_0 t + \omega_0 \tau + 2\theta)] = \frac{1}{2\pi} \int_0^{2\pi} \cos(2\omega_0 t + \omega_0 \tau + 2\theta) d\theta = 0$$

$$R_{XX}(t, t + \tau) = \frac{1}{6} \cos \omega \tau$$

6.70 Find the autocorrelation function of a random process $X(t) = A \cos(\omega_0 t + \theta)$, where ω_0 , A and θ are mutually independent random variables distributed over $(0,1)$, $(0,1)$ and $(0, \pi)$ respectively.

Solution Given $X(t) = A \cos(\omega_0 t + \theta)$

$$R_{XX}(t, t + \tau) = E[A \cos(\omega_0 t + \theta) A \cos(\omega_0(t + \tau) + \theta)]$$

ω_0 , A and θ are uniformly distributed over $(0,1)$, $(0,1)$ and $(0, \pi)$ respectively. Therefore,

$$f(\omega_0) = 1 \text{ for } 0 \leq \omega_0 \leq 1$$

$$= 0 \text{ otherwise}$$

$$f(A) = 1 \text{ for } 0 \leq A \leq 1$$

$$= 0 \text{ otherwise}$$

$$f(\theta) = \frac{1}{\pi} \text{ for } 0 \leq \theta \leq \pi$$

$$= 0 \text{ otherwise}$$

$$\text{Also, } E[\omega_0] = \frac{1+0}{2} = 0.5; \sigma_{\omega_0}^2 = \frac{(1-0)^2}{12} = \frac{1}{12}$$

$$E[A] = \frac{1+0}{2} = 0.5; \sigma_A^2 = \frac{(1-0)^2}{12} = \frac{1}{12}$$

$$E[\theta] = \frac{\pi+0}{2} = \frac{\pi}{2}; \sigma_{\theta}^2 = \frac{(\pi-0)^2}{12} = \frac{\pi^2}{12}$$

$$R_{XX}(t, t + \tau) = E[A^2 \cos(\omega_0 t + \theta) \cos(\omega_0 t + \omega_0 \tau + \theta)]$$

Since ω_0 , A and θ are independent,

$$R_{XX}(t, t + \tau) = E[A^2] E[\cos(\omega_0 t + \theta) \cos(\omega_0 t + \omega_0 \tau + \theta)]$$

$$= \frac{1}{2} E[A^2] \{E(\cos \omega_0 \tau) + E[\cos(2\omega_0 t + \omega_0 \tau) \cos 2\theta - \sin(2\omega_0 t + \omega_0 \tau) \sin 2\theta]\}$$

$$\begin{aligned}
E[\cos \omega_0 \tau] &= \int_{-\infty}^{\infty} \cos(\omega_0 \tau) f(\omega_0) d\omega_0 \\
&= \int_0^1 \cos(\omega_0 \tau) d\omega_0 = \sin \omega_0 \tau \Big|_0^1 = \sin \tau \\
E[A^2] &= \sigma_A^2 + \{E[A]\}^2 = \frac{1}{12} + \left(\frac{1}{2}\right)^2 = \frac{4}{12} = \frac{1}{3} \\
E[\cos 2\theta] &= \int_{-\infty}^{\infty} \cos 2\theta f(\theta) d\theta = \frac{1}{\pi} \int_0^{\pi} \cos 2\theta d\theta = \frac{1}{\pi} \left[\frac{\sin 2\theta}{2} \right]_0^{\pi} = 0 \\
E[\sin 2\theta] &= \int_{-\infty}^{\infty} \sin 2\theta f(\theta) d\theta = \frac{1}{\pi} \int_0^{\pi} \sin 2\theta d\theta = \frac{-1}{\pi} \left[\frac{\cos 2\theta}{2} \right]_0^{\pi} = 0 \\
R_{XX}(t, t + \tau) &= \frac{1}{2} \left(\frac{1}{3} \right) \sin \tau = \frac{1}{6} \sin \tau
\end{aligned}$$



Practice Problems

6.27 Repeat the above problem with $\omega \sim U(1, 2)$, $A \sim U(0, 1)$ and $\theta \sim U\left(0, \frac{\pi}{2}\right)$

6.28 Verify whether the sine-wave process $X(t) = A \sin(\omega t + \theta)$ where A is uniformly distributed in the interval -1 to 1 is WSS or not. (Ans. not – WSS)



BERNOULLI PROCESS 6.19

The Bernoulli random variable is used when the outcome of an experiment has only two outcomes. For example in tossing a coin experiment, the outcome can be either head or tail and in hitting a target, the outcome is either hit or miss. Similarly, the experiments that has outcomes like failure/success, on/off are represented by Bernoulli random variables.

For a Bernoulli random variable if the probability of failure/success is p then the probability of success/failure is $1 - p$. Let X_n be the random variable that denotes, the outcome of n th trial. The value of $X_n = 0$ if the outcome is failure and $X_n = 1$ if the outcome is success. Then $\{X_n\}$ $n = 0, 1, 2, 3, \dots$ is a Bernoulli process. The pmf of X_n is given by

$$p_X(X_n) = \begin{cases} p & x = 1 \\ 1 - p & x = 0 \end{cases} \quad (6.131)$$

The mean of Bernoulli process X_n is

$$E[X_n] = \mu_{X_n} = 1 \cdot P(X_n = 1) + 0 \cdot P(X_n = 0) = p \quad (6.132)$$

and

Variance is

$$\begin{aligned}\text{Var}(X_n) &= E[X_n]^2 - \mu_{X_n}^2 \\ &= (1)^2 p + (0)^2 (1-p) - p^2 = p(1-p)\end{aligned}\quad (6.133)$$

The autocorrelation is given by

$$\begin{aligned}E[X_{n_1} X_{n_2}] &= (1)^2 \cdot p + 0^2(1-p) = p \text{ for } n_1 = n_2 \\ E[X_{n_1}] E[X_{n_2}] &= p^2 \text{ for } n_1 \neq n_2\end{aligned}\quad (6.134)$$

A sample function of Bernoulli process is shown in Fig. 6.21

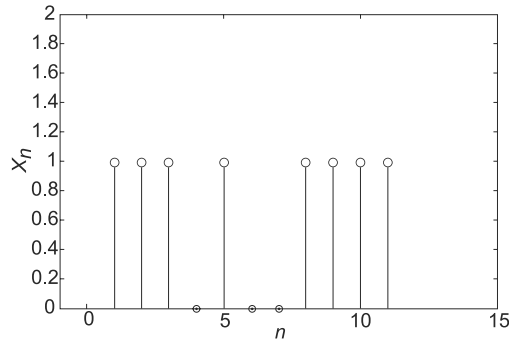


Fig. 6.21 Bernoulli random process

The autocovariance is

$$\begin{aligned}C_{XX}(n_1, n_2) &= R_{XX}(n_1, n_2) - \mu_{X_n}^2 \\ &= \begin{cases} p - p^2 = p(1-p) & \text{if } n_1 = n_2 \\ p^2 - p^2 = 0 & \text{if } n_1 \neq n_2 \end{cases}\end{aligned}\quad (6.135)$$

6.19.1 Binomial Process

Consider a process Y_n which is the sum of the Bernoulli random variables $\{X_i\}$. The process is known as binomial process and is represented as

$$Y_n = \sum_{i=1}^n X_i \quad n > 0 \quad (6.136)$$

The random variable Y_n denotes the number of successes in n Bernoulli trials. The pmf of Y_n is given by

$$\begin{aligned}p_{Y_n}(k) &= \binom{n}{k} p^k (1-p)^{n-k} \quad k = 0, 1, \dots, n \\ E[Y_n] &= E\left[\sum_{i=1}^n X_i\right] = \sum_{i=1}^n E[X_i] = \sum_{i=1}^n p = np\end{aligned}\quad (6.137)$$

$$\begin{aligned}
\text{Var}[Y_n] &= E[Y_n^2] - \{E[Y_n]\}^2 \\
E[Y_n^2] &= E\left[\left(\sum_{i=1}^n X_i\right)^2\right] = \sum_{i=1}^n E[X_i]^2 + \sum_{i=1}^n \sum_{j=1}^n E[X_i X_j] \\
&= np + n(n-1)p^2 \\
\text{Var}[Y_n] &= np + n(n-1)p^2 - n^2 p^2 \\
&= np(1-p)
\end{aligned} \tag{6.138}$$

Solved Problems



6.71 A random process is defined by $S_n = 2X_{n+1}$ where X_n is a Bernoulli process. Find the mean and variance of S_n .

Solution Since X_n is a Bernoulli process, it has two outcomes, one with probability p and the other with probability $(1-p)$. The mean and variance of Bernoulli process is

$$\begin{aligned}
E[X_n] &= p; \quad \text{Var}(X_n) = pq \\
E[S_n] &= 2E[X_n] + 1 = 2p + 1 \\
\text{Var}[S_n] &= \text{Var}[2X_{n+1}] = 4 \text{Var}[X_n] \\
&= 4pq
\end{aligned}$$

6.72 The probability that a student pass in an examination is 0.4. If 10 students have taken the examination, find the following probabilities

(a) At least 5 pass, (b) Exactly 6 students pass.

Solution Let X denote the number of students who pass the examination. Then X has the binomial distribution with $p = 0.4$ and pmf

$$p_X(x) = \binom{10}{n} (0.4)^n (0.6)^{10-n} \quad n = 0, 1, \dots, 10$$

The probability that at least 5 students pass is given by

$$\begin{aligned}
P(X \geq 5) &= \sum_{n=5}^{10} p_X(x) \\
&= \binom{10}{5} (0.4)^5 (0.6)^5 + \binom{10}{6} (0.4)^6 (0.6)^4 + \binom{10}{4} (0.4)^7 (0.6)^3 \\
&\quad + \binom{10}{8} (0.4)^8 (0.6)^2 + \binom{10}{9} (0.4)^9 (0.6) + \binom{10}{10} (0.4)^{10} \\
&= 0.398
\end{aligned}$$

The probability that exactly 6 students pass

$$p_X(X=6) = \binom{10}{6} (0.4)^6 (0.6)^4 = 0.1115$$



Practice Problem

6.29 The diodes produced by a certain company will be defective with probability 0.02. The company sells the diodes in packages of 20 and offers a money-back guarantee that at most 2 of the 50 diodes is defective. What proportional of packages sold must the company replace?



GAUSSIAN RANDOM PROCESS 6.20

Gaussian processes are important in the study of random processes, since many random processes can be modeled as Gaussian process. Any random process can be considered as the cumulative effect of many random phenomena. According to central limit theorem, the linear combination of random processes with different distributions tends to random process with normal distribution. Examples of random processes that can be modeled as Gaussian process include the effect of thermal noise in electronic circuits used in communication systems, the error in measurement of quantity of interest, quantization in digital systems.

If samples of a random process $X(t_1), X(t_2), \dots, X(t_n)$ taken at arbitrary points in time $t_1, t_2, \dots, t_n$ form a set of jointly Gaussian random variables for any $n = 1, 2, 3, \dots$ then the real-valued random process $\{X(t)\}$ is called a Gaussian random process. The random variables $X(t_1), X(t_2), \dots, X(t_n)$ will have a joint pdf given by

$$f_{X_1, X_2, \dots, X_n}(x_1, x_2, \dots, x_n) = \frac{1}{(2\pi)^{n/2} |C_{XX}|^{n/2}} \exp \left\{ -\frac{1}{2} (X - \bar{X})^T C_{XX}^{-1} (X - \bar{X}) \right\} \quad (6.139)$$

where C_{XX} is known as covariance matrix and $\bar{X}$ is the vector of mean functions of $X(t_k)$; X is a vector of $X(t_k)$.

$$\bar{X} = \begin{bmatrix} \bar{X}(t_1) \\ \bar{X}(t_2) \\ \vdots \\ \bar{X}(t_n) \end{bmatrix} \quad X = \begin{bmatrix} X(t_1) \\ X(t_2) \\ \vdots \\ X(t_n) \end{bmatrix} \quad (6.140)$$

$$C_{XX} = \begin{bmatrix} C_{XX}(t_1, t_1) & C_{XX}(t_1, t_2) & \cdots & C_{XX}(t_1, t_n) \\ C_{XX}(t_2, t_1) & C_{XX}(t_2, t_2) & \cdots & C_{XX}(t_2, t_n) \\ \vdots & \vdots & \ddots & \vdots \\ C_{XX}(t_n, t_1) & C_{XX}(t_n, t_2) & \cdots & C_{XX}(t_n, t_n) \end{bmatrix}$$

$$= \begin{bmatrix} C_{11} & C_{12} & \cdots & C_{1n} \\ C_{21} & C_{22} & \cdots & C_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ C_{n1} & C_{n2} & \cdots & C_{nn} \end{bmatrix} \quad (6.141)$$

If the random variables $X(t_1), X(t_2), \dots, X(t_n)$ are uncorrelated then

$$C_{XX}(t_i, t_j) = \begin{cases} \sigma_i^2 & \text{for } i = j \\ 0 & \text{for } i \neq j \end{cases} \quad (6.142)$$

That is C_{XX} is ~ diagonal matrix with elements in the principal diagonal equal to σ_i^2 . That is,

$$C_{XX} = \begin{bmatrix} \sigma_1^2 & 0 & 0 & \cdots & 0 \\ 0 & \sigma_2^2 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \cdots & \vdots \\ 0 & 0 & 0 & \cdots & \sigma_n^2 \end{bmatrix} \quad (6.143)$$

$$C_{XX}^{-1} = \begin{bmatrix} \frac{1}{\sigma_1^2} & 0 & 0 & \cdots & 0 \\ 0 & \frac{1}{\sigma_2^2} & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \cdots & \vdots \\ 0 & 0 & 0 & \cdots & \frac{1}{\sigma_n^2} \end{bmatrix} \quad (6.144)$$

$$C_{XX}^{1/2} = \begin{bmatrix} \sigma_1 & 0 & 0 & \cdots & 0 \\ 0 & \sigma_2 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \cdots & \vdots \\ 0 & 0 & 0 & \cdots & \sigma_n \end{bmatrix} \quad (6.145)$$

$$|C_{XX}^{1/2}| = \sigma_1 \sigma_2 \cdots \sigma_n = \prod_{i=1}^n \sigma_i \quad (6.146)$$

$$(X - \bar{X}) C_{XX}^{-1} (X - \bar{X}) = \sum_{k=1}^n \frac{x_k - \bar{X}(t_k)}{\sigma_k^2} \quad (6.147)$$

$$\begin{aligned} f_{X(t_1), \dots, X(t_n)}(x_1, x_2 \cdots x_n) \\ = \frac{1}{(2\pi)^{n/2} \prod_{i=1}^n \sigma_i} \exp \left[- \sum_{k=1}^n \frac{x_k - \bar{X}(t_k)}{2\sigma_k^2} \right] \end{aligned} \quad (6.148)$$

Properties of Gaussian Processes

1. A Gaussian process that is wide-sense stationary is also SSS.
2. If the random variables $X(t_1), X(t_2), \dots, X(t_n)$ of a Gaussian process $X(t)$ are uncorrelated then they are independent.
3. If the input to a linear system is a Gaussian process, the output is also a Gaussian process.

Solved Problems



6.73 $X(t)$ is a WSS process with ACF

$$R_{XX}(t) = 9 + e^{-2|t|}$$

Determine the covariance matrix for the random variables $X(0), X(1), X(2)$ and $X(3)$.

Solution

$$R_{XX}(\tau) = 9 + e^{-2|\tau|}$$

$$E[X(t)] = \pm \sqrt{\lim_{|\tau| \rightarrow \infty} R_{XX}(\tau)} = \pm \sqrt{9} = \pm 3$$

Let $X(t_1) = X(0)$; $X(t_2) = X(1)$, $X(t_3) = X(2)$

and $X(t_4) = X(3)$

The elements of the covariance matrix are given by

$$\begin{aligned} \text{Cov}[X(t_i) X(t_j)] &= R_{XX}(t_i, t_j) - E[X(t_i)] E[X(t_j)] \\ &= R_{XX}(t_j - t_i) - E[X(t_i)] E[X(t_j)] \\ &= R_{XX}(t_j - t_i) - (\pm 3)^2 \\ &= R_{XX}(t_j - t_i) - 9 \end{aligned}$$

$$\begin{aligned} C_{ij} &= R_{XX}(j - i) - 9 \\ &= 9 + e^{-2|j-i|} - 9 = e^{-2|j-i|} \end{aligned}$$

$$\Rightarrow C_{XX} = \begin{bmatrix} 1 & e^{-2} & e^{-4} & e^{-6} \\ e^{-2} & 1 & e^{-2} & e^{-4} \\ e^{-4} & e^{-2} & 1 & e^{-2} \\ e^{-6} & e^{-4} & e^{-2} & 1 \end{bmatrix}$$

6.74 $X(t)$ is a Gaussian process with mean $E[X(t)] = 0$ and ACF $R_X(t) = e^{-4|t|}$. Let us define a random process

$$Y = \int_0^A X(t) dt$$

where A is a uniformly distributed random variable over 1 and 4 and is independent of the random process $X(t)$. Determine $E[Y]$, σ_Y^2 .

Solution Given: $A \sim U(1, 4)$

$$E[Y] = E\left[\int_0^1 X(t) dt\right] = \int_0^1 E[X(t)] dt = 0$$

$$\Rightarrow E[Y] = 0$$

$$\sigma_Y^2 = E[Y^2]$$

$$E[Y^2] = E\left[\int_0^1 X(t) dt \int_0^1 X(s) ds\right]$$

$$= \int_0^1 \int_0^1 E[X(t) X(s)] dt ds$$

$$= \int_0^1 \int_0^1 R_{XX}(t-s) dt ds = \int_0^1 \int_0^1 e^{-4|s-t|} dt ds$$

We have $e^{-4|s-t|} = \begin{cases} e^{-4|s-t|} & \text{for } s \geq t \\ e^{-4|t-s|} & \text{for } t > s \end{cases}$

$$\sigma_y^2 = \int_0^1 \int_0^1 e^{-4|s-t|} dt ds$$

$$= \int_0^1 \int_0^s e^{-4(s-t)} dt ds + \int_0^1 \int_s^1 e^{-4(t-s)} ds dt$$

$$= 2 \int_{s=0}^1 \int_0^s e^{-4(s-t)} dt ds = 2 \int_{s=0}^1 e^{-4s} \left. \frac{e^{4t}}{4} \right|_0^s ds$$

$$= 2 \int_{s=0}^1 e^{-4s} \left(\frac{e^{4s} - 1}{4} \right) ds$$

$$= \frac{2}{4} \int_{s=0}^1 (1 - e^{-4s}) ds = \frac{1}{2} \int_0^1 (1 - e^{-4s}) ds$$

$$= \frac{1}{2} \left[s \Big|_0^1 - \frac{e^{-4s}}{-4} \Big|_0^1 \right] = \frac{1}{2} \left[1 + \frac{1}{4} (e^{-4} - 1) \right] = 0.377$$

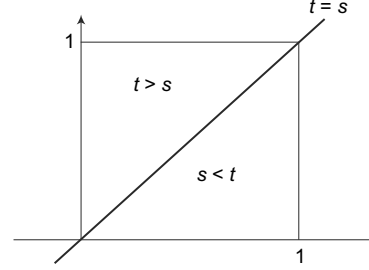


Fig. 6.22

6.75 Let $X(t)$ be a zero-mean Gaussian random process with autocovariance function given by

$$C_{XX}(t_1, t_2) = 4 e^{-|t_1 - t_2|}$$

Find the joint pdf of $X(t)$ and $X(t + \lambda)$.

Solution

$$f_{X(t), X(t+\lambda)}(x_1, x_2) = \frac{1}{(2\pi)^{1/2} |C_{XX}|^{1/2}} \exp \left\{ -\frac{1}{2} (x - \bar{X})^T C_{XX}^{-1} (x - \bar{X}) \right\}$$

$$C_{XX}(t_1, t_2) = 4e^{-|t_1 - t_2|} \Rightarrow C_{XX}(t, t + \lambda) = 4e^{-|\lambda|}$$

The covariance matrix

$$\begin{aligned} C_{XX} &= \begin{bmatrix} C_{XX}(t, t) & C_{XX}(t, t + \lambda) \\ C_{XX}(t + \lambda, t) & C_{XX}(t + \lambda, t + \lambda) \end{bmatrix} \\ &= \begin{bmatrix} 4 & 4e^{-|\lambda|} \\ 4e^{-|\lambda|} & 4 \end{bmatrix} \end{aligned}$$

$$C_{XX}^{-1} = \frac{1}{16 - 16 e^{-2|\lambda|}} \begin{bmatrix} 4 & -4 e^{-|\lambda|} \\ -4 e^{-|\lambda|} & 4 \end{bmatrix}$$

$$|C_{XX}| = \frac{1}{16(1 - e^{-2|\lambda|})}$$

$$f_{X(t), X(t+\lambda)}(x_1, x_2) = \frac{\exp\left\{-\frac{1}{2} X^T C_{XX}^{-1} X\right\}}{2\pi |C_{XX}|^{1/2}}$$

$$\begin{aligned} X^T C_{XX}^{-1} X &= [x_1 \ x_2] \begin{bmatrix} \frac{4}{16(1 - e^{-2|\lambda|})} & \frac{-4 e^{-|\lambda|}}{16(1 - e^{-2|\lambda|})} \\ \frac{-4 e^{-|\lambda|}}{16(1 - e^{-2|\lambda|})} & \frac{4}{16(1 - e^{-2|\lambda|})} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \\ &= \frac{1}{16(1 - e^{-2|\lambda|})} \{4x_1^2 + 4x_2^2 - 8 e^{-2|\lambda|} x_1 x_2\} \end{aligned}$$

$$f_{X(t), X(t+\lambda)}(x_1, x_2) = \frac{1}{8\pi \sqrt{1 - e^{-2|\lambda|}}} \exp\{-(4x_1^2 + 4x_2^2 - 8 e^{-2|\lambda|} x_1 x_2)/16(1 - e^{-2|\lambda|})\}$$



Practice Problem

6.30 A Gaussian process $X(t)$ has an ACF

$$R_{XX}(t) = \frac{9 \sin \pi \tau}{\pi \tau}$$

Determine covariance matrix for random variables $X(t)$, $X(t+2)$, $X(t+4)$ and $X(t+6)$.

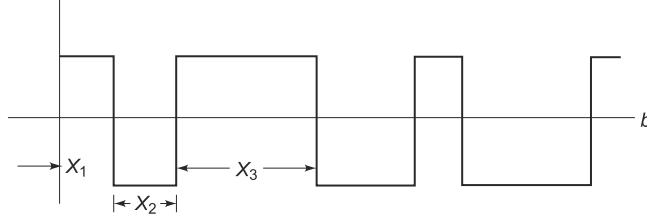
$$\left(\begin{array}{c} \text{Ans.} \\ \begin{bmatrix} 9 & 0 & 0 & 0 \\ 0 & 9 & 0 & 0 \\ 0 & 0 & 9 & 0 \\ 0 & 0 & 0 & 9 \end{bmatrix} \end{array} \right)$$

6.31 If $X(t)$ is a Gaussian random process, prove that $Y(t) = X(t+\lambda) - X(t)$ is a Gaussian random process.



RANDOM TELEGRAPH PROCESS 6.21

Consider a random process $X(t)$ that assumes the value ± 1 with probability $1/2$. Suppose that $X(t)$ changes polarity with each occurrence of an event in a Poisson process of rate λ . The sample function of $X(t)$ is shown in Fig. 6.23. At any time t in a sample function, if the number of switches N between 0 and t is even then $X(0)$ and $X(t)$ have same polarity. On the other hand if the numbering of switches between 0 and t is odd then $X(0)$ and $X(t)$ have opposite polarity.

**Fig. 6.23** Random telegraph process

The pmf of $X(t)$ is given by

$$\begin{aligned} P[X(t) = \pm 1] &= P[X(t) = \pm 1/X(0) = 1] P[X(0) = 1] \\ &\quad + P[X(t) = \pm 1/X(0) = -1] P[X(0) = -1] \end{aligned} \quad (6.149)$$

From Fig. 6.21, we observe that if $X(0) = 1$ then $X(t) = 1$ when even number of events occurs. Similarly, if $X(0) = 1$, then $X(t) = -1$, when odd number of events occur. Similarly, if $X(0) = -1$, then $X(t) = -1$ when even number of events occur, and $X(t) = 1$ when odd numbers of events occur.

$$P[X(t) = 1/X(0) = 1] = P[N(t) = \text{even integer}] \quad (6.150)$$

For a Poisson process,

$$\begin{aligned} P[N(t) = k] &= \frac{(\lambda t)^k e^{-\lambda t}}{k!} \text{ for } k = 0, 1, \dots \\ P[N(t) = \text{even}] &= P[N(t) = 2j] = \sum_{j=0}^{\infty} \frac{(\lambda t)^{2j}}{(2j)!} e^{-\lambda t} \end{aligned} \quad (6.151)$$

$$\begin{aligned} &= e^{-\lambda t} \left[1 + \frac{(\lambda t)^2}{2!} + \frac{(\lambda t)^4}{4!} + \dots \right] \\ &= e^{-\lambda t} \left[\frac{e^{\lambda t} + e^{-\lambda t}}{2} \right] = \frac{1}{2} [1 + e^{-2\lambda t}] \end{aligned} \quad (6.152)$$

Similarly,

$$P[X(t) = -1/X(0) = -1] = P[N(t) = \text{even}] = \frac{1}{2} (1 + e^{-2\lambda t}) \quad (6.153)$$

$$\Rightarrow P[X(t) = \pm 1/X(0) = \pm 1] = \frac{1}{2} (1 + e^{-2\lambda t}) \quad (6.154)$$

$$\begin{aligned} P[X(t) = 1/X(0) = -1] &= P[N(t) = \text{odd}] \\ &= \sum_{j=0}^{\infty} \frac{(\lambda t)^{2j+1}}{(2j+1)!} e^{-\lambda t} \end{aligned}$$

$$\begin{aligned}
&= e^{-\lambda t} \left[\frac{\lambda t}{1!} + \frac{(\lambda t)^3}{3!} + \dots \right] \\
&= e^{-\lambda t} \left[\frac{e^{\lambda t} - e^{-\lambda t}}{2} \right] \\
&= \frac{1}{2} (1 - e^{-2\lambda t})
\end{aligned} \tag{6.155}$$

Similarly,

$$P[X(t) = -1 | X(0) = 1] = P[N(t) = \text{odd}] = \frac{1}{2} (1 - e^{-2\lambda t}) \tag{6.156}$$

$$\Rightarrow P[X(t) = \pm 1 | X(0) = \mp 1] = \frac{1}{2} (1 - e^{-2\lambda t}) \tag{6.157}$$

$$\begin{aligned}
P[X(t) = 1] &= P[X(t) = 1 | X(0) = 1] P[X(0) = 1] + P[X(t) = 1 | X(0) = -1] P[X(0) = -1] \\
&= \frac{1}{2} [1 - e^{-2\lambda t}] \cdot \frac{1}{2} + \frac{1}{2} [1 - e^{-2\lambda t}] \cdot \frac{1}{2} \\
&= \frac{1}{2}
\end{aligned} \tag{6.158}$$

$$P[X(t) = -1] = 1 - P[X(t) = 1] = \frac{1}{2} \tag{6.159}$$

The mean of $X(t)$ is given by

$$\begin{aligned}
\mu_X(t) &= 1 \cdot P[X(t) = 1] + (-1) P[X(t) = -1] \\
&= 1 \left(\frac{1}{2} \right) + (-1) \left(\frac{1}{2} \right) = 0
\end{aligned} \tag{6.160}$$

$\text{Var}[X(t)] = E[X^2(t)]$

$$\begin{aligned}
&= P[X(t) = 1] (1)^2 + P[X(t) = -1] (-1)^2 \\
&= \frac{1}{2} + \frac{1}{2} = 1
\end{aligned} \tag{6.161}$$

The autocovariance is given by

$$\begin{aligned}
C_{XX}(t_1, t_2) &= E[X(t_1) X(t_2)] - E[X(t_1)] E[X(t_2)] \\
&= E[X(t_1) X(t_2)]
\end{aligned} \tag{6.162}$$

$$\begin{aligned}
E[X(t_1) X(t_2)] &= E[X(t_1) = -1; X(t_2) = -1] + E[X(t_1) = -1; X(t_2) = 1] \\
&\quad + E[X(t_1) = 1; X(t_2) = -1] + E[X(t_1) = 1; X(t_2) = 1] \\
&= (-1)(-1) P[X(t_1) = -1; X(t_2) = -1] \\
&\quad + (-1)(1) P[X(t_1) = -1; X(t_2) = 1] \\
&\quad + (1)(-1) P[X(t_1) = 1; X(t_2) = -1]
\end{aligned} \tag{6.163}$$

$$\begin{aligned}
& + (1) (1) P[X(t_1) = 1; X(t_2) = 1] \\
P[X(t_1) = -1; X(t_2) = -1] &= P[X(t_2) = -1 | X(t_1) = 1] P[X(t_1) = -1] \\
&= \frac{1}{2} \left[1 + e^{-2\lambda(t_2 - t_1)} \right] \cdot \frac{1}{2} \\
&= \frac{1}{4} \left[1 + e^{-2\lambda t_2 - t_1} \right]
\end{aligned} \tag{6.164}$$

Similarly,

$$P[X(t_1) = 1; X(t_2) = 1] = \frac{1}{4} [1 + e^{-2\lambda t_2 - t_1}] \tag{6.165}$$

$$\begin{aligned}
P[X(t_1) = -1; X(t_2) = 1] &= P[X(t_2) = 1 | X(t_1) = -1] \cdot P[X(t_1) = -1] \\
&= \frac{1}{2} \left[1 - e^{-2\lambda t_2 - t_1} \right] \cdot \frac{1}{2} \\
&= \frac{1}{4} \left[1 - e^{-2\lambda t_2 - t_1} \right]
\end{aligned} \tag{6.166}$$

Similarly,

$$P[X(t_1) = 1; X(t_2) = -1] = \frac{1}{4} [1 - e^{-2\lambda t_2 - t_1}] \tag{6.167}$$

$$\begin{aligned}
E[X(t_1) X(t_2)] &= \frac{1}{4} [1 + e^{-2\lambda |t_2 - t_1|}] + \frac{1}{4} [1 - e^{-2\lambda t_2 - t_1}] \\
&\quad + \frac{1}{4} [1 - e^{-2\lambda |t_2 - t_1|}] + \frac{1}{4} [1 - e^{-2\lambda t_2 - t_1}] \\
&= \frac{1}{2} [1 + e^{-2\lambda |t_2 - t_1|}] + \frac{1}{2} [1 - e^{-2\lambda t_2 - t_1}] \\
&= e^{-2\lambda |t_2 - t_1|} \\
R_{XX}(t, t + \tau) &= E[X(t) X(t + \tau)] \\
&= e^{-2\lambda \tau}
\end{aligned} \tag{6.168}$$

The plot of autocorrelation function is shown in Fig. 6.24.

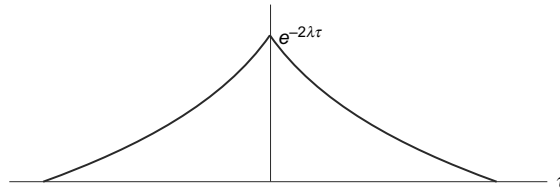


Fig. 6.24 Autocorrelation function

Solved Problems

6.76 A sample realization of a random telegraph signal is shown in Fig. 6.25.

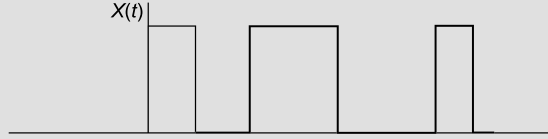


Fig. 6.25

The probability of $X(t)$ changes from 0 to 1 with each occurrence of an event in Poisson process of rate λ . Find the mean, variance, autocorrelation, autocovariance.

Solution $X(t)$ assumes 1 or 0 with equal probability

$$E[X(t)] = 0 \cdot P(X = 1) + 1 \cdot P(X = 0)$$

$$= 0 \left(\frac{1}{2} \right) + 1 \left(\frac{1}{2} \right) = \frac{1}{2}$$

$$E[X^2(t)] = (0)^2 P(X = 1) + (1)^2 P(X = 0)$$

$$= (0) \left(\frac{1}{2} \right) + 1 \left(\frac{1}{2} \right) = \frac{1}{2}$$

$$\text{Var}[X(t)] = E[X^2(t)] - \{E[X(t)]\}^2$$

$$= \frac{1}{2} - \left(\frac{1}{2} \right)^2 = \frac{1}{4}$$

Autocorrelation

$$\begin{aligned} E[X(t_1) X(t_2)] &= E[X(t_1) = 0; X(t_2) = 0] + E[X(t_1) = 0; X(t_2) = 1] \\ &\quad + E[X(t_1) = 1; X(t_2) = 0] + E[X(t_1) = 1; X(t_2) = 1] \\ &= 0.0 P[X(t_1) = 0; X(t_2) = 0] + 0(1) P[X(t_1) = 0, X(t_2) = 1] \\ &\quad + 1(0) P[X(t_1) = 1; X(t_2) = 0] + 1.1 P[X(t_1) = 1; X(t_2) = 1] \\ &= P[X(t_1) = 1; X(t_2) = 1] \end{aligned}$$

We have $P[X(t_1) = 1; X(t_2) = 1] = P[X(t_2) = 1 | X(t_1) = 1] \cdot P[X(t_1) = 1]$

If $X(t_1) = 1$ then it requires even number changes to have a value $X(t_1) = 1$

Therefore, using Eq. (6.113) we can get

$$\begin{aligned} P[X(t_2) = 1/X(t_1) = 1] &= \sum_{\substack{k=0 \\ (k=\text{even})}}^{\infty} \frac{[\lambda(t_2 - t_1)]^k}{k!} e^{-\lambda(t_2 - t_1)} \\ &= \sum_{k=0}^{\infty} \frac{\{\lambda(t_2 - t_1)\}^{2k}}{2k!} e^{-\lambda(t_2 - t_1)} \end{aligned}$$

$$\begin{aligned}
&= e^{-\lambda(t_2-t_1)} \left[\sum_{k=0}^{\infty} \frac{\lambda^{2k} (t_2-t_1)^{2k}}{2k!} \right] \\
&= e^{-\lambda(t_2-t_1)} \left[1 + \frac{\lambda^2 (t_2-t_1)^2}{2!} + \dots \right] \\
&= e^{-\lambda(t_2-t_1)} \left[\frac{e^{\lambda(t_2-t_1)} + e^{-\lambda(t_2-t_1)}}{2} \right] \\
&= \frac{1}{2} [1 + e^{-2\lambda(t_2-t_1)}] \\
R_{XX}(t_1, t_2) &= P[X(t_2) = 1/X(t_1) = 1] P[X(t_1) = 1] \\
&= \frac{1}{2} [1 + e^{-2\lambda(t_2-t_1)}] \cdot \frac{1}{2} \\
&= \frac{1}{4} [1 + e^{-2\lambda(t_2-t_1)}] \quad \text{if } t_2 > t_1
\end{aligned}$$

If $t_2 < t_1$

$$R_{XX}(t_1, t_2) = \frac{1}{4} [1 + e^{-2\lambda(t_1-t_2)}]$$

Therefore,

$$R_{XX}(t_1, t_2) = \frac{1}{4} [1 + e^{-2\lambda|t_1-t_2|}]$$

$$\Rightarrow R_{XX}(t, t + \tau) = \frac{1}{4} [1 + e^{-2\lambda|\tau|}]$$

Autocovariance

$$\begin{aligned}
C_{XX}(t, t + \tau) &= R_{XX}(t, t + \tau) - E[X(t)] E[X(t_2)] \\
&= \frac{1}{4} [1 + e^{-2\lambda|\tau|}] - \left(\frac{1}{2}\right)^2 \\
&= \frac{1}{4} e^{-2\lambda|\tau|}
\end{aligned}$$

6.77 For the random process $X(t)$ with autocorrelation function

$$\begin{aligned}
R_{XX}(\tau) &= 1 + \tau, -1 \leq \tau \leq 0 \\
&= 1 - \tau, 0 \leq \tau \leq 1
\end{aligned}$$

Find the autocorrelation function of $Y(t) = X(t) + X(t - 5)$.

Solution Given: $Y(t) = X(t) + X(t - 5)$

$$R_{YY}(t) = E[Y(t) Y(t + \tau)]$$

$$\begin{aligned}
&= E[\{X(t) + X(t-5)\} \{X(t+\tau) + X(t+\tau-5)\}] \\
&= E[X(t)X(t+\tau) + X(t-5)X(t+\tau) + X(t+\tau-5)X(t) + X(t-5)X(t+\tau-5)] \\
&= E[X(t)X(t+\tau) + E[X(t-5)X(t+\tau)] + E[(t+\tau-5)X(t)] \\
&\quad + E[X(t-5)X(t+\tau-5)]] \\
&= R_{XX}(t) + R_{XX}(\tau+5) + R_{XX}(\tau-5) + R_{XX}(\tau) \\
&= 2R_{XX}(\tau) + R_{XX}(\tau+5) + R_{XX}(\tau-5)
\end{aligned}$$

6.78 Two stationary zero-mean random processes have variance of 25 each and a cross-correlation function $R_{XX}(\tau) = 25 e^{-(\tau-4)^2}$. A new random process $Z(t) = X(t) Y(t+t_x)$ is formed. Find the value of t_x for which $Z(t)$ will have largest variance.

Solution Given: $Z(t) = X(t) + Y(t+t_x)$; $E[X(t)] = 0$ and $E[Y(t)] = 0$

$$\begin{aligned}
\sigma_{z(t)}^2 &= E[Z^2(t)] - \{E[Z(t)]\}^2 \\
E[Z(t)] &= E[X(t) + Y(t+t_x)] \\
&= E[X(t)] + E[Y(t+t_x)] \\
&= 0 \\
\Rightarrow \sigma_{z(t)}^2 &= E[Z^2(t)] \\
&= E[\{X(t) + Y(t+t_x)\}^2] \\
&= E[X^2(t) + Y^2(t+t_x) + 2X(t)Y(t+t_x)] \\
&= E[X^2(t)] + E[Y^2(t+t_x)] + 2E[X(t)Y(t+t_x)]
\end{aligned}$$

We have $E[X^2(t)] = 25$

$$E[Y^2(t+t_x)] = 25$$

$$E[X(t)Y(t+t_x)] = R_{XY}(t_x)$$

Given: $R_{XY}(\tau) = 25 e^{-(\tau-4)^2}$

$$\Rightarrow R_{XY}(t_x) = 25 e^{-(t_x-4)^2}$$

$$\begin{aligned}
\sigma_{z(t)}^2 &= 25 + 25 + 2[25 e^{-(t_x-4)^2}] \\
&= 50 + 50 e^{-(t_x-4)^2}
\end{aligned}$$

$$\frac{d\sigma_{z(t)}^2}{dt_x} = 50 \{-2(t_x-4) e^{-(t_x-4)^2}\} = 0$$

$$\Rightarrow \boxed{t_x = 4}$$

Since $E[X(t)] = 0$

$E[Y(t)] = 0$

REVIEW QUESTIONS



34. Suppose X and Y are two random variables, when do you say that X and Y are (a) orthogonal? (b) uncorrelated?
35. What is the difference between a random variable and random process?
36. Explain with suitable examples: continuous, discrete and mixed-type random process.
37. Define wide-sense stationary process.
38. Find the mean of the stationary process $X(t)$, whose ACF is given by $R_{XX}(\tau) = 16 + \frac{9}{1 + 16\tau^2}$.
39. State any two properties of cross-correlation function.
40. Find the variance of stationary process $X(t)$ whose autocorrelation function is given by $R_{XX}(\tau) = 2 + 4e^{-2|\tau|}$.
41. State the postulates of a Poisson process.
42. Consider the random process $X(t) = \cos(t + \phi)$, where ϕ is a random variable with pdf

$$f(\phi) = \frac{1}{\pi}, -\frac{\pi}{2} < \phi < \frac{\pi}{2}$$

Check whether or not the process is wide-sense stationary.

43. Prove that the sum of two independent Poisson processes is a Poisson process.
44. Prove that for a WSS process $X(t)$, $R_{XX}(t, t + \tau)$ is an even function of τ .
45. When is a random process said to be mean ergodic?
46. Define cross-correlation function of two random processes $X(t)$ and $Y(t)$. State the properties of cross-correlation functions.
47. List the properties of auto-correlation functions.
48. Discuss in detail about
 - (a) First-order stationary random process
 - (b) Second-order and wide-sense stationary random process
49. Define Poisson process stating the assumptions involved and obtain the probability distribution for that. Find the autocorrelation function of the process.
50. Show that the semi-telegraph process is not wide-sense stationary. Construct a random telegraph process which is wide-sense stationary.
51. Find the variance of the stationary process $X(t)$ whose auto-correlation function is given by

$$R_{XX}(\tau) = 2 + 4e^{-2|\tau|}$$



EXERCISES

Problems

1. If a random process $X(t)$ is given by $X(t) = 10 \cos(100t + \theta)$ whose θ is uniformly distributed over $(-\pi, \pi)$, prove that $X(t)$ is correlation ergodic.
2. Consider a random process $X(t)$ that assumes the values ± 1 . Suppose that $X(0) = \pm 1$ with probability $1/2$, and suppose that $X(t)$ then changes polarity with each occurrence of an event in a Poisson process of rate α . Find the mean variance and auto covariance of $X(t)$.
3. Examine whether the random process $X(t) = A \cos(\omega t + \theta)$ is a wide-sense stationary if A and ω are constants and θ is a uniformly distributed random variable in $(0, 2\pi)$.

4. Assume that the number of messages input to a communication channel in an interval of duration 5 seconds is a Poisson process with mean $\lambda = 0.3$. Compute.
 - (a) The probability that exactly 3 messages will arrive during 10 second interval.
 - (b) The probability that the number of message arrivals in an interval of duration 5 seconds is between 3 and 7.
5. If the process $X(t)$ is a Poisson process with parameter λ , obtain $P[X(t) = n]$. Is the process first-order stationary?
6. Prove that a random telegraph signal process $Y(t) = \alpha X(t)$ is a wide-sense stationary process when α is a random variable which is independent of $X(t)$, assumes values -1 and $+1$ with equal probability and $R_{XX}(t_1, t_2) = e^{-2\lambda|t_1 - t_2|}$.
7. If $X(t)$ and $Y(t)$ are two random processes with autocorrelation function $R_{XX}(t)$ and $R_{YY}(t)$ respectively then prove that $|R_{XY}(\tau)| \leq \sqrt{R_{XX}(0) R_{YY}(0)}$.
8. The process $X(t)$ whose probability distribution under certain conditions is given by

$$P\{X(t) = n\} = \begin{cases} \frac{(\alpha t)^{n-1}}{(1 + \alpha t)^{n+1}} & n = 1, 2, \dots \\ \frac{\alpha t}{1 + \alpha t} & n = 0 \end{cases}$$

Find the means and variance of the process. Is the process is first order stationary?

9. Show that the random process $X(t) = A \cos(\omega t + \theta)$ is wide-sense stationary, if A and ω are constants and θ is a uniformly distributed in $(0, 2\pi)$.
10. A stationary random process has an autocorrelation function of

$$R_{XX}(\tau) = \begin{cases} 10 \left(1 - \frac{|\tau|}{0.05} \right); & |\tau| < 0.05 \\ 0 & \text{elsewhere} \end{cases}$$

Determine the mean and variance of the process $X(t)$.

11. Consider random variables Y_1 and Y_2 related to arbitrary random variables X and Y by the coordinate rotation.

$$Y_1 = X \cos \theta + Y \sin \theta; Y_2 = -X \sin \theta + Y \cos \theta$$

- (a) Find the covariance of Y_1 and Y_2
- (b) For what value of θ is the random variables Y_1 and Y_2 uncorrelated
12. Let two random processes $X(t)$ and $Y(t)$ be denoted by

$$X(t) = A \cos \omega_0 t + \beta \sin \omega_0 t$$

$$Y(t) = B \cos \omega_0 t - A \sin \omega_0 t$$

where A and B are random variables and ω_0 is a constant. Assume A and B are uncorrelated, zero mean random variables with same variance. Find the cross-correlation function $R_{XY}(t, t + \tau)$ and show that $X(t)$ and $Y(t)$ are jointly wide-sense stationary.

13. Let $X(t)$ be a stationary continuous random process that is differentiable. Denote its time derivative $\dot{X}(t)$.
 - (a) Show that $E[\dot{X}(t)] = 0$
 - (b) Find $R_{\dot{X}\dot{X}}(\tau)$ in terms of $R_{XX}(\tau)$

14. Statistically independent zero mean random processes $X(t)$ and $Y(t)$ have autocorrelation functions $R_{XX}(\tau) = e^{-|\tau|}$ and $R_{YY}(\tau) = \cos(2\pi\tau)$ respectively

(a) Find the autocorrelation function of the sum

$$W_1(t) = X(t) + Y(t)$$

(b) Find the autocorrelation function of difference

$$W_2(t) = X(t) - Y(t)$$

(c) Find the cross-correlation function of $W_1(t)$ and $W_2(t)$

15. For two random variables X and Y ,

$$f_{X,Y}(x, y) = 0.05 \delta(x+1) \delta(y) + 0.1 \delta(x) \delta(y) + 0.1 \delta(x) \delta(y-2) \\ + 0.4 \delta(x-1) \delta(y+2) + 0.2 \delta(x-1) + 0.15 (\delta(x-1) \delta(y-3))$$

Find (a) the correlation,

(b) the covariance, and

(c) the correlation coefficient of X and Y .

(d) Are X and Y either uncorrelated or orthogonal?

16. If customer arrives at a Counter in accordance with a Poisson process with a mean rate of 2 per minutes, find the probability that the interval between 2 consecutive arrivals is

(a) more than 1.5 minutes

(b) between 2 and 3 minutes

(c) 4 minutes or less

17. The particles are emitted from a radioactive source at the rate of 10 per hour. Find the probability that exactly 4 particles are emitted during a 25 minute period.

18. If the customers arrive at a bank according to a Poisson process with mean rate of 5 per minute, find the probability that during a 1 minute interval no customer arrives.

19. The random binary transmission process $X(t)$ is a wide-sense process with zero-mean and autocorrelation function $R_{XX}(\tau) = 1 - \frac{|\tau|}{T}$ where T is a constant. Find mean and variance of time

average of $X(t)$ over $(0, T)$. Is $X(t)$ mean ergodic?

20. Consider the random process $X(t) = 5 \sin(\omega t + \phi)$ where ω is a constant and ϕ is a random variable uniformly distributed in $(0, 2\pi)$. Show that $X(t)$ is autocorrelation ergodic. Verify any two properties of the auto-correlation function of $X(t)$.

21. Show that the random process $X(t) = A \cos \omega t + B \sin \omega t$ where ω is a constant, A and B are uncorrelated random variables with zero-mean and common variance is stationary in the wide-sense.

22. Establish the property $|R_{XX}(\tau)| \leq \sqrt{R_{XX}(0) R_{YY}(0)}$ for stationary processes $X(t)$ and $Y(t)$. Verify this property for the processes $X(t) = A \cos(\omega t + \phi)$ and $Y(t) = B \sin(\omega t + \phi)$ where A, B and ω are constants while ϕ is a random variable uniformly distributed in $(0, 2\pi)$.

23. The autocorrelation function of a random process $X(t)$ is given by

$$R_{XX}(\tau) = \begin{cases} 0 & ; |\tau| > 1 \\ 5(1 - |\tau|) & ; |\tau| \leq 1 \end{cases}$$

Is the process is mean ergodic?

24. Calculate the autocorrelation function of the rectangular pulse shown in Fig. 6.26.

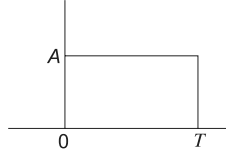


Fig. 6.26

25. A random process $X(t)$ is given by

$$X(t) = A \cos(2\pi t) \quad t \geq 0$$

where A is a random variable uniformly distributed between 0 and 1. Find autocorrelation function of $X(t)$.

26. A random process $X(t)$ is given by

$$X(t) = A \sin t + (B + 1) \cos t \quad -\infty < t < \infty$$

where A and B are random variables with $E[A] = E[B] = 0$ and $\sigma_A^2 = \sigma_B^2 = 1$. Find the auto-covariance function of $X(t)$.

27. Explain why the following matrices can or cannot be a valid auto-correlation matrices of a zero-mean wide-sense stationary random process $X(t)$.

$$A = \begin{bmatrix} 1 & 0.5 & 0.6 & 0.9 \\ 0.5 & 1 & 0.4 & 0.1 \\ 0.6 & 0.4 & 1 & 1.3 \\ 0.9 & 0.1 & 1.3 & 1 \end{bmatrix} \quad B = \begin{bmatrix} 2 & 0.2 & 0.6 & 0.1 \\ 0.1 & 2 & 0.6 & 0.8 \\ 0.5 & 0.6 & 2 & 0.1 \\ 0.2 & 0.7 & 0.1 & 2 \end{bmatrix}$$

28. Two jointly stationary random processes $X(t)$ and $Y(t)$ are defined as follows:

$$X(t) = 10 \sin(10t + \phi)$$

$$Y(t) = 5 \cos(105t + \phi)$$

where ϕ is a random variable that is uniformly distributed between 0 and 2π . Find $R_{XY}(\tau)$ and $R_{YX}(\tau)$.

29. State why each of the functions $X(t)$, $Y(t)$, $Z(t)$ shown in Fig. 6.27 can or cannot be a valid auto-correlation function of a wide-sense stationary process.

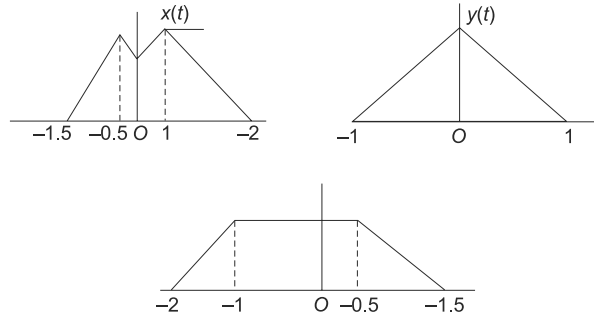


Fig. 6.27

30. A wide-sense stationary random process $X(t)$ has a mean sense value $E[X^2(t)] = 9$. Give reasons why the functions given below can or cannot be its auto-correlation function

$$(a) \quad R_{XX}(t) = \frac{9 \cos 2\tau}{1 + \tau^2} \quad (b) \quad R_{XX}(\tau) = \frac{9\tau}{1 + 2\tau^2 + 4\tau^4}$$

$$(c) \quad R_{XX}(\tau) = \frac{\tau^2 + 36}{\tau^2 + 4} \quad (d) \quad R_{XX}(\tau) = \frac{9 \cos(\tau)}{1 + 2\tau^2 + 4\tau^4}$$

31. A stationary process has an autocorrelation function

$$R_{XX}(\tau) = \begin{cases} 20(1 - |\tau|) & |\tau| \leq k \\ 0 & |\tau| > k \end{cases}$$

Find the largest value of k for which $R_{XX}(t)$ could be a valid auto-correlation function. (Ans. $k = 1$)

Multiple-choice Questions

- Which of the following statements is/are correct?
 - Cross-correlation function is an even function.
 - Autocorrelation function is an even function.
 - Cross-correlation has maximum value at the origin.
 - Autocorrelation function is an odd function
- The autocovariance function of $X(t)$ is given by
 - $E[X(t_1) X(t_2)]$
 - $E[X(t_1) X(t_2)] - E[X(t_1)] E[X(t_2)]$
 - $E[X(t_1)] E[X(t_2)]$
 - $E[X(t_1) + X(t_2)]$
- For a random process $X(t) = \cos(\lambda t + \theta)$, where θ is uniformly distributed in $(-\pi, \pi)$. The ACF is given by
 - $\cos \frac{\lambda \tau}{2}$
 - $\frac{1}{2} \cos \lambda \tau$
 - $\frac{1 - \cos \lambda \tau}{2}$
 - $\frac{1 - \cos^2 \lambda \tau}{2}$
- If $X(t)$ is a Poisson process then the correlation coefficient between $X(t)$ and $X(t + \tau)$ is
 - $\sqrt{\frac{t + \tau}{t}}$
 - $\frac{t}{t + \tau}$
 - $\sqrt{\frac{t}{t + \tau}}$
 - $\frac{t + \tau}{t}$
- $R_{XX}(\tau) =$
 - $R_{XX}(-\tau)$
 - $-R_{XX}(\tau)$
 - $|R_{XX}(\tau)|$
 - None of these
- The variance of the stationary process whose ACF as $\frac{25\tau^2 + 36}{6.25\tau^2 + 4}$ is
 - 6
 - 5
 - 10
 - 9
- A random process is defined as

$$X(t) = \begin{cases} A & \text{for } 0 \leq t \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

$A \sim U(-\theta, \theta)$. The autocorrelation function is

- (a) $\frac{\theta^2}{2}$ (b) $\frac{\theta^2}{3}$ (c) $\frac{\theta^3}{3}$ (d) $\frac{\theta^2}{4}$

8. The variance of a random process whose autocorrelation function

$$R_{XX}(t) = 36 + \frac{4}{1 + 3\tau^2} \text{ is given by}$$

- (a) 2 (b) 4 (c) 40 (d) 36

9. Which of the following the ACF of $X(t)$ does not satisfy

- (a) $R_{XX}(\tau) = R_{XX}(-\tau)$ (b) $R_{XX}(0) = E[X^2]$
 (c) $R_X(0) \leq |R_{XX}(\tau)|$ (d) $R_{XX}(\tau) = E[X(t) X(t + \tau)]$

9. $R_{XX}(\tau) =$

- (a) $\frac{d}{d\tau} R_{XX}(\tau)$ (b) $\frac{d^2}{d\tau^2} R_{XX}(\tau)$ (c) $-\frac{d}{d\tau} R_{XX}(\tau)$ (d) $-\frac{d^2}{d\tau^2} R_{XX}(\tau)$

10. Two random processes $X(t)$ and $Y(t)$ are orthogonal if

- (a) $R_{XY}(t_1, t_2) = R_{XY}(t_2, t_1)$ (b) $R_{XY}(t, t + \tau) = R_{XY}(\tau)$
 (c) $R_{XY}(t_1, t_2) = 0 \forall t_1, t_2$ (d) $R_{XY}(t_1, t_2) = E[X(t_1) Y(t_2)]$

11. Two random processes $X(t)$ and $Y(t)$ are uncorrelated

- (a) $R_{XY}(t_1, t_2) = 0 \forall t_1, t_2$ (b) $R_{XY}(t_1, t_2) = E[X(t_1) Y(t_2)]$
 (c) $R_{XY}(t_1, t_2) = R_{XY}(t_2, t_1)$ (d) $R_{XY}(t, t + \tau) = R_{XY}(\tau)$

12. $R_{XY}(\tau) =$

- (a) $R_{XY}(-\tau)$ (b) $R_{YX}(\tau)$ (c) $-R_{YX}(\tau)$ (d) $R_{YX}(-\tau)$

Answers to Multiple-Choice Questions

1. (c) 2. (b) 3. (b) 4. (c) 5. (a) 6. (b) 7. (b) 8. (b) 9. (c) 10. (d)
 11. (b) 12. (d)

7

RANDOM PROCESSES—SPECTRAL CHARACTERISTICS



INTRODUCTION 7.1

In the previous chapter we studied the characterization of a random process in time domain by its mean, autocorrelation function and covariance functions. In this chapter, we will study the spectral description of the random process using the Power Spectral Density (PSD).

For a deterministic signal, it is well-known that the spectral information can be obtained using Fourier transform. However, the Fourier transform cannot be obtained for random signals because for such type of signals Fourier transform does not exist. Therefore, the frequency domain analysis can be carried out on a random signal by transforming the autocorrelation function into frequency domain. That is, the Fourier transforms of autocorrelation function known as power density spectrum can be used to obtain the frequency-domain representation of random process. Since it requires the knowledge of Fourier transform, a brief introduction to Fourier transform is given in the next section.

Fourier Transform

Fourier transform of a continuous-time deterministic signal $x(t)$ is given by

$$X(j\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t} dt \quad (7.1)$$

The function $X(j\omega)$ is called the *spectrum* of $x(t)$. Since $X(j\omega)$ is a complex quantity, it has magnitude $|X(j\omega)|$ and phase $\angle X(j\omega)$. The plot between $|X(j\omega)|$ versus ω is known as *magnitude spectrum* and the plot between $\angle X(j\omega)$ versus ω is known as *phase spectrum*. The Fourier transform for a signal exists only if it is absolutely integrable. That is

$$\int_{-\infty}^{\infty} |x(t)| dt < \infty \quad (7.2)$$

The inverse Fourier transform of $X(j\omega)$ is given by

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) e^{j\omega t} d\omega \quad (7.3)$$



POWER SPECTRAL DENSITY (PSD) 7.2

In the previous section, we studied that Fourier transform exists for a signal if it is absolutely integrable. The Fourier transform does not exist for a random process, since we assume the ensemble of a random process exists for all time. Therefore, frequency-domain representation of a random process cannot be obtained using the Fourier transform of $x(t)$. Hence, we have to find some other way of obtaining frequency-domain representation of random process. For that purpose first we will consider a portion of a sample function $x(t)$ that exists between $-T$ and T for our analysis. Let

$$x_T(t) = \begin{cases} x(t) & -T \leq t \leq T \\ 0 & \text{otherwise} \end{cases} \quad (7.4)$$

Since $x_T(t)$ is a finite interval signal, it satisfies the condition

$$\int_{-T}^T x_T(t) dt < \infty$$

and Fourier transform can be obtained using the formula

$$X_T(j\omega) = \int_{-T}^T x_T(t) e^{-j\omega t} dt = \int_{-T}^T x(t) e^{-j\omega t} dt$$

Using Parseval's theorem, we can write the energy of $x(t)$ as

$$E(T) = \int_{-T}^T x^2(t) dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} |X_T(j\omega)|^2 d\omega \quad (7.5)$$

By dividing the above expression by $2T$, we obtain the average power $P(T)$. That is,

$$P(T) = \frac{1}{2T} \int_{-T}^T x^2(t) dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{|X_T(j\omega)|^2}{2T} d\omega \quad (7.6)$$

Note that the above equation does not represent the power in an entire sample function. The power in the entire sample function can be obtained by letting T to infinity. Then it represents the power in one sample function but does not represent the average power. Therefore, to get the average power for the random process we take expected value in Eq. (7.6). That is,

$$P_{XX} = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T E[X^2(t)] dt = A\{E[X^2(t)]\} \quad (7.7)$$

where $A\{E[X^2(t)]\}$ is the time average of $E[X^2(t)]$.

Note that

$$A[g(t)] = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T g(t) dt \quad (7.8)$$

$$\Rightarrow P_{XX} = \frac{1}{2\pi} \int_{-\infty}^{\infty} \lim_{T \rightarrow \infty} \frac{E[|X_T(j\omega)|^2]}{2T} d\omega = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_{XX}(\omega) d\omega = \int_{-\infty}^{\infty} S_X(f) df \quad (7.9)$$

where

$$S_{XX}(\omega) = \lim_{T \rightarrow \infty} \frac{E\left[|X_T(j\omega)|^2\right]}{2T} \quad (7.10)$$

is known as power spectral density. $S_{XX}(f)$ represents the average power per hertz. Therefore, the term Power Spectral Density (PSD).

Definition: For a random process $X(t)$, the Power Spectral Density is defined as

$$S_{XX}(\omega) = \lim_{T \rightarrow \infty} \frac{E\left[|X_T(j\omega)|^2\right]}{2T}$$

where $X_T(j\omega)$ is Fourier transform of the truncated version of the process.

Properties of Power Density Spectrum

Property 1: The power density spectrum $S_{XX}(\omega)$ is a real value and non-negative function.

That is $S_{XX}(\omega) \geq 0$

Property 2: The power density spectrum is an even function in ω . That is

$$S_{XX}(\omega) = S_{XX}(-\omega) \quad (7.11)$$

$$S_{XX}(\omega) = \int_{-\infty}^{\infty} R_{XX}(\tau) e^{-j\omega\tau} d\tau$$

$$S_{XX}(-\omega) = \int_{-\infty}^{\infty} R_{XX}(\tau) e^{j\omega\tau} d\tau = \int_{-\infty}^{\infty} R_{XX}(-\tau) e^{-j\omega\tau} d\tau$$

Since $R_{XX}(-\tau) = R_{XX}(\tau)$

$$\begin{aligned} S_{XX}(-\omega) &= \int_{-\infty}^{\infty} R_{XX}(\tau) e^{-j\omega\tau} d\tau \\ &= S_{XX}(\omega) \end{aligned}$$

Property 3: The power density spectrum is a real function if $X(t)$ is real.

$$\begin{aligned} S_{XX}(\omega) &= \int_{-\infty}^{\infty} R_{XX}(\tau) e^{-j\omega\tau} d\tau \\ &= \int_{-\infty}^{\infty} R_{XX}(\tau) [\cos \omega\tau - j \sin \omega\tau] d\tau \\ &= \int_{-\infty}^{\infty} R_{XX}(\tau) \cos \omega\tau d\tau - j \int_{-\infty}^{\infty} R_{XX}(\tau) \sin \omega\tau d\tau \end{aligned} \quad (7.12)$$

We know $S_{XX}(\omega)$ is an even function. In the above equation, $R_{XX}(\tau) \cos \omega\tau$ is an even function of τ and $R_{XX}(\tau) \sin \omega\tau$ is an odd function of τ . The integral of an odd function over $(-\infty, \infty)$ is zero. Therefore, the imaginary part vanishes. Thus,

$$S_{XX}(\omega) = \int_{-\infty}^{\infty} R_{XX}(\tau) \cos \omega\tau d\tau = 2 \int_0^{\infty} R_{XX}(\tau) \cos \omega\tau d\tau \quad (7.13)$$

Property 4: Average power is equal to the integral of $S_{XX}(\omega)$.

$$P_{XX} = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_{XX}(\omega) d\omega \quad (7.14a)$$

Average power in the frequency band $[\omega_1, \omega_2]$ is given by

$$P_{XX}[\omega_1, \omega_2] = \frac{1}{\pi} \int_{\omega_1}^{\omega_2} S_{XX}(\omega) d\omega \quad (7.14b)$$

Property 5: $S_{XX}^*(\omega) = S_{XX}(\omega)$ where $S_{XX}^*(\omega)$ is the complex conjugate of $S_{XX}(\omega)$. This means that $S_{XX}(\omega)$ cannot be a complex function.

Property 6: If $\int_{-\infty}^{\infty} |R_{XX}(\tau)| d\tau < \infty$, then $S_{XX}(\omega)$ is a continuous function of ω .

Property 7: $S_{\dot{X}\dot{X}}(\omega) = \omega^2 S_{XX}(\omega)$ (7.15)

The power density spectrum of the derivation $\dot{X}(t) = \frac{dX(t)}{dt}$ is ω^2 times the power spectrum of $X(t)$.

Property 8: If $S_{XX}(\omega)$ is the power density spectrum of $X(t)$ then the power density spectrum of $\alpha X(t)$ is $|\alpha|^2 S_{XX}(\omega)$.

Property 9: If $S_{XX}(\omega)$ is the power density spectrum of $X(t)$ then the power density spectrum of $X(t) + b$ with $E[X(t)] = 0$ is

$$S_{XX}(\omega) + 2\pi |b|^2 \delta(\omega)$$

Property 10: Let $S_{XX}^1(\omega)$ and $S_{XX}^2(\omega)$ are PSDs of $X_1(t)$ and $X_2(t)$ respectively. If $X_1(t) \perp X_2(t)$ then PSD of $X_1(t) + X_2(t)$ is $S_{XX}^1(\omega) + S_{XX}^2(\omega)$.

The above property indicates that if $X_1(t)$ and $X_2(t)$ are orthogonal then the power of their sum is simply the sum of their powers.

Property 11: If $S_{XX}(\omega)$ is PSD of $X(t)$ then PSD of $\frac{d^n X(t)}{dt^n}$ is equal to $\omega^{2n} S_{XX}(\omega)$

The above property can be obtained by a repeated application of the property

$$R_{\dot{X}\dot{X}}(\tau) = -\frac{d^2}{d\tau^2} R_{XX}(\tau)$$

and the differentiation property of Fourier transform.

Property 12: If $S_{XX}(\omega)$ is PSD of $X(t)$ then the PSD of $X(t)e^{j\omega_0 t}$ with non-random ω_0 is $S_{XX}(\omega - \omega_0)$.

Property 13: If $S_{XX}(\omega)$ is PSD of $X(t)$ then PSD of $X(t) \cos(\omega_0 t + \theta)$ with $\theta \sim U(0, 2\pi)$ is

$$\frac{1}{2} [S_{XX}(\omega + \omega_0) + S_{XX}(\omega - \omega_0)]$$

Table 7.1 Some common Fourier-transform pairs

$x(t)$	$X(j\omega)$
$e^{-at}; a > 0; t \geq 0$	$\frac{1}{a + j\omega}$
$e^{bt}; b > 0; t < 0$	$\frac{1}{b - j\omega}$
$e^{-a t }; a > 0$	$\frac{2a}{a^2 + \omega^2}$
1	$2\pi \delta(\omega)$
$\delta(t)$	1
$te^{-at}; a > 0; t \geq 0$	$\frac{1}{(a + j\omega)^2}$
$e^{j\omega_0 t}$	$2\pi \delta(\omega - \omega_0)$
$\cos \omega_0 t$	$\pi [\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$
$\sin \omega_0 t$	$-j\pi [\delta(\omega - \omega_0) - \delta(\omega + \omega_0)]$
$e^{-a t } \cos \omega_0 t$	$\frac{a}{a^2 + (\omega - \omega_0)^2} + \frac{a}{a^2 + (\omega + \omega_0)^2}$
$\begin{cases} 1 & -T/2 < t < T/2 \\ 0 & \text{otherwise} \end{cases}$	$\frac{T \sin\left(\frac{\omega T}{2}\right)}{\left(\frac{\omega T}{2}\right)}$
$\begin{cases} 1 - t /T & t < T \\ 0 & \text{otherwise} \end{cases}$	$T \left[\frac{\sin\left(\frac{\omega T}{2}\right)}{\left(\frac{\omega T}{2}\right)} \right]^2$

REVIEW QUESTIONS

- 7.1 Define power spectral density.
- 7.2 Define Fourier transform pair,
- 7.3 Explain in detail the properties of power density spectrum.
- 7.4 Prove that power density spectrum is real and even function in ω .
- 7.5 What is the condition for the existence of Fourier transform?

Solved Problems

7.1 Determine which of the following functions are valid PSDs.

$$(a) S_{XX}(\omega) = \frac{\omega^2}{\omega^6 + 3\omega^2 + 3}$$

$$(b) S_{XX}(\omega) = e^{-(\omega-1)^2}$$

$$(c) S_{XX}(\omega) = \frac{\omega^2}{\omega^4 + 1} - \delta(\omega)$$

$$(d) \frac{\omega^4}{1 + \omega^2 + j\omega^6}$$

Solution

$$(a) S_{XX}(\omega) = \frac{\omega^2}{\omega^6 + 3\omega^2 + 3}$$

$$S_{XX}(-\omega) = \frac{(-\omega)^2}{(-\omega)^6 + 3(-\omega)^2 + 3} = \frac{\omega^2}{\omega^6 + 3\omega^2 + 3}$$

$$S_{XX}(\omega) = S_{XX}(-\omega).$$

Therefore, $S_{XX}(\omega)$ is an even function of ω . $S_{XX}(\omega)$ is a valid PSD.

$$(b) S_{XX}(\omega) = e^{-(\omega-1)^2}$$

$$S_{XX}(-\omega) = e^{-(-\omega-1)^2} = e^{-(\omega+1)^2}$$

$$S_{XX}(\omega) \neq S_{XX}(-\omega)$$

Since $S_{XX}(\omega)$ is not an even function of ' ω '. Therefore, $S_{XX}(\omega)$ is not a valid PSD.

$$(c) S_{XX}(\omega) = \frac{\omega^2}{\omega^4 + 1} - \delta(\omega)$$

For $\omega = 0$, $S_{XX}(\omega) = -1$. Since $S_{XX}(\omega)$ is non-negative, it is not a valid PSD.

$$(d) S_{XX}(\omega) = \frac{\omega^4}{1 + \omega^2 + j\omega^6}$$

The $S_{XX}(\omega)$ is not a power spectrum because it is not real valued.

7.2 Determine which of the following functions are power spectrum and, which are not.

$$(a) S_{XX}(\omega) = \frac{\cos 2\omega}{1 + 2\omega^2 + \omega^4}$$

$$(b) S_{XX}(\omega) = \frac{\omega^2 + 3}{\omega^4 + 2\omega^2 - 3}$$

Solution

$$(a) S_{XX}(\omega) = \frac{\cos 2\omega}{1 + 2\omega^2 + \omega^4}$$

$$S_{XX}(-\omega) = \frac{\cos 2(-\omega)}{1 + 2(-\omega)^2 + (-\omega)^4} = \frac{\cos 2\omega}{1 + 2\omega^2 + \omega^4}$$

It satisfies $S_{XX}(\omega) = S_{XX}(-\omega)$. But $S_{XX}(\omega)$ is not always non-negative. For $\omega = \frac{(2k+1)\pi}{2}$, $k = 0, 1, \dots$, the value $\cos 2\omega$ is -1 . Hence, $S_{XX}(\omega)$ is not a valid PSD.

$$\begin{aligned}
 (b) \quad S_{XX}(\omega) &= \frac{\omega^2 + 3}{\omega^2 + 2\omega^2 - 3} \\
 S_{XX}(-\omega) &= \frac{(-\omega)^2 + 3}{(-\omega)^4 + 2(-\omega)^2 + 3} = \frac{\omega^2 + 3}{\omega^4 + 2\omega^2 - 3} \\
 S_{XX}(\omega) &= S_{XX}(-\omega) \\
 S_{XX}(\omega) &\text{ is an even function of } \omega. \\
 S_{XX}(\omega) &= \frac{\omega^2 + 3}{(\omega^4 + 2\omega^2 - 3)} = \frac{\omega^2 + 3}{(\omega^2 - 1)(\omega^2 + 3)} \\
 S_{XX}(\omega) &\text{ is not always non-negative. Hence, } S_{XX}(\omega) \text{ is not a valid PSD.}
 \end{aligned}$$



Practice Problem

7.1 Determine which of the following functions are valid PSDs.

$$\begin{aligned}
 (a) \quad S_{XX}(\omega) &= \frac{1}{\sqrt{1 - \omega^2}} & (b) \quad S_{XX}(\omega) &= \frac{j\omega^3}{1 + 2\omega^2 + \omega^4} \\
 (c) \quad S_{XX}(\omega) &= e^{-\omega} & (d) \quad S_{XX}(\omega) &= \frac{\omega^2 + 6}{1 + \omega^2 - 2\omega^4} \\
 (e) \quad \frac{\cos(\omega)}{\omega} & & (f) \quad \frac{4\omega^2}{1 + 3\omega^2 + 4\omega^4} &
 \end{aligned}$$

(Ans: (a) not valid (b) not valid (c) valid (d) not valid (e) not valid (f) valid)



RELATIONSHIP BETWEEN PSD AND ACF 7.3

Let $x(t)$ be an ensemble of the random process $X(t)$. Let us define $x_T(t)$ as

$$x_T(t) = \begin{cases} x(t) & \text{for } -T \leq t \leq T \\ 0 & \text{otherwise} \end{cases} \quad (7.16)$$

The Fourier transform of $x_T(t)$ is given by

$$X_T(j\omega) = \int_{-T}^T x_T(t) e^{-j\omega t} dt \quad (7.17)$$

Using Parseval's theorem, we can write

$$\int_{-T}^T x^2(t) dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} |X_T(j\omega)|^2 d\omega \quad (7.18)$$

The average power is given by

$$P(T) = \frac{1}{2T} \int_{-T}^T x^2(t) dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{|X_T(j\omega)|^2}{2T} d\omega \quad (7.19)$$

To find the average power of the random process, we take the expected value with T tending to infinity in the above equation.

$$\begin{aligned}
 P_{XX} &= \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T E[X^2(t)] dt \\
 &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \lim_{T \rightarrow \infty} \frac{E[|X_T(j\omega)|^2]}{2T} d\omega \\
 &= \frac{1}{2\pi} \int_{-\infty}^{\infty} S_{XX}(\omega) d\omega
 \end{aligned} \tag{7.20}$$

where
$$S_{XX}(\omega) = \lim_{T \rightarrow \infty} \frac{E[|X_T(j\omega)|^2]}{2T}$$

We know

$$X_T(j\omega) = \int_{-T}^T x_T(t) e^{-j\omega t} dt$$

Using the above equation, we can obtain

$$S_{XX}(\omega) = \lim_{T \rightarrow \infty} E \left[\frac{1}{2T} \int_{-T}^T X(t_1) e^{j\omega t_1} dt_1 \int_{-T}^T X(t_2) e^{-j\omega t_2} dt_2 \right] \tag{7.21}$$

$$= \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T \int_{-T}^T E[X(t_1) X(t_2)] e^{-j\omega(t_2 - t_1)} dt_2 dt_1 \tag{7.22}$$

We know $E[X(t_1) X(t_2)] = R_{XX}(t_1, t_2)$

$$\Rightarrow S_{XX}(\omega) = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T \int_{-T}^T R_{XX}(t_1, t_2) e^{-j\omega(t_2 - t_1)} dt_2 dt_1 \tag{7.23}$$

Now the inverse Fourier transform of $S_{XX}(\omega)$ is

$$\begin{aligned}
 F^{-1}[S_{XX}(\omega)] &= \frac{1}{2\pi} \int_{-\infty}^{\infty} S_{XX}(\omega) e^{j\omega\tau} d\omega \\
 &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T \int_{-T}^T R_{XX}(t_1, t_2) e^{-j\omega(t_2 - t_1)} e^{j\omega\tau} dt_2 dt_1 d\omega \\
 &= \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T \int_{-T}^T R_{XX}(t_1, t_2) \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-j\omega(t_2 - t_1 - \tau)} d\omega dt_2 dt_1 \\
 &= \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T \int_{-T}^T R_{XX}(t_1, t_2) \frac{1}{2\pi} [2\pi \delta(t_2 - t_1 - \tau)] dt_2 dt_1
 \end{aligned}$$

$$\begin{aligned}
&= \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T \int_{-T}^T R_{XX}(t_1, t_2) \delta(t_2 - t_1 - \tau) dt_2 dt_1 \\
&= \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T R_{XX}(t_1, t_1 + \tau) dt_1 \\
&= \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T R_{XX}(t, t + \tau) dt \\
&= A[R_{XX}(t, t + \tau)]
\end{aligned} \tag{7.24}$$

which is the time average of the process' autocorrelation function.

The above equation says that the inverse Fourier transform of the power density spectrum is the time average of the autocorrelation function of the process. Therefore, we can write

$$S_{XX}(\omega) = \int_{-\infty}^{\infty} A[R_{XX}(t, t + \tau)] e^{-j\omega\tau} d\tau \tag{7.25}$$

For a wide-sense stationary process,

$$A[R_{XX}(t, t + \tau)] = R_{XX}(\tau)$$

Therefore,

$$S_{XX}(\omega) = \int_{-\infty}^{\infty} R_{XX}(\tau) e^{-j\omega\tau} d\tau \tag{7.26}$$

and

$$R_{XX}(\tau) = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_{XX}(\omega) e^{j\omega\tau} d\omega \tag{7.27}$$

That is, power spectral density is Fourier transform of autocorrelation function. The Eqs. (7.26) and (7.27) are known as *Wiener–Khinchine* relations.

REVIEW QUESTIONS



- 7.6 Prove that PSD and autocorrelation function of a random process form a Fourier transform pair
- 7.7 Define autocorrelation function.
- 7.8 State and prove *Wiener–Khinchin* relations.

Solved Problems



7.3 If $R_{XX}(\tau) = ae^{-b|\tau|}$, find the spectral density function, where a and b are constants.

Solution

Given: $R_{XX}(\tau) = ae^{-b|\tau|}$

The PSD $S_{XX}(\omega) = F[R_{XX}(\tau)]$

$$\begin{aligned}
&= \int_{-\infty}^{\infty} a e^{-b|\tau|} e^{-j\omega\tau} d\tau \\
&= a \left[\int_{-\infty}^0 a e^{b\tau} e^{-j\omega\tau} d\tau + \int_0^{\infty} e^{-b\tau} e^{-j\omega\tau} d\tau \right] \\
&= a \left[\int_{-\infty}^0 e^{(b-j\omega)\tau} d\tau + \int_0^{\infty} e^{-(b+j\omega)\tau} d\tau \right] \\
&= \frac{a}{b-j\omega} e^{(b-j\omega)\tau} \Big|_{-\infty}^0 - \frac{a}{b+j\omega} e^{-(b+j\omega)\tau} \Big|_0^{\infty} \\
&= \frac{a}{b-j\omega} [1-0] - \frac{a}{b+j\omega} [0-1] \\
&= \frac{a}{b-j\omega} + \frac{a}{b+j\omega} = \frac{2ab}{b^2 + \omega^2}
\end{aligned}$$

$$S_{XX}(\omega) = \frac{2ab}{b^2 + \omega^2}$$



Practice Problem

7.2 A stationary random process $X(t)$ has an autocorrelation function given by

$$R_{XX}(\tau) = 3e^{-|\tau|} + 5e^{-4|\tau|}$$

Find the power spectral density of the process.

$$\left(\text{Ans: } \frac{46\omega^2 + 136}{(\omega^2 + 1)(\omega^2 + 16)} \right)$$

Solved Problems



7.4 If $X(t)$ is a wide-sense stationary process with PSD $S_{XX}(\omega)$ then find the power spectrum of $Y(t) = X(t) + X(t - t_0)$.

Solution

Given: $Y(t) = X(t) + X(t - t_0)$

$$R_{YY}(\tau) = E[Y(t) Y(t + \tau)]$$

$$= E[\{X(t) + X(t - t_0)\} \{X(t + \tau) + X(t - t_0 + \tau)\}]$$

$$= E[X(t) X(t + \tau)] + E[X(t - t_0) X(t + \tau)] + E[X(t) X(t - t_0 + \tau)] + E[X(t - t_0) X(t - t_0 + \tau)]$$

$$R_{XX}(\tau) + R_{XX}(t_0 + \tau) + R_{XX}(\tau - t_0) + R_{XX}(\tau) = 2R_{XX}(\tau) + R_{XX}(t_0 + \tau) + R_{XX}(\tau - t_0)$$

Apply Fourier transform on both sides.

$$\begin{aligned}
 F[R_{YY}(\tau)] &= 2 F[R_{XX}(\tau)] + F[R_{XX}(\tau + t_0)] + F[R_{XX}(\tau - t_0)] & F[x(t - t_0)] &= e^{-j\omega t_0} X(j\omega) \\
 &= 2 S_{XX}(\omega) + e^{j\omega t_0} S_{XX}(\omega) + e^{-j\omega t_0} S_{XX}(\omega). \\
 &= 2S_{XX}(\omega) + 2\cos \omega t_0 S_{XX}(\omega) \\
 &= 2S_{XX}(\omega) [1 + \cos \omega t_0]
 \end{aligned}$$

7.5 The power spectral density of a stationary random process is given by

$$S_{XX}(\omega) = \begin{cases} A & -K \leq \omega \leq K \\ 0 & \text{otherwise} \end{cases}$$

Find the auto-correlation function.

Solution We know $R_{XX}(\tau) = F^{-1}[S_{XX}(\omega)]$

$$\begin{aligned}
 &= \frac{1}{2\pi} \int_{-\infty}^{\infty} S_{XX}(\omega) e^{j\omega\tau} d\omega \\
 &= \frac{1}{2\pi} \int_{-K}^K A e^{j\omega\tau} d\omega = \frac{A}{2\pi} \left. \frac{e^{j\omega\tau}}{j\tau} \right|_{-K}^K \\
 &= \frac{A}{2\pi} \left[\frac{e^{jK\tau} - e^{-jK\tau}}{j\tau} \right] = \frac{A}{\pi\tau} \left[\frac{e^{jK\tau} - e^{-jK\tau}}{2j} \right] \\
 &= \frac{A}{\pi\tau} \sin(K\tau)
 \end{aligned}$$

7.6 A WSS random process $X(t)$ has PSD

$$S_{XX}(\omega) = \frac{\omega^2}{\omega^4 + 10\omega^2 + 9}$$

Find autocorrelation function and mean square value of the process.

Solution Given:

$$\begin{aligned}
 S_{XX}(\omega) &= \frac{\omega^2}{\omega^4 + 10\omega^2 + 9} \\
 &= \frac{\omega^2}{(\omega^2 + 1)(\omega^2 + 9)} \\
 &= \frac{A}{\omega^2 + 1} + \frac{B}{\omega^2 + 9} \\
 A &= (\omega^2 + 1) \left. \frac{\omega^2}{(\omega^2 + 1)(\omega^2 + 9)} \right|_{\omega^2 = -1} = \frac{-1}{8}
 \end{aligned}$$

$$\begin{aligned}
 B &= (\omega^2 + 9) \left. \frac{\omega^2}{(\omega^2 + 1)(\omega^2 + 9)} \right|_{\omega^2 = -9} = \frac{-9}{-8} = \frac{9}{8} \\
 S_{XX}(\omega) &= -\frac{1}{8} \left(\frac{1}{\omega^2 + 1} \right) + \frac{9}{8} \left(\frac{1}{\omega^2 + 9} \right) \\
 R_{XX}(\tau) &= -\frac{1}{16} F^{-1} \left(\frac{2}{\omega^2 + 1} \right) + \frac{9}{8(6)} F^{-1} \left[\frac{2(3)}{\omega^2 + 9} \right] \\
 &= -\frac{1}{16} e^{-|\tau|} + \frac{3}{16} e^{-3|\tau|}
 \end{aligned}$$

7.7 For a random process with power spectrum

$$S_{XX}(\omega) = \begin{cases} 1 - \frac{\omega^2}{4} & |\omega| \leq 2 \\ 0 & \text{elsewhere} \end{cases}$$

find the autocorrelation function of the process.

Solution

$$\begin{aligned}
 R_{XX}(\tau) &= F^{-1}[S_{XX}(\omega)] \\
 &= \frac{1}{2\pi} \int_{-\infty}^{\infty} S_{XX}(\omega) e^{j\omega\tau} d\omega \\
 &= \frac{1}{2\pi} \int_{-2}^2 \left(1 - \frac{\omega^2}{4} \right) e^{j\omega\tau} d\omega \\
 &= \frac{1}{2\pi} \left\{ \left. \frac{e^{j\omega\tau}}{j\tau} \right|_{-2}^2 - \frac{1}{4} \left[\left. \frac{\omega^2 e^{j\omega\tau}}{j\tau} + \frac{2\omega e^{j\omega\tau}}{\tau^2} - \frac{2e^{j\omega\tau}}{j\tau^3} \right] \right|_{-2}^2 \right\} \\
 &= \frac{1}{2\pi} \left(\frac{e^{j2\tau} - e^{-j2\tau}}{j\tau} \right) - \frac{1}{4} \left[\frac{4e^{j2\tau} - 4e^{-j2\tau}}{j\tau} + \frac{4e^{j2\tau} + 4e^{-j2\tau}}{\tau^2} - \frac{2}{j\tau^3} (e^{j2\tau} - e^{-j2\tau}) \right] \\
 &= \frac{1}{\pi} \left\{ \frac{\sin 2\tau}{\tau} - \frac{1}{4} \left[\frac{4}{\tau} \sin 2\tau + \frac{4}{\tau^2} \cos 2\tau - \frac{2}{\tau^3} \sin 2\tau \right] \right\} \\
 &= \frac{1}{\pi} \left\{ \frac{1}{2\tau^3} \sin 2\tau - \frac{1}{\tau^2} \cos 2\tau \right\}
 \end{aligned}$$



Practice Problem

7.3 For a random process with power spectrum

$$S_{XX}(\omega) = 1 + \omega^2 \quad \text{for } |\omega| < 1$$

$$= 0 \quad \text{otherwise}$$

find the autocorrelation function of the process.

$$\left(\text{Ans: } \frac{2}{\pi} \left\{ \frac{\sin \tau}{\tau} + \frac{\cos \tau}{\tau^2} - \frac{\sin \tau}{\tau^3} \right\} \right)$$

Solved Problem



7.8 Consider the random process $X(t) = A \cos(\omega_0 t + \theta)$ where A and ω_0 are real constants and θ is a random variable uniformly distributed on the interval $\left(0, \frac{\pi}{2}\right)$. Find the average power P_{XX} in $X(t)$.

Solution: Given:

$$X(t) = A \cos(\omega_0 t + \theta)$$

' θ ' is uniformly distributed on the interval $\left(0, \frac{\pi}{2}\right)$

$$\Rightarrow f_{\theta}(\theta) = \begin{cases} \frac{2}{\pi} & 0 \leq \theta \leq \frac{\pi}{2} \\ 0 & \text{otherwise} \end{cases}$$

$$P_{XX}(T) = E[X^2(t)]$$

$$= \int_{-\infty}^{\infty} X^2(t) f_{\theta}(\theta) d\theta = \int_0^{\pi/2} A^2 \cos^2(\omega_0 t + \theta) \frac{2}{\pi} d\theta$$

$$= \frac{2A^2}{\pi} \int_0^{\pi/2} \frac{1 + \cos 2(\omega_0 t + \theta)}{2} d\theta$$

$$= \frac{A^2}{\pi} \left[\theta \Big|_0^{\pi/2} + \frac{\sin 2(\omega_0 t + \theta)}{2} \Big|_0^{\pi/2} \right]$$

$$= \frac{A^2}{\pi} \left[\frac{\pi}{2} + \frac{1}{2} \{ \sin(2\omega_0 t + \pi) - \sin 2\omega_0 t \} \right]$$

$$= \frac{A^2}{\pi} \left[\frac{\pi}{2} - \sin(2\omega_0 t) \right]$$

$$= \frac{A^2}{2} - \frac{A^2}{\pi} \sin(2\omega_0 t)$$

The time average power is

$$P_{XX} = A[P_{XX}(T)]$$

$$\begin{aligned}
&= \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T P_{XX}(T) dt \\
&= \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T \left(\frac{A^2}{2} - \frac{A^2}{\pi} \sin 2\omega_0 t \right) dt \\
&= \frac{A^2}{2}
\end{aligned}$$

$$P_{XX} = \frac{A^2}{2}$$

Practice Problems

7.4 Repeat Solved Problem (7.8) for

(a) $X(t) = A \cos(\omega_0 t + \theta) u(t)$

(b) $X(t) = A \sin(\omega_0 t + \theta)$

where A and ω_0 are constants and θ is a random variable uniformly distributed on the interval $(0, \pi)$

$$\left(\text{Ans: (a) } \frac{A^2}{4} \text{ (b) } \frac{A^2}{2} \right)$$

7.5 Consider a random process $X(t) = \cos(\omega t + \theta)$ where ω is a real constant and θ is a uniform random variable in $\left(0, \frac{\pi}{2}\right)$. Find the average power in the process.

(Ans. 1/2)

Solved Problem

7.9 The rectangular function $\Pi\left(\frac{\tau}{T}\right)$ is shown in Fig. 7.1. Is it a valid autocorrelation function?

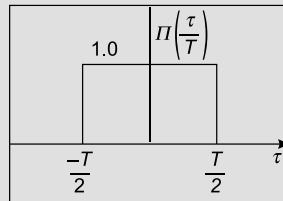


Fig. 7.1

Solution The power spectral density is

$$S_{XX}(\omega) = F[R_{XX}(\tau)] = F\left[\Pi\left(\frac{\tau}{T}\right)\right]$$

$$\begin{aligned}
 &= \int_{-T/2}^{T/2} e^{-j\omega t} dt = \frac{e^{\frac{j\omega T}{2}} - e^{-\frac{j\omega T}{2}}}{j\omega} \\
 &= T \frac{\sin\left(\frac{\omega T}{2}\right)}{\left(\frac{\omega T}{2}\right)}
 \end{aligned}$$

From the above equation, we can observe that $S_{XX}(\omega)$ takes negative value for some ω . Since PSD is always non-negative, $\Pi\left(\frac{\tau}{T}\right)$ is not a valid autocorrelation function.



Practice Problems

7.6 Find the ACF corresponding to the PSD shown below.

(Ans: $4\sin c 2\omega + 2\sin c \omega$)

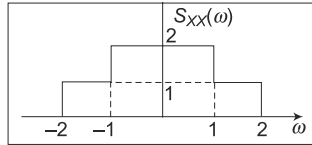


Fig. 7.2

7.7 If $R_{XX}(\tau) = 9\left(\frac{1}{3}\right)^{-|\tau|} + 16\left(\frac{1}{4}\right)^{-|\tau|}$, find $S_{XX}(\omega)$.

(Ans: $\frac{6}{\omega^2 + \frac{1}{9}} + \frac{8}{\omega^2 + \frac{1}{16}}$)

Solved Problem



7.10 Given a zero-mean stationary process $X(t)$ and its auto-correlation function $R_{XX}(\tau)$ and power spectrum $S_{XX}(\omega)$, find the autocorrelation and power spectrum of $Y(t) = X(t) + a$, where a is a non-zero non-random real constant.

Solution

$$\begin{aligned}
 R_{YY}(t, t + \tau) &= E[Y(t) Y(t + \tau)] \\
 &= E\{[X(t) + a] [X(t + \tau) + a]\} \\
 &= E[X(t) X(t + \tau)] + a E[X(t + \tau)] + a E[X(t)] + E[a^2] \\
 &= E[X(t) X(t + \tau)] + E[a^2] \\
 &= R_{XX}(\tau) + a^2
 \end{aligned}$$

If a is negative,

$$R_{YY}(t, t + \tau) = E\{[X(t) - a] [X(t + \tau) - a]\}$$

$$\begin{aligned}
 &\because E[X(t)] = 0 \\
 &E[X(t + \tau)] = 0
 \end{aligned}$$

$$\begin{aligned}
&= E[X(t) X(t + \tau)] - a E[X(t + \tau)] - a E[X(t)] + a^2 \\
&= R_{XX}(\tau) + a^2
\end{aligned}$$

Therefore, $R_{YY}(t, t + \tau) = R_{YY}(\tau) = R_{XX}(\tau) + |a|^2$

$$\begin{aligned}
S_{YY}(\omega) &= S_{XX}(\omega) + |a|^2 F[1] \\
&= S_{XX}(\omega) + |a|^2 2\pi \delta(\omega)
\end{aligned}$$



Practice Problem

7.8 If $X(t)$ is a stationary process, with autocorrelation function $R_{XX}(\tau)$ and power spectrum $S_{XX}(\omega)$, find the power spectrum of $Y(t) = A + B X(t)$.

Hint: Proceed in the same way as the above problem. Since $E[X(t)] \neq 0$, we get $R_{YY}(\tau) = B^2 R_{XX}(\tau) + 2AB\mu_X + A^2$ and $S_{YY}(\omega) = B^2 S_{XX}(\omega) + (2AB\mu_X + A^2) 2\pi \delta(\omega)$.

Solved Problems



7.11 Given a stationary random process $X(t)$ and its autocorrelation function $R_{XX}(\tau)$ and power spectrum $S_{XX}(\omega)$, find the autocorrelation and power spectrum of $Y(t) = X(t) e^{j\omega_0 t}$ where ω_0 is a non-random real constant.

Solution Since $Y(t)$ is complex,

$$\begin{aligned}
R_{YY}(\tau) &= E[Y(t) Y^*(t + \tau)] \\
&= E[X(t) e^{j\omega_0 t} X(t + \tau) e^{-j\omega_0(t + \tau)}] \\
&= E[X(t) X(t + \tau)] e^{-j\omega_0 \tau} \\
&= e^{-j\omega_0 \tau} R_{XX}(\tau)
\end{aligned}$$

7.12 The given power spectrum is of a random process $X(t)$ with a non-random dc term

$$S_{XX}(\omega) = 6\pi \delta(\omega) + 1.5\pi \delta(\omega - 2.5\pi) + 1.5\pi \delta(\omega + 2.5\pi) + 2\pi \delta(\omega - 10) + 2\pi \delta(\omega + 10)$$

- What frequencies are included in $X(t)$? How large are these frequencies in magnitude?
- Find the mean value and average power of $X(t)$.
- Find the variance of $X(t)$.

Solution Given:

$$(a) \quad S_{XX}(\omega) = 6\pi \delta(\omega) + 1.5\pi \delta(\omega - 2.5\pi) + 1.5\pi \delta(\omega + 2.5\pi) + 2\pi \delta(\omega - 10) + 2\pi \delta(\omega + 10)$$

Taking inverse Fourier transform on both sides, we get

$$R_{XX}(\tau) = 3 + 1.5 \cos(2.5\pi\tau) + 2 \cos(10\tau)$$

Thus, the dc term in $R_{XX}(\tau) = 3$

That is, dc power = 3, hence $X(t)$ has dc term $\sqrt{3}$

The frequencies are $\omega_1 = 2.5\pi$ and $\omega_2 = 10$ from which $f_1 = 1.25$ Hz and $f_2 = 1.592$ Hz

The frequency component $f_1 = 1.25$ Hz has a power of

$$\frac{A^2}{2} = 1.5 \Rightarrow A = 1.732$$

The frequency component $f_2 = 1.592$ Hz has a power of

$$\frac{A^2}{2} = 2 \Rightarrow A = 2$$

(b) The mean value = dc value = 3

Average power = $R_{XX}(0) = 3 + 1.5 + 2 = 6.5$

(c) $\text{Var}(X) = E[X^2] - \{E[X]\}^2$
 $= 6.5 - (\sqrt{3})^2 = 3.5$

7.13 A wide-sense stationary process $X(t)$ has power spectrum $S_{XX}(\omega) = \frac{9}{\omega^2 + 4}$. (a) Find the average power of $X(t)$. (b) Find the average power over the frequency band $0 \leq \omega \leq 4$.

Solution Given:

$$S_{XX}(\omega) = \frac{9}{\omega^2 + 4} = \frac{9}{4} \left(\frac{4}{\omega^2 + 4} \right)$$

$$R_{XX}(\tau) = \frac{9}{4} F^{-1} \left[\frac{4}{\omega^2 + 4} \right] = \frac{9}{4} e^{-2|\tau|}$$

$$\text{Average power } P_{XX} = R_{XX}(\tau) \Big|_{\tau=0} = \frac{9}{4}$$

Power over frequency band $0 \leq \omega \leq 4$ is equal to

$$\begin{aligned} P_{XX[0, 4]} &= \frac{1}{\pi} \int_0^4 S_{XX}(\omega) d\omega && [\text{Refer Eq. (7.14b)}] \\ &= \frac{1}{\pi} \int_0^4 \frac{9}{\omega^2 + 4} d\omega = \frac{9}{\pi} \int_0^4 \frac{1}{\omega^2 + 4} d\omega \\ &= \frac{9}{4\pi} \int_0^4 \frac{1}{1 + \left(\frac{\omega}{2}\right)^2} d\omega \\ &= \frac{9}{2\pi} \tan^{-1} \left(\frac{\omega}{2} \right) \Big|_0^4 \\ &= \left[\frac{9}{2\pi} \tan^{-1} (2) \right] \end{aligned}$$

7.14 Assume that the ergodic random process $X(t)$ has an autocorrelation function

$$R_{XX}(\tau) = 18 + \frac{2}{6 + \tau^2} [1 + 4 \cos(12\tau)]$$

What is the average power?

Solution Given:

$$R_{XX}(\tau) = 18 + \frac{2}{6 + \tau^2} [1 + 4 \cos(12\tau)]$$

Average power

$$\begin{aligned} P_{XX} = R_{XX}(0) &= 18 + \frac{2}{6} [1 + 4] \\ &= 18 + \frac{5}{3} = 19.67 \text{ watts} \end{aligned}$$

7.15 The power density spectrum of a WSS process is

$$S_{XX}(\omega) = 4\pi \delta(\omega) + 3\pi \delta(\omega - 5\pi) + 3\pi \delta(\omega + 5\pi) + 2\pi \delta(\omega - 4) + 2\pi \delta(\omega + 4)$$

- (a) What are the frequencies in $X(t)$?
 (b) Find out mean, variance and average power of $X(t)$.

Solution

(a) Given:

$$S_{XX}(\omega) = 4\pi \delta(\omega) + 3\pi \delta(\omega - 5\pi) + 3\pi \delta(\omega + 5\pi) + 2\pi \delta(\omega - 4) + 2\pi \delta(\omega + 4)$$

Taking inverse Fourier transform on both sides, we get

$$R_{XX}(\tau) = 2 + 3 \cos(5\pi\tau) + 2 \cos(4\tau)$$

The frequencies are $\omega_1 = 5\pi \Rightarrow f_1 = 2.5 \text{ Hz}$

$$\text{and } \omega_2 = 4 \Rightarrow f_2 = \frac{4}{2\pi} \text{ Hz}$$

$$\begin{aligned} F^{-1}[2\pi \delta(\omega)] &= 1 \\ F^{-1}[\pi \delta(\omega - \omega_0) + \pi \delta(\omega + \omega_0)] &= \cos \omega_0 t \end{aligned}$$

(c) The mean value = dc value = 2

That is dc power = 2, hence, $X(t)$ has a dc term $\sqrt{2}$

$$\Rightarrow \mu_X = E[X(t)] = \sqrt{2}$$

$$\text{Average power } E[X^2(t)] = R_{XX}(0) = 2 + 3 + 2 = 7$$

$$\begin{aligned} \sigma_X^2 &= E[X^2] - \{E[X]\}^2 \\ &= 7 - (\sqrt{2})^2 = 5 \end{aligned}$$



Practice Problems

7.9 A random process $X(t)$ has a power spectral density given by

$$S_{XX}(\omega) = \begin{cases} \frac{9}{4} - \frac{\omega^2}{16} & |\omega| \leq 6 \\ 0 & \text{otherwise} \end{cases}$$

$$\left(\text{Ans: (a) } \frac{9}{\pi}; \text{ (b) } \frac{\sin 6\tau - 6\tau \cos 6\tau}{8\pi \tau^3} \right)$$

Determine (a) the average power, and (b) the autocorrelation function.

7.10 A random process $X(t)$ has the power spectral density

$$S_{XX}(\omega) = \frac{16}{\omega^2 + 49} \quad \left(\text{Ans: (a) } \frac{8}{7} w; \text{ (b) } \frac{8}{7} e^{-7|\tau|} \right)$$

Find (a) average power in the process, and (b) autocorrelation function.

Solved Problems



7.16 A stationary random process $X(t)$ has auto-correlation function

$$R_{XX}(\tau) = 10 + 5 \cos(2\tau) + 10e^{-2|\tau|}$$

Find the dc and ac average power of $X(t)$.

Solution

$$R_{XX}(\tau) = 10 + 5 \cos(2\tau) + 10e^{-2|\tau|}$$

$R_{XX}(\tau)$ consists of periodic and non-periodic components

$$R_{XX}(t) = R_1(\tau) + R_2(\tau)$$

where $R_1(\tau) = 5 \cos 2\tau$ is periodic component and $R_2(\tau)$ is the non-periodic component given by

$$R_2(\tau) = 10 + 10e^{-2|\tau|}$$

The autocorrelation function of $A \cos(2\tau + \theta)$ is $\frac{A^2}{2} \cos 2\tau$.

Comparing with $R_1(\tau)$, we get $\frac{A^2}{2} = 5$

Since $X(t) = A \cos(2t + \theta)$ has zero mean, the average power is $R_1(0) = \frac{A^2}{2} = 5$

The average power of the non-periodic component is

$$R_2(0) = 10 + 10 = 20$$

The average power of $X(t)$ is

$$R_{XX}(0) = 10 + 5 + 10 = 25$$



CROSS-POWER DENSITY SPECTRUM 7.4

Definition 1: Consider the jointly wide-sense stationary process $X(t)$ and $Y(t)$, their sum $W(t) = X(t) + Y(t)$.

The autocorrelation function of $W(t)$ is

$$\begin{aligned} R_{WW}(t, t + \tau) &= E\{[X(t) + Y(t)][X(t + \tau) + Y(t + \tau)]\} \\ &= E[X(t)X(t + \tau)] + E[Y(t)X(t + \tau)] + E[X(t)Y(t + \tau)] + E[Y(t)Y(t + \tau)] \\ &= R_{XX}(\tau) + R_{YX}(\tau) + R_{XY}(\tau) + R_{YY}(\tau) \end{aligned}$$

Applying Fourier transform on both sides, we get

$$S_{WW}(\omega) = S_{XX}(\omega) + S_{YX}(\omega) + S_{XY}(\omega) + S_{YY}(\omega)$$

The second and third terms in the above equation are known as cross-power density spectrum defined as

$$S_{XY}(\omega) = F[R_{XY}(\tau)] = \int_{-\infty}^{\infty} R_{XY}(\tau) e^{-j\omega\tau} d\tau \quad (7.28)$$

$$S_{YX}(\omega) = F[R_{YX}(\tau)] = \int_{-\infty}^{\infty} R_{YX}(\tau) e^{-j\omega\tau} d\tau \quad (7.29)$$

By taking inverse Fourier transform, we can obtain cross-correlation function

$$R_{XY}(\tau) = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_{XY}(\omega) e^{j\omega\tau} d\omega \quad (7.30)$$

$$R_{YX}(\tau) = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_{YX}(\omega) e^{j\omega\tau} d\omega \quad (7.31)$$

Definition 2: Consider two real random processes $X(t)$ and $Y(t)$. Let $x_T(t)$ and $y_T(t)$ be truncated ensembles of the random processes defined as

$$\begin{aligned} x_T(t) &= x(t) \quad \text{for } -T \leq t \leq T \\ &= 0 \quad \text{otherwise} \end{aligned}$$

and

$$\begin{aligned} y_T(t) &= y(t) \quad \text{for } -T \leq t \leq T \\ &= 0 \quad \text{otherwise} \end{aligned}$$

The Fourier transforms of $x_T(t)$ and $y_T(t)$ are given by

$$X_T(j\omega) = \int_{-T}^T x_T(t) e^{-j\omega t} dt$$

and
$$Y_T(j\omega) = \int_{-T}^T y_T(t) e^{-j\omega t} dt$$

The cross power between two processes is given by

$$\begin{aligned} P_{xy}(T) &= \frac{1}{2T} \int_{-T}^T x_T(t) y_T(t) dt \\ &= \frac{1}{2T} \int_{-T}^T x(t) y(t) dt \end{aligned} \quad (7.32)$$

The cross power between two random processes can be obtained by taking expected value of the above equations

$$P_{XX} = E \left[\lim_{T \rightarrow \infty} \frac{1}{2\pi} \int_{-T}^T x(t) y(t) dt \right] \quad (7.33)$$

Using Parseval's theorem,

$$\int_{-T}^T x(t) y(t) dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} X_T(j\omega) Y_T^*(j\omega) d\omega \quad (7.34)$$

We can write

$$\begin{aligned} P_{XY} &= E \left[\lim_{T \rightarrow \infty} \frac{1}{2T} \frac{1}{2\pi} \int_{-\infty}^{\infty} X_T(j\omega) Y_T^*(j\omega) d\omega \right] \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \lim_{T \rightarrow \infty} \frac{1}{2T} E [X_T(j\omega) Y_T^*(j\omega)] d\omega \end{aligned} \quad (7.35)$$

where $X_T(j\omega)$, $Y_T(j\omega)$ are Fourier transforms of truncated version of $x(t)$ and $y(t)$ respectively.

Equation (7.35) can be written as

$$P_{XY} = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_{XY}(\omega) d\omega \quad (7.36)$$

$$\text{where } S_{XY}(\omega) = \lim_{T \rightarrow \infty} \frac{E[X_T(j\omega) Y_T^*(j\omega)]}{2T} \quad (7.37)$$

and is known as cross-power density spectrum.

From Eq. (7.37)

$$\begin{aligned} S_{XY}(\omega) &= E \left[\lim_{T \rightarrow \infty} \frac{X_T(j\omega) Y_T^*(j\omega)}{2T} \right] \\ &= E \left[\lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T X(t) Y(t + \tau) e^{-j\omega t} dt \right] \\ &= F \{A[E(X(t) Y(t + \tau))]\} \\ &= F \{A[R_{XY}(t, t + \tau)]\} \end{aligned} \quad (7.38)$$

That is, the cross-power density spectrum can be defined as the Fourier transform of the time average of the cross-correlation function.

$$\begin{aligned} S_{XY}(\omega) &= F(\text{time average of } R_{XY}(t, t + \tau)) \\ \Rightarrow S_{XY}(\omega) &= F \left\{ \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T R_{XY}(t, t + \tau) dt \right\} \\ &= \int_{-\infty}^{\infty} \left\{ \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T R_{XY}(t, t + \tau) dt \right\} e^{-j\omega \tau} d\tau \end{aligned} \quad (7.39)$$

For a jointly stationary process,

$$S_{XY}(\omega) = F[R_{XY}(\tau)] \text{ and } S_{YX}(\omega) = F[R_{YX}(\tau)] \quad (7.40)$$

Properties of Cross Power Density Spectrum

Property 1: Unlike auto-power spectrum, the cross-power spectrum need not be real, non-negative or an even function of ω .

Property 2: For a real valued random process $X(t)$ and $Y(t)$,

$$S_{YX}(\omega) = S_{XY}^*(\omega) = S_{XY}(-\omega) \quad (7.41)$$

Proof

$$\begin{aligned} S_{YX}(\omega) &= \int_{-\infty}^{\infty} R_{YX}(\tau) e^{-j\omega\tau} d\tau \\ &= \int_{-\infty}^{\infty} E[Y(t) X(t + \tau)] e^{-j\omega\tau} d\tau \\ &= \int_{-\infty}^{\infty} E[X(t) Y(t - \tau)] e^{-j\omega\tau} d\tau \\ &= \int_{-\infty}^{\infty} R_{XY}(-\tau) e^{-j\omega\tau} d\tau = \int_{-\infty}^{\infty} R_{XY}(\tau) e^{j\omega\tau} d\tau \\ &= S_{XY}(-\omega) \\ &= \left[\int_{-\infty}^{\infty} R_{XY}(\tau) e^{-j\omega\tau} d\tau \right]^* = S_{XY}^*(\omega) \end{aligned}$$

Property 3: The real part of $S_{XY}(\omega)$ and real part of $S_{YX}(\omega)$ are even functions of ω

$$\begin{aligned} S_{XY}(\omega) &= \int_{-\infty}^{\infty} R_{XY}(\tau) e^{-j\omega\tau} d\tau \\ &= \int_{-\infty}^{\infty} R_{XY}(\tau) [\cos \omega\tau - j \sin \omega\tau] d\tau \\ \text{Re}[S_{XY}(\omega)] &= \int_{-\infty}^{\infty} R_{XY}(\tau) \cos \omega\tau d\tau \\ \text{Re}[S_{XY}(-\omega)] &= \int_{-\infty}^{\infty} R_{XY}(\tau) \cos(-\omega)\tau d\tau \\ &= \int_{-\infty}^{\infty} R_{XY}(\tau) \cos \omega\tau d\tau \end{aligned}$$

Real Part of $S_{XY}(\omega)$ is even function of ω ,

Similarly, $\text{Re}[S_{YX}(\omega)]$ is also an even function.

From Eq. (7.41), we have

$$\begin{aligned} S_{YX}(\omega) &= S_{XY}^*(\omega) = S_{XY}(-\omega) \\ \Rightarrow \quad \text{Re}[S_{YX}(\omega)] &= \text{Re}[S_{XY}(\omega)] \\ \text{Im}[S_{YX}(\omega)] &= -\text{Im}[S_{XY}^*(\omega)] \end{aligned}$$

$$\Rightarrow \operatorname{Re}[S_{XY}(\omega)] = \operatorname{Re}[S_{XY}^*(\omega)] = \operatorname{Re}[S_{XY}(-\omega)] \quad (7.42)$$

Property 4: Imaginary parts of $S_{XY}(\omega)$ and $S_{YX}(\omega)$ are odd functions of ω .

$$\begin{aligned} S_{XY}(\omega) &= \int_{-\infty}^{\infty} R_{XY}(\tau) e^{-j\omega\tau} d\tau \\ &= \int_{-\infty}^{\infty} R_{XY}(\tau) [\cos \omega\tau - j \sin \omega\tau] d\tau \end{aligned} \quad (7.43)$$

$$\operatorname{Im}[S_{XY}(\omega)] = - \int_{-\infty}^{\infty} R_{XY}(\tau) \sin \omega\tau d\tau \quad (7.44)$$

$$\begin{aligned} \operatorname{Im}[S_{XY}(-\omega)] &= - \int_{-\infty}^{\infty} R_{XY}(\tau) \sin(-\omega)\tau d\tau \\ &= \int_{-\infty}^{\infty} R_{XY}(\tau) \sin \omega\tau d\tau = -\operatorname{Im}[S_{XY}(-\omega)] \end{aligned} \quad (7.45)$$

Therefore, Imaginary part of $S_{XY}(\omega)$ is an odd function of ω .

From Eq. (7.41),

$$S_{YX}(\omega) = S_{XY}^*(-\omega) = S_{XY}(-\omega) \quad (7.46)$$

$$\operatorname{Im}[S_{YX}(\omega)] = -\operatorname{Im}[S_{XY}^*(\omega)] = -\operatorname{Im}[S_{XY}(-\omega)]$$

So odd function in ω .

Property 5: If $X(t)$ and $Y(t)$ are uncorrelated and have constant means, then, $S_{XY}(\omega) = S_{YX}(\omega) = 2\pi E[X] E[Y] \delta(\omega)$

Property 5: If $X(t)$ and $Y(t)$ are orthogonal

$$S_{XY}(\omega) = S_{YX}(\omega) = 0$$

REVIEW QUESTIONS



7.9 Define cross power density spectrum.

7.10 Explain the properties of cross-power density spectrum.

7.11 Prove that $S_{XY}(\omega) = S_{YX}^*(\omega)$.

Solved Problems



7.17 (a) For $X(t)$ and $Y(t)$ random processes, prove that $S_{XY}(\omega) = \lim_{T \rightarrow \infty} \frac{E[X_T^*(j\omega)Y_T(j\omega)]}{2T}$

(b) If $R_{XY}(\tau) = 4u(\tau)e^{-\alpha\tau}$, find $S_{XY}(\omega)$.

Solution

(a) Let $x_T(t) = x(t)$ for $-T \leq x \leq T$
 $= 0$ otherwise

$$\begin{aligned} \text{and } y_T(t) &= y(t) & \text{for } -T \leq t \leq T \\ &= 0 & \text{otherwise} \end{aligned}$$

$$\text{Let } F[x_T(t)] = X_T(j\omega)$$

$$\text{and } F[y_T(t)] = Y_T(j\omega)$$

The cross power $P_{XY}(T)$ is given by

$$\begin{aligned} P_{XY}(T) &= \frac{1}{2T} \int_{-T}^T x_T(t)y_T(t)dt \\ &= \frac{1}{2T} \int_{-T}^T x(t)y(t)dt \end{aligned} \quad (7.47)$$

Since Fourier transform of $x_T(t)$ and $y_T(t)$ exist,

we can apply Parseval's theorem to Eq. (7.47)

$$P_{XY}(T) = \frac{1}{2T} \int_{-T}^T x(t)y(t)dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{X_T^*(j\omega)Y_T(j\omega)}{2T} d\omega$$

The expected value of cross power is

$$\bar{P}_{XY}(T) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{E[X_T^*(j\omega)Y_T(j\omega)]}{2T} d\omega$$

The total average power is

$$\begin{aligned} P_{XY} &= \lim_{T \rightarrow \infty} \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{E[X_T^*(j\omega)Y_T(j\omega)]}{2T} d\omega \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \lim_{T \rightarrow \infty} \frac{E[X_T^*(j\omega)Y_T(j\omega)]}{2T} d\omega \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} S_{XY}(\omega) d\omega \end{aligned}$$

where

$$S_{XY}(\omega) = \lim_{T \rightarrow \infty} \frac{E[X_T^*(j\omega)Y_T(j\omega)]}{2T}$$

(b) Given:

$$R_{XY}(\tau) = 4u(\tau)e^{-\alpha\tau}$$

$$S_{XY}(\omega) = F[R_{XY}(\tau)]$$

$$= \int_{-\infty}^{\infty} 4u(\tau)e^{-\alpha\tau} e^{-j\omega\tau} d\tau = 4 \int_0^{\infty} e^{-(\alpha + j\omega)\tau} d\tau$$

$$= \frac{4}{-(\alpha + j\omega)} e^{-(\alpha + j\omega)\tau} \bigg|_0^{\infty} = \frac{-4}{\alpha + j\omega} [0 - 1] = \frac{4}{\alpha + j\omega}$$

$$S_{XY}(\omega) = \frac{4}{\alpha + j\omega}$$



Practice Problems

7.11 If $Y(t)$ is moving average of $X(t)$ over $(t - T, t + T)$, express $S_{YY}(\omega)$ in terms of $S_{XX}(\omega)$. Hence, find the autocorrelation function of $Y(t)$ in terms of autocorrelation function of $X(t)$.

7.12 The cross-power spectrum of real random processes $X(t)$ and $Y(t)$ is

$$S_{XY}(\omega) = \frac{6}{(9 + \omega^2)(3 + j\omega)^2}$$

Find the cross-correlation function.

$$\left(\text{Ans: } R_{XY}(\tau) = \frac{e^{-3|\tau|}}{6} \{ (9\tau - 1)u(\tau) + u(-\tau) \} \right)$$

Solved Problem



7.18 If $X(t)$ and $Y(t)$ are real random processes, determine which of the following functions can be valid?

$$(a) S_{XX}(\omega) = \frac{5}{(3 + 6\omega^3)} \quad (b) S_{XY}(\omega) = 6 + j\omega^2 \quad (c) S_{XX}(\omega) = \frac{2e^{-2|\tau|}}{1 + 2\omega^2}$$

Solution

$$(a) S_{XX}(-\omega) = \frac{5}{3 - 6\omega^3}$$

Since $S_{XX}(-\omega) \neq S_{XX}(\omega)$, it is not a valid PSD.

(b) From Property 4 of cross power spectral density, we know the imaginary part is an odd function of ω .

In the given problem, $S_{XY}(\omega)$ is an even function of ω . Hence, $S_{XY}(\omega)$ is not a valid PSD.

(c) Since $S_{XX}(\omega)$ depends on ' τ ', it is not a valid PSD.



Practice Problem

7.13 If $X(t)$ and $Y(t)$ are real random processes, determine which of the following functions can be valid?

$$(a) S_{XY}(\omega) = \omega^3 + j\omega^2 \quad (b) S_{XX}(\omega) = \frac{1}{1 - \omega^2}$$

$$(c) S_{XY}(\omega) = 7\delta(\omega)$$

(Ans: (a) no, imaginary part must be odd

(b) no (c) yes)

Solved Problems



7.19 Find the cross-spectral density function of two random processes $X(t) = A \cos(\omega_0 t) + B \sin(\omega_0 t)$ and $Y(t) = -A \sin(\omega_0 t) + B \cos(\omega_0 t)$. The mean values $\mu_X = \mu_Y = 0$. The variance of these processes are σ^2 .

Solution

$$\begin{aligned}
R_{XY}(t, t + \tau) &= E[X(t)Y(t + \tau)] \\
&= E\{[A \cos(\omega_0 t) + B \sin(\omega_0 t)] [-A \sin(\omega_0 t + \omega_0 \tau) + B \cos(\omega_0 t + \omega_0 \tau)]\} \\
&= E[-A^2 \cos(\omega_0 t) \sin(\omega_0 t + \omega_0 \tau)] + E[(-AB) \sin(\omega_0 t) \sin(\omega_0 t + \omega_0 \tau)] \\
&\quad + E[AB \cos(\omega_0 t) \cos(\omega_0 t + \omega_0 \tau)] + E[B^2 \sin(\omega_0 t) \cos(\omega_0 t + \omega_0 \tau)]
\end{aligned}$$

Given: $E[A] = E[B] = 0$

$$E[AB] = E[A]E[B] = 0$$

and $E[A^2] = E[B^2] = \sigma^2$

$$\begin{aligned}
\Rightarrow R_{XY}(t, t + \tau) &= \sigma^2 [\sin(\omega_0 t) \cos(\omega_0 t + \omega_0 \tau) - \cos(\omega_0 t) \sin(\omega_0 t + \omega_0 \tau)] \\
&= -\sigma^2 \sin(\omega_0 \tau) \\
S_{XY}(\omega) &= F[R_{XY}(\tau)] \\
&= F[-\sigma^2 \sin(\omega_0 \tau)] \\
&= -\sigma^2 j\pi [\delta(\omega + \omega_0) - \delta(\omega - \omega_0)]
\end{aligned}$$

**Practice Problem**

7.14 Two jointly stationary random processes $X(t)$ and $Y(t)$ have the cross-correlation function given by

$$R_{XY}(\tau) = 2e^{-2\tau} \tau \geq 0.$$

Find $S_{XY}(\omega)$ $S_{YX}(\omega)$

$$\left(\text{Ans: } \frac{2}{2 + j\omega}, \frac{2}{2 - j\omega} \right)$$

Solved Problems

7.20 Two independent stationary random processes $X(t)$ and $Y(t)$ have power spectrum densities $S_{XX}(\omega) = \frac{16}{\omega^2 + 16}$ and $S_{YY}(\omega) = \frac{\omega^2}{\omega^2 + 16}$ respectively. Assume $X(t)$ and $Y(t)$ are of zero mean. Find (a) PSD of $U(t) = X(t) + Y(t)$, and (b) $S_{XY}(\omega)$ and $S_{XU}(\omega)$.

Solution

(a) Given $S_{XX}(\omega) = \frac{16}{\omega^2 + 16}$ and $S_{YY}(\omega) = \frac{\omega^2}{\omega^2 + 16}$

$$U(t) = X(t) + Y(t)$$

$$S_{UU}(\omega) = S_{XX}(\omega) + S_{YY}(\omega)$$

$$= \frac{16}{\omega^2 + 16} + \frac{\omega^2}{\omega^2 + 16} = 1$$

(b) $S_{XY}(\omega) = F[R_{XY}(\tau)]$
 $= F[E(X(t)Y(t + \tau))]$
 $= F\{E[X(t)] \cdot E[Y(t + \tau)]\}$

Since $X(t)$ and $Y(t)$ are independent,

$$\begin{aligned}
 S_{XY}(\omega) &= 0 \\
 S_{XU}(\omega) &= F[R_{XU}(\tau)] \\
 &= F\{E[X(t)U(t + \tau)]\} \\
 &= F\{E[X(t) \cdot (X(t + \tau) + Y(t + \tau))]\} \\
 &= F\{E[X(t)X(t + \tau)] + E[X(t)Y(t + \tau)]\} \\
 &= F[R_{XX}(\tau) + 0] \\
 &= F[R_{XX}(\tau)] = S_{XX}(\omega) \\
 &= \frac{16}{\omega^2 + 16}
 \end{aligned}$$

7.21 For a random process, $X(t) = A \cos \omega_0 t + B \sin \omega_0 t$, find the power density spectrum if A and B are uncorrelated random variables with zero mean and same variance.

Solution Given: $X(t) = A \cos \omega_0 t + B \sin \omega_0 t$

$$\begin{aligned}
 R_{XX}(t, t + \tau) &= E\{[A \cos \omega_0 t + B \sin \omega_0 t] [A \cos(\omega_0 t + \omega_0 \tau) + B \sin(\omega_0 t + \omega_0 \tau)]\} \\
 &= E[A^2 \cos(\omega_0 t) \cos(\omega_0 t + \omega_0 \tau)] + E[B \sin(\omega_0 t) A \cos(\omega_0 t + \omega_0 \tau)] \\
 &\quad + E[A \cos(\omega_0 t) B \sin(\omega_0 t + \omega_0 \tau)] + E[B^2 \sin(\omega_0 t) \sin(\omega_0 t + \omega_0 \tau)]
 \end{aligned}$$

Since A and B are uncorrelated with zero mean and same variance, we can write

$$E[AB] = E[A] E[B]$$

$$E[A] = E[B] = 0 \text{ and}$$

$$E[A^2] = E[B^2] = \sigma^2$$

$$\begin{aligned}
 R_{XX}(t, t + \tau) &= \sigma^2 [\cos(\omega_0 t) \cos(\omega_0 t + \omega_0 \tau) + \sin(\omega_0 t) \sin(\omega_0 t + \omega_0 \tau)] \\
 &= \sigma^2 \cos \omega_0 \tau
 \end{aligned}$$

The power density spectrum

$$F[R_{XX}(t, t + \tau)] = F[R_{XX}(\tau)] = S_{XX}(\omega)$$

$$S_{XX}(\omega) = \pi \sigma^2 [\delta(\omega + \omega_0) + \delta(\omega - \omega_0)]$$

7.22 Let $Z(t) = X(t) \cos(\omega_0 t + y)$ where $X(t)$ is a zero-mean stationary Gaussian random process with $E[X^2(t)] = \sigma_X^2$ (a) If Y is constant, say zero, find $E[Z^2(t)]$; $Z(t)$ is stationary. (b) If y is random variable with a uniform pdf in the interval $(-\pi, \pi)$, find $R_{ZZ}(t, t + \tau)$. Is $Z(t)$ wide-sense stationary? If so, find the PSD of $Z(t)$.

Solution

(a) Given $y = 0$; $Z(t) = X(t) \cos \omega_0 t$

$$E[Z^2(t)] = E[X^2(t) \cos^2(\omega_0 t)] = \sigma_X^2 \cos^2 \omega_0 t$$

(b) 'y' is a random variable with a uniform pdf

$$f_Y(y) = \frac{1}{2\pi} \quad \text{for } -\pi \leq y \leq \pi$$

$$= 0 \quad \text{otherwise}$$

$$R_{ZZ}(t, t + \tau) = E[Z(t)Z(t + \tau)]$$

$$\begin{aligned}
&= E[X(t) \cos(\omega_0 t + y) X(t + \tau) \cos(\omega_0 t + \omega_0 \tau + y)] \\
&= E[X(t) X(t + \tau)] E[\cos(\omega_0 t + y) \cos(\omega_0 t + \omega_0 \tau + y)] \\
&= R_{XX}(\tau) E[\cos(\omega_0 t + y) \cos(\omega_0 t + \omega_0 \tau + y)]
\end{aligned}$$

$$\begin{aligned}
E[\cos(\omega_0 t + y) \cos(\omega_0 t + \omega_0 \tau + y)] &= \frac{1}{2\pi} \int_{-\pi}^{\pi} \cos(\omega_0 t + y) \cos(\omega_0 t + \omega_0 \tau + y) dy \\
&= \frac{1}{2\pi} \cdot \frac{1}{2} \left[\int_{-\pi}^{\pi} \cos \omega_0 \tau dy + \underbrace{\int_{-\pi}^{\pi} \cos(2\omega_0 t + 2y + \omega_0 \tau) dy}_{=0} \right] \\
&= \frac{1}{4\pi} [2\pi \cos \omega_0 \tau] = \frac{1}{2} \cos \omega_0 \tau
\end{aligned}$$

$$R_{ZZ}(t, t + \tau) = \frac{1}{2} R_{XX}(\tau) \cos \omega_0 \tau$$

Since $R_{ZZ}(t, t + \tau)$ depends on τ alone $Z(t)$ is wide-sense stationary

$$\begin{aligned}
S_{ZZ}(\omega) &= F(R_{ZZ}(\tau)) = F\left[\frac{1}{2} R_{XX}(\tau) \cos \omega_0 \tau\right] \\
&= \frac{1}{4} [S_{XX}(\omega + \omega_0) + S_{XX}(\omega - \omega_0)]
\end{aligned}$$



WHITE NOISE 7.5

A zero-mean, wide-sense stationary noise process $N(t)$ is said to be white noise if the power density spectrum is constant at all frequencies.

$$\text{That is, } S_{NN}(\omega) = S_0 \quad (7.48)$$

where S_0 is called its *intensity*. By taking inverse Fourier transform for Eq. (7.48), the autocorrelation function of $N(t)$ is equal to

$$\begin{aligned}
R_{NN}(t) &= F^{-1}[S_{NN}(\omega)] \\
&= F^{-1}[S_0] = S_0 F^{-1}[1] \\
&= S_0 \delta[\tau] \\
R_{NN}(\tau) &= S_0 \delta(\tau) \quad (7.49)
\end{aligned}$$

The autocorrelation function and power density spectrum of white noise are shown in Fig. 7.3.

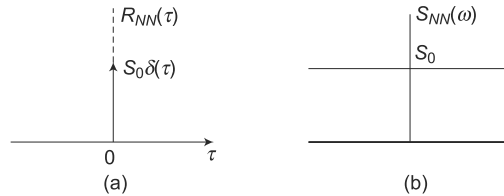


Fig. 7.3 (a) ACF of white noise; (b) PSD of white noise

The term *white noise* is derived from “white” light, which has a spectrum, that is constant over all variable light frequencies. White noise is a popular mathematical model of many physical noises. For example, thermal noise generated by thermal agitation of electrons in any electric conductor can be modelled as white noise.

White noise has the following properties.

- (i) The autocorrelation function is an impulse of intensity S_o .
- (ii) The power density spectrum is constant at all frequencies.
- (iii) It is physically unrealizable because it possesses infinite average power.

$$P_{NN} = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_{NN}(\omega) d\omega = \infty$$

- (iv) It is completely unpredictable because its future values are uncorrelated with present and past values.
- (v) It is easier to handle because the value at one time is uncorrelated with any other time.

7.5.1 Bandlimited White Noise

Noise having a non-zero and constant power spectral density over a finite frequency band and zero outside this frequency band is called bandlimited white noise. That is,

$$S_{NN}(\omega) = \begin{cases} S_o & -W \leq \omega \leq W \\ 0 & \text{elsewhere} \end{cases} \quad (7.50)$$

The autocorrelation function is given by

$$\begin{aligned} R_{NN}(\tau) &= \frac{1}{2\pi} \int_{-W}^W S_o e^{j\omega\tau} d\omega = \frac{S_o}{2\pi} \frac{e^{j\omega\tau}}{j\tau} \Big|_{-W}^W = \frac{S_o}{\pi\tau} \frac{e^{jW\tau} - e^{-jW\tau}}{2j} \\ &= \frac{S_o}{\pi\tau} \sin W\tau = \frac{WS_o}{\pi} \sin(W\tau) \end{aligned} \quad (7.51)$$

The bandlimited noise and its power spectral density is shown in Fig. 7.4.

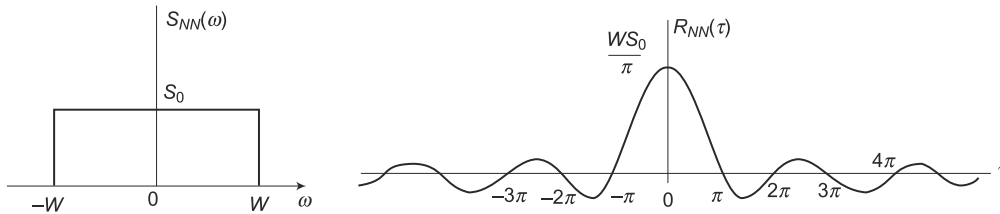


Fig. 7.4 Band limited white noise and its ACF

7.5.2 Coloured Noise

A noise process that is not white is called coloured noise. An example for coloured noise with autocorrelation function is

$$R_{NN}(\tau) = S_o e^{-\alpha|\tau|} \quad (7.52)$$

The power spectral density

$$S_{NN}(\omega) = \frac{2\alpha S_o}{\alpha^2 + \omega^2} \quad (7.53)$$

The autocorrelation function and its PSD is shown in Fig. 7.5.

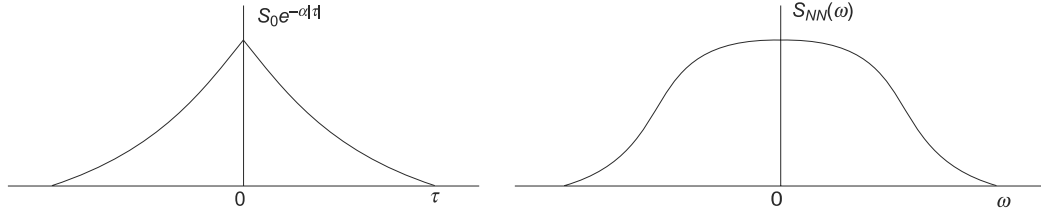


Fig. 7.5 ACF and PSD of coloured noise

REVIEW QUESTIONS



- 7.12 Define white noise.
- 7.13 What is PSD of white noise?
- 7.14 What are the properties of white noise?
- 7.15 What is coloured noise?
- 7.16 Draw the PSD of white noise and coloured noise.
- 7.17 Draw the PSD of bandlimited white noise.
- 7.18 What are the properties of white noise?

Solved Problem



7.23 For a wide-sense stationary random process, $Y(t) = X(t) + N(t)$, find the power spectral density, where $X(t)$ is the actual signal and $N(t)$ is a zero-mean noise process with variance σ_N^2 . $N(t)$ is independent of $X(t)$.

Solution Given: $Y(t) = X(t) + N(t)$

$$\begin{aligned} R_{YY}(\tau) &= E[Y(t)Y(t + \tau)] = E\{[X(t) + N(t)][X(t + \tau) + N(t + \tau)]\} \\ &= E[X(t)X(t + \tau)] + E[N(t)X(t + \tau)] + E[X(t)N(t + \tau)] + E[N(t)N(t + \tau)] \end{aligned}$$

Since $N(t)$ is independent of $X(t)$,

$$E[X(t)N(t + \tau)] = E[N(t)X(t + \tau)] = 0$$

$$\begin{aligned} R_{YY}(t) &= R_{XX}(\tau) + R_{NN}(\tau) \\ &= R_{XX}(\tau) + \sigma_N^2 \delta(\tau) \end{aligned}$$

The PSD is

$$F[R_{YY}(\tau)] = S_{YY}(\omega) = S_{XX}(\omega) + \sigma_N^2$$



PRODUCT DEVICE RESPONSE TO A RANDOM SIGNAL 7.6

Consider a product device with two inputs as shown in Fig. 7.6. The first input is a random waveform $X(t)$, which is either a signal or noise or the sum of signal and noise and the second input is cosine or sine wave carrier. The output of the product device will be

$$Y(t) = X(t) A \cos \omega t \quad (7.54)$$

where A and ω_0 are constants.

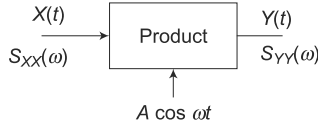


Fig. 7.6 A product device

The power spectral density can be obtained by applying Fourier transform to the autocorrelation function

The autocorrelation function of $Y(t)$ is

$$\begin{aligned} R_{YY}(t, t + \tau) &= E[Y(t)Y(t + \tau)] \\ &= E[A^2 X(t) \cos(\omega_0 t) X(t + \tau) \cos(\omega_0 t + \omega_0 \tau)] \\ &= \frac{A^2}{2} E\{X(t)X(t + \tau) [\cos(\omega_0 \tau) + \cos(2\omega_0 t + \omega_0 \tau)]\} \end{aligned} \quad (7.55)$$

If $X(t)$ is wide-sense stationary, we can write

$$E[X(t)X(t + \tau)] = R_{XX}(\tau)$$

which is the autocorrelation function of $X(t)$. Hence,

$$R_{YY}(t, t + \tau) = \frac{A^2}{2} R_{XX}(\tau) [\cos(\omega_0 \tau) + \cos(2\omega_0 t + \omega_0 \tau)] \quad (7.56)$$

However, the autocorrelation function of $Y(t)$ is not WSS since $R_{YY}(t, t + \tau)$ is a function of t . Hence, we take time average of $R_{YY}(t, t + \tau)$. Then

$$\begin{aligned} A[R_{YY}(t, t + \tau)] &= \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T A^2 R_{XX}(t, t + \tau) [\cos \omega_0 \tau + \cos(2\omega_0 t + \omega_0 \tau)] dt \\ &= \frac{A^2}{2} R_{XX}(\tau) \cos(\omega_0 \tau) \\ A[R_{YY}(t, t + \tau)] &= \frac{A^2}{2} R_{XX}(\tau) \cos(\omega_0 \tau) \end{aligned} \quad (7.57)$$

If $S_{XX}(\omega)$ is PSD of $X(t)$ then the PSD of $Y(t)$ is $F\{A[R_{YY}(t, t + \tau)]\}$

$$F\{A[R_{YY}(t, t + \tau)]\} = S_{YY}(\omega)$$

$$= F\left[\frac{A^2}{2} R_{XX}(\tau) \cos(\omega_0 \tau)\right]$$

$$= \frac{A^2}{2} \{F[R_{XX}(\tau) \cos(\omega_0 \tau)]\}$$

Since $F[R_{XX}(\tau)] = S_{XX}(\omega)$, we can apply Fourier transform properties to get $S_{YY}(\omega)$.

$$S_{YY}(\omega) = \frac{A^2}{2} [S_{XX}(\omega - \omega_0) + S_{XX}(\omega + \omega_0)] \quad (7.58)$$

REVIEW QUESTION



7.19 Find $R_{YY}(t)$ and hence, $S_{YY}(\omega)$ in terms of $S_{XX}(\omega)$ for the product device shown in Fig. 7.7, if $X(t)$ is WSS.

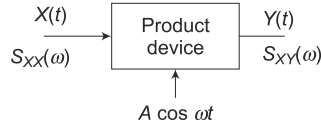


Fig. 7.7

Solved Problems



7.24 Find the PSD of a random process

$Z(t) = X(t) + Y(t)$ where $X(t)$ and $Y(t)$ are zero mean, independent random processes.

Solution Given: $Z(t) = X(t) + Y(t)$

The autocorrelation function of $Z(t)$ is

$$\begin{aligned} R_{ZZ}(t) &= E[Z(t)Z(t + \tau)] \\ &= E[(X(t) + Y(t)) (X(t + \tau) + Y(t + \tau))] \\ &= E[X(t)X(t + \tau)] + E[Y(t)X(t + \tau)] + E[X(t)Y(t + \tau)] + E[Y(t)Y(t + \tau)] \\ &= R_{XX}(\tau) + R_{YX}(\tau) + R_{XY}(\tau) + R_{YY}(\tau) \end{aligned}$$

Since $X(t)$ and $Y(t)$ are zero mean, independent random processes

$$R_{YX}(\tau) = R_{XY}(\tau) = 0.$$

Hence, $R_{ZZ}(\tau) = R_{XX}(\tau) + R_{YY}(\tau)$

Taking Fourier transform on both sides,

$$S_{ZZ}(\omega) = S_{XX}(\omega) + S_{YY}(\omega)$$

7.25 Find the PSD of random process $X(t)$ if $E[X(t)] = 1$ and $R_{XX}(\tau) = 1 + e^{-\alpha|\tau|}$

Solution $S_{XX}(\omega) = F[R_{XX}(\tau)]$

Given: $R_{XX}(t) = 1 + e^{-\alpha|t|}$

$$\begin{aligned}
 S_{XX}(\omega) &= \int_{-\infty}^{\infty} \{1 + e^{-\alpha|\tau|}\} e^{-j\omega\tau} d\tau \\
 &= \underbrace{\int_{-\infty}^{\infty} (1) e^{-j\omega\tau} d\tau}_{2\pi\delta(\omega)} + \int_{-\infty}^{\infty} e^{-\alpha|\tau|} e^{-j\omega\tau} d\tau \\
 &= 2\pi\delta(\omega) + \int_{-\infty}^{\infty} e^{-\alpha|\tau|} e^{-j\omega\tau} d\tau
 \end{aligned}$$

Inverse Fourier transform of $\delta(\omega)$ is

$$\begin{aligned}
 &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \delta(\omega) e^{j\omega t} d\omega \\
 &= \frac{1}{2\pi}
 \end{aligned}$$

$\Rightarrow$ Inverse Fourier transforms of $2\pi\delta(\omega) = 1$

Or

Fourier transforms of 1 is $2\pi\delta(\omega)$

Consider

$$\begin{aligned}
 \int_{-\infty}^{\infty} e^{-\alpha|\tau|} e^{-j\omega\tau} d\tau &= \int_{-\infty}^0 e^{\alpha\tau} e^{-j\omega\tau} d\tau + \int_0^{\infty} e^{-\alpha\tau} e^{-j\omega\tau} d\tau \\
 &= \frac{1}{\alpha - j\omega} e^{(\alpha - j\omega)\tau} \Big|_{-\infty}^0 + \frac{1}{-(\alpha + j\omega)} e^{-(\alpha + j\omega)\tau} \Big|_0^{\infty} \\
 &= \frac{1}{\alpha - j\omega} + \frac{1}{\alpha + j\omega} = \frac{2\alpha}{\alpha^2 + \omega^2}
 \end{aligned}$$

$$\Rightarrow S_{XX}(\omega) = 2\pi\delta(\omega) + \frac{2\alpha}{\alpha^2 + \omega^2}$$



Practice Problem

7.15 A triangular pulse shown in Fig. 7.8 is given by

$$x(t) = \Delta\left(\frac{t}{\tau}\right)$$

- Find the power spectral density.
- Find the autocorrelation function corresponding to the power spectral density

$$S_{XX}(\omega) = \Delta\left(\frac{\omega}{W}\right)$$

Hint: Use duality property of Fourier transform)

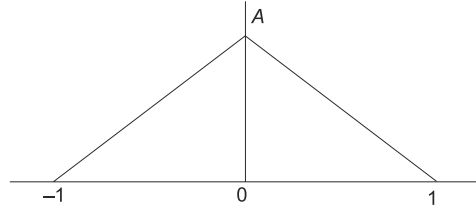


Fig. 7.8

$$\left(\begin{array}{l} \text{Ans: (a) } S_{XX}(\omega) = A\tau \left(\frac{\sin \frac{\omega\tau}{2}}{\frac{\omega\tau}{2}} \right)^2 \\ \text{(b) } R_{XX}(\tau) = AW \left(\frac{\sin \frac{W\tau}{2}}{\frac{W\tau}{2}} \right)^2 \end{array} \right)$$

Solved Problems

7.26 Find the power spectral density if the autocorrelation function of a WSS process is $R_{XX}(\tau) = ke^{-k|\tau|}$

Solution

$$\begin{aligned}
 S_{XX}(\omega) &= F[R_{XX}(\tau)] \\
 &= F[k e^{-k|\tau|}] = kF[e^{-k|\tau|}] \\
 &= k \int_{-\infty}^{\infty} e^{-k|\tau|} e^{-j\omega\tau} d\tau = k \left[\int_{-\infty}^0 e^{k\tau} e^{-j\omega\tau} d\tau + \int_0^{\infty} e^{-k\tau} e^{-j\omega\tau} d\tau \right] \\
 &= k \left[\frac{1}{k-j\omega} e^{(k-j\omega)\tau} \Big|_{-\infty}^0 + \frac{1}{-(k+j\omega)} e^{-(k+j\omega)\tau} \Big|_0^{\infty} \right] \\
 &= k \left[\frac{1}{k-j\omega} (1) - \frac{1}{k+j\omega} (-1) \right] \\
 &= k \frac{2k}{k^2 + \omega^2} = \frac{2k^2}{k^2 + \omega^2} = \frac{2}{1 + \left(\frac{\omega}{k}\right)^2}
 \end{aligned}$$

7.27 Find the cross-power spectral density if (a) $R_{XY}(\tau) = \frac{A^2}{2} \sin \omega_0 \tau$, and (b) $R_{XY}(\tau) = \frac{A^2}{2} \cos \omega_0 \tau$.

Solution

(a) Given $R_{XY}(\tau) = \frac{A^2}{2} \sin \omega_0 \tau$

$$\begin{aligned}
 S_{XY}(\omega) &= F[R_{XY}(\tau)] = F\left[\frac{A^2}{2} \sin \omega_0 \tau\right] \\
 &= \frac{A^2}{2} F[\sin \omega_0 \tau] \\
 &= \frac{A^2}{2} F\left[\frac{e^{j\omega_0 \tau} - e^{-j\omega_0 \tau}}{2j}\right] \\
 &= \frac{A^2}{4j} [F(e^{j\omega_0 \tau}) - F(e^{-j\omega_0 \tau})] \\
 &= \frac{A^2}{4j} [2\pi \delta(\omega - \omega_0) - 2\pi \delta(\omega + \omega_0)] \\
 &= \frac{jA^2}{2} [\pi \delta(\omega + \omega_0) - \pi \delta(\omega - \omega_0)]
 \end{aligned}$$

$$\begin{aligned}
 F^{-1}[\delta(\omega - \omega_0)] &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \delta(\omega - \omega_0) e^{j\omega\tau} d\omega \\
 &= \frac{e^{j\omega_0 \tau}}{2\pi} \\
 \Rightarrow F[e^{j\omega_0 \tau}] &= 2\pi \delta(\omega - \omega_0)
 \end{aligned}$$

$$\begin{aligned}
 (b) \quad R_{XX}(\tau) &= \frac{A^2}{2} \cos(\omega_0 \tau) \\
 S_{XX}(\omega) &= F \left[\frac{A^2}{2} \cos \omega_0 \tau \right] \\
 &= \frac{A^2}{2} F \left[\frac{e^{j\omega_0 \tau} + e^{-j\omega_0 \tau}}{2} \right] \\
 &= \frac{A^2}{4} [F(e^{j\omega_0 \tau}) + F(e^{-j\omega_0 \tau})] \\
 &= \frac{A^2}{4} [2\pi \delta(\omega - \omega_0) + 2\pi \delta(\omega + \omega_0)] \\
 &= \frac{A^2 \pi}{2} [\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]
 \end{aligned}$$



BANDWIDTH OF THE POWER SPECTRAL DENSITY 7.7

The bandwidth of a random process is measured by considering the positive frequency range over which the PSD is non-zero. A random process is said to be a lowpass process, if its spectral components are concentrated around $\omega = 0$. That is, for a lowpass process most of the power is near dc. The bandwidth of such a process is the lowest frequency for which the PSD is non-zero. The PSD of a lowpass process is shown in Fig. 7.9.

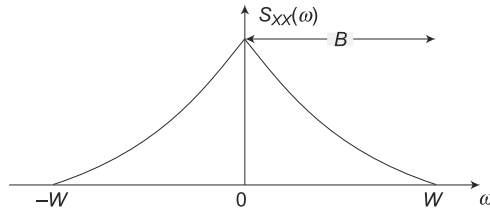


Fig. 7.9 PSD of a lowpass process

A process is said to be bandpass process for which the power is centred around some non-zero frequency $\omega = \omega_0$. For a bandpass process, the bandwidth is given by $B = B_R - B_L$, where B_R is the smallest value of B such that $S_{XX}(\omega) = 0$ for all $B < \omega$, and B_L be the largest value of B such that $S_{XX}(\omega) = 0$ for all $0 < \omega < B$.

The PSD of a bandpass process is shown in Fig. 7.10.

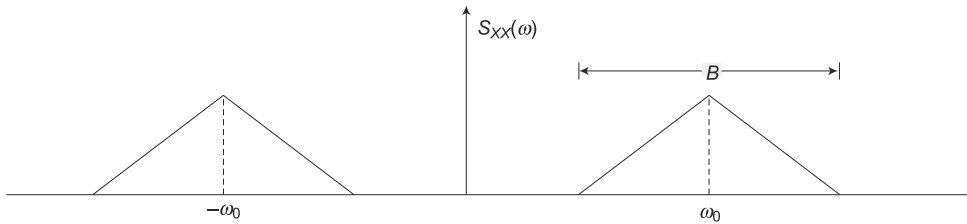


Fig. 7.10 PSD of a bandpass process

7.7.1 rms Bandwidth

The rms bandwidth is the spread of normalized power spectral density. The spread of the power spectral density is equal to

$$\int_{-\infty}^{\infty} \omega^2 S_{XX}(\omega) d\omega$$

The area enclosed by the PSD is given by

$$A = \int_{-\infty}^{\infty} S_{XX}(\omega) d\omega$$

Therefore, the rms bandwidth is equal to

$$\text{rms bandwidth} = \frac{\int_{-\infty}^{\infty} \omega^2 S_{XX}(\omega) d\omega}{\int_{-\infty}^{\infty} S_{XX}(\omega) d\omega} \quad (7.59)$$

Solved Problems



7.28 Find the rms bandwidth of the power density spectrum

$$S_{XX}(\omega) = \begin{cases} \frac{A}{1 + \left(\frac{\omega}{W}\right)^2} & |\omega| < kW \\ 0 & |\omega| > kW \end{cases}$$

Solution

$$\text{rms bandwidth} = \frac{\int_{-\infty}^{\infty} \omega^2 S_{XX}(\omega) d\omega}{\int_{-\infty}^{\infty} S_{XX}(\omega) d\omega} = \frac{\int_{-kW}^{kW} \frac{A\omega^2}{1 + \left(\frac{\omega}{W}\right)^2} d\omega}{\int_{-kW}^{kW} \frac{A}{1 + \left(\frac{\omega}{W}\right)^2} d\omega}$$

Let $N = \int_{-kW}^{kW} \frac{A\omega^2}{1 + \left(\frac{\omega}{W}\right)^2} d\omega$

and $D = \int_{-kW}^{kW} \frac{A}{1 + \left(\frac{\omega}{W}\right)^2} d\omega$

Let $\frac{\omega}{W} = p$

$\Rightarrow d\omega = (dp)W$

$\omega^2 = p^2 W^2$

If $\omega = kW, p = k$

and $\omega = -kW, p = -k$

$$D = \int_{-kW}^{kW} \frac{A\omega^2}{1 + \left(\frac{\omega}{W}\right)^2} d\omega$$

$$N = \int_{-k}^k \frac{Ap^2W^2}{1 + p^2} (Wdp)$$

$$= AW^3 \int_{-k}^k \frac{p^2}{1 + p^2} dp$$

$$= AW^3 \left[\int_{-k}^k \left(1 - \frac{1}{1 + p^2} \right) dp \right]$$

$$= AW^3 \left[p \Big|_{-k}^k - \tan^{-1}(p) \Big|_{-k}^k \right]$$

$$= AW^3 \left[2k - 2 \tan^{-1} k \right]$$

$$D = \int_{-kW}^{kW} \frac{A}{1 + \left(\frac{\omega}{W}\right)^2} d\omega = \int_{-k}^k \frac{AW}{1 + p^2} dp = AW \tan^{-1} p \Big|_{-k}^k$$

$$= 2AW \tan^{-1} k$$

$$\text{rms bandwidth} = \frac{N}{D} = \frac{AW^3(2k - 2 \tan^{-1} k)}{2AW \tan^{-1} k} = W^2 \left[\frac{k - \tan^{-1} k}{\tan^{-1} k} \right]$$

7.29 Find rms bandwidth for the PSD $S_{XX}(\omega) = \frac{1}{\left[1 + \left(\frac{\omega}{W}\right)^2\right]^3}$.

Solution

$$\text{rms bandwidth} = \frac{\int_{-\infty}^{\infty} \omega^2 S_{XX}(\omega) d\omega}{\int_{-\infty}^{\infty} S_{XX}(\omega) d\omega}$$

$$\int_{-\infty}^{\infty} \omega^2 S_{XX}(\omega) d\omega = \int_{-\infty}^{\infty} \frac{\omega^2}{\left[1 + \left(\frac{\omega}{W}\right)^2\right]^3} d\omega$$

Let	$\frac{\omega}{W} = p$
Then	$d\omega = Wdp$

$$\int_{-\infty}^{\infty} \frac{W^2 p^2}{(1+p^2)^3} W dp = W^3 \int_{-\infty}^{\infty} \frac{p^2}{(1+p^2)} dp$$

$$= W^3 \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\tan^2 \theta \sec^2 \theta d\theta}{(1+\tan^2 \theta)}$$

$$= W^3 \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin^2 \theta \cos^2 \theta d\theta$$

$$= \frac{W^3}{4} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin^2 2\theta d\theta = \frac{W^3}{4} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left(\frac{1-\cos 2\theta}{2} \right) d\theta = \frac{W^3}{8} \pi$$

$$\int_{-\infty}^{\infty} S_{XX}(\omega) d\omega = \int_{-\infty}^{\infty} \frac{1}{\left[1 + \left(\frac{\omega}{W} \right)^2 \right]^3} d\omega = \int_{-\infty}^{\infty} \frac{W dp}{(1+p^2)^3}$$

$$= W \int_{-\infty}^{\infty} \frac{dp}{(1+p^2)^3}$$

$$= W \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\sec^2 \theta d\theta}{(1+\tan^2 \theta)^3}$$

$$= W \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos^4 \theta d\theta$$

$$= W \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left(\frac{3}{8} + \frac{1}{8} \cos 4\theta + \frac{1}{2} \cos 2\theta \right) d\theta$$

$$= W \left(\frac{3}{8} \pi \right)$$

$$\text{rms bandwidth} = \frac{(W^3 \pi / 8)}{(2W \pi / 8)} = \frac{W^2}{3}$$

Let $p = \tan \theta$

$$dp = \sec^2 \theta d\theta$$

when $p = \infty, \theta = \frac{\pi}{2}$

$$p = -\infty, \theta = -\frac{\pi}{2}$$

$$\cos^4 \theta = \frac{1}{4} [1 + \cos 2\theta]^2$$

$$= \frac{1}{4} [1 + \cos^2 2\theta + 2 \cos 2\theta]$$

$$= \frac{3}{8} + \frac{1}{8} \cos 4\theta + \frac{1}{2} \cos 2\theta$$

7.30 Find rms bandwidth and average power for the power spectrum

$$S_{XX} = \begin{cases} 4 - \frac{\omega^2}{9} & \text{for } |\omega| < 6 \\ 0 & \text{otherwise} \end{cases}$$

Solution

$$\begin{aligned} \text{rms bandwidth} &= \frac{\int_{-\infty}^{\infty} \omega^2 S_{XX}(\omega) d\omega}{\int_{-\infty}^{\infty} S_{XX}(\omega) d\omega} \\ \int_{-\infty}^{\infty} \omega^2 S_{XX}(\omega) d\omega &= \int_{-6}^6 \omega^2 \left(4 - \frac{\omega^2}{9} \right) d\omega \\ &= \frac{4}{3} (2 \times 6^3) - \frac{1}{45} (2 \times 6^5) \\ &= 2 \times 6^3 \left(\frac{4}{3} - \frac{36}{45} \right) = \frac{8}{15} (2 \times 6^3) \\ \int_{-\infty}^{\infty} S_{XX}(\omega) d\omega &= \int_{-6}^6 \left(4 - \frac{\omega^2}{9} \right) d\omega = 4(12) - \frac{1}{27} (2 \times 6^3) = 32 \\ \text{rms bandwidth} &= \frac{\frac{8}{15} (2 \times 6^3)}{32} = \frac{36}{5} \end{aligned}$$

Average power

$$P_{XX} = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_{XX}(\omega) d\omega = \frac{32}{2\pi} = \frac{16}{\pi}$$

7.31 Prove that the rms bandwidth can be obtained using the relation

$$-\frac{1}{R_{XX}(0)} \left. \frac{d^2 R_{XX}(\tau)}{d\tau^2} \right|_{\tau=0} \quad (7.60)$$

Solution We know

$$\text{rms bandwidth} = \frac{\int_{-\infty}^{\infty} \omega^2 S_{XX}(\omega) d\omega}{\int_{-\infty}^{\infty} S_{XX}(\omega) d\omega}$$

Also,
$$R_{XX}(\tau) = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_{XX}(\omega) e^{j\omega\tau} d\omega$$

Differentiating the above equation with respect to τ ,

$$\frac{dR_{XX}(\tau)}{d\tau} = \frac{1}{2\pi} \int_{-\infty}^{\infty} (j\omega) S_{XX}(\omega) e^{j\omega\tau} d\omega$$

Similarly,

$$\frac{d^2 R_{XX}(\tau)}{d\tau^2} = \frac{1}{2\pi} \int_{-\infty}^{\infty} (-\omega^2) S_{XX}(\omega) e^{j\omega\tau} d\omega$$

$$\left. \frac{d^2 R_{XX}(\tau)}{d\tau^2} \right|_{\tau=0} = \frac{1}{2\pi} \int_{-\infty}^{\infty} (-\omega^2) S_{XX}(\omega) d\omega$$

$$R_{XX}(0) = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_{XX}(\omega) d\omega$$

$$\Rightarrow \frac{-\left. \frac{d^2 R_{XX}(\tau)}{d\tau^2} \right|_{\tau=0}}{R_{XX}(0)} = \frac{\int_{-\infty}^{\infty} \omega^2 S_{XX}(\omega) d\omega}{\int_{-\infty}^{\infty} S_{XX}(\omega) d\omega}$$

which is equal to the rms bandwidth.

Hence,

$$\text{rms bandwidth} = -\frac{1}{R_{XX}(0)} \left. \frac{d^2 R_{XX}(\tau)}{d\tau^2} \right|_{\tau=0}$$



Practice Problems

7.16 A random process has a power spectrum

$$S_{XX}(\omega) = \frac{6|\omega|}{(W^2 + \omega^2)^2}$$

which can be treated as a bandpass process. Find the mean frequency and rms bandwidth

$$\left(\text{Ans. } \frac{\pi W}{2}, \infty \right)$$

7.17 Find the rms bandwidth of the power spectrum

$$S_{XX}(\omega) = \frac{\omega^2}{\left[1 + \left(\frac{\omega}{W} \right)^2 \right]^4} \quad (\text{Ans. } W^2)$$



PSD OF DISCRETE-TIME RANDOM PROCESS 7.8

The Discrete-Time Fourier Transforms (DTFT) of a deterministic signal $x(n)$ is given by

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x(n)e^{-j\omega n} \quad (7.61)$$

The inverse Fourier transform of $X(e^{j\omega})$ is given by

$$x(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega}) e^{j\omega n} d\omega \quad (7.62)$$

The DTFT exists for a sequence $x(n)$ if it is absolutely summable. That is,

$$\sum_{n=-\infty}^{\infty} |x(n)| < \infty \quad (7.63)$$

An ensemble of random process $X(n)$ exists for all values of n . Hence, DTFT does not exist for a random process. Therefore, to obtain spectral information of a random process, we transform autocorrelation function into frequency domain. That is, the PSD of a discrete-time random process is

$$S_{XX}(\Omega) = \sum_{m=-\infty}^{\infty} R_{XX}(m) e^{-j\Omega m} \quad (7.64)$$

Since $e^{-j\Omega m}$ is periodic with 2π , we can prove that $S_{XX}(\Omega)$ is also periodic with a period 2π .

$$\begin{aligned} S_{XX}(\Omega + 2\pi) &= \sum_{m=-\infty}^{\infty} R_{XX}(m) e^{-j(\Omega + 2\pi)m} \\ &= \sum_{m=-\infty}^{\infty} R_{XX}(m) e^{-j\Omega m} \\ &= S_{XX}(\Omega) \end{aligned} \quad (7.65)$$

Therefore, it is sufficient to define $S_{XX}(\Omega)$ only in the range $(-\pi, \pi)$. The autocorrelation function is given by

$$R_{XX}(m) = \frac{1}{2\pi} \int_{-\pi}^{\pi} S_{XX}(\Omega) e^{j\Omega m} d\Omega \quad (7.66)$$

The properties of $S_{XX}(\Omega)$ are

- (i) $S_{XX}(\Omega)$ is periodic with period 2π .
- (ii) $S_{XX}(-\Omega) = S_{XX}(\Omega)$ which means that $S_{XX}(\Omega)$ is an even function

$$S_{XX}(\Omega) = \sum_{m=-\infty}^{\infty} R_{XX}(m) e^{-j\Omega m}$$

$$S_{XX}(-\Omega) = \sum_{m=-\infty}^{\infty} R_{XX}(m) e^{-j(-\Omega)m} = \sum_{m=-\infty}^{\infty} R_{XX}(m) e^{j\Omega m}$$

$$= \sum_{m=-\infty}^{\infty} R_{XX}(m) e^{-j\Omega(-m)}$$

Let $m = -p$

$$= \sum_{p=-\infty}^{\infty} R_{XX}(-p) e^{-j\Omega p}$$

Since $R_{XX}(-p) = R_{XX}(p)$

$$\begin{aligned} S_{XX}(-\Omega) &= \sum_{p=-\infty}^{\infty} R_{XX}(p) e^{-j\Omega p} \\ &= S_{XX}(\Omega) \end{aligned}$$

- (i) $S_{XX}(\Omega)$ is real, which means that $S_{XX}(\Omega) > 0$.
- (ii) The average power of the process is

$$E[X^2(n)] = R_{XX}(0) = \frac{1}{2\pi} \int_{-\pi}^{\pi} S_{XX}(\Omega) d\Omega \quad (7.67)$$

REVIEW QUESTIONS



- 20. Define PSD of a discrete-time random process.
- 21. Prove that $S_{XX}(\Omega)$ is a periodic with period 2π .

Solved Problems



7.32 A random process $X(n)$ has auto-correlation function $R_{XX}(m) = a^{|m|}$. Find the power spectral density.

Solution

$$\begin{aligned} S_{XX}(\Omega) &= \sum_{m=-\infty}^{\infty} R_{XX}(m) e^{-j\Omega m} \\ &= \sum_{m=-\infty}^{-1} R_{XX}(m) e^{-j\Omega m} + \sum_{m=0}^{\infty} R_{XX}(m) e^{-j\Omega m} \\ &= \sum_{m=-\infty}^{-1} a^{-m} e^{-j\Omega m} + \sum_{m=0}^{\infty} a^m e^{-j\Omega m} \\ &= \sum_{m=1}^{\infty} a^m e^{j\Omega m} + \sum_{m=0}^{\infty} a^m e^{-j\Omega m} \\ &= \frac{ae^{j\Omega}}{1 - ae^{j\Omega}} + \frac{1}{1 - ae^{-j\Omega}} \end{aligned}$$

$$\begin{aligned}
&= \frac{ae^{j\Omega} - a^2 + 1 - ae^{-j\Omega}}{1 - ae^{j\Omega} - ae^{-j\Omega} - a^2} \\
&= \frac{1 - a^2}{1 - 2a \cos \Omega + a^2}
\end{aligned}$$

7.33 The ACF of a random process $X(n)$ is given by

$$R_{XX}(m) = \begin{cases} \sigma_x^2 & m = 0 \\ \frac{4\sigma_x^2}{m^2\pi^2} & m = \text{odd} \\ 0 & m = \text{even} \end{cases}$$

Solution

$$\begin{aligned}
S_{XX}(\Omega) &= \sum_{m=-\infty}^{\infty} R_{XX}(m) e^{-j\Omega m} \\
&= \dots - \frac{4\sigma_x^2}{3^2\pi^2} e^{j3\Omega} + \frac{4\sigma_x^2}{\pi^2} e^{j\Omega} + \sigma_x^2 + \frac{4\sigma_x^2}{\pi^2} e^{-j\Omega} + \frac{4\sigma_x^2}{3^2\pi^2} e^{-3\Omega} + \dots \\
&= \sigma_x^2 + \frac{4\sigma_x^2}{\pi^2} \left[2 \cos \Omega + \frac{2}{3^2} \cos 3\Omega + \dots \right] \\
&= \sigma_x^2 + \frac{8\sigma_x^2}{\pi^2} \left[\cos \Omega + \frac{1}{3^2} \cos 3\Omega + \frac{1}{5^2} \cos 5\Omega + \dots \right]
\end{aligned}$$

7.34 Find $S_{XX}(\Omega)$, if $R_{XX}(k) = 9(\alpha)^{|k|} + 25(\beta)^{|k|}$, $\alpha < 1$, $\beta < 1$.

Solution Given:

$$\begin{aligned}
R_{XX}(k) &= 9(\alpha)^{|k|} + 25(\beta)^{|k|} \\
S_{XX}(\Omega) &= F[R_{XX}(k)] \\
&= F[9(\alpha)^{|k|} + 25(\beta)^{|k|}] \\
F[(\alpha)^{|k|}] &= \sum_{k=\alpha}^{\infty} (\alpha)^{|k|} e^{-j\Omega k} = \sum_{k=-\alpha}^{-1} \alpha^{-k} e^{-j\Omega k} + \sum_{k=0}^{\infty} \alpha^k e^{-j\Omega k} \\
&= \sum_{k=1}^{\infty} (\alpha)^k e^{j\Omega k} + \sum_{k=0}^{\infty} (\alpha e^{-j\Omega})^k \\
&= \frac{\alpha e^{j\Omega}}{1 - \alpha e^{j\Omega}} + \frac{1}{1 - \alpha e^{-j\Omega}} = \frac{1 - \alpha^2}{1 - 2\alpha \cos \Omega + \alpha^2}
\end{aligned}$$

$$S_{XX}(\Omega) = 9 \left[\frac{1 - \alpha^2}{1 - 2\alpha \cos \Omega + \alpha^2} \right] + 25 \left[\frac{1 - \beta^2}{1 - 2\beta \cos \Omega + \beta^2} \right]$$

7.35 A discrete-time random process $Y(n)$ is defined by $Y(n) = X(n) + \alpha X(n-1)$, where $X(n)$ is the white noise process with zero mean and variance σ_X^2 . Find the mean, autocorrelation and the PSD of the random process $Y(n)$.

Solution Given:

$$Y(n) = X(n) + \alpha X(n-1)$$

$$E[X(n)] = 0$$

$$\text{Var}[X(n)] = \sigma_X^2$$

Since $X(n)$ is a white-noise process,

$$R_{XX}(m) = \begin{cases} \sigma_X^2, & m = 0 \\ 0 & m \neq 0 \end{cases}$$

The mean of $Y(n)$ is

$$\begin{aligned} E[Y(n)] &= E[X(n) + \alpha X(n-1)] \\ &= E[X(n)] + \alpha E[X(n-1)] \\ &= 0 \end{aligned}$$

The variance of $Y(n)$ is

$$\begin{aligned} \sigma_Y^2 &= \text{Var}[X(n) + \alpha X(n-1)] \\ &= \sigma_X^2 + \alpha^2 \sigma_X^2 = \sigma_X^2 (1 + \alpha^2) \end{aligned}$$

The autocorrelation

$$\begin{aligned} R_{YY}(m) &= E[Y(n)Y(n+m)] = E\{[X(n) + \alpha X(n-1)][X(n+m) + \alpha X(n-1+m)]\} \\ &= E[X(n)X(n+m)] + \alpha E[X(n-1)X(n+m)] + \alpha E[X(n)X(n-1+m)] \\ &\quad + \alpha^2 E[X(n-1)X(n-1+m)] \end{aligned}$$

$$\text{For } m = 0, R_{YY}(m) = \sigma_X^2 + \alpha^2 \sigma_X^2$$

$$\text{For } m = 1, R_{YY}(m) = \alpha R_{XX}(m) = \alpha \sigma_X^2$$

$$\text{For } m = -1, R_{YY}(m) = \alpha R_{XX}(m) = \alpha \sigma_X^2$$

$$R_{XX}(m) = \begin{cases} \sigma_X^2 (1 + \alpha^2) & \text{for } m = 0 \\ \alpha \sigma_X^2 & \text{for } m = \pm 1 \\ 0 & \text{otherwise} \end{cases}$$

The PSD is given by

$$\begin{aligned} S_{XX}(\Omega) &= \sum_{m=-\infty}^{\infty} R_{XX}(m) e^{-j\Omega m} \\ &= \alpha \sigma_X^2 e^{j\Omega m} + \sigma_X^2 (1 + \alpha^2) + \alpha \sigma_X^2 e^{-j\Omega m} \\ &= \sigma_X^2 [(1 + \alpha) + 2\alpha \cos \Omega] \end{aligned}$$

7.36 Consider a discrete-time random process $X(n) = \cos(\Omega_0 n + \theta)$ where θ is a uniformly distributed random variable in the interval $(0, 2\pi)$. Find $S_{XX}(\Omega)$.

Solution Given $X(n) = \cos(\Omega_0 n + \theta)$

$$\begin{aligned} R_{XX}(n, n+m) &= E[X(n) X(n+m)] \\ &= E[\cos(\Omega_0 n + \theta) \cos(\Omega_0 n + \Omega_0 m + \theta)] \\ &= \frac{1}{2} E[\cos(2\Omega_0 n + \Omega_0 m + 2\theta) + \cos(\Omega_0 m)] \end{aligned}$$

Since θ is uniformly distributed in the interval $(0, 2\pi)$,

$$\begin{aligned} E[\cos(2\Omega_0 n + \Omega_0 m + 2\theta)] &= \int_0^{2\pi} \cos(2\Omega_0 n + \Omega_0 m + 2\theta) \frac{1}{2\pi} d\theta \\ &= \frac{1}{2\pi} \left[\frac{\sin(2\Omega_0 n + \Omega_0 m + 2\theta)}{2} \right]_0^{2\pi} \\ &= 0 \end{aligned}$$

$$\Rightarrow R_{XX}(n, n+m) = \frac{1}{2} \cos(\Omega_0 m)$$

$$\begin{aligned} S_{XX}(\Omega) &= F \left[\frac{1}{2} \cos \Omega_0 m \right] \\ &= \frac{1}{2} F [\cos \Omega_0 m] \\ &= \frac{1}{2} [\pi \delta(\Omega + \Omega_0) + \pi \delta(\Omega - \Omega_0)] \end{aligned}$$



Practice Problems

7.18 If ACF of white noise $N(t)$ is $R_{NN}(KT) = \begin{cases} \sigma_N^2 & K=0 \\ 0 & K \neq 0 \end{cases}$, then find PSD. (Ans. σ_N^2)

7.19 Let $Y(n) = X(n) - X(n-T)$. If $X(n)$ is a zero mean WSS, find $R_{YY}(m)$, $S_{YY}(\Omega)$.

$$\begin{aligned} \text{(Ans. } R_{YY}(m) &= 2R_{XX}(m) - R_{XX}(m+T) - R_{XX}(m-T) \\ S_{YY}(\Omega) &= 2S_{XX}(\Omega) (1 - \cos \Omega T) \end{aligned}$$



POWER SPECTRUM OF COMPLEX PROCESS 7.9

Consider a WSS complex process $Z(t)$ with autocorrelation function $R_{ZZ}(\tau)$. The power spectrum of such a process is defined as

$$S_{ZZ}(\omega) = \int_{-\infty}^{\infty} R_{ZZ}(\tau) e^{-j\omega\tau} d\tau \quad (7.68)$$

And the autocorrelation function is given by

$$R_{zz}(\tau) = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_{ZZ}(\omega) e^{j\omega\tau} d\omega \quad (7.69)$$

If $Z_m(t)$ and $Z_n(t)$ are two jointly wide-sense stationary complex processes then their cross-power density spectrum and cross-correlation functions are

$$S_{z_m z_n}(\omega) = \int_{-\infty}^{\infty} R_{z_m z_n}(\tau) e^{-j\omega\tau} d\tau \quad (7.70)$$

$$R_{z_m z_n} = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_{z_m z_n}(\omega) e^{j\omega\tau} d\omega \quad (7.71)$$

Solved Problems



7.37 Consider a random process $Z(t) = \sum_{i=1}^N z_i \cdot e^{j\omega_i t}$, z_i , $i = 1, \dots, N$ be N complex zero-mean uncorrelated random variables with variance $\sigma_{z_i}^2$, and ω_i are the angular frequencies. Find the power spectral density.

Solution Given:

$$\begin{aligned} Z(t) &= \sum_{i=1}^N z_i \cdot e^{j\omega_i t} \\ E[z_i(t)] &= 0; E[z_i z_j] = 0, i \neq j \\ &= \sigma_{z_i}^2 \quad i = j \\ R_{ZZ}(\tau) &= E[Z^*(t) Z(t + \tau)] \\ &= E \left[\sum_{i=1}^N z_i^* e^{-j\omega_i t} \sum_{j=1}^N z_j e^{j\omega_j(t + \tau)} \right] \\ &= \sum_{i=1}^N \sum_{j=1}^N E[z_i^* z_j] e^{j(\omega_j - \omega_i)t} e^{j\omega_i \tau} \\ &= \sum_{i=1}^N \sigma_{z_i}^2 e^{j\omega_i \tau} \\ S_{ZZ}(\omega) &= F \left[\sum_{i=1}^N \sigma_{z_i}^2 e^{j\omega_i \tau} \right] \\ &= \sum_{i=1}^N \sigma_{z_i}^2 2\pi \delta(\omega - \omega_i) = 2\pi \sum_{i=1}^N \sigma_{z_i}^2 \delta(\omega - \omega_i) \end{aligned}$$

7.38 Let $X(t)$ and $Y(t)$ be statistically independent processes with power spectrum

$$S_{XX}(\omega) = 2\delta(\omega) + \frac{1}{1 + \left(\frac{\omega}{9}\right)^2} \text{ and } S_{YY}(\omega) = \frac{6}{1 + \left(\frac{\omega}{3}\right)^2}$$

A complex process $Z(t) = [X(t) + jY(t)]e^{j\omega_0 t}$ is defined where ω_0 is constant much larger than 9. Find ACF and PSD.

Solution Given:

$$S_{XX}(\omega) = 2\delta(\omega) + \frac{1}{1 + \left(\frac{\omega}{9}\right)^2}$$

We know

$$F[1] = 2\pi\delta(\omega)$$

$$\Rightarrow F^{-1}[2\delta(\omega)] = \frac{1}{\pi}$$

$$\text{Also, } F[e^{-\alpha|\tau|}] = \frac{2\alpha}{\alpha^2 + \omega^2}$$

from which

$$F^{-1}\left[\frac{1}{1 + \left(\frac{\omega}{9}\right)^2}\right] = \frac{9}{2}e^{-9|\tau|} \text{ and } F^{-1}\left[\frac{6}{1 + \left(\frac{\omega}{3}\right)^2}\right] = 9e^{-3|\tau|}$$

$$\Rightarrow R_{XX}(\tau) = \frac{1}{\pi} + \frac{9}{2}e^{-9|\tau|} \text{ and } R_{YY}(\tau) = 9e^{-3|\tau|}$$

The complex process $Z(t) = [X(t) + jY(t)]e^{j\omega_0 t}$

$$\begin{aligned} R_{ZZ}(\tau) &= E[Z^*(t)Z(t+\tau)] \\ &= E\{[X(t) - jY(t)]e^{-j\omega_0 t}[X(t+\tau) + jY(t+\tau)]e^{j\omega_0(t+\tau)}\} \\ &= \{E[X(t)X(t+\tau)] - jE[Y(t)X(t+\tau)] + jE[X(t)Y(t+\tau)] + E[Y(t)Y(t+\tau)]\}e^{j\omega_0\tau} \end{aligned}$$

Since $X(t)$ and $Y(t)$ are independent,

$$\begin{aligned} R_{ZZ}(\tau) &= [R_{XX}(\tau) + R_{YY}(\tau)]e^{j\omega_0\tau} \\ &= \left[\frac{1}{\pi} + \frac{9}{2}e^{-9|\tau|} + 9e^{-3|\tau|}\right]e^{j\omega_0\tau} \end{aligned}$$

The PSD is given by

$$\begin{aligned} S_{ZZ}(\omega) &= 2 \delta(\omega - \omega_0) + \frac{9}{2} \frac{18}{81 + (\omega - \omega_0)^2} + 9 \frac{6}{9 + (\omega - \omega_0)^2} \Big] \\ &= 2 \delta(\omega - \omega_0) + \frac{81}{81 + (\omega - \omega_0)^2} + \frac{54}{9 + (\omega - \omega_0)^2} \end{aligned}$$

7.39 Find the auto-correlation function if the power spectral density of a stationary random process is given by

$$S_{XX}(\omega) = \begin{cases} A & -k < \omega < k \\ 0 & \text{otherwise} \end{cases}$$

Solution $S_{XX}(\omega) = \begin{cases} A & -k < \omega < k \\ 0 & \text{otherwise} \end{cases}$

$$\begin{aligned} R_{XX}(\tau) &= F^{-1}[S_{XX}(\omega)] \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} S_{XX}(\omega) e^{j\omega\tau} d\omega = \frac{1}{2\pi} \int_{-k}^k A e^{j\omega\tau} d\omega \\ &= \frac{A}{2\pi} \left[\frac{e^{j\omega\tau}}{j\tau} \right]_{-k}^k = \frac{A}{2\pi} \left(\frac{e^{j\tau k} - e^{-j\tau k}}{j\tau} \right) \\ &= \frac{Ak}{\pi} \frac{e^{j\tau k} - e^{-j\tau k}}{2j\tau k} \\ &= \frac{Ak}{\pi} \frac{\sin(k\tau)}{k\tau} \end{aligned}$$

7.40 The cross-spectral density of two random process $X(t)$ and $Y(t)$ is given by

$$S_{XY}(\omega) = \frac{1}{-\omega^2 + j4\omega + 4}$$

Find the cross-correlation function.

Solution $S_{XY}(\omega) = \frac{1}{-\omega^2 + j4\omega + 4} = \frac{1}{(2 + j\omega)^2}$

$$\begin{aligned} R_{XY}(\tau) &= F^{-1}[S_{XY}(\omega)] \\ &= F^{-1} \left[\frac{1}{(2 + j\omega)^2} \right] \\ &= \tau e^{-2\tau} u(\tau) \end{aligned}$$

$$R_{XY}(\tau) = \tau e^{-2\tau} \text{ for } \tau > 0$$

7.41 Find the average power of the random process with the following power spectrum

$$S_{XX}(\omega) = \frac{\omega^2 + 14}{(\omega^2 + 25)(\omega^2 + 36)}$$

Solution $S_{XX}(\omega) = \frac{\omega^2 + 14}{(\omega^2 + 25)(\omega^2 + 36)}$

$$\begin{aligned} R_{XX}(\tau) &= F^{-1}[S_{XX}(\omega)] \\ &= F^{-1}\left[\frac{\omega^2 + 14}{(\omega^2 + 25)(\omega^2 + 36)}\right] = F^{-1}\left[\frac{-1}{\omega^2 + 25} + \frac{2}{\omega^2 + 36}\right] \\ &= \frac{-1}{10}\left[F^{-1}\left(\frac{10}{\omega^2 + 25}\right)\right] + \frac{2}{12}F^{-1}\left(\frac{12}{\omega^2 + 36}\right) \\ &= \frac{-1}{10}e^{-5|\tau|} + \frac{1}{6}e^{-6|\tau|} \end{aligned}$$

The average power

$$R_{XX}(0) = \frac{-1}{10} + \frac{1}{6} = \frac{1}{15} \text{ W}$$

7.42 The spectral density of a random process $X(t)$ is given by

$$S_{XX}(\omega) = \frac{\omega^2}{\omega^4 + 13\omega^2 + 36}$$

Find autocorrelation function and average power.

Solution $R_{XX}(\tau) = F^{-1}[S_{XX}(\omega)]$

$$\begin{aligned} &= F^{-1}\left[\frac{\omega^2}{\omega^4 + 13\omega^2 + 36}\right] \\ &= F^{-1}\left[\frac{\omega^2}{(\omega^2 + 9)(\omega^2 + 4)}\right] \\ &= F^{-1}\left[\frac{-4/5}{\omega^2 + 4} + \frac{9/5}{\omega^2 + 9}\right] \end{aligned}$$

Using $F[e^{-a|t|}] = \frac{2a}{a^2 + \omega^2}$ we can find

$$R_{XX}(\tau) = -\frac{1}{5}e^{-4|\tau|} + \frac{3}{10}e^{-3|\tau|}$$

The average power

$$R_{XX}(0) = -\frac{1}{5} + \frac{3}{10} = \frac{1}{10} W$$



Practice Problems

7.20 Given the power spectral density of a continuous process as $S_{XX}(\omega) = \frac{\omega^2 + 9}{\omega^4 + 5\omega^2 + 4}$. Find the mean square value of the process.

$$\left[\text{Ans. } \frac{4}{3}e^{-|\tau|} - \frac{5}{20}e^{-2|\tau|} \right]$$

7.21 Given the power spectral density $S_{XX}(\omega) = \frac{1}{4 + \omega^2}$. Find the average power of the process. (Ans. 1/4)

Solved Problem



7.43 The power spectral density of a WSS process is given by

$$S_{XX}(\omega) = \begin{cases} \frac{b}{a}(a - |\omega|) & |\omega| < a \\ 0 & \text{otherwise} \end{cases}$$

Find the autocorrelation function of the process.

Solution The autocorrelation function is given by

$$\begin{aligned} R_{XX}(\tau) &= F^{-1}[S_{XX}(\omega)] = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_{XX}(\omega) e^{j\omega\tau} d\omega \\ &= \frac{1}{2\pi} \int_{-a}^a \frac{b}{a}(a - |\omega|) e^{j\omega\tau} d\omega = \frac{b}{2a\pi} \left[\int_{-a}^0 (a + \omega) e^{j\omega\tau} d\omega + \int_0^a (a - \omega) e^{j\omega\tau} d\omega \right] \\ &= \frac{b}{2a\pi} \left[\frac{a}{j\tau} e^{j\omega\tau} \Big|_{-a}^0 + \frac{\omega e^{j\omega\tau}}{j\tau} \Big|_{-a}^0 + \frac{e^{j\omega\tau}}{\tau^2} \Big|_{-a}^0 + \frac{a}{j\tau} e^{j\omega\tau} \Big|_0^a - \frac{\omega e^{j\omega\tau}}{j\tau} \Big|_0^a - \frac{e^{j\omega\tau}}{\tau^2} \Big|_0^a \right] \\ &= \frac{b}{2a\pi} \left[\frac{a}{j\tau} (1 - e^{-ja\tau}) + \frac{a}{j\tau} e^{-ja\tau} + \frac{1}{\tau^2} (1 - e^{-ja\tau}) + \frac{a}{j\tau} (e^{ja\tau} - 1) - \frac{a}{j\tau} e^{ja\tau} - \frac{1}{\tau^2} (e^{ja\tau} - 1) \right] \\ &= \frac{b}{2a\pi} \left[\frac{2}{\tau^2} - \frac{1}{\tau^2} (e^{ja\tau} + e^{-ja\tau}) \right] \\ &= \frac{b}{2a\pi\tau^2} [2(1 - \cos a\tau)] \end{aligned}$$

$$= \frac{2b}{a\pi\tau^2} \sin^2 \frac{a\tau}{2}$$



Practice Problem

7.22 Find the autocorrelation function of a stationary random process whose power spectral density is given by

$$S_{xx}(\omega) = \begin{cases} \omega^2 & \text{for } |\omega| \leq 1 \\ 0 & \text{for } |\omega| > 1 \end{cases} \quad \left[\text{Ans. } \frac{1}{\pi} \left(\frac{\sin \tau}{\tau} + \frac{2 \cos \tau}{\tau^2} - \frac{2 \sin \tau}{\tau^3} \right) \right]$$

Solved Problems



7.44 Find the PSD of a stationary random process for which auto-correlation is $R_{XX}(\tau) = e^{-\alpha|\tau|}$.

Solution The PSD is given by

$$\begin{aligned} S_{XX}(\omega) &= F[R_{XX}(\tau)] \\ &= F[e^{-\alpha|\tau|}] = \int_{-\infty}^{\infty} e^{-\alpha|\tau|} e^{-j\omega\tau} d\tau \\ &= \int_{-\infty}^0 e^{\alpha\tau} e^{-j\omega\tau} d\tau + \int_0^{\infty} e^{-\alpha\tau} e^{-j\omega\tau} d\tau \\ &= \frac{1}{\alpha - j\omega} + \frac{1}{\alpha + j\omega} = \frac{2\alpha}{\alpha^2 + \omega^2} \end{aligned}$$

7.45 A random process has the power density spectrum $S_{XX}(\omega) = \frac{\omega^2}{1 + \omega^2}$. Find the average power in the process.

Solution Given:

$$\begin{aligned} S_{XX}(\omega) &= \frac{\omega^2}{1 + \omega^2} = 1 - \frac{1}{1 + \omega^2} \\ &= 1 - \frac{1}{2} \cdot \frac{2}{1 + \omega^2} \\ R_{XX}(\tau) &= F^{-1}[S_{XX}(\omega)] \\ &= F^{-1} \left[1 - \frac{1}{2} \cdot \frac{2}{1 + \omega^2} \right] \\ &= \delta(\tau) - \frac{1}{2} e^{-|\tau|} \end{aligned}$$

$$\text{Average power} = R_{XX}(0) = 1 - \frac{1}{2} = \frac{1}{2}$$



Practice Problem

7.23 A random process has the power density spectrum $S_{XX}(\omega) = \frac{6\omega^2}{1 + \omega^4}$. Find average power in the process.

$$\left(\text{Ans: } \frac{3}{\sqrt{2}} \right)$$

Solved Problems



7.46 If the random process $X(t)$ represents a white normal noise, find the auto-correlation function of

$$Y(t) = \int_0^t X(\tau) d\tau \quad (7.72)$$

Solution For white noise, the autocorrelation function is an impulse given by

$$R_{XX}(t_1, t_2) = \sigma^2 \delta(t_1 - t_2) \quad (7.73)$$

The autocorrelation function of $Y(t)$ is given by

$$\begin{aligned} R_{YY}(t_1, t_2) &= E \left[\int_0^{t_1} X(\tau) d\tau \int_0^{t_2} X(s) ds \right] \\ &= \int_0^{t_1} \int_0^{t_2} E[X(\tau)X(s)] ds d\tau \\ &= \int_0^{t_1} \int_0^{t_2} R_{XX}(\tau, s) ds d\tau \end{aligned} \quad (7.74)$$

Substituting Eq. (7.73) in Eq. (7.74), we get

$$\begin{aligned} R_{YY}(t_1, t_2) &= \int_0^{t_1} \int_0^{t_2} \sigma^2 \delta(\tau - s) d\tau ds \\ &= \sigma^2 \int_0^{t_2} u(\tau - s) d\tau \\ &= \sigma^2 \int_0^{\min(t_1, t_2)} d\tau = \sigma^2 \min(t_1, t_2) \end{aligned}$$

7.47 A stationary random process $X(t)$ has power spectrum density

$$S_{XX}(\omega) = \frac{25}{\omega^2 + 25}$$

Find the mean square value of the process.

Solution

First method The average or mean square value of the process is given by

$$\begin{aligned} E[X^2(t)] &= R_{XX}(0) = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_{XX}(\omega) d\omega \\ E[X^2(t)] &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{25}{\omega^2 + 25} d\omega \\ &= \frac{1}{2\pi} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{25}{25(1 + \tan^2 \theta)} 5 \sec^2 \theta d\theta \\ &= \frac{1}{2\pi} 5 \theta \Big|_{-\frac{\pi}{2}}^{\frac{\pi}{2}} = \frac{5}{2} = 2.5 \end{aligned}$$

$$\begin{aligned} \text{Let } \omega &= 5 \tan \theta \\ d\omega &= 5 \sec^2 \theta d\theta \end{aligned}$$

$$\begin{aligned} \text{At } \omega = \infty; \theta &= \frac{\pi}{2} \\ \omega = -\infty; \theta &= -\frac{\pi}{2} \end{aligned}$$

Second method We know

$$F[e^{-a|\tau|}] = \frac{2a}{a^2 + \omega^2}$$

$$\begin{aligned} \text{Given: } S_{XX}(\omega) &= \frac{25}{\omega^2 + 25} \\ &= \frac{25}{10} \left(\frac{10}{\omega^2 + 25} \right) = \frac{5}{2} \cdot \frac{2(5)}{\omega^2 + 25} \end{aligned}$$

$$\begin{aligned} R_{XX}(\tau) &= F^{-1} \left[\frac{5}{2} \frac{2(5)}{\omega^2 + 25} \right] \\ &= \frac{5}{2} e^{-5|\tau|} \end{aligned}$$

$$R_{XX}(0) = \frac{5}{2}$$

7.48 The cross-power spectrum of a real random processes $X(t)$ and $Y(t)$ is given by

$$S_{XY}(\omega) = \begin{cases} a + jb\omega & |\omega| < 1 \\ 0 & \text{otherwise} \end{cases}$$

Find the cross-correlation function.

Solution We know

$$\begin{aligned} R_{XX}(\tau) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} S_{XY}(\omega) e^{j\omega\tau} d\omega \\ &= \frac{1}{2\pi} \int_{-1}^1 (a + jb\omega) e^{j\omega\tau} d\omega \\ &= \frac{1}{2\pi} \left\{ \frac{a}{j\tau} e^{j\omega\tau} \Big|_{-1}^1 + jb \left[\frac{\omega}{j\tau} e^{j\omega\tau} \Big|_{-1}^1 + \frac{1}{\tau^2} e^{j\omega\tau} \Big|_{-1}^1 \right] \right\} \\ &= \frac{1}{2\pi} \left\{ \frac{a}{j\tau} (e^{j\tau} - e^{-j\tau}) + jb \left[\frac{1}{j\tau} (e^{j\tau} + e^{-j\tau}) \right] + \frac{jb}{\tau^2} [e^{j\tau} - e^{-j\tau}] \right\} \\ &= \frac{1}{2\pi} \left[\frac{2a}{\tau} \sin \tau + \frac{2b}{\tau} \cos \tau - \frac{2b}{\tau^2} \sin \tau \right] \\ &= \frac{1}{\pi\tau^2} [a\tau \sin \tau - b \sin \tau + b\tau \cos \tau] \end{aligned}$$

7.49 A WSS noise process $N(t)$ has ACF $R_{NN}(\tau) = Pe^{-3|\tau|}$. Find PSD and plot both ACF and PSD.

Solution Given: $R_{NN}(\tau) = Pe^{-3|\tau|}$

$$\begin{aligned} S_{XX}(\omega) &= \int_{-\infty}^{\infty} R_{XX}(\tau) e^{-j\omega\tau} d\tau \\ &= \int_{-\infty}^0 Pe^{3\tau} e^{-j\omega\tau} d\tau + \int_0^{\infty} Pe^{-3\tau} e^{-j\omega\tau} d\tau \\ &= P \left[\int_{-\infty}^0 e^{(3-j\omega)\tau} d\tau + \int_0^{\infty} e^{-(3+j\omega)\tau} d\tau \right] \\ &= P \left[\frac{1}{3-j\omega} e^{(3-j\omega)\tau} \Big|_{-\infty}^0 + \frac{1}{-(3+j\omega)} e^{-(3+j\omega)\tau} \Big|_0^{\infty} \right] \\ &= P \left[\frac{1}{3-j\omega} + \frac{1}{3+j\omega} \right] = \frac{6P}{\omega^2 + 9} \end{aligned}$$



Practice Problem

- 7.24** Find the power spectral density of a random process whose autocorrelation is given by $R_{XX}(\tau) = e^{-\tau^2/2\alpha^2}$, $\alpha > 0$.
(Ans: $\alpha\sqrt{2\pi} e^{-\omega^2\alpha^2/2}$)

Solved Problems



- 7.50** The PSD of a random process is given by

$$S_{XX}(\omega) = \pi, |\omega| < 1, \\ = 0; \text{ elsewhere}$$

Find its autocorrelation function.

Solution We know

$$\begin{aligned} R_{XX}(\tau) &= F^{-1}[S_{XX}(\omega)] \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} S_{XX}(\omega) e^{j\omega\tau} d\omega \\ &= \frac{1}{2\pi} \int_{-1}^1 \pi e^{j\omega\tau} d\omega = \frac{1}{2} \left. \frac{e^{j\omega\tau}}{j\tau} \right|_{-1}^1 \\ &= \frac{1}{2j\tau} (e^{j\tau} - e^{-j\tau}) \\ &= \frac{\sin \tau}{\tau} \\ R_{XX}(\tau) &= \frac{\sin \tau}{\tau} \end{aligned}$$

- 7.51** A random process has autocorrelation function $R_{XX}(\tau) = 5e^{-4|\tau|} - e^{-2|\tau|} \cos(3\pi\tau) + \cos(4\pi\tau)$. Find its power spectrum.

Solution Given: $R_{XX}(\tau) = 5e^{-4|\tau|} - e^{-2|\tau|} \cos(3\pi\tau) + \cos(4\pi\tau)$

$$\begin{aligned} S_{XX}(\omega) &= F[R_{XX}(\tau)] \\ &= 5F[e^{-4|\tau|}] - F[e^{-2|\tau|} \cos 3\pi\tau] + F[\cos 4\pi\tau] \end{aligned}$$

$$\text{We know } F[e^{-a|\tau|}] = \frac{2a}{a^2 + \omega^2}$$

$$F[\cos \omega_0\tau] = \pi[\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$$

$$F[x(t) \cos \omega_0 t] = \frac{1}{2} \{X[j(\omega - \omega_0)] + X[j(\omega + \omega_0)]\}$$

Using above relations,

$$S_{XX}(\omega) = 5 \frac{8}{\omega^2 + 16} - \frac{1}{2} \left[\frac{4}{(\omega - 3\pi)^2 + 4} + \frac{4}{(\omega + 3\pi)^2 + 4} \right] + \pi [\delta(\omega - 4\pi) + \delta(\omega + 4\pi)]$$

7.52 Find the autocorrelation and power spectrum of random process $X(t) = Y \sin \omega_0 t + Z \cos \omega_0 t$, where Y and Z are independent zero-mean random variables with the same variance σ^2 .

Solution

$$\begin{aligned} R_{XX}(\tau) &= E[X(t) X(t + \tau)] \\ &= E\{[Y \sin \omega_0 t + Z \cos \omega_0 t] [Y \sin \omega_0(t + \tau) + Z \cos \omega_0(t + \tau)]\} \\ &= E[Y^2 \sin \omega_0 t \sin \omega_0(t + \tau)] + E[ZY \cos(\omega_0 t) \sin \omega_0(t + \tau)] \\ &\quad + E[(Y \sin \omega_0 t) Z \cos \omega_0(t + \tau)] + E[Z^2 \cos(\omega_0 t) \cos \omega_0(t + \tau)] \\ &= E[Y^2] \sin \omega_0 t \sin \omega_0(t + \tau) + E[YZ] \{\cos \omega_0 t \sin \omega_0(t + \tau) \\ &\quad + \sin(\omega_0 t) \cos \omega_0(t + \tau)\} + E[Z^2] \cos \omega_0 t \cos \omega_0(t + \tau) \\ E[YZ] &= E[Y]E[Z] = 0 \\ E[Y^2] &= E[Z^2] = \sigma^2 \\ R_{XX}(\tau) &= \sigma^2 [\sin \omega_0 t \sin \omega_0(t + \tau) + \cos \omega_0 t \cos \omega_0(t + \tau)] \\ &= \sigma^2 \cos \omega_0 \tau \\ S_{XX}(\omega) &= F[R_{XX}(\tau)] \\ &= \sigma^2 \pi [\delta(\omega - \omega_0) + \delta(\omega + \omega_0)] \end{aligned}$$

7.53 Find the power spectrum $S_{XX}(\omega)$ and average power P_{XX} of the random process $X(t) = A + B \cos(\omega_0 t + \theta)$. A and B are independent random variables, $\omega_0 > 0$ is a constant and θ is a uniformly distributed random variable in the interval 0 to 2π .

Solution Given:

$$\begin{aligned} X(t) &= A + B \cos(\omega_0 t + \theta) \\ R_{XX}(t, t + \tau) &= E[X(t)X(t + \tau)] \\ &= E\{[A + B \cos(\omega_0 t + \theta)] [A + B \cos(\omega_0 t + \omega_0 \tau + \theta)]\} \\ &= E[A^2] + E[A] E[B] E[\cos(\omega_0 t + \theta)] + E[A] E[B] E[\cos(\omega_0 t + \omega_0 \tau + \theta)] \\ &\quad + E[B^2] E\{[\cos(\omega_0 t + \theta)] [\cos(\omega_0 t + \omega_0 \tau + \theta)]\} \end{aligned}$$

Since θ is a uniformly distributed random variable in the interval $(0, 2\pi)$, we have

$$\begin{aligned} E[\cos \omega_0 t + \theta] &= 0 \\ E[\cos(\omega_0 t + \omega_0 \tau + \theta)] &= 0 \\ E\{[\cos(\omega_0 t + \theta)] [\cos(\omega_0 t + \omega_0 \tau + \theta)]\} \\ &= \frac{1}{2} E[\cos(\omega_0 \tau) + \cos(2\omega_0 t + \omega_0 \tau + 2\theta)] \end{aligned}$$

$$= \frac{1}{2} E [\cos(\omega_0 \tau)] + \frac{1}{2} \underbrace{E [\cos(2\omega_0 t + \omega_0 \tau + 2\theta)]}_{\text{zero}}$$

Therefore,

$$R_{XX}(t, t + \tau) = E[A^2] + \frac{1}{2} E[B^2] \cos \omega_0 \tau = R_{XX}(\tau)$$

$$S_{XX}(\omega) = F[R_{XX}(\tau)]$$

$$= F\{E[A^2] + \frac{1}{2} E[B^2] \cos(\omega_0 \tau)\}$$

$$= E[A^2] F[1] + \frac{1}{2} E[B^2] F[\cos(\omega_0 \tau)]$$

$$= 2\pi \delta(\omega) E[A^2] + \frac{\pi}{2} E[B^2] [\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$$

$$P_{XX} = R_{XX}(0) = E[A^2] + \frac{1}{2} E[B^2]$$

7.54 For a random process $X(t)$, assume that $R_{XX}(\tau) = P e^{-\tau^2/2a^2}$ where $P > 0$ and $a > 0$ are constants. Find the power density spectrum of $X(t)$.

Solution $R_{XX}(\tau) = P e^{-\tau^2/2a^2}$

$$\begin{aligned} S_{XX}(\omega) &= \int_{-\infty}^{\infty} P e^{-\tau^2/2a^2} e^{-j\omega\tau} d\tau \\ &= P \int_{-\infty}^{\infty} e^{-\left(\frac{\tau^2}{2a^2} + j\omega\tau\right)} d\tau \\ &= P e^{-\omega^2 a^2/2} \int_{-\infty}^{\infty} e^{-\left(\frac{\tau^2}{2a^2} + j\omega\tau - \frac{\omega^2 a^2}{2}\right)} d\tau \\ &= P e^{-\omega^2 a^2/2} \int_{-\infty}^{\infty} e^{-\left(\frac{\tau}{\sqrt{2}a} + \frac{j\omega a}{\sqrt{2}}\right)^2} d\tau \end{aligned}$$

Let $\frac{\tau}{\sqrt{2}a} + \frac{j\omega a}{\sqrt{2}} = z$

Then $d\tau = \sqrt{2} a dz$

$$S_{XX}(\omega) = P e^{-\omega^2 a^2/2} \sqrt{2} a \int_{-\infty}^{\infty} e^{-z^2} dz$$

$$\int_{-\infty}^{\infty} e^{-z^2} dz = \sqrt{\pi}$$

$$S_{XX}(\omega) = \sqrt{2\pi} Pa e^{-\omega^2 a^2 / 2}$$

7.55 Given a random process $Y(t) = X(t) \cos \omega_0 t$, where ω_0 is constant and $X(t)$ is a stationary random process with power spectrum $S_{XX}(\omega)$, find the power spectrum of $Y(t)$.

Solution The autocorrelation function of $Y(t)$ is

$$\begin{aligned} E[Y(t)Y(t + \tau)] &= E[X(t) \cos \omega_0 t X(t + \tau) \cos \omega_0 (t + \tau)] \\ &= E[X(t)X(t + \tau)] \cos \omega_0 t \cos \omega_0 (t + \tau) \end{aligned}$$

$\therefore$ since ω_0 is constant

$$R_{YY}(t, t + \tau) = R_{XX}(\tau) \cos \omega_0 t \cos \omega_0 (t + \tau)$$

The time average of the $R_{YY}(t, t + \tau)$ is

$$\begin{aligned} A[R_{YY}(t, t + \tau)] &= \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T R_{YY}(t, t + \tau) dt \\ &= \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T R_{XX}(\tau) \cos \omega_0 t \cos \omega_0 (t + \tau) dt \\ &= \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T R_{XX}(\tau) \left[\frac{\cos \omega_0 \tau + \cos(2\omega_0 t + \omega_0 \tau)}{2} \right] dt \\ &= \frac{1}{2} R_{XX}(\tau) \cos \omega_0 \tau + \underbrace{\lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T \frac{1}{2} \cos(2\omega_0 t + \omega_0 \tau) dt}_{= \text{zero}} \\ &= \frac{1}{2} R_{XX}(\tau) \cos \omega_0 \tau \end{aligned}$$

The power spectrum

$$\begin{aligned} S_{YY}(\omega) &= F \left[\frac{1}{2} R_{XX}(\tau) \cos \omega_0 \tau \right] \\ &= \frac{1}{2} F [R_{XX}(\tau) \cos \omega_0 \tau] \\ &= \frac{1}{4} [S_{XX}(\omega - \omega_0) + S_{XX}(\omega + \omega_0)] \end{aligned}$$

7.56 Find the ACF of the following PSDs.

$$(a) S_{XX}(\omega) = \frac{157 + 12\omega^2}{(16 + \omega^2)(9 + \omega^2)} \quad (b) S_{XX}(\omega) = \frac{8}{(9 + \omega^2)^2}$$

Solution

$$\begin{aligned}
 \text{(a) Given: } S_{XX}(\omega) &= \frac{157 + 12\omega^2}{(16 + \omega^2)(9 + \omega^2)} \\
 R_{XX}(\tau) &= F^{-1}[S_{XX}(\omega)] \\
 S_{XX}(\omega) &= \frac{157 + 12\omega^2}{(16 + \omega^2)(9 + \omega^2)} \\
 &= \frac{A}{16 + \omega^2} + \frac{B}{9 + \omega^2} \\
 &= \frac{5}{16 + \omega^2} + \frac{7}{9 + \omega^2} \\
 R_{XX}(\tau) &= F^{-1}\left[\frac{5}{16 + \omega^2}\right] + F^{-1}\left[\frac{7}{9 + \omega^2}\right]
 \end{aligned}$$

We have

$$\begin{aligned}
 F[e^{-a|\tau|}] &= \frac{2a}{a^2 + \omega^2} \\
 S_{XX}(\omega) &= \frac{5}{8} \left(\frac{8}{\omega^2 + 16} \right) + \frac{7}{6} \frac{6}{9 + \omega^2} \\
 R_{XX}(\tau) &= \frac{5}{8} F^{-1}\left[\frac{8}{\omega^2 + 16}\right] + \frac{7}{6} F^{-1}\left[\frac{6}{9 + \omega^2}\right] \\
 &= \frac{5}{8} e^{-4|\tau|} + \frac{7}{6} e^{-3|\tau|} \\
 \text{(b) } S_{XX}(\omega) &= \frac{8}{(9 + \omega^2)^2} \\
 &= \frac{8}{36} \left[\frac{6^2}{(9 + \omega^2)^2} \right] = \frac{2}{9} \left[\left(\frac{6^2}{9 + \omega^2} \right)^2 \right] \\
 F^{-1}\left[\frac{6}{9 + \omega^2}\right] &= e^{-3|\tau|} \\
 R_{XX}(\tau) &= \frac{2}{9} [e^{-3|\tau|} * e^{-3|\tau|}]
 \end{aligned}$$

7.57 Consider two random processes $X(t) = 3 \cos(\omega t + \theta)$ and $Y(t) = 2 \cos\left(\omega t + \theta - \frac{\pi}{2}\right)$ where θ is a random variable uniformly distributed in $(0, 2\pi)$. Prove that $\sqrt{R_{XX}(0) R_{YY}(0)} \geq |R_{XY}(\tau)|$.

Solution Given: $X(t) = 3 \cos(\omega t + \theta)$

$$Y(t) = 2 \cos\left(\omega t + \theta - \frac{\pi}{2}\right)$$

Since θ is uniformly distributed in $(0, 2\pi)$,

$$f_{\Theta}(\theta) = \begin{cases} \frac{1}{2\pi} & \text{for } 0 \leq \theta \leq 2\pi \\ 0 & \text{otherwise} \end{cases}$$

$$\begin{aligned} R_{XX}(\tau) &= E[X(t)X(t + \tau)] \\ &= E[3 \cos(\omega t + \theta) 3 \cos(\omega t + \omega\tau + \theta)] \\ &= \frac{9}{2} E[\cos(2\omega t + \omega\tau + 2\theta) + \cos \omega\tau] \end{aligned}$$

$$\begin{aligned} E[\cos(2\omega t + \omega\tau + 2\theta)] &= \int_{-\infty}^{\infty} \cos(2\omega t + \omega\tau + 2\theta) f_{\Theta}(\theta) d\theta \\ &= \int_0^{2\pi} \cos(2\omega t + \omega\tau + 2\theta) \frac{1}{2\pi} d\theta \\ &= \frac{1}{2\pi} \left. \frac{\sin(2\omega t + \omega\tau + 2\theta)}{2} \right|_0^{2\pi} = 0 \end{aligned}$$

$$\Rightarrow R_{XX}(\tau) = \frac{9}{2} E[\cos \omega\tau] = \frac{9}{2} \cos \omega\tau$$

$$\begin{aligned} R_{YY}(\tau) &= E[Y(t)Y(t + \tau)] \\ &= E\left[4 \cos\left(\omega t + \theta - \frac{\pi}{2}\right) \cos\left(\omega t + \omega\tau + \theta - \frac{\pi}{2}\right)\right] \\ &= \underbrace{2E[\cos(2\omega\tau + 2\theta + \omega\tau - \pi)]}_{\text{zero}} + 2E[\cos \omega\tau] \\ &= 2 \cos \omega\tau \end{aligned}$$

$$R_{XX}(0) = \frac{9}{2} \text{ and } R_{YY}(0) = 2$$

$$\begin{aligned} R_{XY}(\tau) &= E\left[3 \cos(\omega t + \theta) 2 \cos\left(\omega t + \omega\tau + \theta - \frac{\pi}{2}\right)\right] \\ &= \frac{6}{2} \left\{ \underbrace{E \cos\left[2\omega t + \omega\tau + 2\theta - \frac{\pi}{2}\right]}_{\text{zero}} + E\left[\cos\left(\omega\tau - \frac{\pi}{2}\right)\right] \right\} \\ &= 3 \sin \omega\tau \end{aligned}$$

The maximum value of $\sin \omega\tau = 1$.

Hence, $R_{XY}(\tau)$ has peak value as 1.

Hence, we can write

$$R_{XY}(\tau) \leq \sqrt{R_{XX}(0) R_{YY}(0)}$$

7.58 Given that a process $X(t)$ has the auto-correlation function $R_{XX}(\tau) = Ae^{-\alpha|\tau|} \cos(\omega_0 \tau)$ where $A > 0$, $\alpha > 0$ and ω_0 are real constants, find the power spectrum of $X(t)$.

Solution $R_{XX}(\tau) = Ae^{-\alpha|\tau|} \cos(\omega_0 \tau)$

$$= \frac{Ae^{-\alpha|\tau|}}{2} [e^{j\omega_0 \tau} + e^{-j\omega_0 \tau}]$$

$$S_{XX}(\omega) = F[R_{XX}(\tau)] = \int_{-\infty}^{\infty} R_{XX}(\tau) e^{-j\omega \tau} d\tau$$

We know $F[e^{-\alpha|\tau|}] = \frac{2\alpha}{\alpha^2 + \omega^2}$

Using frequency shifting property of Fourier transform, we have

$$F[e^{j\omega_0 \tau} e^{-\alpha|\tau|}] = \frac{2\alpha}{\alpha^2 + (\omega - \omega_0)^2}$$

and

$$F[e^{-j\omega_0 \tau} e^{-\alpha|\tau|}] = \frac{2\alpha}{\alpha^2 + (\omega + \omega_0)^2}$$

$$\Rightarrow S_{XX}(\omega) = \frac{A}{2} \left[\frac{2\alpha}{\alpha^2 + (\omega - \omega_0)^2} + \frac{2\alpha}{\alpha^2 + (\omega + \omega_0)^2} \right]$$

7.59 An ergodic random process is known to have an auto-correlation function of the form

$$R_{XX}(\tau) = \begin{cases} 1 - |\tau|, & |\tau| \leq 1 \\ 0, & |\tau| > 1 \end{cases}$$

Show that the spectral density is given by

$$S_{XX}(\omega) = \left(\frac{\sin \frac{\omega}{2}}{\frac{\omega}{2}} \right)^2$$

Solution $S_{XX}(\omega) = F[R_{XX}(\tau)]$

$$= \int_{-\infty}^{\infty} R_{XX}(\tau) e^{-j\omega \tau} d\tau$$

$$\begin{aligned}
&= \int_{-1}^1 (1 - |\tau|) e^{-j\omega\tau} d\tau \\
&= \int_{-1}^0 (1 + \tau) e^{-j\omega\tau} d\tau + \int_0^1 (1 - \tau) e^{-j\omega\tau} d\tau \\
&= (1 + \tau) \frac{e^{-j\omega\tau}}{-j\omega} \Big|_{-1}^0 - \frac{e^{-j\omega\tau}}{-\omega^2} \Big|_{-1}^0 + (1 - \tau) \frac{e^{-j\omega\tau}}{-j\omega} \Big|_0^1 - (-1) \frac{e^{-j\omega\tau}}{-\omega^2} \Big|_0^1 \\
&= -\frac{1}{j\omega} + \frac{1}{\omega^2} - \frac{e^{j\omega}}{\omega^2} + \frac{1}{j\omega} - \frac{e^{-j\omega}}{\omega^2} + \frac{1}{\omega^2} \\
&= \frac{2}{\omega^2} - \frac{1}{\omega^2} (e^{j\omega} + e^{-j\omega}) \\
&= \frac{2}{\omega^2} - \frac{2}{\omega^2} \cos \omega = \frac{2}{\omega^2} [1 - \cos \omega] = \frac{2}{\omega^2} \left[2 \sin^2 \frac{\omega}{2} \right] \\
&= \frac{4 \sin^2 \frac{\omega}{2}}{\omega^2} = \frac{\sin^2 \frac{\omega}{2}}{\frac{\omega^2}{4}} = \left(\frac{\sin \frac{\omega}{2}}{\frac{\omega}{2}} \right)^2
\end{aligned}$$



Practice Problem

7.25 Find the power spectrum of the triangular autocorrelation function

$$R_{XX}(\tau) = \begin{cases} 1 - \frac{|\tau|}{T} & \text{for } |\tau| < T \\ 0 & \text{for } |\tau| \geq T \end{cases}$$

$$\left(\text{Ans: } \frac{4 \sin^2 \frac{\omega T}{2}}{\omega^2} \right)$$

Solved Problems



7.60 The autocorrelation function of a WSS random process is $R_{XX}(\tau) = a e^{-\tau^2/b^2}$. Find the power spectral density and average power.

Solution $R_{XX}(\tau) = a e^{-\tau^2/b^2}$

$$\begin{aligned}
S_{XX}(\omega) &= \int_{-\infty}^{\infty} R_{XX}(\tau) e^{-j\omega\tau} d\tau = \int_{-\infty}^{\infty} a e^{-\tau^2/b^2} e^{-j\omega\tau} d\tau \\
&= a \int_{-\infty}^{\infty} e^{-\left(\frac{\tau^2}{b^2} + j\omega\tau\right)} d\tau = a e^{-\frac{\omega^2 b^2}{4}} \int_{-\infty}^{\infty} e^{-\left(\frac{\tau^2}{b^2} + j\omega\tau - \frac{\omega^2 b^2}{4}\right)} d\tau
\end{aligned}$$

$$= a e^{-\frac{\omega^2 b^2}{4}} \int_{-\infty}^{\infty} e^{-\left(\frac{\tau}{b} + \frac{j\omega b}{2}\right)^2} d\tau$$

$$\text{Let } \frac{\tau}{b} + \frac{j\omega b}{2} = z \quad \Rightarrow d\tau = b dz$$

$$\int_{-\infty}^{\infty} e^{-z^2} dz = \sqrt{\pi}$$

$$\begin{aligned} \Rightarrow S_{XX}(\omega) &= a b e^{-\omega^2 b^2/4} \int_{-\infty}^{\infty} e^{-z^2} dz \\ &= \sqrt{\pi} a b e^{-\omega^2 b^2/4} \end{aligned}$$

Average power

$$R_{XX}(0) = a e^{-\tau^2/b^2} \Big|_{\tau=0} = a \text{ watts}$$



Practice Problems

7.26 The autocorrelation function of a periodic random process is $R_{XX}(\tau) = e^{-\tau^2/2\sigma^2}$. Find PSD and average power of the signal.
(Ans: $\sigma\sqrt{2\pi} e^{-\sigma^2\omega^2/2}$)

7.27 Find the power density spectrum corresponding to the autocorrelation function (a) $R_{XX}(\tau) = e^{-\alpha\tau^2} \cos\beta\tau$
(b) $R_{XX}(\tau) = 1 + e^{-2|\tau|}$.

$$\left(\begin{array}{l} \text{(a) } \frac{1}{2} \left[\sqrt{\frac{\pi}{\alpha}} e^{-(\omega-\beta)^2/4\alpha} + \sqrt{\frac{\pi}{\alpha}} e^{-(\omega+\beta)^2/4\alpha} \right] \\ \text{(b) } 2\pi\delta(\omega) + \frac{4}{\omega^2 + 4} \end{array} \right)$$

Solved Problems



7.61 The cross-correlation function of two processes $X(t)$ and $Y(t)$ is

$$R_{XY}(t, t + \tau) = \frac{AB}{2} [\sin(\omega_0 \tau) + \cos(\omega_0 (2t + \tau))]$$

where A , B and ω_0 are constants. Find the cross power spectrum.

Solution The time average of $R_{XY}(\tau)$ is given by

$$\begin{aligned} &= \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T R_{XY}(t, t + \tau) dt \\ &= \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T \left[\frac{AB}{2} \sin \omega_0 \tau + \cos(2\omega_0 t + \omega_0 \tau) \right] dt \end{aligned}$$

$$\begin{aligned}
&= \frac{AB}{2} \sin \omega_0 \tau + \lim_{T \rightarrow \infty} \frac{1}{4T} \sin(2\omega_0 t + \omega_0 \tau) \Big|_{-T}^T \\
&= \frac{AB}{2} \sin \omega_0 \tau
\end{aligned}$$

The cross-power spectrum is

$$\begin{aligned}
S_{XY}(\omega) &= F \left[\frac{AB}{2} \sin \omega_0 \tau \right] \\
&= \frac{AB}{2} F[\sin \omega_0 \tau] = \frac{AB}{2} \frac{1}{2j} F[e^{j\omega_0 \tau} - e^{-j\omega_0 \tau}]
\end{aligned}$$

$$F[e^{j\omega_0 \tau}] = 2\pi \delta(\omega - \omega_0)$$

and $F[e^{-j\omega_0 \tau}] = 2\pi \delta(\omega + \omega_0)$

$$\begin{aligned}
\Rightarrow S_{XY}(\omega) &= \frac{-jAB}{4} 2\pi [\delta(\omega - \omega_0) + \delta(\omega + \omega_0)] \\
&= \frac{-j\pi AB}{2} [\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]
\end{aligned}$$

7.62 Consider a random process $Y(t) = X(t - T)$ where T is the delay. If $X(t)$ is a WSS random process find $R_{YY}(\tau)$, $S_{YY}(\omega)$, $R_{YX}(\tau)$ and $S_{YX}(\omega)$.

Solution Given:

$$Y(t) = X(t - T)$$

We know $R_{YY}(\tau) = E[Y(t)Y(t + \tau)]$

$$\begin{aligned}
&= E[X(\underbrace{t - T}_{t_1}) X(\underbrace{t - T + \tau}_{t_2})] \\
&= R_{XX}(t_2 - t_1)
\end{aligned}$$

Since $X(t)$ is WSS,

$$\begin{aligned}
R_{YY}(\tau) &= R_{XX}(\tau) \\
S_{YY}(\omega) &= F[R_{YY}(\tau)] = F[R_{XX}(\tau)] = S_{XX}(\omega) \\
R_{YX}(\tau) &= E[Y(t)X(t + \tau)] \\
&= E[X(\underbrace{t - T}_{t_1}) X(\underbrace{t + \tau}_{t_2})] \\
&= R_{XX}(\tau + T)
\end{aligned}$$

Using the time shifting property of Fourier transform,

$$S_{YX}(\omega) = e^{j\omega T} S_{XX}(\omega)$$



Practice Problem

7.28 Let $Y(t) = X(t) + X(t - T)$. If $X(t)$ is a WSS random process, find $R_{XY}(\tau)$, $S_{XY}(\omega)$.

Solved Problems



7.63 Find the cross-spectral density function of two random processes, $X(t) = A \cos(\omega_0 t) + B \sin(\omega_0 t)$ and $Y(t) = -A \sin(\omega_0 t) + B \cos(\omega_0 t)$. The mean values $\mu_X = \mu_Y = 0$. The variance of these processes are σ^2 .

Solution $R_{XY}(t, t + \tau) = E[X(t) Y(t + \tau)]$

$$\begin{aligned} &= E\{[A \cos(\omega_0 t) + B \sin(\omega_0 t)] [-A \sin(\omega_0 t + \omega_0 \tau) + B \cos(\omega_0 t + \omega_0 \tau)]\} \\ &= E[-A^2 \cos(\omega_0 t) \sin(\omega_0 t + \omega_0 \tau)] - E[AB \sin \omega_0 t \sin(\omega_0 t + \omega_0 \tau)] \\ &\quad + E[AB \cos(\omega_0 t) \cos(\omega_0 t + \omega_0 \tau)] + E[B^2 \sin \omega_0 t \cos \omega_0 t + \omega_0 \tau] \end{aligned}$$

$$E[AB] = E[A] E[B] = 0$$

and $E[A^2] = E[B^2] = \sigma^2$

$$\Rightarrow R_{XY}(t, t + \tau) = \sigma^2 (\sin(-\omega_0 \tau)) = -\sigma^2 (\sin \omega_0 \tau) = -\sigma^2 (\sin \omega_0 \tau)$$

$$S_{XY}(\omega) = -j\sigma^2 \pi [\delta(\omega + \omega_0) - \delta(\omega - \omega_0)]$$

7.64 Find the average power of the random process $X(t)$ with power density spectrum

$$S_{XX}(\omega) = \frac{6\omega^2}{(1 + \omega^2)^3}$$

Solution The average power

$$\begin{aligned} P_{XX} &= \frac{1}{2\pi} \int_{-\infty}^{\infty} S_{XX}(\omega) d\omega \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{6\omega^2}{(1 + \omega^2)^3} d\omega \end{aligned}$$

Let $\omega = \tan \theta$ then $d\omega = \sec^2 \theta d\theta$. Also, the limits of integration changes from $-\frac{\pi}{2}$ to $\frac{\pi}{2}$. Therefore, we can write

$$\begin{aligned} P_{XX} &= \frac{1}{2\pi} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{6 \tan^2 \theta \sec^2 \theta d\theta}{(1 + \tan^2 \theta)^3} = \frac{6}{2\pi} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin^2 \theta \cos^2 \theta d\theta \\ &= \frac{6}{2\pi} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left(\frac{\sin 2\theta}{2} \right)^2 d\theta = \frac{3}{4\pi} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left(\frac{1 - \cos 4\theta}{2} \right) d\theta = \frac{3}{8} \end{aligned}$$

$$\Rightarrow \text{Average power } P_{XX} = \frac{3}{8}$$

7.65 A WSS random process $X(t)$ has PSD $S_{XX}(\omega) = \frac{\omega^2}{\omega^4 + 10\omega^2 + 9}$. Find ACF and mean square value of the process.

Solution Given:

$$S_{XX}(\omega) = \frac{\omega^2}{\omega^4 + 10\omega^2 + 9}$$

$$\text{The ACF} = F^{-1}[S_{XX}(\omega)]$$

$$= F^{-1}\left[\frac{\omega^2}{\omega^4 + 10\omega^2 + 9}\right] = F^{-1}\left[\frac{\omega^2}{(\omega^2 + 9)(\omega^2 + 1)}\right]$$

$$S_{XX}(\omega) = \frac{-1/8}{\omega^2 + 1} + \frac{9/8}{\omega^2 + 9}$$

$$R_{XX}(\tau) = F^{-1}[S_{XX}(\omega)] = F^{-1}\left[\frac{-1/8}{\omega^2 + 1} + \frac{9/8}{\omega^2 + 9}\right]$$

$$\text{We know } F[e^{-a|\tau|}] = \frac{2a}{a^2 + \omega^2}$$

$$\begin{aligned} R_{XX}(\tau) &= -\frac{1}{16} F^{-1}\left[\frac{2(1)}{\omega^2 + 1}\right] + \frac{9}{8(6)} F^{-1}\left[\frac{2(3)}{\omega^2 + 9}\right] \\ &= -\frac{1}{16} e^{-|\tau|} + \frac{3}{16} e^{-3|\tau|} \end{aligned}$$

The mean square value is

$$E[X^2(t)] = R_{XX}(0) = -\frac{1}{16} + \frac{3}{16} = \frac{1}{8} \text{ watts}$$

7.66 For a random process $X(t) = A \cos \omega_0 t$ where ω_0 is a constant and A is uniformly distributed with mean 5 and variance 2, find the average power of $X(t)$.

Solution Given $X(t) = A \cos \omega_0 t$; $A \sim U(5, 2)$

$$E[A] = 5; \sigma_A^2 = 2$$

$$\begin{aligned} R_{XX}(\tau) &= E[X(t)X(t + \tau)] \\ &= E[A^2 \cos \omega_0 t \cos \omega_0(t + \tau)] \\ &= E[A^2] \cos \omega_0 t \cos \omega_0(t + \tau) \end{aligned}$$

$$\begin{aligned} \text{We know } E[A^2] &= \sigma_A^2 + \{E[A]\}^2 \\ &= 2 + (5)^2 = 27 \end{aligned}$$

$$\Rightarrow R_{XX}(\tau) = 27 \cos \omega_0 t \cos \omega_0(t + \tau)$$

$$E\{X^2(t)\} = R_{XX}(0) = 27 \cos^2 \omega_0 t$$

The average power

$$\begin{aligned} P_{XX} &= A\{E[X^2(t)]\} \\ &= \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T E[X^2(t)] dt \\ &= \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T 27 \cos^2 \omega_0 t dt \\ &= \lim_{T \rightarrow \infty} \frac{27}{2T} \int_{-T}^T \left(\frac{1 + \cos 2\omega_0 t}{2} \right) dt \\ &= \frac{27}{4} \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T}^T (1 + \cos 2\omega_0 t) dt \\ &= \frac{27}{4} \left\{ \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T}^T dt + \underbrace{\lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T}^T \cos 2\omega_0 t dt}_{\text{zero}} \right\} \\ &= \frac{27}{4} \left[\frac{1}{T} (2T) \right] = \frac{27}{2} = 13.5 \text{ watts} \end{aligned}$$

7.67 Find the average power of the WSS random process $X(t)$ has power spectral density

$$S_{XX}(\omega) = \frac{\omega^2 - 17}{(\omega^2 + 49)(\omega^2 + 16)}.$$

Solution Given:

$$\begin{aligned} S_{XX}(\omega) &= \frac{\omega^2 - 17}{(\omega^2 + 49)(\omega^2 + 16)} \\ &= \frac{A}{\omega^2 + 49} + \frac{B}{\omega^2 + 16} \\ A &= (\omega^2 + 49) \frac{(\omega^2 - 17)}{(\omega^2 + 49)(\omega^2 + 16)} \Big|_{\omega^2 = -49} \\ &= \frac{-66}{-33} = 2 \\ B &= (\omega^2 + 16) \frac{(\omega^2 - 17)}{(\omega^2 + 49)(\omega^2 + 16)} \Big|_{\omega^2 = -16} \end{aligned}$$

$$\begin{aligned}
&= \frac{-33}{33} = -1 \\
\Rightarrow S_{XX}(\omega) &= \frac{2}{\omega^2 + 49} - \frac{1}{\omega^2 + 16} \\
&= \frac{2}{14} \frac{2(7)}{\omega^2 + 49} - \frac{1}{8} \frac{2(4)}{\omega^2 + 16} \\
R_{XX}(\tau) &= \frac{2}{14} F^{-1} \left[\frac{2(7)}{\omega^2 + 49} \right] - \frac{1}{8} F^{-1} \left[\frac{2(4)}{\omega^2 + 16} \right] \\
&= \frac{2}{14} e^{-7|\tau|} - \frac{1}{8} e^{-4|\tau|} \\
P_{XX} = R_{XX}(0) &= \frac{2}{14} - \frac{1}{8} = \frac{8-7}{56} = \frac{1}{56} \text{ watts}
\end{aligned}$$

7.68 Find the average power of the random process $X(t)$ which has the power spectral density

$$S_{XX}(\omega) = \begin{cases} 1 - \frac{\omega}{4\pi} & |\omega| \leq 4\pi \\ 0 & \text{otherwise} \end{cases}$$

Solution Given:

$$S_{XX}(\omega) = \begin{cases} 1 - \frac{\omega}{4\pi} & |\omega| \leq 4\pi \\ 0 & \text{otherwise} \end{cases}$$

The average power

$$\begin{aligned}
P_{XX} &= \frac{1}{2\pi} \int_{-\infty}^{\infty} S_{XX}(\omega) d\omega \\
&= \frac{1}{2\pi} \int_{-4\pi}^{4\pi} \left(1 - \frac{\omega}{4\pi} \right) d\omega \\
&= \frac{1}{2\pi} \left\{ \omega \Big|_{-4\pi}^{4\pi} - \frac{1}{4\pi} \frac{\omega^2}{2} \Big|_{-4\pi}^{4\pi} \right\} \\
&= \frac{1}{2\pi} \left\{ 8\pi - \frac{1}{4\pi} (16\pi^2 - 16\pi^2) \right\} \\
&= 4 \text{ watts}
\end{aligned}$$

7.69 If $R_{XX}(\tau) = e^{-2\lambda|\tau|}$ is the auto-correlation function of a random process $X(t)$, obtain the spectral density.

Solution The spectral density

$$\begin{aligned}
 S_{XX}(\omega) &= F[R_{XX}(\tau)] \\
 &= F[e^{-2\lambda|\tau|}] \\
 \Rightarrow S_{XX}(\omega) &= \int_{-\infty}^{\infty} R_{XX}(\tau) e^{-j\omega\tau} d\tau = \int_{-\infty}^{\infty} e^{-2\lambda|\tau|} e^{-j\omega\tau} d\tau \\
 &= \int_{-\infty}^0 e^{2\lambda\tau} e^{-j\omega\tau} d\tau + \int_0^{\infty} e^{-2\lambda\tau} e^{-j\omega\tau} d\tau \\
 &= \frac{1}{2\lambda - j\omega} + \frac{1}{2\lambda + j\omega} = \frac{4\lambda}{4\lambda^2 + \omega^2}
 \end{aligned}$$

7.70 Given the power spectral density $S_{XX}(\omega) = \frac{1}{4 + \omega^2}$, find the average power of the process.

Solution The average power

$$\begin{aligned}
 P_{XX} &= \frac{1}{2\pi} \int_{-\infty}^{\infty} S_{XX}(\omega) d\omega \\
 &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{1}{4 + \omega^2} d\omega \\
 &= \frac{1}{2\pi} \left[\frac{1}{2} \tan^{-1} \frac{\omega}{2} \right]_{-\infty}^{\infty} \quad \boxed{\int \frac{1}{x^2 + a^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}} \\
 &= \frac{1}{4\pi} [\tan^{-1}(\infty) - \tan^{-1}(-\infty)] \\
 &= \frac{1}{4\pi} \left[\frac{\pi}{2} - \left(-\frac{\pi}{2} \right) \right] = \frac{1}{4}
 \end{aligned}$$

7.71 Find the power spectral density of the random process $X(t)$ if $E[X(t)] = 1$ and $R_{XX}(\tau) = 1 + e^{-a|\tau|}$.

Solution The power spectral density

$$\begin{aligned}
 S_{XX}(\omega) &= \int_{-\infty}^{\infty} R_{XX}(\tau) e^{-j\omega\tau} d\tau \\
 &= \int_{-\infty}^{\infty} (1 + e^{-a|\tau|}) e^{-j\omega\tau} d\tau
 \end{aligned}$$

$$\begin{aligned}
&= \int_{-\infty}^{\infty} e^{-j\omega\tau} d\tau + \int_{-\infty}^{\infty} e^{-|\tau|} e^{-j\omega\tau} d\tau \\
\int_{-\infty}^{\infty} e^{-a|\tau|} e^{-j\omega\tau} d\tau &= \int_{-\infty}^0 e^{a\tau} e^{-j\omega\tau} d\tau + \int_0^{\infty} e^{-a\tau} e^{-j\omega\tau} d\tau \\
&= \frac{1}{a - j\omega} + \frac{1}{a + j\omega} = \frac{2a}{a^2 + \omega^2}
\end{aligned}$$

Consider inverse Fourier transform of $\delta(\omega)$

$$\begin{aligned}
F^{-1}[\delta(\omega)] &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \delta(\omega) e^{j\omega\tau} d\omega = \frac{1}{2\pi} \\
\Rightarrow F\left[\frac{1}{2\pi}\right] &= \delta(\omega) \text{ or } F[1] = 2\pi \delta(\omega) \\
F[1] &= \int_{-\infty}^{\infty} 1 \cdot e^{-j\omega\tau} d\tau = \int_{-\infty}^{\infty} e^{-j\omega\tau} d\tau = 2\pi \delta(\omega) \\
\Rightarrow S_{XX}(\omega) &= 2\pi \delta(\omega) + \frac{2a}{a^2 + \omega^2}
\end{aligned}$$

7.72 If $X(t)$ is a WSS process with auto-correlation function $R_{XX}(\tau)$ and if $Y(\tau) = X(t+a) - X(t-a)$, prove that $S_{YY}(\omega) = 4 S_{XX}(\omega) \sin^2(a\omega)$.

Solution Given:

$$\begin{aligned}
Y(t) &= X(t+a) - X(t-a) \\
R_{YY}(\tau) &= E[Y(t)Y(t+\tau)] \\
&= E\{[X(t+a) - X(t-a)][X(t+a+\tau) - X(t-a+\tau)]\} \\
&= E[X(t+a)X(t+a+\tau)] - E[X(t-a)X(t-a+\tau)] \\
&\quad - E[X(t+a)X(t-a+\tau)] + E[X(t-a)X(t-a+\tau)] \\
&= R_{XX}(\tau) - R_{XX}(2a+\tau) - R_{XX}(\tau-2a) + R_{XX}(\tau) \\
&= 2 R_{XX}(\tau) - R_{XX}(\tau+2a) - R_{XX}(\tau-2a)
\end{aligned}$$

Taking Fourier transform on both sides, we get

$$S_{YY}(\omega) = 2S_{XX}(\omega) - F[R_{XX}(\tau+2a)] - F[R_{XX}(\tau-2a)]$$

Using time-shifting property of Fourier transforms, we can write

$$\begin{aligned}
F[R_{XX}(\tau+2a)] &= e^{j\omega 2a} S_{XX}(\omega) \\
F[R_{XX}(\tau-2a)] &= e^{-j\omega 2a} S_{XX}(\omega) \\
\Rightarrow S_{YY}(\omega) &= 2S_{XX}(\omega) - S_{XX}(\omega)[e^{j2a\omega} + e^{-j2a\omega}] \\
&= 2S_{XX}(\omega) - 2S_{XX}(\omega) \cos 2\omega a \\
&= 2S_{XX}(\omega) [1 - \cos 2\omega a] \\
&= 4 S_{XX}(\omega) \sin^2(a\omega)
\end{aligned}$$

7.73 Find the auto-correlation function of the process $X(t)$ for which the PSD is given by $S_{XX}(\omega) = 1 + \omega^2$ for $|\omega| < 1$ and $S_{XX}(\omega) = 0$ for $|\omega| > 1$.

Solution Given:

$$S_{XX}(\omega) = 1 + \omega^2 \text{ for } |\omega| < 1$$

$$= 0 \text{ otherwise}$$

$$\begin{aligned} R_{XX}(\tau) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} S_{XX}(\omega) e^{j\omega\tau} d\omega = \frac{1}{2\pi} \left[\int_{-1}^1 (1 + \omega^2) e^{j\omega\tau} d\omega \right] \\ &= \frac{1}{2\pi} \left[\frac{e^{j\omega\tau}}{j\tau} \Big|_{-1}^1 + \frac{\omega^2}{j\tau} e^{j\omega\tau} \Big|_{-1}^1 + \frac{2\omega}{\tau^2} e^{j\omega\tau} \Big|_{-1}^1 - \frac{2}{j\tau^3} e^{j\omega\tau} \Big|_{-1}^1 \right] \\ &= \frac{1}{2\pi} \left[\frac{2 \sin \tau}{\tau} + \frac{2 \sin \tau}{\tau} + \frac{2}{\tau^2} (2 \cos \tau) - \frac{2}{\tau^3} (2 \sin \tau) \right] \\ &= \frac{2}{\pi \tau^3} [\tau^2 \sin \tau + \tau \cos \tau - \sin \tau] \end{aligned}$$

7.74 Find the mean sanane value of the process whose power spectral density is

$$S_{XX}(\omega) = \frac{1}{\omega^4 + 10\omega^2 + 9}.$$

Solution Given:

$$\begin{aligned} S_{XX}(\omega) &= \frac{1}{\omega^4 + 10\omega^2 + 9} = \frac{1}{(\omega^2 + 9)(\omega^2 + 1)} \\ &= \frac{A}{\omega^2 + 9} + \frac{B}{\omega^2 + 1} \\ &= -\frac{1}{8} \frac{1}{\omega^2 + 9} + \frac{1}{8} \frac{1}{(\omega^2 + 1)} \\ S_{XX}(\omega) &= -\frac{1}{8} \left(\frac{1}{6} \right) \frac{6}{\omega^2 + 9} + \frac{1}{8} \left(\frac{1}{2} \right) \frac{2}{\omega^2 + 1} \\ R_{XX}(\tau) &= -\frac{1}{8} \left(\frac{1}{6} \right) e^{-3|\tau|} + \frac{1}{8} \left(\frac{1}{2} \right) e^{-2|\tau|} \end{aligned}$$

The mean square value

$$R_{XX}(0) = -\frac{1}{48} + \frac{1}{16} = \frac{-1+3}{48} = \frac{1}{24}$$

7.75 The auto-correlation function for a stationary process $X(t)$ is given by $R_{XX}(\tau) = 9 + 2e^{-|\tau|}$. Find the mean of the random variable $Y = \int_0^2 X(t) dt$ and variance of $X(t)$.

Solution Given:

$$R_{XX}(\tau) = 9 + 2e^{-|\tau|}$$

$$\mu_X^2 = \{E[X(t)]\}^2 = \lim_{\tau \rightarrow \infty} R_{XX}(\tau)$$

$$= \lim_{\tau \rightarrow \infty} [9 + e^{-2|\tau|}] = 9$$

$$\Rightarrow \mu_X = 3$$

$$E[X^2(t)] = R_{XX}(0) = [9 + 2e^{-|\tau|}]_{\tau=0} = 11$$

$$\text{Var}[X(t)] = E[X^2(t)] - \{E[X(t)]\}^2$$

$$= 11 - (3)^2 = 2$$

$$E[Y(t)] = E\left[\int_0^2 X(t) dt\right] = \int_0^2 E[X(t)] dt$$

$$= \int_0^2 3 dt = 3t \Big|_0^2 = 3(2 - 0)$$

$$= 6$$



Practice Problems

7.29 Given that the autocorrelation function for a stationary ergodic process with no periodic components is $R_{XX}(\tau) = 25 + \frac{4}{1 + 6\tau^2}$. Find the mean and variance of the process.

(Ans. $\mu_X = 5$, $\text{Var}(X) = 4$)

7.30 Find the mean and variance of a stationary process whose autocorrelation function is given by $R_{XX}(\tau) = 18 + \frac{2}{6 + \tau^2}$

$$\mu_X^2 = \lim_{\tau \rightarrow \infty} R_{XX}(\tau) = 18 \Rightarrow \mu_X = 3\sqrt{2}$$

$$E[X^2(t)] = R_{XX}(0) = \frac{55}{3}$$

$$\text{Var}[X(t)] = \frac{1}{3}$$

7.31 A stationary random process has an autocorrelation function and is given by $R_{XX}(\tau) = \frac{25\tau^2 + 36}{6.25\tau^2 + 4}$. Find the mean and variance of the process.

$$\mu_X^2 = \lim_{\tau \rightarrow \infty} R_{XX}(\tau) = \frac{25}{6.25} = 4; E[X^2(t)] = R_{XX}(0) = 9$$

$$\text{Var}(x) = 9 - 4 = 5$$

Solved Problem

7.76 Find the average power of a random process $X(t) = A \cos(\omega_0 t + \theta)$ if

- (a) $\omega_0 \sim U(50, 100)$ and nothing else in $X(t)$ is random.
- (b) $A \sim N(5, 1)$ and nothing else in $X(t)$ is random.
- (c) $\theta \sim U(0, 2\pi)$ and nothing else in $X(t)$ is random.

Solution

- (a) Given $X(t) = A \cos(\omega_0 t + \theta)$ and $\omega_0 \sim U(50, 100)$

$$f(\omega_0) = \frac{1}{100 - 50} = \frac{1}{50} \text{ for } 50 \leq \omega_0 \leq 100$$

$$= 0 \text{ otherwise}$$

$$\begin{aligned} R_{XX}(t, t + \tau) &= E[X(t) X(t + \tau)] \\ &= E\{[A \cos(\omega_0 t + \theta)] [A \cos(\omega_0 t + \omega_0 \tau + \theta)]\} \\ &= \frac{A^2}{2} [E[\cos \omega_0 \tau + \cos(2\omega_0 t + \omega_0 \tau + 2\theta)]] \\ &= \frac{A^2}{2} \int_{-\infty}^{\infty} [\cos \omega_0 \tau + \cos(2\omega_0 t + \omega_0 \tau + 2\theta)] f(\omega_0) d\omega_0 \\ &= \frac{A^2}{100} \int_{50}^{100} [\cos \omega_0 \tau + \cos(2\omega_0 t + \omega_0 \tau + 2\theta)] d\omega_0 \\ &= \frac{A^2}{100} \left[\frac{\sin \omega_0 \tau}{\tau} \Big|_{50}^{100} + \frac{\sin(2\omega_0 t + \omega_0 \tau + 2\theta)}{2t + \tau} \Big|_{50}^{100} \right] \\ &= \frac{A^2}{100} \left[\frac{\sin 100 \tau - \sin 50 \tau}{\tau} + \frac{\sin[100(2t + \tau) + 2\theta] - \sin[50(2t + \tau) + 2\theta]}{2t + \tau} \right] \\ E[X^2(t)] &= \lim_{\tau \rightarrow 0} R_{XX}(t, t + \tau) \\ &= \left\{ \frac{A^2}{100} \lim_{\tau \rightarrow 0} \left[\frac{100 \sin 100 \tau}{100 \tau} - 50 \frac{\sin 50 \tau}{50} \right] + \frac{\sin(200t + 2\theta) - \sin(100t + 2\theta)}{2t} \right\} \\ &= \frac{A^2}{100} \left[(100 - 50) + \frac{\sin(200t + 2\theta) - \sin(100t + 2\theta)}{2t} \right] \end{aligned}$$

The average power

$$\begin{aligned} R_{XX} &= \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T E[X^2(t)] dt \\ &= \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T \frac{A^2}{100} \left[50 + \frac{\sin(200t + 2\theta) - \sin(100t + 2\theta)}{2t} \right] dt \end{aligned}$$

$$= \frac{A^2}{100}(50) = \frac{A^2}{2}$$

$$(b) \quad R_{XX}(t, t + \tau) = E\{[A \cos(\omega_0 t + \theta)] [A \cos(\omega_0 t + \omega_0 \tau + \theta)]\}$$

$$= E[A^2] \cos(\omega_0 t + \theta) \cos(\omega_0 t + \omega_0 \tau + \theta)$$

$$E[A^2] = \sigma_A^2 + \{E[A]\}^2$$

Given: $A \sim N(5, 1)$

$$\Rightarrow E[A] = 5; \sigma_A = 1$$

$$E[A^2] = \sigma_A^2 + \{E[A]\}^2 = 1 + 5^2 = 26$$

$$R_{XX}(t, t + \tau) = 26 \cos(\omega_0 t + \theta) \cos(\omega_0 t + \omega_0 \tau + \theta)$$

$$E[X^2(t)] = R_{XX}(t, t) = 26 \cos^2(\omega_0 t + \theta)$$

$$P_{XX} = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T E[X^2(t)] dt = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T 26 \cos^2(\omega_0 t + \theta) dt$$

$$= \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T \frac{26}{2} [1 + \cos 2(\omega_0 t + \theta)] d\theta$$

$$= \frac{26}{2} = 13$$

$$(c) \quad R_{XX}(t, t + \tau) = E\{[A \cos(\omega_0 t + \theta)] [A \cos(\omega_0 t + \omega_0 \tau + \theta)]\}$$

$$\theta \sim U(0, 2\pi)$$

$$\Rightarrow f(\theta) = \begin{cases} \frac{1}{2\pi} & \text{for } 0 \leq \theta \leq 2\pi \\ 0 & \text{otherwise} \end{cases}$$

$$R_{XX}(t, t + \tau) = \frac{A^2}{2} \{E[\cos \omega_0 \tau + \cos(2\omega_0 t + \omega_0 \tau + 2\theta)]\}$$

$$= \frac{A^2}{2} \int_{-\infty}^{\infty} [\cos \omega_0 \tau + \cos(2\omega_0 t + \omega_0 \tau + 2\theta)] f(\theta) d\theta$$

$$= \frac{A^2}{2(2\pi)} \int_0^{2\pi} [\cos \omega_0 \tau + \cos(2\omega_0 t + \omega_0 \tau + 2\theta)] d\theta$$

$$= \frac{A^2}{4\pi} \left[\int_0^{2\pi} \cos \omega_0 \tau d\theta + \underbrace{\int_0^{2\pi} \cos(2\omega_0 t + \omega_0 \tau + 2\theta) d\theta}_{\text{Zero}} \right]$$

$$= \frac{A^2}{4\pi} 2\pi \cos \omega_0 \tau$$

$$= \frac{A^2}{2} \cos \omega_0 \tau$$

The average power

$$P_{XX} = \frac{A^2}{2} \cos \omega_0 t \Big|_{\tau=0} = \frac{A^2}{2}$$

7.77 A random process $X(t)$ has only a dc component A . A is a random variable with zero mean and unit variance. Find the auto-correlation function and power spectrum.

Solution Given: $X(t) = A$

$$\begin{aligned} R_{XX}(\tau) &= E[X(t)X(t + \tau)] \\ &= E[(A)(A)] = E[A^2] \end{aligned}$$

$$E[A^2] = \sigma_A^2 + \{E[A]\}^2 = 1$$

$$\Rightarrow R_{XX}(\tau) = 1$$

$$\begin{aligned} S_{XX}(\omega) &= F[R_{XX}(\tau)] = F[1] \\ &= 2\pi\delta(\omega) \end{aligned}$$

7.78 Given a stationary random process $X(t)$ and its auto-correlation function $R_{XX}(\tau)$ and power spectrum $S_{XX}(\omega)$. Find the auto-correlation and power spectrum of $Y(t) = \alpha X(t)$, where α is a constant.

Solution Given:

$$\begin{aligned} Y(t) &= \alpha X(t) \\ R_{YY}(\tau) &= E[Y(t)Y(t + \tau)] \\ &= E[\alpha X(t)\alpha X(t + \tau)] \\ &= E[\alpha^2 X(t)X(t + \tau)] \\ &= \alpha^2 E[X(t)X(t + \tau)] \\ &= \alpha^2 R_{XX}(\tau) \end{aligned}$$

If α is negative,

$$\begin{aligned} R_{YY}(\tau) &= E[(-\alpha x(t))(-\alpha x(t + \tau))] \\ &= E[\alpha^2 x(t)x(t + \tau)] \\ &= \alpha^2 R_{XX}(\tau) \end{aligned}$$

Hence, for any α

$$R_{YY}(\tau) = |\alpha|^2 R_{XX}(\tau)$$

The power spectrum

$$\begin{aligned} S_{YY}(\omega) &= F[R_{YY}(\tau)] \\ &= |\alpha|^2 S_{XX}(\omega) \end{aligned}$$

7.79 If $S_{XX}(\omega)$ and $S_{YY}(\omega)$ are power density spectra of two random process $X(t)$ and $Y(t)$ respectively, under what condition the power density spectrum of $X(t) + Y(t)$ is equal to $S_{XX}(\omega) + S_{YY}(\omega)$.

Solution Given: PSD of $X(t)$ is $S_{XX}(\omega)$ and PSD of $Y(t)$ is $S_{YY}(\omega)$.

Let $Z(t) = X(t) + Y(t)$, then

$$S_{ZZ}(\omega) = F[R_{ZZ}(\tau)]$$

$$R_{ZZ}(\tau) = E[Z(t) Z(t + \tau)]$$

$$= E\{(X(t) + Y(t)) [X(t + \tau) + Y(t + \tau)]\}$$

$$= E[X(t)X(t + \tau)] + E[Y(t)X(t + \tau)] + E[X(t)Y(t + \tau)] + E[Y(t)Y(t + \tau)]$$

$$= R_{XX}(\tau) + R_{YX}(\tau) + R_{XY}(\tau) + R_{YY}(\tau)$$

$$\Rightarrow S_{ZZ}(\omega) = S_{XX}(\omega) + S_{YX}(\omega) + S_{XY}(\omega) + S_{YY}(\omega)$$

If $X(t)$ and $Y(t)$ are orthogonal then

$$R_{XY}(\tau) = R_{YX}(\tau) = 0.$$

Under this condition,

$$S_{ZZ}(\omega) = S_{XX}(\omega) + S_{YY}(\omega)$$

7.80 Let $X(t)$ and $Y(t)$ be independent wide-sense stationary random processes. Let $Z(t) = X(t) Y(t)$.

- (a) Prove that $Z(t)$ is wide-sense stationary.
- (b) Find $R_{ZZ}(\tau)$ and $S_{ZZ}(\omega)$.

Solution Given: $X(t)$ and $Y(t)$ are independent wide-sense stationary processes. Therefore,

$$R_{XX}(t, t + \tau) = R_{XX}(\tau)$$

and $R_{YY}(t, t + \tau) = R_{YY}(\tau)$

Also, $Z(t) = X(t) Y(t)$

The mean of $Z(t)$ is

$$E[Z(t)] = E[X(t) Y(t)]$$

Since $X(t)$ and $Y(t)$ are independent,

$$\mu_z = E[X(t) Y(t)]$$

$$= \mu_X \mu_Y$$

The autocorrelation function

$$R_{ZZ}(\tau) = E[Z(t) Z(t + \tau)]$$

$$= E[X(t)Y(t)X(t + \tau)Y(t + \tau)]$$

$$= E[X(t)X(t + \tau)] E[Y(t)Y(t + \tau)]$$

$$= R_{XX}(\tau) R_{YY}(\tau)$$

The PSD of $Z(t)$ is

$$F[R_{ZZ}(\tau)] = F[R_{XX}(\tau) R_{YY}(\tau)]$$

Using convolution property of Fourier transforms, we can write

$$S_{ZZ}(\omega) = S_{XX}(\omega) * S_{YY}(\omega)$$

7.81 If $X(t)$ and $Y(t)$ are uncorrelated and have constant means $\bar{X}$ and $\bar{Y}$ then

$$S_{XY}(\omega) = S_{YX}(\omega) = 2\pi \bar{X} \bar{Y} \delta(\omega).$$

Solution If $X(t)$ and $Y(t)$ are uncorrelated,

$$R_{XY}(t, t + \tau) = \bar{X} \bar{Y}$$

$$\begin{aligned} S_{XY}(\omega) &= \int_{-\infty}^{\infty} \left\{ \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T R_{XY}(t, t + \tau) dt \right\} e^{-j\omega\tau} d\tau \\ &= \int_{-\infty}^{\infty} \left\{ \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T \bar{X} \bar{Y} dt \right\} e^{-j\omega\tau} d\tau \\ &= \bar{X} \bar{Y} \int_{-\infty}^{\infty} e^{-j\omega\tau} d\tau \end{aligned}$$

we have

$$\boxed{\begin{aligned} F[1] &= \int_{-\infty}^{\infty} e^{-j\omega\tau} d\tau \\ &= 2\pi \delta(\omega) \end{aligned}}$$

Therefore,

$$S_{XY}(\omega) = 2\pi \bar{X} \bar{Y} \delta(\omega)$$

7.82 The cross-correlation function of the processes $X(t)$ and $Y(t)$ is given by

$$R_{XY}(t, t + \tau) = \frac{AB}{2} [\sin(\omega_0 T) + \cos \omega_0 (2t + \tau)]$$

where A and B are constants. Find $S_{XY}(\omega)$.

Solution

$$\begin{aligned} S_{XY}(\omega) &= F\{A[R_{XY}(t, t + \tau)]\} \\ &= F\left\{ \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T}^T R_{XY}(t, t + \tau) dt \right\} \\ &= F\left\{ \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T \left[\frac{AB}{2} \sin \omega_0 \tau + \cos \omega_0 (2t + \tau) \right] dt \right\} \\ &= F\left\{ \frac{AB}{2} \sin \omega_0 \tau \left(\frac{2T}{2T} \right) + \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T \cos \omega_0 (2t + \tau) dt \right\} \\ &= F\left[\frac{AB}{2} \sin \omega_0 \tau \right] + \frac{AB}{2} F\left[\lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T \cos (\omega_0 (2t + \tau)) d\tau \right] \\ &= \frac{AB}{2} F[\sin \omega_0 \tau] + \frac{AB}{2} F\left[\lim_{T \rightarrow \infty} \frac{1}{2T} \left(\frac{\sin \omega_0 (2t + \tau)}{\omega_0} \right) \Big|_{-T}^T \right] \end{aligned}$$

The second term in the above equation becomes zero. Therefore,

$$S_{XY}(\omega) = \frac{AB}{2} F[\sin \omega_0 \tau]$$

$$\begin{aligned}
&= \frac{AB}{2} \{-j\pi [\delta(\omega - \omega_0) - \delta(\omega + \omega_0)]\} \\
&= \frac{-j\pi AB}{2} [\delta(\omega - \omega_0) - \delta(\omega + \omega_0)]
\end{aligned}$$

7.83 The power spectral density of zero mean process $X(t)$ is given by

$$S_{XX}(\omega) = \begin{cases} 1 & |\omega| \leq \omega_0 \\ 0 & \text{elsewhere} \end{cases}$$

Find $R_{XX}(\tau)$ and prove that $X(t)$ and $X\left(t + \frac{\pi}{\omega_0}\right)$ are uncorrelated.

Solution

$$\begin{aligned}
R_{XX}(\tau) &= F^{-1} [S_{XX}(\omega)] \\
&= \frac{1}{2\pi} \int_{-\infty}^{\infty} S_{XX}(\omega) e^{j\omega\tau} d\omega = \frac{1}{2\pi} \int_{-\omega_0}^{\omega_0} e^{j\omega\tau} d\omega \\
&= \frac{1}{2\pi j\tau} e^{j\omega\tau} \Big|_{-\omega_0}^{\omega_0} = \frac{1}{\pi\tau} \left(\frac{e^{j\omega_0\tau} - e^{-j\omega_0\tau}}{2j} \right) \\
&= \frac{\sin \omega_0 \tau}{\pi \tau}
\end{aligned}$$

Since $X(t)$ is a zero-mean process,

$$\begin{aligned}
\text{Cov} \left[X(t) X\left(t + \frac{\pi}{\omega_0}\right) \right] &= E \left[X(t) X\left(t + \frac{\pi}{\omega_0}\right) \right] = R_{XX} \left(\frac{\pi}{\omega_0} \right) \\
&= \frac{\sin \omega_0 \left(\frac{\pi}{\omega_0} \right)}{\pi \left(\frac{\pi}{\omega_0} \right)} = 0
\end{aligned}$$

Therefore, $X(t)$ and $X\left(t + \frac{\pi}{\omega_0}\right)$ are uncorrelated.

7.84 A random process has an auto-correlation function

$$P_{XX}(t) = \begin{cases} A[1 - (2\tau/T)] & 0 < \tau \leq T/2 \\ A[1 + (2\tau/T)] & -T/2 \leq \tau < 0 \\ 0 & \tau < -T/2 \text{ and } \tau > T/2 \end{cases}$$

Find and sketch power density spectrum.

Solution The plot of autocorrelation function is shown in Fig. 7.11.

The power density spectrum is given

$$S_{XX}(\omega) = F[R_{XX}(\tau)]$$

$$= \int_{-\infty}^{\infty} R_{XX}(\tau) e^{-j\omega\tau} d\tau$$

$$= A \left[\int_{-T/2}^0 \left(1 + \frac{2\tau}{T}\right) e^{-j\omega\tau} d\tau + \int_0^{T/2} \left(1 - \frac{2\tau}{T}\right) e^{-j\omega\tau} d\tau \right]$$

$$= A \left\{ \frac{2}{\tau} \left[\frac{\tau e^{-j\omega\tau}}{-j\omega} - \frac{1}{(j\omega)^2} e^{-j\omega\tau} \right] \right\}_{-T/2}^0 + \frac{A e^{-j\omega\tau}}{-j\omega} \Big|_{-T/2}^0$$

$$+ \frac{-2A}{\tau} \left\{ \left[\frac{\tau e^{-j\omega\tau}}{-j\omega} - \frac{1}{(j\omega)^2} e^{-j\omega\tau} \right] \right\}_0^{T/2} + \frac{A e^{-j\omega\tau}}{-j\omega} \Big|_0^{T/2}$$

$$= \frac{2A}{\tau} \left[\frac{-T e^{j\omega T/2}}{2j\omega} + \frac{T e^{j\omega T/2}}{2j\omega} - \frac{2}{(j\omega)^2} + \frac{e^{j\omega T/2}}{(j\omega)^2} + \frac{e^{-j\omega T/2}}{(j\omega)^2} \right] + \frac{A}{j\omega} [e^{j\omega T/2} - e^{-j\omega T/2}]$$

$$= \frac{2A}{\tau} \left[\frac{e^{j\omega T/2}}{(j\omega)^2} + \frac{e^{-j\omega T/2}}{(j\omega)^2} - \frac{2}{(j\omega)^2} \right]$$

$$= \frac{2A}{\tau} \left(\frac{e^{j\omega T/4} - e^{-j\omega T/4}}{j\omega} \right)^2 = \frac{8A}{T\omega^2} \sin^2 \left(\frac{\omega T}{4} \right)$$

$$= \frac{2\tau^2}{16} \cdot \frac{8}{T} \frac{\sin^2 \frac{\omega T}{4}}{\left(\frac{\omega T}{4} \right)^2} = \frac{AT}{2} \left(\frac{\sin \frac{\omega T}{4}}{\frac{\omega T}{4}} \right)^2$$

$$= \frac{AT}{2} \sin^2 \left(\frac{\omega T}{4} \right)$$

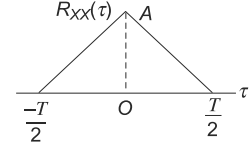


Fig. 7.11

7.85 The ACF of a random process $X(t)$ is given by

$$R_{XX}(\tau) = e^{-\alpha|\tau|} \cos(\omega_0 \tau)$$

where $\alpha > 0$ and ω_0 are real constants. Find $S_{XX}(\omega)$.

Solution $S_{XX}(\omega) = F[R_{XX}(\tau)]$

$$= F[e^{-\alpha|\tau|} (\cos \omega_0 \tau)]$$

We have $F[e^{-\alpha|\tau|}] = \frac{2\alpha}{\alpha^2 + \omega^2}$

Using the modulation theorem of Fourier transform,

we have $F[f(t) \cos \omega_0 t] = \frac{1}{2} [F(\omega + \omega_0) + F(\omega - \omega_0)]$

Therefore,

$$F[e^{-\alpha|\tau|} \cos(\omega_0 t)] = \frac{1}{2} \left[\frac{2\alpha}{\alpha^2 + (\omega + \omega_0)^2} + \frac{2\alpha}{\alpha^2 + (\omega - \omega_0)^2} \right]$$

$$\Rightarrow S_{XX}(\omega) = \frac{\alpha}{\alpha^2 + (\omega + \omega_0)^2} + \frac{\alpha}{\alpha^2 + (\omega - \omega_0)^2}$$

REVIEW QUESTIONS



22. State any two properties of cross-correlation function.
23. Define bandlimited white noise.
24. Find the mean of the stationary process $X(t)$, whose autocorrelation function is given by

$$R_{XX}(\tau) = 16 + \frac{9}{1 + 6\tau^2}$$

25. Find the power spectral density function of the stationary process whose autocorrelation function is given by $e^{-|\tau|}$.
26. What is autocorrelation function of the white noise?
27. If $X(t)$ is a normal process with $\mu_X(t) = 10$ and $C_{XX}(t_1, t_2) = 16 e^{-|t_1 - t_2|}$, find the variance of $X(10) - X(6)$.
28. The autocorrelation function of a stationary random process is

$$R_{XX}(t) = 25 + \frac{4}{1 + 6\tau^2}. \text{ Find the mean and variance of the process.}$$

29. Explain power spectral density function. State its important properties.
30. Given the power spectral density of a stationary process as $S_{XX}(\omega) = \frac{1}{4 + \omega^2}$, find the average power of the process.
31. State and prove Wiener–Khinchin relations.
32. State and prove any three properties of cross-power density spectrum.
33. Derive the relation between cross-power spectrum and cross-correlation function.



EXERCISES

Problems

1. Find the autocorrelation function of the periodic time function $X(t) = A \sin \omega t$.
2. The autocorrelation function of the random binary transmission $X(t)$ is given by $R_{XX}(\tau) = 1 - \frac{|\tau|}{T}$ for $|\tau| < T$ and $R_{XX}(\tau) = 0$ for $|\tau| > T$. Find the power spectrum of the process $X(t)$.

3. $X(t)$ and $Y(t)$ are zero-mean and stochastically independent random processes having autocorrelation functions $R_{XX}(\tau) = e^{-|\tau|}$ and $R_{YY}(\tau) = \cos 2\pi\tau$ respectively. Find (a) the autocorrelation function of $W(t) = X(t) + Y(t)$ and $Z(t) = X(t) - Y(t)$, (b) The cross-correlation function of $W(t)$ and $Z(t)$.
4. Find the ACF of the process $X(t)$ for which the PDS is given by $S_{XX}(\omega) = 1 + \omega^2$ for $|\omega| < 1$ and $S_{XX}(\omega) = 0$ for $|\omega| > 1$.
5. Find the variance of the stationary process $X(t)$ whose autocorrelation function is given by $R_{XX}(\tau) = 2 + 4e^{-2|\tau|}$.
6. If $X(t)$ and $Y(t)$ are two random process with autocorrelation function $R_{XX}(\tau)$ and $R_{YY}(\tau)$ respectively then prove that $|R_{XY}(\tau)| \leq \sqrt{R_{XX}(0) R_{YY}(0)}$.
7. Find the power spectral density of the random process whose autocorrelation function is

$$R_{XX}(\tau) = \begin{cases} 1 - |\tau| & \text{for } |\tau| \leq 1 \\ 0 & \text{elsewhere} \end{cases}$$

8. The ACF of a random process is given by

$$R_{XX}(\tau) = \begin{cases} \lambda^2 & i|\tau| > \varepsilon \\ \lambda^2 + \frac{\lambda}{\varepsilon} \left(1 - \frac{|\tau|}{\varepsilon}\right); & |\tau| \leq \varepsilon \end{cases}$$

Find the power spectral density of the process.

9. Given the power spectral density of a continuous process as

$$S_{XX}(\omega) = \frac{\omega^2 + 9}{\omega^4 + 5\omega^2 + 4}$$

find the mean square value of the process.

10. The cross-power spectrum of real random processes $X(t)$ and $Y(t)$ is given by

$$S_{XY}(\omega) = \begin{cases} a + jb\omega, & \text{for } |\omega| < 1 \\ 0 & \text{elsewhere} \end{cases}$$

Find the cross-correlation function.

11. If the PSD of $X(t)$ is $S_{XX}(\omega)$, find the PSD of $\frac{dX(t)}{dt}$.
12. Prove that $S_{XX}(\omega) = S_{XX}(-\omega)$.
13. If $R_{YY}(\tau) = ae^{-b|\tau|}$, find the spectral density function, where a and b are constants.
14. Given $R_{XX}(\tau) = \frac{A_0^2}{2} \sin \omega_0 \tau$. Find $S_{XX}(\omega)$.
15. Let $X(t)$ be a stationary continuous random process that is differentiable. Denote its time derivative $\dot{X}(t)$ show that $E[\dot{X}(t)] = 0$
Find $F[R_{\dot{X}\dot{X}}(\tau)]$ in terms of $S_{XX}(\omega)$.
16. Find the ACF of the following PSDs.

$$S_{XX}(\omega) = \frac{157 + 2\omega^2}{(16 + \omega^2)(9 + \omega^2)}$$

$$S_{XX}(\omega) = \frac{8}{(9 + \omega^2)^2}$$

17. Find the mean square value of the process whose power spectral density is

$$S_{XX}(\omega) = \frac{1}{\omega^4 + 10\omega^2 + 9} \quad (\text{Ans. } 1/24)$$

18. Check whether $S_{XX}(\omega) = \frac{2\omega^2 + 6}{8\omega^2 + 3\omega^2 + 4}$ can represent a valid power spectral density function. (Ans. A valid PSD)

19. Find the mean square value of the process whose power density spectrum is

$$S_{XX}(\omega) = \frac{\omega^2 + 2}{\omega^4 + 5\omega^2 - 36} \quad (\text{Ans. } 1/2)$$

20. Find the power spectral density for the stationary process $X(t)$ with autocorrelation function

$$R_{XX}(\tau) = \sigma^2 e^{-\alpha|\tau|}$$

21. The power spectral density of a stationary random process is $S_{XX}(\omega) = \frac{\omega^2 + 1}{\omega^4 + 4\omega^2 + 4}$. Find the autocorrelation function and average power of the process.

$$(\text{Ans. } R_{XX}(\tau) = \frac{1}{2} e^{2|\tau|} - \tau e^{-|\tau|}; R_{XX}(0) = 1/2)$$

22. If the power spectral density of a WSS process is given by

$$S_{XX}(\omega) = \begin{cases} \frac{b}{a} [a - |\omega|] & ; \quad |\omega| \leq a \\ 0 & ; \quad |\omega| > a \end{cases}$$

find the autocorrelation of the process.

23. A stationary random process $X(t)$ has an ACF given by

$$R_{XX}(\tau) = 2e^{-2|\tau|} + 3e^{-3|\tau|}$$

Find the power spectral density of the process.

24. A zero-mean wide-sense stationary random process $X(t)$, $-\infty < t < \infty$, has the following power spectral density

$$S_{XX}(\omega) = \frac{2}{1 + \omega^2} \quad -\infty < \omega < \infty$$

The random process $Y(t)$ is defined by

$$Y(t) = X(t) + X(t - 2)$$

Find the mean and variance of $Y(t)$.

25. Two jointly stationary random processes $X(t)$ and $Y(t)$ have the cross-correlation function given by

$$R_{XY}(\tau) = 16e^{-4\tau} \quad \tau \geq 0.$$

Find $S_{XY}(\omega)$ and $S_{YX}(\omega)$.

26. A random process has a power spectral density function given by

$$S_{XX}(\omega) = \frac{1}{\left[1 + \left(\frac{\omega}{B}\right)^2\right]^3}$$

Find the bandwidth.

27. A random process has a power spectral density function given by

$$S_{XX}(\omega) = \frac{\omega^2}{\left[1 + \left(\frac{\omega}{\beta}\right)^2\right]^3}$$

Find rms bandwidth.

28. Determine which of the following functions can and cannot be valid power density spectrum.

(a) $\frac{\cos 5\omega}{1 + \omega^2}$

(b) $\frac{|\omega|}{1 + 3\omega + \omega^2}$

(c) $\frac{1}{\sqrt{1 - 4\omega^2}}$

(d) $\delta(\omega - 3) + \delta(\omega + 3) - e^{-|\omega|} \cos^2(8\omega)$

29. The random process $X(t) = A_0 \sin(\omega_0 t + \theta)$ where A_0 and ω_0 are constants and θ is a random variable uniformly distributed on the intervals $(0, \pi)$. (a) Is $X(t)$ wide-sense stationary? (b) Find the power density spectrum.

30. Find the rms bandwidth of the power spectrum

$$S_{XX}(\omega) = \begin{cases} A \cos\left(\frac{\pi\omega}{2W}\right); & |\omega| \leq W \\ 0 & |\omega| > W \end{cases}$$

where $W > 0$ and $A > 0$ are constants.

31. The autocorrelation function of a random process $X(t)$ is

$$R_{XX}(t) = 1 + 4 \exp(-2t^2)$$

- (a) Find the power spectrum of $X(t)$
 (b) What is the average power in $X(t)$?

32. State whether or not each of the following functions can be a valid power-density spectrum.

(a) $\frac{|\omega| e^{-\omega^2}}{1 + j\omega}$

(b) $\frac{\omega^4}{(8 + \omega^2)^4}$

(c) $\cos^2(\omega) e^{-\omega^2}$

(d) $e^{-2 \sin(2\omega)}$

33. A random process has a power spectrum

$$S_{XX}(\omega) = \begin{cases} 8 - \frac{\omega^2}{16}; & |\omega| \leq 8 \\ 0 & \text{elsewhere} \end{cases}$$

Find (a) the average power, and (b) the rms bandwidth.

34. Find the power spectrum corresponding to the autocorrelation function

$$R_{XX}(t, t + \tau) = e^{-\tau^2} \cos^2(8t)$$

35. A random process is given by

$$W(t) = AX(t) + B(Y(t))$$

If $X(t)$ and $Y(t)$ are uncorrelated, find the power spectrum of $W(t)$ in terms of power spectrum of $X(t)$ and $Y(t)$.

36. A sample function of a random binary waveform can take two values 1 or -1 . The transition from one values to another occurs once every T_0 seconds. The probability of transition from one state to the other is equally likely. Find power spectral density of the process.

Multiple-Choice Questions

- For a signal $x(t)$, Fourier transform exists if it satisfies the condition
 - $\int_{-\infty}^{\infty} |x(t)|^2 dt < \infty$
 - $\int_{-\infty}^{\infty} |x(t)| dt < \infty$
 - $\int_{-\infty}^{\infty} x(t) dt < \infty$
 - $\int_{-\infty}^{\infty} x(t^2) dt < \infty$
- The power density spectrum satisfies the following condition:
 - $S_{XX}(\omega) = -S_{XX}(-\omega)$
 - $S_{XX}(\omega) = S_{XX}(-\omega)$
 - $S_{XX}(\omega) = \infty$ at $\omega = 0$
 - $S_{XX}(\omega) = S_{XX}(\omega^2)$
- $S_{\dot{X}\dot{X}}(\omega) =$
 - $\omega S_{XX}(\omega)$
 - $-\omega^2 S_{XX}(\omega)$
 - $\omega^2 S_{XX}(\omega)$
 - $\omega^2 S_{\dot{X}\dot{X}}(\omega)$
- If $S_{XX}(\omega)$ is the power density spectrum of $X(t)$ then the power density spectrum of $\alpha X(t)$ is
 - $\alpha S_{XX}(\omega)$
 - $\alpha^2 S_{XX}(\omega)$
 - $-\alpha S_{XX}(\omega)$
 - $|\alpha|^2 S_{XX}(\omega)$
- Which of the following power density spectrum is/are valid?
 - $\frac{\omega}{\omega^2 + 16}$
 - $\frac{1}{3\omega^4 + 9\omega^2}$
 - $\frac{|\omega| e^{-8\omega^2}}{1 + j2\omega}$
 - $\frac{\omega + 2}{\omega^4 + \omega^2}$
- The mean square value of the process whose power density spectrum $\frac{4}{4 + \omega^2}$ is
 - 1
 - $\frac{1}{2}$
 - $\frac{1}{4}$
 - 2
- The rms bandwidth of the process whose ACF is $2e^{-bt}$ is
 - 1 Hz
 - $\frac{3}{4}$ Hz
 - $\frac{5}{4}$ Hz
 - 2 Hz
- The PSD of a random process whose $R_{XX}(\tau) = e^{-bt}$ is
 - $\frac{1}{b^2 + \omega^2}$
 - $\frac{2b}{b^2 + \omega^2}$
 - $\frac{b}{2(b^2 + \omega^2)}$
 - $\frac{b}{b^2 + \omega^2}$
- If $X(t)$ and $Y(t)$ are orthogonal then
 - $S_{XY}(\omega) = 1$
 - $S_{XY}(\omega) = 0$
 - $S_{XY}(\omega) = S_{YX}(\omega)$
 - $S_{XY}(\omega) \leq 1$
- The cross-power spectrum $S_{XY}(\omega) =$
 - $S_{XY}(-\omega)$
 - $-S_{XY}(-\omega)$
 - $S_{YX}(-\omega)$
 - $-S_{YX}(-\omega)$
- If $S_{XX}(\omega) = \frac{\omega^2 + 4}{\omega^4 + 8\omega^2 + 7}$ the ACF is
 - $\frac{1}{2} e^{-|\tau|} + \frac{1}{2\sqrt{7}} e^{-\sqrt{7}|\tau|}$
 - $\frac{1}{4} e^{-|\tau|} + \frac{1}{4\sqrt{7}} e^{-\sqrt{7}|\tau|}$
 - $e^{-|\tau|} + \frac{1}{4\sqrt{7}} e^{-\sqrt{7}|\tau|}$
 - $\frac{1}{4} e^{-|\tau|} + \frac{1}{2\sqrt{7}} e^{-\sqrt{7}|\tau|}$
- The average power of the random process with power spectrum $S_{XX}(\omega) = \frac{3\omega^2 + 4}{2\omega^2 + 6\omega^2 + 4}$ is
 - $\frac{1}{4} + \frac{1}{2\sqrt{2}}$
 - $\frac{1}{2} + \frac{1}{\sqrt{2}}$
 - $\frac{1}{2} + \frac{1}{2\sqrt{2}}$
 - $\frac{1}{4} + \frac{1}{\sqrt{2}}$

13. The power spectral density of a stationary process whose autocorrelation function is $e^{-|\tau|}$ is
 (a) $\frac{1}{1+\omega^2}$ (b) $\frac{2}{1+\omega^2}$ (c) $\frac{2}{(1+\omega^2)^2}$ (d) $\frac{1}{(1+\omega^2)^2}$
14. The power spectral density of the random process whose ACF is $R_{XX}(\tau) = \begin{cases} 1-|\tau|; & |\tau| \leq 1 \\ 0 & \text{elsewhere} \end{cases}$
 (a) $\frac{2}{\omega^2} (1 - \sin \omega)$ (b) $\frac{2}{\omega^2} (1 - \cos \omega)$ (c) $\frac{2}{\omega^2} \sin^2 \frac{\omega}{2}$ (d) $\frac{2}{\omega^2} \cos^2 \frac{\omega}{2}$
15. The cross-power density spectrum of $X(t)$ and $Y(t)$ is the $S_{XY}(\omega) = \frac{8}{(\alpha + j\omega)^3}$ cross-relation function is
 (a) $\tau e^{-\alpha\tau}$ (b) $2\tau e^{-\alpha\tau} u(\tau)$ (c) $4\tau^2 e^{-\alpha\tau} u(\tau)$ (d) $\tau^2 e^{-\alpha\tau} u(\tau)$
16. The variance of the process whose ACF is given below is $R_{XX}(t) = 2 + 4e^{-2|t|}$
 (a) 4 (b) 2 (c) 16 (d) 8
17. If $X(t)$ and $Y(t)$ are wide-sense stationary random processes which is true,
 (a) $|R_{XX}(\tau)| \leq \sqrt{R_{XX}(0) R_{YY}(0)}$ (b) $R_{XX}(\tau) \leq [R_{XX}(0) + R_{YY}(0)]$
 (c) $|R_{XX}(\tau)| = R_{XX}(0) R_{YY}(0)$ (d) none of the above
18. If $X(t)$ and $Y(t)$ are un-correlated and have constant means μ_X and μ_Y then $S_{XX}(\omega) =$
 (a) $XY \mu_X \mu_Y$ (b) $2\pi \mu_X \mu_Y$ (c) $2\pi \mu_X \mu_Y \delta(\omega)$ (d) $(\mu_X + \mu_Y) \delta(\omega)$
19. The rms bandwidth W_{rms}^2 in terms of power spectral density is
 (a) $\frac{\int_{-\infty}^{\infty} \omega^2 S_{XX}(\omega) d\omega}{\int_{-\infty}^{\infty} S_{XX}(\omega) d\omega}$ (b) $\frac{\int_{-\infty}^{\infty} \omega S_{XX}(\omega) d\omega}{\int_{-\infty}^{\infty} S_{XX}(\omega) d\omega}$ (c) $\frac{\int_{-\infty}^{\infty} S_{XX}(\omega) d\omega}{\int_{-\infty}^{\infty} \omega^2 S_{XX}(\omega) d\omega}$ (d) $\frac{\int_{-\infty}^{\infty} \omega^2 S_{XX}(\omega) d\omega}{\int_{-\infty}^{\infty} \omega S_{XX}(\omega) d\omega}$
20. The PSD of a bandlimited white noise is given by

$$S_{NN}(\omega) = \begin{cases} \frac{A\pi}{W}; & -W < \omega < W \\ 0 & \text{elsewhere} \end{cases}$$

Its autocorrelation function is

- (a) $\frac{A \sin^2(W\tau)}{W\tau}$ (b) $A \frac{\sin(W\tau)}{W\tau}$ (c) $A \frac{\sin^2(W\tau)}{(W\tau)^2}$ (d) $A \frac{\sin \frac{W\tau}{2}}{\frac{\omega\tau}{2}}$
21. The average power of a random process with power spectral density $S_{XX}(\omega)$ is
 (a) $\int_{-\infty}^{\infty} S_{XX}(\omega) d\omega$ (b) $\int_0^{\infty} S_{XX}(\omega) d\omega$ (c) $\frac{1}{2\pi} \int_0^{\infty} S_{XX}(\omega) d\omega$ (d) $\frac{1}{2\pi} \int_{-\infty}^{\infty} S_{XX}(\omega) d\omega$

22. The power spectral density of a WSS process is given by $S_{XX}(\omega) = 1; |\omega| < \omega_0$
The autocorrelation function is given by
- (a) $\frac{\sin \omega_0 \tau}{\pi \tau}$ (b) $\frac{\sin^2 \omega_0 \tau}{\pi \tau}$ (c) $\frac{\sin^2 \omega_0 \tau}{\pi^2 \tau^2}$ (d) $\frac{\sin^2 \omega_0 \tau/2}{\pi \tau/2}$
23. The PSD of a wide-sense process is always
- (a) finite (b) zero (c) negative (d) non-negative
24. The average power of a waveform $X(t) = A \cos(\omega_0 t + \theta)$ is
- (a) $\frac{A}{\sqrt{2}}$ (b) $\frac{A^2}{4}$ (c) $\frac{A^2}{2}$ (d) A^2
25. The autocorrelation function of white noise is given by
- (a) $R_{NN}(\tau) = S_0$ (b) $R_{NN}(\tau) = S_0 \delta(\tau)$ (c) $R_{NN}(\tau) = S_0 u(\tau)$ (d) $R_{NN}(\tau) = S_0^2 \delta(\tau)$

Answers to Multiple-Choice Questions

- | | | | | | | | | | |
|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| 1. (b) | 2. b | 3. (c) | 4. (d) | 5. (b) | 6. (a) | 7. (b) | 8. (b) | 9. (b) | 10. (c) |
| 11. (b) | 12. (a) | 13. (b) | 14. (b) | 15. (c) | 16. (a) | 17. (a) | 18. (c) | 19. (a) | 20. (b) |
| 21. (d) | 22. (a) | 23. (d) | 24. (c) | 25. (b) | | | | | |

LINEAR SYSTEMS WITH RANDOM INPUTS



INTRODUCTION 8.1

In the previous chapters, we studied about random signals and random processes. We know that a random signal can be modelled as a sample function of a random process. Such random signals are described mathematically using time-domain statistical measures like mean and autocorrelation. The frequency-domain method used to describe a random signal includes power density spectrum which describes how the average power is distributed across frequencies. Communication and control system engineers deal with noise which is random. When a signal containing noise is applied to a communication system, it is necessary to know the effect of noise on the performance of the system. Therefore, it is necessary to study the output of a system for a random input. In this chapter, we will concentrate on this aspect. First a brief introduction to system analysis is introduced in the following section.



FUNDAMENTALS OF LINEAR SYSTEMS 8.2

A system is defined as a physical device that performs an operation on an input signal and produces another signal as output. Examples for systems include amplifier, filter and mass-spring, dashpot system etc. The box shown in Fig. (8.1) represents a system. An arrow entering the box is the input signal which is represented by $x(t)$ and the arrow leaving the box represents the output signal which is denoted by $y(t)$. We can say that the output $y(t)$ is the transformation of the input $x(t)$.

The relation between the input and corresponding output $y(t)$ of a system has the form

$$y(t) = \text{operation on } x(t)$$

$$\text{Mathematically, } y(t) = T\{x(t)\} \quad (8.1)$$

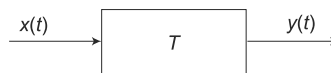


Fig. 8.1 Linear system

Systems can be classified in many ways. The important classification is

1. Causal and non-causal systems
2. Linear and non-linear systems
3. Time-invariant and time-variant systems

8.2.1 Causal and Non-causal Systems

A *causal system* is one for which the output at any time t depends on the present and past inputs but not future inputs. These systems are also known as non-anticipative systems.

Examples for causal systems are

$$y(t) = |x(t)|$$

$$y(n) = x(n) + x(n-1) + 2x(n-2)$$

A non-causal system is one whose output depends on future values.

Examples for non-causal systems are

$$y(t) = x(3t) + x^2(t)$$

$$y(n) = x(n) + x(n+1)$$

8.2.2 Linear and Non-linear Systems

A system is said to be a linear system if it obeys the superposition principle which states that the response to a weighted sum of input signals is equal to the corresponding weighted sum of the outputs of the system to each of the individual input signals.

Let us assume that an arbitrary input $x_1(t)$ produces an output $y_1(t)$ and an arbitrary input $x_2(t)$ produces an output $y_2(t)$. Let us consider the weighted sum of inputs $ax_1(t) + bx_2(t)$ where a and b are constants. If this input produces an output $ay_1(t) + by_2(t)$ then the system is linear. That is, for a linear system

$$T\{ax_1(t) + bx_2(t)\} = aT[x_1(t)] + bT[x_2(t)] \quad (8.2)$$

If the above condition is not satisfied then the system is a non-linear system.

Similarly, for a discrete-time linear system

$$T\{ax_1(n) + bx_2(n)\} = aT[x_1(n)] + bT[x_2(n)] \quad (8.3)$$

From the superposition property, we can find that a zero input results a zero output.

8.2.3 Time-invariant and Time-variant Systems

A system is said to be time-invariant if its input-output characteristics do not change with time. Let us assume that the input signal and output signal of system are $x(t)$ and $y(t)$ respectively. If we delay the input by T seconds, then for a time-invariant system the output will also be delayed by T seconds.

To test the time-invariant property of a system, we apply a signal $x(t)$ and find $y(t)$. Now we delay the input signal by T seconds and then find the output signal and denote it as

$$y(t, T) = T[x(t - T)] \quad (8.4)$$

Delay the output signal by T seconds and denote it as $y(t - T)$.

If

$$y(t, T) = y(t - T) \quad (8.5)$$

for all values of T then the system is time-invariant. On the other hand, if the output

$$y(t, T) \neq y(t - T) \quad (8.6)$$

then the system is time-variant.

To test the time invariance property of the given discrete-time system, first apply an arbitrary input sequence $x(n)$ and find $y(n)$. Delay the input sequence by k samples and find the output sequence, denote it as

$$y(n, k) = T[x(n - k)] \quad (8.7)$$

Delay the output sequence by k samples denote it as $y(n - k)$

$$\text{If } y(n, k) = y(n - k) \quad (8.8)$$

for all possible values of k then the system is time invariant, otherwise the system is time variant.



THE RESPONSE OF AN LTI SYSTEM 8.3

Consider an LTI system shown in Fig. 8.2. If the input to the system is an impulse then the output of the system is known as impulse response denoted by $h(t)$. That is,

$$h(t) = T[\delta(t)] \quad (8.9)$$

Any arbitrary signal $x(t)$ can be represented as linear combination of scaled and shifted unit impulse functions. That is,

$$x(t) = \int_{-\infty}^{\infty} x(\tau) \delta(t - \tau) d\tau \quad (8.10a)$$

The system output is given by $y(t) = T[x(t)]$

Substituting Eq. (8.10a) in Eq. (8.10b) yields

$$y(t) = T \left[\int_{-\infty}^{\infty} x(\tau) \delta(t - \tau) d\tau \right] \quad (8.10b)$$

For a linear system,

$$y(t) = \int_{-\infty}^{\infty} x(\tau) T[\delta(t - \tau)] d\tau \quad (8.11a)$$

If the response of the system due to impulse $\delta(t)$ is $h(t)$ then the response of the system due to delayed impulse is

$$h(t, T) = T[\delta(t - T)] \quad (8.11b)$$

Substituting Eq. (8.11b) in Eq. (8.11a), we get

$$y(t) = \int_{-\infty}^{\infty} x(\tau) h(t, \tau) d\tau \quad (8.11c)$$

For a time-invariant system, the output due to delayed input by T seconds is equal to delayed output by T seconds.



Fig. 8.2 LTI System

That is, $h(t, \tau) = h(t - \tau)$

Therefore, for a linear time-invariant (LTI) system,

$$y(t) = \int_{-\infty}^{\infty} x(\tau)h(t - \tau)d\tau \quad (8.12a)$$

This is called convolution integral, or simply convolution. The convolution of two signals $x(t)$ and $h(t)$ can be represented as

$$y(t) = x(t) * h(t) \quad (8.12b)$$

Properties of Convolution

Consider a system with impulse response $h(t)$ and input $x(t)$. Then the output of the system

$$y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau)h(t - \tau)d\tau$$

1. Commutative Property Convolution obeys commutative property. That is,

$$x(t) * h(t) = h(t) * x(t) \quad (8.13a)$$

2. Distributive Property $x(t) * [h_1(t) + h_2(t)] = x(t) * h_1(t) + x(t) * h_2(t)$ (8.13b)

3. Associative property $x(t) * [h_1(t) * h_2(t)] = [x(t) * h_1(t)] * h_2(t)$ (8.13c)



CAUSAL AND STABLE SYSTEMS 8.4

A system is said to be causal if the output depends on present and past values of input. Consider a continuous-time LTI system whose output $y(t)$ can be obtained using convolution integral given by

$$y(t) = \int_{-\infty}^{\infty} h(\tau)x(t - \tau)d\tau \quad (8.14)$$

In the above equation, the output $y(t)$ depends on past, present and future values of input. For $\tau \geq 0$, the output depends on present and past values of input. For $\tau < 0$, the output depends on future values of input. But for a causal system, the output depends only on present and past values of input which implies that

$$h(\tau) = 0 \text{ for } \tau < 0. \quad (8.15)$$

An LTI continuous system is causal if and only if its impulse response is zero for negative values of t .

Stability

A system is said to be BIBO stable if it produces a bounded output for a bounded input.

Let us consider an input signal $x(t)$ that has a bounded magnitude M . That is,

$$|x(t)| \leq M < \infty$$

Then from convolution integral, the magnitude of the output can be expressed as

$$|y(t)| = \left| \int_{-\infty}^{\infty} h(\tau)x(t - \tau)d\tau \right|$$

$$\begin{aligned} &\leq \int_{-\infty}^{\infty} |h(\tau)| |x(t-\tau)| d\tau \\ &\leq M \int_{-\infty}^{\infty} |h(\tau)| d\tau \end{aligned}$$

The output $|y(t)|$ is bounded when

$$\int_{-\infty}^{\infty} |h(\tau)| d\tau < \infty \quad (8.16)$$

Therefore, the system is stable if the impulse response is absolutely integrable.

REVIEW QUESTIONS



1. Define a system.
2. State superposition principle.
3. Explain in detail about the classification of systems.
4. Derive an expression for output of a system for any arbitrary input.
5. What is convolution integral?
6. State properties of convolution.
7. What is the condition on impulse response for (a) causality, and (b) stability?

Solved Problems



8.1 Determine which of the following systems are linear, time invariant, stable and realizable.

(a) $y(t) = |x(t)|$ (b) $y(t) = x(t/2)$ (c) $y(t) = x(t)$ (d) $y(t) = tx(t)$.

Solution

Linearity Check

(a) Given: $y(t) = |x(t)|$

$$y_1(t) = |x_1(t)| \text{ and } y_2(t) = |x_2(t)|$$

The output due to weighted sum of inputs is

$$y_3(t) = |a_1x_1(t) + b_1x_2(t)|$$

The weighted sum of outputs is

$$a_1y_1(t) + b_1y_2(t) = a_1|x_1(t)| + b_1|x_2(t)|$$

$$y_3(t) \neq a_1y_1(t) + b_1y_2(t)$$

Hence, the system is nonlinear.

Time Invariant Check

Given: $y(t) = |x(t)|$

The output due to delayed input is

$$y(t, T) = |x(t - T)|$$

The delayed output is $y(t - T) = |x(t - T)|$

$$y(t, T) = y(t - T). \text{ Therefore, the system is time invariant.}$$

Stability Check

Let the input $x(t)$ is bounded satisfying $|x(t)| \leq M_x < \infty$. Then the output $y(t) = |x(t)| < \infty$. Therefore, the system is stable.

Reliability Check

The output depends on the present value of the input. Therefore, the system is realizable.

(b) Given: $y(t) = x(t/2)$

Linear Check

$$y_1(t) = x_1(t/2) \text{ and } y_2(t) = x_2(t/2)$$

The output due to weighted sum of inputs is

$$y_3(t) = a_1 x_1(t/2) + b_1 x_2(t/2)$$

The weighted sum of outputs is

$$a_1 y_1(t) + b_1 y_2(t) = a_1 x_1(t/2) + b_1 x_2(t/2)$$

$$y_3(t) = a_1 y_1(t) + b_1 y_2(t)$$

Hence system is linear.

Time Invariant Check

Given: $y(t) = x(t/2)$

The output due to delayed input is

$$y(t, T) = x\left(\frac{t}{2} - T\right)$$

The delayed output is $y(t - T) = x((t - T)/2)$

$$y(t, T) \neq y(t - T). \text{ Therefore the system is time-variant.}$$

Stability Check

Let the input $x(t)$ be bounded satisfying $|x(t)| \leq M_x < \infty$. Then the magnitude of the output

$$|y(t)| = |x(t/2)| < \infty$$

Therefore, the system is stable.

Reliability Check

The output depends on the future values of the input. Therefore, the system is not realizable.

(c) Given: $y(t) = e^{x(t)}$

Linear Check

$$y_1(t) = e^{x_1(t)} \text{ and } y_2(t) = e^{x_2(t)}$$

The output due to weighted sum of inputs is

$$y_3(t) = e^{a_1 x_1(t) + b_1 x_2(t)}$$

The weighted sum of outputs is

$$a_1 y_1(t) + b_1 y_2(t) = a_1 e^{x_1(t)} + b_1 e^{x_2(t)}$$

$y_3(t) \neq a_1 y_1(t) + b_1 y_2(t)$. Hence, the system is nonlinear.

Time Invariant Check

Given: $y(t) = e^{x(t)}$

The output due to delayed input is

$$y(t, T) = e^{x(t-T)}$$

The delayed output is $y(t-T) = e^{x(t-T)}$

$y(t, T) = y(t-T)$. Therefore, the system is time invariant.

Stability Check

Let the input $x(t)$ be bounded satisfying $|x(t)| \leq M_x < \infty$. Then the magnitude of the output $|y(t)| = |e^{x(t)}| < \infty$. Therefore, the system is stable.

Reliability Check

The output depends on present values of the input. Therefore, the system is realizable.

(ii) Given: $y(t) = tx(t)$

$$y_1(t) = tx_1(t) \text{ and } y_2(t) = tx_2(t)$$

The output due to weighted sum of inputs is

$$y_3(t) = a_1 tx_1(t) + b_1 tx_2(t)$$

The weighted sum of outputs is

$$a_1 y_1(t) + b_1 y_2(t) = a_1 tx_1(t) + b_1 tx_2(t)$$

$$y_3(t) = a_1 y_1(t) + b_1 y_2(t). \text{ Hence, the system is linear.}$$

Time Invariant Check

Given: $y(t) = tx(t)$

The output due to delayed input is

$$y(t, T) = t x(t-T)$$

The delayed output is $y(t-T) = (t-T) x(t-T)$

$y(t, T) \neq y(t-T)$. Therefore, the system is time-variant.

Stability Check

Let the input $x(t)$ be bounded satisfying $|x(t)| \leq M_x < \infty$. Then the magnitude of the output $|y(t)| = |tx(t)|$ tends to infinity as $t \rightarrow \infty$.

Therefore, the system is unstable.

Reliability Check

The output depends on present values of the input. Therefore, the system is realizable.

8.2 Find whether the following systems with the given impulse response are stable/realizable.

(a) $h(t) = e^{-|t|}$ (b) $h(t) = tu(t)$ (c) $h(t) = e^t \cos(\omega_0 t)u(t)$.

Solution

Given:

$$(a) \quad h(t) = e^{-|t|}$$

For the system to be stable, the impulse response must be absolutely summable.

$$\text{That is, } \int_{-\infty}^{\infty} |h(t)| dt < \infty.$$

$$\begin{aligned} \int_{-\infty}^{\infty} |h(t)| dt &= \int_{-\infty}^{\infty} e^{-|t|} dt = \int_{-\infty}^0 e^{-|t|} dt + \int_0^{\infty} e^{-|t|} dt = \int_{-\infty}^0 e^t dt + \int_0^{\infty} e^{-t} dt \\ &= e^t \Big|_{-\infty}^0 - e^{-t} \Big|_0^{\infty} = 2 < \infty \end{aligned}$$

Therefore, the system is stable.

The impulse response is not equal to zero for $t < 0$. Therefore, the system is not realizable.

$$(b) \quad h(t) = tu(t)$$

$$\int_{-\infty}^{\infty} |h(t)| dt = \int_{-\infty}^{\infty} |tu(t)| dt = \int_0^{\infty} t dt = \frac{t^2}{2} \Big|_0^{\infty} = \infty$$

Therefore, the system is unstable.

The impulse response is equal to zero for $t < 0$. Therefore, the system is realizable.

$$(c) \quad h(t) = e^t \cos(\omega_0 t) u(t)$$

$$\int_{-\infty}^{\infty} |h(t)| dt = \int_{-\infty}^{\infty} |e^t \cos \omega_0 t u(t)| dt = \int_0^{\infty} |e^t \cos \omega_0 t| dt = \infty$$

Therefore, the system is unstable. The impulse response is equal to zero for $t < 0$. Therefore, the system is realizable.

8.3 Determine which of the following impulse responses do not correspond to a system that is stable or realizable or both and state why?

- (a) $h(t) = u(t + 3)$
- (b) $h(t) = u(t)e^{-t^2}$
- (c) $h(t) = e^t \sin(\omega_0 t)$ ω_0 : real constant
- (d) $h(t) = e^{-3t} \sin(\omega_0 t) u(t)$ ω_0 : real constant

Solution A system is said to be stable if its impulse response is absolutely integrable. That is,

$$\int_{-\infty}^{\infty} |h(t)| dt < \infty$$

Similarly, a system is physically realizable if $h(t) = 0$ for $t < 0$. Using the above two conditions, we will solve the given problem.

- (a) Given: $h(t) = u(t + 3)$
 $u(t + 3) = 0$ for $t < -3$
 $= 1$ for $t \geq -3$

Here, $h(t) \neq 0$ for $t < 0$. Hence, the system is physically not realizable.

$$\text{Also, } \int_{-\infty}^{\infty} |h(t)| dt = \int_{-3}^{\infty} dt = \infty$$

Since the above integral attains an infinity value. Therefore, the system is not stable.

- (b) $h(t) = e^{-t^2} u(t)$

The system is physically realizable since $h(t) = 0$ for $t < 0$.

For, the system to be stable, $\int_{-\infty}^{\infty} |h(t)| dt < \infty$

$$\begin{aligned} \int_{-\infty}^{\infty} |h(t)| dt &= \int_0^{\infty} e^{-t^2} dt \\ &= \frac{1}{2} \left\{ \sqrt{2\pi\sigma^2} \left[\frac{1}{\sqrt{2\pi\sigma^2}} \int_{-\infty}^{\infty} e^{-t^2} dt \right] \right\} \end{aligned}$$

The above equation follows a Gaussian distribution with $\sigma = \frac{1}{\sqrt{2}}$ and $\mu = 0$.

Therefore, $\frac{1}{\sqrt{2\pi\sigma^2}} \int_{-\infty}^{\infty} e^{-t^2} dt = 1$. Hence

$$\int_{-\infty}^{\infty} |h(t)| dt = \frac{\sqrt{\pi}}{2}$$

$\int_{-\infty}^{\infty} |h(t)| dt < \infty$. Hence, the system is stable.

- (c) $h(t) = e^t \sin(\omega_0 t)$, ω_0 : real constant

Since $h(t) \neq 0$ for $t < 0$, the system is not physically realizable.

$$\begin{aligned} \int_{-\infty}^{\infty} h(t) dt &= \int_{-\infty}^{\infty} e^t [\sin(\omega_0 t)] dt \\ \int_{-\infty}^{\infty} e^t \sin(\omega_0 t) dt &= \frac{-e^t \cos(\omega_0 t)}{\omega_0} + \frac{e^t \sin(\omega_0 t)}{\omega_0^2} - \int_{-\infty}^{\infty} \frac{e^t \sin(\omega_0 t)}{\omega_0^2} dt \\ \left(1 + \frac{1}{\omega_0^2} \right) \int_{-\infty}^{\infty} e^t \sin(\omega_0 t) dt &= \frac{-e^t \cos(\omega_0 t)}{\omega_0} \Big|_{-\infty}^{\infty} + \frac{e^t \sin(\omega_0 t)}{\omega_0^2} \Big|_{-\infty}^{\infty} = \infty \end{aligned}$$

Hence, the system is unstable.



Practice Problems

8.1 Test whether the following systems are linear/nonlinear, time-invariant/time variant, causal/non-causal, stable/unstable

Ans: (a) Linear, time invariant, causal, unstable
 (b) Nonlinear, time-invariant, causal, stable
 (c) Nonlinear, time-invariant, causal, stable.

$$(a) y(t) = \int_{-\infty}^t x(\tau) d\tau \quad (b) y(t) = \log [x(t)] \quad (c) y(t) = x^2(t)$$

8.2 Test whether the following systems are linear/nonlinear, time-invariant/time variant, causal/non-causal, stable/unstable

Ans: (a) Linear, time invariant, noncausal, stable
 (b) Linear, time-invariant, noncasual stable

$$(a) y(n) = x(n+1) + \frac{1}{2}x(n-1) \quad (b) y(n) = x(n^2)$$

FREQUENCY-DOMAIN

CHARACTERIZATION OF A SYSTEM 8.5



So far we studied the time-domain representation of a system. In this section, we will study the frequency domain representation of a system using Fourier transform.

The Fourier transform of a signal $x(t)$ is given by

$$X(\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t} dt \text{ for all } \omega \quad (8.17)$$

and inverse Fourier transform of $X(\omega)$ is given by

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega)e^{j\omega t} d\omega \text{ for all } t \quad (8.18)$$

The Fourier transform of a signal $x(t)$ exists if $x(t)$ is absolutely integrable over $(-\infty, \infty)$. That is,

$$\int_{-\infty}^{\infty} |x(t)| dt < \infty \quad (8.19)$$

Consider an LTI continuous-time system with impulse response $h(t)$. If the input to the system is $x(t)$ then output $y(t)$ of the system can be obtained using convolution integral. That is,

$$\begin{aligned} y(t) &= \int_{-\infty}^{\infty} x(\tau)h(t-\tau) d\tau \\ &= \int_{-\infty}^{\infty} h(\tau)x(t-\tau) d\tau \end{aligned}$$

If $x(t)$ is a complex exponential signal $x(t) = e^{j\omega t}$ then the output

$$y(t) = \int_{-\infty}^{\infty} h(\tau)e^{j\omega(t-\tau)} d\tau$$

$$\begin{aligned}
&= e^{j\omega t} \int_{-\infty}^{\infty} h(\tau) e^{-j\omega\tau} d\tau \\
&= e^{j\omega t} H(\omega)
\end{aligned} \tag{8.20}$$

where
$$H(\omega) = \int_{-\infty}^{\infty} h(\tau) e^{-j\omega\tau} d\tau \tag{8.21}$$

is a Fourier transform of $h(t)$.

If $Y(\omega)$ is a Fourier transform of $y(t)$ then

$$\begin{aligned}
Y(\omega) &= \int_{-\infty}^{\infty} y(t) e^{-j\omega t} dt = \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} x(\tau) h(t-\tau) d\tau \right] e^{-j\omega t} dt \\
&= \int_{-\infty}^{\infty} x(\tau) \left[\int_{-\infty}^{\infty} h(t-\tau) e^{-j\omega(t-\tau)} dt \right] e^{-j\omega\tau} d\tau \\
&= \int_{-\infty}^{\infty} x(\tau) \left[\int_{-\infty}^{\infty} h(p) e^{-j\omega(p)} dp \right] e^{-j\omega\tau} d\tau \\
&= \int_{-\infty}^{\infty} x(\tau) [H(\omega)] e^{-j\omega\tau} d\tau \\
&= H(\omega) \int_{-\infty}^{\infty} x(\tau) e^{-j\omega\tau} d\tau \\
&= H(\omega) X(\omega)
\end{aligned} \tag{8.22}$$

The function $H(\omega)$ is called the transfer function of the system. That is,

$$H(\omega) = \frac{Y(\omega)}{X(\omega)} \tag{8.23}$$

which is the ratio of Fourier transform of the output $y(t)$ to the Fourier transform of the input $x(t)$.

Solved Problem



8.4 Find the transfer function of the RC network shown in Fig. 8.3.

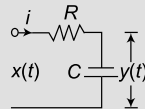


Fig. 8.3

Solution Applying Kirchhoff's voltage law we can write

$$x(t) = Ri(t) + y(t) \quad (8.24)$$

$$\text{The output } y(t) = \frac{1}{C} \int i dt \Rightarrow i(t) = C \frac{dy(t)}{dt} \quad (8.25)$$

$$\text{If } x(t) = e^{j\omega t} \text{ then the output } y(t) = H(\omega) x(t) = H(\omega) e^{j\omega t} \quad (8.26)$$

The differentiation of the output.

$$\frac{dy(t)}{dt} = j\omega H(\omega) e^{j\omega t} = j\omega H(\omega) x(t) \quad (8.27)$$

Substituting Eq. (8.26) in Eq. (8.25) we get

$$i(t) = j\omega CH(\omega) x(t)$$

Now substituting the values of $x(t)$, $y(t)$ and $i(t)$ in Eq. (8.24)

$$x(t) = j\omega RCH(\omega) x(t) + H(\omega) x(t) \quad (8.28)$$

$$\Rightarrow H(\omega) = \frac{1}{1 + j\omega RC} \quad (8.29)$$



Practice Problem

8.3 Find the transfer function of RLC circuit shown in Fig. 8.4.

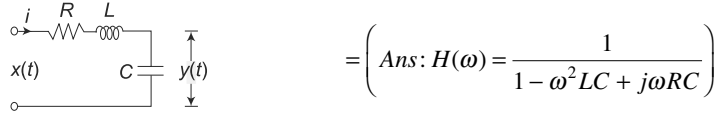


Fig. 8.4



LTI SYSTEM WITH RANDOM INPUT SIGNALS 8.6

In the previous section we studied that the output $y(t)$ of an LTI system is the convolution of its input $x(t)$ and impulse response $h(t)$

$$y(t) = x(t) * h(t)$$

$$\begin{aligned} &= \int_{-\infty}^{\infty} x(\tau) h(t - \tau) d\tau \\ &= \int_{-\infty}^{\infty} x(t - \tau) h(\tau) d\tau \end{aligned} \quad (8.30)$$

If $X(t)$ is a WSS random process then the output of the system can be expressed as $Y(t) = X(t) * h(t)$

By definition,

$$Y(t) = \int_{-\infty}^{\infty} X(\tau)h(t-\tau)d\tau \quad (8.31)$$

8.6.1 Mean Value Function

The mean value of the output process is given by

$$\begin{aligned} E[Y(t)] &= E[X(t) * h(t)] \\ &= E \left[\int_{-\infty}^{\infty} X(t-\tau)h(\tau)d\tau \right] \\ &= \int_{-\infty}^{\infty} E[X(t-\tau)]h(\tau)d\tau \\ &= \int_{-\infty}^{\infty} \mu_X(t-\tau)h(\tau)d\tau \end{aligned}$$

where $\mu_X(t)$ is the mean function of the process $X(t)$. Since $X(t)$ is stationary in the wide-sense,

$$\mu_X(t-\tau) = \mu_X(t) = \text{constant}$$

Hence, the mean function $\mu_Y(t)$ of the process $Y(t)$ is

$$\mu_Y(t) = E[Y(t)] = \mu_X \int_{-\infty}^{\infty} h(t)dt \quad (8.32)$$

We have

$$H(\omega) = \int_{-\infty}^{\infty} h(t)e^{-j\omega t}dt$$

When $\omega = 0$,

$$H(0) = \int_{-\infty}^{\infty} h(t)dt$$

where $H(0)$ is the dc gain of the system. Therefore,

$$\begin{array}{ccccc} \mu_Y & = & \mu_X & H(0) & \\ \text{Output dc component} & & \text{Input dc component} & \text{System's dc gain} & \end{array} \quad (8.33)$$

Since the mean of the input random process is constant, from the above expression, we can observe that the mean value of the output process $Y(t)$ is also a constant.

8.6.2 Autocorrelation Function of Output

The autocorrelation function of $Y(t)$ is

$$R_{YY}(t, t + \tau) = E[Y(t)Y(t + \tau)]$$

$$\begin{aligned}
&= E \left[\int_{-\infty}^{\infty} h(\tau_1) X(t - \tau_1) d\tau_1 \int_{-\infty}^{\infty} X(t + \tau - \tau_2) h(\tau_2) d\tau_2 \right] \\
&= E \left[\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} X(t - \tau_1) X(t + \tau - \tau_2) h(\tau_1) h(\tau_2) d\tau_1 d\tau_2 \right] \\
&= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} E[X(t - \tau_1) X(t + \tau - \tau_2)] h(\tau_1) h(\tau_2) d\tau_1 d\tau_2 \\
&= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} R_{XX}(\tau + \tau_1 - \tau_2) h(\tau_1) h(\tau_2) d\tau_1 d\tau_2 \tag{8.34}
\end{aligned}$$

Since $R_{XX}(\tau + \tau_1 - \tau_2)$ is a function of time difference, $R_{YY}(t, t + \tau)$ will also be a function of time difference.

So we can say that the output process $Y(t)$ is also a WSS random process.

Note: If the input an LTI system is a WSS random process, the output obtained is also a WSS.

The Mean Square Value

The mean square value of the output process is

$$\begin{aligned}
E[Y^2(t)] &= E \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} X(t - t_1) X(t - t_2) h(t_1) h(t_2) dt_1 dt_2 \\
&= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} E[X(t - t_1) X(t - t_2)] h(t_1) h(t_2) dt_1 dt_2 \\
E[Y^2(t)] &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} R_{XX}(t_1, t_2) h(t - t_1) h(t - t_2) dt_1 dt_2
\end{aligned}$$

If $X(t)$ is stationary in the wide-sense and $\alpha = t - t_1$ and $\beta = t - t_2$

$$E[Y^2(t)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} R_{XX}(\alpha - \beta) h(\alpha) h(\beta) d\alpha d\beta \tag{8.35}$$

The average power of the response $Y(t)$ to a stationary input is

$$P_Y = E[Y^2(t)] = R_{YY}(0) \tag{8.36}$$

8.6.3 Cross-correlation between input and output

The correlation function between output $Y(t)$ and stationary input $X(t)$ is

$$\begin{aligned}
R_{YX}(t, t + \tau) &= E[X(t + \tau) Y(t)] \\
&= E \left[X(t + \tau) \int_{-\infty}^{\infty} h(\tau_1) X(t - \tau_1) d\tau_1 \right]
\end{aligned}$$

$$\begin{aligned}
&= \int_{-\infty}^{\infty} h(\tau_1) E[X(t+\tau)X(t-\tau_1)] d\tau_1 \\
&= \int_{-\infty}^{\infty} h(\tau_1) R_{XX}(\tau+\tau_1) d\tau_1
\end{aligned}$$

Let $t = -\tau_1$

$$R_{YX} = \int_{-\infty}^{\infty} h(-t) R_{XX}(\tau-t) dt \quad (8.37)$$

Let $g(t) = h(-t)$

$$\begin{aligned}
R_{YX} &= \int_{-\infty}^{\infty} g(t) R_{XX}(\tau-t) dt \\
R_{YX} &= R_{XX}(\tau) * g(\tau) = R_{XX}(\tau) * h(-\tau)
\end{aligned} \quad (8.38)$$

$$\begin{aligned}
R_{YX} &= \int_{-\infty}^{\infty} h(-t) R_{XX}(\tau-t) dt \\
&= \int_{-\infty}^{\infty} h(t) R_{XX}(\tau+t) dt \\
R_{XY}(\tau) &= R_{YX}(-\tau) = R_{XX}(-\tau) * h(t) \quad (8.39) \\
&= R_{XX}(\tau) * h(\tau) \quad (8.40)
\end{aligned}$$

8.6.4 Autocorrelation Function and Spectral Output

The autocorrelation function of the response $Y(t)$ to an input $X(t)$ is

$$\begin{aligned}
R_{YY}(t, t+\tau) &= E[Y(t)Y(t+\tau)] \\
&= E\left[\left(\int_{-\infty}^{\infty} h(\tau_1)X(t+\tau-\tau_1)d\tau_1\right)Y(t)\right] \\
&= \int_{-\infty}^{\infty} h(\tau_1) E[X(t+\tau-\tau_1)Y(t)] d\tau_1 \\
&= \int_{-\infty}^{\infty} h(\tau_1) R_{YX}(t, t+\tau-\tau_1) d\tau_1
\end{aligned}$$

For wide-sense stationary input,

$$\begin{aligned} R_{YY}(\tau) &= \int_{-\infty}^{\infty} h(\tau_1) R_{YX}(\tau - \tau_1) d\tau_1 \\ &= R_{YX}(\tau) * h(\tau) \end{aligned} \quad (8.41)$$

$$R_{YX}(\tau) = R_{XY}(\tau) * h(-\tau) \quad (8.42)$$

$$\Rightarrow R_{YY}(\tau) = R_{XX}(\tau) * h(\tau) * h(-\tau) \quad (8.43)$$

REVIEW QUESTIONS



8. Prove that if the input to a time invariant stable linear system is a WSS process, then the output also is a WSS process.
9. If $X(t)$ is a WSS process and if $Y(t) = \int_{-\infty}^{\infty} X(\tau) h(t - \tau) d\tau$ then prove
 - (a) $R_{XY}(\tau) = R_{XX}(\tau) * h(\tau)$
 - (b) $R_{YX}(\tau) = R_{XX}(\tau) * h(-\tau)$
 - (c) $R_{YY}(\tau) = R_{XX}(\tau) * h(-\tau) * h(\tau)$

8.6.5 Spectral Density at the System Output

The relationship between the power spectral density $S_{XX}(\omega)$ and the autocorrelation function is given by

$$S_{XX}(\omega) = F[R_{XX}(\tau)] \quad (8.44)$$

We have

$$\begin{aligned} R_{YY}(\tau) &= R_{XX}(\tau) * h(\tau) * h(-\tau) \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} R_{XX}(\tau_2 - \tau_1 - \tau) h(\tau_1) h(\tau_2) d\tau_1 d\tau_2 \end{aligned} \quad (8.45)$$

$$\begin{aligned} S_{YY}(\omega) &= F[R_{YY}(\tau)] \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} R_{XX}(\tau_2 - \tau_1 - \tau) h(\tau_1) h(\tau_2) e^{-j\omega\tau} d\tau d\tau_1 d\tau_2 \\ &= \int_{-\infty}^{\infty} d\tau_1 \int_{-\infty}^{\infty} h(\tau_1) h(\tau_2) d\tau_2 \int_{-\infty}^{\infty} R_{XX}(\tau_2 - \tau_1 - \tau) e^{-j\omega\tau} d\tau \\ &= \int_{-\infty}^{\infty} d\tau_1 \int_{-\infty}^{\infty} h(\tau_1) h(\tau_2) S_{XX}(\omega) e^{-j\omega(\tau_2 - \tau_1)} d\tau_2 \\ &= S_{XX}(\omega) \int_{-\infty}^{\infty} h(\tau_1) e^{j\omega\tau_1} d\tau_1 \int_{-\infty}^{\infty} h(\tau_2) e^{-j\omega\tau_2} d\tau_2 \end{aligned}$$

$$\begin{aligned}
&= S_{XX}(\omega) H(-\omega)H(\omega) \\
&= S_X(\omega)|H(\omega)|^2 \\
S_{YY}(\omega) &= |H(\omega)|^2 S_{XX}(\omega)
\end{aligned} \tag{8.46}$$

In terms of f

$$S_{YY}(f) = |H(f)|^2 S_{XX}(f) \tag{8.47}$$

Equation (8.47) implies that a system may be viewed as a filter that allows selectively certain frequency components of the input process to pass. Since autocorrelation and power spectrum are a Fourier transform pair

$$R_{YY}(\tau) = F^{-1}[|H(\omega)|^2 S_{XX}(\omega)] \tag{8.48}$$

8.6.6 Cross-spectral Density between Input and Output

Let $X(t)$ and $Y(t)$ be two real and jointly wide-sense stationary random processes. Their cross-spectral densities are defined as

$$S_{XY}(\omega) = \int_{-\infty}^{\infty} R_{XY}(\tau) e^{-j\omega\tau} d\tau$$

and

$$S_{YX}(\omega) = \int_{-\infty}^{\infty} R_{YX}(\tau) e^{-j\omega\tau} d\tau$$

REVIEW QUESTION



10. Show that $S_{YY}(\omega) = |H(\omega)|^2 S_{XX}(\omega)$ where $S_{XX}(\omega)$ and $S_{YY}(\omega)$ are the power spectral density functions of the input $X(t)$ and output $Y(t)$ and $H(\omega)$ is the output transfer function.

Solved Problems



8.5 If $X(t)$ is a differentiable WSS random process and $Y(t) = \frac{dX(t)}{dt}$, find an expression for $S_{YY}(\omega)$ and $R_{YY}(\tau)$.

Solution We know,

$$S_{YY}(\omega) = |H(\omega)|^2 S_{XX}(\omega)$$

Given:

$$Y(t) = \frac{dX(t)}{dt}$$

Taking Fourier transform on both sides, we get

$$Y(\omega) = j\omega X(\omega)$$

$$\begin{aligned}
H(\omega) &= \frac{Y(\omega)}{X(\omega)} = j\omega \\
S_{YY}(\omega) &= |H(\omega)|^2 S_{XX}(\omega) \\
&= |j\omega|^2 S_{XX}(\omega) = \omega^2 S_{XX}(\omega) \\
R_{YY}(\tau) &= F^{-1}[S_{YY}(\omega)] \\
&= F^{-1}[\omega^2 S_{XX}(\omega)] = \frac{-d^2}{d\tau^2} R_{XX}(\tau)
\end{aligned}$$

8.6 A zero mean white noise with noise power density $\frac{N_0}{2}$ is applied to a filter with transfer function

$$H(\omega) = \frac{1}{1 + j\omega}$$

- (a) Find $S_{YX}(\omega)$ and $R_{YX}(\tau)$.
- (b) Find $S_{YY}(\omega)$ and $R_{YY}(\tau)$.
- (c) What is the average power of the output.

Solution We know

$$\begin{aligned}
S_{YX}(\omega) &= H(\omega) S_{XX}(\omega) \\
&= \frac{N_0/2}{1 + j\omega} \\
R_{YX}(\tau) &= F^{-1}\left(\frac{N_0/2}{1 + j\omega}\right) = \frac{N_0}{2} e^{-\tau} \quad \tau > 0 \\
S_{YY}(\omega) &= |H(\omega)|^2 S_{XX}(\omega) = \frac{N_0/2}{1 + \omega^2} \\
R_{YY}(\tau) &= F^{-1}[S_{YY}(\omega)] = \frac{N_0}{4} e^{-|\tau|}
\end{aligned}$$

$$(c) \text{ Average power} = R_{YY}(0) = \frac{N_0}{4}$$

8.7 A random process $X(t)$ is applied to a linear system whose impulse response is $h(t) = e^{-2t} t \geq 0$. If ACF of the process is $R_{XX}(\tau) = e^{-|\tau|}$, find the PSD of the output process $Y(t)$.

Solution The PSD of the input process is

$$\begin{aligned}
S_{XX}(\omega) &= F[R_{XX}(\tau)] \\
&= \int_{-\infty}^{\infty} R_{XX}(\tau) e^{-j\omega\tau} d\tau
\end{aligned}$$

$$\begin{aligned}
&= \int_{-\infty}^{\infty} e^{-|\tau|} e^{-j\omega\tau} d\tau = \int_{-\infty}^0 e^{\tau} e^{-j\omega\tau} d\tau + \int_0^{\infty} e^{-\tau} e^{-j\omega\tau} d\tau \\
&= \int_{-\infty}^0 e^{(1-j\omega)\tau} d\tau + \int_0^{\infty} e^{-(1+j\omega)\tau} d\tau \\
&= \frac{1}{1-j\omega} e^{(1-j\omega)\tau} \Big|_{-\infty}^0 + \frac{1}{-(1+j\omega)} e^{-(1+j\omega)\tau} \Big|_0^{\infty} \\
&= \frac{1}{1-j\omega} + \frac{1}{1+j\omega} = \frac{2}{1+\omega^2}
\end{aligned}$$

The transfer function of the linear system is

$$H(\omega) = F(h(t)) = \int_{-\infty}^{\infty} h(t) e^{-j\omega t} dt = \int_0^{\infty} e^{-2t} e^{-j\omega t} dt = \frac{1}{-(2+j\omega)} e^{-(2+j\omega)t} \Big|_0^{\infty} = \frac{1}{2+j\omega}$$

The PSD of the output process is

$$\begin{aligned}
S_{YY}(\omega) &= |H(\omega)|^2 S_{XX}(\omega) \\
&= \left| \frac{1}{2+j\omega} \right|^2 \frac{1}{1+\omega^2}
\end{aligned}$$

8.8 White noise $n(t)$ with $G(f) = \frac{\eta}{2}$ is passed through a low-pass RC network with a 3 dB frequency f_c .

- Find the autocorrelation $R(\tau)$ of the output noise of the network.
- Find $P(\tau) = R(\tau)/R(0)$.
- Find τ such that $P(\tau) \leq 0.1$.

Solution The RC circuit is shown in Fig. 8.5.

In solved problem (8.4), we obtained the transfer function of the circuit as

$$H(\omega) = \frac{Y(\omega)}{X(\omega)} = \frac{1}{1+j\omega RC}$$

Given: $S_{NN}(\omega) = \frac{\eta}{2}$

$$\begin{aligned}
S_{YY}(\omega) &= |H(\omega)|^2 S_{NN}(\omega) \\
&= \frac{1}{1+\omega^2 R^2 C^2} \left(\frac{\eta}{2} \right)
\end{aligned}$$

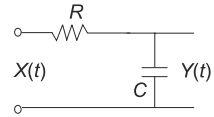


Fig. 8.5

$$R(\tau) = F^{-1} \left[\frac{\eta/2}{1 + \omega^2 C^2 R^2} \right] = \frac{\eta}{2} \frac{1/R^2 C^2}{\frac{1}{R^2 C^2} + \omega^2}$$

$$= \frac{\eta}{4RC} F^{-1} \left\{ \frac{2(1/RC)}{(1/RC)^2 + \omega^2} \right\}$$

$$= \frac{\eta}{4RC} e^{-|\tau|/RC}$$

$$\begin{aligned} \text{(b)} \quad P(\tau) &= \frac{R(\tau)}{R(0)} \\ &= \frac{\frac{\eta}{4RC} e^{-|\tau|/RC}}{\frac{\eta}{4RC}} = e^{-|\tau|/RC} \end{aligned}$$

$$\begin{aligned} \text{(c)} \quad P(\tau) &\leq 0.1, \tau = ? \\ e^{-|\tau|/RC} &\leq 0.1 \Rightarrow e^{|\tau|/RC} \geq 10 \\ \frac{|\tau|}{RC} &\geq \ln 10 \Rightarrow |\tau| \geq RC \ln(10) \end{aligned}$$

8.9 The input voltage to an RL, LPF circuit is a stationary random process $X(t)$ with $E[X(t)] = 2$ and $R_{XX}(\tau) = 4 + e^{-2|\tau|}$. Let $Y(t)$ be the voltage across the resistor. Find $E[Y(t)]$ and $S_{YY}(\omega)$.

Solution The RL circuit is shown in Fig. 8.6.

Applying KVL for the circuit,

$$L \frac{di}{dt} + Ri = X(t)$$

$$\text{and} \quad Y(t) = Ri(t)$$

Applying Fourier transform for above equations, we obtain

$$Lj\omega I(\omega) + RI(\omega) = X(\omega)$$

$$\text{and} \quad Y(\omega) = RI(\omega)$$

The transfer function

$$H(\omega) = \frac{Y(\omega)}{X(\omega)} = \frac{R}{R + j\omega L} = \frac{1}{1 + j\omega \frac{L}{R}}$$

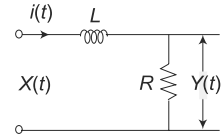


Fig. 8.6

$$|H(\omega)|^2 = \frac{1}{1 + \frac{\omega^2 L^2}{R^2}}$$

$$E[Y(t)] = H(0) E[X(t)] = E[X(t)] = 2 \quad (\because H(0) = 1)$$

$$S_{XX}(\omega) = F[R_{XX}(\tau)] = F[4 + e^{-2|\tau|}]$$

$$= 4F(1) + F[e^{-2|\tau|}]$$

$$= 4[2\pi\delta(\omega)] + \frac{4}{4 + \omega^2}$$

$$= 8\pi\delta(\omega) + \frac{4}{4 + \omega^2}$$

$$S_{YY}(\omega) = |H(\omega)|^2 S_{XX}(\omega)$$

$$= \frac{1}{1 + \left(\frac{\omega L}{R}\right)^2} \left\{ 8\pi\delta(\omega) + \frac{4}{4 + \omega^2} \right\}$$

8.10 A WSS random process $X(t)$ with PSD $S_{XX}(\omega)$ is applied as input to the following system. Find $S_{YY}(\omega)$.

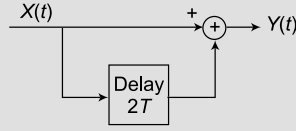


Fig. 8.7

Solution From Fig. 8.7, we can find the output.

$$Y(t) = X(t) + X(t - 2T)$$

Taking Fourier transform on both sides,

$$Y(\omega) = X(\omega) + e^{-j2\omega T} X(\omega)$$

$$= X(\omega) [1 + e^{-j2\omega T}]$$

$$H(\omega) = \frac{Y(\omega)}{X(\omega)} = 1 + e^{-j2\omega T}$$

$$|H(\omega)|^2 = (1 + \cos 2\omega T)^2 + (\sin 2\omega T)^2$$

$$= 2 + 2 \cos 2\omega T$$

$$= 2 \left[1 + \cos 2\omega T \right] = 2 \left(2 \cos^2 \frac{\omega T}{2} \right)$$

$$= 4 \cos^2 \frac{\omega T}{2}$$

The PSD

$$\begin{aligned} S_{YY}(\omega) &= |H(\omega)|^2 S_{XX}(\omega) \\ &= 4 \cos^2(\omega T/2) S_{XX}(\omega) \end{aligned}$$

8.11 A signal $x(t) = u(t) e^{-\alpha t}$ is applied to a network having impulse response $h(t) = Wu(t) e^{-Wt}$. Here, α and W are real positive constants. Find the network response.

Solution Given: $x(t) = u(t) e^{-\alpha t}$ and $h(t) = Wu(t) e^{-Wt}$

$$\begin{aligned} y(t) &= x(t) * h(t) \\ &= \int_{-\infty}^{\infty} x(\tau) h(t - \tau) d\tau \\ &= \int_{-\infty}^{\infty} u(\tau) e^{-\alpha \tau} Wu(t - \tau) e^{-W(t - \tau)} d\tau \end{aligned}$$

Since $u(\tau) = 0$ for $\tau < 0$

and $u(t - \tau) = 0$ for $\tau > t$

$u(\tau) = 1$	for $\tau \geq 0$
$= 0$	for $\tau < 0$
$u(t - \tau) = 1$	for $t - \tau \geq 0$ or $\tau \leq t$
$= 0$	for $\tau > t$

$$\begin{aligned} y(t) &= W \int_0^t e^{-\alpha \tau} e^{-W(t - \tau)} d\tau \\ &= W e^{-Wt} \int_0^t e^{-(\alpha - W)\tau} d\tau \\ &= W \frac{e^{-Wt}}{-(\alpha - W)} e^{-(\alpha - W)\tau} \Big|_0^t = \frac{W e^{-Wt}}{W - \alpha} \left[e^{-(\alpha - W)t} - 1 \right] \\ &= \frac{W}{W - \alpha} \left[e^{-\alpha t} - e^{-Wt} \right] \\ y(t) &= \frac{W}{W - \alpha} \left[e^{-\alpha t} - e^{-Wt} \right] \end{aligned}$$

8.12 The input to an RLC series circuit is a stationary random process $X(t)$ with $E[X(t)] = 2$ and $R_{XX}(\tau) = 4 + e^{-2|\tau|}$. Let $Y(t)$ be the voltage across the capacitor. Find $E[Y(t)]$ and PSD of $Y(t)$.

Solution Let $i(t)$ be the current flowing through the circuit show in Fig. 8.8.

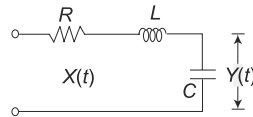


Fig. 8.8

Applying KVL, we get

$$Ri(t) + L \frac{di}{dt} + \frac{1}{C} \int i dt = X(t)$$

Taking Fourier transform on both sides, we get

$$RI(\omega) + Lj\omega I(\omega) + \frac{1}{C} \frac{I(\omega)}{j\omega} = X(\omega)$$

The voltage across the capacitor $Y(t) = \frac{1}{C} \int i(t) dt$

$$\begin{aligned} \Rightarrow Y(\omega) &= \frac{1}{j\omega C} I(\omega) \\ H(\omega) &= \frac{Y(\omega)}{X(\omega)} = \frac{I(\omega)/j\omega C}{I(\omega) \left\{ R + j\omega L + \frac{1}{j\omega C} \right\}} \\ &= \frac{1}{1 - \omega^2 LC + j\omega CR} \\ |H(\omega)|^2 &= \frac{1}{(1 - \omega^2 LC)^2 + \omega^2 C^2 R^2} \end{aligned}$$

The PSD of $Y(t)$ is

$$\begin{aligned} S_{YY}(\omega) &= |H(\omega)|^2 S_{XX}(\omega) \\ S_{XX}(\omega) &= F[R_{XX}(\tau)] = F[4 + e^{-2|\tau|}] \\ &= 4 \left[2\pi\delta(\omega) + \frac{2(2)}{2^2 + \omega^2} \right] \\ &= 8\pi\delta(\omega) + \frac{4}{4 + \omega^2} \end{aligned}$$

$$E[Y(t)] = \mu_X H(0)$$

$$H(0) = 1; \mu_Y = E[X(t)] = 2$$

$$\Rightarrow E[Y(t)] = 2$$

8.13 A WSS random process $X(t)$ is applied to the input of an LTI system whose impulse response is $5te^{-2t}$. The mean of $X(t)$ is 3. Find the mean output of the system.

Solution Given: $h(t) = 5te^{-2t}$

$$H(\omega) = F[h(t)] = \frac{5}{(j\omega + 2)^2}$$

$$H(0) = \frac{5}{4}$$

The mean value of $Y(t) = E[Y(t)] = \mu_X H(0) = 3(5/4) = \frac{15}{4}$.

8.14 A white Gaussian noise process of zero mean and PSD $\frac{\eta}{2}$ is applied to a high pass filter as shown in Fig. 8.9.

- Find the ACF and its PSD.
- Find the mean and variance of $Y(t)$.

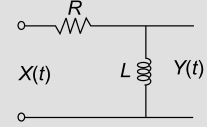


Fig. 8.9

Solution Applying KVL for the input,

$$Ri(t) + L \frac{di}{dt} = X(t)$$

Taking Fourier transform on both sides, we get

$$RI(\omega) + Lj\omega I(\omega) = X(\omega)$$

The output

$$Y(t) = L \frac{di}{dt}$$

Applying Fourier transform on both sides,

$$Y(\omega) = j\omega LI(\omega)$$

The transfer function is

$$\frac{Y(\omega)}{X(\omega)} = \frac{j\omega L I(\omega)}{(R + j\omega L)I(\omega)} = \frac{j\omega L}{R + j\omega L}$$

$$|H(\omega)|^2 = \frac{\omega^2 L^2}{R^2 + \omega^2 L^2}$$

The PSD of the output

$$S_{YY}(\omega) = \frac{\omega^2 L^2}{R^2 + \omega^2 L^2} \left(\frac{\eta}{2} \right)$$

The autocorrelation function

$$\begin{aligned} R_{YY}(t) &= F^{-1}[S_{YY}(\omega)] \\ &= F^{-1} \left[\frac{\eta}{2} \frac{\omega^2 L^2}{R^2 + \omega^2 L^2} \right] \\ &= \frac{\eta}{2} F^{-1} \left[\frac{\omega^2 L^2}{R^2 + \omega^2 L^2} \right] \end{aligned}$$

$$\begin{aligned}
 &= \frac{\eta}{2} F^{-1} \left\{ 1 - \frac{R^2}{R^2 + \omega^2 L^2} \right\} \\
 &= \frac{\eta}{2} \left[\delta(t) - \frac{1}{2} e^{-R|t|/L} \right] \\
 &= \frac{\eta}{2} \delta(t) - \frac{\eta}{4} e^{-R|t|/L} \\
 \mu_Y(t) &= H(0) \mu_X(t)
 \end{aligned}$$

Since $\mu_X(t) = 0$; $\mu_Y(t) = 0$

$$\begin{aligned}
 \text{Var}[Y(t)] &= E[Y^2(t)] - \mu_Y^2(t) \\
 &= E[Y^2(t)] \\
 E[Y^2(t)] &= R_{YY}(\tau)|_{\tau=0} \\
 &= \left\{ \frac{\eta}{2} \delta(\tau) - \frac{\eta}{4} e^{-R|\tau|/L} \right\} \Big|_{\tau=0} \\
 &= \frac{\eta}{2} - \frac{\eta}{4} = \frac{\eta}{4}
 \end{aligned}$$

8.15 A stationary random process $X(t)$ having an autocorrelation function $R_{XX}(\tau) = e^{-2|\tau|}$ is applied to the network as shown in Fig. 8.10.

Find (a) $S_{XX}(\omega)$ (b) $S_{YY}(\omega)$.

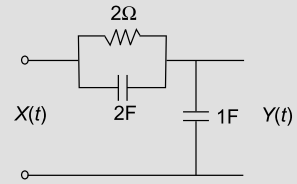


Fig. 8.10

Solution

(a) Given: $R_{XX}(\tau) = e^{-2|\tau|}$

$$S_{XX}(\omega) = F[R_{XX}(\tau)] = F[e^{-2|\tau|}] = \frac{2(2)}{\omega^2 + 2^2} = \frac{4}{\omega^2 + 4}$$

(b) The phasor form of the network is shown in Fig. 8.11. From Fig. 8.12 we find

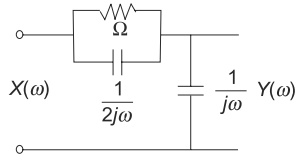


Fig. 8.11

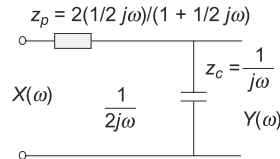


Fig. 8.12

$$Y(\omega) = \frac{X(\omega)Z_C}{Z_p + Z_C}$$

$$Z_p = \frac{2 \left(\frac{1}{\sqrt{2j\omega}} \right)}{\left(2 + \frac{1}{2j\omega} \right)} = \frac{2}{1 + j4\omega}$$

$$\begin{aligned} Z_p + Z_C &= \frac{2}{1 + j4\omega} + \frac{1}{j\omega} \\ &= \frac{1 + j6\omega}{j\omega(1 + j4\omega)} \end{aligned}$$

$$H(\omega) = \frac{Y(\omega)}{X(\omega)} = \frac{Z_C}{Z_p + Z_C} = \frac{\left(\frac{1}{j\omega} \right)}{\frac{(1 + j6\omega)}{j\omega(1 + j4\omega)}} = \frac{1 + j4\omega}{1 + j6\omega}$$

$$\begin{aligned} S_{YY}(\omega) &= |H(\omega)|^2 S_{XX}(\omega) \\ &= \frac{1 + 16\omega^2}{1 + 36\omega^2} S_{XX}(\omega) \\ &= \frac{1 + 16\omega^2}{1 + 36\omega^2} \left(\frac{4}{\omega^2 + 4} \right) \end{aligned}$$

8.16 A random process $X(t)$ has an autocorrelation function $R_{XX}(\tau) = A^2 + Be^{-|\tau|}$, where A and B are positive constants. Find the mean value of the response of a system having an impulse response.

$$\begin{aligned} h(t) &= e^{-kt} \quad \text{for } t > 0 \\ &= 0 \quad \text{for } t < 0 \end{aligned}$$

where k is a real positive constant for which $X(t)$ is its input.

Solution Given:

$$R_{XX}(\tau) = A^2 + Be^{-|\tau|}$$

The mean value of the response is

$$\mu_Y(t) = \mu_X(t) H(0)$$

We know that

$$\begin{aligned} \{E[X(t)]\}^2 &= \lim_{\tau \rightarrow \infty} R_{XX}(\tau) \\ &= \lim_{\tau \rightarrow \infty} [A^2 + Be^{-2|\tau|}] = A^2 \end{aligned}$$

$$\Rightarrow E[X(t)] = A$$

The transfer function

$$H(\omega) = F[h(t)]$$

$$= \int_0^{\infty} e^{-at} e^{-j\omega t} dt = \left. \frac{e^{-(a+j\omega)t}}{-(a+j\omega)} \right|_0^{\infty} = \frac{1}{a+j\omega}$$

$$\Rightarrow H(0) = \frac{1}{a}$$

$$\mu_Y(t) = \mu_X(t) H(0) = \frac{A}{a}$$

8.17 A WSS process $X(t)$, with mean value 10 and power spectrum

$$S_{XX}(\omega) = 25\pi\delta(\omega) + \frac{4}{1 + \left(\frac{\omega}{4}\right)^2} \text{ is applied to a network with impulse response } h(t) = 2e^{-2|t|}$$

(a) Find $H(\omega)$ of the network. (b) Find the mean and power spectrum of the response $Y(t)$.

Solution

$$(a) \quad H(\omega) = F[2e^{-2|t|}] = 2 \left\{ \frac{2(2)}{\omega^2 + 2^2} \right\} = \frac{8}{\omega^2 + 4}$$

(b) Mean of the response $Y(t)$ is

$$\mu_Y(t) = H(0) \mu_X(t)$$

$$\mu_X(t) = E[X(t)] = 10. \text{ Also } H(0) = \frac{8}{4} = 2$$

$$\mu_Y(t) = 2(10) = 20$$

The power spectrum

$$\begin{aligned} S_{YY}(\omega) &= |H(\omega)|^2 S_{XX}(\omega) \\ &= \frac{64}{(\omega^2 + 4^2)} \left[25\pi\delta(\omega) + \frac{4}{1 + \frac{\omega^2}{16}} \right] \end{aligned}$$

8.18 A random process $X(t) = A \sin(\omega_0 t + \theta)$ where A and ω_0 are real positive constants and θ is a random variable uniformly distributed in the interval $(-\pi, \pi)$ is applied to the network shown in Fig. 8.13. Find the expression for the network response.

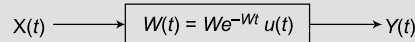


Fig. 8.13

Solution Given: $X(t) = A \sin(\omega_0 t + \theta)$ and $h(t) = We^{-Wt} u(t)$

The output response

$$\begin{aligned}
 Y(t) &= X(t) * h(t) \\
 &= \int_{-\infty}^{\infty} X(t-\tau)h(\tau)d\tau \\
 &= \int_{-\infty}^{\infty} A \sin(\omega_0\tau + \theta) W e^{-W(t-\tau)} u(t-\tau) d\tau
 \end{aligned}$$

$$\begin{aligned}
 u(t-\tau) &= 1 \quad \text{for } t-\tau \geq 0 \\
 &= 0 \quad \text{for } t-\tau < 0 \text{ or } \tau > t
 \end{aligned}$$

$$\text{Hence} \quad Y(t) = AW \int_{-\infty}^{\tau} \sin(\omega_0\tau + \theta) e^{-W(t-\tau)} d\tau = AW e^{-Wt} \int_{-\infty}^{\tau} \sin(\omega_0\tau + \theta) e^{W\tau} d\tau$$

$$\text{We know } \int e^{ax} \sin bx = \frac{e^{ax}}{a^2 + b^2} [a \sin bx - b \cos bx]$$

$$\begin{aligned}
 \Rightarrow \quad Y(t) &= AW e^{-Wt} \left\{ \frac{e^{W\tau}}{W^2 + \omega_0^2} [W \sin(\omega_0\tau + \theta) - \omega_0 \cos(\omega_0\tau + \theta)] \right\} \Bigg|_{-\infty}^t \\
 &= \frac{AW}{W^2 + \omega_0^2} [W \sin(\omega_0t + \theta) - \omega_0 \cos(\omega_0t + \theta)] \\
 &= \frac{AW}{\sqrt{W^2 + \omega_0^2}} \left[\frac{W}{\sqrt{W^2 + \omega_0^2}} \sin(\omega_0t + \theta) - \frac{\omega_0}{\sqrt{W^2 + \omega_0^2}} \cos(\omega_0t + \theta) \right]
 \end{aligned}$$

$$\text{Let} \quad \cos \phi = \frac{W}{\sqrt{W^2 + \omega_0^2}} \text{ and } \sin \phi = \frac{\omega_0}{\sqrt{W^2 + \omega_0^2}}$$

$$\text{then} \quad \phi = \tan^{-1} \left(\frac{\omega_0}{W} \right)$$

$$Y(t) = \frac{AW}{\sqrt{W^2 + \omega_0^2}} [\sin(\omega_0t + \theta - \phi)]$$

8.19 A random noise process $X(t)$ having power spectrum $S_{XX}(\omega) = \frac{3}{49 + \omega^2}$ is applied to a network for which $h(t) = t^2 e^{-7t} u(t)$. The network response is denoted by $Y(t)$.

- What is the average power of $X(t)$?
- Find the power spectrum of $Y(t)$.
- Find the average power of $Y(t)$.

Solution Given: $S_{XX}(\omega) = \frac{3}{49 + \omega^2}$ and $h(t) = t^2 e^{-7t} u(t)$

The ACF is given by

$$R_{XX}(\tau) = F^{-1}[S_{XX}(\omega)] = F^{-1}\left[\frac{3}{49 + \omega^2}\right] = F^{-1}\left[\frac{3}{7^2 + \omega^2}\right]$$

We have

$$F^{-1}\left[\frac{2a}{a^2 + \omega^2}\right] = e^{-a|\tau|}$$

$$R_{XX}(\tau) = F^{-1}\left[\frac{1}{2(7)} \frac{3(2 \times 7)}{7^2 + \omega^2}\right] = \frac{3}{14} F^{-1}\left[\frac{2(7)}{7^2 + \omega^2}\right] = \frac{3}{14} [e^{-7|\tau|}]$$

The average power of $X(t)$ is

$$R_{XX}(0) = \frac{3}{14}$$

(b) Given: $h(t) = t^2 e^{-7t} u(t)$; $F[e^{-7t} u(t)] = \frac{1}{7 + j\omega}$

$$F[t^n x(t)] = (-1)^n \frac{d^n X(\omega)}{d\omega^n}$$

$$F[t^2 e^{-7t} u(t)] = (-1)^2 \frac{d^2}{d\omega^2} \left(\frac{1}{7 + j\omega} \right) = \frac{d}{d\omega} \left(\frac{-j}{(7 + j\omega)^2} \right) = \frac{-2}{(7 + j\omega)^3}$$

The power spectrum of $Y(t)$ is $S_{YY}(\omega) = |H(\omega)|^2 S_{XX}(\omega)$

$$|H(\omega)|^2 = \frac{4}{(49 + \omega^2)^3}; S_{YY}(\omega) = \frac{4}{(49 + \omega^2)^3} \left[\frac{3}{49 + \omega^2} \right] = \frac{12}{(49 + \omega^2)^4}$$

The average power of $S_{YY}(\omega)$ is

$$P_Y = \frac{1}{2\pi} \int_{-\pi}^{\pi} S_{YY}(\omega) d\omega = \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{12}{(49 + \omega^2)^4} d\omega$$

Let $\omega = 7 \tan \theta$; $d\omega = 7 \sec^2 \theta d\theta$

For $\omega = \infty$; $\theta = \frac{\pi}{2}$ and for $\omega = -\infty$; $\theta = -\frac{\pi}{2}$

$$\begin{aligned} P_Y &= \frac{1}{2\pi} \int_{-\pi/2}^{\pi/2} \frac{12(7 \sec^2 \theta)}{(49 + 49 \tan^2 \theta)^4} d\theta = \frac{1}{2\pi} \int_{-\pi/2}^{\pi/2} \frac{12(7 \sec^2 \theta)}{7^8 \sec^8 \theta} d\theta \\ &= \frac{12}{2\pi(7)^7} \int_{-\pi/2}^{\pi/2} \cos^6 \theta d\theta = \frac{12}{2\pi(7)^7} \int_{-\pi/2}^{\pi/2} \left(\frac{1 + \cos 2\theta}{2} \right)^3 d\theta \end{aligned}$$

$$= \frac{12}{16\pi(7)^7} \int_{-\pi/2}^{\pi/2} (1 + \cos 2\theta)^3 d\theta$$

$$(1 + \cos 2\theta)^3 = 1 + \cos^2 2\theta + 3 \cos^2 2\theta + 3 \cos 2\theta$$

$$\cos^3 2\theta = \cos^2 2\theta \cos 2\theta = \left(\frac{1 + \cos 4\theta}{2} \right) \cos 2\theta$$

$$= \frac{1}{2} \cos 2\theta + \frac{1}{2} \cos 4\theta \cos 2\theta$$

$$= \frac{1}{2} \cos 2\theta + \frac{1}{4} (\cos 2\theta - \cos 6\theta) = \frac{3}{4} \cos 2\theta - \frac{1}{4} \cos 6\theta$$

$$\begin{aligned} \Rightarrow (1 + \cos 2\theta)^3 &= 1 + \frac{3}{4} \cos 2\theta - \frac{1}{4} \cos 6\theta + \frac{3}{2} + \frac{3}{2} \cos 4\theta + 3 \cos 2\theta \\ &= \frac{5}{2} + \frac{15}{4} \cos 2\theta + \frac{3}{2} \cos 4\theta - \frac{1}{4} \cos 6\theta \end{aligned}$$

$$\begin{aligned} P_Y &= \frac{3}{4\pi(7)^7} \int_{-\pi/2}^{\pi/2} (2.5 + 3.75 \cos 2\theta + 1.5 \cos 4\theta - 0.25 \cos 6\theta) d\theta \\ &= \frac{3}{4\pi(7)^7} \{2.5\pi\} = \frac{15}{8(7)^7} \\ P_Y &= \frac{15}{8(7)^7} \end{aligned}$$

8.20 $X(t)$ is a stationary random process with zero mean and autocorrelation $R_{XX}(\tau) = e^{-2|\tau|}$ is applied to a system with transfer function $H(\omega) = \frac{1}{j\omega + 2}$. Find the mean and PSD of its output.

Solution Given:

$$R_{XX}(\tau) = e^{-2|\tau|} \text{ and } H(\omega) = \frac{1}{j\omega + 2}$$

$$S_{XX}(\omega) = F[e^{-2|\tau|}] = \frac{2(2)}{4 + \omega^2} = \frac{4}{4 + \omega^2}$$

The PSD of the output

$$\begin{aligned} S_{YY}(\omega) &= |H(\omega)|^2 S_{XX}(\omega) \\ &= \frac{1}{4 + \omega^2} \left(\frac{4}{4 + \omega^2} \right) = \frac{4}{(4 + \omega^2)^2} \end{aligned}$$

$$S_{YY}(\omega) = \frac{4}{(4 + \omega^2)^2}$$

Since mean of input $X(t)$ is zero, the mean of output is also equal to zero.

$$\mu_Y = H(0) \mu_X = 0$$

8.21 A random process $X(t)$ is applied to a network with impulse response $h(t) = u(t) te^{-bt}$ where $b > 0$ is a constant. The cross-correlation of $X(t)$ with the output $Y(t)$ is known to have the same form

$$R_{XY}(\tau) = \tau e^{-b\tau} u(\tau)$$

(a) Find the autocorrelation of $Y(t)$. (b) What is the average power in $Y(t)$?

Solution Given:

$$h(t) = u(t) te^{-bt}$$

$$S_{XY}(\omega) = H(\omega) S_{XX}(\omega)$$

$$H(\omega) = F[te^{-bt}u(t)] = \frac{1}{(b + j\omega)^2}$$

$$S_{XY}(\omega) = F[te^{-bt}u(t)] = \frac{1}{(b + j\omega)^2}$$

$$S_{XX}(\omega) = \frac{S_{XY}(\omega)}{H(\omega)} = 1$$

$$\text{The PSD of the output } S_{YY}(\omega) = |H(\omega)|^2 S_{XX}(\omega) = \frac{1}{(b^2 + \omega^2)^2}.$$

8.22 A random process $n(t)$ has a power spectral density $G(f) = \frac{\eta}{2}$ for $-\infty \leq f \leq \infty$. The random process is passed through a low-pass filter which has transfer function $H(f) = 2$ for $-f_m \leq f \leq f_m$ and $H(f) = 0$ otherwise. Find the PSD of the waveform at the output of the filter.

Solution Given: $G(f) = \frac{\eta}{2}$ for $-\infty \leq f \leq \infty$

$$\begin{aligned} H(f) &= 2 & \text{for } -f_m \leq f \leq f_m \\ &= 0 & \text{otherwise} \end{aligned}$$

$$\begin{aligned} S_{YY}(f) &= |H(f)|^2 G(f) \\ &= 2\eta \text{ for } |f| \leq f_m \\ &= 0 & \text{elsewhere.} \end{aligned}$$

8.23 Find the input correlation function, output correlation and output spectral density of an RC low-pass filter, where the filter is subjected to a white noise of spectral density $N_0/2$.

Solution The transfer function of an RC low-pass filter is

$$H(\omega) = \frac{1}{1 + j\omega RC}$$

Given: $S_{NN}(\omega) = \frac{N_0}{2}$

$$S_{YY}(\omega) = |H(\omega)|^2 S_{NN}(\omega) = \left(\frac{1}{1 + \omega^2 R^2 C^2} \right) \frac{N_0}{2}$$

$$\text{Input correlation} = F^{-1} \left[\frac{N_0}{2} \right] = \frac{N_0}{2} \delta(\tau)$$

$$\begin{aligned} \text{Output correlation} &= F^{-1} [S_{YY}(\omega)] = F^{-1} \left[\frac{1}{1 + \omega^2 R^2 C^2} \frac{N_0}{2} \right] \\ &= \frac{N_0}{2} F^{-1} \left[\frac{1}{R^2 C^2} \left(\frac{1}{\omega^2 + \frac{1}{R^2 C^2}} \right) \right] \\ &= \frac{N_0}{4RC} e^{-|t|/RC} \end{aligned}$$

8.24 A wide sense stationary random process $X(t)$ with autocorrelation function $R_{XX}(\tau) = Ae^{-\alpha|\tau|}$ where A and α are real positive constants, is applied to the input of a linear transmission input system with impulse response $h(t) = e^{-bt}u(t)$ where b is a real positive constant. Find the autocorrelation of the output $Y(t)$ of the system.

Solution Given: $R_{XX}(\tau) = Ae^{-\alpha|\tau|}$; $h(t) = e^{-bt}u(t)$

$$S_{XX}(\omega) = F[R_{XX}(\tau)] = F[Ae^{-\alpha|\tau|}] = \frac{2A\alpha}{\alpha^2 + \omega^2}$$

$$h(t) = e^{-bt}u(t); H(\omega) = F[e^{-bt}u(t)] = \frac{1}{b + j\omega}$$

$$|H(\omega)|^2 = \frac{1}{b^2 + \omega^2}$$

$$\begin{aligned} S_{YY}(\omega) &= |H(\omega)|^2 S_{XX}(\omega) \\ &= \frac{1}{b^2 + \omega^2} \left(\frac{2A\alpha}{\alpha^2 + \omega^2} \right) \end{aligned}$$



Practice Problems

8.4 If $X(t)$ is a band limited process such that $S_{XX}(\omega) = 0$ where $|\omega| > \sigma$, prove that $[2R_{XX}(0) - R_{XX}(\tau)] \leq \sigma^2 \tau^2 R_{XX}(0)$.

Solved Problem



8.25 Assume a random process $X(t)$ is given as input to a system with transfer function $H(\omega) = 1$ for $-\omega_0 \leq \omega \leq \omega_0$. If the autocorrelation function of the input process is $\frac{N_0}{2} \delta(\tau)$. Find the autocorrelation function of the output process.

Solution Given: $R_{XX}(\tau) = \frac{N_0}{2} \delta(\tau)$ and $H(\omega) = 1$ for $-\omega_0 \leq \omega \leq \omega_0$

$$S_{XX}(\omega) = F[R_{XX}(\tau)] = \frac{N_0}{2}$$

$$\begin{aligned} S_{YY}(\omega) &= |H(\omega)|^2 S_{XX}(\omega) \\ &= (1)^2 \frac{N_0}{2} = \frac{N_0}{2} \end{aligned}$$

The output ACF $R_{YY}(\tau) = F^{-1}[S_{YY}(\omega)]$

$$\begin{aligned} &= \frac{1}{2\pi} \int_{-\omega_0}^{\omega_0} \frac{N_0}{2} e^{j\omega\tau} d\omega \\ &= \frac{N_0}{2} \frac{e^{j\omega\tau}}{2\pi j\tau} \Big|_{-\omega_0}^{\omega_0} = \frac{N_0}{2} \left(\frac{e^{j\omega_0\tau} - e^{-j\omega_0\tau}}{2\pi j\tau} \right) = N_0 \frac{\sin \omega_0\tau}{2\pi\tau} \end{aligned}$$



Practice Problems

8.5 If $Y(t)$ be the derivative of $X(t)$, and limited white noise process with $S_{XX}(\omega) = \frac{N_0}{2}$ for $|\omega| < W$

(a) Find $S_{YY}(\omega)$ and $R_{YY}(\tau)$, (b) What is the average of the output?

8.6 A system has an impulse response $h(t) = e^{-\beta t} u(t)$. Find the power spectral density of the output $Y(t)$ corresponding to the input $X(t)$.

8.7 A bandlimited white noise process with $S_{XX}(\omega) = \frac{N_0}{2}$ for $-\omega_0 \leq \omega \leq \omega_0$ is input to a filter with transfer function

$H(\omega) = 1 + j\omega$ (a) Find $S_{YX}(\omega)$, $R_{YX}(\tau)$, $S_{YY}(\omega)$, $R_{YY}(\tau)$ and average power.

Solved Problems



8.26 Two networks with identical impulse response $h(t) = W e^{-Wt} u(t)$ are connected in cascade. The input to first network is $X(t) = e^{-\alpha t} u(t)$. Find the output of second network.

Solution Given: $h(t) = We^{-Wt} u(t)$
 and $x(t) = e^{-\alpha t} u(t)$

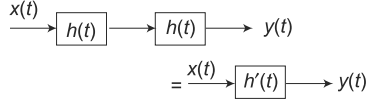


Fig. 8.14

$$h'(t) = h(t) * h(t)$$

$$= \int_0^t We^{-W\tau} We^{-W(t-\tau)} d\tau$$

$$= W^2 \int_0^t e^{-W\tau} e^{-Wt} e^{W\tau} d\tau = W^2 \left[\int_0^t d\tau \right] e^{-Wt}$$

$$= W^2 t e^{-Wt} u(t)$$

$$y(t) = x(t) * h'(t) = \int_{-\infty}^{\infty} h'(\tau) x(t-\tau) d\tau$$

$$= \int_0^t W^2 \tau e^{-W\tau} e^{-\alpha(t-\tau)} d\tau$$

$$= W^2 e^{-\alpha t} \int_0^t \tau e^{-(W-\alpha)\tau} d\tau$$

$$= W^2 e^{-\alpha t} \left\{ -\tau \frac{e^{-(W-\alpha)\tau}}{W-\alpha} \Big|_0^t - \frac{e^{-(W-\alpha)\tau}}{(W-\alpha)^2} \Big|_0^t \right\}$$

$$= \frac{W^2}{(W-\alpha)^2} \left\{ t e^{-Wt} (W-\alpha) - e^{-Wt} + e^{-\alpha t} \right\}$$

$$= \frac{W^2}{(W-\alpha)^2} \left\{ e^{-\alpha t} - e^{-Wt} (1 + (W-\alpha)t) \right\}$$

8.27 The impulse response of a system is

$$h(t) = \begin{cases} t^3 e^{-t} & t > 0 \\ 0 & t < 0 \end{cases}$$

$$\text{If } x(t) = \begin{cases} A & 0 < t < T \\ 0 & \text{elsewhere} \end{cases}$$

Find the response of the system.

Solution

$$\text{Given: } x(t) = \begin{cases} A & 0 < t < T \\ 0 & \text{elsewhere} \end{cases}$$

$$h(t) = \begin{cases} t^3 e^{-t} & t > 0 \\ 0 & t < 0 \end{cases}$$

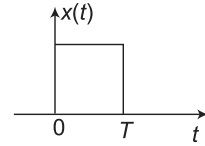


Fig. 8.15

The response $y(t) = x(t) * h(t)$

The signal $x(t)$ is shown in Fig. 8.15.

Mathematical expression for $x(t)$ is

$$x(t) = A\{u(t) - u(t - T)\}$$

$$y(t) = Au(t) * [t^3 e^{-t} u(t)] - Au(t - T) * [t^3 e^{-t} u(t)]$$

$$= A \int_0^t \tau^3 e^{-\tau} d\tau - A \int_0^{t-T} \tau^3 e^{-\tau} d\tau$$

$$= A\{6 - t^3 e^{-t} - 3t^2 e^{-t} - 6te^{-t} - 6e^{-t}\}u(t) \\ - A\{6 - (t - T)^3 e^{-(t-T)} - 3(t - T)^2 e^{-(t-T)} - 6(t - T)e^{-(t-T)} - 6e^{-(t-T)}\}u(t - T)$$

8.28 For the system shown in Fig 8.16 (a). Find the impulse response $h(t)$ and (b) $H(\omega)$.

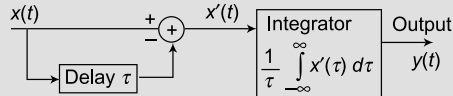


Fig. 8.16

Solution From Fig. 8.16 $x'(t) = x(t) - x(t - \tau)$

If $x(t) = \delta(t)$

$x'(t) = \delta(t) - \delta(t - \tau)$

$$\begin{aligned} \text{Output } y(t) &= \frac{1}{\tau} \int_{-\infty}^{\tau} x'(\tau) d\tau \\ &= \frac{1}{\tau} \int_{-\infty}^{\tau} (\delta(t) - \delta(t - \tau)) d\tau \\ &= \frac{1}{\tau} (u(t) - u(t - \tau)) \end{aligned}$$

$$\Rightarrow h(t) = \frac{1}{\tau} (u(t) - u(t - \tau))$$

The Fourier transform of $h(t)$ is

$$\begin{aligned} H(\omega) &= F[h(t)] = \frac{1}{\tau} \int_0^{\tau} e^{-j\omega t} dt = \frac{1}{\tau} \left(\frac{e^{-j\omega t}}{-j\omega} \right) \Bigg|_0^{\tau} = \frac{1}{-j\omega\tau} [e^{-j\omega\tau} - 1] \\ &= \frac{1 - e^{j\omega\tau}}{j\omega\tau} = \frac{e^{-j\omega\tau/2} [e^{j\omega\tau/2} - e^{-j\omega\tau/2}]}{j\omega\tau} \\ &= e^{-j\omega\tau/2} \frac{\sin(\omega\tau/2)}{\omega\tau/2} \end{aligned}$$

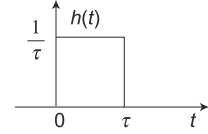


Fig. 8.17

8.29 Find the transfer function of the network shown in Fig. 8.18.

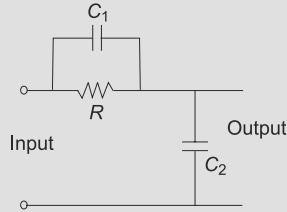
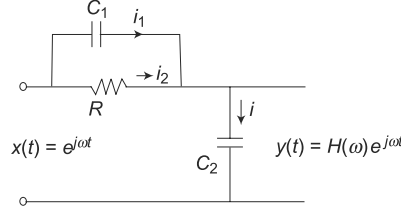


Fig. 8.18

Solution If input $x(t) = e^{j\omega t}$, then the output $y(t) = H(\omega) e^{j\omega t} \Rightarrow y(t) = H(\omega) x(t)$

$$\begin{aligned} \Rightarrow \frac{dy(t)}{dt} &= j\omega H(\omega) e^{j\omega t} \\ &= j\omega H(\omega) x(t) \end{aligned}$$

**Fig. 8.19**

Let the current i_1 and i_2 flow through C_1 and R respectively.

Then we can write

$$i = i_1 + i_2$$

where
$$i = C_2 \frac{dy(t)}{dt}$$

The voltage across C_1 and R is equal to $x(t) - y(t)$ from which

$$i_2 = \frac{x(t) - y(t)}{R}$$

$$i_1 = C_1 \frac{d}{dt} \{x(t) - y(t)\}$$

$$C_2 \frac{dy(t)}{dt} = \frac{x(t) - y(t)}{R} + C_1 \frac{d}{dt} \{x(t) - y(t)\}$$

$$\begin{aligned} C_2 j\omega H(\omega)x(t) &= \frac{x(t)}{R} - \frac{H(\omega)x(t)}{R} + C_1 \left\{ \frac{d}{dt} x(t) - \frac{d}{dt} y(t) \right\} \\ &= \frac{x(t)}{R} - \frac{H(\omega)x(t)}{R} + C_1 j\omega x(t) - C_1 j\omega H(\omega)x(t) \end{aligned}$$

$$j\omega C_2 H(\omega) = \frac{1}{R} - \frac{H(\omega)}{R} + C_1 j\omega - C_1 j\omega H(\omega)$$

$$H(\omega) \left[j\omega C_2 + \frac{1}{R} + j\omega C_1 \right] = \frac{1 + j\omega C_1 R}{R}$$

$$H(\omega) = \frac{1 + j\omega C_1 R}{1 + j\omega R(C_1 + C_2)}$$

or

The phasor form of the circuit is shown in Fig. 8.20.

From Fig. 8.20, $Z_p = \frac{1}{j\omega C_1} \parallel R = \frac{R/j\omega C_1}{R + 1/j\omega C_1} = \frac{R}{1 + j\omega RC_1}$

$$I(\omega) = \frac{X(\omega)}{Z_p + Z_s}, \text{ and } Y(\omega) = \frac{X(\omega)Z_s}{Z_p + Z_s}$$

$$H(\omega) = \frac{Y(\omega)}{X(\omega)} = \frac{Z_s}{Z_p + Z_s} = \frac{1/j\omega C_2}{\frac{R}{1 + j\omega RC_1} + 1/j\omega C_2}$$

$$= \frac{1 + j\omega RC_1}{1 + j\omega RC_1 + j\omega RC_2}$$

$$H(\omega) = \frac{1 + j\omega RC_1}{1 + j\omega R(C_1 + C_2)}$$

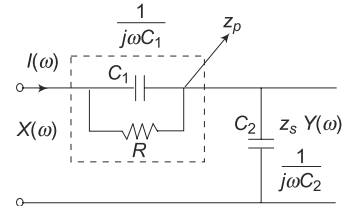


Fig. 8.20



Practice Problem

8.8 Find the transfer function of the following networks.

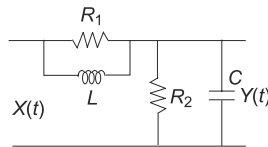


Fig. 8.21

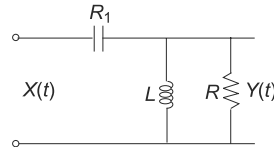


Fig. 8.22

Solved Problems



8.30 For the network shown in Fig. 8.23, find transfer function.

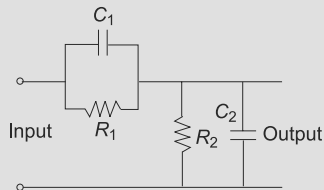


Fig. 8.23

Solution The phasor form of the circuit is shown in Fig. 8.24.

From Fig. 8.24,

$$I(\omega) = \frac{V_i(\omega)}{\frac{R_1}{1 + j\omega C_1 R_1} + \frac{R_2}{1 + j\omega C_2 R_2}}$$

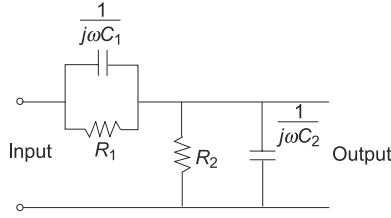


Fig. 8.24

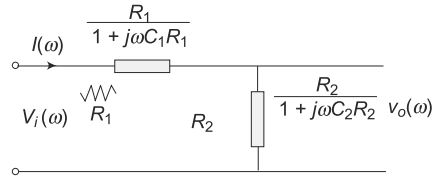


Fig. 8.25

$$\begin{aligned} V_o(\omega) &= \frac{I(\omega) R_2}{1 + j\omega C_2 R_2} \\ &= \frac{R_2 / \frac{R_1}{1 + j\omega C_1 R_1}}{\frac{R_1}{1 + j\omega C_1 R_1} + \frac{R_2}{1 + j\omega C_2 R_2}} \\ &= \frac{R_2 (1 + j\omega C_1 R_1)}{R_1 (1 + j\omega C_2 R_2) + R_2 (1 + j\omega C_1 R_1)} \\ &= \frac{R_2 (1 + j\omega C_1 R_1)}{R_1 + R_2 + j\omega R_1 R_2 (C_1 + C_2)} \end{aligned}$$

8.31 The PSD of a random process is given by $S_{XX}(\omega) = \frac{4}{\omega^2 + 4}$. Find whether this is a valid spectral density or not. If it is transmitted through a system shown in Fig. 8.26, find out the PSD of output.

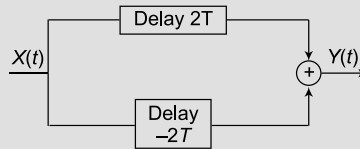


Fig. 8.26

Solution Given:

$$S_{XX}(\omega) = \frac{4}{\omega^2 + 4}$$

We can find that $S_{XX}(\omega) > 0$ for any value of ω and $S_{XX}(-\omega) = S_{XX}(\omega)$. Hence it is a valid PSD.

From Fig. 8.26, $Y(t) = X(t + 2T) + X(t - 2T)$

Applying Fourier transform on both the sides, we get

$$Y(\omega) = e^{j2\omega T} X(\omega) + e^{-j2\omega T} X(\omega)$$

$$= 2X(\omega) \cos 2\omega T$$

$$H(\omega) = \frac{Y(\omega)}{X(\omega)} = 2 \cos 2\omega T$$

$$S_{YY}(\omega) = |H(\omega)|^2 S_{XX}(\omega)$$

$$= 4 \cos^2 \omega T \left(\frac{4}{\omega^2 + 4} \right)$$

8.32 A zero mean random process is applied as input to two filters as shown in Fig. 8.27.

The PSD of the input is $S_{XX}(\omega) = \frac{\eta}{2}$

The impulse response of filters is

$$h_1(t) = e^{-t} u(t)$$

$$h_2(t) = \begin{cases} 1 & \text{for } 0 \leq t \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

Find $H_1(e^{j\omega})$, $H_2(e^{j\omega})$, $E[Y_1(t)]$ and $E[Y_2(t)]$

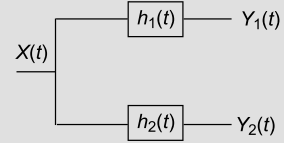


Fig. 8.27

Solution Given:

$$S_{XX}(\omega) = \frac{\eta}{2}$$

$$h_1(t) = e^{-t} u(t);$$

$$h_2(t) = \begin{cases} 1 & \text{for } 0 \leq t \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

$$H_1(\omega) = F[e^{-t} u(t)] = \frac{1}{1 + j\omega}$$

$$H_2(\omega) = \int_{-\infty}^{\infty} h_2(t) e^{-j\omega t} dt = \int_0^1 e^{-j\omega t} dt = -\frac{1}{j\omega} e^{-j\omega t} \Big|_0^1$$

$$= \frac{1 - e^{-j\omega}}{j\omega}$$

The mean $E[Y(t)] = \mu_X H(0)$

Since the mean of $X(t)$ is zero,

$$E[Y_1(t)] = E[Y_2(t)] = 0$$

8.33 The power transfer function of the system

$$|H(\omega)|^2 = \frac{64}{(16 + \omega^2)^2}$$

Find $h(t)$.

Solution

$$|H(\omega)|^2 = \frac{64}{(16 + \omega^2)^2}$$

$$|H(\omega)| = \frac{8}{16 + \omega^2} = H(\omega)$$

$$h(t) = ?$$

We know $F[e^{-a|\tau|}] = \frac{2a}{a^2 + \omega^2}$

If $a = 4$

$$F[e^{-4|\tau|}] = \frac{8}{16 + \omega^2} \Rightarrow h(t) = F^{-1}\left[\frac{8}{16 + \omega^2}\right] = e^{-4|t|}$$

8.34 A WSS process has the ACF given by

$$R_{XX}(\tau) = \cos(\omega_0 \tau)$$

The process is input to the system with the power transfer function

$$|H(\omega)|^2 = \frac{64}{(16 + \omega^2)^2}$$

- (a) Find the PSD of the output.
- (b) If $Y(t)$ is the output of the process, find $S_{XX}(\omega)$.

Solution Given:

$$R_{XX}(\tau) = \cos(\omega_0 \tau)$$

$$S_{XX}(\omega) = F[\cos \omega_0 t]$$

$$= \pi[\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$$

$$S_{YY}(\omega) = |H(\omega)|^2 S_{XX}(\omega)$$

$$= \frac{64}{(16 + \omega^2)^2} \pi[\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$$

$$\begin{aligned}
 S_{YY}(\omega) &= H(\omega)S_{XX}(\omega) \\
 &= \frac{8\pi}{16 + \omega^2} \left\{ \delta(\omega - \omega_0) + \delta(\omega + \omega_0) \right\}
 \end{aligned}$$

8.35 $X(t)$ is a WSS process which is input to a linear system with the transfer function.

$$H(\omega) = \frac{1}{a + j\omega}$$

where $a > 0$. If $X(\omega)$ is a zero mean white noise with PSD $N_0/2$, determine the following:

- (a) $h(t)$ (b) $S_{XY}(\omega)$ (c) $R_{XY}(\tau)$ (d) $R_{YX}(\tau)$ (e) $S_{YX}(\omega)$ (f) $S_{YY}(\omega)$

Solution Given:

$$H(\omega) = \frac{1}{a + j\omega}$$

$$h(t) = F^{-1} \left[\frac{1}{a + j\omega} \right] = e^{-at} u(t)$$

$$S_{XX}(\omega) = N_0/2$$

$$S_{XY}(\omega) = H(\omega)S_{XX}(\omega) = \frac{1}{a + j\omega} \left(\frac{N_0}{2} \right)$$

$$R_{XY}(\tau) = F^{-1} \left[\frac{N_0}{2} \frac{1}{a + j\omega} \right] = \frac{N_0}{2} e^{-a\tau} u(\tau)$$

$$S_{YX}(\omega) = H^*(\omega)S_{XX}(\omega) = \frac{1}{a - j\omega} \left(\frac{N_0}{2} \right)$$

$$R_{YX}(\tau) = F^{-1} \left[\frac{N_0}{2} \frac{1}{a - j\omega} \right] = \frac{N_0}{2} e^{a\tau} u(\tau)$$

$$S_{YY}(\omega) = |H(\omega)|^2 S_{XX}(\omega)$$

$$= \frac{1}{a^2 + \omega^2} \left(\frac{N_0}{2} \right)$$

8.36 White noise with power density 10 W/Hz is applied to a system with impulse response

$$h(t) = \begin{cases} te^{-Wt} & 0 < t \\ 0 & t < 0 \end{cases}$$

Find the mean square value of the response.

Solution Given: $S_{NN}(\omega) = 10 \text{ W/Hz}$

$$R_{NN}(\tau) = 10\delta(\tau)$$

Mean square value of the system response is

$$\begin{aligned}
 E[Y^2(t)] &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} R_{XX}(\tau_1 - \tau_2) h(\tau_1) h(\tau_2) d\tau_1 d\tau_2 \\
 &= 10 \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \delta(\tau_1 - \tau_2) \tau_1 e^{-W\tau_1} \tau_2 e^{-W\tau_2} d\tau_1 d\tau_2 \\
 &= 10 \int_0^{\infty} \tau_2^2 e^{-2W\tau_2} d\tau_2 \\
 &= 10 \left[\frac{\tau_2^2 e^{-2W\tau_2}}{-2W} \Big|_0^{\infty} - \frac{2\tau_2 e^{-2W\tau_2}}{4W^2} \Big|_0^{\infty} - \frac{e^{-2W\tau_2}}{4W^3} \Big|_0^{\infty} \right] \\
 &= \frac{10}{4W^3} = \frac{2.5}{W^3}
 \end{aligned}$$

8.37 A stationary random process $X(t)$ is applied to the input of a system for which $h(t) = t^2 e^{-8t} u(t)$. If $E[X(t)] = 5$. What is the mean value of the system's response $Y(t)$?

Solution Given: $h(t) = t^2 e^{-8t} u(t)$; $\mu_X = 5$

The mean value

$$\begin{aligned}
 \mu_Y &= \mu_X \int_{-\infty}^{\infty} h(t) dt = 5 \int_0^{\infty} t^2 e^{-8t} dt \\
 &= 5 \left[\frac{-t^2 e^{-8t}}{8} \Big|_0^{\infty} - \frac{2te^{-8t}}{64} \Big|_0^{\infty} + \frac{2e^{-8t}}{-8(64)} \Big|_0^{\infty} \right] = \frac{5}{256} \\
 \mu_Y &= \frac{5}{256}
 \end{aligned}$$



Practice Problem

8.9 A random process $X(t)$ is applied to a network with impulse response $h(t) = e^{-bt} u(t)$, where $b > 0$ is a constant. The cross correlation of $X(t)$ with the output $Y(t)$ is known to have the same form

$$R_{XY}(\tau) = te^{-bt} u(t)$$

- Find the autocorrelation of $Y(t)$.
- What is the average power of $Y(t)$?

Solved Problems

8.38 A random process $X(t)$ is applied to a network with impulse response.

$$h(t) = te^{-bt} u(t)$$

where $b > 0$ is a constant. This cross-correlation of $X(t)$ with the output is known to have the same form

$$R_{XY}(\tau) = \tau e^{-b\tau} u(\tau)$$

- (a) Find the autocorrelation of $Y(t)$.
- (b) What is the average power in $Y(t)$?

Solution Given:

$$h(t) = te^{-bt} u(t)$$

and $R_{XY}(\tau) = \tau e^{-b\tau} u(\tau)$

We know

$$\begin{aligned} R_{YY}(\tau) &= R_{XY}(\tau) * h(-\tau) \\ &= \int_{-\infty}^{\infty} R_{XY}(\tau + \tau_1) h(\tau_1) d\tau_1 \\ &= \int_0^{\infty} (\tau + \tau_1) e^{-b(\tau + \tau_1)} \tau_1 e^{-b\tau_1} d\tau_1 \\ &= e^{-b\tau} \int_0^{\infty} (\tau_1^2 + \tau\tau_1) e^{-2b\tau_1} d\tau_1 \\ R_{YY}(\tau) &= e^{-b\tau} \left\{ \left. \frac{-\tau_1^2 e^{-2b\tau_1}}{2b} \right|_0^{\infty} - \left. \frac{2\tau_1 e^{-2b\tau_1}}{4b^2} \right|_0^{\infty} - \left. \frac{e^{-2b\tau_1}}{4b^3} \right|_0^{\infty} \right\} + e^{-b\tau} \tau \left\{ \left. \frac{-\tau_1 e^{-2b\tau_1}}{2b} \right|_0^{\infty} - \left. \frac{e^{-2b\tau_1}}{4b^2} \right|_0^{\infty} \right\} \\ &= e^{-b\tau} \left\{ \frac{1}{4b^3} + \frac{\tau}{4b^2} \right\} = e^{-b\tau} \left(\frac{1+b\tau}{4b^3} \right) \end{aligned}$$

Since $R_{YY}(\tau)$ is an even function of τ , for any τ ,

$$R_{YY}(\tau) = \frac{1+b|\tau|}{4b^3} e^{-b|\tau|}$$

$$\text{Power} = E[Y^2(t)] = R_{YY}(0) = \frac{1}{4b^3}$$

8.39 Two identical networks each with impulse response $h(t) = te^{-2t} u(t)$ are cascaded. A wide-sense stationary process $X(t)$ is applied to the system. Find $Y(t)$ and $E[Y(t)]$ if $E[X(t)] = 10$.

Solution The system is shown in Fig. 8.28,

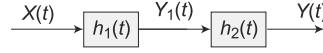


Fig. 8.28

Using convolution, we can obtain

$$Y_1(t) = \int_{-\infty}^{\infty} h(\tau_1) X(t - \tau_1) d\tau_1$$

and

$$Y(t) = Y_1(t) * h(t)$$

$$= \int_{-\infty}^{\infty} h(\tau_2) Y_1(t - \tau_2) d\tau_2$$

$$= \int_{-\infty}^{\infty} h(\tau_1) \int_{-\infty}^{\infty} h(\tau_2) X(t - \tau_1 - \tau_2) d\tau_1 d\tau_2$$

$$= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} X(t - \tau_1 - \tau_2) h(\tau_1) h(\tau_2) d\tau_1 d\tau_2$$

$$= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} X(t - \tau_1 - \tau_2) \tau_1 e^{-2\tau_1} \tau_2 e^{-2\tau_2} u(\tau_1) u(\tau_2) d\tau_1 d\tau_2$$

$$= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} X(t - \tau_1 - \tau_2) \tau_1 \tau_2 e^{-2(\tau_1 + \tau_2)} d\tau_1 d\tau_2$$

$$E[Y(t)] = E \left[\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} X(t - \tau_1 - \tau_2) \tau_1 \tau_2 e^{-2(\tau_1 + \tau_2)} d\tau_1 d\tau_2 \right]$$

$$= E[X] \left\{ \int_0^{\infty} (\tau e^{-2\tau}) d\tau \right\}^2$$

$$= 10 \left\{ -\frac{\tau e^{-2\tau}}{2} \Big|_0^{\infty} + \frac{e^{-2\tau}}{4} \Big|_0^{\infty} \right\}^2$$

$$E[Y] = 10 \left(\frac{1}{4} \right)^2 = \frac{10}{16} = \frac{5}{8}$$

8.40 A wide-sense stationary process $X(t)$ with mean value 10 and power spectrum

$$S_{XX}(\omega) = 10\pi\delta(\omega) + 4/[1 + (\omega/4)^2]$$

is applied to a network with impulse response $h(t) = 2e^{-2|t|}$

- (a) Find $H(\omega)$ for the network.
 (b) Find $E[Y]$ and (c) $S_{YY}(\omega)$

Solution Given: $h(t) = 2e^{-2|t|}$

$$H(\omega) = F[2e^{-2|t|}] = 2F[e^{-2|t|}] = 2 \frac{2(2)}{2^2 + \omega^2} = \frac{8}{4 + \omega^2}$$

(b) $E[Y] = E[X] H(0)$

$$= 10 \frac{8}{4 + \omega^2} \Big|_{\omega=0} = 20$$

(c) $S_{YY}(\omega) = |H(\omega)|^2 S_{XX}(\omega)$

$$= |H(\omega)|^2 \{10\pi\delta(\omega) + 4/[1 + (\omega/4)^2]\}$$

$$= 10\pi\delta(\omega) |H(0)|^2 + \{4/[1 + (\omega/4)^2]\} \left(\frac{8}{4 + \omega^2} \right)^2$$

$$= 40\pi\delta(\omega) + \{4/[1 + (\omega/4)^2]\} \left(\frac{8}{4 + \omega^2} \right)^2$$

8.41 A white noise with $R_{NN}(\tau) = 0.1 \delta(\tau)$ is applied to a network with impulse response

$$h(t) = te^{-2t} u(t)$$

- (a) Find the network's output noise power in a 1-ohm resistor.
 (b) Obtain an expression for the output power spectrum.

Solution Given: $R_{NN}(\tau) = 0.1 \delta(\tau)$

$$h(t) = te^{-2t} u(t)$$

$$H(\omega) = F[h(t)] = \frac{1}{(2 + \omega^2)}$$

$$|H(\omega)|^2 = \frac{1}{(4 + \omega^2)^2}$$

Output noise power

$$E[Y^2(t)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} R_{NN}(\tau_1 - \tau_2) h(\tau_1) h(\tau_2) d\tau_1 d\tau_2$$

Given: $R_{NN}(\tau) = 0.1 \delta(\tau)$

$$\begin{aligned}
 E[Y^2(t)] &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} 0.1 \delta(\tau_1 - \tau_2) h(\tau_1) h(\tau_2) d\tau_1 d\tau_2 \\
 &= 0.1 \int_0^{\infty} \tau^2 e^{-4\tau} d\tau \\
 &= 0.1 \left\{ \left. \frac{-\tau^2 e^{-4\tau}}{4} \right|_0^{\infty} - \frac{2\tau e^{-4\tau}}{16} \right|_0^{\infty} - \frac{e^{-4\tau}}{32} \right|_0^{\infty} \right\} = \frac{1}{320} \\
 S_{YY}(\omega) &= |H(\omega)|^2 S_{NN}(\omega) = \frac{0.1}{(4 + \omega^2)^2}
 \end{aligned}$$

8.42 A stationary random signal $X(t)$ has an autocorrelation function $R_{XX}(\tau) = 5e^{-2|\tau|}$. It is added to a white noise for which $N_0/2 = 10^{-2}$ and sum is applied to a filter having a transfer function

$$H(\omega) = \frac{2}{(2 + j\omega)^2}$$

If the noise is independent of $X(t)$,

- Find the signal component of the output power spectrum and the average power in the output signal.
- Find the power spectrum of, and average power in the output noise, and
- What is the ratio of the output signal's power to the output average noise power?

Solution Given: $R_{XX}(\tau) = 5e^{-2|\tau|}$

$$S_{XX}(\omega) = 5 \frac{2(2)}{2^2 + \omega^2} = \frac{20}{4 + \omega^2}$$

$$H(\omega) = \frac{2}{(2 + j\omega)^2}$$

$$\text{Power transfer function } |H(\omega)|^2 = \frac{4}{(4 + \omega^2)^2}$$

Signal component of output power spectrum

$$= |H(\omega)|^2 S_{XX}(\omega) = \frac{4}{(4 + \omega^2)^2} \cdot \frac{20}{4 + \omega^2} = \frac{80}{(4 + \omega^2)^3}$$

Average power in the output signal

$$P = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{80}{(4 + \omega^2)^3} d\omega$$

Let $\omega = 2 \tan \theta$,

then $d\omega = 2 \sec^2 \theta d\theta$

$\theta = -\pi/2$ when $\omega = -\infty$

$\theta = \pi/2$ when $\omega = \infty$

$$\begin{aligned}
 P_s &= \frac{1}{2\pi} \int_{-\pi/2}^{\pi/2} \frac{80 (2 \sec^2 \theta)}{4^3 (\sec^2 \theta)^3} d\theta \\
 &= \frac{5}{4\pi} \int_{-\pi/2}^{\pi/2} \cos^4 \theta d\theta = \frac{5}{4\pi} \int_{-\pi/2}^{\pi/2} \left(\frac{1 + \cos 2\theta}{2} \right)^2 d\theta \\
 &= \frac{5}{16\pi} \int_{-\pi/2}^{\pi/2} (1 + \cos^2 2\theta + 2 \cos 2\theta) d\theta \\
 &= \frac{5}{16\pi} \left(\frac{3}{2} \pi \right) = \frac{15}{32}
 \end{aligned}$$

The noise output power

$$S_{NN}(\omega) = \frac{N_0}{2} \frac{4}{(4 + \omega^2)^2} = \frac{2N_0}{(4 + \omega^2)^2}$$

Average noise power

$$\begin{aligned}
 P_N &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{2N_0}{(4 + \omega^2)^2} d\omega \\
 &= \frac{N_0}{\pi} \int_{-\infty}^{\infty} \frac{1}{(4 + \omega^2)^2} d\omega \\
 &= \frac{N_0}{\pi} \int_{-\pi/2}^{\pi/2} \frac{1}{4^2 (\sec^2 \theta)^2} 2 \sec^2 \theta d\theta \\
 &= \frac{N_0}{8\pi} \int_{-\pi/2}^{\pi/2} \cos^2 \theta d\theta = \frac{N_0}{8\pi} \left(\frac{\pi}{2} \right) = \frac{N_0}{16}
 \end{aligned}$$

Given, $\frac{N_0}{2} = 10^{-2}$

$$P_N = \frac{10^{-2}}{8}$$

$$\begin{aligned}
 \frac{\text{Output signal's power}}{\text{Output average noise power}} &= \frac{15/32}{(10^{-2}/8)} \\
 &= \frac{15}{32} \cdot \frac{8}{10^{-2}} = 375
 \end{aligned}$$

8.43 A white noise with power density $N_0/2$ is applied to a low-pass network for which $|H(0)| = 2$; it has a noise bandwidth of 2 MHz. If the average output noise power is 0.1 W in a 1 Ω resistor, what is N_0 ?

Solution Given: $|H(0)| = 2$; $W_N = 2$ MHz; $P_{YY} = 0.2$

We know

$$P_{YY} = \frac{N_0}{2\pi} |H(\omega_0)|^2 W_N$$

For a low-pass filter,

$$P_{YY} = \frac{N_0}{2\pi} |H(0)|^2 W_N$$

That is,

$$P_{YY} = \frac{N_0}{2\pi} |H(0)|^2 W_N = 0.1$$

$$N_0 = \frac{0.1(2\pi)}{|H(0)|^2 W_N}$$

$$= \frac{0.1(2\pi)}{4 \times 2\pi \times 2 \times 10^6} = 1.25 \times 10^{-8} \text{ W/Hz}$$

$$|H(0)|^2 = 4$$

$$W_N = 2\pi(2 \times 10^6)$$



MEASUREMENT OF POWER DENSITY SPECTRUM 8.7

The schematic to measure the power density spectrum of a low-pass process $X(t)$ is shown in Fig. (8.29). The random process $X(t)$ with power spectrum shown in Fig. (8.29(b)) is applied as input to a tunable narrowband, band-pass filter which is tuned to centre frequency $\omega = \omega_f$. The centre frequency of the band-pass filter can be adjusted from near $\omega = 0$ to angular frequency W . The output of the filter $Y(t)$ is applied to a power meter. The function of the power meter is to measure the average power in $Y(t)$. The following assumptions are made in the measurement of the spectrum.

1. The input process $X(t)$ and output $Y(t)$ are stationary, ergodic processes.
2. The power meter averages $Y^2(t)$ over a long time such that the fluctuation in its readings are relative to the measured power.
3. The spectral extent, W_f of the filter is very small compared to W .

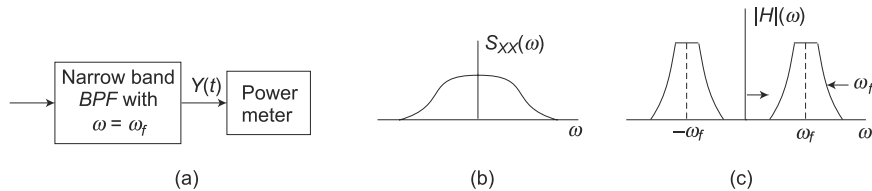


Fig. 8.29 (a) An arrangement to measure power density spectrum of a low-pass process
(b) Power spectrum of $X(t)$ (c) Power transfer function of BPF

The power

$$P_{YY}(\omega_f) = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_{YY}(\omega_f) d\omega \quad (8.49)$$

$$S_{YY}(\omega) = |H(\omega)|^2 S_{XX}(\omega) \quad (8.50)$$

$$\Rightarrow P_{YY}(\omega_f) = \frac{1}{2\pi} \int_{-\infty}^{\infty} |H(\omega)|^2 S_{XX}(\omega) d\omega \quad (8.51)$$

For real filters and real $X(t)$, the PSD of $X(t)$ and the power transfer function are even functions of ω .

Therefore, we can write

$$\begin{aligned} P_{YY}(\omega_f) &= \frac{1}{2\pi} \left\{ 2 \int_0^{\infty} |H(\omega)|^2 S_{XX}(\omega) d\omega \right\} \\ &= \frac{1}{\pi} \int_0^{\infty} S_{XX}(\omega_f) |H(\omega)|^2 d\omega \\ &\simeq \frac{1}{\pi} S_{XX}(\omega_f) \int_0^{\infty} |H(\omega)|^2 d\omega \\ &= \frac{S_{XX}(\omega_f) |H(\omega_f)|^2 W_N}{\pi} \end{aligned} \quad (8.52)$$

$$W_N = \frac{\int_0^{\infty} |H(\omega)|^2 d\omega}{|H(\omega_f)|^2} \quad (8.53)$$

is known as noise bandwidth.

Rearranging Eq. (8.52) and substituting Eq. (8.53), we get

$$S_{XX}(\omega_f) = \frac{\pi P_{YY}(\omega_f)}{W_N |H(\omega_f)|^2} \quad (8.54)$$

That is, the approximate $S_{XX}(\omega)$ can be obtained by measuring the power at the output of the filter when the filter is tuned to $\omega = \omega_f$ and multiplying the same with a constant $\pi \{W_N |H(\omega_f)|^2\}$.

Noise Bandwidth

Consider a system with low-pass characteristics. Let the system transfer function be $H(\omega)$. If the system input is white noise with PSD $N_0/2$, then the average power at the output of the system is

$$P_{YY} = \frac{1}{2\pi} \int_{-\infty}^{\infty} \left(\frac{N_0}{2} \right) |H(\omega)|^2 d\omega \quad (8.55)$$

If the system impulse response is real then the power transfer function $|H(\omega)|^2$ will be an even function of ω . Therefore, Eq. (8.55) can be written as

$$P_{YY} = \frac{N_0}{2\pi} \int_0^{\infty} |H(\omega)|^2 d\omega \quad (8.56)$$

Now consider an idealized system that produces the same average power as that of an actual system when it is excited by the same white-noise source. The power transfer function of such a system is

$$|H_I(\omega)|^2 = \begin{cases} |H(0)|^2 & |\omega| < W_N \\ 0 & |\omega| > W_N \end{cases} \quad (8.57)$$

where W_N is a constant that makes the output powers in the two systems equal and $|H(0)|^2$ is the power transfer function value at midband.

The output power in the idealized system is

$$P'_{YY} = \frac{1}{2\pi} \int_{-\infty}^{\infty} |H_I(\omega)|^2 S_{XX}(\omega) d\omega = \frac{1}{2\pi} \int_{-W_N}^{W_N} |H(0)|^2 S_{XX}(\omega) d\omega$$

Since $|H(0)|^2$ is even and $S_{XX}(\omega) = \frac{N_0}{2}$,

$$\begin{aligned} P'_{YY} &= \frac{2}{2\pi} \int_0^{W_N} |H(0)|^2 \left(\frac{N_0}{2} \right) d\omega \\ &= \frac{N_0}{2\pi} \int_0^{W_N} |H(0)|^2 d\omega = \frac{N_0 |H(0)|^2 W_N}{2\pi} \end{aligned}$$

from which

$$W_N = \frac{\int_0^{\infty} |H(\omega)|^2 d\omega}{|H(0)|^2} \quad (8.58)$$

W_N is called the *noise bandwidth* of the system.

Let the input be applied to an idealized system with bandpass transfer function having a centre band frequency ω_0 . The power transfer function of the system is shown in Fig. (8.30).

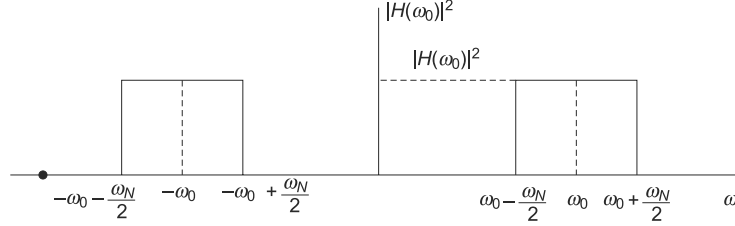


Fig. 8.30 The power transfer function of bandpass system

Mathematically,

$$|H(\omega)|^2 = \begin{cases} |H(\omega_0)|^2 & \text{for } \omega_0 - \frac{\omega_N}{2} \leq \omega \leq \omega_0 + \frac{\omega_N}{2} \text{ and} \\ & -\omega_0 - \frac{\omega_N}{2} \leq \omega \leq -\omega_0 + \frac{\omega_N}{2} \\ 0 & \text{elsewhere} \end{cases}$$

Therefore, we can write

$$\begin{aligned} \frac{N_0}{2\pi} \int_0^\infty |H(\omega)|^2 d\omega &= \frac{1}{2\pi} \left\{ \int_{-\omega_0 - \omega_N/2}^{-\omega_0 + \omega_N/2} \frac{N_0}{2} |H(\omega_0)|^2 d\omega \right\} + \frac{1}{2\pi} \int_{\omega_0 - \omega_N/2}^{\omega_0 + \omega_N/2} \frac{N_0}{2} |H(\omega_0)|^2 d\omega \\ &= \frac{1}{2\pi} \left(\frac{N_0}{2} |H(\omega_0)|^2 \right) [2W_N] \\ W_N &= \frac{\int_0^\infty |H(\omega)|^2 d\omega}{|H(\omega_0)|^2} \end{aligned}$$

Solved Problems



8.44 Find the noise bandwidth of the system having the power transfer function

$$|H(\omega)|^2 = \frac{1}{[1 + (\omega/W)^2]^2}$$

Solution

$$\begin{aligned} \text{Noise bandwidth } W_N &= \frac{\int_0^\infty |H(\omega)|^2 d\omega}{|H(0)|^2} \\ |H(0)|^2 &= 1 \end{aligned}$$

$$\begin{aligned}
 \int_0^{\infty} |H(\omega)|^2 d\omega &= \int_0^{\infty} \frac{1}{[1 + (\omega/W)^2]^2} d\omega \\
 &= \int_0^{\pi/2} \frac{W \sec^2 \theta}{(1 + \tan^2 \theta)^2} d\theta \\
 &= W \int_0^{\pi/2} \cos^2 \theta d\theta = W \int_0^{\pi/2} \left(\frac{1 + \cos 2\theta}{2} \right) d\theta = \frac{\pi W}{4}
 \end{aligned}$$

Let $\omega = W \tan \theta$ $d\omega = W \sec^2 \theta d\theta$

8.45 The power transfer function of a network is given by

$$|H(\omega)|^2 = \frac{\omega^4}{[1 + (\omega/W)^2]^4}$$

Treat the network as a band-pass function and

- Find ω_0 , the value of ω where $|H(\omega)|^2$ is maximum
- Find the network's noise bandwidth.

Solution To find ω_0 for ω at which $|H(\omega)|^2$ is maximum, find $\frac{d}{d\omega} [|H(\omega)|^2]$

$$\begin{aligned}
 \frac{d}{d\omega} \left\{ \frac{\omega^4}{[1 + (\omega/W)^2]^4} \right\} \\
 = \frac{[1 + (\omega/W^2)^2]^4 4\omega^3 - 4\omega^4 [1 + (\omega/W^2)^2]^3 2(\omega/W)(1/W)}{[1 + (\omega/W^2)^2]^8}
 \end{aligned}$$

Equating the above term to zero and solving for ω_0 ,

$$\Rightarrow [1 + (\omega/W)^2] 4\omega^3 = 4\omega^4 (2\omega/W^2)$$

$$1 + (\omega/W)^2 = 2\omega^2/W^2$$

$$1 + (\omega^2/W^2) + 2(\omega/W) = 2\omega^2/W^2$$

$$(\omega/W)^2 - 2(\omega/W) - 1 = 0$$

$$\omega_0 = (1 + \sqrt{2})W$$

$$\begin{aligned}
 \text{(b)} \quad |H(\omega_0)|^2 &= \frac{\omega^4}{[1 + (\omega/W)^2]^4} \bigg|_{\omega=\omega_0} \\
 &= \frac{(1 + \sqrt{2})^4 W^4}{[1 + (1 + \sqrt{2})^2]^4} = \frac{33.97W^4}{2174.116} = 0.0156W^4
 \end{aligned}$$

$$W_N = \frac{\int_0^\infty |H(\omega)|^2 d\omega}{|H(\omega_0)|^2}$$

Let $\omega = W \tan \theta$ $d\omega = W \sec^2 \theta d\theta$
--

$$\begin{aligned} \int_0^\infty |H(\omega)|^2 d\omega &= \int_0^\infty \frac{\omega^4}{[1 + (\omega/W)^2]^4} d\omega \\ &= \int_0^{\pi/2} W^5 \frac{\tan^4 \theta \sec^2 \theta d\theta}{(\sec^2 \theta)^4} \\ &= \int_0^{\pi/2} W^5 \sin^4 \theta \cos^2 \theta d\theta \end{aligned}$$

Using trigonometric identities,

$$\begin{aligned} \int_0^{\pi/2} |H(\omega)|^2 d\omega &= W^5 \int_0^{\pi/2} \frac{1}{8} (0.5 - 0.5 \cos 4\theta + 0.5 \cos 6\theta) d\theta \\ &= \frac{\pi}{32} W^5 \\ W_N &= \frac{(\pi/32)W^5}{0.0156 W^4} = 6.29 W \end{aligned}$$

8.46 Find the noise bandwidth of a low-pass RC filter.

Solution The transfer function of a low-pass RC filter is $H(\omega) = \frac{1}{1 + j\omega RC}$

The magnitude is $|H(\omega)| = \frac{1}{\sqrt{1 + \omega^2 R^2 C^2}}$

$$W_N = \frac{\int_0^\infty |H(\omega)|^2 d\omega}{|H(0)|^2}$$

$$\int_0^\infty |H(\omega)|^2 d\omega = \int_0^\infty \frac{1}{\sqrt{1 + \omega^2 R^2 C^2}} d\omega$$

$$\begin{aligned}
 \int_0^{\infty} |H(\omega)|^2 d\omega &= \int_0^{\infty} \frac{1}{\sqrt{1+\omega^2 R^2 C^2}} d\omega \\
 &= \frac{1}{RC} \int_0^{\infty} \frac{1}{\sqrt{1+u^2}} du \quad \text{Let } \omega RC = u, \text{ then } d\omega = \frac{1}{RC} du \\
 &= \frac{1}{RC} \tan^{-1} u \Big|_0^{\infty} = \frac{\pi}{2RC}
 \end{aligned}$$

8.47 What is the noise equivalent bandwidth of an ideal BPF with bandwidth W ?

Solution

$$\begin{aligned}
 W_N &= \frac{\int_0^{\infty} |H(\omega)|^2 d\omega}{|H(\omega_0)|^2} \\
 \int_0^{\infty} |H(\omega)|^2 d\omega &= \int_{\omega_0 - W/2}^{\omega_0 + W/2} |H(\omega)|^2 d\omega = W |H(\omega_0)|^2 = W \\
 |H(\omega_0)|^2 &= 1 \quad \text{for ideal BPF.}
 \end{aligned}$$

8.48 The output of a linear filter is C times the input $X(t)$. Show that

$$R_{YY}(\tau) = C^2 R_{XX}(\tau)$$

Solution Given: $Y(t) = CX(t)$

$$\begin{aligned}
 R_{YY}(t, t + \tau) &= E[Y(t)Y(t + \tau)] \\
 &= E[CX(t) CX(t + \tau)] \\
 &= C^2 E[X(t)X(t + \tau)] \\
 &= C^2 R_{XX}(\tau)
 \end{aligned}$$

and $R_{YY}(\tau) = C^2 R_{XX}(\tau)$

8.49 An LTI system has an input-output relationship given by

$$Y(t) = \frac{X(t) - X(t - t_0)}{t_0}$$

- Find $R_{YY}(\tau)$ in terms of $R_{XX}(\tau)$
- Find output PSD $S_{YY}(\omega)$ in terms of the input PSD $S_{XX}(\omega)$.
- Does your answer to Part (b), make sense in the limit as $t_0 \rightarrow 0$?

Solution

$$\begin{aligned}
 (a) \quad R_{YY}(\tau) &= E[Y(t)Y(t + \tau)] \\
 &= E \left[\left\{ \frac{X(t) - X(t - t_0)}{t_0} \right\} \left\{ \frac{X(t + \tau) - X(t + \tau - t_0)}{t_0} \right\} \right]
 \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{t_0^2} \left\{ E[X(t)X(t+\tau)] - E[X(t-t_0)X(t+\tau)] \right. \\
&\quad \left. - E[X(t)X(t-t_0+\tau)] + E[X(t-t_0)X(t-t_0+\tau)] \right\} \\
&= \frac{1}{t_0^2} [R_{XX}(\tau) - R_{XX}(\tau+t_0) - R_{XX}(\tau-t_0) + R_{XX}(\tau)] \\
&= \frac{1}{t_0^2} [2R_{XX}(\tau) - R_{XX}(\tau+t_0) - R_{XX}(\tau-t_0)]
\end{aligned}$$

$$\begin{aligned}
(b) \quad S_{YY}(\omega) &= F[R_{YY}(\tau)] \\
&= \frac{1}{t_0^2} [2S_{XX}(\omega) - S_{XX}(\omega)\{e^{j\omega t_0} + e^{-j\omega t_0}\}] \\
&= \frac{1}{t_0^2} [2S_{XX}(\omega) - 2\cos\omega t_0 S_{XX}(\omega)] \\
&= \frac{4S_{XX}(\omega)\sin^2(\omega t_0/2)}{t_0^2} \\
S_{YY}(\omega) &= \frac{4S_{XX}(\omega)}{t_0^2} \sin^2\left(\frac{\omega t_0}{2}\right) \\
&= S_{XX}(\omega) \left[\frac{2\sin(\omega t_0/2)}{t_0} \right] \left[\frac{2\sin(\omega t_0/2)}{t_0} \right]
\end{aligned}$$

$$\lim_{t_0 \rightarrow 0} \frac{X(t) - X(t-t_0)}{t_0} = \frac{dX(t)}{dt}$$

$$\text{That is, } Y(t) = \frac{dX(t)}{dt}$$

$$\begin{aligned}
S_{YY}(\omega) &= \lim_{t_0 \rightarrow 0} S_{XX}(\omega) \left(\frac{2\sin(\omega t_0/2)}{t_0} \right) \left(\frac{2\sin(\omega t_0/2)}{t_0} \right) \\
&= S_{XX}(\omega) 2(\omega/2) 2(\omega/2) \\
&= \omega^2 S_{XX}(\omega)
\end{aligned}$$

$$\lim_{\tau_0 \rightarrow 0} \frac{\sin(\omega \tau_0/2)}{\omega \tau_0/2} = 1$$

(c) $\Rightarrow$ It is proved that for a differentiate process,

$$\frac{dY(t)}{dt} \quad \text{the } S_{YY}(\omega) = \omega^2 S_{XX}(\omega)$$

8.50 Let $Y(t)$ be a short-term integration of $X(t)$

$$Y(t) = \frac{1}{T} \int_{t-T}^t x(t') dt'$$

- (a) Find $h(t)$ and $H(\omega)$.
(b) $S_{YY}(\omega)$ in term of $S_{XX}(\omega)$.

Solution

$$\begin{aligned}
 h(t) &= \frac{1}{T} \int_{t-T}^t x(t') dt' = \frac{1}{T} \int_{t-T}^t x(t') dt' \\
 &= \frac{1}{T} \int_{-\infty}^t \delta(t) dt - \frac{1}{T} \int_{-\infty}^{t-T} \delta(t) dt = u(t) - u(t-T) \\
 H(\omega) &= F(h(t)) = \int_{-\infty}^{\infty} h(t) e^{-j\omega t} dt = \frac{1}{T} \int_0^T e^{-j\omega t} dt \\
 &= \frac{1}{T(-j\omega)} e^{-j\omega t} \Big|_0^T = \frac{1 - e^{-j\omega T}}{j\omega T} = e^{-j\omega T/2} \frac{\sin(\omega T/2)}{(\omega T/2)} \\
 |H(\omega)| &= \left| \frac{\sin(\omega T/2)}{\omega T/2} \right| \Rightarrow |H(\omega)|^2 = \frac{\sin^2(\omega T/2)}{(\omega T/2)^2} \\
 S_{YY}(\omega) &= |H(\omega)|^2 S_{XX}(\omega) \\
 &= \frac{\sin^2(\omega T/2)}{(\omega T/2)^2} S_{XX}(\omega)
 \end{aligned}$$

8.51 A white-noise process with a PSD of $S_{XX}(\omega) = \frac{N_0}{2}$ is passed through a finite time integrator whose output is given by

$$Y(t) = \frac{1}{T} \int_{t-T}^t X(u) du$$

- Find (a) The PSD of the output process.
 (b) The total power in the output process.
 (c) The noise equivalent bandwidth of the integrator.

Solution

$$S_{YY}(\omega) = \frac{\sin^2(\omega T/2)}{(\omega T/2)^2} \left(\frac{N_0}{2} \right)$$

The noise equivalent bandwidth

$$W_N = \frac{\int_0^{\infty} |H(\omega)|^2 d\omega}{|H(0)|^2}$$

$$|H(0)|^2 = \lim_{\omega \rightarrow 0} \frac{\sin^2(\omega T / 2)}{(\omega T / 2)^2} = 1$$

$$\begin{aligned} \int_0^\infty |H(\omega)|^2 d\omega &= \int_0^\infty \frac{\sin^2(\omega T / 2)}{(\omega T / 2)^2} d\omega \\ &= \frac{2}{T} \left[\frac{1}{2} \int_{-\infty}^\infty \frac{\sin^2 \theta}{\theta^2} d\theta \right] \\ &= \frac{1}{T} [\pi] = \frac{\pi}{T} \end{aligned}$$

Let $\omega T / 2 = \theta$
 $d\omega = \frac{2}{T} d\theta$



MODELING OF NOISE SOURCES 8.8

In this chapter so far we discussed about the response of a linear system for random signal. The randomness in the signal may be due to nature of the signal itself or it may be noise affecting a deterministic signal. In finding output of a system due to noise, we assumed that the system is free from noise. But in the real world, such an assumption is never justified because all systems generate one or more types of noise internally. One such example is an electronic circuit that generate thermal noise due to random movement of electrons as a result of thermal agitation. The thermal noise can be closely modeled by a Gaussian process because the current produced by movement of electrons in a circuit is the sum of small currents produced by a very large number of sources. If we assume that these sources are i.i.d. then the total current is the sum of a large number of i.i.d. random sources. By the central limit theorem thus the total current has a Gaussian distribution.

If a system or network contains such type of internal noise then suitable modeling techniques are used to replace the noisy network with a noise-free network that is driven by external noise source.



THERMAL NOISE SOURCE 8.9

Thermal noise source is due to the random movement of free electrons in any conductor. We know that any conductive medium is characterized by the resistor R . When the temperature of the conductor is above absolute zero, the free electrons in the resistor can wander randomly through the entire volume of the resistor. This random motion of electrons results in non-uniform charge distribution throughout the conductor which results in a noise voltage across the terminals of the resistor. Such a noise voltage is called thermal noise or resistor noise. At absolute zero, there is no motion of electrons and the noise voltage is zero. When the temperature increases, the noise voltage also increases. That is, thermal noise depends mainly on temperature.

The noise power delivered by a resistor is proportions to temperature and bandwidth B_n of the system. Therefore,

$$P_n = KT B_n \quad (8.59)$$

where T = temperature in K

B_n = noise bandwidth in Hz and

K = Boltzman constant = 1.38×10^{-23} J/°K

The spectral density is defined as power per unit bandwidth. Therefore, the PSD of thermal noise is

$$S_N = \frac{P_n}{B} = KT \text{ watts/Hz} \quad (8.60)$$

The noisy resistor R can be modelled as a noise source $V_n(t)$ in series with a resistor R .

The average or mean noise voltage across the resistor is zero, but the rms value is finite. Therefore, we consider rms value of V_n for noise calculations. Let the noisy resistor R is driving a load resistance R_L as shown in Fig. 8.31.

According to maximum power transfer theorem, the circuit delivers maximum power when $R = R_L$. In such a case, the load is matched to the source and the maximum power delivered to the load is

$$\frac{V_n^2}{4R} \text{ or } \frac{I_n^2 R}{4} \text{ watts}$$

where V_n^2 and I_n^2 are rms values of voltage and current respectively. Equating the expression for thermal noise power to the maximum power delivered results

$$\frac{V_n^2}{4R} = KT B_n \quad (8.61)$$

$$V_n^2 = 4 R KT B_n \quad (8.62)$$

Thus, a noisy resistor R can be represented as a noise-free resistor R in series with noise voltage $v_n(t)$ with mean square value.

$$V_n^2 = 4 RKT B_n \text{ volt}^2 \quad (8.63)$$

The above relation is known as Nyquist theorem.

The equivalent circuit of noisy resistance is shown in Fig. 8.32.

We have

$$\begin{aligned} I_n^2 &= \frac{V_n^2}{R^2} = \frac{4 RKT B_n}{R^2} \\ &= 4 GKT B_n \text{ amp}^2 \end{aligned} \quad (8.64)$$

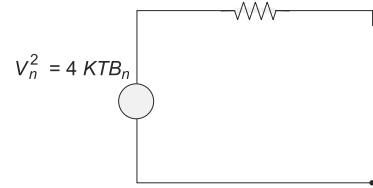


Fig. 8.32 Noisy resistance

Thus, a noisy resistance can be replaced by a noiseless resistance in parallel with noise current source with mean square value

$$I_n^2 = 4 GKT B_n$$

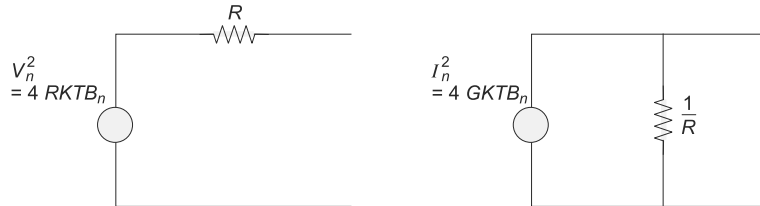


Fig. 8.33 Replacement of noisy resistance with noiseless resistance

8.9.1 Resistors in Series

Consider a circuit with N resistors connected in series. Then the equivalent resistance

$$R_{\text{ser}} = R_1 + R_2 + \dots + R_N \quad (8.65)$$

The noise voltage of the equivalent series resistance is

$$\begin{aligned} V_n^2 &= 4R_{\text{ser}} KT B_n \\ &= 4(R_1 + R_2 + \dots + R_N) KT B_n \\ &= 4R_1 KT B_n + 4R_2 KT B_n + \dots + 4R_N KT B_n \\ &= V_{n_1}^2 + V_{n_2}^2 + \dots + V_{nN}^2 \end{aligned} \quad (8.66)$$

That is, the mean square values are added to obtain total mean square voltage

$$V_n = \sqrt{V_{n_1}^2 + V_{n_2}^2 + \dots + V_{nN}^2} \quad (8.67)$$

8.9.2 Resistors in Parallel

Consider a circuit with N resistors connected in parallel. If we use conductance term to find the equivalent circuit, we obtain the equivalent conductance as

$$G_{\text{par}} = G_1 + G_2 + \dots + G_N \quad (8.68)$$

$$\begin{aligned} I_n^2 &= 4 G_{\text{par}} KT B_n = 4(G_1 + G_2 + \dots + G_N) KT B_n \\ &= I_{n_1}^2 + I_{n_2}^2 + \dots + I_{nN}^2 \end{aligned} \quad (8.69)$$

That is, the mean square values are added to obtain the total mean current

$$I_n = \sqrt{I_{n_1}^2 + I_{n_2}^2 + \dots + I_{nN}^2} \quad (8.70)$$

If we use the relation

$$\frac{1}{R_{\text{par}}} = \frac{1}{R_1} + \frac{1}{R_2} + \dots + \frac{1}{R_N} \quad (8.71)$$

Then the mean square value of voltage is

$$V_n^2 = 4 R_{\text{par}} KT B_n$$

8.9.3 Effective Noise Temperature

Consider two different resistors at different physical temperatures which are placed in series as shown in Fig. 8.34. Each resistor can be replaced by a noise voltage source $V_n^2(t)$ in series with R_n . The noise sources V_1^2 and V_2^2 are in series, therefore can be replaced by a single noise source V_n^2 whose mean square value is sum of mean square value of V_1^2 and V_2^2 .

$$\begin{aligned} V_n^2 &= V_1^2 + V_2^2 \\ V_n^2 &= 4 RKT B_n \\ \Rightarrow V_n^2 &= 4 R_1 KT_1 B_n + 4 R_2 KT_2 B_n \end{aligned}$$

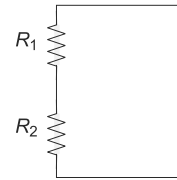


Fig. 8.34 Resistors in series

$$\begin{aligned}
\Rightarrow \quad V_n^2 &= 4(R_1 + R_2) KTs B_n = 4 R_1 K T_1 B_n + 4 R_2 K T_2 B_n \\
\Rightarrow \quad (R_1 + R_2) T_s &= R_1 T_1 + R_2 T_2 \\
T_s &= \frac{R_1 T_1 + R_2 T_2}{R_1 + R_2}
\end{aligned} \tag{8.72}$$



NOISE ANALYSIS OF A TWO-PORT NETWORK 8.10

Consider a linear two-port network that contain resistors and active devices which are sources of noise. Let the input impedance, output impedance and output voltage of the network be z_1 , z_0 and V_0 respectively. An example of such a network is an amplifier.

Let the available signal power from the source be p_{si} and the available signal power at the output be p_{s0} then

$$p_{s0} = G_a p_{si} \tag{8.73}$$

The available noise power denoted by N_{as} of the source is

$$N_{as} = \frac{v_i^2(t)}{4R_s} = \frac{V_i^2 \text{ rms}}{4 R_s} \tag{8.74}$$

where R_s is the real part of z_s .

$$\text{The available noise output power } N_{a0s} = \frac{v_0^2(t)}{4 R_0} \tag{8.75}$$

where R_0 is the real part of z_0 .

The available power gain G_a of a two-port network is defined as

$$\begin{aligned}
G_a &= \frac{\text{Available noise power at the output of network due to source}}{\text{Available noise power of the source}} \\
&= \frac{N_{a0s}}{N_{as}} = \frac{v_0^2(t)/4 R_0}{v_i^2(t)/4 R_s} \\
&= \frac{v_0^2(t)}{R_0} \cdot \frac{R_s}{v_i^2(t)}
\end{aligned} \tag{8.76}$$

when N two-port networks with individual gains $G_1, G_2 \dots G_N$ are connected then the available power gain

$$G_a = \prod_{i=1}^N G_i \tag{8.77}$$



PSD OF THERMAL NOISE 8.11

The PSD of thermal noise is given by

$$S_R(f) = \frac{2R h(f)}{e^{h(f)/KT} - 1} \tag{8.78}$$

where h is Planck's constant equal to 6.62×10^{-34} J-S, K is Boltzman's constant equal to 1.38×10^{-23} J/°K and T is the temperature in degree Kelvin.

$$\text{Approximating, } e^{h(f)/KT} + 1 + \frac{h(f)}{KT} \quad (8.79)$$

$$S_R(f) = \frac{2Rh(f)}{1 + \frac{h(f)}{KT} - 1} = 2 RKT \text{ V}^2 \text{ Hz} \quad (8.80)$$

Therefore, we can say that the mean square voltage spectral density of thermal noise is constant at

$$S_R(f) = 2RK T \text{ V}^2/\text{Hz} \quad (8.81)$$

A noise resistance with spectral density $S_R(f)$ can be replaced by a noiseless resistance of the same value with noise source having PSD = $2RK T$. If this noise source is connected to a resistive load, then maximum power density will be delivered which is equal to

$$S_n(f) = \frac{S_R(f)}{4R} = \frac{2R KT}{4R} = \frac{KT}{2} \text{ W/Hz} \quad (8.82)$$

$$= \frac{N_0}{2} \text{ W/Hz} \quad (8.83)$$



EQUIVALENT INPUT NOISE TEMPERATURE 8.12

From Eq. (8.76) The output noise power is

$$N_{aos} = G_a N_{as} \quad (8.84)$$

Since the two-part network also noisy it introduces some noise. Therefore, the total noise power

$$N_{ao} = G_a N_{as} + N_{an} \quad (8.85)$$

where N_{an} is the noise power due to network.

The noise power due to source is

$$N_{as} = K T_s B_N \quad (8.86)$$

where T_s is the operating temperature, B_N = Bandwidth

$$N_{ao} = G_a K T_s B_N + N_{an} \quad (8.87)$$

$$= G_a K B_N \left[T_s + \frac{N_{an}}{G_a K B_N} \right] \quad (8.88)$$

$$= G_a K B_N [T_s + T_e] \quad (8.89)$$

where $T_e = \frac{N_{an}}{G_a K B_N} \quad (8.90)$

known as effective noise temperature of the two-port network.

A signal source with power p_{si} at the input of the amplifier will produce an output with power

$$p_{so} = G_a p_{si}$$

The signal-to-noise ratio

$$\left(\frac{S}{N}\right)_0 = \frac{P_{so}}{N_{ao}} = \frac{G_a P_{si}}{G_a K T_s B_N + G_a K B_N T_e} \quad (8.91)$$

$$= \frac{P_{si}}{K T_s B_N \left(1 + \frac{T_e}{T_s}\right)} = \frac{P_{si}}{N_{as} \left(1 + \frac{T_e}{T_s}\right)} = \left(\frac{S}{N}\right)_i \frac{1}{1 + \frac{T_e}{T_s}} = \frac{1}{F} \left(\frac{S}{N}\right)_i \quad (8.92)$$

where $\left(\frac{S}{N}\right)_i$ is the SNR at input and F is known as noise figure.

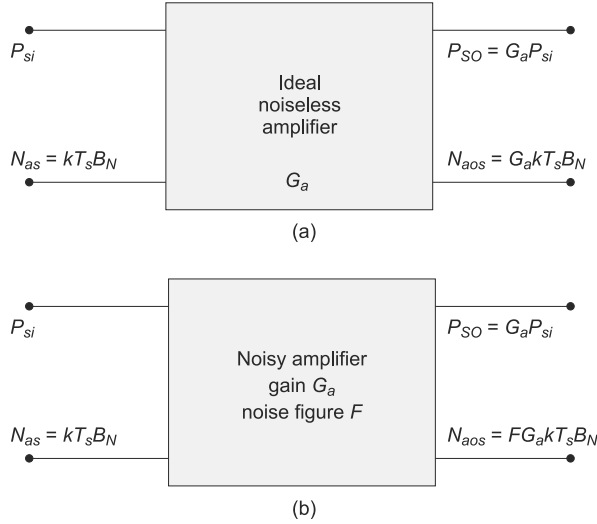


Fig. 8.35 (a) Ideal noiseless amplifier (b) Noisy amplifier

Thus, T_e is a measure of the noisiness of the network. For an ideal network, $T_e = 0$; so $F = 1$

For any real network, F is larger than unity.



SPOT NOISE FIGURE 8.13

The noise figure corresponding to a particular frequency is known as spot noise figure. In evaluating spot noise figure, the gain of the network at that frequency is used. When $T_s = T_0 = 290$ K, the factor $1 + \frac{T_e}{T_0}$ is called *standard spot noise figure* of the network.

The standard spot noise figure of a two-part network is defined as the ratio of the output noise power to the output noise power of a noiseless two-port network at room temperature ($T_0 = 290^\circ\text{K}$).

$$\Rightarrow F = \frac{N_{ao}}{N_{aos}} = \frac{N_{an} + G_a N_{as}}{N_{aos}} = \frac{N_{an} + G_a N_{as}}{N_{aos}} = \frac{N_{an} + G_a N_{as}}{G_a N_{as}}$$

$$= 1 + \frac{N_{an}}{G_a N_{as}} = 1 + \frac{K T_e B_n G_a}{K T_o B_n G_a} \quad (8.93a)$$

$$\Rightarrow F_0 = 1 + \frac{T_e}{T_o} \quad (8.93b)$$

$$\left(\frac{S}{N}\right)_0 = \frac{1}{F_0} \left(\frac{S}{N}\right)_i \quad (8.94)$$

Taking logarithm on both sides,

$$10 \log \left(\frac{S}{N}\right)_0 = -10 \log F_0 + 10 \log \left(\frac{S}{N}\right)_i \quad (8.95)$$

when a network is used with the source for which, it is intended to operate, F will be called the operating noise figure

$$\text{For } F_{op} = 1 + \frac{T_e}{T_s} \quad (8.96)$$



NOISE FIGURE IN TERMS OF PSD 8.14

The noise figure F of a two-port network can also be defined as the ratio of total available noise PSD at the output to the noise PSD at the noiseless two-port network.

$$F = \frac{G_{ao}}{G_{aos}} \quad (8.97)$$

Average Noise Figure

The spot noise figure corresponds to a particular frequency in the frequency spectrum. But for a realistic network, the average noise figure is obtained by considering a band of frequencies or frequencies from 0 to ∞ .

The average noise figure $\bar{F}$ is defined as the total output available noise power N_{as} from a network divided by the total output available noise power N_{aos} due to source. that is,

$$\bar{F} = \frac{N_{ao}}{N_{aos}} \quad (8.98)$$

If G_a is the network's available power gain, the available output noise power due to source alone is

$$N_{aos} = G_a N_{as} = G_a K T_s B_N \quad (8.99)$$

For a band of frequencies from 0 to ∞ ,

$$N_{aos} = \frac{k}{2\pi} \int_0^\infty T_s G_a d\omega \quad (8.100)$$

Similarly, we can write N_{ao} for a band of frequencies from 0 to ∞ as

$$N_{as} = \frac{k}{2\pi} \int_0^\infty F \cdot T_s G_a d\omega \quad (8.101)$$

Therefore, the average noise figure is

$$\bar{F} = \frac{\int_0^{\infty} F \cdot T_s G_a d\omega}{\int_0^{\infty} T_s G_a d\omega} \quad (8.102)$$

If T_s is constant,

$$\bar{F} = \frac{\int_0^{\infty} F_s G_a d\omega}{\int_0^{\infty} G_a d\omega} \quad (8.103)$$

For $T = 290^\circ\text{K}$,

$$\bar{F}_0 = \frac{\int_0^{\infty} F_0 G_a d\omega}{\int_0^{\infty} G_a d\omega} \quad (8.104)$$



AVERAGE NOISE TEMPERATURE 8.15

The available noise power from a network with available power gain G_a and with source temperature T_s is

$$N_{ao} = G_a K(T_s + T_e) B_N \quad (8.105)$$

For a band of frequencies from 0 to ∞ , the total available power is

$$N_{ao} = \frac{k}{2\pi} \int_0^{\infty} G_a (T_e + T_s) d\omega \quad (8.106)$$

If average effective temperature is $\bar{T}_s$ and average effective noise temperature is $\bar{T}_e$ then

$$N_{ao} = \frac{k}{2\pi} (\bar{T}_s + \bar{T}_e) \int_0^{\infty} G_a d\omega \quad (8.107)$$

$$\bar{T}_s = \frac{\int_0^{\infty} T_s G_a d\omega}{\int_0^{\infty} G_a d\omega} \quad (8.108)$$

and

$$\bar{T}_e = \frac{\int_0^{\infty} T_e G_a d\omega}{\int_0^{\infty} G_a d\omega} \quad (8.109)$$

Now average noise figure at room temperature

$$\bar{F}_0 = 1 + \frac{\bar{T}_e}{T_0} \quad (8.110)$$

and average noise figure

$$\bar{F} = 1 + \frac{\bar{T}_e}{T_s} \quad (8.111)$$

NOISE FIGURE OF TWO-PORT NETWORKS IN CASCADE

8.16



Consider two two-port networks in cascade as shown in Fig. 8.36. Let the noise figure of individual networks be F_1 and F_2 and gain are G_1 and G_2 . Now the problem is to determine the overall noise figure in terms of the individual noise figures and available power gains.

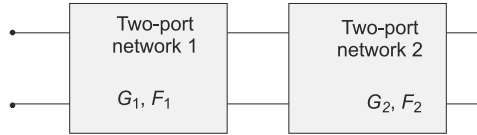


Fig. 8.36 Noise figure of two-port network in cascade

The total output noise power is computed as follows:

Amplifier Source Noise

The noise power of the source is given by $KT_s B_n$. This noise power is amplified by both stages and the amplifier source noise is given by $G_2 G_1 KT_s B_n$.

Internal Noise from the First Stage is Amplified by the second Stage

Internal noise generated by the first stage source can be obtained as follows:

$$N_{ao1} = G_1 N_{as1} + N_{an1} = G_1 KT_s B_n + N_{an1} \quad (8.112)$$

$$\begin{aligned} \frac{N_{ao1}}{G_1 KT_s B_n} &= 1 + \frac{N_{an1}}{G_1 KT_s B_n} \\ \Rightarrow N_{an1} &= (F_1 - 1) G_1 KT_s B_n = G_1 KT_{e1} B_n \end{aligned} \quad (8.113)$$

This internal noise is amplified by the second stage and is given by $G_1 G_2 KT_{e1} B_n$.

Internal Noise Generated by Second Stage

$$N_{an2} = G_2 KT_{e2} B_n \quad (8.114)$$

Total available noise power at the output of second stage is

$$N_{ao2} = G_1 G_2 KT_s B_n + G_1 G_2 KT_{e1} B_n + G_2 KT_{e2} B_n \quad (8.115)$$

If the equivalent noise figure is F then it is defined as

$$F = \frac{N_{ao2}}{N_{aos}} = \frac{N_{ao2}}{G_1 G_2 k T_s B_n} \quad (8.116)$$

$$= \frac{G_1 G_2 K T_s B_n + G_1 G_2 K T_{e_1} B_n + G_2 K T_{e_2} B_n}{G_1 G_2 K T_s B_n} \quad (8.117)$$

$$= 1 + \frac{T_{e_1}}{T_s} + \frac{T_{e_2}}{G_1 T_s} \quad (8.118)$$

We have the relation

$$F_1 = 1 + \frac{T_{e_1}}{T_s} \Rightarrow T_{e_1} = (F_1 - 1) T_s \quad (8.119)$$

and $F_2 = 1 + \frac{T_{e_2}}{T_s} \Rightarrow T_{e_2} = (F_2 - 1) T_s \quad (8.120)$

Substituting Eq. (8.119) and Eq. (8.120) in Eq. (8.118)

$$\begin{aligned} F &= 1 + (F_1 - 1) + \frac{(F_2 - 1)}{G_1} \\ &= F_1 + \frac{F_2 - 1}{G_1} \end{aligned} \quad (8.121)$$

The generalization of this result to arbitrary number of stages is known as Friis' formula

$$F = F_1 + \frac{F_2 - 1}{G_1} + \frac{F_3 - 1}{G_1 G_2} + \dots \quad (8.122)$$

The above formula can be expressed in terms of noise equivalent temperature as

$$1 + \frac{T_e}{T_s} = \left(1 + \frac{T_{e_1}}{T_s} \right) + \frac{T_{e_2}}{T_s G_1} + \frac{T_{e_3}}{T_s G_1 G_2} + \dots \quad (8.123)$$

$$\Rightarrow T_e = T_{e_1} + \frac{T_{e_2}}{G_1} + \frac{T_{e_3}}{G_1 G_2} + \dots \quad (8.124)$$

NOISE FIGURE OF AN ATTENUATOR (LOSSY NETWORK)

8.17



In an attenuator, the available noise power at the output and available noise power at the input is related by

$$N_{aos} = \frac{N_{as}}{L} = \frac{K T_s B_n}{L} \quad (8.125)$$

where L is known as loss factor or insertion loss. The internal noise generated by the attenuator is

$$N_{an} = \frac{k T_e B_n}{L} \quad (8.126)$$

The total noise power at the output of the attenuator is

$$N_{ao} = \frac{k T_s B_n}{L} + \frac{K T_e B_n}{L} \quad (8.127)$$

$$= \frac{K}{L} \{T_s + T_e\} B_n \quad (8.128)$$

Since the attenuator is resistive and assumed to be at the same temperature T_s , as the equivalent resistance at its input, the available output power is

$$N_{ao} = KT_s B_n \quad (8.129)$$

Equating Eq. (8.125) and Eq. (8.129), we get

$$\begin{aligned} KT_s B_n &= \frac{K}{L} [T_s + T_e] B_n \\ \Rightarrow T_s &= T_s + T_e \\ T_e &= T_s(L - 1) \end{aligned} \quad (8.130)$$

$$\text{At room temperature, } T_e = T_0(L - 1) \quad (8.131)$$

$$F_0 = 1 + \frac{T_e}{T_0} \Rightarrow T_e = T_0(F_0 - 1) \quad (8.132)$$

Comparing Eq. (8.131) and Eq. (8.132), we get

$$F_0 = L$$

REVIEW QUESTIONS



11. Define (a) noise figure, and (b) Spot noise figure.
12. Derive the mathematical expression of noise figure
13. Explain the concept of effective input noise temperature.
14. For three amplifiers in cascade, derive an expression of overall noise figure in terms of the power gain and noise figures of the individual amplifiers.
15. Bring out the importance of Friis' formula.

Solved Problems



8.52 An electronic system has an amplifier followed by a mixer stage. The noise figures of the amplifier and mixer are 30 dB and 20 dB respectively. If the power gain of the amplifier is 10 dB; calculate the overall noise figure referred to the input.

Solution Given: $F_A = 30 \text{ dB} \Rightarrow 1000:1$ power ratio

$$F_M = 20 \text{ dB} \Rightarrow 100:1 \text{ power ratio}$$

$$G_A = 10 \text{ dB} = 10:1 \text{ power ratio}$$

$$\begin{aligned} F &= F_A + \frac{F_m - 1}{G_A} \\ &= 1000 + \frac{100 - 1}{10} = 1009.9 \end{aligned}$$

$$(F)_{\text{dB}} = 10 \log(1009.9) = 30.043 \text{ dB}$$

8.53 The noise present at the input of a two-part network is $5 \mu\text{W}$. The noise figure is $F = 0.75 \text{ dB}$. The receiver gain is 10^7 . Find

- (a) Available noise power by two ports.
- (b) Output available noise power.

Solution We have, $N_{ao} = N_{an} + N_{aos}$

$$N_{aos} = G_a N_{as}$$

$$\text{Given: } N_{as} = 5 \mu\text{W}, G_a = 10^7 \Rightarrow N_{aos} = 5 \times 10^{-6} \times 10^7 = 50 \text{ W}$$

$$F = \frac{N_{ao}}{N_{aos}} \Rightarrow N_{ao} = F(N_{aos})$$

$$F = 0.75 \text{ dB} = 1.188 : 1$$

- (a) $N_{ao} = (1.188) 50 = 59.425 \text{ W}$
- (b) $N_{an} = N_{ao} - N_{aos} = 59.425 - 50 = 9.425 \text{ W}$

8.54 A low-noise receiver for satellite ground station consists of the following stages:

Antenna with $T_i = 125^\circ\text{K}$

Wave guide with a loss of 0.5 dB

Power amplifier with $G_a = 30 \text{ dB}$, $T_e = 6^\circ\text{K}$, $B_n = 20 \text{ MHz}$

TWT amplifier with $G_a = 16 \text{ dB}$, $F = 6 \text{ dB}$, $B_n = 20 \text{ MHz}$

Calculate effective noise temperature of the system.

Solution Given Antenna temperature = 125°K

Loss of wave guide $L = 0.5 \text{ dB} = 1.12 : 1$

$$T_{eA} = 6^\circ\text{K}; F_{TWT} = 6 \text{ dB} = 3.98 : 1$$

Effective noise temperature of waveguide

$$T_e = (L - 1) T_0 = (1.12 - 1) 290 = 35.385$$

$$G_{AP} = 30 \text{ dB} = 1000 : 1; G_{AT} = 16 \text{ dB} = 39.81 : 1$$

The total effective noise temperature

$$\begin{aligned} T_s &= 125 + 35.385 + L[T_{eA} + T_{TWT}] \\ &= 125 + 35.385 + 1.12 \left(6 + \frac{(F_{TWT} - 1) 290}{1000} \right) \\ &= 125 + 35.385 + 1.12 \left(6 + \frac{2.98 \times 290}{1000} \right) \\ &= 168.07 \end{aligned}$$

8.55 Two resistors with resistance R_1 and R_2 are connected in parallel and have physical temperatures T_1 and T_2 respectively. Find the effective noise temperature of T_s of an equivalent resistor with resistance equal to the parallel combination of R_1 and R_2 .

Solution The noise equivalent circuit is shown in Fig. 8.37.

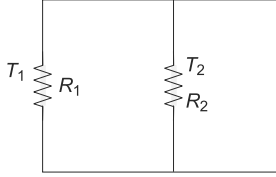


Fig. 8.37

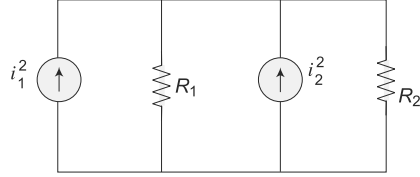


Fig. 8.38

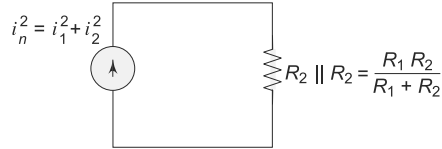


Fig. 8.39

$$i_n^2 = i_1^2 + i_2^2$$

We have, $i_1^2 = 4 k T G_1 B_n$ and $i_2^2 = 4 K T G_2 B_n$

$$i_n^2 = \frac{4 K T_1 B_n}{R_1} + \frac{4 K T_2 B_n}{R_2}$$

In terms of physical temperature

$$\begin{aligned} i_n^2 &= 4 K T_s B_n \left(\frac{1}{R_1} + \frac{1}{R_2} \right) \\ &= 4 K T_s B_n \left(\frac{1}{R_1} + \frac{1}{R_2} \right) = 4 K B_n \left(\frac{T_1 R_2 + T_2 R_1}{R_1 R_2} \right) \end{aligned}$$

$$\Rightarrow T_s = \frac{T_1 R_2 + T_2 R_1}{R_1 R_2}$$



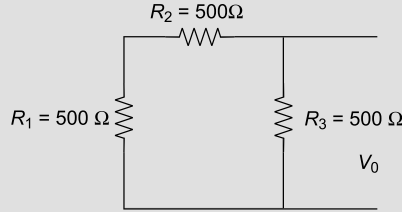
Practice Problems

8.10 Repeat Solved Problem 8.55. If three resistors R_1 , R_2 and R_3 are connected in parallel.

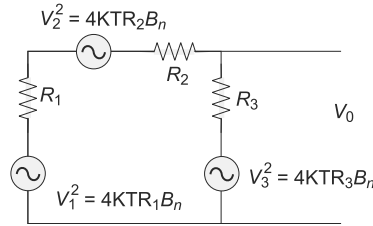
$$\left(\text{Ans. } T_s = \frac{T_1 R_2 R_3 + T_2 R_1 R_3 + T_3 R_1 R_2}{R_1 R_2 + R_2 R_3 + R_1 R_3} \right)$$

Solved Problems

8.56 For the resistive network shown in Fig. 8.40, find the rms noise voltage appearing at the output terminals if bandwidth = 50 kHz and $T = 290^\circ\text{K}$.

**Fig. 8.40**

Solution Let us replace each resistors with a noise voltage source in series with resistance and then find noise voltage at the output due to each resistor. The noise equivalent circuit is shown in Fig. 8.41.

**Fig. 8.41**

The total mean-square voltage

$$V_0^2 = V_{01}^2 + V_{02}^2 + V_{03}^2$$

V_{01} is the output noise voltage due to V_1 , V_{02} is the output noise voltage due to V_2 and V_{03} is the output noise voltage due to V_3 .

$$V_{01} = V_1 \left(\frac{R_3}{R_1 + R_2 + R_3} \right) = \sqrt{4 K T R_1 B_n} \left(\frac{R_3}{R_1 + R_2 + R_3} \right)$$

$$V_{02} = V_2 \left(\frac{R_3}{R_1 + R_2 + R_3} \right) = \sqrt{4 K T R_2 B_n} \left(\frac{R_3}{R_1 + R_2 + R_3} \right)$$

$$V_{03} = V_3 \left(\frac{R_1 + R_2}{R_1 + R_2 + R_3} \right) = \sqrt{4 K T R_3 B_n} \left(\frac{R_1 + R_2}{R_1 + R_2 + R_3} \right)$$

$$V_1 = V_2 = V_3 = 6.326 \times 10^{-7} \text{ V}$$

Substituting the value $R_1 = R_2 = R_3 = 500 \Omega$,

$$V_{01} = V_{02} = 2.108 \times 10^{-7} \text{ V}; V_{03} = 4.217 \times 10^{-7} \text{ V}$$

$$\begin{aligned}
 V_0 &= \sqrt{V_9^2 + V_{02}^2 + V_{03}^2} \\
 &= 5.164 \times 10^{-7} V_{(\text{rms})}
 \end{aligned}$$

8.57 Two resistors of 50 k Ω and 100 k Ω are at room temperature. For a bandwidth of 500 kHz, calculate the thermal noise voltage generated by (a) each resistor, (b) the two resistors in series, and (c) the two resistors in parallel.

Solution Let $R_1 = 50 \text{ k}\Omega$; $R_2 = 100 \text{ k}\Omega$

Thermal noise voltage due to 50 k Ω resistor is

$$\begin{aligned}
 V_1 &= \sqrt{4 k T R_1 B_n} \\
 &= \sqrt{4 \times 1.38 \times 10^{-23} \times 290 \times 50 \times 1000 \times 500 \times 10^3} \\
 &= 20 \mu\text{V}
 \end{aligned}$$

Thermal noise voltage due to 100 k Ω resistor is

$$\begin{aligned}
 V_2 &= \sqrt{4 K T R_2 B_n} \\
 &= \sqrt{4 \times 1.38 \times 10^{-23} \times 290 \times 100 \times 1000 \times 500 \times 10^3} \\
 &= 2.829 \times 10^{-5} \text{ volts}
 \end{aligned}$$

For series combination,

$$\begin{aligned}
 R_s &= R_1 + R_2 = 150 \text{ k}\Omega \\
 V_s &= \sqrt{4 K T R_s B_n} = 34.65 \mu\text{V}
 \end{aligned}$$

For parallel combination,

$$\begin{aligned}
 R_p &= R_1 \parallel R_2 = \frac{(100)(50)}{150} = 33.33 \text{ k}\Omega \\
 V_p &= \sqrt{4 K T R_p B_n} = 16.33 \mu\text{V}
 \end{aligned}$$

8.58 An amplifier has three stages for which $T_{e1} = 250 \text{ K}$, $T_{e2} = 400 \text{ K}$ and $T_{e3} = 900 \text{ K}$. If the available power gain of the second stage is 10, what gain must the first stage have to guarantee an effective input noise temperature of 280 K.

Solution Given: $T_{e1} = 250 \text{ K}$, $T_{e2} = 400 \text{ K}$ and $T_{e3} = 900 \text{ K}$ and $G_2 = 10$

$$T_e = 280$$

We have
$$T_e = T_{e1} + \frac{T_{e2}}{G_1} + \frac{T_{e3}}{G_1 G_2}$$

$$280 = 250 + \frac{400}{G_1} + \frac{900}{10G_1}$$

$$30 = \frac{1}{G_1} [400 + 90]$$

$$G_1 = \frac{400}{30} = 16.33$$

8.59 An amplifier has a standard spot noise figure $F_0 = 10$ dB. An engineer uses the amplifier to amplify the output of antenna whose temperature $T_s = 200$ K

- (a) What is the effective input noise temperature of the amplifier?
- (b) What is the operating noise figure?

Solution Given: $F_0 = 10$ dB $\Rightarrow 10 : 1$ ratio

$$T_e = (F_0 - 1) T_0 = 290 (10 - 1) = 2610 \text{ K}$$

$$F_{op} = 1 + \frac{T_e}{T_s} = 1 + \frac{2610}{200} = 14.05$$



Practice Problem

8.11 The noise figure of an amplifier at room temperature is 0.5 dB. Find the equivalent temperature. (Ans. 35.38 K)

REVIEW QUESTIONS



16. Derive the relation between PSDs of input and output random process of an LTI system.
17. Discuss the significance of noise equivalent temperature of an electronic system.
18. Derive the expression for noise figure.
19. A Gaussian random process $X(t)$ is applied to a stable linear filter. Show that the random process $Y(t)$ developed at the output of the filter is also Gaussian.
20. Explain the methods of finding the following properties of a system
 - (i) Linearity
 - (ii) Time invariant
 - (iii) Causality
21. Which of the following noise parameters is true representation of noise in electrical circuits?
 - (i) Noise figure
 - (ii) Noise temperature
 - (iii) Noise bandwidth
 Support your answer with the help of suitable examples
22. Write short note on
 - (i) Noise spectral density
 - (ii) Noise figure
23. Explain available power of a noise source.
24. What are the important parameters that determine the overall noise figure of a multistage filtering?
25. For three stage amplifier, derive an expression for overall noise figure in terms of Gains and noise figures of the individual amplifiers.



EXERCISES

Problems

1. A signal $x(t) = e^{-bt} u(t)$ is applied to a network having impulse response $h(t) = u(t)$. Find the system's response.
2. A rectangular pulse of amplitude A and duration T is applied to a system with impulse response $h(t) = u(t)$. Find $y(t)$.

The rectangular pulse is defined as

$$x(t) = \begin{cases} A & \text{for } 0 < t < T \\ 0 & \text{elsewhere} \end{cases}$$

3. The impulse response of a system is

$$h(t) = \begin{cases} te^{-t} & \text{for } t \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

Find the response of the network to the pulse

$$x(t) = \begin{cases} A & \text{for } 0 < t < T \\ 0 & \text{elsewhere} \end{cases}$$

4. Repeat the exercise 3 if the network's impulse response is

$$h(t) = \begin{cases} t^2 e^{-t} & \text{for } t \geq 0 \\ 0 & \text{for } t < 0 \end{cases}$$

5. Find the transfer function of the network of Fig. 8.42.

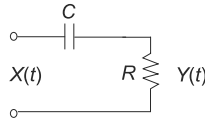


Fig. 8.42

6. Find the transfer function of the network shown in Fig. 8.43.

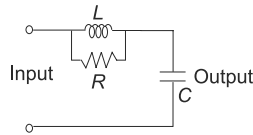


Fig. 8.43

7. Determine which of the following impulse responses do not correspond to a system that is stable, or realizable, or both, and state why?

(a) $h(t) = u(t - 2)$	(b) $h(t) = te^{-t} u(t)$
(c) $h(t) = e^t u(-t) + e^{-t} u(t)$	(d) $h(t) = \sin \omega_0 t$
(e) $h(t) = e^{-t^2}$	
8. A random process $X(t)$ is applied to a linear time-invariant system. If the response $Y(t) = X(t) - X(t - T)$. Find the system's transfer function.

9. A stationary random process $X(t)$, having an autocorrelation function $R_{XX}(\tau) = 2 \exp(-2|\tau|)$ is applied to a network of Fig. 8.44

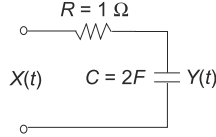


Fig. 8.44

- Find (a) $S_{XX}(\omega)$ (b) $|H(\omega)|^2$ (c) $S_{YY}(\omega)$
 10. White noise with autocorrelation function $R_{XX}(\tau) = 0.1 \delta(\tau)$ is applied to a network with impulse response

$$h(t) = 2e^{-2t} u(t)$$

obtain an expression for the output power spectrum.

11. Find the noise bandwidth of the system having the power transfer function

$$|H(\omega)|^2 = \frac{1}{1 + (\omega/W)^4}$$

12. A system's power transfer function is

$$|H(\omega)|^2 = \frac{8}{(64 + \omega^2)}$$

What is its noise bandwidth?

13. An antenna is connected to a receiver having an equivalent noise temperature $T_e = 200^\circ\text{K}$. The available gain of the receiver is 10^6 and its noise bandwidth is 5 MHz. If the available noise power is $5 \mu\text{W}$, find the antenna temperature.
 14. White noise with power density $\frac{N_o}{2} = 6 \times 10^{-6} \text{ W/Hz}$ is applied to an ideal filter with bandwidth $W(\text{rad/s})$. Find W so that output average noise power is 15 W.

Multiple-Choice Questions

- A system that obeys superposition principle is said to be a
 - Causal system
 - Linear system
 - Static system
 - Time-invariant system
- The system $y(t) = \sin h[x(t)]$ is a
 - Linear system
 - Time-invariant system
 - Causal system
 - Non-causal system
- The convolution integral is given by

$$(a) \quad y(t) = \int_{-\infty}^{\infty} x(-\tau) h(t + \tau) d\tau$$

$$(b) \quad y(t) = \int_{-\infty}^{\infty} x(\tau) h(t - \tau) d\tau$$

$$(c) \quad y(t) = \int_{-\infty}^{\infty} x(\tau) h(t + \tau) d\tau$$

$$(d) \quad y(t) = \int_{-\infty}^{\infty} x(t + \tau) h(\tau) d\tau$$

4. The system is said to be stable if its impulse response $h(t)$ satisfies the condition

(a) $h(t) = 0$ for $t < 0$

(b) $\int_{-\infty}^{\infty} h(\tau) d\tau = \infty$

(c) $\int_{-\infty}^{\infty} |h(\tau)| d\tau < \infty$

(d) $h(t) \neq 0$ for $t < 0$

5. The transfer function of the RC network shown in Fig. 8.45 is

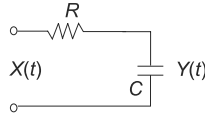


Fig. 8.45

(a) $j\omega RC + 1$

(b) $\frac{1}{1 + j\omega RC}$

(c) $j\omega RC$

(d) $\frac{1}{1 - j\omega RC}$

6. Which of the following is/are correct?

(a) $R_{YX}(\tau) = R_{XX}(\tau) * h(\tau)$

(b) $R_{XY}(\tau) = R_{XX}(\tau) * h(\tau)$

(c) $R_{YX}(\tau) = R_{XX}(\tau) * h(-\tau)$

(d) $R_{YX}(\tau) = R_{XX}(\tau) * h(\tau)$

7. The noise bandwidth of a low-pass RC filter is

(a) $\frac{\pi}{RC}$

(b) $\frac{\pi}{2RC}$

(c) πRC

(d) $2\pi RC$

8. Which of the following is/are correct?

(a) $T_o = T_e(F_o - 1)$

(b) $T_e = \frac{T_o}{F_o - 1} + 1$

(c) $T_e = T_o(F_o - 1)$

(d) $T_e + T_o = F_o - 1$

9. The noise power delivered by a resistor is

(a) $P_n = KT B_n$

(b) $P_n = \frac{KT}{B_n}$

(c) $P_n = \frac{K B_n}{T}$

(d) $P_n = \frac{1}{KT B_n}$

10. The relation between PSD of output and PSD of input is

(a) $S_{YY}(\omega) = |H(\omega)| S_{XX}(\omega)$

(b) $S_{YY}(\omega) = |H(-\omega)| S_{XX}(\omega)$

(c) $S_{YY}(\omega) = H(\omega) S_{XX}(\omega)$

(d) $S_{YY}(\omega) = |H(\omega)|^2 S_{XX}(\omega)$

Answers to Multiple-Choice Questions

1. (b) 2. (b) 3. (b) 4. (c) 5. (b) 6. (b) and (c) 7. (b) 8. (c) 9. (a)
10. (d)

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